

Gaia DR4: preliminary catalogue data model

Draft

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Abstract

The preliminary data model for Gaia Data Release 4 is described in the following pages. Please note that this is a draft only: the final published catalogue data model is subject to change with respect to that described herein without any public notification. Note also that this draft contains links to external resources (e.g. the body of the full documentation set for DR4) that are inactive presently. The document is provided for advance information and will be supported on a best–efforts basis only prior to the formal release.

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Datamodel description

Data model(s) included in this draft are

`gaia.cu9.archive 21.7.branch, revision SB-21.7.0-RC-r0`

`gaia.cu9.directdelivery 21.6.branch revision SB-21.6.0-RC-r0`

1 Main source catalogue

As indicated in the below subsection titles by the labels [TAP] and [DataLink], these products are served from the ESA Gaia Archive using two different protocols, the Table Access Protocol (TAP) for tabular products and the DataLink protocol for other products such as spectra, light curves, MCMC samples, etc. For more details please see Section **??**.

1.1 GAIA_SOURCE [TAP]

This table has an entry for each source observed by Gaia which is considered to have both high-quality astrometry and high-quality photometry from at least one module of the Gaia Data Processing and Analysis Consortium (DPAC).

As such, it only contains those sources from the tables `all_source_astrometry`, `all_source_photometry` and `all_source_rvs` that meet those quality criteria. The detailed definition of the criteria is given in Chapter 18 of the DR4 online documentation. Compared to the `gaia_source` table published in earlier releases, the parameters provided in the DR4 `gaia_source` table correspond to a consolidation of the best results obtained across DPAC, and as such they do not correspond to those of a homogeneous module (in contrast with what is tabulated in tables `all_source_astrometry`, `all_source_photometry` and `all_source_rvs`, respectively).

For each source in this table, dedicated flags indicate which module did provide the astrometry (`gaia_source.astrometry_origin`), the G-band photometry (`gaia_source.g_photometry_origin`), the BP and RP-band photometry (`gaia_source.colour_origin`), the radial velocity (`gaia_source.radial_velocity_origin`), the broadening/rotational velocity (`gaia_source.vbroad_origin`), the stellar parameters (`gaia_source.ap_origin`) and the extinction parameters (`gaia_source.extinction_ori`
In some cases, these solutions correspond to non-single stars, or more sophisticated star models, for which complementary information is given

in other tables from the data release. These tables can be found directly in the flags listed above.

Besides, like in previous releases, further array data types (spectra, epoch measurements) are presented separately via 'DataLink' resources. The availability of specific 'DataLink' products is indicated with flags in the table `all_source_flags`.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use (to chase up the causes in case problems are found with the public data) but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID, you may visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

DESIGNATION : Unique source designation (unique across all Data Releases) (string)

A source designation, unique across all Gaia Data Releases, that is constructed from the prefix 'Gaia DRx ' followed by a string of digits corresponding to `gaia_source.source_id` (3 space-separated words in total). Note that the integer source identifier `source_id` is **not** guaranteed to be unique across Data Releases; moreover it is not guaranteed that the same astronomical source will always have the same `source_id` in different Data Releases. Hence the only safe way to compare source records between different Data Releases in general is to check the records of proximal source(s) in the same small part of the sky.

Nevertheless, for convenience, in table `dr3_neighbourhood` DPAC provides a cross-match for those sources that already existed in DR3.

SOURCE_ID : Unique source identifier (unique within a particular Data Release) (long)

A unique numerical identifier of the source, encoding the approximate position of the source (roughly to the nearest arcmin), the provenance (data processing centre where it was created), a running number, and a component number.

The approximate equatorial (ICRS) position is encoded using the nested HEALPix scheme at level 12 (Nside = 4096), which divides the sky into $\simeq$ 200 million pixels of about 0.7 arcmin^2.

The source ID consists of a 64-bit integer, least significant bit = 1 and most significant bit = 64, comprising:

- a HEALPix index number (sky pixel) in bits 36 - 63; by definition the smallest HEALPix index number is zero.
- a 3-bit Data Processing Centre code in bits 33 - 35; for example MOD(`source_id` / 4294967296, 8) can be used to distinguish between sources initialised via the Initial Gaia Source List (Smart & Nicastro 2014) by the Torino DPC (code = 0) and sources otherwise detected and assigned by Gaia observations (code > 0)
- a 25-bit plus 7 bit sequence number within the HEALPix pixel in bits 1 - 32 split into:
  - a 25 bit running number in bits 8 - 32; the running numbers are defined to be positive, i.e. never zero
  - a 7-bit component number in bits 1 - 7

This means that the HEALPix index at level 12 of a given source is contained in the most significant bits. HEALPix index of level 12 and lower can thus be retrieved as follows:

- index at level 12 = `source_id` / 34359738368
- index at level 11 = `source_id` / 137438953472
- index level 10 = `source_id` / 549755813888
- index at level n = `source_id` / $(2^{35} \times 4^{(12-n)})$ = `source_id` / $2^{(59-2n)}$

Additional details can be found in Bastian & Portell (2020). A tool to decode a `source_id` into its components is provided in [https://gaia.esac.esa.int/decoder/srcidDecoder.jsp].

Note that the `source_id` of successive Gaia Data Releases (DR) are independent in principle. Thus, the DR must always be specified along with the numerical content of a `source_id`, i.e. the numerical part must be prefixed by Gaia DR1, DR2, DR3, DR4 or DR5. Consistency of the `source_id` from DR to DR is aimed at, but can in no way be guaranteed. Post-hoc splits of initially unrecognized multiple stars, as well as the merging of source pairs that turn out to be duplicates are unavoidable. In principle, other, more serious reasons may turn up and enforce a more extended revision of the `source_id` assignment. One conceivable reason is a counter overflow due to the higher-than-expected total number of sources detected by Gaia. As of DR4,

no such revisions are expected.

WINNING_SOURCE_ID_ASTRO : Identifier of the source that provided the astrometry (long)

The identifier of the source that provided the astrometry. Used in combination with the `astrometry_origin` flag, it allows to access the full set of parameters derived for the winning source in the table specified by this flag.

This field is necessary because the astrometric parameters of a given object will not always coincide with those of the source identified as `source_id` in other tables. This will happen only when the astrometry stems from the SEAPipe or Crowded Field modules, which correspond to `astrometry_origin = gaia_source_environment` and `astrometry_origin = crowded_field_source` respectively. In this case, the source identifier from e.g. `all_source_astrometry` will be maintained, instead of using the distinct identifier from `gaia_source_environment` or `crowded_field_source`. The latter identifier is the one stored in `winning_source_id_astro`.

When the astrometry is not provided by any of these two modules, `winning_source_id_astro` will be equal to `source_id`.

WINNING_SOURCE_ID_PHOT : Identifier of the source that provided the G-band photometry (long)

The identifier of the source that provided the G-band photometry. Used in combination with the `g_photometry_origin` flag, it allows to access the full set of parameters derived for the winning source in the table specified by this flag.

This field is necessary because the G-band photometry parameters of a given object will not always coincide with those of the source identified as `source_id` in other tables. This will happen only when the G-band photometry stems from the SEAPipe or Crowded Field modules, which correspond to `g_photometry_origin = gaia_source_environment` and `g_photometry_origin = crowded_field_source` respectively. In this case, the source identified from e.g. `all_source_astrometr` will be maintained, instead of using the distinct source identifier from `gaia_source_environment` or `crowded_field_source`. The latter source identififer is the one stored in `winning_source_id_phot`.

When the G-band photometry is not provided by any of these two modules, `winning_source_id_phot` will be equal to `source_id`.

This field only concerns the G-band photometry because in the case of the BP and RP photometry, the `source_id` in `gaia_source` will always match that used in the table storing the results from any of the modules susceptible to provide that photometry.

RANDOM_INDEX : Random index for use when selecting subsets (long)

A random index which can be used to select smaller subsets of the data that are still representative. The column contains a random permutation of the numbers from 0 to $N-1$, where N is the number of sources in the table.

The random index can be useful for validation (testing on 10 different random subsets), visualization (displaying 1% of the data), and statistical exploration of the data, without the need to download all the data.

ASTROMETRY_ORIGIN : Flag indicating which module provided the astrometry for this source (string)

This flag indicates which of the DPAC modules provided the astrometry that got consolidated for this source. It is provided as a string giving the name of the Gaia DR4 table where the full set of parameters associated with this astrometric solution are stored. The following values can be found:

1. `all_source_astrometry` for single-star astrometric solutions from the AGIS module
2. `gaia_source_environment` for single-star astrometric solutions from the source environment analysis module (SEAPipe)
3. `crowded_field_source` for single-star astrometric solutions from the crowded field analysis module
4. Non-single star astrometric solutions
   - `nss_acceleration_astro` for binary solutions with acceleration and jerk terms
   - `nss_two_body_orbit` for binary solutions from orbital models and combinations thereof, e.g. with spectroscopic binaries or variability-induced movers (VIM)
   - `nss_vim_fl` for binary solutions with VIM models
   - `nss_multiple_orbits` for binary solutions with multiple orbits models

Note that there are additional DPAC modules producing sources and their astrometric parameters, namely the special bright-stars treatment and the gravitational-lenses processing of the extended-objects task. These have not been included in the construction of `gaia_source`. They can

be found in the dedicated tables bright_source_astrometry and grav_lens_component, respectively.

This field, astrometry_origin, should not be confused with astro_pipeline in table all_source_astro which informs about the processing chain (only AGIS or IDU) that provided the astrometry in that table.

UNRELIABLE_ASTROMETRY : Unreliable AGIS astrometry (boolean)

An unreliable AGIS solution is defined as follows:

It is in all_source_astrometry with astro_pipeline = 3 and quality_flag = 0 or it fulfils any of the following conditions:

- R > 1.2
- visibility_periods_used < max(10, min(25, 5 + 40 * R))
- duplicated_source & !blended & abs(parallax) > 90 & abs(ecl_lat) < 15

where:

R = astrometric_sigma5d_max / gamma4(phot_g_mean_mag),

gamma4(G) = min(10, pow(10.0, 0.2 * max(6 - G, 0, G - 18)))

The flag is set to 'true' for 414 million sources in this table.

For another 338 million sources in this table, with 142 million 2-parameter solutions being the biggest part among them, and the rest mostly being non-single-star solutions, the value of unreliable_astrometry is set to null. The reason in all cases is that some of the defining quantitites are unavailable.

UNRELIABLE_G_PHOTOMETRY : Unreliable G-band mean flux (boolean)

The G-band mean flux is considered unreliable when phot_g_n_obs < 10.

Note that this criterion is different than that applied for all_source_photometry.unreliable_g.

REF_EPOCH : Reference epoch (double, Time[Julian Years])

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Reference epoch to which the astrometric source parameters are referred, expressed as a Julian Year in TCB (Temps Coordonnée Barycentrique = Barycentric Coordinate Time).

RA : Right ascension (double, Angle[deg])

Barycentric right ascension α at the reference epoch `ref_epoch` of the source in the ICRS (International Celestial Reference System).

RA_ERROR : Standard error of right ascension (float, Angle[mas])

Standard error $\sigma_{\alpha*} \equiv \sigma_\alpha \cos\delta$ of the right ascension of the source in ICRS at the reference epoch `ref_epoch`.

DEC : Declination (double, Angle[deg])

Barycentric declination δ at the reference epoch `ref_epoch` of the source in ICRS (International Celestial Reference System).

DEC_ERROR : Standard error of declination (float, Angle[mas])

Standard error σ_δ of the declination of the source in ICRS at the reference epoch `ref_epoch`.

PARALLAX : Parallax (double, Angle[mas])

Absolute stellar parallax ϖ of the source at the reference epoch `ref_epoch`.

PARALLAX_ERROR : Standard error of parallax (float, Angle[mas])

Standard error σ_ϖ of the stellar parallax at the reference epoch `ref_epoch`.

PARALLAX_OVER_ERROR : Parallax divided by its standard error (float)

Parallax divided by its standard error.

PM : Total proper motion (float, Angular Velocity[mas yr^{-1}])

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The total proper motion pm, calculated as the magnitude of the resultant vector of the proper motion component vectors pmra and pmdec, i.e. $\mathtt{pm}^2 = \mathtt{pmra}^2 + \mathtt{pmdec}^2$.

PMRA : Proper motion in right ascension direction (double, Angular Velocity[mas yr^{-1}])

Proper motion in right ascension $\mu_{\alpha*} \equiv \mu_\alpha \cos\delta$ of the source in ICRS at the reference epoch ref_epoch. This is the local tangent plane projection of the proper motion vector in the direction of increasing right ascension.

PMRA_ERROR : Standard error of proper motion in right ascension direction (float, Angular Velocity[mas yr^{-1}])

Standard error $\sigma_{\mu\alpha*}$ of the local tangent plane projection of the proper motion vector in the direction of increasing right ascension at the reference epoch ref_epoch.

PMDEC : Proper motion in declination direction (double, Angular Velocity[mas yr^{-1}])

Proper motion in declination μ_δ of the source at the reference epoch ref_epoch. This is the projection of the proper motion vector in the direction of increasing declination.

PMDEC_ERROR : Standard error of proper motion in declination direction (float, Angular Velocity[mas yr^{-1}])

Standard error $\sigma_{\mu\delta}$ of the proper motion component in declination at the reference epoch ref_epoch.

PSEUDOCOLOUR : Astrometrically estimated pseudocolour of the source (float, Misc[μm^{-1}])

Astrometrically determined colour of the source, represented as an effective wavelength in μm^{-1}. This quantity can be used as a proxy for the actual effective wavenumber (i.e. broad-band colour) of the source if the latter cannot be computed from CU5 photometry. It is estimated in AGIS from the chromatic displacements of image centroids. Technically, it is the sum of gaia_source.nu_eff_used_in and a correction to that value derived by AGIS.

For details see **?**.

PSEUDOCOLOUR_ERROR : Standard error of the pseudocolour of the source (float, Misc[μm^{-1}])

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Standard error $\sigma_{\texttt{pseudocolour}}$ of the astrometrically determined `pseudocolour` of the source.

ASTROMETRIC_PARAMS : Bitwise code describing which astrometric parameters are provided (short)

The bit–wise encoding in `astrometric_params` indicates which parameters have been estimated in AGIS (primary or secondary solution), or in a non-single star solution, and have been exported for this source. A set bit means the parameter was updated, an unset bit means the parameter was not updated. The least-significant bit corresponds to α=`ra`. The table below shows the values for relevant combinations of the parameters.

The radial proper motion (μ_r) is formally considered to be one of the astrometric parameters of a source, and the sixth bit is therefore reserved for it. It is also in principle updatable in AGIS, but in practice it will always be computed from a spectroscopic radial velocity and the estimated parallax, in which case the bit is not set.

C is the `gaia_source.pseudocolour` of the source, i.e. the astrometrically estimated effective wavenumber.

Additional bits are added when an acceleration or its derivative are fitted. Three more bits are used to indicate that the astrometry has been estimated with a non-single star model: orbital models, Variability-Induced Mover (VIM) models, or resolved-binary models.

astrometric_params	α	δ	ϖ	$\mu_{\alpha*}$	μ_δ	μ_r	C	$\dot{\mu}_{\alpha*}$	$\dot{\mu}_\delta$	$\ddot{\mu}_{\alpha*}$	$\ddot{\mu}_\delta$	Orbital	VIM	Resolved
$0000000000000011_2 = 3$	✓	✓												
$0000000000000111_2 = 7$	✓	✓	✓											
$0000000000011011_2 = 27$	✓	✓		✓	✓									
$0000000000011111_2 = 31$	✓	✓	✓	✓	✓									
$0000000001111111_2 = 63$	✓	✓	✓	✓	✓	✓								
$0000000010111111_2 = 95$	✓	✓	✓	✓	✓		✓							
$0000000111011111_2 = 479$	✓	✓	✓	✓	✓		✓	✓	✓					
$0000011111011111_2 = 2015$	✓	✓	✓	✓	✓		✓	✓	✓	✓	✓			
$0000100000011111_2 = 2079$	✓	✓	✓	✓	✓							✓		
$0000100110011111_2 = 2463$	✓	✓	✓	✓	✓			✓	✓			✓		
$0000111110011111_2 = 3999$	✓	✓	✓	✓	✓			✓	✓	✓	✓	✓		
$0001000000011111_2 = 4127$	✓	✓	✓	✓	✓								✓	
$0001000110011111_2 = 4575$	✓	✓	✓	✓	✓			✓	✓				✓	
$0001011110011111_2 = 6111$	✓	✓	✓	✓	✓			✓	✓	✓	✓		✓	
$0001000000011111_2 = 6175$	✓	✓	✓	✓	✓							✓	✓	
$0010000000011111_2 = 8223$	✓	✓	✓	✓	✓									✓
$0010000110011111_2 = 8607$	✓	✓	✓	✓	✓			✓	✓					✓
$0010011110011111_2 = 10143$	✓	✓	✓	✓	✓			✓	✓	✓	✓			✓
$0010100000011111_2 = 10271$	✓	✓	✓	✓	✓							✓		✓

TENTATIVE_PARALLAX_BIAS : Parallax bias correction (double, Angle[mas])

This is the parallax bias correction computed based on the recipe in **?**.

This correction is to be **subtracted** from `parallax` to get the corrected parallax.

BLENDED : Source update flag indicating use of blended transits (byte)

This field indicates whether blended transits have been used in the astrometric solution:

- when set to 0, it indicates that no blended transit was used.
- when set to 1, it indicates that only blended transits have been used.
- when set to 2, it indicates that a mix of non-blended and blended transits has been used. This happens when neither the number of non-blended nor the number of blended transits are high enough to attempt a six-parameter source update.

A blended transit occurs when in an observation of a known close source pair (or close mutiple source) the scanning direction is such that the individual members of the pair (or multiple) are not separated in the CCD window.

For more details see Section 4.3 of the DR4 astrometry paper (**?**).

RA_DEC_CORR : Correlation between right ascension and declination (float)

Correlation coefficient $\rho(\alpha, \delta)$ between right ascension and declination. This is a dimensionless quantity in the range [-1,+1].

RA_PARALLAX_CORR : Correlation between right ascension and parallax (float)

Correlation coefficient $\rho(\alpha, \varpi)$ between right ascension and parallax, a dimensionless quantity in the range [-1,+1].

RA_PMRA_CORR : Correlation between RA and proper motion in RA (float)

Correlation coefficient $\rho(\alpha, \mu_{\alpha*})$ between right ascension and proper motion in right ascension, a dimensionless quantity in the range [-1,+1].

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RA_PMDEC_CORR : Correlation between right ascension and proper motion in declination (float)

Correlation coefficient $\rho(\alpha, \mu_\delta)$ between right ascension and proper motion in declination, a dimensionless quantity in the range [-1,+1].

DEC_PARALLAX_CORR : Correlation between declination and parallax (float)

Correlation coefficient $\rho(\delta, \varpi)$ between declination and parallax, a dimensionless quantity in the range [-1,+1].

DEC_PMRA_CORR : Correlation between declination and proper motion in right ascension (float)

Correlation coefficient $\rho(\delta, \mu_{\alpha*})$ between declination and proper motion in right ascension, a dimensionless quantity in the range [-1,+1].

DEC_PMDEC_CORR : Correlation between declination and proper motion in declination (float)

Correlation coefficient $\rho(\delta, \mu_\delta)$ between declination and proper motion in declination, a dimensionless quantity in the range [-1,+1].

PARALLAX_PMRA_CORR : Correlation between parallax and proper motion in right ascension (float)

Correlation coefficient $\rho(\varpi, \mu_{\alpha*})$ between parallax and proper motion in right ascension, a dimensionless quantity in the range [-1,+1].

PARALLAX_PMDEC_CORR : Correlation between parallax and proper motion in declination (float)

Correlation coefficient $\rho(\varpi, \mu_\delta)$ between parallax and proper motion in declination, a dimensionless quantity in the range [-1,+1].

PMRA_PMDEC_CORR : Correlation between proper motion in RA and proper motion in Dec (float)

Correlation coefficient $\rho(\mu_{\alpha*}, \mu_\delta)$ between proper motion in right ascension and proper motion in declination, a dimensionless quantity in the range [-1,+1].

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RA_PSEUDOCOLOUR_CORR : Correlation between right ascension and pseudocolour (float)

Correlation coefficient $\rho(\alpha, \texttt{pseudocolour})$ between right ascension `ra` and `pseudocolour`, a dimensionless quantity in the range [-1,+1]

DEC_PSEUDOCOLOUR_CORR : Correlation between declination and pseudocolour (float)

Correlation coefficient $\rho(\delta, \texttt{pseudocolour})$ between declination `dec` and `pseudocolour`, a dimensionless quantity in the range [-1,+1]

PARALLAX_PSEUDOCOLOUR_CORR : Correlation between parallax and pseudocolour (float)

Correlation coefficient $\rho(\varpi, \texttt{pseudocolour})$ between `parallax` and `pseudocolour`, a dimensionless quantity in the range [-1,+1]

PMRA_PSEUDOCOLOUR_CORR : Correlation between proper motion in right asension and pseudocolour (float)

Correlation coefficient $\rho(\mu_{\alpha*}, \texttt{pseudocolour})$ between proper motion in right ascension `pmra` and `pseudocolour`, a dimensionless quantity in the range [-1,+1]

PMDEC_PSEUDOCOLOUR_CORR : Correlation between proper motion in declination and pseudocolour (float)

Correlation coefficient $\rho(\mu_{\delta}, \texttt{pseudocolour})$ between proper motion in declination `pmdec` and `pseudocolour`, a dimensionless quantity in the range [-1,+1]

ASTROMETRIC_N_OBS_AL : Total number of observations in the along-scan (AL) direction (short)

Total number of AL observations (= CCD transits) used in the astrometric solution of the source, independent of their weight. Note that some observations may be strongly downweighted and their number can be derived from `astrometric_n_obs_al` - `astrometric_n_good_obs_al`.

ASTROMETRIC_N_OBS_AC : Total number of observations in the across-scan (AC) direction

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(short)

Total number of AC observations (= CCD transits) used in the astrometric solution of the source, independent of their weight (note that some observations may be strongly downweighted). Nearly all sources having G < 13 will have AC observations from 2D windows, while fainter than that limit only ∼ 1% of transit observations (known as 'calibration faint stars') are assigned 2D windows resulting in AC observations.

ASTROMETRIC_N_GOOD_OBS_AL : Number of good observations in the along-scan (AL) direction (short)

Number of AL observations (= CCD transits) that were not strongly downweighted in the astrometric solution of the source. The number of strongly downweighted observations (with downweighting factor $w < 0.2$) can instead be derived from
`astrometric_n_obs_al` - `astrometric_n_good_obs_al`,
where `astrometric_n_obs_al` is the total number of AL observations used in the astrometric solution of the source.

ASTROMETRIC_GOF_AL : Goodness of fit statistic of model wrt along-scan observations (float)

Goodness-of-fit statistic of the astrometric solution for the along-scan measurements of the source. This is a rescaled version of the 'gaussianized chi-square', which for good fits should approximately follow a normal distribution with zero mean value and unit standard deviation. Values exceeding, say, +3 thus indicate a significantly bad fit to the data.

Starting with DR4, `astrometric_gof_al` is probably more useful than `ruwe` as a general goodness-of-fit measure, because it takes into account the dependence on the degree of freedom and has a simple statistical interpretation.

Note that the `astrometric_gof_al` is not simply the 'gaussianized' version of the original `astrometric_chi2_al` but an empirically rescaled version. The rescaling procedure is described and motivated in detail in **?** which also gives an explanation and discussion of other astrometric quality indicators.

ASTROMETRIC_CHI2_AL : AL chi-square value (float)

Weighted sum of the squared astrometric residuals (χ^2) in the AL direction.

χ^2 values were computed using the 'good' AL observations of the source, without taking into account the `astrometric_excess_noise` (if any) of the source. They do however take into account the calibration excess noise (if any) of each observation.

See **?** for a discussion of other astrometric quality indicators.

ASTROMETRIC_EXCESS_NOISE : Rescaled Excess noise of the source (float, Angle[mas])

This is the rescaled source excess noise ϵ_i of the source. It is not the same as the source excess noise in Gaia DR3 and earlier releases, but re-scaled in order to be consistent with the (also redefined) unit weight error (`ruwe`). But, like the original quantity in earlier releases, it measures the disagreement, expressed as an angle, between the observations of a source and the best-fitting standard astrometric model (using six astrometric parameters). The assumed observational noise in each observation is quadratically increased by ϵ_i in order to statistically match the residuals in the astrometric solution.

A value of $\epsilon_i = 0$ thus signifies that the source is astrometrically well-behaved, i.e. that the residuals of the fit statistically agree with the assumed observational noise. Conversely, a positive ϵ_i signifies that the residuals are statistically larger than expected. This may happen for sources that deviate from the standard six-parameter astrometric model (e.g. unresolved binaries, exoplanet systems, etc.). The statistical significance of the excess noise is given by `astrometric_gof_al`.

Although the rescaling makes ϵ_i much less sensitive to instrumental effects than in previous releases, it may still absorb residual modelling errors not accounted for by the observational noise (image centroiding error) and the calibration excess noise. This includes LSF and PSF calibration errors, saturation effects, geometric instrument calibration errors, and part of the high-frequency attitude noise. Such modelling errors may be particularly important for very bright sources and sources of extreme extreme colours. Given the many other possible contributions to the excess noise, the user is advised to study the empirical distributions of ϵ_i and `astrometric_gof_al` for their particular samples in order to make sensible choices of filtering limits.

The source excess noise is further explained in **?** and in Section **??** of the DR4 online documentation.

ASTROMETRIC_PRIMARY_FLAG : Primary or secondary (boolean)

Flag indicating if this source was used as a primary source (`true`) or secondary source (`false`) in the AGIS core solution for DR4. Only primary sources contribute to the estimation of attitude, calibration, and global parameters. The estimation of source parameters is otherwise done in exactly the same way for primary and secondary sources (using the attitude, calibration, and global parameters from the core solution).

RV_USED_IN_ASTROMETRY : Assumed radial velocity used in the astrometric solution (float, Velocity[km s^{-1}])

This value is the one used to account for possible perspective acceleration effects. This correction is applied even when the expected perspective acceleration is negligible.

Due to cyclic processing of the astrometry and the RVS data, this radial velocity might be different from the one computed using the latest available RVS information, and published under `all_source_rvs.radial_velocity_single`. Moreover, if no reliable radial velocity was available at the time of the astrometric processing, radial velocities from the litterature have been used when relevant, and their value is reported in `rv_used_in_astrometry`. In all other cases, zero radial velocity was assumed.

NU_EFF_USED_IN_ASTROMETRY : Effective wavenumber of the source used in the astrometric solution (float, Misc[μm^{-1}])

This effective wavenumber $\nu_{\rm eff}$, in units of $\mu{\rm m}^{-1}$, is the assumed value used for the selection of the appropriate LSF/PSF model in the image parameter determination, as well as for the selection of the astrometric calibration parameters in the astrometric adjustment, for the given source. If no BP and RP photometry of the given source was available, `nu_eff_used_in_astrometry` is set to the default value $\nu_{\rm eff} = 1.43\mu{\rm m}^{-1}$. Else, if reliable BP and RP photometry was available, it is set to the weighted mean of the default value and the actual photon-flux weighted inverse wavelength as estimated from the BP and RP bands. If highly precise BP and RP data are available, the `nu_eff_used_in_astrometry` is very close to the value from the BP and RP bands. However, due to the cyclic processing of the Gaia astrometry and photometry, this effective wavenumber will even in these cases be (slightly) different from the one computed using the latest available photometry for DR4.

Some details on the selection of the default value and the construction of the weighted mean:
The default value is chosen as the ensemble average of the actually measured values for all Gaia sources, and its standard error for the weighted mean is set to $0.085\mu{\rm m}^{-1}$, which is close to the actual rms scatter of the measured values.
For the measured values, the formal errors assigned by the photometry processing were used in the construction of the weighted mean. However, the original values delivered by CU5 (from the DR3 photometry processing) were found to be strongly overestimated. Therefore, they were rescaled to give realistic values before being used for the weighted mean.

ASTROMETRIC_MATCHED_TRANSITS : Matched FOV transits used in the AGIS solution (short)

The number of field–of–view transits matched to this source, counting only the transits containing CCD observations actually used to compute the astrometric solution.

This number will always be equal to or smaller than `matched_transits`, the difference being the FOV transits that were not used in the astrometric solution because of bad data or excluded time intervals.

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VISIBILITY_PERIODS_USED : Number of visibility periods used in Astrometric solution (short)

Number of visibility periods used in the astrometric solution.

A visibility period is a group of observations separated from other groups by a gap of at least 4 days. A source may have from one to tens of field-of-view transits in a visibility period, but with a small spread in time, direction of scanning, and parallax factor. From one visibility period to the next these variables have usually changed significantly. A high number of visibility periods is therefore a better indicator of an astrometrically well-observed source than a large number of field-of-view transits (`matched_transits` or `astrometric_matched_transits`) or CCD observations (`astrometric_n_obs_al`). A small value (e.g. less than 10) indicates that the calculated parallax could be more vulnerable to errors, e.g. from the calibration model, not reflected in the formal uncertainties. See **?** for a discussion of this and other astrometric quality indicators.

This field is always zero for sources with astrometric parameters from Source Environment analysis and from non-single star solutions, as the corresponding processing pipelines do not compute this quantity. It is filled, however, for non-single star acceleration solutions, as these are produced by the AGIS pipeline.

ASTROMETRIC_SIGMA5D_MAX : The longest semi-major axis of the 5-d error ellipsoid (float, Angle[mas])

The longest principal axis in the 5-dimensional error ellipsoid spanned by `ra, dec, parallax, pmra, pmdec`.

This is a 5-dimensional equivalent to the semi-major axis of the position error ellipse and is therefore useful for filtering out cases where one of the five parameters, or some linear combination of several parameters, is particularly ill-determined. It is measured in mas and computed as the square root of the largest singular value of the scaled 5×5 covariance matrix of the astrometric parameters. The matrix is scaled so as to put the five parameters on a comparable scale, taking into account the maximum along-scan parallax factor for the parallax and the time coverage of the observations for the proper motion components. If C is the unscaled covariance matrix, the scaled matrix is SCS, where $S = \mathrm{diag}(1, 1, \sin\xi, T/2, T/2)$, $\xi = 45°$ is the solar aspect angle in the nominal scanning law, and T the time coverage of the data used in the solution.

`astrometric_sigma5d_max` is given for all the solutions that are derived using AGIS. Its size is one of the criteria for accepting or rejecting the 6-parameter solution. In case of a 2-parameter solution (i.e. `astrometric_params` = 3) it gives the value for the rejected 6-parameter solution, and can then be arbitrarily large.

In case of acceleration&jerk solutions (10p-solutions) in the table, i.e. in cases where the 10p solution supersedes the 6p AGIS solution, `astrometric_sigma5d_max` actually is the longest principal axis in the 9-dimensional error ellipsoid (again, like in the 6-parameter case, omitting the `pseudocolour` dimension).

The value of `astrometric_sigma5d_max` is NULL for all cases except AGIS (2p or 6p), SIFCF (2p or 6p) and acceleration&jerk solutions (10p). In other words, the non-NULL cases are those with field `astrometry_origin` being either `all_source_astrometry`, `crowded_field_source` or `nss_acceleration_astro`.

IPD_GOF_HARMONIC_AMPLITUDE : Amplitude of the IPD GoF versus position angle of scan (float)

This statistic measures the amplitude of the variation of the Image Parameter Determination (IPD) goodness–of–fit (GoF; reduced chi-square) as function of the position angle of the scan direction. A large amplitude indicates that the source has some non-isotropic spatial structure, for example a binary or galaxy, that is at least partially resolved by Gaia. The phase of the variation is given by the parameter `ipd_gof_harmonic_phase`.

Let ψ be the position angle of the scan direction. The following expression is fitted to the IPD GoF for the accepted AF observations of the source:

$$\ln(\mathrm{GoF}) = c_0 + c_2 \cos(2\psi) + s_2 \sin(2\psi)$$

The amplitude and phase of the variation are calculated as

$$\texttt{ipd_gof_harmonic_amplitude} = \sqrt{c_2^2 + s_2^2}$$

$$\texttt{ipd_gof_harmonic_phase} = \frac{1}{2}\,\mathrm{atan2}(s_2, c_2) \quad (+\,180^\circ)$$

where atan2 returns the angle in degrees. In the last expression 180 is added for negative values, so that `ipd_gof_harmonic_phase` is always between 0 and 180°. Only the AF observations accepted by the astrometric solution are used to compute the amplitude and phase, thus for example outliers and observations in the early Ecliptic Pole Scanning Law phase are not used.

The GoF variation is modelled as a periodic function of 2ψ because a source with fixed structure is normally expected to give fits of similar quality when scanned in opposite directions (ψ differing by 180°). See `ipd_gof_harmonic_phase` for the interpretation of the phase.

IPD_GOF_HARMONIC_PHASE : Phase of the IPD GoF versus position angle of scan (float, Angle[deg])

This statistic measures the phase of the variation of the IPD GoF (reduced chi-square) as function of the position angle of the scan direction. See the description of `ipd_gof_harmonic_amplitude` for details on the computation of the phase.

The interpretation of this parameter is non-trivial because of the complex interaction between the source structure and the IPD. At least the following different scenarios could occur:

- For a binary with separation $\lesssim 0.1$ arcsec the GoF is expected to be higher when the scan is along the arc joining the components than in the perpendicular direction, in which case `ipd_gof_harmonic_phase` should indicate the position angle of the binary modulo 180°. Such a binary will normally have negligible `ipd_frac_multi_peak` (less than a few per cent).
- For a resolved binary the GoF may instead have a minimum when the scan is along the arc joining the two components, in which case `ipd_gof_harmonic_phase` differs from the position angle of the binary (modulo 180°) by approximately ±90°. Such a binary will normally have a large `ipd_frac_multi_peak`.
- For a bright binary ($G \lesssim 13$) the GoF refers to the fitting of a two-dimensional PSF, which could further complicate the interpretation.
- For a galaxy with elongated intensity distribution, the IPD may give a smaller GoF when the scan is along the major axis of the image, resulting in an offset of approximately ±90° between the `ipd_gof_harmonic_phase` and the position angle of the major axis (modulo 180°).

IPD_FRAC_MULTI_PEAK : Percent of successful-IPD windows with more than one peak (byte)

This field provides information on the raw windows used for the astrometric processing of this source coming from the Image Parameters Determination (IPD) module in the core processing. It provides the fraction of windows (having a successful IPD result), as percentage (from 0 to 100), for which the IPD algorithm has identified a double peak, meaning that the detection may be a visually resolved double star (either just visual double or real binary). The quantity was computed using all transits where the IPD was successful.

Note: Some transits with more than one peak are produced by superposition of unrelated sources from the other field of view.

IPD_FRAC_ODD_WIN : Percent of transits with truncated windows or multiple gate (byte)

This field is calculated during AGIS and provides information on the raw windows used for the astrometric processing of this source. It provides the fraction (as a percentage, from 0 to 100)

of transits having either truncation or multiple gates flagged in one or more windows. Such a situation invariably means that the on-board video processing unit (VPU) detected some nearby source (which may be just a spurious detection, but typically could be some real nearby source — having another distinct transit and most probably assigned to a different source). So in general a non–zero fraction indicates that this source may be contaminated by another nearby source. The quantity was computed using all transits where the IPD was successful.

Note: Some odd-window transits are produced by superposition of unrelated sources from the other field of view.

IPD_FRAC_HIGH_GOF : Percent of successful-IPD windows with only one peak but high GoF (byte)

This field provides information (coming from IDU IPD) on the raw windows used for the astrometric processing of this source. It provides the fraction of windows (as percentage, from 0 to 100), having a successful IPD result and *without* any 'double peak' detected, for which the IPD GoF (goodness-of-fit) was 'too high' — meaning that the image did not fit very well with a point-like PSF. Thus, a high fraction means that the source is probably a slightly extended object, or a partially resolved double star (or blended binary), etc. The quantity was computed using all transits for the given source where the IPD was successful.

The exact IPD GoF threshold considered to be 'too high' for a given window is determined from a statistical analysis of a larger sample of IPD inputs to AGIS, with separate thresholds in bins per magnitude, FOV, strip, row and gate. Within this parameter space, the threshold is determined as the 95th percentile of IPD GoF values in a representative (random) input sample of sources. This ensures that this threshold is really meaningful, although of course being arbitrary to some extent.

A certain fraction of high GoF values will always be produced by superposition of unrelated sources from the other field of view, by cosmic-ray events, etc.

RUWE : Renormalised unit weight error (float)

The Renormalised Unit Weight Error is a re-scaled version of the Unit Weight Error

$$\texttt{uwe} = \sqrt{\texttt{astrometric_chi2_al}/\nu}$$

where $\nu = \texttt{astrometric_n_good_obs_al} - N$ is the number of degrees of freedom in the AGIS source update, and $N = 6$ is the number of parameters solved.

The purpose of the re-scaling is to make the `ruwe` behave in the statistically expected manner of the `uwe`, when there is a good fit of the standard 6-parameter model to the astrometric data. A

well-behaved source is expected to have `ruwe` $\simeq$ 1.0, while a significantly greater value indicates a bad fit. The re-scaling procedure is described in **?**.

The degree of deviation from the expected `ruwe` $\simeq$ 1.0 is most conveniently measured by the Gaussianized statistic `astrometric_gof_al`. Starting with DR4, `astrometric_gof_al` is probably more useful than `ruwe` as a general goodness-of-fit measure, because it takes into account the dependence on ν and has a simple statistical interpretation. For continuity, `ruwe` is retained with the same name and nominal meaning as in earlier releases, though more accurately computed.

The `ruwe` is set to null for sources that only have a two-parameter solution ($N = 2$), as well as sources for which the astrometry is stemming from gaia_source_environment, crowded_field_source, nss_two_body_orbit, NssVimfl, nss_multiple_orbits, and for non-Jerk solutions stemming from table nss_acceleration_astro.

G_PHOTOMETRY_ORIGIN : Flag indicating which module provided the G photometry (string)

This flag indicates which of the DPAC modules provided the G-band photometry that got consolidated for this source. It is provided as a string giving the name of the Gaia DR4 table where the full set of parameters associated with this photometric solution are stored. The following values can be found:

1. `all_source_photometry` for single-star photometric solutions from the PhotPipe module
2. `gaia_source_environment` for single-star photometric solutions from the source environment analysis module (SEAPipe)
3. `crowded_field_source` for single-star photometric solutions from the crowded field analysis module
4. Variability modules
   - `vari_cepheid` for photometric solutions based on a Cepheid variability model
   - `vari_rrlyrae` for photometric solutions based on a RR Lyrae variability model
   - `vari_eclipsing_binary` for photometric solutions based on an Eclipsing Binary variability model
5. `all_source_astrometry` for DPAC-internal approximate source magnitudes (IDU or onboard estimates) when the single-star photometric solutions from the PhotPipe module have odd values:
   - `phot_proc_flag` is 4 or 8 AND

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- `all_source_photometry.bp_rp` < -1 OR `all_source_photometry.nu_eff` > 3 OR `all_source_photometry.phot_g_mean_mag` > 23

PHOT_G_N_OBS : Number of observations contributing to G photometry (short)

Number of observations (CCD transits) that contributed to the G mean flux (`phot_g_mean_flux`) and mean flux error (`phot_g_mean_flux_error`).

In case of a flux resulting from the modelling of variable stars, this is the number of G FoV epochs used in the fitting algorithm. The exception is for G-band photometry stemming from the EclipsingBinaries table, for which no information about this number of observations is provided (the field is then set to 0).

There will also be some few cases where this field is set to 0 when the photometry stems from gaia_source_environment. These correspond to cases of unmatched SEAPipe sources for which no calibrated photometry could be producted. Those will hold a photometry derived from Image Segregation processing (see AstometryOrigin in gaia_source_environment), for which the information on the number of observations used is not available.

PHOT_G_MEAN_FLUX : G-band mean flux (double, Flux[e^- s^{-1}])

Weighted mean flux in the G-band. See Chapter **??** of the online documentation regarding the photometric accumulations for details about this calculation.

The origin of the mean G-band flux is given in `g_photometry_origin`.

PHOT_G_MEAN_FLUX_ERROR : Error on G-band mean flux (float, Flux[e^- s^{-1}])

The error on the weighted mean flux in the G-band `phot_g_mean_flux`. See Chapter **??** of the online documentation regarding the photometric accumulations for details about this calculation.

PHOT_G_MEAN_FLUX_OVER_ERROR : G-band mean flux divided by its error (float)

Mean flux in the G-band `phot_g_mean_flux` divided by its error `phot_g_mean_flux_error`.

PHOT_G_MEAN_MAG : G-band mean magnitude (float, Magnitude[mag])

Mean magnitude in the G band. The origin of the mean G-band magnitude is given in `g_photometry_origin`.

For G-band photometry stemming from `all_source_photometry`, this is computed from the G-band mean flux (`phot_g_mean_flux`) applying the magnitude zero-point in the Vega scale (see Section **??**).

No error is provided for this quantity as the error distribution is only symmetric in flux space. This converts to an asymmetric error distribution in magnitude space which cannot be represented by a single error value.

COLOUR_ORIGIN : Flag indicating which module provided the XP photometry (string)

This flag indicates which of the DPAC modules provided the BP- and RP-band photometry that got consolidated for this source. It is provided as a string giving the name of the Gaia DR4 table where the full set of parameters associated with this astrometric solution are stored. The following values can be found:

1. `all_source_photometry` for single-star photometric solutions from the PhotPipe module
2. Variability modules
   - `vari_cepheid` for photometric solutions based on a Cepheid variability model
   - `vari_rrlyrae` for photometric solutions based on a RR Lyrae variability model

PHOT_BP_N_OBS : Number of observations contributing to BP photometry (short)

Number of observations (CCD transits) that contributed to the integrated BP mean flux (`phot_bp_mean_flux`) and its mean flux error (`phot_bp_mean_flux_error`). In case of a flux resulting from the modelling of variable stars, this is the number of BP epochs used in the fitting algorithm.

When no BP-band photometry is available, the field will be set to 0. When the BP-band photometry is considered unreliable as per `unreliable_bp` from `all_source_photometry`, the field will be set to null.

PHOT_BP_MEAN_FLUX : Integrated BP mean flux (double, Flux[e^- s^{-1}])

Weighted mean flux in the integrated blue photometer (BP) band. See Chapter **??** of the online documentation regarding the photometric accumulations for details about this calculation.

The origin of the mean BP-band flux is given in `colour_origin`.

PHOT_BP_MEAN_FLUX_ERROR : Error on the integrated BP mean flux (float, Flux[e^- s^{-1}])

The error on the weighted mean flux in the integrated blue photometer (BP) band `phot_bp_mean_flux`. See Chapter **??** of the online documentation regarding the photometric accumulations for details about this calculation.

PHOT_BP_MEAN_FLUX_OVER_ERROR : Integrated BP mean flux divided by its error (float)

Integrated BP mean flux `phot_bp_mean_flux` divided by its error `phot_bp_mean_flux_error`.

PHOT_BP_MEAN_MAG : Integrated BP mean magnitude (float, Magnitude[mag])

Mean magnitude in the integrated BP band. The origin of the mean BP-band magnitude is given in `colour_origin`.

For BP-band photometry stemming from `all_source_photometry`, this is computed from the BP-band mean flux (`phot_bp_mean_flux`) applying the magnitude zero-point in the Vega scale (see Section **??**).

No error is provided for this quantity as the error distribution is only symmetric in flux space. This converts to an asymmetric error distribution in magnitude space which cannot be represented by a single error value.

PHOT_RP_N_OBS : Number of observations contributing to RP photometry (short)

Number of observations (CCD transits) that contributed to the integrated RP mean flux (`phot_rp_mean_flux`) and mean flux error (`phot_rp_mean_flux_error`). In case of a flux resulting from the modelling of variable stars, this is the number of RP epochs used in the fitting algorithm.

When no RP-band photometry is available, the field will be set to 0. When the RP-band photometry is considered unreliable as per `unreliable_rp` from `all_source_photometry`, the field will be set to null.

PHOT_RP_MEAN_FLUX : Integrated RP mean flux (double, Flux[e^- s^{-1}])

Weighted mean flux in the integrated red photometer (RP) band. See Chapter **??** of the online documentation regarding the photometric accumulations for details about this calculation.

The origin of the mean RP-band flux is given in `colour_origin`.

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PHOT_RP_MEAN_FLUX_ERROR : Error on the integrated RP mean flux (float, Flux[e^- s^{-1}])

The error on the weighted mean flux in the integrated red photometer (RP) band `phot_rp_mean_flux`. See Chapter **??** of the online documentation regarding the photometric accumulations for details about this calculation.

PHOT_RP_MEAN_FLUX_OVER_ERROR : Integrated RP mean flux divided by its error (float)

Integrated RP mean flux `phot_rp_mean_flux` divided by its error `phot_rp_mean_flux_error`.

PHOT_RP_MEAN_MAG : Integrated RP mean magnitude (float, Magnitude[mag])

Mean magnitude in the integrated RP band. The origin of the mean RP-band magnitude is given in `colour_origin`.

For RP-band photometry stemming from `all_source_photometry`, this is computed from the RP-band mean flux (`phot_rp_mean_flux`) applying the magnitude zero-point in the Vega scale (see Section **??**).

No error is provided for this quantity as the error distribution is only symmetric in flux space. This converts to an asymmetric error distribution in magnitude space which cannot be represented by a single error value.

PHOT_PROC_FLAG : Source-level processing information for photometry (byte)

Source-level processing information. These flags apply to all bands.

Each byte will bit-wise encode the following values:

- Bit 0: This source was used as calibrator for the photometry.
- Bit 1: This source was a primary source.
- Bit 2: '1' means 'Nominal photometric treatment'. This requires full colour information to be available from synthetic photometry. The additional application of specific calibrations is reported in other bits/flags.
- Bit 3: '1' means 'Incomplete colour photometric treatment'. A special treatment was

required for this source because of incomplete synthetic photometry. The additional application of specific calibrations is reported in other bits/flags.

- Bit 4: '1' means 'Default colour photometric treatment'. A default colour was adopted for this source. The additional application of specific calibrations is reported in other bits/flags.

If a bit is not set, then the corresponding description does not apply, e.g. if bit 0 is not set, then this source was not a calibrator.

Other bits are available for future extensions.

Finally, for G-band photometry resulting from the modelling of variable stars, or stemming from `gaia_source_environment` or `crowded_field_source`, this field is set to 0.

BP_RP : BP - RP colour (float, Magnitude[mag])

BP−RP colour (`phot_bp_mean_mag` – `phot_rp_mean_mag`).

RADIAL_VELOCITY_ORIGIN : Flag indicating which module provided the radial velocity (string)

This flag indicates which of the DPAC modules provided the radial velocity that got consolidated for this source. It is provided as a string giving the name of the Gaia DR4 table where the full set of parameters associated with this astrometric solution are stored. The following values can be found:

1. `all_source_rvs` for single-star solutions from the RVS Multi-Transit Analysis module
2. Non-single star solutions
   - `nss_two_body_orbit` for radial velocity solution from spectroscopic binary models combinations thereof (with eclipsing binaries or orbital solutions)
   - `nss_non_linear_spectro` for radial velocity solution from non-linear spectroscopic binary models
   - `nss_multiple_orbits` for radial velocity solution from multiple orbits binary models
   - `rvs_two_body_orbit` for radial velocity solution from binary models based on the RVS Multi-Transit Analysis module (LIA)

3. Variability modules
    - `vari_cepheid` for radial velocity based on a Cepheid variability model
    - `vari_rrlyrae` for radial velocity based on a RR Lyrae variability model

RADIAL_VELOCITY : Radial velocity (float, Velocity[km s^{-1}])

Spectroscopic radial velocity in the Solar system barycentric reference frame. The origin of this radial velocity is given in `radial_velocity_origin`.

The `radial_velocity` is the multi-epoch (or multi-transit) value obtained by combining the epoch (transit) data of these stars. The total number of epochs used is stored in `rv_nb_transits`. Both non-blended and deblended spectra are used, and the number of epochs having deblended spectra is stored in `rv_nb_deblended_transits`.

The majority of the `radial_velocity` values originate from `all_source_rvs.radial_velocity_single` This value is not provided for stars assumed to be SB2 (`rv_assumed_sb2`=true), stars assumed to have emission lines (`fraction_emission_lines`> 0.4), and any star with fewer than two valid transits. Four methods are used to obtain `radial_velocity_single`, depending on the magnitude `external_apparent_grvs` of the star. Information about which method has been used is stored in `all_source_rvs.rv_single_method_used`.

For bright stars (`external_apparent_grvs` $\leq$ 11), `radial_velocity_single` is calculated as the median value of the epoch radial velocities. For fainter stars with less precise epoch radial velocities, the epoch cross-correlation functions obtained by comparing the RVS spectrum and the associated template spectrum are combined to derive `radial_velocity_single`.

For stars with magnitude between 11 and 14 mag (11 $<$ `external_apparent_grvs` $\leq$ 14), the epoch cross-correlation function is a combination of the CCD cross-correlation functions obtained by comparing each CCD spectrum with the template.

For stars with magnitude greater than 14 mag (14 $<$ `external_apparent_grvs`), the CCD spectra of the transit are combined first and then compared with the template to estimate the epoch cross-correlation function.

For the few stars that exhibit a scan angle signal in the radial velocity, `radial_velocity_single` is taken from `radial_velocity_scan_angle`.

See Section **??** for the complete list of parameters containing information on the `radial_velocity` measurements stemming from `all_source_rvs` or `rvs_two_body_orbit`.

In Gaia DR4, a `radial_velocity` is provided for about 85 million of stars. See **?** for a descrip-

tion of the validation of the radial_velocity measurements from all_source_rvs, and **?** for the specific procedures used to derive and validate the radial velocities of hot stars.

RADIAL_VELOCITY_ERROR : Radial velocity error (float, Velocity[km s^{-1}])

The uncertainty associated with radial_velocity. The origin of radial_velocity_error is given in radial_velocity_origin.

The majority of the radial_velocity_error originates from all_source_rvs.radial_velocity_singl Four methods are employed to derive radial_velocity_single and radial_velocity_single_error, depending on the magnitude external_apparent_grvs of the star. The specific method utilized is indicated by the rv_single_method_used field within all_source_rvs. Additionally, a constant shift of 0.112 km s^{-1} is applied to account for calibration floor contributions.

For stars with external_apparent_grvs ≤ 11, radial_velocity_single_error is the uncertainty on the median of the epoch radial velocities ($\sigma_{\rm med}$):

$$\texttt{radial_velocity_error} = \sqrt{\sigma_{\rm med}^2 + 0.112^2}$$

$$\sigma_{\rm med} = \sqrt{\frac{\pi}{2}} \cdot \frac{\sigma}{\sqrt{\texttt{rv_nb_transits}}}$$

where σ is the standard deviation of the epoch radial velocity measurements, and rv_nb_transits is the number of transits for which an epoch radial velocity has been obtained.

For fainter stars (external_apparent_grvs > 11), the radial_velocity_error is derived from the combined cross-correlation functions:

$$\texttt{radial_velocity_error}^2 = \frac{C^2 - 1}{MNC''C} + 0.112^2$$

where C and C'' are the values of the cross-correlation function and its second derivative at the maximum, respectively. N is the number of bins of the input spectrum, and M is the number of transits (rv_nb_transits) (see Zucker (2003), Sec. 2.3).

For the few stars that exhibit a scan angle signal in the radial velocity, radial_velocity_single_error is taken from radial_velocity_scan_angle_error.

RV_NB_TRANSITS : Number of transits used to compute the radial velocity (short)

The total number of epochs (transits) used to compute `radial_velocity`. In general, one transit in the RVS field-of-view includes three RVS CCD observations.

In the case of a `radial_velocity` resulting from the modelling of variable or binary stars (i.e., `radial_velocity_origin` other than `all_source_rvs`), this represents the number of radial velocity epochs utilised in the fitting algorithm.

RV_EXPECTED_SIG_TO_NOISE : Expected signal to noise in the spectrum used for radial velocity (float)

The expected signal to noise ratio is obtained by combining the information from the CCD spectra used for estimating the `radial_velocity` obtained from `all_source_rvs`,

$$\texttt{rv_expected_sig_to_noise} = \frac{S}{\sqrt{S + BCK + RN}}$$

where:

$$S = \frac{medianFlux \times N_{ValidStrips} \times \text{EXP_TIME} \times \text{PIXEL_WIDTH_AL}}{\text{BAND_WIDTH}}$$

$$BCK = medianBackground \times N_{ValidStrips} \times N_{ACpixels}$$

$$RN = RON^2 \times N_{ValidStrips} \times N_{ACsamples}$$

where:

• $medianFlux$ is the median of the integrated flux of the CCD spectra used to obtain the radial velocity for `external_apparent_grvs` ≤ 12; otherwise, the $medianFlux$ is obtained from `external_apparent_grvs` [$e^- s^{-1}$]
• $N_{ValidStrips}$ = the total number of CCD spectra used to obtain the `radial_velocity` (in general there are 3 CCD spectra per transit)
• BAND_WIDTH = 870 - 846 [nm]
• EXP_TIME = 4.4167032 [s]
• PIXEL_WIDTH_AL = 0.02453 [nm]
• $medianBackground$ = the median background in one exposure of the CCD spectra used to obtain the `radial_velocity` [$e^- pixel^{-1}$]

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• $N_{ACpixels}$ = 10 is the number of pixels in the AC direction of the non-truncated RVS window
• $N_{ACsamples}$ is the size in the AC direction of the non-truncated RVS window. $N_{ACsamples}$ = 10 for the 2D RVS windows (Window Class=0) and $N_{ACsamples}$ = 1 for the 1D windows (Window Class=1)
• *RON* (Read Out Noise) = 3.2 [e^-]

The SNR of the spectra combination used for `vbroad` determination can be estimated from:

$$\texttt{rv_expected_sig_to_noise}\sqrt{\frac{\texttt{vbroad_nb_transits}}{\texttt{rv_nb_transits}}},$$

and similarly for `grvs_mag` from

$$\texttt{rv_expected_sig_to_noise}\sqrt{\frac{\texttt{grvs_mag_nb_transits}}{\texttt{rv_nb_transits}}}.$$

Note: The SNR measured directly on the combined spectrum `rvs_mean_spectrum` is stored in `rvs_spec_sig_to_noise`.

RV_RENORMALISED_GOF : Radial velocity renormalised goodness of fit (float)

The renormalised goodness of fit (see Perryman 1997, p. 112) is an empirical value computed using this source and all the other sources having `all_source_rvs.rv_template_teff` and `external_apparent_grvs` in a given range. The scatter of the source epoch radial velocities is compared to the typical epoch uncertainty for the appropriate `external_apparent_grvs` and `all_source_rvs.rv_template_teff` range.

$$\texttt{rv_renormalised_gof} = \sqrt{4.5\nu}\left(RUWE^{2/3} + \frac{2}{9\nu} - 1\right)$$

where:

$$\nu = \texttt{rv_nb_transits} - 1$$

$$RUWE = \frac{UWE}{\text{mode}(UWE)}$$

$$UWE = \sqrt{\frac{1}{\nu}\sum_j \left(\frac{V_j - \texttt{radial_velocity}}{\epsilon_j}\right)^2}$$

• `radial_velocity` is the median of the epoch radial velocities V_j (`all_source_rvs.rv_median`);

• V_j are the epoch radial velocities (`rvs_epoch_parameters_single`);

• ϵ_j is the uncertainty on V_j (`rvs_epoch_parameters_single`);

• mode(UWE) is the mode of the UWE distribution in the appropriate bin of the UWE (`all_source_rvs.rv_external_apparent_grvs`) look-up table obtained by an offline analysis (see Section **??**);

• $RUWE$: Renormalized Unit Weight Error, UWE: Unit Weight Error.

`rv_renormalised_gof` is provided for bright stars with $5.5 \leq$ `external_apparent_grvs` ≤ 12 and `all_source_rvs.rv_template_teff` $< 14{,}500$ K. It is null for stars with `external_apparent_grv` and `all_source_rvs.rv_template_teff` outside this range.

See Section **??** for more details on how this field is derived.

RV_INVALID_REASON : The reason why the radial velocity is set as invalid (long)

This 64-bit field encodes the reasons for the invalidation of the `radial_velocity` data originating from `all_source_rvs`. Each bit represents a specific reason, set to 1 if true and 0 if false.

Post-processing assigns these flags based on validation analysis. Invalidation reasons are determined based on various conditions, and may apply even when a radial velocity is not provided.

A radial velocity may be invalidated for multiple reasons, leading to the activation of multiple bits. See Table **??** and Table **??** for the occurrence of each invalid reason in bright (`external_apparent_grvs` ≤ 14) and faint stars, respectively.

Only the valid data are published in `gaia_source`, while both valid and invalid data are published in `all_source_rvs`.

- Bit 0: Hot star. This invalidation reason is set for stars with `external_apparent_grvs` ≤ 12 mag and `rv_template_teff` $> 14{,}500$ K, as well as for those with `external_apparent_gr` > 12 mag and `rv_template_teff` $> 8{,}500$ K.
- Bit 1: [Not Used].
- Bit 2: Suspected SB2 (double-lined spectroscopic binary). Radial velocity is invalidated for stars with `fraction_double_lined` ≥ 0.1.

- Bit 3: Suspected emission-line star. This invalidation reason is set for stars identified as emission-line stars by the pipeline (`fraction_emission_lines` > 0.4) for which no radial velocity is provided. It is also set to approximately 2,700 stars detected as emission-line stars during post-processing, even if their radial velocity was measured.
- Bit 4: [Not Used].
- Bit 5: Faint star. This invalidation reason is set when `external_apparent_grvs` > 16 mag.
- Bits 6-7: [Not Used].
- Bit 8: Excessive uncertainty. Radial velocity is invalidated when `radial_velocity_error` > 1000 km s^{-1}.
- Bit 9: Invalid cross-correlation function. This flag is set for stars with `external_apparent_grvs` ≤ 14 mag, where the cross-correlation function exhibits a spurious secondary peak that is higher than the primary (correct) peak. As a result, the radial velocity is erroneously derived from this spurious peak. These problematic stars were identified using the UMAP dimensionality reduction technique (see **?**).
- Bits 10-11: [Not Used].
- Bit 12: Radial velocity outside the allowed range. Radial velocity is invalidated for stars with `radial_velocity` > 1000 km s^{-1} or `radial_velocity` < −1000 km s^{-1}.
- Bits 13-19: [Not Used].
- Bit 20: Template mismatch. This invalidation reason is set for a list of stars with `external_apparent_grvs` ≤ 14 mag, identified using the UMAP dimensionality reduction tool, and to the stars having `rv_atm_param_origin` = 888.
- Bit 21: [Not Used].
- Bit 22: Contamination from a bright source. This invalidation reason is set for a list of stars identified as contaminated using the method described in Katz et al. (2023).
- Bit 23: High probability of being an outlier. Radial velocity is invalidated if `rv_single_outlier_probability` = 2. It is also invalidated for faint stars with `external_apparent_grvs` > 14 mag that satisfy the condition:

  `rv_single_outlier_probability` > 1 − 0.5 × (`external_apparent_grvs` − 14)

  where `rv_single_outlier_probability` is provided only for stars with 3700 K ≤ `rv_template_teff` ≤ 6500 K.

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- Bit 24: No radial velocity provided by the pipeline. This typically occurs for stars with only one valid transit available during processing (with validity stored in `rvs_transit_description.is_ccd_valid`), stars marked as `rv_assumed_sb2`, assumed emission-line stars, or cases where the pipeline failed.
- Bit 25: Excessive standard deviation in epoch radial velocities. Radial velocity is invalidated for stars with `external_apparent_grvs` ≤ 11 mag and `rv_std_dev` ≥ 577 km s^{-1}.
- Bit 26: Faint star with no `rv_single_outlier_probability`. This flag is assigned to faint stars (`external_apparent_grvs` > 14 mag) whose `rv_template_teff` falls outside the range used for `rv_single_outlier_probability` computation (i.e., `rv_template_teff` < 3700 K or > 6500 K). Due to time constraints, their radial velocities could not be analyzed with alternative methods. Given that most faint-star radial velocities are invalid, they are systematically marked as such.
- Bit 27: No associated atmospheric parameters. This flag is set when `rv_atm_param_star_teff` $= 0$.
- Bit 28: Only one valid transit remaining. Radial velocity is invalidated for a set of 68 stars with `external_apparent_grvs` ≤ 11 mag that have only one valid transit left after post-processing.
- Bits 29-63: [Not Used].

RV_SINGLE_OUTLIER_PROBABILITY : Probability for radial velocity to be an outlier (float)

This field provides an estimate of whether `radial_velocity` originating from `all_source_rvs` is erroneous. It is computed using machine learning techniques, specifically a balanced random forest trained on parameters that describe the shape of the Cross-Correlation Function, and is available only for stars that satisfy the following criteria:

- `external_apparent_grvs` > 11
- $3700 \leq$ `rv_template_teff` ≤ 6500 K

Outside this validity range, the field is not populated. It is also not populated when `rv_single_method_used` $= 5$ in `all_source_rvs`. Values usually lie between 0 and 1, acting as a probability-like score where larger values indicate a higher likelihood that the radial velocity is erroneous.

In some cases, the field is set to 2 to indicate that the method could not compute a valid score, typically when `radial_velocity_single` itself is erroneous. In such cases, the radial velocity is invalidated and `rv_invalid_reason` has bit 23 set.

For a detailed description of how `rv_single_outlier_probability` is derived, see **?**.

HAS_SYSTEMIC_RV : Indicates whether the source has a systemic radial velocity (boolean)

This flag indicates whether the star has a systemic radial velocity. If set to True, an orbital model is provided and stored in `rvs_two_body_orbit`. The systemic velocity is tabulated both in `all_source_rvs` as `radial_velocity_sys` and in `rvs_two_body_orbit` as `center_of_mass_velocity`

VBROAD_ORIGIN : Flag indicating which module provided the vbroad for this source (string)

This flag indicates which of the DPAC modules provided the spectral line broadening that got consolidated for this source. It is provided as a string giving the name of the Gaia DR4 table where the full set of parameters associated with this astrometric solution are stored. The following values can be found:

1. `all_source_rvs` for single-star solutions from the RVS Multi-Transit Analysis module
2. `ap_hot` for `vbroad` derived by ESP-HS module from the Apsis processing chain. Note that in those cases, the parameter corresponds to a rotational velocity (see `ap_hot.vsini_esphs` for further details).

VBROAD : Spectral line broadening parameter (float, Velocity[km s^{-1}])

In the case of `vbroad_origin` = `all_source_rvs` :

The spectral line broadening parameter, `vbroad`, is measured using a rotational broadening kernel; its value therefore includes $v \sin i$, but also any other source of line broadening (macroturbulence, residual instrumental effects, etc.).

`vbroad` is computed as the median of the epoch `vbroad` measurements stored in `rvs_epoch_parameters_si` Measurements are excluded if they fall outside the allowed range (Bit 2 in `rvs_epoch_parameters_single.` if they are obtained from deblended spectra (Bit 4), if they were measured at the beginning of Gaia operations when the optics were contaminated by water ice, which caused an artificial broadening of the spectral lines (Bit 5), or if they are associated with a reported algorithm failure (Bit 15). The total number of transits used in the median computation is stored in `vbroad_nb_transits`.

`all_source_rvs.vbroad` is provided for about 7 million stars with `external_apparent_grvs`≤ 12, and it is deemed valid (`vbroad_is_valid`=true) for about 2.6 million stars. See **?** for a

description of the validation of the vbroad measurements.

VBROAD_ERROR : Uncertainty on the spectral line broadening (float, Velocity[km s^{-1}])

vbroad_error is the uncertainty associated with vbroad.

In the case of vbroad_origin = all_source_rvs :

It is the uncertainty on the median of the epoch measurements (rvs_epoch_parameters_single.vbroad) used to obtain vbroad:

$$\texttt{vbroad_error} = \sqrt{\frac{\pi}{2}} \cdot \frac{\sigma}{\sqrt{\texttt{vbroad_nb_transits}}}$$

where σ is the standard deviation of the epoch $\texttt{vbroad}_j$ measurements used, and vbroad_nb_transits is the number of epochs used.

Note: When vbroad_error > vbroad, the star's vbroad estimation is: 0 ≤ vbroad ≤ vbroad + vbroad_error.

VBROAD_NB_TRANSITS : Number of transits used to compute vbroad (short)

The number of transits (epochs) used to compute vbroad.

In the case of vbroad_origin = all_source_rvs :

The epoch measurements excluded are those with values outside the allowed range (Bit 2 and Bit 3 in rvs_epoch_parameters_single.vbroad_invalid_reason), those obtained using de-blended spectra (Bit 4 in rvs_epoch_parameters_single.vbroad_invalid_reason), and those obtained at the beginning of the Gaia operations (before OBMT 1317 revolutions) when the optics were contaminated with water ice resulting in broadening of the spectral lines (Bit 5 in rvs_epoch_parameters_single.vbroad_invalid_reason).

VBROAD_INVALID_REASON : The reason why vbroad is set as invalid (short)

This 16-bit field specifies the reasons for invalidating the vbroad data originating from all_source_rvs. Each bit represents a specific reason, set to 1 if true and 0 if false.

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Post-processing assigns these flags based on validation analysis. Invalidation reasons are determined based on various conditions, and may apply even when a vbroad is not provided.

A vbroad may be invalidated for multiple reasons, leading to the activation of multiple bits. See Table **??** for the occurrence of each invalid reason.

Only the valid data are published in gaia_source, while both valid and invalid data are published in all_source_rvs.

- Bit 0: Invalid radial velocity (this includes all rv_invalid_reason cases).
- Bit 1: Too faint star. vbroad is invalidated for stars with external_apparent_grvs > 11.
- Bit 2: [Not Used].
- Bit 3: [Not Used].
- Bit 4: Cool star. vbroad is invalidated for stars with rv_template_teff $\leq 3400K$.
- Bit 5: [Not Used].
- Bit 6: Too few valid transits. vbroad is invalidated for stars with vbroad_nb_transits ≤ 5.
- Bit 7: The pipeline did not provide a vbroad value.
- Bit 8: Hot star. vbroad is invalidated for stars with rv_template_teff > 25000.
- Bits 9-15: [Not Used].

GRVS_MAG : Integrated Grvs magnitude (float, Magnitude[mag])

The magnitude grvs_mag is estimated using the flux of the star in the RVS spectra and the zero-point magnitude for grvs_mag ($ZP_{\mathrm{grvs}} = 21.31847$) (see **?**):

$$\mathtt{grvs_mag} = -2.5 \times \log(\mathtt{grvs_flux}) + ZP_{\mathrm{grvs}}$$

The flux of the star is stored in the field grvs_flux of the table all_source_rvs. It is computed as the median value of the calibrated epoch grvs_flux measurements stored in the DataLink table rvs_epoch_parameters_single, excluding all epoch values obtained from deblended spectra, spectra contaminated by a nearby bright source, duplicated transits (Bit 2, Bit 3, or Bit

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8 activated in rvs_epoch_parameters_single.grvs_invalid_reason), and spectra with emission lines (rvs_epoch_parameters_single.emission_lines = True).

This field is empty when rv_assumed_sb2 is true, when grvs_flux < 0 (which generally occurs for faint stars with an overestimated subtracted background) or when no grvs_flux value is available at all.

For stars where no grvs_mag is provided, an estimate of the magnitude in the RVS band is obtained using the magnitudes phot_g_mean_mag and phot_rp_mean_mag (see Sartoretti et al. (2023)), and it is stored in the field external_apparent_grvs.

GRVS_MAG_ERROR : Grvs magnitude uncertainty (float, Magnitude[mag])

The grvs_mag_error is calculated by adding in quadrature the uncertainty derived from the flux, denoted as σ, with a constant term of 0.002 mag. This constant accounts for a calibration floor contribution:

$$\texttt{grvs_mag_error} = \sqrt{\sigma^2 + 0.002^2}$$

The value of σ is expected to be computed as:

$$\sigma = \texttt{grvs_flux_error}/\texttt{grvs_flux} \times 2.5/\ln(10)$$

Where grvs_flux_error and grvs_flux are stored in all_source_rvs.

However, due to a coding error in the post-processing of faint sources (those with external_apparent_grvs > 14), the published value of σ is instead:

$$\sigma = \texttt{grvs_flux_error}/(1.016 \times \texttt{grvs_flux} + 25) \times 2.5/\ln(10)$$

As a result, the magnitude uncertainties for faint sources may be slightly underestimated. For more information, see **?**.

GRVS_MAG_NB_TRANSITS : Number of transits used to compute Grvs (short)

The number of transits (epochs) used to compute grvs_mag.

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Transits considered valid during pipeline processing and used for the radial velocity determination are included in the combination (see `rvs_transit_description.is_ccd_valid`). Consequently, transits flagged as `emission_lines` (`rvs_epoch_parameters_single`) and double-lined transits (`rvs_epoch_parameters_double`) are excluded.

In addition, transits with deblended spectra or identified as contaminated by a background source are also excluded, as indicated by Bit 2 and Bit 3 in `rvs_epoch_parameters_single.grvs_invalid_reas`

GRVS_MAG_INVALID_REASON : The reason why Grvs magnitude is set as invalid (short)

This 16-bit field specifies the reasons for invalidating the `grvs_mag` data. Each bit represents a specific reason, set to 1 if true and 0 if false.

Post-processing assigns these flags based on validation analysis. Invalidation reasons are determined based on various conditions, and may apply even when a `grvs_mag` is not provided.

A `grvs_mag` may be invalidated for multiple reasons, leading to the activation of multiple bits. See Table **??** for the occurrence of each invalid reason.

Only the valid data are published in `gaia_source`, while both valid and invalid data are published in `all_source_rvs`.

- Bit 0: [Not used].
- Bit 1: Too faint star: `grvs_mag` is invalidated for stars with `external_apparent_grvs` > 16 mag.
- Bit 2: Contamination by a nearby star. This invalidation reason applies to both `grvs_mag` and `radial_velocity` for a list of stars identified as contaminated using the method described in Katz et al. (2023).
- Bit 3: The pipeline did not provide a `grvs_mag` value.
- Bit 4: Too few valid transits. `grvs_mag` is invalidated if `grvs_mag` $<$ 13 and `grvs_mag_nb_transits` $<$ 2 or `all_source_rvs.nb_transits_total` $\leq$ 5; or if `grvs_mag` $\geq$ 13 and `grvs_mag_nb_transits` $<$ 3 or `all_source_rvs.nb_transits_total` $\leq$ 5.
- Bit 5: Excessive uncertainty. `grvs_mag` is invalidated if `grvs_mag_error` > 16 mag.
- Bit 6: Value outside allowed range. `grvs_mag` is invalidated if `all_source_rvs.grvs_flux` $<$ $MinFlux$.

$$\mathrm{MinFlux} = \frac{\mathrm{MinFlux_{CCD}}}{\sqrt{3 \times \texttt{grvs_mag_nb_transits}}}$$

where $MinFlux_{CCD}$ = 1024 e^-/s represents twice the estimated average precision of straylight flux integrated over a single CCD spectrum in a typical Calibration Unit.

- Bits 7-15: Not used.

RVS_SPEC_SIG_TO_NOISE : Signal to noise ratio in the mean RVS spectrum (float)

The signal to noise ratio in the mean spectrum is calculated using the median of the ratio of the fluxes and flux uncertainties $\bar{f}_{\mathrm{bin}}/\sigma_{f,\mathrm{bin}}$ on all the individual bins. It is set to null when it results in a negative value (this may happen for some very faint spectra).

The flux per wavelength bin in the mean spectrum is:

$$\bar{f}_{\mathrm{bin}} = \frac{\sum_{j=1}^{n} f_j}{n}$$

where j is the jth CCD spectrum being combined out of a total of n in a particular wavelength bin.

The uncertainty on the flux per wavelength bin is:

$$\sigma_{f,\mathrm{bin}} = \frac{\sigma_{f_n}}{\sqrt{n}}$$

where σ_{f_n} is the standard deviation of all the flux values contributing to an individual wavelength bin:

$$\sigma_{f_n} = \sqrt{\frac{1}{n}\sum_{j=1}^{n}(f_j - \bar{f}_{\mathrm{bin}})^2}$$

This field is provided only for stars with a published `rvs_mean_spectrum`, which can be selected using the flag `has_rvs` in the table `all_source_flags`.

RV_ASSUMED_SB2 : the star is assumed to be an SB2 (boolean)

An SB2 star (double-lined spectroscopic binary) is a binary system in which both stellar components contribute visible spectral lines, whose radial velocities vary with orbital motion and can therefore be measured separately.

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This boolean is set to True when double lines are detected in at least 10 RVS epoch spectra of the source. The detection of double lines is performed for stars brighter than `external_apparent_grvs`=12.

When `rv_assumed_sb2` is True, the source is excluded from the multi-transit analysis in the RVS pipeline. As a result, no combined radial velocity, line broadening velocity, mean spectrum, or associated parameters are produced by the RVS pipeline.

However, the epoch results obtained by the RVS pipeline are published. Sources may have both types of transits. Those identified as double-lined are provided in the table `rvs_epoch_parameters_double`, while those not identified as double-lined are provided in the table `rvs_epoch_parameters_single`. The corresponding epoch spectra are available in `rvs_epoch_spectrum`.

Note: Only about one third of these stars are subsequently confirmed as SB2 by the NSS pipeline; see Chapter **??**. Confirmed SB2 systems correspond to NSS solutions of type:
`nss_two_body_orbit.solution_type` = SB2ScanAngleK1, SB2ScanAngleK12, AstroSpectroSB2, or EclipsingSpectroSB2;
`nss_non_linear_spectro.solution_type` = TrendSB2;
`nss_multiple_orbits.solution_type` = SB2SB1 or SB2TrendSB1.

FRACTION_DOUBLE_LINED : Ratio of occurrences of multiple components in radial velocity (float)

The ratio of the number of transits where double lines are detected in the spectrum to the total number of transits analysed. The presence of double lines is estimated for bright stars with `external_apparent_grvs` ≤ 12.

Note: the total number of analysed transits for a source ($N_{\rm tot}$) is given by:

$$N_{\rm tot} = \frac{\texttt{rv_nb_transits}}{1 - \texttt{fraction_double_lined}}$$

Epoch data extracted from spectra in which two sources are detected (double-lined spectra) are provided in the DataLink table `rvs_epoch_parameters_double`, while those extracted from spectra in which only one source is detected are stored in the DataLink table `rvs_epoch_parameters_single`

FRACTION_EMISSION_LINES : ratio of occurrences of emission lines (float)

The ratio of the number of transits where emission lines were detected to the total number of transits analyzed. Transits with detected emission lines are flagged in the epoch DataLink tables `rvs_epoch_parameters_single` via the field `emission_lines`. The presence of emission

lines is estimated using a deep-learning technique for transits of stars with external_apparent_grvs ≤ 13. See **?**.

For these transits, neither the epoch radial velocity nor the epoch vbroad is computed. However, the epoch spectrum is published in rvs_epoch_spectrum, and the epoch grvs_mag is computed. These transits are excluded from the computation of the mean quantities.

Stars with fraction_emission_lines > 0.4 are considered emission-line stars. For these sources, the mean spectrum is published, but no other mean parameters are computed.

EXTERNAL_APPARENT_GRVS : Apparent Grvs obtained from non-RVS data (float, Magnitude[mag])

The magnitude external_apparent_grvs is obtained from non-RVS data. There are two possible origins of external_apparent_grvs, which are stored in external_apparent_grvs_origin.

For most sources, it is obtained using the magnitudes measured from the DR3 Gaia data obtained with the astrometric instrument (phot_g_mean_mag) and the Red Photometer (phot_rp_mean_mag), with the transformation described in Sartoretti et al. (2023).

For the sources for which the red photometer magnitude from DR3 is not available, external_apparent_grv is the onboard magnitude obtained by the onboard software from Red Photometer spectra when these are not saturated, and otherwise from the astrometric images.

EXTERNAL_APPARENT_GRVS_ORIGIN : Origin of external apparent Grvs (short)

The origin of the external_apparent_grvs can be:

3: Onboard: Obtained by the onboard software from Red Photometer spectra when these are not saturated, and otherwise from the astrometric images.

4: Using the DR3 measurements of phot_g_mean_mag and phot_rp_mean_mag, and the transformation described in Sartoretti et al. (2023).

Origins 1 and 2 were never used and are therefore not documented here.

PHOT_VARIABLE_FLAG : Photometric variability flag (string)

Flag indicating if variability was identified in the photometric data:

- 'VARIABLE': source identified and processed as variable, see variability tables and Chapter **??**;
- 'POSSIBLE': source identified as possible variable, see variability tables and Chapter **??**;
- 'NON_VARIABLE': source not identified as variable;
- empty: source not processed by the variability analyses.

PHOT_VARIABLE_SCORE : Photometric variability score between 0 and 1 (float)

Variability score coming from General Variability Detection and set for all sources processed by the variability analyses having sufficient observations.

PHOT_VARIABLE_BEST_CLASS : Photometric best variability class (string)

Best variability 'class' being a merger of `classifier_best_class_name` in `vari_classifier_result`, and the occurence in specific variable object tables. The latter may take priority when there are disagreements (should be a small number), but this is a merge, so not every source with a best class name corresponding to a specific variable object table is included in that table.

This field is not directly related to the `phot_variable_score`, and is only available for a subset of sources that appear in the `vari_summary` table (and is otherwise null). See Chapter **??** for further details.

CLASSPROB_DSC_COMBMOD_QUASAR : Probability from DSC-Combmod of being a quasar (float)

Probability that the object is of the named class. This is the overall probability for this class, computed by combining the class probabilities from DSC-Specmod (which classifies objects using BP/RP spectra) and DSC-Allosmod (which classifies objects using several astrometric and photometric features). See Section **??** of the online documentation for details. It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSPROB_DSC_COMBMOD_GALAXY : Probability from DSC-Combmod of being a galaxy (float)

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Probability that the object is of the named class. This is the overall probability for this class, computed by combining the class probabilities from DSC-Specmod (which classifies objects using BP/RP spectra) and DSC-Allosmod (which classifies objects using several astrometric and photometric features). See Section **??** of the online documentation for details. It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSPROB_DSC_COMBMOD_STAR : Probability from DSC-Combmod of being a single star (float)

Probability that the object is of the named class. This is the overall probability for this class, computed by combining the class probabilities from DSC-Specmod (which classifies objects using BP/RP spectra) and DSC-Allosmod (which classifies objects using several astrometric and photometric features). See Section **??** of the online documentation for details. It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

DISTANCE_GSPPHOT : Distance from GSP-Phot Aeneas best library using BP/RP spectra (float, Length & Distance[pc])

Distance (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax (see Section **??** of the online documentation). This is the median of the MCMC samples. Taken from best library that achieves the highest goodness-of-fit value. NB: The actual fit parameter is $\log_{10} d$ and a prior is imposed to ensure a value between [0,5], thus the minimum possible distance is 1 pc and the maximum is 100 kpc.

AP_ORIGIN : Origin of effective temperature, surface gravity, and metallicity (string)

This flag indicates which of the DPAC modules provided the stellar effective temperature `teff`, surface gravity `logg`, and metallicity `mh`. It is provided as a string giving the name of the Gaia DR4 table where the full set of parameters associated with the source are stored. The following values can be found:

1. `vari_rrlyrae` for metallicities derived from a dedicated RR Lyrae modelling of the photometric light curves. In this case, only metallicity is provided (no effective temperature or surface gravity are provided).
2. `ap_hot` for astrophysical parameters provided by the ESP-HS Apsis module.

3. ap_rvs for astrophysical parameters provided by the GSP-Spec Apsis module.
4. ap_xp for astrophysical parameters provided by the GSP-Phot or the ESP-UCD Apsis modules. In the case of ESP-UCD, only effective temperature is provided (no surface gravity or metallicity are provided).
5. ap_emission for astrophysical parameters provided by the ESP-CS Apsis module.

TEFF : Stellar effective temperature (float, Temperature[K])

Effective temperature (assuming source is a single star) inferred by the module indicated in ap_origin.

In case the metallicity results from the modelling of variable stars, no stellar temperature is provided even though it might have been computed by another module.

LOGG : Stellar surface gravity (float, GravitySurface[log cgs])

Surface gravity (assuming source is a single star) inferred by the module indicated in ap_origin.

In case the effective temperature is provided by the module ESP-UCD, no surface gravity is provided even though it might have been computed by another module. The same applies when the metallicity results from the modelling of variable stars.

MH : Stellar metallicity (float, Abundances[dex])

Decimal logarithm of the ratio of the number abundance of iron to the number abundance of hydrogen relative to the same ratio of solar abundances inferred by the module indicated in ap_origin.

In case the effective temperature is provided by the module ESP-UCD, no stellar metallicity is provided even though it might have been computed by another module.

EXTINCTION_ORIGIN : Origin of extinction in G band and reddening (string)

This flag indicates which of the DPAC modules provided the extinction in the G band ag and the BP-RP reddening ebpminrp. It is provided as a string giving the name of the Gaia DR4 table where the full set of parameters associated with the source are stored. The following values can be found:

1. vari_rrlyrae for ag derived from a dedicated RR Lyrae modelling of the photometric light curves. In this case, no reddening is provided.
2. ap_hot for extinction parameters provided by the ESP-HS Apsis module.
3. ap_xp for extinction parameters provided by the GSP-Phot Apsis module.

AG : Extinction in G band (float, Magnitude[mag])

Broadband extinction in G band (assuming source is a single star) inferred by the module indicated in extinction_origin.

EBPMINRP : Reddening in BP minus RP colour (float, Magnitude[mag])

Reddening $E(G_{\rm BP} - G_{\rm RP})$ (assuming source is a single star) inferred by the module indicated in extinction_origin.

In case the G-band extinction results from the modelling of variable stars, no reddenning is provided even though it might have been computed by another module.

L : Galactic longitude (double, Angle[deg])

Galactic Longitude of the object at reference epoch ref_epoch, see Section **??** of the release documentation for conversion details.

B : Galactic latitude (double, Angle[deg])

Galactic Latitude of the object at reference epoch ref_epoch, see Section **??** of the release documentation for conversion details.

ECL_LON : Ecliptic longitude (double, Angle[deg])

Ecliptic Longitude of the object at reference epoch ref_epoch, obtained from the equatorial coordinates using the transformation defined in Volume 1, Section 1.5.3 of ESA (1997).

Note that in the transformation applied here the ICRS origin is shifted in the equatorial plane from Γ by $\phi = 0.05542$ arcsec, positive from Γ to the ICRS origin (Chapront et al. 2002). The ICRS has an unambiguous definition with an origin in the ICRF equator defined by the realisation of the ICRF. The ecliptic system is less well-defined, potentially depending on additional

conventions in dynamical theories. The transformation employed here corresponds to the inertial mean ecliptic with obliquity and Γ defined by reference to the ICRS equator. Both the obliquity and the position of Γ on the ICRS equator with respect to the ICRS origin have been obtained from Lunar Laser Ranging measurements. This has no time dependence – there is no secular variation of the obliquity and no precession – and it simply defines the relative situation of the various planes at J2000.

ECL_LAT : Ecliptic latitude (double, Angle[deg])

Ecliptic Latitude of the object at reference epoch `ref_epoch`. For further details see the description for attribute `ecl_lon`.

HEALPIX_ID : Level 12 HEALPix identification string (string)

Level 12 HEALPix identification string consisting of 12 characters and based on the unique identifier (nested numbering scheme) encoded in the `source_id` (Bastian & Portell 2020). The first character in the string is a hexadecimal digit in the range 0 to B to identify the top level. The following 11 characters are quaternary digits in the range 0 to 3, each encoding the HEALPixel at the next level down that contains the source. The HEALPixel at level n (where $n \leq 12$) that contains the source is then simply the first n characters of this string.

HEALPIX29 : Level 29 HEALPix index for HATS (long)

A level–29 HEALPix spatial index for compliance with the Hierarchical Adaptive Tiling Scheme standard in the International Virtual Observatory Alliance – see `https://www.ivoa.net/documents/Notes/HATS/` for further information.

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1.2 SPURIOUS_SIGNAL [TAP]

This table contains a variety of statistics and models which can help in the identification of scan-angle related (spurious) signals. They have been computed for nearly all sources with published photometric and astrometric time series.

This table is an evolution of the DR3 'vari_spurious_signals' table associated with Holl et al. (2023) where it provided statistics for all photometric variable objects published in Gaia DR3.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

PHOT_G_N_OBS : Number of selected observations of G FoV transits, used in the photometry frequency fields (short)

Number of photometric FoV observations in the *G*-band.

The observations counted in this field are used in the computation of the various photometric frequency fields.

PHOT_G_FREQ : Frequency identified by Generalised Least Squares period search on G FoV timeseries (float, Frequency[day^{-1}])

Using the selected *G*-band FoV observations counted in `phot_g_n_obs` an unweighted periodogram is made using Generalised Least-Squares (Heck et al. 1985; Cumming et al. 1999; Zechmeister & Kürster 2009). This is an extension of the Fourier periodogram on uneven sam-

pled data that is independent of the mean of the data. The periodogram is computed between 25 cycles day^{-1} (about 1 h) and $7 \cdot 10^{-4}$ cycles day^{-1} (1700 d), with a step size of typically 10^{-5} cycles day^{-1}.

Value only provided when `phot_g_n_obs` ≥ 5.

PHOT_G_FREQ_AMPL : Normalised amplitude of the frequency identified by Generalised Least Squares period search on G FoV timeseries (float)

The normalised amplitude associated with `phot_g_freq`.

Value only provided when `phot_g_n_obs` ≥ 5.

PHOT_G_FREQ_SDE : Signal Detection Efficiency (SDE) of the frequency identified by Generalised Least Squares period search on G FoV timeseries (float)

The signal detection efficiency (SDE, see Alcock et al. 2000; Kovács et al. 2002) associated with `phot_g_freq`.

Value only provided when `phot_g_n_obs` ≥ 5.

PHOT_G_FREQ_FAP : Log10 of the False Alarm Probability (FAP) of the frequency identified by Generalised Least Squares period search on G FoV timeseries (float)

Log10 value of the Baluev false alarm probability (FAP, see Baluev 2008) associated with `phot_g_freq`.

Value only provided when `phot_g_n_obs` ≥ 5.

PHOT_G_N_OBS_COMMON_XP : Number of observations in common between FoV-G, BP and RP bands (short)

Number of photometric FoV observations in which all of the G, $G_{\rm BP}$ and $G_{\rm RP}$ were available.

The observations counted in this field are used in the computation of the field `phot_g_corr_exf`.

PHOT_G_CORR_EXF : G-band FoV photometry Spearman correlation with corrected flux excess factor (float)

Correlation statistic based on the Spearman correlation ρ between the G-band magnitude and excess flux C^* introduced in Distefano et al. (2023):

$$\rho = f_{\mathrm{SpearmanCorr}}\left(\{ G[\mathrm{mag}](i),\ C^*(i) \mid i \in 1, ..., N \ \} \right)$$

using the corrected excess flux C^* from Riello et al. (2021):

$$C^* = C - f(G_{\mathrm{BP}} - G_{\mathrm{RP}})$$

where $C = (I_{BP} + I_{RP})/I_G$ is the excess flux factor, I_{BP}, I_{RP} and I_G are the cumulative fluxes in the G_{BP}, G_{RP} and G bands, and $f(G_{\mathrm{BP}} - G_{\mathrm{RP}})$ is a correction function defined in [TO BE UPDATED FOR DR4!].

Values close to 1 can be due to a variety of intrinsic, extrinsic and calibration reasons as further detailed in Distefano et al. (2023):

- the star has strong emission lines whose intensity is correlated with G;
- the star is a partially resolved binary;
- the star has some calibration issue, e.g., due to the transition from a window-gate configuration to another.

Values around 0 can be considered 'well-behaved' in terms of excess flux.

Because of the dependency on availability of transit flux from all three G_{BP}, G_{RP} and G bands, only transits for which all three were available are used in the computation of `phot_g_corr_exf`. This number of used observations N (listed in `phot_g_n_obs_common_xp`) is occasionally quite low, i.e., less than ten. Empirically we see that correlations computed on at least 11 points could be useful, see Holl et al. (2023) for further details.

Value only provided when `phot_g_n_obs_common_xp` ≥ 6.

PHOT_G_N_OBS_EXCL_EPSL : Number of G-band FoV photometry observations excluding EPSL (short)

Number of G-band FoV photometric observations excluding EPSL, here defined as all observation after BJD 1693.14 (offset by 2455197.5).

The observations counted in this field are used in the computation of the `phot_g_corr_ipd` and photometric scan angle fields.

PHOT_G_CORR_IPD : G-band FoV photometry Spearman correlation with IPD model (float)

The IPD GoF model $M_{\mathrm{ipd}}(\psi)$ as detailed in the description of `ipd_gof_harmonic_amplitude` in `gaia_source` provides information about the phase and amplitude of a potential scan-angle dependent signal in the data. In order to quantify how much the derived G-band photometry is in reality affected by a scan-angle dependent signal, we can assess how much the photometry correlates in phase with this IPD model. To do so we compute the Spearman correlation ρ between the G-band time series and the IPD GoF model sampled at the scan angles of the time series observations:

$$\rho = f_{\mathrm{SpearmanCorr}}\left(\{ G_i[\mathrm{flux}](\psi_i),\ M_{\mathrm{ipd}}(\psi_i) \mid i \in 1, ..., N \ \} \right)$$

with i being the observation index in the time series of a source having a total of N observations (excluding observations during the EPSL), as listed in `phot_g_n_obs_excl_epsl`. ψ_i is the associated scan angle, which can be found in table `commanded_scan_law` for the time and FoV associated with each observation. Note that when using photometry expressed in magnitude instead of flux, the sign of the correlation should be flipped to produce the exact same `phot_g_corr_ipd`.

Values close to 1 generally show strong scan-angle dependent signals and should thus be treated with most care. Values around 0 are uncorrelated with the IPD GoF scan-angle model and thus should be free from any scan-angle dependent signals. Values close to -1 would have an anti-correlation with IPD GoF model and should thus also be treated with care but do not appear systematically in the data.

The number of observations N is occasionally quite low, i.e., less than ten. Empirically we see that correlations computed on at least 11 points could be useful. Value only provided when `phot_g_n_obs_excl_epsl` ≥ 6.

For further details on this parameter and how to use it see Holl et al. (2023).

PHOT_G_SCANANGLE_AMPL24 : Combined K2 and K4 component amplitude of the K024 scan angle model fit to G-band FoV photometry (float, Magnitude[mag])

TODO: add all details of the model and subsequent parameter fields provided.

PHOT_G_SCANANGLE_AMPL2 : Amplitude of the K2 component of the K024 scan angle model fit to G-band FoV photometry (float, Magnitude[mag])

See the detailed description of `phot_g_scanangle_ampl24` for more details.

PHOT_G_SCANANGLE_AMPL4 : Amplitude of the K4 component of the K024 scan angle model fit to G-band FoV photometry (float, Magnitude[mag])

See the detailed description of `phot_g_scanangle_ampl24` for more details.

PHOT_G_SCANANGLE_AMPL24_OVER_ERROR : Combined K2 and K4 component amplitude of the K024 scan angle model divided by its error (float)

See the detailed description of `phot_g_scanangle_ampl24` for more details.

PHOT_G_SCANANGLE_AMPL2_OVER_ERROR : Amplitude of the K2 component of the K024 scan angle model divided by its error (float)

See the detailed description of `phot_g_scanangle_ampl24` for more details.

PHOT_G_SCANANGLE_AMPL4_OVER_ERROR : Amplitude of the K4 component of the K024 scan angle model divided by its error (float)

See the detailed description of `phot_g_scanangle_ampl24` for more details.

PHOT_G_SCANANGLE_SIG24 : Significance of the combined K2 and K4 component of the K024 scan angle model fit to G-band FoV photometry (float)

See the detailed description of `phot_g_scanangle_ampl24` for more details.

PHOT_G_SCANANGLE_SIG2 : Significance of the K2 component of the K024 scan angle model fit to G-band FoV photometry (float)

See the detailed description of `phot_g_scanangle_ampl24` for more details.

PHOT_G_SCANANGLE_SIG4 : Significance of the K4 component of the K024 scan angle model fit to G-band FoV photometry (float)

See the detailed description of `phot_g_scanangle_ampl24` for more details.

PHOT_G_SCANANGLE_RED_CHI2 : Reduced Chi2 of the K024 scan angle model fit to G-band

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FoV photometry, used to rescale the covariances (float)

See the detailed description of `phot_g_scanangle_ampl24` for more details.

PHOT_G_SCANANGLE_F2 : F2 of the K024 scan angle model fit to G-band FoV photometry (float)

See the detailed description of `phot_g_scanangle_ampl24` for more details.

ASTRO_N_OBS : Number of selected astrometric along-scan observations (short)

Number of selected astrometric along-scan observations, only provided when ≥ 15.

The observations counted in this field are used in the computation of all the other astrometric fields.

ASTRO_CIRC_FREQ : Frequency identified by the circular orbital model periodogram on astrometric along-scan timeseries (float, Frequency[day^{-1}])

TODO description. Only provided when `astro_n_obs` ≥ 15.

ASTRO_CIRC_FREQ_FAP : Log10 of the False Alarm Probability (FAP) of the frequency identified by the circular orbital model periodogram (float)

See the detailed description of `astro_circ_freq` for more details.

ASTRO_CIRC_SCANANGLE_FREQ_OFFSET : Frequency offset of the circular orbit+K35 scan angle model periodogram, with respect to the circular-only model frequency (float, Frequency[day^{-1}])

See the detailed description of `astro_circ_freq` for more details.

ASTRO_CIRC_SCANANGLE_FREQ_FAP : Log10 of the False Alarm Probability (FAP) of the frequency identified by the circular orbit +K35 scan-angle model periodogram (float)

See the detailed description of `astro_circ_freq` for more details.

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ASTRO_SCANANGLE_AMPL35 : Combined K3 and K5 component amplitude of the K35 scan angle model fit to astrometric along-scan timeseries (float, Angle[mas])

TODO: add all details of the model and subsequent parameter fields provided. Only provided when `astro_n_obs` $\geq$ 15.

ASTRO_SCANANGLE_AMPL3 : Amplitude of the K3 component of the K35 scan angle model fit to astrometric along-scan timeseries (float, Angle[mas])

See the detailed description of `astro_scanangle_ampl35` for more details.

ASTRO_SCANANGLE_AMPL5 : Amplitude of the K5 component of the K35 scan angle model fit to astrometric along-scan timeseries (float, Angle[mas])

See the detailed description of `astro_scanangle_ampl35` for more details.

ASTRO_SCANANGLE_AMPL35_OVER_ERROR : Combined K3 and K5 component amplitude of the K35 scan angle model divided by its error (float)

See the detailed description of `astro_scanangle_ampl35` for more details.

ASTRO_SCANANGLE_AMPL3_OVER_ERROR : Amplitude of the K3 component of the K35 scan angle model divided by its error (float)

See the detailed description of `astro_scanangle_ampl35` for more details.

ASTRO_SCANANGLE_AMPL5_OVER_ERROR : Amplitude of the K5 component of the K35 scan angle model divided by its error (float)

See the detailed description of `astro_scanangle_ampl35` for more details.

ASTRO_SCANANGLE_SIG35 : Significance of the combined K3 and K5 component of the K35 scan angle model fit to astrometric along-scan timeseries (float)

See the detailed description of `astro_scanangle_ampl35` for more details.

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ASTRO_SCANANGLE_SIG3 : Significance of the K3 component of the K35 scan angle model fit to astrometric along-scan timeseries (float)

See the detailed description of `astro_scanangle_ampl35` for more details.

ASTRO_SCANANGLE_SIG5 : Significance of the K5 component of the K35 scan angle model fit to astrometric along-scan timeseries (float)

See the detailed description of `astro_scanangle_ampl35` for more details.

ASTRO_SCANANGLE_RED_CHI2 : Reduced Chi2 of the K35 scan angle model fit to astrometric along-scan timeseries, used to rescale the covariances (float)

See the detailed description of `astro_scanangle_ampl35` for more details.

ASTRO_SCANANGLE_F2 : F2 of the K35 scan angle model fit astrometric along-scan timeseries (float)

See the detailed description of `astro_scanangle_ampl35` for more details.

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2 All–source catalogues

As indicated in the below subsection titles by the labels [TAP] and [DataLink], these products are served from the ESA Gaia Archive using two different protocols, the Table Access Protocol (TAP) for tabular products and the DataLink protocol for other products such as spectra, light curves, MCMC samples, etc. For more details please see Section **??**.

2.1 ALL_SOURCE_ASTROMETRY [TAP]

This table contains the astrometric parameters obtained for nearly all sources occurring in DR4. The sole exceptions are the sources only recognized in the special crowded-fields processing (see the crowded field source and related tables), and the sources only recognized in the special Source Environment Analysis (SEApipe).

This table is complemented by `all_source_photometry`, `all_source_rvs` and `all_source_flags`, which provide the corresponding information on the photometry, radial velocity (and related parameters), and a variety of flags of interest.

The astrometric data in this table is not homogeneous in the sense that it can originate from different processing modules, and even from different processing cycles. While the majority of it stems from AGIS processing over the full 66 months of DR4 data, a small number of sources contain astrometry obtained from other processing modules and/or are based on the data acquired on the shorter time baselines of DR1, DR2 or DR3. Finally, a very small number of sources will have neither of these available, and will fall back to the IGSL astrometry. The information about the origin of the astrometry is given in field `astro_pipeline`.

The described fall-back scheme is necessitated by the decision to publish epoch astrometry for all sources detected on board by Gaia, regardless of their fate in the subsequent data reduction chain. This decision, in turn, is motivated by the desire not to lose scientifically valuable astrophysical transients detected by Gaia only once or twice. While the provision of epoch astrometry via DataLink needs a reference `source_id` in some source-oriented table, the data processing for DR4 has not produced an astrometric solution for all detected sources.

If, for instance, all detections of a source fall into one of the AGIS gaps (i.e. time intervals not usable for the high-precision astrometric adjustment), they may fail to get an AGIS solution. On the other hand, the IDU processing for DR4 has produced astrometric parameters only for new sources, i.e. those sources that had not yet been recognised for DR3

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or earlier. This situation will be avoided for DR5, so that no fallback to earlier processing cycles will be needed in DR5.

The sources only recognized in the crowded-field processing need not be included in `all_source_astrometry`, as they have their dedicated source table and epoch astrometry table. Similarly, the SEApipe sources need not be included — in this case because there will be no epoch astrometry at all.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Source Identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

DESIGNATION : Unique source designation (unique across all Data Releases) (string)

A source designation, unique across all Gaia Data Releases, that is constructed from the prefix 'Gaia DRx ' followed by a string of digits corresponding to `source_id` (3 space–separated words in total). Note that the integer source identifier `source_id` is **not** guaranteed to be unique across Data Releases; moreover it is not guaranteed that the same astronomical source will always have the same `source_id` in different Data Releases. Hence the only safe way to compare source records between different Data Releases in general is to check the records of proximal source(s) in the same small part of the sky.

QUALITY_FLAG : Bitwise code indicating the quality of the astrometry of this source (byte)

This field indicates whether the source astrometry is considered of high or low confidence. It is set as a bitwise code mapping the different reasons for which the source astrometry could be considered of a lower level of confidence. A value of 0 therefore means that the source astrometry is considered of high confidence. Note, however, that a non-zero value of this flag

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does not necessarily implies that the source was not eligible for `gaia_source` because it could well be that a low confidence astrometry in `all_source_astrometry` gets superseded by a higher quality one in `gaia_source`.

- bit 0 (LSB): set to 1 if the semi-major axis of the error ellipse in position is >100 mas, and to 0 otherwise.
- bit 1: set to 1 if the source astrometry stems from an older processing cycle of AGIS, or from the IDU processing, whichever cycle it applies to. This happens when the AGIS solution for DR4 was non-converged, or was not attempted at all.
- bit 2: set to 1 if the number of field-of-view transits used by AGIS in DR4 is <5, and to 0 otherwise.
- bit 3: set to 1 if the source was considered a loser of a duplication scenario (i.e. `source_id` != `astrometric_duplicate_source_id`).
- bit 4: set to 1 if the source is considered a spurious candidate. The spurious categories applying to the source are given in `spurious_class`.

SPURIOUS_CLASS : Bitwise code indicating the spurious category for this source (byte)

This field flags whether the `source_id` of interest was classified as potential spurious candidate, and indicates in which of the spurious categories it was identified. When this flag is set to 0, the source is not considered a spurious candidate. The following coding is used:

- bit 0 (LSB): set to 1 for sources with no valid observations available in AGIS due to attitude gaps.
- bit 1: set to 1 for sources affected by processing issues i.e. XM code limitations, DIPD spurious detections, etc.
- bit 2: set to 1 for sources with no AGIS solution in DR4. Note that this bit will be redundant with bit 1 from `quality_flag`
- bit 3: set to 1 for sources with just one observation (i.e. one on-board detection) and without on-ground detection solution in IDU.
- bit 4: set to 1 for sources falling over the spike patterns around a bright source.
- bit 5: set to 1 for sources having all their matches associated with SSO transits.

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UNRELIABLE_ASTROMETRY : Unreliable AGIS astrometry (boolean)

An unreliable AGIS solution is defined as follows:

It is in `all_source_astrometry` with `astro_pipeline` = 3 and `quality_flag` = 0 or it fulfils any of the following conditions:

- R > 1.2
- `visibility_periods_used` < max(10, min(25, 5 + 40 * R))
- `duplicated_source` & !`blended` & abs(`parallax`) > 90 & abs(`ecl_lat`) < 15

where:

R = `astrometric_sigma5d_max` / gamma4(`phot_g_mean_mag`),

gamma4(G) = min(10, pow(10.0, 0.2 * max(6 - G, 0, G - 18)))

The flag is set to 'true' for 440 million sources in this table.

For another 958 million sources in this table, the value of `unreliable_astrometry` is set to null. Among these are the 865 million 2-parameter solutions, about 20 million 6-parameter solutions, and the 73 million without an AGIS solution. The reason in all cases is that some of the defining quantitites are unavailable.

RANDOM_INDEX_AS : Random index for use when selecting subsets (long)

A random index which can be used to select smaller subsets of the data that are still representative. The column contains a random permutation of the numbers from 0 to $N - 1$, where N is the number of sources in the catalogue. Please note that the random index in the four 'all source' tables is the same, i.e. the same `source_id` in `all_source_astrometry`, `all_source_photometry`, `all_source_flags` and `all_source_rvs` will have the same `random_index_a` (while noting that `all_source_rvs` contains fewer sources than N). However this is not so for `random_index` in `gaia_source` which has it's own assigned random index values.

The random index can be useful for validation (testing on 10 different random subsets), visualization (displaying 1% of the data), and statistical exploration of the data, without the need to download all the data.

IN_GAIA_SOURCE : Flag indicating presence in the main Gaia source catalogue (boolean)

This flag indicates that this source was eligible to be tabulated in `gaia_source`.

ASTRO_PIPELINE : Pipeline module that provided the astrometry for this source (short)

This field indicates the pipeline module that provided the astrometry for the source of interest. If no AGIS solution is available from the on-going cycle processing (i.e. cycle no. 4 for Gaia DR4), the astrometry will be populated by that of either an earlier version of AGIS, or that of another module (IDU), which can also be of an earlier processing cycle. Finally, a very small number of sources will have neither of these available, and will fall back to the IGSL (Initial Gaia Source List) astrometry (Gaia Archive table `public.igsl_source`; see Smart & Nicastro (2014)).

- Values 0, 1, 2, 3 indicate that the astrometry comes from AGIS cycle 1, 2, 3, 4 resp.
- Values 5, 6, 7, 8 indicate that the astrometry comes from IDU cycle 1, 2, 3, 4 resp.
- Value 10 indicates that the astrometry comes from IGSL

For more remarks on this fall-back scheme see the table description.

This field should not be confused with field `gaia_source.astrometry_origin` in table `gaia_source`, which informs about the processing chain that provided the astrometry in that table. In particular, the present table `all_source_astrometry` does not invoke any DPAC processing pipelines other than AGIS and IDU.

REF_EPOCH : Reference epoch (double, Time[Julian Years])

Reference epoch to which the astrometric source parameters are referred, expressed as a Julian Year in TCB.

RA : Right ascension (double, Angle[deg])

Barycentric right ascension α of the source in ICRS at the reference epoch `ref_epoch`.

RA_ERROR : Standard error of right ascension (float, Angle[mas])

Standard error $\sigma_{\alpha*} \equiv \sigma_\alpha \cos\delta$ of the right ascension of the source in ICRS at the reference epoch `ref_epoch`.

DEC : Declination (double, Angle[deg])

Barycentric declination δ of the source in ICRS at the reference epoch `ref_epoch`.

DEC_ERROR : Standard error of declination (float, Angle[mas])

Standard error σ_δ of the declination of the source in ICRS at the reference epoch `ref_epoch`.

PARALLAX : Parallax (double, Angle[mas])

Absolute stellar parallax ϖ of the source at the reference epoch `ref_epoch`.

PARALLAX_ERROR : Standard error of parallax (float, Angle[mas])

Standard error σ_ϖ of the stellar parallax at the reference epoch `ref_epoch`.

PARALLAX_OVER_ERROR : Parallax divided by its standard error (float)

Parallax divided by its standard error.

PMRA : Proper motion in right ascension direction (double, Angular Velocity[mas yr^{-1}])

Proper motion in right ascension $\mu_{\alpha*} \equiv \mu_\alpha \cos\delta$ of the source in ICRS at the reference epoch `ref_epoch`. This is the local tangent plane projection of the proper motion vector in the direction of increasing right ascension.

PMRA_ERROR : Standard error of proper motion in right ascension direction (float, Angular Velocity[mas yr^{-1}])

Standard error $\sigma_{\mu\alpha*}$ of the local tangent plane projection of the proper motion vector in the direction of increasing right ascension at the reference epoch `ref_epoch`.

PMDEC : Proper motion in declination direction (double, Angular Velocity[mas yr^{-1}])

Proper motion in declination μ_δ of the source at the reference epoch `ref_epoch`. This is the projection of the proper motion vector in the direction of increasing declination.

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PMDEC_ERROR : Standard error of proper motion in declination direction (float, Angular Velocity[mas yr^{-1}])

Standard error $\sigma_{\mu\delta}$ of the proper motion component in declination at the reference epoch `ref_epoch`.

PM : Total proper motion (float, Angular Velocity[mas yr^{-1}])

The total proper motion `pm`, calculated as the magnitude of the resultant vector of the proper motion component vectors `pmra` and `pmdec`, i.e. `pm`^{2} = `pmra`^{2} + `pmdec`^{2}.

RA_DEC_CORR : Correlation between right ascension and declination (float)

Correlation coefficient $\rho(\alpha, \delta)$ between right ascension and declination. This is a dimensionless quantity in the range [-1,+1].

RA_PARALLAX_CORR : Correlation between right ascension and parallax (float)

Correlation coefficient $\rho(\alpha, \varpi)$ between right ascension and parallax, a dimensionless quantity in the range [-1,+1].

RA_PMRA_CORR : Correlation between RA and proper motion in RA (float)

Correlation coefficient $\rho(\alpha, \mu_{\alpha*})$ between right ascension and proper motion in right ascension, a dimensionless quantity in the range [-1,+1].

RA_PMDEC_CORR : Correlation between right ascension and proper motion in declination (float)

Correlation coefficient $\rho(\alpha, \mu_{\delta})$ between right ascension and proper motion in declination, a dimensionless quantity in the range [-1,+1].

DEC_PARALLAX_CORR : Correlation between declination and parallax (float)

Correlation coefficient $\rho(\delta, \varpi)$ between declination and parallax, a dimensionless quantity in the range [-1,+1].

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DEC_PMRA_CORR : Correlation between declination and proper motion in right ascension (float)

Correlation coefficient $\rho(\delta, \mu_{\alpha*})$ between declination and proper motion in right ascension, a dimensionless quantity in the range [-1,+1].

DEC_PMDEC_CORR : Correlation between declination and proper motion in declination (float)

Correlation coefficient $\rho(\delta, \mu_{\delta})$ between declination and proper motion in declination, a dimensionless quantity in the range [-1,+1].

PARALLAX_PMRA_CORR : Correlation between parallax and proper motion in right ascension (float)

Correlation coefficient $\rho(\varpi, \mu_{\alpha*})$ between parallax and proper motion in right ascension, a dimensionless quantity in the range [-1,+1].

PARALLAX_PMDEC_CORR : Correlation between parallax and proper motion in declination (float)

Correlation coefficient $\rho(\varpi, \mu_{\delta})$ between parallax and proper motion in declination, a dimensionless quantity in the range [-1,+1].

PMRA_PMDEC_CORR : Correlation between proper motion in RA and proper motion in Dec (float)

Correlation coefficient $\rho(\mu_{\alpha*}, \mu_{\delta})$ between proper motion in right ascension and proper motion in declination, a dimensionless quantity in the range [-1,+1].

ASTROMETRIC_N_OBS_AL : Total number of observations in the along-scan (AL) direction (short)

Total number of AL observations (= CCD transits) used in the astrometric solution of the source, independent of their weight. Note that some observations may be strongly downweighted and their number can be derived from `astrometric_n_obs_al` - `astrometric_n_good_obs_al`.

ASTROMETRIC_N_OBS_AC : Total number of observations in the across-scan (AC) direction (short)

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Total number of AC observations (= CCD transits) used in the astrometric solution of the source, independent of their weight (note that some observations may be strongly downweighted). Nearly all sources having G < 13 will have AC observations from 2D windows, while fainter than that limit only ∼ 1% of transit observations (known as 'calibration faint stars') are assigned 2D windows resulting in AC observations.

ASTROMETRIC_N_GOOD_OBS_AL : Number of good observations in the along-scan (AL) direction (short)

Number of AL observations (= CCD transits) that were not strongly downweighted in the astrometric solution of the source. Strongly downweighted observations (with downweighting factor $w < 0.2$) can instead be derived from `astrometric_n_obs_al` - `astrometric_n_good_obs_al`, where `astrometric_n_obs_al` is the total number of AL observations used in the astrometric solution of the source.

ASTROMETRIC_GOF_AL : Goodness of fit statistic of model wrt along-scan observations (float)

Goodness-of-fit statistic of the astrometric solution for the along-scan measurements of the source. This is a rescaled version of the 'gaussianized chi-square', which for good fits should approximately follow a normal distribution with zero mean value and unit standard deviation. Values exceeding, say, +3 thus indicate a significantly bad fit to the data.

Starting with DR4, `astrometric_gof_al` is probably more useful than `ruwe` as a general goodness-of-fit measure, because it takes into account the dependence on the degree of freedom and has a simple statistical interpretation.

Note that the `astrometric_gof_al` is not simply the 'gaussianized' version of the original `astrometric_chi2_al` but an empirically rescaled version. The rescaling procedure is described and motivated in detail in **?** where an explanation and discussion of other astrometric quality indicators is provided.

ASTROMETRIC_CHI2_AL : AL chi-square value (float)

Weighted sum of the squared astrometric residuals (χ^2) in the AL direction.

χ^2 values were computed using the 'good' AL observations of the source, without taking into account the `astrometric_excess_noise` (if any) of the source. They do however take into account the calibration excess noise (if any) of each observation.

See **?** for a discussion of other astrometric quality indicators.

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MEAN_PARALLAX_FACTOR_AL : Mean Parallax factor AL (float)

Mean parallax factor in the AL direction, computed from all the good observations of the source processed in the astrometry.

The AL parallax factor for an individual observation is defined as $\partial\eta/\partial\varpi$, where η is the AL field angle of the source and ϖ its parallax, and is constrained to $[-0.73, +0.73]$ by the scanning law. The value given in this field is typically in the range $[-0.23, +0.31]$ (10th and 90th percentiles). A value outside this range indicates a distribution of observations that is unfavourable for the determination of the parallax, and the calculated parallax could then be more vulnerable to errors, e.g. from the calibration model, not reflected in the formal uncertainties. See **?** for a discussion of other astrometric quality indicators.

ASTROMETRIC_EXCESS_NOISE : Rescaled excess noise of the source (float, Angle[mas])

This is the rescaled source excess noise ϵ_i of the source. It is not the same as the source excess noise in Gaia DR3 and earlier releases, but re-scaled in order to be consistent with the (also redefined) unit weight error (`ruwe`). But, like the original quantity in earlier releases, it measures the disagreement, expressed as an angle, between the observations of a source and the fit to the standard six-parameter astrometric model. The assumed observational noise in each observation is quadratically increased by ϵ_i in order to statistically match the residuals in the astrometric solution.

A value of $\epsilon_i = 0$ thus signifies that the source is astrometrically well-behaved, i.e. that the residuals of the fit statistically agree with the assumed observational noise. Conversely, a positive ϵ_i signifies that the residuals are statistically larger than expected. This may happen for sources that deviate from the standard six-parameter astrometric model (e.g. unresolved binaries, exoplanet systems, etc.). The statistical significance of the excess noise is given by `astrometric_gof_al`.

Although the rescaling makes ϵ_i much less sensitive to instrumental effects than in previous releases, it may still absorb residual modelling errors not accounted for by the observational noise (image centroiding error) and the calibration excess noise. This includes LSF and PSF calibration errors, saturation effects, geometric instrument calibration errors, and part of the high-frequency attitude noise. Such modelling errors may be particularly important for very bright sources and sources of extreme extreme colours. Given the many other possible contributions to the excess noise, the user is advised to study the empirical distributions of ϵ_i and `astrometric_gof_al` for their particular samples in order to make sensible choices of filtering limits.

The source excess noise is further explained in **?** and in Section **??** of the DR4 online documentation.

ASTROMETRIC_PARAMS : Bitwise code describing which astrometric parameters are provided

(short)

The bit–wise encoding in `astrometric_params` indicates which parameters have been estimated in AGIS (primary or secondary solution) and exported for this source. A set bit means the parameter was updated, an unset bit means the parameter was not updated. The least-significant bit corresponds to α. The table below shows the values for relevant combinations of the parameters.

The radial proper motion (μ_r) is formally considered to be one of the astrometric parameters of a source, and the sixth bit is therefore reserved for it. It is also in principle updatable in AGIS, but in practice it will always be computed from a spectroscopic radial velocity and the estimated parallax, in which case the bit is not set.

C is the pseudo-colour of the source, i.e. the astrometrically estimated effective wavenumber.

`astrometric_params`	α	δ	ϖ	$\mu_{\alpha*}$	μ_δ	μ_r	C	$\dot{\mu}_{\alpha*}$	$\dot{\mu}_\delta$	$\ddot{\mu}_{\alpha*}$	$\ddot{\mu}_\delta$
$000000000000_2 = 0$											
$000000000011_2 = 3$	✓	✓									
$000001011111_2 = 95$	✓	✓	✓	✓	✓		✓				

TENTATIVE_PARALLAX_BIAS : Parallax bias correction (double, Angle[mas])

This is the parallax bias correction computed based on the recipe in **?**.

This correction is to be **subtracted** from `parallax` to get the corrected parallax.

BLENDED : Source update flag indicating use of blended transits (byte)

This field indicates whether blended transits have been used in the astrometric solution:

- when set to 0, it indicates that no blended transit was used.
- when set to 1, it indicates that only blended transits have been used.
- when set to 2, it indicates that a mix of non-blended and blended transits has been used. This happens when neither the number of non-blended nor the number of blended transits are high enough to attempt a six-parameter source update.

A blended transit occurs when in an observation of a known close source pair (or close mutiple source) the scanning direction is such that the individual members of the pair (or multiple) are not separated in the CCD window.

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For more details see Section 4.3 of the DR4 astrometry paper, **?**.

ASTROMETRIC_PRIMARY_FLAG : Primary or secondary (boolean)

Flag indicating if this source was used as a primary source (`true`) or secondary source (`false`). Only primary sources contribute to the estimation of attitude, calibration, and global parameters. The estimation of source parameters is otherwise done in exactly the same way for primary and secondary sources.

RV_USED_IN_ASTROMETRY : Assumed radial velocity used in the astrometric solution (float, Velocity[km s^{-1}])

This value is the one used to account for possible perspective acceleration effects. This correction is applied even when the expected perspective acceleration is negligible.

Due to cyclic processing of the astrometry and the RVS data, this radial velocity might be different from the one computed using the latest available RVS information, and published under `all_source_rvs.radial_velocity_single`. Moreover, if no reliable radial velocity was available at the time of the astrometric processing, radial velocities from the literature have been used when relevant, and their value is reported in `rv_used_in_astrometry`. In all other cases, zero radial velocity was assumed.

NU_EFF_USED_IN_ASTROMETRY : Effective wavenumber of the source used in the astrometric solution (float, Misc[μm^{-1}])

This effective wavenumber $\nu_{\rm eff}$, in units of $\mu{\rm m}^{-1}$, is the assumed value used for the selection of the appropriate LSF/PSF model in the image parameter determination, as well as for the selection of the astrometric calibration parameters in the astrometric adjustment, for the given source. If no BP and RP photometry of the given source was available, `nu_eff_used_in_astrometry` is set to the default value $\nu_{\rm eff} = 1.43\mu{\rm m}^{-1}$. Else, if reliable BP and RP photometry was available, it is set to the weighted mean of the default value and the actual photon-flux weighted inverse wavelength as estimated from the BP and RP bands. If highly precise BP and RP data are available, the `nu_eff_used_in_astrometry` is very close to the value from the BP and RP bands. However, due to the cyclic processing of the Gaia astrometry and photometry, this effective wavenumber will even in these cases be (slightly) different from the one computed using the latest available photometry for DR4.

Concerning the selection of the default value and the construction of the weighted mean: the default value is chosen as the ensemble average of the actually measured values for all Gaia sources, and its standard error for the weighted mean is set to $0.085\mu{\rm m}^{-1}$, which is close to the

actual rms scatter of the measured values; for the measured values, the formal errors assigned by the photometry processing were used in the construction of the weighted mean. However, the original values delivered by CU5 (from the DR3 photometry processing) were found to be strongly overestimated. Therefore, they were rescaled to give realistic values before being used for the weighted mean.

PSEUDOCOLOUR : Astrometrically estimated pseudocolour of the source (float, Misc[μm^{-1}])

Astrometrically determined colour of the source, represented as an effective wavelength in μm^{-1}. This quantity can be used as a proxy for the actual effective wavenumber (i.e. broad-band colour) of the source if the latter cannot be computed from CU5 photometry. It is estimated in AGIS from the chromatic displacements of image centroids. Technically, it is the sum of `nu_eff_used_in_astrometry` and a correction to that value derived by AGIS.

PSEUDOCOLOUR_ERROR : Standard error of the pseudocolour of the source (float, Misc[μm^{-1}])

Standard error $\sigma_{\texttt{pseudocolour}}$ of the astrometrically determined `pseudocolour` of the source.

RA_PSEUDOCOLOUR_CORR : Correlation between right ascension and pseudocolour (float)

Correlation coefficient $\rho(\alpha, \texttt{pseudocolour})$ between right ascension `ra` and `pseudocolour`, a dimensionless quantity in the range [-1,+1]

DEC_PSEUDOCOLOUR_CORR : Correlation between declination and pseudocolour (float)

Correlation coefficient $\rho(\delta, \texttt{pseudocolour})$ between declination `dec` and `pseudocolour`, a dimensionless quantity in the range [-1,+1]

PARALLAX_PSEUDOCOLOUR_CORR : Correlation between parallax and pseudocolour (float)

Correlation coefficient $\rho(\varpi, \texttt{pseudocolour})$ between `parallax` and `pseudocolour`, a dimensionless quantity in the range [-1,+1]

PMRA_PSEUDOCOLOUR_CORR : Correlation between proper motion in right asension and pseudocolour (float)

Correlation coefficient $\rho(\mu_{\alpha*}, \texttt{pseudocolour})$ between proper motion in right ascension `pmra` and `pseudocolour`, a dimensionless quantity in the range [-1,+1]

PMDEC_PSEUDOCOLOUR_CORR : Correlation between proper motion in declination and pseudocolour (float)

Correlation coefficient $\rho(\mu_\delta, \texttt{pseudocolour})$ between proper motion in declination `pmdec` and `pseudocolour`, a dimensionless quantity in the range [-1,+1]

ASTROMETRIC_MATCHED_TRANSITS : Matched FOV transits used in the AGIS solution (short)

The number of field–of–view transits matched to this source, counting only the transits containing CCD observations actually used to compute the astrometric solution.

This number will always be equal to or smaller than `matched_transits`, the difference being the FOV transits that were not used in the astrometric solution because of bad data or excluded time intervals.

VISIBILITY_PERIODS_USED : Number of visibility periods used in Astrometric solution (short)

Number of visibility periods used in the astrometric solution.

A visibility period is a group of observations separated from other groups by a gap of at least 4 days. A source may have from one to tens of field-of-view transits in a visibility period, but with a small spread in time, direction of scanning, and parallax factor. From one visibility period to the next these variables have usually changed significantly. A high number of visibility periods is therefore a better indicator of an astrometrically well-observed source than a large number of field-of-view transits (`matched_transits` or `astrometric_matched_transits`) or CCD observations (`astrometric_n_obs_al`). A small value (e.g. less than 10) indicates that the calculated parallax could be more vulnerable to errors, e.g. from the calibration model, not reflected in the formal uncertainties. See **?** for a discussion of this and other astrometric quality indicators.

ASTROMETRIC_SIGMA5D_MAX : The longest semi-major axis of the 5-d error ellipsoid (float, Angle[mas])

The longest principal axis in the 5-dimensional error ellipsoid.

This is a 5-dimensional equivalent to the semi-major axis of the position error ellipse and is therefore useful for filtering out cases where one of the five parameters, or some linear combination of several parameters, is particularly ill-determined. It is measured in mas and computed as the square root of the largest singular value of the scaled 5×5 covariance matrix of the astrometric parameters. The matrix is scaled so as to put the five parameters on a comparable scale, taking into account the maximum along-scan parallax factor for the parallax and the time coverage of

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the observations for the proper motion components. If C is the unscaled covariance matrix, the scaled matrix is SCS, where $S = \mathrm{diag}(1, 1, \sin\xi, T/2, T/2)$, $\xi = 45°$ is the solar aspect angle in the nominal scanning law, and T the time coverage of the data used in the solution.

`astrometric_sigma5d_max` is given for all the solutions, as its size is one of the criteria for accepting or rejecting the 6-parameter solution. In case of a 2-parameter solution (`astrometric_params` = 3) it gives the value for the rejected 6-parameter solution, and can then be arbitrarily large.

MATCHED_TRANSITS : Number of transits matched to this source (short)

Total number of field-of-view transits matched to this source. The number includes blended transits and transits where the AF images could not be processed.

This count was made at the cross-match stage and based on early AF image processing in IDU, which may differ slightly from the final image processing.

MATCHED_MULTI_PEAK_TRANSITS : Number of transits matched having more than one DIPD peak (short)

Number of transits where this source was detected on ground as either a blended or a resolved peak, but only counting transits where more than one peak was detected.

This count was made at the cross-match stage and based on early AF image processing in IDU, which may differ slightly from the final image processing.

MATCHED_SEMESTERS : Number of semesters having at least one peak matched (byte)

This field indicates the constancy (timewise) in the detection of this source throughout the time range covered by the applicable Data Release.

In DR4 (which covers 66 months), the ideal value here would be 11, meaning that an onboard or onground detection for this source exists, at least, once every 6 months through the whole time span). In DR5, the ideal value will be 21, as it will cover 10.5 years.

Some sources may have fewer matched semesters, which may happen for very faint sources (e.g. being detected or not depending on the instrumental response at a given time), for sources with some extension (e.g. being detected only before some changes were done in the on-board VPU detection parameters), or more interestingly, sources with astrophysical changes: supernovae, long-interval cataclysmics, GRBs, FRBs, NS mergers, long-period variables close to the detection limit, short-period resolved binaries with a wrong crossmatch solution, etc. Some

sources with just one matched semester may be spurious, but on the other hand, if they have several matches (at least around 5) during that semester, it may actually be an interesting astrophysical transient.

The semesters are determined from the `transit_id` of the matched peaks, obtaining the corresponding OBMT, and from that, then compute the semester. It is calculated since the start of nominal operations (OBMT rev. 1078.0), then truncating it to the lowest integer value. More specifically, we count from OBMT 23292998211919880 (in nanoseconds), or 1078.379547 (in revolutions), or 2014-07-25T10:29:59.99 (in UTC). From this, each semester adds 15778800000000000 nanoseconds (730.5 revolutions).

WHICH_MATCHED_SEMESTERS : Bitwise info on which semesters have at least one matched peak (int)

This is a bitwise value, with 11 (DR4) or 21 (DR5) meaningful bits (starting from the least-significant bit). Each bit indicates whether that semester had a peak matched to this source. See `matched_semesters` for some details. Specifically:

- If at least one of the `matched_transits` of the source has its epoch during the first semester, the least-significant bit (LSB, i.e., bit 0) will be 1. Otherwise (if none of the `matched_transits` has its epoch during the first semester), it will be 0.
- If the source has at least one of the `matched_transits` during the second semester, bit 1 will be 1; if there is none during that semester, it will be 0.

And so on, up to the last semester. If there is at least one of the `matched_transits` in the last semester, bit 10 (in DR4) will be 1, else it will be 0. The following are the exact time ranges of the 11 semesters in DR4:

Semester	Start (OBMT)	Start (rev.)	Start (UTC)
0	23292998211919880	1078.379547	2014-07-25T10:29:59.99
1	39071798211919880	1808.879547	2015-01-24T01:29:59.97
2	54850598211919880	2539.379547	2015-07-25T16:29:58.96
3	70629398211919880	3269.879547	2016-01-24T07:29:58.93
4	86408198211919880	4000.379547	2016-07-24T22:29:58.91
5	102186998211919880	4730.879547	2017-01-23T13:29:57.89
6	117965798211919880	5461.379547	2017-07-25T04:29:57.86
7	133744598211919880	6191.879547	2018-01-23T19:29:57.84
8	149523398211919880	6922.379547	2018-07-25T10:29:57.81
9	165302198211919880	7652.879547	2019-01-24T01:29:57.78
10	181080998211919880	8383.379547	2019-07-25T16:29:57.75
11	196859798211919880	9113.879547	2020-01-24T07:29:57.72

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Each user should determine how to best use this field. Just to give some examples (in DR4):

- If `which_matched_semesters` has a very low value such as 63 or less, the source was matched only during the first half of the mission.
- If it takes a value of 1024, it had a match only during the last semester.
- If it takes a value of 1536, just in the last 2 semesters. If it is 1792, then only in the last 3 semesters.
- A value of 2046 would mean detections in all of the semesters except the first one.
- With some simple function, it is also possible to get the total time span (with a resolution of one semester) during which a given source was observed by Gaia.

The ideal case in DR4 would be a value of 2047, meaning a match in all of the semesters.

VERIFIED_TRANSITS : Number of processable transits matched to this source (short)

Number of field of view transits where AF image processing identified a peak matched to this source. This is a subset of `matched_transits`. Note that a peak may be matched to more than one source, cf. `blended_transits`.

This count was made at the cross-match stage and based on early AF image processing in IDU, which may differ slightly from the final image processing.

BLENDED_TRANSITS : Number of transits where this source blended with another source (short)

Number of field of view transits where an image peak was matched to this source and also to another source. This is a subset of `verified_transits`.

This count was made at the cross-match stage and based on early AF image processing in IDU, which may differ slightly from the final image processing.

MATCHED_TRANSITS_SSO : Number of transits that are also matched to a Solar System Object (short)

This field indicates the number of detected transits that have been matched to a given source during the last cross-match revision, and that have also been considered as matches to a Solar-System Object as tabulated in `sso_source`.

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IPD_GOF_HARMONIC_AMPLITUDE : Amplitude of the IPD GoF versus position angle of scan (float)

This statistic measures the amplitude of the variation of the Image Parameter Determination (IPD) goodness–of–fit (GoF; reduced chi-square) as function of the position angle of the scan direction. A large amplitude indicates that the source has some non-isotropic spatial structure, for example a binary or galaxy, that is at least partially resolved by Gaia. The phase of the variation is given by the parameter `ipd_gof_harmonic_phase`.

Let ψ be the position angle of the scan direction. The following expression is fitted to the IPD GoF for the accepted AF observations of the source:

$$\ln(\mathrm{GoF}) = c_0 + c_2 \cos(2\psi) + s_2 \sin(2\psi)$$

The amplitude and phase of the variation are calculated as

$$\texttt{ipd_gof_harmonic_amplitude} = \sqrt{c_2^2 + s_2^2}$$

$$\texttt{ipd_gof_harmonic_phase} = \frac{1}{2}\,\mathrm{atan2}(s_2, c_2) \quad (+\,180^\circ)$$

where atan2 returns the angle in degrees. In the last expression 180 is added for negative values, so that `ipd_gof_harmonic_phase` is always between 0 and 180°. Only the AF observations accepted by the astrometric solution are used to compute the amplitude and phase, thus for example outliers and observations in the early Ecliptic Pole Scanning Law phase are not used.

The GoF variation is modelled as a periodic function of 2ψ because a source with fixed structure is normally expected to give fits of similar quality when scanned in opposite directions (ψ differing by 180°). See `ipd_gof_harmonic_phase` for the interpretation of the phase.

IPD_GOF_HARMONIC_PHASE : Phase of the IPD GoF versus position angle of scan (float, Angle[deg])

This statistic measures the phase of the variation of the IPD GoF (reduced chi-square) as function of the position angle of the scan direction. See the description of `ipd_gof_harmonic_amplitude` for details on the computation of the phase.

The interpretation of this parameter is non-trivial because of the complex interaction between the source structure and the IPD. At least the following different scenarios could occur:

- For a binary with separation $\lesssim 0.1$ arcsec the GoF is expected to be higher when the scan is along the arc joining the components than in the perpendicular direction, in

which case `ipd_gof_harmonic_phase` should indicate the position angle of the binary modulo 180°. Such a binary will normally have negligible `ipd_frac_multi_peak` (less than a few per cent).

- For a resolved binary the GoF may instead have a minimum when the scan is along the arc joining the two components, in which case `ipd_gof_harmonic_phase` differs from the position angle of the binary (modulo 180°) by approximately ±90°. Such a binary will normally have a large `ipd_frac_multi_peak`.
- For a bright binary ($G \lesssim 13$) the GoF refers to the fitting of a two-dimensional PSF, which could further complicate the interpretation.
- For a galaxy with elongated intensity distribution, the IPD may give a smaller GoF when the scan is along the major axis of the image, resulting in an offset of approximately ±90° between the `ipd_gof_harmonic_phase` and the position angle of the major axis (modulo 180°).

IPD_FRAC_MULTI_PEAK : Percent of successful-IPD windows with more than one peak (byte)

This field provides information on the raw windows used for the astrometric processing of this source coming from the Image Parameters Determination (IPD) module in the core processing. It provides the fraction of windows (having a successful IPD result), as percentage (from 0 to 100), for which the IPD algorithm has identified a double peak, meaning that the source may be a visually resolved double star (either just visual double or real binary). The quantity was computed using all transits where the IPD was successful.

Note: Some transits with more than one peak are produced by superposition of unrelated sources from the other field of view.

IPD_FRAC_ODD_WIN : Percent of transits with truncated windows or multiple gate (byte)

This field is calculated during AGIS and provides information on the raw windows used for the astrometric processing of this source. It provides the fraction (as a percentage, from 0 to 100) of transits having either truncation or multiple gates flagged in one or more windows. Such a situation invariably means that the on-board video processing unit (VPU) detected some nearby source (which may be just a spurious detection, but typically could be some real nearby source having another distinct transit and most probably assigned to a different source). So in general a non-zero fraction indicates that this source may be contaminated by another nearby source. The quantity was computed using all transits where the IPD was successful.

Note: Some odd-window transits are produced by superposition of unrelated sources from the other field of view.

Draft

IPD_FRAC_HIGH_GOF : Percent of successful-IPD windows with only one peak but high GoF (byte)

This field provides information (coming from IDU IPD) on the raw windows used for the astrometric processing of this source. It provides the fraction of windows (as percentage, from 0 to 100), having a successful IPD result and *without* any 'double peak' detected, for which the IPD GoF (goodness-of-fit) was 'too high' — meaning that the image did not fit very well with a point-like PSF. Thus, a high fraction means that the source is probably a slightly extended object, or a partially resolved double star (or blended binary), etc. The quantity was computed using all transits for the given source where the IPD was successful.

The exact IPD GoF threshold considered to be 'too high' for a given window is determined from a statistical analysis of a larger sample of IPD inputs to AGIS, with separate thresholds in bins per magnitude, FOV, strip, row and gate. Within this parameter space, the threshold is determined as the 95th percentile of IPD GoF values in a representative (random) input sample of sources. This ensures that this threshold is really meaningful, although of course being arbitrary to some extent.

A certain fraction of high GoF values will always be produced by superposition of unrelated sources from the other field of view, by cosmic-ray events, etc.

RUWE : Renormalised unit weight error (float)

The Renormalised Unit Weight Error is a re-scaled version of the Unit Weight Error

$$\texttt{uwe} = \sqrt{\texttt{astrometric_chi2_al}/\nu}$$

where $\nu = \texttt{astrometric_n_good_obs_al} - N$ is the number of degrees of freedom in the AGIS source update, and $N = 6$ is the number of parameters solved.

The purpose of the re-scaling is to make the `ruwe` behave in the statistically expected manner of the `uwe`, when there is a good fit of the standard 6-parameter model to the astrometric data. A well-behaved source is expected to have `ruwe` $\simeq 1.0$, while a significantly greater value indicates a bad fit. The re-scaling procedure is described in **?**.

The degree of deviation from the expected `ruwe` $\simeq 1.0$ is most conveniently measured by the Gaussianized statistic `astrometric_gof_al`. Starting with DR4, `astrometric_gof_al` is probably more useful than `ruwe` as a general goodness-of-fit measure, because it takes into account the dependence on ν and has a simple statistical interpretation. For continuity, `ruwe` is retained with the same name and nominal meaning as in earlier releases, though more accurately computed.

The `ruwe` is set to null for sources that only have a two-parameter solution ($N = 2$).

SCAN_DIRECTION_STRENGTH_K1 : Degree of concentration of scan directions across the source (float)

The scan_direction_strength_k1...4 and scan_direction_mean_k1...4 quantify the distribution of AL scan directions across the source. scan_direction_strength_k1 (and similarly 2,3,4) equals $\left|\langle\exp(ik\theta)\rangle\right|$ for $k = 1, 2, 3, 4$ where θ is the position angle of the scan and the mean value is taken over the astrometric_n_good_obs_al observations contributing to the astrometric parameters of the source. θ is defined in the usual astronomical sense: $\theta = 0°$ when the FoV is moving towards local North, and $\theta = 90°$ towards local East.

The scan_direction_strength_k1...4 are numbers between 0 and 1, where 0 means that the scan directions are well spread out in different directions, while 1 means that they are concentrated in a single direction (given by the corresponding scan_direction_mean_k1...4).

The different orders k are statistics of the scan directions modulo $360°/k$. For example, at first order ($k = 1$), $\theta = 10°$ and $\theta = 190°$ count as different directions, but at second order ($k = 2$) they are the same. Thus, scan_direction_strength_k1 is the degree of concentration when the sense of direction is taken into account, while scan_direction_strength_k2 is the degree of concentration without regard to the sense of direction. A large value of scan_direction_strength_k4 indicates that the scans are concentrated in two nearly orthogonal directions.

SCAN_DIRECTION_STRENGTH_K2 : Degree of concentration of scan directions across the source (float)

The scan_direction_strength_k1...4 and scan_direction_mean_k1...4 attributes quantify the distribution of AL scan directions across the source.

See the description for attribute scan_direction_strength_k1 for further details.

SCAN_DIRECTION_STRENGTH_K3 : Degree of concentration of scan directions across the source (float)

The scan_direction_strength_k1...4 and scan_direction_mean_k1...4 attributes quantify the distribution of AL scan directions across the source.

See the description for attribute scan_direction_strength_k1 for further details.

SCAN_DIRECTION_STRENGTH_K4 : Degree of concentration of scan directions across the source (float)

The scan_direction_strength_k1...4 and scan_direction_mean_k1...4 attributes quantify the distribution of AL scan directions across the source.

See the description for attribute scan_direction_strength_k1 for further details.

SCAN_DIRECTION_MEAN_K1 : Mean position angle of scan directions across the source (float, Angle[deg])

The scan_direction_strength_k1...4 and scan_direction_mean_k1...4 attributes quantify the distribution of AL scan directions across the source. scan_direction_mean_k1 (and similarly for $k = 2, 3, 4$) equals $\arg(\langle\exp(ik\theta)\rangle)/k$ for k = 1,2,3,4 where θ is the position angle of the scan and the mean value is taken over the astrometric_n_good_obs_al observations contributing to the astrometric parameters of the source. θ is defined in the usual astronomical sense: $\theta = 0°$ when the FoV is moving towards local North, and $\theta = 90°$ towards local East.

scan_direction_mean_k1 (and similarly for $k = 2, 3, 4$) is an angle between $-180°/k$ and $+180°/k$, giving the mean position angle of the scans at order k.

The different orders k are statistics of the scan directions modulo $360°/k$. For example, at first order ($k = 1$), $\theta = 10°$ and $\theta = 190°$ count as different directions, but at second order ($k = 2$) they are the same. Thus, scan_direction_mean_k1 is the mean direction when the sense of direction is taken into account, while scan_direction_mean_k2 is the mean direction without regard to the sense of the direction. For example, scan_direction_mean_k1 = 0 means that the scans preferentially go towards North, while scan_direction_mean_k2 = 0 means that they preferentially go in the North-South direction, and scan_direction_mean_k4 = 0 that they preferentially go either in the North-South or in the East-West direction.

SCAN_DIRECTION_MEAN_K2 : Mean position angle of scan directions across the source (float, Angle[deg])

The scan_direction_strength_k1...4 and scan_direction_mean_k1...4 attributes quantify the distribution of AL scan directions across the source.

See the description for attribute scan_direction_mean_k1 for further details.

SCAN_DIRECTION_MEAN_K3 : Mean position angle of scan directions across the source (float, Angle[deg])

The scan_direction_strength_k1...4 and scan_direction_mean_k1...4 attributes quantify the distribution of AL scan directions across the source.

See the description for attribute `scan_direction_mean_k1` for further details.

SCAN_DIRECTION_MEAN_K4 : Mean position angle of scan directions across the source (float, Angle[deg])

The `scan_direction_strength_k1...4` and `scan_direction_mean_k1...4` attributes quantify the distribution of AL scan directions across the source.

See the description for attribute `scan_direction_mean_k1` for further details.

DUPLICATED_SOURCE : Source with multiple source identifiers (boolean)

During data processing, this source happened to be duplicated and only one source identifier has been kept. Observations assigned to the discarded source identifier(s) were not used. This may indicate observational, cross-matching or processing problems, or stellar multiplicity, and probable astrometric or photometric problems in all cases. The duplicity criterion used for Gaia DR4 is an angular distance of 0.12 arcsec. For the record, a limit of 0.18 arcsec was used for Gaia (E)DR3, while a limit of 0.4 arcsec was used for Gaia DR2.

Known issue: There are cases of duplicates being separated by more than the duplication radius of 0.12 arcsec. This is a consequence of the cascade of source links involved in the duplication logics: Condider three sources, of which the seond one is already labelled as duplicate of the first. If the third source is less than the duplication radius from a second, it will also become a duplicate of the first. In a few extreme cases, this mechanism has led to sizable 'chains' of duplicates reaching up to more than an arcsec in angular extent.

ASTROMETRIC_DUPLICATE_SOURCE_ID : Duplicate source identifier (long)

This field provides the `source_id` of a source which has been identified as a duplicate of the present source astrometrically (i.e. it is very close to another source).

If the field is identical to the present `source_id`, it means that the source has other duplicates and that the present one is the preferred source that should be kept in case of duplicate-sources removal in a release preparation or in the IDU processing, for example.

If the source has no duplicates identified, this field will be null.

L : Galactic longitude (double, Angle[deg])

Galactic Longitude of the object at reference epoch `ref_epoch`, see Section **??** of the release

documentation for conversion details.

B : Galactic latitude (double, Angle[deg])

Galactic Latitude of the object at reference epoch `ref_epoch`, see Section **??** of the release documentation for conversion details.

ECL_LON : Ecliptic longitude (double, Angle[deg])

Ecliptic Longitude of the object at reference epoch `ref_epoch`, obtained from the ICRS coordinates (`ra,dec`) using the transformation defined in Volume 1, Section 1.5.3 of ESA (1997).

Note that in the transformation applied here the ICRS origin is shifted in the equatorial plane from Γ by $\phi = 0.05542$ arcsec, positive from Γ to the ICRS origin (Chapront et al. 2002). The ICRS has an unambiguous definition with an origin in the ICRF equator defined by the realisation of the ICRF. The ecliptic system is less well-defined, potentially depending on additional conventions in dynamical theories. The transformation employed here corresponds to the inertial mean ecliptic with obliquity and Γ defined by reference to the ICRS equator. Both the obliquity and the position of Γ on the ICRS equator with respect to the ICRS origin have been obtained from Lunar Laser Ranging measurements. This has no time dependence – there is no secular variation of the obliquity and no precession – and it simply defines the relative situation of the various planes at J2000.0.

ECL_LAT : Ecliptic latitude (double, Angle[deg])

Ecliptic Latitude of the object at reference epoch `ref_epoch`. For further details see the description for attribute `ecl_lon`.

SEA_ENV_FLAG : Flag indicating the nature of the source environment (byte)

Flag indicating the nature of the Source Environment Analysis (SEA) result for this source. The following cases are foreseen:

- 0: the data are consistent with a single point source
- 1: the data are consistent with multiple sources
- 2: the data are not consistent with the presence of just point sources, the primary source is extended

- 3: the primary source is extended, and additional sources are found in its surrounding

For further details see Chapter **??** in the online documentation for the release.

SEA_NUM_DETECTIONS : Number of components found by the image segregation (int)

Number of sources found by the Image Segregation step of the Source Environment Analysis (SEA). This will always be greater than or equal to the number of sources for which parameters are provided. (This is supplied as it indicates that there may be other low signal-to-noise sources in the vicinity, which were rejected at the image parameter analysis stage).

For further details see Chapter **??** in the online documentation for the release.

SEA_NUM_COMPONENTS : Number of components for which parameters are provided (int)

Number of sources for which parameters are provided (includes primary). For further details see Chapter **??** in the online documentation for the release.

SEA_POWER_RATIO : Ratio between the flux levels measured in two apertures by SEAPipe (float)

Ratio between the flux levels measured in two apertures centred on the primary source position in the reconstructed image (used in determining whether the primary source is extended). For further details see Chapter **??** in the online documentation for the release.

SEA_POWER_FRACTION : Fraction of the larger aperture above detection threshold in SEAPipe (float)

Fraction of the larger aperture where the flux level is above the threshold representing the background level for detection of sources in the image segregation. Used in determining whether the primary source is extended. For further details see Chapter **??** in the online documentation for the release.

SEA_MAX_GAP : Maximum angular gap between useable transits of SEAPipe (float, Angle[deg])

Maximum angular gap between the available scans in degrees. For further details see Chapter **??** in the online documentation for the release.

SEA_NUM_TRANSITS : Number of transits which were useable by SEAPipe (int)

Number of FoV transits that were useable by SEAPipe. For further details see Chapter **??** in the online documentation for the release.

NUM_COMPONENTS : The number of consolidated components found by SEAPipe (short)

The number of consolidated components found around this source by SEAPipe. For further details see Chapter **??** in the online documentation for the release.

CONSOLIDATED_NUM_INPUT_CPTS : Number of SEAPipe sources forming each of the components reported (short[`num_components`] array)

Each SEAPipe component reported in the surrounding of a `source_id` (and listed in `component_ids`) may result from the consolidation of several other SEAPipe sources considered duplicates of this component. This number indicates how many such SEAPipe sources were involved in this consolidation for each component in `component_ids`.

Mostly this will be '1', occasionally '2' if two components were consolidated.

For further details see Chapter **??** in the online documentation for the release.

COMPONENT_IDS : Source identifiers for the consolidated components found by SEAPipe (long[`num_compone` array)

An array of the full source identifiers (including the component bits) for the consolidated components found around this source by SEAPipe. For further details see Chapter **??** in the online documentation for the release.

SOURCE_IDS_SHARING_TRANSITS : Source identifier related to this one in data processing (long[] array)

The list of sources that have been processed in conjunction with this source during the cross-match. They have been detected together on ground in at least one transit either as blended or resolved peaks.

Draft

HEALPIX_ID : Level 12 HEALPix identification string (string)

Level 12 HEALPix identification string consisting of 12 characters and based on the unique identifier (nested numbering scheme) encoded in the `source_id` (Bastian & Portell 2020). The first character in the string is a hexadecimal digit in the range 0 to B to identify the top level. The following 11 characters are quaternary digits in the range 0 to 3, each encoding the HEALPixel at the next level down that contains the source. The HEALPixel at level n (where $n \leq 12$) that contains the source is then simply the first n characters of this string.

RAW_G_MEAN_MAG : Approximate G magnitude (float, Magnitude[mag])

Approximate magnitude of the source. In most of the cases, it is obtained in IDU from the mean G–band flux of the source estimated from the AF2 to AF9 IPD measurements when avaiable. In the computation of this flux any observations flagged as blended are not used.

For a very small proportion of sources where the IPD flux is missing, the approximate magnitude corresponds instead to the mean on–board VPU G magnitude estimated using all FOV observations not flagged as blended. In such cases the quantity is limited in resolution to 0.015625 magnitudes as well as being limited in accuracy.

HEALPIX29 : Level 29 HEALPix index for HATS (long)

A level–29 HEALPix spatial index for compliance with the Hierarchical Adaptive Tiling Scheme standard in the International Virtual Observatory Alliance – see `https://www.ivoa.net/documents/Notes/HATS/` for further information.

Draft

2.2 ALL_SOURCE_FLAGS [TAP]

This table compiles flags of interest for each source published in the Gaia catalogue. It allows in particular to perform cuts in queries targetting the products served through the DataLink protocol (e.g. spectra or epoch data).

This table is complemented by `all_source_astrometry`, `all_source_photometry` and `all_source_rvs`, which provide the corresponding information on the astrometry, photometry and the radial velocity and other related parameters.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Source Identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

DESIGNATION : Unique source designation (unique across all Data Releases) (string)

A source designation, unique across all Gaia Data Releases, that is constructed from the prefix 'Gaia DRx ' followed by a string of digits corresponding to `source_id` (3 space–separated words in total). Note that the integer source identifier `source_id` is **not** guaranteed to be unique across Data Releases; moreover it is not guaranteed that the same astronomical source will always have the same `source_id` in different Data Releases. Hence the only safe way to compare source records between different Data Releases in general is to check the records of proximal source(s) in the same small part of the sky.

RANDOM_INDEX_AS : Random index for use when selecting subsets (long)

A random index which can be used to select smaller subsets of the data that are still rep-

resentative. The column contains a random permutation of the numbers from 0 to $N-1$, where N is the number of sources in the catalogue. Please note that the random index in the four 'all source' tables is the same, i.e. the same `source_id` in `all_source_astrometry`, `all_source_photometry`, `all_source_flags` and `all_source_rvs` will have the same `random_index_a` (while noting that `all_source_rvs` contains fewer sources than N). However this is not so for `random_index` in `gaia_source` which has it's own assigned random index values.

The random index can be useful for validation (testing on 10 different random subsets), visualization (displaying 1% of the data), and statistical exploration of the data, without the need to download all the data.

IN_GAIA_SOURCE : Flag indicating presence in the main Gaia source catalogue (boolean)

This flag indicates that this source was eligible to be tabulated in `gaia_source`.

PHOT_VARIABLE_FLAG : Photometric variability flag (string)

Flag indicating if variability was identified in the photometric data:

- 'VARIABLE': source identified and processed as variable, see variability tables and Chapter **??**;
- 'POSSIBLE': source identified as possible variable, see variability tables and Chapter **??**;
- 'NON_VARIABLE': source not identified as variable;
- empty: source not processed by the variability analyses.

PHOT_VARIABLE_SCORE : Photometric variability score between 0 and 1 (float)

Variability score coming from General Variability Detection and set for all sources processed by the variability analyses having sufficient observations.

PHOT_VARIABLE_BEST_CLASS : Photometric best variability class (string)

Best variability 'class' being a merger of `classifier_best_class_name` in `vari_classifier_result`, and the occurence in specific variable object tables. The latter may take priority when there are disagreements (should be a small number), but this is a merge, so not every source with a best class name corresponding to a specific variable object table is included in that table.

Draft

This field is not directly related to the `phot_variable_score`, and is only available for a subset of sources that appear in the `vari_summary` table (and is otherwise null). See Chapter **??** for further details.

IN_QSO_CANDIDATES : Flag indicating presence in the QSO candidates table (boolean)

This flag indicates that this source has been identified and/or processed as a QSO candidate by some of the DPAC processing chains. The information derived from this processing is provided in the separate `qso_candidates` table (see Chapter **??**).

IN_GALAXY_CANDIDATES : Flag indicating presence in the galaxy candidates table (boolean)

This flag indicates that this source has been identified and/or processed as a galaxy candidate by some of the DPAC processing chains. The information derived from this processing is provided in the separate `galaxy_candidates` table (see Chapter **??**).

LENS_NUM_COMPONENTS : Number of components found in the lens candidates table (int)

Number of components found by the gravitional lense module around this source. These components are tabulated in `grav_lens_component`.

NON_SINGLE_STAR : Flag indicating presence in the various Non-Single Star tables (short)

This flag indicates that this source has been identified as a non-single star by some of the DPAC processing chains (see Chapter **??**). The additional parameters derived from this processing are provided separately in one of the tables from the Non-Single Star section.

The flag is organised in three bits informing about the nature of the non-single star model.

- bit 0 (least-significant bit) is set to 1 in case of an astrometric binary
- bit 1 is set to 1 in case of a spectroscopic binary
- bit 2 is set to 1 in case of an eclisping binary
- bit 3 is set to 1 in case of resolved astrometric binaries

Note that some models can be combinations of two of the non-single star natures coded in this

flag.

SEA_ENV_FLAG : Flag indicating the nature of the source environment (byte)

Flag indicating the nature of the source environment. The following cases are foreseen:

- 0: the data are consistent with a single point source
- 1: the data are consistent with multiple sources
- 2: the data are not consistent with the presence of just point sources, the primary source is extended
- 3: the primary source is extended, and additional sources are found in its surrounding

HAS_XP : Flag indicating the availability of mean BP/RP spectrum (boolean)

This flag indicates that a mean BP/RP spectrum is available for this source in continuous representation. These will be served through the DataLink protocol.
For more information about this representation, see the table `xp_continuous_mean_spectrum` and **??** of the documentation.

HAS_ALL_SOURCE_RVS_RV : Flag indicating the availability of mean RV obtained from RVS (boolean)

This flag indicates that a mean Radial Velocity as computed by the RVS pipeline is available for this source. In this case, the radial velocity is provided as `radial_velocity_single` in table `all_source_rvs`, and it may be valid or invalid.

HAS_ALL_SOURCE_RVS_VBROAD : Flag indicating the availability of mean vbroad obtained from RVS (boolean)

This flag indicates that a mean vbroad as computed by the RVS pipeline is available for this source. In this case, the vbroad is provided as `vbroad` in table `all_source_rvs`.

HAS_RVS : Flag indicating the availability of mean RVS spectrum (byte)

This flag indicates that a mean RVS spectrum is available for this source and can be accessed through the DataLink protocol. The coding of this flag is as follows:

- 0: No spectrum is available
- 1: extremely weak spectrum: `all_source_rvs.rvs_spec_sig_to_noise` < 1 or set to null
- 2: weak spectrum: 1 ≤ `all_source_rvs.rvs_spec_sig_to_noise` < 3.5
- 3: moderate to strong spectrum: `all_source_rvs.rvs_spec_sig_to_noise` ≥ 3.5

HAS_EPOCH_ASTROMETRY : Flag indicating the availability of epoch astrometry (boolean)

This flag indicates that epoch astrometry is available for the source. These will be served through the DataLink protocol.

HAS_EPOCH_IMAGE : Flag indicating the availability of epoch images (boolean)

This flag indicates that epoch image is available for the source. These will be served through the DataLink protocol.

HAS_EPOCH_PHOTOMETRY : Flag indicating the availability of epoch photometry (boolean)

This flag indicates that epoch photometry is available for the source. Epoch photometry always contains G band integrated photometry (at averaged FoV level, and individual CCD level), together with BP and/or RP integrated photometry when available. These are served through the DataLink protocol (see the product descriptions for `epoch_photometry` and `epoch_photometry_ccd`).

HAS_EPOCH_RV : Flag indicating the availability of epoch radial velocity (boolean)

This flag indicates that epoch radial velocities are provided for the source. For a subset of those, epoch spectral line broadening and epoch Grvs will also be provided in the same product, which will be served through the DataLink protocol.

HAS_EPOCH_XP : Flag indicating the availability of epoch BP/RP spectrum (boolean)

This flag indicates that an epoch BP/RP spectrum is provided for the source. These will be served through the DataLink protocol.

Draft

HAS_EPOCH_RVS : Flag indicating the availability of epoch RVS spectrum (byte)

This flag indicates that an epoch RVS spectrum is provided for the source and that it can be accessed through the DataLink protocol. The coding of this flag is as follows:

- 0: No spectrum is available. This includes transits of stars not processed by the RVS pipeline, as well as those excluded in the initial steps. The reasons for exclusion of CCD spectra are recorded in `rvs_transit_description.ccd_invalid_reason` and correspond to values ≤ 53, equal to 62, or > 90.
- 1: Very weak to extremely weak signal (these spectra cannot be used independently). This applies to epoch spectra of stars with `all_source_rvs.external_apparent_grvs` > 14. For these stars, the pipeline did not compute any epoch radial velocities.
- 2: Weak signal. This applies to epoch spectra of stars with $12 <$ `all_source_rvs.external_app` ≤ 14. For these stars, the pipeline computed epoch radial velocities.
- 3: Moderate to strong signal. This applies to epoch spectra of stars with `all_source_rvs.extern` ≤ 12. For these stars, the pipeline computed both epoch radial velocities and broadening velocities. Note that the threshold of `external_apparent_grvs` = 12 also determines the type of normalisation applied to the CCD spectra before combination.

HAS_MCMC_GSPPHOT : Flag indicating the availability of GSP-Phot MCMC samples (boolean)

This flag indicates that MCMC samples from the GSP-Phot processing are available for this source. These will be served through the DataLink protocol.

IN_AP_CLASS : Flag for class astrophysical parameters from a Gaia-only processing (boolean)

This flag indicates that results for this source are available in table `ap_class`.

IN_AP_CLASS_IR : Flag for class astrophysical parameters from a Gaia+IR processing (boolean)

This flag indicates that results for this source are available in table `ap_class_ir`.

IN_AP_XP : Flag indicating the availability of XP astrophysical parameters (boolean)

This flag indicates that results for this source are available in table `ap_xp`.

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IN_AP_XP_SUPP : Flag for supplementary astrophysical parameters (boolean)

This flag indicates that results for this source are available in table `ap_xp_supp`.

IN_AP_RVS : Flag indicating the availability of RVS astrophysical parameters (boolean)

This flag indicates that results for this source are available in table `ap_rvs_epoch`.

IN_AP_CHEM : Flag indicating the availability of chemo astrophysical parameters (boolean)

This flag indicates that results for this source are available in table `ap_chem`.

IN_AP_EMISSION : Flag for emission star astrophysical parameters (boolean)

This flag indicates that results for this source are available in table `ap_emission`.

IN_AP_HOT : Flag for hot star astrophysical parameters (boolean)

This flag indicates that results for this source are available in table `ap_hot`.

IN_AP_RVS_EPOCH : Flag for RVS epoch astrophysical parameters (boolean)

This flag indicates that results for this source are available in table `ap_rvs_epoch`.

IN_AP_XP_EPOCH : Flag for XP epoch astrophysical parameters (boolean)

This flag indicates that results for this source are available in table `ap_xp_epoch`.

HAS_RESIDUAL_IMAGE : Flag indicating presence of reconstructed 2d image data (boolean)

This flag indicates the presence of 2d reconstructed image data. These images are delivered via the archive DataLink functionality (see the description of `residual_image` for further details).

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2.3 ALL_SOURCE_MATCH [TAP]

Global cross-match table between the main catalogue superset of all sources and the individual sets from the special Source Environment Analysis (SEA) and Crowded Field (CF) pipelines.

The table compiles all source match pairs between the `all_source_astrometry` (named `source_id`), SEAPipe (named `sea_source_id`) and Crowded Field image analysis sources (named `cf_source_id`).

For any given entry, `source_id` and another identifier will be valid, while the other will be null (only one to one comparisons with respect to `all_source_astrometry`).

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier for IDU-AGIS (long)

A unique single numerical identifier of the source obtained from `all_source_astrometry` (for a detailed description see `gaia_source.source_id`).

SEA_SOURCE_ID : Unique source identifier for Gaia Source Environment (SEA) (long)

Unique identifier for the Gaia Source Environment source.

CF_SOURCE_ID : Unique source identifier for crowded fields (long)

Unique identifier for the crowded field image analysis source.

WINNER : Match winner (string)

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Match winner:

- AGIS: IDU-AGIS.
- SEA: Gaia Source Environment Analysis (SEA).
- SIFCF: Crowded-field Sky Mapper (SM) SIF observations.
- NONE: No winner scenario.

RULE : Rule applied to the astrometric solution selection for this source (string)

A short string describing the rule applied to decide on whether the astrometric solution of this source is the recommended one. The possible values that can be taken by this field are described below:

For source pairs resulting from a cross-match and consolidation between an AGIS `source_id` and a SEAPipe `sea_source_id`, the possible alues are:

`sea_source_id` solution type	`source_id` solution type	Winner	Rule number	
5p	0p	SEAPipe	1A	SEA
5p	0p	AGIS	1B	SEAPi
5p	2p	SEAPipe	2A	SEA
5p	2p	AGIS	2B	SEAPi
5p	6p	SEAPipe	3A	QUA
5p	6p	AGIS	3B	QUA
2p	0p	SEAPipe	6A	SEA
2p	0p	AGIS	6B	SEAPi
2p	2p	AGIS	7	
2p	6p	AGIS	8	
Any	Any	SEAPipe	100	Uncompared pairs of multiple
Any	Any	AGIS	200	Uncompared pairs of multiple
Any	Any	AGIS	1000	Uncompared pairs of Multiple SEA
Any	Any	SEAPipe	2000	Uncompared pairs of Multiple SEA
Any	Any	AGIS/SEAPipe	10000	Remaining of uncompared pairs (winner cho

SEAPipe primaries are SEAPipe components matched to their parent source, i.e. the source around which the environment analysis was performed. SEAPipe primaries are those unmatched to their parent source.

For source pairs resulting from a cross-match and consolidation between an AGIS `source_id` and a Crowded Field (CF) `cf_source_id`, the possible values are:

`cf_source_id` solution type	`source_id` solution type	Status	Rule number	Comment
6p	0p	CF	1C	
	0p	AGIS	1B	
6p	2p	CF	2C	
	2p	AGIS	2B	
6p	6p	CF	3C	QUADSUM
	6p	AGIS	3B	QUADSUM
2p	0p	CF	6C	
	0p	AGIS	6B	
2p	2p	CF	7C	
	2p	AGIS	7B	
2p	6p	CF	8C	
	6p	AGIS	8B	
Any	Any	None	32	No comparison was performed for this pair

The so-called QUADSUM comparison is explained in the description of `astrometric_error_ratio`. See also Section **??** of the on-line documentation.

ASTROMETRIC_ERROR_RATIO : Ratio between the combined astrometric errors (float)

This field provides the ratio of the square root of the quadratic sums of the astrometric formal errors of the respective AGIS source with respect to Gaia Source Environment (SEA) or Crowded Fields. This ratio was used to decide which of the two solutions to recommend.

The quadratic sums of these formal errors are defined as:

$$\text{quadsum} = \sqrt{(0.445 \times \sigma_{\text{parallax}})^2 + (\sigma_{\text{pmra}})^2 + (\sigma_{\text{pmdec}})^2}$$

with

$$\texttt{astrometric_error_ratio} = \text{quadsum}_{\texttt{source_id}} / \text{quadsum}_{\texttt{sea_source_id or cf_source_id}}$$

The factor 0.445 applied to the parallax formal error is here to take account that, for long mission durations, the standard error of a parallax is normally larger than that of a proper motion component. Gaia DR4, with a mission duration of 5.5 years, is already well within this regime. Thus simply quadratically adding the three terms (parallax, proper motion in RA and Dec) would

overemphasise the parallax, compared to the other two dimensions in the space of unknowns.

SEPARATION : Distance between sources (float, Angle[mas])

Distance between the matched sources.

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2.4 ALL_SOURCE_PHOTOMETRY [TAP]

This table contains the photometry in the G and, where applicable, BP and/or RP band information for all the sources published in the Gaia catalogue. This photometry is generated exclusively by the PhotPipe module.

For a full description of the photometric processing please see Section **??** in the release documentation.

The majority of the sources in this table feature calibrated G band photometry, and those are also part of the `gaia_source` table, unless their astrometry is of insufficient quality (see also the description of the `all_source_astrometry` table). When no calibrated G band photometry from PhotPipe is available, an approximate on-board magnitude is provided as an indication of the source brightness. Such sources may, or many not, feature calibrated photometry in the BP and/or RP bands.

This table is complemented by `all_source_astrometry`, `all_source_rvs` and `all_source_flags`, which provide the corresponding information on the astrometry, radial velocity (and related parameters), and a variety of flags of interest.

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Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Source Identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

DESIGNATION : Unique source designation (unique across all Data Releases) (string)

A source designation, unique across all Gaia Data Releases, that is constructed from the prefix ‘Gaia DRx ’ followed by a string of digits corresponding to `source_id` (3 space–separated

words in total). Note that the integer source identifier `source_id` is **not** guaranteed to be unique across Data Releases; moreover it is not guaranteed that the same astronomical source will always have the same `source_id` in different Data Releases. Hence the only safe way to compare source records between different Data Releases in general is to check the records of proximal source(s) in the same small part of the sky.

RANDOM_INDEX_AS : Random index for use when selecting subsets (long)

A random index which can be used to select smaller subsets of the data that are still representative. The column contains a random permutation of the numbers from 0 to $N - 1$, where N is the number of sources in the catalogue. Please note that the random index in the four 'all source' tables is the same, i.e. the same `source_id` in `all_source_astrometry`, `all_source_photometry`, `all_source_flags` and `all_source_rvs` will have the same `random_index_a` (while noting that `all_source_rvs` contains fewer sources than N). However this is not so for `random_index` in `gaia_source` which has it's own assigned random index values.

The random index can be useful for validation (testing on 10 different random subsets), visualization (displaying 1% of the data), and statistical exploration of the data, without the need to download all the data.

IN_GAIA_SOURCE : Flag indicating presence in the main Gaia source catalogue (boolean)

This flag indicates that this source was eligible to be tabulated in `gaia_source`.

RAW_G_MEAN_MAG : Approximate G magnitude (float, Magnitude[mag])

Approximate magnitude of the source. In most of the cases, it is obtained in IDU from the mean G–band flux of the source estimated from the AF2 to AF9 IPD measurements when avaiable. In the computation of this flux any observations flagged as blended are not used.

For a very small proportion of sources where the IPD flux is missing, the approximate magnitude corresponds instead to the mean on–board VPU G magnitude estimated using all FOV observations not flagged as blended. In such cases the quantity is limited in resolution to 0.015625 magnitudes as well as being limited in accuracy.

PHOT_G_N_OBS : Number of observations contributing to G photometry (short)

Number of observations (CCD transits) that contributed to the G mean flux (`phot_g_mean_flux`) and mean flux error (`phot_g_mean_flux_error`).

Note that for just under 43 million sources there is no calibrated photometry from PhotPipe. In those cases only `raw_g_mean_mag` is populated with an estimated raw magnitude from core processing in IDU. The value of `phot_g_n_obs` will be zero in these cases since all photometric fields are either empty (null), set to zero (integer fields) or false (booleans) to indicate this situation. Note also that there are a few of these cases where there is BP and/or RP information despite the lack of information in the G fields.

PHOT_G_MEAN_FLUX : G-band mean flux (double, Flux[e^- s^{-1}])

Weighted mean flux in the G-band. See Chapter **??** of the online documentation regarding the photometric accumulations for details about this calculation.

PHOT_G_MEAN_FLUX_ERROR : Error on G-band mean flux (float, Flux[e^- s^{-1}])

The error on the weighted mean flux in the G-band `phot_g_mean_flux`. See Chapter **??** of the online documentation regarding the photometric accumulations for details about this calculation.

PHOT_G_MEAN_FLUX_OVER_ERROR : G-band mean flux divided by its error (float)

Mean flux in the G-band `phot_g_mean_flux` divided by its error `phot_g_mean_flux_error`.

PHOT_G_MEAN_MAG : G-band mean magnitude (float, Magnitude[mag])

Mean magnitude in the G band. This is computed from the G-band mean flux (`phot_g_mean_flux`) applying the magnitude zero-point in the Vega scale (see Section **??**).

No error is provided for this quantity as the error distribution is only symmetric in flux space. This converts to an asymmetric error distribution in magnitude space which cannot be represented by a single error value.

UNRELIABLE_G : Unreliable G-band mean flux (boolean)

The G-band mean flux is considered unreliable when:

(`phot_proc_flag` == 4) or (`phot_proc_flag` == 8)

and

((bp_rp < -1 or nu_eff > 3) or phot_g_mean_mag > 23).

PHOT_BP_N_OBS : Number of observations contributing to BP photometry (short)

Number of observations (CCD transits) used as input for the calculation of the integrated BP mean flux (phot_bp_mean_flux) and its mean flux error (phot_bp_mean_flux_error). It is possible that some of these epochs have been filtered a priori from the generation of the mean photometry because they were considered to be of insufficient quality based on acquisition and processing flags or rejected during the accumulation.

Note that

- a transit can be both contaminated and blended
- not all blended transits have been treated. The selection of cases to be de-blended adopted a set of criteria that were different from the criteria used when flagging all epochs for blends (the magnitude difference between the blended and blending source were assessed in the G band or in the BP/RP band in the two cases). This led to a small number of inconsistencies between the number of blended and de-blended cases.

PHOT_BP_MEAN_FLUX : Integrated BP mean flux (double, Flux[e^- s^{-1}])

Weighted mean flux in the integrated blue photometer (BP) band. See Chapter **??** of the online documentation regarding the photometric accumulations for details about this calculation.

PHOT_BP_MEAN_FLUX_ERROR : Error on the integrated BP mean flux (float, Flux[e^- s^{-1}])

The error on the weighted mean flux in the integrated blue photometer (BP) band phot_bp_mean_flux. See Chapter **??** of the online documentation regarding the photometric accumulations for details about this calculation.

PHOT_BP_MEAN_FLUX_OVER_ERROR : Integrated BP mean flux divided by its error (float)

Integrated BP mean flux phot_bp_mean_flux divided by its error phot_bp_mean_flux_error.

PHOT_BP_MEAN_MAG : Integrated BP mean magnitude (float, Magnitude[mag])

Mean magnitude in the integrated BP band. This is computed from the BP-band mean flux (phot_bp_mean_flux) applying the magnitude zero-point in the Vega scale (see Section **??**).

No error is provided for this quantity as the error distribution is only symmetric in flux space. This converts to an asymmetric error distribution in magnitude space which cannot be represented by a single error value.

UNRELIABLE_BP : Unreliable BP mean flux (boolean)

The BP-band mean flux is considered unreliable when:

(phot_proc_flag == 8)

and

phot_bp_n_obs < 4

and

phot_bp_mean_mag < 14.

PHOT_RP_N_OBS : Number of observations contributing to RP photometry (short)

Number of observations (CCD transits) that contributed to the integrated RP mean flux (phot_rp_mean_flux) and mean flux error (phot_rp_mean_flux_error).

PHOT_RP_MEAN_FLUX : Integrated RP mean flux (double, Flux[e^- s^{-1}])

Weighted mean flux in the integrated red photometer (RP) band. See Chapter **??** of the online documentation regarding the photometric accumulations for details about this calculation.

PHOT_RP_MEAN_FLUX_ERROR : Error on the integrated RP mean flux (float, Flux[e^- s^{-1}])

The error on the weighted mean flux in the integrated red photometer (RP) band phot_rp_mean_flux. See Chapter **??** of the online documentation regarding the photometric accumulations for details about this calculation.

PHOT_RP_MEAN_FLUX_OVER_ERROR : Integrated RP mean flux divided by its error (float)

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Integrated RP mean flux phot_rp_mean_flux divided by its error phot_rp_mean_flux_error.

PHOT_RP_MEAN_MAG : Integrated RP mean magnitude (float, Magnitude[mag])

Mean magnitude in the integrated RP band. This is computed from the RP-band mean flux (phot_rp_mean_flux) applying the magnitude zero-point in the Vega scale (see Section **??**).

No error is provided for this quantity as the error distribution is only symmetric in flux space. This converts to an asymmetric error distribution in magnitude space which cannot be represented by a single error value.

UNRELIABLE_RP : Unreliable RP mean flux (boolean)

The RP-band mean flux is considered unreliable when:

(phot_proc_flag == 8)

and

phot_rp_n_obs < 4

and

phot_rp_mean_mag < 14.

PHOT_PROC_FLAG : Source-level processing information (byte)

Source-level processing information. These flags apply to all bands. Flags that are specific to a given photometric band (G, integrated BP or RP) are reported in the processing flag in the corresponding mean photometry object.

Each byte will bit-wise encode the following values:

- Bit 0: This source was used as calibrator for the photometry.
- Bit 1: This source was a primary source.
- Bit 2: '1' means 'Nominal photometric treatment'. This requires full colour information to be available from synthetic photometry. The additional application of specific calibrations is reported in other bits/flags.

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- Bit 3: '1' means 'Incomplete colour photometric treatment'. A special treatment was required for this source because of incomplete synthetic photometry. The additional application of specific calibrations is reported in other bits/flags.
- Bit 4: '1' means 'Default colour photometric treatment'. A default colour was adopted for this source. The additional application of specific calibrations is reported in other bits/flags.

If a bit is not set, then the corresponding description does not apply, e.g. if bit 0 is not set, then this source was not a calibrator.

Other bits are available for future extensions.

BP_RP : BP - RP colour (float, Magnitude[mag])

BP−RP colour (`phot_bp_mean_mag` – `phot_rp_mean_mag`).

BP_G : BP - G colour (float, Magnitude[mag])

BP−G colour (`phot_bp_mean_mag` – `phot_g_mean_mag`).

G_RP : G - RP colour (float, Magnitude[mag])

G−RP colour (`phot_g_mean_mag` – `phot_rp_mean_mag`).

NU_EFF : Effective wavenumber (float, Misc[μm^{-1}])

Effective wavenumber.

NU_EFF_ERROR : Uncertainty on the effective wavenumber (float, Misc[μm^{-1}])

Estimated uncertainty on the effective wavenumber.

COL_CONST_PROB : Colour constancy probability (float)

This describes how constant a source is in terms of colour. This is a probability-based statistic with respect to the null hypothesis that the source is constant. P-values will be used for this. See

Evans (2008) for more details.

NUM_NOISY_CCD : Number of noisy G-band CCD transits (short)

This flag indicates that a higher than expected noise level has been detected within a number of G-band Field Transits. This flag will be the number of such CCD transits detected as noisy.

COLOUR_PARAMS : Colour parameters used to calibrate the epoch photometry (float[] array)

Colour parameters used to calibrate the epoch photometry. The content of this field depends on the definition of the calibration model which is available in the official Gaia data release photometry paper, online documentation, and in the DPCI operations outline document for the DRC.

CORRELATION_METRIC_R : Correlation metric R for the G, BP and RP photometry (float)

The correlation metric R for the G, BP and RP photometry. This is described in Evans et al. (2023).

CORRELATION_METRIC_DIAG_DIST : Correlation metric diagonal distance for the G, BP and RP photometry (float, Angle[deg])

The correlation metric diagonal distance for the G, BP and RP photometry. This is described in Evans et al. (2023).

PHOT_BP_N_CONTAMINATED_TRANSITS : Number of BP contaminated transits (short)

Number of BP transits that were considered to be contaminated by one or more nearby sources. The contaminating sources may come from the other field of view. It is possible that some of these epochs have been filtered a priori from the generation of the mean photometry because they were considered to be of insufficient quality based on acquisition and processing flags or rejected during the accumulation.

PHOT_BP_N_BLENDED_TRANSITS : Number of BP blended transits (short)

Number of BP transits affected by blends (i.e. cases where more than one source is present in the observed window). It is possible that some of these epochs have been filtered a priori from the generation of the mean photometry because they were considered to be of insufficient quality

based on acquisition and processing flags or rejected during the accumulation. Because of differences in the criteria used to generate this counter and to select the cases that were de-blended, in a small number of cases it can happen that the number of de-blended transits is larger than the number of blended transits.

PHOT_BP_N_DEBLENDED_TRANSITS : Number of BP deblended transits (short)

Number of BP transits that have been de-blended (i.e. an attempt has been made to remove the flux of a non-target blending source from the window). It is possible that some of these epochs have been filtered a priori from the generation of the mean photometry because they were considered to be of insufficient quality based on acquisition and processing flags or rejected during the accumulation. Because of differences in the criteria used to generate the counter of all sources affected by blends and to select the cases that were de-blended, in a small number of cases it can happen that the number of de-blended transits is larger than the number of blended transits.

PHOT_BP_N_RECONSTRUCTED_TRANSITS : Number of BP reconstructed transits (short)

Number of truncated BP epoch transits that were reconstructed.

PHOT_RP_N_CONTAMINATED_TRANSITS : Number of RP contaminated transits (short)

Number of RP transits that were considered to be contaminated by one or more nearby sources. The contaminating sources may come from the other field of view. It is possible that some of these epochs have been filtered a priori from the generation of the mean photometry because they were considered to be of insufficient quality based on acquisition and processing flags or rejected during the accumulation.

PHOT_RP_N_BLENDED_TRANSITS : Number of RP blended transits (short)

Number of RP transits affected by blends (i.e. cases where more than one source is present in the observed window). It is possible that some of these epochs have been filtered a priori from the generation of the mean photometry because they were considered to be of insufficient quality based on acquisition and processing flags or rejected during the accumulation. Because of differences in the criteria used to generate this counter and to select the cases that were de-blended, in a small number of cases it can happen that the number of de-blended transits is larger than the number of blended transits.

PHOT_RP_N_DEBLENDED_TRANSITS : Number of RP deblended transits (short)

Number of RP transits that have been de-blended (i.e. an attempt has been made to remove the flux of a non-target blending source from the window). It is possible that some of these epochs have been filtered a priori from the generation of the mean photometry because they were considered to be of insufficient quality based on acquisition and processing flags or rejected during the accumulation. Because of differences in the criteria used to generate the counter of all sources affected by blends and to select the cases that were de-blended, in a small number of cases it can happen that the number of de-blended transits is larger than the number of blended transits.

PHOT_RP_N_RECONSTRUCTED_TRANSITS : Number of RP reconstructed transits (short)

Number of truncated RP epoch transits that were reconstructed.

PHOT_G_N_OUTLIER : Number of rejected observations for the G band (int)

Number of epochs that were rejected as outliers during the computation of the mean G photometry. The rejection process is based on sigma clipping.

PHOT_G_N_CAL : Number of calibrated epochs for the G band (int)

Number of epochs that could be calibrated for the G-band. It is possible that some of these epochs have been filtered a priori from the generation of the mean photometry because they were considered to be of insufficient quality based on acquisition and processing flags. The number of filtered observations can be derived as:

```
nFiltered = phot_g_n_cal - phot_g_n_obs - phot_g_n_outlier
```

The total number of epochs that were processed can be derived as:

```
numEpochs = phot_g_n_cal + phot_g_n_uncal
```

This figure does not include the epochs in the time ranges that were excluded from the processing (gaps).

PHOT_G_N_UNCAL : Number of uncalibrated epochs for the G band (int)

Number of epochs that could not be calibrated in the G-band because of a problem. Possible problems:

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- The application of the calibration failed (e.g. produced NaN).
- A suitable calibration for the gate and window class configuration was not available.
- The epoch was acquired in a non-nominal configuration (e.g. complex gate, truncation, AOCS) and therefore a suitable integrated flux could not be produced.

PHOT_G_FLUX_STANDARD_DEVIATION : Standard deviation of the flux distribution for the G band (float, Flux[e^- s^{-1}])

Measure of the standard deviation of the flux distribution for the G-band.

PHOT_G_FLUX_SKEWNESS : Measure of the skewness of the flux distribution for the G band (float)

Measure of the skewness of the flux distribution for the G-band.

When an insufficient number of observations are available to compute these quantities, the corresponding field will be set to Float.NaN.

PHOT_G_FLUX_KURTOSIS : Measure of the kurtosis of the flux distribution for the G band (float)

Kurtosis of the flux distribution for the G-band.

When an insufficient number of observations are available to compute these quantities, the corresponding field will be set to Float.NaN.

PHOT_G_PC05 : 5th percentile of the flux distribution for the G band (float, Flux[e^- s^{-1}])

The 5th percentile of the flux distribution for the G-band.

When an insufficient number of observations are available to compute these quantities, the corresponding field will be set to Float.NaN.

PHOT_G_PC95 : 95th percentile of the flux distribution for the G band (float, Flux[e^- s^{-1}])

The 95th percentile of the flux distribution for the G-band.

When an insufficient number of observations are available to compute these quantities, the cor-

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responding field will be set to Float.NaN.

PHOT_G_FLUX_UWV : Unit weight variance of fluxes for the G band (double)

The unit weight variance of the flux values fpr the G-band. See Evans (2008) for a rigorous mathematical definition.

PHOT_G_FLUX_MEDIAN : Median flux in the G band (float, Flux[e^- s^{-1}])

Median of all available AF calibrated flux measurements in the G-band.

When an insufficient number of observations are available to compute this quantity, the value will be set to Float.NaN.

PHOT_G_FLUX_MAD : Median absolute deviation of the G-band flux distribution (float, Flux[e^- s^{-1}])

Median Absolute Deviation (MAD) of the G-band flux distribution.

When an insufficient number of observations are available to compute this quantity, the value will be set to Float.NaN.

PHOT_G_FLUX_MIN : Minimum flux value for the G band (float, Flux[e^- s^{-1}])

Minimum flux value for the G-band.

PHOT_G_FLUX_MAX : Maximum flux value for the G band (float, Flux[e^- s^{-1}])

Maximum flux value for the G-band.

PHOT_G_FLUX_FIRST_QUARTILE : First quartile of the G-band flux distribution (float, Flux[e^- s^{-1}])

First quartile of the G-band flux distribution.

When an insufficient number of observations are available to compute this quantity, the value will be set to Float.NaN.

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PHOT_G_FLUX_THIRD_QUARTILE : Third quartile of the G-band flux distribution (float, Flux[e^- s^{-1}])

Third quartile of the G-band flux distribution.

When an insufficient number of observations are available to compute this quantity, the value will be set to Float.NaN.

PHOT_G_STETSON_INDEX : Stetson variability index for the G band (float)

This is still to be defined.

PHOT_G_CONST_PROB : Constant source probability for the G band (float)

This describes how constant a source is in the G-band. This is a probability-based statistic with respect to the null hypothesis that the source is constant. P-values will be used for this. See DWE-010 for more details.

PHOT_G_ABBE_VALUE : Abbé value for the G band (float)

This is the Abbé variability statistic calculated in flux space for the G-band.

PHOT_G_HARMONIC_AMPLITUDE : Amplitude of G photometric residual versus position angle of scan (float)

This statistic measures the amplitude of the variation of the G photometric residual (in magnitudes) as a function of the position angle of the scan direction. A large amplitude indicates that the source is double, in which case the phase indicates the position angle of the pair modulo 180 degrees.

Let ψ be the position angle of the scan direction. The following expression is fitted to the photometric residual for all the AF observations of the source:

$$\delta mag = c_0 + c_2 \cos(2\psi) + s_2 \sin(2\psi)$$

The amplitude and phase of the variation are calculated as

$$\texttt{phot_g_harmonic_amplitude} = \sqrt{c_2^2 + s_2^2}$$

$$\texttt{phot_g_harmonic_phase} = \frac{1}{2}\,\mathrm{atan2}(s_2, c_2) \quad (+\,180^\circ)$$

where atan2 returns the angle in degrees, and 180 is added for negative values.

PHOT_G_HARMONIC_PHASE : Phase of the G photometric residual versus position angle of scan (float, Angle[deg])

This statistic measures the phase of the variation of the G photometric residual (in magnitudes) as a function of the position angle of the scan direction. A large amplitude indicates that the source is double, in which case the phase indicates the position angle of the pair modulo 180 degrees.

Let ψ be the position angle of the scan direction. The following expression is fitted to the photometric residual for all the AF observations of the source:

$$\delta mag = c_0 + c_2 \cos(2\psi) + s_2 \sin(2\psi)$$

The amplitude and phase of the variation are calculated as

$$\texttt{phot_g_harmonic_amplitude} = \sqrt{c_2^2 + s_2^2}$$

$$\texttt{phot_g_harmonic_phase} = \frac{1}{2}\,\mathrm{atan2}(s_2, c_2)$$

where atan2 returns the angle in degrees.

PHOT_G_PROC_FLAGS : Photometric processing flags in the G band (byte)

Processing flags applicable to the G-band at source-level.

Each byte will bit-wise encode the following values:

- Bit 0: When set it indicates that a correction for the use of a default $\nu_{\rm eff}$ in DIPD to select the PSF/LSF was applied to the epochs that contributed to the mean G-band photometry. When not set, it indicates that the correction was not needed or not applied. The propagated IDU flags will report for each transit if a default $\nu_{\rm eff}$ was used or not. This only applies to *G*-band and will never be set for integrated BP/RP photometry.
- Bit 1: When set it indicates that one or more epochs affected by saturation contributed to the mean *G* photometry. When not set it indicates that they were not included, were corrected for or were not present. The CCD-level processing flag will report on the need and application of a saturation correction. This only applies to *G*-band and will never be set for integrated BP/RP photometry.
- Bit 2 and 3: Filtering level applied to the epoch measurements to select those contributing to the mean photometry (outlier rejection is applied in addition to this in the accumulation stage). Four different levels are envisaged: primary (both Bit 2 and 3 are 0), secondary (Bit 2 is 1 and Bit 3 is 0), sparse (Bit 2 is 0 and Bit 3 is 1) and no filtering (both Bit 2 and 3 are 1). Further details about the filtering criteria and rules will be provided in the documentation.

Other bits are available for future extensions.

PHOT_BP_N_OUTLIER : Number of rejected observations for the BP band (int)

Number of epochs that were rejected as outliers during the computation of the mean BP photometry. The rejection process is based on sigma clipping.

PHOT_BP_N_CAL : Number of calibrated epochs for the BP band (int)

Number of epochs that could be calibrated for the BP-band. It is possible that some of these epochs have been filtered a priori from the generation of the mean photometry because they were considered to be of insufficient quality based on acquisition and processing flags. The number of filtered observations can be derived as:

```
nFiltered = phot_bp_n_cal - phot_bp_n_obs - phot_bp_n_outlier
```

The total number of epochs that were processed can be derived as:

```
numEpochs = phot_bp_n_cal + phot_bp_n_uncal
```

This figure does not include the epochs in the time ranges that were excluded from the processing

(gaps).

PHOT_BP_N_UNCAL : Number of uncalibrated epochs for the BP band (int)

Number of epochs that could not be calibrated in the BP-band because of a problem. Possible problems:

- The application of the calibration failed (e.g. produced NaN).
- A suitable calibration for the gate and window class configuration was not available.
- The epoch was acquired in a non-nominal configuration (e.g. complex gate, truncation, AOCS) and therefore a suitable integrated flux could not be produced.

PHOT_BP_NEG : Number of observations with negative fluxes for the BP band (int)

Number of BP-band epochs with negative fluxes.

PHOT_BP_FLUX_STANDARD_DEVIATION : Standard deviation of the flux distribution for the BP band (float, Flux[e^- s^{-1}])

Measure of the standard deviation of the flux distribution for the BP-band.

PHOT_BP_FLUX_SKEWNESS : Measure of the skewness of the flux distribution for the BP band (float)

Measure of the skewness of the flux distribution for the BP-band.

When an insufficient number of observations are available to compute these quantities, the corresponding field will be set to Float.NaN.

PHOT_BP_FLUX_KURTOSIS : Measure of the kurtosis of the flux distribution for the BP band (float)

Kurtosis of the flux distribution for the BP-band.

When an insufficient number of observations are available to compute these quantities, the corresponding field will be set to Float.NaN.

Draft

PHOT_BP_PC05 : 5th percentile of the flux distribution for the BP band (float, Flux[e^- s^{-1}])

The 5th percentile of the flux distribution for the BP-band.

When an insufficient number of observations are available to compute these quantities, the corresponding field will be set to Float.NaN.

PHOT_BP_PC95 : 95th percentile of the flux distribution for the BP band (float, Flux[e^- s^{-1}])

The 95th percentile of the flux distribution for the BP-band.

When an insufficient number of observations are available to compute these quantities, the corresponding field will be set to Float.NaN.

PHOT_BP_FLUX_UWV : Unit weight variance of fluxes for the BP-band (double)

The unit weight variance of the flux values fpr the BP-band. See Evans (2008) for a rigorous mathematical definition.

PHOT_BP_FLUX_MEDIAN : Median flux in the BP band (float, Flux[e^- s^{-1}])

Median of all available calibrated flux measurements in the BP-band.

When an insufficient number of observations are available to compute this quantity, the value will be set to Float.NaN.

PHOT_BP_FLUX_MAD : Median absolute deviation of the BP-band flux distribution (float, Flux[e^- s^{-1}])

Median Absolute Deviation (MAD) of the BP-band flux distribution.

When an insufficient number of observations are available to compute this quantity, the value will be set to Float.NaN.

PHOT_BP_FLUX_MIN : Minimum flux value for the BP band (float, Flux[e^- s^{-1}])

Minimum flux value for the BP-band.

Draft

PHOT_BP_FLUX_MAX : Maximum flux value for the BP band (float, Flux[e^- s^{-1}])

Maximum flux value for the BP-band.

PHOT_BP_FLUX_FIRST_QUARTILE : First quartile of the BP-band flux distribution (float, Flux[e^- s^{-1}])

First quartile of the BP-band flux distribution.

When an insufficient number of observations are available to compute this quantity, the value will be set to Float.NaN.

PHOT_BP_FLUX_THIRD_QUARTILE : Third quartile of the BP-band flux distribution (float, Flux[e^- s^{-1}])

Third quartile of the BP-band flux distribution.

When an insufficient number of observations are available to compute this quantity, the value will be set to Float.NaN.

PHOT_BP_STETSON_INDEX : Stetson variability index for the BP band (float)

This is still to be defined.

PHOT_BP_CONST_PROB : Constant source probability for the BP band (float)

This describes how constant a source is in the BP-band. This is a probability-based statistic with respect to the null hypothesis that the source is constant. P-values will be used for this. See DWE-010 for more details.

PHOT_BP_ABBE_VALUE : Abbé value for the BP band (float)

This is the Abbé variability statistic calculated in flux space for the BP-band.

PHOT_BP_HARMONIC_AMPLITUDE : Amplitude of BP photometric residual versus position angle of scan (float)

This statistic measures the amplitude of the variation of the BP photometric residual (in magni-

tudes) as a function of the position angle of the scan direction. A large amplitude indicates that the source is double, in which case the phase indicates the position angle of the pair modulo 180 degrees.

Let ψ be the position angle of the scan direction. The following expression is fitted to the photometric residual for all the BP observations of the source:

$$\delta mag = c_0 + c_2 \cos(2\psi) + s_2 \sin(2\psi)$$

The amplitude and phase of the variation are calculated as

$$\texttt{phot_bp_harmonic_amplitude} = \sqrt{c_2^2 + s_2^2}$$

$$\texttt{phot_bp_harmonic_phase} = \frac{1}{2}\,\mathrm{atan2}(s_2, c_2) \quad (+\,180^\circ)$$

where atan2 returns the angle in degrees, and 180 is added for negative values.

PHOT_BP_HARMONIC_PHASE : Phase of BP photometric residual versus position angle of scan (float, Angle[deg])

This statistic measures the phase of the variation of the BP photometric residual (in magnitudes) as a function of the position angle of the scan direction. A large amplitude indicates that the source is double, in which case the phase indicates the position angle of the pair modulo 180 degrees.

Let ψ be the position angle of the scan direction. The following expression is fitted to the photometric residual for all the BP observations of the source:

$$\delta mag = c_0 + c_2 \cos(2\psi) + s_2 \sin(2\psi)$$

The amplitude and phase of the variation are calculated as

$$\texttt{phot_bp_harmonic_amplitude} = \sqrt{c_2^2 + s_2^2}$$

$$\texttt{phot_bp_harmonic_phase} = \frac{1}{2}\,\mathrm{atan2}(s_2, c_2)$$

where atan2 returns the angle in degrees.

PHOT_BP_PROC_FLAGS : Photometric processing flags in the BP band (byte)

Processing flags applicable to the BP-band at source-level.

Each byte will bit-wise encode the following values:

- Bit 0: When set it indicates that a correction for the use of a default ν_{eff} in DIPD to select the PSF/LSF was applied to the epochs that contributed to the mean G-band photometry. When not set, it indicates that the correction was not needed or not applied. The propagated IDU flags will report for each transit if a default ν_{eff} was used or not. This only applies to G-band and will never be set for integrated BP/RP photometry.
- Bit 1: When set it indicates that one or more epochs affected by saturation contributed to the mean G photometry. When not set it indicates that they were not included, were corrected for or were not present. The CCD-level processing flag will report on the need and application of a saturation correction. This only applies to G-band and will never be set for integrated BP/RP photometry.
- Bit 2 and 3: Filtering level applied to the epoch measurements to select those contributing to the mean photometry (outlier rejection is applied in addition to this in the accumulation stage). Four different levels are envisaged: primary (both Bit 2 and 3 are 0), secondary (Bit 2 is 1 and Bit 3 is 0), sparse (Bit 2 is 0 and Bit 3 is 1) and no filtering (both Bit 2 and 3 are 1). Further details about the filtering criteria and rules will be provided in the documentation.

Other bits are available for future extensions.

PHOT_RP_N_OUTLIER : Number of rejected observations for the RP band (int)

Number of epochs that were rejected as outliers during the computation of the mean RP photometry. The rejection process is based on sigma clipping.

PHOT_RP_N_CAL : Number of calibrated epochs for the RP band (int)

Number of epochs that could be calibrated for the RP-band. It is possible that some of these epochs have been filtered a priori from the generation of the mean photometry because they were considered to be of insufficient quality based on acquisition and processing flags. The number of filtered observations can be derived as:

```
nFiltered = phot_rp_n_cal - phot_rp_n_obs - phot_rp_n_outlier
```

The total number of epochs that were processed can be derived as:

```
numEpochs = phot_rp_n_cal + phot_rp_n_uncal
```

This figure does not include the epochs in the time ranges that were excluded from the processing (gaps).

PHOT_RP_N_UNCAL : Number of uncalibrated epochs for the RP band (int)

Number of epochs that could not be calibrated in the RP-band because of a problem. Possible problems:

- The application of the calibration failed (e.g. produced NaN).
- A suitable calibration for the gate and window class configuration was not available.
- The epoch was acquired in a non-nominal configuration (e.g. complex gate, truncation, AOCS) and therefore a suitable integrated flux could not be produced.

PHOT_RP_NEG : Number of observations with negative fluxes for the RP band (int)

Number of RP-band epochs with negative fluxes.

PHOT_RP_FLUX_STANDARD_DEVIATION : Standard deviation of the flux distribution for the RP band (float, Flux[e^{-} s^{-1}])

Measure of the standard deviation of the flux distribution for the RP-band.

PHOT_RP_FLUX_SKEWNESS : Measure of the skewness of the flux distribution for the RP band (float)

Measure of the skewness of the flux distribution for the RP-band.

When an insufficient number of observations are available to compute these quantities, the corresponding field will be set to Float.NaN.

PHOT_RP_FLUX_KURTOSIS : Measure of the kurtosis of the flux distribution for the RP band (float)

Kurtosis of the flux distribution for the RP-band.

When an insufficient number of observations are available to compute these quantities, the corresponding field will be set to Float.NaN.

PHOT_RP_PC05 : 5th percentile of the flux distribution for the RP band (float, Flux[e^- s^{-1}])

The 5th percentile of the flux distribution for the RP-band.

When an insufficient number of observations are available to compute these quantities, the corresponding field will be set to Float.NaN.

PHOT_RP_PC95 : 95th percentile of the flux distribution for the RP band (float, Flux[e^- s^{-1}])

The 95th percentile of the flux distribution for the RP-band.

When an insufficient number of observations are available to compute these quantities, the corresponding field will be set to Float.NaN.

PHOT_RP_FLUX_UWV : Unit weight variance of fluxes for the RP band (double)

The unit weight variance of the flux values fpr the RP-band. See Evans (2008) for a rigorous mathematical definition.

PHOT_RP_FLUX_MEDIAN : Median flux in the RP band (float, Flux[e^- s^{-1}])

Median of all available calibrated flux measurements in the RP-band.

When an insufficient number of observations are available to compute this quantity, the value will be set to Float.NaN.

Draft

PHOT_RP_FLUX_MAD : Median absolute deviation of the RP-band flux distribution (float, Flux[e^- s^{-1}])

Median Absolute Deviation (MAD) of the RP-band flux distribution.

When an insufficient number of observations are available to compute this quantity, the value will be set to Float.NaN.

PHOT_RP_FLUX_MIN : Minimum flux value for the RP band (float, Flux[e^- s^{-1}])

Minimum flux value for the RP-band.

PHOT_RP_FLUX_MAX : Maximum flux value for the RP band (float, Flux[e^- s^{-1}])

Maximum flux value for the RP-band.

PHOT_RP_FLUX_FIRST_QUARTILE : First quartile of the RP-band flux distribution (float, Flux[e^- s^{-1}])

First quartile of the RP-band flux distribution.

When an insufficient number of observations are available to compute this quantity, the value will be set to Float.NaN.

PHOT_RP_FLUX_THIRD_QUARTILE : Third quartile of the RP-band flux distribution (float, Flux[e^- s^{-1}])

Third quartile of the RP-band flux distribution.

When an insufficient number of observations are available to compute this quantity, the value will be set to Float.NaN.

PHOT_RP_STETSON_INDEX : Stetson variability index for the RP band (float)

This is still to be defined.

PHOT_RP_CONST_PROB : Constant source probability for the RP band (float)

Draft

This describes how constant a source is in the RP-band. This is a probability-based statistic with respect to the null hypothesis that the source is constant. P-values will be used for this. See DWE-010 for more details.

PHOT_RP_ABBE_VALUE : Abbe value for the RP band (float)

This is the Abbé variability statistic calculated in flux space for the RP-band.

PHOT_RP_HARMONIC_AMPLITUDE : Amplitude of RP photometric residual versus position angle of scan (float)

This statistic measures the amplitude of the variation of the RP photometric residual (in magnitudes) as a function of the position angle of the scan direction. A large amplitude indicates that the source is double, in which case the phase indicates the position angle of the pair modulo 180 degrees.

Let ψ be the position angle of the scan direction. The following expression is fitted to the photometric residual for all the RP observations of the source:

$$\delta mag = c_0 + c_2 \cos(2\psi) + s_2 \sin(2\psi)$$

The amplitude and phase of the variation are calculated as

$$\texttt{phot_rp_harmonic_amplitude} = \sqrt{c_2^2 + s_2^2}$$

$$\texttt{phot_rp_harmonic_phase} = \frac{1}{2}\,\mathrm{atan2}(s_2, c_2) \quad (+\ 180^\circ)$$

where atan2 returns the angle in degrees, and 180 is added for negative values.

PHOT_RP_HARMONIC_PHASE : Phase of RP photometric residual versus position angle of scan (float, Angle[deg])

This statistic measures the phase of the variation of the RP photometric residual (in magnitudes) as a function of the position angle of the scan direction. A large amplitude indicates that the

source is double, in which case the phase indicates the position angle of the pair modulo 180 degrees.

Let ψ be the position angle of the scan direction. The following expression is fitted to the photometric residual for all the RP observations of the source:

$$\delta mag = c_0 + c_2 \cos(2\psi) + s_2 \sin(2\psi)$$

The amplitude and phase of the variation are calculated as

$$\texttt{phot_rp_harmonic_amplitude} = \sqrt{c_2^2 + s_2^2}$$

$$\texttt{phot_rp_harmonic_phase} = \frac{1}{2}\,\mathrm{atan2}(s_2, c_2)$$

where atan2 returns the angle in degrees.

PHOT_RP_PROC_FLAGS : Photometric processing flags in the RP band (byte)

Processing flags applicable to the RP-band at source-level.

Each byte will bit-wise encode the following values:

- Bit 0: When set it indicates that a correction for the use of a default $\nu_{\rm eff}$ in DIPD to select the PSF/LSF was applied to the epochs that contributed to the mean G-band photometry. When not set, it indicates that the correction was not needed or not applied. The propagated IDU flags will report for each transit if a default $\nu_{\rm eff}$ was used or not. This only applies to G-band and will never be set for integrated BP/RP photometry.
- Bit 1: When set it indicates that one or more epochs affected by saturation contributed to the mean G photometry. When not set it indicates that they were not included, were corrected for or were not present. The CCD-level processing flag will report on the need and application of a saturation correction. This only applies to G-band and will never be set for integrated BP/RP photometry.
- Bit 2 and 3: Filtering level applied to the epoch measurements to select those contributing to the mean photometry (outlier rejection is applied in addition to this in the

accumulation stage). Four different levels are envisaged: primary (both Bit 2 and 3 are 0), secondary (Bit 2 is 1 and Bit 3 is 0), sparse (Bit 2 is 0 and Bit 3 is 1) and no filtering (both Bit 2 and 3 are 1). Further details about the filtering criteria and rules will be provided in the documentation.

Other bits are available for future extensions.

BP_INTEGRATED_SPECTRUM : Integral of the BP spectrum (float, Flux[e^- s^{-1}])

Integrated BP flux computed from the BP mean spectrum. This is different from the integrated BP photometry provided via CalPhotSource and is only meant to be used for validation.

RP_INTEGRATED_SPECTRUM : Integral of the RP spectrum (float, Flux[e^- s^{-1}])

Integrated RP flux computed from the RP mean spectrum. This is different from the integrated RP photometry provided via CalPhotSource and is only meant to be used for validation.

Draft

2.5 ALL_SOURCE_RVS [TAP]

This table contains information derived from the Radial Velocity Spectrometer (RVS), including radial velocities, spectral line broadening, GRVS magnitude and related parameters, along with various flags and ancillary data.

As it is limited to sources observed by Gaia's RVS, this dataset contains fewer sources than `gaia_source`, `all_source_astrometry`, `all_source_photometry`, or `all_source_flags`.

It is generated for all stars processed by the RVS pipeline. However, only a small subset of the data produced is considered valid; these can be identified using the validity flags. For a more detailed description of this table see Section **??**.

The processing of the RVS data is described in **?**; the validation of `radial_velocity_single` in **?**, `radial_velocity_sys` in **?**, `vbroad` in **?**, `grvs_mag` in **?**, and the various statistical parameters in **?**.

Columns description:

SOLUTION_ID : Solution identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

RADIAL_VELOCITY_SINGLE_IS_VALID : Radial velocity validity flag for the single star model (boolean)

This flag indicates if `radial_velocity_single` is deemed valid or not.

RV_INVALID_REASON : The reason why the radial velocity is set as invalid (long)

Draft

This 64-bit field encodes the reasons for the invalidation of `radial_velocity_single`. Each bit represents a specific reason, set to 1 if true and 0 if false.

Post-processing assigns these flags based on validation analysis. Invalidation reasons are determined based on various conditions, and may apply even when a radial velocity is not provided.

A radial velocity may be invalidated for multiple reasons, resulting in the activation of multiple bits. See Table **??** and Table **??** for the occurrence of each invalid reason in bright (`external_apparent_grvs` $\leq$ 14) and faint stars, respectively.

Both valid and invalid data are published in `all_source_rvs`.

- Bit 0: Hot star. This invalidation reason is set for stars with `external_apparent_grvs` $\leq$ 12 mag and `rv_template_teff` > 14,500 K, as well as for those with `external_apparent_gr` > 12 mag and `rv_template_teff` > 8,500 K.
- Bit 1: [Not Used].
- Bit 2: Suspected SB2 (double-lined spectroscopic binary). Radial velocity is invalidated for stars with `fraction_double_lined` $\geq$ 0.1.
- Bit 3: Suspected emission-line star. This invalidation reason is set for stars identified as emission-line stars by the pipeline (`fraction_emission_lines` > 0.4) for which no radial velocity is provided. It is also set to approximately 2,700 stars detected as emission-line stars during post-processing, even if their radial velocity was measured.
- Bit 4: [Not Used].
- Bit 5: Faint star. This invalidation reason is set when `external_apparent_grvs` > 16 mag.
- Bits 6-7: [Not Used].
- Bit 8: Excessive uncertainty. Radial velocity is invalidated when `radial_velocity_single_err` > 1000 km s^{-1}.
- Bit 9: Invalid cross-correlation function. This flag is set for stars with `external_apparent_grvs` $\leq$ 14 mag, where the cross-correlation function exhibits a spurious secondary peak that is higher than the primary (correct) peak. As a result, the radial velocity is erroneously derived from this spurious peak. These problematic stars were identified using the UMAP dimensionality reduction technique (see **?**).
- Bits 10-11: [Not Used].

Draft

- Bit 12: Radial velocity outside the allowed range. Radial velocity is invalidated for stars with `radial_velocity_single` > 1000 km s^{-1} or `radial_velocity_single` < −1000 km s^{-1}.
- Bits 13-19: [Not Used].
- Bit 20: Template mismatch. This invalidation reason is set for a list of stars with `external_apparent_grvs` ≤ 14 mag, identified using the UMAP dimensionality reduction tool (see **?**), and to the stars having `rv_atm_param_origin` = 888.
- Bit 21: [Not Used].
- Bit 22: Contamination from a bright source. This invalidation reason is set for a list of stars identified as contaminated using the method described in Katz et al. (2023).
- Bit 23: High probability of being an outlier. Radial velocity is invalidated if `rv_outlier_probabi` = 2. It is also invalidated for faint stars with `external_apparent_grvs` > 14 mag that satisfy the condition:

  `rv_outlier_probability` > 1 − 0.5 × (`external_apparent_grvs` − 14)

  where `rv_outlier_probability` is provided only for stars with 3700 K ≤ `rv_template_teff` ≤ 6500 K.
- Bit 24: No radial velocity provided by the pipeline. This typically occurs for stars with only one valid transit available during processing (with validity stored in `rvs_transit_desc` stars marked as `rv_assumed_sb2`, assumed emission-line stars, or cases where the pipeline failed.
- Bit 25: Excessive standard deviation in epoch radial velocities. Radial velocity is invalidated for stars with `external_apparent_grvs` ≤ 11 mag and `rv_std_dev` ≥ 577 km s^{-1}.
- Bit 26: Faint star with no `rv_outlier_probability`. This flag is assigned to faint stars (`external_apparent_grvs` > 14 mag) whose `rv_template_teff` falls outside the range used for `rv_outlier_probability` computation (i.e., `rv_template_teff` < 3700 K or > 6500 K). Due to time constraints, their radial velocities could not be analyzed with alternative methods. Given that most faint-star radial velocities are invalid, they are systematically marked as such.
- Bit 27: No associated atmospheric parameters. This flag is set when `rv_atm_param_star_teff` = 0.
- Bit 28: Only one valid transit remaining. Radial velocity is invalidated for a set of 68 stars with `external_apparent_grvs` ≤ 11 mag that have only one valid transit left after post-processing.
- Bits 29-63: [Not Used].

Draft

VBROAD_IS_VALID : vbroad validity flag (boolean)

This flag indicates if `vbroad` is deemed valid or not.

VBROAD_INVALID_REASON : The reason why vbroad is set as invalid (short)

This 16-bit field specifies the reasons for invalidating the `vbroad` data. Each bit represents a specific reason, set to 1 if true and 0 if false.

Post-processing assigns these flags based on validation analysis. Invalidation reasons are determined based on various conditions, and may apply even when a `vbroad` is not provided.

A `vbroad` may be invalidated for multiple reasons, leading to the activation of multiple bits. See Table **??** for the occurrence of each invalid reason.

Both valid and invalid data are published in `all_source_rvs`.

- Bit 0: Invalid radial velocity (this includes all `rv_invalid_reason` cases).
- Bit 1: Too faint star. `vbroad` is invalidated for stars with `external_apparent_grvs` > 11.
- Bit 2: [Not Used].
- Bit 3: [Not Used].
- Bit 4: Cool star. `vbroad` is invalidated for stars with `rv_template_teff` $\leq 3400K$.
- Bit 5: [Not Used].
- Bit 6: Too few valid transits. `vbroad` is invalidated for stars with `vbroad_nb_transits` ≤ 5.
- Bit 7: The pipeline did not provide a `vbroad` value.
- Bit 8: Hot star. `vbroad` is invalidated for stars with `rv_template_teff` > 25000.
- Bits 9-15: [Not Used].

GRVS_MAG_IS_VALID : Validity flag for the Grvs magnitude (boolean)

This flag indicates if `grvs_mag` is valid or not.

Draft

GRVS_MAG_INVALID_REASON : The reason why the Grvs magnitude is set as invalid (short)

This 16-bit field specifies the reasons for invalidating the `grvs_mag` data. Each bit represents a specific reason, set to 1 if true and 0 if false.

Post-processing assigns these flags based on validation analysis. Invalidation reasons are determined based on various conditions, and may apply even when a `grvs_mag` is not provided.

A `grvs_mag` may be invalidated for multiple reasons, leading to the activation of multiple bits. See Table **??** for the occurrence of each invalid reason.

Both valid and invalid data are published in `all_source_rvs`.

- Bit 0: [Not used].
- Bit 1: Too faint star: `grvs_mag` is invalidated for stars with `external_apparent_grvs` $>$ 16 mag.
- Bit 2: Contamination by a nearby star. This invalidation reason applies to both `grvs_mag` and `radial_velocity_single` for a list of stars identified as contaminated using the method described in Katz et al. (2023).
- Bit 3: The pipeline did not provide a `grvs_mag` value.
- Bit 4: Too few valid transits. `grvs_mag` is invalidated if `grvs_mag` $<$ 13 and `grvs_mag_nb_transits` $<$ 2 or `nb_transits_total` $\leq$ 5; or if `grvs_mag` $\geq$ 13 and `grvs_mag_nb_transits` $<$ 3 or `nb_transits_total` $\leq$ 5.
- Bit 5: Excessive uncertainty. `grvs_mag` is invalidated if `grvs_mag_error` $>$ 16 mag.
- Bit 6: Value outside allowed range. `grvs_mag` is invalidated if `grvs_flux` $<$ $MinFlux$.

$$\mathrm{MinFlux} = \frac{\mathrm{MinFlux_{CCD}}}{\sqrt{3 \times \texttt{grvs_mag_nb_transits}}}$$

where $MinFlux_{CCD}$ = 1024 e^-/s represents twice the estimated average precision of straylight flux integrated over a single CCD spectrum in a typical Calibration Unit.

- Bits 7-15: Not used.

DESIGNATION : Unique source designation (unique across all Data Releases) (string)

A source designation, unique across all Gaia Data Releases, that is constructed from the prefix 'Gaia DRx ' followed by a string of digits corresponding to `source_id` (3 space–separated words in total). Note that the integer source identifier `source_id` is **not** guaranteed to be unique across Data Releases; moreover it is not guaranteed that the same astronomical source will always have the same `source_id` in different Data Releases. Hence the only safe way to compare source records between different Data Releases in general is to check the records of proximal source(s) in the same small part of the sky.

RANDOM_INDEX_AS : Random index for use when selecting subsets (long)

A random index which can be used to select smaller subsets of the data that are still representative. The column contains a random permutation of the numbers from 0 to $N - 1$, where N is the number of sources in the catalogue. Please note that the random index in the four 'all source' tables is the same, i.e. the same `source_id` in `all_source_astrometry`, `all_source_photometry`, `all_source_flags` and `all_source_rvs` will have the same `random_index_a` (while noting that `all_source_rvs` contains fewer sources than N). However this is not so for `random_index` in `gaia_source` which has it's own assigned random index values.

The random index can be useful for validation (testing on 10 different random subsets), visualization (displaying 1% of the data), and statistical exploration of the data, without the need to download all the data.

IN_GAIA_SOURCE : Flag indicating presence in the main Gaia source catalogue (boolean)

This flag indicates that this source is eligible to be tabulated in `gaia_source`, based solely on astrometric and photometric criteria. Consequently, `in_gaia_source` may be True even for sources with invalid radial velocity, Grvs, or `vbroad`, and vice versa. If a source with an invalid CU6 entry is included in `gaia_source`, the CU6 data is nullified, while the reason for its invalidity is still provided.

NB_TRANSITS_TOTAL : Total number of transits available (int)

The total number of RVS transits (including the invalid ones). This is the number of entries for a given source in the `rvs_transit_description` table.

RV_ASSUMED_SB2 : The star is assumed to be an SB2 (boolean)

An SB2 star (double-lined spectroscopic binary) is a binary system in which both stellar components contribute visible spectral lines, whose radial velocities vary with orbital motion and can

therefore be measured separately.

This boolean is set to True when double lines are detected in at least 10 RVS epoch spectra of the source. The detection of double lines is performed for stars brighter than `external_apparent_grvs`=12.

When `rv_assumed_sb2` is True, the source is excluded from the multi-transit analysis in the RVS pipeline. As a result, no combined radial velocity, line broadening velocity, mean spectrum, or associated parameters are produced by the RVS pipeline.

However, the epoch results obtained by the RVS pipeline are published. Sources may have both types of transits. Those identified as double-lined are provided in the table `rvs_epoch_parameters_double`, while those not identified as double-lined are provided in the table `rvs_epoch_parameters_single`. The corresponding epoch spectra are available in `rvs_epoch_spectrum`.

Note: Only about one third of these stars are subsequently confirmed as SB2 by the NSS pipeline; see Chapter **??**. Confirmed SB2 systems correspond to NSS solutions of type:
`nss_two_body_orbit.solution_type` = SB2ScanAngleK1, SB2ScanAngleK12, AstroSpectroSB2, or EclipsingSpectroSB2;
`nss_non_linear_spectro.solution_type` = TrendSB2;
`nss_multiple_orbits.solution_type` = SB2SB1 or SB2TrendSB1.

FRACTION_DOUBLE_LINED : Ratio of occurrences of multiple components in radial velocity (float)

The ratio of the number of transits where double lines are detected in the spectrum to the total number of transits analysed. The presence of doublelines is estimated for bright stars with `external_apparent_grvs`≤12.

Note: the total number of analysed transits for a source ($N_{\rm tot}$) is given by:

$$N_{\rm tot} = \frac{\texttt{rv_nb_transits}}{1 - \texttt{fraction_double_lined}}$$

Epoch data extracted from spectra in which two sources are detected (double-lined spectra) are provided in the DataLink table `rvs_epoch_parameters_double`, while those extracted from spectra in which only one source is detected are stored in the DataLink table `rvs_epoch_parameters_single`

FRACTION_EMISSION_LINES : ratio of occurrences of emission lines (float)

The ratio of the number of transits where emission lines were detected to the total number of

transits analyzed. Transits with detected emission lines are flagged in the epoch DataLink tables `rvs_epoch_parameters_single` via the field `emission_lines`. The presence of emission lines is estimated using a deep-learning technique for transits of stars with `external_apparent_grvs` ≤ 13. See **?**.

For these transits, neither the epoch radial velocity nor the epoch `vbroad` is computed. However, the epoch spectrum is published in `rvs_epoch_spectrum`, and the epoch `grvs_mag` is computed. These transits are excluded from the computation of the mean quantities.

Stars with `fraction_emission_lines` > 0.4 are considered emission-line stars. For these sources, the mean spectrum is published, but no other mean parameters are computed.

RADIAL_VELOCITY_SINGLE : Radial velocity for the single star model (float, Velocity[km s^{-1}])

Spectroscopic radial velocity in the Solar system barycentric reference frame.

The `radial_velocity_single` is the multi-epoch (or multi-transit) value obtained by combining the epoch (transit) data of these stars. The total number of epochs used is stored in `rv_nb_transits`. Both non-blended and deblended spectra are used, and the number of epochs having deblended spectra is stored in `rv_nb_deblended_transits`.

There are four methods used to obtain `radial_velocity_single`, depending on the magnitude `external_apparent_grvs` of the star. Information about which method has been used is stored in `rv_single_method_used`.

For bright stars (`external_apparent_grvs` ≤ 11; `rv_single_method_used`=1) `radial_velocity_sing` is calculated as the median value of the epoch radial velocities. For fainter stars with less precise epoch radial velocities, the epoch cross-correlation functions obtained by comparing the RVS spectrum and the associated template spectrum are combined to derive `radial_velocity_single`.

For stars with magnitude between 11 and 14 mag (11 $<$ `external_apparent_grvs` $\leq$ 14 ; `rv_single_method_used`=2), the epoch cross-correlation function is a combination of the CCD cross-correlation functions obtained by comparing each CCD spectrum with the template.

For stars with magnitude greater than 14 mag (14 $<$ `external_apparent_grvs` `rv_single_method_used`= the CCD spectra of the transit are combined first and then compared with the template to estimate the epoch cross-correlation function.

For the few stars that exhibit a scan angle signal in the radial velocity `radial_velocity_single` is taken from `radial_velocity_scan_angle` (`rv_single_method_used`=5). Note that `rv_single_meth` is never used.

See Section **??** for the complete list of parameters containing information on the radial_velocity_single measurements

In Gaia DR4, radial_velocity_single is provided for about 250 million stars and is considered valid for approximately 85 million of them. It is not provided for stars flagged as SB2 (rv_assumed_sb2=true), stars suspected of having emission lines (fraction_emission_lines> 0.4), or any star with fewer than two valid transits. See **?** for a description of the validation of the radial_velocity_single measurements, **?** for the validation of the radial_velocity_sys, and **?** for the specific treatments to obtain and validate radial velocities for hot stars.

RADIAL_VELOCITY_SINGLE_ERROR : Radial velocity error for the single star model (float, Velocity[km s^{-1}])

The uncertainty associated with radial_velocity_single.

Four methods are employed to derive radial_velocity_single and radial_velocity_single_error, depending on the magnitude external_apparent_grvs of the star. The specific method utilized is indicated by the rv_single_method_used field. Additionally, a constant shift of 0.112 km s^{-1} is applied to account for calibration floor contributions.

For stars with external_apparent_grvs ≤ 11 (rv_single_method_used=1), radial_velocity_singl is the uncertainty on the median of the epoch radial velocities ($\sigma_{\rm med}$):

$$\texttt{radial_velocity_error} = \sqrt{\sigma^2_{\rm med} + 0.112^2}$$

$$\sigma_{\rm med} = \sqrt{\frac{\pi}{2}} \cdot \frac{\sigma}{\sqrt{\texttt{rv_nb_transits}}}$$

where σ is the standard deviation of the epoch radial velocity measurements, and rv_nb_transits is the number of transits for which an epoch radial velocity has been obtained.

For fainter stars (external_apparent_grvs > 11; rv_single_method_used = 2 or 3), the radial_velocity_error is derived from the combined cross-correlation functions:

$$\texttt{radial_velocity_error}^2 = \frac{C^2 - 1}{MNC''C} + 0.112^2$$

where C and C'' are the values of the cross-correlation function and its second derivative at the maximum, respectively. N is the number of bins of the input spectrum, and M is the number of transits (rv_nb_transits) (see Zucker (2003), Sec. 2.3).

For the few stars that exhibit a scan angle signal in the radial velocity (rv_single_method_used=5), radial_velocity_single_error is taken from radial_velocity_scan_angle_error.

RAW_RAD_VEL_SINGLE_ERROR : The uncertainty provided by the model (float, Velocity[km s^{-1}])

The raw uncertainty is the uncertainty directly provided by the model used to compute radial_velocity_sin with no correction applied.

It corresponds to the uncorrected version of radial_velocity_single_error, i.e., without the addition in quadrature of 0.112 km s^{-1}.

RV_SINGLE_METHOD_USED : Method used to obtain the radial velocity for the single star model (byte)

rv_single_method_used contains information on the method used to obtain radial_velocity_single and its associated uncertainty, radial_velocity_single_error.

The methods used are:

- rv_single_method_used = 0: No model applied (no radial_velocity_single is obtained for this star).
- rv_single_method_used = 1: Combination of the epoch radial velocities. This method is used for bright stars (external_apparent_grvs ≤ 11). The radial_velocity_sing value is the median of the epoch radial velocities stored in rvs_epoch_parameters_single, after excluding those flagged with rv_invalid_reason as 'Contaminated by a bright neighbour' (bit 8) or 'Duplicated transit' (bit 15), also stored in rvs_epoch_parameters_s The associated uncertainty, radial_velocity_single_error, is the uncertainty on the median.
- rv_single_method_used = 2: Combination of the epoch cross-correlation functions (CCFs). The epoch CCFs are obtained in two different ways depending on the magnitude of the star. For stars with 11 < external_apparent_grvs ≤ 14, the CCFs obtained for each CCD spectrum are combined to produce the epoch CCF. For stars with 14 < external_apparent_grvs, the epoch CCFs are obtained directly from the epoch-combined spectrum. The epoch CCFs are then combined to obtain a final combined CCF, which is used to estimate radial_velocity_single and radial_velocity_single_error.
- rv_single_method_used = 5: This method is used for the few stars that exhibit a scan-angle signal in the radial velocity. radial_velocity_single is copied

from `radial_velocity_scan_angle`, and `radial_velocity_single_error` is copied from `radial_velocity_scan_angle`.

A constant of 0.112 km s^{-1} is added to `radial_velocity_single_error` in all methods to account for a calibration floor.

Note that `rv_single_method_used` = 3 and 4 were never used and are therefore not described.

RV_NB_TRANSITS : Number of transits used to compute the radial velocity (short)

The total number of epochs (transits) used to compute `radial_velocity_single` and `radial_velocity_sy`

The transits used in the combination are those deemed valid during the pipeline processing. Information on which transits are valid is stored in the table `rvs_transit_description.is_ccd_valid` (at least one of the three CCDs should be valid).

`rv_nb_transits` = 0 when `radial_velocity_single` is not provided. This typically occurs when the number of valid transits is <2, when the star is assumed to be an SB2 (`rv_assumed_sb2` = true), or when the star is assumed to be an emission-line star (`fraction_emission_lines` > 0.4).

RV_NB_DEBLENDED_TRANSITS : Number of valid transits that have undergone deblending (short)

The number of epochs (transits) among `rv_nb_transits` for which at least one of the 3 RVS CCD spectra has undergone deblending. The deblending of the spectra is decribed in Seabroke et al. (2026).

RV_VISIBILITY_PERIODS_USED : Number of visibility periods used to estimate the radial velocity (short)

Number of visibility periods used to obtain `radial_velocity_single` and `radial_velocity_sys`. A visibility period is a group of transits separated from other groups by a gap of at least 4 days.

RV_EXPECTED_SIG_TO_NOISE : Expected signal to noise in spectrum used to obtain radial velocity (float)

The expected signal to noise ratio is obtained by combining the information from the CCD spec-

tra used for estimating the `radial_velocity_single` and `radial_velocity_sys`,

$$\texttt{rv_expected_sig_to_noise} = \frac{S}{\sqrt{S + BCK + RN}}$$

where:

$$S = \frac{f_{\rm med} \times N_{ValidStrips} \times \mathrm{EXP_TIME} \times \mathrm{PIXEL_WIDTH_AL}}{\mathrm{BAND_WIDTH}}$$

$$BCK = b_{\rm med} \times N_{ValidStrips} \times N_{ACpixels}$$

$$RN = RON^2 \times N_{ValidStrips} \times N_{ACsamples}$$

where:

- $f_{\rm med}$ is the median of the integrated flux of the CCD spectra used to obtain the radial velocity for `external_apparent_grvs` ≤ 12; otherwise it is obtained from `external_apparent_grvs` [$e^- s^{-1}$]
- $N_{ValidStrips}$ = the total number of CCD spectra used to obtain the `radial_velocity` (in general there are 3 CCD spectra per transit)
- BAND_WIDTH = 870 - 846 [nm]
- EXP_TIME = 4.4167032 [s]
- PIXEL_WIDTH_AL = 0.02453 [nm]
- $b_{\rm med}$ = the median background in one exposure of the CCD spectra used to obtain the radial velocity [e^-pixel^{-1}]
- $N_{ACpixels}$ = 10 is the number of pixels in the AC direction of the non-truncated RVS window
- $N_{ACsamples}$ is the size in the AC direction of the non-truncated RVS window. $N_{ACsamples}$ = 10 for the 2D RVS windows (Window Class=0) and $N_{ACsamples}$ = 1 for the 1D windows (Window Class=1)
- RON (Read Out Noise) = 3.2 [e^-]

The SNR of the spectra combination used for `vbroad` determination can be estimated from

$$\texttt{rv_expected_sig_to_noise} \sqrt{\frac{\texttt{vbroad_nb_transits}}{\texttt{rv_nb_transits}}},$$

and similarly for `grvs_mag` from

$$\texttt{rv_expected_sig_to_noise} \sqrt{\frac{\texttt{grvs_mag_nb_transits}}{\texttt{rv_nb_transits}}}.$$

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rv_expected_sig_to_noise is not provided when rv_assumed_sb2 = True, when fraction_emission_
> 0.4, or when fewer than two valid transits are available (rvs_transit_description).

Note: The SNR measured directly on the combined spectrum rvs_mean_spectrum is stored in rvs_spec_sig_to_noise.

RV_SINGLE_OUTLIER_PROBABILITY : Probability for radial velocity to be an outlier (float)

This field provides an estimate of whether radial_velocity_single is erroneous. It is computed using machine learning techniques, specifically a balanced random forest trained on parameters that describe the shape of the Cross-Correlation Function, and is available only for stars that satisfy the following criteria:

- external_apparent_grvs > 11
- $3700 \leq$ rv_template_teff ≤ 6500 K

Outside this validity range, the field is not populated. It is also not populated when rv_single_method_used = 5 or when radial_velocity_single is unavailable. Values usually lie between 0 and 1, acting as a probability-like score where larger values indicate a higher likelihood that the radial velocity is erroneous.

In some cases, the field is set to 2 to indicate that the method could not compute a valid score, typically when radial_velocity_single itself is erroneous. In such cases, the radial velocity is invalidated and rv_invalid_reason has bit 23 set.

For a detailed description of how rv_single_outlier_probability is derived, see **?**.

RV_RENORMALISED_GOF : Radial velocity renormalised goodness of fit (float)

The renormalised goodness of fit (see Perryman 1997, p. 112) is an empirical value computed using this source and all the other sources having rv_template_teff and external_apparent_grvs in a given range. The scatter of the source epoch radial velocities is compared to the typical epoch uncertainty for the appropriate external_apparent_grvs and rv_template_teff range.

$$\texttt{rv_renormalised_gof} = \sqrt{4.5\nu}\left(RUWE^{2/3} + \frac{2}{9\nu} - 1\right)$$

where:

$$\nu = \texttt{rv_nb_transits} - 1$$

$$RUWE = \frac{UWE}{\mathrm{mode}(UWE)}$$

$$UWE = \sqrt{\frac{1}{\nu}\sum_j \left(\frac{V_j - \texttt{radial_velocity}}{\epsilon_j}\right)^2}$$

• `radial_velocity` is the median of the epoch radial velocities V_j (`rv_median`);

• V_j are the epoch radial velocities (`rvs_epoch_parameters_single`);

• ϵ_j is the uncertainty on V_j (`rvs_epoch_parameters_single`);

• mode(UWE) is the mode of the UWE distribution in the appropriate bin of the UWE (`rv_template_teff`, `external_apparent_grvs`) look-up table obtained by an offline analysis (see Section **??**);

• $RUWE$: Renormalized Unit Weight Error, UWE: Unit Weight Error.

`rv_renormalised_gof` is provided for bright stars with $5.5 \leq$ `external_apparent_grvs` ≤ 12 and `rv_template_teff` $< 14{,}500$ K. It is empty for stars with `external_apparent_grvs` and `rv_template_teff` outside this range.

See Section **??** for more details on how this field is derived.

RV_CHISQ_PVALUE : P-value for radial velocity constancy using a chi-squared criterion (float)

`rv_chisq_pvalue` is the P-value for radial velocity constancy. It is defined as the probability that the χ^2 value of the radial velocity time series will exceed the expected value from the chi-square probability distribution function with significance level α and having $k = N - 1$ degrees of freedom. P takes values between [0,1] assuming known radial velocity uncertainties that are normally distributed. It ranges from 0 (zero) for very low probability of radial velocity constancy to 1 (one) for very strong probability of radial velocity constancy. V_j is the time series of size N (N=`rv_nb_stat_transits_robust`), with $N > 1$ of valid epoch radial velocity values for which the large statistical outliers are excluded (see below). The radial velocity values have uncertainties ϵ_j and weights $w_j \equiv 1/(\epsilon_j)^2$. The weighted chi-squared test statistic value χ^2 is then

defined as:

$$\chi^2 = \frac{1}{N-1}\frac{W\sum_{j=1}^{N} w_j V_j^2 - \left(\sum_{j=1}^{N} w_j V_j\right)^2}{W}$$

where $W \equiv \sum_{j=1}^{N} w_j$ is the sum of weights. The P-value is defined as the cumulative probability of χ^2 from the chi-square probability distribution function with k degrees of freedom:

$$P \equiv P(\chi^2, k) = 1 - \frac{2^{-k/2}}{\Gamma(k/2)}\int_0^{\chi^2} t^{k/2-1}\exp(-t/2)\,dt$$

where $\Gamma(k/2)$ is the gamma function which has closed-form values for integer k.

The P-value is a probability-based statistic with respect to the null hypothesis that the source is constant in radial velocity. It assumes that ϵ_j are normally distributed uncertainties (that may not always be known or available). If one selects for example all sources having $P < 0.05$ from the sample (i.e., α is set equal to 0.05), this will yield ~5% of radial velocity constant sources that can still contaminate the selected subset of radial velocity variable sources. In case one wants a subset of radial velocity variables less contaminated with constant sources, choose a smaller value of α, but which will also decrease the size of the selected subset of variables.

The outlier transits are those for which the field `outlier_status` in `rvs_epoch_parameters_single` is set to one of the following values: 1,4,5,6,9,11,13 or 15. To find potential radial velocity variable stars we suggest using also `rv_nb_stat_transits_robust` and `rv_renormalised_gof`. The following selection is suggested:

`rv_chisq_pvalue` < 0.01 and `rv_renormalised_gof` > 4 and `rv_nb_stat_transits_robust` ≥ 10.

Note: `rv_chisq_pvalue` is calculated in DR4 for the stars with `external_apparent_grvs` ≤ 14 and `rv_nb_stat_transits_robust`>1. However, for `external_apparent_grvs` ≥ 13.5, due to the large dispersion of the data and the underestimation of the uncertainties of outlying transits, `rv_chisq_pvalue` often takes the value below 1E-16.

See **?** for more information on `rv_chisq_pvalue` and on other statistical parameters.

RV_TEMPLATE_TEFF : Teff of the template used to compute the radial velocity (float, Temperature[K])

Effective temperature of the synthetic spectrum template used to determine the mean and epoch `radial_velocity` and `vbroad` (in `all_source_rvs` and in `rvs_epoch_parameters_single`).

The synthetic spectrum associated with the star is the one with the atmospheric parameters closest to those associated with the star. However, when `redetermine_ap_hot_stars` is set to 'True',

the template parameters are redetermined and may be very different from those associated with the star.

N.B. the purpose of this parameter is to provide information on the synthetic template spectrum used, not to provide an estimate of the stellar effective temperature of this source. The available synthetic spectra are listed in Section **??**. More information on how the template is produced can be found in Section **??**.

RV_TEMPLATE_LOGG : Logg of the template used to compute the radial velocity (float, GravitySurface[log cgs])

Surface gravity log g of the synthetic spectrum template used to determine the mean and epoch `radial_velocity` and `vbroad` (in `all_source_rvs` and in `rvs_epoch_parameters_single`).

The synthetic spectrum associated with the star is the one with the atmospheric parameters closest to those associated with the star. However, when `redetermine_ap_hot_stars` is set to 'True', the template parameters are redetermined and may be very different from those associated with the star.

N.B. the purpose of this parameter is to provide information on the synthetic template spectrum used, not to provide an estimate of the surface gravity of this source. The available synthetic spectra are listed in Section **??**. More information on how the template is produced can be found in Section **??**.

RV_TEMPLATE_FE_H : [Fe/H] of the template used to compute the radial velocityy (float, Abundances[dex])

Metallicity (compared to solar) [Fe/H] of the synthetic spectrum template used to determine the mean and epoch `radial_velocity` and `vbroad` (in `all_source_rvs` and in `rvs_epoch_parameters_sin`

The synthetic spectrum associated with the star is the one with the atmospheric parameters closest to those associated with the star. However, when `redetermine_ap_hot_stars` is set to 'True', the template parameters are redetermined and may be very different from those associated with the star.

N.B. the purpose of this parameter is to provide information on the synthetic template spectrum used, not to provide an estimate of the metallicity of this source. The available synthetic spectra are listed in Section **??**. More information on how the template is produced can be found in Section **??**.

RV_TEMPLATE_VSINI : vsini of the template used to estimate the radial velocity (float, Velocity[km s^{-1}])

Projected rotational velocity, $v \sin i$, with which the synthetic template used to determine the radial velocity is convolved.

The $v \sin i$ value is taken from a pre-specified $v \sin i$ law (5 km s^{-1} below log(`rv_template_teff`)=3.8, 100 km s^{-1} above log(`rv_template_teff`)=3.9, linear between 3.8 and 3.9), or from an appropriate redetermination when `redetermine_ap_hot_stars` is set to True. More information on how the template is produced can be found in Section **??**.

RV_ATM_PARAM_ORIGIN : Origin of the atmospheric parameters associated with the template (short)

`rv_atm_param_origin` is a three-digit code that encodes the origin of the three atmospheric parameters (APs) associated with a star, namely `rv_atm_param_star_teff`, `rv_atm_param_star_logg`, and `rv_atm_param_star_fe_h`, in that order, corresponding to effective temperature ($T_{\rm eff}$), surface gravity ($\log g$), and metallicity ([Fe/H]).

The possible origins for each atmospheric parameter are:

- 0: Default. The APs are set to default values when no information is available and no estimation is attempted by the spectroscopic pipeline. The default values are: $T_{\rm eff}$ = 5500 K, $\log g$ = 4.5 dex, and [Fe/H] = 0 dex. In a few cases (when Teff origin = 1), [Fe/H] = −2 dex is used instead.
- 1: GSP-Phot. APs obtained from the DR3 version of the GSP-Phot pipeline.
- 2: Ground-based. APs obtained from ground-based catalogues. The catalogue name is provided in `rv_atm_param_gb_catalogue`.
- 3: DetermineAP. APs derived using the internal *DetermineAP* method of the spectroscopic pipeline.
- 4: GSP-Spec. APs obtained from the DR3 version of the GSP-Spec pipeline.
- 5: ESP-HS. APs derived from the DR3 version of the ESP-HS pipeline.
- 6: ESP-UCD. APs derived from the DR3 version of the ESP-UCD pipeline.
- 7: APs taken from Andrae et al. (2023a).

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- 8: Post-pipeline correction. APs modified in the final pipeline step due to large inconsistencies with the DR3 spectral type, which would result in an incorrect radial velocity.

The possible values of `rv_atm_param_origin` are:

- -1: Parameters are not provided.
- 000: All parameters use default values: $T_{\rm eff}$ = 5500 K, $\log g$ = 4.5 dex, [Fe/H] = 0 dex.
- 111: All parameters from GSP-Phot.
- 110: $T_{\rm eff}$ and $\log g$ from GSP-Phot; [Fe/H] = 0 (default).
- 101: $T_{\rm eff}$ and [Fe/H] from GSP-Phot; $\log g$ = 4.5 (default).
- 100: $T_{\rm eff}$ from GSP-Phot; $\log g$ = 4.5, [Fe/H] = 0 (default).
- 222: All parameters from ground-based catalogues.
- 220: $T_{\rm eff}$ and $\log g$ from ground-based data; [Fe/H] = default. ([Fe/H] = −1 if `rv_atm_param_gb_catalogue` = 26, otherwise [Fe/H] = 0.)
- 200: $T_{\rm eff}$ from ground-based data; $\log g$ and [Fe/H] are default.
- 202: $T_{\rm eff}$ and [Fe/H] from ground-based data; $\log g$ is default.
- 333: All parameters from DetermineAP.
- 444: All parameters from GSP-Spec.
- 440: $T_{\rm eff}$ and $\log g$ from GSP-Spec; [Fe/H] = 0 (default).
- 550: $T_{\rm eff}$ and $\log g$ from ESP-HS; [Fe/H] = 0 (default).
- 600: $T_{\rm eff}$ from ESP-UCD; $\log g$ = 4.5, [Fe/H] = 0 (default).
- 777: All parameters from Andrae et al. (2023b).
- 888: All parameters were modified in the last pipeline step due to inconsistency with DR3 inputs, resulting in incorrect radial velocity.

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RV_ATM_PARAM_GB_CATALOGUE : The catalogue from where the ground based params are extracted (byte)

This field, `rv_atm_param_gb_catalogue`, contains information about the catalogue from which the ground-based atmospheric parameters are extracted (see Section **??**).

It is filled only when `rv_atm_param_origin` is 222, 220, 202, 200, or 777. For all other origins, its value is 0.

Origin of the Ground-Based Atmospheric Parameters:

- 0 : No GB catalogue

GBOG determinations: fundamental laws and/or spectroscopy

- 1 : Benchmark stars from Soubiran et al. (2014)
- 2 : iSpec, NEP stars by L. Chemin from Soubiran et al. (2014)
- 3 : Bacchus by G. J. (21 SEP stars) from Soubiran et al. (2014)
- 4 : South Ecliptic Pole stars from Frémat et al. (2017)
- 5 : Hot star sample; Frémat (private comm.)

Literature: asteroseismology or spectroscopy

- 10 : Seismic log g from Soubiran et al. (2014)
- 11 : PASTEL from Soubiran et al. (2014)
- 12 : OBA compilation from Soubiran et al. (2014)
- 13 : AMBRE from Soubiran et al. (2014)
- 14 : APOGEE DR17
- 15 : RAVE DR5
- 16 : MILES from Soubiran et al. (2014)
- 17 : CFLIB library from Soubiran et al. (2014)

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- 18 : GES DR6
- 19 : GALAH DR3
- 20 : LAMOST DR7
- 21 : SDSS DR14
- 22 : Kepler; Mathur et al. (2017)
- 23 : OB star selection from Simón-Díaz et al. (2017)
- 24 : Not used
- 25 : APOGEE red clump stars from Bovy et al. (2014)
- 26 : Cepheids and RR Lyrae from Clementini et al. (2023) and Ripepi et al. (2023)
- 27 : Not used
- 28 : OB star selection from Bragança et al. (2019)
- 29 : Not used
- 30 : SEGUE selection from Yanny et al. (2009)

GBOG determinations from photometry

- 39 : *uvby* photometry of OBA stars by Lobel & Frémat; Soubiran et al. (2014)

Literature: photometry

- 40 : Pinsonneault et al. (2012)
- 41 : Casagrande et al. (2011)
- 42 : Robinson et al. (2007)
- 43–44 : Not used
- 45 : Caffau (private comm.)
- 50 : Gaia XP; Andrae et al. (2023b)

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RV_ATM_PARAM_STAR_TEFF : Teff associated with the star (float, Temperature[K])

Effective temperature of the star. The origin of the atmospheric parameters associated with the star is in `rv_atm_param_origin`.

RV_ATM_PARAM_STAR_LOGG : logg associated with the star (float, GravitySurface[log cgs])

Surface gravity of the star. The origin of the atmospheric parameters associated with the star is in `rv_atm_param_origin`.

RV_ATM_PARAM_STAR_FE_H : Fe/H associated with the star (float, Abundances[dex])

Metallicity of the star. The origin of the atmospheric parameters associated with the star is in `rv_atm_param_origin`.

REDETERMINE_AP_HOT_STARS : Are the template atmospheric parameters redetermined? (boolean)

This field contains the information on if the template atmospheric parameters have been determined by the spectroscopic pipleine internal method 'RedetermineAPHS'(Blomme et al. (2023)).
In this case the template parameters (`rv_template_teff`, `rv_template_logg` and `rv_template_fe_h`) may be inconsistent with the star parameters (`rv_atm_param_star_teff`, `rv_atm_param_star_logg` and `rv_atm_param_star_fe_h`).

ASTROMETRIC_MOTION_CORRECTED : The epoch radial velocities are corrected by astrometric motion (boolean)

This flag indicates whether the star is among the approximately 1.46 million stars identified as astrometric binaries, for which the astrometric motion has been estimated and corrected in each transit. The correction is applied by shifting the wavelength grid of the CCD spectra. For more information on the processing of these stars, see the Non-Single Star mode described in Section **??**.

HAS_SYSTEMIC_RV : Indicates whether the source has a systemic radial velocity (boolean)

This flag indicates whether the star has a systemic radial velocity. If set to True, an orbital model is provided and stored in `rvs_two_body_orbit`. The systemic velocity is tabulated both in `all_source_rvs` as `radial_velocity_sys` and in `rvs_two_body_orbit` as `center_of_mass_velocity`

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RADIAL_VELOCITY_SYS : Radial velocity for the System (float, Velocity[km s^{-1}])

The centre of mass radial velocity in the Solar system barycentric reference frame.

This field is populated in the case the star is found to be part of a spectroscopic binary system (`has_systemic_rv`=True) and is not an assumed SB2: `rv_assumed_sb2`=False. The binary solution obtained is tabulated in `rvs_two_body_orbit`. See **?**.

RADIAL_VELOCITY_SYS_ERROR : Systemic radial velocity error (float, Velocity[km s^{-1}])

The uncertainty associated with `radial_velocity_sys`. See **?** for further details.

RADIAL_VELOCITY_SCAN_ANGLE : Spectroscopic barycentric velocity from the scan angle signal (float, Velocity[km s^{-1}])

Radial velocity from the scan angle signal.

If the observed source is a resolved or partially resolved binary, and if it is not flagged as `rv_assumed_sb2`, the epoch radial velocities may be affected by the presence of the secondary. The effect depends on the projected separation between the two components along the scan direction and can be modelled by a sinusoidal signal as function of the scan angle.

The pipeline fits the epoch radial velocities with a sinusoidal model as a function of the scan angle, estimate the signal's significance and store its phase and amplitude.

The single-lined epoch radial velocities (`radial_velocity`) from `rvs_epoch_parameters_single` are used, excluding cases where the radial velocity is invalid due to contamination from a brighter neighbour, duplication, or detection of emission lines. These velocities are fitted as a function of the scan angle, ψ, which is the position angle of the scan direction, using the following expression:

$$V(\psi) = V_0 + c \cdot \cos(\psi) + s \cdot \sin(\psi) \tag{1}$$

where:

$$\texttt{radial_velocity_scan_angle} = V_0$$

The amplitude and phase of the signal are calculated as:

$$\texttt{scan_angle_signal_amplitude} = \sqrt{c^2 + s^2}$$

$$\texttt{scan_angle_signal_phase} = \mathrm{atan2}(s, c)$$

The scan_angle_signal_significance gives the significance of the fit (see Sec. 2.3.2 in Halbwachs et al. (2023)). A large amplitude and significance may indicate that the source is a candidate for having some non-isotropic spatial structure, for example a binary or galaxy that is at least partially resolved by Gaia.

The scan angle signal is considered significant if:

abs(radial_velocity_scan_angle) < 150
AND
(scan_angle_signal_significance > 25
OR
(scan_angle_signal_significance > 15 AND ipd_frac_multi_peak > 50))

where ipd_frac_multi_peak is stored in all_source_astrometry.

When the scan angle signal is significant:

- radial_velocity_single is copied from radial_velocity_scan_angle, and radial_velocity_single_error from radial_velocity_scan_angle_error.
- rv_single_method_used is set to 5.
- The epoch radial_velocity in rvs_epoch_parameters_single is set as invalid.
- rv_outlier_probability is set as NaN.

User warning: The scan-angle derived quantities can be affected by spurious or contaminated epoch radial velocities, as well as by poor sampling. In particular:
radial_velocity_scan_angle may be biased if some epoch velocities are invalid or outliers, and may not reflect the true systemic velocity of the source.
scan_angle_signal_amplitude can be inflated or distorted by a few extreme epochs, giving the impression of a stronger scan-angle effect than truly present.
scan_angle_signal_significance may be over- or under-estimated, since it depends on the fit quality relative to the scatter of the epoch velocities.

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Users are strongly advised to verify all sources with large amplitudes or high significance using the original epoch radial velocities before drawing physical conclusions.

RADIAL_VELOCITY_SCAN_ANGLE_ERROR : Error on `radial_velocity_scan_angle` (float, Velocity[km s^{-1}])

The uncertainty associated with `radial_velocity_scan_angle`.

SCAN_ANGLE_SIGNAL_SIGNIFICANCE : Scan angle signal significance (float, Velocity[km s^{-1}])

Significance of the scan angle signal.

The single-lined epoch radial velocities (`radial_velocity`) from `rvs_epoch_parameters_single` are used, excluding cases where the radial velocity is invalid due to contamination from a brighter neighbour, duplication, or detection of emission lines. These velocities are fitted as a function of the scan angle, ψ, which is the position angle of the scan direction, using the following model:

$$V(\psi) = V_0 + c \cdot \cos(\psi) + s \cdot \sin(\psi) \tag{2}$$

where:

$$\texttt{radial_velocity_scan_angle} = V_0$$

$$\texttt{scan_angle_signal_amplitude} = \sqrt{c^2 + s^2}$$

$$\texttt{scan_angle_signal_phase} = \text{atan2}(s, c)$$

The `scan_angle_signal_significance` gives the significance of the fit (see Sec. 2.3.2 in Halbwachs et al. (2023)).

The scan angle signal is considered significant if:

abs(`radial_velocity_scan_angle`) < 150
AND
(`scan_angle_signal_significance` > 25

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OR
(scan_angle_signal_significance > 15 AND ipdfracmultipeak > 50))

where ipd_frac_multi_peak is stored in all_source_astrometry.

When the scan angle signal is considered significant:

- radial_velocity_single is copied from radial_velocity_scan_angle, and radial_velocity_single_error from radial_velocity_scan_angle_error.
- rv_single_method_used is set to 5.
- The epoch radial_velocity in rvs_epoch_parameters_single is set as invalid.
- rv_outlier_probability is set as NaN.

User warning: The scan-angle derived quantities can be affected by spurious or contaminated epoch radial velocities, as well as by poor sampling. In particular:
radial_velocity_scan_angle may be biased if some epoch velocities are invalid or outliers, and may not reflect the true systemic velocity of the source.
scan_angle_signal_amplitude can be inflated or distorted by a few extreme epochs, giving the impression of a stronger scan-angle effect than truly present.
scan_angle_signal_significance may be over- or under-estimated, since it depends on the fit quality relative to the scatter of the epoch velocities.
Users are strongly advised to verify all sources with large amplitudes or high significance using the original epoch radial velocities before drawing physical conclusions.

SCAN_ANGLE_SIGNAL_AMPLITUDE : Scan angle signal amplitude (float, Velocity[km s^{-1}])

Amplitude of the scan angle signal.

The single-lined epoch radial velocities (radial_velocity) from rvs_epoch_parameters_single are used, excluding cases where the radial velocity is invalid due to contamination from a brighter neighbour, duplication, or detection of emission lines. These velocities are fitted as a function of the scan angle, ψ, which is the position angle of the scan direction, using the following model:

$$V(\psi) = V_0 + c \cdot \cos(\psi) + s \cdot \sin(\psi) \quad (3)$$

where:

$$\texttt{radial_velocity_scan_angle} = V_0$$

The amplitude and phase of the signal are calculated as:

$$\texttt{scan_angle_signal_amplitude} = \sqrt{c^2 + s^2}$$

$$\texttt{scan_angle_signal_phase} = \text{atan2}(s, c)$$

The `scan_angle_signal_significance` gives the significance of the fit (see Sec. 2.3.2 in Halbwachs et al. (2023)).

User warning: The scan-angle derived quantities can be affected by spurious or contaminated epoch radial velocities, as well as by poor sampling. In particular:
`radial_velocity_scan_angle` may be biased if some epoch velocities are invalid or outliers, and may not reflect the true systemic velocity of the source.
`scan_angle_signal_amplitude` can be inflated or distorted by a few extreme epochs, giving the impression of a stronger scan-angle effect than truly present.
`scan_angle_signal_significance` may be over- or under-estimated, since it depends on the fit quality relative to the scatter of the epoch velocities.
Users are strongly advised to verify all sources with large amplitudes or high significance using the original epoch radial velocities before drawing physical conclusions.

SCAN_ANGLE_SIGNAL_PHASE : Scan angle signal phase (float, Angle[deg])

Phase of the scan angle signal.

The single-lined epoch radial velocities (`radial_velocity`) from `rvs_epoch_parameters_single` are used, excluding cases where the radial velocity is invalid due to contamination from a brighter neighbour, duplication, or detection of emission lines. These velocities are fitted as a function of the scan angle, ψ, which is the position angle of the scan direction, using the following model:

$$V(\psi) = V_0 + c \cdot \cos(\psi) + s \cdot \sin(\psi) \tag{4}$$

where:

$$\texttt{radial_velocity_scan_angle} = V_0$$

The amplitude and phase of the signal are calculated as:

$$\texttt{scan_angle_signal_amplitude} = \sqrt{c^2 + s^2}$$

$$\texttt{scan_angle_signal_phase} = \mathrm{atan2}(s, c)$$

User warning: The scan-angle derived quantities can be affected by spurious or contaminated epoch radial velocities, as well as by poor sampling. In particular:
`radial_velocity_scan_angle` may be biased if some epoch velocities are invalid or outliers, and may not reflect the true systemic velocity of the source.
`scan_angle_signal_amplitude` can be inflated or distorted by a few extreme epochs, giving the impression of a stronger scan-angle effect than truly present.
`scan_angle_signal_significance` may be over- or under-estimated, since it depends on the fit quality relative to the scatter of the epoch velocities.
Users are strongly advised to verify all sources with large amplitudes or high significance using the original epoch radial velocities before drawing physical conclusions.

VBROAD : Spectral line broadening parameter (float, Velocity[km s^{-1}])

The spectral line broadening parameter, `vbroad`, is measured using a rotational broadening kernel; its value therefore includes $v \sin i$, as well as contributions from other sources of line broadening such as macroturbulence and residual instrumental effects.

`vbroad` is computed as the median of the epoch `vbroad` measurements stored in `rvs_epoch_parameters_si`
Measurements are excluded if they fall outside the allowed range (Bit 2 in `rvs_epoch_parameters_single.` if they are obtained from deblended spectra (Bit 4), if they were measured at the beginning of Gaia operations when the optics were contaminated by water ice, which caused an artificial broadening of the spectral lines (Bit 5), or if they are associated with a reported algorithm failure (Bit 15). The total number of transits used in the median computation is stored in `vbroad_nb_transits`.

A value of `vbroad` is provided for approximately 7 million stars with `external_apparent_grvs` $\leq$ 12, and it is deemed valid (`vbroad_is_valid` = true) for about 2.6 million stars. See **?** for a description of the validation of the `vbroad` measurements.

VBROAD_ERROR : Uncertainty on the spectral line broadening (float, Velocity[km s^{-1}])

`vbroad_error` is the uncertainty associated with `vbroad`. It is the uncertainty on the median of

the epoch measurements (rvs_epoch_parameters_single.vbroad) used to obtain vbroad:

$$\texttt{vbroad_error} = \sqrt{\frac{\pi}{2}} \cdot \frac{\sigma}{\sqrt{\texttt{vbroad_nb_transits}}}$$

where σ is the standard deviation of the epoch measurements used, and vbroad_nb_transits is the number of epochs used.

Note: When vbroad_error > vbroad, the star's vbroad estimation is: 0 ≤ vbroad ≤ vbroad + vbroad_error.

VBROAD_NB_TRANSITS : Number of transits used to compute vbroad (short)

The number of transits (epochs) used to compute vbroad.

vbroad is the median value of the transit vbroad measurements stored in rvs_epoch_parameters_single. The transits excluded from the computation of the median are those for which the vbroad value is outside the allowed range (Bit 2 and Bit 3 in rvs_epoch_parameters_single.vbroad_invalid_reason), those obtained using deblended spectra (Bit 4 in rvs_epoch_parameters_single.vbroad_invalid_reas and those obtained at the beginning of the Gaia operations when the optics were contaminated with water ice resulting in broadening of the spectral lines (Bit 5 in rvs_epoch_parameters_single.vbroa

vbroad_nb_transits = 0 when vbroad is not provided.

GRVS_FLUX : Integrated flux (float, Flux[e^- s^{-1}])

grvs_flux is computed as the median of the calibrated epoch grvs_flux measurements from the DataLink table rvs_epoch_parameters_single, excluding all epochs obtained from deblended spectra, spectra contaminated by a nearby bright source, duplicated transits (Bit 2, Bit 3, or Bit 8 activated in rvs_epoch_parameters_single.grvs_invalid_reason), or spectra with emission lines (rvs_epoch_parameters_single.emission_lines = True).

In some cases, grvs_flux < 0, which generally occurs for faint stars with an overestimated subtracted background. In these cases, grvs_flux_error is not provided.

The field grvs_flux is not provided when rv_assumed_sb2 = True or when all epoch observations are excluded according to the above criteria.

The total number of epoch transits used to compute grvs_flux is stored in grvs_mag_nb_transits.

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GRVS_FLUX_ERROR : Error on the integrated flux (float, Flux[e^- s^{-1}])

grvs_flux_error represents the uncertainty associated with grvs_flux. It is the uncertainty on the median of the epoch measurements (rvs_epoch_parameters_single.grvs_flux) used to obtain grvs_flux:

$$\texttt{grvs_flux_error} = \sqrt{\frac{\pi}{2}}.\frac{\sigma}{\sqrt{\texttt{grvs_mag_nb_transits}}}$$

where σ is the standard deviation of the epoch flux measurements used and grvs_mag_nb_transits is the number of epochs.

It is not provided in cases when grvs_flux ≤ 0.

GRVS_MAG : Grvs magnitude (float, Magnitude[mag])

The magnitude grvs_mag is estimated using the flux of the star in the RVS spectra, stored in the field grvs_flux, and the zero-point magnitude for grvs_mag ($ZP_{\mathrm{grvs}} = 21.31847$):

$$\texttt{grvs_mag} = -2.5 \times \log(\texttt{grvs_flux}) + ZP_{\mathrm{grvs}}$$

The value of grvs_flux is computed as the median of the calibrated epoch grvs_flux measurements from the DataLink table rvs_epoch_parameters_single, excluding all epochs flagged as deblended, contaminated by a nearby bright source, duplicated transits (Bit 2, Bit 3, or Bit 8 activated in rvs_epoch_parameters_single.grvs_invalid_reason), or with emission lines (rvs_epoch_parameters_single.emission_lines = True).

The field grvs_mag is empty when rv_assumed_sb2 is true, when all epoch measurements are excluded by the above criteria, or when grvs_flux < 0 (which generally occurs for faint stars with an overestimated subtracted background).

For stars where no grvs_mag is provided, an estimate of the magnitude in the RVS band based on non-RVS photometry (phot_g_mean_mag phot_rp_mean_mag)(see Sartoretti et al. (2023)), and is stored in the field external_apparent_grvs.

GRVS_MAG_ERROR : Grvs magnitude uncertainty (float, Magnitude[mag])

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The grvs_mag_error is calculated by adding in quadrature the uncertainty derived from the flux, denoted as σ, with a constant term of 0.002 mag. This constant accounts for a calibration floor contribution:

$$\texttt{grvs_mag_error} = \sqrt{\sigma^2 + 0.002^2}$$

The value of σ is expected to be computed as:

$$\sigma = \texttt{grvs_flux_error}/\texttt{grvs_flux} \times 2.5/\ln(10)$$

However, due to a coding error in the post-processing of faint sources (those with external_apparent_grvs > 14), the published value of σ is instead:

$$\sigma = \texttt{grvs_flux_error}/(1.016 \times \texttt{grvs_flux} + 25) \times 2.5/\ln(10)$$

As a result, the magnitude uncertainties for faint sources may be slightly underestimated. For more information, see **?**.

GRVS_MAG_NB_TRANSITS : Number of transits used to compute Grvs (short)

The number of transits (epochs) used to compute grvs_mag and grvs_flux.

Transits considered valid during pipeline processing and used for the radial velocity determination are included in the combination (see rvs_transit_description.is_ccd_valid). Consequently, transits flagged as emission_lines (rvs_epoch_parameters_single) and double-lined transits (rvs_epoch_parameters_double) are excluded.

In addition, transits with deblended spectra or identified as contaminated by a background source are also excluded, as indicated by Bit 2 and Bit 3 in rvs_epoch_parameters_single.grvs_invalid_reas

grvs_mag_nb_transits = 0 when grvs_flux is not provided.

EXTERNAL_APPARENT_GRVS : Apparent Grvs obtained from non-RVS data (float, Magnitude[mag])

The magnitude external_apparent_grvs is obtained from non-RVS data. There are two possible origins of external_apparent_grvs, which are stored in external_apparent_grvs_origin.

For most sources, it is obtained using the magnitudes measured from the DR3 Gaia data obtained with the astrometric instrument (`phot_g_mean_mag`) and the Red Photometer (`phot_rp_mean_mag`), with the transformation described in Sartoretti et al. (2023).

For the sources for which the red photometer magnitude from DR3 is not available, `external_apparent_grv` is the onboard magnitude obtained by the onboard software from Red Photometer spectra when these are not saturated, and otherwise from the astrometric images.

EXTERNAL_APPARENT_GRVS_ORIGIN : Origin of external_apparent_grvs (short)

The origin of the `external_apparent_grvs` can be:

0: Not provided.

3: Onboard: Obtained by the onboard software from Red Photometer spectra when these are not saturated, and otherwise from the astrometric images.

4: Using the DR3 measurements of `phot_g_mean_mag` and `phot_rp_mean_mag`, and the transformation described in Sartoretti et al. (2023).

RVS_SPEC_SIG_TO_NOISE : Signal to noise ratio in the mean RVS spectrum (float)

The signal to noise ratio in the mean spectrum is calculated using the median of the ratio of the fluxes and flux uncertainties $\bar{f}_{\rm bin}/\sigma_{f,\rm bin}$ on all the individual bins. It is set to null when it results in a negative value (this may happen for some very faint spectra).

The flux per wavelength bin in the mean spectrum is:

$$\bar{f}_{\rm bin} = \frac{\sum_{j=1}^{n} f_j}{n}$$

where j is the jth CCD spectrum being combined out of a total of n in a particular wavelength bin.

The uncertainty on the flux per wavelength bin is:

$$\sigma_{f,\rm bin} = \frac{\sigma_{f_n}}{\sqrt{n}}$$

where σ_{f_n} is the standard deviation of all the flux values contributing to an individual wavelength bin:

$$\sigma_{f_n} = \sqrt{\frac{1}{n}\sum_{j=1}^{n}(f_j - \bar{f}_{\rm bin})^2}$$

This field is provided only for stars with a published `rvs_mean_spectrum`, which can be selected using the flag `has_rvs` in the table `all_source_flags`.

RV_TIME_DURATION : Time coverage of the radial velocity time series (float, Time[day])

The difference between the first and last transit mean observing time. The transits are those used to compute `radial_velocity_single` and `radial_velocity_sys`, and their number is `rv_nb_transits`.

RV_NB_STAT_TRANSITS : Number of transits of the RV time series (int)

The number of transits (epochs) used to determine the statistical parameters.

Note: This field pertains to epoch data. It can have non-zero values even when `rv_nb_transits` equals 0 if the pipeline fails to compute `radial_velocity_single`.

RV_MIN : Minimum value of the RV time series (float, Velocity[km s^{-1}])

The minimum value of all values in the radial velocity time series.
V_j is the time series of size N of valid epoch values at times T_j, with $j = 1$ to N. The minimum value `rv_min` is defined as:

$$\texttt{rv_min} = min(V_j, \forall j \in (1, N))$$

RV_MAX : Maximum value of the RV time series (float, Velocity[km s^{-1}])

The maximum value of all values in the radial velocity time series.
V_j is the time series of size N of valid epoch values at times T_j, with $j = 1$ to N. The maximum value `rv_max` is defined as:

$$\texttt{rv_max} = max(V_j, \forall j \in (1, N))$$

RV_AMPLITUDE : Amplitude of the RV time series (float, Velocity[km s^{-1}])

The full (also peak-to-peak) amplitude of all values in the radial velocity time series.
V_j is the time series of size N of valid epoch values at times T_j, with $j = 1$ to N. The full

amplitude `rv_amplitude` is defined as:

$$\texttt{rv_amplitude} = max(V_j, \forall j \in (1, N)) - min(V_j, \forall j \in (1, N))$$

RV_MEAN : Mean value of the RV time series (float, Velocity[km s^{-1}])

The mean of all values in the radial velocity time series.

V_j is the time series of size N of valid epoch values. The arithmetic mean RV $\overline{V}$ is defined as:

$$\overline{V} = \frac{1}{N} \sum_{j=1}^{N} V_j$$

RV_MEDIAN : Median value of the RV time series (float, Velocity[km s^{-1}])

Median value obtained from the time series of spectroscopic barycentric radial velocities. It is calculated in both cases of `has_systemic_rv`.

This field is populated using only the single-lined transits.

RV_MEDIAN_ERROR : Uncertainty on the median of the RV time series (float, Velocity[km s^{-1}])

Uncertainty on `rv_median`. It is calculated in both cases of `has_systemic_rv`.

This field is populated using only the single-lined transits.

RV_MAD : Median absolute deviation of the RV time series (float, Velocity[km s^{-1}])

The absolute deviation on the median value of all values in the radial velocity time series.
V_j is the time series of size N of valid epoch values. The median absolute deviation RV `rv_mad` is defined as:

$$\texttt{rv_mad} = 1.4826 \times median\{|V_j - median\{V_k, \forall k \in (1, N)\}|, \forall j \in (1, N)\}$$

The scale factor of 1.4826 is used so that the expectation of the scaled median towards large N equals the standard deviation of a normal distribution. The scale factor for asymptotic normal

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consistency equals $1/\Phi^{-1}(3/4) \approx 1.4826$, where Φ^{-1} is the inverse of the cumulative distribution function for the standard normal distribution.

RV_STD_DEV : Standard deviation of the RV time series (float, Velocity[km s^{-1}])

The standard deviation of all values in the radial velocity time series.
V_j is the time series of size N, with $N > 1$ of valid epoch values. The Bessel-corrected standard deviation σ is defined as:

$$\sigma = \left(\frac{1}{N-1} \sum_{j=1}^{N} (V_j - \overline{V})^2 \right)^{1/2}$$

where $\overline{V}$ denotes the mean value `rv_mean`.

RV_SKEWNESS : Skewness of the RV time series (float, Velocity[km s^{-1}])

The skewness of all values in the radial velocity time series.
V_j is the time series of size N, with $N > 2$ of valid epoch values. The unbiased sample skewness γ is defined as:

$$\gamma = \frac{N}{(N-1)(N-2)} \sum_{j=1}^{N} \left(\frac{V_j - \overline{V}}{\sigma} \right)^3$$

where $\overline{V}$ is the mean `rv_mean` and σ is the Bessel-corrected standard deviation `rv_std_dev`.

RV_KURTOSIS : Kurtosis of the RV time series (float, Velocity[km s^{-1}])

The kurtosis of all values in the radial velocity time series.
V_j is the time series of size N, with $N > 3$ of valid epoch values. The unbiased sample kurtosis K is defined as:

$$K = \frac{N(N+1)}{(N-1)(N-2)(N-3)} \sum_{j=1}^{N} \left(\frac{V_j - \overline{V}}{\sigma} \right)^4 - \frac{3\,(N-1)^2}{(N-2)(N-3)}$$

where $\overline{V}$ is the mean `rv_mean` and σ is the Bessel-corrected standard deviation `rv_std_dev`.

RV_ABBE_VALUE : Abbe value of the RV time series (float, Velocity[km s^{-1}])

The Abbe-value for radial velocity variability based on the Abbe criterion.

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$\{T_j, V_j\}$ is the time series of size N, with $N > 2$ of valid epoch values. V_j is ordered so that $T_j < T_{j+1}$ for all $j < N$ (mandatory at input). The Abbe-value `rv_abbe_value` is defined as:

$$\texttt{rv_abbe_value} = \frac{1}{2}\frac{\sum_{j=1}^{N-1}(V_{j+1} - V_j)^2}{\sum_{j=1}^{N}(V_j - \bar{V})^2}$$

where $\bar{V}$ is the mean `rv_mean`.

The Abbe value quantifies the degree of smoothness in a time series. It approaches zero for very smooth, slowly varying time series, and tends toward unity for purely random (white noise) series.

RV_INTERQUARTILE_RANGE : Interquartile range of the RV time series (float, Velocity[km s^{-1}])

The inter-quartile range of all values in the radial velocity time series.

V_j is the time series of size N of valid epoch RV values, and $V_{(j)}$ the series of ordered values with $V_{(1)} \leq V_{(2)} \leq \cdots \leq V_{(N)}$.

The interquartile range IQR is defined as the difference between 75th and 25th percentile values. It is the first quartile Q1 subtracted from the third quartile Q3, or IQR = Q3 − Q1:

$$\mathrm{IQR} = P_{75}\left(V_{(j)}, \forall j \in (1, N)\right) - P_{25}\left(V_{(j)}, \forall j \in (1, N)\right)$$

The IQR is a measure of statistical dispersion in the time series.

RV_UWE : Unit weight error of the RV time series (float)

The unit-weight uncertainty of all values in the radial velocity time series.

V_j is the time series of size N, with $N > 1$ of valid epoch values with uncertainties ϵ_j. The unit-weight error `rv_uwe` is defined as:

$$\texttt{rv_uwe} = \left(\frac{\sum_{j=1}^{N}\left(\frac{V_j - median(V_j)}{\epsilon_j}\right)^2}{N-1}\right)^{1/2}$$

where $N =$ `rv_nb_transits` and $median(V_j)$ denotes the median RV `rv_median`.

RV_PC05 : 5th percentile of the RV time series (float, Velocity[km s^{-1}])

The 5th percentile of the radial velocity time series computed without excluding outliers.

RV_PC95 : 95th percentile of the RV time series (float, Velocity[km s^{-1}])

The 95th percentile of the radial velocity time series computed without excluding outliers.

RV_MEAN_OF_ERRORS : Mean of the errors of the RV time series (float, Velocity[km s^{-1}])

The mean of the internal uncertainties of all values in the radial velocity time series.
V_j is the time series of size N of valid epoch RV values with uncertainties ϵ_j. The mean of errors $\overline{\epsilon}$ is defined as:

$$\overline{\epsilon} = \frac{1}{N} \sum_{j=1}^{N} \epsilon_j$$

RV_MEAN_CCF_MAX : Mean value of CCF maximum of the RV time series (float)

Mean value of the maximum of the cross-correlation function in the radial velocity time series from the RvDir method.

RV_NB_STAT_TRANSITS_ROBUST : Number of transits of the RV time series after outlier removal (int)

The total number of transits in the radial velocity time series after removing outlier transits. The outlier transits are those for which the field `outlier_status` in `rvs_epoch_parameters_single` is set to 1, 4, 5, 6, 9, 11, 13, or 15.

Note:
1) The outlier transits are not removed from the computation of `radial_velocity_single` or from `radial_velocity_sys`.
2) This field pertains to epoch data. It can have non-zero values even when `rv_nb_transits` equals 0 if the pipeline fails to compute `radial_velocity_single`.

RV_MIN_ROBUST : Minimum value of the RV time series after outliers removal (float, Velocity[km s^{-1}])

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The minimum value of all values in the radial velocity time series after removing the outliers.

V_j is the time series of size N of valid epoch values at times T_j, with $j = 1$ to N after removing the outliers. The robust minimum RV `rv_min_robust` is defined as:

$$\texttt{rv_min_robust} = min(V_j, \forall j \in (1, N))$$

The outlier transits are those for which the field `outlier_status` in `rvs_epoch_parameters_single` is set to one of the following values: 1,4,5,6,9,11,13 or 15.

The outlier transits are not removed from the computation of `radial_velocity_single` or from `radial_velocity_sys`.

RV_MAX_ROBUST : Maximum value of the RV time series after outliers removal (float, Velocity[km s^{-1}])

The maximum value of all values in the radial velocity time series after removing the outliers.

V_j is the time series of size N of valid epoch values at times T_j, with $j = 1$ to N after removing the outliers. The robust maximum `rv_max_robust` is defined as:

$$\texttt{rv_max_robust} = max(V_j, \forall j \in (1, N))$$

The outlier transits are those for which the field `outlier_status` in `rvs_epoch_parameters_single` is set to one of the following values: 1,4,5,6,9,11,13 or 15.

The outlier transits are not removed from the computation of `radial_velocity_single` or from `radial_velocity_sys`.

RV_AMPLITUDE_ROBUST : Amplitude of the RV time series after outlier removal (float, Velocity[km s^{-1}])

The total amplitude `rv_max_robust−rv_min_robust` in the radial velocity time series after outlier removal.

The outlier transits are those for which the field `outlier_status` in `rvs_epoch_parameters_single` is set to one of the following values: 1,4,5,6,9,11,13 or 15.

This field is provided for the bright stars for which `rv_nb_stat_transits_robust` ≥ 2 and `external_apparent_grvs`≤ 14 it is empty otherwise.

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RV_MEAN_ROBUST : Mean value of the radial velocity after outlier removal (float, Velocity[km s^{-1}])

The mean of all values in the radial velocity time series after removing the outliers. Unlike the other statistical parameters, which require at least two transits, `rv_mean_robust` can be computed from a single transit, in which case its value is equal to the corresponding single-transit measurement.

Let V_j be the time series of N valid epoch radial velocity values after outlier rejection. The robust arithmetic mean `rv_mean_robust` is defined as:

$$\texttt{rv_mean_robust} = \frac{1}{N}\sum_{j=1}^{N} V_j$$

Outlier transits are identified as those for which the field `outlier_status` in `rvs_epoch_parameters_sing` is set to one of the following values: 1,4,5,6,9,11,13 or 15.

These outlier transits are excluded from the computation of `rv_mean_robust`, but are retained in the computation of `radial_velocity_single` and `radial_velocity_sys`.

RV_MEDIAN_ROBUST : Median value of the radial velocity after outlier removal (float, Velocity[km s^{-1}])

The median value of all values in the radial velocity time series after removing outlier transits.

V_j is the time series of size N of valid epoch values, and $V_{(j)}$ the series of ordered values with $V_{(1)} \leq V_{(2)} \leq \cdots \leq V_{(N)}$ after removing the outliers.

The robust median `rv_median_robust` is defined as the 50th percentile value:

$$\texttt{rv_median_robust} = P_{50}\left(V_{(j)}, \forall j \in (1, N)\right)$$

The outlier transits are those for which the field `outlier_status` in `rvs_epoch_parameters_single` is set to one of the following values: 1,4,5,6,9,11,13 or 15.

Note that these outlier transits are not removed from the computation of `radial_velocity_single` or `radial_velocity_sys`.

RV_STD_DEV_ROBUST : Standard deviation of the radial velocity after outlier removal (float, Velocity[km s^{-1}])

The standard deviation of all values in the radial velocity time series after removing the outliers.

V_j is the time series of size N, with $N > 1$ of valid epoch RV values after removing the outliers. The robust Bessel-corrected standard deviation `rv_std_dev_robust` is defined as:

$$\texttt{rv_std_dev_robust} = \left(\frac{1}{N-1} \sum_{j=1}^{N} (V_j - \texttt{rv_mean_robust})^2 \right)^{1/2}$$

where `rv_mean_robust` is the robust mean value.

The NonConstancyDetection (NCD) module calculates the Bessel-corrected sample robust standard deviation using $N - 1$ in the denominator.

Outlier transits are those for which the field `outlier_status` in `rvs_epoch_parameters_single` is set to one of the following values: 1, 4, 5, 6, 9, 11, 13, or 15.

Note that these outlier transits are not excluded from the computation of `radial_velocity_single` or `radial_velocity_sys`.

RV_INTERQUARTILE_RANGE_ROBUST : Interquartile range of the radial velocity after outlier removal (float, Velocity[km s^{-1}])

The inter-quartile range of all values in the radial velocity time series after removing the outliers.

V_j is the time series of size N of valid epoch values, and $V_{(j)}$ the series of ordered values with $V_{(1)} \leq V_{(2)} \leq \cdots \leq V_{(N)}$ after removing the outliers.

The robust interquartile range `rv_interquartile_range_robust` is defined as the difference between 75th and 25th percentile values. It is the first quartile Q1 subtracted from the third quartile Q3, or Q3 − Q1:

$$\texttt{rv_interquartile_range_robust} = P_{75}\left(V_{(j)}, \forall j \in (1, N)\right) - P_{25}\left(V_{(j)}, \forall j \in (1, N)\right)$$

The `rv_interquartile_range_robust` is a measure of statistical dispersion in the time series.

Outlier transits are those for which the field `outlier_status` in `rvs_epoch_parameters_single` is set to one of the following values: 1, 4, 5, 6, 9, 11, 13, or 15.

Note that these outlier transits are not excluded from the computation of `radial_velocity_single` or `radial_velocity_sys`.

RV_UWE_ROBUST : Unit weight error of the radial velocity after outlier removal (float, Velocity[km s^{-1}])

The unit-weight uncertainty of all values in the radial velocity time series after removing the outliers.

V_j is the time series of size N, with $N > 1$ of valid epoch values with uncertainties ϵ_j after removing the outliers. The robust unit-weight error `rv_uwe_robust` is defined as:

$$\texttt{rv_uwe_robust} = \left(\frac{\sum_{j=1}^{N} \left(\frac{V_j - medianRobust(V_j)}{\epsilon_j} \right)^2}{N-1} \right)^{1/2}$$

where N = `rv_nb_stat_transits_robust` and $medianRobust(V_j)$ denotes the robust median RV `rv_median_robust`.

Outlier transits are those for which the field `outlier_status` in `rvs_epoch_parameters_single` is set to one of the following values: 1, 4, 5, 6, 9, 11, 13, or 15.

Note that these outlier transits are not excluded from the computation of `radial_velocity_single` or `radial_velocity_sys`.

RV_PC05_ROBUST : 5th percentile of radial velocity values after outlier removal (float, Velocity[km s^{-1}])

The 5th percentile of the radial velocity time series computed excluding outliers.

Outlier transits are those for which the field `outlier_status` in `rvs_epoch_parameters_single` is set to one of the following values: 1, 4, 5, 6, 9, 11, 13, or 15.

Note that these outlier transits are not excluded from the computation of `radial_velocity_single` or `radial_velocity_sys`.

RV_PC95_ROBUST : 95th percentile of radial velocity values after outlier removal (float, Velocity[km s^{-1}])

95th percentile of robust radial velocity time series.

Outlier transits are those for which the field `outlier_status` in `rvs_epoch_parameters_single` is set to one of the following values: 1, 4, 5, 6, 9, 11, 13, or 15.

Note that these outlier transits are not excluded from the computation of `radial_velocity_single`

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or radial_velocity_sys.

RV_MEAN_OF_ERRORS_ROBUST : Mean error of the radial velocity time series after outlier removal (float, Velocity[km s^{-1}])

The mean of the internal uncertainties of all values in the radial velocity time series after removing the outliers.

V_j is the time series of size N of valid epoch values with uncertainties ϵ_j after removing the outliers. The robust mean of errors rv_mean_of_errors_robust is defined as:

$$\texttt{rv_mean_of_errors_robust} = \frac{1}{N}\sum_{j=1}^{N}\epsilon_j$$

Outlier transits are those for which the field outlier_status in rvs_epoch_parameters_single is set to one of the following values: 1, 4, 5, 6, 9, 11, 13, or 15.

Note that these outlier transits are not excluded from the computation of radial_velocity_single or radial_velocity_sys.

RV_MEAN_CCF_MAX_ROBUST : Mean of the CCF maximum values after outlier removal (float)

Mean value of the maximum of the cross-correlation function after outliers removal.

The outlier transits are not removed from the computation of radial_velocity_single or from radial_velocity_sys.

VBROAD_MEAN : Mean value of the vbroad time series (float, Velocity[km s^{-1}])

The mean of all values in the vbroad time series.

V_j is the time series of size N of valid epoch values. The arithmetic mean vbroad $\overline{V}$ is defined as:

$$\overline{V} = \frac{1}{N}\sum_{j=1}^{N} V_j$$

N=vbroad_nb_transits.

VBROAD_MAD : Median absolute deviation of the vbroad time series (float, Velocity[km s^{-1}])

The absolute deviation on the median value of all values in the `vbroad` time series.
V_j is the time series of size N=`vbroad_nb_transits`. The median absolute deviation `vbroad_mad` is defined as:

$$\texttt{vbroad_mad} = 1.4826 \times median\{|V_j - median\{V_k, \forall k \in (1,N)\}|, \forall j \in (1,N)\}$$

The scale factor of 1.4826 is used so that the expectation of the scaled median towards large N equals the standard deviation of a normal distribution. The scale factor for asymptotic normal consistency equals $1/\Phi^{-1}(3/4) \approx 1.4826$, where Φ^{-1} is the inverse of the cumulative distribution function for the standard normal distribution.

VBROAD_STD_DEV : Standard deviation of the vbroad time series (float, Velocity[km s^{-1}])

The standard deviation of all values in the `vbroad` time series.
V_j is the time series of size N, with $N > 1$ of valid epoch values. The Bessel-corrected standard deviation σ is defined as:

$$\sigma = \left(\frac{1}{N-1} \sum_{j=1}^{N} (V_j - \overline{V})^2 \right)^{1/2}$$

where $\overline{V}$ denotes the mean value `vbroad_mean`.

N=`vbroad_nb_transits`.

VBROAD_PC05 : 5th percentile of the vbroad time series (float, Velocity[km s^{-1}])

The 5th percentile of the `vbroad` time series computed without excluding outliers.

VBROAD_PC95 : 95th percentile of the vbroad time series (float, Velocity[km s^{-1}])

The 95th percentile of the `vbroad` time series computed without excluding outliers.

GRVS_MAG_MEAN : Mean value of the Grvs time series (float, Magnitude[mag])

The mean of the values in the `grvs_mag` time series (`rvs_epoch_parameters_single.grvs_mag`)

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G_j is the time series of size N =grvs_mag_nb_transits. The arithmetic mean is defined as:

$$\texttt{grvs_mag_mean} = \frac{1}{N}\sum_{j=1}^{N} G_j$$

Notes:
1. Due to a coding error, this value is incorrect when grvs_mag_nb_transits=0 and must be ignored, as it does not represent a valid measurement in that case.
2. The statistical parameters are computed only for external_apparent_grvs≤14.

GRVS_MAG_MEDIAN : Median value of the Grvs time series (float, Magnitude[mag])

The median of the values in the grvs_mag time series (rvs_epoch_parameters_single.grvs_mag).

Notes:
1. Due to a coding error, this value is incorrect when grvs_mag_nb_transits=0 and must be ignored, as it does not represent a valid measurement in that case.
2. The statistical parameters are computed only for external_apparent_grvs≤14.

GRVS_MAG_MAD : Median absolute deviation of the Grvs time series (float, Magnitude[mag])

The median of the absolute deviations from the median value of the grvs_mag values in the time series stored in the table rvs_epoch_parameters_single.

G_j is the time series of size N=grvs_mag_nb_transits. The median absolute deviation grvs_mag_mad is defined as:

$$\texttt{grvs_mag_mad} = median\{|V_j - \texttt{grvs_mag_median}|, \forall j \in (1, N)\}$$

Notes:

1. Unlike vbroad_mad and rv_mad, this value is not normalised; that is, it is **not** multiplied by the consistency constant (1.4826), which is typically used to make the MAD a consistent estimator of the standard deviation under a normal distribution.
2. Due to a coding error, this value is incorrect when grvs_mag_nb_transits = 0 and must be ignored, as it does not represent a valid measurement in that case.
3. The statistical parameters are computed only for external_apparent_grvs≤14

GRVS_MAG_STD_DEV : Standard deviation of the Grvs time series (float, Magnitude[mag])

The standard deviation of the values in the `grvs_mag` time series stored in the table `rvs_epoch_parameters_`

G_j is the time series of size N=`grvs_mag_nb_transits`, the Bessel-corrected standard deviation σ is defined as:

$$\sigma = \left(\frac{1}{N-1} \sum_{j=1}^{N} (G_j - \texttt{grvs_mag_mean})^2 \right)^{1/2}$$

Notes:
1. Due to a coding error, this value is incorrect when `grvs_mag_nb_transits`=0 and must be ignored, as it does not represent a valid measurement in that case.
2. The statistical parameters are computed only for `external_apparent_grvs`≤14.

GRVS_MAG_PC05 : 5th percentile of the Grvs time series (float, Magnitude[mag])

The 5th percentile of the `grvs_mag` time series.

Notes:
1. Due to a coding error, this value is incorrect when `grvs_mag_nb_transits`=0 and must be ignored, as it does not represent a valid measurement in that case.
2. The statistical parameters are computed only for `external_apparent_grvs`≤14.

GRVS_MAG_PC95 : 95th percentile of the Grvs time series (float, Magnitude[mag])

The 95th percentile of the `grvs_mag` time series.

Notes:
1. Due to a coding error, this value is incorrect when `grvs_mag_nb_transits`=0 and must be ignored, as it does not represent a valid measurement in that case.
2. The statistical parameters are computed only for `external_apparent_grvs`≤14.

GRVS_MAG_AMPLITUDE_ROBUST : Amplitude of the Grvs time series after outlier removal (float, Magnitude[mag])

The total amplitude (max−min) in the `grvs_mag` time series after outlier removal.

Outlier transits are those for which the field `outlier_status` in `rvs_epoch_parameters_single` is set to one of the following values: 3, 4, 6, 7, 12, 13, 14, or 15.

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Notes:
1. Due to a coding error, this value is incorrect when `grvs_mag_nb_transits`=0 and must be ignored, as it does not represent a valid measurement in that case.
2. The statistical parameters are computed only for `external_apparent_grvs`≤14.

GRVS_MAG_NB_STAT_TRANSITS_ROBUST : Number of transits of the Grvs time series after outlier removal (int)

The total number of transits in the `grvs_mag` time series after removing outlier transits. The outlier transits are those for which the field `outlier_status` in `rvs_epoch_parameters_single` is set to 3, 4, 6, 7, 12, 13, 14, 15.

Notes:
1. Due to a coding error, this value is incorrect when `grvs_mag_nb_transits`=0 and must be ignored, as it does not represent a valid measurement in that case.
2. The statistical parameters are computed only for `external_apparent_grvs`≤14.
3. The outlier transits are not removed from the computation of `grvs_mag`. This field pertains to epoch data. It can have non-zero values even when `grvs_mag_nb_transits` equals 0 if the pipeline fails to compute `grvs_mag`.

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2.6 CROWDED_FIELD_SOURCE_ENVIRONMENT_MATCH [TAP]

A cross-match table between the individual source lists coming from the Crowded Field and Source Environment analysis pipelines.

This contains all matches between valid `crowded_field_source` and `gaia_source_environment` sources within a radius of 120 mas. The tentative winner of a one-to-one astrometric comparison is provided, together with the corresponding rule. A flag showing when both sources are in `gaia_source` can be used as an indication of possible duplicates.

For more details of the procedure used to create this crossmatch, including an equivalent ADQL query capable to reproduce it, see Section **??** in the online documentation.

Columns description:

SEA_SOURCE_ID : Unique source identifier for Gaia Source Environment (SEA) (long)

Unique identifier for the SEAPipe source.

CF_SOURCE_ID : Unique source identifier for crowded field analysis (long)

Unique identifier coming from the crowded field analysis pipeline.

WINNER : Match winner (string)

Match winner:

- SEA: Gaia Source Environment Analysis (SEA).
- CF: Crowded-field Sky Mapper (SM) observations.
- NONE: No winner scenario.

RULE : Rule applied to the astrometric solution selection for this source (string)

A short string describing the rule applied to decide on whether the astrometric solution of this source is the recommended one. This field can take the following values:

- 2A: SEA wins. 5 vs 2 parameters solution for SEAPipe primary components.

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- 2C: CF wins. 5 vs 2 parameters solution.
- 3A: SEA wins. 5 vs 6 parameters solution, lower quadsum (see `astrometric_error_ratio`) and SEAPipe primary components.
- 3C: CF wins. 6 vs 5 parameters solution, lower quadsum and/or SEAPipe secondary component.
- 7C: CF wins. 2 vs 2 parameters solution.
- 8C: CF wins. 6 vs 2 parameters solution.

ASTROMETRIC_ERROR_RATIO : Ratio between the combined astrometric errors (float)

This field provides the ratio of the square root of the quadratic sums of the Gaia Source Environment (SEA) versus Crowded Field analysis astrometric formal errors. This ratio decides the winner for rule 3.

The quadratic sums of these formal errors are defined as:

$$\text{quadsum} = \sqrt{(0.445 \times \sigma^2_{\texttt{parallax}} + \sigma^2_{\texttt{pmra}} + \sigma^2_{\texttt{pmdec}}}$$

with

$$\texttt{astrometric_error_ratio} = \text{quadsum}_{\texttt{sea_source_id}} / \text{quadsum}_{\texttt{cf_source_id}}$$

The factor 0.445 applied to the parallax formal error is here to take account that, for long mission durations, the standard error of a parallax is normally larger than that of a proper motion component. Gaia DR4, with a mission duration of 5.5 years, is already well within this regime. Thus simply quadratically adding the three uncertainties (for parallax and proper motion in RA and Dec) would overemphasise the parallax, compared to the other two dimensions in the space of unknowns.

SEPARATION : Distance between sources (float, Angle[mas])

Distance between the matched sources.

BOTH_IN_GAIA_SOURCE : Are both source identifiers in Gaia source? (boolean)

Possible duplication indicator.

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If both source identifiers are in `gaia_source` then this means they have been considered as new sources not in `all_source_astrometry` but are close enough to require an individual analysis beyond the scope of DR4.

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3 Astrometry

As indicated in the below subsection titles by the labels [TAP] and [DataLink], these products are served from the ESA Gaia Archive using two different protocols, the Table Access Protocol (TAP) for tabular products and the DataLink protocol for other products such as spectra, light curves, MCMC samples, etc. For more details please see Section **??**.

3.1 EPOCH_ASTROMETRY [DataLink]

This table provides the epoch astrometry, i.e. the individual astrometric measurements, for all sources treated in the main astrometric processing.

The epoch astrometry data is given in the form of local plane coordinates, both for the two-dimensional and for the one-dimensional astrometric Gaia measurements. See Chapter **??** of the online documentation for more explanation and **?** for full details.

The table `epoch_astrometry` contains one row per `source_id-transit_id` matching pair. The same `source_id` occurs as many times as there are FoV transits matched with this source. Any combination of `source_id` and `transit_id` occurs exactly once in the table.

Each row contains three per-source fields (`ra0`, `dec0` and `agis_source_excess_noise`); for ease of handling and usage, these are repeated as many times as there are transits matched to the given source. Similarly, some of the per-transit and per-CCD data are repeated for the same reasons.

Note: this table is not available through the main archive TAP interface. Data are delivered via the Massive Data service indexed by the VO DataLink protocol and described in Chapter **??**. For example this can be actioned in the archive user interface by joining queries with the table `all_source_flags` and selecting `has_epoch_astrometry = 'true'`. The DataLink retrieval type to specify for this product in programmatic archive access via the Astroquery Gaia Python package is `EPOCH_ASTROMETRY`.

We note that in many cases the fields `scan_pos_angle`, `parallax_factor_al` and `parallax_factor_ac` for a given `transit_id` differ between the `epoch_image` and `epoch_astrometry` tables. These differences are small (up to 3.8E-4 degrees, 1.8E-5 and 9.5E-6, respectively), and they are largely unavoidable. Partly, they are due to the — necessarily — different calibration models used: In `epoch_image`, no unique magnitude or colour can be assigned to any sample, nor to the window centre. On the contrary, for `epoch_astrometry` the full, chromatic and magnitude-dependent models for the underlying celestial source are used, of course.

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Additionally, the effective observation times are different: In epoch_astrometry, the obvious and optimal choice is the mean exposure time for the image centroid. However, this is not appropriate for epoch_image, in particular for multi-peak windows. The reasonable choice here would be the mean exposure time for the window centre. In the DR4 tables of epoch_image a mixture between centroid times and window centre times is implemented. For DR5, efforts will be taken to always use the window centre.

We furthermore note that the number of records in the epoch_image table is about 128 million higher than in the epoch_astrometry table. The difference is primarily due to time intervals known as AGIS gaps, where no final high-precision attitude is available, and thus the transits were not used in the astrometric source solutions. A minority of the cases is caused by multiple (secondary) peaks detected in the early DIPD processing for DR4 that were not recovered in the latest DIPD runs.

DataLink products are served with a choice of two different data structures:

- DATAMODEL_STANDARD: the product is serialised following a standard data model (for example an IVOA-compliant data model in applicable cases);
- DATAMODEL_GAIA: the product is serialised following the Gaia DPAC(-internal) data model.

The DATAMODEL_GAIA format is described here whereas the DATAMODEL_STANDARD format is described in https://www.cosmos.esa.int/web/gaia-users/archive/datalink-products#datalink_serialisation

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution identifier, please visit

https://gaia.esac.esa.int/decoder/solnDecoder.jsp

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source.

It is copied from `all_source_astrometry`. For a detailed description see `gaia_source.source_id`.

TRANSIT_ID : Transit Identifier (long)

The `transit_id` is a unique identifier assigned to each detected (and confirmed) source as it transits the Gaia focal plane. Each time a given source is detected as Gaia scans and re-scans the sky a new `transit_id` will be created to badge that apparition. Hence the along-scan time and the across-scan position along with the telescope in which the source was detected are used to form a unique integer with which to label the transit. The detection of transits is done using the SkyMapper CCD data, and the confirmation is done using the AF1 CCD data. Both happens on board.

The several features of a transit that are encoded in `transit_id` can be easily retrieved using bit masks (&) and shifts (>>) as follows:

- On-Board Mission Time line (OBMT) [ns]
  = 204800 * ((`transit_id` >> 17) & (0x000003FFFFFFFFFF))
- Field-of-view = 1 + (`transit_id` >> 15) & 0x03 [1 for 'preceding' and 2 for 'following' fields-of-view respectively]
- CCD row = (`transit_id` >> 12) & 0x07 [dimensionless, in the range 1 to 7]
- Across-scan 'reference acquisition pixel' in strip AF1 = (`transit_id`) & 0x0FFF [pixels] (this is the across-scan centre of the AF1 window and is odd if immediately below the mid-point of the window and even if immediately above)

where the bit mask prefix '0x' denotes hexadecimal.

For further details see Portell et al. (2020). For convenience a decoder for `transit_id` is available on-line at

`https://gaia.esac.esa.int/decoder/transitidDecoder.jsp`

RA0 : Right ascension of reference point (double, Angle[deg])

Right ascension α_0 of the tangent point on the plane tangential to the celestial sphere in which the coordinate direction of the object is locally modelled, i.e. of the celestial reference point from which the local plane coordinates `centroid_pos_al` and `centroid_pos_ac` are counted. The specific value will be taken from the valid/relevant source catalogue (which in turn will be defined implicitly by the `solution_id`).

Note: `ra0` and `dec0` are always the same for all transits of a given `source_id`.

For more details see **?**.

DEC0 : Declination of reference point (double, Angle[deg])

Declination δ_0 of the tangent point on the plane tangential to the celestial sphere in which the coordinate direction of the object is locally modelled, i.e. of the celestial reference point from which the local plane coordinates `centroid_pos_al` and `centroid_pos_ac` are counted. The specific value will be taken from the valid/relevant astrometric source catalogue (which in turn will be defined implicitly by the `solution_id`).

Note: `ra0` and `dec0` are always the same for all transits of a given `source_id`.

For more details see **?**.

AGIS_SOURCE_EXCESS_NOISE : the unscaled AGIS Source Excess Noise (float, Angle[mas])

This is the unscaled AGIS source excess noise.

The actual weight used by AGIS per CCD is $(1mas^2/\sigma^2_{tot})$, where $\sigma^2_{tot} = \sigma^2_{IPD} + excess^2_{source} + excess^2_{cal.}$. This is the weight to be used if normal AGIS solutions shall be reproduced from epoch astrometry data.

The quantity `centroid_pos_error_al` contains the first and third terms in σ^2_{tot}. The separate `agis_source_excess_noise` item is needed to enable external users to change the source model, whence $excess^2_{source}$ from the normal AGIS solution becomes irrelevant, while the other two components of σ^2_{tot} remain valid.

For more explanations see the AGIS paper for the relevant Gaia Data Release, here **?**.

Note that, contrary to the `gaia_source` and `all_source_astrometry` tables, this is the original source excess noise, not the rescaled excess noise according to **?**. Reason: The rescaled value is the one which is useful for the quality criteria in the source astrometry tables, but the original value is needed to enable users to reproduce the published AGIS solutions from epoch

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astrometry data.

OBS_TIME_TCB : effective observing time as TCB (Long[10] array, Time[ns])

Effective observation time for each CCD in this FoV transit. Centre of the actual exposure time for the given window/CCD, depending on readout time and gate. In units of nanoseconds since J2010.0(TCB).

The presentation as long integer (rather than as double float) is chosen for the sake of keeping the full resolution of the time coordinates used in Gaia/DPAC, and for the sake of invertability of transformations from OBMT to TCB and vice versa.

Such transformations are done using HATT (High-Accuracy Time Transformations), which have the resolution of 1 nanosecond. But users should be aware that the absolute accuracy of them is in the order of 150 nanoseconds.

OBS_TIME_BARY_CORR : barycentric correction to obs_time_tcb (float, Time[ns])

Barycentric correction to the observation time, in the sense of TCB(barycentric)-TCB(at Gaia), calculated for the obsTime of the AF4 CCD, i.e. at the middle of the FoV transit.

For the correct production and usage of the epoch astrometry, both the true and the barycentrically corrected observation times are needed. It is, however, not needed to have barycentric corrections for each CCD of a transit. One value is sufficient. Explanation: The value can change at most by about 30 km/s/c*30s=3ms from the middle of a transit to either the SkyMapper or AF9 CCDs. Anyone who needs it even more precisely, will also need the complete 3d location of Gaia (not just the 1d location along the line of sight) and should thus consult the orbit anyway. Therefore it is specified that `obs_time_bary_corr` be computed for the AF4 CCD transit time (or for an immediately adjacent CCD in case the AF4 IPD should happen to fail).

The maximum values that the quantity can reach is about ±370 s (not the canonical ± 500s because the scan axis of Gaia always points at 45 degrees from the sun). The resolution of the float thus is around 50 microseconds or better. This is clearly sufficient in view of the reasoning above.

SCAN_POS_ANGLE : Position angle of the scan (double[10] array, Angle[deg])

Position angle θ of the scan for each of the CCD transits.

This angle uniquely determines the transformation between the local celestial coordinates (a, d) and the local scan coordinates (w, z) in the tangential plane touching the celestial sphere at `(ra0,dec0)`. See **?** for details. The scan angle varies along a FoV transit. In contrast, the parallax factors are sufficiently constant over a transit.

The sine and cosine of θ are the derivatives of the along-scan local plane coordinate w with respect to right ascension and declination, respectively.

Please note the remark in the 6th paragraph of the table description.

ZETA : across-scan field angle zeta of the reference point (ra0, dec0) (float, Angle[deg])

This is the AC field angle ζ of the reference point (`ra0,dec0`). It needs not be an array of values per CCD, as it is sufficiently constant over a FoV transit. Thus it is specified that the `zeta` field angle be computed for the AF4 CCD transit time (or for an immediately adjacent CCD in case the AF4 IPD should happen to fail).

This quantity is needed for the following reason: When analysing a non-single system, it is necessary that the epoch astrometry of each component is expressed with respect to the same reference point. But that will not be the case for resolved systems, where the components have different AGIS solutions, and therefore different reference points. Hence there is a need to be able to transform the epoch data from one reference point to another (within an arcmin or so). But this is not possible - at least not rigorously - unless we also know ζ. The error in the transformed epoch astrometry is marginal (< 1 muas) for shifts less than about 5 arcsec, but increases quadratically for larger separations. ζ could of course be computed to sufficient accuracy if the FoV, CCD row, and AC coordinate `mu` (μ) are known, but to give it explicitly is a convenience for users.

Adding ζ to the table also brings a nice symmetry to the information in `epoch_astrometry`: with the AL&AC parallax factors plus the Roemer delay we have complete (3D) knowledge of Gaia's barycentric position at the time of observation (albeit only to float precision per transit, and in strange units); with `scan_pos_angle` and `zeta` we similarly have complete knowledge of the spin (SRS z) axis at the time of observation. Both pieces of information (Gaia's 3D position and the SRS z axis) are needed to transform epoch data to a different reference point - and back.

PARALLAX_FACTOR_AL : Parallax factor AL (float)

Parallax factor in the AL direction w, needed to use the local plane coordinates for detailed object modelling.

The parallax factor f_w is the derivative of the along-scan local plane coordinate w with respect to the parallax. See **?** for details.

The parallax factors need not be an array of values per CCD, as they are sufficiently constant over a FoV transit. Thus it was decided that the parallax factors are computed for the AF4 CCD

transit time (or for an immediately adjacent CCD in case the AF4 IPD should happen to fail).

Please note the remark in the 6th paragraph of the table description.

PARALLAX_FACTOR_AC : Parallax factor AC (float)

Parallax factor in the AC direction z needed to use the local plane coordinates for detailed object modelling.

The parallax factor f_z is the derivative of the across-scan local plane coordinate z with respect to the parallax. See **?** for details.

The parallax factors need not be an array of values per CCD, as they are sufficiently constant over a FoV transit. Thus it is specified that the parallax factors be computed for the AF4 CCD transit time (or for an immediately adjacent CCD in case the AF4 IPD should happen to fail).

Please note the remark in the 6th paragraph of the table description.

COLOUR_FACTOR_AL : colour factor AL (float[10] array)

The colour factor in AL direction w.

This is the negative of the derivative of w with respect to colour parameter ν_{eff} (effective wavenumber in units of inverse micrometers). The unit of the quantity is mas*micrometer.

COLOUR_FACTOR_AC : colour factor AC (float[10] array)

The colour factor in AC direction z.

This is the derivative of z with respect to colour parameter ν_{eff} (effective wavenumber in units of inverse micrometers). The unit of the quantity is mas*micrometer.

This field will be empty in DR4.

NU_EFF_USED_IN_ASTROMETRY : colour (i.e. effective wavenumber) used in AGIS (and in IPD) (float, Misc[μm^{-1}])

Effective wavenumber of the source used as starting value in the astrometric solution.

This effective wavenumber $\nu_{\rm eff}$, in units of μm^{-1}, is the assumed value used for the selection of the

appropriate LSF/PSF model in the image parameter determination, as well as for the selection of the astrometric calibration parameters in the astrometric adjustment, for the given source. If no BP and RP photometry of the given source was available, `nu_eff_used_in_astrometry` is set to the default value $\nu_{\rm eff} = 1.43\mu{\rm m}^{-1}$. Else, if reliable BP and RP photometry was available, it is set to the weighted mean of the default value and the actual photon-flux weighted inverse wavelength as estimated from the BP and RP bands. If highly precise BP and RP data are available, the `nu_eff_used_in_astrometry` is very close to the value from the BP and RP bands. However, due to the cyclic processing of the Gaia astrometry and photometry, this effective wavenumber will even in these cases be (slightly) different from the one computed using the latest available photometry for DR4.

Some details on the selection of the default value and the construction of the weighted mean:
The default value is chosen as the ensemble average of the actually measured values for all Gaia sources, and its standard error for the weighted mean is set to $0.085\mu{\rm m}^{-1}$, which is close to the actual rms scatter of the measured values.
For the measured values, the formal errors assigned by the photometry processing were used in the construction of the weighted mean. However, the original values delivered (from the DR3 photometry processing) were found to be strongly overestimated. Therefore, they were rescaled to give realistic values before being used for the weighted mean. For details see the DR4 astrometry paper, **?**, Section 2.3.

NU_EFF_ERROR : standard error of nu_eff_used_in_astrometry (float, Misc[μm^{-1}])

This is the standard error of the input colour parameter used for a given source in the AGIS source parameter adjustment. The original values delivered (from the DR3 photometry processing) were found to be strongly overestimated. Therefore, a constant value of 0.085 inverse micrometers was used instead.

For more details see the DR4 astrometry paper, **?**, Section 2.3.

CENTROID_POS_AL : Local plane coordinate (w) of image centroid AL (double[10] array, Angle[mas])

AL local plane coordinate w of the image centroid (one per CCD transit). See **?** for details.

There is a known issue: Among the $\simeq 10^{12}$ entries of `centroid_pos_al` in the table, there is a very small portion with grossly deviating values. Such values may be caused by excursions of the attitude modelling at the start and end of data gaps. There are a total of 16,437,562 cases where `centroid_pos_al` is beyond the range allowed by the window size, and there are a few cases taking exceptionally large values. Notably, there are 208 transits where the max value of `centroid_pos_al` exceeds 10^9 mas. On the user side, the obvious way to avoid the problem is to simply ignore the very few records where `centroid_pos_al` is unreasonably large.

CENTROID_POS_AC : Local plane coordinate (z) of image centroid AC (double[10] array, Angle[mas])

AC local plane coordinate z of the image centroid position (one per CCD transit). See **?** for details.

This is based on the AC centroiding and therefore it is only available for 2D windows. In DR4 this field and its associated error `centroid_pos_error_ac` will not be available.

CALCULATED_POS_AC : Calculated Local plane coordinate (z) (double[10] array, Angle[mas])

Calculated AC local plane coordinate z.

This is not based on a real measurement like the `centroid_pos_ac` but has been calculated using the AL measurement and the source astrometry to compute the geometric source position at the time of the observation, then using the attitude to get the calculated AC local plane coordinate z.

CENTROID_POS_ERROR_AL : standard error of centroid_pos_al (float[10] array, Angle[mas])

Standard errors of AL local plane coordinates w of the image centroid positions.

This quantity represents the standard errors used by AGIS, or more precisely: together with the `agis_source_excess_noise` and the `used_by_agis_al` flag it represents the actual weight for the AL coordinate used in the final AGIS source update (if `used_by_agis_al` is 'true').

This quantity contains two components, quadratically added via

$(\texttt{centroid_pos_error_al})^2 = \sigma^2_{IPD} + excess^2_{cal}$

The two components are:

- The IDU IPD centroid's standard error,
- The AGIS calibration excess noise.

The former is given explicitly in the table (`ipd_error_al`), while the second can be computed by quadratically subtracting the other one from `centroid_pos_error_al`.

To be very clear: if a user wants to reproduce the released AGIS solution for a given source they should use weights derived from quadratically adding `agis_source_excess_noise` and

centroid_pos_error_al so that the result contains all three relevant components of the weighting used by AGIS:

$\sigma^2_{tot} = \sigma^2_{IPD} + excess^2_{source} + excess^2_{cal}$

Special case of SM data: The centroid_pos_error_al is significantly underestimated, especially for bright stars. For details see Section **??** of the online documentation.

CENTROID_POS_ERROR_AC : standard error of centroid_pos_ac (float[10] array, Angle[mas])

Standard errors of the AC local plane coordinates z of the image centroid positions. For 1d windows it is NaN.

This quantity represents the standard errors used by AGIS, or more precisely: together with the agis_source_excess_noise and the used_by_agis_ac flag it represents the actual weight for the AC coordinate used in the final AGIS source update (if used_by_agis_ac is 'true', which never happens for DR4). For 1d windows, the value will always be set to NaN.

This quantity contains two components, quadratically added via

$(\texttt{centroid_pos_error_ac})^2 = \sigma^2_{IPD} + excess^2_{cal.}$

The two components are:

- The IDU IPD centroid's standard error,
- The AGIS calibration excess noise.

The former is given explicitly in the table (ipd_error_ac), while the second can be computed by quadratically subtracting the other one from centroid_pos_error_ac.

To be very clear: if a user wants to reproduce the released AGIS solution for a given source they should use weights derived from quadratically adding agis_source_excess_noise and centroid_pos_error_ac so that the result contains all three relevant components of the weighting used by AGIS:

$\sigma^2_{tot} = \sigma^2_{IPD} + excess^2_{source} + excess^2_{cal.}$

This field is not available in DR4.

USED_BY_AGIS_AL : whether the CCD transit was used in AL by AGIS-DR4 (Boolean[10]

array)

Flag saying whether the CCD transit was used by AGIS-DR4 in AL direction. This assumes that the AGIS DR4 downweight factors are either zero or one.

USED_BY_AGIS_AC : whether the CCD transit was used in AC by AGIS-DR4 (Boolean[10] array)

Flag saying whether the AC information from this CCD transit was used by AGIS for producing the released astrometric parameters. It means that the AGIS downweight factors in DR4 are either zero or one. For DR4 this will always be 'false'.

Note: This flag, like the corresponding one for AL, refers only to the usage of the AC measurement in the final AGIS source update producing the released astrometric source parameters. It ignores the usage of the AC measurements for the subset of (bright) primary stars needed to set up the overall astrometric system. In particular, these are indispensable for deriving the AC attitude (which is needed for a correct usage of the AL measurements), and this in turn requires an AC astrometric calibration.

TRANSIT_ACQ_FLAGS : Transit-level information on the on-board acquisition (short)

This field contains a set of bitwise flags (from Bit 15 = msb to Bit 0 = lsb) related to the data acquisition. It has some oddities for historical reasons.

Windows may suffer truncation or trimming. Here, truncation refers to a reduction in size - often irregular - caused by an intra-field conflict with another window, while trimming refers to the permanent use of a reduced window size across scan. Truncation and trimming only affect 1D windows and never affect SM and AF1 windows. Trimming was only used for AF2-9 and only for a short period early in the mission (around OMBT revolutions 2103 to 2238).

The shape of all 1D AF2-9 windows is detailed in `epoch_image.shape_af`.

The field is coded as follows:

- Bits 15 and 14: unused
- Bit 13 (0x2000): Always set.
- Bit 12 (0x1000): The AF2-9 windows are trimmed AC but have not suffered truncation.

- Bits 11-9: unused
- Bits 7 and 8 (0x0180): `xp_offset`. This is the value of a continuous two-bit TDI counter. It is needed for the reconstruction of the XP windows.
- Bit 6 (0x0040): AF2-9 windows are 2D.
- Bit 5 (0x0020): AF2-9 windows are 1D (complement of Bit 6).
- Bit 4 (0x0010): AF2-9 windows are 1D and with the full AC size of 12 pixels.
- Bit 3 (0x0008): No BP window acquired.
- Bit 2 (0x0004): No RP window acquired.
- Bit 1 (0x0002): BP window truncated.
- Bit 0 (0x0001): RP window truncated.

TRANSIT_PROC_FLAGS : Transit-level information on the processing results (short)

This field contains a set of bitwise flags, coded as follows:

- Bits 3 to 0 (`transit_proc_flags` AND 0x000F): **IPD strategy and findings**. As of IDT 15.1 and until IDT 21.0, as well as in IDU until DRC-02, this field was not used (0 was always stored here). As of IDU DRC-03, the following information is provided:

  '0': Normal IPD. Ambiguity not tested or not detected. Default value for all records produced until IDT 21.0 and IDU until DRC-02

  '1': Multiple Source IPD: As of IDT 21.1, this value is used to indicate that the earlier object in time detected in the window has been selected. In case of 2D windows with larger AC than AL separation between detected peaks: higher μ selected.

  '2': Multiple Source IPD: As of IDT 21.1, this value is used to indicate that the later object in time detected in the window has been selected. In case of 2D windows with larger AC than AL separation between detected peaks: lower μ selected.

  '3': Multiple Source IPD: As of IDU DRC-03, this value is used to indicate that at least one of the windows has a multiple source and that these disturbing sources were masked for the IPD processing. `ccd_proc_flags` can be checked to know which and how many windows have been affected.

  '4': DIPD: As of IDU 21.0, this value is used to indicate that this object has been obtained using DIPD implementation and thus include a window model template and combined AF solution.

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'5': TukeyBiWeight IPD: this value is used to indicate that this object has been obtained using TukeyBiWeight IPD initialisation for the internal processing subsystem IDU-AttAc; such initial values do not propagate into the published data.

Values 6-15 are reserved for future extensions.

- Bits 6 to 4 (0x0070): **Bias and PEM OI strategy baseline**: this field indicates the Bias mitigation mode requested to process this transit. The consumer must check the corresponding `ccd_proc_flags` field to determine the actual mode used for each individual CCD as sometimes some modes can not be applied due to missing input data or calibration limitations.

  Value '0': Default initialisation value for the Bias strategy. If found in an AstroElementary it means that something went wrong and the Bias (or maybe just its flagging) was not done appropriately. Thus, '0' means 'UNDEFINED/FAILURE'.

  Value '1': Only PreScan bias used.

  Value '2': PreScan + common NU baseline.

  Value '3': Full PEM NU.

  Values 4-7 are reserved for future extensions.

- Bits 10 to 7, **colour info**:

  Bit 7 (0x0080), **default colour**: '1' means 'a default colour used for IPD processing', which can be caused by some problem in the colour determination or retrieval (e.g. no BP/RP data in IDT or no valid PhotPipe fluxes in IDU).

  Bit 8 (0x0100), **epoch colour**: '1' means 'epoch (transit-level) colour used for IPD processing', '0' means 'mean (source-level) colour used'.

  Bit 9 (0x0200), **single-param colour**: '1' means 'single-parameter colour used for IPD processing' (such as `nu_eff_used_in_astrometry`), '0' means 'multi-parameter colour used' (such as SSCs).

  Bit 10 (0x0400), **colour problems**: '1' means 'missing colour information or dubious colour quatiliy' (e.g. problematic/missing BP/RP data in IDT, NaN or rejected PhotPipe colour in IDU, etc).

  Please note some remarkable combinations of this bit with Bit 7:

  - Non-default + no problems (Bit 7 = 0, Bit 10 = 0): Good colour from catalogue or epoch data
  - Non-default + problems (Bit 7 = 0, Bit 10 = 1): Dubious colour from catalogue or best approximation from partial or problematic epoch data. Implemented only in IDT.
  - Default + no problems (Bit 7 = 1, Bit 10 = 0): Default colour has been used because no colour information was available (i.e. for blacklisted observations). Implemented only in IDU.

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  - Default + problems (Bit 7 = 1, Bit 10 = 1): Default colour used instead of the available colour from catalogue or epoch data (i.e. dubious colour rejected).

- Bit 11 (0x0800), **cosmetic effects** flagging: '1' means 'dead pixel columns affecting some of the windows'.
- Bit 12 (0x1000), **cosmetic effects** flagging: '1' means 'anomalously low samples found in some of the windows'.
- Bit 13 (0x2000), **cosmetic effects** flagging: '1' means 'anomalously high samples found in some of the windows'.
- Bit 14 (0x4000), **problems in initial centroid**: '1' means that problems have been found in the Initial Centroids determination. In the case of IDT, combined with the 'some problem found in the colour determination' flag (bit 10), it may indicate problems in the TransitPredictor (e.g. predicted position outside window, or no attitude available) in which case a default position is used for colour determination.
- Bit 15, **attitude issues**: '1' indicates that some attitude problem was found affecting some of the windows i.e. missing attitude or problems computing AL/AC rates. This flag is only set by IDU which depends on the attitude to access the ELSF and for the combine AF processing in DIPD.

See Chapter 3 of the DR4 online documentation for more information.

We note that in some cases, the `transit_proc_flags` and `ccd_proc_flags` for a given transit differ between `epoch_image` and `epoch_astrometry`. This is due to differing processing configurations in the production of these two tables. In case of doubt, the version in `epoch_astrometry` should be considered the more relevant one. Efforts will be taken to avoid such differences in Gaia DR5.

CCD_PROC_FLAGS : CCD-level information on the processing results (Short[10] array)

This field contains information about the findings in the IPD processing for the SM and AF CCD data. It is defined as an array, with one or more sets of 10 components (one per CCD). The indexes of the componenents for a given CCD meet the following expression:

$$s + 10k \quad \begin{cases} s & [0...9] & \text{CCD index [SM...AF9]} \\ k & [0...n] & \text{CCD set index} \end{cases} \tag{5}$$

In the present table, i.e. `epoch_astrometry`, only the first set of 10 components is included. The second set is given in table `epoch_image` only. Nevertheless, the description of that second set

is provided here for completeness. If a user of `epoch_astrometry` wants to utilize the second set, too, then a table join with the corresponding `epoch_image` records is recommended.

The first set of 10 components (SM has index 0, AF9 index 9) of this array contains a set of bitwise flags, coded as follows:

- Bits 3 to 0 (`ccd_proc_flags` AND 0x000F): **IPD result**. It is coded as follows:
  - '0': **Success**; the fitting has succeeded without any problem.
  - '1': **Illegal Amplitude**; invalid amplitude obtained after the initial fitting. It may happen for windows with wrong background determination or similar.
  - '2': **Illegal Coordinate**; invalid AL or/and AC locations obtained in the fitting i.e. locations outside the window. It may happen for windows close to a charge injection, multiple sources, strongly affected by some cosmic ray or other similar issues.
  - '3': **DEPRECATED: Illegal Output**; location estimation outside window.
  - '4': **Maximum Iteration Exceeded**; convergence not reached, but results achieved so far stored in AstroElem. The IPD result may still be reliable enough.
  - '5': **Failed**; diverging iteration instead of converging.
  - '6': **Pending**; calculations in progress; IPD output values not reliable; maybe the processing failed for some reason before finishing the calculations, or the IPD was just initialised, and no processing has occurred yet
  - '7': **No LSF/PSF data**; no result available due to no LSF/PSF data available.
  - '8': **Not Enough Samples**; IPD was attempted but no result is available because at a given iteration it runs out of samples (there are not enough inputs to the equations/matrix system).
  - '9': **Odd LSF/PSF Success**; the fitting has succeeded but using an odd LSF/PSF. The LSF/PSF is considered odd when it is not well defined for some samples. The IPD result may still be reliable enough similar as with the 'Maximum Iteration Exceeded' case.
  - '14': **Not Attempted**; IPD has not been attempted on purpose due to missing gate information, missing CI information or similar or when not enough usable input samples are available due to heavy truncation, CCD edge, complex AOCS update, multiple gates or similar.
  - '15': **Crashed**; calculation finished and crashed for some reason.
  - Values '3' and '10' to '13': Available for future extensions.
- Bit 4 (0x0010): **IPD nominal/non nominal**. '1' indicates the IPD is non-nominal in the sense it has been performed over a reduced window area (details in `ccd_proc_geom`). This flag is only set for AF windows.

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- Bit 5 (0x0020): **IPD not available**. '1' indicates that IPD result is not available and thus its output values are not valid. This flag set to true and IPD result equals to NOT_ATTEMPTED (value '14' in bits 3-0) indicates IPD was not attempted on purpose, and with IPD result equals to PENDING (value '6' in bits 3-0) indicates unexpected IPD termination. This flag set to true with any other IPD result should not be possible.
- Bit 6 (0x0040): **Some cosmetic issue**. '1' indicates that there is at least one contaminated or discarded sample in the selected window area. This flag collects the presence of additional discarded or contaminated sample without an explicit CCD level flag thus excluding saturation, cosmics, odd debiased, etc.
- Bit 7 (0x0080): **Cosmic removed**. '1' means that at least one sample in the selected window area has been discarded for being a suspected cosmic.
- Bit 8 (0x0100): **Saturation removed**. '1' means that at least one sample in the selected window area has been discarded due to saturation.
- Bit 9 (0x0200): **Part of window discarded**. '1' means that at least one sample has been discarded in the full window area.
- Bit 10 (0x0400): **No window**. '1' means 'no samples available' (i.e. all the samples in this window were discarded on board, or this is e.g. AF 9 Row 4, or a deactivated CCD, etc).
- Bit 11 (0x0800): **Non-success InitialCentroid**. '1' means 'the Initial Centroid was not successful, and thus we have used the centre of the window as the start condition of IPD'.
- Bit 12 (0x1000): **Odd Background found**. '1' indicates that the background determined for this window (for at least one of its samples) was odd i.e negative, too large or NaN. This flag may be used to indicate that the fluxes determined for this window are potentially wrong (although its AL/AC centroid may still be useful).
- Bit 13 (0x2000): **fall-back to lower Bias Mitigation mode**. '1' indicates that 'fall-back' to lower mode has occurred for this window (for at least one of its samples). This flag will always be activated for AF1 when FULL_MITIGATION mode is enabled because this mode is not supported in this CCD.
- Bit 14 (0x4000): **non-target source removed**. '1' indicates that non-target sources have been detected and masked in this window.
- Bit 15 (0x8000): **LSF/PSF observation/source parameters clamped**. '1' indicates that the observation/source parameters used to access the LSF/PSF have been clamped to fit the ELSF parameter space.

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The second set of 10 components (SM has index 10, AF9 index 19) of this array contains a set of bitwise flags, coded as follows:

- Bit 0 (0x0001): **Odd debiased sample**. '1' indicates that there is at least one debiased sample below $-TDN^2$ limit.
- Bit 1 (0x0002): **Numerical overflow recovery**: '1' indicates that there is at least one sample for which the numerical overflow has been corrected (only applicable in SM and AF1).
- Bit 2 (0x0004): **ELSF reliability mask**: '1' indicates that at least one sample has been discarded in IPD because the corresponding ELSF model is sub-optimal in that region.
- Bits 3 to 5 (0x0038); **Number of saturated samples discarded**. For AF it codes from 0 up to 7 or more masked samples and for SM it codes the position of the most significant bit of the masked sample count and thus coding 0, 1, 2 or 3, 4 to 7, up to 64 samples or more.
- Bit 6 (0x0040): **Background Mode**: '0' indicates background has been fitted along with location and flux parameters whereas '1' indicates that a different overall background estimation has been used i.e. KNN background in regular IPD or specific background module for SIF CF.
- Bit 7, 0x0080), **Charge injection (CI) distance updated**: indicating that the ciDist parameter has been redetermined and it does not match the original one in the raw data.
- Bit 8, 0x0100). **Non-gated CI line**: indicating that at least one non-gated charge injection is within the window.

See Chapter 3 of the DR4 online documentation for more information.

We note that in some cases, the `transit_proc_flags` and `ccd_proc_flags` for a given transit differ between `epoch_image` and `epoch_astrometry`. This is due to differing processing configurations in the production of these two tables. In case of doubt, the version in `epoch_astrometry` should be considered the more relevant one. Efforts will be taken to avoid such differences in Gaia DR5.

MULTIPEAK : True if the AstroElementary is multipeak; false otherwise (boolean)

If true, the transit is considered as a multi-peak observation, i.e. as having more than one astronomical light source within its CCD windows.

BLENDED : True if the peak for this source is also matched to another source (boolean)

The blended flag indicates the fact that a different source has been matched to the same peak as the present source, no matter if the transit has one or more peaks.

IPD_ERROR_AL : standard error of IDU IPD centroid AL (float[10] array, Angle[mas])

Standard errors from the calculation of the AL image centroids in SM and AF1-9, as estimated by the centroiding (IPD/DIPD) algorithm in the cyclic preprocessing of astrometric CCD observations. For details see Chapter 3 in the DR4 online documentation.

When the IPD algorithm has given some problem (and thus a result could not be obtained), the error is NaN.

IPD_ERROR_AC : standard error of IDU IPD centroid AC (NaN for 1d windows) (float[10] array, Angle[mas])

Standard errors from the calculation of the AC image centroids in SM and AF1-9, as estimated by the centroiding (IPD/DIPD) algorithm in the cyclic preprocessing of astrometric CCD observations. For details see Chapter 3 in the DR4 online documentation.

When the IPD algorithm has given some problem (and thus a result could not be obtained), the error is NaN. Also, the value is NaN for all 1d CCD windows.

G_MAG : G Magnitude, either from CU5 or from on-board estimate (float, Magnitude[mag])

This field is only needed in order to re-apply or re-determine the AGIS calibrations, so it was put together with the block containing the other parameters used to evaluate the calibration.

It is the G magnitude of the source used in the IPD centroiding: possibly the mean magnitude of the source, possibly the magnitude at the epoch of this particular observation, possibly the on-board magnitude estimate.

In most cases this will be the mean G magnitude from the main photometry processing. Only when that is not available and assuming that we also generate epoch astrometry for sources without G magnitude, then it would be an on-board magnitude. For variable sources it may also happen that in the future this will be the epoch G magnitude from the photometry processing.

If the field contains the magnitude estimated on board, the value is taken from the raw AstroObservation record, but after transformation from the technical telemetry format coding into a straight magnitude.

There is a very small number of sources (slightly over 4000, corresponding to only 20 000 transits) for which the AGIS processing has received no input `g_mag`. In the epoch astrometry table, these sources and their transits have `g_mag` set to NaN.

G_CLASS : Window class used in SM, AF and XP (byte)

This field is only needed in order to re-apply or re-determine the AGIS calibrations, so it was put together with the block containing the other parameters used to evaluate the calibration.

Window class used in the sampling of SM, AF and BP/RP windows. It can take the following values:

- 0 (window class 0a), G<11.5: full-resolution (2D) windows for AF and BP/RP, and high-resolution (2x2 pixels) window for SM.
- 1 (window class 0b) 11.5<G<13: full-resolution (2D) windows for AF, high-resolution (2x2 pixels) window for SM, and 1D windows for BP/RP.
- 2 (window class 1), 13<G<16: large 1D windows for AF (18 or 12 samples), low-resolution (2x2 samples, or 4x4 pixels) window for SM, and 1D windows for BP/RP.
- 3 (window class 2), G>16: small 1D windows for AF (12 or 6 samples), low-resolution (2x2 samples, or 4x4 pixels) window for SM, and 1D windows for BP/RP.

GATES : Gate information for all CCDs (Byte[10] array)

This field is only needed in order to re-apply or re-determine the AGIS calibrations, so it was put together with the block containing the other parameters used to evaluate the calibration.

This field contains an array, with one component per CCD (SM has index 0, AF9 index 9) indicating the gate used for the computation of the image parameters: flux and observing time.

Each of the components of the array contains a set of bitwise flags, coded as follows:

- Bit 7: The gate used in IPD is different from the one reported in the raw AstroObservation record. Basically, the true gate value was reconstructed by the Pre-processing (see Chapter **??** of the DR4 online documentation) according to the on-board software (CDB) configuration and the surrounding observations. This only happens when the preprocessing detects an inconsistency between the raw AstroObservation record gate and the expected value from the CDB configuration.
- Bit 6: The window has been acquired with a different gate with respect to the CDB. This flag could be tricky for windows close to the AC boundaries of a CCD, as we can

not fully reconstruct the AC coordinates used on board, but we could safely identify 99% of the cases.

- Bit 5 (0x20): '1' means 'multiple gates in this window'.
- Bit 4 (0x10): '1' means 'window affected by a gate release'.
- Bits 3 to 0 (0x0F): these bits indicate the gate number of the central sample of the window (e.g. '0' in case of no gate).

Please note that some of these flags are not applicable for SM (bits 5 and 4), whereas bits 3 to 0 indicate the permanent SM gate configuration, gate 12.

Convention for bits 3 to 0: '1' means the shortest gate, i.e. 2 CCD lines in length, '12' means the longest gate, i.e. 2900 lines in length, and '0' means no gate, i.e. 4500 lines.

See Section **??**, Table **??**, and throughout Chapter **??** of the DR4 online documentation for more information on gates.

SOURCE_DIST_TO_LAST_CI : source distance to last charge injection (float[10] array, Length & Distance[pixel])

Source distance to last charge injection (CI). This field is only needed in order to re-apply or re-determine the AGIS calibrations, so it was put together with the block containing the other parameters used to evaluate the calibration.

Distance in TDI time intervals from the trailing edge of the contiguous set of injection samples that include or precede the AL centroid of the source image in this window. If the CI to which these injection samples belong has suffered gate encounter(s), a flag is set in `ccd_acq_flags`. If any sample in the second half of the window contains charge injection, a flag is set in `ccd_acq_flags`.

If CI is not activated, a value of −32768 (MIN_SHORT) is given.

The first component of the array is for SM, which will not have CIs in principle. Thus, the reserved value −32768 will be stored there. The rest will contain the information for AF1 ([1]), etc. until AF9 ([9]). The reserved value (−32768) will also be used there in case the distance to the last CI cannot be calculated (which should never happen — and thus this value should indicate that something has gone wrong in the observation reconstruction).

A flag mentioned in `ccd_acq_flags` indicates if there is a CI in the 2nd half of the window. If there is a conflict between the CI and a gate, an appropriate bit in `ccd_acq_flags` will also be set.

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AC_RATE : AstroElementary AC rates for the transit time (float, Angular Velocity[pix/s])

This field is only needed in order to re-apply or re-determine the AGIS calibrations, so it was put together with the block containing the other parameters used to evaluate the calibration.

It is the speed of the AC motion of the image centroid on the focal plane, calculated as the mean AC rate over all CCDs (only one value per FoV transit).

SUB_PIXEL_COORD : Subpixel coordinate AL (float[10] array, Length & Distance[pixel])

This field is only needed in order to re-apply or re-determine the AGIS calibrations, so it was put together with the block containing the other parameters used to evaluate the calibration.

The subpixel AL coordinate for DR4 is computed as:

subpixel = mod(((transit time (OBMT) in seconds)/(TDI period in seconds)), 1.0).

MU : Ac coordinate of the image (or window centre) on a CCD (float[10] array, Length & Distance[pixel])

This field is only needed in order to re-apply or re-determine the AGIS calibrations, so it was put together with the block containing the other parameters used to evaluate the calibration.

This is the AC pixel coordinate of the image on the given CCD. Precisely, for DR4 it is the window centre (both for 1D and 2D windows), truncated to the nearest integer pixel. For DR5 it will be the 'true' image centroid AC, as predicted from source parameters, geometric calibration and attitude.

Note that the μ coordinate contained in the transit identifier is only for AF1, and it is always a window coordinate, not an actual or predicted image centroid.

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3.2 EPOCH_IMAGE [DataLink]

This table provides the image data from the Sky Mapper CCDs and from the CCDs of the main astrometric field, for all sources treated in the main astrometric processing. It provides the individual CCD sample values (not the original data, but pre-processed by various electronic calibration corrections), plus their precise projection onto the celestial sphere, and all relevant flags describing the properties of each image.

The table `epoch_image` contains one row per `source_id-transit_id` matching pair. The same `source_id` occurs as many times as there are FoV transits matched with this source. The same `transit_id` occurs as many times as there are individual peaks in multi-peak FoV transits. Any combination of `source_id` and `transit_id` occurs exactly once in the table.

Each row contains two per-source fields (`ra0` and `dec0`); for ease of handling and usage, these are repeated as many times as there are transits matched to the given source. Similarly, some of the per-transit and per-CCD data are repeated for the same reasons.

The precise projection of the individual CCD samples onto the celestial sphere is given in the form of local plane coordinates (see **?** for details), both for the two-dimensional and for the one-dimensional astrometric Gaia measurements.

Note: this table is not available through the main archive TAP interface. Data are delivered via the Massive Data service indexed by the VO DataLink protocol and described in Chapter **??**. For example this can be actioned in the archive user interface by joining queries with the table `all_source_flags` and selecting `has_epoch_image = 'true'`. The DataLink retrieval type to specify for this product in programmatic archive access via the Astroquery Gaia Python package is `EPOCH_IMAGE`.

We note that in many cases the fields `scan_pos_angle`, `parallax_factor_al` and `parallax_factor_ac` for a given `transit_id` differ between the `epoch_image` and `epoch_astrometry` tables. These differences are small (up to 3.8E-4 degrees, 1.8E-5 and 9.5E-6, respectively), and they are largely unavoidable. Partly, they are due to the — necessarily — different calibration models used: In `epoch_image`, no unique magnitude or colour can be assigned to any sample, nor to the window centre. On the contrary, for `epoch_astrometry` the full, chromatic and magnitude-dependent models for the underlying celestial source are used, of course. Additionally, the effective observation times are different: In `epoch_astrometry`, the obvious and optimal choice is the mean exposure time for the image centroid. However, this is not appropriate for `epoch_image`, in

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particular for multi-peak windows. The reasonable choice here would be the mean exposure time for the window centre. In the DR4 tables of epoch_image a mixture between centroid times and window centre times is implemented. For DR5, efforts will be taken to always use the window centre.

We furthermore note that the number of records in the epoch_image table is about 128 million higher than in the epoch_astrometry table. The difference is primarily due to time intervals known as AGIS gaps, where no final high-precision attitude is available, and thus the transits were not used in the astrometric source solutions. A minority of the cases is caused by multiple (secondary) peaks detected in the early DIPD processing for DR4 that were not recovered in the latest DIPD runs.

DataLink products are served with a choice of two different data structures:

- DATAMODEL_STANDARD: the product is serialised following a standard data model (for example an IVOA-compliant data model in applicable cases);
- DATAMODEL_GAIA: the product is serialised following the Gaia DPAC(-internal) data model.

The DATAMODEL_GAIA format is described here whereas the DATAMODEL_STANDARD format is described in https://www.cosmos.esa.int/web/gaia-users/archive/datalink-products#datalink_serialisation

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution identifier, please visit https://gaia.esac.esa.int/decoder/solnDecoder.jsp

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source.

The general format and construction of the identifier is described in `gaia_source.source_id`.

TRANSIT_ID : Transit Identifier (long)

The `transit_id` is a unique identifier assigned to each detected (and confirmed) source as it transits the Gaia focal plane. Each time a given source is detected as Gaia scans and re-scans the sky a new transit identifier will be created to badge that apparition. Hence the along-scan time and the across-scan position along with the telescope in which the source was detected are used to form a unique integer with which to label the transit. The detection of transits is done using the SkyMapper CCD data, and the confirmation is done using the AF1 CCD data. Both happens on board.

The several features of a detection that are encoded in `transit_id` can be easily retrieved using bit masks (&) and shifts (>>) as follows:

- On-Board Mission Time line [ns]
  = 204800 * ((`transit_id` >> 17) & (0x000003FFFFFFFFFF))
- Field-of-view = 1 + (`transit_id` >> 15) & 0x03 [1 for 'preceding' and 2 for 'following' fields-of-view respectively]
- CCD row = (`transit_id` >> 12) & 0x07 [dimensionless, in the range 1 to 7]
- Across-scan 'reference acquisition pixel' in strip AF1 = (`transit_id`) & 0x0FFF [pixels] (this is the across-scan centre of the AF1 window and is odd if immediately below the mid-point of the window and even if immediately above)

where the bit mask prefix '0x' denotes hexadecimal.

For further details see Portell et al. (2020). For convenience a decoder for `transit_id` is available on-line at

`https://gaia.esac.esa.int/decoder/transitidDecoder.jsp`

RA0 : Right ascension of reference point (double, Angle[deg])

Right ascension α_0 of the tangent point on the plane tangential to the celestial sphere in which the coordinate direction of the object is locally modelled, i.e. of the celestial reference point from which the local plane coordinates `sample_pos_al`, `sample_pos_ac` and the corresponding sample size and shear terms are counted. The specific value will be taken from the valid/relevant source catalogue (which in turn will be defined implicitly by the `solution_id`).

Note: ra0 and dec0 are always the same for all transits of a given source_id.

For more details see **?**.

DEC0 : Declination of reference point (double, Angle[deg])

Declination δ_0 of the tangent point of the plane tangential to the celestial sphere in which the coordinate direction of the object is locally modelled, i.e. of the celestial reference point from which the local plane coordinates sample_pos_al, sample_pos_ac and the corresponding sample size and shear terms are counted. Specific value will be taken from the valid/relevant astrometric source catalogue (which in turn will be defined implicitly by the solution_id).

Note: ra0 and dec0 are always the same for all transits of a given source_id.

For more details see **?**.

CCD_ROW : CCD Row (byte)

CCD Row, from 1 to 7. Although the transit_id contains this, we include it for performance issues.

RES_REF_TIME : Residual (high-resolution) reference time (short, Time[OBMT50])

Residual (high-resolution) AF1 reference acquisition time.

The transit_id is defined in a way that it contains the reference acquisition time (from the AF1 acquired window) without its 12 least-significant bits (LSB). That is, if we use only the time contained in the transit_id we lose resolution.

This data field, res_ref_time, contains these 12 bits removed from the OBMT before the composition of transit_id. Hence, res_ref_time ranges from 0 to 204.75μs, in steps of 50ns (the same resolution as OBT).

The coding of res_ref_time follows these same steps; that is, 1 LSB of res_ref_time is 50ns. Hence the maximum code is 4095 (equal to 204.75μs).

Adding this value to the extracted time from transit_id (with the adequate units) will lead to the full OBMT resolution of the AF1 acquisition window.

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OBJECT_TYPE : Object type (byte)

Object type flag assigned by the VPU:

- '0': Star ('S')
- '1': Star sent to AOCS ('AOCS')
- '2': Calibration faint star ('CFS')
- '3': Suspected Moving Object ('SMO')
- '4': Virtual Star ('VS')
- '5': Star acquired with gate ('SG')
- '6': Star acquired with Charge Injection ('CI')

G_CLASS : Window class used in SM, AF and XP (byte)

Window class used in the sampling of SM, AF and BP/RP windows. It can take the following values:

- 0 (window class 0a), G<11.5: full-resolution (2D) windows for AF and BP/RP, and high-resolution (2x2 pixels) window for SM.
- 1 (window class 0b) 11.5<G<13: full-resolution (2D) windows for AF, high-resolution (2x2 pixels) window for SM, and 1D windows for BP/RP.
- 2 (window class 1), 13<G<16: large 1D windows for AF (18 or 12 samples), low-resolution (2x2 samples, or 4x4 pixels) window for SM, and 1D windows for BP/RP.
- 3 (window class 2), G>16: small 1D windows for AF (12 or 6 samples), low-resolution (2x2 samples, or 4x4 pixels) window for SM, and 1D windows for BP/RP.

AF19_CENTRING : Centring information for AF1-9 (byte[5] array)

This array always has a length of 1. It indicates the on-board trimming ('centring') of the AF1 window for `g_class`=2,3 in AL direction in order to reduce it to the most relevant samples centred on the actual star image. A centring value of zero means that the first 12 or 6 samples of an AF1 readout window are transmitted to the ground for `g_class`=2,3, respectively, and a value of 6 means that the last 12 or 6 samples are transmitted to the ground. A value of 3 indicates that the central samples are transmitted, thus discarding the 3 heading and the 3 trailing samples.

The value of 7 is used for `g_class`=0,1 where the full acquired AF1 window (18 samples in AL direction) is always transmitted.

This field is probably of no real use for any application of the public DR4 data.

SHAPE_AF : Shape (truncation) information for AF2-9 windows (byte[] array)

This array details the shape of the AF2-9 windows and is important if the windows are truncated or trimmed. It is only relevant for 1D windows and therefore not given for 2D windows. SM and AF1 windows are never truncated or trimmed.

Here, truncation refers to a reduction in size - often irregular - caused by an intra-field conflict with another window, while trimming refers to the permanent use of a reduced window size across scan. Trimming was only used for AF2-9 and only for a short period early in the mission (around OMBT revolutions 2103 to 2238).

The array will have the following possible sizes:

- NULL, for 2D windows, i.e. for `g_class`=0 or 1.
- 18 elements, for `g_class`=2.
- 12 elements, for `g_class`=3.

Each of the 12 or 18 elements corresponds to the same AL sample position in the 1D windows in AF2 to AF9. That is: this truncation geometry is identical for all AF2-9 windows in a transit.

This is because truncation is activated when another nearby source is detected *in the same field of view*, and thus following the same AL and AC propagation through the AF2-9 CCDs during the transit.

The meaning of each of the array elements is the following:

- The four most-significant bits (4 MSB) indicate the AC offset of the first pixel of the sample.
- The 4 LSB indicate the AC length (minus 1) of the sample.
- A value of -1 means that the sample was eliminated.

Some example values:

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- The most typical case, for non-truncated 1D windows, is a value of 11 in each of the elements. It means a nominal 1D window of AF2-9 (with an AC size of 12 pixels), that is, an offset of 0 and a length of 11 + 1 = 12 pixels.
- A special case of -1 means that all the samples at that AL position were completely eliminated. Thus, we can expect zeroes in the corresponding samples for all AF2-9 windows.
- A value of 10 would mean an offset of 0 and a length of just 11 pixels AC.
- A value of 86 would mean an offset of 5 pixels and a length of 7 pixels AC.
- A value of 0 would mean an offset of 0 pixels and a length of just 1 pixel AC.

AOCS_UPDATE : Information on AOCS updates occurred in SM + AF1-9 (byte[10] array)

This field indicates if some samples of a given window (SM and AF1-9) have been affected by an AOCS update, and hence if the window is not rectangular. The information is coded in a special way:

- '0' means 'no update', i.e. the window is rectangular (all the samples have the same AC coordinate).
- A value of '-1' means that more than 1 AC Shift update has occurred within the window (this should never happen), or that some other unexpected thing has occurred.
- In the other cases, i.e. when a 'normal' AOCS update has occured during the window acquisition:

  Bits 0 to 4 (the least-significant ones) indicate the first line (1...17), after `al_win_coord`, for which the AC coordinate has changed. Note that value '0' should not be used here (anyway it would mean 'no AC update affecting the window').

  Bits 5 to 7 indicate the update (-3 to +3 pixels AC), but coded in a special way to avoid negative values: '+4' is added to the actual update. Thus, code '1' in these three bits actually means '-3 pixels AC', and code '7' means '+3 pixels AC'. Please note that code '4' (i.e. '0 pixels AC') should never be used to avoid confusion. Also note that, in case of larger updates, code '0' in these three bits will be generated, and then the actual shift shall be retrieved from the AC shifts table; in other words, an `aocs_update` value between 1 and 31 (i.e. 0<`aocs_update`<32) means that a large update has occurred.

Please note that, for simplicity, we cover all of the CCDs — also including SM and AF1, for which we will never have AOCS updates (and thus their corresponding components will always

be zero).

TRANSIT_ACQ_FLAGS : Transit-level information on the on-board acquisition (short)

This field contains a set of bitwise flags (from Bit 15 = msb to Bit 0 = lsb) related to the data acquisition. It has some oddities for historical reasons.

Windows may suffer truncation or trimming. Here, truncation refers to a reduction in size - often irregular - caused by an intra-field conflict with another window, while trimming refers to the permanent use of a reduced window size across scan. Truncation and trimming only affect 1D windows and never affect SM and AF1 windows. Trimming was only used for AF2-9 and only for a short period early in the mission (around OMBT revolutions 2103 to 2238).

The shape of all 1D AF2-9 windows is detailed in `shape_af`.

The field is coded as follows:

- Bits 15 and 14: unused
- Bit 13 (0x2000): Always set.
- Bit 12 (0x1000): The AF2-9 windows are trimmed AC but have not suffered truncation.
- Bits 11-9: unused
- Bits 7 and 8 (0x0180): `xp_offset`. This is the value of a continuous two-bit TDI counter. It is needed for the reconstruction of the XP windows.
- Bit 6 (0x0040): AF2-9 windows are 2D.
- Bit 5 (0x0020): AF2-9 windows are 1D (complement of Bit 6).
- Bit 4 (0x0010): AF2-9 windows are 1D and with the full AC size of 12 pixels.
- Bit 3 (0x0008): No BP window acquired.
- Bit 2 (0x0004): No RP window acquired.
- Bit 1 (0x0002): BP window truncated.
- Bit 0 (0x0001): RP window truncated.

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CCD_ACQ_FLAGS : CCD-level information on the on-board acquisition (byte[10] array)

This field contains an array, with one component per CCD (SM has index 0, AF9 index 9).

Each of the components of the array contains a set of bitwise flags, coded as follows:

- Bit 7 (AND bit mask 0x80): '1' means 'window missing' (i.e. all samples are zero). It will happen for AF9 Row 4 (which does not exist), and for switched off CCDs, full overlapping in an inter-field conflict, etc.
- Bit 6 (0x40): '1' means 'wide window', '0' means 'narrow window'. This distinction is only applicable to AF windows of Class 2 or 3. For the rest of the cases, that is, SM (any class), and for AF windows of Class 0 or 1 (i.e. 2D windows), we can consider them as wide windows, and thus the flag will be '1'. It will also be '1' for truncated AF windows obviously (since they all become wide).
- Bit 5 (0x20): '1' means 'window at the CCD edge', i.e. this window has been acquired with fewer AC pixels than usual.
- Bit 4 (0x10): '1' means 'some samples of this window have been eliminated, e.g. due to an inter-field conflict'.
- Bit 3 (0x08): '1' means 'this window is affected by an AOCS update'.
- Bit 2 (0x04): '1' means 'this window is affected by more than one gate (multiple gate)'.
- Bit 1 (0x02): '1' means 'there is a Charge Injection in the last half of the window'.
- Bit 0 (0x01): '1' means 'the Charge Injection preceding this window is abnormal', mainly meaning that the preceding CI has interacted with a Gate i.e. the injection charge has been held in a gate line and therefore its length has been shortened and the TDI distance to the previous injection has been extended.

Please note that some of these flags are not applicable for SM (such as bits 3, 2, 1 and 0) and for AF1 (such as 3).

See Chapter 3 of the DR4 online documentation for more information.

GATES : Gate information for all CCDs (byte[10] array)

This field is only needed in order to re-apply the AGIS calibrations, so it was put together with the block containing the other parameters used to evaluate the calibration.

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This field contains an array, with one component per CCD (SM has index 0, AF9 index 9) indicating the gate used for the computation of the image parameters: flux and observing time.

Each of the components of the array contains a set of bitwise flags, coded as follows:

- Bit 7: The gate used in IPD is different from the one reported in the raw AstroObservation record. Basically, the true gate value was reconstructed by the Pre-processing (see Chapter **??** of the DR4 online documentation) according to the on-board software (CDB) configuration and the surrounding observations. This only happens when the preprocessing detects an inconsistency between the raw AstroObservation record gate and the expected value from the CDB configuration.
- Bit 6: The window has been acquired with a different gate with respect to the CDB. This flag could be tricky for windows close to the AC boundaries of a CCD, as we can not fully reconstruct the AC coordinates used on board, but we could safely identify 99% of the cases.
- Bit 5 (0x20): '1' means 'multiple gates in this window'.
- Bit 4 (0x10): '1' means 'window affected by a gate release'.
- Bits 3 to 0 (0x0F): these bits indicate the gate number of the central sample of the window (e.g. '0' in case of no gate).

Please note that some of these flags are not applicable for SM (bits 5 and 4), whereas bits 3 to 0 indicate the permanent SM gate configuration, gate 12.

Convention for bits 3 to 0: '1' means the shortest gate, i.e. 2 CCD lines in length, '12' means the longest gate, i.e. 2900 lines in length, and '0' means no gate, i.e. 4500 lines.

See Section **??**, Table **??**, and throughout Chapter **??** of the DR4 online documentation for more information on gates.

AL_WIN_COORD : Along Scan Coordinate of transmitted SM and AF windows wrt ref (short[10] array, Time[TDI Period])

AL coordinates of the transmitted SM window and of the AF1-9 windows with respect to the reference acquisition time (of AF1, included in `transit_id` and `res_ref_time`). That is, these are the window coordinates (following Bastian (2020)), not the coordinates of the central pixels (which is the reference system used by Astrium in the raw telemetry).

Note: These are the absolute AL coordinates of the windows w.r.t. the reference time, that is, they already include the operational AL Phasing Table correction. However the following modification was done to avoid exceeding the range of a 'short':

al_win_coord_coded[i] = al_win_coord_true[i] - (1 << 14) = al_win_coord_true[i] - 16384.

The approximate range of al_win_coord_coded[i] is -27260 TDIs to +23137 TDIs.

AC_WIN_COORD : AC coordinate of the transmitted SM and AF windows (byte[10] array, Length & Distance[pixel])

AC coordinates of the transmitted SM window and of the AF1-9 windows with respect to the reference acquisition pixel (of AF1, included in transit_id). That is, these are the window coordinates (following Bastian (2020)), not the coordinates of the central pixels (which is the reference system used in the raw telemetry).

SM_SAMPLES : De-biased sample values from the sky mapper (float[] array, Charge[e-])

Samples of the SM window, in electrons, using the nominal PEM gain for conversion from ADUs (Analog-to-Digital Units) to electrons.

The number of samples included in this window depends on g_class:

- g_class<2 (G≤13) leads to 40×6=240 samples, each from 2×2 pixels.
- g_class>1 (G>13) leads to 20×3=60 samples, each from 4×4 pixels, namely from 2×2 software-binned read-out samples of 2×2 pixels each.

That is, the SM window always covers the same area on the sky, but with window class 0 (0a or 0b) it has more resolution.

Regarding the sample order (SM windows are always 2D windows), it follows the original order as provided by the VPU on board — that is, first in AC order, then in AL. This means, e.g. for g_class<2, the array is composed of 40 AL 'blocks' (with the TDI value increasing with the array index), and each 'block' composed of 6 AC samples (with the AC coordinate increasing with the array index). This is in agreement with Bastian (2020).

AF19_SAMPLES : De-biased sample values from the Astrometric Field (AF1-AF9) (float[] array, Charge[e-])

The samples of the AF windows, bias corrected and converted to electrons using the nominal gain for conversion from ADUs (Analog-to-Digital Units) to electrons. This is a two dimensional array listing the samples for each AF window, beginning with AF1 and ending with AF9 (or AF8 in case of row 4).

The two-dimensional array is served by the Gaia ESA Archive as a flattened (one-dimensional)

array. The number of samples per FoV transit depends on `g_class`, and it is different for AF2-9 as compared to AF1.

For the faintest (G>16) stars, i.e. for `g_class`=3, the AF1 window contains 6 samples in the AL direction, while the AF2-9 windows contain 12 samples each.

For `g_class`=2 (13<G<16), all windows are 6 samples wider with 12 samples for AF1 and 18 for the rest.

Finally, for `g_class`<2 (G≤13), all AF windows are 2-dimensional and with 18 samples in the AL direction. AF1 has a total of 18×6 samples and AF2-9 have 18×12. AF1 has only 6 samples AC because each sample contains 2 pixels binned in the AC direction, whereas there is no binning for AF2-9.

Within the array, samples are given in the same order as read off the CCDs. For 1D windows (`g_class`>1) they come simply in TDI (time) order. For 2D windows (`g_class`<2), the samples for each TDI line come in increasing column (AC coordinate) order, listing all samples for that TDI line before continuing with the next TDI line. This is in agreement with Bastian (2020).

Note: in case of truncated windows (which can only occur for `g_class`≥2), windows of AF2-9 have smaller AC sample sizes.

BACKGROUND : Mean background derived by IPD for these SM and AF windows (float[10] array, Flux[e^- pix^{-1} s^{-1}])

The values of the astrophysical background, usually derived by the IPD process, for the SM window and each AF window of the present transit. Sometimes IPD uses the low-resolution background, for instance when DIPD has detected multiple peaks.

The background is not subtracted from the de-biased sample values in `af19_samples`. The reasoning here is that, contrary to the de-biasing components of the sample values, this one is not an unprejudiced general property of the Gaia CCDs and electronics, but it is part of the scientific interpretation of the samples in the given window. Thus, a user might wish to derive the background in a way that is different from what IDU IPD has done. On the other hand, users not wanting to do that can simply subtract the given `background` from the sample values to get a 'pure' star signal.

BACKGROUND_ERR : Background formal error at the centre of these SM and AF windows (float[10] array, Flux[e^- pix^{-1} s^{-1}])

Standard errors of `background` determined for the SM and AF window centres, derived in the on-ground processing by IDU.

Please note that observations close to a charge injection may be strongly affected by the delayed charge release from the saturated pixels, and thus the background may have a significant variation along the window.

TRANSIT_PROC_FLAGS : Transit-level information on the processing results (short)

This field contains a set of bitwise flags, coded as follows:

- Bits 3 to 0 (`transit_proc_flags` AND 0x000F): **IPD strategy and findings**. As of IDT 15.1 and until IDT 21.0, as well as in IDU until DRC-02, this field was not used (0 was always stored here). As of IDU DRC-03, the following information is provided:

  '0': Normal IPD. Ambiguity not tested or not detected. Default value for all records produced until IDT 21.0 and IDU until DRC-02

  '1': Multiple Source IPD: As of IDT 21.1, this value is used to indicate that the earlier object in time detected in the window has been selected. In case of 2D windows with larger AC than AL separation between detected peaks: higher mu selected.

  '2': Multiple Source IPD: As of IDT 21.1, this value is used to indicate that the later object in time detected in the window has been selected. In case of 2D windows with larger AC than AL separation between detected peaks: lower mu selected.

  '3': Multiple Source IPD: As of IDU DRC-03, this value is used to indicate that at least one of the windows has a multiple source and that these disturbing sources were masked for the IPD processing. `ccd_proc_flags` can be checked to know which and how many windows have been affected.

  '4': DIPD: As of IDU 21.0, this value is used to indicate that this object has been obtained using DIPD implementation and thus include a window model template and combined AF solution.

  '5': TukeyBiWeight IPD: this value is used to indicate that this object has been obtained using TukeyBiWeight IPD initialisation for the internal processing subsystem IDU-AttAc; such initial values do not propagate into the published data.

  Values 6-15 are reserved for future extensions.

- Bits 6 to 4 (0x0070): **Bias and PEM OI strategy baseline**: this field indicates the Bias mitigation mode requested to process this transit. The consumer must check the corresponding `ccd_proc_flags` field to determine the actual mode used for each individual CCD as sometimes some modes can not be applied due to missing input data or calibration limitations.

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Value ‘0’: Default initialisation value for the Bias strategy. If found in an AstroElementary it means that something went wrong and the Bias (or maybe just its flagging) was not done appropriately. Thus, ‘0’ means ‘UNDEFINED/FAILURE’.

Value ‘1’: Only PreScan bias used.

Value ‘2’: PreScan + common NU baseline.

Value ‘3’: Full PEM NU.

Values 4-7 are reserved for future extensions.

- Bits 10 to 7, **colour info**:

  Bit 7 (0x0080), **default colour**: ‘1’ means ‘a default colour used for IPD processing’, which can be caused by some problem in the colour determination or retrieval (e.g. no BP/RP data in IDT or no valid PhotPipe fluxes in IDU).

  Bit 8 (0x0100), **epoch colour**: ‘1’ means ‘epoch (transit-level) colour used for IPD processing’, ‘0’ means ‘mean (source-level) colour used’.

  Bit 9 (0x0200), **single-param colour**: ‘1’ means ‘single-parameter colour used for IPD processing’ (such as `nu_eff_used_in_astrometry`), ‘0’ means ‘multi-parameter colour used’ (such as SSCs).

  Bit 10 (0x0400), **colour problems**: ‘1’ means ‘missing colour information or dubious colour quatiliy’ (e.g. problematic/missing BP/RP data in IDT, NaN or rejected PhotPipe colour in IDU, etc).

  Please note some remarkable combinations of this bit with Bit 7:

  – Non-default + no problems (Bit 7 = 0, Bit 10 = 0): Good colour from catalogue or epoch data
  – Non-default + problems (Bit 7 = 0, Bit 10 = 1): Dubious colour from catalogue or best approximation from partial or problematic epoch data. Implemented only in IDT.
  – Default + no problems (Bit 7 = 1, Bit 10 = 0): Default colour has been used because no colour information was available (i.e. for blacklisted observations). Implemented only in IDU.
  – Default + problems (Bit 7 = 1, Bit 10 = 1): Default colour used instead of the available colour from catalogue or epoch data (i.e. dubious colour rejected).

- Bit 11 (0x0800), **cosmetic effects** flagging: ‘1’ means ‘dead pixel columns affecting some of the windows’.

- Bit 12 (0x1000), **cosmetic effects** flagging: ‘1’ means ‘anomalously low samples found in some of the windows’.

- Bit 13 (0x2000), **cosmetic effects** flagging: ‘1’ means ‘anomalously high samples found in some of the windows’.

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- Bit 14 (0x4000), **problems in initial centroid**: '1' means that problems have been found in the Initial Centroids determination. In the case of IDT, combined with the 'some problem found in the colour determination' flag (bit 10), it may indicate problems in the TransitPredictor (e.g. predicted position outside window, or no attitude available) in which case a default position is used for colour determination.
- Bit 15, **attitude issues**: '1' indicates that some attitude problem was found affecting some of the windows i.e. missing attitude or problems computing AL/AC rates. This flag is only set by IDU which depends on the attitude to access the ELSF and for the combine AF processing in DIPD.

See Chapter 3 of the DR4 online documentation for more information.

We note that in some cases, the `transit_proc_flags` and `ccd_proc_flags` for a given transit differ between `epoch_image` and `epoch_astrometry`. This is due to differing processing configurations in the production of these two tables. In case of doubt, the version in `epoch_astrometry` should be considered the more relevant one. Efforts will be taken to avoid such differences in Gaia DR5.

CCD_PROC_FLAGS : CCD-level information on the processing results (short[20] array)

This field contains information about the findings in the IPD processing of the SM and AF CCD data. It is included here although no IPD processing has actually been performed on the CCD data given here. The reason is that the IPD findings constitute diagnostic information relevant for any usage of the image data from the CCDs.

The field is defined as an array, with one or more sets of 10 components (one per CCD). The indexes of the componenents for a given CCD meet the following expression:

$$s + 10k \quad \begin{cases} s & [0...9] & \text{CCD index [SM...AF9]} \\ k & [0...n] & \text{CCD set index} \end{cases} \tag{6}$$

The first set of 10 components (SM has index 0, AF9 index 9) of this array contains a set of bitwise flags, coded as follows:

- Bits 3 to 0 (`ccd_proc_flags` AND 0x000F): **IPD result**. It is coded as follows:
  - '0': **Success**; the fitting has succeeded without any problem.
  - '1': **Illegal Amplitude**; invalid amplitude obtained after the initial fitting. It may happen for windows with wrong background determination or similar.

  - '2': **Illegal Coordinate**; invalid AL or/and AC locations obtained in the fitting i.e. locations outside the window. It may happen for windows close to a charge injection, multiple sources, strongly affected by some cosmic ray or other similar issues.
  - '3': **DEPRECATED: Illegal Output**; location estimation outside window.
  - '4': **Maximum Iteration Exceeded**; convergence not reached, but results achieved so far stored in the 'astro–elementaries'. The IPD result may still be reliable enough.
  - '5': **Failed**; diverging iteration instead of converging.
  - '6': **Missing or unreliable**; maybe the processing failed for some reason before finishing the calculations, or the IPD was initialised, but no actual processing has occurred.
  - '7': **No LSF/PSF data**; no result available due to no LSF/PSF data available.
  - '8': **Not Enough Samples**; IPD was attempted but no result is available because at a given iteration it runs out of samples (there are not enough inputs to the equations/matrix system).
  - '9': **Odd LSF/PSF Success**; the fitting has succeeded but using an odd LSF/PSF. The LSF/PSF is considered odd when it is not well defined for some samples. The IPD result may still be reliable enough similar as with the 'Maximum Iteration Exceeded' case.
  - '14': **Not Attempted**; IPD has not been attempted on purpose due to missing gate information, missing CI information or similar or when not enough usable input samples are available due to heavy truncation, CCD edge, complex AOCS update, multiple gates or similar.
  - '15': **Crashed**; calculation finished and crashed for some reason.
  - Values '3' and '10' to '13': Available for future extensions.

- Bit 4 (0x0010): **IPD nominal/non nominal**. '1' indicates the IPD is non-nominal in the sense it has been performed over a reduced window area (details in `ccd_proc_geom`). This flag is only set for AF windows.

- Bit 5 (0x0020): **IPD not available**. '1' indicates that IPD result is not available and thus its output values are not valid. This flag set to true and IPD result equals to NOT_ATTEMPTED (value '14' in bits 3-0) indicates IPD was not attempted on purpose, and with IPD result equals to PENDING (value '6' in bits 3-0) indicates unexpected IPD termination. This flag set to true with any other IPD result should not be possible.

- Bit 6 (0x0040): **Some cosmetic issue**. '1' indicates that there is at least one contaminated or discarded sample in the selected window area. This flag collects the

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presence of additional discarded or contaminated sample without an explicit CCD level flag thus excluding saturation, cosmics, odd debiased, etc.

- Bit 7 (0x0080): **Cosmic removed**. '1' means that at least one sample in the selected window area has been discarded for being a suspected cosmic.
- Bit 8 (0x0100): **Saturation removed**. '1' means that at least one sample in the selected window area has been discarded due to saturation.
- Bit 9 (0x0200): **Part of window discarded**. '1' means that at least one sample has been discarded in the full window area.
- Bit 10 (0x0400): **No window**. '1' means 'no samples available' (i.e. all the samples in this window were discarded on board, or this is e.g. AF 9 Row 4, or a deactivated CCD, etc).
- Bit 11 (0x0800): **Non-success InitialCentroid**. '1' means 'the Initial Centroid was not successful, and thus we have used the centre of the window as the start condition of IPD'.
- Bit 12 (0x1000): **Odd Background found**. '1' indicates that the background determined for this window (for at least one of its samples) was odd i.e negative, too large or NaN. This flag may be used to indicate that the fluxes determined for this window are potentially wrong (although its AL/AC centroid may still be useful).
- Bit 13 (0x2000): **fall-back to lower Bias Mitigation mode**. '1' indicates that 'fall-back' to lower mode has occurred for this window (for at least one of its samples). This flag will always be activated for AF1 when FULL_MITIGATION mode is enabled because this mode is not supported in this CCD.
- Bit 14 (0x4000): **non-target source removed**. '1' indicates that non-target sources have been detected and masked in this window.
- Bit 15 (0x8000): **LSF/PSF observation/source parameters clamped**. '1' indicates that the observation/source parameters used to access the LSF/PSF have been clamped to fit the ELSF parameter space.

The second set of 10 components (SM has index 10, AF9 index 19) of this array contains a set of bitwise flags, coded as follows:

- Bit 0 (0x0001): **Odd debiased sample**. '1' indicates that there is at least one debiased sample below $-TDN^2$ (total detection noise) limit.
- Bit 1 (0x0002): **Numerical overflow recovery**: '1' indicates that there is at least one sample for which the numerical overflow has been corrected (only applicable in SM and AF1).

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- Bit 2 (0x0004): **ELSF reliability mask**: '1' indicates that at least one sample has been discarded in IPD because the corresponding ELSF model is sub-optimal in that region.
- Bits 3 to 5 (0x0038); **Number of saturated samples discarded**. For AF it codes from 0 up to 7 or more masked samples and for SM it codes the position of the most significant bit of the masked sample count and thus coding 0, 1, 2 or 3, 4 to 7, up to 64 samples or more.
- Bit 6 (0x0040): **Background Mode**: '0' indicates background has been fitted along with location and flux parameters whereas '1' indicates that a different overall background estimation has been used i.e. KNN background in regular IPD or specific background module for SIF CF.
- Bit 7, 0x0080), **Charge injection (CI) distance updated**: indicating that the charge injection distance parameter has been redetermined and it does not match the original one in the raw data.
- Bit 8, 0x0100). **Non-gated CI line**: indicating that at least one non-gated charge injection is within the window.

See Chapter 3 of the DR4 online documentation for more information.

We note that in some cases, the `transit_proc_flags` and `ccd_proc_flags` for a given transit differ between `epoch_image` and `epoch_astrometry`. This is due to differing processing configurations in the production of these two tables. In case of doubt, the version in `epoch_astrometry` should be considered the more relevant one. Efforts will be taken to avoid such differences in Gaia DR5.

CCD_PROC_GEOM : CCD-level information on the processing geometry used (int[10] array)

This field contains an array, with one component per CCD (SM has index 0, AF9 index 9). It indicates, per CCD, the 'geometry' of the area actually processed by the IPD.

Each of the components of the array contains a set of bitwise fields, coded as follows:

- Bits 5-0 (0x0000003F): width (samples) of the sub-window used in the IPD.
- Bits 9-6 (0x000003C0): height (samples) of the sub-window used in the IPD.
- Bits 16-10 (0x0001FC00): AL offset (samples) of the sub-window used in the IPD.
- Bits 20-17 (0x001E0000): AC offset (samples) of the sub-window used in the IPD.

- Bits 31-21: spare bits, available for user-defined flags or future extensions.

This field always contains the actual dimensions of the SM sub-window used in IPD (8 AL samples for window class 0 and 4 AL samples for window class 1 by default). Note that, contrary to AF cases, the `ccd_proc_flags`[] bit 4 is never set to true for SM.

See Chapter 3 of the DR4 online documentation for more details.

SCAN_POS_ANGLE : Position angle of the scan (double[10] array, Angle[deg])

Position angle θ of the scan for each of the CCD transits.

This angle uniquely determines the transformation between the local celestial coordinates (a, d) and the local scan coordinates (w, z) in the tangential plane touching the celestial sphere at `(ra0,dec0)`. See **?** for details. The scan angle varies along a FoV transit. In contrast, the parallax factors are sufficiently constant over a transit.

The sine and cosine of θ are the derivatives of the along-scan local plane coordinate w with respect to right ascension and declination, respectively.

Please note the remark in the 6th paragraph of the table description.

PARALLAX_FACTOR_AL : Parallax factor AL (float)

Parallax factor in the AL direction (w), needed to use the local plane coordinates for detailed object modelling.

The parallax factor f_w is the derivative of the along-scan local plane coordinate w with respect to the parallax. See **?** for details.

The parallax factors need not be an array of values per CCD, as they are sufficiently constant over a FoV transit. Thus it was decided that the parallax factors are computed for the AF4 CCD transit time (or for an immediately adjacent CCD in case the AF4 IPD should happen to fail).

Please note the remark in the 6th paragraph of the table description.

PARALLAX_FACTOR_AC : Parallax factor AC (float)

Parallax factor in the AC direction (z) needed to use the local plane coordinates for detailed

object modelling.

The parallax factor f_z is the derivative of the across-scan local plane coordinate z with respect to the parallax. See **?** for details.

The parallax factors need not be an array of values per CCD, as they are sufficiently constant over a FoV transit. Thus it was decided that the parallax factors are computed for the AF4 CCD transit time (or for an immediately adjacent CCD in case the AF4 IPD should happen to fail).

Please note the remark in the 6th paragraph of the table description.

N_SAMP_AL : Number of samples AL (byte[10] array)

Array with numbers of samples per window in AC direction. It is to be noted that the value always refers to the normal window size according to `g_class`, i.e. it does not take into account any truncation of the windows.

N_SAMP_AC : Number of samples AC (byte[10] array)

Array with numbers of samples per window in AC direction. It is to be noted that the value always refers to the normal window size according to `g_class`, i.e. it does not take into account any truncation of the windows.

SAMPLE_POS_AL : Local plane coord. of samples (0, 0) AL (float[10] array, Angle[mas])

For each CCD transit the AL local coordinate w_0 of the first sample (0,0) of the window; see **?** for details.

There is a known issue: Among the $\simeq 10^{12}$ entries of `sample_pos_al` in the table, there is a very small portion with grossly deviating values. Such values may be caused by excursions of the attitude modelling at the start and end of data gaps. For more information see the description of `sample_size_al`. On the user side, the obvious way to avoid the problem is to simply ignore the very few records where `sample_pos_al` is unreasonably large.

SAMPLE_POS_AC : Local plane coord. of samples (0, 0) AC (float[10] array, Angle[mas])

For each CCD transit the AC local coordinate z_0 of the first sample (0,0) of the window; see **?** for details.

There is a known issue: Among the $\simeq 10^{12}$ entries of `sample_pos_ac` in the table, there is a

very small portion with grossly deviating values. Such values may be caused by excursions of the attitude modelling at the start and end of data gaps. For more information see the description of `sample_size_al`. On the user side, the obvious way to avoid the problem is to simply ignore the very few records where `sample_pos_ac` is unreasonably large.

SAMPLE_SIZE_AL : Sample sizes AL (float[10] array, Angle[mas])

For each CCD transit the AL sample size Δw. See **?** for details.

There is a known issue: Among the $\simeq 10^{12}$ entries of `sample_size_al` in the table, there may be a very small portion with grossly deviating values. The normal values are around 58.9 mas, with a very small scatter due to the tiny variations of the scan speed and to the image rotation caused by the precession of the spin axis. Grossly deviating values may be caused by excursions of the attitude modelling at the start and end of data gaps. The same effect created the very small number of extremely large values of `centroid_pos_al` in `epoch_astrometry`: There are a total of 16,437,562 cases where `centroid_pos_al` is beyond the range allowed by the window size, and a few taking exceptionally large values. Notably, there are 208 transits where the max value of `epoch_astrometry.centroid_pos_al` exceeds 10^9 mas.
On the user side, the obvious way to avoid the problem is to simply ignore the very few records where `sample_size_al` deviates from the usual values by more than, say, 0.1 mas.

SAMPLE_SIZE_AC : Sample sizes AC (float[10] array, Angle[mas])

For each CCD transit the AC sample size Δz. See **?** for details.

There is a known issue: Among the $\simeq 10^{12}$ entries of `sample_size_ac` in the table, there is a small portion with grossly deviating values. While the normal values are around 177 mas (with small variations due to focal-length changes and to optical distortion in the fields of view), the grossly deviating values reach up to several thousand mas, and there are 82986 transits with negative values even (only one of the 10 values in each case). These deviating values are caused by occasionally big discontinuities of the calibration model at the CCD stitch boundaries. The majority of the cases concern SM CCDs, because their calibration is less precise and less reliable than for AF CCDs. For DR5, efforts will be taken to mitigate the problem. A complete solution will probably not be possible, however.
On the user side, there are several conceivable ways to avoid the problem, with increasing complexity: 1) Ignore all records where `sample_size_ac` deviates from the usual values by more than, say, 1 mas. — 2) If any problematic record is of especially high scientific interest, try using it by replacing the deviating `sample_size_ac` by the nominal value of 176.8 mas. Note that for AF1 twice this value is applicable, and for SM twice and four times this value, depending on `g_class`. — 3) Collect the `sample_size_ac` from neighbouring transits (same CCD, similar μ coordinate but with normal `sample_size_ac`, and within a few seconds in time) and use their median value. This latter method takes focal-length variations and the local optical distortion

into account.

SHEAR_TERM_AL : Shear terms AL (float[10] array, Angle[mas])

For each CCD transit the AL shear term c_w. See **?** for details.

SHEAR_TERM_AC : Shear terms AC (float[10] array, Angle[mas])

For each CCD transit the AC shear term v_z. See **?** for details.

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4 Astrophysical parameters

As indicated in the below subsection titles by the labels [TAP] and [DataLink], these products are served from the ESA Gaia Archive using two different protocols, the Table Access Protocol (TAP) for tabular products and the DataLink protocol for other products such as spectra, light curves, MCMC samples, etc. For more details please see Section **??**.

4.1 AP_CHEM [TAP]

This table contains the abundances of chemical elements of stars derived from mean RVS spectra by GSP-Spec as part of the Apsis processing chain.

See Section **??** of the online documentation for details on the GSP-Spec module.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source. For a detailed description see `gaia_source.source_id`.

FEM_GSPSPEC : Abundance of neutral iron [Fe/M] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Median abundance of neutral iron (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `fem_gspspec_nlines`. The neutral iron abundance [Fe/H] is obtained by [Fe/H]=[Fe/M]+[M/H].

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FEM_GSPSPEC_LOWER : 16th percentile of the abundance of neutral iron [Fe/M] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of neutral iron (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval. The neutral iron abundance [Fe/H] is obtained by [Fe/H]=[Fe/M]+[M/H].

FEM_GSPSPEC_UPPER : 84th percentile of the abundance of neutral iron [Fe/M] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of neutral iron (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval. The neutral iron abundance [Fe/H] is obtained by [Fe/H]=[Fe/M]+[M/H].

FEM_GSPSPEC_NLINES : Number of lines used for [Fe/M] abundance estimation (int)

Number of lines used to compute the [Fe/M] abundance. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

FEM_GSPSPEC_LINESCATTER : Uncertainty estimation of [Fe/M] abundance (float, Abundances[dex])

Standard deviation of the individual N lines abundance results.

The number of lines is given by `fem_gspspec_nlines`.

FEM_GSPSPEC_ULIM : Minimum value of [Fe/M] that can be measured. (float, Abundances[dex])

Upper limit for the [Fe/M] abundance. Minimum value of [Fe/M] that can be measured, averaged over all the estimations from the individual lines.

FEM_GSPSPEC_CALIB : Calibrated abundance of neutral iron [Fe/M] from GSP-Spec MatisseGauguin (float, Abundances[dex])

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Calibrated median abundance of neutral iron (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element

The number of lines is given in `fem_gspspec_nlines`. The neutral iron abundance [Fe/H] is obtained by [Fe/H]=[Fe/M]+[M/H].

FEM_GSPSPEC_CALIB_LOWER : Calibrated 16th percentile of the abundance of neutral iron [Fe/M] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated lower confidence level (16%) of the median abundance of neutral iron (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval. The neutral iron abundance [Fe/H] is obtained by [Fe/H]=[Fe/M]+[M/H].

FEM_GSPSPEC_CALIB_UPPER : Calibrated 84th percentile of the abundance of neutral iron [Fe/M] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated upper confidence level (84%) of the median abundance of neutral iron (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval. The neutral iron abundance [Fe/H] is obtained by [Fe/H]=[Fe/M]+[M/H].

SIFE_GSPSPEC : Abundance of silicon [Si/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Median abundance of silicon (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `sife_gspspec_nlines`.

SIFE_GSPSPEC_LOWER : 16th percentile of the abundance of silicon [Si/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of silicon (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

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SIFE_GSPSPEC_UPPER : 84th percentile of the abundance of silicon [Si/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of silicon (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

SIFE_GSPSPEC_NLINES : Number of lines used for [Si/Fe] abundance estimation (int)

Number of lines used to compute the [Si/Fe] abundance. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

SIFE_GSPSPEC_LINESCATTER : Uncertainty estimation of [Si/Fe] abundance (float, Abundances[dex])

Standard deviation of the individual N lines abundance results.

The number of lines is given by (`sife_gspspec_nlines`).

SIFE_GSPSPEC_ULIM : Minimum value of [Si/Fe] that can be measured. (float, Abundances[dex])

Upper limit for the [Si/Fe] abundance. Minimum value of [Si/Fe] that can be measured, averaged over all the estimations from the individual lines.

SIFE_GSPSPEC_CALIB : Calibrated abundance of silicon [Si/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated median abundance of silicon (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `sife_gspspec_nlines`.

SIFE_GSPSPEC_CALIB_LOWER : Calibrated 16th percentile of the abundance of silicon [Si/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated lower confidence level (16%) of the median abundance of silicon (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS

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spectra. Lower and upper levels include 68% confidence interval.

SIFE_GSPSPEC_CALIB_UPPER : Calibrated 84th percentile of the abundance of silicon [Si/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated upper confidence level (84%) of the median abundance of silicon (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

CAFE_GSPSPEC : Abundance of calcium [Ca/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Median abundance of calcium (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `cafe_gspspec_nlines`.

CAFE_GSPSPEC_LOWER : 16th percentile of the abundance of calcium [Ca/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of calcium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

CAFE_GSPSPEC_UPPER : 84th percentile of the abundance of calcium [Ca/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of calcium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

CAFE_GSPSPEC_NLINES : Number of lines used for [Ca/Fe] abundance estimation (int)

Number of lines used to compute the [Ca/Fe] abundance. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

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CAFE_GSPSPEC_LINESCATTER : Uncertainty estimation of [Ca/Fe] abundance (float, Abundances[dex])

Standard deviation of the individual N lines abundance results.

The number of lines is given by `cafe_gspspec_nlines`.

CAFE_GSPSPEC_ULIM : Minimum value of [Ca/Fe] that can be measured. (float, Abundances[dex])

Upper limit for the [Ca/Fe] abundance. Minimum value of [Ca/Fe] that can be measured, averaged over all the estimations from the individual lines.

CAFE_GSPSPEC_CALIB : Calibrated abundance of calcium [Ca/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated median abundance of calcium (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given by `cafe_gspspec_nlines`.

CAFE_GSPSPEC_CALIB_LOWER : Calibrated 16th percentile of the abundance of calcium [Ca/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated lower confidence level (16%) of the median abundance of calcium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

CAFE_GSPSPEC_CALIB_UPPER : Calibrated 84th percentile of the abundance of calcium [Ca/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated upper confidence level (84%) of the median abundance of calcium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

TIFE_GSPSPEC : Abundance of titanium [Ti/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

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Median abundance of titanium (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `tife_gspspec_nlines`.

TIFE_GSPSPEC_LOWER : 16th percentile of the abundance of titanium [Ti/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of titanium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

TIFE_GSPSPEC_UPPER : 84th percentile of the abundance of titanium [Ti/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of titanium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

TIFE_GSPSPEC_NLINES : Number of lines used for [Ti/Fe] abundance estimation (int)

Number of lines used to compute the [Ti/Fe] abundance. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

TIFE_GSPSPEC_LINESCATTER : Uncertainty estimation of [Ti/Fe] abundance (float, Abundances[dex])

Standard deviation of the individual N lines abundance results.

The number of lines is given by `tife_gspspec_nlines`.

TIFE_GSPSPEC_ULIM : Minimum value of [Ti/Fe] that can be measured. (float, Abundances[dex])

Upper limit for the [Ti/Fe] abundance. Minimum value of [Ti/Fe] that can be measured, averaged over all the estimations from the individual lines.

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TIFE_GSPSPEC_CALIB : Calibrated abundance of titanium [Ti/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated median abundance of titanium (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `tife_gspspec_nlines`.

TIFE_GSPSPEC_CALIB_LOWER : Calibrated 16th percentile of the abundance of titanium [Ti/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated lower confidence level (16%) of the median abundance of titanium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

TIFE_GSPSPEC_CALIB_UPPER : Calibrated 84th percentile of the abundance of titanium [Ti/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated upper confidence level (84%) of the median abundance of titanium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

MGFE_GSPSPEC : Abundance of magnesium [Mg/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Median abundance of magnesium (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `mgfe_gspspec_nlines`.

MGFE_GSPSPEC_LOWER : 16th percentile of the abundance of magnesium [Mg/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of magnesium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

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MGFE_GSPSPEC_UPPER : 84th percentile of the abundance of magnesium [Mg/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of magnesium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

MGFE_GSPSPEC_NLINES : Number of lines used for [Mg/Fe] abundance estimation (int)

Number of lines used to compute the [Mg/Fe] abundance. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

MGFE_GSPSPEC_LINESCATTER : Uncertainty estimation of [Mg/Fe] abundance (float, Abundances[dex])

Standard deviation of the individual N lines abundance results.

The number of lines is given by `mgfe_gspspec_nlines`.

MGFE_GSPSPEC_ULIM : Minimum value of [Mg/Fe] that can be measured. (float, Abundances[dex])

Upper limit for the [Mg/Fe] abundance. Minimum value of [Mg/Fe] that can be measured, averaged over all the estimations from the individual lines.

MGFE_GSPSPEC_CALIB : Calibrated abundance of magnesium [Mg/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated median abundance of magnesium (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `mgfe_gspspec_nlines`.

MGFE_GSPSPEC_CALIB_LOWER : Calibrated 16th percentile of the abundance of magnesium [Mg/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated lower confidence level (16%) of the median abundance of magnesium (assuming

source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

MGFE_GSPSPEC_CALIB_UPPER : Calibrated 84th percentile of the abundance of magnesium [Mg/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated upper confidence level (84%) of the median abundance of magnesium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

NDFE_GSPSPEC : Abundance of Neodymium [Nd/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Median abundance of neodymium (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `ndfe_gspspec_nlines`.

NDFE_GSPSPEC_LOWER : 16th percentile of the abundance of Neodymium [Nd/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of neodymium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

NDFE_GSPSPEC_UPPER : 84th percentile of the abundance of Neodymium [Nd/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of neodymium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

NDFE_GSPSPEC_NLINES : Number of lines used for [Nd/Fe] abundance estimation (int)

Number of lines used to compute the [Nd/Fe] abundance. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

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NDFE_GSPSPEC_LINESCATTER : Uncertainty estimation of [Nd/Fe] abundance (float, Abundances[dex])

Standard deviation of the individual N lines abundance results.

The number of lines is given by `ndfe_gspspec_nlines`.

NDFE_GSPSPEC_ULIM : Minimum value of [Nd/Fe] that can be measured. (float, Abundances[dex])

Upper limit for the [Nd/Fe] abundance. Minimum value of [Nd/Fe] that can be measured, averaged over all the estimations from the individual lines.

NDFE_GSPSPEC_CALIB : Calibrated abundance of Neodymium [Nd/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated median abundance of neodymium (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `ndfe_gspspec_nlines`.

NDFE_GSPSPEC_CALIB_LOWER : Calibrated 16th percentile of the abundance of Neodymium [Nd/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated lower confidence level (16%) of the median abundance of neodymium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

NDFE_GSPSPEC_CALIB_UPPER : Calibrated 84th percentile of the abundance of Neodymium [Nd/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated upper confidence level (84%) of the median abundance of neodymium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

FEIIM_GSPSPEC : Abundance of ionised iron [FeII/M] from GSP-Spec MatisseGauguin (float,

Abundances[dex])

Median abundance of ionised iron (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `feiim_gspspec_nlines`. The ionised iron abundance [FeII/H] is obtained by [FeII/H]=[FeII/M]+[M/H].

FEIIM_GSPSPEC_LOWER : 16th percentile of the abundance of ionised iron [FeII/M] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of ionised iron (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval. The ionised iron abundance [FeII/H] is obtained by [FeII/H]=[FeII/M]+[M/H].

FEIIM_GSPSPEC_UPPER : 84th percentile of the abundance of ionised iron [FeII/M] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of ionised iron (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval. The ionised iron abundance [FeII/H] is obtained by [FeII/H]=[FeII/M]+[M/H].

FEIIM_GSPSPEC_NLINES : Number of lines used for [FeII/M] abundance estimation (int)

Number of lines used to compute the [FeII/M] abundance. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

FEIIM_GSPSPEC_LINESCATTER : Uncertainty estimation of [FeII/M] abundance (float, Abundances[dex])

Standard deviation of the individual N lines abundance results.

The number of lines is given by `feiim_gspspec_nlines`.

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FEIIM_GSPSPEC_ULIM : Minimum value of [FeII/M] that can be measured. (float, Abundances[dex])

Upper limit for the ionised iron abundance. Minimum value of [FeII/M] that can be measured. The ionised iron abundance [FeII/H] is obtained by [FeII/H]=[FeII/M]+[M/H].

FEIIM_GSPSPEC_CALIB : Calibrated abundance of ionised iron [FeII/M] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated median abundance of ionised iron (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element. The number of lines is given in `feiim_gspspec_nlines`. The ionised iron abundance [FeII/H] is obtained by [FeII/H]=[FeII/M]+[M/H].

FEIIM_GSPSPEC_CALIB_LOWER : Calibrated 16th percentile of the abundance of ionised iron [FeII/M] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated lower confidence level (16%) of the median abundance of ionised iron (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval. The ionised iron abundance [FeII/H] is obtained by [FeII/H]=[FeII/M]+[M/H].

FEIIM_GSPSPEC_CALIB_UPPER : Calibrated 84th percentile of the abundance of ionised iron [FeII/M] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated upper confidence level (84%) of the median abundance of ionised iron (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval. The ionised iron abundance [FeII/H] is obtained by [FeII/H]=[FeII/M]+[M/H].

SFE_GSPSPEC : Abundance of sulphur [S/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Median abundance of sulphur (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `sfe_gspspec_nlines`.

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SFE_GSPSPEC_LOWER : 16th percentile of the abundance of sulphur [S/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of sulphur (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

SFE_GSPSPEC_UPPER : 84th percentile of the abundance of sulphur [S/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of sulphur (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

SFE_GSPSPEC_NLINES : Number of lines used for [S/Fe] abundance estimation (int)

Number of lines used to compute the [S/Fe] abundance. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

SFE_GSPSPEC_LINESCATTER : Uncertainty estimation of [S/Fe] abundance (float, Abundances[dex])

Standard deviation of the individual N lines abundance results.

The number of lines is given by `sfe_gspspec_nlines`.

SFE_GSPSPEC_ULIM : Minimum value of [S/Fe] that can be measured. (float, Abundances[dex])

Upper limit for the [S/Fe] abundance. Minimum value of [S/Fe] that can be measured, averaged over all the estimations from the individual lines.

SFE_GSPSPEC_CALIB : Calibrated abundance of sulphur [S/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated median abundance of sulphur (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric

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parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `sfe_gspspec_nlines`.

SFE_GSPSPEC_CALIB_LOWER : Calibrated 16th percentile of the abundance of sulphur [S/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated lower confidence level (16%) of the median abundance of sulphur (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

SFE_GSPSPEC_CALIB_UPPER : Calibrated 84th percentile of the abundance of sulphur [S/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated upper confidence level (84%) of the median abundance of sulphur (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

ZRFE_GSPSPEC : Abundance of zirconium [Zr/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Median abundance of zirconium (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `zrfe_gspspec_nlines`.

ZRFE_GSPSPEC_LOWER : 16th percentile of the abundance of zirconium [Zr/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of zirconium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

ZRFE_GSPSPEC_UPPER : 84th percentile of the abundance of zirconium [Zr/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of zirconium (assuming source is a sin-

gle star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

ZRFE_GSPSPEC_NLINES : Number of lines used for [Zr/Fe] abundance estimation (int)

Number of lines used to compute the [Zr/Fe] abundance. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

ZRFE_GSPSPEC_LINESCATTER : Uncertainty estimation of [Zr/Fe] abundance (float, Abundances[dex])

Standard deviation of the individual N lines abundance results.

The number of lines is given by `zrfe_gspspec_nlines`.

ZRFE_GSPSPEC_ULIM : Minimum value of [Zr/Fe] that can be measured. (float, Abundances[dex])

Upper limit for the [Zr/Fe] abundance. Minimum value of [Zr/Fe] that can be measured, averaged over all the estimations from the individual lines.

ZRFE_GSPSPEC_CALIB : Calibrated abundance of zirconium [Zr/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated median abundance of zirconium (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `zrfe_gspspec_nlines`.

ZRFE_GSPSPEC_CALIB_LOWER : Calibrated 16th percentile of the abundance of zirconium [Zr/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated lower confidence level (16%) of the median abundance of zirconium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

ZRFE_GSPSPEC_CALIB_UPPER : Calibrated 84th percentile of the abundance of zirconium [Zr/Fe]

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from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated upper confidence level (84%) of the median abundance of zirconium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

NFE_GSPSPEC : Abundance of nitrogen [N/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Median abundance of nitrogen (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `nfe_gspspec_nlines`.

NFE_GSPSPEC_LOWER : 16th percentile of the abundance of nitrogen [N/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of nitrogen (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

NFE_GSPSPEC_UPPER : 84th percentile of the abundance of nitrogen [N/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of nitrogen (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

NFE_GSPSPEC_NLINES : Number of lines used for [N/Fe] abundance estimation (int)

Number of lines used to compute the [N/Fe] abundance. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

NFE_GSPSPEC_LINESCATTER : Uncertainty estimation of [N/Fe] abundance (float, Abundances[dex])

Standard deviation of the individual N lines abundance results.

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The number of lines is given by `nfe_gspspec_nlines`.

NFE_GSPSPEC_ULIM : Minimum value of [N/Fe] that can be measured. (float, Abundances[dex])

Upper limit for the [N/Fe] abundance. Minimum value of [N/Fe] that can be measured, averaged over all the estimations from the individual lines.

NFE_GSPSPEC_CALIB : Calibrated abundance of nitrogen [N/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated median abundance of nitrogen (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `nfe_gspspec_nlines`.

NFE_GSPSPEC_CALIB_LOWER : Calibrated 16th percentile of the abundance of nitrogen [N/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated lower confidence level (16%) of the median abundance of nitrogen (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

NFE_GSPSPEC_CALIB_UPPER : Calibrated 84th percentile of the abundance of nitrogen [N/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated upper confidence level (84%) of the median abundance of nitrogen (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

CRFE_GSPSPEC : Abundance of chromium [Cr/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Median abundance of chromium (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `crfe_gspspec_nlines`.

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CRFE_GSPSPEC_LOWER : 16th percentile of the abundance of chromium [Cr/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of chromium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

CRFE_GSPSPEC_UPPER : 84th percentile of the abundance of chromium [Cr/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of chromium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

CRFE_GSPSPEC_NLINES : Number of lines used for [Cr/Fe] abundance estimation (int)

Number of lines used to compute the [Cr/Fe] abundance. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

CRFE_GSPSPEC_LINESCATTER : Uncertainty estimation of [Cr/Fe] abundance (float, Abundances[dex])

Standard deviation of the individual N lines abundance results.

The number of lines is given by `crfe_gspspec_nlines`.

CRFE_GSPSPEC_ULIM : Minimum value of [Cr/Fe] that can be measured. (float, Abundances[dex])

Upper limit for the [Cr/Fe] abundance. Minimum value of [Cr/Fe] that can be measured, averaged over all the estimations from the individual lines.

CRFE_GSPSPEC_CALIB : Calibrated abundance of chromium [Cr/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated median abundance of chromium (assuming source is a single star) from RVS spectra

and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `crfe_gspspec_nlines`.

CRFE_GSPSPEC_CALIB_LOWER : Calibrated 16th percentile of the abundance of chromium [Cr/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated lower confidence level (16%) of the median abundance of chromium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

CRFE_GSPSPEC_CALIB_UPPER : Calibrated 84th percentile of the abundance of chromium [Cr/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated upper confidence level (84%) of the median abundance of chromium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

CEFE_GSPSPEC : Abundance of cerium [Ce/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Median abundance of cerium (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `cefe_gspspec_nlines`.

CEFE_GSPSPEC_LOWER : 16th percentile of the abundance of cerium [Ce/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of cerium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

CEFE_GSPSPEC_UPPER : 84th percentile of the abundance of cerium [Ce/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Draft

Upper confidence level (84%) of the median abundance of cerium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

CEFE_GSPSPEC_NLINES : Number of lines used for [Ce/Fe] abundance estimation (int)

Number of lines used to compute the [Ce/Fe] abundance. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

CEFE_GSPSPEC_LINESCATTER : Uncertainty estimation of [Ce/Fe] abundance (float, Abundances[dex])

Standard deviation of the individual N lines abundance results.

The number of lines is given by `cefe_gspspec_nlines`.

CEFE_GSPSPEC_ULIM : Minimum value of [Ce/Fe] that can be measured. (float, Abundances[dex])

Upper limit for the [Ce/Fe] abundance. Minimum value of [Ce/Fe] that can be measured, averaged over all the estimations from the individual lines.

CEFE_GSPSPEC_CALIB : Calibrated abundance of cerium [Ce/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated median abundance of cerium (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `cefe_gspspec_nlines`.

CEFE_GSPSPEC_CALIB_LOWER : Calibrated 16th percentile of the abundance of cerium [Ce/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated lower confidence level (16%) of the median abundance of cerium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

Draft

CEFE_GSPSPEC_CALIB_UPPER : Calibrated 84th percentile of the abundance of cerium [Ce/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated upper confidence level (84%) of the median abundance of cerium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

NIFE_GSPSPEC : Abundance of nickel [Ni/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Median abundance of nickel (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `nife_gspspec_nlines`.

NIFE_GSPSPEC_LOWER : 16th percentile of the abundance of nickel [Ni/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of nickel (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

NIFE_GSPSPEC_UPPER : 84th percentile of the abundance of nickel [Ni/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of nickel (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

NIFE_GSPSPEC_NLINES : Number of lines used for [Ni/Fe] abundance estimation (int)

Number of lines used to compute the [Ni/Fe] abundance. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

NIFE_GSPSPEC_LINESCATTER : Uncertainty estimation of [Ni/Fe] abundance (float, Abundances[dex])

Draft

Standard deviation of the individual N lines abundance results.

The number of lines is given by `nife_gspspec_nlines`.

NIFE_GSPSPEC_ULIM : Minimum value of [Ni/Fe] that can be measured. (float, Abundances[dex])

Upper limit for the nickel abundance. Minimum value of [Ni/Fe] that can be measured, averaged over all the estimations from the individual lines.

NIFE_GSPSPEC_CALIB : Calibrated abundance of nickel [Ni/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated median abundance of nickel (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `nife_gspspec_nlines`.

NIFE_GSPSPEC_CALIB_LOWER : Calibrated 16th percentile of the abundance of nickel [Ni/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated lower confidence level (16%) of the median abundance of nickel (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

NIFE_GSPSPEC_CALIB_UPPER : Calibrated 84th percentile of the abundance of nickel [Ni/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated upper confidence level (84%) of the median abundance of nickel (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

NAFE_GSPSPEC : Abundance of Sodium [Na/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Median abundance of Sodium [Na/Fe](assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `nafe_gspspec_nlines`.

NAFE_GSPSPEC_LOWER : 16th percentile of the abundance of Sodium [Na/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of Sodium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

NAFE_GSPSPEC_UPPER : 84th percentile of the abundance of Sodium [Na/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of Sodium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

NAFE_GSPSPEC_NLINES : Number of lines used for [Na/Fe] abundance estimation (int)

Number of lines used to compute the [Na/Fe] abundance. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

NAFE_GSPSPEC_LINESCATTER : Uncertainty estimation of [Na/Fe] abundance (float, Abundances[dex])

Standard deviation of the individual N lines abundance results.

The number of lines is given by `nafe_gspspec_nlines`.

NAFE_GSPSPEC_ULIM : Minimum value of [Na/Fe] that can be measured. (float, Abundances[dex])

Upper limit for the Sodium abundance. Minimum value of [Na/Fe] that can be measured, averaged over all the estimations from the individual lines.

NAFE_GSPSPEC_CALIB : Calibrated abundance of Sodium [Na/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Draft

Calibrated median abundance of Sodium (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `nafe_gspspec_nlines`.

NAFE_GSPSPEC_CALIB_LOWER : Calibrated 16th percentile of the abundance of Sodium [Na/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated lower confidence level (16%) of the median abundance of Sodium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

NAFE_GSPSPEC_CALIB_UPPER : Calibrated 84th percentile of the abundance of Sodium [Na/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated upper confidence level (84%) of the median abundance of Sodium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

ALFE_GSPSPEC : Abundance of Aluminum [Al/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Median abundance of Aluminum (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `alfe_gspspec_nlines`.

ALFE_GSPSPEC_LOWER : 16th percentile of the abundance of Aluminum [Al/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of Aluminum (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

ALFE_GSPSPEC_UPPER : 84th percentile of the abundance of Aluminum [Al/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Draft

Upper confidence level (84%) of the median abundance of Aluminum (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

ALFE_GSPSPEC_NLINES : Number of lines used for [Al/Fe] abundance estimation (int)

Number of lines used to compute the [Al/Fe] abundance. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

ALFE_GSPSPEC_LINESCATTER : Uncertainty estimation of [Al/Fe] abundance (float, Abundances[dex])

Standard deviation of the individual N lines abundance results.

The number of lines is given by `alfe_gspspec_nlines`.

ALFE_GSPSPEC_ULIM : Minimum value of [Al/Fe] that can be measured. (float, Abundances[dex])

Upper limit for the Aluminum abundance. Minimum value of [Al/Fe] that can be measured, averaged over all the estimations from the individual lines.

ALFE_GSPSPEC_CALIB : Calibrated abundance of Aluminum [Al/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated median abundance of Aluminum (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `alfe_gspspec_nlines`.

ALFE_GSPSPEC_CALIB_LOWER : Calibrated 16th percentile of the abundance of Aluminum [Al/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated lower confidence level (16%) of the median abundance of Aluminum (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

Draft

ALFE_GSPSPEC_CALIB_UPPER : Calibrated 84th percentile of the abundance of Aluminum [Al/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated upper confidence level (84%) of the median abundance of Aluminum (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

PFE_GSPSPEC : Abundance of Phosphorus [P/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Median abundance of Phosphorus (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `pfe_gspspec_nlines`.

PFE_GSPSPEC_LOWER : 16th percentile of the abundance of Phosphorus [P/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of Phosphorus (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

PFE_GSPSPEC_UPPER : 84th percentile of the abundance of Phosphorus [P/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of Phosphorus (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

PFE_GSPSPEC_NLINES : Number of lines used for [P/Fe] abundance estimation (int)

Number of lines used to compute the [P/Fe] abundance. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

PFE_GSPSPEC_LINESCATTER : Uncertainty estimation of [P/Fe] abundance (float, Abundances[dex])

Draft

Standard deviation of the individual N lines abundance results.

The number of lines is given by `pfe_gspspec_nlines`.

PFE_GSPSPEC_ULIM : Minimum value of [P/Fe] that can be measured. (float, Abundances[dex])

Upper limit for the Phosphorus abundance. Minimum value of [P/Fe] that can be measured, averaged over all the estimations from the individual lines.

PFE_GSPSPEC_CALIB : Calibrated abundance of Phosphorus [P/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated median abundance of Phosphorus (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `pfe_gspspec_nlines`.

PFE_GSPSPEC_CALIB_LOWER : Calibrated 16th percentile of Phosphorus [P/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated lower confidence level (16%) of the median abundance of Phosphorus (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

PFE_GSPSPEC_CALIB_UPPER : Calibrated 84th percentile of Phosphorus [P/Fe] element [X3/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated upper confidence level (84%) of the median abundance of Phosphorus (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

FEM_GSPSPEC_EPOCH_MEAN : Mean abundance of neutral iron [Fe/M] from GSP-Spec MatisseGauguin using epoch RVS spectra (float, Abundances[dex])

Mean abundance of neutral iron (assuming source is a single star) from epoch RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm. The neutral iron abundance [Fe/H] is obtained

by [Fe/H]=[Fe/M]+[M/H].

FEM_GSPSPEC_EPOCH_STD : Standard deviation of the abundance of neutral iron [Fe/M] from GSP-Spec MatisseGauguin using epoch RVS spectra (float, Abundances[dex])

Standard deviation of the abundance of neutral iron (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. The neutral iron abundance [Fe/H] is obtained by [Fe/H]=[Fe/M]+[M/H].

SIFE_GSPSPEC_EPOCH_MEAN : Mean abundance of silicon [Si/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra (float, Abundances[dex])

Mean abundance of silicon (assuming source is a single star) from epoch RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm.

SIFE_GSPSPEC_EPOCH_STD : Standard deviation of the abundance of silicon [Si/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra (float, Abundances[dex])

Standard deviation of the abundance of silicon (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra.

CAFE_GSPSPEC_EPOCH_MEAN : Mean abundance of calcium [Ca/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra (float, Abundances[dex])

Mean abundance of calcium (assuming source is a single star) from epoch RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm.

CAFE_GSPSPEC_EPOCH_STD : Standard deviation of the abundance of calcium [Ca/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra (float, Abundances[dex])

Standard deviation of the abundance of calcium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra.

TIFE_GSPSPEC_EPOCH_MEAN : Mean abundance of titanium [Ti/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra (float, Abundances[dex])

Draft

Mean abundance of titanium (assuming source is a single star) from epoch RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm.

TIFE_GSPSPEC_EPOCH_STD : Standard deviation of the abundance of titanium [Ti/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra (float, Abundances[dex])

Standard deviation of the abundance of titanium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra.

MGFE_GSPSPEC_EPOCH_MEAN : Mean abundance of magnesium [Mg/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra (float, Abundances[dex])

Mean abundance of magnesium (assuming source is a single star) from epoch RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm.

MGFE_GSPSPEC_EPOCH_STD : Standard deviation of the abundance of magnesium [Mg/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra (float, Abundances[dex])

Standard deviation of the abundance of magnesium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra.

NDFE_GSPSPEC_EPOCH_MEAN : Mean abundance of neodymium [Nd/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra (float, Abundances[dex])

Mean abundance of Neodymium (assuming source is a single star) from epoch RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm.

NDFE_GSPSPEC_EPOCH_STD : Standard deviation of the abundance of Neodymium [Nd/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra (float, Abundances[dex])

Standard deviation of the abundance of Neodymium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra.

FEIIM_GSPSPEC_EPOCH_MEAN : Mean abundance of ionised iron [FeII/M] from GSP-Spec

MatisseGauguin using epoch RVS spectra (float, Abundances[dex])

Mean abundance of ionised iron (assuming source is a single star) from epoch RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm. The ionised iron abundance [FeII/H] is obtained by [FeI-I/H]=[FeII/M]+[M/H].

FEIIM_GSPSPEC_EPOCH_STD : Standard deviation of the abundance of ionised iron [FeII/M] from GSP-Spec MatisseGauguin using epoch RVS spectra (float, Abundances[dex])

Standard deviation of the abundance of ionised iron (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. The ionised iron abundance [FeII/H] is obtained by [FeII/H]=[FeII/M]+[M/H].

SFE_GSPSPEC_EPOCH_MEAN : Mean abundance of sulphur [S/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra (float, Abundances[dex])

Mean abundance of sulphur (assuming source is a single star) from epoch RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm.

SFE_GSPSPEC_EPOCH_STD : Standard deviation of the abundance of sulphur [S/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra (float, Abundances[dex])

Standard deviation of the abundance of sulphur (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra.

ZRFE_GSPSPEC_EPOCH_MEAN : Mean abundance of zirconium [Zr/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra (float, Abundances[dex])

Mean abundance of zirconium (assuming source is a single star) from epoch RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm.

ZRFE_GSPSPEC_EPOCH_STD : Standard deviation of the abundance of zirconium [Zr/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra (float, Abundances[dex])

Standard deviation of the abundance of zirconium (assuming source is a single star) inferred by

GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra.

NFE_GSPSPEC_EPOCH_MEAN : Mean abundance of nitrogen [N/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra (float, Abundances[dex])

Mean abundance of nitrogen (assuming source is a single star) from epoch RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm.

NFE_GSPSPEC_EPOCH_STD : Standard deviation of the abundance of nitrogen [N/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra (float, Abundances[dex])

Standard deviation of the abundance of nitrogen (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra.

CRFE_GSPSPEC_EPOCH_MEAN : Mean abundance of chromium [Cr/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra (float, Abundances[dex])

Mean abundance of chromium (assuming source is a single star) from epoch RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm.

CRFE_GSPSPEC_EPOCH_STD : Standard deviation of the abundance of chromium [Cr/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra (float, Abundances[dex])

Standard deviation of the abundance of chromium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra.

CEFE_GSPSPEC_EPOCH_MEAN : Mean abundance of cerium [Ce/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra (float, Abundances[dex])

Mean abundance of cerium (assuming source is a single star) from epoch RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm.

CEFE_GSPSPEC_EPOCH_STD : Standard deviation of the abundance of cerium [Ce/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra (float, Abundances[dex])

Draft

Standard deviation of the abundance of cerium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra.

NIFE_GSPSPEC_EPOCH_MEAN : Mean abundance of nickel [Ni/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra (float, Abundances[dex])

Mean abundance of nickel (assuming source is a single star) from epoch RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm.

NIFE_GSPSPEC_EPOCH_STD : Standard deviation of the abundance of nickel [Ni/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra (float, Abundances[dex])

Standard deviation of the abundance of nickel (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra.

NAFE_GSPSPEC_EPOCH_MEAN : Mean abundance of Sodium [Na/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra (float, Abundances[dex])

Mean abundance of Sodium (assuming source is a single star) from epoch RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm.

NAFE_GSPSPEC_EPOCH_STD : Standard deviation of the abundance of Sodium [Na/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra (float, Abundances[dex])

Standard deviation of the abundance of Sodium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra.

ALFE_GSPSPEC_EPOCH_MEAN : Mean abundance of Aluminum [Al/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra (float, Abundances[dex])

Mean abundance of Aluminum (assuming source is a single star) from epoch RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm.

ALFE_GSPSPEC_EPOCH_STD : Standard deviation of the abundance of Aluminum [Al/Fe] from

Draft

GSP-Spec MatisseGauguin using epoch RVS spectra (float, Abundances[dex])

Standard deviation of the abundance of Aluminum (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra.

PFE_GSPSPEC_EPOCH_MEAN : Mean abundance of Phosphorus [P/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra (float, Abundances[dex])

Mean abundance of Phosphorus (assuming source is a single star) from epoch RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm.

PFE_GSPSPEC_EPOCH_STD : Standard deviation of the abundance of Phosphorus [P/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra (float, Abundances[dex])

Standard deviation of the abundance of Phosphorus (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra.

CLFE_GSPSPEC : Abundance of Chlorine [Cl/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Median abundance of Chlorine (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `clfe_gspspec_nlines`.

CLFE_GSPSPEC_LOWER : 16th percentile of the abundance of Chlorine [Cl/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of Chlorine (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

CLFE_GSPSPEC_UPPER : 84th percentile of the abundance of Chlorine [Cl/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of Chlorine (assuming source is a sin-

gle star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

CLFE_GSPSPEC_NLINES : Number of lines used for [Cl/Fe] abundance estimation (int)

Number of lines used to compute the [Cl/Fe] abundance. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

CLFE_GSPSPEC_LINESCATTER : Uncertainty estimation of [Cl/Fe] abundance (float, Abundances[dex])

Standard deviation of the individual N lines abundance results.

The number of lines is given by `clfe_gspspec_nlines`.

CLFE_GSPSPEC_ULIM : Minimum value of [Cl/Fe] that can be measured. (float, Abundances[dex])

Upper limit for the Chlorine abundance. Minimum value of [Cl/Fe] that can be measured, averaged over all the estimations from the individual lines.

CLFE_GSPSPEC_CALIB : Calibrated abundance of Chlorine [Cl/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated median abundance of Chlorine (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `clfe_gspspec_nlines`.

CLFE_GSPSPEC_CALIB_LOWER : Calibrated 16th percentile of the abundance of Chlorine [Cl/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated lower confidence level (16%) of the median abundance of Chlorine (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

CLFE_GSPSPEC_CALIB_UPPER : Calibrated 84th percentile of the abundance of Chlorine [Cl/Fe]

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from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated upper confidence level (84%) of the median abundance of Chlorine (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

KFE_GSPSPEC : Abundance of Potassium [K/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Median abundance of Potassium (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `kfe_gspspec_nlines`.

KFE_GSPSPEC_LOWER : 16th percentile of the abundance of Potassium [K/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of Potassium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

KFE_GSPSPEC_UPPER : 84th percentile of the abundance of Potassium [K/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of Potassium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

KFE_GSPSPEC_NLINES : Number of lines used for [K/Fe] abundance estimation (int)

Number of lines used to compute the [K/Fe] abundance. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

KFE_GSPSPEC_LINESCATTER : Uncertainty estimation of [K/Fe] abundance (float, Abundances[dex])

Standard deviation of the individual N lines abundance results.

The number of lines is given by `kfe_gspspec_nlines`.

KFE_GSPSPEC_ULIM : Minimum value of [K/Fe] that can be measured. (float, Abundances[dex])

Upper limit for the Potassium abundance. Minimum value of [K/Fe] that can be measured, averaged over all the estimations from the individual lines.

KFE_GSPSPEC_CALIB : Calibrated abundance of Potassium [K/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated median abundance of Potassium (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `kfe_gspspec_nlines`.

KFE_GSPSPEC_CALIB_LOWER : Calibrated 16th percentile of the abundance of Potassium [K/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated lower confidence level (16%) of the median abundance of Potassium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

KFE_GSPSPEC_CALIB_UPPER : Calibrated 84th percentile of the abundance of Potassium [K/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated upper confidence level (84%) of the median abundance of Potassium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

SRFE_GSPSPEC : Abundance of Strontium [Sr/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Median abundance of Strontium (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `srfe_gspspec_nlines`.

SRFE_GSPSPEC_LOWER : 16th percentile of the abundance of Strontium [Sr/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of Strontium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

SRFE_GSPSPEC_UPPER : 84th percentile of the abundance of Strontium [Sr/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of Strontium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

SRFE_GSPSPEC_NLINES : Number of lines used for [Sr/Fe] abundance estimation (int)

Number of lines used to compute the [Sr/Fe] abundance. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

SRFE_GSPSPEC_LINESCATTER : Uncertainty estimation of [Sr/Fe] abundance (float, Abundances[dex])

Standard deviation of the individual N lines abundance results.

The number of lines is given by `srfe_gspspec_nlines`.

SRFE_GSPSPEC_ULIM : Minimum value of [Sr/Fe] that can be measured. (float, Abundances[dex])

Upper limit for the Strontium abundance. Minimum value of [Sr/Fe] that can be measured, averaged over all the estimations from the individual lines.

SRFE_GSPSPEC_CALIB : Calibrated abundance of Strontium [Sr/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated median abundance of Strontium (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmo-

spheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `srfe_gspspec_nlines`.

SRFE_GSPSPEC_CALIB_LOWER : Calibrated 16th percentile of the abundance of Strontium [Sr/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated lower confidence level (16%) of the median abundance of Strontium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

SRFE_GSPSPEC_CALIB_UPPER : Calibrated 84th percentile of the abundance of Strontium [Sr/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated upper confidence level (84%) of the median abundance of Strontium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

SMFE_GSPSPEC : Abundance of Samarium [Sm/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Median abundance of Samarium (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `smfe_gspspec_nlines`.

SMFE_GSPSPEC_LOWER : 16th percentile of the abundance of Samarium [Sm/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of Samarium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

SMFE_GSPSPEC_UPPER : 84th percentile of the abundance of Samarium [Sm/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of Samarium (assuming source is a sin-

gle star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

SMFE_GSPSPEC_NLINES : Number of lines used for [Sm/Fe] abundance estimation (int)

Number of lines used to compute the [Sm/Fe] abundance. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

SMFE_GSPSPEC_LINESCATTER : Uncertainty estimation of [Sm/Fe] abundance (float, Abundances[dex])

Standard deviation of the individual N lines abundance results.

The number of lines is given by `smfe_gspspec_nlines`.

SMFE_GSPSPEC_ULIM : Minimum value of [Sm/Fe] that can be measured. (float, Abundances[dex])

Upper limit for the Samarium abundance. Minimum value of [Sm/Fe] that can be measured, averaged over all the estimations from the individual lines.

SMFE_GSPSPEC_CALIB : Calibrated abundance of Samarium [Sm/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated median abundance of Samarium (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `smfe_gspspec_nlines`.

SMFE_GSPSPEC_CALIB_LOWER : Calibrated 16th percentile of the abundance of Samarium [Sm/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated lower confidence level (16%) of the median abundance of Samarium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

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SMFE_GSPSPEC_CALIB_UPPER : Calibrated 84th percentile of the abundance of Samarium [Sm/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated upper confidence level (84%) of the median abundance of Samarium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

HFFE_GSPSPEC : Abundance of Hafnium [Hf/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Median abundance of Hafnium (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `hffe_gspspec_nlines`.

HFFE_GSPSPEC_LOWER : 16th percentile of the abundance of Hafnium [Hf/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of Hafnium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

HFFE_GSPSPEC_UPPER : 84th percentile of the abundance of Hafnium [Hf/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of Hafnium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

HFFE_GSPSPEC_NLINES : Number of lines used for [Hf/Fe] abundance estimation (int)

Number of lines used to compute the [Hf/Fe] abundance. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

HFFE_GSPSPEC_LINESCATTER : Uncertainty estimation of [Hf/Fe] abundance (float, Abundances[dex])

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Standard deviation of the individual N lines abundance results.

The number of lines is given by `hffe_gspspec_nlines`.

HFFE_GSPSPEC_ULIM : Minimum value of [Hf/Fe] that can be measured. (float, Abundances[dex])

Upper limit for the Hafnium abundance. Minimum value of [Hf/Fe] that can be measured, averaged over all the estimations from the individual lines.

HFFE_GSPSPEC_CALIB : Calibrated abundance of Hafnium [Hf/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated median abundance of Hafnium (assuming source is a single star) from RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element.

The number of lines is given in `hffe_gspspec_nlines`.

HFFE_GSPSPEC_CALIB_LOWER : Calibrated 16th percentile of the abundance of Hafnium [Hf/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated lower confidence level (16%) of the median abundance of Hafnium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

HFFE_GSPSPEC_CALIB_UPPER : Calibrated 84th percentile of the abundance of Hafnium [Hf/Fe] from GSP-Spec MatisseGauguin (float, Abundances[dex])

Calibrated upper confidence level (84%) of the median abundance of Hafnium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using RVS spectra. Lower and upper levels include 68% confidence interval.

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4.2 AP_CLASS [TAP]

This table contains the classification probabilities and parameters of sources derived by DSC, OA, and UCA from the Apsis processing chain.

See Section **??**, Section **??**, and Section **??** of the online documentation for details on the DSC, OA, and UCA modules.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

CLASSPROB_DSC_COMBMOD_QUASAR : Probability from DSC-Combmod of being a quasar using BP/RP spectra, photometry, and astrometry (float)

Probability that the object is of the named class. This is the overall probability for this class, computed by combining the class probabilities from DSC-Specmod (which classifies objects using BP/RP spectra) and DSC-Allosmod (which classifies objects using several astrometric and photometric features). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSPROB_DSC_COMBMOD_GALAXY : Probability from DSC-Combmod of being a galaxy using BP/RP spectra, photometry, and astrometry (float)

Probability that the object is of the named class. This is the overall probability for this class, computed by combining the class probabilities from DSC-Specmod (which classifies objects using BP/RP spectra) and DSC-Allosmod (which classifies objects using several astrometric and

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photometric features). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSPROB_DSC_COMBMOD_STAR : Probability from DSC-Combmod of being a star using BP/RP spectra, photometry, and astrometry (float)

Probability that the object is of the named class. This is the overall probability for this class, computed by combining the class probabilities from DSC-Specmod (which classifies objects using BP/RP spectra) and DSC-Allosmod (which classifies objects using several astrometric and photometric features). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSPROB_DSC_SPECMOD_QUASAR : Probability from DSC-Specmod of being a quasar using BP/RP spectra (float)

Probability that the object is of the named class. This is the probability from a classifier that uses the BP/RP spectrum (module DSC-Specmod). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSPROB_DSC_SPECMOD_GALAXY : Probability from DSC-Specmod of being a galaxy using BP/RP spectra (float)

Probability that the object is of the named class. This is the probability from a classifier that uses the BP/RP spectrum (module DSC-Specmod). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSPROB_DSC_SPECMOD_STAR : Probability from DSC-Specmod of being a star using BP/RP spectra (float)

Probability that the object is of the named class. This is the probability from a classifier that uses the BP/RP spectrum (module DSC-Specmod). It is important to realise that the DSC classes are

defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSPROB_DSC_ALLOSMOD_QUASAR : Probability from DSC-Allosmod of being a quasar using photometry and astrometry (float)

Probability that the object is of the named class. This is the probability from a classifier that uses various astrometric and photometric features (module DSC-Allosmod). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSPROB_DSC_ALLOSMOD_GALAXY : Probability from DSC-Allosmod of being a galaxy using photometry and astrometry (float)

Probability that the object is of the named class. This is the probability from a classifier that uses various astrometric and photometric features (module DSC-Allosmod). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSPROB_DSC_ALLOSMOD_STAR : Probability from DSC-Allosmod of being a star using photometry and astrometry (float)

Probability that the object is of the named class. This is the probability from a classifier that uses various astrometric and photometric features (module DSC-Allosmod). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSPROB_DSC_COMBMODALPHA_QUASAR : Probability from DSC-CombmodAlpha of being a quasar using BP/RP spectra, photometry, and astrometry (float)

Probability that the object is of the named class. This is the overall probability for this class, computed using a parametric combination of the class probabilities from DSC-Specmod (which classifies objects using BP/RP spectra) and DSC-Allosmod (which classifies objects using several astrometric and photometric features). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class

than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSPROB_DSC_COMBMODALPHA_GALAXY : Probability from DSC-CombmodAlpha of being a galaxy using BP/RP spectra, photometry, and astrometry (float)

Probability that the object is of the named class. This is the overall probability for this class, computed using a parametric combination of the class probabilities from DSC-Specmod (which classifies objects using BP/RP spectra) and DSC-Allosmod (which classifies objects using several astrometric and photometric features). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSPROB_DSC_COMBMODALPHA_STAR : Probability from DSC-CombmodAlpha of being a star using BP/RP spectra, photometry, and astrometry (float)

Probability that the object is of the named class. This is the overall probability for this class, computed using a parametric combination of the class probabilities from DSC-Specmod (which classifies objects using BP/RP spectra) and DSC-Allosmod (which classifies objects using several astrometric and photometric features). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSPROB_DSC_SPECMODSUPP_QUASAR : Probability from DSC-SpecmodSupp of being a quasar using BP/RP spectra, photometry, and astrometry (float)

Probability that the object is of the named class. This is the probability from a classifier that jointly uses the BP/RP spectrum and various astrometric and photometric features (module DSC-SpecmodSupp). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSPROB_DSC_SPECMODSUPP_GALAXY : Probability from DSC-SpecmodSupp of being a galaxy using BP/RP spectra, photometry, and astrometry (float)

Probability that the object is of the named class. This is the probability from a classifier that

jointly uses the BP/RP spectrum and various astrometric and photometric features (module DSC-SpecmodSupp). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSPROB_DSC_SPECMODSUPP_STAR : Probability from DSC-SpecmodSupp of being a star using BP/RP spectra, photometry, and astrometry (float)

Probability that the object is of the named class. This is the probability from a classifier that jointly uses the BP/RP spectrum and various astrometric and photometric features (module DSC-SpecmodSupp). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSPROB_DSC_COMBMODSUPP_QUASAR : Probability from DSC-CombmodSupp of being a quasar using BP/RP spectra, photometry, and astrometry (float)

Probability that the object is of the named class. This is the overall probability for this class, using the class probabilities from DSC-SpecmodSupp (which classifies objects jointly using BP/RP spectra and several astrometric and photometric features) when available, otherwise uses DSC-Specmod (which classifies objects using BP/RP spectra). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSPROB_DSC_COMBMODSUPP_GALAXY : Probability from DSC-CombmodSupp of being a galaxy using BP/RP spectra, photometry, and astrometry (float)

Probability that the object is of the named class. This is the overall probability for this class, using the class probabilities from DSC-SpecmodSupp (which classifies objects jointly using BP/RP spectra and several astrometric and photometric features) when available, otherwise uses DSC-Specmod (which classifies objects using BP/RP spectra). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSPROB_DSC_COMBMODSUPP_STAR : Probability from DSC-CombmodSupp of being a star

using BP/RP spectra, photometry, and astrometry (float)

Probability that the object is of the named class. This is the overall probability for this class, using the class probabilities from DSC-SpecmodSupp (which classifies objects jointly using BP/RP spectra and several astrometric and photometric features) when available, otherwise uses DSC-Specmod (which classifies objects using BP/RP spectra). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSLABEL_DSC_JOINT : Class label from DSC based on the probabilities of both the Specmod and Allosmod classifiers (string)

Class label. The definition of the joint class label is based on the posterior probabilities of the DSC classifiers trained on the BP/RP spectra (module DSC-Specmod) and on the collection of features from astrometry and photometry (module DSC-Allosmod). The joint class label identifies a source as a quasar or a galaxy if the two classifiers predict a maximum probability higher than 0.50 for the same target class. Otherwise, the joint class label is declared as unclassified.

FLAGS_DSC : Flags indicating the processing status and scientific quality from DSC (string)

Processing flag array. The field is a collection of numerical indicators referring to the status of DSC processing.

- The 1st digit is a default value.
- The 2nd digit indicates if a source is an outlier candidate. An outlier is declared based on the probabilities of the combined DSC classifiers. The value is equal to 0 for non-outliers when the combined classifiers separately give a maximum probability higher than 0.999, otherwise value is equal to 1 for outliers. Note that the predicted classes of the combined classifiers are not required to be the same in this analysis. An outlier is also declared when the number of observations contributing to the BP and RP photometry is below a defined threshold (5).
- The 4th digit indicates if a source is an outlier candidate based on the probabilities of the Combmod module.
- The 5th digit indicates if a source is an outlier candidate based on the probabilities of the Combmod-alpha module.
- The 7th digit indicates if a source is an outlier candidate based on the probabilities of the CombmodSupp module.

- The 9th digit indicates the status of data processing.

Values of the 4th, 5th and 7th digits are integers from 0 to 3, based on the maximum predicted probability. Value is equal to 3 when the maximum probability is lower than 0.90, equal to 2 when the maximum probability is within 0.90 and 0.95, equal to 1 when the maximum probability is within 0.95 and 0.999, and equal to 0 if the maximum probability is higher than 0.999.
Values of the 9th digit are integers from 1 to 3. Value is equal to 1 when the Allosmod and the Specmod-Supp probabilities are N/A due to missing astrometric and photometric features. Value is equal to 2 when the required inputs are available to the classifiers. Value is equal to 3 when the astrometric and photometric features, the BP/RP spectra and the infrared photometry are available to the classifiers.

CLASSLABEL_OA : Class assigned by OA in the neuron that represents the source (string)

Class label of the neuron that represents the source, as assigned by the OA module in Apsis. If the source was not considered as an outlier according to DSC or UCA classifiers or if it was discarded by OA module, then this field will be set to unknown. See Section **??** for further details.

NEURON_OA_ID : Identifier of the OA SOM map neuron that represents the source (long)

A unique identifier for the neuron that represents the source in the SOM map produced by the OA module. If the source was not considered as an outlier according to DSC or UCA classifications or if it was discarded by OA module, then this field will be null. See Section **??** for further details.

NEURON_OA_DIST : Distance between the source XP spectra and the OA neuron XP prototype that represents the source (float)

Squared Euclidean distance between the source XP spectra (preprocessed) and the neuron XP prototype (`xp_spectrum_prototype_flux` in table `oa_neuron_xp_spectra`) that represents such a source.If the source was not considered as an outlier according to DSC or UCA classifiers or if it was discarded by the OA module, then this field will be null. See Section **??** for further details.

NEURON_OA_DIST_PERCENTILE_RANK : Percentile rank according to the distance distribution of the OA neuron that represents the source (int)

Percentile rank according to the squared Euclidean distance distribution of the OA neuron that represents the source. If the source was not considered as an outlier according to DSC or UCA

classifiers or if it was discarded by OA module, then this field will be null. See Section **??** for further details.

FLAGS_OA : Flags indicating the processing status and scientific quality from OA (string)

Processing flags related to the quality of the classification and source processing performed by the OA module, which is encoded as follows:

$$ABCD$$

where:

Code	Description	Value range	
A	Number of gaps with negative fluxes in BP spectrum	[0, 2]	*
B	Number of gaps with negative fluxes in RP spectrum	[0, 2]	*
C	Source classification quality category	[0, 5]	
D	Neuron quality category	[0, 6]	

If the source was not considered as an outlier according to DSC or UCA classifiers or if it was discarded by OA module, then this field will be null.

(*) *A* and *B* flags taking value 2 means two or more gaps taking negative flux values.

CLASSLABEL_UCA : Class assigned by UCA in the neuron that represents the source (string)

Class label of the neuron that represents the source, as assigned by the UCA module in Apsis. There will be four main label types: star, galaxy, quasar and outlier. The last one is assigned when the source cannot be classified as any of the other types. This value is null if the source was not processed by UCA. See Section **??** for further details.

SOM_UCA_ID : Identifier of the UCA SOM map where the source belongs to (long)

Identifier of the UCA SOM map to which the source belongs. If the source is either not processed by UCA or classified as an outlier, then this field will be set to null. See table `uca_neuron_information` and Section **??** for further details.

NEURON_UCA_ID : Identifier of the UCA SOM map neuron that represents the source (long)

Identifier of the UCA SOM map neuron that represents the source. If the source is either not processed by UCA or classified as outlier, then this field will be set to null. See Section **??** for further details and table `uca_neuron_information` for more detailed information about the neurons.

NEURON_UCA_DIST : Distance between the source XP spectra and the UCA neuron XP prototype that represents the source (float)

Distance between the source XP spectra and the UCA neuron XP prototype that represents the source. If the source is either not processed by UCA or classified as outlier, then this field will be set to null. See Section **??** for further details, and tables `uca_neuron_information` and `uca_neuron_xp_spectra` for more detailed information about the neurons and their prototypes, respectively.

FLAGS_UCA : Flags indicating the processing status from UCA (string)

Processing flags related to the source processing performed by the UCA module, which is encoded as follows:

$$ABC$$

where:

Code	Description	Value range	
A	Number of gaps with negative fluxes in BP spectrum	[0, 2]	*
B	Number of gaps with negative fluxes in RP spectrum	[0, 2]	*
C	Classification quality flag	[0, 1]	**

If the source is either not processed by UCA or classified as outlier, then this field will be set to null.

(*) *A* and *B* flags taking value 2 means two or more gaps taking negative flux values. (**) See Section **??** for more details about *C*.

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4.3 AP_CLASS_IR [TAP]

This table contains the classification probabilities and parameters of sources derived by DSC from the Apsis processing chain by combining Gaia data with infrared photometry.

See Section **??** of the online documentation for details on the DSC module.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

CLASSPROB_DSC_COMBMOD_QUASAR : Probability from DSC-Combmod of being a quasar using BP/RP spectra, photometry, astrometry, supplemented with infrared photometry (float)

Probability that the object is of the named class. This is the overall probability for this class, computed by combining the class probabilities from DSC-Specmod (which classifies objects using BP/RP spectra) and DSC-Allosmod (which classifies objects using several astrometric and photometric features supplemented with infrared photometry). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSPROB_DSC_COMBMOD_GALAXY : Probability from DSC-Combmod of being a galaxy using BP/RP spectra, photometry, and astrometry, supplemented with infrared photometry (float)

Probability that the object is of the named class. This is the overall probability for this class, computed by combining the class probabilities from DSC-Specmod (which classifies objects us-

ing BP/RP spectra) and DSC-Allosmod (which classifies objects using several astrometric and photometric features supplemented with infrared photometry). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSPROB_DSC_COMBMOD_STAR : Probability from DSC-Combmod of being a star using BP/RP spectra, photometry, and astrometry, supplemented with infrared photometry (float)

Probability that the object is of the named class. This is the overall probability for this class, computed by combining the class probabilities from DSC-Specmod (which classifies objects using BP/RP spectra) and DSC-Allosmod (which classifies objects using several astrometric and photometric features supplemented with infrared photometry). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSPROB_DSC_SPECMOD_QUASAR : Probability from DSC-Specmod of being a quasar using BP/RP spectra (float)

Probability that the object is of the named class. This is the probability from a classifier that uses the BP/RP spectrum (module DSC-Specmod). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSPROB_DSC_SPECMOD_GALAXY : Probability from DSC-Specmod of being a galaxy using BP/RP spectra (float)

Probability that the object is of the named class. This is the probability from a classifier that uses the BP/RP spectrum (module DSC-Specmod). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSPROB_DSC_SPECMOD_STAR : Probability from DSC-Specmod of being a star using BP/RP spectra (float)

Probability that the object is of the named class. This is the probability from a classifier that uses

the BP/RP spectrum (module DSC-Specmod). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSPROB_DSC_ALLOSMOD_QUASAR : Probability from DSC-Allosmod of being a quasar using photometry, astrometry, supplemented with infrared photometry (float)

Probability that the object is of the named class. This is the probability from a classifier that uses various astrometric and photometric features supplemented with infrared photometry (module DSC-Allosmod). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSPROB_DSC_ALLOSMOD_GALAXY : Probability from DSC-Allosmod of being a galaxy using photometry, astrometry, supplemented with infrared photometry (float)

Probability that the object is of the named class. This is the probability from a classifier that uses various astrometric and photometric features supplemented with infrared photometry (module DSC-Allosmod). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSPROB_DSC_ALLOSMOD_STAR : Probability from DSC-Allosmod of being a star using photometry, astrometry, supplemented with infrared photometry (float)

Probability that the object is of the named class. This is the probability from a classifier that uses various astrometric and photometric features supplemented with infrared photometry (module DSC-Allosmod). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSPROB_DSC_COMBMODALPHA_QUASAR : Probability from DSC-CombmodAlpha of being a quasar using BP/RP spectra, photometry, astrometry, supplemented with infrared photometry (float)

Probability that the object is of the named class. This is the overall probability for this class, computed using a parametric combination of the class probabilities from DSC-Specmod (which classifies objects using BP/RP spectra) and DSC-Allosmod (which classifies objects using several astrometric and photometric features supplemented with infrared photometry). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSPROB_DSC_COMBMODALPHA_GALAXY : Probability from DSC-CombmodAlpha of being a galaxy using BP/RP spectra, photometry, astrometry, supplemented with infrared photometry (float)

Probability that the object is of the named class. This is the overall probability for this class, computed using a parametric combination of the class probabilities from DSC-Specmod (which classifies objects using BP/RP spectra) and DSC-Allosmod (which classifies objects using several astrometric and photometric features supplemented with infrared photometry). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSPROB_DSC_COMBMODALPHA_STAR : Probability from DSC-CombmodAlpha of being a star using BP/RP spectra, photometry, astrometry, supplemented with infrared photometry (float)

Probability that the object is of the named class. This is the overall probability for this class, computed using a parametric combination of the class probabilities from DSC-Specmod (which classifies objects using BP/RP spectra) and DSC-Allosmod (which classifies objects using several astrometric and photometric features supplemented with infrared photometry). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSPROB_DSC_SPECMODSUPP_QUASAR : Probability from DSC-SpecmodSupp of being a quasar using BP/RP spectra, photometry, astrometry, supplemented with infrared photometry (float)

Probability that the object is of the named class. This is the probability from a classifier that jointly uses the BP/RP spectrum and various astrometric and photometric features supplemented with infrared photometry (module DSC-SpecmodSupp). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes

the global class prior, given in the documentation.

CLASSPROB_DSC_SPECMODSUPP_GALAXY : Probability from DSC-SpecmodSupp of being a galaxy using BP/RP spectra, photometry, astrometry, supplemented with infrared photometry (float)

Probability that the object is of the named class. This is the probability from a classifier that jointly uses the BP/RP spectrum and various astrometric and photometric features supplemented with infrared photometry (module DSC-SpecmodSupp). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSPROB_DSC_SPECMODSUPP_STAR : Probability from DSC-SpecmodSupp of being a star using BP/RP spectra, photometry, astrometry, supplemented with infrared photometry (float)

Probability that the object is of the named class. This is the probability from a classifier that jointly uses the BP/RP spectrum and various astrometric and photometric features supplemented with infrared photometry (module DSC-SpecmodSupp). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSPROB_DSC_COMBMODSUPP_QUASAR : Probability from DSC-CombmodSupp of being a quasar using BP/RP spectra, photometry, astrometry, supplemented with infrared photometry (float)

Probability that the object is of the named class. This is the overall probability for this class, using the class probabilities from DSC-SpecmodSupp (which classifies objects jointly using BP/RP spectra and several astrometric and photometric features supplemented with infrared photometry) when available, otherwise uses DSC-Specmod (which classifies objects using BP/RP spectra). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSPROB_DSC_COMBMODSUPP_GALAXY : Probability from DSC-CombmodSupp of being a galaxy using BP/RP spectra, photometry, astrometry, supplemented with infrared photometry (float)

Probability that the object is of the named class. This is the overall probability for this class, using the class probabilities from DSC-SpecmodSupp (which classifies objects jointly using BP/RP spectra and several astrometric and photometric features supplemented with infrared photometry) when available, otherwise uses DSC-Specmod (which classifies objects using BP/RP spectra). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSPROB_DSC_COMBMODSUPP_STAR : Probability from DSC-CombmodSupp of being a star using BP/RP spectra, photometry, astrometry, supplemented with infrared photometry (float)

Probability that the object is of the named class. This is the overall probability for this class, using the class probabilities from DSC-SpecmodSupp (which classifies objects jointly using BP/RP spectra and several astrometric and photometric features supplemented with infrared photometry) when available, otherwise uses DSC-Specmod (which classifies objects using BP/RP spectra). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSLABEL_DSC_JOINT : Class label from DSC based on the probabilities of both the Specmod and Allosmod classifiers (string)

Class label. The definition of the joint class label is based on the posterior probabilities of the DSC classifiers trained on the BP/RP spectra (module DSC-Specmod) and on the collection of features from astrometry and photometry (module DSC-Allosmod). The joint class label identifies a source as a quasar or a galaxy if the two classifiers predict a maximum probability higher than 0.50 for the same target class. Otherwise, the joint class label is declared as unclassified.

FLAGS_DSC : Flags indicating the processing status and scientific quality from DSC (string)

Processing flag array. The field is a collection of numerical indicators referring to the status of DSC processing.

- The 1st digit is a default value.
- The 2nd digit indicates if a source is an outlier candidate. An outlier is declared based on the probabilities of the combined DSC classifiers. The value is equal to 0 for non-outliers when the combined classifiers separately give a maximum probability higher than 0.999, otherwise value is equal to 1 for outliers. Note that the predicted classes of the combined classifiers are not required to be the same in this analysis.

An outlier is also declared when the number of observations contributing to the BP and RP photometry is below a defined threshold (5).

- The 4th digit indicates if a source is an outlier candidate based on the probabilities of the Combmod module.
- The 5th digit indicates if a source is an outlier candidate based on the probabilities of the Combmod-alpha module.
- The 7th digit indicates if a source is an outlier candidate based on the probabilities of the CombmodSupp module.
- The 9th digit indicates the status of data processing.

Values of the 4th, 5th and 7th digits are integers from 0 to 3, based on the maximum predicted probability. Value is equal to 3 when the maximum probability is lower than 0.90, equal to 2 when the maximum probability is within 0.90 and 0.95, equal to 1 when the maximum probability is within 0.95 and 0.999, and equal to 0 if the maximum probability is higher than 0.999.
Values of the 9th digit are integers from 1 to 3. Value is equal to 1 when the Allosmod and the Specmod-Supp probabilities are N/A due to missing astrometric and photometric features. Value is equal to 2 when the required inputs are available to the classifiers. Value is equal to 3 when the astrometric and photometric features, the BP/RP spectra and the infrared photometry are available to the classifiers.

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4.4 AP_EMISSION [TAP]

This table contains the astrophysical properties of emission-line stars, active stars, and pre-main sequence mass-accreting stars, as derived by the ESP-CS and ESP-ELS modules from the Apsis processing chain. These properties comprise H-alpha emission and Ca triplet chromospheric emission, classification of emission-line stars, and astrophysical parameters of pre-main sequence mass accreting stars.

See Section **??** and Section **??** of the online documentation for details on the ESP-CS and ESP-ELS modules.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

CLASSLABEL_ESPELS : Adopted ELS class label from ESP-ELS (string)

The emission-line star (ELS) class label is based on the analysis of the BP/RP spectra. Two random forest algorithms were used to classify the ELS targets. The first classifier identifies Wolf-Rayet stars and planetary nebulæ, and was trained on a subset of Gaia spectra representative of each class. The second classifier identifies Be, Herbig Ae/Be, T Tauri, and active M dwarf stars. It is applied only if the source was not identified as WC, WN, or PN, and if significant Hα emission was detected. This classifier was trained on Gaia BP/RP spectra obtained for known members of these classes, together with their corresponding astrophysical parameters. The astrophysical parameters adopted during both training and processing are those obtained by GSP-Phot (best library value, taken before any post-processing filter or calibration was applied). In both classifiers, the class label corresponds to the one having the highest probability and can take one of the following values:

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- **beStar** (Be star)
- **HerbigStar** (Herbig Ae/Be star)
- **PlanetaryNebula** (Planetary Nebula)
- **RedDwarfEmStar** (active M dwarf star)
- **TTauri** (T Tauri star)
- **wC** (Wolf-Rayet star of type WC)
- **wN** (Wolf-Rayet star of type WN)

When no emission was detected or when the spectrum could not be classified, the field is left empty. ESP-ELS was applied only on targets brighter than $G = 18.25$ (i.e. within the first three data chunks). Further details on the module are provided in Section **??** and Section **??**.

FLAGS_ESPELS_CLASSLABEL : Quality flag of the adopted ELS class label from ESP-ELS (string)

The quality flag provided with the ELS class label indicates better quality for lower flag values. The quality assessment is based on the difference (Δp) between the two highest class probabilities. The flag therefore takes the following values:

$\Delta p \geq 0.8$: 0

$\Delta p \geq 0.6$: 1

$\Delta p \geq 0.4$: 2

$\Delta p \geq 0.2$: 3

$\Delta p < 0.2$: 4

This assessment is performed on the set of probabilities provided by the classifier (RFC4 or RFC5) that produced the class label. ESP-ELS was applied only to targets brighter than $G = 18.25$ (i.e. within the first three data chunks). Further details on the module are given in Section **??** and Section **??**.

CLASSLABEL_ESPELS_MODE : Processing mode adopted by ESP-ELS to derive the class label (string)

Mode adopted by ESP-ELS to derive the ELS class probabilities. It takes the value '1' when the label is assigned directly by classifier RFC4 (i.e. the source is a candidate WC, WN, or planetary nebula), or '2' when the classification chain RFC3 + RFC4 + RFC5 is used (i.e. the source is a candidate Be star, Herbig Ae/Be, T Tauri, or active M dwarf star). The two modes provide independent sets of class probabilities:

- Mode '1':

  `classprob_espels_wcstar`

  `classprob_espels_wnstar`

  `classprob_espels_pne`

- Mode '2':

  `classprob_espels_bestar`

  `classprob_espels_ttauristar`

  `classprob_espels_herbigstar`

  `classprob_espels_dmestar`

In mode '1', the complement of the sum of the published probabilities corresponds to the probability of the source belonging to any class other than WC, WN, or PN. Further details on the module are provided in Section **??**.

CLASSPROB_ESPELS_WCSTAR : Probability from ESP-ELS of being a Wolf-Rayet star of type WC (float)

The probability of being a Wolf-Rayet star of type WC is derived on the BP/RP spectra from processing mode '1'. ESP-ELS was only applied on targets brighter than $G = 18.25$. Further details on the module are provided in Section **??**.

CLASSPROB_ESPELS_WNSTAR : Probability from ESP-ELS of being a Wolf-Rayet star of type WN (float)

The probability of being a Wolf-Rayet star of type WN is derived on the BP/RP spectra from processing mode '1'. ESP-ELS was only applied on targets brighter than $G = 18.25$. Further details on the module are provided in Section **??**.

CLASSPROB_ESPELS_BESTAR : Probability from ESP-ELS of being a Be Star (float)

The probability of being a Be star is derived on the BP/RP spectra from processing mode '2'. If the star was labelled 'wC', 'wN', or 'PlanetaryNebula', the field remains empty. ESP-ELS was only applied on targets brighter than $G = 18.25$. Further details on the module are provided in Section **??**.

CLASSPROB_ESPELS_TTAURISTAR : Probability from ESP-ELS of being a T Tauri Star (float)

The probability of being a T Tauri star is derived on the BP/RP spectra from processing mode '2'. If the star was labelled 'wC', 'wN', or 'PlanetaryNebula', the field remains empty. ESP-ELS was only applied on targets brighter than $G = 18.25$. Further details on the module are provided in Section **??**.

CLASSPROB_ESPELS_HERBIGSTAR : Probability from ESP-ELS of being a Herbig Ae/Be Star (float)

The probability of being a Herbig Ae/Be star is derived on the BP/RP spectra from processing mode '2'. If the star was labelled 'wC', 'wN', or 'PlanetaryNebula', the field remains empty. ESP-ELS was only applied on targets brighter than $G = 18.25$. Further details on the module are provided in Section **??**.

CLASSPROB_ESPELS_DMESTAR : Probability from ESP-ELS of being an active M dwarf Star (float)

The probability of being an active M dwarf star is derived on the BP/RP spectra from processing mode '2'. If the star was labelled 'wC', 'wN', or 'PlanetaryNebula', the field remains empty. ESP-ELS was only applied on targets brighter than $G = 18.25$. Further details on the module are provided in Section **??**.

CLASSPROB_ESPELS_PNE : Probability from ESP-ELS of being a planetary nebula (float)

The probability of being a planetary nebula is derived on the BP/RP spectra from processing mode '1'. ESP-ELS was only applied on targets brighter than $G = 18.25$. Further details on the module are provided in Section **??**.

ACTIVITYINDEX : Chromospheric activity index from ESP-CS, measured on the calcium triplet using RVS spectra (float, Length & Distance[nm])

The activity index from the Apsis module ESP-CS is computed by comparing the observed RVS

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spectrum with a purely photospheric template spectrum obtained by interpolating in a grid of synthetic spectra. The atmospheric parameters (APs) adopted in the interpolation are taken from the output of either GSP-Spec or GSP-Phot. GSP-Spec APs are adopted when all three parameters `teff_gspspec`, `logg_gspspec`, and `mh_gspspec` are provided. Otherwise `teff_gspphot`, `logg_gspphot`, and `mh_gspphot` are adopted, if they are all provided. The activity index gives the excess equivalent width factor in the cores of the Ca II infrared triplet lines with respect to the template inactive spectrum. The excess equivalent width is computed for each of the Ca II infrared triplet lines and the three values obtained are averaged. When the projected rotational velocity `vbroad` in table `gaia_source` is provided, rotational broadening is taken into account. See Section **??** for details.

ACTIVITYINDEX_UNCERTAINTY : Uncertainty in the chromospheric activity index from ESP-CS (float, Length & Distance[nm])

Uncertainty in the activity index from ESP-CS (`activityindex`). The uncertainty is computed by considering the standard deviation of the excess equivalent width factor in each of the Ca II infrared triplet lines, taking spectrum noise into account, and applying error propagation. See Section **??** for details.

ACTIVITYINDEX_INPUT : Source of input stellar parameters for the computation of the activity index by ESP-CS (string)

Flag indicating the source of the stellar atmospheric parameters used by ESP-CS in deriving the activity index. The flag has the value 'M1' if the source is GSP-Spec, and 'M2' if the source is GSP-Phot. See the description of `activityindex`.

ACTIVITYINDEX_T1_SPEC : Activity index measured on the CaII (849.8nm) line by ESP-CS with input APs from GSP-Spec (float, Length & Distance[nm])

Activity index measured on the CaII(849.8nm) line by ESP-CS derived by comparison with a synthetic spectrum generated with input APs from GSP-Spec. The activity index is defined as the excess equivalent width with respect to the purely photospheric value assumed to be that given by the synthetic spectrum computed in radiative equilibrium.

ACTIVITYINDEX_T1_SPEC_UNCERTAINTY : Uncertainty in the activity index measured on the CaII (849.8nm) with input APs from GSP-Spec (float, Length & Distance[nm])

Uncertainty in the chromospheric activity index measured on the CaII(849.8nm) derived by comparison with a synthetic spectrum generated with input APs from GSP-Spec.

ACTIVITYINDEX_T2_SPEC : Activity index measured on the CaII (854.2nm) line by ESP-CS with input APs from GSP-Spec (float, Length & Distance[nm])

Activity index measured on the CaII(854.2nm) line by ESP-CS derived by comparison with a synthetic spectrum generated with input APs from GSP-Spec. The activity index is defined as the excess equivalent width with respect to the purely photospheric value assumed to be that given by the synthetic spectrum computed in radiative equilibrium.

ACTIVITYINDEX_T2_SPEC_UNCERTAINTY : Uncertainty in the activity index measured on the CaII (854.2nm) with input APs from GSP-Spec (float, Length & Distance[mm])

Uncertainty in the chromospheric activity index measured on the CaII(854.2nm) derived by comparison with a synthetic spectrum generated with input APs from GSP-Spec.

ACTIVITYINDEX_T3_SPEC : Activity index measured on the CaII (866.2nm) line by ESP-CS with input APs from GSP-Spec (float, Length & Distance[nm])

Activity index measured on the CaII(866.2nm) line by ESP-CS derived by comparison with a synthetic spectrum generated with input APs from GSP-Spec. The activity index is defined as the excess equivalent width with respect to the purely photospheric value assumed to be that given by the synthetic spectrum computed in radiative equilibrium.

ACTIVITYINDEX_T3_SPEC_UNCERTAINTY : Uncertainty in the activity index measured on the CaII (866.2nm) line with input APs from GSP-Spec (float, Length & Distance[nm])

Uncertainty in the chromospheric activity index measured on the CaII(866.2nm) line derived by comparison with a synthetic spectrum generated with input APs from GSP-Spec.

ACTIVITYINDEX_T1_PHOT : Activity index measured on the CaII (849.8nm) line by ESP-CS with input APs from GSP-Phot (float, Length & Distance[nm])

Activity index measured on the CaII(849.8nm) line by ESP-CS derived by comparison with a synthetic spectrum generated with input APs from GSP-Phot. The activity index is defined as the excess equivalent width with respect to the purely photospheric value assumed to be that given by the synthetic spectrum computed in radiative equilibrium.

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ACTIVITYINDEX_T1_PHOT_UNCERTAINTY : Uncertainty in the activity index measured on the CaII (849.8nm) with input APs from GSP-Phot (float, Length & Distance[nm])

Uncertainty in the chromospheric activity index measured on the CaII(849.8nm) derived by comparison with a synthetic spectrum generated with input APs from GSP-Phot.

ACTIVITYINDEX_T2_PHOT : Activity index measured on the CaII (854.2nm) line by ESP-CS with input APs from GSP-Phot (float, Length & Distance[nm])

Activity index measured on the CaII(854.2nm) line by ESP-CS derived by comparison with a synthetic spectrum generated with input APs from GSP-Phot. The activity index is defined as the excess equivalent width with respect to the purely photospheric value assumed to be that given by the synthetic spectrum computed in radiative equilibrium.

ACTIVITYINDEX_T2_PHOT_UNCERTAINTY : Uncertainty in the activity index measured on the CaII (854.2nm) with input APs from GSP-Phot (float, Length & Distance[mm])

Uncertainty in the chromospheric activity index measured on the CaII(854.2nm) derived by comparison with a synthetic spectrum generated with input APs from GSP-Phot.

ACTIVITYINDEX_T3_PHOT : Activity index measured on the CaII (866.2nm) line by ESP-CS with input APs from GSP-Phot (float, Length & Distance[nm])

Activity index measured on the CaII(866.2nm) line by ESP-CS derived by comparison with a synthetic spectrum generated with input APs from GSP-Phot. The activity index is defined as the excess equivalent width with respect to the purely photospheric value assumed to be that given by the synthetic spectrum computed in radiative equilibrium.

ACTIVITYINDEX_T3_PHOT_UNCERTAINTY : Uncertainty in the activity index measured on the CaII (866.2nm) line with input APs from GSP-Phot (float, Length & Distance[nm])

Uncertainty in the chromospheric activity index measured on the CaII(866.2nm) line derived by comparison with a synthetic spectrum generated with input APs from GSP-Phot.

ACTIVITYINDEX_T1_ESPCS : Activity index measured on the CaII (849.8nm) line by ESP-CS with input APs from ESP-CS (float, Length & Distance[nm])

Activity index measured on the CaII(849.8nm) line by ESP-CS derived by comparison with a

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synthetic spectrum generated with input APs from ESP-CS. The activity index is defined as the excess equivalent width with respect to the purely photospheric value assumed to be that given by the synthetic spectrum computed in radiative equilibrium.

ACTIVITYINDEX_T1_ESPCS_UNCERTAINTY : Uncertainty in the activity index measured on the CaII (849.8nm) with input APs from ESP-CS (float, Length & Distance[nm])

Uncertainty in the chromospheric activity index measured on the CaII(849.8nm) derived by comparison with a synthetic spectrum generated with input APs from ESP-CS.

ACTIVITYINDEX_T2_ESPCS : Activity index measured on the CaII (854.2nm) line by ESP-CS with input APs from ESP-CS (float, Length & Distance[nm])

Activity index measured on the CaII(854.2nm) line by ESP-CS derived by comparison with a synthetic spectrum generated with input APs from ESP-CS. The activity index is defined as the excess equivalent width with respect to the purely photospheric value assumed to be that given by the synthetic spectrum computed in radiative equilibrium.

ACTIVITYINDEX_T2_ESPCS_UNCERTAINTY : Uncertainty in the activity index measured on the CaII (854.2nm) with input APs from ESP-CS (float, Length & Distance[mm])

Uncertainty in the chromospheric activity index measured on the CaII(854.2nm) derived by comparison with a synthetic spectrum generated with input APs from ESP-CS.

ACTIVITYINDEX_T3_ESPCS : Activity index measured on the CaII (866.2nm) line by ESP-CS with input APs from ESP-CS (float, Length & Distance[nm])

Activity index measured on the CaII(866.2nm) line by ESP-CS derived by comparison with a synthetic spectrum generated with input APs from ESP-CS. The activity index is defined as the excess equivalent width with respect to the purely photospheric value assumed to be that given by the synthetic spectrum computed in radiative equilibrium.

ACTIVITYINDEX_T3_ESPCS_UNCERTAINTY : Uncertainty in the activity index measured on the CaII (866.2nm) line with input APs from ESP-CS (float, Length & Distance[nm])

Uncertainty in the chromospheric activity index measured on the CaII(866.2nm) line derived by comparison with a synthetic spectrum generated with input APs from ESP-CS.

ACTIVITYFLUX : Activity flux measured on the CaII triplet (float)

Activity flux measured on the CaII(849.8nm) line by ESP-CS derived by comparison with a synthetic spectrum generated with input APs from GSP-Spec. The activity flux is defined as the excess flux in the line with respect to the purely photospheric value assumed to be that given by the synthetic spectrum computed in radiative equilibrium. The excess flux is given in bolometric flux units, i.e. $\sigma T_{\rm eff}^4$.

ACTIVITYFLUX_UNCERTAINTY : Uncertainty in activityfluxEspcs (float)

Activity flux measured on the CaII(849.8nm) line by ESP-CS derived by comparison with a synthetic spectrum generated with input APs from GSP-Spec. The activity flux is defined as the excess flux in the line with respect to the purely photospheric value assumed to be that given by the synthetic spectrum computed in radiative equilibrium. The excess flux is given in bolometric flux units, i.e. $\sigma T_{\rm eff}^4$.

ACTIVITYFLUX_T1_SPEC : Activity flux measured on the CaII (849.8nm) line by ESP-CS with input APs from GSP-Spec (float)

Activity flux measured on the CaII(849.8nm) line by ESP-CS derived by comparison with a synthetic spectrum generated with input APs from GSP-Spec. The activity flux is defined as the excess flux in the line with respect to the purely photospheric value assumed to be that given by the synthetic spectrum computed in radiative equilibrium. The excess flux is given in bolometric flux units, i.e. $\sigma T_{\rm eff}^4$.

ACTIVITYFLUX_T1_SPEC_UNCERTAINTY : Uncertainty in the activity flux measured on the CaII (849.8nm) with input APs from GSP-Spec (float)

Uncertainty in the activity flux measured on the CaII(849.8nm) derived by comparison with a synthetic spectrum generated with input APs from GSP-Spec.

ACTIVITYFLUX_T2_SPEC : Activity flux measured on the CaII (854.2nm) line by ESP-CS with input APs from GSP-Spec (float)

Activity flux measured on the CaII(854.2nm) line by ESP-CS derived by comparison with a synthetic spectrum generated with input APs from GSP-Spec. The activity flux is defined as the excess flux in the line with respect to the purely photospheric value assumed to be that given by the synthetic spectrum computed in radiative equilibrium. The excess flux is given in bolometric flux units, i.e. $\sigma T_{\rm eff}^4$.

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ACTIVITYFLUX_T2_SPEC_UNCERTAINTY : Uncertainty in the activity flux measured on the CaII (854.2nm) with input APs from GSP-Spec (float)

Uncertainty in the activity flux measured on the CaII(854.2nm) derived by comparison with a synthetic spectrum generated with input APs from GSP-Spec.

ACTIVITYFLUX_T3_SPEC : Activity flux measured on the CaII (866.2nm) line by ESP-CS with input APs from GSP-Spec (float)

Activity flux measured on the CaII(866.2nm) line by ESP-CS derived by comparison with a synthetic spectrum generated with input APs from GSP-Spec. The activity flux is defined as the excess flux in the line with respect to the purely photospheric value assumed to be that given by the synthetic spectrum computed in radiative equilibrium. The excess flux is given in bolometric flux units, i.e. $\sigma T_{\rm eff}^4$.

ACTIVITYFLUX_T3_SPEC_UNCERTAINTY : Uncertainty in the activity flux measured on the CaII (866.2nm) line with input APs from GSP-Spec (float)

Uncertainty in the activity flux measured on the CaII(866.2nm) line derived by comparison with a synthetic spectrum generated with input APs from GSP-Spec.

ACTIVITYFLUX_T1_PHOT : Activity flux measured on the CaII (849.8nm) line by ESP-CS with input APs from GSP-Phot (float)

Activity flux measured on the CaII(849.8nm) line by ESP-CS derived by comparison with a synthetic spectrum generated with input APs from GSP-Phot. The activity flux is defined as the excess flux in the line with respect to the purely photospheric value assumed to be that given by the synthetic spectrum computed in radiative equilibrium. The excess flux is given in bolometric flux units, i.e. $\sigma T_{\rm eff}^4$.

ACTIVITYFLUX_T1_PHOT_UNCERTAINTY : Uncertainty in the activity flux measured on the CaII (849.8nm) with input APs from GSP-Phot (float)

Uncertainty in the activity flux measured on the CaII(849.8nm) derived by comparison with a synthetic spectrum generated with input APs from GSP-Phot.

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ACTIVITYFLUX_T2_PHOT : Activity flux measured on the CaII (854.2nm) line by ESP-CS with input APs from GSP-Phot (float)

Activity flux measured on the CaII(854.2nm) line by ESP-CS derived by comparison with a synthetic spectrum generated with input APs from GSP-Phot. The activity flux is defined as the excess flux in the line with respect to the purely photospheric value assumed to be that given by the synthetic spectrum computed in radiative equilibrium. The excess flux is given in bolometric flux units, i.e. $\sigma T_{\rm eff}^4$.

ACTIVITYFLUX_T2_PHOT_UNCERTAINTY : Uncertainty in the activity flux measured on the CaII (854.2nm) with input APs from GSP-Phot (float)

Uncertainty in the activity flux measured on the CaII(854.2nm) derived by comparison with a synthetic spectrum generated with input APs from GSP-Phot.

ACTIVITYFLUX_T3_PHOT : Activity flux measured on the CaII (866.2nm) line by ESP-CS with input APs from GSP-Phot (float)

Activity flux measured on the CaII(866.2nm) line by ESP-CS derived by comparison with a synthetic spectrum generated with input APs from GSP-Phot. The activity flux is defined as the excess flux in the line with respect to the purely photospheric value assumed to be that given by the synthetic spectrum computed in radiative equilibrium. The excess flux is given in bolometric flux units, i.e. $\sigma T_{\rm eff}^4$.

ACTIVITYFLUX_T3_PHOT_UNCERTAINTY : Uncertainty in the activity flux measured on the CaII (866.2nm) line with input APs from GSP-Phot (float)

Uncertainty in the activity flux measured on the CaII(866.2nm) line derived by comparison with a synthetic spectrum generated with input APs from GSP-Phot.

ACTIVITYFLUX_T1_ESPCS : Activity flux measured on the CaII (849.8nm) line by ESP-CS with input APs from ESP-CS (float)

Activity flux measured on the CaII(849.8nm) line by ESP-CS derived by comparison with a synthetic spectrum generated with input APs from ESP-CS. The activity flux is defined as the excess flux in the line with respect to the purely photospheric value assumed to be that given by the synthetic spectrum computed in radiative equilibrium. The excess flux is given in bolometric flux units, i.e. $\sigma T_{\rm eff}^4$.

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ACTIVITYFLUX_T1_ESPCS_UNCERTAINTY : Uncertainty in the activity flux measured on the CaII (849.8nm) with input APs from ESP-CS (float)

Uncertainty in the activity flux measured on the CaII(849.8nm) derived by comparison with a synthetic spectrum generated with input APs from ESP-CS.

ACTIVITYFLUX_T2_ESPCS : Activity flux measured on the CaII (854.2nm) line by ESP-CS with input APs from ESP-CS (float)

Activity flux measured on the CaII(854.2nm) line by ESP-CS derived by comparison with a synthetic spectrum generated with input APs from ESP-CS. The activity flux is defined as the excess flux in the line with respect to the purely photospheric value assumed to be that given by the synthetic spectrum computed in radiative equilibrium. The excess flux is given in bolometric flux units, i.e. $\sigma T_{\rm eff}^4$.

ACTIVITYFLUX_T2_ESPCS_UNCERTAINTY : Uncertainty in the activity flux measured on the CaII (854.2nm) with input APs from ESP-CS (float)

Uncertainty in the activity flux measured on the CaII(854.2nm) derived by comparison with a synthetic spectrum generated with input APs from ESP-CS.

ACTIVITYFLUX_T3_ESPCS : Activity flux measured on the CaII (866.2nm) line by ESP-CS with input APs from ESP-CS (float)

Activity flux measured on the CaII(866.2nm) line by ESP-CS derived by comparison with a synthetic spectrum generated with input APs from ESP-CS. The activity flux is defined as the excess flux in the line with respect to the purely photospheric value assumed to be that given by the synthetic spectrum computed in radiative equilibrium. The excess flux is given in bolometric flux units, i.e. $\sigma T_{\rm eff}^4$.

ACTIVITYFLUX_T3_ESPCS_UNCERTAINTY : Uncertainty in the activity flux measured on the CaII (866.2nm) line with input APs from ESP-CS (float)

Uncertainty in the activity flux measured on the CaII(866.2nm) line derived by comparison with a synthetic spectrum generated with input APs from ESP-CS.

TEFF_ESPCS : Effective temperature from ESP-CS assuming a magnetospheric accretion model (float, Temperature[K])

Effective temperature of the pre-main sequence star according to the magnetospheric accretion model from ESP-CS.

TEFF_ESPCS_LOWER : Lower confidence level (16%) of effective temperature from ESP-CS (float, Temperature[K])

Lower confidence level (16%) of effective temperature of the pre-main sequence star according to the magnetospheric accretion model from ESP-CS.

TEFF_ESPCS_UPPER : Upper confidence level (84%) of effective temperature from ESP-CS (float, Temperature[K])

Upper confidence level (84%) of effective temperature of pre-main sequence star according to the magnetospheric accretion model from ESP-CS.

LOGG_ESPCS : Surface gravity from ESP-CS assuming a magnetospheric accretion model (float, GravitySurface[log cgs])

Surface gravity of the pre-main sequence star according to the magnetospheric accretion model from ESP-CS.

LOGG_ESPCS_LOWER : Lower confidence level (16%)of surface gravity from ESP-CS (float, GravitySurface[log cgs])

Lower confidence level (16%) of surface gravity of the pre-main sequence star according to the magnetospheric accretion model from ESP-CS.

LOGG_ESPCS_UPPER : Upper confidence level (84%) of surface gravity from ESP-CS (float, GravitySurface[log cgs])

Upper confidence level of surface gravity of the pre-main sequence star according to the magnetospheric accretion model from ESP-CS.

FEH_ESPCS : Global iron abundance from ESP-CS assuming a magnetospheric accretion model (float, Abundances[dex])

Global iron abundance of the pre-main sequence star according to the magnetospheric accretion

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model from ESP-CS.

FEH_ESPCS_LOWER : Lower confidence level (16%) of global iron abundance from ESP-CS (float, Abundances[dex])

Lower confidence level of global iron abundance of the pre-main sequence star according to the magnetospheric accretion model from ESP-CS.

FEH_ESPCS_UPPER : Upper confidence level (84%) of global iron abundance from ESP-CS (float, Abundances[dex])

Lower confidence level (84%) of global iron abundance of the pre-main sequence star according to the magnetospheric accretion model from ESP-CS.

AZERO_ESPCS : Monochromatic extinction at 542.9 nm from ESP-CS assuming a magnetospheric accretion model (float, Magnitude[mag])

Monochromatic extinction A_0 at 542.9 nm according to the magnetospheric accretion model from ESP-CS.

AZERO_ESPCS_LOWER : Lower confidence level (16%) of monochromatic extinction at 542.9 nm from ESP-CS (float, Magnitude[mag])

Lower condidence level (16%) of monochromatic extinction A_0 at 542.9 nm according to the magnetospheric accretion model from ESP-CS.

AZERO_ESPCS_UPPER : Upper confidence level (84%) of monochromatic extinction at 542.9 nm from ESP-CS (float, Magnitude[mag])

Upper condidence level (84%) of monochromatic extinction A_0 at 542.9 nm according to the magnetospheric accretion model from ESP-CS.

ACCRFILLFACTOR_ESPCS : Filling factor of the hot spot due to mass accretion from ESP-CS (float)

Filling factor of the hot spot due to mass accretion according to the magnetospheric accretion model from ESP-CS.

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ACCRFILLFACTOR_ESPCS_LOWER : Lower confidence level (16%) of filling factor of the hot spot due to mass accretion from ESP-CS (float)

Lower confidence level (16%) of filling factor of the hot spot due to mass accretion according to the magnetospheric accretion model from ESP-CS.

ACCRFILLFACTOR_ESPCS_UPPER : Upper confidence level (84%) of filling factor of the hot spot due to mass accretion from ESP-CS (float)

Upper confidence level (84%) of filling factor of the hot spot due to mass accretion according to the magnetospheric accretion model from ESP-CS.

MASSACCRRATE_ESPCS : Log10 of Mass accretion rate from ESP-CS (float, Mass[SolarMass.year-1])

Log10 of mass accretion rate of the pre-main sequence star according to the magnetospheric accretion model from ESP-CS.

MASSACCRRATE_ESPCS_LOWER : Lower confidence level (16%) of Log10 of mass accretion rate from ESP-CS (float, Mass[SolarMass.year-1])

Log10 of lower confidence level (16%) of mass accretion rate of the pre-main sequence star according to the magnetospheric accretion model from ESP-CS.

MASSACCRRATE_ESPCS_UPPER : Upper confidence level(84%) of Log10 of mass accretion rate from ESP-CS (float, Mass[SolarMass.year-1])

Log10 of upper confidence level (84%) of mass accretion rate of the pre-main sequence star according to the magnetospheric accretion model from ESP-CS.

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4.5 AP_HOT [TAP]

This table contains the astrophysical parameters of hot stars derived by the ESP-HS module as part of the Apsis processing chain. The parameters are derived from mean XP spectra, and when available, combined with the mean RVS spectra.

See Section **??** of the online documentation for details on the ESP-HS module.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

TEFF_ESPHS : Effective temperature from ESP-HS (float, Temperature[K])

Effective temperature derived by fitting the BP/RP spectra and, when available, the RVS spectra with synthetic spectra. ESP-HS processes targets brighter than $G = 18.25$ that are labelled O, B, or A in the `spectraltype_espels` field of the `ap_xp` table. A more detailed description of the module is provided in Section **??** and Section **??**. A quality assessment was performed on all parameter estimates. No entries were removed based on their values or uncertainties; instead, a flagging scheme taking these quantities into account has been adopted.

TEFF_ESPHS_UNCERTAINTY : Uncertainty on the effective temperature provided by ESP-HS (float, Temperature[K])

Uncertainty on the effective temperature derived by ESP-HS. Uncertainties were estimated from Monte Carlo noise realisations of the data; the reported value corresponds to half the interquantile range (15.87th – 84.13th percentiles) of the parameter distribution. A more detailed description

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of ESP-HS is provided in Section **??** and Section **??**. A quality assessment was performed on all parameter estimates. No entries were removed based on their values or uncertainties; instead, a flagging scheme taking these quantities into account has been adopted.

LOGG_ESPHS : Surface gravity from ESP-HS (float, GravitySurface[log cgs])

Surface gravity derived by fitting the BP/RP spectra and, when available, the RVS spectra with synthetic spectra. ESP-HS processes targets brighter than $G = 18.25$ that are labelled O, B, or A in the `spectraltype_espels` field of the `ap_xp` table. A more detailed description of the module is provided in Section **??** and Section **??**. A quality assessment was performed on all parameter estimates. No entries were removed based on their values or uncertainties; instead, a flagging scheme taking these quantities into account has been adopted.

LOGG_ESPHS_UNCERTAINTY : Uncertainty on the surface gravity provided by ESP-HS (float, GravitySurface[log cgs])

Uncertainty on the surface gravity derived by ESP-HS. Uncertainties were estimated from Monte Carlo noise realisations of the data; the reported value corresponds to half the interquantile range (15.87th – 84.13th percentiles) of the parameter distribution. A more detailed description of ESP-HS is provided in Section **??** and Section **??**. A quality assessment was performed on all parameter estimates. No entries were removed based on their values or uncertainties; instead, a flagging scheme taking these quantities into account has been adopted.

FEH_ESPHS : Global iron abundance derived by ESP-HS (float, Abundances[dex])

Global iron abundance (as defined by the synthetic spectra) derived by fitting the BP/RP spectra and, when available, the RVS spectra with synthetic spectra. This value corresponds to the abundance used in the synthetic spectra, which, for the A and OB libraries, is a global scaling of all metal abundances rather than iron alone. Unlike the other astrophysical parameters, for which different starting values were tested, the initial value was set at zero (Solar chemical abundance). ESP-HS processes sources brighter than $G = 18.25$ that are tagged as O, B, or A in the `spectraltype_espels` field of the `ap_xp` table. A more detailed description of the module is provided in Section **??** and Section **??**. A quality assessment was performed on all parameter estimates. No entries were removed based on their values or uncertainties; instead, a flagging scheme taking these quantities into account has been adopted.

FEH_ESPHS_UNCERTAINTY : Uncertainty on the global iron abundance derived by ESP-HS (float, Abundances[dex])

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Uncertainty on the global iron abundance derived by ESP-HS. Uncertainties were estimated from Monte Carlo noise realisations of the data; the reported value corresponds to half the interquantile range (15.87th – 84.13th percentiles) of the parameter distribution. A more detailed description of ESP-HS is provided in Section **??** and Section **??**. A quality assessment was performed on all parameter estimates. No entries were removed based on their values or uncertainties; instead, a flagging scheme taking these quantities into account has been adopted.

VSINI_ESPHS : Projected rotational velocity from ESP-HS (float, Velocity[km s^{-1}])

Line broadening parameter derived from the co-added RVS spectra. The line broadening is assumed to be due to stellar rotation only. To make a clear distinction with the value obtained by CU6 on the transit RVS spectra, we named it $v \sin i$. A value is provided only in the BP/RP+RVS mode (see ESP-HS flags). ESP-HS processes sources brighter than $G = 18.25$ that are tagged as O, B, or A in the `spectraltype_espels` field of the `ap_xp` table. A more detailed description of the module is provided in Section **??** and Section **??**. A quality assessment was performed on all parameter estimates. No entries were removed based on their values or uncertainties; instead, a flagging scheme taking these quantities into account has been adopted.

VSINI_ESPHS_UNCERTAINTY : Uncertainty on the projected rotational velocity provided by ESP-HS (float, Velocity[km s^{-1}])

Uncertainty on the $v \sin i$ parameter derived by ESP-HS. Uncertainties were estimated from Monte Carlo noise realisations of the data; the reported value corresponds to half the interquantile range (15.87th – 84.13th percentiles) of the parameter distribution. A more detailed description of ESP-HS is provided in Section **??** and Section **??**. A quality assessment was performed on all parameter estimates. No entries were removed based on their values or uncertainties; instead, a flagging scheme taking these quantities into account has been adopted.

AZERO_ESPHS : Monochromatic extinction from ESP-HS (float, Magnitude[mag])

Extinction parameter A_0 corresponding to the monochromatic extinction at a specific reference wavelength (see Section **??** of the online documentation) inferred by ESP-HS from the simultaneous fit of BP/RP spectra and, when available, the RVS spectra. A_0 is the extinction parameter in the adopted extinction curve from Gordon et al. (2023). ESP-HS processes sources brighter than $G = 18.25$ that are tagged as O, B, or A in the `spectraltype_espels` field of the `ap_xp` table. A more detailed description of the module is provided in Section **??** and Section **??**. A quality assessment was performed on all parameter estimates. No entries were removed based on their values or uncertainties; instead, a flagging scheme taking these quantities into account has been adopted.

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AZERO_ESPHS_UNCERTAINTY : Uncertainty on the monochromatic extinction derived by ESP-HS (float, Magnitude[mag])

Uncertainty on the A_0 parameter derived by ESP-HS. Uncertainties were estimated from Monte Carlo noise realisations of the data; the reported value corresponds to half the interquantile range (15.87th – 84.13th percentiles) of the parameter distribution. The extinction curve adopted in the Apsis pipeline is the one described in Gordon et al. (2023). A more detailed description of ESP-HS is provided in Section **??**, Section **??** and Section **??**. A quality assessment was performed on all parameter estimates. No entries were removed based on their values or uncertainties; instead, a flagging scheme taking these quantities into account has been adopted.

AG_ESPHS : Interstellar extinction in G band from ESP-HS (float, Magnitude[mag])

Broadband interstellar extinction in the G photometric band derived by ESP-HS. The value is obtained to ensure consistency with the fitted parameter A_0 by adopting the extinction curve described by Gordon et al. (2023). ESP-HS processes sources brighter than $G = 18.25$ that are tagged as O, B, or A in the `spectraltype_espels` field of the `ap_xp` table. A more detailed description of the module is provided in Section **??** and Section **??**. A quality assessment was performed on all parameter estimates. No entries were removed based on their values or uncertainties; instead, a flagging scheme taking these quantities into account has been adopted.

AG_ESPHS_UNCERTAINTY : Uncertainty on the interstellar extinction in the G band derived by ESP-HS (float, Magnitude[mag])

Uncertainty on $A_{\rm G}$ derived by ESP-HS. The value is obtained by propagating the uncertainty on the fitted parameter A_0 through the adopted extinction curve. A more detailed description of ESP-HS is provided in Section **??** and Section **??**. A quality assessment was performed on all parameter estimates. No entries were removed based on their values or uncertainties; instead, a flagging scheme taking these quantities into account has been adopted.

EBPMINRP_ESPHS : Reddening in BP minus RP colour from ESP-HS (float, Magnitude[mag])

Interstellar reddening, $E(G_{\rm BP} - G_{\rm RP})$, derived by ESP-HS. The value is obtained to ensure consistency with the fitted parameter A_0 by adopting the extinction curve described by Gordon et al. (2023). ESP-HS processes sources brighter than $G = 18.25$ that are tagged as O, B, or A in the `spectraltype_espels` field of the `ap_xp` table. A more detailed description of the module is provided in Section **??** and Section **??**. A quality assessment was performed on all parameter estimates. No entries were removed based on their values or uncertainties; instead, a flagging scheme taking these quantities into account has been adopted.

EBPMINRP_ESPHS_UNCERTAINTY : Uncertainty on the E(BP-RP) reddening provided by ESP-HS (float, Magnitude[mag])

Uncertainty on $E(G_{BP} - G_{RP})$ derived by ESP-HS. The value is obtained by propagating the uncertainty on the fitted parameter A_0 through the adopted extinction curve. A more detailed description of ESP-HS is provided in Section **??** and Section **??**. A quality assessment was performed on all parameter estimates. No entries were removed based on their values or uncertainties; instead, a flagging scheme taking these quantities into account has been adopted.

FLAGS_ESPHS : Processing/quality flags of the ESP-HS parametrisation (string)

Processing flags. When an incident occurred during the processing or post-processing, it was recorded in a 64-bit integer used as a bitmask and stored as a string. The following bits were initially defined:

1: G magnitude is not available.

20-26: Convergence issues.

27: The spectral type tag derived by ESP-ELS, and used for the target selection disagrees with the astrophysical parameters from GSP-Phot.

28: The algorithm run twice on the same target after adapting the processing mode.

29: The wavelength offset derived on the RVS is, in radial velocity units, larger than 50 $\mathrm{km\,s^{-1}}$.

30: The astrophysical parameters are reaching the limits of the spectra libraries.

31: Poor RVS goodness of fit.

32: Poor BP/RP goodness of fit.

34: Issues with the RVS re-normalisation.

35: Astrophysical parameters not valid.

36: One of the astrophysical parameters have non realistic values.

37: The solution is not compatible with the expected $(\mathrm{GBP\text{-}GRP})_0$ versus Teff relation.

39: The spectral type tag used for the target selection `spectraltype_espels_flags`> 4.

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When no issue was reported, the flag is set to '0'. A value of '0' indicates that the parameters satisfy all quality criteria and are eligible for inclusion in `gaia_source`. The value '1' is used for what is expected to be the next-best parameter estimates. For the remaining cases, the processing flag was left in its original signed integer format. These parameter values should be treated with caution. More details can be found in Section **??**.

AV_ESPHS : Broadband interstellar extinction in the Johnson V band from ESP-HS (float, Magnitude[mag])

Broadband interstellar extinction in the Johnson V band, A_V, derived by ESP-HS. The value is obtained to ensure consistency with the fitted parameter A_0 by adopting the extinction curve described by Gordon et al. (2023). ESP-HS processes sources brighter than $G = 18.25$ that are tagged as O, B, or A in the `spectraltype_espels` field of the `ap_xp` table. A more detailed description of the module is provided in Section **??** and Section **??**. A quality assessment was performed on all parameter estimates. No entries were removed based on their values or uncertainties; instead, a flagging scheme taking these quantities into account has been adopted.

AV_ESPHS_UNCERTAINTY : Uncertainty on the interstellar extinction in the Johnson V band provided by ESP-HS (float, Magnitude[mag])

Uncertainty on A_V derived by ESP-HS. The value is obtained by propagating the uncertainty on the fitted parameter A_0 through the adopted extinction curve. A more detailed description of ESP-HS is provided in Section **??** and Section **??**. A quality assessment was performed on all parameter estimates. No entries were removed based on their values or uncertainties; instead, a flagging scheme taking these quantities into account has been adopted.

EBMINV_ESPHS : Colour excess in Johnson B and V bands due to interstellar reddening from ESP-HS (float, Magnitude[mag])

Interstellar reddening on the Johnson $B - V$ colour, $E(B - V)$, derived by ESP-HS. The value is obtained to ensure consistency with the fitted parameter A_0 by adopting the extinction curve described by Gordon et al. (2023). ESP-HS processes sources brighter than $G = 18.25$ that are tagged as O, B, or A in the `spectraltype_espels` field of the `ap_xp` table. A more detailed description of the module is provided in Section **??** and Section **??**. A quality assessment was performed on all parameter estimates. No entries were removed based on their values or uncertainties; instead, a flagging scheme taking these quantities into account has been adopted.

EBMINV_ESPHS_UNCERTAINTY : Uncertainty on the colour excess B minus V due to interstellar reddening measured by ESP-HS (float, Magnitude[mag])

Draft

Uncertainty on $E(B-V)$ derived by ESP-HS. The value is obtained by propagating the uncertainty on the fitted parameter A_0 through the adopted extinction curve. A more detailed description of ESP-HS is provided in Section **??** and Section **??**. A quality assessment was performed on all parameter estimates. No entries were removed based on their values or uncertainties; instead, a flagging scheme taking these quantities into account has been adopted.

DELTA_RV_ESPHS : Radial velocity offset from published value imposed during the fitting of the RVS spectrum (float, Velocity[km s^{-1}])

Value of the wavelength offset, expressed in velocity units, that the procedure took into account during the fitting of the RVS spectra in order to optimise the alignment between the observed and synthetic spectra. The value can be used as a quality indicator of the fit (the closer to zero, the best). There is no uncertainty associated with this quantity. A quality assessment was performed on all parameter estimates. No entries were removed based on their values or uncertainties; instead, a flagging scheme taking these quantities into account has been adopted.

PROCESSMODE_ESPHS : Processing mode adopted by ESP-HS to derive the astrophysical parameters (string)

Processing mode adopted by ESP-HS to derive the astrophysical parameters. The processing mode can have the following values:

4: Both BP/RP and RVS data were used. The metallicity, the total-to-selective extinction and projected rotational velocity parameters were fitted together with the main astrophysical parameters.

5: Only BP/RP data were used. The metallicity and the total-to-selective extinction parameters were fitted together with the main astrophysical parameters.

6: BP/RP and RVS data, as well as the apparent G magnitude and parallax, were used. The metallicity, the total-to-selective extinction, and projected rotational velocity parameters were fitted together with the main astrophysical parameters.

7: BP/RP data, along with the apparent G magnitude and parallax, were used. The metallicity and the total-to-selective extinction parameters were fitted together with the main astrophysical parameters.

A more detailed description of the module is provided in Section **??** and Section **??**.

RZEROMAG_ESPHS : Total-to-selective extinction ratio to be used in the extinction curve (float)

Total-to-selective interstellar extinction ratio derived from the extinction curve given by Gordon et al. (2023). R_0 is the value to be used as input in the extinction curve, and is derived by ESP-HS from the comparison between the observed BP/RP spectra and, when available, the RVS spectra with the corresponding simulations. ESP-HS processes targets brighter than $G = 18.25$ that are tagged as O, B, or A in the `spectraltype_espels` field of the `ap_xp` table. A more detailed description of the module is provided in Section **??** and Section **??**. A quality assessment was performed on all parameter estimates. No entries were removed based on their values or uncertainties; instead, a flagging scheme taking these quantities into account has been adopted.

RZEROMAG_ESPHS_UNCERTAINTY : Uncertainty on the total-to-selective extinction ratio derived by ESP-HS (float)

Uncertainty on the total-to-selective interstellar extinction ratio inferred by adopting the extinction curve described in Gordon et al. (2023). Uncertainties were estimated from Monte Carlo noise realisations of the data; the reported value corresponds to half the interquantile range (15.87th – 84.13th percentiles) of the parameter distribution. A more detailed description of ESP-HS is provided in Section **??** and Section **??**. A quality assessment was performed on all parameter estimates. No entries were removed based on their values or uncertainties; instead, a flagging scheme taking these quantities into account has been adopted.

RVMAG_ESPHS : Broadband total-to-selective extinction ratio in the Johnson V band (float)

Broadband total-to-selective interstellar extinction ratio in the Johnson V band derived by adopting the extinction curve described in Gordon et al. (2023). The value is defined as $\frac{A_V}{E(B-V)}$. ESP-HS processes sources brighter than $G = 18.25$ that are tagged as O, B, or A in the `spectraltype_espels` field of the `ap_xp` table. A more detailed description of the module is provided in Section **??** and Section **??**. A quality assessment was performed on all parameter estimates. No entries were removed based on their values or uncertainties; instead, a flagging scheme taking these quantities into account has been adopted.

RVMAG_ESPHS_UNCERTAINTY : Uncertainty on the broadband total-to-selective extinction ratio measured by ESP-HS (float)

Uncertainty on total-to-selective interstellar extinction ratio derived in the Johnson V band by adopting the extinction function described in Gordon et al. (2023). The value is obtained by propagating the uncertainty on the fitted parameter R_0 through the relation that defines R_V. A more detailed description of ESP-HS is provided in Section **??** and Section **??**. A quality assessment was performed on all parameter estimates. No entries were removed based on their values or uncertainties; instead, a flagging scheme taking these quantities into account has been adopted.

4.6 AP_RVS [TAP]

This table contains the astrophysical parameters of stars derived from mean RVS spectra by GSP-Spec and FLAME as part of the Apsis processing chain.

See Sections **??** and **??** of the online documentation for details.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

TEFF_GSPSPEC_CALIB : Calibrated effective temperature from GSP-Spec MatisseGauguin using RVS spectra and Monte Carlo realisations (float, Temperature[K])

Calibrated value of the median effective temperature (assuming the source is a single star), as determined by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from RVS spectra, using literature references.

TEFF_GSPSPEC_LOWER_CALIB : Calibrated 16th percentile of effective temperature from GSP-Spec MatisseGauguin using RVS spectra and Monte Carlo realisations (float, Temperature[K])

Lower confidence level (16%) of the calibrated effective temperature (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from RVS spectra. Lower and upper levels include 68% confidence interval.

TEFF_GSPSPEC_UPPER_CALIB : Calibrated 84th percentile of effective temperature from GSP-Spec MatisseGauguin using RVS spectra and Monte Carlo realisations (float, Temperature[K])

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Upper confidence level (84%) of the calibrated effective temperature (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from RVS spectra. Lower and upper levels include 68% confidence interval.

LOGG_GSPSPEC_CALIB : Calibrated logarithm of the stellar surface gravity from GSP-Spec MatisseGauguin using RVS spectra and Monte Carlo realisations (float, GravitySurface[log cgs])

Calibrated value of the median logarithm of the stellar surface gravity (assuming the source is a single star), as determined by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from RVS spectra, using literature references.

LOGG_GSPSPEC_LOWER_CALIB : Calibrated 16th percentile of the logarithm of the stellar surface gravity from GSP-Spec MatisseGauguin using RVS spectra and Monte Carlo realisations (float, GravitySurface[log cgs])

Lower confidence level (16%) of the calibrated value of logarithm of the stellar surface gravity (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from RVS spectra. Lower and upper levels include 68% confidence interval.

LOGG_GSPSPEC_UPPER_CALIB : Calibrated 84th percentile of the logarithm of the stellar surface gravity from GSP-Spec MatisseGauguin using RVS spectra and Monte Carlo realisations (float, GravitySurface[log cgs])

Upper confidence level (84%) of the calibrated value of logarithm of the stellar surface gravity (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from RVS spectra. Lower and upper levels include 68% confidence interval.

MH_GSPSPEC_CALIB : Calibrated global metallicity [M/H] from GSP-Spec MatisseGauguin using RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Calibrated value of the median global metallicity (assuming the source is a single star), as determined by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from RVS spectra, using literature references.

MH_GSPSPEC_LOWER_CALIB : Calibrated 16th percentile of global metallicity [M/H] from GSP-Spec MatisseGauguin using RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Lower confidence level (16%) of the calibrated global metallicity (assuming source is a single

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star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from RVS spectra. Lower and upper levels include 68% confidence interval.

MH_GSPSPEC_UPPER_CALIB : Calibrated 84th percentile of global metallicity [M/H] from GSP-Spec MatisseGauguin using RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Upper confidence level (84%) of the calibrated global metallicity (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from RVS spectra. Lower and upper levels include 68% confidence interval.

ALPHAFE_GSPSPEC_CALIB : Calibrated abundance of alpha-elements [alpha/Fe] with respect to iron from GSP-Spec MatisseGauguin using RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Calibrated value of the median abundance of alpha-elements with respect to iron (assuming the source is a single star), as determined by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from RVS spectra, using literature references. The considered alpha elements are: O, Ne, Mg, Si, S, Ar, Ca, Ti.

ALPHAFE_GSPSPEC_LOWER_CALIB : Calibrated 16th percentile of [alpha/Fe] from GSP-Spec MatisseGauguin using RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Lower confidence level (16%) of the calibrated abundance of alpha-elements with respect to iron (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from RVS spectra. Lower and upper levels include 68% confidence interval.

ALPHAFE_GSPSPEC_UPPER_CALIB : Calibrated 84th percentile of [alpha/Fe] from GSP-Spec MatisseGauguin using RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Upper confidence level (84%) of the calibrated abundance of alpha-elements with respect to iron (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from RVS spectra. Lower and upper levels include 68% confidence interval.

TEFF_GSPSPEC : Effective temperature from GSP-Spec MatisseGauguin using RVS spectra and Monte Carlo realisations (float, Temperature[K])

Median value of the effective temperature (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from RVS spectra and Monte Carlo realisations.

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TEFF_GSPSPEC_LOWER : 16th percentile of effective temperature from GSP-Spec MatisseGauguin using RVS spectra and Monte Carlo realisations (float, Temperature[K])

Lower confidence level (16%) of the median effective temperature (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from RVS spectra. Lower and upper levels include 68% confidence interval.

TEFF_GSPSPEC_UPPER : 84th percentile of effective temperature from GSP-Spec MatisseGauguin using RVS spectra and Monte Carlo realisations (float, Temperature[K])

Upper confidence level (84%) of the median effective temperature (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from RVS spectra. Lower and upper levels include 68% confidence interval.

LOGG_GSPSPEC : Logarithm of the stellar surface gravity from GSP-Spec MatisseGauguin using RVS spectra and Monte Carlo realisations (float, GravitySurface[log cgs])

Median value of logarithm of the stellar surface gravity (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from RVS spectra.

LOGG_GSPSPEC_LOWER : 16th percentile of the logarithm of the stellar surface gravity from GSP-Spec MatisseGauguin using RVS spectra and Monte Carlo realisations (float, GravitySurface[log cgs])

Lower confidence level (16%) of the median value of logarithm of the stellar surface gravity (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from RVS spectra. Lower and upper levels include 68% confidence interval.

LOGG_GSPSPEC_UPPER : 84th percentile of the logarithm of the stellar surface gravity from GSP-Spec MatisseGauguin using RVS spectra and Monte Carlo realisations (float, GravitySurface[log cgs])

Upper confidence level (84%) of the median value of logarithm of the stellar surface gravity (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from RVS spectra. Lower and upper levels include 68% confidence interval.

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MH_GSPSPEC : Global metallicity [M/H] from GSP-Spec MatisseGauguin using RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Median global metallicity (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from RVS spectra.

MH_GSPSPEC_LOWER : 16th percentile of global metallicity [M/H] from GSP-Spec MatisseGauguin using RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Lower confidence level (16%) of the median global metallicity (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from RVS spectra. Lower and upper levels include 68% confidence interval.

MH_GSPSPEC_UPPER : 84th percentile of global metallicity [M/H] from GSP-Spec MatisseGauguin using RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Upper confidence level (84%) of the median global metallicity (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from RVS spectra. Lower and upper levels include 68% confidence interval.

ALPHAFE_GSPSPEC : Abundance of alpha-elements [alpha/Fe] with respect to iron from GSP-Spec MatisseGauguin using RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Median abundance of alpha-elements with respect to iron (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from RVS spectra. The considered alpha elements are: O, Ne, Mg, Si, S, Ar, Ca, Ti.

ALPHAFE_GSPSPEC_LOWER : 16th percentile of [alpha/Fe] from GSP-Spec MatisseGauguin using RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of alpha-elements with respect to iron (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from RVS spectra. Lower and upper levels include 68% confidence interval.

ALPHAFE_GSPSPEC_UPPER : 84th percentile of [alpha/Fe] from GSP-Spec MatisseGauguin using RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of alpha-elements with respect to iron

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(assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from RVS spectra. Lower and upper levels include 68% confidence interval.

FEH_GSPSPEC : Global iron abundance [Fe/H] from GSP-Spec MatisseGauguin using RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Median iron abundance [Fe/H] (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from RVS spectra.

FEH_GSPSPEC_LOWER : 16th percentile of global iron abundance [Fe/H] from GSP-Spec MatisseGauguin using RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Lower confidence level (16%) of the median iron abundance (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from RVS spectra. Lower and upper levels include 68% confidence interval.

FEH_GSPSPEC_UPPER : 84th percentile of global iron abundance [Fe/H] from GSP-Spec MatisseGauguin using RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Upper confidence level (84%) of the median iron abundance (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from RVS spectra. Lower and upper levels include 68% confidence interval.

FLAGS_GSPSPEC : Catalogue flags for GSP-Spec MatisseGauguin (string)

Definitions of each character in the GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) quality flag chain. In this chain, value ‘0’ is the best, and ‘9’ is the worst. Flag names are split in two categories: parameter flags (green) and individual abundance flags (blue). In the following, the term ‘gof’ refers to the chi-square value beteween the input spectrum fluxes array and the solution spectrum fluxes array.

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Chain character number - name	Considered quality aspect	Possible adopted values
1 **vbroadT**	vbroad induced bias in $T_{\rm eff}$	0,1,2,9
2 **vbroadG**	vbroad induced bias in $\log g$	0,1,2,9
3 **vbroadM**	vbroad induced bias in [M/H]	0,1,2,9
4 **vradT**	vrad induced bias in $T_{\rm eff}$	0,1,2,9
5 **vradG**	vrad induced bias in $\log g$	0,1,2,9
6 **vradM**	vrad induced bias in [M/H]	0,1,2,9
7 **fluxNoise**	flux noise uncertainties	0,1,2,3,4,5,9
8 **extrapol**	extrapolation	0,1,2,3,4,9
9 **neg_flux**	negative flux pixels	0,1,9
10 **nanFlux**	NaN flux pixels	0,9
11 **emission**	emission line	0,9
12 **nullFluxErr**	null uncertainties	0,9
13 **KMgiantPar**	KM-type giant stars	0,1,2
14 **NUpLim**	nitrogen abundance upper limit	0,1,2,9
15 **NUncer**	nitrogen abundance uncertainty quality	0,1,2,9
16 **MgUpLim**	magnesium abundance upper limit	0,1,2,9
17 **MgUncer**	magnesium abundance uncertainty quality	0,1,2,9
18 **SiUpLim**	silicon abundance upper limit	0,1,2,9
19 **SiUncer**	silicon abundance uncertainty quality	0,1,2,9
20 **SUpLim**	sulphur abundance upper limit	0,1,2,9
21 **SUncer**	sulphur abundance uncertainty quality	0,1,2,9
22 **CaUpLim**	calcium abundance upper limit	0,1,2,9
23 **CaUncer**	calcium abundance uncertainty quality	0,1,2,9
24 **TiUpLim**	titanium abundance upper limit	0,1,2,9
25 **TiUncer**	titanium abundance uncertainty quality	0,1,2,9
26 **CrUpLim**	chromium abundance upper limit	0,1,2,9
27 **CrUncer**	chromium abundance uncertainty quality	0,1,2,9
28 **FeUpLim**	neutral iron abundance upper limit	0,1,2,9
29 **FeUncer**	neutral iron abundance uncertainty quality	0,1,2,9
30 **FeIIUpLim**	ionised iron abundance upper limit	0,1,2,9
31 **FeIIUncer**	ionised iron abundance uncertainty quality	0,1,2,9
32 **NiUpLim**	nickel abundance upper limit	0,1,2,9
33 **NiUncer**	nickel abundance uncertainty quality	0,1,2,9
34 **ZrUpLim**	zirconium abundance upper limit	0,1,2,9
35 **ZrUncer**	zirconium abundance uncertainty quality	0,1,2,9
36 **CeUpLim**	cerium abundance upper limit	0,1,2,9
37 **CeUncer**	cerium abundance uncertainty quality	0,1,2,9
38 **NdUpLim**	neodymium abundance upper limit	0,1,2,9
39 **NdUncer**	neodymium abundance uncertainty quality	0,1,2,9

Draft

Definition of parameter flags considering potential biases due to rotational velocity and macro-turbulence:

Flag name	Condition	Flag value
vbroadT	$\Delta T_{\rm eff}$>2000 K	Filter all Flag = 9
	500< $\Delta T_{\rm eff}$≤2000 K	Flag = 2
	250< $\Delta T_{\rm eff}$≤500 K	Flag = 1
	$\Delta T_{\rm eff}$≤250 K	Flag = 0
vbroadG	$\Delta\log g$>2 dex	Filter all *except* $T_{\rm eff}$ *if* $T_{\rm eff}$>7000 *K* Flag = 9
	1< $\Delta\log g$≤2 dex	Flag = 2
	0.5< $\Delta\log g$≤1 dex	Flag = 1
	$\Delta\log g$≤0.5 dex	Flag = 0
vbroadM	Δ[M/H]>2 dex	Filter [M/H] and [X/Fe] Flag = 9
	0.5< Δ[M/H]≤2 dex	Flag = 2
	0.25< Δ[M/H]≤0.5 dex	Flag = 1
	Δ[M/H]≤0.25 dex	Flag = 0

Definition of parameter flags considering potential biases due to uncertainties in the radial velocity shift correction:

Flag name	Condition	Flag value
vradT	$\Delta T_{\rm eff}$>2000 K	Filter all Flag = 9
	500< $\Delta T_{\rm eff}$≤2000 K	Flag = 2
	250< $\Delta T_{\rm eff}$≤500 K	Flag = 1
	$\Delta T_{\rm eff}$≤250 K	Flag = 0
vradG	$\Delta\log g$>2 dex	Filter all *except* $T_{\rm eff}$ *if* $T_{\rm eff}$>7000 *K* Flag = 9
	1< $\Delta\log g$≤2 dex	Flag = 2
	0.5< $\Delta\log g$≤1 dex	Flag = 1
	$\Delta\log g$≤0.5 dex	Flag = 0
vradM	Δ[M/H]>2 dex	Filter [M/H] and [X/Fe] Flag = 9
	0.5< Δ[M/H]≤2 dex	Flag = 2
	0.25< Δ[M/H]≤0.5 dex	Flag = 1
	Δ[M/H]≤0.25 dex	Flag = 0

Draft

Definition of parameter flags considering potential biases due to uncertainties in the RVS flux:

Flag name	Condition	Flag value
fluxNoise	$\sigma T_{\rm eff}$>2000 K or σlog g>2 dex	Filter all Flag = 9
	$\sigma T_{\rm eff}$≤2000 K and σlog g≤2 dex and σ[M/H]>2 dex	Filter [M/H], [X/Fe] Flag = 5
	$\sigma T_{\rm eff}$≤2000 K and σlog g≤2 dex and σ[M/H]≤2 dex and σ[α/Fe]>0.8 dex	Filter [α/Fe], [X/Fe] Flag = 4
	500< $\sigma T_{\rm eff}$≤2000 K and 1< σlog g≤2 dex and 0.5< σ[M/H]≤2 dex and 0.2< σ[α/Fe]≤0.8 dex	Flag = 3
	250< $\sigma T_{\rm eff}$≤500 K and 0.5< σlog g≤1 dex and 0.25< σ[M/H]≤0.5 dex and 0.1< σ[α/Fe]≤0.2 dex	Flag = 2
	100< $\sigma T_{\rm eff}$≤250 K and 0.2< σlog g≤0.5 dex and 0.1< σ[M/H]≤0.25 dex and 0.05< σ[α/Fe]≤0.1 dex	Flag = 1
	$\sigma T_{\rm eff}$≤100 K and σlog g≤0.2 dex and σ[M/H]≤0.1 dex and σ[α/Fe]≤0.05 dex	Flag = 0

Definition of parameter flags considering potential biases due to extrapolated parameters:

Draft

Flag name	Condition	Flag value
extrapol	gof=NaN and ($T_{\rm eff}$>9000 K or $T_{\rm eff}$<2500 K or $\log g$>6 dex or $\log g$< −1 dex)	Filter all *if $T_{\rm eff}$>7000 K* Flag = 9
	gof=NaN and 2500≤$T_{\rm eff}$≤9000 K and −1 ≤$\log g$≤6 dex and ([M/H]< −6 dex or [M/H]>1.5 dex)	Filter [M/H],[X/Fe] Flag = 4
	gof=NaN and 2500≤$T_{\rm eff}$<9000 K and −1 ≤$\log g$≤6 dex and −6 ≤[M/H]≤1.5 dex and [α/Fe] out from standard by ± 0.8	Filter [X/Fe] Flag = 3
	gof=NaN and 2500≤$T_{\rm eff}$≤9000 K and −1 ≤$\log g$≤6 dex and −6 ≤[M/H]≤1.5 dex and [α/Fe] within ± 0.8 from standard	Flag = 2
	gof≠NaN and ($T_{\rm eff}$≥7625 or $T_{\rm eff}$≤3500 K or $\log g$≥4.75 or $\log g$≤0.25 dex or [M/H]≤ −3 or [M/H]≥0.75 dex or [α/Fe] out from standard by ± 0.35)	Flag = 1
	gof≠NaN and 3500<$T_{\rm eff}$<7625 K and 0.25<$\log g$<4.75 dex and −3 <[M/H]<0.75 dex and [α/Fe] within ± 0.35 from standard	Flag = 0

Definition of parameter flags considering RVS flux problems or emission line probability:

Draft

Flag name	Condition	Flag value
nanFlux	Flux=NaN	Filter all *if* $T_{\rm eff}$>7000 K Flag = 9
	no Nans in Flux	Flag = 0
emission	*CU6_is_emission*=True	Filter all *if* $T_{\rm eff}$>7000 K Flag = 9
	CU6_is_emission=False	Flag = 0
neg_flux	> 2 pixels with flux<0	Filter all *if* $T_{\rm eff}$>7000 K Flag = 9
	1 or 2 pixels with flux<0	Flag = 1
	flux>0	Flag = 0
nullFluxErr	$\sigma T_{\rm eff}$=0 K or σlog g=0 dex or σ[M/H]=0 dex or σ[α/Fe]=0 dex	Filter all Flag = 9
	no null uncertainties for all pixel flux	Flag = 0

Definition of parameter flags considering problems in the parameterisation of KM-type giants. $F_{\rm min}$ is the minumum flux value in the corresponding RVS spectrum:

Flag name	Condition	Flag value
KMgiantPar	$T_{\rm eff}$<4000 K and log g<3.5 and (gof> −3.0 or $F_{\rm min}$>0.22)	Filter [α/Fe] Flag = 2
	$T_{\rm eff}$<4000 K and log g<3.5 and (−3.4 <gof< −3.0 or (gof> −3.0 and $F_{\rm min}$<0.22) or (gof< −3.4 and $F_{\rm min}$>0.22))	Filter [α/Fe] Flag = 1
	($T_{\rm eff}$<4000 K and log g<3.5 and (gof< −3.4 or $F_{\rm min}$<0.22)) or $T_{\rm eff}$<4000 or log g>3.5	Flag = 0

Definition of individual abundance upper limit flags. Xfe_gspspec_upper is the upper confidence value of the abundance, corresponding to the 84th quantile of the Monte Carlo distribution. σ[X/Fe] is the 84th−16th interquantile abundance uncertainty. XfeUpperLimit is the mean value of the abundance upper limit for the considered lines of element X in the spectrum, depending on the mean signal-to-noise ratio (SNR) in the line pixels and on the stellar parameters. X_MAD_UpperLimit is the median absolute deviation of the upper limit in the line pixels; the coefficients c1 to c8 are reported in a table further below:

Draft

Flag name	Condition	Flag value
XUpLim	vbroadT≥2 or vbroadG≥2 or vbroadM≥2 or σ[X/Fe]=0 or (Xfe_gspspec_upper−Xfe_gspspec)= 0 or $T_{\rm eff}$≤c1 or $T_{\rm eff}$≥c2 or log g≤c3 or log g≥c4 or ((2− XfeUpperLimit)/σ[X/Fe])≤c5 or (SNR≤c6 and gof≥c7) or (Xfe_gspspec+[M/H])≤c8	Filter [X/Fe] Flag = 9
	vbroadT<2 and vbroadG<2 and vbroadM<2 and σ[X/Fe]≠0 and (Xfe_gspspec_upper− Xfe_gspspec)≠ 0 and c1<$T_{\rm eff}$<c2 and c3<log g<c4 and ((2− XfeUpperLimit)/σ[X/Fe])>c5 and (SNR>c6 or (SNR≤c6 and gof<c7)) and (Xfe_gspspec+[M/H])<c8 and ((Xfe_gspspec− XfeUpperLimit)/(1.48·X_MAD_UpperLimit))<1.5	Flag = 2
	vbroadT<2 and vbroadG<2 and vbroadM<2 and σ[X/Fe]≠0 and (Xfe_gspspec_upper− Xfe_gspspec)≠ 0 and c1<$T_{\rm eff}$<c2 and c3<log g<c4 and ((2− XfeUpperLimit)/σ[X/Fe])>c5 and (SNR>c6 or (SNR≤c6 and gof<c7)) and (Xfe_gspspec+[M/H])<c8 and 1.5≤((Xfe_gspspec− XfeUpperLimit)/(1.48·X_MAD_UpperLimit))<2.5	Flag = 1
	vbroadT<2 and vbroadG<2 and vbroadM<2 and σ[X/Fe]≠0 and (Xfe_gspspec_upper− Xfe_gspspec)≠ 0 and c1<$T_{\rm eff}$<c2 and c3<log g<c4 and ((2− XfeUpperLimit)/σ[X/Fe])>c5 and (SNR>c6 or (SNR≤c6 and gof<c7)) and (Xfe_gspspec+[M/H])<c8 and ((Xfe_gspspec− XfeUpperLimit)/(1.48·X_MAD_UpperLimit))≥2.5	Flag = 0

Definition of individual abundance uncertainty flags. Xfe_gspspec_upper is the upper confidence value of the abundance, corresponding to the 84th quantile of the Monte Carlo distribution. σ[X/Fe] is the 84th−16th interquantile abundance uncertainty. XfeUpperLimit is the mean value of the abundance upper limit for the considered lines of element X in the spectrum, depending on the mean SNR in the line pixels and the stellar parameters. X_MAD_UpperLimit is the median absolute deviation of the upper limit in the line pixels; the coefficients c1 to c8 are reported in a table further below:

Flag name	Condition	Flag value
XUncer	vbroadT≥2 or vbroadG≥2 or vbroadM≥2 or σ[X/Fe]=0 or (Xfe_gspspec_upper−Xfe_gspspec)= 0 or $T_{\rm eff}$≤c1 or $T_{\rm eff}$≥c2 or log g≤c3 or log g≥c4 or ((2−XfeUpperLimit)/σ[X/Fe])≤c5 or (SNR≤c6 and gof≥c7) or (Xfe_gspspec+[M/H])≤c8	Filter [X/Fe] Flag = 9
	vbroadT<2 and vbroadG<2 and vbroadM<2 and σ[X/Fe]≠0 and (Xfe_gspspec_upper−Xfe_gspspec)≠ 0 and c1<$T_{\rm eff}$<c2 and c3<log g<c4 and c5<((2−XfeUpperLimit)/σ[X/Fe])<7 and (SNR>c6 or (SNR≤c6 and gof<c7)) and (Xfe_gspspec+[M/H])<c8	Flag = 2
	vbroadT<2 and vbroadG<2 and vbroadM<2 and σ[X/Fe]≠0 and (Xfe_gspspec_upper−Xfe_gspspec)≠ 0 and c1<$T_{\rm eff}$<c2 and c3<log g<c4 and 7≤((2−XfeUpperLimit)/σ[X/Fe])<10 and (SNR>c6 or (SNR≤c6 and gof<c7)) and (Xfe_gspspec+[M/H])<c8	Flag = 1
	vbroadT<2 and vbroadG<2 and vbroadM<2 and σ[X/Fe]≠0 and (Xfe_gspspec_upper−Xfe_gspspec)≠ 0 and c1<$T_{\rm eff}$<c2 and c3<log g<c4 and ((2−XfeUpperLimit)/σ[X/Fe])≥10 and (SNR>c6 or (SNR≤c6 and gof<c7)) and (Xfe_gspspec+[M/H])<c8	Flag = 0

Coefficients for individual chemical abundance filtering used in the previous two tables:

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Chemical abundance	c1	c2	c3	c4	c5	c6	c7	c8
[N/Fe]	4200	8000	0.0	5.5	4.5	100	−3.6	99
[Mg/Fe]	3500	8000	−1.0	5.5	5.5	80	−3.5	99
[Si/Fe]	4000	8000	−1.0	5.5	6.0	110	−3.8	99
[S/Fe]	5500	8000	3.0	5.5	5.0	120	−3.7	99
[Ca/Fe]	3500	8000	−1.0	5.5	10.0	60	−3.2	99
[Ti/Fe]	4000	6500	−1.0	5.5	6.0	110	−3.65	99
[Cr/Fe]	3500	6000	−1.0	5.5	6.0	1000	−3.65	1.5
[Fe/M]	3500	8000	−1.0	5.5	5.0	1000	−3.4	1.5
[FeII/M]	5700	8000	3.5	5.5	5.0	70	−3.5	1.5
[Ni/Fe]	4000	6500	−1.0	5.5	6.0	100	−3.6	1.5
[Zr/Fe]	3500	8000	−1.0	5.5	1.0	100	−3.4	99
[Ce/Fe]	3500	8000	−1.0	5.5	5.0	100	−3.5	99
[Nd/Fe]	3500	5500	−1.0	5.5	2.0	100	−3.5	99

LOGCHISQ_GSPSPEC : Logarithm of the goodness-of-fit for the GSP-Spec MatisseGauguin parameters (float)

Logarithm to the base 10 of the chi-squared between input spectrum rebinned to 800 pixels and solution synthetic spectrum computed from GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023).

CONTPOLYFIT0_GSPSPEC : Independent term of the continuum normalization polynomial used by GSP-Spec (float)

Independent term p_0 of the polynomial used by GSP-Spec for the continuum normalization: $p(x) = p_0 + p_1 \cdot x + p_2 \cdot x^2 + p_3 \cdot x^3 + p_4 \cdot x^4$.

CONTPOLYFIT1_GSPSPEC : Coefficient of the first degree term in the continuum normalization polynomial used by GSP-Spec (float)

Coefficient of the first degree term p_1 in the continuum normalization polynomial used by GSP-Spec: $p(x) = p_0 + p_1 \cdot x + p_2 \cdot x^2 + p_3 \cdot x^3 + p_4 \cdot x^4$.

CONTPOLYFIT2_GSPSPEC : Coefficient of the second degree term in the continuum normalization polynomial used by GSP-Spec (float)

Coefficient of the second degree term p_2 in the continuum normalization polynomial used by

GSP-Spec: $p(x) = p_0 + p_1 \cdot x + p_2 \cdot x^2 + p_3 \cdot x^3 + p_4 \cdot x^4$.

CONTPOLYFIT3_GSPSPEC : Coefficient of the third degree term in the continuum normalization polynomial used by GSP-Spec (float)

Coefficient of the third degree term p_3 in the continuum normalization polynomial used by GSP-Spec: $p(x) = p_0 + p_1 \cdot x + p_2 \cdot x^2 + p_3 \cdot x^3 + p_4 \cdot x^4$.

CONTPOLYFIT4_GSPSPEC : Coefficient of the fourth degree term in the continuum normalization polynomial used by GSP-Spec (float)

Coefficient of the fourth degree term p_4 in the continuum normalization polynomial used by GSP-Spec: $p(x) = p_0 + p_1 \cdot x + p_2 \cdot x^2 + p_3 \cdot x^3 + p_4 \cdot x^4$.

RADIUS_FLAME_SPEC : Radius of the star from FLAME using effective temperature from GSP-Spec and luminosity from FLAME (float, Length & Distance[Solar Radius])

The radius of the star from FLAME, derived from `teff_gspspec` and `lum_flame_spec` using the Stefan-Boltzmann law. The solar effective temperature is 5772 K, see Section **??**. It is defined as the median value (50th percentile) of the distribution from sampling.

RADIUS_FLAME_SPEC_LOWER : Lower confidence level (16%) of FLAME radius (float, Length & Distance[Solar Radius])

Lower confidence level of `radius_flame_spec`. It is defined as the 16th percentile value of the distribution from sampling. Lower and upper levels include 68% confidence interval.

RADIUS_FLAME_SPEC_UPPER : Upper confidence level (84%) of FLAME radius (float, Length & Distance[Solar Radius])

Upper confidence level of `radius_flame_spec`. It is defined as the 84th percentile value of the distribution from sampling. Lower and upper levels include 68% confidence interval.

LUM_FLAME_SPEC : Luminosity of the star from FLAME using G band magnitude, extinction, parallax and a bolometric correction (float, Luminosity[Solar Luminosity])

Luminosity of the star from FLAME using G-band magnitude, extinction from GSP-Phot `ap_xp.ag_gspphot`

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parallax and a bolometric correction `bc_flame_spec`. It is defined as the median value (50th percentile) of the distribution from sampling. The bolometric correction depends on the effective temperature, metallicity and surface gravity, and these are based on GSP-Spec values. The bolometric magnitude of the Sun = 4.74 mag, see Section **??** and the reference absolute G-band magnitude of the Sun is 4.66 mag, see Section **??**.

LUM_FLAME_SPEC_LOWER : Lower confidence level (16%) of FLAME luminosity (float, Luminosity[Solar Luminosity])

Lower confidence level of `lum_flame_spec`. It is defined as the 16th percentile value of the distribution from sampling. Lower and upper levels include 68% confidence interval.

LUM_FLAME_SPEC_UPPER : Upper confidence level (84%) of FLAME luminosity (float, Luminosity[Solar Luminosity])

Upper confidence level of `lum_flame_spec`. It is defined as the 84th percentile value of the distribution from sampling. Lower and upper levels include 68% confidence interval.

MASS_FLAME_SPEC : Mass of the star from FLAME using stellar models, the FLAME luminosity and GSP-Spec parameters (float, Mass[Solar Mass])

Mass of the star from FLAME using GSP-Spec based parameters as input. It is defined as the median value (50th percentile) of the 1D projected distribution from sampling in mass, age, and metallicity. It is derived by comparing `teff_gspspec`, `lum_flame_spec`, `mh_gspspec`, and `logg_gspspec` to BASTI stellar evolution models (Hidalgo et al. 2018). The input metallicity to the model-based inference has been corrected for differences in the reference solar parameters between the atmospheric and evolution models, and when available, the alpha abundance is also included, see **?** and Section **??** in the online documentation for details.

MASS_FLAME_SPEC_LOWER : Lower confidence level (16%) of FLAME mass (float, Mass[Solar Mass])

Lower confidence level of `mass_flame_spec`. It is defined as the 16th percentile value of the 1D projected distribution from sampling in mass, age, and metallicity. Lower and upper levels include 68% confidence interval.

MASS_FLAME_SPEC_UPPER : Upper confidence level (84%) of FLAME mass (float, Mass[Solar Mass])

Upper confidence level of `mass_flame_spec`. It is defined as the 84th percentile value of the 1D projected distribution from sampling in mass, age, and metallicity. Lower and upper levels include 68% confidence interval.

AGE_FLAME_SPEC : Age of the star from FLAME using stellar models, see FLAME mass for details (float, Time[Gyr])

Age of the star from FLAME using GSP-Spec based parameters as input. It is defined as the median value (50th percentile) of the 1D projected distribution from sampling in mass, age, and metallicity, see `mass_flame_spec` for details.

AGE_FLAME_SPEC_LOWER : Lower confidence level (16%) of FLAME age (float, Time[Gyr])

Lower confidence level of `age_flame_spec`. It is defined as the 16th percentile value of the 1D projected distribution from sampling in mass, age, and metallicity. Lower and upper levels include 68% confidence interval.

AGE_FLAME_SPEC_UPPER : Upper confidence level (84%) of FLAME age (float, Time[Gyr])

Upper confidence level of `age_flame_spec`. It is defined as the 84th percentile value of the 1D projected distribution from sampling in mass, age, and metallicity. Lower and upper levels include 68% confidence interval.

FLAGS_FLAME_SPEC : Flag indicating quality of parameters from FLAME (string)

This field contains the quality and processing flag for the data products from FLAME derived using GSP-Spec atmospheric parameters. It is a string that consists of four digits 'ABCD'. A value of '0' generally indicates nothing to report.

Details of the description of the flag is given in Section **??** and the user is referred to this section for details. Example uses are also made available there. The four digits provide information on the quality of the input data to FLAME for each source along with an indication of quality of the FLAME parameters and some processing information.

A provides quality information on the non-Apsis input data (0, 1, 2, or 3)

B warning information on the Apsis input data (0 or 1)

C warning information on some FLAME parameters (0, 1, 2 or 3)

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D provides additional processing information in particular if a parameter is missing (0 – 6)

See Section **??** of the online documentation for further details.

EVOLSTAGE_FLAME_SPEC : Evolutionary stage of the star from FLAME using stellar models, see FLAME mass for details (float)

Evolution stage of the star from FLAME. It is a float value typically between 100 and 1300 and defined using the median value (50th percentile) of the 1D projected distribution from sampling in mass, age, and metallicity, see `mass_flame_spec` for details. The value is adapted from the BASTI model grid (Hidalgo et al. 2018) which adopts the following convention:

- 100 = zero age main sequence (ZAMS)
- 300 = first minimum of $T_{\rm eff}$ for massive stars or central hydrogen mass fraction = 0.30 for low-mass stars
- 360 = main sequence turn-off
- 420 = central hydrogen mass fraction = 0.00
- 490 = base of the red giant branch (RGB)
- 860 = maximum $\mathcal{L}$ along the RGB bump
- 890 = minimum $\mathcal{L}$ along the RGB bump
- 1290 = tip of the RGB

EVOLSTAGE_FLAME_SPEC_LOWER : Lower confidence level (16%) of the FLAME evolutionary stage (float)

Lower confidence level of `evolstage_flame_spec`. It is defined as the 16th percentile value of the distribution from sampling. Lower and upper levels include 68% confidence interval.

EVOLSTAGE_FLAME_SPEC_UPPER : Upper confidence level (84%) of the FLAME evolutionary stage (float)

Upper confidence level of `evolstage_flame_spec`. It is defined as the 84th percentile value of the distribution from sampling. Lower and upper levels include 68% confidence interval.

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GRAVREDSHIFT_FLAME_SPEC : Radial velocity correction due to the gravitational redshift from FLAME using GSP-Spec logg and FLAME radius (float, Velocity[km s^{-1}])

Radial velocity correction due to gravitational redshift from FLAME derived using `radius_flame_spec` and `logg_gspspec`.

GRAVREDSHIFT_FLAME_SPEC_LOWER : Lower confidence level (16%) of the FLAME parameter gravredshift (float, Velocity[km s^{-1}])

Lower confidence level of `gravredshift_flame_spec`. It is defined as the 16th percentile value of the distribution from sampling. Lower and upper levels include 68% confidence interval.

GRAVREDSHIFT_FLAME_SPEC_UPPER : Upper confidence level (84%) of the FLAME parameter gravredshift (float, Velocity[km s^{-1}])

Upper confidence level of `gravredshift_flame_spec`. It is defined as the 84th percentile value of the distribution from sampling. Lower and upper levels include 68% confidence interval.

BC_FLAME_SPEC : Bolometric correction applied to derive the FLAME luminosity using GSP-Spec parameters (float, Magnitude[mag])

Bolometric correction for the G-band magnitude used to derive `lum_flame_spec`. It is defined as the median value (50th percentile) of the distribution from sampling. It is a function of effective temperature, surface gravity, and metallicity, and has been derived from MARCS models, see Section **??** of the online documentation, using GSP-Spec-based parameters. The bolometric correction for the Sun is defined as +0.062 mag, see Section **??**, where $M_{bol,\odot}$ = 4.74 mag, see Section **??**, i.e. $M_{G\odot}$ = 4.678 mag.

BC_FLAME_SPEC_LOWER : Lower confidence level (16%) of the FLAME bolometric correction (float, Magnitude[mag])

Lower confidence level of `bc_flame_spec`. It is defined as the 16th percentile value of the distribution from sampling. Lower and upper levels include 68% confidence interval.

BC_FLAME_SPEC_UPPER : Upper confidence level (84%) of the FLAME bolometric correction (float, Magnitude[mag])

Upper confidence level of `bc_flame_spec`. It is defined as the 84th percentile value of the distribution from sampling. Lower and upper levels include 68% confidence interval.

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MH_FLAME_MODEL_SPEC : Model metallicity from FLAME using stellar models, see FLAME mass for details (float, Abundances[dex])

Model metallicity from FLAME using GSP-Spec and FLAME input parameters. It is the derived model metallicity corresponding to the mass and age determination. It is defined as the median value (50th percentile) of the 1D projected distribution from sampling in mass, age, and metallicity, see `mass_flame_spec` for details.

MH_FLAME_MODEL_SPEC_LOWER : Lower confidence level (16%) of the model metallicity from FLAME (float, Abundances[dex])

Lower confidence level of `mh_flame_model_spec`. It is defined as the 16th percentile value of the 1D projected distribution from sampling in mass, age, and metallicity. Lower and upper levels include 68% confidence interval.

MH_FLAME_MODEL_SPEC_UPPER : Upper confidence level (84%) of the model metallicity from FLAME (float, Abundances[dex])

Upper confidence level of `mh_flame_model_spec`. It is defined as the 84th percentile value of the 1D projected distribution from sampling in mass, age, and metallicity. Lower and upper levels include 68% confidence interval.

LOGG_FLAME_MODEL_SPEC : Model surface gravity from FLAME (float, GravitySurface[log cgs])

Model surface gravity from FLAME using GSP-Spec and FLAME input parameters. It is the derived model surface gravity corresponding to the mass and age determination. It is defined as the median value (50th percentile) of the 1D projected distribution from sampling in mass, age, and metallicity, see `mass_flame_spec` for details.

LOGG_FLAME_MODEL_SPEC_LOWER : Lower confidence level (16%) of the model surface gravity from FLAME (float, GravitySurface[log cgs])

Lower confidence level of `logg_flame_model_spec`. It is defined as the 16th percentile value of the distribution from sampling. Lower and upper levels include 68% confidence interval.

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LOGG_FLAME_MODEL_SPEC_UPPER : Upper confidence level (84%) of the model surface gravity from FLAME (float, GravitySurface[log cgs])

Upper confidence level of `logg_flame_model_spec`. It is defined as the 84th percentile value of the distribution from sampling. Lower and upper levels include 68% confidence interval.

TEFF_FLAME_MODEL_SPEC : Model effective temperature from FLAME (float, Temperature[K])

Model effective temperature from FLAME using GSP-Spec and FLAME input parameters. It is the derived model effective temperature corresponding to the mass and age determination. It is defined as the median value (50th percentile) of the 1D projected distribution from sampling in mass, age, and metallicity, see `mass_flame_spec` for details.

TEFF_FLAME_MODEL_SPEC_LOWER : Lower confidence level (16%) of the model effective temperature from FLAME (float, Temperature[K])

Lower confidence level of `teff_flame_model_spec`. It is defined as the 16th percentile value of the distribution from sampling. Lower and upper levels include 68% confidence interval.

TEFF_FLAME_MODEL_SPEC_UPPER : Upper confidence level (84%) of the model effective temperature from FLAME (float, Temperature[K])

Upper confidence level of `teff_flame_model_spec`. It is defined as the 84th percentile value of the distribution from sampling. Lower and upper levels include 68% confidence interval.

LUM_FLAME_MODEL_SPEC : Model luminosity from FLAME (float, Luminosity[Solar Luminosity])

Model luminosity from FLAME using GSP-Spec and FLAME input parameters. It is the derived model luminosity corresponding to the mass and age determination. It is defined as the median value (50th percentile) of the 1D projected distribution from sampling in mass, age, and metallicity, see `mass_flame_spec` for details.

LUM_FLAME_MODEL_SPEC_LOWER : Lower confidence level (16%) of the model luminosity from FLAME (float, Luminosity[Solar Luminosity])

Lower confidence level of `lum_flame_model_spec`. It is defined as the 16th percentile value of the distribution from sampling. Lower and upper levels include 68% confidence interval.

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LUM_FLAME_MODEL_SPEC_UPPER : Upper confidence level (84%) of the model luminosity from FLAME (float, Luminosity[Solar Luminosity])

Upper confidence level of `lum_flame_model_spec`. It is defined as the 84th percentile value of the distribution from sampling. Lower and upper levels include 68% confidence interval.

LOGPOSTERIOR_FLAME_SPEC : Median log-posterior of the MCMC samples from FLAME inference on model parameters (float)

Goodness-of-fit score defined as the median log-posterior of all MCMC samples of FLAME for the model parameter determination (mass, age, etc.).

TEFF_GSPSPEC_ANN : Effective temperature from GSP-Spec ANN using RVS spectra and Monte Carlo realisations (float, Temperature[K])

Median effective temperature (assuming source is a single star) inferred by GSP-Spec ANN (Recio-Blanco et al. 2023) from RVS spectra.

TEFF_GSPSPEC_ANN_LOWER : Lower confidence level (16%) of effective temperature from GSP-Spec ANN using RVS spectra and Monte Carlo realisations (float, Temperature[K])

Lower confidence level (16%) of median effective temperature (assuming source is a single star) inferred by GSP-Spec ANN (Recio-Blanco et al. 2023) from RVS spectra. Lower and upper levels include 68% confidence interval.

TEFF_GSPSPEC_ANN_UPPER : Upper confidence level (84%) of effective temperature from GSP-Spec ANN using RVS spectra and Monte Carlo realisations (float, Temperature[K])

Upper confidence level (84%) of median effective temperature (assuming source is a single star) inferred by GSP-Spec ANN (Recio-Blanco et al. 2023) from RVS spectra. Lower and upper levels include 68% confidence interval.

LOGG_GSPSPEC_ANN : Surface gravity from GSP-Spec ANN using RVS spectra and Monte Carlo realisations (float, GravitySurface[log cgs])

Median surface gravity (assuming source is a single star) inferred by GSP-Spec ANN (Recio-

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Blanco et al. 2023) from RVS spectra.

LOGG_GSPSPEC_ANN_LOWER : Lower confidence level (16%) of surface gravity from GSP-Spec ANN using RVS spectra and Monte Carlo realisations (float, GravitySurface[log cgs])

Lower confidence level (16%) of median surface gravity (assuming source is a single star) inferred by GSP-Spec ANN (Recio-Blanco et al. 2023) from RVS spectra. Lower and upper levels include 68% confidence interval.

LOGG_GSPSPEC_ANN_UPPER : Upper confidence level (84%) of surface gravity from GSP-Spec ANN using RVS spectra and Monte Carlo realisations (float, GravitySurface[log cgs])

Upper confidence level (84%) of median surface gravity (assuming source is a single star) inferred by GSP-Spec ANN (Recio-Blanco et al. 2023) from RVS spectra. Lower and upper levels include 68% confidence interval.

MH_GSPSPEC_ANN : Global metallicity from GSP-Spec ANN using RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Median global metallicity (assuming source is a single star) inferred by GSP-Spec ANN (Recio-Blanco et al. 2023) from RVS spectra.

MH_GSPSPEC_ANN_LOWER : Lower confidence level (16%) of global metallicity from GSP-Spec ANN using RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Lower confidence level (16%) of median global metallicity (assuming source is a single star) inferred by GSP-Spec ANN (Recio-Blanco et al. 2023) from RVS spectra. Lower and upper levels include 68% confidence interval.

MH_GSPSPEC_ANN_UPPER : Upper confidence level (84%) of global metallicity from GSP-Spec ANN using RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Upper confidence level (84%) of median global metallicity (assuming source is a single star) inferred by GSP-Spec ANN (Recio-Blanco et al. 2023) from RVS spectra. Lower and upper levels include 68% confidence interval.

ALPHAFE_GSPSPEC_ANN : Abundance of alpha-elements with respect to iron from GSP-Spec

ANN using RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Median abundance of alpha-elements with respect to iron (assuming source is a single star) inferred by GSP-Spec ANN (Recio-Blanco et al. 2023) from RVS spectra. The considered alpha elements are: O, Ne, Mg, Si, S, Ar, Ca, Ti.

ALPHAFE_GSPSPEC_ANN_LOWER : Lower confidence level (16%) of [alpha/Fe] from GSP-Spec ANN using RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Lower confidence level (16%) of median abundance of alpha-elements with respect to iron (assuming source is a single star) inferred by GSP-Spec ANN (Recio-Blanco et al. 2023) from RVS spectra. Lower and upper levels include 68% confidence interval.

ALPHAFE_GSPSPEC_ANN_UPPER : Upper confidence level (84%) of [alpha/Fe] from GSP-Spec ANN using RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Upper confidence level (84%) of median abundance of alpha-elements with respect to iron (assuming source is a single star) inferred by GSP-Spec ANN (Recio-Blanco et al. 2023) from RVS spectra. Lower and upper levels include 68% confidence interval.

LOGCHISQ_GSPSPEC_ANN : Logarithm of the goodness-of-fit for the GSP-Spec ANN parameters (float)

Logarithm to the base 10 of the chi-squared between input spectrum rebinned to 800 pixels and solution synthetic spectrum computed from GSP-Spec ANN (Recio-Blanco et al. 2023).

FLAGS_GSPSPEC_ANN : Catalogue flags for GSP-Spec ANN (string)

Catalogue flags for GSP-Spec ANN (Recio-Blanco et al. 2023), used to indicate the quality of the estimated parameters of this source. In this chain, value ‘0’ represents the best quality, and ‘9’ the worst:

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Chain character number - name	Considered quality aspect	Possible adopted values
1 **vbroadT**	vbroad induced bias in $T_{\rm eff}$	0,1,2,9
2 **vbroadG**	vbroad induced bias in $\log g$	0,1,2,9
3 **vbroadM**	vbroad induced bias in [M/H]	0,1,2,9
4 **vradT**	vrad induced bias in $T_{\rm eff}$	0,1,2,9
5 **vradG**	vrad induced bias in $\log g$	0,1,2,9
6 **vradM**	vrad induced bias in [M/H]	0,1,2,9
7 **fluxNoise**	flux noise uncertainties	0,1,2,3,9
8 **extrapol**	extrapolation	0,1,9
9 **neg_flux**	negative flux pixels	0,1,9
10 **nanFlux**	NaN flux pixels	0,9
11 **emission**	emission line	0,1,9
12 **nullFluxErr**	null uncertainties	0,1,9

Definition of parameter flags considering potential biases due to rotational velocity and macro-turbulence:

Flag name	Condition	Flag value
vbroadT	$\Delta T_{\rm eff}$>2000 K	Filter all Flag=9
	500< $\Delta T_{\rm eff}\leq$2000 K	Flag=2
	250< $\Delta T_{\rm eff}\leq$500 K	Flag=1
	$\Delta T_{\rm eff}\leq$250 K	Flag=0
vbroadG	$\Delta\log g$>2 dex	Filter all Flag=9
	1< $\Delta\log g\leq$2 dex	Flag=2
	0.5< $\Delta\log g\leq$1 dex	Flag=1
	$\Delta\log g\leq$0.5 dex	Flag=0
vbroadM	Δ[M/H]>2 dex	Filter [M/H] and [X/Fe] Flag=9
	0.5< Δ[M/H]$\leq$2 dex	Flag=2
	0.25< Δ[M/H]$\leq$0.5 dex	Flag=1
	Δ[M/H]$\leq$0.25 dex	Flag=0

Definition of parameter flags considering potential biases due to uncertainties in the radial velocity shift correction:

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Flag name	Condition	Flag value
vradT	$\Delta T_{\rm eff}$>2000 K	Filter all Flag=9
	500< $\Delta T_{\rm eff}$≤2000 K	Flag=2
	250< $\Delta T_{\rm eff}$≤500 K	Flag=1
	$\Delta T_{\rm eff}$≤250 K	Flag=0
vradG	$\Delta\log g$>2 dex	Filter all Flag=9
	1< $\Delta\log g$≤2 dex	Flag=2
	0.5< $\Delta\log g$≤1 dex	Flag=1
	$\Delta\log g$≤0.5 dex	Flag=0
vradM	Δ[M/H]>2 dex	Filter [M/H] and [X/Fe] Flag=9
	0.5< Δ[M/H]≤2 dex	Flag=2
	0.25< Δ[M/H]≤0.5 dex	Flag=1
	Δ[M/H]≤0.25 dex	Flag=0

Definition of parameter flags considering potential biases due to uncertainties in the RVS flux:

Flag name	Condition	Flag value
fluxNoise	$\sigma T_{\rm eff}$>527 K or $\sigma\log g$>0.916 dex or σ[M/H]>0.498 dex or $\sigma[\alpha/{\rm Fe}]$>0.163 dex	Filter all Flag = 9
	224< $\sigma T_{\rm eff}$≤527 K and 0.427< $\sigma\log g$≤0.916 dex or 0.185< σ[M/H]≤0.498 dex or 0.093< $\sigma[\alpha/{\rm Fe}]$≤0.163 dex	Flag = 3
	138< $\sigma T_{\rm eff}$≤224 K and 0.3< $\sigma\log g$≤0.427 dex or 0.119< σ[M/H]≤0.185 dex or 0.064< $\sigma[\alpha/{\rm Fe}]$≤0.093 dex	Flag = 2
	102< $\sigma T_{\rm eff}$≤138 K and 0.24< $\sigma\log g$≤0.3 dex or 0.09< σ[M/H]≤0.119 dex or 0.05< $\sigma[\alpha/{\rm Fe}]$≤0.064 dex	Flag = 1
	$\sigma T_{\rm eff}$≤102 K and $\sigma\log g$≤0.24 dex and σ[M/H]≤0.09 dex and $\sigma[\alpha/{\rm Fe}]$≤0.05 dex	Flag = 0

Definition of parameter flags considering potential biases due to extrapolated parameters:

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Flag name	Condition	Flag value
extrapol	$T_{\rm eff}$>8318 K or $T_{\rm eff}$<3682 K or log g>6.06 dex or log g< −1.06 dex or [M/H]>1.38 dex or [M/H]< −5.38 dex or [α/Fe]>0.96 dex or [α/Fe]< −0.56 dex	Filter all Flag = 9
	7682<$T_{\rm eff}$≤8318 K or 3682≤$T_{\rm eff}$<4318 K or 4.94<log g≤6.06 dex or −1.06 ≤log g<0.06 dex or 0.62<[M/H]≤1.38 dex or −5.38 ≤[M/H]< −4.62 dex or 0.64<[α/Fe]≤0.96 dex or −0.56 ≤[α/Fe]< −0.24 dex	Flag = 1
	4318≤$T_{\rm eff}$≤7682 K and 0.06≤log g≤4.94 dex and −4.62 ≤[M/H]≤0.62 dex and −0.24 ≤[α/Fe]≤0.64 dex	Flag = 0

Definition of parameter flags considering RVS flux problems or emission line probability:

Flag name	Condition	Flag value
nanFlux	Flux=NaN	Filter all Flag = 9
	no Nans in Flux	Flag = 0
emission	*CU6_is_emission*=True	Filter all Flag = 9
	CU6_is_emission=False	Flag = 0
neg_flux	> 2 pixels with flux<0	Filter all Flag = 9
	1 or 2 pixels with flux<0	Flag = 1
	flux>0	Flag = 0
nullFluxErr	$\sigma T_{\rm eff}$=0 K or σlog g=0 dex or σ[M/H]=0 dex or σ[α/Fe]=0 dex	Filter all Flag = 9
	no null uncertainties for all pixel flux	Flag = 0

TEFF_GSPSPEC_EPOCH_MEAN : Mean effective temperature from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Temperature[K])

Mean value of the effective temperatures (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from epoch RVS spectra and Monte Carlo realisations.

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TEFF_GSPSPEC_EPOCH_STD : Standard deviation of the effective temperature from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Temperature[K])

Standard deviation of the epoch estimations of the effective temperature inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra and Monte Carlo realisations.

LOGG_GSPSPEC_EPOCH_MEAN : Mean logarithm of the stellar surface gravity from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, GravitySurface[log cgs])

Mean value of the epoch estimations of the logarithm of the stellar surface gravity (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from epoch RVS spectra and Monte Carlo realisations.

LOGG_GSPSPEC_EPOCH_STD : Standard deviation of the logarithm of the stellar surface gravity from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, GravitySurface[log cgs])

Standard deviation of the epoch estimations of the logarithm of the stellar surface gravity (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from epoch RVS spectra and Monte Carlo realisations.

MH_GSPSPEC_EPOCH_MEAN : Mean global metallicity [M/H] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Mean value of the epoch estimations of the global metallicity [M/H] (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from epoch RVS spectra and Monte Carlo realisations.

MH_GSPSPEC_EPOCH_STD : Standard deviation of the global metallicity from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Standard deviation of the epoch estimations of the global metallicity [M/H] (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from epoch RVS spectra and Monte Carlo realisations.

MH_GSPSPEC_EPOCH_NTRANSITS : Number of transits used in the computation of the mean

metallicity (int)

Number of transits used in the computation of the mean metallicity.

ALPHAFE_GSPSPEC_EPOCH_MEAN : Mean abundance of [alpha/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Mean value of the epoch estimations of the abundance of alpha-elements with respect to iron [α/Fe] (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from epoch RVS spectra and Monte Carlo realisations. The considered alpha elements are: O, Ne, Mg, Si, S, Ar, Ca, Ti.

ALPHAFE_GSPSPEC_EPOCH_STD : Standard deviation of [alpha/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Standard deviation of the epoch estimations of the abundance of alpha-elements with respect to iron [α/Fe] (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from epoch RVS spectra and Monte Carlo realisations. The considered alpha elements are: O, Ne, Mg, Si, S, Ar, Ca, Ti.

ALPHAFE_GSPSPEC_EPOCH_NTRANSITS : Number of transits used in the computation of the mean [alpha/Fe] (int)

Number of transits used in the computation of the mean [alpha/Fe].

RV_GSPSPEC_EPOCH_MEAN : Mean radial velocity from GSP-Spec using epoch RVS spectra (float, Velocity[km s^{-1}])

Mean value of the radial velocities inferred by GSP-Spec (Recio-Blanco et al. 2023) from epoch RVS spectra.

RV_GSPSPEC_EPOCH_STD : Standard deviation of the radial velocity of the epoch RVS spectra measured by GSP-Spec (float, Velocity[km s^{-1}])

Standard deviation of the radial velocities inferred by GSP-Spec (Recio-Blanco et al. 2023) from epoch RVS spectra.

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LOGCHISQ_GSPSPEC_EPOCH_MEAN : Mean logarithm of the goodness-of-fit for the GSP-Spec MatisseGauguin parameters from epoch spectra (float)

Mean logarithm to the base 10 of the chi-squared between input epoch spectrum rebinned to 800 pixels and solution synthetic spectrum computed from GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023).

LOGCHISQ_GSPSPEC_EPOCH_STD : Standard deviation of the logarithm of the goodness-of-fit for the GSP-Spec MatisseGauguin parameters from epoch spectra (float)

Standard deviation of the logarithm to the base 10 of the chi-squared between input epoch spectrum rebinned to 800 pixels and solution synthetic spectrum computed from GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023).

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4.7 AP_RVS_EPOCH [TAP]

This table contains the astrophysical parameters, and in some cases the abundances, derived from the epoch RVS spectra for RRLyrae and Cepheids by GSP-Spec as part of the Apsis processing chain.

See Section **??** of the online documentation for details on the GSP-Spec module.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

TRANSIT_ID : Transit Identifier (long)

The `transit_id` is a unique identifier assigned to each detected (and confirmed) source as it transits the Gaia focal plane. Each time a given source is detected as Gaia scans and re-scans the sky a new `transit_id` will be created to badge that apparition. Hence the along-scan time and the across-scan position along with the telescope in which the source was detected are used to form a unique integer with which to label the transit. The detection of transits is done using the SkyMapper CCD data, and the confirmation is done using the AF1 CCD data. Both happens on board.

The several features of a transit that are encoded in `transit_id` can be easily retrieved using bit masks (&) and shifts (>>) as follows:

- On-Board Mission Time line (OBMT) [ns]
  = 204800 * ((`transit_id` >> 17) & (0x000003FFFFFFFFFF))

- Field-of-view = 1 + (`transit_id` >> 15) & 0x03 [1 for 'preceding' and 2 for 'following' fields-of-view respectively]
- CCD row = (`transit_id` >> 12) & 0x07 [dimensionless, in the range 1 to 7]
- Across-scan 'reference acquisition pixel' in strip AF1 = (`transit_id`) & 0x0FFF [pixels] (this is the across-scan centre of the AF1 window and is odd if immediately below the mid-point of the window and even if immediately above)

where the bit mask prefix '0x' denotes hexadecimal.

For further details see Portell et al. (2020). For convenience a decoder for `transit_id` is available on-line at

`https://gaia.esac.esa.int/decoder/transitidDecoder.jsp`

OBSERVATION_TIME : Barycentric light-travel time corrected timestamp for each observation (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

DELTA_RV_GSPSPEC : Radial velocity deviation from CU6 solution from GSP-Spec (double, Velocity[km s^{-1}])

Radial velocity deviation from CU6 solution.

TEFF_GSPSPEC : Effective temperature from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Temperature[K])

Median value of the effective temperature (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from epoch RVS spectra and Monte Carlo realisations.

TEFF_GSPSPEC_LOWER : 16th percentile of effective temperature from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Temperature[K])

Lower confidence level (16%) of the median effective temperature (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from epoch RVS spectra. Lower and upper levels include 68% confidence interval.

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TEFF_GSPSPEC_UPPER : 84th percentile of effective temperature from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Temperature[K])

Upper confidence level (84%) of the median effective temperature (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from epoch RVS spectra. Lower and upper levels include 68% confidence interval.

LOGG_GSPSPEC : Logarithm of the stellar surface gravity from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, GravitySurface[log cgs])

Median value of logarithm of the stellar surface gravity (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from epoch RVS spectra.

LOGG_GSPSPEC_LOWER : 16th percentile of the logarithm of the stellar surface gravity from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, GravitySurface[log cgs])

Lower confidence level (16%) of the median value of logarithm of the stellar surface gravity (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from epoch RVS spectra. Lower and upper levels include 68% confidence interval.

LOGG_GSPSPEC_UPPER : 84th percentile of the logarithm of the stellar surface gravity from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, GravitySurface[log cgs])

Upper confidence level (84%) of the median value of logarithm of the stellar surface gravity (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from epoch RVS spectra. Lower and upper levels include 68% confidence interval.

MH_GSPSPEC : Global metallicity [M/H] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Median global metallicity (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from epoch RVS spectra.

MH_GSPSPEC_LOWER : 16th percentile of global metallicity [M/H] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Draft

Lower confidence level (16%) of the median global metallicity (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from epoch RVS spectra. Lower and upper levels include 68% confidence interval.

MH_GSPSPEC_UPPER : 84th percentile of global metallicity [M/H] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Upper confidence level (84%) of the median global metallicity (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from epoch RVS spectra. Lower and upper levels include 68% confidence interval.

ALPHAFE_GSPSPEC : Abundance of alpha-elements [alpha/Fe] with respect to iron from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Median abundance of alpha-elements with respect to iron (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from epoch RVS spectra. The considered alpha elements are: O, Ne, Mg, Si, S, Ar, Ca, Ti.

ALPHAFE_GSPSPEC_LOWER : 16th percentile of the abundance of alpha-elements [alpha/Fe] with respect to iron from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of alpha-elements with respect to iron (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from epoch RVS spectra. Lower and upper levels include 68% confidence interval.

ALPHAFE_GSPSPEC_UPPER : 84th percentile of the abundance of alpha-elements [alpha/Fe] with respect to iron from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of alpha-elements with respect to iron (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) from epoch RVS spectra. Lower and upper levels include 68% confidence interval.

FEM_GSPSPEC : Abundance of neutral iron [Fe/M] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations, applied to the individual N lines of the element (float, Abundances[dex])

Draft

Median abundance of neutral iron (assuming source is a single star) from epoch RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element, where the number of lines is given in `fem_gspspec_nlines`. The neutral iron abundance [Fe/H] is obtained by [Fe/H]=[Fe/M]+[M/H].

FEM_GSPSPEC_LOWER : 16th percentile of the abundance of neutral iron [Fe/M] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of neutral iron (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval. The neutral iron abundance [Fe/H] is obtained by [Fe/H]=[Fe/M]+[M/H].

FEM_GSPSPEC_UPPER : 84th percentile of the abundance of neutral iron [Fe/M] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of neutral iron (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval. The neutral iron abundance [Fe/H] is obtained by [Fe/H]=[Fe/M]+[M/H].

FEM_GSPSPEC_NLINES : Number of lines used for [Fe/M] abundance estimation from epoch RVS spectra (int)

Number of lines used to compute the [Fe/M] abundance from epoch RVS spectra. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

FEM_GSPSPEC_LINESCATTER : Uncertainty estimation of [Fe/M] abundance determined from epoch RVS spectra using N lines of the element (float, Abundances[dex])

Standard deviation of the individual N lines (`fem_gspspec_nlines`) abundance results from epoch RVS spectra.

FEM_GSPSPEC_ULIM : Minimum value of [Fe/M] that can be measured. (float, Abundances[dex])

Upper limit for the [Fe/M] abundance extracted from epoch RVS spectra. Minimum value of

[Fe/M] that can be measured, averaged over all the estimations from the individual lines.

SIFE_GSPSPEC : Abundance of silicon [Si/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations, applied to the individual N lines of the element (float, Abundances[dex])

Median abundance of silicon (assuming source is a single star) from epoch RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element, where the number of lines is given in `sife_gspspec_nlines`.

SIFE_GSPSPEC_LOWER : 16th percentile of the abundance of silicon [Si/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of silicon (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval.

SIFE_GSPSPEC_UPPER : 84th percentile of the abundance of silicon [Si/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of silicon (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval.

SIFE_GSPSPEC_NLINES : Number of lines used for [Si/Fe] abundance estimation from epoch RVS spectra (int)

Number of lines used to compute the [Si/Fe] abundance from epoch RVS spectra. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

SIFE_GSPSPEC_LINESCATTER : Uncertainty estimation of [Si/Fe] abundance determined from epoch RVS spectra using N lines of the element (float, Abundances[dex])

Standard deviation of the individual N lines (`sife_gspspec_nlines`) abundance results from epoch RVS spectra.

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SIFE_GSPSPEC_ULIM : Minimum value of [Si/Fe] that can be measured. (float, Abundances[dex])

Upper limit for the [Si/Fe] abundance extracted from epoch RVS spectra. Minimum value of [Si/Fe] that can be measured, averaged over all the estimations from the individual lines.

CAFE_GSPSPEC : Abundance of calcium [Ca/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations, applied to the individual N lines of the element (float, Abundances[dex])

Median abundance of calcium (assuming source is a single star) from epoch RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element, where the number of lines is given in `cafe_gspspec_nlines`.

CAFE_GSPSPEC_LOWER : 16th percentile of the abundance of calcium [Ca/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of calcium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval.

CAFE_GSPSPEC_UPPER : 84th percentile of the abundance of calcium [Ca/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of calcium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval.

CAFE_GSPSPEC_NLINES : Number of lines used for [Ca/Fe] abundance estimation from epoch RVS spectra (int)

Number of lines used to compute the [Ca/Fe] abundance from epoch RVS spectra. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

CAFE_GSPSPEC_LINESCATTER : Uncertainty estimation of [Ca/Fe] abundance determined from epoch RVS spectra using N lines of the element (float, Abundances[dex])

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Standard deviation of the individual N lines (`cafe_gspspec_nlines`) abundance results from epoch RVS spectra.

CAFE_GSPSPEC_ULIM : Minimum value of [Ca/Fe] that can be measured. (float, Abundances[dex])

Upper limit for the [Ca/Fe] abundance extracted from epoch RVS spectra. Minimum value of [Ca/Fe] that can be measured, averaged over all the estimations from the individual lines.

TIFE_GSPSPEC : Abundance of titanium [Ti/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations, applied to the individual N lines of the element (float, Abundances[dex])

Median abundance of titanium (assuming source is a single star) from epoch RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element, where the number of lines is given in `tife_gspspec_nlines`.

TIFE_GSPSPEC_LOWER : 16th percentile of the abundance of titanium [Ti/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of titanium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval.

TIFE_GSPSPEC_UPPER : 84th percentile of the abundance of titanium [Ti/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of titanium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval.

TIFE_GSPSPEC_NLINES : Number of lines used for [Ti/Fe] abundance estimation from epoch RVS spectra (int)

Number of lines used to compute the [Ti/Fe] abundance from epoch RVS spectra. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

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TIFE_GSPSPEC_LINESCATTER : Uncertainty estimation of [Ti/Fe] abundance determined from epoch RVS spectra using N lines of the element (float, Abundances[dex])

Standard deviation of the individual N lines (`tife_gspspec_nlines`) abundance results from epoch RVS spectra.

TIFE_GSPSPEC_ULIM : Minimum value of [Ti/Fe] that can be measured. (float, Abundances[dex])

Upper limit for the [Ti/Fe] abundance extracted from epoch RVS spectra. Minimum value of [Ti/Fe] that can be measured, averaged over all the estimations from the individual lines.

MGFE_GSPSPEC : Abundance of magnesium [Mg/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations, applied to the individual N lines of the element (float, Abundances[dex])

Median abundance of magnesium (assuming source is a single star) from epoch RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element, where the number of lines is given in `mgfe_gspspec_nlines`.

MGFE_GSPSPEC_LOWER : 16th percentile of the abundance of magnesium [Mg/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of magnesium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval.

MGFE_GSPSPEC_UPPER : 84th percentile of the abundance of magnesium [Mg/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of magnesium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval.

MGFE_GSPSPEC_NLINES : Number of lines used for [Mg/Fe] abundance estimation from epoch RVS spectra (int)

Number of lines used to compute the [Mg/Fe] abundance from epoch RVS spectra. Lines with

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interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

MGFE_GSPSPEC_LINESCATTER : Uncertainty estimation of [Mg/Fe] abundance determined from epoch RVS spectra using N lines of the element (float, Abundances[dex])

Standard deviation of the individual N lines (`mgfe_gspspec_nlines`) abundance results from epoch RVS spectra.

MGFE_GSPSPEC_ULIM : Minimum value of [Mg/Fe] that can be measured. (float, Abundances[dex])

Upper limit for the [Mg/Fe] abundance extracted from epoch RVS spectra. Minimum value of [Mg/Fe] that can be measured, averaged over all the estimations from the individual lines.

NDFE_GSPSPEC : Abundance of neodymium [Nd/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations, applied to the individual N lines of the element (float, Abundances[dex])

Median abundance of Neodymium (assuming source is a single star) from epoch RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element, where the number of lines is given in `ndfe_gspspec_nlines`.

NDFE_GSPSPEC_LOWER : 16th percentile of the abundance of neodymium [Nd/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of Neodymium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval.

NDFE_GSPSPEC_UPPER : 84th percentile of the abundance of neodymium [Nd/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of Neodymium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval.

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NDFE_GSPSPEC_NLINES : Number of lines used for [Nd/Fe] abundance estimation from epoch RVS spectra (int)

Number of lines used to compute the [Nd/Fe] abundance from epoch RVS spectra. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

NDFE_GSPSPEC_LINESCATTER : Uncertainty estimation of [Nd/Fe] abundance determined from epoch RVS spectra using N lines of the element (float, Abundances[dex])

Standard deviation of the individual N lines (`ndfe_gspspec_nlines`) abundance results from epoch RVS spectra.

NDFE_GSPSPEC_ULIM : Minimum value of [Nd/Fe] that can be measured. (float, Abundances[dex])

Upper limit for the [Nd/Fe] abundance extracted from epoch RVS spectra. Minimum value of [Nd/Fe] that can be measured, averaged over all the estimations from the individual lines.

FEIIM_GSPSPEC : Abundance of ionised iron [FeII/M] from GSP-Spec MatisseGauguin using RVS spectra and Monte Carlo realisations, applied to the individual N lines of the element (float, Abundances[dex])

Median abundance of ionised iron (assuming source is a single star) from epoch RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element, where the number of lines is given in `feiim_gspspec_nlines`. The ionised iron abundance [FeII/H] is obtained by [FeII/H]=[FeII/M]+[M/H].

FEIIM_GSPSPEC_LOWER : 16th percentile of the abundance of ionised iron [FeII/M] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of ionised iron (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval. The ionised iron abundance [FeII/H] is obtained by [FeII/H]=[FeII/M]+[M/H].

FEIIM_GSPSPEC_UPPER : 84th percentile of the abundance of ionised iron [FeII/M] from GSP-

Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of ionised iron (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval. The ionised iron abundance [FeII/H] is obtained by [FeII/H]=[FeII/M]+[M/H].

FEIIM_GSPSPEC_NLINES : Number of lines used for [FeII/M] abundance estimation from epoch RVS spectra (int)

Number of lines used to compute the [FeII/M] abundance from epoch RVS spectra. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

FEIIM_GSPSPEC_LINESCATTER : Uncertainty estimation of [FeII/M] abundance determined from epoch RVS spectra using N lines of the element (float, Abundances[dex])

Standard deviation of the individual N lines (`feiim_gspspec_nlines`) abundance results from epoch RVS spectra.

FEIIM_GSPSPEC_ULIM : Minimum value of [FeII/M] that can be measured. (float, Abundances[dex])

Upper limit for the ionised iron abundance extracted from epoch RVS spectra. Minimum value of [FeII/M] that can be measured. The ionised iron abundance [FeII/H] is obtained by [FeII/H]=[FeII/M]+[M/H].

SFE_GSPSPEC : Abundance of sulphur [S/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations, applied to the individual N lines of the element (float, Abundances[dex])

Median abundance of sulphur (assuming source is a single star) from epoch RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element, where the number of lines is given in `sfe_gspspec_nlines`.

SFE_GSPSPEC_LOWER : 16th percentile of the abundance of sulphur [S/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

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Lower confidence level (16%) of the median abundance of sulphur (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval.

SFE_GSPSPEC_UPPER : 84th percentile of the abundance of sulphur [S/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of sulphur (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval.

SFE_GSPSPEC_NLINES : Number of lines used for [S/Fe] abundance estimation from epoch RVS spectra (int)

Number of lines used to compute the [S/Fe] abundance from epoch RVS spectra. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

SFE_GSPSPEC_LINESCATTER : Uncertainty estimation of [S/Fe] abundance determined from epoch RVS spectra using N lines of the element (float, Abundances[dex])

Standard deviation of the individual N lines (`sfe_gspspec_nlines`) abundance results from epoch RVS spectra.

SFE_GSPSPEC_ULIM : Minimum value of [S/Fe] that can be measured. (float, Abundances[dex])

Upper limit for the [S/Fe] abundance extracted from epoch RVS spectra. Minimum value of [S/Fe] that can be measured, averaged over all the estimations from the individual lines.

ZRFE_GSPSPEC : Abundance of zirconium [Zr/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations, applied to the individual N lines of the element (float, Abundances[dex])

Median abundance of zirconium (assuming source is a single star) from epoch RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element, where the number of lines is given in `zrfe_gspspec_nlines`.

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ZRFE_GSPSPEC_LOWER : 16th percentile of the abundance of zirconium [Zr/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of zirconium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval.

ZRFE_GSPSPEC_UPPER : 84th percentile of the abundance of zirconium [Zr/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of zirconium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval.

ZRFE_GSPSPEC_NLINES : Number of lines used for [Zr/Fe] abundance estimation from epoch RVS spectra (int)

Number of lines used to compute the [Zr/Fe] abundance from epoch RVS spectra. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

ZRFE_GSPSPEC_LINESCATTER : Uncertainty estimation of [Zr/Fe] abundance determined from epoch RVS spectra using N lines of the element (float, Abundances[dex])

Standard deviation of the individual N lines (`zrfe_gspspec_nlines`) abundance results from epoch RVS spectra.

ZRFE_GSPSPEC_ULIM : Minimum value of [Zr/Fe] that can be measured. (float, Abundances[dex])

Upper limit for the [Zr/Fe] abundance extracted from epoch RVS spectra. Minimum value of [Zr/Fe] that can be measured, averaged over all the estimations from the individual lines.

NFE_GSPSPEC : Abundance of nitrogen [N/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations, applied to the individual N lines of the element (float, Abundances[dex])

Median abundance of nitrogen (assuming source is a single star) from epoch RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric

parameters and the Gauguin algorithm, applied to the individual N lines of the element, where the number of lines is given in `nfe_gspspec_nlines`.

NFE_GSPSPEC_LOWER : 16th percentile of the abundance of nitrogen [N/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of nitrogen (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval.

NFE_GSPSPEC_UPPER : 84th percentile of the abundance of nitrogen [N/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of nitrogen (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval.

NFE_GSPSPEC_NLINES : Number of lines used for [N/Fe] abundance estimation from epoch RVS spectra (int)

Number of lines used to compute the [N/Fe] abundance from epoch RVS spectra. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

NFE_GSPSPEC_LINESCATTER : Uncertainty estimation of [N/Fe] abundance determined from epoch RVS spectra using N lines of the element (float, Abundances[dex])

Standard deviation of the individual N lines (`nfe_gspspec_nlines`) abundance results from epoch RVS spectra.

NFE_GSPSPEC_ULIM : Minimum value of [N/Fe] that can be measured. (float, Abundances[dex])

Upper limit for the [N/Fe] abundance extracted from epoch RVS spectra. Minimum value of [N/Fe] that can be measured, averaged over all the estimations from the individual lines.

CRFE_GSPSPEC : Abundance of chromium [Cr/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations, applied to the individual N lines of the element (float,

Abundances[dex])

Median abundance of chromium (assuming source is a single star) from epoch RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element, where the number of lines is given in `crfe_gspspec_nlines`.

CRFE_GSPSPEC_LOWER : 16th percentile of the abundance of chromium [Cr/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of chromium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval.

CRFE_GSPSPEC_UPPER : 84th percentile of the abundance of chromium [Cr/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of chromium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval.

CRFE_GSPSPEC_NLINES : Number of lines used for [Cr/Fe] abundance estimation from epoch RVS spectra (int)

Number of lines used to compute the [Cr/Fe] abundance from epoch RVS spectra. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

CRFE_GSPSPEC_LINESCATTER : Uncertainty estimation of [Cr/Fe] abundance determined from epoch RVS spectra using N lines of the element (float, Abundances[dex])

Standard deviation of the individual N lines (`crfe_gspspec_nlines`) abundance results from epoch RVS spectra.

CRFE_GSPSPEC_ULIM : Minimum value of [Cr/Fe] that can be measured. (float, Abundances[dex])

Upper limit for the [Cr/Fe] abundance extracted from epoch RVS spectra. Minimum value of [Cr/Fe] that can be measured, averaged over all the estimations from the individual lines.

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CEFE_GSPSPEC : Abundance of cerium [Ce/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations, applied to the individual N lines of the element (float, Abundances[dex])

Median abundance of cerium (assuming source is a single star) from epoch RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element, where the number of lines is given in `cefe_gspspec_nlines`.

CEFE_GSPSPEC_LOWER : 16th percentile of the abundance of cerium [Ce/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of cerium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval.

CEFE_GSPSPEC_UPPER : 84th percentile of the abundance of cerium [Ce/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of cerium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval.

CEFE_GSPSPEC_NLINES : Number of lines used for [Ce/Fe] abundance estimation from epoch RVS spectra (int)

Number of lines used to compute the [Ce/Fe] abundance from epoch RVS spectra. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

CEFE_GSPSPEC_LINESCATTER : Uncertainty estimation of [Ce/Fe] abundance determined from epoch RVS spectra using N lines of the element (float, Abundances[dex])

Standard deviation of the individual N lines (`cefe_gspspec_nlines`) abundance results from epoch RVS spectra.

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CEFE_GSPSPEC_ULIM : Minimum value of [Ce/Fe] that can be measured. (float, Abundances[dex])

Upper limit for the [Ce/Fe] abundance extracted from epoch RVS spectra. Minimum value of [Ce/Fe] that can be measured, averaged over all the estimations from the individual lines.

NIFE_GSPSPEC : Abundance of nickel [Ni/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations, applied to the individual N lines of the element (float, Abundances[dex])

Median abundance of nickel (assuming source is a single star) from epoch RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element, where the number of lines is given in `nife_gspspec_nlines`.

NIFE_GSPSPEC_LOWER : 16th percentile of the abundance of nickel [Ni/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of nickel (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval.

NIFE_GSPSPEC_UPPER : 84th percentile of the abundance of nickel [Ni/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of nickel (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval.

NIFE_GSPSPEC_NLINES : Number of lines used for [Ni/Fe] abundance estimation from epoch RVS spectra (int)

Number of lines used to compute the [Ni/Fe] abundance from epoch RVS spectra. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

NIFE_GSPSPEC_LINESCATTER : Uncertainty estimation of [Ni/Fe] abundance using N lines of the element using epoch RVS spectra (float, Abundances[dex])

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Standard deviation of the individual N lines (`nife_gspspec_nlines`) abundance results from epoch RVS spectra.

NIFE_GSPSPEC_ULIM : Minimum value of [Ni/Fe] that can be measured. (float, Abundances[dex])

Upper limit for the [Ni/Fe] abundance extracted from epoch RVS spectra. Minimum value of [Ni/Fe] that can be measured, averaged over all the estimations from the individual lines.

ALFE_GSPSPEC : Abundance of Aluminum [Al/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations, applied to the individual N lines of the element (float, Abundances[dex])

Median abundance of Aluminum (assuming source is a single star) from epoch RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element, where the number of lines is given in `alfe_gspspec_nlines`.

ALFE_GSPSPEC_LOWER : 16th percentile of the abundance of Aluminum [Al/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of Aluminum (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval.

ALFE_GSPSPEC_UPPER : 84th percentile of the abundance of Aluminum [Al/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of Aluminum (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval.

ALFE_GSPSPEC_NLINES : Number of lines used for [Al/Fe] abundance estimation from epoch RVS spectra (int)

Number of lines used to compute the [Al/Fe] abundance from epoch RVS spectra. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

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ALFE_GSPSPEC_LINESCATTER : Uncertainty estimation of [Al/Fe] abundance using N lines of the element using epoch RVS spectra (float, Abundances[dex])

Standard deviation of the individual N lines (`alfe_gspspec_nlines`) abundance results from epoch RVS spectra.

ALFE_GSPSPEC_ULIM : Minimum value of [Al/Fe] that can be measured. (float, Abundances[dex])

Upper limit for the [Al/Fe] abundance extracted from epoch RVS spectra. Minimum value of [Al/Fe] that can be measured, averaged over all the estimations from the individual lines.

CLFE_GSPSPEC : Abundance of Chlorine [Cl/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations, applied to the individual N lines of the element (float, Abundances[dex])

Median abundance of Chlorine(assuming source is a single star) from epoch RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element, where the number of lines is given in `clfe_gspspec_nlines`.

CLFE_GSPSPEC_LOWER : 16th percentile of the abundance of Chlorine [Cl/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of Chlorine (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval.

CLFE_GSPSPEC_UPPER : 84th percentile of the abundance of Chlorine [Cl/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of Chlorine (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval.

CLFE_GSPSPEC_NLINES : Number of lines used for [Cl/Fe] abundance estimation from epoch RVS spectra (int)

Number of lines used to compute the [Cl/Fe] abundance from epoch RVS spectra. Lines with

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interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

CLFE_GSPSPEC_LINESCATTER : Uncertainty estimation of [Cl/Fe] abundance using N lines of the element using epoch RVS spectra (float, Abundances[dex])

Standard deviation of the individual N lines (`clfe_gspspec_nlines`) abundance results from epoch RVS spectra.

CLFE_GSPSPEC_ULIM : Minimum value of [Cl/Fe] that can be measured. (float, Abundances[dex])

Upper limit for the [Cl/Fe] abundance extracted from epoch RVS spectra. Minimum value of [Cl/Fe] that can be measured, averaged over all the estimations from the individual lines.

HFFE_GSPSPEC : Abundance of Hafnium [Hf/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations, applied to the individual N lines of the element (float, Abundances[dex])

Median abundance of Hafnium (assuming source is a single star) from epoch RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element, where the number of lines is given in `hffe_gspspec_nlines`.

HFFE_GSPSPEC_LOWER : 16th percentile of the abundance of Hafnium [Hf/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of Hafnium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval.

HFFE_GSPSPEC_UPPER : 84th percentile of the abundance of Hafnium [Hf/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of Hafnium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval.

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HFFE_GSPSPEC_NLINES : Number of lines used for [Hf/Fe] abundance estimation from epoch RVS spectra (int)

Number of lines used to compute the [Hf/Fe] abundance from epoch RVS spectra. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

HFFE_GSPSPEC_LINESCATTER : Uncertainty estimation of [Hf/Fe] abundance using N lines of the element using epoch RVS spectra (float, Abundances[dex])

Standard deviation of the individual N lines (`hffe_gspspec_nlines`) abundance results from epoch RVS spectra.

HFFE_GSPSPEC_ULIM : Minimum value of [Hf/Fe] that can be measured. (float, Abundances[dex])

Upper limit for the [Hf/Fe] abundance extracted from epoch RVS spectra. Minimum value of [Hf/Fe] that can be measured, averaged over all the estimations from the individual lines.

KFE_GSPSPEC : Abundance of Potassium [K/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations, applied to the individual N lines of the element (float, Abundances[dex])

Median abundance of Potassium (assuming source is a single star) from epoch RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element, where the number of lines is given in `kfe_gspspec_nlines`.

KFE_GSPSPEC_LOWER : 16th percentile of the abundance of Potassium [K/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of Potassium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval.

KFE_GSPSPEC_UPPER : 84th percentile of the abundance of Potassium [K/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of Potassium (assuming source is a sin-

gle star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval.

KFE_GSPSPEC_NLINES : Number of lines used for [K/Fe] abundance estimation from epoch RVS spectra (int)

Number of lines used to compute the [K/Fe] abundance from epoch RVS spectra. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

KFE_GSPSPEC_LINESCATTER : Uncertainty estimation of [K/Fe] abundance using N lines of the element using epoch RVS spectra (float, Abundances[dex])

Standard deviation of the individual N lines (`kfe_gspspec_nlines`) abundance results from epoch RVS spectra.

KFE_GSPSPEC_ULIM : Minimum value of [K/Fe] that can be measured. (float, Abundances[dex])

Upper limit for the [K/Fe] abundance extracted from epoch RVS spectra. Minimum value of [K/Fe] that can be measured, averaged over all the estimations from the individual lines.

NAFE_GSPSPEC : Abundance of Sodium [Na/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations, applied to the individual N lines of the element (float, Abundances[dex])

Median abundance of Sodium (assuming source is a single star) from epoch RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element, where the number of lines is given in `nafe_gspspec_nlines`.

NAFE_GSPSPEC_LOWER : 16th percentile of the abundance of Sodium [Na/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of Sodium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval.

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NAFE_GSPSPEC_UPPER : 84th percentile of the abundance of Sodium [Na/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of Sodium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval.

NAFE_GSPSPEC_NLINES : Number of lines used for [Na/Fe] abundance estimation from epoch RVS spectra (int)

Number of lines used to compute the [Na/Fe] abundance from epoch RVS spectra. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

NAFE_GSPSPEC_LINESCATTER : Uncertainty estimation of [Na/Fe] abundance using N lines of the element using epoch RVS spectra (float, Abundances[dex])

Standard deviation of the individual N lines (`nafe_gspspec_nlines`) abundance results from epoch RVS spectra.

NAFE_GSPSPEC_ULIM : Minimum value of [Na/Fe] that can be measured. (float, Abundances[dex])

Upper limit for the [Na/Fe] abundance extracted from epoch RVS spectra. Minimum value of [Na/Fe] that can be measured, averaged over all the estimations from the individual lines.

PFE_GSPSPEC : Abundance of Phosphorus [P/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations, applied to the individual N lines of the element (float, Abundances[dex])

Median abundance of Phosphorus (assuming source is a single star) from epoch RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element, where the number of lines is given in `pfe_gspspec_nlines`.

PFE_GSPSPEC_LOWER : 16th percentile of the abundance of Phosphorus [P/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of Phosphorus (assuming source is a

single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval.

PFE_GSPSPEC_UPPER : 84th percentile of the abundance of Phosphorus [P/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of Phosphorus (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval.

PFE_GSPSPEC_NLINES : Number of lines used for [P/Fe] abundance estimation from epoch RVS spectra (int)

Number of lines used to compute the [P/Fe] abundance from epoch RVS spectra. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

PFE_GSPSPEC_LINESCATTER : Uncertainty estimation of [P/Fe] abundance using N lines of the element using epoch RVS spectra (float, Abundances[dex])

Standard deviation of the individual N lines (`pfe_gspspec_nlines`) abundance results from epoch RVS spectra.

PFE_GSPSPEC_ULIM : Minimum value of [P/Fe] that can be measured. (float, Abundances[dex])

Upper limit for the [P/Fe] abundance extracted from epoch RVS spectra. Minimum value of [P/Fe] that can be measured, averaged over all the estimations from the individual lines.

SMFE_GSPSPEC : Abundance of Samarium [Sm/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations, applied to the individual N lines of the element (float, Abundances[dex])

Median abundance of Samarium (assuming source is a single star) from epoch RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric parameters and the Gauguin algorithm, applied to the individual N lines of the element, where the number of lines is given in `smfe_gspspec_nlines`.

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SMFE_GSPSPEC_LOWER : 16th percentile of the abundance of Samarium [Sm/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of Samarium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval.

SMFE_GSPSPEC_UPPER : 84th percentile of the abundance of Samarium [Sm/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of Samarium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval.

SMFE_GSPSPEC_NLINES : Number of lines used for [Sm/Fe] abundance estimation from epoch RVS spectra (int)

Number of lines used to compute the [Sm/Fe] abundance from epoch RVS spectra. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

SMFE_GSPSPEC_LINESCATTER : Uncertainty estimation of [Sm/Fe] abundance using N lines of the element using epoch RVS spectra (float, Abundances[dex])

Standard deviation of the individual N lines (`smfe_gspspec_nlines`) abundance results from epoch RVS spectra.

SMFE_GSPSPEC_ULIM : Minimum value of [Sm/Fe] that can be measured. (float, Abundances[dex])

Upper limit for the [Sm/Fe] abundance extracted from epoch RVS spectra. Minimum value of [Sm/Fe] that can be measured, averaged over all the estimations from the individual lines.

SRFE_GSPSPEC : Abundance of Strontium [Sr/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations, applied to the individual N lines of the element (float, Abundances[dex])

Median abundance of Strontium (assuming source is a single star) from epoch RVS spectra and Monte Carlo realisations derived using MatisseGauguin (Recio-Blanco et al. 2023) atmospheric

parameters and the Gauguin algorithm, applied to the individual N lines of the element, where the number of lines is given in `srfe_gspspec_nlines`.

SRFE_GSPSPEC_LOWER : 16th percentile of the abundance of Strontium [Sr/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Lower confidence level (16%) of the median abundance of Strontium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval.

SRFE_GSPSPEC_UPPER : 84th percentile of the abundance of Strontium [Sr/Fe] from GSP-Spec MatisseGauguin using epoch RVS spectra and Monte Carlo realisations (float, Abundances[dex])

Upper confidence level (84%) of the median abundance of Strontium (assuming source is a single star) inferred by GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) using epoch RVS spectra. Lower and upper levels include 68% confidence interval.

SRFE_GSPSPEC_NLINES : Number of lines used for [Sr/Fe] abundance estimation from epoch RVS spectra (int)

Number of lines used to compute the [Sr/Fe] abundance from epoch RVS spectra. Lines with interquartile difference (84th quantile value - 16th quantile value) in the Monte Carlo line abundance distribution higher than 0.5 dex have been excluded.

SRFE_GSPSPEC_LINESCATTER : Uncertainty estimation of [Sr/Fe] abundance using N lines of the element using epoch RVS spectra (float, Abundances[dex])

Standard deviation of the individual N lines (`srfe_gspspec_nlines`) abundance results from epoch RVS spectra.

SRFE_GSPSPEC_ULIM : Minimum value of [Sr/Fe] that can be measured. (float, Abundances[dex])

Upper limit for the [Sr/Fe] abundance extracted from epoch RVS spectra. Minimum value of [Sr/Fe] that can be measured, averaged over all the estimations from the individual lines.

FLAGS_GSPSPEC : Catalogue flags for the GSP-Spec MatisseGauguin analysis of epoch RVS spectra (string)

Definitions of each character in the GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023) quality flag chain. In this chain, value ‘0’ is the best, and ‘9’ is the worst. Flag names are split in two categories: parameter flags (green) and individual abundance flags (blue). In the following, the term ‘gof’ refers to the chi-square value beteween the input epoch spectrum fluxes array and the solution spectrum fluxes array.

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Chain character number - name	Considered quality aspect	Possible adopted values
1 **vbroadT**	vbroad induced bias in $T_{\rm eff}$	0,1,2,9
2 **vbroadG**	vbroad induced bias in $\log g$	0,1,2,9
3 **vbroadM**	vbroad induced bias in [M/H]	0,1,2,9
4 **vradT**	vrad induced bias in $T_{\rm eff}$	0,1,2,9
5 **vradG**	vrad induced bias in $\log g$	0,1,2,9
6 **vradM**	vrad induced bias in [M/H]	0,1,2,9
7 **fluxNoise**	flux noise uncertainties	0,1,2,3,4,5,9
8 **extrapol**	extrapolation	0,1,2,3,4,9
9 **neg_flux**	negative flux pixels	0,1,9
10 **nanFlux**	NaN flux pixels	0,9
11 **emission**	emission line	0,9
12 **nullFluxErr**	null uncertainties	0,9
13 **KMgiantPar**	KM-type giant stars	0,1,2
14 **NUpLim**	nitrogen abundance upper limit	0,1,2,9
15 **NUncer**	nitrogen abundance uncertainty quality	0,1,2,9
16 **MgUpLim**	magnesium abundance upper limit	0,1,2,9
17 **MgUncer**	magnesium abundance uncertainty quality	0,1,2,9
18 **SiUpLim**	silicon abundance upper limit	0,1,2,9
19 **SiUncer**	silicon abundance uncertainty quality	0,1,2,9
20 **SUpLim**	sulphur abundance upper limit	0,1,2,9
21 **SUncer**	sulphur abundance uncertainty quality	0,1,2,9
22 **CaUpLim**	calcium abundance upper limit	0,1,2,9
23 **CaUncer**	calcium abundance uncertainty quality	0,1,2,9
24 **TiUpLim**	titanium abundance upper limit	0,1,2,9
25 **TiUncer**	titanium abundance uncertainty quality	0,1,2,9
26 **CrUpLim**	chromium abundance upper limit	0,1,2,9
27 **CrUncer**	chromium abundance uncertainty quality	0,1,2,9
28 **FeUpLim**	neutral iron abundance upper limit	0,1,2,9
29 **FeUncer**	neutral iron abundance uncertainty quality	0,1,2,9
30 **FeIIUpLim**	ionised iron abundance upper limit	0,1,2,9
31 **FeIIUncer**	ionised iron abundance uncertainty quality	0,1,2,9
32 **NiUpLim**	nickel abundance upper limit	0,1,2,9
33 **NiUncer**	nickel abundance uncertainty quality	0,1,2,9
34 **ZrUpLim**	zirconium abundance upper limit	0,1,2,9
35 **ZrUncer**	zirconium abundance uncertainty quality	0,1,2,9
36 **CeUpLim**	cerium abundance upper limit	0,1,2,9
37 **CeUncer**	cerium abundance uncertainty quality	0,1,2,9
38 **NdUpLim**	neodymium abundance upper limit	0,1,2,9
39 **NdUncer**	neodymium abundance uncertainty quality	0,1,2,9

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Definition of parameter flags considering potential biases due to rotational velocity and macro-turbulence:

Flag name	Condition	Flag value
vbroadT	$\Delta T_{\rm eff}$>2000 K	Filter all Flag = 9
	500< $\Delta T_{\rm eff}$≤2000 K	Flag = 2
	250< $\Delta T_{\rm eff}$≤500 K	Flag = 1
	$\Delta T_{\rm eff}$≤250 K	Flag = 0
vbroadG	$\Delta\log g$>2 dex	Filter all *except* $T_{\rm eff}$ *if* $T_{\rm eff}$>7000 *K* Flag = 9
	1< $\Delta\log g$≤2 dex	Flag = 2
	0.5< $\Delta\log g$≤1 dex	Flag = 1
	$\Delta\log g$≤0.5 dex	Flag = 0
vbroadM	Δ[M/H]>2 dex	Filter [M/H] and [X/Fe] Flag = 9
	0.5< Δ[M/H]≤2 dex	Flag = 2
	0.25< Δ[M/H]≤0.5 dex	Flag = 1
	Δ[M/H]≤0.25 dex	Flag = 0

Definition of parameter flags considering potential biases due to uncertainties in the radial velocity shift correction:

Flag name	Condition	Flag value
vradT	$\Delta T_{\rm eff}$>2000 K	Filter all Flag = 9
	500< $\Delta T_{\rm eff}$≤2000 K	Flag = 2
	250< $\Delta T_{\rm eff}$≤500 K	Flag = 1
	$\Delta T_{\rm eff}$≤250 K	Flag = 0
vradG	$\Delta\log g$>2 dex	Filter all *except* $T_{\rm eff}$ *if* $T_{\rm eff}$>7000 *K* Flag = 9
	1< $\Delta\log g$≤2 dex	Flag = 2
	0.5< $\Delta\log g$≤1 dex	Flag = 1
	$\Delta\log g$≤0.5 dex	Flag = 0
vradM	Δ[M/H]>2 dex	Filter [M/H] and [X/Fe] Flag = 9
	0.5< Δ[M/H]≤2 dex	Flag = 2
	0.25< Δ[M/H]≤0.5 dex	Flag = 1
	Δ[M/H]≤0.25 dex	Flag = 0

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Definition of parameter flags considering potential biases due to uncertainties in the RVS flux:

Flag name	Condition	Flag value
fluxNoise	$\sigma T_{\rm eff}$>2000 K or σlog g>2 dex	Filter all Flag = 9
	$\sigma T_{\rm eff}\leq$2000 K and σlog $g\leq$2 dex and σ[M/H]>2 dex	Filter [M/H], [X/Fe] Flag = 5
	$\sigma T_{\rm eff}\leq$2000 K and σlog $g\leq$2 dex and σ[M/H]$\leq$2 dex and σ[α/Fe]>0.8 dex	Filter [α/Fe], [X/Fe] Flag = 4
	500< $\sigma T_{\rm eff}\leq$2000 K and 1< σlog $g\leq$2 dex and 0.5< σ[M/H]$\leq$2 dex and 0.2< σ[α/Fe]$\leq$0.8 dex	Flag = 3
	250< $\sigma T_{\rm eff}\leq$500 K and 0.5< σlog $g\leq$1 dex and 0.25< σ[M/H]$\leq$0.5 dex and 0.1< σ[α/Fe]$\leq$0.2 dex	Flag = 2
	100< $\sigma T_{\rm eff}\leq$250 K and 0.2< σlog $g\leq$0.5 dex and 0.1< σ[M/H]$\leq$0.25 dex and 0.05< σ[α/Fe]$\leq$0.1 dex	Flag = 1
	$\sigma T_{\rm eff}\leq$100 K and σlog $g\leq$0.2 dex and σ[M/H]$\leq$0.1 dex and σ[α/Fe]$\leq$0.05 dex	Flag = 0

Definition of parameter flags considering potential biases due to extrapolated parameters:

Draft

Flag name	Condition	Flag value
extrapol	gof=NaN and ($T_{\rm eff}$>9000 K or $T_{\rm eff}$<2500 K or $\log g$>6 dex or $\log g$< −1 dex)	Filter all *if $T_{\rm eff}$>7000 K* Flag = 9
	gof=NaN and 2500≤$T_{\rm eff}$≤9000 K and −1 ≤$\log g$≤6 dex and ([M/H]< −6 dex or [M/H]>1.5 dex)	Filter [M/H],[X/Fe] Flag = 4
	gof=NaN and 2500≤$T_{\rm eff}$<9000 K and −1 ≤$\log g$≤6 dex and −6 ≤[M/H]≤1.5 dex and [α/Fe] out from standard by ± 0.8	Filter [X/Fe] Flag = 3
	gof=NaN and 2500≤$T_{\rm eff}$≤9000 K and −1 ≤$\log g$≤6 dex and −6 ≤[M/H]≤1.5 dex and [α/Fe] within ± 0.8 from standard	Flag = 2
	gof≠NaN and ($T_{\rm eff}$≥7625 or $T_{\rm eff}$≤3500 K or $\log g$≥4.75 or $\log g$≤0.25 dex or [M/H]≤ −3 or [M/H]≥0.75 dex or [α/Fe] out from standard by ± 0.35)	Flag = 1
	gof≠NaN and 3500<$T_{\rm eff}$<7625 K and 0.25<$\log g$<4.75 dex and −3 <[M/H]<0.75 dex and [α/Fe] within ± 0.35 from standard	Flag = 0

Definition of parameter flags considering RVS flux problems or emission line probability:

Draft

Flag name	Condition	Flag value
nanFlux	Flux=NaN	Filter all *if* $T_{\rm eff}$>7000 K Flag = 9
	no Nans in Flux	Flag = 0
emission	*CU6_is_emission*=True	Filter all *if* $T_{\rm eff}$>7000 K Flag = 9
	CU6_is_emission=False	Flag = 0
neg_flux	> 2 pixels with flux<0	Filter all *if* $T_{\rm eff}$>7000 K Flag = 9
	1 or 2 pixels with flux<0	Flag = 1
	flux>0	Flag = 0
nullFluxErr	$\sigma T_{\rm eff}$=0 K or σlog g=0 dex or σ[M/H]=0 dex or σ[α/Fe]=0 dex	Filter all Flag = 9
	no null uncertainties for all pixel flux	Flag = 0

Definition of parameter flags considering problems in the parameterisation of KM-type giants. $F_{\rm min}$ is the minumum flux value in the corresponding RVS spectrum:

Flag name	Condition	Flag value
KMgiantPar	$T_{\rm eff}$<4000 K and log g<3.5 and (gof> −3.0 or $F_{\rm min}$>0.22)	Filter [α/Fe] Flag = 2
	$T_{\rm eff}$<4000 K and log g<3.5 and (−3.4 <gof< −3.0 or (gof> −3.0 and $F_{\rm min}$<0.22) or (gof< −3.4 and $F_{\rm min}$>0.22))	Filter [α/Fe] Flag = 1
	($T_{\rm eff}$<4000 K and log g<3.5 and (gof< −3.4 or $F_{\rm min}$<0.22)) or $T_{\rm eff}$<4000 or log g>3.5	Flag = 0

Definition of individual abundance upper limit flags. Xfe_gspspec_upper is the upper confidence value of the abundance, corresponding to the 84th quantile of the Monte Carlo distribution. σ[X/Fe] is the 84th−16th interquantile abundance uncertainty. XfeUpperLimit is the mean value of the abundance upper limit for the considered lines of element X in the spectrum, depending on the mean signal-to-noise ratio (SNR) in the line pixels and on the stellar parameters. X_MAD_UpperLimit is the median absolute deviation of the upper limit in the line pixels; the coefficients c1 to c8 are reported in a table further below:

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Flag name	Condition	Flag value
XUpLim	vbroadT≥2 or vbroadG≥2 or vbroadM≥2 or σ[X/Fe]=0 or (Xfe_gspspec_upper−Xfe_gspspec)= 0 or $T_{\rm eff}$≤c1 or $T_{\rm eff}$≥c2 or log g≤c3 or log g≥c4 or ((2− XfeUpperLimit)/σ[X/Fe])≤c5 or (SNR≤c6 and gof≥c7) or (Xfe_gspspec+[M/H])≤c8	Filter [X/Fe] Flag = 9
	vbroadT<2 and vbroadG<2 and vbroadM<2 and σ[X/Fe]≠0 and (Xfe_gspspec_upper− Xfe_gspspec)≠ 0 and c1<$T_{\rm eff}$<c2 and c3<log g<c4 and ((2− XfeUpperLimit)/σ[X/Fe])>c5 and (SNR>c6 or (SNR≤c6 and gof<c7)) and (Xfe_gspspec+[M/H])<c8 and ((Xfe_gspspec− XfeUpperLimit)/(1.48·X_MAD_UpperLimit))<1.5	Flag = 2
	vbroadT<2 and vbroadG<2 and vbroadM<2 and σ[X/Fe]≠0 and (Xfe_gspspec_upper− Xfe_gspspec)≠ 0 and c1<$T_{\rm eff}$<c2 and c3<log g<c4 and ((2− XfeUpperLimit)/σ[X/Fe])>c5 and (SNR>c6 or (SNR≤c6 and gof<c7)) and (Xfe_gspspec+[M/H])<c8 and 1.5≤((Xfe_gspspec− XfeUpperLimit)/(1.48·X_MAD_UpperLimit))<2.5	Flag = 1
	vbroadT<2 and vbroadG<2 and vbroadM<2 and σ[X/Fe]≠0 and (Xfe_gspspec_upper− Xfe_gspspec)≠ 0 and c1<$T_{\rm eff}$<c2 and c3<log g<c4 and ((2− XfeUpperLimit)/σ[X/Fe])>c5 and (SNR>c6 or (SNR≤c6 and gof<c7)) and (Xfe_gspspec+[M/H])<c8 and ((Xfe_gspspec− XfeUpperLimit)/(1.48·X_MAD_UpperLimit))≥2.5	Flag = 0

Definition of individual abundance uncertainty flags. Xfe_gspspec_upper is the upper confidence value of the abundance, corresponding to the 84th quantile of the Monte Carlo distribution. σ[X/Fe] is the 84th−16th interquantile abundance uncertainty. XfeUpperLimit is the mean value of the abundance upper limit for the considered lines of element X in the spectrum, depending on the mean SNR in the line pixels and the stellar parameters. X_MAD_UpperLimit is the median absolute deviation of the upper limit in the line pixels; the coefficients c1 to c8 are reported in a table further below:

Flag name	Condition	Flag value
XUncer	vbroadT≥2 or vbroadG≥2 or vbroadM≥2 or σ[X/Fe]=0 or (Xfe_gspspec_upper−Xfe_gspspec)= 0 or $T_{\rm eff}$≤c1 or $T_{\rm eff}$≥c2 or log g≤c3 or log g≥c4 or ((2−XfeUpperLimit)/σ[X/Fe])≤c5 or (SNR≤c6 and gof≥c7) or (Xfe_gspspec+[M/H])≤c8	Filter [X/Fe] Flag = 9
	vbroadT<2 and vbroadG<2 and vbroadM<2 and σ[X/Fe]≠0 and (Xfe_gspspec_upper−Xfe_gspspec)≠ 0 and c1<$T_{\rm eff}$<c2 and c3<log g<c4 and c5<((2−XfeUpperLimit)/σ[X/Fe])<7 and (SNR>c6 or (SNR≤c6 and gof<c7)) and (Xfe_gspspec+[M/H])<c8	Flag = 2
	vbroadT<2 and vbroadG<2 and vbroadM<2 and σ[X/Fe]≠0 and (Xfe_gspspec_upper−Xfe_gspspec)≠ 0 and c1<$T_{\rm eff}$<c2 and c3<log g<c4 and 7≤((2−XfeUpperLimit)/σ[X/Fe])<10 and (SNR>c6 or (SNR≤c6 and gof<c7)) and (Xfe_gspspec+[M/H])<c8	Flag = 1
	vbroadT<2 and vbroadG<2 and vbroadM<2 and σ[X/Fe]≠0 and (Xfe_gspspec_upper−Xfe_gspspec)≠ 0 and c1<$T_{\rm eff}$<c2 and c3<log g<c4 and ((2−XfeUpperLimit)/σ[X/Fe])≥10 and (SNR>c6 or (SNR≤c6 and gof<c7)) and (Xfe_gspspec+[M/H])<c8	Flag = 0

Coefficients for individual chemical abundance filtering used in the previous two tables:

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Chemical abundance	c1	c2	c3	c4	c5	c6	c7	c8
[N/Fe]	4200	8000	0.0	5.5	4.5	100	−3.6	99
[Mg/Fe]	3500	8000	−1.0	5.5	5.5	80	−3.5	99
[Si/Fe]	4000	8000	−1.0	5.5	6.0	110	−3.8	99
[S/Fe]	5500	8000	3.0	5.5	5.0	120	−3.7	99
[Ca/Fe]	3500	8000	−1.0	5.5	10.0	60	−3.2	99
[Ti/Fe]	4000	6500	−1.0	5.5	6.0	110	−3.65	99
[Cr/Fe]	3500	6000	−1.0	5.5	6.0	1000	−3.65	1.5
[Fe/M]	3500	8000	−1.0	5.5	5.0	1000	−3.4	1.5
[FeII/M]	5700	8000	3.5	5.5	5.0	70	−3.5	1.5
[Ni/Fe]	4000	6500	−1.0	5.5	6.0	100	−3.6	1.5
[Zr/Fe]	3500	8000	−1.0	5.5	1.0	100	−3.4	99
[Ce/Fe]	3500	8000	−1.0	5.5	5.0	100	−3.5	99
[Nd/Fe]	3500	5500	−1.0	5.5	2.0	100	−3.5	99

LOGCHISQ_GSPSPEC : Logarithm of the goodness-of-fit for the GSP-Spec MatisseGauguin epoch parameters (float)

Logarithm to the base 10 of the chi-squared between input epoch spectrum rebinned to 800 pixels and solution synthetic spectrum computed from GSP-Spec MatisseGauguin (Recio-Blanco et al. 2023).

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4.8 AP_XP [TAP]

This table contains the astrophysical parameters of stars derived from mean XP spectra by ESP-UCD, ESP-ELS, GSP-Phot, and FLAME as part of the Apsis processing chain.

See Section **??**, Section **??**, Section **??**, and Section **??** of the online documentation for details on the ESP-UCD, ESP-ELS, GSP-Phot, and FLAME modules.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

TEFF_ESPUCD : Effective temperature estimate from ESP-UCD based on the RP spectrum (float, Temperature[K])

Effective temperature estimate from ESP-UCD inferred from the RP spectrum. The prediction module is based on a Gaussian Process trained with empirical examples. See Section **??** for details.

TEFF_ESPUCD_UNCERTAINTY : Uncertainty of the effective temperature estimate produced by ESP-UCD (float, Temperature[K])

Standard deviation of 10 effective temperature predictions obtained by the ESP-UCD module for 10 RP spectra generated using random sampling from Gaussian distributions centred at the observed fluxes and with standard deviations given by the flux uncertainties. The value thus obtained is rescaled to match the root-mean-square error of the module predictions for a set of well-known ultracool dwarfs (see Section **??** for further details).

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FLAGS_ESPUCD : Quality flags of the ESP-UCD parameter estimates (string)

ESP-UCD quality flag. A digit based on a weighted average of several quality diagnostics (of the astrometric solution and of the RP spectrum). See Section **??** for a quantitative definition of the quality categories.

YOUTHINDEX_ESPUCD : Youth index for ultracool dwarfs (float)

Youth index based on the Diffusion Map coordinates of the RP spectrum derived using the Nyquist approximation for a pre-existing map constructed with a selection of ultracool dwarfs in star forming regions and young clusters and in the Main Sequence. The value goes from zero (for Main Sequence typical diffusion coordinates) to 1 for UCDs in the youngest star forming regions.

SPECTRALTYPE_ESPELS : Spectral type tag from ESP-ELS (string)

Spectral type tag derived by ESP-ELS from the analysis of the BP/RP data and (reddened) absolute magnitude. The recognised classes are O, B, A, F, G, K, M, UCD (Ultra-Cool Dwarf), AGBC (carbon-rich AGB star), WD (White Dwarf). This label is used to help ESP-HS determine which sources to process as ESP-HS processes only those tagged as A, B and O. ESP-ELS processes sources brighter than $G = 18.25$. More information on the module can be found in Section **??** and Section **??**.

FLAGS_ESPELS_SPECTRALTYPE : Quality flag of the spectraltype classification made by ESP-ELS (string)

The quality flag is assigned based on the analysis of the probability distribution. It can take integer values from 1 to 6, with 1 and 6 representing the highest and lowest quality, respectively. Values of 5 and 6 should be considered cautiously. The flag 0 is assigned to AGBC sources that are the most likely to be carbon-rich AGB stars (the *golden sample*), following the criteria defined by Gaia Collaboration et al. (2023a). ESP-ELS was applied only to sources brighter than $G = 18.25$. During validation, missing spectral-type labels were reported. During post-processing, the GSP-Phot results were therefore checked, and the field was updated for sources with effective temperatures above 7500 K. For these sources, which also have parameter estimates from ESP-HS, the quality flag was set to 7. More information on the module and quality flag can be found in Section **??** and Section **??**.

SPECTRALTYPE_ESPELS_PROCESSMODE : Processing mode adopted to derive the spectral type

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tag (string)

Processing mode used to assign the spectral type. A set of two classifiers (RFC1 and RFC2) is used to determine the spectral type. If the parallax and G magnitude are not available, the module switches from RFC1 (processing mode '0') to RFC2 ('1'). During validation, missing spectral-type labels were reported. The post-processing module therefore checked the GSP-Phot results, and the field was updated for sources with effective temperatures above 7500 K. For these sources, which also have parameter estimates from ESP-HS, the processing mode was set to '2'. ESP-ELS was applied only to sources brighter than $G = 18.25$. More information on the module can be found in Section **??** and Section **??**.

EW_ESPELS_HALPHA : Halpha pseudo-equivalent width from ESP-ELS (float, Length & Distance[nm])

Pseudo-equivalent width (pEW) of the Hα line measured on the RP spectra. Unlike in Gaia DR3, the model estimate was not subtracted. ESP-ELS was applied only to targets brighter than $G = 18.25$. More information on the module can be found in Section **??** and Section **??**.

EW_ESPELS_HALPHA_UNCERTAINTY : Uncertainty of the Halpha pseudo-equivalent width from ESP-ELS (float, Length & Distance[nm])

Uncertainty on the pseudo-equivalent width (pEW) of the Hα line. It is computed by propagating the RP flux uncertainties through the integration over the Hα window. Correlations between samples are ignored. ESP-ELS was applied only to targets brighter than $G = 18.25$. Further information on the module is provided in Section **??** and Section **??**.

FLAGS_ESPELS_HALPHA : Quality flag of the Halpha pseudo-equivalent width from ESP-ELS (string)

Quality flag indicating the processing mode adopted to measure the Hα pseudo-equivalent width. This flag is always set to 0. ESP-ELS was applied only on targets brighter than $G = 18.25$. Further information on the module is provided in Section **??** and Section **??**.

EW_ESPELS_HALPHA_MODEL : Halpha pseudo-equivalent width from ESP-ELS measured on the synthetic spectrum (float, Length & Distance[nm])

Hα pseudo-equivalent width (pEW) measured on the synthetic spectrum having the nearest astrophysical parameters to those derived by GSP-Phot (best library value, before post-processing). The value can be used as an attempt to remove the photospheric contribution from the measured

pEW. ESP-ELS was applied only to targets brighter than $G = 18.25$. More information on the module can be found in Section **??** and Section **??**, as well as in **?**.

EW_ESPELS_HALPHA_PROBABILITY : Probability that H-alpha is in emission (float, Dimensionless[percentage/100])

Probability that the Hα line shows significant emission. The probability is obtained from a random-forest classifier trained on the Hα line extracted from the simulated spectra. The probability value ranges from 0 (no Hα emission) to 1 (significant emission in the line is expected). ESP-ELS was applied only to targets brighter than $G = 18.25$. More information on the module can be found in Section **??** and Section **??**, as well as in **?**.

TEFF_GSPPHOT : Effective temperature from GSP-Phot Aeneas best library using BP/RP spectra (float, Temperature[K])

Effective temperature (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax (see Section **??** of the online documentation). This is the median of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation).

TEFF_GSPPHOT_LOWER : Lower confidence level (16%) of effective temperature from GSP-Phot Aeneas best library using BP/RP spectra (float, Temperature[K])

Lower confidence level (16%) of effective temperature (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

TEFF_GSPPHOT_UPPER : Upper confidence level (84%) of effective temperature from GSP-Phot Aeneas best library using BP/RP spectra (float, Temperature[K])

Upper confidence level (84%) of effective temperature (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

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LOGG_GSPPHOT : Surface gravity from GSP-Phot Aeneas best library using BP/RP spectra (float, GravitySurface[log cgs])

Surface gravity (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax (see Section **??** of the online documentation). This is the median of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation).

LOGG_GSPPHOT_LOWER : Lower confidence level (16%) of surface gravity from GSP-Phot Aeneas best library using BP/RP spectra (float, GravitySurface[log cgs])

Lower confidence level (16%) of surface gravity (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

LOGG_GSPPHOT_UPPER : Upper confidence level (84%) of surface gravity from GSP-Phot Aeneas best library using BP/RP spectra (float, GravitySurface[log cgs])

Upper confidence level (84%) of surface gravity (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

MH_GSPPHOT : Global metallicity from GSP-Phot Aeneas best library using BP/RP spectra (float, Abundances[dex])

Decimal logarithm of the ratio of the number abundance of metals to the number abundance of hydrogen relative to the same ratio of solar abundances inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax, assuming source is a single star (see Section **??** of the online documentation). This is the median of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation).

MH_GSPPHOT_LOWER : Lower confidence level (16%) of global metallicity from GSP-Phot Aeneas best library using BP/RP spectra (float, Abundances[dex])

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Lower confidence level (16%) of decimal logarithm of the ratio of the number abundance of metals to the number abundance of hydrogen relative to the same ratio of solar abundances inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax, assuming source is a single star (see Section **??** of the online documentation). This is the 16th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

MH_GSPPHOT_UPPER : Upper confidence level (84%) of global metallicity from GSP-Phot Aeneas best library using BP/RP spectra (float, Abundances[dex])

Upper confidence level (84%) of decimal logarithm of the ratio of the number abundance of metals to the number abundance of hydrogen relative to the same ratio of solar abundances inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax, assuming source is a single star (see Section **??** of the online documentation). This is the 84th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

FEH_GSPPHOT : Iron abundance from GSP-Phot Aeneas best library using BP/RP spectra (float, Abundances[dex])

Decimal logarithm of the ratio of the number abundance of iron to the number abundance of hydrogen relative to the same ratio of solar abundances inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax, assuming source is a single star (see Section **??** of the online documentation). This is the median of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation).

FEH_GSPPHOT_UPPER : Upper confidence level (84%) of iron abundance from GSP-Phot Aeneas best library using BP/RP spectra (float, Abundances[dex])

Upper confidence level (84%) of decimal logarithm of the ratio of the number abundance of iron to the number abundance of hydrogen relative to the same ratio of solar abundances inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax, assuming source is a single star (see Section **??** of the online documentation). This is the 84th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

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FEH_GSPPHOT_LOWER : Lower confidence level (16%) of iron abundance from GSP-Phot Aeneas best library using BP/RP spectra (float, Abundances[dex])

Lower confidence level (16%) of decimal logarithm of the ratio of the number abundance of iron to the number abundance of hydrogen relative to the same ratio of solar abundances inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax, assuming source is a single star (see Section **??** of the online documentation). This is the 16th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

DISTANCE_GSPPHOT : Distance from GSP-Phot Aeneas best library using BP/RP spectra (float, Length & Distance[pc])

Distance (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax(see Section **??** of the online documentation). This is the median of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). NB: The actual fit parameter is $\log_{10} d$ and a prior is imposed to ensure a value between [0,5], thus the minimum possible distance is 1 pc and the maximum is 100 kpc.

DISTANCE_GSPPHOT_LOWER : Lower confidence level (16%) of distance from GSP-Phot Aeneas best library using BP/RP spectra (float, Length & Distance[pc])

Lower confidence level (16%) of distance (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval. NB: The actual fit parameter is $\log_{10} d$ and a prior is imposed to ensure a value between [0,5], thus the minimum possible distance is 1 pc and the maximum is 100 kpc.

DISTANCE_GSPPHOT_UPPER : Upper confidence level (84%) of distance from GSP-Phot Aeneas best library using BP/RP spectra (float, Length & Distance[pc])

Upper confidence level (84%) of distance (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval. NB: The actual fit parameter is $\log_{10} d$ and a prior is

imposed to ensure a value between [0,5], thus the minimum possible distance is 1 pc and the maximum is 100 kpc.

AZERO_GSPPHOT : Extinction parameter from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Extinction parameter A_0 corresponding to a monochromatic extinction at a specific reference wavelength (see Section **??** of the online documentation) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). NB: This is the extinction parameter in the adopted extinction curve from Gordon et al. (2023).

AZERO_GSPPHOT_LOWER : Lower confidence level (16%) of extinction parameter from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of extinction parameter A_0 corresponding to a monochromatic extinction at a specific reference wavelength (see Section **??** of the online documentation) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). NB: This is the extinction parameter in the adopted extinction curve from Gordon et al. (2023).

AZERO_GSPPHOT_UPPER : Upper confidence level (84%) of extinction parameter from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of extinction parameter A_0 corresponding to a monochromatic extinction at a specific reference wavelength (see Section **??** of the online documentation) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). NB: This is the extinction parameter in the adopted extinction curve from Gordon et al. (2023).

AG_GSPPHOT : Extinction in G band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Broadband extinction in G band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the

MCMC chain (see Section **??** of the online documentation).

AG_GSPPHOT_LOWER : Lower confidence level (16%) of extinction in G band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of broadband extinction in G band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

AG_GSPPHOT_UPPER : Upper confidence level (84%) of extinction in G band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of broadband extinction in G band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

ABP_GSPPHOT : Extinction in BP band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Broadband extinction in $G_{\rm BP}$ band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation).

ABP_GSPPHOT_LOWER : Lower confidence level (16%) of extinction in BP band band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of broadband extinction in $G_{\rm BP}$ band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

ABP_GSPPHOT_UPPER : Upper confidence level (84%) of extinction in BP band from GSP-Phot

Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of broadband extinction in G_{BP} band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

ARP_GSPPHOT : Extinction in RP band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Broadband extinction in G_{RP} band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation).

ARP_GSPPHOT_LOWER : Lower confidence level (16%) of extinction in RP band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of broadband extinction in G_{RP} band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

ARP_GSPPHOT_UPPER : Upper confidence level (84%) of extinction in RP band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of broadband extinction in G_{RP} band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

AV_GSPPHOT : Extinction in Johnson V band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Broadband extinction in Johnson V band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the

MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation).

AV_GSPPHOT_LOWER : Lower confidence level (16%) of extinction in Johnson V band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of broadband extinction in Johnson V band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

AV_GSPPHOT_UPPER : Upper confidence level (84%) of extinction in Johnson V band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of broadband extinction in Johnson V band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

AJ_GSPPHOT : Extinction in 2MASS J band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Broadband extinction in 2MASS J band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation).

AJ_GSPPHOT_LOWER : Lower confidence level (16%) of extinction in 2MASS J band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of broadband extinction in 2MASS J band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

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AJ_GSPPHOT_UPPER : Upper confidence level (84%) of extinction in 2MASS J band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of broadband extinction in 2MASS J band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

AH_GSPPHOT : Extinction in 2MASS H band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Broadband extinction in 2MASS H band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation).

AH_GSPPHOT_LOWER : Lower confidence level (16%) of extinction in 2MASS H band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of broadband extinction in 2MASS H band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

AH_GSPPHOT_UPPER : Upper confidence level (84%) of extinction in 2MASS H band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of broadband extinction in 2MASS H band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

AKS_GSPPHOT : Extinction in 2MASS Ks band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Broadband extinction in 2MASS Ks band (assuming source is a single star) inferred by GSP-

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Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation).

AKS_GSPPHOT_LOWER : Lower confidence level (16%) of extinction in 2MASS Ks band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of broadband extinction in 2MASS Ks band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

AKS_GSPPHOT_UPPER : Upper confidence level (84%) of extinction in 2MASS Ks band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of broadband extinction in 2MASS Ks band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

AW1_GSPPHOT : Extinction in WISE W1 band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Broadband extinction in WISE W1 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation).

AW1_GSPPHOT_LOWER : Lower confidence level (16%) of extinction in WISE W1 band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of broadband extinction in WISE W1 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

AW1_GSPPHOT_UPPER : Upper confidence level (84%) of extinction in WISE W1 band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of broadband extinction in WISE W1 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

AW2_GSPPHOT : Extinction in WISE W2 band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Broadband extinction in WISE W2 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation).

AW2_GSPPHOT_LOWER : Lower confidence level (16%) of extinction in WISE W2 band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of broadband extinction in WISE W2 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

AW2_GSPPHOT_UPPER : Upper confidence level (84%) of extinction in WISE W2 band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of broadband extinction in WISE W2 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

EBPMINRP_GSPPHOT : Reddening in BP minus RP colour from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Reddening $E(G_{\rm BP} - G_{\rm RP})$ (assuming source is a single star) inferred by GSP-Phot Aeneas from

BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Note that while $E(G_{\rm BP} - G_{\rm RP}) = A_{\rm BP} - A_{\rm RP}$, this was computed at the level of MCMC samples. Hence, this relation is not exactly true for the median values.

EBPMINRP_GSPPHOT_LOWER : Lower confidence level (16%) of reddening in BP minus RP colour from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of reddening $E(G_{\rm BP} - G_{\rm RP})$ (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval. Note that while $E(G_{\rm BP} - G_{\rm RP}) = A_{\rm BP} - A_{\rm RP}$, this was computed at the level of MCMC samples. Hence, this relation is not exactly true for the lower confidence levels.

EBPMINRP_GSPPHOT_UPPER : Upper confidence level (84%) of reddening in BP minus RP colour from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of reddening $E(G_{\rm BP} - G_{\rm RP})$ (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval. Note that while $E(G_{\rm BP} - G_{\rm RP}) = A_{\rm BP} - A_{\rm RP}$, this was computed at the level of MCMC samples. Hence, this relation is not exactly true for the upper confidence levels.

EBMINV_GSPPHOT : Reddening in B minus V colour from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Reddening $E(B-V)$ (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation).

EBMINV_GSPPHOT_LOWER : Lower confidence level (16%) of reddening in B minus V colour from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

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Lower confidence level (16%) of reddening $E(B-V)$ (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

EBMINV_GSPPHOT_UPPER : Upper confidence level (84%) of reddening in B minus V colour from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of reddening $E(B-V)$ (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

ABSMAG_G_GSPPHOT : Absolute magnitude in G band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Absolute magnitude M_{G} (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation).

ABSMAG_G_GSPPHOT_LOWER : Lower confidence level (16%) of absolute magnitude in G band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of absolute magnitude M_{G} (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

ABSMAG_G_GSPPHOT_UPPER : Upper confidence level (84%) of absolute magnitude G band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of absolute magnitude M_{G} (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documenta-

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tion). Lower and upper levels include 68% confidence interval.

ABSMAG_J_GSPPHOT : Absolute magnitude in 2MASS J band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Absolute magnitude M_J in 2MASS J band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation).

ABSMAG_J_GSPPHOT_LOWER : Lower confidence level (16%) of absolute magnitude in 2MASS J band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of absolute magnitude M_J in 2MASS J band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

ABSMAG_J_GSPPHOT_UPPER : Upper confidence level (84%) of absolute magnitude in 2MASS J band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of absolute magnitude M_J in 2MASS J band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

ABSMAG_H_GSPPHOT : Absolute magnitude in 2MASS H band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Absolute magnitude M_H in 2MASS H band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation).

ABSMAG_H_GSPPHOT_LOWER : Lower confidence level (16%) of absolute magnitude in 2MASS H band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of absolute magnitude $M_{\rm H}$ in 2MASS H band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

ABSMAG_H_GSPPHOT_UPPER : Upper confidence level (84%) of absolute magnitude in 2MASS H band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of absolute magnitude $M_{\rm H}$ in 2MASS H band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

ABSMAG_KS_GSPPHOT : Absolute magnitude in 2MASS Ks band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Absolute magnitude $M_{\rm Ks}$ in 2MASS Ks band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation).

ABSMAG_KS_GSPPHOT_LOWER : Lower confidence level (16%) of absolute magnitude in 2MASS Ks band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of absolute magnitude $M_{\rm Ks}$ in 2MASS Ks band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

ABSMAG_KS_GSPPHOT_UPPER : Upper confidence level (84%) of absolute magnitude in 2MASS Ks band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of absolute magnitude $M_{\rm Ks}$ in 2MASS Ks band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online doc-

umentation). Lower and upper levels include 68% confidence interval.

ABSMAG_W1_GSPPHOT : Absolute magnitude in WISE W1 band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Absolute magnitude M_{W1} in WISE W1 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation).

ABSMAG_W1_GSPPHOT_LOWER : Lower confidence level (16%) of absolute magnitude in WISE W1 band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of absolute magnitude M_{W1} in WISE W1 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

ABSMAG_W1_GSPPHOT_UPPER : Upper confidence level (84%) of absolute magnitude in WISE W1 band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of absolute magnitude M_{W1} in WISE W1 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

ABSMAG_W2_GSPPHOT : Absolute magnitude in WISE W2 band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Absolute magnitude M_{W2} in WISE W2 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation).

ABSMAG_W2_GSPPHOT_LOWER : Lower confidence level (16%) of absolute magnitude in WISE W2 band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of absolute magnitude M_{W2} in WISE W2 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

ABSMAG_W2_GSPPHOT_UPPER : Upper confidence level (84%) of absolute magnitude in WISE W2 band from GSP-Phot Aeneas best library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of absolute magnitude M_{W2} in WISE W2 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

RADIUS_GSPPHOT : Radius from GSP-Phot Aeneas best library using BP/RP spectra (float, Length & Distance[Solar Radius])

Stellar radius (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation).

RADIUS_GSPPHOT_LOWER : Lower confidence level (16%) of radius from GSP-Phot Aeneas best library using BP/RP spectra (float, Length & Distance[Solar Radius])

Lower confidence level (16%) of stellar radius (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

RADIUS_GSPPHOT_UPPER : Upper confidence level (84%) of radius from GSP-Phot Aeneas best library using BP/RP spectra (float, Length & Distance[Solar Radius])

Upper confidence level (84%) of stellar radius (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper

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levels include 68% confidence interval.

MASS_GSPPHOT : Mass from GSP-Phot Aeneas best library using BP/RP spectra (float, Mass[Solar Mass])

Stellar mass (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation).

MASS_GSPPHOT_LOWER : Lower confidence level (16%) of mass from GSP-Phot Aeneas best library using BP/RP spectra (float, Mass[Solar Mass])

Lower confidence level (16%) of stellar mass (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

MASS_GSPPHOT_UPPER : Upper confidence level (84%) of mass from GSP-Phot Aeneas best library using BP/RP spectra (float, Mass[Solar Mass])

Upper confidence level (84%) of stellar mass (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

AGE_GSPPHOT : Age from GSP-Phot Aeneas best library using BP/RP spectra (float, Time[Gyr])

Stellar age (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation).

AGE_GSPPHOT_LOWER : Lower confidence level (16%) of age from GSP-Phot Aeneas best library using BP/RP spectra (float, Time[Gyr])

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Lower confidence level (16%) of stellar age (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

AGE_GSPPHOT_UPPER : Upper confidence level (84%) of age from GSP-Phot Aeneas best library using BP/RP spectra (float, Time[Gyr])

Upper confidence level (84%) of stellar age (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from best library that achieves the highest mean log-posterior averaged over the MCMC chain (see Section **??** of the online documentation). Lower and upper levels include 68% confidence interval.

LOGPOSTERIOR_GSPPHOT : mean log-posterior of MCMC of GSP-Phot Aeneas MCMC best library (float)

Goodness-of-fit score defined as the mean log-posterior of all MCMC samples of GSP-Phot Aeneas MCMC for best library. The higher the mean log-posterior, the better the fit. Values are usually negative. Subtracting the mean log-likelihood from the mean log-posterior yields the mean prior. NB: This is not a Bayesian evidence!

LOGLIKELIHOOD_GSPPHOT : mean likelihood of MCMC of GSP-Phot Aeneas MCMC best library (float)

Mean log-likelihood of all MCMC samples of GSP-Phot Aeneas MCMC for best library. The higher the mean log-likelihood, the better the fit. Values are usually negative. Subtracting the mean log-likelihood from the mean log-posterior yields the mean prior. NB: This is not a Bayesian evidence!

LOGPOST_SPREAD_GSPPHOT : maximum minus minimum of log-posterior throughout MCMC chain for best library (float)

Difference of maximum log-posterior minus minimum log-posterior in MCMC chain (excluding burn-in phase) for best library. If this spread is much larger than 25, say > 100, it suggests that the MCMC chain has not fully converged.

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MCMCACCEPT_GSPPHOT : MCMC acceptance rate of GSP-Phot Aeneas MCMC best library (float)

MCMC acceptance rate of GSP-Phot Aeneas MCMC best library. This is computed from all MCMC samples (before thinning the chain to 2000 or 100 samples).

LIBNAME_GSPPHOT : Name of library that achieves the highest mean log-posterior in MCMC samples and was used to derive GSP-Phot parameters in this table (string)

Name of library of synthetic stellar spectra (one of A, MARCS, OB, PHOENIX) for which GSP-Phot achieves the highest goodness-of-fit score, i.e. the highest mean log-posterior in its MCMC samples, referred to as 'best library' (see Section **??** of the online documentation). This is the library used to derive GSP-Phot parameters in this table. For more information on the synthetic libraries see Section **??**.

RADIUS_FLAME : Radius of the star from FLAME using effective temperature from GSP-Phot and luminosity from FLAME (float, Length & Distance[Solar Radius])

The radius of the star from FLAME, derived from `teff_gspphot` and `lum_flame` using the Stefan-Boltzmann law. The solar effective temperature is 5772 K, see Section **??**. It is defined as the median value (50th percentile) of the distribution from sampling.

RADIUS_FLAME_LOWER : Lower confidence level (16%) of FLAME radius (float, Length & Distance[Solar Radius])

Lower confidence level of `radius_flame`. It is defined as the 16th percentile value of the distribution from sampling. Lower and upper levels include 68% confidence interval.

RADIUS_FLAME_UPPER : Upper confidence level (84%) of FLAME radius (float, Length & Distance[Solar Radius])

Upper confidence level of `radius_flame`. It is defined as the 84th percentile value of the distribution from sampling. Lower and upper levels include 68% confidence interval.

LUM_FLAME : Luminosity of the star from FLAME using G band magnitude, extinction, parallax or distance, and a bolometric correction (float, Luminosity[Solar Luminosity])

Luminosity of the star from FLAME using G-band magnitude, extinction from GSP-Phot `ag_gspphot`,

a distance estimate, and a bolometric correction `bc_flame`. It is defined as the median value (50th percentile) of the distribution from sampling. The bolometric correction depends on the effective temperature, metallicity and surface gravity, and these are based on GSP-Phot values. The bolometric magnitude of the Sun = 4.74 mag, see Section **??** and the reference absolute G-band magnitude of the Sun is 4.66 mag, see Section **??**.

LUM_FLAME_LOWER : Lower confidence level (16%) of FLAME luminosity (float, Luminosity[Solar Luminosity])

Lower confidence level of `lum_flame`. It is defined as the 16th percentile value of the distribution from sampling. Lower and upper levels include 68% confidence interval.

LUM_FLAME_UPPER : Upper confidence level (84%) of FLAME luminosity (float, Luminosity[Solar Luminosity])

Upper confidence level of `lum_flame`. It is defined as the 84th percentile value of the distribution from sampling. Lower and upper levels include 68% confidence interval.

MASS_FLAME : Mass of the star from FLAME using stellar models, the FLAME luminosity and GSP-Phot parameters (float, Mass[Solar Mass])

Mass of the star from FLAME using GSP-Phot based parameters as input. It is defined as the median value (50th percentile) of the 1D projected distribution from sampling in mass, age, and metallicity. It is derived by comparing `teff_gspphot`, `lum_flame`, `mh_gspphot` and `logg_gspphot` to BASTI stellar evolution models (Hidalgo et al. 2018). The input metallicity to the model-based inference has been corrected for differences in the reference solar parameters between the atmospheric and evolution models, see **?** and Section **??** in the online documentation for details.

MASS_FLAME_LOWER : Lower confidence level (16%) of FLAME mass (float, Mass[Solar Mass])

Lower confidence level of `mass_flame`. It is defined as the 16th percentile value of the 1D projected distribution from sampling in mass and age. Lower and upper levels include 68% confidence interval.

MASS_FLAME_UPPER : Upper confidence level (84%) of FLAME mass (float, Mass[Solar Mass])

Upper confidence level of `mass_flame`. It is defined as the 84th percentile value of the 1D projected distribution from sampling in mass and age. Lower and upper levels include 68% confidence interval.

AGE_FLAME : Age of the star from FLAME using stellar models, see FLAME mass for details (float, Time[Gyr])

Age of the star from FLAME using GSP-Phot based parameters as input. It is defined as the median value (50th percentile) of the 1D projected distribution from sampling in mass, age, and metallicity, see `mass_flame` for details.

AGE_FLAME_LOWER : Lower confidence level (16%) of FLAME age (float, Time[Gyr])

Lower confidence level of `age_flame`. It is defined as the 16th percentile value of the 1D projected distribution from sampling in mass, age, and metallicity. Lower and upper levels include 68% confidence interval.

AGE_FLAME_UPPER : Upper confidence level (84%) of FLAME age (float, Time[Gyr])

Upper confidence level of `age_flame`. It is defined as the 84th percentile value of the 1D projected distribution from sampling in mass, age, and metallicity. Lower and upper levels include 68% confidence interval.

FLAGS_FLAME : Flags indicating quality and processing information from FLAME (string)

This field contains the quality and processing flag for the data products from FLAME derived using GSP-Phot atmospheric parameters. It is a string that consists of four digits 'ABCD'. A value of '0' generally indicates nothing to report.

Details of the description of the flag is given in Section **??** and the user is referred to this section for details. Example uses are also made available there. The four digits provide information on the quality of the input data to FLAME for each source along with an indication of quality of the FLAME parameters and some processing information.

A provides quality information on the non-Apsis input data (0, 1, 2, or 3)

B warning information on the Apsis input data (0 or 1)

C warning information on some FLAME parameters (0, 1, 2 or 3)

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D provides additional processing information in particular if a parameter is missing (0 – 6)

See Section **??** of the online documentation for further details.

EVOLSTAGE_FLAME : Evolutionary stage of the star from FLAME using stellar models, see FLAME mass for details (float)

Evolution stage of the star from FLAME. It is a float value typically between 100 and 1300 and defined using the median value (50th percentile) of the 1D projected distribution from sampling in mass, age, and metallicity, see `mass_flame` for details. The value is adapted from the BASTI model grid (Hidalgo et al. 2018) which adopts the following convention:

- 100 = zero age main sequence (ZAMS)
- 300 = first minimum of $T_{\rm eff}$ for massive stars or central hydrogen mass fraction = 0.30 for low-mass stars
- 360 = main sequence turn-off
- 420 = central hydrogen mass fraction = 0.00
- 490 = base of the red giant branch (RGB)
- 860 = maximum $\mathcal{L}$ along the RGB bump
- 890 = minimum $\mathcal{L}$ along the RGB bump
- 1290 = tip of the RGB

EVOLSTAGE_FLAME_LOWER : Lower confidence level (16%) of the FLAME evolutionary stage (float)

Lower confidence level of `evolstage_flame`. It is defined as the 16th percentile value of the distribution from sampling. Lower and upper levels include 68% confidence interval.

EVOLSTAGE_FLAME_UPPER : Upper confidence level (84%) of the FLAME evolutionary stage (float)

Upper confidence level of `evolstage_flame`. It is defined as the 84th percentile value of the distribution from sampling. Lower and upper levels include 68% confidence interval.

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GRAVREDSHIFT_FLAME : Radial velocity correction due to the gravitational redshift from FLAME using GSP-Phot logg and FLAME radius (float, Velocity[km s^{-1}])

Radial velocity correction due to gravitational redshift from FLAME derived using `radius_flame` and `logg_gspphot`.

GRAVREDSHIFT_FLAME_LOWER : Lower confidence level (16%) of the FLAME parameter gravredshift (float, Velocity[km s^{-1}])

Lower confidence level of `gravredshift_flame`. It is defined as the 16th percentile value of the distribution from sampling. Lower and upper levels include 68% confidence interval.

GRAVREDSHIFT_FLAME_UPPER : Upper confidence level (84%) of the FLAME parameter gravredshift (float, Velocity[km s^{-1}])

Upper confidence level of `gravredshift_flame`. It is defined as the 84th percentile value of the distribution from sampling. Lower and upper levels include 68% confidence interval.

BC_FLAME : Bolometric correction applied to derive the FLAME luminosity using GSP-Phot parameters (float, Magnitude[mag])

Bolometric correction for the G-band magnitude used to derive `lum_flame`. It is defined as the median value (50th percentile) of the distribution from sampling. It is a function of effective temperature, surface gravity, and metallicity, and has been derived from MARCS models, see Section **??** of the online documentation, using GSP-Phot-based parameters. The bolometric correction for the Sun is defined as +0.062 mag, see Section **??**, where $M_{bol,\odot}$ = 4.74 mag, see Section **??**, i.e. $M_{G\odot}$ = 4.678 mag.

BC_FLAME_LOWER : Lower confidence level (16%) of the FLAME bolometric correction (float, Magnitude[mag])

Lower confidence level of `bc_flame`. It is defined as the 16th percentile value of the distribution from sampling. Lower and upper levels include 68% confidence interval.

BC_FLAME_UPPER : Upper confidence level (84%) of the FLAME bolometric correction (float, Magnitude[mag])

Upper confidence level of `bc_flame`. It is defined as the 84th percentile value of the distribution from sampling. Lower and upper levels include 68% confidence interval.

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MH_FLAME_MODEL : Model metallicity from FLAME using stellar models, see FLAME mass for details (float, Abundances[dex])

Model metallicity from FLAME using GSP-Phot and FLAME input parameters. It is the derived model metallicity corresponding to the mass and age determination. It is defined as the median value (50th percentile) of the 1D projected distribution from sampling in mass, age, and metallicity, see `mass_flame` for details.

MH_FLAME_MODEL_LOWER : Lower confidence level (16%) of the model metallicity from FLAME (float, Abundances[dex])

Lower confidence level of `mh_flame_model`. It is defined as the 16th percentile value of the 1D projected distribution from sampling in mass, age, and metallicity. Lower and upper levels include 68% confidence interval.

MH_FLAME_MODEL_UPPER : Upper confidence level (84%) of the model metallicity from FLAME (float, Abundances[dex])

Upper confidence level of `mh_flame_model`. It is defined as the 84th percentile value of the 1D projected distribution from sampling in mass, age, and metallicity. Lower and upper levels include 68% confidence interval.

LOGG_FLAME_MODEL : Model surface gravity from FLAME (float, GravitySurface[log cgs])

Model surface gravity from FLAME using GSP-Phot and FLAME input parameters. It is the derived model surface gravity corresponding to the mass and age determination. It is defined as the median value (50th percentile) of the 1D projected distribution from sampling in mass, age, and metallicity, see `mass_flame` for details.

LOGG_FLAME_MODEL_LOWER : Lower confidence level (16%) of the model surface gravity from FLAME (float, GravitySurface[log cgs])

Lower confidence level of `logg_flame_model`. It is defined as the 16th percentile value of the distribution from sampling. Lower and upper levels include 68% confidence interval.

LOGG_FLAME_MODEL_UPPER : Upper confidence level (84%) of the model surface gravity

from FLAME (float, GravitySurface[log cgs])

Upper confidence level of `logg_flame_model`. It is defined as the 84th percentile value of the distribution from sampling. Lower and upper levels include 68% confidence interval.

TEFF_FLAME_MODEL : Model effective temperature from FLAME (float, Temperature[K])

Model effective temperature from FLAME using GSP-Phot and FLAME input parameters. It is the derived model effective temperature corresponding to the mass and age determination. It is defined as the median value (50th percentile) of the 1D projected distribution from sampling in mass, age, and metallicity, see `mass_flame` for details.

TEFF_FLAME_MODEL_LOWER : Lower confidence level (16%) of the model effective temperature from FLAME (float, Temperature[K])

Lower confidence level of `teff_flame_model`. It is defined as the 16th percentile value of the distribution from sampling. Lower and upper levels include 68% confidence interval.

TEFF_FLAME_MODEL_UPPER : Upper confidence level (84%) of the model effective temperature from FLAME (float, Temperature[K])

Upper confidence level of `teff_flame_model`. It is defined as the 84th percentile value of the distribution from sampling. Lower and upper levels include 68% confidence interval.

LUM_FLAME_MODEL : Model luminosity from FLAME (float, Luminosity[Solar Luminosity])

Model luminosity from FLAME using GSP-Phot and FLAME input parameters. It is the derived model luminosity corresponding to the mass and age determination. It is defined as the median value (50th percentile) of the 1D projected distribution from sampling in mass, age, and metallicity, see `mass_flame` for details.

LUM_FLAME_MODEL_LOWER : Lower confidence level (16%) of the model luminosity from FLAME (float, Luminosity[Solar Luminosity])

Lower confidence level of `lum_flame_model`. It is defined as the 16th percentile value of the distribution from sampling. Lower and upper levels include 68% confidence interval.

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LUM_FLAME_MODEL_UPPER : Upper confidence level (84%) of the model luminosity from FLAME (float, Luminosity[Solar Luminosity])

Upper confidence level of `lum_flame_model`. It is defined as the 84th percentile value of the distribution from sampling. Lower and upper levels include 68% confidence interval.

LOGPOSTERIOR_FLAME : Median log-posterior of the MCMC samples from FLAME inference on model parameters (float)

Goodness-of-fit score defined as the median log-posterior of all MCMC samples of FLAME for the model parameter determination (mass, age, etc.).

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4.9 AP_XP_EPOCH [TAP]

This table contains the astrophysical parameters derived from the epoch RP spectra for ultra-cool dwarfs by ESP-UCD as part of the Apsis processing chain.

See Section **??** of the online documentation for details on the ESP-UCD module.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

TRANSIT_ID : Transit Identifier (long)

The `transit_id` is a unique identifier assigned to each detected (and confirmed) source as it transits the Gaia focal plane. Each time a given source is detected as Gaia scans and re-scans the sky a new `transit_id` will be created to badge that apparition. Hence the along-scan time and the across-scan position along with the telescope in which the source was detected are used to form a unique integer with which to label the transit. The detection of transits is done using the SkyMapper CCD data, and the confirmation is done using the AF1 CCD data. Both happens on board.

The several features of a transit that are encoded in `transit_id` can be easily retrieved using bit masks (&) and shifts (>>) as follows:

- On-Board Mission Time line (OBMT) [ns]
  = 204800 * ((`transit_id` >> 17) & (0x000003FFFFFFFFFF))

- Field-of-view = 1 + (`transit_id` >> 15) & 0x03 [1 for 'preceding' and 2 for 'following' fields-of-view respectively]
- CCD row = (`transit_id` >> 12) & 0x07 [dimensionless, in the range 1 to 7]
- Across-scan 'reference acquisition pixel' in strip AF1 = (`transit_id`) & 0x0FFF [pixels] (this is the across-scan centre of the AF1 window and is odd if immediately below the mid-point of the window and even if immediately above)

where the bit mask prefix '0x' denotes hexadecimal.

For further details see Portell et al. (2020). For convenience a decoder for `transit_id` is available on-line at

`https://gaia.esac.esa.int/decoder/transitidDecoder.jsp`

OBSERVATION_TIME : Barycentric light-travel time corrected timestamp for each observation (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Barycentric light-travel time corrected timestamp for each observation. It is provided in Julian Day in TCB at the Barycentre, minus 2 455 197.5 days.

TEFF_ESPUCD : Effective temperature estimated from the epoch RP spectrum (float, Temperature[K])

Effective temperature estimate from ESP-UCD inferred from the epoch RP spectrum. The prediction module is based on a Gaussian Process trained with empirical examples. See Section **??** for details.

TEFF_ESPUCD_UNCERTAINTY : Effective temperature uncertainty for the corresponding epoch RP spectrum (float, Temperature[K])

Standard deviation of 10 effective temperature predictions obtained by the ESP-UCD module for 10 RP spectra generated using random sampling from Gaussian distributions in the space of coefficients, the parameters of which are derived from the forward model plus ridge regression fits to the partially calibrated epoch RP spectra. The value thus obtained is rescaled to match the root-mean-square error of the module predictions for a set of well-known ultracool dwarfs (see Section **??** for further details).

Draft

EW_ESPUCD_HALPHA : Halpha pseudo-equivalent width estimated from epoch RP spectrum (float, Length & Distance[nm])

Pseudo-equivalent width (pEW) estimate derived from a comparison with templates with EW=0. First the best matching template is chosen using the RP wavelength range to the right of 700.0 nm (longer wavelengths). The pEW is calculated using a linear continuum using the fluxes at the SMSGen wavelengths closest to 646 and 670 nm (different linear continua for the epoch RP spectrum and for the template). The pEW is calculated as the difference between the the value for the epoch RP spectrum and the template, and calibrated as a function of pEW and effective temperature. See Section **??** for details.

EW_ESPUCD_HALPHA_UNCERTAINTY : Halpha pseudo-equivalent width uncertainty estimated from epoch RP spectrum (float, Length & Distance[nm])

Estimated uncertainty of the pseudo-equivalent width measured for each epoch RP spectrum derived from random replicates of the RP spectrum using the coefficients uncertainties.

FLAGS_ESPUCD : Flag describing the quality of the epoch RP spectra (int)

One-digit flag in the [0-4] range, with 0 representing the best quality class. The flag is mostly determined by the uncertainties in the Halpha pseudo-equivalent width estimate, but also by the fraction of the epoch RP spectrum with negative fluxes and by the normalized residuals of the ridge regression used to calibrate the RP epoch spectra, as described in the documentation.

Draft

4.10 AP_XP_SUPP [TAP]

This table contains the astrophysical parameters of stars derived from mean XP spectra by GSP-Phot as part of the Apsis processing chain. Each source has been analysed with three different stellar librarires (MARCS, A, and OB), and all of these results are published in this table.

For each source a 'best' library is chosen and the results from the best library for each source are found in the `ap_xp` table. The name of the best library is given in `ap_xp.libname_gspphot`. See Section **??** of the online documentation for details on the GSP-Phot module.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

TEFF_GSPPHOT_MARCS : Effective temperature from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Temperature[K])

Effective temperature (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax (see Section **??** of the online documentation). This is the median of the MCMC samples. Taken from MARCS library.

TEFF_GSPPHOT_MARCS_LOWER : Lower confidence level (16%) of effective temperature from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Temperature[K])

Lower confidence level (16%) of effective temperature (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include

68% confidence interval.

TEFF_GSPPHOT_MARCS_UPPER : Upper confidence level (84%) of effective temperature from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Temperature[K])

Upper confidence level (84%) of effective temperature (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

LOGG_GSPPHOT_MARCS : Surface gravity from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, GravitySurface[log cgs])

Surface gravity (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax (see Section **??** of the online documentation). This is the median of the MCMC samples. Taken from MARCS library.

LOGG_GSPPHOT_MARCS_LOWER : Lower confidence level (16%) of surface gravity from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, GravitySurface[log cgs])

Lower confidence level (16%) of surface gravity (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

LOGG_GSPPHOT_MARCS_UPPER : Upper confidence level (84%) of surface gravity from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, GravitySurface[log cgs])

Upper confidence level (84%) of surface gravity (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

MH_GSPPHOT_MARCS : Global metallicity from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Abundances[dex])

Decimal logarithm of the ratio of the number abundance of metals to the number abundance of hydrogen relative to the same ratio of solar abundances inferred by GSP-Phot Aeneas from

BP/RP spectra, apparent G magnitude and parallax, assuming source is a single star (see Section **??** of the online documentation). This is the median of the MCMC samples. Taken from MARCS library.

MH_GSPPHOT_MARCS_LOWER : Lower confidence level (16%) of global metallicity from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Abundances[dex])

Lower confidence level (16%) of decimal logarithm of the ratio of the number abundance of metals to the number abundance of hydrogen relative to the same ratio of solar abundances inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax, assuming source is a single star (see Section **??** of the online documentation). This is the 16th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

MH_GSPPHOT_MARCS_UPPER : Upper confidence level (84%) of global metallicity from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Abundances[dex])

Upper confidence level (84%) of decimal logarithm of the ratio of the number abundance of metals to the number abundance of hydrogen relative to the same ratio of solar abundances inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax, assuming source is a single star (see Section **??** of the online documentation). This is the 84th percentile of the MCMC samples. Taken from PHOENIX library. Lower and upper levels include 68% confidence interval.

FEH_GSPPHOT_MARCS : Iron abundance from GSP-Phot Aeneas for MARCS library best library using BP/RP spectra (float, Abundances[dex])

Decimal logarithm of the ratio of the number abundance of iron to the number abundance of hydrogen relative to the same ratio of solar abundances inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax, assuming source is a single star (see Section **??** of the online documentation). This is the median of the MCMC samples. Taken from MARCS library.

FEH_GSPPHOT_MARCS_UPPER : Upper confidence level (84%) of iron abundance from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Abundances[dex])

Upper confidence level (84%) of decimal logarithm of the ratio of the number abundance of iron to the number abundance of hydrogen relative to the same ratio of solar abundances inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax, assuming source is

a single star (see Section **??** of the online documentation). This is the 84th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

FEH_GSPPHOT_MARCS_LOWER : Lower confidence level (16%) of iron abundance from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Abundances[dex])

Lower confidence level (16%) of decimal logarithm of the ratio of the number abundance of iron to the number abundance of hydrogen relative to the same ratio of solar abundances inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax, assuming source is a single star (see Section **??** of the online documentation). This is the 16th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

DISTANCE_GSPPHOT_MARCS : Distance from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Length & Distance[pc])

Distance (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax (see Section **??** of the online documentation). This is the median of the MCMC samples. Taken from MARCS library. NB: The actual fit parameter is $\log_{10} d$ and a prior is imposed to ensure a value between [0,5], thus the minimum possible distance is 1 pc and the maximum is 100 kpc.

DISTANCE_GSPPHOT_MARCS_LOWER : Lower confidence level (16%) of distance from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Length & Distance[pc])

Lower confidence level (16%) of distance (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval. NB: The actual fit parameter is $\log_{10} d$ and a prior is imposed to ensure a value between [0,5], thus the minimum possible distance is 1 pc and the maximum is 100 kpc.

DISTANCE_GSPPHOT_MARCS_UPPER : Upper confidence level (84%) of distance from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Length & Distance[pc])

Upper confidence level (84%) of distance (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval. NB: The actual fit parameter is $\log_{10} d$ and a prior is imposed to ensure a value

between [0,5], thus the minimum possible distance is 1 pc and the maximum is 100 kpc.

AZERO_GSPPHOT_MARCS : Extinction parameter from GSP-Phot for MARCS library using BP/RP spectra (float, Magnitude[mag])

Extinction parameter A_0 corresponding to a monochromatic extinction at a specific reference wavelength (see Section **??** of the online documentation) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from MARCS library. NB: This is the extinction parameter in the adopted extinction curve from Gordon et al. (2023).

AZERO_GSPPHOT_MARCS_LOWER : Lower confidence level (16%) of extinction parameter from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of extinction parameter A_0 corresponding to a monochromatic extinction at a specific reference wavelength (see Section **??** of the online documentation) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from MARCS library. NB: This is the extinction parameter in the adopted extinction curve from Gordon et al. (2023).

AZERO_GSPPHOT_MARCS_UPPER : Upper confidence level (84%) of extinction parameter from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of extinction parameter A_0 corresponding to a monochromatic extinction at a specific reference wavelength (see Section **??** of the online documentation) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from MARCS library. NB: This is the extinction parameter in the adopted extinction curve from Gordon et al. (2023).

AG_GSPPHOT_MARCS : Extinction in G band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Broadband extinction in G band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from MARCS library.

AG_GSPPHOT_MARCS_LOWER : Lower confidence level (16%) of extinction in G band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Draft

Lower confidence level (16%) of broadband extinction in G band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

AG_GSPPHOT_MARCS_UPPER : Upper confidence level (84%) of extinction in G band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of broadband extinction in G band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

ABP_GSPPHOT_MARCS : Extinction in BP band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Broadband extinction in G_{BP} band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from MARCS library.

ABP_GSPPHOT_MARCS_LOWER : Lower confidence level (16%) of extinction in BP band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of broadband extinction in G_{BP} band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

ABP_GSPPHOT_MARCS_UPPER : Upper confidence level (84%) of extinction in BP band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of broadband extinction in G_{BP} band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

ARP_GSPPHOT_MARCS : Extinction in RP band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Draft

Broadband extinction in G_{RP} band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from MARCS library.

ARP_GSPPHOT_MARCS_LOWER : Lower confidence level (16%) of extinction in RP band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of broadband extinction in G_{RP} band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

ARP_GSPPHOT_MARCS_UPPER : Upper confidence level (84%) of extinction in RP band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of broadband extinction in G_{RP} band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

AV_GSPPHOT_MARCS : Extinction in Johnson V band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Broadband extinction in Johnson V band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from MARCS library.

AV_GSPPHOT_MARCS_LOWER : Lower confidence level (16%) of extinction in Johnson V band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of broadband extinction in Johnson V band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

AV_GSPPHOT_MARCS_UPPER : Upper confidence level (84%) of extinction in Johnson V band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Draft

Upper confidence level (84%) of broadband extinction in Johnson V band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

AJ_GSPPHOT_MARCS : Extinction in 2MASS J band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Broadband extinction in 2MASS J band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from MARCS library.

AJ_GSPPHOT_MARCS_LOWER : Lower confidence level (16%) of extinction in 2MASS J band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of broadband extinction in 2MASS J band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

AJ_GSPPHOT_MARCS_UPPER : Upper confidence level (84%) of extinction in 2MASS J band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of broadband extinction in 2MASS J band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

AH_GSPPHOT_MARCS : Extinction in 2MASS H band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Broadband extinction in 2MASS H band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from MARCS library.

AH_GSPPHOT_MARCS_LOWER : Lower confidence level (16%) of extinction in 2MASS H band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Draft

Lower confidence level (16%) of broadband extinction in 2MASS H band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

AH_GSPPHOT_MARCS_UPPER : Upper confidence level (84%) of extinction in 2MASS H band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of broadband extinction in 2MASS H band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

AKS_GSPPHOT_MARCS : Extinction in 2MASS Ks band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Broadband extinction in 2MASS Ks band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from MARCS library.

AKS_GSPPHOT_MARCS_LOWER : Lower confidence level (16%) of extinction in 2MASS Ks band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of broadband extinction in 2MASS Ks band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

AKS_GSPPHOT_MARCS_UPPER : Upper confidence level (84%) of extinction in 2MASS Ks band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of broadband extinction in 2MASS Ks band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

AW1_GSPPHOT_MARCS : Extinction in WISE W1 band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Broadband extinction in WISE W1 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from MARCS library.

AW1_GSPPHOT_MARCS_LOWER : Lower confidence level (16%) of extinction in WISE W1 band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of broadband extinction in WISE W1 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

AW1_GSPPHOT_MARCS_UPPER : Upper confidence level (84%) of extinction in WISE W1 band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of broadband extinction in WISE W1 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

AW2_GSPPHOT_MARCS : Extinction in WISE W2 band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Broadband extinction in WISE W2 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from MARCS library.

AW2_GSPPHOT_MARCS_LOWER : Lower confidence level (16%) of extinction in WISE W2 band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of broadband extinction in WISE W2 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

AW2_GSPPHOT_MARCS_UPPER : Upper confidence level (84%) of extinction in WISE W2 band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Draft

Upper confidence level (84%) of broadband extinction in WISE W2 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

EBPMINRP_GSPPHOT_MARCS : Reddening in BP minus RP colour from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Reddening $E(BP-RP)$ (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from MARCS library. Note that while $E(BP-RP) = A_{BP} - A_{RP}$, this was computed at the level of MCMC samples. Hence, this relation is not exactly true for the median values.

EBPMINRP_GSPPHOT_MARCS_LOWER : Lower confidence level (16%) of reddening in BP minus RP colour from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of reddening $E(BP-RP)$ (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval. Note that while $E(BP-RP) = A_{BP} - A_{RP}$, this was computed at the level of MCMC samples. Hence, this relation is not exactly true for the lower confidence levels.

EBPMINRP_GSPPHOT_MARCS_UPPER : Upper confidence level (84%) of reddening in BP minus RP colour from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of reddening $E(BP-RP)$ (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval. Note that while $E(BP-RP) = A_{BP} - A_{RP}$, this was computed at the level of MCMC samples. Hence, this relation is not exactly true for the upper confidence levels.

EBMINV_GSPPHOT_MARCS : Reddening in B minus V colour from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Reddening $E(B-V)$ (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP

spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from MARCS library.

EBMINV_GSPPHOT_MARCS_LOWER : Lower confidence level (16%) of reddening in B minus V colour from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of reddening $E(B-V)$ (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples.Taken from MARCS library. Lower and upper levels include 68% confidence interval.

EBMINV_GSPPHOT_MARCS_UPPER : Upper confidence level (84%) of reddening in B minus V colour from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of reddening $E(B-V)$ (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

ABSMAG_G_GSPPHOT_MARCS : Absolute magnitude in G band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Absolute magnitude M_G (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from MARCS library.

ABSMAG_G_GSPPHOT_MARCS_LOWER : Lower confidence level (16%) of absolute magnitude in G band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of absolute magnitude M_G (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

ABSMAG_G_GSPPHOT_MARCS_UPPER : Upper confidence level (84%) of absolute magnitude in G band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

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Upper confidence level (84%) of absolute magnitude M_G (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

ABSMAG_J_GSPPHOT_MARCS : Absolute magnitude in 2MASS J band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Absolute magnitude M_J in 2MASS J band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from MARCS library.

ABSMAG_J_GSPPHOT_MARCS_LOWER : Lower confidence level (16%) of absolute magnitude in 2MASS J band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of absolute magnitude M_J in 2MASS J band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

ABSMAG_J_GSPPHOT_MARCS_UPPER : Upper confidence level (84%) of absolute magnitude in 2MASS J band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of absolute magnitude M_J in 2MASS J band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

ABSMAG_H_GSPPHOT_MARCS : Absolute magnitude in 2MASS H band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Absolute magnitude M_H in 2MASS H band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from MARCS library.

ABSMAG_H_GSPPHOT_MARCS_LOWER : Lower confidence level (16%) of absolute magnitude

in 2MASS H band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of absolute magnitude M_{H} in 2MASS H band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

ABSMAG_H_GSPPHOT_MARCS_UPPER : Upper confidence level (84%) of absolute magnitude in 2MASS H band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of absolute magnitude M_{H} in 2MASS H band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

ABSMAG_KS_GSPPHOT_MARCS : Absolute magnitude in 2MASS Ks band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Absolute magnitude M_{Ks} in 2MASS Ks band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from MARCS library.

ABSMAG_KS_GSPPHOT_MARCS_LOWER : Lower confidence level (16%) of absolute magnitude in 2MASS Ks band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of absolute magnitude M_{Ks} in 2MASS Ks band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

ABSMAG_KS_GSPPHOT_MARCS_UPPER : Upper confidence level (84%) of absolute magnitude in 2MASS Ks band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of absolute magnitude M_{Ks} in 2MASS Ks band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and

parallax. This is the 84th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

ABSMAG_W1_GSPPHOT_MARCS : Absolute magnitude in WISE W1 band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Absolute magnitude M_{W1} in WISE W1 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from MARCS library.

ABSMAG_W1_GSPPHOT_MARCS_LOWER : Lower confidence level (16%) of absolute magnitude in WISE W1 band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of absolute magnitude M_{W1} in WISE W1 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

ABSMAG_W1_GSPPHOT_MARCS_UPPER : Upper confidence level (84%) of absolute magnitude in WISE W1 band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of absolute magnitude M_{W1} in WISE W1 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

ABSMAG_W2_GSPPHOT_MARCS : Absolute magnitude in WISE W2 band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Absolute magnitude M_{W2} in WISE W2 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from MARCS library.

ABSMAG_W2_GSPPHOT_MARCS_LOWER : Lower confidence level (16%) of absolute magnitude in WISE W2 band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of absolute magnitude M_{W2} in WISE W2 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

ABSMAG_W2_GSPPHOT_MARCS_UPPER : Upper confidence level (84%) of absolute magnitude in WISE W2 band from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of absolute magnitude M_{W2} in WISE W2 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

RADIUS_GSPPHOT_MARCS : Radius from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Length & Distance[Solar Radius])

Stellar radius (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from MARCS library.

RADIUS_GSPPHOT_MARCS_LOWER : Lower confidence level (16%) of radius from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Length & Distance[Solar Radius])

Lower confidence level (16%) of stellar radius (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

RADIUS_GSPPHOT_MARCS_UPPER : Upper confidence level (84%) of radius from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Length & Distance[Solar Radius])

Upper confidence level (84%) of stellar radius (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

MASS_GSPPHOT_MARCS : Mass from GSP-Phot Aeneas for MARCS library using BP/RP spec-

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tra (float, Mass[Solar Mass])

Stellar mass (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from MARCS library.

MASS_GSPPHOT_MARCS_LOWER : Lower confidence level (16%) of mass from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Mass[Solar Mass])

Lower confidence level (16%) of stellar mass (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

MASS_GSPPHOT_MARCS_UPPER : Upper confidence level (84%) of mass from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Mass[Solar Mass])

Upper confidence level (84%) of stellar mass (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

AGE_GSPPHOT_MARCS : Age from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Time[Gyr])

Stellar age (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from MARCS library.

AGE_GSPPHOT_MARCS_LOWER : Lower confidence level (16%) of age from GSP-Phot Aeneas for MARCS library using BP/RP spectra (float, Time[Gyr])

Lower confidence level (16%) of stellar age (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from MARCS library. Lower and upper levels include 68% confidence interval.

AGE_GSPPHOT_MARCS_UPPER : Upper confidence level (84%) of age from GSP-Phot Aeneas

for MARCS library using BP/RP spectra (float, Time[Gyr])

Upper confidence level (84%) of stellar age (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from the MARCS library. Lower and upper levels include 68% confidence interval.

LOGPOSTERIOR_GSPPHOT_MARCS : mean log-posterior of MCMC of GSP-Phot Aeneas MCMC for MARCS library (float)

Goodness-of-fit score defined as the mean log-posterior of all MCMC samples of GSP-Phot Aeneas MCMC for MARCS library. The higher the goodness-of-fit score, the better the fit. Values are usually negative. NB: This is not a Bayesian evidence!

LOGLIKELIHOOD_GSPPHOT_MARCS : mean likelihood of MCMC of GSP-Phot Aeneas MCMC for MARCS library (float)

Mean log-likelihood of all MCMC samples of GSP-Phot Aeneas MCMC for MARCS library. The higher the mean log-likelihood, the better the fit. Values are usually negative. Subtracting the mean log-likelihood from the mean log-posterior yields the mean prior. NB: This is not a Bayesian evidence!

LOGPOST_SPREAD_GSPPHOT_MARCS : maximum minus minimum of log-posterior throughout MCMC chain for MARCS library (float)

Difference of maximum log-posterior minus minimum log-posterior in MCMC chain (excluding burn-in phase) for MARCS library. If this spread is much larger than 25, say > 100, it suggests that the MCMC chain has not fully converged.

MCMCACCEPT_GSPPHOT_MARCS : MCMC acceptance rate of GSP-Phot Aeneas MCMC for MARCS library (float)

MCMC acceptance rate of GSP-Phot Aeneas MCMC for MARCS library. This is computed from all MCMC samples (before thinning the chain to 2000 or 100 samples).

TEFF_GSPPHOT_OB : Effective temperature from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Temperature[K])

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Effective temperature (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax (see Section **??** of the online documentation). This is the median of the MCMC samples. Taken from OB library.

TEFF_GSPPHOT_OB_LOWER : Lower confidence level (16%) of effective temperature from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Temperature[K])

Lower confidence level (16%) of effective temperature (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

TEFF_GSPPHOT_OB_UPPER : Upper confidence level (84%) of effective temperature from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Temperature[K])

Upper confidence level (84%) of effective temperature (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

LOGG_GSPPHOT_OB : Surface gravity from GSP-Phot Aeneas for OB library using BP/RP spectra (float, GravitySurface[log cgs])

Surface gravity (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax (see Section **??** of the online documentation). This is the median of the MCMC samples. Taken from OB library.

LOGG_GSPPHOT_OB_LOWER : Lower confidence level (16%) of surface gravity from GSP-Phot Aeneas for OB library using BP/RP spectra (float, GravitySurface[log cgs])

Lower confidence level (16%) of surface gravity (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

LOGG_GSPPHOT_OB_UPPER : Upper confidence level (84%) of surface gravity from GSP-Phot Aeneas for OB library using BP/RP spectra (float, GravitySurface[log cgs])

Draft

Upper confidence level (84%) of surface gravity (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

FEH_GSPPHOT_OB : Iron abundance from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Abundances[dex])

Decimal logarithm of the ratio of the number abundance of iron to the number abundance of hydrogen relative to the same ratio of solar abundances inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax, assuming source is a single star (see Section **??** of the online documentation). This is the median of the MCMC samples. Taken from OB library.

FEH_GSPPHOT_OB_LOWER : Lower confidence level (16%) of iron abundance from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Abundances[dex])

Lower confidence level (16%) of decimal logarithm of the ratio of the number abundance of iron to the number abundance of hydrogen relative to the same ratio of solar abundances inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax, assuming source is a single star (see Section **??** of the online documentation). This is the 16th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

FEH_GSPPHOT_OB_UPPER : Upper confidence level (84%) of iron abundance from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Abundances[dex])

Upper confidence level (84%) of decimal logarithm of the ratio of the number abundance of iron to the number abundance of hydrogen relative to the same ratio of solar abundances inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax, assuming source is a single star (see Section **??** of the online documentation). This is the 84th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

DISTANCE_GSPPHOT_OB : Distance from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Length & Distance[pc])

Distance (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples (see Section **??** of the online documentation). Taken from OB library. NB: The actual fit parameter is $\log_{10} d$ and a prior is imposed to ensure a value between [0,5], thus the minimum possible distance is 1 pc and the maximum is 100 kpc.

DISTANCE_GSPPHOT_OB_LOWER : Lower confidence level (16%) of distance from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Length & Distance[pc])

Lower confidence level (16%) of distance (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval. NB: The actual fit parameter is $\log_{10} d$ and a prior is imposed to ensure a value between [0,5], thus the minimum possible distance is 1 pc and the maximum is 100 kpc.

DISTANCE_GSPPHOT_OB_UPPER : Upper confidence level (84%) of distance from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Length & Distance[pc])

Upper confidence level (84%) of distance (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval. NB: The actual fit parameter is $\log_{10} d$ and a prior is imposed to ensure a value between [0,5], thus the minimum possible distance is 1 pc and the maximum is 100 kpc.

AZERO_GSPPHOT_OB : Extinction parameter from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Extinction parameter A_0 corresponding to a monochromatic extinction at a specific reference wavelength (see Section **??** of the online documentation) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from OB library. NB: This is the extinction parameter in the adopted extinction curve from Gordon et al. (2023).

AZERO_GSPPHOT_OB_LOWER : Lower confidence level (16%) of extinction parameter from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of extinction parameter A_0 corresponding to a monochromatic extinction at a specific reference wavelength (see Section **??** of the online documentation) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from OB library. NB: This is the extinction parameter in the adopted extinction curve from Gordon et al. (2023).

AZERO_GSPPHOT_OB_UPPER : Upper confidence level (84%) of extinction parameter from

GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of extinction parameter A_0 corresponding to a monochromatic extinction at a specific reference wavelength (see Section **??** of the online documentation) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from OB library. NB: This is the extinction parameter in the adopted extinction curve from Gordon et al. (2023).

AG_GSPPHOT_OB : Extinction in G band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Broadband extinction in G band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from OB library.

AG_GSPPHOT_OB_LOWER : Lower confidence level (16%) of extinction in G band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of broadband extinction in G band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

AG_GSPPHOT_OB_UPPER : Upper confidence level (84%) of extinction in G band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of broadband extinction in G band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

ABP_GSPPHOT_OB : Extinction in BP band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Broadband extinction in G_{BP} band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from OB library.

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ABP_GSPPHOT_OB_LOWER : Lower confidence level (16%) of extinction in BP band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of broadband extinction in G_{BP} band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

ABP_GSPPHOT_OB_UPPER : Upper confidence level (84%) of extinction in BP band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of broadband extinction in G_{BP} band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

ARP_GSPPHOT_OB : Extinction in RP band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Broadband extinction in G_{RP} band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from OB library.

ARP_GSPPHOT_OB_LOWER : Lower confidence level (16%) of extinction in RP band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of broadband extinction in G_{RP} band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

ARP_GSPPHOT_OB_UPPER : Upper confidence level (84%) of extinction in RP band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of broadband extinction in G_{RP} band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

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AV_GSPPHOT_OB : Extinction in Johnson V band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Broadband extinction in Johnson V band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from OB library.

AV_GSPPHOT_OB_LOWER : Lower confidence level (16%) of extinction in Johnson V band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of broadband extinction in Johnson V band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

AV_GSPPHOT_OB_UPPER : Upper confidence level (84%) of extinction in Johnson V band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of broadband extinction in Johnson V band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

AJ_GSPPHOT_OB : Extinction in 2MASS J band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Broadband extinction in 2MASS J band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from OB library.

AJ_GSPPHOT_OB_LOWER : Lower confidence level (16%) of extinction in 2MASS J band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of broadband extinction in 2MASS J band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

Draft

AJ_GSPPHOT_OB_UPPER : Upper confidence level (84%) of extinction in 2MASS J band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of broadband extinction in 2MASS J band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

AH_GSPPHOT_OB : Extinction in 2MASS H band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Broadband extinction in 2MASS H band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from OB library.

AH_GSPPHOT_OB_LOWER : Lower confidence level (16%) of extinction in 2MASS H band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of broadband extinction in 2MASS H band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

AH_GSPPHOT_OB_UPPER : Upper confidence level (84%) of extinction in 2MASS H band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of broadband extinction in 2MASS H band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

AKS_GSPPHOT_OB : Extinction in 2MASS Ks band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Broadband extinction in 2MASS Ks band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from OB library.

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AKS_GSPPHOT_OB_LOWER : Lower confidence level (16%) of extinction in 2MASS Ks band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of broadband extinction in 2MASS Ks band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

AKS_GSPPHOT_OB_UPPER : Upper confidence level (84%) of extinction in 2MASS Ks band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of broadband extinction in 2MASS Ks band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

AW1_GSPPHOT_OB : Extinction in WISE W1 band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Broadband extinction in WISE W1 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from OB library.

AW1_GSPPHOT_OB_LOWER : Lower confidence level (16%) of extinction in WISE W1 band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of broadband extinction in WISE W1 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

AW1_GSPPHOT_OB_UPPER : Upper confidence level (84%) of extinction in WISE W1 band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of broadband extinction in WISE W1 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

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AW2_GSPPHOT_OB : Extinction in WISE W2 band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Broadband extinction in WISE W2 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from OB library.

AW2_GSPPHOT_OB_LOWER : Lower confidence level (16%) of extinction in WISE W2 band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of broadband extinction in WISE W2 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

AW2_GSPPHOT_OB_UPPER : Upper confidence level (84%) of extinction in WISE W2 band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of broadband extinction in WISE W2 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

EBPMINRP_GSPPHOT_OB : Reddening in BP minus RP colour from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Reddening $E(BP-RP)$ (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from OB library. Note that while $E(BP-RP) = A_{BP} - A_{RP}$, this was computed at the level of MCMC samples. Hence, this relation is not exactly true for the median values.

EBPMINRP_GSPPHOT_OB_LOWER : Lower confidence level (16%) of reddening in BP minus RP colour from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of reddening $E(BP-RP)$ (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval. Note that while $E(BP-RP) = A_{BP} - A_{RP}$, this was computed at the level of MCMC samples. Hence, this relation is not exactly true for the lower confidence levels.

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EBPMINRP_GSPPHOT_OB_UPPER : Upper confidence level (84%) of reddening in BP minus RP colour from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of reddening $E(BP-RP)$ (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval. Note that while $E(BP-RP) = A_{BP} - A_{RP}$, this was computed at the level of MCMC samples. Hence, this relation is not exactly true for the upper confidence levels.

EBMINV_GSPPHOT_OB : Reddening in B minus V colour from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Reddening $E(B-V)$ (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from OB library.

EBMINV_GSPPHOT_OB_LOWER : Lower confidence level (16%) of reddening in B minus V colour from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of reddening $E(B-V)$ (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples.Taken from OB library. Lower and upper levels include 68% confidence interval.

EBMINV_GSPPHOT_OB_UPPER : Upper confidence level (84%) of reddening in B minus V colour from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of reddening $E(B-V)$ (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

ABSMAG_G_GSPPHOT_OB : Absolute magnitude in G band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Absolute magnitude M_G (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples.

Taken from OB library.

ABSMAG_G_GSPPHOT_OB_LOWER : Lower confidence level (16%) of absolute magnitude in G band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of absolute magnitude M_G (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

ABSMAG_G_GSPPHOT_OB_UPPER : Upper confidence level (84%) of absolute magnitude in G band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of absolute magnitude M_G (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

ABSMAG_J_GSPPHOT_OB : Absolute magnitude in 2MASS J band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Absolute magnitude M_J in 2MASS J band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from OB library.

ABSMAG_J_GSPPHOT_OB_LOWER : Lower confidence level (16%) of absolute magnitude in 2MASS J band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of absolute magnitude M_J in 2MASS J band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

ABSMAG_J_GSPPHOT_OB_UPPER : Upper confidence level (84%) of absolute magnitude in 2MASS J band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

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Upper confidence level (84%) of absolute magnitude M_J in 2MASS J band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

ABSMAG_H_GSPPHOT_OB : Absolute magnitude in 2MASS H band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Absolute magnitude M_H in 2MASS H band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from OB library.

ABSMAG_H_GSPPHOT_OB_LOWER : Lower confidence level (16%) of absolute magnitude in 2MASS H band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of absolute magnitude M_H in 2MASS H band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

ABSMAG_H_GSPPHOT_OB_UPPER : Upper confidence level (84%) of absolute magnitude in 2MASS H band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of absolute magnitude M_H in 2MASS H band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

ABSMAG_KS_GSPPHOT_OB : Absolute magnitude in 2MASS Ks band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Absolute magnitude M_{Ks} in 2MASS Ks band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from OB library.

ABSMAG_KS_GSPPHOT_OB_LOWER : Lower confidence level (16%) of absolute magnitude in

2MASS Ks band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of absolute magnitude M_{Ks} in 2MASS Ks band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

ABSMAG_KS_GSPPHOT_OB_UPPER : Upper confidence level (84%) of absolute magnitude in 2MASS Ks band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of absolute magnitude M_{Ks} in 2MASS Ks band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

ABSMAG_W1_GSPPHOT_OB : Absolute magnitude in WISE W1 band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Absolute magnitude M_{W1} in WISE W1 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from OB library.

ABSMAG_W1_GSPPHOT_OB_LOWER : Lower confidence level (16%) of absolute magnitude in WISE W1 band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of absolute magnitude M_{W1} in WISE W1 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

ABSMAG_W1_GSPPHOT_OB_UPPER : Upper confidence level (84%) of absolute magnitude in WISE W1 band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of absolute magnitude M_{W1} in WISE W1 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and

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parallax. This is the 84th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

ABSMAG_W2_GSPPHOT_OB : Absolute magnitude in WISE W2 band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Absolute magnitude M_{W2} in WISE W2 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from OB library.

ABSMAG_W2_GSPPHOT_OB_LOWER : Lower confidence level (16%) of absolute magnitude in WISE W2 band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of absolute magnitude M_{W2} in WISE W2 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

ABSMAG_W2_GSPPHOT_OB_UPPER : Upper confidence level (84%) of absolute magnitude in WISE W2 band from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of absolute magnitude M_{W2} in WISE W2 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

RADIUS_GSPPHOT_OB : Radius from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Length & Distance[Solar Radius])

Stellar radius (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from OB library.

RADIUS_GSPPHOT_OB_LOWER : Lower confidence level (16%) of radius from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Length & Distance[Solar Radius])

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Lower confidence level (16%) of stellar radius (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

RADIUS_GSPPHOT_OB_UPPER : Upper confidence level (84%) of radius from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Length & Distance[Solar Radius])

Upper confidence level (84%) of stellar radius (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

MASS_GSPPHOT_OB : Mass from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Mass[Solar Mass])

Stellar mass (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from OB library.

MASS_GSPPHOT_OB_LOWER : Lower confidence level (16%) of mass from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Mass[Solar Mass])

Lower confidence level (16%) of stellar mass (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

MASS_GSPPHOT_OB_UPPER : Upper confidence level (84%) of mass from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Mass[Solar Mass])

Upper confidence level (84%) of stellar mass (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

AGE_GSPPHOT_OB : Age from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Time[Gyr])

Stellar age (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from OB library.

AGE_GSPPHOT_OB_LOWER : Lower confidence level (16%) of age from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Time[Gyr])

Lower confidence level (16%) of stellar age (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from OB library. Lower and upper levels include 68% confidence interval.

AGE_GSPPHOT_OB_UPPER : Upper confidence level (84%) of age from GSP-Phot Aeneas for OB library using BP/RP spectra (float, Time[Gyr])

Upper confidence level (84%) of stellar age (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from the OB library. Lower and upper levels include 68% confidence interval.

LOGPOSTERIOR_GSPPHOT_OB : mean log-posterior of MCMC of GSP-Phot Aeneas MCMC for OB library (float)

Goodness-of-fit score defined as the mean log-posterior of all MCMC samples of GSP-Phot Aeneas MCMC for OB library. The higher the goodness-of-fit score, the better the fit. Values are usually negative. NB: This is not a Bayesian evidence!

LOGLIKELIHOOD_GSPPHOT_OB : mean likelihood of MCMC of GSP-Phot Aeneas MCMC for OB library (float)

Mean log-likelihood of all MCMC samples of GSP-Phot Aeneas MCMC for OB library. The higher the mean log-likelihood, the better the fit. Values are usually negative. Subtracting the mean log-likelihood from the mean log-posterior yields the mean prior. NB: This is not a Bayesian evidence!

LOGPOST_SPREAD_GSPPHOT_OB : maximum minus minimum of log-posterior throughout MCMC chain for OB library (float)

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Difference of maximum log-posterior minus minimum log-posterior in MCMC chain (excluding burn-in phase) for OB library. If this spread is much larger than 25, say > 100, it suggests that the MCMC chain has not fully converged.

MCMCACCEPT_GSPPHOT_OB : MCMC acceptance rate of GSP-Phot Aeneas MCMC for OB library (float)

MCMC acceptance rate of GSP-Phot Aeneas MCMC for OB library. This is computed from all MCMC samples (before thinning the chain to 2000 or 100 samples).

TEFF_GSPPHOT_A : Effective temperature from GSP-Phot Aeneas for A library using BP/RP spectra (float, Temperature[K])

Effective temperature (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax (see Section **??** of the online documentation). This is the median of the MCMC samples. Taken from A library.

TEFF_GSPPHOT_A_LOWER : Lower confidence level (16%) of effective temperature from GSP-Phot Aeneas for A library using BP/RP spectra (float, Temperature[K])

Lower confidence level (16%) of effective temperature (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

TEFF_GSPPHOT_A_UPPER : Upper confidence level (84%) of effective temperature from GSP-Phot Aeneas for A library using BP/RP spectra (float, Temperature[K])

Upper confidence level (84%) of effective temperature (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

LOGG_GSPPHOT_A : Surface gravity from GSP-Phot Aeneas for A library using BP/RP spectra (float, GravitySurface[log cgs])

Surface gravity (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax (see Section **??** of the online documentation). This

is the median of the MCMC samples. Taken from A library.

LOGG_GSPPHOT_A_LOWER : Lower confidence level (16%) of surface gravity from GSP-Phot Aeneas for A library using BP/RP spectra (float, GravitySurface[log cgs])

Lower confidence level (16%) of surface gravity (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

LOGG_GSPPHOT_A_UPPER : Upper confidence level (84%) of surface gravity from GSP-Phot Aeneas for A library using BP/RP spectra (float, GravitySurface[log cgs])

Upper confidence level (84%) of surface gravity (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

FEH_GSPPHOT_A : Iron abundance from GSP-Phot Aeneas for A library using BP/RP spectra (float, Abundances[dex])

Decimal logarithm of the ratio of the number abundance of iron to the number abundance of hydrogen relative to the same ratio of solar abundances inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax, assuming source is a single star (see Section **??** of the online documentation). This is the median of the MCMC samples. Taken from A library.

FEH_GSPPHOT_A_LOWER : Lower confidence level (16%) of iron abundance from GSP-Phot Aeneas for A library using BP/RP spectra (float, Abundances[dex])

Lower confidence level (16%) of decimal logarithm of the ratio of the number abundance of iron to the number abundance of hydrogen relative to the same ratio of solar abundances inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax, assuming source is a single star (see Section **??** of the online documentation). This is the 16th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

FEH_GSPPHOT_A_UPPER : Upper confidence level (84%) of iron abundance from GSP-Phot Aeneas for A library using BP/RP spectra (float, Abundances[dex])

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Upper confidence level (84%) of decimal logarithm of the ratio of the number abundance of iron to the number abundance of hydrogen relative to the same ratio of solar abundances inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax, assuming source is a single star (see Section **??** of the online documentation). This is the 84th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

DISTANCE_GSPPHOT_A : Distance from GSP-Phot Aeneas for A library using BP/RP spectra (float, Length & Distance[pc])

Distance (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax (see Section **??** of the online documentation). This is the median of the MCMC samples. Taken from A library. NB: The actual fit parameter is $\log_{10} d$ and a prior is imposed to ensure a value between [0,5], thus the minimum possible distance is 1 pc and the maximum is 100 kpc.

DISTANCE_GSPPHOT_A_LOWER : Lower confidence level (16%) of distance from GSP-Phot Aeneas for A library using BP/RP spectra (float, Length & Distance[pc])

Lower confidence level (16%) of distance (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval. NB: The actual fit parameter is $\log_{10} d$ and a prior is imposed to ensure a value between [0,5], thus the minimum possible distance is 1 pc and the maximum is 100 kpc.

DISTANCE_GSPPHOT_A_UPPER : Upper confidence level (84%) of distance from GSP-Phot Aeneas for A library using BP/RP spectra (float, Length & Distance[pc])

Upper confidence level (84%) of distance (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval. NB: The actual fit parameter is $\log_{10} d$ and a prior is imposed to ensure a value between [0,5], thus the minimum possible distance is 1 pc and the maximum is 100 kpc.

AZERO_GSPPHOT_A : Extinction parameter from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Extinction parameter A_0 corresponding to a monochromatic extinction at a specific reference wavelength (see Section **??** of the online documentation) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples.

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Taken from A library. NB: This is the extinction parameter in the adopted extinction curve from Gordon et al. (2023).

AZERO_GSPPHOT_A_LOWER : Lower confidence level (16%) of extinction parameter from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of extinction parameter A_0 corresponding to a monochromatic extinction at a specific reference wavelength (see Section **??** of the online documentation) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from A library. NB: This is the extinction parameter in the adopted extinction curve from Gordon et al. (2023).

AZERO_GSPPHOT_A_UPPER : Upper confidence level (84%) of extinction parameter from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of extinction parameter A_0 corresponding to a monochromatic extinction at a specific reference wavelength (see Section **??** of the online documentation) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from A library. NB: This is the extinction parameter in the adopted extinction curve from Gordon et al. (2023).

AG_GSPPHOT_A : Extinction in G band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Broadband extinction in G band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from A library.

AG_GSPPHOT_A_LOWER : Lower confidence level (16%) of extinction in G band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of broadband extinction in G band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

AG_GSPPHOT_A_UPPER : Upper confidence level (84%) of extinction in G band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of broadband extinction in G band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

ABP_GSPPHOT_A : Extinction in BP band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Broadband extinction in G_{BP} band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from A library.

ABP_GSPPHOT_A_LOWER : Lower confidence level (16%) of extinction in BP band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of broadband extinction in G_{BP} band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

ABP_GSPPHOT_A_UPPER : Upper confidence level (84%) of extinction in BP band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of broadband extinction in G_{BP} band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

ARP_GSPPHOT_A : Extinction in RP band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Broadband extinction in G_{RP} band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from A library.

ARP_GSPPHOT_A_LOWER : Lower confidence level (16%) of extinction in RP band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Draft

Lower confidence level (16%) of broadband extinction in G_{RP} band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

ARP_GSPPHOT_A_UPPER : Upper confidence level (84%) of extinction in RP band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of broadband extinction in G_{RP} band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

AV_GSPPHOT_A : Extinction in Johnson V band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Broadband extinction in Johnson V band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from A library.

AV_GSPPHOT_A_LOWER : Lower confidence level (16%) of extinction in Johnson V band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of broadband extinction in Johnson V band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

AV_GSPPHOT_A_UPPER : Upper confidence level (84%) of extinction in Johnson V band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of broadband extinction in Johnson V band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

AJ_GSPPHOT_A : Extinction in 2MASS J band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Draft

Broadband extinction in 2MASS J band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from A library.

AJ_GSPPHOT_A_LOWER : Lower confidence level (16%) of extinction in 2MASS J band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of broadband extinction in 2MASS J band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

AJ_GSPPHOT_A_UPPER : Upper confidence level (84%) of extinction in 2MASS J band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of broadband extinction in 2MASS J band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

AH_GSPPHOT_A : Extinction in 2MASS H band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Broadband extinction in 2MASS H band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from A library.

AH_GSPPHOT_A_LOWER : Lower confidence level (16%) of extinction in 2MASS H band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of broadband extinction in 2MASS H band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

AH_GSPPHOT_A_UPPER : Upper confidence level (84%) of extinction in 2MASS H band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Draft

Upper confidence level (84%) of broadband extinction in 2MASS H band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

AKS_GSPPHOT_A : Extinction in 2MASS Ks band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Broadband extinction in 2MASS Ks band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from A library.

AKS_GSPPHOT_A_LOWER : Lower confidence level (16%) of extinction in 2MASS Ks band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of broadband extinction in 2MASS Ks band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

AKS_GSPPHOT_A_UPPER : Upper confidence level (84%) of extinction in 2MASS Ks band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of broadband extinction in 2MASS Ks band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

AW1_GSPPHOT_A : Extinction in WISE W1 band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Broadband extinction in WISE W1 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from A library.

AW1_GSPPHOT_A_LOWER : Lower confidence level (16%) of extinction in WISE W1 band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of broadband extinction in WISE W1 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

AW1_GSPPHOT_A_UPPER : Upper confidence level (84%) of extinction in WISE W1 band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of broadband extinction in WISE W1 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

AW2_GSPPHOT_A : Extinction in WISE W2 band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Broadband extinction in WISE W2 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from A library.

AW2_GSPPHOT_A_LOWER : Lower confidence level (16%) of extinction in WISE W2 band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of broadband extinction in WISE W2 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

AW2_GSPPHOT_A_UPPER : Upper confidence level (84%) of extinction in WISE W2 band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of broadband extinction in WISE W2 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

EBPMINRP_GSPPHOT_A : Reddening in BP minus RP colour from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Reddening $E(BP-RP)$ (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from A library. Note that while $E(BP-RP) = A_{BP} - A_{RP}$, this was computed at the level of MCMC samples. Hence, this relation is not exactly true for the median values.

EBPMINRP_GSPPHOT_A_LOWER : Lower confidence level (16%) of reddening in BP minus RP colour from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of reddening $E(BP-RP)$ (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval. Note that while $E(BP-RP) = A_{BP} - A_{RP}$, this was computed at the level of MCMC samples. Hence, this relation is not exactly true for the lower confidence levels.

EBPMINRP_GSPPHOT_A_UPPER : Upper confidence level (84%) of reddening in BP minus RP colour from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of reddening $E(BP-RP)$ (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval. Note that while $E(BP-RP) = A_{BP} - A_{RP}$, this was computed at the level of MCMC samples. Hence, this relation is not exactly true for the upper confidence levels.

EBMINV_GSPPHOT_A : Reddening in B minus V colour from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Reddening $E(B-V)$ (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from A library.

EBMINV_GSPPHOT_A_LOWER : Lower confidence level (16%) of reddening in B minus V colour from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of reddening $E(B-V)$ (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples.Taken from A library. Lower and upper levels include 68% confidence interval.

Draft

EBMINV_GSPPHOT_A_UPPER : Upper confidence level (84%) of reddening in B minus V colour from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of reddening $E(B-V)$ (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

ABSMAG_G_GSPPHOT_A : Absolute magnitude in G band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Absolute magnitude M_G (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from A library.

ABSMAG_G_GSPPHOT_A_LOWER : Lower confidence level (16%) of absolute magnitude in G band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of absolute magnitude M_G (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

ABSMAG_G_GSPPHOT_A_UPPER : Upper confidence level (84%) of absolute magnitude in G band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of absolute magnitude M_G (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

ABSMAG_J_GSPPHOT_A : Absolute magnitude in 2MASS J band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Absolute magnitude M_J in 2MASS J band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from A library.

Draft

ABSMAG_J_GSPPHOT_A_LOWER : Lower confidence level (16%) of absolute magnitude in 2MASS J band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of absolute magnitude M_{J} in 2MASS J band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

ABSMAG_J_GSPPHOT_A_UPPER : Upper confidence level (84%) of absolute magnitude in 2MASS J band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of absolute magnitude M_{J} in 2MASS J band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

ABSMAG_H_GSPPHOT_A : Absolute magnitude in 2MASS H band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Absolute magnitude M_{H} in 2MASS H band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from A library.

ABSMAG_H_GSPPHOT_A_LOWER : Lower confidence level (16%) of absolute magnitude in 2MASS H band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of absolute magnitude M_{H} in 2MASS H band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

ABSMAG_H_GSPPHOT_A_UPPER : Upper confidence level (84%) of absolute magnitude in 2MASS H band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of absolute magnitude M_{H} in 2MASS H band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

Draft

ABSMAG_KS_GSPPHOT_A : Absolute magnitude in 2MASS Ks band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Absolute magnitude M_{Ks} in 2MASS Ks band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from A library.

ABSMAG_KS_GSPPHOT_A_LOWER : Lower confidence level (16%) of absolute magnitude in 2MASS Ks band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of absolute magnitude M_{Ks} in 2MASS Ks band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

ABSMAG_KS_GSPPHOT_A_UPPER : Upper confidence level (84%) of absolute magnitude in 2MASS Ks band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of absolute magnitude M_{Ks} in 2MASS Ks band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

ABSMAG_W1_GSPPHOT_A : Absolute magnitude in WISE W1 band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Absolute magnitude M_{W1} in WISE W1 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from A library.

ABSMAG_W1_GSPPHOT_A_LOWER : Lower confidence level (16%) of absolute magnitude in WISE W1 band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of absolute magnitude M_{W1} in WISE W1 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

Draft

ABSMAG_W1_GSPPHOT_A_UPPER : Upper confidence level (84%) of absolute magnitude in WISE W1 band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of absolute magnitude M_{W1} in WISE W1 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

ABSMAG_W2_GSPPHOT_A : Absolute magnitude in WISE W2 band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Absolute magnitude M_{W2} in WISE W2 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from A library.

ABSMAG_W2_GSPPHOT_A_LOWER : Lower confidence level (16%) of absolute magnitude in WISE W2 band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Lower confidence level (16%) of absolute magnitude M_{W2} in WISE W2 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

ABSMAG_W2_GSPPHOT_A_UPPER : Upper confidence level (84%) of absolute magnitude in WISE W2 band from GSP-Phot Aeneas for A library using BP/RP spectra (float, Magnitude[mag])

Upper confidence level (84%) of absolute magnitude M_{W2} in WISE W2 band (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

RADIUS_GSPPHOT_A : Radius from GSP-Phot Aeneas for A library using BP/RP spectra (float, Length & Distance[Solar Radius])

Stellar radius (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from A library.

Draft

RADIUS_GSPPHOT_A_LOWER : Lower confidence level (16%) of radius from GSP-Phot Aeneas for A library using BP/RP spectra (float, Length & Distance[Solar Radius])

Lower confidence level (16%) of stellar radius (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

RADIUS_GSPPHOT_A_UPPER : Upper confidence level (84%) of radius from GSP-Phot Aeneas for A library using BP/RP spectra (float, Length & Distance[Solar Radius])

Upper confidence level (84%) of stellar radius (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

MASS_GSPPHOT_A : Mass from GSP-Phot Aeneas for A library using BP/RP spectra (float, Mass[Solar Mass])

Stellar mass (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from A library.

MASS_GSPPHOT_A_LOWER : Lower confidence level (16%) of mass from GSP-Phot Aeneas for A library using BP/RP spectra (float, Mass[Solar Mass])

Lower confidence level (16%) of stellar mass (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

MASS_GSPPHOT_A_UPPER : Upper confidence level (84%) of mass from GSP-Phot Aeneas for A library using BP/RP spectra (float, Mass[Solar Mass])

Upper confidence level (84%) of stellar mass (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence

Draft

interval.

AGE_GSPPHOT_A : Age from GSP-Phot Aeneas for A library using BP/RP spectra (float, Time[Gyr])

Stellar age (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the median of the MCMC samples. Taken from A library.

AGE_GSPPHOT_A_LOWER : Lower confidence level (16%) of age from GSP-Phot Aeneas for A library using BP/RP spectra (float, Time[Gyr])

Lower confidence level (16%) of stellar age (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 16th percentile of the MCMC samples. Taken from A library. Lower and upper levels include 68% confidence interval.

AGE_GSPPHOT_A_UPPER : Upper confidence level (84%) of age from GSP-Phot Aeneas for A library using BP/RP spectra (float, Time[Gyr])

Upper confidence level (84%) of stellar age (assuming source is a single star) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. This is the 84th percentile of the MCMC samples. Taken from the A library. Lower and upper levels include 68% confidence interval.

LOGPOSTERIOR_GSPPHOT_A : mean log-posterior of MCMC of GSP-Phot Aeneas MCMC for A library (float)

Goodness-of-fit score defined as the mean log-posterior of all MCMC samples of GSP-Phot Aeneas MCMC for A library. The higher the goodness-of-fit score, the better the fit. Values are usually negative. NB: This is not a Bayesian evidence!

LOGLIKELIHOOD_GSPPHOT_A : mean likelihood of MCMC of GSP-Phot Aeneas MCMC for A library (float)

Mean log-likelihood of all MCMC samples of GSP-Phot Aeneas MCMC for A library. The higher the mean log-likelihood, the better the fit. Values are usually negative. Subtracting the mean log-likelihood from the mean log-posterior yields the mean prior. NB: This is not a Bayesian evidence!

Draft

LOGPOST_SPREAD_GSPPHOT_A : maximum minus minimum of log-posterior throughout MCMC chain for A library (float)

Difference of maximum log-posterior minus minimum log-posterior in MCMC chain (excluding burn-in phase) for A library. If this spread is much larger than 25, say > 100, it suggests that the MCMC chain has not fully converged.

MCMCACCEPT_GSPPHOT_A : MCMC acceptance rate of GSP-Phot Aeneas MCMC for A library (float)

MCMC acceptance rate of GSP-Phot Aeneas MCMC for A library. This is computed from all MCMC samples (before thinning the chain to 2000 or 100 samples).

Draft

4.11 INTERSTELLAR_MEDIUM_PARAMS [TAP]

This table contains the diffuse interstellar band parameters derived from stacked RVS spectra by the DIB-Spec module as part of the Apsis processing chain. The parameters are derived for each 3D bin, classified by central Galactic longitude, central Galactic latitude, and distance.

See Section **??** of the online documentation and Gaia Collaboration et al. (2023b) for details on the DIB-Spec module. Table `interstellar_medium_spectra` contains the associated stacked RVS spectra.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

HEALPIX : Pixel number determined by a level 6 HEALPix scheme (int)

The pixel number determined by a level 6 HEALPix scheme, which is one of the two identifiers of the DIB-Spec bin voxel. Each voxel is defined by a HEALPix number (level 6) made in the equatorial system and the central heliocentric distance (dc).

LC : Central galactic longitude of voxel (float, Angle[deg])

Galactic longitude of the centre of the DIB-Spec voxel, converted from the central equatorial coordinate.

BC : Central galactic latitude of voxel (float, Angle[deg])

Galactic latitude of the centre of the DIB-Spec voxel, converted from the central equatorial coordinate.

DC : Central heliocentric distance of voxel (float, Length & Distance[kpc])

Heliocentric distance of the centre of the DIB-Spec voxel. The distance is defined as the inverse of parallax.

N_TARGETS : Number of target stars in a voxel (int)

Number of target stars used in each DIB-Spec voxel to make the stacked interstellar medium (ISM) spectrum.

SNR : SNR of stacked ISM spectrum (float)

Signal-to-noise ratio measured in the stacked ISM spectrum in each DIB-Spec voxel between 860.2 and 861.2 nm, defined as the mean value of flux divided by the standard deviation of the flux.

EW8621 : Equivalent width of the DIB at 862.1 nm (float, Length & Distance[nm])

Equivalent width of the DIB at 862.1 nm in each DIB-Spec voxel.

It is defined from a Gaussian model $f(\lambda, p_0, p_1, p_2) = p_0 \times \exp\left(-\frac{(\lambda-p_1)^2}{2p_2^2}\right)$ where p_0 and p_2 are the depth and width of the DIB profile, p_1 is the central wavelength and λ is the wavelength. The latter is the parameter `lambda` in the interstellar medium spectra table.

EW8621_LOWER : Lower confidence level (16%) of DIB equivalent width at 862.1 nm (float, Length & Distance[nm])

Lower confidence level (16%) of the equivalent width of the DIB at 862.1 nm in each DIB-Spec voxel, estimated by the distributions of p_0 and p_2 drawn from the MCMC posterior samplings.

EW8621_UPPER : Upper confidence level (84%) of DIB equivalent width at 862.1 nm (float, Length & Distance[nm])

Upper confidence level (84%) of the equivalent width of the DIB at 862.1 nm in each DIB-Spec voxel, estimated by the distributions of p_0 and p_2 drawn from the MCMC posterior samplings.

FLAGS8621 : Quality flag of DIB parameters at 862.1 nm (int)

Draft

Quality flag of the DIB parameters at 862.1 nm in each DIB-Spec voxel. 0 is the best value and 5 is the worst. The exact definition of the quality flags can be found in Gaia Collaboration et al. (2023b).

P08621 : Depth of the DIB profile at 862.1 nm (float)

Depth parameter of the DIB at 862.1 nm in each DIB-Spec voxel, see parameter ew8621 for details.

P08621_LOWER : Lower confidence level (16%) of p0 parameter at 862.1 nm (float)

Lower confidence level (16%) of the p0 parameter (depth) of the DIB at 862.1 nm in each DIB-Spec voxel, see parameter `p08621` for details.

P08621_UPPER : Upper confidence level (84%) of p0 parameter at 862.1 nm (float)

Upper confidence level (84%) of the p0 parameter (depth) of the DIB at 862.1 nm in each DIB-Spec voxel, see parameter `p08621` for details.

P18621 : Central wavelength (in vacuum) of the DIB profile at 862.1 nm (float, Length & Distance[nm])

Central wavelength (in vacuum) of the DIB at 862.1 nm in each DIB-Spec voxel, see parameter `ew8621` for details.

P18621_LOWER : Lower confidence level (16%) of p1 parameter at 862.1 nm (float, Length & Distance[nm])

Lower confidence level (16%) of the p1 parameter (central wavelength) of the DIB at 862.1 nm in each DIB-Spec voxel, see parameter `p18621` for details.

P18621_UPPER : Upper confidence level (84%) of p1 parameter at 862.1 nm (float, Length & Distance[nm])

Upper confidence level (84%) of the p1 parameter (central wavelength) of the DIB at 862.1 nm in each DIB-Spec voxel, see parameter `p18621` for details.

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p28621 : Gaussian width of the DIB profile at 862.1 nm (float, Length & Distance[nm])

Width (p2 parameter) of the DIB at 862.1 nm in each DIB-Spec voxel, see parameter `ew8621` for details.

p28621_lower : Lower confidence level (16%) of p2 parameter at 862.1 nm (float, Length & Distance[nm])

Lower confidence level (16%) of the p2 parameter (width) of the DIB at 862.1 nm in each DIB-Spec voxel, see parameter `p28621` for details.

p28621_upper : Upper confidence level (84%) of p2 parameter at 862.1 nm (float, Length & Distance[nm])

Upper confidence level (84%) of the p2 parameter (width) of the DIB at 862.1 nm in each DIB-Spec voxel, see parameter `p28621` for details.

ew8648 : Equivalent width of the DIB at 864.8 nm (float, Length & Distance[nm])

Equivalent width of the DIB at 864.8 nm in each DIB-Spec voxel.

It is defined from a Lorentzian model $f(\lambda, p_0, p_1, p_2) = \frac{-(p_0 p_2^2)}{(\lambda - p_1)^2 + p_2^2}$ where p_0 and p_2 are the depth and width of the DIB profile, p_1 is the central wavelength and λ is the wavelength. The latter is the parameter `lambda` in the interstellar medium spectra table.

ew8648_lower : Lower confidence level (16%) of DIB equivalent width at 864.8 nm (float, Length & Distance[nm])

Lower confidence level (16%) of the equivalent width of the DIB at 864.8 nm in each DIB-Spec voxel, estimated by the distributions of p_0 and p_2 drawn from the MCMC posterior samplings.

ew8648_upper : Upper confidence level (84%) of DIB equivalent width at 864.8 nm (float, Length & Distance[nm])

Upper confidence level (84%) of the equivalent width of the DIB at 864.8 nm in each DIB-Spec voxel, estimated by the distributions of p_0 and p_2 drawn from the MCMC posterior samplings.

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FLAGS8648 : Quality flag of DIB parameters at 864.8 nm (int)

Quality flag of the DIB parameters at 864.8 nm in each DIB-Spec voxel. 0 is the best value and 5 is the worst. The exact definition of the quality flags can be found in Gaia Collaboration et al. (2023b).

P08648 : Depth of the DIB profile at 864.8 nm (float)

Depth (p0 parameter) of the DIB at 864.8 nm in each DIB-Spec voxel, see parameter `ew8648` for details.

P08648_LOWER : Lower confidence level (16%) of p0 parameter at 864.8 nm (float)

Lower confidence level (16%) of the p0 parameter (depth) of the DIB at 864.8 nm in each DIB-Spec voxel, see parameter `p08648` for details.

P08648_UPPER : Upper confidence level (84%) of p0 parameter at 864.8 nm (float)

Upper confidence level (84%) of the p0 parameter (depth) of the DIB at 864.8 nm in each DIB-Spec voxel, see parameter `p08648` for details.

P18648 : Central wavelength (in vacuum) of the DIB profile at 864.8 nm (float, Length & Distance[nm])

Central wavelength (in vacuum) of the DIB at 864.8 nm in each DIB-Spec voxel, see parameter `ew8648` for details.

P18648_LOWER : Lower confidence level (16%) of p1 parameter at 864.8 nm (float, Length & Distance[nm])

Lower confidence level (16%) of the p1 parameter (central wavelength) of the DIB at 864.8 nm in each DIB-Spec voxel, see parameter `p18648` for details.

P18648_UPPER : Upper confidence level (84%) of p1 parameter at 864.8 nm (float, Length & Distance[nm])

Upper confidence level (84%) of the p1 parameter (central wavelength) of the DIB at 864.8 nm

Draft

in each DIB-Spec voxel, see parameter `p18648` for details.

P28648 : Lorentzian width of the DIB profile at 864.8 nm (float, Length & Distance[nm])

Lorentzian width of the DIB at 864.8 nm in each DIB-Spec voxel, see parameter `ew8648` for details.

P28648_LOWER : Lower confidence level (16%) of p2 parameter at 864.8 nm (float, Length & Distance[nm])

Lower confidence level (16%) of the p2 parameter (width) of the DIB at 864.8 nm in each DIB-Spec voxel, see parameter `p28648` for details.

P28648_UPPER : Upper confidence level (84%) of p2 parameter at 864.8 nm (float, Length & Distance[nm])

Upper confidence level (84%) of the p2 parameter (width) of the DIB at 864.8 nm of the DIB-Spec bin, see parameter `p28648` for details.

DIBCONT_A0 : Slope a0 of the global continuum fit of the full stacked ISM spectrum (float)

The coefficient a_0 (slope) of the global fit to the continuum of the stacked spectrum. The continuum is defined as $a_0\lambda + a_1$ where λ is the wavelength given in parameter `lambda` in the `interstellar_medium_spectra` table.

DIBCONT_A0_LOWER : Lower confidence level (16%) of the slope a0 of the global continuum fit (float)

Lower confidence level (16%) of the `dibcont_a0` parameter in each DIB-Spec voxel, see parameter `dibcont_a0` for details.

DIBCONT_A0_UPPER : Upper confidence level (84%) of the slope a0 of the global continuum fit (float)

Upper confidence level (84%) of the `dibcont_a0` parameter in each DIB-Spec voxel, see parameter `dibcont_a0` for details.

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DIBCONT_A1 : Intercept a1 of the global continuum fit of the full stacked ISM spectrum (float)

The coefficient a_1 (intercept) of the global fit to the continuum of the stacked spectrum, see parameter `dibcont_a0` for details.

DIBCONT_A1_LOWER : Lower confidence level (16%) of the intercept a1 of the global continuum fit (float)

The coefficient a_1 (intercept) of the global fit to the continuum of the stacked spectrum, see parameter `dibcont_a0` for details.

DIBCONT_A1_UPPER : Upper cofidence level (84%) of the intercept a1 of the global continuum fit (float)

The coefficient a_1 (intercept) of the global fit to the continuum of the stacked spectrum, see parameter `dibcont_a0` for details.

EW8621INT : Integrated equivalent width of DIB at 862.1 nm (float, Length & Distance[nm])

Integrated Equivalent width of the DIB at 862.1 nm in each DIB-Spec voxel. We refer to the FPR paper (Gaia Collaboration et al. 2023b) for more details.

EW8621INT_LOWER : Lower confidence level (16%) of integrated equivalent width of DIB at 862.1 nm (float, Length & Distance[nm])

Lower confidence level (16%) of integrated equivalent width of DIB at 862.1 nm.

EW8621INT_UPPER : Upper confidence level (84%) of integrated equivalent width of DIB at 862.1 nm (float, Length & Distance[nm])

Upper confidence level (84%) of integrated equivalent width of DIB at 862.1 nm.

EW8648INT : Integrated equivalent width of DIB at 864.8 nm (float, Length & Distance[nm])

Integrated equivalent width of the DIB at 864.8 nm in each DIB-Spec voxel. We refer to the FPR paper (Gaia Collaboration et al. 2023b) for more details.

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EW8648INT_LOWER : Lower confidence level (16%) of integrated equivalent width of DIB at 864.8 nm (float, Length & Distance[nm])

Lower confidence level (16%) of integrated equivalent width of DIB at 864.8 nm.

EW8648INT_UPPER : Upper confidence level (84%) of integrated equivalent width of DIB at 864.8 nm (float, Length & Distance[nm])

Upper confidence level (84%) of integrated equivalent width of DIB at 864.8 nm.

Draft

4.12 INTERSTELLAR_MEDIUM_SPECTRA [TAP]

This table contains the stacked interstellar medium RVS spectra derived by DIB-Spec as part of the Apsis processing chain. The spectra are derived for each 3D bin, classified by central Galactic longitude, central Galactic latitude, and distance.

See Section **??** of the online documentation and Gaia Collaboration et al. (2023b) for details on the DIB-Spec module. Table `interstellar_medium_params` contains the associated interstellar band parameters.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

HEALPIX : Pixel number determined by a level 6 HEALPix scheme (int)

The pixel number determined by a level 6 HEALPix scheme, which is one of the two identifiers of the DIB-Spec bin voxel. Each voxel is defined by a HEALPix number (level 6) made in the equatorial system and the central heliocentric distance (dc).

LC : Central galactic longitude of voxel (float, Angle[deg])

Galactic longitude of the centre of the DIB-Spec voxel, converted from the central equatorial coordinate.

BC : Central galactic latitude of voxel (float, Angle[deg])

Galactic latitude of the centre of the DIB-Spec voxel, converted from the central equatorial coordinate.

DC : Central heliocentric distance of voxel (float, Length & Distance[kpc])

Draft

Heliocentric distance of the centre of the DIB-Spec voxel. The distance is defined as the inverse of parallax.

LAMBDA : Wavelength in vacuum (float, Length & Distance[Å])

Wavelength in vacuum of the stacked interstellar medium spectrum.

FLUX : Normalised flux (float)

Normalised flux of the stacked interstellar medium spectrum corresponding to wavelength `lambda`.

FLUX_UNCERTAINTY : Uncertainty of the normalised flux (float)

Uncertainty value of the parameter `flux`. It is defined as the mean value of the flux errors on the individual spectra used to construct the stacked spectrum, divided by the square root of the number of spectra.

Draft

4.13 MCMC_SAMPLES_GSP_PHOT [DataLink]

This is the DataLink table hosting Markov Chain Monte Carlo (MCMC) samples for the posterior probability distribution of all parameters derived from the General Stellar Parametrizer from Photometry (GSP-Phot).

For all sources, 100 samples are provided drawn randomly from the full MCMC chain (see Section **??** in the online documentation).

The table considers the following parameters:

- effective temperature $T_{\rm eff}$ in K
- monochromatic extinction at 542.9 nm A_0 in mag
- surface gravity $\log g$
- metallicity [M/H]
- G-band extinction in mag
- $G_{\rm BP}$ band extinction $A_{\rm BP}$ in mag
- $G_{\rm RP}$ band extinction $A_{\rm RP}$ in mag
- Johnson V band extinction $A_{\rm V}$ in mag
- 2MASS J band extinction $A_{\rm J}$ in mag
- 2MASS H band extinction $A_{\rm H}$ in mag
- 2MASS Ks band extinction $A_{\rm Ks}$ in mag
- WISE W1 band extinction $A_{\rm W1}$ in mag
- WISE W2 band extinction $A_{\rm W2}$ in mag
- absolute magnitude $M_{\rm G}$ in mag
- absolute magnitude in 2MASS $M_{\rm J}$ in mag
- absolute magnitude in 2MASS $M_{\rm H}$ in mag
- absolute magnitude in 2MASS $M_{\rm Ks}$ in mag
- absolute magnitude in WISE $M_{\rm W1}$ in mag
- absolute magnitude in WISE $M_{\rm W2}$ in mag
- distance in pc
- reddening $E(G_{\rm BP} - G_{\rm RP}) = A_{\rm BP} - A_{\rm RP}$ in mag
- logarithmic posterior probability
- logarithmic likelihood
- stellar radius in $R_\odot$
- stellar mass in $M_\odot$

Draft

- logarithmic age in years

Note this table is not available through the main archive TAP interface. Data are delivered via the Massive Data service indexed by the VO DataLink protocol and described in Chapter **??**. For example this can be actioned in the archive user interface via ADQL queries joining on `source_id` with the table `all_source_flags` and selecting `has_mcmc_gspphot = 'true'`. The DataLink retrieval type to specify for this product in programmatic archive access via the Astroquery Gaia Python package is `MCMC_GSP_PHOT`.

DataLink products are served with a choice of two different data structures:

- `DATAMODEL_STANDARD`: the product is serialised following a standard data model (for example an IVOA-compliant data model in applicable cases);
- `DATAMODEL_GAIA`: the product is serialised following the Gaia DPAC(-internal) data model.

The `DATAMODEL_GAIA` format is described here whereas the `DATAMODEL_STANDARD` format is described in `https://www.cosmos.esa.int/web/gaia-users/archive/datalink-products#datalink_serialisation`

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (unique within a particular Data Release) (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

NSAMPLES : Number of samples in the chain from GSP-Phot (short)

The sample size is 100 for most sources, and is set to 2000 for 1) all sources brighter than G=12, 2) a random sample of 0.1% of the sources fainter than G=12.

TEFF : MCMC samples for effective temperature from GSP-Phot (float[`nsamples`] array, Temperature[K])

MCMC samples for posterior probability distribution of the stellar effective temperature from GSP-Phot.

AZERO : MCMC samples for monochromatic extinction from GSP-Phot (float[`nsamples`] array, Magnitude[mag])

MCMC samples for posterior probability distribution of the extinction parameter A_0 corresponding to a monochromatic extinction at a specific reference wavelength (see Section **??** of the online documentation) inferred by GSP-Phot Aeneas from BP/RP spectra, apparent G magnitude and parallax. NB: This is the extinction parameter in the adopted extinction curve from Gordon et al. (2023).

LOGG : MCMC samples for surface gravity from GSP-Phot (float[`nsamples`] array, GravitySurface[log cgs])

MCMC samples for posterior probability distribution of the stellar surface gravity $\log g$ from GSP-Phot.

MH : MCMC samples for the metallicity from GSP-Phot (float[`nsamples`] array, Abundances[dex])

MCMC samples for posterior probability distribution of metallicity [M/H] from GSP-Phot.

AG : MCMC samples for extinction in G band from GSP-Phot (float[`nsamples`] array, Magnitude[mag])

MCMC samples for posterior probability distribution of the extinction in the G band from GSP-Phot.

ABP : MCMC samples for extinction in BP band from GSP-Phot (float[`nsamples`] array, Magnitude[mag])

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MCMC samples for posterior probability distribution of the extinction in the G_{BP} band from GSP-Phot.

ARP : MCMC samples for extinction in RP band from GSP-Phot (float[`nsamples`] array, Magnitude[mag])

MCMC samples for posterior probability distribution of the extinction in the G_{RP} band from GSP-Phot.

AV : MCMC samples for extinction in Johnson V band from GSP-Phot (float[`nsamples`] array, Magnitude[mag])

MCMC samples for posterior probability distribution of the extinction in the Johnson V band from GSP-Phot.

AJ : MCMC samples for extinction in 2MASS J band from GSP-Phot (float[`nsamples`] array, Magnitude[mag])

MCMC samples for posterior probability distribution of the extinction in the 2MASS J band from GSP-Phot.

AH : MCMC samples for extinction in 2MASS H band from GSP-Phot (float[`nsamples`] array, Magnitude[mag])

MCMC samples for posterior probability distribution of the extinction in the 2MASS H band from GSP-Phot.

AKS : MCMC samples for extinction in 2MASS Ks band from GSP-Phot (float[`nsamples`] array, Magnitude[mag])

MCMC samples for posterior probability distribution of the extinction in the 2MASS Ks band from GSP-Phot.

AW1 : MCMC samples for extinction in WISE W1 band from GSP-Phot (float[`nsamples`] array, Magnitude[mag])

MCMC samples for posterior probability distribution of the extinction in the WISE W1 band

from GSP-Phot.

AW2 : MCMC samples for extinction in WISE W2 band from GSP-Phot (float[nsamples] array, Magnitude[mag])

MCMC samples for posterior probability distribution of the extinction in the WISE W2 band from GSP-Phot.

ABSMAG_G : MCMC samples for absolute magnitude in G band from GSP-Phot (float[nsamples] array, Magnitude[mag])

MCMC samples for posterior probability distribution of the absolute magnitude in the G band M_G from GSP-Phot. The absolute magnitude is obtained from isochrones (for details see Section **??** in the online documentation).

ABSMAG_J : MCMC samples for absolute magnitude in 2MASS J band from GSP-Phot (float[nsamples] array, Magnitude[mag])

MCMC samples for posterior probability distribution of the absolute magnitude in the 2MASS J band M_J from GSP-Phot. The absolute magnitude is obtained from isochrones (for details see Section **??** in the online documentation).

ABSMAG_H : MCMC samples for absolute magnitude in 2MASS H band from GSP-Phot (float[nsamples] array, Magnitude[mag])

MCMC samples for posterior probability distribution of the absolute magnitude in the 2MASS H band M_H from GSP-Phot. The absolute magnitude is obtained from isochrones (for details see Section **??** in the online documentation).

ABSMAG_KS : MCMC samples for absolute magnitude in 2MASS Ks band from GSP-Phot (float[nsamples] array, Magnitude[mag])

MCMC samples for posterior probability distribution of the absolute magnitude in the 2MASS Ks band M_{Ks} from GSP-Phot. The absolute magnitude is obtained from isochrones (for details see Section **??** in the online documentation).

ABSMAG_W1 : MCMC samples for absolute magnitude in WISE W1 band from GSP-Phot

(float[nsamples] array, Magnitude[mag])

MCMC samples for posterior probability distribution of the absolute magnitude in the WISE W1 band M_{W1} from GSP-Phot. The absolute magnitude is obtained from isochrones (for details see Section **??** in the online documentation).

ABSMAG_W2 : MCMC samples for absolute magnitude in WISE W2 band from GSP-Phot (float[nsamples] array, Magnitude[mag])

MCMC samples for posterior probability distribution of the absolute magnitude in the WISE W2 band M_{W2} from GSP-Phot. The absolute magnitude is obtained from isochrones (for details see Section **??** in the online documentation).

DISTANCEPC : MCMC samples for distance from GSP-Phot (float[nsamples] array, Length & Distance[pc])

MCMC samples for posterior probability distribution of the distance from GSP-Phot.

EBPMINRP : MCMC samples for reddening in BP minus RP colour from GSP-Phot (float[nsamples] array, Magnitude[mag])

MCMC samples for posterior probability distribution of the reddening $E(G_{BP}-G_{RP}) = A_{BP}-A_{RP}$ from GSP-Phot.

LOG_POS : MCMC samples for the log-posterior from GSP-Phot (float[nsamples] array)

Logarithmic posterior probability of MCMC samples from GSP-Phot. This can be used in order to replace the GSP-Phot priors by a user-chosen prior using importance sampling (for details see Section **??** in the online documentation).

LOG_LIK : MCMC samples for the log-likelihood from GSP-Phot (float[nsamples] array)

Logarithmic likelihood of MCMC samples from GSP-Phot. This can be used in order to replace the GSP-Phot priors by a user-chosen prior using importance sampling (for details see Section **??** in the online documentation).

RADIUS : MCMC samples for stellar radius from GSP-Phot (float[nsamples] array, Length &

Distance[Solar Radius])

MCMC samples for posterior probability distribution of the stellar radius from GSP-Phot.

MASS : MCMC samples for stellar mass from GSP-Phot (float[`nsamples`] array, Mass[Solar Mass])

MCMC samples for posterior probability distribution of the stellar mass from GSP-Phot.

LOGAGE : MCMC samples for logarithmic age from GSP-Phot (float[`nsamples`] array, Time[year])

MCMC samples for posterior probability distribution of the logarithmic age from GSP-Phot.

Draft

4.14 OA_NEURON_INFORMATION [TAP]

This is the table hosting the content of a Self-Organised Map calculated from a dataset composed of outliers processed by the Apsis OA module, using normalised BP/RP spectra as input. Each entry corresponds to parameters estimated for one particular neuron of the map.

The prototype BP/RP spectrum for a particular neuron is available in the table `oa_neuron_xp_spectra` and the information of the sources composing the dataset is available in the table `ap_class`. See Section **??** of the online documentation for further details.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOM_ID : Self-Organised Map identifier (long)

A unique single numerical identifier of the Self-Organised Map. The first three digits of the identifier are reserved, with the leading digit set to '1' (1xx).

NEURON_ID : Neuron identifier (long)

A unique single numerical identifier of the neuron.

NEURON_ROW_INDEX : Row index of the neuron in the Self-Organised Map lattice (short)

A row index which determines the position of the neuron within the Self-Organised Map lattice, together with `neuron_column_index`.

NEURON_COLUMN_INDEX : Column index of the neuron in the Self-Organised Map lattice (short)

Draft

A column index which determines the position of the neuron within the Self-Organised Map lattice, together with `neuron_row_index`.

HITS : Number of sources populating the neuron (int)

Number of sources that have been assigned to the neuron according to the squared Euclidean distance between the sources XP spectra and the neuron XP prototype (`ap_class.neuron_oa_dist`), see Section **??** for further details.

CLASSLABEL : Astronomical class estimated for the neuron (string)

Astronomical class estimated for the neuron by means of a template matching procedure among a set of reference templates. This field is set to "UNKNOWN" if no template is assigned. See Section **??** for further details.

CENTROID_ID : Identifier of the Gaia source that minimizes the classification distance to the neuron (long)

Identifier of the Gaia source (`gaia_source.source_id`) that minimizes the squared Euclidean distance between the source XP spectrum and the neuron XP prototype. See `centroid_distance`.

CENTROID_DISTANCE : Squared Euclidean distance between the centroid XP spectrum and the neuron XP prototype (float)

Squared Euclidean distance between the XP spectrum of the centroid source and the neuron XP prototype (`xp_spectrum_prototype_flux` in table `oa_neuron_xp_spectra`).

TEMPLATE_DISTANCE : Squared Euclidean distance between the reference XP template and the neuron XP prototype (float)

Squared Euclidean distance between the XP spectrum of the reference template (`xp_spectrum_template_fl` in table `oa_neuron_xp_spectra`) and the neuron XP prototype (`xp_spectrum_prototype_flux` in table `oa_neuron_xp_spectra`). This field can be null if no template is assigned. See Section **??** for further details.

G_MAG_MEAN : Mean of the G-band magnitudes for the sources that belong to the neuron (float, Magnitude[mag])

Draft

Mean G magnitude value for those sources that belong to the neuron. The mean G magnitude (`phot_g_mean_mag` in table `gaia_source`) for each source is used, and if such a value is not finite, then it is not taken into account.

G_MAG_STD_DEV : Standard deviation of the G-band magnitudes for the sources that belong to the neuron (float, Magnitude[mag])

Standard deviation of G magnitude values for those sources that belong to the neuron. The mean G magnitude (`phot_g_mean_mag` in table `gaia_source`) for each source is used, and if such a value is not finite, then it is not taken into account.

G_MAG_MIN : Minimum G-band magnitude value for the sources that belong to the neuron (float, Magnitude[mag])

Minimum G value for those sources that belong to the neuron. The mean G magnitude (`phot_g_mean_mag` in table `gaia_source`) for each source is used, and if such a value is not finite, then it is not taken into account.

G_MAG_MAX : Maximum G-band magnitude value for the sources that belong to the neuron (float, Magnitude[mag])

Maximum G value for those sources that belong to the neuron. The mean G magnitude (`phot_g_mean_mag` in table `gaia_source`) for each source is used, and if such a value is not finite, then it is not taken into account.

BP_MAG_MEAN : Mean of the BP-band magnitudes for the sources that belong to the neuron (float, Magnitude[mag])

Mean $G_{\rm BP}$ value for those sources that belong to the neuron. The integrated mean $G_{\rm BP}$ magnitude (`phot_bp_mean_mag` in table `gaia_source`) for each source is used, and if such a value is not finite, then it is not taken into account.

BP_MAG_STD_DEV : Standard deviation of the BP-band magnitudes for the sources that belong to the neuron (float, Magnitude[mag])

Standard deviation of $G_{\rm BP}$ values for those sources that belong to the neuron. The integrated mean $G_{\rm BP}$ magnitude (`phot_bp_mean_mag` in table `gaia_source`) for each source is used, and if such a value is not finite, then it is not taken into account.

BP_MAG_MIN : Minimum BP-band magnitude value for the sources that belong to the neuron (float, Magnitude[mag])

Minimum value of $G_{\rm BP}$ for those sources that belong to the neuron. The integrated mean $G_{\rm BP}$ magnitude (`phot_bp_mean_mag` in table `gaia_source`) for each source is used, and if such a value is not finite, then it is not taken into account.

BP_MAG_MAX : Maximum BP-band magnitude value for the sources that belong to the neuron (float, Magnitude[mag])

Maximum value of $G_{\rm BP}$ for those sources that belong to the neuron. The integrated mean $G_{\rm BP}$ magnitude (`phot_bp_mean_mag` in table `gaia_source`) for each source is used, and if such a value is not finite, then it is not taken into account.

RP_MAG_MEAN : Mean of the RP-band magnitudes for the sources that belong to the neuron (float, Magnitude[mag])

Mean $G_{\rm RP}$ value for those sources that belong to the neuron. The integrated mean $G_{\rm RP}$ magnitude (`phot_rp_mean_mag` in table `gaia_source`) for each source is used, and if such a value is not finite, then it is not taken into account.

RP_MAG_STD_DEV : Standard deviation of the RP-band magnitudes for the sources that belong to the neuron (float, Magnitude[mag])

Standard deviation of $G_{\rm RP}$ values for those sources that belong to the neuron. The integrated mean $G_{\rm RP}$ magnitude (`phot_rp_mean_mag` in table `gaia_source`) for each source is used, and if such a value is not finite, then it is not taken into account.

RP_MAG_MIN : Minimum RP-band magnitude value for the sources that belong to the neuron (float, Magnitude[mag])

Minimum value of $G_{\rm RP}$ for those sources that belong to the neuron. The integrated mean $G_{\rm RP}$ magnitude (`phot_rp_mean_mag` in table `gaia_source`) for each source is used, and if such a value is not finite, then it is not taken into account.

RP_MAG_MAX : Maximum RP-band magnitude value for the sources that belong to the neuron

Draft

(float, Magnitude[mag])

Maximum value of G_{RP} for those sources that belong to the neuron. The integrated mean G_{RP} magnitude (`phot_rp_mean_mag` in table `gaia_source`) for each source is used, and if such a value is not finite, then it is not taken into account.

PM_RA_MEAN : Mean value of the proper motion in right ascension for the sources that belong to the neuron (float, Angular Velocity[mas yr^{-1}])

Mean value of the proper motion in right ascension direction for those sources that belong to the neuron. The proper motion in right ascension (`gaia_source.pmra`) for each source is used, and if such a value is not finite, then it is not taken into account.

PM_RA_STD_DEV : Standard deviation of the proper motion in right ascension for the sources that belong to the neuron (float, Angular Velocity[mas yr^{-1}])

Standard deviation of the proper motion in right ascension direction for those sources that belong to the neuron. The proper motion in right ascension (`gaia_source.pmra`) for each source is used, and if such a value is not finite, then it is not taken into account.

PM_RA_MIN : Minimum value of the proper motion in right ascension for the sources that belong to the neuron (float, Angular Velocity[mas yr^{-1}])

Minimum value of the proper motion in right ascension direction for those sources that belong to the neuron. The proper motion in right ascension (`gaia_source.pmra`) for each source is used, and if such a value is not finite, then it is not taken into account.

PM_RA_MAX : Maximum value of the proper motion in right ascension for the sources that belong to the neuron (float, Angular Velocity[mas yr^{-1}])

Maximum value of the proper motion in right ascension direction for those sources that belong to the neuron. The proper motion in right ascension (`gaia_source.pmra`) for each source is used, and if such a value is not finite, then it is not taken into account.

PM_DEC_MEAN : Mean value of the proper motion in declination for the sources that belong to the neuron (float, Angular Velocity[mas yr^{-1}])

Mean value of the proper motion in declination direction for those sources that belong to the

neuron. The proper motion in declination (`gaia_source.pmdec`) for each source is used, and if such a value is not finite, then it is not taken into account.

PM_DEC_STD_DEV : Standard deviation of the proper motion in declination for the sources that belong to the neuron (float, Angular Velocity[mas yr^{-1}])

Standard deviation of the proper motion in declination direction for those sources that belong to the neuron. The proper motion in declination (`gaia_source.pmdec`) for each source is used, and if such a value is not finite, then it is not taken into account.

PM_DEC_MIN : Minimum value of the proper motion in declination for the sources that belong to the neuron (float, Angular Velocity[mas yr^{-1}])

Minimum value of the proper motion in declination direction for those sources that belong to the neuron. The proper motion in declination (`gaia_source.pmdec`) for each source is used, and if such a value is not finite, then it is not taken into account.

PM_DEC_MAX : Maximum value of the proper motion in declination for the sources that belong to the neuron (float, Angular Velocity[mas yr^{-1}])

Maximum value of the proper motion in declination direction for those sources that belong to the neuron. The proper motion in declination (`gaia_source.pmdec`) for each source is used, and if such a value is not finite, then it is not taken into account.

PARALLAX_MEAN : Mean parallax value for the sources that belong to the neuron (float, Angle[mas])

Mean parallax value for those sources that belong to the neuron. The parallax (`gaia_source.parallax`) for each source is used, and if such a value is not finite, then it is not taken into account.

PARALLAX_STD_DEV : Standard deviation of the parallax values for the sources that belong to the neuron (float, Angle[mas])

Standard deviation of the parallax values for those sources that belong to the neuron. The parallax (`gaia_source.parallax`) for each source is used, and if such a value is not finite, then it is not taken into account.

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PARALLAX_MIN : Minimum parallax value for the sources that belong to the neuron (float, Angle[mas])

Minimum parallax value for those sources that belong to the neuron. The parallax (`gaia_source.parallax`) for each source is used, and if such a value is not finite, then it is not taken into account.

PARALLAX_MAX : Maximum parallax value for the sources that belong to the neuron (float, Angle[mas])

Maximum parallax value for those sources that belong to the neuron. The parallax (`gaia_source.parallax`) for each source is used, and if such a value is not finite, then it is not taken into account.

GAL_LATITUDE_MEAN : Mean galactic latitude for the sources that belong to the neuron (float, Angle[deg])

Mean value of the galactic latitude for those sources that belong to the neuron. The galactic latitude (`gaia_source.b`) for each source is used, and if such a value is not finite, then it is not taken into account.

GAL_LATITUDE_STD_DEV : Standard deviation of the galactic latitude values for the sources that belong to the neuron (float, Angle[deg])

Standard deviation of the galactic latitude values for those sources that belong to the neuron. The galactic latitude (`gaia_source.b`) for each source is used, and if such a value is not finite, then it is not taken into account.

GAL_LATITUDE_MIN : Minimum galactic latitude for the sources that belong to the neuron (float, Angle[deg])

Minimum value of the galactic latitude for those sources that belong to the neuron. The galactic latitude (`gaia_source.b`) for each source is used, and if such a value is not finite, then it is not taken into account.

GAL_LATITUDE_MAX : Maximum galactic latitude for the sources that belong to the neuron (float, Angle[deg])

Maximum value of the galactic latitude for those sources that belong to the neuron. The galactic latitude (`gaia_source.b`) for each source is used, and if such a value is not finite, then it is not

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taken into account.

INTRA_NEURON_DISTANCE_MEAN : Mean value of the squared Euclidean distance between each of the XP sources in the neuron and the neuron prototype (float)

Mean value of the squared Euclidean distance between each of the XP spectra of the sources that belong to the neuron and the XP prototype of the neuron (`oa_neuron_xp_spectra.xp_spectrum_prototyp`
See Section **??** for further details.

INTRA_NEURON_DISTANCE_STD_DEV : Standard deviation of the squared Euclidean distance between each of the XP sources in the neuron and the neuron prototype (float)

Standard deviation of the squared Euclidean distance between each of the XP spectra of the sources that belong to the neuron and the XP prototype of the neuron (`oa_neuron_xp_spectra.xp_spectru`
See Section **??** for further details.

INTRA_NEURON_DISTANCE_MIN : Minimum squared Euclidean distance between each of the XP sources in the neuron and the neuron prototype (float)

Minimum value of the squared Euclidean distance between each of the XP spectra of the sources that belong to the neuron and the XP prototype of the neuron (`oa_neuron_xp_spectra.xp_spectrum_proto`
See Section **??** for further details.

INTRA_NEURON_DISTANCE_MAX : Maximum squared Euclidean distance between each of the XP sources in the neuron and the neuron prototype (float)

Maximum value of the squared Euclidean distance between each of the XP spectra of the sources that belong to the neuron and the XP prototype of the neuron (`oa_neuron_xp_spectra.xp_spectrum_proto`
See Section **??** for further details.

INTER_NEURON_DISTANCE_MEAN : Mean value of the squared Euclidean distance between the neuron XP prototype and the XP prototypes of its immediate neighbours (float)

Mean value of the squared Euclidean distance between the neuron XP prototype (`xp_spectrum_prototype_f`
in table `oa_neuron_xp_spectra`) and the XP prototypes of its immediate neighbours. See Section **??** for further details.

INTER_NEURON_DISTANCE_STD_DEV : Standard deviation of the squared Euclidean distance between the neuron XP prototype and the XP prototypes of its immediate neighbours (float)

Standard deviation of the squared Euclidean distance between the neuron XP prototype (`xp_spectrum_proto` in table `oa_neuron_xp_spectra`) and the XP prototypes of its immediate neighbours. See Section **??** for further details.

INTER_NEURON_DISTANCE_MIN : Minimum value of the squared Euclidean distance between the neuron XP prototype and the XP prototypes of its immediate neighbours (float)

Minimum value of the squared Euclidean distance between the neuron XP prototype (`xp_spectrum_prototy` in table `oa_neuron_xp_spectra`) and the XP prototypes of its immediate neighbours. See Section **??** for further details.

INTER_NEURON_DISTANCE_MAX : Maximum value of the squared Euclidean distance between the neuron XP prototype and the XP prototypes of its immediate neighbours (float)

Maximum value of the squared Euclidean distance between the neuron XP prototype (`xp_spectrum_prototy` in table `oa_neuron_xp_spectra`) and the XP prototypes of its immediate neighbours. See Section **??** for further details.

TEMPLATE_NAME : Name of the template used to describe the neuron (string)

Name of the template that has been assigned to the neuron. Templates were built from observed XP spectra of representative astronomical classes and allow to assign an astronomical type or class label to the objects populating a neuron. This field can be null if no template is assigned. See Section **??** for further details.

DISTANCE_PERCENTILE25 : 25th percentile value for the intra-neuron distance distribution (float)

25th percentile value for the intra-neuron distance distribution, which includes the 25% of the classified objects above this threshold. That is, the 25% of objects with the highest squared Euclidean distances to their neuron XP prototype (`oa_neuron_xp_spectra.xp_spectrum_prototype_flux`) lie above this value. See Section **??** for further details.

DISTANCE_PERCENTILE50 : 50th percentile value for the intra-neuron distance distribution (float)

50th percentile value for the intra-neuron distance distribution, which includes the 50% of the classified objects above this threshold. That is, the 50% of objects with the highest squared Euclidean distances to their neuron XP prototype (`oa_neuron_xp_spectra.xp_spectrum_prototype_flux`) lie above this value. See Section **??** for further details.

DISTANCE_PERCENTILE68 : 68th percentile value for the intra-neuron distance distribution

(float)

68th percentile value for the intra-neuron distance distribution, which includes the 68% of the classified objects above this threshold. That is, the 68% of objects with the highest squared Euclidean distances to their neuron XP prototype (`oa_neuron_xp_spectra.xp_spectrum_prototype_flux`) lie above this value. See Section **??** for further details.

DISTANCE_PERCENTILE75 : 75th percentile value for the intra-neuron distance distribution

(float)

75th percentile value for the intra-neuron distance distribution, which includes the 75% of the classified objects above this threshold. That is, the 75% of objects with the highest squared Euclidean distances to their neuron XP prototype (`oa_neuron_xp_spectra.xp_spectrum_prototype_flux`) lie above this value. See Section **??** for further details.

DISTANCE_PERCENTILE95 : 95th percentile value for the intra-neuron distance distribution

(float)

95th percentile value for the intra-neuron distance distribution, which includes the 95% of the classified objects above this threshold. That is, the 95% of objects with the highest squared Euclidean distances to their neuron XP prototype (`oa_neuron_xp_spectra.xp_spectrum_prototype_flux`) lie above this value. See Section **??** for further details.

DISTANCE_FWHM : Full Width at Half Maximum value for the intra-neuron distance distribution (float)

FWHM is given by the distance between two points on a neuron squared Euclidean distance distribution at which the function reaches half its maximum value. Neurons showing low values for this quantity contain objects displaying similar spectra, and can be considered to have good clustering quality. See Section **??** for further details.

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DISTANCE_SKEW : Skewness value for the intra-neuron distance distribution (float)

Skewness is the third standardized moment of a distribution. It measures the distortion or asymmetry on a neuron squared Euclidean distance distribution. If the skewness value is positive, then the distribution is shifted to the left, meaning that most values are concentrated on shorter distances. If the value is negative, the function is shifted to the right, with values located at larger distances. In both cases the function is said to be skewed. See Section **??** for further details.

DISTANCE_KURTOSIS : Kurtosis value for the intra-neuron distance distribution (float)

Kurtosis is the ratio of the fourth moment μ^4 to the square of the variance. It is a measure of the heaviness of the tails of the neuron squared Euclidean distance distribution. Better quality neurons tend to exhibit kurtosis values close or slightly below that of a normal distribution (excess kurtosis near zero or negative), indicating a distribution of distances that is more concentrated with fewer extreme values. See Section **??** for further details.

DISTANCE_I_Q_R : Inter-Quartile Range value for the intra-neuron distance distribution (float)

Difference between the 75th (`distance_percentile75`) and 25th (`distance_percentile25`) percentile values for the intra-neuron distance distribution. Low values indicate a focused distance distribution. See Section **??** for further details.

DISTANCE_FWHM_NORM : Normalized FWHM value for the intra-neuron distance distribution (float)

FWHM (`distance_fwhm`) is given by the distance between two points on a neuron squared Euclidean distance distribution at which the function reaches half its maximum value. It is calculated as a normalised value, by dividing the FWHM mean value obtained for the 10% of neurons with the lowest values in the SOM map by the FWHM in the neuron (`distance_fwhm`). Neurons showing values near or above one for this quantity contain objects displaying similar spectra, and can be considered to have good clustering quality. See Section **??** for further details.

QUALITY_CATEGORY : Quality category assigned to the neuron, where 0 corresponds to the most homogeneous neurons and 6 to the most heterogeneous ones (short)

In order to provide a unified interpretation of the goodness of the clustering in a neuron, a general index has been calculated that takes into account the results obtained for the three intra-neuron

distance distribution measurements: FWHM (`distance_fwhm`), skewness (`distance_skew`) and kurtosis (`distance_kurtosis`). To do so, the following percentile values of the three indices have been set: 10th, 32th, 50th, 75th, 90th and 95th. For each of the indices it is possible to obtain seven quality categories depending on the interquartile area where the measures are grouped, and then assign the neuron a global index named quality category (QC). Quality category can take any of the following values: 0 in case all three indices are in percentile level 95th, value 1 if they are in level 90th, and so on to finish with level 6 when all quality indices are outside percentile 10th. Quality category 0 corresponds to the most homogeneous neurons and 6 to the most heterogeneous ones. If no category could be assigned then it takes value -1. See Section **??** for further details.

BP_TRANSITS_MEAN : Mean value of the number of BP transits for the sources that belong to the neuron (float)

Mean value of the number of BP transits for those sources that belong to the neuron. The number of BP transits (`xp_summary.bp_n_transits`) for each source is used, and if such a value is not finite, then it is not taken into account.

BP_TRANSITS_STD_DEV : Standard deviation of the number of BP transits for the sources that belong to the neuron (float)

Standard deviation of the number of BP transits for those sources that belong to the neuron. The number of BP transits (`xp_summary.bp_n_transits`) for each source is used, and if such a value is not finite, then it is not taken into account.

BP_TRANSITS_MIN : Minimum value of the number of BP transits for the sources that belong to the neuron

(float)

Minimum value of the number of BP transits for those sources that belong to the neuron. The number of BP transits (`xp_summary.bp_n_transits`) for each source is used, and if such a value is not finite, then it is not taken into account.

BP_TRANSITS_MAX : Maximum value of the number of BP transits for the sources that belong to the neuron (float)

Maximum value of the number of BP transits for those sources that belong to the neuron. The number of BP transits (`xp_summary.bp_n_transits`) for each source is used, and if such a

value is not finite, then it is not taken into account.

RP_TRANSITS_MEAN : Mean value of the number of RP transits for the sources that belong to the neuron

(float)

Mean value of the number of RP transits for those sources that belong to the neuron. The number of RP transits (`xp_summary.rp_n_transits`) for each source is used, and if such a value is not finite, then it is not taken into account.

RP_TRANSITS_STD_DEV : Standard deviation of the number of RP transits for the sources that belong to the neuron (float)

Standard deviation of the number of RP transits for those sources that belong to the neuron. The number of RP transits (`xp_summary.rp_n_transits`) for each source is used, and if such a value is not finite, then it is not taken into account.

RP_TRANSITS_MIN : Minimum value of the number of RP transits for the sources that belong to the neuron (float)

Minimum value of the number of RP transits for those sources that belong to the neuron. The number of RP transits (`xp_summary.rp_n_transits`) for each source is used, and if such a value is not finite, then it is not taken into account.

RP_TRANSITS_MAX : Maximum value of the number of RP transits for the sources that belong to the neuron (float)

Maximum value of the number of RP transits for those sources that belong to the neuron. The number of RP transits (`xp_summary.rp_n_transits`) for each source is used, and if such a value is not finite, then it is not taken into account.

RUWE_MEAN : Mean value of the renormalised unit weight error for the sources that belong to the neuron (float)

Mean value of the renormalised unit weight error for those sources that belong to the neuron. The renormalised unit weight error (`gaia_source.ruwe`) for each source is used, and if such a value is not finite, then it is not taken into account.

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RUWE_STD_DEV : Standard deviation of the renormalised unit weight error for the sources that belong to the neuron (float)

Standard deviation of the renormalised unit weight error for those sources that belong to the neuron. The renormalised unit weight error (`gaia_source.ruwe`) for each source is used, and if such a value is not finite, then it is not taken into account.

RUWE_MIN : Minimum value of the renormalised unit weight error for the sources that belong to the neuron (float)

Minimum value of the renormalised unit weight error for those sources that belong to the neuron. The renormalised unit weight error (`gaia_source.ruwe`) for each source is used, and if such a value is not finite, then it is not taken into account.

RUWE_MAX : Maximum value of the renormalised unit weight error for the sources that belong to the neuron

(float)

Maximum value of the renormalised unit weight error for those sources that belong to the neuron. The renormalised unit weight error (`gaia_source.ruwe`) for each source is used, and if such a value is not finite, then it is not taken into account.

BPRP_COLOUR_MEAN : Mean of the BP minus RP colours for the sources that belong to the neuron (float, Magnitude[mag])

Mean value of the $G_{\rm BP} - G_{\rm RP}$ colour for those sources that belong to the neuron. The integrated mean $G_{\rm BP}$ magnitude (`gaia_source.phot_bp_mean_mag`) and mean $G_{\rm RP}$ magnitude (`gaia_source.phot_rp_mean_mag`) for each source are used, and if such values are not finite, then it is not taken into account.

BPRP_COLOUR_STD_DEV : Standard deviation of the BP minus RP colours for the sources that belong to the neuron (float, Magnitude[mag])

Standard deviation of the $G_{\rm BP} - G_{\rm RP}$ colour for those sources that belong to the neuron. The integrated mean $G_{\rm BP}$ magnitude (`gaia_source.phot_bp_mean_mag`) and mean $G_{\rm RP}$ magnitude (`gaia_source.phot_rp_mean_mag`) for each source are used, and if such values are not finite,

then it is not taken into account.

BPRP_COLOUR_MIN : Minimum BP minus RP colour value for the sources that belong to the neuron (float, Magnitude[mag])

Minimum value of the $G_{BP} - G_{RP}$ colour for those sources that belong to the neuron. The integrated mean G_{BP} magnitude (`gaia_source.phot_bp_mean_mag`) and mean G_{RP} magnitude (`gaia_source.phot_rp_mean_mag`) for each source are used, and if such values are not finite, then it is not taken into account.

BPRP_COLOUR_MAX : Maximum BP minus RP colour value for the sources that belong to the neuron (float, Magnitude[mag])

Maximum value of the $G_{BP} - G_{RP}$ colour for those sources that belong to the neuron. The integrated mean G_{BP} magnitude (`gaia_source.phot_bp_mean_mag`) and mean G_{RP} magnitude (`gaia_source.phot_rp_mean_mag`) for each source are used, and if such values are not finite, then it is not taken into account.

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4.15 OA_NEURON_XP_SPECTRA [TAP]

This is the table hosting the prototype BP/RP spectrum corresponding to each of the neurons of the Self-Organised Map produced by the Apsis module OA. It is a flat table where each row represents the spectral information at a specific wavelength for a given neuron.

Other neuron attributes, such as statistics on various parameters, are available in another table: `oa_neuron_information`. See Section **??** of the online documentation for further details.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`.

SOM_ID : Self-Organised Map identifier (long)

A unique single numerical identifier of the Self-Organised Map.

NEURON_ID : Neuron identifier (long)

A unique single numerical identifier of the neuron.

NEURON_ROW_INDEX : Row index of the neuron in the Self-Organised Map lattice (short)

A row index which determines the position of the neuron within the Self-Organised Map lattice, together with `neuron_column_index`.

NEURON_COLUMN_INDEX : Column index of the neuron in the Self-Organised Map lattice (short)

A column index which determines the position of the neuron within the Self-Organised Map lat-

tice, together with neuron_row_index.

XP_SPECTRUM_PROTOTYPE_FLUX : Normalised flux for the preprocessed XP spectrum that best represents the neuron (float)

Flux at wavelength value specified by xp_spectrum_prototype_wavelength for the preprocessed XP spectrum (including normalisation) that best represents the sources that belong to the neuron. This is the prototype spectrum for the neuron. See Section **??** for further details.

XP_SPECTRUM_PROTOTYPE_WAVELENGTH : Wavelength associated with the XP spectrum flux values (float, Length & Distance[nm])

Wavelength value associated with the xp_spectrum_prototype_flux and xp_spectrum_template_flux fluxes.

XP_SPECTRUM_TEMPLATE_FLUX : Normalised flux for the preprocessed XP spectrum corresponding to the neuron template (float)

Flux at wavelength value specified by xp_spectrum_prototype_wavelength for the preprocessed XP spectrum (including normalisation) of the reference template assigned to the neuron. This field can be null if no template is assigned. See Section **??** for further details.

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4.16 TOTAL_GALACTIC_EXTINCTION_MAP [TAP]

This table provides the total Galactic extinction map of the extinction at a specific reference wavelength as a HEALPix map, appropriate for extragalactic sources.

The table contains the monochromatic extinction A_0 at a specific reference wavelength (see Section **??** of the online documentation) describing the effective total Galactic extinction and related uncertainties at six separate HEALPix levels, namely levels 6 to 11. The estimated extinction is based on the individual extinction estimates of a selected set of Gaia sources outside the ISM layer of the Milky Way in the HEALPix, which are used as extinction tracers. For further details see Section **??** of the online documentation.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`.

HEALPIX_ID : HEALPix identification (long)

The unique HEALPix index number for which the total Galactic extinction parameters are estimated. For further details on the HEALPix scheme used by Gaia, see Bastian & Portell (2020).

HEALPIX_LEVEL : HEALPix level used (byte)

The HEALPix level of the HEALPix. The Total Galactic Extinction map is provided at HEALPix levels 6 through 11. For further details on the HEALPix scheme used by Gaia, see Bastian & Portell (2020).

A0_MEDIAN : Median extinction parameter (float, Magnitude[mag])

The median A_0 of the selected extinction tracers as estimated by GSP-Phot from BP/RP spectra,

apparent G magnitude and parallax. A_0 is the extinction parameter corresponding to a monochromatic extinction at a specific reference wavelength (see Section **??** of the online documentation).

A0_UNCERTAINTY : Uncertainty on the mean extinction parameter (float, Magnitude[mag])

The uncertainty of the mean A_0 for selected extinction tracers in the HEALPix.

A0_MIN : Minimum value of the extinction parameter used for the HEALPix of interest (float, Magnitude[mag])

Minimum A_0 value of the selected extinction tracers used for the HEALPix of interest.

A0_MAX : Maximum value of the extinction parameter used for the HEALPix of interest (float, Magnitude[mag])

Maximum A_0 value of the selected extinction tracers used for the HEALPix of interest.

NUM_TRACERS_USED : Number of tracers used (int)

The number of extinction tracers used for calculating the extinction parameters for the HEALPix.

OPTIMUM_HPX_FLAG : Flag to indicate whether a given HEALPix level is the optimum (True) or not (False) (boolean)

The Total Galactic Extinction maps are provided at six different HEALPix levels (i.e. 6 different resolutions, levels 6 through 11). This flag indicates whether a given HEALPix level is the optimum level (True), or not (False).

STATUS : Exit status for TGE (short)

Indicates the exit status of TGE.

0: Nominal processing.

1: Insufficient number of tracers, after validation of input for each source (Step 0).

2: Insufficient number of tracers, after selection on astrophysical parameters and apparent mag-

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nitude (Step 1).

3: Insufficient number of tracers, after check on the distance (Step 2).

4: Insufficient number of tracers, after outlier rejection (Step 3).

99: Unknown error flag.

A0_MEAN : Mean extinction parameter (float, Magnitude[mag])

The mean A_0 of the selected extinction tracers as estimated by GSP-Phot from BP/RP spectra, apparent G magnitude and parallax. A_0 is the extinction parameter corresponding to a monochromatic extinction at a specific reference wavelength (see Section **??** of the online documentation).

A0_MEAN_ABSOLUTE_DEVIATION : Mean Absolute Deviation of the extinction parameter (float, Magnitude[mag])

The mean absolute deviation of A_0 from the median value (`a0_median`) of the selected extinction tracers.

A0_MODE : Mode of the distribution of the extinction parameter (float, Magnitude[mag])

The mode of A_0 distribution of the selected extinction tracers. This is only calculated if the number of tracers in the HEALPix is greater than 9, otherwise it will be a NaN.

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4.17 TOTAL_GALACTIC_EXTINCTION_MAP_OPT [TAP]

This table provides an optimum version of the total Galactic extinction map at a specific reference wavelength as a HEALPix map, appropriate for extragalactic sources.

The table contains the monochromatic extinction A_0 at a specific reference wavelength (see Section **??** of the online documentation). It is derived from the table `total_galactic_extinction_map` at HEALPix level 9. For this, the total Galactic extinction and related uncertainties were selected from the optimal HEALPix flagged in `total_galactic_extinction_map` with the boolean flag
tt optimum_hpx_flag. For further details see Section **??** of the online documentation.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`.

HEALPIX_ID : HEALPix identification (long)

The unique HEALPix index number for which the total Galactic extinction parameters are estimated. For further details on the HEALPix scheme used by Gaia, see Bastian & Portell (2020).

A0_MEDIAN : Median extinction parameter (float, Magnitude[mag])

The median A_0 of the selected extinction tracers as estimated by GSP-Phot from BP/RP spectra, apparent G magnitude and parallax. A_0 is the extinction parameter corresponding to a monochromatic extinction at a specific reference wavelength (see Section **??** of the online documentation). Here the A_0 value is taken from the optimum HEALPix in table `total_galactic_extinction_map`.

A0_UNCERTAINTY : Uncertainty on the mean extinction parameter (float, Magnitude[mag])

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The uncertainty of the mean A_0 for selected extinction tracers in the HEALPix. Here the value is taken from the optimum HEALPix in table `total_galactic_extinction_map`.

NUM_TRACERS_USED : Number of tracers used (int)

The number of extinction tracers used for calculating the extinction parameters for this HEALPix, taken from the optimum HEALPix in table `total_galactic_extinction_map`.

STATUS : Exit status for TGE (short)

Indicates the exit status of TGE.

0: Nominal processing.

1: Insufficient number of tracers, after validation of input for each source (Step 0).

2: IInsufficient number of tracers, after selection on astrophysical parameters and apparent magnitude (Step 1).

3: Insufficient number of tracers, after check on the distance (Step 2).

4: Insufficient number of tracers, after outlier rejection (Step 3).

99: Unknown error flag.

OPTIMUM_HPX_LEVEL : Number indicating which HEALPix level was chosen to populate this HEALPix (byte)

The HEALPix level (6-9) of the optimum HEALPix: The `total_galactic_extinction_map` HEALPix at highest resolution (highest HEALPix level) with at least 4 tracers in the direction of this HEALPix, as indicated by the `optimum_hpx_flag`, and used to populate the values in this HEALPix. As the optimum HEALPix may be at a lower HEALPix level than this map (level 9), the values in this level 9 HEALPix will be the same as neighbouring HEALPix pixels if `optimum_hpx_level` is less than 9.

A0_MEAN : Mean extinction parameter (float, Magnitude[mag])

The mean A_0 of the selected extinction tracers as estimated by GSP-Phot from BP/RP spectra, apparent G magnitude and parallax. A_0 is the extinction parameter corresponding to a monochro-

matic extinction at a specific reference wavelength (see Section **??** of the online documentation). The value here is taken from the optimum HEALPix in the table `total_galactic_extinction`.

A0_MEAN_ABSOLUTE_DEVIATION : Mean Absolute Deviation of the extinction parameter (float, Magnitude[mag])

The mean absolute deviation of A_0 from the median value (`a0_median`) of the selected extinction tracers. The value here is taken from the optimum HEALPix in the table `total_galactic_extinction`.

A0_MODE : Mode of the distribution of the extinction parameter (float, Magnitude[mag])

The mode of A_0 distribution of the selected extinction tracers. The value here is taken from the optimum HEALPix in the table `total_galactic_extinction`.

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4.18 TRAINING_SOURCES_DSC [TAP]

This table provides information about the sources used in training the Apsis/DSC (Discrete Source Classifier) module and can be used in conjunction with the apClass and apClassIr tables.

This table provides the Gaia sources used as the reference information in supervised classification by DSC to identify a source as a quasar, a galaxy or a star. Two modes are distinguished by a boolean flag indicating whether the source was used in training and testing the DSC models using Gaia data or using the Gaia data supplemented with infrared photometry. The class labels fall into one of the three DSC classes (quasar, galaxy of star). The quasar and galaxy classes result from a crossmatch at a 1 arcsecond radius to SDSS DR17 spectroscopically-confirmed quasars and galaxies. The star class corresponds to a random selection from Gaia data excluding the extragalactic sample. Further quality cuts are applied to result in the controlled training and testing datasets.

More information on DSC can be found at Section **??**.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution identifier, please visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (unique within a particular Data Release) (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

MODE_GAIA_ONLY : Boolean flag indicating if a source was used in the training and testing of DSC models using Gaia data only (boolean)

Boolean flag indicating if a source was used in the training and testing of DSC models using Gaia data only.

MODE_GAIA_IR : Boolean flag indicating if a source was used in the training and testing of DSC models using Gaia data and infrared photometry (boolean)

Boolean flag indicating if a source was used in the training and testing of DSC models using Gaia data and infrared photometry.

TRUE_CLASS : Class label (string)

The assigned true class label of this source used in the training of the DSC module.

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4.19 TRAINING_SOURCES_UCA_OA [TAP]

This table provides the combined list of reference sources used by the UCA and OA modules as part of the Apsis processing chain. In the UCA module, these sources are employed to train the classification maps (for stars, galaxies, and quasars). In contrast, the OA module uses them to generate the neuron templates.

For each module, the label assigned to the source is provided. In some cases, one of the modules may use a more specific label than the other. If a source is not used for training by a given module, its corresponding label, either `uca_label` or `oa_label`, is set to `null`. For further details, see Section **??** and Section **??** of the online documentation.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution identifier, please visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (unique within a particular Data Release) (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

UCA_LABEL : Source label used by the UCA module for training (string)

Source label used by the UCA module for training.

OA_LABEL : Source label used by the OA module to generate neauron templates (string)

Source label used by the OA module to generate neauron templates.

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4.20 UCA_NEURON_INFORMATION [TAP]

This table provides information about the neurons that compose the pre-trained Self-Organised Maps used by the UCA module from the Apsis processing chain for the classification of BP/RP spectra. The input dataset was built using normalised BP/RP spectra.

The prototype BP/RP spectrum for a particular neuron is available in table `uca_neuron_xp_spectra`. See Section **??** of the online documentation for further details.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOM_ID : Self-Organised Map identifier (long)

A unique single numerical identifier of the Self-Organized Map. The first three digits are reserved for identifying the type of objects in the training map:

2xx Stars map.

3xx Galaxies map.

4xx Quasars map.

See Section **??** for further details.

NEURON_ID : Neuron identifier (long)

A unique single numerical identifier of the neuron.

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NEURON_ROW_INDEX : Row index of the neuron in the Self-Organised Map lattice (short)

A row index which determines the position of the neuron within the Self-Organised Map lattice, together with `neuron_column_index`.

NEURON_COLUMN_INDEX : Column index of the neuron in the Self-Organised Map lattice (short)

A column index which determines the position of the neuron within the Self-Organised Map lattice, together with `neuron_row_index`.

HITS : Number of sources populating the neuron (int)

Number of training sources that have been assigned to the neuron according to the squared Euclidean distance between the sources XP spectra and the neuron XP prototype. See Section **??** and table `uca_neuron_xp_spectra` for further details.

CLASSLABEL : Astronomical class estimated for the neuron (string)

Astronomical class estimated for the neuron is derived from the most frequent astronomical class of the training sources assigned to that neuron. See table `training_sources_uca_oa` for more information on the training labels.

CENTROID_ID : Identifier of the Gaia source that minimizes the classification distance to the neuron (long)

Identifier of the training Gaia source (`gaia_source.source_id`) that minimizes the squared Euclidean distance between the source XP spectrum and the neuron XP prototype. See `centroid_distance`.

CENTROID_DISTANCE : Squared Euclidean distance between the centroid XP spectrum and the neuron XP prototype (float)

Squared Euclidean distance between the XP spectrum of the centroid source and the neuron XP prototype (`xp_spectrum_prototype_flux` in table `uca_neuron_xp_spectra`).

INTRA_NEURON_DISTANCE_MEAN : Mean value of the squared Euclidean distance between each of the XP sources in the neuron and the neuron prototype (float)

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Mean value of the squared Euclidean distance between each of the XP spectra of the sources that belong to the neuron and the XP prototype of the neuron (uca_neuron_xp_spectra.xp_spectrum_prototy

INTRA_NEURON_DISTANCE_STD_DEV : Standard deviation of the squared Euclidean distance between each of the XP sources in the neuron and the neuron prototype (float)

Standard deviation of the squared Euclidean distance between each of the XP spectra of the sources that belong to the neuron and the XP prototype of the neuron (uca_neuron_xp_spectra.xp_spectr
See Section **??** for further details.

INTRA_NEURON_DISTANCE_MIN : Minimum squared Euclidean distance between each of the XP sources in the neuron and the neuron prototype (float)

Minimum value of the Squared Euclidean distance between each of the XP spectra of the sources that belong to the neuron and the XP prototype of the neuron (uca_neuron_xp_spectra.xp_spectrum_pro

INTRA_NEURON_DISTANCE_MAX : Maximum squared Euclidean distance between each of the XP sources in the neuron and the neuron prototype (float)

Maximum value of the Squared Euclidean distance between each of the XP spectra of the sources that belong to the neuron and the XP prototype of the neuron (uca_neuron_xp_spectra.xp_spectrum_pro

INTER_NEURON_DISTANCE_MEAN : Mean value of the squared Euclidean distance between the neuron XP prototype and the XP prototypes of its immediate neighbours (float)

Mean value of the squared Euclidean distance between the neuron XP prototype (xp_spectrum_prototype_f
in table uca_neuron_xp_spectra) and the XP prototypes of its immediate neighbours.

INTER_NEURON_DISTANCE_STD_DEV : Standard deviation of the squared Euclidean distance between the neuron XP prototype and the XP prototypes of its immediate neighbours (float)

Standard deviation of the squared Euclidean distance between the neuron XP prototype (xp_spectrum_proto
in table uca_neuron_xp_spectra) and the XP prototypes of its immediate neighbours.

INTER_NEURON_DISTANCE_MIN : Minimum value of the squared Euclidean distance between the neuron XP prototype and the XP prototypes of its immediate neighbours (float)

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Minimum value of the squared Euclidean distance between the neuron XP prototype (xp_spectrum_prototyp
in table uca_neuron_xp_spectra) and the XP prototypes of its immediate neighbours.

INTER_NEURON_DISTANCE_MAX : Maximum value of the squared Euclidean distance between the neuron XP prototype and the XP prototypes of its immediate neighbours (float)

Maximum value of the squared Euclidean distance between the neuron XP prototype (xp_spectrum_prototy
in table uca_neuron_xp_spectra) and the XP prototypes of its immediate neighbours.

DISTANCE_PERCENTILE25 : 25th percentile value for the intra-neuron distance distribution (float)

25th percentile value for the intra-neuron distance distribution, which includes the 25% of the classified objects above this threshold. That is, the 25% of objects with the highest squared Euclidean distances to their neuron XP prototype (uca_neuron_xp_spectra.xp_spectrum_prototype_flux
lie above this value. See Section **??** for further details.

DISTANCE_PERCENTILE50 : 50th percentile value for the intra-neuron distance distribution (float)

50th percentile value for the intra-neuron distance distribution, which includes the 50% of the classified objects above this threshold. That is, the 50% of objects with the highest squared Euclidean distances to their neuron XP prototype (uca_neuron_xp_spectra.xp_spectrum_prototype_flux
lie above this value. See Section **??** for further details.

DISTANCE_PERCENTILE68 : 68th percentile value for the intra-neuron distance distribution

(float)

68th percentile value for the intra-neuron distance distribution, which includes the 68% of the classified objects above this threshold. That is, the 68% of objects with the highest squared Euclidean distances to their neuron XP prototype (uca_neuron_xp_spectra.xp_spectrum_prototype_flux
lie above this value. See Section **??** for further details.

DISTANCE_PERCENTILE75 : 75th percentile value for the intra-neuron distance distribution

(float)

75th percentile value for the intra-neuron distance distribution, which includes the 75% of the classified objects above this threshold. That is, the 75% of objects with the highest squared Euclidean distances to their neuron XP prototype (`uca_neuron_xp_spectra.xp_spectrum_prototype_flux`) lie above this value. See Section **??** for further details.

DISTANCE_PERCENTILE95 : 95th percentile value for the intra-neuron distance distribution

(float)

95th percentile value for the intra-neuron distance distribution, which includes the 95% of the classified objects above this threshold. That is, the 95% of objects with the highest squared Euclidean distances to their neuron XP prototype (`uca_neuron_xp_spectra.xp_spectrum_prototype_flux`) lie above this value. See Section **??** for further details.

DISTANCE_FWHM : Full Width at Half Maximum value for the intra-neuron distance distribution (float)

FWHM is given by the distance between two points on a neuron squared Euclidean distance distribution at which the function reaches half its maximum value. Those neurons showing low values for this quantity contain objects displaying similar spectra, and can be considered as neurons with a good clustering quality.

DISTANCE_SKEW : Skewness value for the intra-neuron distance distribution (float)

Skewness is the third standardized moment of a distribution. It measures the distortion or asymmetry on a neuron squared Euclidean distance distribution. If the skewness value is positive, then the distribution is shifted to the left, meaning that most values are concentrated on shorter distances. If the value is negative, the function is shifted to the right, with values located at larger distances. In both cases the function is said to be skewed. See Section **??** for further details.

DISTANCE_KURTOSIS : Kurtosis value for the intra-neuron distance distribution (float)

Kurtosis is the ratio of the fourth moment μ^4 to the square of the variance. It is a measure of the heaviness of the tails of the neuron squared Euclidean distance distribution. Those better quality neurons tend to exhibit kurtosis values close or slightly below that of a normal distribution (excess kurtosis near zero or negative), indicating a distribution of distances that is more concentrated with fewer extreme values. See Section **??** for further details.

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DISTANCE_I_Q_R : Inter-Quartile Range value for the intra-neuron distance distribution (float)

Difference between the 75th (`distance_percentile75`) and 25th (`distance_percentile25`) percentile values for the intra-neuron distance distribution. Low values indicate a focused distance distribution.

DISTANCE_FWHM_NORM : Normalized FWHM value for the intra-neuron distance distribution (float)

FWHM (`distance_fwhm`) is given by the distance between two points on a neuron squared Euclidean distance distribution at which the function reaches half its maximum value. We calculate it as a normalized value, by dividing the FWHM mean value obtained for the 10% of neurons with the lowest values in the SOM map by the FWHM in the neuron (`distance_fwhm`). Those neurons showing values near or above one for this quantity contain objects displaying similar spectra, and can be considered as neurons with a good clustering quality.

QUALITY_CATEGORY : Quality category assigned to the neuron, where 0 corresponds to the most homogeneous neurons and 6 to the most heterogeneous ones (short)

In order to provide a unified interpretation of the goodness of the clustering in a neuron, a general index has been calculated that takes into account the results obtained for the three intra-neuron distance distribution measurements: FWHM (`distance_fwhm`), skewness (`distance_skew`) and kurtosis (`distance_kurtosis`). To do so, the following percentile values of the three indices have been set: 10th, 32th, 50th, 75th, 90th and 95th. For each of the indices it is possible to obtain seven quality categories depending on the interquartile area where the measures are grouped, and then assign the neuron a global index named quality category (QC). Quality category can take any of the following values: 0 in case all three indices are in percentile level 95th, value 1 if they are in level 90th, and so on to finish with level 6 when all quality indices are outside percentile 10th. Quality category 0 corresponds to the most homogeneous neurons and 6 to the most heterogeneous ones. If no category could be assigned then it takes value -1.

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4.21 UCA_NEURON_XP_SPECTRA [TAP]

This is the table hosting the prototype BP/RP spectrum corresponding to each of the neurons that compose the pre-trained Self-Organised Maps used by the UCA module from the Apsis processing chain for the classification of BP/RP spectra. It is a flat table where each row represents the spectral information at a specific wavelength for a given neuron.

Other neuron attributes, such as statistics on various parameters, are available in table `uca_neuron_information`. See Section **??** of the online documentation for further details.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOM_ID : Self-Organised Map identifier (long)

A unique single numerical identifier of the Self-Organised Map. The first three digits are reserved for identifying the type of objects in the training map:

2xx Stars map.

3xx Galaxies map.

4xx Quasars map.

See Section **??** for further details.

NEURON_ID : Neuron identifier (long)

A unique single numerical identifier of the neuron.

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NEURON_ROW_INDEX : Row index of the neuron in the Self-Organised Map lattice (short)

A row index which determines the position of the neuron within the Self-Organised Map lattice, together with `neuron_column_index`.

NEURON_COLUMN_INDEX : Column index of the neuron in the Self-Organised Map lattice (short)

A column index which determines the position of the neuron within the Self-Organised Map lattice, together with `neuron_row_index`.

XP_SPECTRUM_PROTOTYPE_FLUX : Normalised flux for the preprocessed XP spectrum that best represents the neuron (float)

Flux at wavelength value specified by `xp_spectrum_prototype_wavelength` for the preprocessed XP spectrum (including normalisation) that best represents the sources that belong to the neuron. This is the prototype spectrum for the neuron.

XP_SPECTRUM_PROTOTYPE_WAVELENGTH : Wavelength associated with the XP spectrum flux values (float, Length & Distance[nm])

Wavelength value associated with the `xp_spectrum_prototype_flux` flux.

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5 Auxiliary

As indicated in the below subsection titles by the labels [TAP] and [DataLink], these products are served from the ESA Gaia Archive using two different protocols, the Table Access Protocol (TAP) for tabular products and the DataLink protocol for other products such as spectra, light curves, MCMC samples, etc. For more details please see Section **??**.

5.1 BAM_OBSERVATION [TAP]

This table provides the original data from the Basic-Angle Monitor (BAM), i.e. from the on-board metrology interferometer designed to continually measure the variations of the effective angle between the viewing directions of the two Gaia fields of view. Specifically, the table provides the BAM interference patterns, i.e. the BAM CCD images, along with some datation parameters.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

BAM_ID : Unique identifier of the BAM Measurement (long)

The `bam_id` is constructed in the same way as `transit_id` in the scientific data records, that is, a bit-wise encoding of the following fields from the original BAM observation (SP4) telemetry packets:

- OBMT50 (On-Board Mission Time, 42 bits), which is `acq_time`, correcting the Clock Resets, removing the 10 MSB (they are zero anyway), and the 12 LSB (leading to a resolution of 204.8μs);
- FOV (2 bits), determined from the across–scan coordinate according to the uploaded VPU parameters (and as available in the on–ground spacecraft Configuration Data Base, a.k.a. CDB). We will only use '0' and '1', despite the availability of 2 bits;

- CCD row (3 bits), directly copied from the original telemetry packet;
- across–scan coordinate (12 bits), directly copied from the original telemetry packet.

This yields a 'long' `bam_id` of 59 bits used out of 64.

Note that, in case of multi-packet observations (i.e. joining several SP4 packets into a single `bam_observation`), the `bam_id` corresponds to the earliest packet, that is the packet with the smaller time tag. Finally, note that all these fields are also available separately in the `bam_observation` table. Those are useful for querying, whereas the `bam_id` is used for indexing and identification purposes.

ACQ_TIME : Acquisition time (OBMT) (long, Time[OBMT])

Acquisition time, in the form of OBMT (on-board mission timeline), the strictly monotonic technical time-tagging coordinate used for all Gaia data.

While for all astronomical Gaia data, this OBMT coordinate is transformed to the standard physical time coordinate TCB, this is neither needed nor useful for purely instrumental data, like the `bam_observation` table.

SW_CMD_ACK : Commands acknowledged (byte)

This field this field has no meaning (and is always zero). It was meant to record specific problems in the execution of BAM observations, but no such problems ever occurred. In DR5 it will no longer be present.

NUM_PACKETS : Number of SP4 packets composing this Observation (byte)

The BAM device can be commanded to acquire very large windows which do not fit into one telemetry packet. In such cases the total BAM interference pattern was joined into one single `bam_observation`.

The `num_packets` parameter is the number of joined packets for this `bam_observation`. It is a purely technical data item which is kept in the data release table for possible calibration purposes.

CCD_ROW : Row of BAM CCD (byte)

CCD row where this BAM measurement has been taken. There are two CCDs on the focal plane of Gaia which can record BAM patterns. They are situated in CCD rows 1 and 2.

AC : AC position of the window (short, Length & Distance[pixel])

Across-scan pixel coordinate of the BAM window on the CCD used. It corresponds to the first pixel of the first sample of the window (which is different from the convention used for astronomical CCD windows from Gaia).

Note that the value of `ac` determines the FOV associated with this BAM measurement (combined adequately with the CDB).

FOV : FoV associated with this BA measurement (byte)

Field of view (0 or 1) corresponding to this BAM observation (i.e. to this BAM CCD window).

SAMPLES : BAM pattern (int[40000] array, Flux[ADU])

Pixel sample values from the BAM CCDs. Note that the samples are coded as 4-byte integers (32 bits), despite the original raw samples being represented in unsigned short (16 bits).

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5.2 COMMANDED_SCAN_LAW [TAP]

This table provides a representation of the Gaia scanning law over the 66 month time period covered by the Gaia Data Release 4 including the Ecliptic Pole Scanning at the beginning of the mission. Note that this is the commanded attitude of the spacecraft, the actual attitude could deviate from it by up to about 30 arcsec.

The scanning law has been sampled at a 10 second interval, in which the satellite rotates about 10 arcminutes (the target spin rate is actually 59.9641857803 arcsec/sec). Note that this is several times shorter than a typical field-of-view transit and the scan position angle will be practically constant during this interval. Note also the table contains records every 10 seconds irrespective of any interruptions that occurred in the course of the real mission, of which the main ones are listed in Chapter **??**, Section **??**.

Notes:

- The times in columns `jd_time`, `bjd_fov1` and `bjd_fov2` are in Julian days in TCB with time origin 2010-01-01T00:00 (JD 2 455 197.5), following the time coordinate convention used in the Gaia archive. TCB stands for Barycentric Coordinate Time and is the time standard used in Gaia processing, equivalent to the proper time experienced by a clock at rest in a coordinate frame co-moving with the barycentre of the Solar system but outside its gravity well, therefore not influenced by the gravitational time dilation caused by the Sun and the rest of the Solar system;
- `jd_time` is the reference time for the spacecraft attitude, while `bjd_fov1` and `bjd_fov2` give the times with the (relativistic) corrections applied for the light-travel time to the Solar system barycentre, corresponding to an infinitely distant source at the RA, DEC at the centres of FOV 1 and 2, respectively;
- FOV1 and FOV2 correspond to the preceding (PFOV) and following (FFOV) fields-of-view, respectively;
- The centres of the fields of view are separated by the basic angle of 106.5 deg, see Fig. 2 of Lindegren et al. (2012). Their origin in the focal plane is illustrated in Fig. 3 of the same paper: both originate in the astrometric field (AF) 7, with FOV1 in row 3 and FOV2 in row 5;
- The scan angle, theta, is the position angle of the direction in which the FOV is moving (also called 'along-scan' direction), and is defined in the usual astronomical sense: theta = 0 when the FoV is

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moving towards local North, and theta = 90 degrees towards local East;

- Time values are given at double precision but this should not be interpreted as representing the accuracy of the data.

Columns description:

JD_TIME : Time [Julian Date in TCB at Gaia − 2 455 197.5] (double, Time[Gaia-Centric JD in TCB − 2 455 197.5 (day)])

Time at Gaia in units of JD (in TCB) in days −2 455 197.5. The time at which the scan angles and FoV angles are evaluated in TCB (Temps Coordonnée Barycentrique) with an offset of 2 455 197.5 days applied (corresponding to a reference time T_0 at 2010-01-01T00:00:00) to have a conveniently small numerical value.

BJD_FOV1 : Time [Julian Date in TCB at barycentre for FOV1 − 2 455 197.5] (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Observation time in units of Barycentric JD (in TCB) in days −2 455 197.5, computed as follows. First the observation time is converted from On-board Mission Time (OBMT) into Julian date in TCB (Temps Coordonnée Barycentrique). Next a correction is applied for the light-travel time to the Solar system barycentre corresponding to an infinitely distant source at (`ra_fov1`, `dec_fov1`), resulting in Barycentric Julian Date (BJD). Finally, an offset of 2 455 197.5 days is applied (corresponding to a reference time T_0 at 2010-01-01T00:00:00) to have a conveniently small numerical value.

BJD_FOV2 : Time [Julian Date in TCB at barycentre for FOV2 − 2 455 197.5] (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Observation time in units of Barycentric JD (in TCB) in days −2 455 197.5, computed as follows. First the observation time is converted from On-board Mission Time (OBMT) into Julian date in TCB (Temps Coordonnée Barycentrique). Next a correction is applied for the light-travel time to the Solar system barycentre corresponding to an infinitely distant source at (`ra_fov2`, `dec_fov2`), resulting in Barycentric Julian Date (BJD). Finally, an offset of 2 455 197.5 days is applied (corresponding to a reference time T_0 at 2010-01-01T00:00:00) to have a conveniently small numerical value.

OBMT_TIME : Time at Gaia (OBMT) (long, Time[OBMT])

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Observation time at Gaia converted to OBMT using the HATT (High Accuracy Time Transformation; see Section **??** and Section **??**).

RA_FOV1 : Right Ascension of FOV1 centre (float, Angle[deg])

Barycentric Right Ascension α of Field of View 1 (preceding) in ICRS at given time.

DEC_FOV1 : Declination of FOV1 centre (float, Angle[deg])

Barycentric Declination δ of Field of View 1 (preceding) in ICRS at given time.

HEAL_PIX_FOV1 : FOV1 HEALPix level 12 (int)

Level 12 nested scheme HEALPix containing the Field of View 1 (preceding) right ascension and declination.

This field can be used in conjunction with `source_id`, whose most significant bits contain HEALPix information.

SCAN_ANGLE_FOV1 : Scan position angle of FOV1 (float, Angle[deg])

Field of View 1 (preceding) scan angle.

PARALLAX_FACTOR_AL_FOV1 : FOV1 parallax factor AL (float)

Parallax factor f_w in the AL direction w of Field of View 1 (preceding). For further details please see 'Mathematical definition of Local Plane Coordinates' in Section **??** of the online release documentation.

PARALLAX_FACTOR_AC_FOV1 : FOV1 parallax factor AC (float)

Parallax factor f_z in the AC direction z of Field of View 1 (preceding). For further details please see 'Mathematical definition of Local Plane Coordinates' in Section **??** of the online release documentation.

RA_FOV2 : Right ascension of FOV2 centre (float, Angle[deg])

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Barycentric Right Ascension α of Field of View 2 (following) in ICRS at given time.

DEC_FOV2 : Declination of FOV2 centre (float, Angle[deg])

Barycentric Declination δ of Field of View 2 (preceding) in ICRS at given time.

HEAL_PIX_FOV2 : FOV2 HEALPix level 12 (int)

Level 12 nested scheme HEALPix containing the Field of View 2 (following) right ascension and declination.

This field can be used in conjunction with `source_id`, whose most significant bits contain HEALPix information.

SCAN_ANGLE_FOV2 : Scan position angle of FOV2 (float, Angle[deg])

Field of View 2 (following) scan angle.

PARALLAX_FACTOR_AL_FOV2 : FOV2 parallax factor AL (float)

Parallax factor f_w in the AL direction w of Field of View 2 (following).

PARALLAX_FACTOR_AC_FOV2 : FOV2 parallax factor AC (float)

Parallax factor f_z in the AC direction z of Field of View 2 (following).

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

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5.3 GAIA_GAP [TAP]

This table lists all time intervals excluded from the astrometric, photometric and/or spectroscopic data processing of the Gaia observations. Gaps are time periods where such observations have been ignored in the various processing pipelines. Each of those pipelines might have separate gaps.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

START_TIME : Start OBMT of the interval (long, Time[OBMT])

Start time of the interval. These times refer either to the actual gap or to the AF1 acquisition time of the affected transits, depending on which pipeline has recorded the gap.

END_TIME : End OBMT of the interval (long, Time[OBMT])

End time of the interval; the `end_time` is included in the range. These times refer either to the actual gap or to the AF1 acquisition time of the affected transits, depending on which pipeline has recorded the gap.

CCD_ROW : CCD Row in which the FOV observation occurred (byte)

CCD Row in which the FOV transit occurred. When all rows are affected by the gap, the value 0 is used. Otherwise, this field can take values between 1 to 7 for the applicable rows.

FOV : Field of view affected by the gap (byte)

Only data from this Field of View is affected by this gap. If this is null, then the Field of View

was not considered when selecting the data to be excluded.

STRIP : Strip affected by the gap (byte)

Only data observed in this strip is affected by this gap. If this is null, then the strip was not considered when selecting the data to be excluded.

The coding convention for this field is: '0' is used for the Sky Mapper (SM), '1' to '9' is used for Astrometric Field (AF) strips, '10' and '11' are used for BP and RP respectively.

WINDOW_CLASS : Window class affected by the gap (byte)

Only data observed with this Window Class is affected by this gap. If this is null, then Window Class was not considered when selecting the data to be excluded.

This field can take the following values:

- 0 (class 0a), G<11: full-resolution (2D) windows for AF and BP/RP, and high-resolution (2x2 pixels) window for SM.
- 1 (class 0b) 11<G<13: full-resolution (2D) windows for AF, high-resolution (2x2 pixels) window for SM, and 1D windows for BP/RP.
- 2 (class 1), 13<G<16: large 1D windows for AF (18 or 12 samples), low-resolution (2x2 samples, or 4x4 pixels) window for SM, and 1D windows for BP/RP.
- 3 (class 2), G>16: small 1D windows for AF (12 or 6 samples), low-resolution (2x2 samples, or 4x4 pixels) window for SM, and 1D windows for BP/RP.

GATE : Gate configuration affected by the gap (byte)

Only data observed with this gate is affected by this gap. If this is null, then gate was not considered when selecting the data to be excluded.

The coding convention used for this field is: '1' means 2 lines in length, '12' means 2900 lines in length. '0' is used for the 'No Gate' case.

REASON : The reason why this time interval was excluded (string)

This field gives the reasons why this time interval was excluded from the processing module indicated in `processing_module`.

In the case of gaps used by AGIS (`processing_module` = ASTRO), the details about the reasons applying to these flags are given in **?**.

In the case of gaps used by the photometric processing ((`processing_module` = PHOTO), there are only two possible reasons tabulated:

- AGIS Gap reasons (485 cases): these cover the gaps present in a preliminary version of the AGIS solution, which is the one that has been by the photometric pipeline.
- AstroCalibration problem (145 cases): these correspond to cases applying to the EPSL periods where the geometric calibration was considered unfit for the photometric processing.

In the case of gaps used by the RVS processing module (`processing_module` = SPECTRO), there are 3966 unique reasons intended for internal DPAC records, which can be grouped into four different categories:

- AGIS Gap reasons (1,940 cases): These cover the gaps present in a preliminary version of the AGIS solution, which is the one that has been by the RVS pipeline, together with an additional AGIS gap (not present in that preliminary version) with the reason labeled 'Removed for C6ACT-485'. The AGIS gaps impact all four CCD rows covered by RVS, i.e. rows 4 to 7. Note that this specific gap list does not fully overlap with the one finally used by AGIS for the DR4 astrometric solution, which is the one labelled as `processing_module` = ASTRO in this table.
- Missing Data reasons (25,062 cases): These gaps are due to the absence of onboard data indispensable for the RVS pipeline. There are 21,282 gaps caused by missing ObjectLog (reasons: DBDOBS-42.ol.gaps and DBOPTSK-28.ol.gaps), and 2,111 gaps due to missing ACShifts (reason: DBOPTSK-28.as.gaps).
- Planet Transit reasons (2,092 cases): These gaps are due to the transits of Jupiter, Saturn, Mars, and Venus across the RVS focal plane. The planet transit gaps affect all four CCD rows covered by RVS, i.e. rows 4 to 7.
- VPU Test/Reset reasons (10 cases): These gaps are due to technical tests dedicated to verifying the performance of the Video Processing Unit 2.8, optimized to mitigate the impact of stray light.

PROCESSING_MODULE : The pipeline to which the gap is applicable (string)

This field indicates the processing module that applied the reported gap interval. The following codes are used:

- ASTRO indicates that the data have been ignored in the AGIS primary solution run for the astrometry processing.
- PHOTO indicates that the data have been ignored in the photometry processsing.
- SPECTRO indicates that the data have been ignored in the RVS spectroscopy processsing.

Draft

5.4 GBOT_OBSERVATION [TAP]

Ground-based position measurements of the Gaia satellite taken by the GBOT project for the purpose of Gaia orbit reconstruction.

For DR4, the table contains the mission time interval which entered the DR4 science data, i.e. from the beginning of the scientific EPSL interval to the end of mission segment 6, i.e. from 2014-07-25 till 2020-01-20.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

OBS_TIME_ISO : middle of the exposure time in ISO format (string, Time[yyyy-mm-ddThh:mm:ss.sssssssss (ISO time string)])

UTC of the centre of the exposure time in the ISO format (yyyy-mm-ddThh:mm:ss.ss) at the observatory. Precision specified as 0.1 seconds, resolution 0.01 seconds.

OBS_TIME_JD : middle of the exposure time as JD (double, Time[Julian Date (day)])

UTC of the centre of the exposure time as Julian Date, JD(UTC) at the observatory. Precision specified as 0.1 seconds, resolution 0.01 seconds.

EXPOSURE_DURATION : duration of the exposure (float, Time[s])

Exposure time in seconds.

PASSBAND : spectral passband filter used (string)

Designation of the filter used; the various values are defined in **?**.

Draft

OBSERVATORY_CODE : IAU observatory code (string)

The IAU observatory of the telescope used. Some multi-telescope observatories have only one code. Hence telescope coordinates must not be taken from the IAU list, but from the corresponding table in **?**.

TELESCOPE : Name of telescope (string)

This column gives the name of the telescope. This is required since the IAU observatory code may only specify the observatory in some cases, e.g. ESO Paranal, meaning that all telescopes of that facility have the same IAU code, e.g. the VST and the VLT unit telescopes.

INSTRUMENT : camera and detector (string)

Identification of the camera and detector used.

ALPHA : measured right ascension (double, Angle[deg])

Right ascension of the measured ICRS astrometric position on the Gaia DR2 optical reference frame.

ALPHA_ERROR : formal standard error of alpha (float, Angle[arcsec])

Formal standard error of `alpha` resulting from image centroiding and astrometric plate reduction.

DELTA : measured declination (double, Angle[deg])

Declination of the measured ICRS astrometric position on the Gaia DR2 optical reference frame.

DELTA_ERROR : formal standard error of delta (float, Angle[arcsec])

Formal standard error of `delta` resulting from image centroiding and astrometric plate reduction.

ERROR_CORRELATION : formal correlation of the errors in alpha and delta (float)

Formal correlation between the errors in `alpha` and `delta`; if unknown a value of 0 is supplied.

Draft

MAGNITUDE : observed magnitude in the given passband (float, Magnitude[mag])

Observed Vega-type magnitude in the given passband; for details see **?**.

DELIVERY_FLAG : true if delivered to ESOC for orbit reconstruction (boolean)

True if submitted to ESOC for Gaia orbit reconstruction. Since for Gaia DR4 only data which were delivered to ESOC are published, this parameter is always true. For Gaia DR5, when all GBOT data is published it can be true or false.

Draft

5.5 ONBOARD_DETECTION_AF_XP [TAP]

This table is derived from the Auxiliary Science Data 7 (ASD7) telemetry packets, aka object logs, for the AF and XP data. It contains all confirmed on-board detections, regardless of whether the data got deleted on board, got lost in transmission to the ground, or otherwise.

The table gives a (very nearly) complete inventory of the confirmed on-board detections (more precisely: the detections that led to an SP1 telemetry packet). It thus includes detections where the SP1 were never downlinked (because they were deliberately deleted or got lost otherwise on board). Also included are detections classified on-ground as spurious (which were omitted from all further ground-based science processing). This table is the only record of these detections, as presently the plan is not to publish the corresponding `epoch_image` and `epoch_astrometry`.

The table also includes the Virtual Objects, i.e. the artificial 'detections' without actual Sky Mapper detections corresponding to them. More information on these is given in the field description of `virtual_object_flag`.

Examples of possible use cases for this table are:

- Sky statistics and Gaia's selection function.
- Studies of diffuse objects, e.g. Cat's Eye, and knotty planetary nebulae in general. Also, knots of Herbig-Haro objects and the like.
- Localisation and identification of astrophysical transients etc.
- Very bright stars and planets. Very probably, there is some useful information in the swarm of spurious Gaia detections around these. And, at least in principle, a number of real astrophysical transients may have been detected within the diffraction spikes and glare of some bright objects, and then been discarded as spurious from the ground processing.
- For any AF/XP obervation, this table contains the information on what is in the other field of view (fov) at the same time and at the same place on the focal plane. This may be very useful for the investigation of outliers in epoch astrometry and/or epoch photometry, and of unexpected features in the epoch images. For this purpose, users might simply check the `onboard_detection_af_xp` information on fov, row and reference aquisition pixel (all embedded in `transit_id`) for transit times close to the `transit_id` under consideration in these other tables.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

TRANSIT_ID : Transit Identifier (long)

The `transit_id` is a unique identifier assigned to each detected (and confirmed) source as it transits the Gaia focal plane. Each time a given source is detected as Gaia scans and re-scans the sky a new `transit_id` will be created to badge that apparition. Hence the along–scan time and the across–scan position along with the telescope in which the source was detected are used to form a unique integer with which to label the transit.

The several features of a detection that are encoded in `transit_id` can be easily retrieved using bit masks (&) and shifts (>>) as follows:

- On-Board Mission Time line [ns]
  = 204800 * ((`transit_id` >> 17) & (0x000003FFFFFFFFFF))
- Field-of-view = 1 + (`transit_id` >> 15) & 0x03 [1 for 'preceding' and 2 for 'following' fields-of-view respectively]. The values 3 and 4 refer to Virtual Object transits in the preceding and following fields of view, respectively. For more explanations see the description of the `virtual_object_flag` field.
- CCD row = (`transit_id` >> 12) & 0x07 [dimensionless, in the range 1 to 7]
- Across-scan 'reference acquisition pixel' in strip AF1 = (`transit_id`) & 0x0FFF [pixels] (this is the across-scan centre of the AF1 window and is odd if immediately below the mid-point of the window and even if immediately above)

where the bit mask prefix '0x' denotes hexadecimal.

For further details see Portell et al. (2020). For convenience a decoder for `transit_id` is available on-line at

`https://gaia.esac.esa.int/decoder/transitidDecoder.jsp`

Draft

RES_REF_TIME : Residual (high-resolution) reference time (short, Time[OBMT50])

The `transit_id` is defined in a way that it contains the OBMT reference acquisition time (from the AF1 window belonging to the given source transit) without its 12 least-significant bits. That is, the time contained in `transit_id` does not have the orginal time resolution of 50 ns of the onboard clock.

This data field contains these 12 bits removed from the OBMT before the composition of `transit_id`. The coding of `res_ref_time` is such that 1 least significant bit corresponds to 50ns and the maximum 12-bit value is 4095 (equal to 204.75μs). Hence `res_ref_time` ranges from 0 to 204.75μs, in steps of 50ns (the same resolution as OBT).

Adding the value of `res_ref_time` to the extracted time from `transit_id` (with the adequate units) will lead to the full-resolution OBMT of the AF1 acquisition window.

G_CLASS : Sampling class (byte)

AF/XP sampling class: the resolution of the acquired SM, AF and XP windows depends on the on-board determination of the G magnitude (see field `g_mag`) of the source under observation. However there are exceptions: calibration-faint stars (CFS), for which 2D windows are always collected regardless of magnitude; and virtual objects (VOs), for which there is no associated G magnitude. VOs may also be assigned 2D windows. The size and resolution of the ordinary windows as transmitted to the ground are:

- 0 means ‘gclass 0, window class 0a’: `g_mag` < 7; SM 40×6 samples of 2×2 pixels, AF1 18×6 samples of 1×2 pixels, AF2-9 18×12 samples of 1×1 pixels, XP 60×12 samples of 1×1 pixels
- 1 means ‘gclass 1, window class 0b’: `g_mag` < 13; SM 40×6 samples of 2×2 pixels, AF1 18×6 samples of 1×2 pixels, AF2-9 18×12 samples of 1×1 pixels, XP 60×1 samples of 1×12 pixels
- 2 means ‘gclass 2, window class 1’: 13 < `g_mag` < 16; SM 20×3 samples of 4×4 pixels, AF1 12×1 samples of 1×12 pixels, AF2-9 18×1 samples of 1×12 pixels, XP 60×1 samples of 1×12 pixels
- 3 means ‘gclass 3, window class 2’: 16 < `g_mag` < 21; SM 20×3 samples of 4×4 pixels, AF1 6×1 samples of 1×12 pixels, AF2-9 12×1 samples of 1×12 pixels, XP 60×1 samples of 1×12 pixels

XP_OFFSET : XP Offset (byte)

Draft

Contrary to the Astrometric field (AF), the XP windows cannot start at every TDI readout line: they need to start at times that are multiples of 4 TDI periods. So the XP windows cannot always be centred on the propagated (predicted) AL position of the given sky source (more precisely: of its XP spectrum) in the XP readout stream. The `xp_offset` field gives, in practice, the offset between the propagated AL position of the source and the TDI line of the middle of the window. It is used in the pipeline to determine the time, for each XP spectral window sample, at which the sample crossed the fiducial line of the relevant CCD: the time of the first sample of the window is the AF1 acquisition time, plus a fixed (but AC-dependent) value stored in the onboard Calibration Database (CDB), minus `xp_offset` multiplied by the TDI period. For further details concerning the fundamental time delay integration operating mode of the focal plane CCDs see Chapter **??** of the online documentation.

G_PRIORITY : Priority of the observation used by the VPU (short)

Priority assigned to this observation by the on-board software. Higher values mean higher priority.

Note that this is the priority used for the on-board window acquisition. It is not the downlink priority, although the latter may depend on this one. It is determined from the on-board G magnitude determination (see field `g_mag`) for normal windows, and set to special values if the window belongs to a calibration faint star (CFS) or to a virtual object (VO). CFS are special windows collected in high resolution despite the detected source being faint (i.e. although their expected G magnitude would give them low-resolution AF and XP windows) – see field `g_class`.

SP_VARIANT : Variant of the Star Packet (byte)

Variant of SP1 telemetry packet corresponding to this detection.

There are two different codings: Values zero to 31 correspond to actually received detection records, according to the bitmap given below (a value of −1 corresponds to reconstructed detection records which got lost in transmission; the value −1 overrides the bitmap):

- Bit 4: standard RP window allocated;
- Bit 3: standard BP window allocated;
- Bit 2: additional SMO (suspected moving object) RP window allocated;
- Bit 1: additional SMO (suspected moving object) BP window allocated;
- Bit 0 (least significant bit): direction of motion of the SMO. ‘0’ means ‘SMO moving downwards’, ‘1’ means ‘SMO moving upwards’. This bit thus states on which side of the standard windows (in the across-scan direction) the additional XP window(s) are placed.

G_MAG : Onboard G magnitude estimate (short)

On-board estimation of the G magnitude of the source under observation, in the original 10-bits on-board coding. This value is estimated from the SM window for this source transit. It is used on board to set the `g_class` and `g_priority` of the corresponding windows. It is coded as $64 \times (21 - G)$. This coding covers the range 5.015625 to 21 mag and offers a resolution of 0.015625 mag. Accordingly, the on-board estimated G value can be calculated in this way: $G = 21 - \texttt{g_mag}/64$.

RA_LOW_PRECISION : Rough right ascension of window centre (float, Angle[deg])

A rough estimate of the right ascension of the source giving rise to this detection. This field, and its declination companion `dec_low_precision`, does not give a precise position of an observed celestial object, but a rough projection of the on-board detected object (precisely: the AF1 reference pixel) onto the sky. The provided precision is sufficient for the main use cases of this table: sky statistics, location and identification of astrophysical transients, etc. Better positions could only be given for Gaia catalogue objects, i.e. not for detection records that are not matched to one of those, which are about 20 percent of all.

The attainable precision is limited by the resolution of the positional information given in `transit_id`, which is one across-scan sample size, i.e. about 0.18 arcsec. This low precision allows the use of 4-byte floats for the given coordinates `ra_low_precision` and `dec_low_precision`. The worst case for the numerical resolution due to this choice is for right ascensions above 256 degrees, where the exponent is 2^9 (=512), and the resolution therefore is 512 degrees / 2^{24} = 110 mas.

It is to be noted that the given coordinates are not the actual viewing direction (technically called proper direction) from which Gaia received the light at the moment of observation. But they are coordinate directions, i.e. the proper directions corrected for aberration and solar-system light bending. This makes it possible to directly match these coordinates with ordinary source catalogue positions and e.g. transient positions measured relative to some source catalogue.

DEC_LOW_PRECISION : Rough declination of window centre (float, Angle[deg])

Rough declination of the on-board detected object. For more information see the description of `ra_low_precision`.

VIRTUAL_OBJECT_FLAG : Flag indicating virtual objects (byte)

Virtual Objects are artificial Gaia CCD windows, i.e. seemingly normal Gaia observations, but they do not correspond to real detections of flux peaks in the Sky Mapper (SM) CCDs. Instead they are wilfully placed into the Gaia data stream, i.e. wilfully placed onto the sky. They are

created for various purposes, and they come in two basic flavours.

The main part of the Virtual Objects are collected for calibration purposes. They are distributed in a more or less random (but deterministic, repeating every two hours) fashion in time, field of view, CCD row and position on the CCDs. They are characterised by artificial values of the field-of-view item in the `transit_id` parameter: 3 for the preceding and 4 for the following field of view, i.e. for SM windows from SM1 and SM2, respectively. They usually contain empty sky. They can for instance be used for the determination of the sky background, of electronic effects in the CCDs and proximity electronics, etc.

The second flavour are the synchronized Virtual Objects. They are wilfully placed on the expected CCD positions of a few dozen of the very brightest stars in the whole sky. Due to extreme overexposure these cannot be detected autonomously on board. They are used in an attempt to get scientifically useful astrometric data on them from Gaia despite the heavy overexposure. The synchronized Virtual Objects are identified by the combination of the `virtual_object_flag` with the normal field-of-view indices in the `transit_id` parameter: 1 for the preceding and 2 for the following field of view. It should be noted that synchronized Virtual Objects have not been produced for significant parts of Gaia's mission duration. For DR4, the total of three time intervals covered is only 1.1 years, scattered over an envelope of about 2.2 years (Dec. 2017 to Jan. 2020).

The scientific analysis of the synchronized Virtual Objects, and the results for DR4, are described in **??** and in **?**. The corresponding data are given in Tables `bright_source_astrometry` and `bright_source_epoch_astrometry`.

(start = 5998.0 rev, 2017-12-06)

Draft

5.6 ONBOARD_DETECTION_RVS [TAP]

This table is derived from the Auxiliary Science Data 7 (ASD7) telemetry packets, a.k.a. object logs, for the RVS data. It contains all RVS observations, regardless of whether the data got deleted on board, got lost in transmission to the ground, or otherwise. This table is the only record of the lost and/or unused observations.

The table also includes the Virtual Objects, i.e. the artificial 'detections' without actual Sky Mapper detections corresponding to them. More information on these is given in the field description of `virtual_object_flag`.

Examples of possible use cases for this table are given in the table description of `onboard_detection_af_xp`.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

TRANSIT_ID : Transit Identifier (long)

The `transit_id` is a unique identifier assigned to each detected (and confirmed) source as it transits the Gaia focal plane. Each time a given source is detected as Gaia scans and re-scans the sky a new `transit_id` will be created to badge that apparition. Hence the along–scan time and the across–scan position along with the telescope in which the source was detected are used to form a unique integer with which to label the transit.

The several features of a detection that are encoded in `transit_id` can be easily retrieved using bit masks (&) and shifts (>>) as follows:

- On-Board Mission Time line [ns]
  = 204800 * ((`transit_id` >> 17) & (0x000003FFFFFFFFF))
- Field-of-view = 1 + (`transit_id` >> 15) & 0x03 [1 for 'preceding' and 2 for 'following' fields-of-view respectively]. The values 3 and 4 refer to Virtual Object tran-

sits in the preceding and following fields of view, respectively. For more explanations see the description of the `virtual_object_flag` field.

- CCD row = (`transit_id` >> 12) & 0x07 [dimensionless, in the range 1 to 7]
- Across-scan 'reference acquisition pixel' in strip AF1 = (`transit_id`) & 0x0FFF [pixels] (this is the across-scan centre of the AF1 window and is odd if immediately below the mid-point of the window and even if immediately above)

where the bit mask prefix '0x' denotes hexadecimal.

For further details see Portell et al. (2020). For convenience a decoder for `transit_id` is available on-line at

`https://gaia.esac.esa.int/decoder/transitidDecoder.jsp`

RES_REF_TIME : Residual (high-resolution) reference time (short, Time[OBMT50])

The `transit_id` is defined in a way that it contains the OBMT reference acquisition time (from the AF1 window belonging to the given source transit) without its 12 least-significant bits. That is, the time contained in `transit_id` does not have the original time resolution of 50 ns of the onboard clock.

This data field contains these 12 bits removed from the OBMT before the composition of `transit_id`. The coding of `res_ref_time` is such that 1 least significant bit corresponds to 50ns and the maximum 12-bit value is 4095 (equal to 204.75μs). Hence `res_ref_time` ranges from 0 to 204.75μs, in steps of 50ns (the same resolution as OBT).

Adding the value of `res_ref_time` to the extracted time from `transit_id` (with the adequate units) will lead to the full-resolution OBMT of the AF1 acquisition window.

G_RVS_CLASS : Sampling class (byte)

This field indicates the RVS sampling class. The size and resolution of the acquired RVS windows depend on the on-board estimate of the RVS magnitude of the source (see field `rvs_mag`). According to this estimated magnitude, the RVS windows transmitted to the ground are configured as follows:

- 0 (`g_rvs_class` 0): for sources with $G_{\rm RVS} < 7$; windows consist of 1260×10 samples of 1×1 pixels.

- 1 (`g_rvs_class` 1): for $7 \leq G_{\rm RVS} < 10$; windows consist of 1260×1 samples of 1×10 pixels.
- 2 (`g_rvs_class` 2): for $10 \leq G_{\rm RVS} < 17$; windows consist of 420×1 samples of 3×10 pixels.

RVS_OFFSET : RVS offset (byte)

Contrary to the Astrometric field (AF), the RVS windows cannot start at every TDI readout line: they need to start at times that are multiples of 108 TDI periods (105 TDI periods for data up to April 2015). So the RVS windows cannot always be centred on the propagated (predicted) AL position of the given sky source (more precisely: of its RVS spectrum) in the RVS readout stream. The `rvs_offset` field gives, in practice, the offset between the propagated AL position of the source and the TDI line of the middle of the window. It is used in the pipeline to determine the time, for each RVS spectral window sample, at which the sample crossed the fiducial line of the relevant CCD: the time of the first sample of the window is the AF1 acquisition time, plus a fixed value (stored in the onboard Calibration Database), minus `rvs_offset` times the TDI period. For further details concerning the fundamental time delay integration operating mode of the focal plane CCDs see Chapter **??** of the on–line documentation.

RVS_PRIORITY : Priority of the observation used by the VPU (short)

Priority assigned to this observation by the on-board software. Note that this is the priority used for the on-board window acquisition. It is not the downlink priority (although the latter may depend on this). A value of 255 means the highest priority, whereas 0 means the lowest priority. It is determined from the on-board RVS magnitude expectation (see field `rvs_mag`) for normal windows, and set to special values if the window belongs to a calibration faint star (CFS) or to a virtual object. CFS are special windows collected in high resolution despite the detected source being faint, i.e. although their expected RVS magnitude would give them a low-resolution RVS window – see field `g_rvs_class`.

RVS_MAG : Special onboard RVS magnitude estimate (short)

On-board expectation of the RVS magnitude of the source under observation, in the original 10-bits on-board coding. This value is estimated from two appropriately selected TDI lines of the RP spectrum for the same source transit. It is used on board to set the `g_rvs_class` and `rvs_priority` of the corresponding RVS window.

It is coded as $\texttt{rvs_mag} = 64 \times (21 - G_{\rm RVS})$. This coding covers the range 5.015625 to 21 mag. It offers a resolution of 0.015625 magnitudes. Accordingly, the on-board $G_{\rm RVS}$ value can be calculated via $G_{\rm RVS} = 21 - \texttt{rvs_mag}/64$

ROTATION_PHASE : Satellite rotation phase, i.e. sun azimuth (short)

The solar azimuthal phase derived from the commanded scanning law and the onboard time, encoded as an 8-bits unsigned integer in the range 0...255. The resolution is about 1.4 degrees (360 degrees divided by 256). The value is computed every 1024 TDI periods, i.e. at a frequency slightly lower than 1 Hz. The value used for a given object is the most recent value available at the time the on-board azimuthal phase filter decision is evaluated.

This field is filled only for the data acquired after OBMT 2103 rev. (2015-04-07; CCD row 5 only), and OBMT 2186 rev. (2015-04-28; all CCD rows), respectively, as it was implemented on board only after launch (starting from onboard VPU software version 2.8).

RA_LOW_PRECISION : Rough right ascension of window centre (float, Angle[deg])

Rough right ascension. This field, and its declination companion `dec_low_precision`, does not give a precise position of an observed celestial object, but a rough projection of the on-board detected object (precisely: the AF1 reference pixel) onto the sky. The provided precision is sufficient for the main use cases of this table: sky statistics, location and identification of astrophysical transients, etc. Better positions could only be given for Gaia catalogue objects, i.e. not for detection records that are not matched to one of those, which are about 20 percent of all.

The attainable precision is limited by the resolution of the positional information given in the `transit_id`, which is one across-scan sample size, i.e. about 0.18 arcsec. This low precision allows use of 4-byte floats for the given coordinates `ra_low_precision` and `dec_low_precision`. The worst case for the numerical resolution due to this choice is for right ascensions above 256 degrees, where the exponent is 2^9 (=512), and the resolution therefore is 512 degrees/2^{24} = 110 mas.

It is to be noted that the given coordinates are not the actual viewing direction (technically called proper direction) from which Gaia received the light at the moment of observation. But they are coordinate directions, i.e. the proper directions corrected for aberration and solar-system light bending. This makes it possible to directly match these coordinates with ordinary source catalogue positions and e.g. transient positions measured relative to some source catalogue.

DEC_LOW_PRECISION : Rough declination of window centre (float, Angle[deg])

Rough declination of the on-board detected object. For more information see the description of `ra_low_precision`.

VIRTUAL_OBJECT_FLAG : Flag indicating virtual objects (byte)

Virtual Objects are artificial Gaia CCD windows, i.e. seemingly normal Gaia observations, but they do not correspond to real detections of flux peaks in the Sky Mapper (SM) CCDs. Instead

they are wilfully placed into the Gaia data stream, i.e. wilfully placed onto the sky. They are created for various purposes, and they come in two basic flavours.

The main part of the Virtual Objects are collected for calibration purposes. They are distributed in a more or less random (but deterministic, repeating every two hours) fashion in time, field of view, CCD row and position on the CCDs. They are characterised by artificial values of the field-of-view item in the `transit_id` parameter: 3 for the preceding and 4 for the following field of view, i.e. for SM windows from SM1 and SM2, respectively. They usually contain empty sky. They can for instance be used for the determination of the sky background, of electronic effects in the CCDs and proximity electronics, etc.

The second flavour are the synchronized Virtual Objects. They are wilfully placed on the expected CCD positions of a few dozen of the very brightest stars in the whole sky. Due to extreme overexposure these cannot be detected autonomously on board. They are used in an attempt to get scientifically useful astrometric data on them from Gaia despite the heavy overexposure. The synchronized Virtual Objects are identified by the combination of the `virtual_object_flag` with the normal field-of-view indices in the `transit_id` parameter: 1 for the preceding and 2 for the following field of view. It should be noted that synchronized Virtual Objects have not been produced for significant parts of Gaia's mission duration. For DR4, the total of three time intervals covered is only 1.1 years, scattered over an envelope of about 2.2 years (Dec. 2017 to Jan. 2020).

(start = 5998.0 rev, 2017-12-06)

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6 Bright stars

As indicated in the below subsection titles by the labels [TAP] and [DataLink], these products are served from the ESA Gaia Archive using two different protocols, the Table Access Protocol (TAP) for tabular products and the DataLink protocol for other products such as spectra, light curves, MCMC samples, etc. For more details please see Section **??**.

6.1 BRIGHT_SOURCE_ASTROMETRY [TAP]

This table contains the astrometric parameters obtained for 49 of the 51 brightest stellar sources in the sky. They are derived from Gaia observations taken in a special mode that was designed only after Gaia's launch, and was introduced into the on-board operational scheme on Dec 6, 2017, i.e. about 2.4 years after the start of Gaia's science operations. Within the mission time range entering Gaia DR4, this special mode was active for three time intervals distributed over 25 months, but covering only 13 months of time in total.

Details on this special observation mode — called 'synchronized Virtual Objects' (VOsync) — and on the special data processing methods used for the data reduction, are given in **?** and partly in the relevant chapter of the DR4 online documentation.

Given the difficult treatment of the heavily saturated Gaia images in the magnitude range of the 51 brightest stars, and given the small portion of the mission covered, the astrometric data in this table are of much lower precision than usually delivered by Gaia. The reader is kindly reminded that the pre-launch specification for the bright limit of Gaia's sky survey was G = 5.7 mag, while the sources in the present table range from G $\simeq$ 2.2 mag to G $\simeq -1.4$ mag. In other words, they are between 25 and 700 times brighter than the originally planned magnitude limit.

The astrometric parameters were produced using the same AGIS and preprocessing (IDU) softwares that were used for the main astrometric catalogue. The various quality diagnostics, selection criteria and the fallback scheme from 6-parameter solutions to 2-parameter solutions are analogous to the AGIS configuration for the main astrometric catalogue. Details are described in **?**. Also, the table format is the same as for the `all_source_astrometry` table, but about 30 columns were omitted which are not relevant for the `bright_source_astrometry` table, and the specific column `matched_source_id` was added.

For two of the selected 51 stars, α Cen A and α Cen B, it was not possible to derive AGIS solutions since the DR4 VOsync CCD windows were not positioned correctly. For Gaia DR5 it is expected that these two stars

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will be recovered, and that 6-parameter solutions will be derived for all 51 stars.

This table is complemented by `bright_source_epoch_astrometry`, providing the individual astrometric observations for the 49 sources, in the same format as the `epoch_astrometry` table does for the main Gaia DR4 source list.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Source Identifier (long)

A unique single numerical identifier of the source. The general format and construction of the identifier is described in `gaia_source.source_id`. Within this general format, the special case of bright source entries is signified by the Data Processing Centre code (in bits 33-35). This is set to the value of 5 (decimal), i.e. 101 (binary). Apart from this, the `source_id` has the usual structure, with the HEALPix index according to the location of the source, the running number counted independently from all other source identifier lists, and the component number always zero.

DESIGNATION : Unique source designation (unique across all Data Releases) (string)

A source designation, unique across all Gaia Data Releases, that is constructed from the prefix 'Gaia DRx ' followed by a string of digits corresponding to `source_id` (3 space–separated words in total). Note that the integer source identifier `source_id` is **not** guaranteed to be unique across Data Releases; moreover it is not guaranteed that the same astronomical source will always have the same `source_id` in different Data Releases. Hence the only safe way to compare source records between different Data Releases in general is to check the records of proximal source(s) in the same small part of the sky.

REF_EPOCH : Reference epoch (double, Time[Julian Years])

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Reference epoch to which the astrometric source parameters are referred, expressed as a Julian Year in TCB (Temps Coordonnée Barycentrique = Barycentric Coordinate Time).

MATCHED_SOURCE_ID : Source identifier from the main source list matched to this source (long)

Source identifier from the source list in `all_source_astrometry` that has been positionally matched to the bright source of interest. When this field is empty, no specific match could be identified.

RA : Right ascension (double, Angle[deg])

Barycentric right ascension α of the source in ICRS at the reference epoch `ref_epoch`.

RA_ERROR : Standard error of right ascension (float, Angle[mas])

Standard error $\sigma_{\alpha*} \equiv \sigma_\alpha \cos\delta$ of the right ascension of the source in ICRS at the reference epoch `ref_epoch`.

DEC : Declination (double, Angle[deg])

Barycentric declination δ of the source in ICRS at the reference epoch `ref_epoch`.

DEC_ERROR : Standard error of declination (float, Angle[mas])

Standard error σ_δ of the declination of the source in ICRS at the reference epoch `ref_epoch`.

PARALLAX : Parallax (double, Angle[mas])

Absolute stellar parallax ϖ of the source at the reference epoch `ref_epoch`.

PARALLAX_ERROR : Standard error of parallax (float, Angle[mas])

Standard error σ_ϖ of the stellar parallax at the reference epoch `ref_epoch`.

PARALLAX_OVER_ERROR : Parallax divided by its standard error (float)

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Parallax divided by its standard error.

PMRA : Proper motion in right ascension direction (double, Angular Velocity[mas yr^{-1}])

Proper motion in right ascension $\mu_{\alpha*} \equiv \mu_\alpha \cos\delta$ of the source in ICRS at the reference epoch `ref_epoch`. This is the local tangent plane projection of the proper motion vector in the direction of increasing right ascension.

PMRA_ERROR : Standard error of proper motion in right ascension direction (float, Angular Velocity[mas yr^{-1}])

Standard error $\sigma_{\mu\alpha*}$ of the local tangent plane projection of the proper motion vector in the direction of increasing right ascension at the reference epoch `ref_epoch`.

PMDEC : Proper motion in declination direction (double, Angular Velocity[mas yr^{-1}])

Proper motion in declination μ_δ of the source at the reference epoch `ref_epoch`. This is the projection of the proper motion vector in the direction of increasing declination.

PMDEC_ERROR : Standard error of proper motion in declination direction (float, Angular Velocity[mas yr^{-1}])

Standard error $\sigma_{\mu\delta}$ of the proper motion component in declination at the reference epoch `ref_epoch`.

PM : Total proper motion (float, Angular Velocity[mas yr^{-1}])

The total proper motion calculated as the magnitude of the resultant vector of the proper motion component vectors `pmra` and `pmdec`, i.e. $\texttt{pm}^2 = \texttt{pmra}^2 + \texttt{pmdec}^2$.

RA_DEC_CORR : Correlation between right ascension and declination (float)

Correlation coefficient $\rho(\alpha, \delta)$ between right ascension and declination. This is a dimensionless quantity in the range [-1,+1].

RA_PARALLAX_CORR : Correlation between right ascension and parallax (float)

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Correlation coefficient $\rho(\alpha, \varpi)$ between right ascension and parallax, a dimensionless quantity in the range [-1,+1].

RA_PMRA_CORR : Correlation between RA and proper motion in RA (float)

Correlation coefficient $\rho(\alpha, \mu_{\alpha*})$ between right ascension and proper motion in right ascension, a dimensionless quantity in the range [-1,+1].

RA_PMDEC_CORR : Correlation between right ascension and proper motion in declination (float)

Correlation coefficient $\rho(\alpha, \mu_{\delta})$ between right ascension and proper motion in declination, a dimensionless quantity in the range [-1,+1].

DEC_PARALLAX_CORR : Correlation between declination and parallax (float)

Correlation coefficient $\rho(\delta, \varpi)$ between declination and parallax, a dimensionless quantity in the range [-1,+1].

DEC_PMRA_CORR : Correlation between declination and proper motion in right ascension (float)

Correlation coefficient $\rho(\delta, \mu_{\alpha*})$ between declination and proper motion in right ascension, a dimensionless quantity in the range [-1,+1].

DEC_PMDEC_CORR : Correlation between declination and proper motion in declination (float)

Correlation coefficient $\rho(\delta, \mu_{\delta})$ between declination and proper motion in declination, a dimensionless quantity in the range [-1,+1].

PARALLAX_PMRA_CORR : Correlation between parallax and proper motion in right ascension (float)

Correlation coefficient $\rho(\varpi, \mu_{\alpha*})$ between parallax and proper motion in right ascension, a dimensionless quantity in the range [-1,+1].

PARALLAX_PMDEC_CORR : Correlation between parallax and proper motion in declination (float)

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Correlation coefficient $\rho(\varpi, \mu_\delta)$ between parallax and proper motion in declination, a dimensionless quantity in the range [-1,+1].

PMRA_PMDEC_CORR : Correlation between proper motion in RA and proper motion in Dec (float)

Correlation coefficient $\rho(\mu_{\alpha*}, \mu_\delta)$ between proper motion in right ascension and proper motion in declination, a dimensionless quantity in the range [-1,+1].

ASTROMETRIC_N_OBS_AL : Total number of observations in the along-scan (AL) direction (short)

Total number of AL observations (= CCD transits) used in the astrometric solution of the source, independent of their weight. Note that some observations may be strongly downweighted and their number can be derived from `astrometric_n_obs_al` − `astrometric_n_good_obs_al`.

ASTROMETRIC_N_OBS_AC : Total number of observations in the across-scan (AC) direction (short)

Total number of AC observations (= CCD transits) used in the astrometric solution of the source, independent of their weight (note that some observations may be strongly downweighted). Nearly all sources having G < 13 will have AC observations from 2D windows, while fainter than that limit only ∼ 1% of transit observations (the so-called 'calibration faint stars') are assigned 2D windows resulting in AC observations.

For DR4, the value is always zero, as even for 2D windows the AC observations are not used by AGIS.

ASTROMETRIC_N_GOOD_OBS_AL : Number of good observations in the along-scan (AL) direction (short)

Number of AL observations (= CCD transits) that were not strongly downweighted in the astrometric solution of the source. Strongly downweighted observations (with downweighting factor $w < 0.2$) can instead be derived from `astrometric_n_obs_al`−`astrometric_n_good_obs_al`, where `astrometric_n_obs_al` is the total number of AL observations used in the astrometric solution of the source.

ASTROMETRIC_GOF_AL : Goodness of fit statistic of model wrt along-scan observations (float)

Goodness-of-fit statistic of the astrometric solution for the along-scan measurements of the source. This is a rescaled version of the 'gaussianized chi-square', which for good fits should approximately follow a normal distribution with zero mean value and unit standard deviation. Values exceeding, say, +3 thus indicate a significantly bad fit to the data.

Starting with DR4, `astrometric_gof_al` is probably more useful than `ruwe` as a general goodness-of-fit measure, because it takes into account the dependence on the degree of freedom and has a simple statistical interpretation.

Note that the `astrometric_gof_al` is not simply the 'gaussianized' version of the original `astrometric_chi2_al` but an empirically rescaled version. The rescaling procedure is described and motivated in detail in **?** where an explanation and discussion of other astrometric quality indicators is provided.

ASTROMETRIC_CHI2_AL : AL chi-square value (float)

Weighted sum of the squared astrometric residuals (χ^2) in the AL direction.

χ^2 values were computed using the 'good' AL observations of the source, without taking into account the `astrometric_excess_noise` (if any) of the source. They do however take into account the calibration excess noise (if any) of each observation.

See **?** for a discussion of other astrometric quality indicators.

MEAN_PARALLAX_FACTOR_AL : Mean Parallax factor AL (float)

Mean parallax factor in the AL direction, computed from all the good observations of the source processed in the astrometry.

The AL parallax factor for an individual observation is defined as $\partial\eta/\partial\varpi$, where η is the AL field angle of the source and ϖ its parallax, and is constrained to $[-0.73, +0.73]$ by the scanning law. The value given in this field is typically in the range $[-0.23, +0.32]$ (1st and 99th percentiles). A value outside this range indicates a distribution of observations that is unfavourable for the determination of the parallax, and the calculated parallax could then be more vulnerable to errors, e.g. from the calibration model, not reflected in the formal uncertainties. See **?** for a discussion of other astrometric quality indicators.

ASTROMETRIC_EXCESS_NOISE : Rescaled excess noise of the source (float, Angle[mas])

This is the rescaled excess noise ϵ_i of the source. It is not the same as the source excess noise in Gaia DR3 and earlier releases, but re-scaled in order to be consistent with the (also rede-

fined) unit weight error (ruwe). But, like the original quantity in earlier releases, it measures the disagreement, expressed as an angle, between the observations of a source and the best-fitting standard astrometric model (using six astrometric parameters). The assumed observational noise in each observation is quadratically increased by ϵ_i in order to statistically match the residuals in the astrometric solution.

A value of $\epsilon_i = 0$ thus signifies that the source is astrometrically well-behaved, i.e. that the residuals of the fit statistically agree with the assumed observational noise. Conversely, a positive ϵ_i signifies that the residuals are statistically larger than expected. This may happen for sources that deviate from the standard six-parameter astrometric model (e.g. unresolved binaries, exoplanet systems, etc.). The statistical significance of the excess noise is given by astrometric_gof_al.

Although the rescaling makes ϵ_i much less sensitive to instrumental effects than in previous releases, it may still absorb residual modelling errors not accounted for by the observational noise (image centroiding error) and the calibration excess noise. This includes LSF and PSF calibration errors, saturation effects, geometric instrument calibration errors, and part of the high-frequency attitude noise. Such modelling errors may be particularly important for very bright sources and sources of extreme extreme colours. Given the many other possible contributions to the excess noise, the user is advised to study the empirical distributions of ϵ_i and astrometric_gof_al for their particular samples in order to make sensible choices of filtering limits.

The source excess noise is further explained in **?** and in Section **??** of the DR4 online documentation.

ASTROMETRIC_PARAMS : Bitwise code describing which astrometric parameters are provided (short)

The bit–wise encoding in astrometric_params indicates which parameters have been estimated in AGIS (primary or secondary solution) and exported for this source. A set bit means the parameter was updated, an unset bit means the parameter was not updated. The least-significant bit corresponds to α. The table below shows the values for relevant combinations of the parameters.

The radial proper motion (μ_r) is formally considered to be one of the astrometric parameters of a source, and the sixth bit is therefore reserved for it. It is also in principle updatable in AGIS, but in practice it will always be computed from a spectroscopic radial velocity and the estimated parallax, in which case the bit is not set.

C is the pseudo-colour of the source, i.e. the astrometrically estimated effective wavenumber.

astrometric_params	α	δ	ϖ	$\mu_{\alpha*}$	μ_δ	μ_r	C	$\dot{\mu}_{\alpha*}$	$\dot{\mu}_\delta$	$\ddot{\mu}_{\alpha*}$	$\ddot{\mu}_\delta$
$000000000011_2 = 3$	✓	✓									
$000000000111_2 = 7$	✓	✓	✓								
$000000011011_2 = 27$	✓	✓		✓	✓						
$000000011111_2 = 31$	✓	✓	✓	✓	✓						
$000000111111_2 = 63$	✓	✓	✓	✓	✓	✓					
$000001011111_2 = 95$	✓	✓	✓	✓	✓		✓				
$000111011111_2 = 479$	✓	✓	✓	✓	✓		✓	✓	✓		
$011111011111_2 = 2015$	✓	✓	✓	✓	✓		✓	✓	✓	✓	✓

ASTROMETRIC_PRIMARY_FLAG : Primary or secondary (boolean)

Flag indicating if this source was used as a primary source (`true`) or secondary source (`false`). Only primary sources contribute to the estimation of attitude, calibration, and global parameters. The estimation of source parameters is otherwise done in exactly the same way for primary and secondary sources.

For the bright sources, this flag is always zero.

RV_USED_IN_ASTROMETRY : Assumed radial velocity used in the astrometric solution (float, Velocity[km s^{-1}])

This value is the one used to account for possible perspective acceleration effects. This correction is applied even when the expected perspective acceleration is negligible.

Due to cyclic processing of the astrometry and the RVS data, this radial velocity might be different from the one computed using the latest available RVS information, and published under `all_source_rvs.radial_velocity_single`. Moreover, if no reliable radial velocity was available at the time of the astrometric processing, radial velocities from the literature have been used when relevant, and their value is reported in `rv_used_in_astrometry`.

NU_EFF_USED_IN_ASTROMETRY : Effective wavenumber of the source used in the astrometric solution (float, Misc[μm^{-1}])

Effective wavenumber of the source used as starting value in the astrometric solution. Due to the unavailability of measured colours for the bright stars, for all sources in this table, the value of this field is identical to the default value.

This effective wavenumber $\nu_{\rm eff}$, in units of μm^{-1}, is the assumed value used for the selection of the appropriate LSF/PSF model in the image parameter determination, as well as for the selection of the astrometric calibration parameters in the astrometric adjustment, for the given source. If no

BP and RP photometry of the given source was available, `nu_eff_used_in_astrometry` is set to the default value $\nu_{\rm eff} = 1.43\mu{\rm m}^{-1}$. Else, if reliable BP and RP photometry was available, it is set to the weighted mean of the default value and the actual photon-flux weighted inverse wavelength as estimated from the BP and RP bands. If highly precise BP and RP data are available, the `nu_eff_used_in_astrometry` is very close to the value from the BP and RP bands. However, due to the cyclic processing of the Gaia astrometry and photometry, this effective wavenumber will even in these cases be (slightly) different from the one computed using the latest available photometry for DR4.

Some details on the selection of the default value and the construction of the weighted mean:
The default value is chosen as the ensemble average of the actually measured values for all Gaia sources, and its standard error for the weighted mean is set to $0.085\mu{\rm m}^{-1}$, which is close to the actual rms scatter of the measured values.
For the measured values, the formal errors assigned by the photometry processing were used in the construction of the weighted mean. However, the original values delivered by CU5 (from the DR3 photometry processing) were found to be strongly overestimated. Therefore, they were rescaled to give realistic values before being used for the weighted mean.

PSEUDOCOLOUR : Astrometrically estimated pseudocolour of the source (float, Misc[μm^{-1}])

Astrometrically determined colour of the source, represented as an effective wavelength in $\mu{\rm m}^{-1}$. This quantity can be used as a proxy for the actual effective wavenumber of the source if the latter cannot be computed from CU5 photometry. It is estimated in AGIS from the chromatic displacements of image centroids. Technically, it is the sum of `nu_eff_used_in_astrometry` and a correction to that value derived by AGIS.

For details see **?**.

PSEUDOCOLOUR_ERROR : Standard error of the pseudocolour of the source (float, Misc[μm^{-1}])

Standard error $\sigma_{\texttt{pseudocolour}}$ of the astrometrically determined `pseudocolour` of the source.

RA_PSEUDOCOLOUR_CORR : Correlation between right ascension and pseudocolour (float)

Correlation coefficient $\rho(\alpha, \texttt{pseudocolour})$ between right ascension `ra` and `pseudocolour`, a dimensionless quantity in the range [-1,+1]

DEC_PSEUDOCOLOUR_CORR : Correlation between declination and pseudocolour (float)

Correlation coefficient $\rho(\delta, \texttt{pseudocolour})$ between declination `dec` and `pseudocolour`, a di-

mensionless quantity in the range [-1,+1]

PARALLAX_PSEUDOCOLOUR_CORR : Correlation between parallax and pseudocolour (float)

Correlation coefficient $\rho(\varpi, \texttt{pseudocolour})$ between `parallax` and `pseudocolour`, a dimensionless quantity in the range [-1,+1]

PMRA_PSEUDOCOLOUR_CORR : Correlation between proper motion in right asension and pseudocolour (float)

Correlation coefficient $\rho(\mu_{\alpha*}, \texttt{pseudocolour})$ between proper motion in right ascension `pmra` and `pseudocolour`, a dimensionless quantity in the range [-1,+1]

PMDEC_PSEUDOCOLOUR_CORR : Correlation between proper motion in declination and pseudocolour (float)

Correlation coefficient $\rho(\mu_{\delta}, \texttt{pseudocolour})$ between proper motion in declination `pmdec` and `pseudocolour`, a dimensionless quantity in the range [-1,+1]

ASTROMETRIC_MATCHED_TRANSITS : Matched FOV transits used in the AGIS solution (short)

The number of field–of–view transits matched to this source, counting only the transits containing CCD observations actually used to compute the astrometric solution.

This number will always be equal to or smaller than `matched_transits`, the difference being the FOV transits that were not used in the astrometric solution because of bad data or excluded time intervals.

VISIBILITY_PERIODS_USED : Number of visibility periods used in Astrometric solution (short)

Number of visibility periods used in the astrometric solution.

A visibility period is a group of observations separated from other groups by a gap of at least 4 days. A source may have from one to tens of field-of-view transits in a visibility period, but with a small spread in time, direction of scanning, and parallax factor. From one visibility period to the next these variables have usually changed significantly. A high number of visibility periods is therefore a better indicator of an astrometrically well-observed source than a large number of field-of-view transits (`matched_transits` or `astrometric_matched_transits`) or CCD

observations (astrometric_n_obs_al). A small value (e.g. less than 10) indicates that the calculated parallax could be more vulnerable to errors, e.g. from the calibration model, not reflected in the formal uncertainties. See **?** for a discussion of this and other astrometric quality indicators.

ASTROMETRIC_SIGMA5D_MAX : The longest semi-major axis of the 5-d error ellipsoid (float, Angle[mas])

The longest principal axis in the 5-dimensional error ellipsoid.

This is a 5-dimensional equivalent to the semi-major axis of the position error ellipse and is therefore useful for filtering out cases where one of the five parameters, or some linear combination of several parameters, is particularly ill-determined. It is measured in mas and computed as the square root of the largest singular value of the scaled 5×5 covariance matrix of the astrometric parameters. The matrix is scaled so as to put the five parameters on a comparable scale, taking into account the maximum along-scan parallax factor for the parallax and the time coverage of the observations for the proper motion components. If C is the unscaled covariance matrix, the scaled matrix is SCS, where $S = \mathrm{diag}(1, 1, \sin\xi, T/2, T/2)$, $\xi = 45°$ is the solar aspect angle in the nominal scanning law, and T the time coverage of the data used in the solution.

astrometric_sigma5d_max is given for all the solutions, as its size is one of the criteria for accepting or rejecting the 6-parameter solution. In case of a 2-parameter solution (astrometric_params = 3) it gives the value for the rejected 6-parameter solution, and can then be arbitrarily large.

MATCHED_TRANSITS : Number of transits matched to this source (short)

Total number of field of view transits matched to this source. The number includes blended transits and transits where the AF images could not be processed.

This count was made at the cross-match stage and based on early AF image processing, which may differ slightly from the final image processing.

RUWE : Renormalised unit weight error (float)

The Renormalised Unit Weight Error is a re-scaled version of the Unit Weight Error

$$\texttt{uwe} = \sqrt{\texttt{astrometric_chi2_al}/\nu}$$

where $\nu = \texttt{astrometric_n_good_obs_al} - N$ is the number of degrees of freedom in the AGIS source update, and $N = 6$ is the number of parameters solved.

The purpose of the re-scaling is to make the ruwe behave in the statistically expected manner of

the uwe, when there is a good fit of the standard 6-parameter model to the astrometric data. A well-behaved source is expected to have ruwe $\simeq$ 1.0, while a significantly greater value indicates a bad fit. The re-scaling procedure is described in **?**.

The degree of deviation from the expected ruwe $\simeq$ 1.0 is most conveniently measured by the Gaussianized statistic astrometric_gof_al. Starting with DR4, astrometric_gof_al is probably more useful than ruwe as a general goodness-of-fit measure, because it takes into account the dependence on ν and has a simple statistical interpretation. For continuity, ruwe is retained with the same name and nominal meaning as in earlier releases, though more accurately computed.

The ruwe is set to null for sources that only have a two-parameter solution ($N = 2$).

SCAN_DIRECTION_STRENGTH_K1 : Degree of concentration of scan directions across the source (float)

The scan_direction_strength_k1...4 and scan_direction_mean_k1...4 quantify the distribution of AL scan directions across the source. scan_direction_strength_k1 (and similarly 2,3,4) equals $\left|\langle \exp(ik\theta)\rangle\right|$ for $k = 1, 2, 3, 4$ where θ is the position angle of the scan and the mean value is taken over the astrometric_n_good_obs_al observations contributing to the astrometric parameters of the source. θ is defined in the usual astronomical sense: $\theta = 0°$ when the FoV is moving towards local North, and $\theta = 90°$ towards local East.

The scan_direction_strength_k1...4 are numbers between 0 and 1, where 0 means that the scan directions are well spread out in different directions, while 1 means that they are concentrated in a single direction (given by the corresponding scan_direction_mean_k1...4).

The different orders k are statistics of the scan directions modulo $360°/k$. For example, at first order ($k = 1$), $\theta = 10°$ and $\theta = 190°$ count as different directions, but at second order ($k = 2$) they are the same. Thus, scan_direction_strength_k1 is the degree of concentration when the sense of direction is taken into account, while scan_direction_strength_k2 is the degree of concentration without regard to the sense of direction. A large value of scan_direction_strength_k4 indicates that the scans are concentrated in two nearly orthogonal directions.

SCAN_DIRECTION_STRENGTH_K2 : Degree of concentration of scan directions across the source (float)

The scan_direction_strength_k1...4 and scan_direction_mean_k1...4 attributes quantify the distribution of AL scan directions across the source.

See the description for attribute scan_direction_strength_k1 for further details.

SCAN_DIRECTION_STRENGTH_K3 : Degree of concentration of scan directions across the source (float)

The scan_direction_strength_k1...4 and scan_direction_mean_k1...4 attributes quantify the distribution of AL scan directions across the source.

See the description for attribute scan_direction_strength_k1 for further details.

SCAN_DIRECTION_STRENGTH_K4 : Degree of concentration of scan directions across the source (float)

The scan_direction_strength_k1...4 and scan_direction_mean_k1...4 attributes quantify the distribution of AL scan directions across the source.

See the description for attribute scan_direction_strength_k1 for further details.

SCAN_DIRECTION_MEAN_K1 : Mean position angle of scan directions across the source (float, Angle[deg])

The scan_direction_strength_k1...4 and scan_direction_mean_k1...4 attributes quantify the distribution of AL scan directions across the source. scan_direction_mean_k1 (and similarly for $k = 2, 3, 4$) equals $\arg(\langle\exp(ik\theta)\rangle)/k$ for k = 1,2,3,4 where θ is the position angle of the scan and the mean value is taken over the astrometric_n_good_obs_al observations contributing to the astrometric parameters of the source. θ is defined in the usual astronomical sense: $\theta = 0°$ when the FoV is moving towards local North, and $\theta = 90°$ towards local East.

scan_direction_mean_k1 (and similarly for $k = 2, 3, 4$) is an angle between $-180°/k$ and $+180°/k$, giving the mean position angle of the scans at order k.

The different orders k are statistics of the scan directions modulo $360°/k$. For example, at first order ($k = 1$), $\theta = 10°$ and $\theta = 190°$ count as different directions, but at second order ($k = 2$) they are the same. Thus, scan_direction_mean_k1 is the mean direction when the sense of direction is taken into account, while scan_direction_mean_k2 is the mean direction without regard to the sense of the direction. For example, scan_direction_mean_k1 = 0 means that the scans preferentially go towards North, while scan_direction_mean_k2 = 0 means that they preferentially go in the North-South direction, and scan_direction_mean_k4 = 0 that they preferentially go either in the North-South or in the East-West direction.

SCAN_DIRECTION_MEAN_K2 : Mean position angle of scan directions across the source (float, Angle[deg])

The scan_direction_strength_k1...4 and scan_direction_mean_k1...4 attributes quantify the distribution of AL scan directions across the source.

See the description for attribute scan_direction_mean_k1 for further details.

SCAN_DIRECTION_MEAN_K3 : Mean position angle of scan directions across the source (float, Angle[deg])

The scan_direction_strength_k1...4 and scan_direction_mean_k1...4 attributes quantify the distribution of AL scan directions across the source.

See the description for attribute scan_direction_mean_k1 for further details.

SCAN_DIRECTION_MEAN_K4 : Mean position angle of scan directions across the source (float, Angle[deg])

The scan_direction_strength_k1...4 and scan_direction_mean_k1...4 attributes quantify the distribution of AL scan directions across the source.

See the description for attribute scan_direction_mean_k1 for further details.

L : Galactic longitude (double, Angle[deg])

Galactic Longitude of the object at reference epoch ref_epoch, see Section **??** of the release documentation for conversion details.

B : Galactic latitude (double, Angle[deg])

Galactic Latitude of the object at reference epoch ref_epoch, see Section **??** of the release documentation for conversion details.

ECL_LON : Ecliptic longitude (double, Angle[deg])

Ecliptic Longitude of the object at reference epoch ref_epoch, obtained from the equatorial coordinates using the transformation defined in Volume 1, Section 1.5.3 of ESA (1997).

Note that in the transformation applied here the ICRS origin is shifted in the equatorial plane from Γ by $\phi = 0.05542$ arcsec, positive from Γ to the ICRS origin (Chapront et al. 2002). The

ICRS has an unambiguous definition with an origin in the ICRF equator defined by the realisation of the ICRF. The ecliptic system is less well-defined, potentially depending on additional conventions in dynamical theories. The transformation employed here corresponds to the inertial mean ecliptic with obliquity and Γ defined by reference to the ICRS equator. Both the obliquity and the position of Γ on the ICRS equator with respect to the ICRS origin have been obtained from Lunar Laser Ranging measurements. This has no time dependence – there is no secular variation of the obliquity and no precession – and it simply defines the relative situation of the various planes at J2000.

ECL_LAT : Ecliptic latitude (double, Angle[deg])

Ecliptic Latitude of the object at reference epoch `ref_epoch`. For further details see the description for attribute `ecl_lon`.

Draft

6.2 BRIGHT_SOURCE_EPOCH_ASTROMETRY [DataLink]

This table provides the epoch astrometry, i.e. the individual astrometric measurements for the same 49 among the 51 brightest stars for which the astrometric parameters are presented in the bright source astrometry table.

Quick information on the special observation mode — called 'synchronized Virtual Objects' (VOsync) — for these stars and on the special data processing methods used for the data reduction, can be found in the description of the `bright_source_astrometry` table. Details are given in **?**, and partly in the relevant chapter of the DR4 online documentation.

The definition and presentation of the data in the present table is the same as in the main `epoch_astrometry` table. This means that the epoch astrometry data is given in the form of local plane coordinates, both for the two-dimensional and for the one-dimensional astrometric Gaia measurements. See the relevant chapter of the online documentation for more explanations, and **?** for full details.

The table `bright_source_epoch_astrometry` contains one row per `source_id-transit_id` matching pair. The same `source_id` occurs as many times as there are FoV transits matched with this source. Any combination of `source_id` and `transit_id` occurs exactly once in the table.

Each row contains three per-source fields (`ra0`, `dec0` and `agis_source_excess_noise`); for ease of handling and usage, these are repeated as many times as there are transits matched to the given source.

Similarly, some of the per-transit and per-CCD data are repeated for the same reasons.

Note this table is not available through the main archive TAP interface. Data are delivered via the Massive Data service indexed by the VO DataLink protocol and described in Chapter **??**. The DataLink retrieval type to specify for this product in programmatic archive access via the Astroquery Gaia Python package is `EPOCH_ASTROMETRY_BRIGHT`.

DataLink products are served with a choice of two different data structures:

- `DATAMODEL_STANDARD`: the product is serialised following a standard data model (for example an IVOA-compliant data model in applicable cases);
- `DATAMODEL_GAIA`: the product is serialised following the Gaia DPAC(-internal) data model.

Draft

The DATAMODEL_GAIA format is described here whereas the DATAMODEL_STANDARD format is described in https://www.cosmos.esa.int/web/gaia-users/archive/datalink-products#datalink_serialisation

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution identifier, please visit

https://gaia.esac.esa.int/decoder/solnDecoder.jsp.

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source.

The general format and construction of the identifier is described in gaia_source.source_id. Within this general format, the special case of bright source entries is signified by the Data Processing Centre code (in bits 33-35). This is set to the value of 5 (decimal), i.e. 101 (binary).

Apart from this, the source_id has the usual structure, with the HEALPix index according to the location of the source, the running number counted independently from all other source identifier lists, and the component number always zero.

TRANSIT_ID : Transit Identifier (long)

The transit_id is a unique identifier assigned to each detected (and confirmed) source as it transits the Gaia focal plane. Each time a given source is detected as Gaia scans and re-scans the sky a new transit_id will be created to badge that apparition. Hence the along-scan time and the across-scan position along with the telescope in which the source was detected are used to form a unique integer with which to label the transit. The detection of transits is done using the SkyMapper CCD data, and the confirmation is done using the AF1 CCD data. Both happens on board.

The several features of a transit that are encoded in transit_id can be easily retrieved using bit

masks (&) and shifts (>>) as follows:

- On-Board Mission Time line (OBMT) [ns]
  = 204800 * ((`transit_id` >> 17) & (0x000003FFFFFFFFFF))
- Field-of-view = 1 + (`transit_id` >> 15) & 0x03 [1 for 'preceding' and 2 for 'following' fields-of-view respectively]
- CCD row = (`transit_id` >> 12) & 0x07 [dimensionless, in the range 1 to 7]
- Across-scan 'reference acquisition pixel' in strip AF1 = (`transit_id`) & 0x0FFF [pixels] (this is the across-scan centre of the AF1 window and is odd if immediately below the mid-point of the window and even if immediately above)

where the bit mask prefix '0x' denotes hexadecimal.

For further details see Portell et al. (2020). For convenience a decoder for `transit_id` is available on-line at

`https://gaia.esac.esa.int/decoder/transitidDecoder.jsp`

Special note for `bright_source_epoch_astrometry`**:** The observations in this table are not based on FoV transits from the on-board SM detection process. Instead they are based on ground-predicted transits of the 51 brightest stars. On board, each of these predictions creates ten windows of 18 pixels AL by 12 pixels AC. On ground, these windows are stitched together into oversized virtual windows of 90 by 24 pixels, and then cropped to artificial CCD windows having the normal dimensions and being centred on the actual star image. The `transit_id` field then is created to have the usual parts and properties as described above, but for the artificial AF1 window.

RA0 : Right ascension of reference point (double, Angle[deg])

Right ascension α_0 of the tangent point on the plane tangential to the celestial sphere in which the coordinate direction of the object is locally modelled, i.e. of the celestial reference point from which the local plane coordinates `centroid_pos_al` and `centroid_pos_ac` are counted. The specific value will be taken from the valid/relevant source catalogue (which in turn will be defined implicitly by the `solution_id`).

Note: `ra0` and `dec0` are always the same for all transits of a given `source_id`.

For more details see **?**.

DEC0 : Declination of reference point (double, Angle[deg])

Declination δ_0 of the tangent point on the plane tangential to the celestial sphere in which the coordinate direction of the object is locally modelled, i.e. of the celestial reference point from which the local plane coordinates `centroid_pos_al` and `centroid_pos_ac` are counted. The specific value will be taken from the valid/relevant astrometric source catalogue (which in turn will be defined implicitly by the `solution_id`).

Note: `ra0` and `dec0` are always the same for all transits of a given `source_id`.

For more details see **?**.

AGIS_SOURCE_EXCESS_NOISE : The unscaled AGIS Source Excess Noise (float, Angle[mas])

This is the unscaled AGIS source excess noise.

The actual weight used by AGIS per CCD for ordinary Gaia astrometric observations (i.e. for those data derived from on-board detections and from AF data) is $(1mas^2/\sigma^2_{tot})$, where $\sigma^2_{tot} = \sigma^2_{IPD} + excess^2_{source} + excess^2_{cal.}$. This is the weight to be used if normal AGIS solutions shall be reproduced from epoch astrometry data.

The quantity `centroid_pos_error_al` contains the first and third terms in σ^2_{tot}. The separate `agis_source_excess_noise` item is needed to enable external users to change the source model, whence $excess^2_{source}$ from the normal AGIS solution becomes irrelevant, while the other two components of σ^2_{tot} remain valid.

For more explanations see the AGIS paper for the relevant Gaia Data Release, here **?**.

Note that, contrary to the `gaia_source` and `all_source_astrometry` tables, this is the original source excess noise, not the rescaled excess noise according to **?**. Reason: The rescaled value is the one which is useful for the quality criteria in the source astrometry tables, but the original value is needed to enable users to reproduce the published AGIS solutions from epoch astrometry data.

For BrightSource observations, no $excess^2_{cal.}$ is derived. Thus, σ^2_{tot} for such observations contains only the first two terms, and `centroid_pos_error_al` contains only the first term in σ^2_{tot}.

OBS_TIME_TCB : Effective observing time as TCB (Long[10] array, Time[ns])

Effective observation time for each CCD in this FoV transit. Centre of the actual exposure time for the given window/CCD, depending on readout time and gate. In units of nanoseconds since

J2010.0(TCB).

The presentation as long integer (rather than as double float) is chosen for the sake of keeping the full resolution of the time coordinates used in Gaia/DPAC, and for the sake of invertability of transformations from OBMT to TCB and vice versa.

Such transformations are done using HATT (High-Accuracy Time Transformations), which have the resolution of 1 nanosecond. But users should be aware that the absolute accuracy of them is in the order of 150 nanoseconds.

OBS_TIME_BARY_CORR : Barycentric correction to obs_time_tcb (float, Time[ns])

Barycentric correction to the observation time, in the sense of TCB(barycentric)-TCB(at Gaia), calculated for the obsTime of the AF4 CCD, i.e. at the middle of the FoV transit.

For the correct production and usage of the epoch astrometry, both the true and the barycentrically corrected observation times are needed. It is, however, not needed to have barycentric corrections for each CCD of a transit. One value is sufficient. Explanation: The value can change by at most by about 30 km/s/c*30s=3ms from the middle of a transit to either the SkyMapper or AF9 CCDs. Anyone who needs it even more precisely, will also need the complete 3d location of Gaia (not just the 1d location along the line of sight) and should thus consult the orbit anyway. Therefore it was decided that `obs_time_bary_corr` shall be computed for the AF4 CCD transit time (or for an immediately adjacent CCD in case the AF4 IPD should happen to fail).

The maximum value that the quantity can reach is about ±370 s (not the canonical ± 500s because the scan axis of Gaia always points at 45 degrees from the sun). The resolution of the float thus is around 50 microseconds or better. This is clearly sufficient in view of the reasoning above.

SCAN_POS_ANGLE : Position angle of the scan (double[10] array, Angle[deg])

Position angle θ of the scan for each of the CCD transits.

This angle uniquely determines the transformation between the local celestial coordinates (a, d) and the local scan coordinates (w, z) in the tangential plane touching the celestial sphere at `(ra0,dec0)`. See **?** for details. The scan angle varies along a FoV transit. In contrast, the parallax factors are sufficiently constant over a transit.

The sine and cosine of θ are the derivatives of the along-scan local plane coordinate w with respect to right ascension and declination, respectively.

ZETA : Across-scan field angle zeta of the reference point (ra0, dec0) (float, Angle[deg])

This is the AC field angle ζ of the reference point (`ra0,dec0`). It needs not be an array of values per CCD, as it is sufficiently constant over a FoV transit. Thus it is specified that the `zeta` field angle be computed for the AF4 CCD transit time (or for an immediately adjacent CCD in case the AF4 IPD should happen to fail).

This quantity is needed for the following reason: When analysing a non-single system, it is necessary that the epoch astrometry of each component is expressed with respect to the same reference point. But that will not be the case for resolved systems, where the components have different AGIS solutions, and therefore different reference points. Hence there is a need to be able to transform the epoch data from one reference point to another (within an arcmin or so). But this is not possible - at least not rigorously - unless we also know ζ. The error in the transformed epoch astrometry is marginal (< 1 muas) for shifts less than about 5 arcsec, but increases quadratically for larger separations. ζ could of course be computed to sufficient accuracy if the FoV, CCD row, and AC coordinate `mu` (μ) are known, but to give it explicitly is a convenience for users.

Adding ζ to the epoch astrometry also brings a nice symmetry to the information in `epoch_astrometry`: with the AL&AC parallax factors plus the Roemer delay we have complete (3D) knowledge of Gaia's barycentric position at the time of observation (albeit only to float precision per transit, and in strange units); with `scan_pos_angle` and `zeta` we similarly have complete knowledge of the spin (SRS z) axis at the time of observation. Both pieces of information (Gaia's 3D position and the SRS z axis) are needed to transform epoch data to a different reference point - and back.

PARALLAX_FACTOR_AL : Parallax factor AL (float)

Parallax factor in the AL direction (w), needed to use the local plane coordinates for detailed object modelling.

The parallax factor f_w is the derivative of the along-scan local plane coordinate w with respect to the parallax. See **?** for details.

The parallax factors need not be an array of values per CCD, as they are sufficiently constant over a FoV transit. Thus it was decided that the parallax factors are computed for the AF4 CCD transit time (or for an immediately adjacent CCD in case the AF4 IPD should happen to fail).

PARALLAX_FACTOR_AC : Parallax factor AC (float)

Parallax factor in the AC direction (z) needed to use the local plane coordinates for detailed object modelling.

The parallax factor f_z is the derivative of the across-scan local plane coordinate z with respect to the parallax. See **?** for details.

The parallax factors need not be an array of values per CCD, as they are sufficiently constant over a FoV transit. Thus it was decided that the parallax factors are computed for the AF4 CCD transit time (or for an immediately adjacent CCD in case the AF4 IPD should happen to fail).

COLOUR_FACTOR_AL : Colour factor AL (float[10] array)

The colour factor in AL direction (w).

This is the derivative of w with respect to colour parameter ν_{eff} (effective wavenumber in units of inverse micrometers). The unit of the quantity is mas*micrometer.

COLOUR_FACTOR_AC : Colour factor AC (float[10] array)

The colour factor in AC direction (z).

This is the derivative of z with respect to colour parameter ν_{eff} (effective wavenumber in units of inverse micrometers). The unit of the quantity is mas*micrometer.

This field will be empty in DR4.

NU_EFF_USED_IN_ASTROMETRY : Colour (i.e. effective wavenumber) used in AGIS (and in IPD) (float, Misc[μm^{-1}])

Effective wavenumber of the source used as starting value in the astrometric solution. Due to the unavailability of measured colours for the bright stars, for all sources in this table, the value of this field is identical to the default value.

This effective wavenumber $\nu_{\rm eff}$, in units of $\mu{\rm m}^{-1}$, is the assumed value used for the selection of the appropriate LSF/PSF model in the image parameter determination, as well as for the selection of the astrometric calibration parameters in the astrometric adjustment, for the given source. If no BP and RP photometry of the given source was available, `nu_eff_used_in_astrometry` is set to the default value $\nu_{\rm eff} = 1.43\mu{\rm m}^{-1}$. Else, if reliable BP and RP photometry was available, it is set to the weighted mean of the default value and the actual photon-flux weighted inverse wavelength as estimated from the BP and RP bands. If highly precise BP and RP data are available, the `nu_eff_used_in_astrometry` is very close to the value from the BP and RP bands. However, due to the cyclic processing of the Gaia astrometry and photometry, this effective wavenumber will even in these cases be (slightly) different from the one computed using the latest available photometry for DR4.

Draft

Some details on the selection of the default value and the construction of the weighted mean:
The default value is chosen as the ensemble average of the actually measured values for all Gaia sources, and its standard error for the weighted mean is set to $0.085\mu m^{-1}$, which is close to the actual rms scatter of the measured values.
For the measured values, the formal errors assigned by the photometry processing were used in the construction of the weighted mean. However, the original values delivered by CU5 (from the DR3 photometry processing) were found to be strongly overestimated. Therefore, they were rescaled to give realistic values before being used for the weighted mean.

NU_EFF_ERROR : Standard error of nu_eff_used_in_astrometry (float, Misc[μm^{-1}])

This is the standard error of the input colour parameter used for a given source in the AGIS source parameter adjustment.

For the bright-source astrometry and crowded-field source astrometry, a constant value of 0.085 inverse micrometers was used, because no measured values were available.

CENTROID_POS_AL : Local plane coordinate (w) of image centroid AL (double[10] array, Angle[mas])

AL local plane coordinate w of the image centroid position (one per CCD transit). See **?** for details.

There is a known issue: Among the $\simeq 10^{12}$ entries of `centroid_pos_al` in the table, there is a very small portion with grossly deviating values. Such values may be caused by excursions of the attitude modelling at the start and end of data gaps. There are a total of 16,437,562 cases where `centroid_pos_al` is beyond the range allowed by the window size, and there are a few cases taking exceptionally large values. Notably, there are 208 transits where the max value of `centroid_pos_al` exceeds 10^9 mas. On the user side, the obvious way to avoid the problem is to simply ignore the very few records where `centroid_pos_al` is unreasonably large.

CENTROID_POS_AC : Local plane coordinate (z) of image centroid AC (double[10] array, Angle[mas])

AC local plane coordinate z of the image centroid position (one per CCD transit). See **?** for details. This is based on the AC centroiding and therefore it is only available for 2D windows. In DR4 this field and its associated error `centroid_pos_error_ac` will not be available.

CALCULATED_POS_AC : Calculated Local plane coordinate (z) (double[10] array, Angle[mas])

Calculated AC local plane coordinate (z).

This is not based on a real measurement like the centroid_pos_ac but has been calculated using the AL measurement and the source astrometry to compute the geometric source position at the time of the observation, then using the attitude to get the calculated AC local plane coordinate z.

CENTROID_POS_ERROR_AL : Standard error of centroid_pos_al (float[10] array, Angle[mas])

Standard errors of AL local plane coordinates (w) of the image centroid positions.

This quantity represents the standard errors used by AGIS, or more precisely: together with the agis_source_excess_noise and the used_by_agis_al flag it represents the actual weight for the AL coordinate used in the final AGIS source update (if used_by_agis_al is 'true').

This quantity contains two components, quadratically added via

$(\texttt{centroid_pos_error_al})^2 = \sigma^2_{IPD} + excess^2_{cal}$

The two components are:

- The IDU IPD centroid's standard error,
- The AGIS calibration excess noise.

The former is given explicitly in the table (ipd_error_al), while the second can be computed by quadratically subtracting the other one from centroid_pos_error_al.

To be very clear: if a user wants to reproduce the released AGIS solution for a given source, they should use weights derived from quadratically adding agis_source_excess_noise and centroid_pos_error_al, so that the result contains all three relevant components of the weighting used by AGIS:

$\sigma^2_{tot} = \sigma^2_{IPD} + excess^2_{source} + excess^2_{cal}$

Note that the centroid_pos_error_al for SM measurements might be significantly underestimated, especially for bright stars. For details see the corresponding column description for the main epoch_astrometry table and Section **??** of the online documentation.

CENTROID_POS_ERROR_AC : Standard error of centroid_pos_ac (float[10] array, Angle[mas])

Standard errors of the AC local plane coordinates z of the image centroid positions. For 1d

windows it is NaN.

This quantity represents the standard errors used by AGIS, or more precisely: together with the `agis_source_excess_noise` and the `used_by_agis_ac` flag it represents the actual weight for the AC coordinate used in the final AGIS source update (if `used_by_agis_ac` is 'true', which never happens for DR4). For 1d windows, the value will always be set to NaN.

This quantity contains two components, quadratically added via

$(\texttt{centroid_pos_error_ac})^2 = \sigma^2_{IPD} + excess^2_{cal}$

The two components are:

- The IDU IPD centroid's standard error,
- The AGIS calibration excess noise.

The former is given explicitly in the table (`ipd_error_ac`), while the second can be computed by quadratically subtracting the other one from `centroid_pos_error_ac`.

To be very clear: If a user wants to reproduce the released AGIS solution for a given source, he/she has to use weights derived from quadratically adding `agis_source_excess_noise` and `centroid_pos_error_ac`, so that the result contains all three relevant components of the weighting used by AGIS:

$\sigma^2_{tot} = \sigma^2_{IPD} + excess^2_{source} + excess^2_{cal.}$

This field is not available in DR4.

USED_BY_AGIS_AL : Whether the CCD transit was used in AL by AGIS-DR4 (Boolean[10] array)

Flag saying whether the CCD transit was used by AGIS-DR4 in AL direction. This assumes that the AGIS DR4 downweight factors are either zero or one.

USED_BY_AGIS_AC : Whether the CCD transit was used in AC by AGIS-DR4 (Boolean[10] array)

Flag saying whether the AC information from this CCD transit was used by AGIS for producing the released astrometric parameters. It means that the AGIS downweight factors in DR4 are

either zero or one. For DR4 this will always be ‘false’.

Note: This flag, like the corresponding one for AL, refers only to the usage of the AC measurement in the final AGIS source update producing the released astrometric source parameters. It ignores the usage of the AC measurements for the subset of (bright) primary stars needed to set up the overall astrometric system. In particular, these are indispensable for deriving the AC attitude (which is needed for a correct usage of the AL measurements), and this in turn requires an AC astrometric calibration.

TRANSIT_ACQ_FLAGS : Transit-level information on the on-board acquisition (short)

This field contains a set of bitwise flags (from Bit 15 = msb to Bit 0 = lsb) related to the data acquisition. It has some oddities for historical reasons.

Windows may suffer truncation or trimming. Here, truncation refers to a reduction in size - often irregular - caused by an intra-field conflict with another window, while trimming refers to the permanent use of a reduced window size across scan. Truncation and trimming only affect 1D windows and never affect SM and AF1 windows. Trimming was only used for AF2-9 and only for a short period early in the mission (around OMBT revolutions 2103 to 2238).

The shape of all 1D AF2-9 windows is detailed in `epoch_image.shape_af`.

The field is coded as follows:

- Bits 15 and 14: unused
- Bit 13 (0x2000): Always set.
- Bit 12 (0x1000): The AF2-9 windows are trimmed AC but have not suffered truncation.
- Bits 11-9: unused
- Bits 7 and 8 (0x0180): `xp_offset`. This is the value of a continuous two-bit TDI counter. It is needed for the reconstruction of the XP windows.
- Bit 6 (0x0040): AF2-9 windows are 2D.
- Bit 5 (0x0020): AF2-9 windows are 1D (complement of Bit 6).
- Bit 4 (0x0010): AF2-9 windows are 1D and with the full AC size of 12 pixels.
- Bit 3 (0x0008): No BP window acquired.

- Bit 2 (0x0004): No RP window acquired.
- Bit 1 (0x0002): BP window truncated.
- Bit 0 (0x0001): RP window truncated.

TRANSIT_PROC_FLAGS : Transit-level information on the processing results (short)

This field contains a set of bitwise flags, coded as follows:

- Bits 3 to 0 (`transit_proc_flags` AND 0x000F): **IPD strategy and findings**. As of IDT 15.1 and until IDT 21.0, as well as in IDU until DRC-02, this field was not used (0 was always stored here). As of IDU DRC-03, the following information is provided:

  '0': Normal IPD. Ambiguity not tested or not detected. Default value for all records produced until IDT 21.0 and IDU until DRC-02.

  '1': Multiple Source IPD: As of IDT 21.1, this value is used to indicate that the earlier object in time detected in the window has been selected. In case of 2D windows with larger AC than AL separation between detected peaks: higher μ selected.

  '2': Multiple Source IPD: As of IDT 21.1, this value is used to indicate that the later object in time detected in the window has been selected. In case of 2D windows with larger AC than AL separation between detected peaks: lower μ selected.

  '3': Multiple Source IPD: As of IDU DRC-03, this value is used to indicate that at least one of the windows has a multiple source and that these disturbing sources were masked for the IPD processing; `ccd_proc_flags` should be checked to find which, and how many, windows have been affected.

  '4': DIPD: As of IDU 21.0, this value is used to indicate that this object has been obtained using DIPD implementation and thus include a window model template and combined AF solution.

  '5': TukeyBiWeight IPD: this value is used to indicate that this object has been obtained using TukeyBiWeight IPD initialisation for the internal processing subsystem IDU-AttAc; such initial values do not propagate into the published data.

  Values 6-15 are reserved for future extensions.

- Bits 6 to 4 (0x0070): **Bias and PEM OI strategy baseline**: this field indicates the Bias mitigation mode requested to process this transit. The consumer must check the corresponding `ccd_proc_flags` field to determine the actual mode used for each individual CCD as sometimes some modes can not be applied due to missing input data or calibration limitations.

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Value ‘0’: Default initialisation value for the Bias strategy. If found in an AstroElementary it means that something went wrong and the Bias (or maybe just its flagging) was not done appropriately. Thus, ‘0’ means ‘UNDEFINED/FAILURE’.

Value ‘1’: Only PreScan bias used.

Value ‘2’: PreScan + common NU baseline.

Value ‘3’: Full PEM NU.

Values 4-7 are reserved for future extensions.

- Bits 10 to 7, **colour info**:

  Bit 7 (0x0080), **default colour**: ‘1’ means ‘a default colour used for IPD processing’, which can be caused by some problem in the colour determination or retrieval (e.g. no BP/RP data in IDT or no valid PhotPipe fluxes in IDU).

  Bit 8 (0x0100), **epoch colour**: ‘1’ means ‘epoch (transit-level) colour used for IPD processing’, ‘0’ means ‘mean (source-level) colour used’.

  Bit 9 (0x0200), **single-param colour**: ‘1’ means ‘single-parameter colour used for IPD processing’ (such as `nu_eff_used_in_astrometry`), ‘0’ means ‘multi-parameter colour used’ (such as SSCs).

  Bit 10 (0x0400), **colour problems**: ‘1’ means ‘missing colour information or dubious colour quality’ (e.g. problematic/missing BP/RP data in IDT, NaN or rejected PhotPipe colour in IDU, etc).

  Please note some remarkable combinations of this bit with Bit 7:

  – Non-default + no problems (Bit 7 = 0, Bit 10 = 0): Good colour from catalogue or epoch data
  – Non-default + problems (Bit 7 = 0, Bit 10 = 1): Dubious colour from catalogue or best approximation from partial or problematic epoch data. Implemented only in IDT.
  – Default + no problems (Bit 7 = 1, Bit 10 = 0): Default colour has been used because no colour information was available (i.e. for blacklisted observations). Implemented only in IDU.
  – Default + problems (Bit 7 = 1, Bit 10 = 1): Default colour used instead of the available colour from catalogue or epoch data (i.e. dubious colour rejected).

- Bit 11 (0x0800), **cosmetic effects** flagging: ‘1’ means ‘dead pixel columns affecting some of the windows’.

- Bit 12 (0x1000), **cosmetic effects** flagging: ‘1’ means ‘anomalously low samples found in some of the windows’.

- Bit 13 (0x2000), **cosmetic effects** flagging: ‘1’ means ‘anomalously high samples found in some of the windows’.

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- Bit 14 (0x4000), **problems in initial centroid**: '1' means that problems have been found in the Initial Centroids determination. In the case of IDT, combined with the 'some problem found in the colour determination' flag (bit 10), it may indicate problems in the TransitPredictor (e.g. predicted position outside window, or no attitude available) in which case a default position is used for colour determination.
- Bit 15, **attitude issues**: '1' indicates that some attitude problem was found affecting some of the windows i.e. missing attitude or problems computing AL/AC rates. This flag is only set by IDU which depends on the attitude to access the ELSF and for the combine AF processing in DIPD.

See Chapter 3 of the DR4 online documentation for more information.

CCD_PROC_FLAGS : CCD-level information on the processing results (Short[10] array)

This field contains information about the processing findings for the SM and AF CCDs. It is defined as an array, with one or more sets of 10 components (one per CCD). The indexes of the componenents for a given CCD meet the following expression:

$$s + 10k \quad \begin{cases} s & [0...9] & \text{CCD index [SM...AF9]} \\ k & [0...n] & \text{CCD set index} \end{cases} \tag{7}$$

In the present table, i.e. `bright_source_epoch_astrometry`, only the first set of 10 components is included. The second set is given in table `epoch_image` only. Nevertheless, the description of that second set is provided here for completeness. If a user of `bright_source_epoch_astrometry` wants to utilize the second set, too, then a table join with the corresponding `epoch_image` records is recommended.

The first set of 10 components (SM has index 0, AF9 index 9) of this array contains a set of bitwise flags, coded as follows:

- Bits 3 to 0 (`ccd_proc_flags` AND 0x000F): **IPD result**. It is coded as follows:
  - '0': **Success**; the fitting has succeeded without any problem.
  - '1': **Illegal Amplitude**; invalid amplitude obtained after the initial fitting. It may happen for windows with wrong background determination or similar.
  - '2': **Illegal Coordinate**; invalid AL or/and AC locations obtained in the fitting i.e. locations outside the window. It may happen for windows close to a charge injection, multiple sources, strongly affected by some cosmic ray or other similar issues.

  - '3': **DEPRECATED: Illegal Output**; location estimation outside window.
  - '4': **Maximum Iteration Exceeded**; convergence not reached, but results achieved so far stored in AstroElem. The IPD result may still be reliable enough.
  - '5': **Failed**; diverging iteration instead of converging.
  - '6': **Pending**; calculations in progress; IPD output values not reliable; maybe the processing failed for some reason before finishing the calculations, or the IPD was just initialised, and no processing has occurred yet
  - '7': **No LSF/PSF data**; no result available due to no LSF/PSF data available.
  - '8': **Not Enough Samples**; IPD was attempted but no result is available because at a given iteration it runs out of samples (there are not enough inputs to the equations/matrix system).
  - '9': **Odd LSF/PSF Success**; the fitting has succeeded but using an odd LSF/PSF. The LSF/PSF is considered odd when it is not well defined for some samples. The IPD result may still be reliable enough similar as with the 'Maximum Iteration Exceeded' case.
  - '14': **Not Attempted**; IPD has not been attempted on purpose due to missing gate information, missing CI information or similar or when not enough usable input samples are available due to heavy truncation, CCD edge, complex AOCS update, multiple gates or similar.
  - '15': **Crashed**; calculation finished and crashed for some reason.
  - Values '3' and '10' to '13': Available for future extensions.

- Bit 4 (0x0010): **IPD nominal/non nominal**. '1' indicates the IPD is non-nominal in the sense it has been performed over a reduced window area (details in `ccd_proc_geom`). This flag is only set for AF windows.

- Bit 5 (0x0020): **IPD not available**. '1' indicates that IPD result is not available and thus its output values are not valid. This flag set to true and IPD result equals to NOT_ATTEMPTED (value '14' in bits 3-0) indicates IPD was not attempted on purpose, and with IPD result equals to PENDING (value '6' in bits 3-0) indicates unexpected IPD termination. This flag set to true with any other IPD result should not be possible.

- Bit 6 (0x0040): **Some cosmetic issue**. '1' indicates that there is at least one contaminated or discarded sample in the selected window area. This flag collects the presence of additional discarded or contaminated sample without an explicit CCD level flag thus excluding saturation, cosmics, odd debiased, etc.

- Bit 7 (0x0080): **Cosmic removed**. '1' means that at least one sample in the selected window area has been discarded for being a suspected cosmic.

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- Bit 8 (0x0100): **Saturation removed**. '1' means that at least one sample in the selected window area has been discarded due to saturation.
- Bit 9 (0x0200): **Part of window discarded**. '1' means that at least one sample has been discarded in the full window area.
- Bit 10 (0x0400): **No window**. '1' means 'no samples available' (i.e. all the samples in this window were discarded on board, or this is e.g. AF 9 Row 4, or a deactivated CCD, etc).
- Bit 11 (0x0800): **Non-success InitialCentroid**. '1' means 'the Initial Centroid was not successful, and thus we have used the centre of the window as the start condition of IPD'.
- Bit 12 (0x1000): **Odd Background found**. '1' indicates that the background determined for this window (for at least one of its samples) was odd i.e negative, too large or NaN. This flag may be used to indicate that the fluxes determined for this window are potentially wrong (although its AL/AC centroid may still be useful).
- Bit 13 (0x2000): **fall-back to lower Bias Mitigation mode**. '1' indicates that 'fall-back' to lower mode has occurred for this window (for at least one of its samples). This flag will always be activated for AF1 when FULL_MITIGATION mode is enabled because this mode is not supported in this CCD.
- Bit 14 (0x4000): **non-target source removed**. '1' indicates that non-target sources have been detected and masked in this window.
- Bit 15 (0x8000): **LSF/PSF observation/source parameters clamped**. '1' indicates that the observation/source parameters used to access the LSF/PSF have been clamped to fit the ELSF parameter space.

The second set of 10 components (SM has index 10, AF9 index 19) of this array contains a set of bitwise flags, coded as follows:

- Bit 0 (0x0001): **Odd debiased sample**. '1' indicates that there is at least one debiased sample below $-TDN^2$ limit.
- Bit 1 (0x0002): **Numerical overflow recovery**: '1' indicates that there is at least one sample for which the numerical overflow has been corrected (only applicable in SM and AF1).
- Bit 2 (0x0004): **ELSF reliability mask**: '1' indicates that at least one sample has been discarded in IPD because the corresponding ELSF model is sub-optimal in that region.

- Bits 3 to 5 (0x0038); **Number of saturated samples discarded**. For AF it codes from 0 up to 7 or more masked samples and for SM it codes the position of the most significant bit of the masked sample count and thus coding 0, 1, 2 or 3, 4 to 7, up to 64 samples or more.
- Bit 6 (0x0040): **Background Mode**: '0' indicates background has been fitted along with location and flux parameters whereas '1' indicates that a different overall background estimation has been used i.e. KNN background in regular IPD or specific background module for SIF CF.
- Bit 7, 0x0080), **Charge injection (CI) distance updated**: indicating that the ciDist parameter has been redetermined and it does not match the original one in the raw data.
- Bit 8, 0x0100). **Non-gated CI line**: indicating that at least one non-gated charge injection is within the window.

See Chapter 3 of the DR4 online documentation for more information.

MULTIPEAK : True if the transit window contains multiple peaks; false otherwise (boolean)

If true, the transit is considered as a multi-peak observation, i.e. as having more than one astronomical light source within its CCD windows.

BLENDED : True if the peak for this source is also matched to another source (boolean)

The blended flag indicates the fact that a different source has been matched to the same peak as the present source, no matter if the transit has one or more peaks.

IPD_ERROR_AL : Standard error of IDU IPD centroid AL (float[10] array, Angle[mas])

Standard errors from the calculation of the AL image centroids in SM and AF1-9, as estimated by the centroiding (IPD/DIPD) algorithm in the cyclic preprocessing of astrometric CCD observations. For details see the preprocessing chapter in the DR4 online documentation.

When the IPD algorithm has given some problem (and thus a result could not be obtained), the error is NaN.

IPD_ERROR_AC : Standard error of IDU IPD centroid AC (NaN for 1d windows) (float[10] array, Angle[mas])

Standard errors from the calculation of the AC image centroids in SM and AF1-9, as estimated by the centroiding (IPD/DIPD) algorithm in the cyclic preprocessing of astrometric CCD observations. For details see the preprocessing chapter in the DR4 online documentation.

When the IPD algorithm has given some problem (and thus a result could not be obtained), the error is NaN. Also, the value is NaN for all 1d CCD windows.

G_MAG : G Magnitude, either from CU5 or from on-board estimate (float, Magnitude[mag])

This field is only needed in order to re-apply the AGIS calibrations, so it was put together with the block containing the other parameters used to evaluate the calibration.

It is the G magnitude of the source used in the IPD centroiding: possibly the mean magnitude of the source, possibly the magnitude at the epoch of this particular observation, possibly the on-board magnitude estimate.

In most cases this will be the mean G magnitude from the main photometry processing. Only when that is not available and assuming that we also generate epoch astrometry for sources without G magnitude, then it would be an on-board magnitude. For variable sources it may also happen that in the future this will be the epoch G magnitude from the photometry processing.

If the field contains the magnitude estimated on board, the value is taken from the raw AstroObservation record, but after transformation from the technical telemetry format coding into a straight magnitude.

G_CLASS : Window class used in SM, AF and XP (byte)

This field is only needed in order to re-apply the AGIS calibrations, so it was put together with the block containing the other parameters used to evaluate the calibration.

Window class used in the sampling of SM, AF and BP/RP windows. It can take the following values:

- 0 (window class 0a), G<11.5: full-resolution (2D) windows for AF and BP/RP, and high-resolution (2x2 pixels) window for SM.
- 1 (window class 0b) 11.5<G<13: full-resolution (2D) windows for AF, high-resolution (2x2 pixels) window for SM, and 1D windows for BP/RP.
- 2 (window class 1), 13<G<16: large 1D windows for AF (18 or 12 samples), low-resolution (2x2 samples, or 4x4 pixels) window for SM, and 1D windows for BP/RP.

- 3 (window class 2), G>16: small 1D windows for AF (12 or 6 samples), low-resolution (2x2 samples, or 4x4 pixels) window for SM, and 1D windows for BP/RP.

GATES : Gate information for all CCDs (Byte[10] array)

This field is only needed in order to re-apply the AGIS calibrations, so it was put together with the block containing the other parameters used to evaluate the calibration.

This field contains an array, with one component per CCD (SM has index 0, AF9 index 9) indicating the gate used for the computation of the image parameters: flux and observing time.

Each of the components of the array contains a set of bitwise flags, coded as follows:

- Bit 7: The gate used in IPD is different from the one reported in the raw AstroObservation record. Basically, the true gate value was reconstructed by the Pre-processing (see Chapter **??** of the DR4 online documentation) according to the on-board software (CDB) configuration and the surrounding observations. This only happens when the preprocessing detects an inconsistency between the raw AstroObservation record gate and the expected value from the CDB configuration.
- Bit 6: The window has been acquired with a different gate with respect to the CDB. This flag could be tricky for windows close to the AC boundaries of a CCD, as we can not fully reconstruct the AC coordinates used on board, but we could safely identify 99% of the cases.
- Bit 5 (0x20): '1' means 'multiple gates in this window'.
- Bit 4 (0x10): '1' means 'window affected by a gate release'.
- Bits 3 to 0 (0x0F): these bits indicate the gate number of the central sample of the window (e.g. '0' in case of no gate).

Please note that some of these flags are not applicable for SM (bits 5 and 4), whereas bits 3 to 0 indicate the permanent SM gate configuration, gate 12.

Convention for bits 3 to 0: '1' means the shortest gate, i.e. 2 CCD lines in length, '12' means the longest gate, i.e. 2900 lines in length, and '0' means no gate, i.e. 4500 lines.

See Section **??**, Table **??**, and throughout Chapter **??** of the DR4 online documentation for more information.

SOURCE_DIST_TO_LAST_CI : Source distance to last charge injection (float[10] array, Length & Distance[pixel])

Source distance to last charge injection (CI). This field is only needed in order to re-apply the AGIS calibrations, so it was put together with the block containing the other parameters used to evaluate the calibration.

Distance in TDI time intervals from the trailing edge of the contiguous set of injection samples that include or precede the AL centroid of the source image in this window. If the CI to which these injection samples belong has suffered gate encounter(s), a flag is set in `ccd_acq_flags`. If any sample in the second half of the window contains charge injection, a flag is set in `ccd_acq_flags`.

If CI is not activated, a value of −32768 (MIN_SHORT) is given.

The first component of the array is for SM, which will not have CIs in principle. Thus, the reserved value −32768 will be stored there. The rest will contain the information for AF1 ([1]), etc. until AF9 ([9]). The reserved value (−32768) will also be used there in case the distance to the last CI cannot be calculated (which should never happen — and thus this value should indicate that something has gone wrong in the observation reconstruction).

See Chapter 3 of the DR4 online documentation for more details and for up-to-date information.

A flag mentioned in `ccd_acq_flags` indicates if there is a CI in the 2nd half of the window. If there is a conflict between the CI and a gate, an appropriate bit in `ccd_acq_flags` will also be set.

AC_RATE : AC rates for the transit time (float, Angular Velocity[pix/s])

This field is only needed in order to re-apply the AGIS calibrations, so it was put together with the block containing the other parameters used to evaluate the calibration.

It is the speed tof the AC motion of the image centroid on the focal plane, calculated as the mean AC rate over all CCDs (only one value per FoV transit).

SUB_PIXEL_COORD : Subpixel coordinate AL (float[10] array, Length & Distance[pixel])

This field is only needed in order to re-apply the AGIS calibrations, so it was put together with the block containing the other parameters used to evaluate the calibration.

The subpixel AL coordinate for DR4 is computed as:

subpixel = mod(((transit time (OBMT) in seconds)/(TDI period in seconds)), 1.0).

MU : AC coordinate of the image (or window centre) on a CCD (float[10] array, Length & Distance[pixel])

This field is only needed in order to re-apply the AGIS calibrations, so it was put together with the block containing the other parameters used to evaluate the calibration.

This is the AC pixel coordinate of the image on the given CCD. Precisely, for DR4 it is the window centre (both for 1D and 2D windows), truncated to the nearest integer pixel. For DR5 it will be the 'true' image centroid AC, as predicted from source parameters, geometric calibration and attitude.

Note that the μ coordinate contained in the transit identifier is only for AF1, and it is always a window coordinate, not an actual or predicted image centroid.

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7 Cross match

As indicated in the below subsection titles by the labels [TAP] and [DataLink], these products are served from the ESA Gaia Archive using two different protocols, the Table Access Protocol (TAP) for tabular products and the DataLink protocol for other products such as spectra, light curves, MCMC samples, etc. For more details please see Section **??**.

7.1 CATWISE2020_BEST_NEIGHBOUR [TAP]

Table `catwise2020_best_neighbour` lists each matched Gaia object with its best neighbour in the external catalogue. The cross-match algorithm is not symmetric and searches for Gaia source counterparts in CatWISE2020. The best neighbour is chosen among good neighbours as the one with the highest value of the figure of merit, which evaluates the ratio between two opposite models/hypotheses: the counterpart candidate is a match or it is found by chance. Good neighbours are nearby objects in the external catalogue whose position is compatible within position errors with the Gaia target. The cross-match algorithm is positional and exploits the full 5 parameter covariance matrix of the Gaia astrometric solution when available and the external catalogue positions and position errors. In addition it takes into account the external catalogue environment using the local density.

Please note that the cross-match algorithm is a trade-off between multiple requirements, in particular between completeness and correctness. It is thus not limited to a simple cone search.

Reference papers:
Marrese et al. (2017)
Marrese et al. (2019)

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Columns description:

SOURCE_ID : Unique Gaia source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

ORIGINAL_EXT_SOURCE_ID : Original External Catalogue source identifier (string)

The unique source identifier in the original external catalogue.

ANGULAR_DISTANCE : Angular Distance between the two sources (float, Angle[arcsec])

Angular distance between a Gaia source and its best neighbour in the external catalogue.

XM_FLAG : Cross-match algorithm flag (short)

This flag is a bitmask indicating the details of the cross-match algorithm used for the source.

`xm_flag` values and descriptions:

- 0 = Initial value; resets all bits.
- 1 = The external catalogue object has one or more multiples. This means that there is at least another object with exactly the same astrometry.
- 2 = The external catalogue object has one or more suspected duplicates. This means that there is at least another object much closer than the catalogue angular resolution.
- 4 = The external catalogue object is resolved in Gaia.
- 8 = The Gaia object has a five parameters astrometric solution.
- 16 = The Gaia object has a two parameters astrometric solution.
- 32 = The external catalogue object is matched only after the special treatment for sources with under-estimated position errors.
- 64 = The external catalogue object is matched only after the special treatment for Gaia sources with large values of `ipd_gof_harmonic_amplitude` or `ruwe`.

For detailed documentation about `xm_flag` and the cross-match algorithm, see Chapter **??**. For a detailed description of `gaia_source` columns `ipd_gof_harmonic_amplitude` and `ruwe`, see Chapter **??**.

NUMBER_OF_NEIGHBOURS : Number of neighbours in External Catalogue (byte)

Number of sources in the external catalogue which match the Gaia source within position errors. The identifiers of all the neighbours can be found in the corresponding neighbourhood table.

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NUMBER_OF_MATES : Number of mates in Gaia Catalogue (byte)

Number of other Gaia sources that have as best neighbour the same external catalogue source. In case there are no other Gaia sources with the same best neighbour in the external catalogue, the number of mates is equal to zero. Given the Gaia high angular resolution, it will be the case that what appears as a single object in an external catalogue may be resolved by Gaia and as such will be the best match of more than one Gaia object.

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7.2 CATWISE2020_NEIGHBOURHOOD [TAP]

Table `catwise2020_neighbourhood` includes all good neighbours for each matched Gaia object. A good neighbour for a given Gaia source is a nearby object in the external catalogue whose position is compatible (within position errors) with the Gaia target. The cross-match algorithm is not symmetric and searches for Gaia source counterparts in CatWISE2020. The cross-match algorithm is positional and exploits the full 5 parameter covariance matrix of the Gaia astrometric solution when available and the external catalogue positions and position errors. In addition it takes into account the external catalogue environment using the local density.

Please note that the cross-match algorithm is a trade-off between multiple requirements, in particular between completeness and correctness. It is thus not limited to a simple cone search.

Reference papers:
Marrese et al. (2017)
Marrese et al. (2019)

Draft

Columns description:

SOURCE_ID : Unique Gaia source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

ORIGINAL_EXT_SOURCE_ID : Original External Catalogue source identifier (string)

The unique source identifier in the original external catalogue.

ANGULAR_DISTANCE : Angular Distance between the two sources (float, Angle[arcsec])

Angular distance between a Gaia source and its good neighbours in the external catalogue.

SCORE : Score of neighbours (double)

Score of a given neighbour. The score is a figure of merit based on geometric distance and local density of the external catalogue: the higher the score, the more probable the match is.

XM_FLAG : Cross-match algorithm flag (short)

This flag is a bitmask indicating the details of the cross-match algorithm used for the source.

`xm_flag` values and descriptions:

- 0 = Initial value; resets all bits.
- 1 = The external catalogue object has one or more multiples. This means that there is at least another object with exactly the same astrometry.
- 2 = The external catalogue object has one or more suspected duplicates. This means that there is at least another object much closer than the catalogue angular resolution.
- 4 = The external catalogue object is resolved in Gaia.
- 8 = The Gaia object has a five parameters astrometric solution.
- 16 = The Gaia object has a two parameters astrometric solution.
- 32 = The external catalogue object is matched only after the special treatment for sources with under-estimated position errors.
- 64 = The external catalogue object is matched only after the special treatment for Gaia sources with large values of `ipd_gof_harmonic_amplitude` or `ruwe`.

For detailed documentation about `xm_flag` and the cross-match algorithm, see Chapter **??**. For a detailed description of `gaia_source` columns `ipd_gof_harmonic_amplitude` and `ruwe`, see Chapter **??**.

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7.3 DR3_NEIGHBOURHOOD [TAP]

This table is intended to help in the identification of which sources in Gaia E/DR3 and DR4 data releases correspond to the same astrophysical object. It provides proximity (separation corrected by proper motion) and photometric (G magnitude difference) similarity proxies.

Within the neighbourhood of a given DR4 source there may be none, one or several possible counterparts in E/DR3 indicated by rows in this table. Most sources associated with the same object can be identified by sharing the same source_id in both releases. However, different indentifiers are assigned in other scenarios. This occasional source confusion is an inevitable consequence of the merging, splitting and deletion of identifiers introduced in previous releases during the DR4 processing.

The contents of this table could be reproduced using the Gaia Archive. An example ADQL query for the first 100,000 random DR4 sources is provided below.

```
with dr4 as (
  select
    source_id,
      ra + 1. / 3600.e3 * to_double(coalesce(pmra, 0)) * -1.5
             / cos(radians(dec)) as ra,
    dec + 1. / 3600.e3 * to_double(coalesce(pmdec, 0)) * -1.5 as dec,
    to_boolean(if_then_else(pmra > -1.e9, 1, 0))
      as proper_motion_propagation,
    phot_g_mean_mag
  from gaiadr4.gaia_source as dr4
  where random_index < 100000
), xmatch as (
  select
    dr4.source_id,
    dr3.source_id as dr3_source_id,
    distance(dr3.ra, dr3.dec, dr4.ra, dr4.dec) * 3600.e3
      as angular_distance,
    dr4.phot_g_mean_mag - dr3.phot_g_mean_mag as magnitude_difference,
    proper_motion_propagation
  from dr4
  join gaiadr3.gaia_source as dr3
    on distance(dr4.ra, dr4.dec, dr3.ra, dr3.dec) < 2. / 3600
)
select
  source_id,
```

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```
  dr3_source_id,
  angular_distance,
  magnitude_difference,
  proper_motion_propagation,
  to_boolean(
    if_then_else(angular_distance < 60,
      if_then_else(-0.25 <= magnitude_difference,
        if_then_else(magnitude_difference <= 0.35, 1, 0),
      0),
    0)
  ) as probable_match
from xmatch
```

For more details of the procedure used to create this crossmatch, see Chapter **??** in the online documentation.

Columns description:

SOURCE_ID : Source identifier in Gaia DR4 (long)

Source identifier assigned during DR4 processing for an astrophysical object in close proximity to one or more (including possibly the same) objects in E/DR3.

DR3_SOURCE_ID : Source identifier in Gaia E/DR3 (long)

Source identifier assigned during E/DR3 processing for an astrophysical object in close proximity to (possibly) the same object in DR4.

ANGULAR_DISTANCE : Angular distance between the two sources (float, Angle[mas])

Angular (great–circle) distance on the sky between the E/DR3 and DR4 coordinates. DR4 proper motions are used to correct DR4 positions when available.

MAGNITUDE_DIFFERENCE : G band magnitude difference between the sources (float, Magnitude[mag])

G band magnitude difference between the sources based on the individual magnitudes materialised in `gaia_source` at the time of the respective data releases. The sense of the difference is

DR4 minus E/DR3.

PROPER_MOTION_PROPAGATION : Have DR4 coordinates been proper motion corrected? (boolean)

This flag indicates whether the DR4 source has proper motion estimates. If yes, they have been used to propagate DR4 coordinates to the E/DR3 epoch (J2016.0). A simple linear correction has been applied, due to the epoch proximity between both catalogues.

PROBABLE_MATCH : Are both sources probably associated with the same astronomical object? (boolean)

Flag indicating whether both sources probably refer to the same astronomical object.

This criterion is probably conservative and has two components:

- Astrometric: separation below 60 mas. DR4 → DR3 proper motion propagation done when possible.
- Photometric: -0.25 ≤ $G_{\mathrm{DR4}} - G_{\mathrm{DR3}}$ ≤ 0.35. Note it is asymmetric, due to the G passband and zero point evolution from DR3 to DR4.

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7.4 GAIA_ESO_SURVEY_BEST_NEIGHBOUR [TAP]

Table `gaia_eso_survey_best_neighbour` lists each matched external catalogue object with its best neighbour in Gaia. The cross-match algorithm is not symmetric and searches Gaia-ESO Survey sources counterparts in Gaia. The best neighbour is chosen among good neighbours as the one with the highest value of the figure of merit, which evaluates the ratio between two opposite models/hypotheses: the counterpart candidate is a match or it is found by chance. Good neighbours are nearby objects in Gaia whose position is compatible within position errors with the external catalogue target. The cross-match algorithm is positional and exploits the full 5 parameter covariance matrix of the Gaia astrometric solution when available and the external catalogue positions and position errors. In addition it takes into account the Gaia environment using the local density.

Please note that the cross-match algorithm is a trade-off between multiple requirements, in particular between completeness and correctness. It is thus not limited to a simple cone search.

Reference papers:
Marrese et al. (2017)
Marrese et al. (2019)

Draft

Columns description:

SOURCE_ID : Unique Gaia source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

ORIGINAL_EXT_SOURCE_ID : Original External Catalogue source identifier (string)

The unique source identifier in the original external catalogue.

ANGULAR_DISTANCE : Angular Distance between the two sources (float, Angle[arcsec])

Angular distance between a Gaia source and its best neighbour in the external catalogue.

XM_FLAG : Cross-match algorithm flag (short)

This flag is a bitmask indicating the details of the cross-match algorithm used for the source.

`xm_flag` values and descriptions:

- 0 = Initial value; resets all bits.
- 1 = The external catalogue object has one or more multiples. This means that there is at least another object with exactly the same astrometry.
- 2 = The external catalogue object has one or more suspected duplicates. This means that there is at least another object much closer than the catalogue angular resolution.
- 4 = The external catalogue object is resolved in Gaia.
- 8 = The Gaia object has a five parameters astrometric solution.
- 16 = The Gaia object has a two parameters astrometric solution.
- 32 = The external catalogue object is matched only after the special treatment for sources with under-estimated position errors.
- 64 = The external catalogue object is matched only after the special treatment for Gaia sources with large values of `ipd_gof_harmonic_amplitude` or `ruwe`.

For detailed documentation about `xm_flag` and the cross-match algorithm, see Chapter **??**. For a detailed description of `gaia_source` columns `ipd_gof_harmonic_amplitude` and `ruwe`, see Chapter **??**.

GAIA_ESO_SURVEY_OID : External Catalogue source identifier (int)

The additional numeric source identifier of the external catalogue, increasing with Declination.

NUMBER_OF_NEIGHBOURS : Number of neighbours in Gaia Catalogue (byte)

Number of sources in the Gaia catalogue which match the external catalogue source within position errors. The identifiers of all the neighbours can be found in the corresponding neighbourhood table.

Draft

7.5 GAIA_ESO_SURVEY_JOIN [TAP]

Convenience table to be used to join Gaia-ESO Survey catalogue with the cross-match results. The table links the external catalogue original source identifier (`original_ext_source_id`) to the corresponding additional numerical identifier (`gaia_eso_survey_oid`).

Both `original_ext_source_id` and `gaia_eso_survey_oid` are present in the cross-match output tables (`gaia_eso_survey_best_neighbour` and `gaia_eso_survey_neighbourhood`).

However, in case there are suspected duplicates in the external catalogue, different `original_ext_source_id` will correspond to the same `gaia_eso_survey_oid`. In the cross-match output table only the `original_ext_source_id` of the source with the best astrometry among the suspected duplicates will be listed.

In practice, users may use the `original_ext_source_id` in the original catalogue to find the matching source with the best astrometry. Users interested to find all matching suspected duplicates should instead use the `gaia_eso_survey_oid` in the join with the cross-match result tables.

See Chapter **??**, for more details on the duplicates in the external catalogues and their treatment in the cross-match computations.

Columns description:

ORIGINAL_EXT_SOURCE_ID : Original External Catalogue source identifier (string)

The unique source identifier in the original external catalogue.

GAIA_ESO_SURVEY_OID : External Catalogue source identifier (int)

The additional numeric source identifier of the external catalogue, increasing with Declination.

Draft

7.6 GAIA_ESO_SURVEY_NEIGHBOURHOOD [TAP]

Table `gaia_eso_survey_neighbourhood` includes all good neighbours for each matched Gaia-ESO Survey object. A good neighbour for a given Gaia-ESO Survey source is a nearby Gaia object whose position is compatible (within position errors) with the Gaia-ESO Survey target. The cross-match algorithm is not symmetric and searches Gaia-ESO Survey sources counterparts in Gaia. The cross-match algorithm is positional and exploits the full 5 parameter covariance matrix of the Gaia astrometric solution when available and the external catalogue positions and position errors. In addition it takes into account the Gaia environment using the local density.

Please note that the cross-match algorithm is a trade-off between multiple requirements, in particular between completeness and correctness. It is thus not limited to a simple cone search.

Reference papers:
Marrese et al. (2017)
Marrese et al. (2019)

Draft

Columns description:

SOURCE_ID : Unique Gaia source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

ORIGINAL_EXT_SOURCE_ID : Original External Catalogue source identifier (string)

The unique source identifier in the original external catalogue.

ANGULAR_DISTANCE : Angular Distance between the two sources (float, Angle[arcsec])

Angular distance between a Gaia source and its good neighbours in the external catalogue.

SCORE : Score of neighbours (double)

Score of a given neighbour. The score is a figure of merit based on geometric distance and local density of the external catalogue: the higher the score, the more probable the match is.

XM_FLAG : Cross-match algorithm flag (short)

This flag is a bitmask indicating the details of the cross-match algorithm used for the source.

`xm_flag` values and descriptions:

- 0 = Initial value; resets all bits.
- 1 = The external catalogue object has one or more multiples. This means that there is at least another object with exactly the same astrometry.
- 2 = The external catalogue object has one or more suspected duplicates. This means that there is at least another object much closer than the catalogue angular resolution.
- 4 = The external catalogue object is resolved in Gaia.
- 8 = The Gaia object has a five parameters astrometric solution.
- 16 = The Gaia object has a two parameters astrometric solution.
- 32 = The external catalogue object is matched only after the special treatment for sources with under-estimated position errors.
- 64 = The external catalogue object is matched only after the special treatment for Gaia sources with large values of `ipd_gof_harmonic_amplitude` or `ruwe`.

For detailed documentation about `xm_flag` and the cross-match algorithm, see Chapter **??**. For a detailed description of `gaia_source` columns `ipd_gof_harmonic_amplitude` and `ruwe`, see Chapter **??**.

GAIA_ESO_SURVEY_OID : External Catalogue source identifier (int)

The additional numeric source identifier of the external catalogue, increasing with Declination.

Draft

7.7 HIPPARCOS2_BEST_NEIGHBOUR [TAP]

Table `hipparcos2_best_neighbour` lists each matched external catalogue object with its best neighbour in Gaia. The cross-match algorithm is not symmetric and searches Hipparcos2 sources counterparts in Gaia. The best neighbour is chosen among good neighbours as the one with the highest value of the figure of merit, which evaluates the ratio between two opposite models/hypotheses: the counterpart candidate is a match or it is found by chance. Good neighbours are nearby objects in Gaia whose position is compatible within position errors with the external catalogue target. The cross-match algorithm is positional and exploits the full 5 parameter covariance matrix of the Gaia astrometric solution when available and the external catalogue positions and position errors. In addition it takes into account the Gaia environment using the local density.

Please note that the cross-match algorithm is a trade-off between multiple requirements, in particular between completeness and correctness. It is thus not limited to a simple cone search.

Reference papers:
Marrese et al. (2017)
Marrese et al. (2019)

Draft

Columns description:

SOURCE_ID : Unique Gaia source identifier (long)

Unique identifier of the Gaia source, the attribute corresponds to gaia_source.source_id

ORIGINAL_EXT_SOURCE_ID : Original External Catalogue source identifier (int)

The unique source identifier in the original External catalogue.

ANGULAR_DISTANCE : Angular Distance between the two sources (float, Angle[arcsec])

Angular distance between a Gaia source and its best neighbour in the External Catalogue

NUMBER_OF_NEIGHBOURS : Number of neighbours in Gaia Catalogue (byte)

Number of sources in the Gaia Catalogue which match the External Catalogue source within

position errors.
The identifiers of all the neighbours can be found in the Neighbourhood table.

XM_FLAG : Cross-match algorithm flag (short)

This flag is a bitmask indicating the details of the cross-match algorithm used for the source.

xm_flag values and descriptions:

- 0 = Initial value; resets all bits.
- 1 = The external catalogue object has one or more multiples. This means that there is at least another object with exactly the same astrometry.
- 2 = The external catalogue object has one or more suspected duplicates. This means that there is at least another object much closer than the catalogue angular resolution.
- 4 = The external catalogue object is resolved in Gaia.
- 8 = The Gaia object has a five parameters astrometric solution.
- 16 = The Gaia object has a two parameters astrometric solution.
- 32 = The external catalogue object is matched only after the special treatment for sources with under-estimated position errors.
- 64 = The external catalogue object is matched only after the special treatment for Gaia sources with large values of `ipd_gof_harmonic_amplitude` or `ruwe`.

For detailed documentation about `xm_flag` and the cross-match algorithm, see Chapter **??**. For a detailed description of `gaia_source` columns `ipd_gof_harmonic_amplitude` and `ruwe`, see Chapter **??**.

Draft

7.8 HIPPARCOS2_NEIGHBOURHOOD [TAP]

Table `hipparcos2_neighbourhood` includes all good neighbours for each matched Hipparcos2 object. A good neighbour for a given Hipparcos2 source is a nearby Gaia object whose position is compatible (within position errors) with the Hipparcos2 target. The cross-match algorithm is not symmetric and searches Hipparcos2 sources counterparts in Gaia. The cross-match algorithm is positional and exploits the full 5 parameter covariance matrix of the Gaia astrometric solution when available and the external catalogue positions and position errors. In addition it takes into account the Gaia environment using the local density.

Please note that the cross-match algorithm is a trade-off between multiple requirements, in particular between completeness and correctness. It is thus not limited to a simple cone search.

Reference papers:
Marrese et al. (2017)
Marrese et al. (2019)

Draft

Columns description:

SOURCE_ID : Unique Gaia source identifier (long)

Unique identifier of the Gaia source, the attribute corresponds to `gaia_source.source_id`.

ORIGINAL_EXT_SOURCE_ID : Original External Catalogue source identifier (int)

The unique source identifier in the original external catalogue.

ANGULAR_DISTANCE : Angular Distance between the two sources (float, Angle[arcsec])

Angular distance between a Gaia source and its good neighbours in the external catalogue.

SCORE : Score of neighbours (double)

The score is a figure of merit based on geometric distance and local density of the external catalogue: the higher the score, the more probable the match is.

XM_FLAG : Cross-match algorithm flag (short)

This flag is a bitmask indicating the details of the cross-match algorithm used for the source.

`xm_flag` values and descriptions:

- 0 = Initial value; resets all bits.
- 1 = The external catalogue object has one or more multiples. This means that there is at least another object with exactly the same astrometry.
- 2 = The external catalogue object has one or more suspected duplicates. This means that there is at least another object much closer than the catalogue angular resolution.
- 4 = The external catalogue object is resolved in Gaia.
- 8 = The Gaia object has a five parameters astrometric solution.
- 16 = The Gaia object has a two parameters astrometric solution.
- 32 = The external catalogue object is matched only after the special treatment for sources with under-estimated position errors.
- 64 = The external catalogue object is matched only after the special treatment for Gaia sources with large values of `ipd_gof_harmonic_amplitude` or `ruwe`.

For detailed documentation about `xm_flag` and the cross-match algorithm, see Chapter **??**. For a detailed description of `gaia_source` columns `ipd_gof_harmonic_amplitude` and `ruwe`, see Chapter **??**.

Draft

7.9 PANSTARRS1_BEST_NEIGHBOUR [TAP]

Table `panstarrs1_best_neighbour` lists each matched Gaia object with its best neighbour in the external catalogue. The cross-match algorithm is not symmetric and searches Gaia sources counterparts in Pan-STARRS1 DR1. The best neighbour is chosen among good neighbours as the one with the highest value of the figure of merit, which evaluates the ratio between two opposite models/hypotheses: the counterpart candidate is a match or it is found by chance. Good neighbours are nearby objects in the external catalogue whose position is compatible within position errors with the Gaia target. The cross-match algorithm is positional and exploits the full 5 parameter covariance matrix of the Gaia astrometric solution when available and the external catalogue positions and position errors. In addition it takes into account the external catalogue environment using the local density.

Please note that the cross-match algorithm is a trade-off between multiple requirements, in particular between completeness and correctness. It is thus not limited to a simple cone search.

Reference papers:
Marrese et al. (2017)
Marrese et al. (2019)

Draft

Columns description:

SOURCE_ID : Unique Gaia source identifier (long)

Unique identifier of the Gaia source, the attribute corresponds to `gaia_source.source_id`.

CLEAN_PANSTARRS1_OID : External Catalogue source identifier (long)

The additional numeric source identifier of the external catalogue, increasing with Declination.

ORIGINAL_EXT_SOURCE_ID : Original External Catalogue source identifier (long)

The unique source identifier in the original external catalogue.

ANGULAR_DISTANCE : Angular Distance between the two sources (float, Angle[arcsec])

Angular distance between a Gaia source and its best neighbour in the external catalogue.

NUMBER_OF_NEIGHBOURS : Number of neighbours in External Catalogue (byte)

Number of sources in the external catalogue which match the Gaia source within position errors. The identifiers of all the neighbours can be found in the corresponding neighbourhood table.

NUMBER_OF_MATES : Number of mates in Gaia Catalogue (byte)

Number of other Gaia sources that have as best neighbour the same external catalogue source. In case there are no other Gaia sources with the same best neighbour in the external catalogue, the number of mates is equal to zero. Given the Gaia high angular resolution, it will be the case that what appears as a single object in an external catalogue may be resolved by Gaia and as such will be the best match of more than one Gaia object.

XM_FLAG : Cross-match algorithm flag (short)

This flag is a bitmask indicating the details of the cross-match algorithm used for the source.

`xm_flag` values and descriptions:

- 0 = Initial value; resets all bits.
- 1 = The external catalogue object has one or more multiples. This means that there is at least another object with exactly the same astrometry.
- 2 = The external catalogue object has one or more suspected duplicates. This means that there is at least another object much closer than the catalogue angular resolution.
- 4 = The external catalogue object is resolved in Gaia.
- 8 = The Gaia object has a five parameters astrometric solution.
- 16 = The Gaia object has a two parameters astrometric solution.
- 32 = The external catalogue object is matched only after the special treatment for sources with under-estimated position errors.
- 64 = The external catalogue object is matched only after the special treatment for Gaia sources with large values of `ipd_gof_harmonic_amplitude` or `ruwe`.

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For detailed documentation about `xm_flag` and the cross-match algorithm, see Chapter **??**. For a detailed description of `gaia_source` columns `ipd_gof_harmonic_amplitude` and `ruwe`, see Chapter **??**.

Draft

7.10 PANSTARRS1_JOIN [TAP]

Convenience table to be used to join Pan-STARRS1 DR1.1 catalogue with the cross-match results. The table links the external catalogue original source identifier (`original_ext_source_id`) to the corresponding additional numerical identifier (`clean_panstarrs1_oid`).

Both `original_ext_source_id` and `clean_panstarrs1_oid` are present in the cross-match output tables (`panstarrs1_best_neighbour` and `panstarrs1_neighbourhood`).

However, in case there are suspected duplicates in the external catalogue, different `original_ext_source_id` will correspond to the same `clean_panstarrs1_oid`. In the cross-match output table only the `original_ext_source_id` of the source with the best astrometry among the suspected duplicates will be listed.

In practice, users may use the `original_ext_source_id` in the original catalogue to find the matching source with the best astrometry. Users interested to find all matching suspected duplicates should instead use the `clean_panstarrs1_oid` in the join with the cross-match result tables.

See Chapter **??**, for more details on the duplicates in the external catalogues and their treatment in the cross-match computations.

Columns description:

ORIGINAL_EXT_SOURCE_ID : Original External Catalogue source identifier (long)

The unique source identifier in the original external catalogue.

CLEAN_PANSTARRS1_OID : External Catalogue source identifier (long)

The additional numeric source identifier of the external catalogue, increasing with Declination.

Draft

7.11 PANSTARRS1_NEIGHBOURHOOD [TAP]

Table `panstarrs1_neighbourhood` includes all good neighbours for each matched Gaia object. A good neighbour for a given Gaia source is a nearby object in the external catalogue whose position is compatible (within position errors) with the Gaia target. The cross-match algorithm is not symmetric and searches Gaia sources counterparts in Pan-STARRS1 DR1. The cross-match algorithm is positional and exploits the full 5 parameter covariance matrix of the Gaia astrometric solution when available and the external catalogue positions and position errors. In addition it takes into account the external catalogue environment using the local density.

Please note that the cross-match algorithm is a trade-off between multiple requirements, in particular between completeness and correctness. It is thus not limited to a simple cone search.

Reference papers:
Marrese et al. (2017)
Marrese et al. (2019)

Draft

Columns description:

SOURCE_ID : Unique Gaia source identifier (long)

Unique identifier of the Gaia source, the attribute corresponds to `gaia_source.source_id`.

CLEAN_PANSTARRS1_OID : External Catalogue source identifier (long)

The additional numeric source identifier of the external catalogue, increasing with Declination.

ORIGINAL_EXT_SOURCE_ID : Original External Catalogue source identifier (long)

The unique source identifier in the original external catalogue.

ANGULAR_DISTANCE : Angular Distance between the two sources (float, Angle[arcsec])

Angular distance between a Gaia source and its good neighbours in the external catalogue.

SCORE : Score of neighbours (double)

The score is a figure of merit based on geometric distance and local density of the external catalogue: the higher the score, the more probable the match is.

XM_FLAG : Cross-match algorithm flag (short)

This flag is a bitmask indicating the details of the cross-match algorithm used for the source.

`xm_flag` values and descriptions:

- 0 = Initial value; resets all bits.
- 1 = The external catalogue object has one or more multiples. This means that there is at least another object with exactly the same astrometry.
- 2 = The external catalogue object has one or more suspected duplicates. This means that there is at least another object much closer than the catalogue angular resolution.
- 4 = The external catalogue object is resolved in Gaia.
- 8 = The Gaia object has a five parameters astrometric solution.
- 16 = The Gaia object has a two parameters astrometric solution.
- 32 = The external catalogue object is matched only after the special treatment for sources with under-estimated position errors.
- 64 = The external catalogue object is matched only after the special treatment for Gaia sources with large values of `ipd_gof_harmonic_amplitude` or `ruwe`.

For detailed documentation about `xm_flag` and the cross-match algorithm, see Chapter **??**. For a detailed description of `gaia_source` columns `ipd_gof_harmonic_amplitude` and `ruwe`, see Chapter **??**.

Draft

7.12 RAVEDR6_BEST_NEIGHBOUR [TAP]

Table `ravedr6_best_neighbour` lists each matched external catalogue object with its best neighbour in Gaia. The cross-match algorithm is not symmetric and searches RAVE DR6 sources counterparts in Gaia. The best neighbour is chosen among good neighbours as the one with the highest value of the figure of merit, which evaluates the ratio between two opposite models/hypotheses: the counterpart candidate is a match or it is found by chance. Good neighbours are nearby objects in Gaia whose position is compatible within position errors with the external catalogue target. The cross-match algorithm is positional and exploits the full 5 parameter covariance matrix of the Gaia astrometric solution when available and the external catalogue positions and position errors. In addition it takes into account the Gaia environment using the local density.

Please note that the cross-match algorithm is a trade-off between multiple requirements, in particular between completeness and correctness. It is thus not limited to a simple cone search.

Reference papers:
Marrese et al. (2017)
Marrese et al. (2019)

Draft

Columns description:

SOURCE_ID : Unique Gaia source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

ORIGINAL_EXT_SOURCE_ID : Original External Catalogue source identifier (string)

The unique source identifier in the original external catalogue.

ANGULAR_DISTANCE : Angular Distance between the two sources (float, Angle[arcsec])

Angular distance between a Gaia source and its best neighbour in the external catalogue.

XM_FLAG : Cross-match algorithm flag (short)

This flag is a bitmask indicating the details of the cross-match algorithm used for the source.

`xm_flag` values and descriptions:

- 0 = Initial value; resets all bits.
- 1 = The external catalogue object has one or more multiples. This means that there is at least another object with exactly the same astrometry.
- 2 = The external catalogue object has one or more suspected duplicates. This means that there is at least another object much closer than the catalogue angular resolution.
- 4 = The external catalogue object is resolved in Gaia.
- 8 = The Gaia object has a five parameters astrometric solution.
- 16 = The Gaia object has a two parameters astrometric solution.
- 32 = The external catalogue object is matched only after the special treatment for sources with under-estimated position errors.
- 64 = The external catalogue object is matched only after the special treatment for Gaia sources with large values of `ipd_gof_harmonic_amplitude` or `ruwe`.

For detailed documentation about `xm_flag` and the cross-match algorithm, see Chapter **??**. For a detailed description of `gaia_source` columns `ipd_gof_harmonic_amplitude` and `ruwe`, see Chapter **??**.

RAVEDR6_OID : External Catalogue source identifier (int)

The additional numeric source identifier of the external catalogue, increasing with Declination.

NUMBER_OF_NEIGHBOURS : Number of neighbours in Gaia Catalogue (byte)

Number of sources in the Gaia catalogue which match the external catalogue source within position errors. The identifiers of all the neighbours can be found in the corresponding neighbourhood table.

Draft

7.13 RAVEDR6_JOIN [TAP]

Convenience table to be used to join RAVE DR6 catalogue with the cross-match results. The table links the external catalogue original source identifier (`original_ext_source_id`) to the corresponding additional numerical identifier (`ravedr6_oid`).

Both `original_ext_source_id` and `ravedr6_oid` are present in the cross-match output tables (`ravedr6_best_neighbour` and `ravedr6_neighbourhood`).

However, in case there are suspected duplicates in the external catalogue, different `original_ext_source_id` will correspond to the same `ravedr6_oid`. In the cross-match output table only the `original_ext_source_id` of the source with the best astrometry among the suspected duplicates will be listed.

In practice, users may use the `original_ext_source_id` in the original catalogue to find the matching source with the best astrometry. Users interested to find all matching suspected duplicates should instead use the `ravedr6_oid` in the join with the cross-match result tables.

See Chapter **??**, for more details on the duplicates in the external catalogues and their treatment in the cross-match computations.

Columns description:

ORIGINAL_EXT_SOURCE_ID : Original External Catalogue source identifier (string)

The unique source identifier in the original external catalogue.

RAVEDR6_OID : External Catalogue source identifier (int)

The additional numeric source identifier of the external catalogue, increasing with Declination.

Draft

7.14 RAVEDR6_NEIGHBOURHOOD [TAP]

Table `ravedr6_neighbourhood` includes all good neighbours for each matched RAVE DR6 object. A good neighbour for a given RAVE DR6 source is a nearby Gaia object whose position is compatible (within position errors) with the RAVE DR6 target. The cross-match algorithm is not symmetric and searches RAVE DR6 sources counterparts in Gaia. The cross-match algorithm is positional and exploits the full 5 parameter covariance matrix of the Gaia astrometric solution when available and the external catalogue positions and position errors. In addition it takes into account the Gaia environment using the local density.

Please note that the cross-match algorithm is a trade-off between multiple requirements, in particular between completeness and correctness. It is thus not limited to a simple cone search.

Reference papers:
Marrese et al. (2017)
Marrese et al. (2019)

Draft

Columns description:

SOURCE_ID : Unique Gaia source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

ORIGINAL_EXT_SOURCE_ID : Original External Catalogue source identifier (string)

The unique source identifier in the original external catalogue.

ANGULAR_DISTANCE : Angular Distance between the two sources (float, Angle[arcsec])

Angular distance between a Gaia source and its good neighbours in the external catalogue.

SCORE : Score of neighbours (double)

Score of a given neighbour. The score is a figure of merit based on geometric distance and local density of the external catalogue: the higher the score, the more probable the match is.

XM_FLAG : Cross-match algorithm flag (short)

This flag is a bitmask indicating the details of the cross-match algorithm used for the source.

`xm_flag` values and descriptions:

- 0 = Initial value; resets all bits.
- 1 = The external catalogue object has one or more multiples. This means that there is at least another object with exactly the same astrometry.
- 2 = The external catalogue object has one or more suspected duplicates. This means that there is at least another object much closer than the catalogue angular resolution.
- 4 = The external catalogue object is resolved in Gaia.
- 8 = The Gaia object has a five parameters astrometric solution.
- 16 = The Gaia object has a two parameters astrometric solution.
- 32 = The external catalogue object is matched only after the special treatment for sources with under-estimated position errors.
- 64 = The external catalogue object is matched only after the special treatment for Gaia sources with large values of `ipd_gof_harmonic_amplitude` or `ruwe`.

For detailed documentation about `xm_flag` and the cross-match algorithm, see Chapter **??**. For a detailed description of `gaia_source` columns `ipd_gof_harmonic_amplitude` and `ruwe`, see Chapter **??**.

RAVEDR6_OID : External Catalogue source identifier (int)

The additional numeric source identifier of the external catalogue, increasing with Declination.

Draft

7.15 SDSSDR13_BEST_NEIGHBOUR [TAP]

Table `sdssdr13_best_neighbour` lists each matched Gaia object with its best neighbour in the external catalogue. The cross-match algorithm is not symmetric and searches Gaia sources counterparts in SDSS DR13. The best neighbour is chosen among good neighbours as the one with the highest value of the figure of merit, which evaluates the ratio between two opposite models/hypotheses: the counterpart candidate is a match or it is found by chance. Good neighbours are nearby objects in the external catalogue whose position is compatible within position errors with the Gaia target. The cross-match algorithm is positional and exploits the full 5 parameter covariance matrix of the Gaia astrometric solution when available and the external catalogue positions and position errors. In addition it takes into account the external catalogue environment using the local density.

Please note that the cross-match algorithm is a trade-off between multiple requirements, in particular between completeness and correctness. It is thus not limited to a simple cone search.

Reference papers:
Marrese et al. (2017)
Marrese et al. (2019)

Draft

Columns description:

SOURCE_ID : Unique Gaia source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

CLEAN_SDSSDR13_OID : External Catalogue source identifier (int)

The additional numeric source identifier of the external catalogue, increasing with Declination.

ORIGINAL_EXT_SOURCE_ID : Original External Catalogue source identifier (long)

The unique source identifier in the original external catalogue.

ANGULAR_DISTANCE : Angular Distance between the two sources (float, Angle[arcsec])

Angular distance between a Gaia source and its best neighbour in the external catalogue.

NUMBER_OF_NEIGHBOURS : Number of neighbours in External Catalogue (byte)

Number of sources in the External catalogue which match the Gaia source within position errorse The identifiers of all the neighbours can be found in the corresponding neighbourhood table.

NUMBER_OF_MATES : Number of mates in Gaia Catalogue (byte)

Number of other Gaia sources that have as best neighbour the same external catalogue source. In case there are no other Gaia sources with the same best neighbour in the external catalogue, the number of mates is equal to zero. Given the Gaia high angular resolution, it will be the case that what appears as a single object in an external catalogue may be resolved by Gaia and as such will be the best match of more than one Gaia object.

XM_FLAG : Cross-match algorithm flag (short)

This flag is a bitmask indicating the details of the cross-match algorithm used for the source.

`xm_flag` values and descriptions:

- 0 = Initial value; resets all bits.
- 1 = The external catalogue object has one or more multiples. This means that there is at least another object with exactly the same astrometry.
- 2 = The external catalogue object has one or more suspected duplicates. This means that there is at least another object much closer than the catalogue angular resolution.
- 4 = The external catalogue object is resolved in Gaia.
- 8 = The Gaia object has a five parameters astrometric solution.
- 16 = The Gaia object has a two parameters astrometric solution.
- 32 = The external catalogue object is matched only after the special treatment for sources with under-estimated position errors.
- 64 = The external catalogue object is matched only after the special treatment for Gaia sources with large values of `ipd_gof_harmonic_amplitude` or `ruwe`.

Draft

For detailed documentation about `xm_flag` and the cross-match algorithm, see Chapter **??**. For a detailed description of `gaia_source` columns `ipd_gof_harmonic_amplitude` and `ruwe`, see Chapter **??**.

Draft

7.16 SDSSDR13_JOIN [TAP]

Convenience table to be used to join SDSS DR13 catalogue with the cross-match results. The table links the external catalogue original source identifier (`original_ext_source_id`) to the corresponding additional numerical identifier (`clean_sdssdr13_oid`).

Both `original_ext_source_id` and `clean_sdssdr13_oid` are present in the cross-match output tables (`sdssdr13_best_neighbour` and `sdssdr13_neighbourhood`).

However, in case there are suspected duplicates in the external catalogue, different `original_ext_source_id` will correspond to the same `clean_sdssdr13_oid`. In the cross-match output table only the `original_ext_source_id` of the source with the best astrometry among the suspected duplicates will be listed.

In practice, users may use the `original_ext_source_id` in the original catalogue to find the matching source with the best astrometry. Users interested to find all matching suspected duplicates should instead use the `clean_sdssdr13_oid` in the join with the cross-match result tables.

See Chapter **??** for more details on the duplicates in the external catalogues and their treatment in the cross-match computations.

Columns description:

ORIGINAL_EXT_SOURCE_ID : Original External Catalogue source identifier (long)

The unique source identifier in the original external catalogue.

CLEAN_SDSSDR13_OID : External Catalogue source identifier (int)

The additional numeric source identifier of the external catalogue, increasing with Declination.

Draft

7.17 SDSSDR13_NEIGHBOURHOOD [TAP]

Table `sdssdr13_neighbourhood` includes all good neighbours for each matched Gaia object. A good neighbour for a given Gaia source is a nearby object in the external catalogue whose position is compatible (within position errors) with the Gaia target. The cross-match algorithm is not symmetric and searches Gaia sources counterparts in SDSS DR13. The cross-match algorithm is positional and exploits the full 5 parameter covariance matrix of the Gaia astrometric solution when available and the external catalogue positions and position errors. In addition it takes into account the external catalogue environment using the local density.

Please note that the cross-match algorithm is a trade-off between multiple requirements, in particular between completeness and correctness. It is thus not limited to a simple cone search.

Reference papers:
Marrese et al. (2017)
Marrese et al. (2019)

Columns description:

SOURCE_ID : Unique Gaia source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

CLEAN_SDSSDR13_OID : External Catalogue source identifier (int)

The additional numeric source identifier of the external catalogue, increasing with Declination.

ORIGINAL_EXT_SOURCE_ID : Original External Catalogue source identifier (long)

The unique source identifier in the original external catalogue.

ANGULAR_DISTANCE : Angular Distance between the two sources (float, Angle[arcsec])

Angular distance between a Gaia source and its good neighbours in the external catalogue.

SCORE : Score of neighbours (double)

Draft

Score of a given neighbour. The score is a figure of merit based on geometric distance and local density of the external catalogue: the higher the score, the more probable the match is.

XM_FLAG : Cross-match algorithm flag (short)

This flag is a bitmask indicating the details of the cross-match algorithm used for the source.

`xm_flag` values and descriptions:

- 0 = Initial value; resets all bits.
- 1 = The external catalogue object has one or more multiples. This means that there is at least another object with exactly the same astrometry.
- 2 = The external catalogue object has one or more suspected duplicates. This means that there is at least another object much closer than the catalogue angular resolution.
- 4 = The external catalogue object is resolved in Gaia.
- 8 = The Gaia object has a five parameters astrometric solution.
- 16 = The Gaia object has a two parameters astrometric solution.
- 32 = The external catalogue object is matched only after the special treatment for sources with under-estimated position errors.
- 64 = The external catalogue object is matched only after the special treatment for Gaia sources with large values of `ipd_gof_harmonic_amplitude` or `ruwe`.

For detailed documentation about `xm_flag` and the cross-match algorithm, see Chapter **??**. For a detailed description of `gaia_source` columns `ipd_gof_harmonic_amplitude` and `ruwe`, see Chapter **??**.

Draft

7.18 SKYMAPPERDR2_BEST_NEIGHBOUR [TAP]

Table `skymapperdr2_best_neighbour` lists each matched Gaia object with its best neighbour in the external catalogue. The cross-match algorithm is not symmetric and searches Gaia sources counterparts in SkyMapper DR2. The best neighbour is chosen among good neighbours as the one with the highest value of the figure of merit, which evaluates the ratio between two opposite models/hypotheses: the counterpart candidate is a match or it is found by chance. Good neighbours are nearby objects in the external catalogue whose position is compatible within position errors with the Gaia target. The cross-match algorithm is positional and exploits the full 5 parameter covariance matrix of the Gaia astrometric solution when available and the external catalogue positions and position errors. In addition it takes into account the external catalogue environment using the local density.

Please note that the cross-match algorithm is a trade-off between multiple requirements, in particular between completeness and correctness. It is thus not limited to a simple cone search.

Reference papers:
Marrese et al. (2017)
Marrese et al. (2019)

Draft

Columns description:

SOURCE_ID : Unique Gaia source identifier (long)

Unique identifier of the Gaia source, the attribute corresponds to `gaia_source.source_id`.

ORIGINAL_EXT_SOURCE_ID : Original External Catalogue source identifier (long)

The unique source identifier in the original external catalogue.

ANGULAR_DISTANCE : Angular Distance between the two sources (float, Angle[arcsec])

Angular distance between a Gaia source and its best neighbour in the external catalogue.

NUMBER_OF_NEIGHBOURS : Number of neighbours in External Catalogue (byte)

Number of sources in the external catalogue which match the Gaia source within position errors. The identifiers of all the neighbours can be found in the corresponding neighbourhood table.

NUMBER_OF_MATES : Number of mates in Gaia Catalogue (byte)

Number of other Gaia sources that have as best neighbour the same external catalogue source. In case there are no other Gaia sources with the same best neighbour in the external catalogue, the number of mates is equal to zero. Given the Gaia high angular resolution, it will be the case that what appears as a single object in an external catalogue may be resolved by Gaia and as such will be the best match of more than one Gaia object.

XM_FLAG : Cross-match algorithm flag (short)

This flag is a bitmask indicating the details of the cross-match algorithm used for the source.

`xm_flag` values and descriptions:

- 0 = Initial value; resets all bits.
- 1 = The external catalogue object has one or more multiples. This means that there is at least another object with exactly the same astrometry.
- 2 = The external catalogue object has one or more suspected duplicates. This means that there is at least another object much closer than the catalogue angular resolution.
- 4 = The external catalogue object is resolved in Gaia.
- 8 = The Gaia object has a five parameters astrometric solution.
- 16 = The Gaia object has a two parameters astrometric solution.
- 32 = The external catalogue object is matched only after the special treatment for sources with under-estimated position errors.
- 64 = The external catalogue object is matched only after the special treatment for Gaia sources with large values of `ipd_gof_harmonic_amplitude` or `ruwe`.

For detailed documentation about `xm_flag` and the cross-match algorithm, see Chapter **??**. For a detailed description of `gaia_source` columns `ipd_gof_harmonic_amplitude` and `ruwe`, see Chapter **??**.

Draft

7.19 SKYMAPPERDR2_JOIN [TAP]

Convenience table to be used to join the SkyMapper DR2 catalogue with the cross-match results. It lists all the SkyMapper DR2 sources which were used in the cross-match. See Chapter **??** for more details on the catalogue.

Columns description:

ORIGINAL_EXT_SOURCE_ID : Original External Catalogue source identifier (long)

The unique source identifier in the original External catalogue.

Draft

7.20 SKYMAPPERDR2_NEIGHBOURHOOD [TAP]

Table `skymapperdr2_neighbourhood` includes all good neighbours for each matched Gaia object. A good neighbour for a given Gaia source is a nearby object in the external catalogue whose position is compatible (within position errors) with the Gaia target. The cross-match algorithm is not symmetric and searches Gaia sources counterparts in SkyMapper DR2. The cross-match algorithm is positional and exploits the full 5 parameter covariance matrix of the Gaia astrometric solution when available and the external catalogue positions and position errors. In addition it takes into account the external catalogue environment using the local density.

Please note that the cross-match algorithm is a trade-off between multiple requirements, in particular between completeness and correctness. It is thus not limited to a simple cone search.

Reference papers:
Marrese et al. (2017)
Marrese et al. (2019)

Draft

Columns description:

SOURCE_ID : Unique Gaia source identifier (long)

Unique identifier of the Gaia source, the attribute corresponds to `gaia_source.source_id`.

ORIGINAL_EXT_SOURCE_ID : Original External Catalogue source identifier (long)

The unique source identifier in the original external catalogue.

ANGULAR_DISTANCE : Angular Distance between the two sources (float, Angle[arcsec])

Angular distance between a Gaia source and its good neighbours in the external catalogue.

SCORE : Score of neighbours (double)

The score is a figure of merit based on geometric distance and local density of the external catalogue: the higher the score, the more probable the match is.

XM_FLAG : Cross-match algorithm flag (short)

This flag is a bitmask indicating the details of the cross-match algorithm used for the source.

`xm_flag` values and descriptions:

- 0 = Initial value; resets all bits.
- 1 = The external catalogue object has one or more multiples. This means that there is at least another object with exactly the same astrometry.
- 2 = The external catalogue object has one or more suspected duplicates. This means that there is at least another object much closer than the catalogue angular resolution.
- 4 = The external catalogue object is resolved in Gaia.
- 8 = The Gaia object has a five parameters astrometric solution.
- 16 = The Gaia object has a two parameters astrometric solution.
- 32 = The external catalogue object is matched only after the special treatment for sources with under-estimated position errors.
- 64 = The external catalogue object is matched only after the special treatment for Gaia sources with large values of `ipd_gof_harmonic_amplitude` or `ruwe`.

For detailed documentation about `xm_flag` and the cross-match algorithm, see Chapter **??**. For a detailed description of `gaia_source` columns `ipd_gof_harmonic_amplitude` and `ruwe`, see Chapter **??**.

Draft

7.21 TMASS_PSC_XSC_BEST_NEIGHBOUR [TAP]

Table `tmass_psc_xsc_best_neighbour` lists each matched Gaia object with its best neighbour in the external catalogue. The cross-match algorithm is not symmetric and searches Gaia sources counterparts in 2MASS. The best neighbour is chosen among good neighbours as the one with the highest value of the figure of merit, which evaluates the ratio between two opposite models/hypotheses: the counterpart candidate is a match or it is found by chance. Good neighbours are nearby objects in the external catalogue whose position is compatible within position errors with the Gaia target. The cross-match algorithm is positional and exploits the full 5 parameter covariance matrix of the Gaia astrometric solution when available and the external catalogue positions and position errors. In addition it takes into account the external catalogue environment using the local density.

Please note that the cross-match algorithm is a trade-off between multiple requirements, in particular between completeness and correctness. It is thus not limited to a simple cone search.

Reference papers:
Marrese et al. (2017)
Marrese et al. (2019)

Draft

Columns description:

SOURCE_ID : Unique Gaia source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

ORIGINAL_EXT_SOURCE_ID : Original External Catalogue source identifier (string)

The unique source identifier in the original external catalogue.

ANGULAR_DISTANCE : Angular Distance between the two sources (float, Angle[arcsec])

Angular distance between a Gaia source and its best neighbour in the external catalogue.

XM_FLAG : Cross-match algorithm flag (short)

This flag is a bitmask indicating the details of the cross-match algorithm used for the source.

`xm_flag` values and descriptions:

- 0 = Initial value; resets all bits.
- 1 = The external catalogue object has one or more multiples. This means that there is at least another object with exactly the same astrometry.
- 2 = The external catalogue object has one or more suspected duplicates. This means that there is at least another object much closer than the catalogue angular resolution.
- 4 = The external catalogue object is resolved in Gaia.
- 8 = The Gaia object has a five parameters astrometric solution.
- 16 = The Gaia object has a two parameters astrometric solution.
- 32 = The external catalogue object is matched only after the special treatment for sources with under-estimated position errors.
- 64 = The external catalogue object is matched only after the special treatment for Gaia sources with large values of `ipd_gof_harmonic_amplitude` or `ruwe`.

For detailed documentation about `xm_flag` and the cross-match algorithm, see Chapter **??**. For a detailed description of `gaia_source` columns `ipd_gof_harmonic_amplitude` and `ruwe`, see Chapter **??**.

CLEAN_TMASS_PSC_XSC_OID : External Catalogue source identifier (int)

The additional numeric source identifier of the external catalogue, increasing with Declination.

NUMBER_OF_NEIGHBOURS : Number of neighbours in External Catalogue (byte)

Number of sources in the external catalogue which match the Gaia source within position errors. The identifiers of all the neighbours can be found in the corresponding neighbourhood table.

NUMBER_OF_MATES : Number of mates in Gaia Catalogue (byte)

Number of other Gaia sources that have as best neighbour the same external catalogue source. In case there are no other Gaia sources with the same best neighbour in the external catalogue,

the number of mates is equal to zero. Given the Gaia high angular resolution, it will be the case that what appears as a single object in an external catalogue will be resolved by Gaia and as such will be the best match of more than one Gaia object.

Draft

7.22 TMASS_PSC_XSC_JOIN [TAP]

Convenience table to be used to join 2MASS PSC+XSC catalogue with the cross-match results. The table links the external catalogue original source identifier (`original_ext_source_id`) to the corresponding additional numerical identifier (`clean_tmass_psc_xsc_oid`).

Both `original_ext_source_id` and `clean_tmass_psc_xsc_oid` are present in the cross-match output tables (`tmass_psc_xsc_best_neighbour` and `tmass_psc_xsc_neighbourhood`). However, in case there are suspected duplicates in the external catalogue, different `original_ext_source_id` will correspond to the same `clean_tmass_psc_xsc_oid`.

In the cross-match output table only the `original_ext_source_id` of the source with the best astrometry among the suspected duplicates will be listed. In practice, users may use the `original_ext_source_id` in the original catalogue to find the matching source with the best astrometry. Users interested to find all matching suspected duplicates should instead use the `clean_tmass_psc_xsc_oid` in the join with the cross-match result tables.

See Chapter **??** for more details on the duplicates in the external catalogues and their treatment in the cross-match computations.

Columns description:

ORIGINAL_PSC_SOURCE_ID : Original 2MASS PSC source identifier (string)

The unique source identifier in 2MASS PSC.

ORIGINAL_XSC_SOURCE_ID : Original 2MASS XSC source identifier (string)

The unique source identifier in 2MASS XSC.

CLEAN_TMASS_PSC_XSC_OID : External Catalogue source identifier (int)

The additional numeric source identifier of the external catalogue, increasing with Declination.

Draft

7.23 TMASS_PSC_XSC_NEIGHBOURHOOD [TAP]

Table `tmass_psc_xsc_neighbourhood` includes all good neighbours for each matched Gaia object. A good neighbour for a given Gaia source is a nearby object in the external catalogue whose position is compatible (within position errors) with the Gaia target. The cross-match algorithm is not symmetric and searches Gaia sources counterparts in 2MASS. The cross-match algorithm is positional and exploits the full 5 parameter covariance matrix of the Gaia astrometric solution when available and the external catalogue positions and position errors. In addition it takes into account the external catalogue environment using the local density.

Please note that the cross-match algorithm is a trade-off between multiple requirements, in particular between completeness and correctness. It is thus not limited to a simple cone search.

Reference papers:
Marrese et al. (2017)
Marrese et al. (2019)

Draft

Columns description:

SOURCE_ID : Unique Gaia source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

ORIGINAL_EXT_SOURCE_ID : Original External Catalogue source identifier (string)

The unique source identifier in the original external catalogue.

ANGULAR_DISTANCE : Angular Distance between the two sources (float, Angle[arcsec])

Angular distance between a Gaia source and its good neighbours in the external catalogue.

SCORE : Score of neighbours (double)

Score of a given neighbour. The score is a figure of merit based on geometric distance and local density of the external catalogue: the higher the score, the more probable the match is.

XM_FLAG : Cross-match algorithm flag (short)

This flag is a bitmask indicating the details of the cross-match algorithm used for the source.

`xm_flag` values and descriptions:

- 0 = Initial value; resets all bits.
- 1 = The external catalogue object has one or more multiples. This means that there is at least another object with exactly the same astrometry.
- 2 = The external catalogue object has one or more suspected duplicates. This means that there is at least another object much closer than the catalogue angular resolution.
- 4 = The external catalogue object is resolved in Gaia.
- 8 = The Gaia object has a five parameters astrometric solution.
- 16 = The Gaia object has a two parameters astrometric solution.
- 32 = The external catalogue object is matched only after the special treatment for sources with under-estimated position errors.
- 64 = The external catalogue object is matched only after the special treatment for Gaia sources with large values of `ipd_gof_harmonic_amplitude` or `ruwe`.

For detailed documentation about `xm_flag` and the cross-match algorithm, see Chapter **??**. For a detailed description of `gaia_source` columns `ipd_gof_harmonic_amplitude` and `ruwe`, see Chapter **??**.

CLEAN_TMASS_PSC_XSC_OID : External Catalogue source identifier (int)

The additional numeric source identifier of the external catalogue, increasing with Declination.

Draft

7.24 TYCHO2TDSC_MERGE_BEST_NEIGHBOUR [TAP]

Table `tycho2tdsc_merge_best_neighbour` lists each matched external catalogue object with its best neighbour in Gaia. The cross-match algorithm is not symmetric and searches `Tycho2tdscMerge` source counterparts in Gaia. The best neighbour is chosen among good neighbours as the one with the highest value of the figure of merit, which evaluates the ratio between two opposite models/hypotheses: the counterpart candidate is a match or it is found by chance. Good neighbours are nearby objects in Gaia whose position is compatible within position errors with the external catalogue target. The cross-match algorithm is positional and exploits the full 5 parameter covariance matrix of the Gaia astrometric solution when available and the external catalogue positions and position errors. In addition it takes into account the Gaia environment using the local density.

Please note that the cross-match algorithm is a trade-off between multiple requirements, in particular between completeness and correctness. It is thus not limited to a simple cone search.

Reference papers:
Marrese et al. (2017)
Marrese et al. (2019)

Draft

Columns description:

SOURCE_ID : Unique Gaia source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

ORIGINAL_EXT_SOURCE_ID : Original External Catalogue source identifier (string)

The unique source identifier in the original external catalogue.

ANGULAR_DISTANCE : Angular Distance between the two sources (float, Angle[arcsec])

Angular distance between a Gaia source and its best neighbour in the external catalogue.

XM_FLAG : Cross-match algorithm flag (short)

This flag is a bitmask indicating the details of the cross-match algorithm used for the source.

`xm_flag` values and descriptions:

- 0 = Initial value; resets all bits.
- 1 = The external catalogue object has one or more multiples. This means that there is at least another object with exactly the same astrometry.
- 2 = The external catalogue object has one or more suspected duplicates. This means that there is at least another object much closer than the catalogue angular resolution.
- 4 = The external catalogue object is resolved in Gaia.
- 8 = The Gaia object has a five parameters astrometric solution.
- 16 = The Gaia object has a two parameters astrometric solution.
- 32 = The external catalogue object is matched only after the special treatment for sources with under-estimated position errors.
- 64 = The external catalogue object is matched only after the special treatment for Gaia sources with large values of `ipd_gof_harmonic_amplitude` or `ruwe`.

For detailed documentation about `xm_flag` and the cross-match algorithm, see Chapter **??**. For a detailed description of `gaia_source` columns `ipd_gof_harmonic_amplitude` and `ruwe`, see Chapter **??**.

TYCHO2TDSC_MERGE_OID : External Catalogue source identifier (int)

The additional numeric source identifier of the external catalogue, increasing with Declination.

NUMBER_OF_NEIGHBOURS : Number of neighbours in Gaia Catalogue (byte)

Number of sources in the Gaia catalogue which match the external catalogue source within position errors. The identifiers of all the neighbours can be found in the corresponding neighbourhood table.

Draft

7.25 TYCHO2TDSC_MERGE_NEIGHBOURHOOD [TAP]

Table `tycho2tdsc_merge_neighbourhood` includes all good neighbours for each matched `Tycho2tdscMerge` object. A good neighbour for a given `Tycho2tdscMerge` source is a nearby Gaia object whose position is compatible (within position errors) with the `Tycho2tdscMerge` target. The cross-match algorithm is not symmetric and searches `Tycho2tdscMerge` source counterparts in Gaia. The cross-match algorithm is positional and exploits the full 5 parameter covariance matrix of the Gaia astrometric solution when available and the external catalogue positions and position errors. In addition it takes into account the Gaia environment using the local density.

Please note that the cross-match algorithm is a trade-off between multiple requirements, in particular between completeness and correctness. It is thus not limited to a simple cone search.

Reference papers:
Marrese et al. (2017)
Marrese et al. (2019)

Columns description:

SOURCE_ID : Unique Gaia source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

ORIGINAL_EXT_SOURCE_ID : Original External Catalogue source identifier (string)

The unique source identifier in the original external catalogue.

ANGULAR_DISTANCE : Angular Distance between the two sources (float, Angle[arcsec])

Angular distance between a Gaia source and its good neighbours in the external catalogue.

SCORE : Score of neighbours (double)

Score of a given neighbour. The score is a figure of merit based on geometric distance and local density of the external catalogue: the higher the score, the more probable the match is.

Draft

XM_FLAG : Cross-match algorithm flag (short)

This flag is a bitmask indicating the details of the cross-match algorithm used for the source.

`xm_flag` values and descriptions:

- 0 = Initial value; resets all bits.
- 1 = The external catalogue object has one or more multiples. This means that there is at least another object with exactly the same astrometry.
- 2 = The external catalogue object has one or more suspected duplicates. This means that there is at least another object much closer than the catalogue angular resolution.
- 4 = The external catalogue object is resolved in Gaia.
- 8 = The Gaia object has a five parameters astrometric solution.
- 16 = The Gaia object has a two parameters astrometric solution.
- 32 = The external catalogue object is matched only after the special treatment for sources with under-estimated position errors.
- 64 = The external catalogue object is matched only after the special treatment for Gaia sources with large values of `ipd_gof_harmonic_amplitude` or `ruwe`.

For detailed documentation about `xm_flag` and the cross-match algorithm, see Chapter **??**. For a detailed description of `gaia_source` columns `ipd_gof_harmonic_amplitude` and `ruwe`, see Chapter **??**.

TYCHO2TDSC_MERGE_OID : External Catalogue source identifier (int)

The additional numeric source identifier of the external catalogue, increasing with Declination.

Draft

8 Crowded fields

As indicated in the below subsection titles by the labels [TAP] and [DataLink], these products are served from the ESA Gaia Archive using two different protocols, the Table Access Protocol (TAP) for tabular products and the DataLink protocol for other products such as spectra, light curves, MCMC samples, etc. For more details please see Section **??**.

8.1 CROWDED_FIELD_COVERAGE_MAP [TAP]

Coverage map for one crowded field region.

The coverage map is provided as a HEALPix table. For each HEALPix cell covering a crowded-field region the number of scans is listed. The number of scans for a given HEALPix cell is obtained from the number of image polygons covering the centre of that cell. It is thus slightly less accurate than the precise number of overlapping image polygons provided for each source.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

REGION_NAME : Crowded field region (string)

A plain text string that identifies the crowded-field region in the sky in which this source is located. A list of the 9 regions observed with crowded-field images can be found in the documentation of `crowded_field_source.region_name`.

HEALPIX_LEVEL : HEALPix level (byte)

The HEALPix level that was used.

Draft

HEALPIX_ID : HEALPix index (int)

The HEALPix cell index for which the number of scans is given.

N_SCANS : Number of scans (short)

The number of scans is the number of bounding polygons that overlap a certain HEALPix cell. It gives the number of times that this cell has been observed. As this number refers to a HEALPix cell rather than a source, it is slightly less precise than the exact number of scans provided for each source.

Draft

8.2 CROWDED_FIELD_EPOCH_ASTROMETRY [DataLink]

This table provides the epoch astrometry, i.e. the individual astrometric measurements for the sources treated in the special crowded-field processing.

For more information see the description of the `crowded_field_source` table and the relevant chapter in the DR4 online documentation. The presentation of the crowded-field epoch astrometry data is basically the same as in the main `epoch_astrometry` table, albeit with some specifics from the crowded-field processing.

The major differences come from the fact that the crowded-field observations use only the SM CCDs, but not the AF CCDs. In particular, many of the table columns that have dimension [10] in the main `epoch_astrometry` table consist of only one value for the crowded-field observations.

The table `crowded_field_epoch_astrometry` contains one row per `source_id-transit_id` matching pair. The same `source_id` occurs as many times as there are FoV transits matched with this source. Any combination of `source_id` and `transit_id` occurs exactly once in the table.

Each row contains three per-source fields (`ra0`, `dec0` and `agis_source_excess_noise`); for ease of handling and usage, these are repeated as many times as there are transits matched to the given source. Similarly, some of the per-transit and per-CCD data are repeated for the same reasons.

Note this table is not available through the main archive TAP interface. Data are delivered via the Massive Data service indexed by the VO DataLink protocol and described in Chapter **??**. For example this can be actioned in the archive user interface by joining queries with the table `crowded_field_source` and selecting `has_epoch_astrometry = 't'`. The DataLink retrieval type to specify for this product in programmatic archive access via the Astroquery Gaia Python package is `EPOCH_ASTROMETRY_CROWDED_FIELD`.

DataLink products are served with a choice of two different data structures:

- `DATAMODEL_STANDARD`: the product is serialised following a standard data model (for example an IVOA-compliant data model in applicable cases);
- `DATAMODEL_GAIA`: the product is serialised following the Gaia DPAC(-internal) data model.

Draft

The DATAMODEL_GAIA format is described here whereas the DATAMODEL_STANDARD format is described in https://www.cosmos.esa.int/web/gaia-users/archive/datalink-products#datalink_serialisation

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution identifier, please visit https://gaia.esac.esa.int/decoder/solnDecoder.jsp.

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source.

A unique numerical identifier of the source, encoding the approximate position of the source (roughly to the nearest arcmin), the provenance (data processing centre where it was created), a running number, and a component number following the construction conventions of those in the main source catalogue (for a detailed description see gaia_source.source_id). Note that the data processing provenance code stored in bits 33 to 35 is 6 for the crowded field image analysis, which means (source_id >> 32) & 7 = 6 for crowded_field_source and crowded_field_epoch_astrometry.

TRANSIT_ID : Transit Identifier (long)

The transit_id is a unique identifier assigned to each detected (and confirmed) source as it transits the Gaia focal plane. Each time a given source is detected as Gaia scans and re-scans the sky, a new transit_id will be created to badge that apparition. For this purpose, the along-scan time of the corresponding AF1 window and its across-scan position in combination with the telescope in which the source was detected are used to form a unique integer with which to label the transit.

Usually, the detection of transits is done using the SkyMapper CCD data, and the confirmation is done using the AF1 CCD data. Both happens on board. For CrowdedField observations, both does not happen. There is no on-board detection, and there is no corresponding AF1 win-

dow. Therefore, an artificial pseudo-AF1 window is predicted from the actual CrowdedField (SkyMapper) image for the given observation, and an artificial `transit_id` is created in the usual coding.

If this process would create a `transit_id` which is identical to a real one, the artificial AF1 time is increased in steps of one unit of the `transit_id`'s time resolution (204.8 microseconds) until a unique value is reached.

The several features of a transit that are encoded in `transit_id` can be easily retrieved using bit masks (&) and shifts (>>) as follows:

- On-Board Mission Time line (OBMT) [ns]
  = 204800 * ((`transit_id` >> 17) & (0x000003FFFFFFFFFF)) as described above
- Field-of-view = 1 + (`transit_id` >> 15) & 0x03 [1 for 'preceding' and 2 for 'following' fields-of-view respectively]
- CCD row = (`transit_id` >> 12) & 0x07 [dimensionless, in the range 1 to 7]
- Artificial across-scan 'reference acquisition pixel' in strip AF1 = (`transit_id`) & 0x0FFF [pixels] (this is the across-scan centre of the artificial AF1 window and is odd if immediately below the mid-point of the window and even if immediately above)

where the bit mask prefix '0x' denotes hexadecimal.

For further details see Portell et al. (2020). For convenience a decoder for `transit_id` is available on-line at

`https://gaia.esac.esa.int/decoder/transitidDecoder.jsp`

RA0 : Right ascension of reference point (double, Angle[deg])

Right ascension α_0 of the tangent point on the plane tangential to the celestial sphere in which the coordinate direction of the object is locally modelled, i.e. of the celestial reference point from which the local plane coordinates `centroid_pos_al` and `centroid_pos_ac` are counted. The specific value will be taken from the valid/relevant source catalogue (which in turn will be defined implicitly by the `solution_id`).

Note: `ra0` and `dec0` are always the same for all transits of a given `source_id`.

For more details see **?**.

DEC0 : Declination of reference point (double, Angle[deg])

Declination δ_0 of the tangent point on the plane tangential to the celestial sphere in which the coordinate direction of the object is locally modelled, i.e. of the celestial reference point from which the local plane coordinates `centroid_pos_al` and `centroid_pos_ac` are counted. The specific value will be taken from the valid/relevant astrometric source catalogue (which in turn will be defined implicitly by the `solution_id`).

Note: `ra0` and `dec0` are always the same for all transits of a given `source_id`.

For more details see **?**.

AGIS_SOURCE_EXCESS_NOISE : the unscaled AGIS Source Excess Noise (float, Angle[mas])

This is the unscaled AGIS source excess noise.

The actual weight used by AGIS per CCD for ordinary Gaia astrometric observations (i.e. for those data derived from on-board detections and from AF data) is $(1mas^2/\sigma^2_{tot})$, where $\sigma^2_{tot} = \sigma^2_{IPD} + excess^2_{source} + excess^2_{cal.}$. This is the weight to be used if normal AGIS solutions shall be reproduced from epoch astrometry data. For more explanations see the AGIS paper for the relevant Gaia Data Release.

The quantity `centroid_pos_error_al` contains the first and third terms in σ^2_{tot}. The separate `agis_source_excess_noise` item is needed to enable external users to change the source model, whence $excess^2_{source}$ from the normal AGIS solution becomes irrelevant, while the other two components of σ^2_{tot} remain valid.

For more explanations see the AGIS paper for the relevant Gaia Data Release, here **?**.

Note that, contrary to the `gaia_source` and `all_source_astrometry` tables, this is the original source excess noise, not the rescaled excess noise according to **?**. Reason: The rescaled value is the one which is useful for the quality criteria in the source astrometry tables, but the original value is needed to enable users to reproduce the published AGIS solutions from epoch astrometry data.

For CrowdedField observations, no $excess^2_{cal.}$ is derived. Thus, σ^2_{tot} for such observations contains only the first two terms, and `centroid_pos_error_al` contains only the first term in σ^2_{tot}.

OBS_TIME_TCB : effective observing time as TCB (Long[1] array, Time[ns])

Effective observation time for each CCD in this FoV transit. Centre of the actual exposure time for the given window/CCD, depending on readout time and gate. In units of nanoseconds since

J2010.0(TCB).

The presentation as long integer (rather than as double float) is chosen for the sake of keeping the full resolution of the time coordinates used in Gaia/DPAC, and for the sake of invertability of transformations from OBMT to TCB and vice versa.

Such transformations are done using HATT (High-Accuracy Time Transformations), which have the resolution of 1 nanosecond. But users should be aware that the absolute accuracy of them is in the order of 150 nanoseconds.

OBS_TIME_BARY_CORR : barycentric correction to effective observing time as TCB (float, Time[ns])

Barycentric correction to the observation time, in the sense of TCB(barycentric)-TCB(at Gaia).

For ordinary Gaia astrometric observations (i.e. for those data derived from on-board detections and from AF data) `obs_time_bary_corr` is calculated for the AF4 CCD, i.e. at the middle of the FoV transit. For the crowded field observations it is computed for the actual observing time on the SkyMapper, i.e. in the middle of the 2.5-second exposure time.

For the correct production and usage of the epoch astrometry, both the true and the barycentrically corrected observation times are needed.

The maximum value that the quantity can reach is about ±370 s (not the canonical ± 500s because the scan axis of Gaia always points at 45 degrees from the sun). The resolution of the float variable thus is around 50 microseconds or better.

SCAN_POS_ANGLE : Position angle of the scan (double[1] array, Angle[deg])

Position angle θ of the scan for each of the CCD transits.

This angle uniquely determines the transformation between the local celestial coordinates (a, d) and the local scan coordinates (w, z) in the tangential plane touching the celestial sphere at `(ra0,dec0)`. See **?** for details. The scan angle varies along a FoV transit. In contrast, the parallax factors are sufficiently constant over a transit.

The sine and cosine of θ are the derivatives of the along-scan local plane coordinate w with respect to right ascension and declination, respectively.

ZETA : across-scan field angle zeta of the reference point (ra0, dec0) (float, Angle[deg])

This is the AC field angle ζ of the reference point `(ra0,dec0)` at the time of observation.

This quantity is needed for the following reason: When analysing a non-single system, it is necessary that the epoch astrometry of each component is expressed with respect to the same reference point. But that will not be the case for resolved systems, where the components have different AGIS solutions, and therefore different reference points. Hence there is a need to be able to transform the epoch data from one reference point to another (within an arcmin or so). But this is not possible - at least not rigorously - unless we also know ζ. The error in the transformed epoch astrometry is marginal (< 1 muas) for shifts less than about 5 arcsec, but increases quadratically for larger separations. ζ could of course be computed to sufficient accuracy if the FoV, CCD row, and AC coordinate `mu` (μ) are known, but to give it explicitly is a convenience for users.

Adding ζ also brings a nice symmetry to the information in `epoch_astrometry`: with the AL&AC parallax factors plus the Roemer delay we have complete (3D) knowledge of Gaia's barycentric position at the time of observation (albeit only to float precision per transit, and in strange units); with `scan_pos_angle` and `zeta` we similarly have complete knowledge of the spin (SRS z) axis at the time of observation. Both pieces of information (Gaia's 3D position and the SRS z axis) are needed to transform epoch data to a different reference point - and back.

PARALLAX_FACTOR_AL : Parallax factor AL (float)

Parallax factor in the AL direction w, needed to use the local plane coordinates for detailed object modelling.

The parallax factor f_w is the derivative of the along-scan local plane coordinate w with respect to the parallax. See **?** for details.

PARALLAX_FACTOR_AC : Parallax factor AC (float)

Parallax factor in the AC direction z, needed to use the local plane coordinates for detailed object modelling.

The parallax factor f_z is the derivative of the across-scan local plane coordinate z with respect to the parallax. See **?** for details.

COLOUR_FACTOR_AL : colour factor AL (float[1] array)

The colour factor in AL direction (w).

This is the negative of the derivative of w with respect to colour parameter ν_{eff} (effective wavenumber in units of inverse micrometers). The unit of the quantity is mas*micrometer.

COLOUR_FACTOR_AC : colour factor AC (float[1] array)

The colour factor in AC direction (z).

This is the derivative of z with respect to colour parameter ν_{eff} (effective wavenumber in units of inverse micrometers). The unit of the quantity is mas*micrometer.

This field will be empty in DR4.

NU_EFF_USED_IN_ASTROMETRY : colour (i.e. effective wavenumber) used in AGIS (and in IPD) (float, Misc[μm^{-1}])

Effective wavenumber of the source used as starting value in the astrometric solution. Due to the unavailability of measured colours for crowded-field stars, for all sources in this table, the value of this field is identical to the default value.

This effective wavenumber $\nu_{\rm eff}$, in units of $\mu{\rm m}^{-1}$, is the assumed value used for the selection of the appropriate LSF/PSF model in the image parameter determination, as well as for the selection of the astrometric calibration parameters in the astrometric adjustment, for the given source. If no BP and RP photometry of the given source was available, `nu_eff_used_in_astrometry` is set to the default value $\nu_{\rm eff} = 1.43\mu{\rm m}^{-1}$. Else, if reliable BP and RP photometry was available, it is set to the weighted mean of the default value and the actual photon-flux weighted inverse wavelength as estimated from the BP and RP bands. If highly precise BP and RP data are available, the `nu_eff_used_in_astrometry` is very close to the value from the BP and RP bands. However, due to the cyclic processing of the Gaia astrometry and photometry, this effective wavenumber will even in these cases be (slightly) different from the one computed using the latest available photometry for DR4.

Some details on the selection of the default value and the construction of the weighted mean:
The default value is chosen as the ensemble average of the actually measured values for all Gaia sources, and its standard error for the weighted mean is set to $0.085\mu{\rm m}^{-1}$, which is close to the actual rms scatter of the measured values.
For the measured values, the formal errors assigned by the photometry processing were used in the construction of the weighted mean. However, the original values delivered by CU5 (from the DR3 photometry processing) were found to be strongly overestimated. Therefore, they were rescaled to give realistic values before being used for the weighted mean.

NU_EFF_ERROR : standard error of nu_eff_used_in_astrometry (float, Misc[μm^{-1}])

This is the standard error of the input colour parameter for a given source in the AGIS source parameter adjustment.

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However, for the bright-source astrometry and crowded-field source astrometry, a constant value of 0.085 inverse micrometers was used, because no measured values were available.

CENTROID_POS_AL : Local plane coordinate (w) of image centroid AL (double[1] array, Angle[mas])

This is the AL local plane coordinate *w* of the image centroid position (one per CCD transit). See **?** for details.

There is a known issue: Among the $\simeq 10^{12}$ entries of `centroid_pos_al` in the table, there is a very small portion with grossly deviating values. Such values may be caused by excursions of the attitude modelling at the start and end of data gaps. There are a total of 16,437,562 cases where `centroid_pos_al` is beyond the range allowed by the window size, and there are a few cases taking exceptionally large values. Notably, there are 208 transits where the max value of `centroid_pos_al` exceeds 10^9 mas. On the user side, the obvious way to avoid the problem is to simply ignore the very few records where `centroid_pos_al` is unreasonably large.

CENTROID_POS_AC : Local plane coordinate (z) of image centroid AC (double[1] array, Angle[mas])

This is the AC local plane coordinate *z* of the image centroid position (one per CCD transit). See **?** for details. This is based on the AC centroiding and therefore it is only available for 2D windows. In DR4 this field and its associated error `centroid_pos_error_ac` will not be available.

CALCULATED_POS_AC : Calculated Local plane coordinate (z) (double[1] array, Angle[mas])

Calculated AC local plane coordinate *z*.

This is not based on a real measurement like the `centroid_pos_ac` but has been calculated using the AL measurement and the source astrometry to compute the geometric source position at the time of the observation, then using the attitude to get the calculated AC local plane coordinate *z*.

CENTROID_POS_ERROR_AL : standard error of AL centroid position (float[1] array, Angle[mas])

Standard error of AL local plane coordinate *w* of the image centroid position.

This quantity represents the standard errors used by AGIS, or more precisely: together with the `agis_source_excess_noise` and the `used_by_agis_al` flag it represents the actual weight

for the AL coordinate used in the final AGIS source update (if `used_by_agis_al` is 'true').

This quantity contains two components, quadratically added via

$(\texttt{centroid_pos_error_al})^2 = \sigma^2_{IPD} + excess^2_{cal}$

The two components are:

- The IDU IPD centroid's standard error,
- The AGIS calibration excess noise.

The former is given explicitly in the table (`ipd_error_al`), while the second can be computed by quadratically subtracting the other one from `centroid_pos_error_al`.

To be very clear: if a user wants to reproduce the released AGIS solution for a given source they should use weights derived from quadratically adding `agis_source_excess_noise` and `centroid_pos_error_al` so that the result contains all three relevant components of the weighting used by AGIS:

$\sigma^2_{tot} = \sigma^2_{IPD} + excess^2_{source} + excess^2_{cal}$.

Note that the `centroid_pos_error_al` for SM measurements might be significantly underestimated, especially for bright stars. For details see the corresponding column description for the main `epoch_astrometry` table and Section **??** of the online documentation.

CENTROID_POS_ERROR_AC : standard error of AC centoid position (float[1] array, Angle[mas])

Standard error of the AC local plane coordinate z of the image centroid position.

This quantity represents the standard errors used by AGIS, or more precisely: together with the `agis_source_excess_noise` and the `used_by_agis_ac` flag it represents the actual weight for the AC coordinate used in the final AGIS source update (if `used_by_agis_ac` is 'true', which never happens for DR4). For 1d windows, the value will always be set to NaN.

This quantity contains two components, quadratically added via

$\texttt{centroid_pos_error_ac}^2 = \sigma^2_{IPD} + excess^2_{cal}$

The two components are:

- The IDU IPD centroid's standard error,
- The AGIS calibration excess noise.

The former is given explicitly in the table (`ipd_error_ac`), while the second can be computed by quadratically subtracting the other one from `centroid_pos_error_ac`.

To be very clear: if a user wants to reproduce the released AGIS solution for a given source they should use weights derived from quadratically adding `agis_source_excess_noise` and `centroid_pos_error_ac` so that the result contains all three relevant components of the weighting used by AGIS:

$\sigma^2_{tot} = \sigma^2_{IPD} + excess^2_{source} + excess^2_{cal.}$

This field is not available in DR4.

USED_BY_AGIS_AL : whether the CCD transit was used in AL by AGIS-DR4 (Boolean[1] array)

Flag saying whether the CCD transit was used by AGIS-DR4 in AL direction. It means that the AGIS downweight factors in DR4 are either zero or one.

USED_BY_AGIS_AC : whether the CCD transit was used in AC by AGIS-DR4 (Boolean[1] array)

Flag saying whether the AC information from this CCD transit was used by AGIS for producing the released astrometric parameters. It means that the AGIS downweight factors in DR4 are either zero or one. For DR4 this will always be 'false'.

Note: This flag, like the corresponding one for AL, refers only to the usage of the AC measurement in the final AGIS source update producing the released astrometric source parameters. It ignores the usage of the AC measurements for the subset of (bright) primary stars needed to set up the overall astrometric system. In particular, these are indispensable for deriving the AC attitude (which is needed for a correct usage of the AL measurements), which in turn requires an AC astrometric calibration.

TRANSIT_PROC_FLAGS : Transit-level information on the processing results (short)

This field contains a set of bitwise flags, coded as follows:

- Bits 3 to 0 (`transit_proc_flags` AND 0x000F): **IPD strategy and findings**. As

of IDT 15.1 and until IDT 21.0, as well as in IDU until DRC-02, this field was not used (0 was always stored here). As of IDU DRC-03, the following information is provided:

'0': Normal IPD. Ambiguity not tested or not detected. Default value for all records produced until IDT 21.0 and IDU until DRC-02

'1': Multiple Source IPD: As of IDT 21.1, this value is used to indicate that the earlier object in time detected in the window has been selected. In case of 2D windows with larger AC than AL separation between detected peaks: higher mu selected.

'2': Multiple Source IPD: As of IDT 21.1, this value is used to indicate that the later object in time detected in the window has been selected. In case of 2D windows with larger AC than AL separation between detected peaks: lower mu selected.

'3': Multiple Source IPD: As of IDU DRC-03, this value is used to indicate that at least one of the windows has a multiple source and that these disturbing sources were masked for the IPD processing. `ccd_proc_flags` can be checked to know which and how many windows have been affected.

'4': DIPD: As of IDU 21.0, this value is used to indicate that this object has been obtained using DIPD implementation and thus include a window model template and combined AF solution.

'5': TukeyBiWeight IPD: this value is used to indicate that this object has been obtained using TukeyBiWeight IPD initialisation for the internal processing subsystem IDU-AttAc; such initial values do not propagate into the published data.

Values 6-15 are reserved for future extensions.

- Bits 6 to 4 (0x0070): **Bias and PEM OI strategy baseline**: this field indicates the Bias mitigation mode requested to process this transit. The consumer must check the corresponding `ccd_proc_flags` field to determine the actual mode used for each individual CCD as sometimes some modes can not be applied due to missing input data or calibration limitations.

  Value '0': Default initialisation value for the Bias strategy. If found in an AstroElementary it means that something went wrong and the Bias (or maybe just its flagging) was not done appropriately. Thus, '0' means 'UNDEFINED/FAILURE'.

  Value '1': Only PreScan bias used.

  Value '2': PreScan + common NU baseline.

  Value '3': Full PEM NU.

  Values 4-7 are reserved for future extensions.

- Bits 10 to 7, **colour info**:

Bit 7 (0x0080), **default colour**: '1' means 'a default colour used for IPD processing', which can be caused by some problem in the colour determination or retrieval (e.g. no BP/RP data in IDT or no valid PhotPipe fluxes in IDU).

Bit 8 (0x0100), **epoch colour**: '1' means 'epoch (transit-level) colour used for IPD processing', '0' means 'mean (source-level) colour used'.

Bit 9 (0x0200), **single-param colour**: '1' means 'single-parameter colour used for IPD processing' (such as `nu_eff_used_in_astrometry`), '0' means 'multi-parameter colour used' (such as SSCs).

Bit 10 (0x0400), **colour problems**: '1' means 'missing colour information or dubious colour quatiliy' (e.g. problematic/missing BP/RP data in IDT, NaN or rejected PhotPipe colour in IDU, etc).

Please note some remarkable combinations of this bit with Bit 7:

- Non-default + no problems (Bit 7 = 0, Bit 10 = 0): Good colour from catalogue or epoch data
- Non-default + problems (Bit 7 = 0, Bit 10 = 1): Dubious colour from catalogue or best approximation from partial or problematic epoch data. Implemented only in IDT.
- Default + no problems (Bit 7 = 1, Bit 10 = 0): Default colour has been used because no colour information was available (i.e. for blacklisted observations). Implemented only in IDU.
- Default + problems (Bit 7 = 1, Bit 10 = 1): Default colour used instead of the available colour from catalogue or epoch data (i.e. dubious colour rejected).

- Bit 11 (0x0800), **cosmetic effects** flagging: '1' means 'dead pixel columns affecting some of the windows'.
- Bit 12 (0x1000), **cosmetic effects** flagging: '1' means 'anomalously low samples found in some of the windows'.
- Bit 13 (0x2000), **cosmetic effects** flagging: '1' means 'anomalously high samples found in some of the windows'.
- Bit 14 (0x4000), **problems in initial centroid**: '1' means that problems have been found in the Initial Centroids determination. In the case of IDT, combined with the 'some problem found in the colour determination' flag (bit 10), it may indicate problems in the TransitPredictor (e.g. predicted position outside window, or no attitude available) in which case a default position is used for colour determination.
- Bit 15, **attitude issues**: '1' indicates that some attitude problem was found affecting some of the windows i.e. missing attitude or problems computing AL/AC rates. This flag is only set by IDU which depends on the attitude to access the ELSF and for the combine AF processing in DIPD.

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See Chapter 3 of the DR4 online documentation for more information.

CCD_PROC_FLAGS : CCD-level information on the processing results (Short[1] array)

This field contains an array, with one component per CCD (SM has index 0, AF9 index 9), where in this case only the SM is present. Note furthermore that there is only one 'short' quantity per observation, while in the main `epoch_astrometry` table there are two, and that some of the bits used in the main `epoch_astrometry` table have not been used here (because they are irrelevant or not available).

Each of the components of the array contains a set of bitwise flags, coded as follows:

- Bits 3 to 0 (`ccd_proc_flags` AND 0x000F): **IPD result**. It is coded as follows:
  - '0': Success
  - '1': Illegal input (negative amplitude; may happen due to a wrong background determination)
  - '2': Illegal input (coordinate outside window; may happen in windows close to a charge injection, or strongly affected by some cosmic ray or other problems)
  - '3': Illegal output (location estimation outside window)
  - '4': Maximum iteration exceeded (convergence not reached, but results achieved so far stored in AstroElem; they may be reliable enough)
  - '5': Failed (diverging iteration instead of converging)
  - '6': Missing or unreliable; maybe the processing failed for some reason before finishing the calculations, or the IPD was initialised, but no actual processing has occurred.
  - '7': IPD crashed (no result available) due to no LSF data available from the LSF Server. AstroElem values not reliable for that CCD.
  - '8': IPD crashed (no result available), because when checking the inputs to the centroiding algorithm (after applying the mask) it was found that there are not enough inputs to the equations/matrix system.
  - '9': 'Odd' LSF model received; with the recent improvements in the handling of the realistic (empirical) LSF, there will be cases when the LSF will not be defined for some samples. NOTE that, despite of this, the IPD result may still be reliable. It should simply be handled with care (similar as with the 'Maximum Iteration Exceeded' case).
  - '15': Crashed (calculation finished and crashed for some reason; IPD output values not reliable)

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  - Values ‘10’ to ‘14’: Reserved for future extensions.

- Bit 4 (0x0010): **IPD nominal/non nominal**. ‘1’ means ‘non-nominal IPD’ (details in `ccd_proc_geom`).
- Bit 5 (0x0020): **IPD not even attempted**. ‘1’ means ‘IPD not attempted’ (and thus the IPD output values are not valid).
- Bit 6 (0x0040): **Some cosmetic issue**. ‘1’ means ‘cosmetic issue’.
- Bit 7 (0x0080): **Cosmic removed**. ‘1’ means ‘suspected cosmic removed’.
- Bit 8 (0x0100): **Saturation removed**. ‘1’ means ‘saturated sample removed’.
- Bit 9 (0x0200): **Part of window discarded**. ‘1’ means ‘part of the window has been discarded in the IPD’.
- Bit 10 (0x0400): **No window**. ‘1’ means ‘no sample data available’ (i.e. all the samples in this window were removed, or this is e.g. AF 9 row 4, or a deactivated CCD, etc).
- Bit 11 (0x0800): **Non-success InitialCentroid**. ‘1’ means ‘the Initial Centroid was not successful, and thus we have used the centre of the window as the start condition of IPD’.
- Bits 15-12: spare bits (for user-defined flags if necessary).

See Chapter 3 of the DR4 online documentation for more information.

IPD_ERROR_AL : standard error of CrowdedField IPD centroid AL (float[1] array, Angle[mas])

Standard error from the calculation of the AL image centroid, as estimated by the centroiding (IPD/DIPD) algorithm in the special crowded-field preprocessing.

For details see the processing paper, **?**.

When the IPD algorithm has given some problem (and thus a result could not be obtained), the error is NaN.

IPD_ERROR_AC : standard error of CrowdedField IPD centroid AC (float[1] array, Angle[mas])

Standard error from the calculation of the AC image centroid, as estimated by the centroiding (IPD/DIPD) algorithm in the special crowded-field preprocessing.

For details see the processing paper, **?**.

When the IPD algorithm has given some problem (and thus a result could not be obtained), the error is NaN.

G_MAG : G Magnitude, either from CU5 or from on-board estimate (float, Magnitude[mag])

This field is only needed in order to re-apply the AGIS calibrations, so it was put together with the block containing the other parameters used to evaluate the calibration.

It is the G magnitude of the source used for IPD: possibly a mean magnitude from the main photometry processing, possibly a magnitude at the epoch of this particular observation.

When a `g_mag` from the main photometry processing is not available, then it is an epoch magnitude.

AC_RATE : source image AC rate for the transit time (float, Angular Velocity[pix/s])

AC rate of the source image during the 2.5-second CCD exposure, computed during the CrowdedField IPD.

This field is only needed in order to re-apply the AGIS calibrations, so it was put together with the block containing the other parameters used to evaluate the calibration.

SUB_PIXEL_COORD : Subpixel coordinate AL (float[1] array, Length & Distance[pixel])

This field is only needed in order to re-apply the AGIS calibrations, so it was put together with the block containing the other parameters used to evaluate the calibration.

The subpixel AL coordinate for DR4 is computed as:

subpixel = mod(((transit time (OBMT) in seconds)/(TDI period in seconds)), 1.0).

MU : Ac coordinate of the image (or window centre) on a CCD (float[1] array, Length & Distance[pixel])

This field is only needed in order to re-apply the AGIS calibrations, so it was put together with the block containing the other parameters used to evaluate the calibration.

This is the AC pixel coordinate of the image on the given CCD. Precisely, for DR4 it is the

window centre (both for 1D and 2D windows), truncated to the nearest integer pixel. For DR5 it will be the 'true' image centroid AC, as predicted from source parameters, geometric calibration and attitude.

Note that the μ coordinate contained in the transit identifier is only for AF1, and it is always a window coordinate, not an actual or predicted image centroid.

IMAGE_ID : Unique crowded field image identifier (long)

The `image_id` relates the present CrowdedField observation to the specific CrowdedField SM image in which it was recorded (see Table `crowded_field_image`).

The `image_id` encodes the OBMT start time (degraded to TDI resolution), the duration of the image (in units of TDI periods), the SIF PACKET CONTENT TYPE index, and the CCD row for the given CrowdedField SM image.

It is encoded on 64 bits with the OBMT start time occupying the most significant bits.

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8.3 CROWDED_FIELD_EPOCH_PHOTOMETRY [DataLink]

This epoch photometry table contains the light curve in the G band for the sources from the crowded field source tables. This photometry is obtained exclusively from the Sky Mapper (SM) CCDs.

Note this table is not available through the main archive TAP interface. Data are delivered via the Massive Data service indexed by the VO DataLink protocol and described in Chapter **??**. For example this can be actioned in the archive user interface by joining queries with the table `crowded_field_source` and selecting `has_epoch_photometry = 't'`. The DataLink retrieval type to specify for this product in programmatic archive access via the Astroquery Gaia Python package is `EPOCH_PHOTOMETRY_CROWDED_FIELD`.

DataLink products are served with a choice of two different data structures:

- `DATAMODEL_STANDARD`: the product is serialised following a standard data model (for example an IVOA-compliant data model in applicable cases);
- `DATAMODEL_GAIA`: the product is serialised following the Gaia DPAC(-internal) data model.

The `DATAMODEL_GAIA` format is described here whereas the `DATAMODEL_STANDARD` format is described in `https://www.cosmos.esa.int/web/gaia-users/archive/datalink-products#datalink_serialisation`

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (unique within a particular Data Release) (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

TRANSIT_ID : Unique transit identifier (long)

The `transit_id` is a unique identifier assigned to each detected (and confirmed) source as it transits the Gaia focal plane. Each time a given source is detected as Gaia scans and re-scans the sky a new `transit_id` will be created to badge that apparition. Hence the along–scan time and the across–scan position along with the telescope in which the source was detected are used to form a unique integer with which to label the transit.

The several features of a detection that are encoded in `transit_id` can be easily retrieved using bit masks (&) and shifts (>>) as follows:

- On-Board Mission Time line [ns]
  = 204800 * ((`transit_id` >> 17) & (0x000003FFFFFFFFFF))
- Field-of-view = 1 + (`transit_id` >> 15) & 0x03 [1 for 'preceding' and 2 for 'following' fields-of-view respectively]
- CCD row = (`transit_id` >> 12) & 0x07 [dimensionless, in the range 1 to 7]
- Across-scan 'reference acquisition pixel' in strip AF1 = (`transit_id`) & 0x0FFF [pixels] (this is the across-scan centre of the AF1 window and is odd if immediately below the mid-point of the window and even if immediately above)

where the bit mask prefix '0x' denotes hexadecimal.

For further details see Portell et al. (2020). For convenience a decoder for `transit_id` is available on-line at

`https://gaia.esac.esa.int/decoder/transitidDecoder.jsp`

G_OBS_TIME : Observing time of the transit in SM (double, Time[Barycentric JD in TCB – 2 455 197.5 (day)])

Sky Mapper observation time in units of Barycentric JD (in TCB), in days minus 2 455 197.5, computed as follows. First the observation time is converted from On-board Mission Time (OBMT) into Julian date in TCB (Temps Coordonnée Barycentrique). Next a correction is applied for the light-travel time to the Solar system barycentre, resulting in Barycentric Julian Date (BJD). Finally an offset of 2 455 197.5 days is applied (corresponding to a reference time T_0 at 2010-01-01T00:00:00) to have a conveniently small numerical value.

G_FLUX : G band flux of the transit in SM (float, Flux[e^- s^{-1}])

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The Sky Mapper flux value of the transit.

G_FLUX_ERROR : G band flux error of the transit in SM (float, Flux[e^{-} s^{-1}])

Uncertainty on the Sky Mapper flux for this transit. This takes into account the standard error on the uncalibrated flux and the error on the calibration factor. The two errors are added in quadrature to form a single value for the uncertainty.

G_FLUX_OVER_ERROR : G band flux divided by its error (float)

Mean flux in the band divided by its error.

G_MAG : G band Vega magnitude of the transit in SM (float, Magnitude[mag])

The Sky Mapper magnitude of the transit.

PHOT_CCD_PROC_FLAGS : CCD-level information on PhotPipe processing (byte)

This flag encodes the following information:

- Bit 0: When set implies that a calibration was missing for this epoch. In this case, no calibrated flux or error will be available.
- Bit 1: When set the application of the calibration failed. In this case, no calibrated flux or error will be available.
- Bit 2: This epoch contributed to the mean photometry for this source. This implies that a calibrated flux was available and the measurement was not rejected when computing the mean source photometry statistics.
- Bit 3 to Bit 6: Origin of colour information for calibration purposes, see the explanation below.

When generating the Crowded Field (CF) photometry, the need for a pseudo-colour value was fulfilled by the following steps:

- If the CF source was crossmatched to a nominal AGIS source from `all_source_astrometry` (as per the information from `all_source_match`), then the AGIS pseudo-colour assigned to the nominal source would be preferred (Bit 3 or 4 would be set in this

case). If no pseudo-colour was available for the nominal source, then the pseudo-colour computed for the CF source as tabulated in `crowded_field_source` would be used (bit 5 would be set in this case). Failing that, no calibration could be applied (Bit 6 will be set in this case).

- If the CF source did not have any match in with a source from `all_source_astrometry`, then the pseudo-colour assigned to the CF source as tabulated in `crowded_field_source` was used if available (Bit 5 will be set in this case). Failing that, no calibration could be applied (Bit 6 will be set in this case).

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8.4 CROWDED_FIELD_IMAGE [DataLink]

This table contains the electronically de-biased and astrometrically calibrated crowded-field images published as FITS files.

The table rows include the calibrated image data, background values estimated on a sparse grid and the positions of the brightest sources. Other data, including WCS transformation parameters, is included in the header of the FITS file.

Note this table is not available through the main archive TAP interface. Data are delivered via the Massive Data service indexed by the VO DataLink protocol and described in Chapter **??**. The column to use to action linking functions in the Gaia archive is `image_id` so, for example, if queries on the associated TAP summary table `crowded_field_image_summary` include `image_id` then `crowded_field_image` data products will be available via the link functionality in the user interface. The DataLink retrieval type to specify for this product in programmatic archive access via the Astroquery Gaia Python package is `CROWDED_FIELD_IMAGE`.

The datamodel described here refers to the `DATAMODEL_GAIA` serialisation used in the Parquet files available from the bulk download repository. The FITS files accessible through the Gaia ESA Archive use the `DATAMODEL_STANDARD` serialisation and are documented in the archive help pages.

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Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

IMAGE_ID : Unique crowded field image identifier (long)

The `image_id` is a unique identifier for the individual crowded-field CCD images. The `image_id` is a structured item, encoding the OBMT start time of the image (degraded to TDI resolution), the duration of the image (i.e. its extent in along-scan direction), the Sky Mapper device used for it (SM1 or SM2), and the CCD row in which the image was taken. The `image_id` is encoded into

64 bits, with the OBMT start time occupying the most significant bits. In detail, the components of the `image_id` and their encoding are as follows:

- On-Board Mission Time of the image acquisition start: The full OBMT (in nanoseconds) can be computed as $204800 * (\texttt{image_id} >> 24)\&0x7FFFFFFFF$)
- Image duration (in TDI intervals): It can be computed as $(\texttt{image_id} >> 3)\&0xFFFF$
- CCD strip index (dimensionless, in the range 0 to 19): It can be computed as $\texttt{image_id} >> 19)\&0x19$. In the present context, i.e. as part of the crowded-field `image_id`, only two values of this quantity are valid: strip index 2 means SM1, and index 3 means SM2.
- CCD row index (dimensionless, in the range 1 to 7): It can be computed as $\texttt{image_id}\&0x07$

REGION_NAME : Crowded field region (string)

A plain text string that identifies the crowded field region in the sky that this source is located in. A list of the 9 regions observed with crowded field images can be found in the documentation of `crowded_field_source.region_name`.

IMAGE_LENGTH : Length of image in samples (int, Length & Distance[sample])

The size of the image in along scan direction in image samples. This value is determined by the length of the acquisition time, that can vary from a few seconds to a minute. Note that in crowded field images, one image sample spans 2×2 CCD pixels.

IMAGE_WIDTH : Width of image in samples (int, Length & Distance[sample])

The size of the image in across scan direction in image samples. This value is constant and determined by the CCD height. Note that in crowded field images, one image sample spans 2×2 CCD pixels.

IMAGE_START_TIME : Time of image acquisition start (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

The time at which the acquisition of this image started in Julian Days since 2010-01-01T00:00:00 observed at the solar system barycentre. It is computed as follows: the On-board Mission Time

(OBMT) is converted into Julian date in TCB (Temps Coordonnée Barycentrique), then a correction is applied for the light-travel time to the Solar system barycentre, resulting in Barycentric Julian Date (BJD), and finally an offset of 2 455 197.5 days is applied (corresponding to a reference time T_0 at 2010-01-01T00:00:00) to have a conveniently small numerical value.

IMAGE_END_TIME : Time of image acquisition end (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

The time at which the acquisition of this image ended in Julian Days since 2010-01-01T00:00:00 observed at the solar system barycentre; further details can be found under `image_start_time`.

WCS_HEADER : World Coordinate System header (string)

The World Coordinate System header. Stored as key-value pairs in JSON, to be later transformed into a FITS HDU header.

IMAGE : Calibrated crowded field image (float[] array, Charge[e-])

The calibrated crowded field image. This is the instrumentally corrected image data as fed into the first software of the crowded field pipeline, the Image Parameter Determination (IPD).

BACKGROUND_KNOTS_X : X position of background knots (int[] array, Length & Distance[sample])

The background is estimated on a grid over the image. The along scan (x-axis) positions of the grid knots are given in this list.

BACKGROUND_KNOTS_Y : Y position of background knots (int[] array, Length & Distance[sample])

The across scan (y-axis) positions of the background grid knots are given in this list.

BACKGROUND_VALUES : Values of background knots (float[] array, Charge[e-])

This array provides the background values for all grid points, as defined in `background_knots_x` and `background_knots_y`. The background should be interpolated between these knots.

GAIN : CCD Gain (float, Conversion[ADU/e-])

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The CCD transfer gain.

TDN : Total detection noise (float, Charge[e-])

The total detection noise.

IMAGE_BOUNDARIES_RA : Right ascension of image polygon corners (float[] array, Angle[deg])

The image boundaries consist of the edge points of the polygon best describing the image shape on the sky. `image_boundaries_ra` provides the right ascensions of these edge points.

The polygon provided here is the one that best describes the full image and differs from the polygon used in the crowded field image processing pipeline. For the pipeline, a two sample wide rim is cut from all polygons in order to avoid source detections at the very border of the image. Additionally, in the pipeline some time ranges are cut from the image, which happens in case the satellite attitude is not available or strong straylight features were detected. In the processing, a single image might therefore be split into several subimages with gaps inbetween. For the affected source locations, the number of scans calculated from the image boundaries provided here would not correspond to the number of scans used in the crowded field pipeline. The latter is provided in the field `n_scans` of the `crowded_field_source` table.

IMAGE_BOUNDARIES_DEC : Declination of image polygon corners (float[] array, Angle[deg])

The declination of the edge points of the polygon best describing the shape of the image on the sky.

SOURCE_IDS : Source identifiers of selected bright stars (long[] array)

A few selected bright sources that are contained in this SIF CF image are provided to facilitate image-to-sky mapping without using the World Coordinate System. This field contains the list of source identifiers for the selected bright stars.

TRANSIT_IDS : Transit identifiers of selected bright stars (long[] array)

This field contains the list of transit identifiers for the selected bright stars contained within the image.

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SOURCE_X : X positions of selected bright stars (double[] array, Length & Distance[sample])

This field contains the list of along scan (or x-axis) positions for the selected bright stars.

SOURCE_Y : Y positions of selected bright stars (double[] array, Length & Distance[sample])

This field contains the list of across scan (or y-axis) positions for the selected bright stars.

SOURCE_RA : Right ascension of selected bright stars (double[] array, Angle[deg])

This field contains the list of right ascensions for the selected bright stars, propagated to the image epoch.

SOURCE_DEC : Declination of selected bright stars (double[] array, Angle[deg])

This field contains the list of declinations for the selected bright stars, propagated to the image epoch.

CCD_ROW : CCD Row (short)

The CCD row specifies the position of the precise CCD detector this image was taken with on the focal plane in across scan direction.

FOV : Field of view (short)

This field specifies which of the two fields of view is contained in this crowded field image. Other than all other *Gaia* observations, the crowded field images contain only one single field of view at a time.

IMAGE_CENTRE_RA : Right ascension of image centre (float, Angle[deg])

The right ascension of the centre of the crowded field image.

The image centre provides an approximate location only, and should therefore be used only as an estimated reference point. More accurate image boundaries are provided via `image_boundaries_ra` and `image_boundaries_dec`.

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IMAGE_CENTRE_DEC : Declination of image centre (float, Angle[deg])

The declination of the centre of the crowded field image.

SCAN_ANGLE : Mean scan angle over the image (float, Angle[deg])

The average scan angle for this image. This is only an approximate value taken at the image centre, and should be used as a reference only.

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8.5 CROWDED_FIELD_IMAGE_SUMMARY [TAP]

This table contains metadata for the electronically de-biased and astrometrically calibrated crowded-field images (also known as SIF CF images) published as FITS files.

The metadata includes the image size, the acquisition time, the CCDs used, the edge points of the bounding polygon (describing the image position on the celestial sphere) and some other properties.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

IMAGE_ID : Unique crowded field image identifier (long)

The `image_id` is a unique identifier for the individual crowded-field CCD images. The `image_id` is a structured item, encoding the OBMT start time of the image (degraded to TDI resolution), the duration of the image (i.e. its extent in along-scan direction), the Sky Mapper device used for it (SM1 or SM2), and the CCD row in which the image was taken. The `image_id` is encoded into 64 bits, with the OBMT start time occupying the most significant bits. In detail, the components of the `image_id` and their encoding are as follows:

- On-Board Mission Time of the image acquisition start: The full OBMT (in nanoseconds) can be computed as $204800 * (\texttt{image_id} >> 24)\&0x7FFFFFFFF$)
- Image duration (in TDI intervals): It can be computed as $(\texttt{image_id} >> 3)\&0xFFFF$
- CCD strip index (dimensionless, in the range 0 to 19): It can be computed as `image_id` $>>$ 19)$\&0x19$. In the present context, i.e. as part of the crowded-field `image_id`, only two values of this quantity are valid: strip index 2 means SM1, and index 3 means SM2.
- CCD row index (dimensionless, in the range 1 to 7): It can be computed as `image_id`$\&0x07$

REGION_NAME : Crowded field region (string)

A plain text string that identifies the crowded-field region in the sky that this image is located in. A list of the 9 regions observed with crowded field images can be found in the documentation of `crowded_field_source.region_name`.

IMAGE_LENGTH : Length of image in samples (int, Length & Distance[sample])

The size of the image in along scan direction in image samples. This value is determined by the length of the acquisition time, that can vary from a few seconds to a minute. In crowded field images one sample spans 2×2 CCD pixels.

IMAGE_WIDTH : Width of image in samples (int, Length & Distance[sample])

The size of the image in across scan direction in image samples. This value is constant and determined by the CCD height. In crowded field images one sample spans 2×2 CCD pixels.

IMAGE_START_TIME : Time of image acquisition start (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

The time at which the acquisition of this image started in Julian Days since 2010-01-01T00:00:00 observed at the solar system barycentre. It is computed as follows: the On-board Mission Timeline (OBMT) is converted into Julian date in TCB (Temps Coordonnée Barycentrique), then a correction is applied for the light-travel time to the Solar system barycentre, resulting in Barycentric Julian Date (BJD), and finally an offset of 2 455 197.5 days is applied (corresponding to a reference time T_0 at 2010-01-01T00:00:00) to have a conveniently small numerical value.

IMAGE_END_TIME : Time of image acquisition end (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

The time at which the acquisition of this image ended in Julian Days since 2010-01-01T00:00:00 observed at the solar system barycentre; for further details see under `image_start_time`.

GAIN : CCD Gain (float, Conversion[ADU/e-])

The CCD transfer gain.

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TDN : Total detection noise (float, Charge[e-])

The total detection noise.

IMAGE_BOUNDARIES_RA : Right ascension of image polygon corners (float[] array, Angle[deg])

The image boundaries consist of the edge points of the polygon best describing the image shape on the sky. `image_boundaries_ra` provides the right ascensions of these edge points.

The polygon provided here is the one that best describes the full image and differs from the polygon used in the crowded-field image processing pipeline. For the pipeline, a two sample wide rim is cut from all polygons in order to avoid source detections at the very border of the image. Additionally, in the pipeline some time ranges are cut from the image, which happens in case the satellite attitude is not available or strong straylight features were detected. In the processing, a single image might therefore be split into several subimages with gaps inbetween. For the affected source locations, the number of scans calculated from the image boundaries provided here would not correspond to the number of scans used in the crowded field pipeline. The latter is provided in the field `n_scans` of the `crowded_field_source` table.

IMAGE_BOUNDARIES_DEC : Declination of image polygon corners (float[] array, Angle[deg])

The declination of the edge points of the polygon best describing the shape of the image on the sky.

CCD_ROW : CCD Row (short)

The CCD row, together with `fov`, specifies the position of the specific CCD detector this image was taken with on the focal plane.

FOV : Field of view (short)

This field specifies which of the two fields of view is contained in this crowded-field image. Values are 0 (zero) for the preceding and 1 (one) for the following field of view.

Note: Other than all other *Gaia* observations, the crowded-field images, being collected on the Sky Mapper CCDs, contain only one single field of view at a time.

IMAGE_CENTRE_RA : Right ascension of image centre (float, Angle[deg])

The right ascension of the centre of the crowded-field image. The image centre provides an approximate location only, and should therefore be used only as an estimated reference. More accurate image boundaries are provided via `image_boundaries_ra` and `image_boundaries_dec`.

IMAGE_CENTRE_DEC : Declination of image centre (float, Angle[deg])

The declination of the centre of the crowded-field image.

SCAN_ANGLE : Mean scan angle over the image (float, Angle[deg])

The average scan angle for this image. This is only an approximate value taken at the image centre, and should be used as a reference only.

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8.6 CROWDED_FIELD_SOURCE [TAP]

Sources created with the crowded-field image analysis pipeline.

The crowded-field images, also called Service Interface Function (SIF) images of crowded fields (CF), were obtained with *Gaia's* Sky Mapper instrument for nine selected regions, indicated by column `region_name` for each source recorded in the table. They were processed with a specially implemented Gaia pipeline that deviates significantly from nominal processing, especially in the source detection. Details of the images and processing are described in Chapter **??** of the online documentation and a dedicated DR4 paper **?**.

The meaning and presentation of the data is largely the same as in the other DR4 source tables, although with some specific modifications, omissions and additions.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

DESIGNATION : Unique source designation (unique across all Data Releases) (string)

A source designation, unique across all Gaia Data Releases, that is constructed from the prefix 'Gaia DRx ' or 'Gaia FPR ' followed by a string of digits corresponding to `source_id` (3 space-separated words in total). Note that the integer source identifier `source_id` is **not** guaranteed to be unique across Data Releases; moreover it is not guaranteed that the same astronomical source will always have the same `source_id` in different Data Releases. Hence the only safe way to compare source records between different Data Releases in general is to check the records of proximal source(s) in the same small part of the sky. For convenience, in table `dr3_neighbourhood` DPAC provides a cross-match for those DR4 sources that already existed in DR3.

REGION_NAME : Crowded field region (string)

A plain text string that identifies the crowded field region in the sky in which this source is located. Gaia observed 9 regions with crowded field images. The corresponding region names are:

BAADE_S_WINDOW: Baade's Window (Bulge region);
SGR_I: Sagittarius window (Bulge region);
LMC: Large Magellanic Cloud, satellite galaxy of the Milky Way;
SMC: Small Magellanic Cloud, dwarf irregular galaxy near the Milky Way;
NGC104: (aka 47 Tucanae) in the constellation Tucana;
NGC4372: (aka Caldwell 108) in the southern constellation of Musca;
NGC5139: (aka Omega Centauri or Caldwell 80) in the constellation of Centaurus;
NGC6121: (aka Messier 4) in the constellation of Scorpius;
NGC6656: (aka Messier 22) in the constellation Sagittarius.

For further details see Chapter **??** of the online documentation.

SOURCE_ID : Unique source identifier (unique within a particular Data Release) (long)

A unique numerical identifier of the source, encoding the approximate position of the source (roughly to the nearest arcmin), the provenance (data processing centre where it was created), a running number, and a component number following the construction conventions of those in the main source catalogue (for a detailed description see `gaia_source.source_id`). Note that the data processing provenance code stored in bits 33 to 35 is 6 for the crowded field image analysis, which means (`source_id` >> 32) & 7 = 6 for `crowded_field_source`.

REF_EPOCH : Reference epoch (double, Time[Julian Years])

Reference epoch to which the astrometric source parameters are referred, expressed as a Julian Year in TCB (Temps Coordonnée Barycentrique = Barycentric Coordinate Time).

ASTRO_PIPELINE : Pipeline module that provided the astrometry for this source (short)

This field indicates the pipeline module that provided the astrometric parameters for the source of interest. When no AGIS solution is available for the given Gaia data release, the astrometric parameters will be populated by that of another module (specifically IDU):

- Value 3 indicates that the astrometric parameters come from AGIS DR4 (from the special processing for crowded fields);
- Value 8 indicates that the astrometric parameters come from IDU DR4.

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Other values are unused at present.

HAS_EPOCH_ASTROMETRY : Flag indicating the availability of epoch astrometry for this source (boolean)

This flag indicates that epoch astrometry is available for the source. These are served through the DataLink protocol – see `crowded_field_epoch_astrometry` for further details.

HAS_EPOCH_PHOTOMETRY : Flag indicating the availability of epoch photometry for this source (boolean)

This flag indicates that epoch photometry is available for the source. Epoch photometry for crowded field sources are limited to G band integrated photometry. These are served through the DataLink protocol – see `crowded_field_epoch_photometry` for more details.

IS_SPURIOUS : Flag indicating whether the source is considered spurious (boolean)

Flag indicating whether the source is considered spurious. The filtering and flagging strategy is further detailed in Chapter **??**.

MATCHED : Flag indicating whether the source is in the all source match table (boolean)

Flag indicating whether the source is in the `all_source_match` table.

RA : Right ascension (double, Angle[deg])

Barycentric right ascension α of the source in ICRS at the reference epoch `ref_epoch`.

RA_ERROR : Standard error of right ascension (float, Angle[mas])

Standard error $\sigma_{\alpha*} \equiv \sigma_\alpha \cos\delta$ of the right ascension of the source in ICRS at the reference epoch `ref_epoch`.

DEC : Declination (double, Angle[deg])

Barycentric declination δ of the source in ICRS at the reference epoch `ref_epoch`.

DEC_ERROR : Standard error of declination (float, Angle[mas])

Standard error σ_δ of the declination of the source in ICRS at the reference epoch `ref_epoch`.

PARALLAX : Parallax (double, Angle[mas])

Absolute stellar parallax ϖ of the source at the reference epoch `ref_epoch`.

PARALLAX_ERROR : Standard error of parallax (float, Angle[mas])

Standard error σ_ϖ of the stellar parallax at the reference epoch `ref_epoch`.

PARALLAX_OVER_ERROR : Parallax divided by its standard error (float)

Parallax divided by its standard error.

PM : Total proper motion (float, Angular Velocity[mas yr^{-1}])

The total proper motion `pm`, calculated as the magnitude of the resultant vector of the proper motion component vectors `pmra` and `pmdec`, i.e. $\texttt{pm}^2 = \texttt{pmra}^2 + \texttt{pmdec}^2$.

PMRA : Proper motion in right ascension direction (double, Angular Velocity[mas yr^{-1}])

Proper motion in right ascension $\mu_{\alpha*} \equiv \mu_\alpha \cos\delta$ of the source in ICRS at the reference epoch `ref_epoch`. This is the local tangent plane projection of the proper motion vector in the direction of increasing right ascension.

PMRA_ERROR : Standard error of proper motion in right ascension direction (float, Angular Velocity[mas yr^{-1}])

Standard error $\sigma_{\mu\alpha*}$ of `pmra`.

PMDEC : Proper motion in declination direction (double, Angular Velocity[mas yr^{-1}])

Proper motion in declination μ_δ of the source at the reference epoch `ref_epoch`. This is the projection of the proper motion vector in the direction of increasing declination.

PMDEC_ERROR : Standard error of proper motion in declination direction (float, Angular Velocity[mas yr^{-1}])

Standard error $\sigma_{\mu\delta}$ of `pmdec`.

RA_DEC_CORR : Correlation between right ascension and declination (float)

Correlation coefficient $\rho(\alpha, \delta)$ between right ascension and declination. This is a dimensionless quantity in the range [-1,+1].

RA_PARALLAX_CORR : Correlation between right ascension and parallax (float)

Correlation coefficient $\rho(\alpha, \varpi)$ between right ascension and parallax, a dimensionless quantity in the range [-1,+1].

RA_PMRA_CORR : Correlation between RA and proper motion in RA (float)

Correlation coefficient $\rho(\alpha, \mu_{\alpha*})$ between right ascension and proper motion in right ascension, a dimensionless quantity in the range [-1,+1].

RA_PMDEC_CORR : Correlation between right ascension and proper motion in declination (float)

Correlation coefficient $\rho(\alpha, \mu_\delta)$ between right ascension and proper motion in declination, a dimensionless quantity in the range [-1,+1].

DEC_PARALLAX_CORR : Correlation between declination and parallax (float)

Correlation coefficient $\rho(\delta, \varpi)$ between declination and parallax, a dimensionless quantity in the range [-1,+1].

DEC_PMRA_CORR : Correlation between declination and proper motion in right ascension (float)

Correlation coefficient $\rho(\delta, \mu_{\alpha*})$ between declination and proper motion in right ascension, a di-

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mensionless quantity in the range [-1,+1].

DEC_PMDEC_CORR : Correlation between declination and proper motion in declination (float)

Correlation coefficient $\rho(\delta, \mu_\delta)$ between declination and proper motion in declination, a dimensionless quantity in the range [-1,+1].

PARALLAX_PMRA_CORR : Correlation between parallax and proper motion in right ascension (float)

Correlation coefficient $\rho(\varpi, \mu_{\alpha*})$ between parallax and proper motion in right ascension, a dimensionless quantity in the range [-1,+1].

PARALLAX_PMDEC_CORR : Correlation between parallax and proper motion in declination (float)

Correlation coefficient $\rho(\varpi, \mu_\delta)$ between parallax and proper motion in declination, a dimensionless quantity in the range [-1,+1].

PMRA_PMDEC_CORR : Correlation between proper motion in RA and proper motion in Dec (float)

Correlation coefficient $\rho(\mu_{\alpha*}, \mu_\delta)$ between proper motion in right ascension and proper motion in declination, a dimensionless quantity in the range [-1,+1].

N_SCANS : Number of scans (short)

The number of scans covering this source location, `n_scans`, is determined as the number of bounding polygons describing crowded field images that cover this source location. It reflects the number of times a source could have been observed. When compared to the number of actually matched detections of this source `matched_transits`, it allows an assessment of the reliability of the source.

ASTROMETRIC_N_OBS_AL : Total number of observations in the along-scan (AL) direction (short)

Total number of AL observations (= CCD transits) used in the astrometric solution of the source,

independent of their weight. Note that some observations may be strongly downweighted (see `astrometric_n_good_obs_al`).

For the specific case of crowded-field sources, each FoV transit creates only one CCD transit at most, viz. the one from the Sky Mapper CCD.

ASTROMETRIC_N_OBS_AC : Total number of observations in the across-scan (AC) direction (short)

Total number of AC observations (= CCD transits) used in the astrometric solution of the source, independent of their weight (note that some observations may be strongly downweighted).

For DR4, the value is always zero, as even for 2D windows the AC observations are not used by AGIS.

For nominal *Gaia* sources, nearly all sources having G $<$ 13 mag will have AC observations from 2D windows, while fainter than that limit only $\sim$ 1% of transit observations (known as 'calibration faint stars') are assigned 2D windows resulting in AC observations. Crowded field sources instead are always based on 2D images.

For the specific case of crowded-field sources, each FoV transit creates only one CCD transit at most, viz. the one from the Sky Mapper CCD.

ASTROMETRIC_N_GOOD_OBS_AL : Number of good observations in the along-scan (AL) direction (short)

Number of AL observations (= CCD transits) that were not strongly downweighted in the astrometric solution of the source. The number of strongly downweighted observations (with downweighting factor $w < 0.2$) can instead be derived from
`astrometric_n_obs_al` - `astrometric_n_good_obs_al`,
where `astrometric_n_obs_al` is the total number of AL observations used in the astrometric solution of the source.

ASTROMETRIC_GOF_AL : Goodness of fit statistic of model wrt along-scan observations (float)

Goodness-of-fit statistic of the astrometric solution for the along-scan measurements of the source. This is a rescaled version of the 'gaussianized chi-square', which for good fits should approximately follow a normal distribution with zero mean value and unit standard deviation. Values exceeding, say, +3 thus indicate a significantly bad fit to the data.

Starting with DR4, `astrometric_gof_al` is probably more useful than `ruwe` as a general goodness-of-fit measure, because it takes into account the dependence on the degree of freedom and has a simple statistical interpretation.

Note that the `astrometric_gof_al` is not simply the 'gaussianized' version of the original `astrometric_chi2_al` but an empirically rescaled version. The rescaling procedure is described and motivated in detail in **?** where an explanation discussion of other astrometric quality indicators is provided.

ASTROMETRIC_CHI2_AL : AL chi-square value (float)

Weighted sum of the squared astrometric residuals (χ^2) in the AL direction.

χ^2 values were computed using the 'good' AL observations of the source, without taking into account the `astrometric_excess_noise` (if any) of the source. They do however take into account the calibration excess noise (if any) of each observation.

See **?** for a discussion of other astrometric quality indicators.

ASTROMETRIC_EXCESS_NOISE : Excess noise of the source (float, Angle[mas])

This is the rescaled excess noise ϵ_i of the source. It is not the same as the source excess noise in Gaia DR3 and earlier releases, but re-scaled in order to be consistent with the (also redefined) unit weight error (`ruwe`). But, like the original quantity in earlier releases, it measures the disagreement, expressed as an angle, between the observations of a source and the best-fitting standard astrometric model (using six astrometric parameters). The assumed observational noise in each observation is quadratically increased by ϵ_i in order to statistically match the residuals in the astrometric solution.

A value of $\epsilon_i = 0$ thus signifies that the source is astrometrically well-behaved, i.e. that the residuals of the fit statistically agree with the assumed observational noise. Conversely, a positive ϵ_i signifies that the residuals are statistically larger than expected. This may happen for sources that deviate from the standard six-parameter astrometric model (e.g. unresolved binaries, exoplanet systems, etc.). The statistical significance of the excess noise is given by `astrometric_gof_al`.

Although the rescaling makes ϵ_i much less sensitive to instrumental effects than in previous releases, it may still absorb residual modelling errors not accounted for by the observational noise (image centroiding error) and the calibration excess noise. This includes LSF and PSF calibration errors, saturation effects, geometric instrument calibration errors, and part of the high-frequency attitude noise. Such modelling errors may be particularly important for very bright sources and sources of extreme extreme colours. Given the many other possible contributions to the excess noise, the user is advised to study the empirical distributions of ϵ_i and `astrometric_gof_al`

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for their particular samples in order to make sensible choices of filtering limits.

The source excess noise is further explained in **?** and in Section **??** of the DR4 online documentation.

ASTROMETRIC_PARAMS : Bitwise code describing which astrometric parameters are provided (byte)

The seven bits of `astrometric_params` indicate which parameters have been estimated in AGIS for this source. A set bit means the parameter was updated, an unset bit means the parameter was not updated. The least-significant bit corresponds to `ra`. The table below shows the values of `astrometric_params` for relevant combinations of the parameters.

The radial proper motion (μ_r) is formally considered to be one of the astrometric parameters of a source, and the sixth bit is therefore reserved for it. It is also in principle updatable in AGIS, but in practice it will always be computed from a spectroscopic radial velocity and the estimated parallax, in which case the bit is not set.

C is the `pseudocolour` of the source, i.e. the astrometrically estimated effective wavenumber.

`astrometric_params`	`ra`	`dec`	`parallax`	`pmra`	`pmdec`	μ_r	C
$0000011_2 = 3$	✓	✓					
$0000111_2 = 7$	✓	✓	✓				
$0011011_2 = 27$	✓	✓		✓	✓		
$0011111_2 = 31$	✓	✓	✓	✓	✓		
$0111111_2 = 63$	✓	✓	✓	✓	✓	✓	
$1011111_2 = 95$	✓	✓	✓	✓	✓		✓

In practice all the sources in DR4 have only values of 3, 31 or 95 for the `astrometric_params`, corresponding to two-parameter (position), five-parameter (position, parallax, and proper motion) and six-parameter (position, parallax, proper motion and astrometrically estimated effective wavenumber) solutions.

NU_EFF_USED_IN_ASTROMETRY : Effective wavenumber of the source used in the astrometric solution (float, Misc[μm^{-1}])

This effective wavenumber $\nu_{\rm eff}$, in units of $\mu{\rm m}^{-1}$, is the assumed value used for the selection of the appropriate LSF/PSF model in the image parameter determination, as well as for the selection of the astrometric calibration parameters in the astrometric adjustment, for the given source. In this table, no BP and RP photometry has been used and `nu_eff_used_in_astrometry` is set to the default value $\nu_{\rm eff} = 1.43\mu{\rm m}^{-1}$.

Additional details are provided in the `all_source_photometry` table `nu_eff_used_in_astrometry` field description.

PSEUDOCOLOUR : Astrometrically estimated pseudocolour of the source (float, Misc[μm^{-1}])

Effective wavenumber of the source estimated in the final astrometric processing.

The `pseudocolour` is the astrometrically estimated effective wavenumber of the photon flux distribution in the astrometric (G) band, measured in μm^{-1}. The value in this field was estimated from the chromatic displacements of image centroids, calibrated by means of the photometrically determined effective wavenumbers ($\nu_{\rm eff}$) of primary sources. The field is empty for sources for which a position-only solution is reported.

For the crowded-field sources, the default prior 1.43μm^{-1} with formal uncertainty 0.08μm^{-1} (see explanation in the DR4 astrometry paper, **?**) has always been used for the `pseudocolour` estimation.

PSEUDOCOLOUR_ERROR : Standard error of the pseudocolour of the source (float, Misc[μm^{-1}])

Standard error $\sigma_{\texttt{pseudocolour}}$ of the astrometrically determined `pseudocolour` of the source.

RA_PSEUDOCOLOUR_CORR : Correlation between right ascension and pseudocolour (float)

Correlation coefficient $\rho(\alpha, \texttt{pseudocolour})$ between right ascension `ra` and `pseudocolour`, a dimensionless quantity in the range [-1,+1].

DEC_PSEUDOCOLOUR_CORR : Correlation between declination and pseudocolour (float)

Correlation coefficient $\rho(\delta, \texttt{pseudocolour})$ between declination `dec` and `pseudocolour`, a dimensionless quantity in the range [-1,+1].

PARALLAX_PSEUDOCOLOUR_CORR : Correlation between parallax and pseudocolour (float)

Correlation coefficient $\rho(\varpi, \texttt{pseudocolour})$ between `parallax` and `pseudocolour`, a dimensionless quantity in the range [-1,+1].

PMRA_PSEUDOCOLOUR_CORR : Correlation between proper motion in right asension and pseu-

docolour (float)

Correlation coefficient $\rho(\mu_{\alpha*}, \texttt{pseudocolour})$ between proper motion in right ascension `pmra` and `pseudocolour`, a dimensionless quantity in the range [-1,+1].

PMDEC_PSEUDOCOLOUR_CORR : Correlation between proper motion in declination and pseudocolour (float)

Correlation coefficient $\rho(\mu_{\delta}, \texttt{pseudocolour})$ between proper motion in declination `pmdec` and `pseudocolour`, a dimensionless quantity in the range [-1,+1].

ASTROMETRIC_MATCHED_TRANSITS : Matched FOV transits used in the AGIS solution (short)

The number of field-of-view (FoV) transits matched to this source, counting only the transits containing CCD observations actually used to compute the astrometric solution.

This number will always be equal to or smaller than `matched_transits`, the difference being the FoV transits that were not used in the astrometric solution because of bad data or excluded time intervals.

VISIBILITY_PERIODS_USED : Number of visibility periods used in Astrometric solution (short)

Number of visibility periods used in the astrometric solution.

A visibility period is a group of observations separated from other groups by a gap of at least 4 days. A source may have from one to tens of field–of–view transits in a visibility period, but with a small spread in time, direction of scanning, and parallax factor. From one visibility period to the next these variables have usually changed significantly. A high number of visibility periods is therefore a better indicator of an astrometrically well–observed source than a large number of field-of-view transits (`matched_transits` or `astrometric_matched_transits`) or CCD observations (`astrometric_n_obs_al`). A small value (e.g. less than 10) indicates that the calculated parallax could be more vulnerable to errors, e.g. from the calibration model, not reflected in the formal uncertainties. See **?** for a discussion of this and other astrometric quality indicators.

ASTROMETRIC_SIGMA5D_MAX : The longest semi-major axis of the 5-d error ellipsoid (float, Angle[mas])

The longest principal axis in the 5-dimensional error ellipsoid.

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This is a 5-dimensional equivalent to the semi-major axis of the position error ellipse and is therefore useful for filtering out cases where one of the five parameters, or some linear combination of several parameters, is particularly ill-determined. It is measured in mas and computed as the square root of the largest singular value of the scaled 5×5 covariance matrix of the astrometric parameters. The matrix is scaled so as to put the five parameters on a comparable scale, taking into account the maximum along-scan parallax factor for the parallax and the time coverage of the observations for the proper motion components. If C is the unscaled covariance matrix, the scaled matrix is SCS, where $S = \mathrm{diag}(1, 1, \sin\xi, T/2, T/2)$, $\xi = 45°$ is the solar aspect angle in the nominal scanning law, and T the time coverage of the data used in the solution.

`astrometric_sigma5d_max` is given for all the solutions, as its size is one of the criteria for accepting or rejecting the 5 or 6-parameter solution. In case of a 2- parameter solution (`astrometric_params` = 3) it gives the value for the rejected 5 or 6-parameter solution, and can then be arbitrarily large.

MATCHED_TRANSITS : The number of transits matched to this source (short)

The total number of field-of-view transits matched to this source.

IPD_GOF_HARMONIC_AMPLITUDE : Amplitude of the IPD GoF versus position angle of scan (float)

This statistic measures the amplitude of the variation of the Image Parameter Determination (IPD) goodness–of–fit (GoF; reduced chi-square) as function of the position angle of the scan direction. A large amplitude indicates that the source has some non-isotropic spatial structure, for example a binary or galaxy, that is at least partially resolved by Gaia. The phase of the variation is given by the parameter `ipd_gof_harmonic_phase`.

Let ψ be the position angle of the scan direction. The following expression is fitted to the IPD GoF for the accepted AF observations of the source:

$$\ln(\mathrm{GoF}) = c_0 + c_2 \cos(2\psi) + s_2 \sin(2\psi)$$

The amplitude and phase of the variation are calculated as

$$\texttt{ipd_gof_harmonic_amplitude} = \sqrt{c_2^2 + s_2^2}$$

$$\texttt{ipd_gof_harmonic_phase} = \frac{1}{2}\,\mathrm{atan2}(s_2, c_2) \quad (+\,180°)$$

where atan2 returns the angle in degrees. In the last expression 180 is added for negative values, so that `ipd_gof_harmonic_phase` is always between 0 and 180°. Only the AF observations accepted by the astrometric solution are used to compute the amplitude and phase, thus for example outliers and observations in the early Ecliptic Pole Scanning Law phase are not used.

The GoF variation is modelled as a periodic function of 2ψ because a source with fixed structure is normally expected to give fits of similar quality when scanned in opposite directions (ψ differing by 180°). See `ipd_gof_harmonic_phase` for the interpretation of the phase.

IPD_GOF_HARMONIC_PHASE : Phase of the IPD GoF versus position angle of scan (float, Angle[deg])

This statistic measures the phase of the variation of the IPD GoF (reduced chi-square) as function of the position angle of the scan direction. See the description of `ipd_gof_harmonic_amplitude` for details on the computation of the phase.

The interpretation of this parameter is non-trivial because of the complex interaction between the source structure and the IPD. For crowded field sources, the GoF always refers to the fitting of a two-dimensional PSF.

SCAN_DIRECTION_STRENGTH_K1 : Degree of concentration of scan directions across the source (float)

The `scan_direction_strength_k1...4` and `scan_direction_mean_k1...4` quantify the distribution of AL scan directions across the source. `scan_direction_strength_k1` (and similarly 2,3,4) equals $\left|\langle\exp(ik\theta)\rangle\right|$ for $k = 1, 2, 3, 4$ where θ is the position angle of the scan and the mean value is taken over the `astrometric_n_good_obs_al` observations contributing to the astrometric parameters of the source. θ is defined in the usual astronomical sense: $\theta = 0°$ when the FoV is moving towards local North, and $\theta = 90°$ towards local East.

The `scan_direction_strength_k1...4` are numbers between 0 and 1, where 0 means that the scan directions are well spread out in different directions, while 1 means that they are concentrated in a single direction (given by the corresponding `scan_direction_mean_k1...4`).

The different orders k are statistics of the scan directions modulo $360°/k$. For example, at first order ($k = 1$), $\theta = 10°$ and $\theta = 190°$ count as different directions, but at second order ($k = 2$) they are the same. Thus, `scan_direction_strength_k1` is the degree of concentration when the sense of direction is taken into account, while `scan_direction_strength_k2` is the degree of concentration without regard to the sense of direction. A large value of `scan_direction_strength_k4` indicates that the scans are concentrated in two nearly orthogonal directions.

SCAN_DIRECTION_STRENGTH_K2 : Degree of concentration of scan directions across the source (float)

The `scan_direction_strength_k1...4` and `scan_direction_mean_k1...4` attributes quan-

tify the distribution of AL scan directions across the source.

See the description for attribute scan_direction_strength_k1 for further details.

SCAN_DIRECTION_STRENGTH_K3 : Degree of concentration of scan directions across the source (float)

The scan_direction_strength_k1...4 and scan_direction_mean_k1...4 attributes quantify the distribution of AL scan directions across the source.

See the description for attribute scan_direction_strength_k1 for further details.

SCAN_DIRECTION_STRENGTH_K4 : Degree of concentration of scan directions across the source (float)

The scan_direction_strength_k1...4 and scan_direction_mean_k1...4 attributes quantify the distribution of AL scan directions across the source.

See the description for attribute scan_direction_strength_k1 for further details.

SCAN_DIRECTION_MEAN_K1 : Mean position angle of scan directions across the source (float, Angle[deg])

The scan_direction_strength_k1...4 and scan_direction_mean_k1...4 attributes quantify the distribution of AL scan directions across the source. scan_direction_mean_k1 (and similarly for $k = 2, 3, 4$) equals $\arg(\langle\exp(ik\theta)\rangle)/k$ for k = 1,2,3,4 where θ is the position angle of the scan and the mean value is taken over the astrometric_n_good_obs_al observations contributing to the astrometric parameters of the source. θ is defined in the usual astronomical sense: $\theta = 0°$ when the FoV is moving towards local North, and $\theta = 90°$ towards local East.

scan_direction_mean_k1 (and similarly for $k = 2, 3, 4$) is an angle between $-180°/k$ and $+180°/k$, giving the mean position angle of the scans at order k.

The different orders k are statistics of the scan directions modulo $360°/k$. For example, at first order ($k = 1$), $\theta = 10°$ and $\theta = 190°$ count as different directions, but at second order ($k = 2$) they are the same. Thus, scan_direction_mean_k1 is the mean direction when the sense of direction is taken into account, while scan_direction_mean_k2 is the mean direction without regard to the sense of the direction. For example, scan_direction_mean_k1 = 0 means that the scans preferentially go towards North, while scan_direction_mean_k2 = 0 means that they preferentially go in the North-South direction, and scan_direction_mean_k4 = 0 that

they preferentially go either in the North-South or in the East-West direction.

SCAN_DIRECTION_MEAN_K2 : Mean position angle of scan directions across the source (float, Angle[deg])

The scan_direction_strength_k1...4 and scan_direction_mean_k1...4 attributes quantify the distribution of AL scan directions across the source.

See the description for attribute scan_direction_mean_k1 for further details.

SCAN_DIRECTION_MEAN_K3 : Mean position angle of scan directions across the source (float, Angle[deg])

The scan_direction_strength_k1...4 and scan_direction_mean_k1...4 attributes quantify the distribution of AL scan directions across the source.

See the description for attribute scan_direction_mean_k1 for further details.

SCAN_DIRECTION_MEAN_K4 : Mean position angle of scan directions across the source (float, Angle[deg])

The scan_direction_strength_k1...4 and scan_direction_mean_k1...4 attributes quantify the distribution of AL scan directions across the source.

See the description for attribute scan_direction_mean_k1 for further details.

ASTROMETRIC_DUPLICATE_SOURCE_ID : Duplicate source identifier (long)

This field provides the source_id of a source which has been identified as a duplicate of the present source astrometrically (i.e. it is very close to another source).

If the field is identical to the present source_id, it means that the source has other duplicates and that the present one is the preferred source that should be kept in case of duplicate-sources removal in CU9 or IDU for example.

If the source has no duplicates identified, this field will be null.

ASTROMETRIC_PRIMARY_FLAG : Primary or secondary (boolean)

Flag indicating if this source was used as a primary source (`true`) or secondary source (`false`). Only primary sources contribute to the estimation of attitude, calibration, and global parameters. The estimation of source parameters is otherwise done in exactly the same way for primary and secondary sources.

PHOT_G_N_OBS : Number of observations contributing to G photometry (short)

Number of observations (CCD transits) that contributed to the G mean flux (`phot_g_mean_flux`) and mean flux error (`phot_g_mean_flux_error`).

PHOT_G_MEAN_FLUX : G-band mean flux (double, Flux[e^- s^{-1}])

Mean flux in the G-band.

PHOT_G_MEAN_FLUX_ERROR : Error on G-band mean flux (float, Flux[e^- s^{-1}])

The error on the weighted mean flux in the G-band `phot_g_mean_flux`. See Chapter **??** of the online documentation regarding the photometric accumulations for details about this calculation.

PHOT_G_MEAN_FLUX_OVER_ERROR : G-band mean flux divided by its error (float)

Mean flux in the G-band `phot_g_mean_flux` divided by its error `phot_g_mean_flux_error`.

PHOT_G_MEAN_MAG : G-band mean magnitude (float, Magnitude[mag])

Mean magnitude in the G band. This is computed from the G-band mean flux (`phot_g_mean_flux`) applying the magnitude zero-point in the Vega scale (see Section **??**).

No error is provided for this quantity as the error distribution is only symmetric in flux space. This converts to an asymmetric error distribution in magnitude space which cannot be represented by a single error value.

PHOT_G_FLUX_UWV : Unit weight variance of fluxes (double)

The unit weight variance of the flux values. See Evans (2008) for a rigorous mathematical definition.

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PHOT_G_FLUX_MEDIAN : Median flux (float, Flux[e^- s^{-1}])

Median of the flux distribution.

When an insufficient number of observations are available to compute these quantities, the corresponding field will be set to Float.NaN.

PHOT_G_FLUX_SKEWNESS : Measure of the skewness of the flux distribution (float)

Measure of the skewness of the flux distribution.

When an insufficient number of observations are available to compute these quantities, the corresponding field will be set to Float.NaN.

PHOT_G_FLUX_KURTOSIS : Measure of the kurtosis of the flux distribution (float)

Kurtosis of the flux distribution.

When an insufficient number of observations are available to compute these quantities, the corresponding field will be set to Float.NaN.

PHOT_G_FLUX_MAD : MAD of the flux distribution (float, Flux[e^- s^{-1}])

Median Absolute Deviation (MAD) of the flux distribution.

When an insufficient number of observations are available to compute these quantities, the corresponding field will be set to Float.NaN.

PHOT_G_FLUX_FIRST_QUARTILE : First quartile of the flux distribution (float, Flux[e^- s^{-1}])

First (25%) quartile of the flux distribution.

When an insufficient number of observations are available to compute these quantities, the corresponding field will be set to Float.NaN.

PHOT_G_FLUX_THIRD_QUARTILE : Third quartile of the flux distribution (float, Flux[e^- s^{-1}])

Third (75%) quartile of the flux distribution.

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When an insufficient number of observations are available to compute these quantities, the corresponding field will be set to Float.NaN.

PHOT_G_FLUX_MIN : Minimum flux value (float, Flux[e^- s^{-1}])

Minimum flux value.

PHOT_G_FLUX_MAX : Maximum flux value (float, Flux[e^- s^{-1}])

Maximum flux value.

PHOT_PROC_MODE : Photometry processing mode (byte)

This flag indicates the photometric calibration process used for the source. For nominal processing (values 0, 1 and 2), this process is determined by the availability of colour information derived from the internally calibrated mean BP and RP source spectra. For crowded field image analysis a dedicated overlap calibration was performed (value 16).

L : Galactic longitude (double, Angle[deg])

Galactic Longitude of the object at reference epoch `ref_epoch`, see Section **??** of the release documentation for conversion details.

B : Galactic latitude (double, Angle[deg])

Galactic Latitude of the object at reference epoch `ref_epoch`, see Section **??** of the release documentation for conversion details.

ECL_LON : Ecliptic longitude (double, Angle[deg])

Ecliptic Longitude of the object at reference epoch `ref_epoch`, obtained from the equatorial coordinates using the transformation defined in Volume 1, Section 1.5.3 of ESA (1997).

Note that in the transformation applied here the ICRS origin is shifted in the equatorial plane from Γ by $\phi = 0.05542$ arcsec, positive from Γ to the ICRS origin (Chapront et al. 2002). The ICRS has an unambiguous definition with an origin in the ICRF equator defined by the realisation of the ICRF. The ecliptic system is less well-defined, potentially depending on additional

conventions in dynamical theories. The transformation employed here corresponds to the inertial mean ecliptic with obliquity and Γ defined by reference to the ICRS equator. Both the obliquity and the position of Γ on the ICRS equator with respect to the ICRS origin have been obtained from Lunar Laser Ranging measurements. This has no time dependence – there is no secular variation of the obliquity and no precession – and it simply defines the relative situation of the various planes at J2000.

ECL_LAT : Ecliptic latitude (double, Angle[deg])

Ecliptic Latitude of the object at reference epoch `ref_epoch`. For further details see the description for attribute `ecl_lon`.

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9 Extra–galactic

As indicated in the below subsection titles by the labels [TAP] and [DataLink], these products are served from the ESA Gaia Archive using two different protocols, the Table Access Protocol (TAP) for tabular products and the DataLink protocol for other products such as spectra, light curves, MCMC samples, etc. For more details please see Section **??**.

9.1 GALAXY_CANDIDATES [TAP]

This table contains parameters derived from various modules dedicated to the classification and characterisation of sources considered as galaxy candidates.

This table has been constructed with the intention to be complete rather than pure and, as such, it will contain a large fraction of non-genuine extragalactic sources. Purer samples can be drawn using dedicated flags or queries. Please refer to Chapter **??** of the on-line documentation for details about how this table was built, its content, and for recommendations regarding its exploitation.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (unique within a particular Data Release) (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

VARI_BEST_CLASS_NAME : Name of photometric variability best class as listed in Gaia source (string)

Best class name as found in `phot_variable_best_class` of tables `gaia_source` and `all_source_flags`. See Chapter **??** of the release documentation for details.

VARI_BEST_CLASS_PROB : Probability from the variability classification result corresponding to the variability best class, when available (float)

It is filled with the maximum of the `classifier_best_class_prob` values of table `vari_classifier_resu` for this `source_id`. Note that this value is only available for a subset of the sources that have a `vari_best_class_name`. See Section **??** of the release documentation for details.

CLASSPROB_DSC_COMBMOD_GALAXY : Probability from DSC-Combmod of being a galaxy (data used: BP/RP spectrum, photometry, astrometry) (float)

Probability that the object is of the named class. This is the overall probability for this class, computed by combining the class probabilities from DSC-Specmod (which classifies objects using BP/RP spectra) and DSC-Allosmod (which classifies objects using several astrometric and photometric features). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSPROB_DSC_COMBMOD_QUASAR : Probability from DSC-Combmod of being a quasar (data used: BP/RP spectrum, photometry, astrometry) (float)

Probability that the object is of the named class. This is the overall probability for this class, computed by combining the class probabilities from DSC-Specmod (which classifies objects using BP/RP spectra) and DSC-Allosmod (which classifies objects using several astrometric and photometric features). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSLABEL_DSC : Class assigned by DSC based on the probability from its Combmod classifier (string)

Class assigned by DSC based on the probability from its Combmod classifier. Probabilities are computed from a combination of the probabilities of two DSC classifiers, DSC-Specmod (which classifies objects using BP/RP spectra) and DSC-Allosmod (which classifies objects using several astrometric and photometric features).

DSC-Combmod provides a normalized posterior probability vector across several classes. Note

that this incorporates the global class prior, as explained in the documentation. This class label is set to that class with the largest probability above 0.5. If no probability is above 0.5, this class label is 'unclassified'. If users want to perform classification using a different threshold, or by adopting a different prior, they should use the DSC probability vectors.

CLASSLABEL_DSC_ALPHA : Class assigned by DSC based on the probability from its CombmodAlpha classifier (string)

Class assigned by DSC based on the probability from its CombmodAlpha classifier. Probabilities are computed from a parametric combination of the probabilities of two DSC classifiers, DSC-Specmod (which classifies objects using BP/RP spectra) and DSC-Allosmod (which classifies objects using several astrometric and photometric features).

DSC-CombmodAlpha provides a normalized posterior probability vector across several classes. Note that this incorporates the global class prior, as explained in the documentation. This class label is set to that class with the largest probability above 0.5. If no probability is above 0.5, this class label is 'unclassified'. If users want to perform classification using a different threshold, or by adopting a different prior, they should use the DSC probability vectors.

CLASSLABEL_DSC_SUPP : Class assigned by DSC based on the probability from its CombmodSupp classifier (string)

Class assigned by DSC based on the probability from its CombmodSupp classifier. Probabilities are computed from a combination of the probabilities of two DSC classifiers, DSC-SpecmodSupp (which classifies objects jointly using BP/RP spectra and several astrometric and photometric features) when available, otherwise DSC-Specmod (which classifies objects using BP/RP spectra).

DSC-CombmodSupp provides a normalized posterior probability vector across several classes. Note that this incorporates the global class prior, as explained in the documentation. This class label is set to that class with the largest probability above 0.5. If no probability is above 0.5, this class label is 'unclassified'. If users want to perform classification using a different threshold, or by adopting a different prior, they should use the DSC probability vectors.

CLASSLABEL_DSC_ALLOS : Class assigned by DSC based on the probability from its Allosmod classifier (string)

Class assigned by DSC based on the probability from its Allosmod classifier. Probabilities are computed from from a classifier that uses various astrometric and photometric features.

CLASSLABEL_DSC_SPEC : Class assigned by DSC based on the probability from its Specmod classifier (string)

Class assigned by DSC based on the probability from its Specmod classifier. Probabilities are computed from from a classifier that uses the BP/RP spectrum.

CLASSLABEL_DSC_SPEC_SUPP : Class assigned by DSC based on the probability from its SpecmodSupp classifier (string)

Class assigned by DSC based on the probability from its SpecmodSupp classifier. Probabilities are computed from a classifier that jointly uses the BP/RP spectrum and various astrometric and photometric features.

CLASSLABEL_DSC_JOINT : Class assigned by DSC based on the probability from its Specmod and Allosmod classifiers (string)

Class label. The definition of the joint class label is based on the posterior probabilities of the DSC classifiers trained on the BP/RP spectra (module DSC-Specmod) and on the collection of features from astrometry and photometry (module DSC-Allosmod).

The joint class label identifies a source as a quasar or a galaxy if the two classifiers predict a maximum probability higher than 0.50 for the same target class. Otherwise, the joint class label is declared as unclassified.

FLAGS_DSC : Flag reporting the DSC processing status per source (string)

Processing flag array. The field is a collection of numerical indicators referring to the status of DSC processing.

- The 1st digit is a default value.
- The 2nd digit indicates if a source is an outlier candidate. An outlier is declared based on the probabilities of the combined DSC classifiers. The value is equal to 0 for non-outliers when the combined classifiers separately give a maximum probability higher than 0.99, otherwise value is equal to 1 for outliers. Note that the predicted classes of the combined classifiers are not required to be the same in this analysis. An outlier is also declared when the number of observations contributing to the BP and RP photometry is below a defined threshold (5).
- The 4th digit indicates if a source is an outlier candidate based on the probabilities

of the Combmod module.

- The 5th digit indicates if a source is an outlier candidate based on the probabilities of the Combmod-alpha module.
- The 7th digit indicates if a source is an outlier candidate based on the probabilities of the CombmodSupp module.
- The 9th digit indicates the status of data processing.

Values of the 4th, 5th and 7th digits are integers from 0 to 3, based on the maximum predicted probability. Value is equal to 3 when the maximum probability is lower than 0.90, equal to 2 when the maximum probability is within 0.90 and 0.95, equal to 1 when the maximum probability is lower than 0.999, and equal to 0 if the maximum probability is higher than 0.999.
Values of the 9th digit are integers from 1 to 3. Value is equal to 1 when the Allosmod and the Specmod-Supp probabilities are N/A due to missing astrometric and photometric features. Value is equal to 2 when the required inputs are available to the classifiers. Value is equal to 3 when the astrometric and photometric features, the BP/RP spectra and the infrared photometry are available to the classifiers.

CLASSLABEL_OA : Class assigned by OA in the neuron that represents the source (string)

Class label of the neuron that represents the source, as assigned by the OA module in Apsis. See Section **??** for further details.

CLASSLABEL_UCA : Class assigned by UCA in the neuron that represents the source (string)

Class label of the neuron that represents the source, as assigned by the UCA module in Apsis. There will be four main label types: star, galaxy, quasar and unknown. The last one is assigned when the source cannot be classified as any of the other types. See Section **??** for further details.

REDSHIFT_UGC : Redshift from UGC (float)

The redshift of the source treated as a galaxy by the Apsis module UGC. The redshift is estimated from the sampled mean BP/RP spectrum applying a regression model based on Support Vector Machines (SVM), a supervised machine learning algorithm. Details are provided in Section **??** of the on-line documentation.

As an estimate of the uncertainty on `redshift_ugc` the value (`redshift_ugc_upper` – `redshift_ugc_low`
may be used. We note that in the ranges `redshift_ugc`< 0.02, 0.28 <`redshift_ugc`< 0.30 and

redshift_ugc> 0.58 the SVM-model performance is particularly low and the derived redshifts lie outside of the achievable prediction range. A very narrow interval at 0.070 <redshift_ugc< 0.071 includes several thousand bright sources which probably have redshifts lower than 0.04, as explained in the on-line documentation (Section **??**).

REDSHIFT_UGC_LOWER : Redshift prediction lower limit from UGC (float)

The performance limits are based on the mean error $predictDiffMean[r_i]$ and its standard deviation $predictDiffStDev[r_i]$ (1σ-level) obtained during the SVM training with a test data set, where r_i=int(redshift_ugc/0.02). The lower limit is defined as redshift_ugc$-predictDiffMean[r_i]$ $-predictDiffStDev[r_i]$. For more details see the on-line documentation, Section **??**.

REDSHIFT_UGC_UPPER : Redshift prediction upper limit from UGC (float)

The performance limits are based on the mean error $predictDiffMean[r_i]$ and its standard deviation $predictDiffStDev[r_i]$ (1σ-level) obtained during the SVM training with a test data set, where r_i=int(redshift_ugc/0.02). The upper limit is defined as redshift_ugc$-predictDiffMean[r_i]$ $+predictDiffStDev[r_i]$. For more details see the on-line documentation, Section **??**.

N_TRANSITS_AF : Number of AF transits used for the morphological analysis (int)

The number of AF transits used to analyse the object morphology.

N_TRANSITS_SM : Number of SM transits used for the morphological analysis (int)

The number of SM transits used to analyse the object morphology.

ANGULAR_COVERAGE_AF : Angular coverage of AF transits used for the morphological analysis (float)

The angular coverage of a source is estimated as the ratio between the area of the polygon created from the union of all the AF observed windows and transit angles over the area of a circle with a diameter equal to the diagonal of the largest observed window. The coverage corresponds to the fraction of the surface of the source covered by the observations.

ANGULAR_COVERAGE_SM : Angular coverage of SM transits used for the morphological analysis (float)

The angular coverage of a source is estimated as the ratio between the area of the polygon created from the union of all the SM observed windows and transit angles over the area of a circle with a diameter equal to the diagonal of the largest observed window. The coverage corresponds to the fraction of the surface of the source covered by the observations.

POSANGLE_SERSIC : Fitted position angle of the source for the Sérsic Profile (double, Angle[deg])

Position angle of the source for fitted Sérsic profile with respect to Celestial North Pole.

POSANGLE_SERSIC_ERROR : Error on the fitted position angle of the source for the Sérsic Profile (double, Angle[deg])

Uncertainty in the position angle of source with respect to Celestial North Pole for the Sérsic profile.

INTENSITY_SERSIC : Fitted intensity of the source for the Sérsic Profile (double, Flux[e^- pix^{-1} s^{-1}])

Intensity of the source at the fitted effective radius (`radius_sersic`) for the Sérsic profile. The unit of this information is e-/(pix*s) with the size of the pixel being 14.733 mas.

INTENSITY_SERSIC_ERROR : Error on the fitted intensity of the source at effective radius radius_sersic (double, Flux[e^- pix^{-1} s^{-1}])

Uncertainty of the light intensity at the fitted effective radius (`radius_sersic`) of the source for the Sérsic profile. The unit of this information is e-/(pix*s) with the size of the pixel being 14.733 mas.

RADIUS_SERSIC : Fitted effective radius of the source for the Sérsic Profile (double, Angle[mas])

Effective radius containing half of the total luminosity of the source as obtained by morphological fitting for the Sérsic profile.

RADIUS_SERSIC_ERROR : Error on the fitted effective radius of the source for the Sérsic Profile (double, Angle[mas])

Uncertainty in the effective radius containing half of the total luminosity of the source for the

Sérsic profile.

ELLIPTICITY_SERSIC : Fitted ellipticity of source for the Sérsic Profile (double)

Ellipticity (defined as 1-(b/a) where b/a is the axis ratio) of the source for the Sérsic profile.

ELLIPTICITY_SERSIC_ERROR : Error on the fitted ellipticity of the source for the Sérsic Profile (double)

Uncertainty in the ellipticity (defined as 1-(b/a) where b/a is the axis ratio) of the source for the Sérsic profile.

N_SERSIC : Fitted Sérsic Index for Sérsic Profile (float)

Sérsic index for the Sérsic profile. The Sérsic profile is a mathematical function that describes how the intensity of a galaxy varies with the distance from its centre. The Sérsic index describes how steep this variation is.

N_SERSIC_ERROR : Error on the fitted Sérsic Index for Sérsic Profile (float)

Uncertainty in the Sérsic index for the Sérsic profile.

L2_SERSIC : L2 norm for the Sérsic Profile (double)

This value represents the mean squared error between the integrated flux of all observed samples (from the Sky Mapper and Astrometric Field) and the integrated flux of synthetic samples produced with the fitted profile.

FLAGS_SERSIC : Flag indicative of processing or scientific quality for the morphological parameters fitting for the Sérsic Profile (byte)

This flag provides information about the processing or scientific quality of the results of the Galaxy morphology analysis chain for the source of interest for the Sérsic profile.

The flag coding is the following (note that not all flag cases apply to a given profile):

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- When the solution is valid and can be used by the user, `flags_sersic` can then take two possible values:

  0: This corresponds to optimal sources with no problem,

  1: This corresponds to sources with good parameters but with a neighbour that enters the SM at < 2.3 arcsec and for which there is a possibility that the solution be perturbed.

- When the solution is not reliable and the parameters are not provided in the table, the reason is given by the value of `flags_sersic`:

  2: Not reliable solution because fitting did not converge correctly, parameters reached the bounds of the permitted domain, or parameters are smaller than their errors.

  3: Not reliable solution due to low signal-to-noise ratio at effective radius.

POSANGLE_DE_VAUCOULEURS : Fitted position angle of the source for the de Vaucouleurs Profile (double, Angle[deg])

Position angle of the source for fitted de Vaucouleurs profile with respect to Celestial North Pole.

POSANGLE_DE_VAUCOULEURS_ERROR : Error on the fitted position angle of the source for the de Vaucouleurs Profile (double, Angle[deg])

Uncertainty in the position angle of source with respect to Celestial North Pole for the de Vaucouleurs profile.

INTENSITY_DE_VAUCOULEURS : Fitted intensity of the source for the de Vaucouleurs Profile (double, Flux[e^- pix^{-1} s^{-1}])

Intensity of the source at the fitted effective radius (`radius_de_vaucouleurs`) for the de Vaucouleurs profile. The unit of this information is e-/(pix*s) with the size of the pixel being 14.733 mas.

INTENSITY_DE_VAUCOULEURS_ERROR : Error on the fitted intensity of the bulge for the de Vaucouleurs Profile (double, Flux[e^- pix^{-1} s^{-1}])

Uncertainty of the light intensity at the fitted effective radius (`radius_de_vaucouleurs`) of the source for the de Vaucouleurs profile. The unit of this information is e-/(pix*s) with the size of

the pixel being 14.733 mas.

RADIUS_DE_VAUCOULEURS : Fitted effective radius of the source for de Vaucouleurs Profile (double, Angle[mas])

Effective radius containing half of the total luminosity of the source as obtained by morphological fitting for the de Vaucouleurs profile.

RADIUS_DE_VAUCOULEURS_ERROR : Error on the fitted effective radius of the source for the de Vaucouleurs Profile (double, Angle[mas])

Uncertainty in the effective radius containing half of the total luminosity of the source for the de Vaucouleurs profile.

ELLIPTICITY_DE_VAUCOULEURS : Fitted ellipticity of source for the de Vaucouleurs Profile (double)

Ellipticity (defined as 1-(b/a) where b/a is the axis ratio) of the source for the de Vaucouleurs profile.

ELLIPTICITY_DE_VAUCOULEURS_ERROR : Error on the fitted ellipticity of the source for the de Vaucouleurs Profile (double)

Uncertainty in the ellipticity (defined as 1-(b/a) where b/a is the axis ratio) of the source for the de Vaucouleurs profile.

L2_DE_VAUCOULEURS : L2 norm for the de Vaucouleurs Profile (double)

This value represents the mean squared error between the integrated flux of all observed samples (from the Sky Mapper and Astrometric Field) and the integrated flux of synthetic samples produced with the fitted profile.

FLAGS_DE_VAUCOULEURS : Flag indicative of processing or scientific quality for the morphological parameters fitting for the de Vaucouleurs Profile (byte)

This flag provides information about the processing or scientific quality of the results of the Galaxy morphology analysis chain for the source of interest for the de Vaucouleurs profile.

Draft

The flag coding is the following (note that not all flag cases apply to a given profile):

- When the solution is valid and can be used by the user, `flags_de_vaucouleurs` can then take two possible values
  - 0: This corresponds to optimal sources with no problem,
  - 1: This corresponds to sources with good parameters but with a neighbour that enters the SM at < 2.3 arcsec and for which there is a possibility that the solution be perturbed.
- When the solution is not reliable and the parameters are not provided in the table, the reason is given by the value of `flags_de_vaucouleurs`:
  - 2: Not reliable solution because fitting did not converge correctly, parameters reached the bounds of the permitted domain, or parameters are smaller than their errors.
  - 3: Not reliable solution due to low signal-to-noise ratio at effective radius.

POSANGLE_COMBINED : Fitted position angle of the source for the combined (bulge+disc) Profile (double, Angle[deg])

Position angle of the source for fitted combined (bulge+disc) profile with respect to Celestial North Pole.

POSANGLE_COMBINED_ERROR : Error on the fitted position angle of the source for the combined (bulge+disc) Profile (double, Angle[deg])

Uncertainty in the position angle of source with respect to Celestial North Pole for the combined (bulge+disc) profile.

INTENSITY_BULGE_COMBINED : Fitted intensity of the bulge for the combined (bulge+disc) Profile (double, Flux[e^{-} pix^{-1} s^{-1}])

Intensity of the bulge at the fitted effective radius (`radius_bulge_combined`) for the combined (bulge+disc) profile. The unit of this information is e-/(pix*s) with the size of the pixel being 14.733 mas.

INTENSITY_BULGE_COMBINED_ERROR : Error on the fitted intensity of the bulge for the combined (bulge+disc) Profile (double, Flux[e^{-} pix^{-1} s^{-1}])

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Uncertainty of the light intensity at the fitted effective radius (`radius_bulge_combined`) of the bulge for the combined (bulge+disc) profile. The unit of this information is e-/(pix*s) with the size of the pixel being 14.733 mas.

RADIUS_BULGE_COMBINED : Fitted effective radius of the bulge for combined (bulge+disc) Profile (double, Angle[mas])

Effective radius containing half of the total luminosity of the bulge as obtained by morphological fitting for the combined (bulge+disc) profile.

RADIUS_BULGE_COMBINED_ERROR : Error on the fitted effective radius of the bulge for the combined (bulge+disc) Profile (double, Angle[mas])

Uncertainty in the effective radius containing half of the total luminosity of the bulge for the combined (bulge+disc) profile.

INTENSITY_DISC_COMBINED : Fitted intensity of the disc for the combined (bulge+disc) Profile (double, Flux[e^{-} pix^{-1} s^{-1}])

Light intensity at the centre of the disc as obtained by fitting an exponential profile for the combined (bulge+disc) profile. The unit of this information is e-/(pix*s) with the size of the pixel being 14.733 mas.

INTENSITY_DISC_COMBINED_ERROR : Error on the fitted intensity of the disc for the combined (bulge+disc) Profile (double, Flux[e^{-} pix^{-1} s^{-1}])

Uncertainty of the light intensity at the centre of the disc as obtained by fitting an exponential profile for the combined (bulge+disc) profile. The unit of this information is e-/(pix*s) with the size of the pixel being 14.733 mas.

RADIUS_DISC_COMBINED : Fitted effective radius of the disc for combined (bulge+disc) Profile (double, Angle[mas])

Disc scale length as obtained by morphological fitting for the combined (bulge+disc) profile.

RADIUS_DISC_COMBINED_ERROR : Error on the fitted effective radius of the disc for the combined (bulge+disc) Profile (double, Angle[mas])

Draft

Uncertainty in the disc scale length as obtained by morphological fitting for the combined (bulge+disc) profile.

ELLIPTICITY_COMBINED : Fitted ellipticity of source for the combined (bulge+disc) Profile (double)

Ellipticity (defined as 1-(b/a) where b/a is the axis ratio) of the source for the combined (bulge+disc) profile.

ELLIPTICITY_COMBINED_ERROR : Error on the fitted ellipticity of the source for the combined (bulge+disc) Profile (double)

Uncertainty in the ellipticity (defined as 1-(b/a) where b/a is the axis ratio) of the source for the combined (bulge+disc) profile.

L2_COMBINED : L2 norm for the combined (bulge+disc) Profile (double)

This value represents the mean squared error between the integrated flux of all observed samples (from the Sky Mapper and Astrometric Field) and the integrated flux of synthetic samples produced with the fitted profile.

FLAGS_COMBINED : Flag indicative of processing or scientific quality for the morphological parameters fitting for the combined (bulge+disc) Profile (byte)

This flag provides information about the processing or scientific quality of the results of the Galaxy morphology analysis chain for the source of interest for the combined (bulge+disc) profile.

The flag coding is the following (note that not all flag cases apply to a given profile):

- When the solution is valid and can be used by the user, `flags_combined` can then take two possible values:

  0: This corresponds to optimal sources with no problem,

  1: This corresponds to sources with good parameters but with a neighbour that enters the SM at < 2.3 arcsec and for which there is a possibility that the solution be perturbed.

Draft

- When the solution is not reliable and the parameters are not provided in the table, the reason is given by the value of `flags_combined`:

  2: Not reliable solution because fitting did not converge correctly, parameters reached the bounds of the permitted domain, or parameters are smaller than their errors.

  3: Not reliable solution due to low signal-to-noise ratio at effective radius.

SOURCE_SELECTION_FLAGS : Bit indicative of whether the input data from a given module met the source list eligibility criteria for the source of interest (int)

Bit field indicative of whether the input data from a given module met the source list eligibility criteria for the source of interest.

The bit is coded as follows:

- bit 0 (LSB): The source meets the eligibility criteria for the output of the Surface brightness analysis module (see Chapter **??**).
- bit 1: The source belongs to the `galaxy_morphology_input_list` table.
- bit 2: The source meets the eligibility criteria for the output of the classification module based on photometric lightcurves (see Chapter **??**).
- bit 3: The source meets the eligibility criteria for the output of the DSC module (see Chapter **??**).
- bit 4: The source meets the eligibility criteria for the redshift determined by the UGC module (see Chapter **??**).
- bit 5: The source meets the eligibility criteria for the classification output of the UGC module (not yet applicable to DR4).
- bit 6: The source meets the eligibility criteria for the classification output of the OA module (see Chapter **??**). In DR4, no source from the OA module will be considered in the input source list of this table.
- bit 7: The source meets the eligibility criteria for the classification output of the UCA module (see Chapter **??**). In DR4, no source from the UCA module will be considered in the input source list of this table.

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9.2 GALAXY_MORPHOLOGY_INPUT_LIST [TAP]

This table provides the list of input catalogues that have been used to select any given source for which morphological analysis parameters are provided in the Galaxy candidates table.

For DR4 one unique catalogue was used to build this source input list; this is why no catalogue ID is used in this table unlike what is done in `qso_morphology_input_list` which is instead based on external quasar catalogues.

The method used to select these sources is described in the chapter on extended object processing (Section **??**). This method also produces a probability to be a galaxy, which is tabulated as `galaxy_probability`.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (unique within a particular Data Release) (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

GALAXY_PROBABILITY : The probability for the source to be a galaxy (float)

This is the probability of the source of interest to be a galaxy based on the algorithm that generated the catalogue used as input list for this processing.

Draft

9.3 GRAV_LENS_COMBINATION [TAP]

This table contains the information relevant at the level of a gravitational lens multiplet system analysed around the lensed source identified in the gravitational lens observation table.

The components forming this multiplet are tabulated in `grav_lens_component`, and their corresponding individual observation data are provided in `grav_lens_observation`. The input list of the sources considered in `grav_lens_component` is given in `grav_lens_input_list`.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

NAME : The name of the gravitational lens candidate (string)

The gravitational lens candidate name, in the form DR4Gaiahhmmss.sss+/-ddmmss.ss, where hhmmss.sss and ddmmss.ss are the right ascension and declination respectively.

LENS_SOURCE_ID : Unique source identifier for the lens multiplet (long)

A unique single numerical identifier of the related quasar whose neighbourhood has been analysed. The `lens_source_id` is obtained from the Gaia main source catalogue (for a detailed description see `gaia_source.source_id`).

FLAG : Binary flags coming from the comparison of the internally calibrated spectra. (long)

Binary flags coming from the comparison of the internally calibrated spectra.

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Bit	Value	Description
0	SPEC_MINOR	Some of the spectra used in this comparison raised one of the EPOCH_ALL_MASKED, EPOCH_NEGATIVE_FLUX, EPOCH_MANY_OUTLIERS, EPOCH_LARGE_NOISE, EPOCH_NEGATIVE_NOISE, or EPOCH_LARGE_MSE flags. If set, the comparison is presumably good but still deserves some minor attention.
1	SPEC_WARNING	Some of the spectra used in this comparison raised one of the LARGE_SCORE, UNDERCONSTRAINED, MANY_OUTLIERS, LOW_NEPOCH, LOW_SNR or NEGATIVE_TOTAL_FLUX flags. If set, the comparison can potentially be wrong.
2	SPEC_ERROR	Some of the spectra used in this comparison raised the NOT_CONVERGED flag. There is a high probability that this comparison is wrong.
3	CHI2R_NOT_CONVERGED	The chi-square solution did not converged. The chi-square comparison is wrong.

N_COMPONENTS : Number of components in this combination (int)

Number of components around the `lens_source_id` obtained from the clustering algorithm applied in a fixed radius.

COMPONENTS : The indices of the components in this combination (int[`n_components`] array)

The indices of the components composing this combination.

CHI2R : The reduced chi-square coming from the comparison of the internally-calibrated spectra. (double)

The reduced chi-square coming from the comparison of the internally-calibrated spectra.

DOF : The degree of freedom associated with the reduced chi-square (int)

The degree of freedom associated with chi2r.

GOF : The goodness of fit associated with reduced chi-square (double)

The goodness of fit associated with chi2r. This is a gaussianized version of the chi-square computed as $gof = \sqrt{4.5 \times dof} \times (\sqrt[3]{chi2r} + 2/9/dof - 1)$.

WASSERSTEIN_DISTANCE : The minimal Wasserstein distance between any two pair of components in this combination. (double)

The minimal Wasserstein distance between any two pair of components in this combination.

Draft

9.4 GRAV_LENS_COMPONENT [TAP]

This table contains the sources identified as possible gravitational lens candidates based on an analysis of the observations in the neighbourhood of a list of sources selected from a compilation of quasar candidates. Information about the catalogues from which these quasars proceed is given in the table of lens catalogue names.

For each `source_id` from `grav_lens_component` a list of components found around that source is tabulated, together with some information about their position and flux as derived from the cluster of observations associated with each of them. These individual observations are provided in `grav_lens_observation`.

A separate table `grav_lens_outlier` compiles all those observations that could not be associated with a component around a given `source_id`, and that have therefore been discarded from the analysis.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

LENS_SOURCE_ID : Unique source identifier for the lens multiplet (long)

A unique single numerical identifier of the related quasar whose neighbourhood has been analysed. The `lens_source_id` is obtained from the Gaia main source catalogue (for a detailed description see `gaia_source.source_id`).

SOURCE_ID : Unique source identifier for the component of the lens multiplet (long)

Unique source identifier for the component of the lens multiplet identified around `lens_source_id`.

DESIGNATION : Unique source designation (unique across all Data Releases) (string)

A source designation, unique across all Gaia Data Releases, that is constructed from the prefix 'Gaia DRx ' followed by a string of digits corresponding to `source_id` (3 space–separated words in total). Note that the integer source identifier `source_id` is **not** guaranteed to be unique across Data Releases; moreover it is not guaranteed that the same astronomical source will always have the same `source_id` in different Data Releases. Hence the only safe way to compare source records between different Data Releases in general is to check the records of proximal source(s) in the same small part of the sky.

NAME : The name of the gravitational lens candidate (string)

The gravitational lens candidate name, in the form DR4Gaiahhmmss.sss+/-ddmmss.ss, where hhmmss.sss and ddmmss.ss are the right ascension and declination respectively.

FLAG : GravLens flag (byte)

This flag provides information about the processing or scientific quality of the results of the GravLens chain for the source of interest. The flag checks two conditions:

1. Bit 0 is raised when the magnitude difference between the brightest and the faintest component of the system is larger than 5 mag.
2. Bit 1 is raised when the number of outlier observations is larger than the number of valid observations.

A flag value of 0 indicates that no known issue applies to the source of interest.

N_COMPONENTS : Number of components (int)

Number of components around the `source_id` obtained from the clustering algorithm applied in a fixed radius.

COMPONENT_ID : Index of the component for this source (int)

An index used to identify the various components found around this `source_id`. This number varies between 1 and `n_components`.

N_OBS : Number of valid observations used for this component (short)

Number of valid observations used for this component_id.

COMPONENT_FLAG : Component object flag (byte)

This flag provides information about the processing or scientific quality of the results of the GravLens chain on the component level. The flag checks two conditions:

1. Bit 0 is raised when ra_std > 100 mas or dec_std > 100 mas.
2. Bit 1 is raised when g_mag_std > 0.4 mag.

A flag value of 0 indicates that no known issue applies to the component of interest.

RA : Mean right ascension of the component (double, Angle[deg])

The mean right ascension of the cluster of observations associated with this component within a fixed radius. The individual right ascensions as decoded from the transit_ids are stored in grav_lens_observation.

RA_STD : Standard deviation of the right ascension of the component (float, Angle[mas])

The standard deviation of right ascension measurements (multiplied by cos (dec_std)) for the cluster of observations associated with this component within a fixed radius. The individual right ascensions as decoded from the transit_ids are stored in grav_lens_observation.

DEC : Mean declination of the clusters of measurements (double, Angle[deg])

The mean declination of the cluster of observations associated with this component within a fixed radius. The individual declinations as decoded from the transit_ids are stored in grav_lens_observation

DEC_STD : Standard deviation of the declination measurements (float, Angle[mas])

The standard deviation of declination measurements for the cluster of observations associated with this component within a fixed radius. The individual declinations as decoded from the transit identifiers are stored in grav_lens_observation.

G_FLUX : Calibrated mean G flux of the component (double, Flux[e^- s^{-1}])

Calibrated mean G-band flux value for all observations belonging to this component. Note that this value may be missing in some cases.

G_FLUX_ERROR : Uncertainty of the calibrated mean flux value for this component (float, Flux[e^- s^{-1}])

Uncertainty of the mean G-band flux value for this component. Note that this value may be missing in some cases.

G_MAG : Calibrated mean G magnitude of the component (float, Magnitude[mag])

Mean calibrated G-band magnitude for the observations belonging to this component.

G_MAG_STD : Standard deviation of the calibrated G magnitude of the component (float, Magnitude[mag])

Standard deviation of the calibrated G-band magnitudes for the observations belonging to this component.

BP_FLUX : The integrated flux in the BP band (double, Flux[e^- s^{-1}])

The integrated flux.

BP_FLUX_ERROR : The uncertainty on the integrated flux in the BP band (float, Flux[e^- s^{-1}])

The uncertainty on the integrated BP flux.

RP_FLUX : The integrated flux in the RP band (double, Flux[e^- s^{-1}])

The integrated flux.

RP_FLUX_ERROR : The uncertainty on the integrated flux in the RP band (float, Flux[e^- s^{-1}])

The uncertainty on the integrated RP flux.

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RAW_G_FLUX : Mean uncalibrated flux value for all measurements belonging to this component (double, Flux[e^- s^{-1}])

Mean uncalibrated flux value for all measurements belonging to this component.

RAW_G_FLUX_ERROR : Uncertainty of the mean uncalibrated flux value for this component (float, Flux[e^- s^{-1}])

Uncertainty of the mean uncalibrated flux value for this component.

RAW_G_MAG : Mean uncalibrated G Magnitude (float, Magnitude[mag])

Mean uncalibrated G Magnitude.

MATCHED_SOURCE_ID : Gaia source identifiers matched to this component source (long)

A cross match in positions is done between the mean position of the current component and all Gaia sources positions. The Gaia `source_id` of the closest component is indicated in this field.

MATCHED_DISTANCE : Distance between the mean position of the component and the Gaia position of the matched source (float, Angle[mas])

Distance in mas between the mean position of the component identified by `lens_source_id` and the Gaia position of the matched source identified in `matched_source_id`.

HESIOD_SCORE_MEAN : The mean value of the HESIOD scores (double)

The mean value of the HESIOD (Heuristical Ensemble Splitting Imputation and Organization of Data) scores (Gaia Collaboration et al. 2024).

HESIOD_SCORE_MODE : The mode of the HESIOD scores (double)

The mode of the HESIOD (Heuristical Ensemble Splitting Imputation and Organization of Data) scores (Gaia Collaboration et al. 2024).

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HESIOD_SCORE_STD : The standard deviation of the HESIOD scores (double)

The standard deviation of the HESIOD (Heuristical Ensemble Splitting Imputation and Organization of Data) scores (Gaia Collaboration et al. 2024).

IS_QUASAR : Component considered as quasar (boolean)

This field is true when the component is identified as the quasar analysed and which was designated as 'quasar or qso candidate' in the catalogue identified by `catalogue_id` in table `grav_lens_input_list`.

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9.5 GRAV_LENS_INPUT_LIST [TAP]

This table provides the list of input catalogues that have been used to select any given source that was analysed by the Gravitational Lens module and published in the lens candidate table. Each catalogue is assigned a `catalogue_id`. For the DR4 the following applies:

- 1: Ducourant+2025 (Ducourant et al. 2025)
- 2: Assef C75 (Assef et al. 2018)
- 3: Assef R90 (Assef et al. 2018)
- 4: Chaussidon+2023 (Chaussidon et al. 2023)
- 5: Shu+2019 (Shu et al. 2019)
- 6: The Milliquas catalogue: v6.4 (Flesch 2019)
- 7: The Milliquas catalogue: v7.0 (no dedicated paper – see reference for v6.4, Flesch (2019))
- 8: The Milliquas catalogue: v7.1b (no dedicated paper – see reference for v6.4, Flesch (2019))
- 9: The Milliquas catalogue: v7.4c (no dedicated paper – see reference for v7.2,Flesch (2021))
- 10: The Milliquas catalogue: v7.5 (no dedicated paper – see reference for v7.2, Flesch (2021))
- 11: The Milliquas catalogue: v7.5b (no dedicated paper – see reference for v7.2, Flesch (2021))
- 12: The Milliquas catalogue: v7.7 (no dedicated paper – see reference for v7.2, Flesch (2021))
- 13: The Milliquas catalogue: v7.7c (no dedicated paper – see reference for v7.2, Flesch (2021))
- 14: The Milliquas catalogue: v8.0 (Flesch 2023)
- 15: Storey-Fisher+2024 (Storey-Fisher et al. 2024)
- 16: Sergei Klioner, private communication, 2022
- 17: Eltvedt+2024 (Eltvedt et al. 2024)
- 18: Wu+2023 (Wu & Shen 2023)
- 19: González-Santamaría+2021 (González-Santamaría et al. 2021)

Further details on these catalogues are given in **?**.

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Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (unique within a particular Data Release) (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

CATALOGUE_ID : The unique identifier for the catalogue(s) used to select the sources in the gravitational lenses analysis (byte)

Each catalogue used to select the sources analysed by the gravitational lens module is assigned a unique ID. The corresponding catalogues are listed in the description of the `grav_lens_input_list` table, and further details are given in the documentation.

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9.6 GRAV_LENS_OBSERVATION [TAP]

This table contains the observations associated with the components found in the lens candidates table.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

LENS_SOURCE_ID : Unique source identifier for the lens multiplet (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

SOURCE_ID : Unique source identifier for the component of the lens multiplet (long)

Unique source identifier for the component of the lens multiplet identified around `lens_source_id`.

TRANSIT_ID : Transit Identifier (long)

Transit ID as defined in JP-011 and received by DPCI from IDT.

COMPONENT_ID : Index of the component for this source_id (int)

An index used to identify the various components found around this `source_id`. This number varies between 1 and `grav_lens_component.n_components`.

OBSERVATION_ID : Counter for the observations of each component (int)

A sequential counter that uniquely identifies the observations belonging to a given component.

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RA : Right ascension of each individual observation belonging to this component (double, Angle[deg])

Right ascension of each individual observation belonging to this component, as decoded from the `transit_id`.

RA_ERROR : Error of the right ascensions of individual measurements (float, Angle[mas])

Error of the right ascension of each individual observation belonging to this component, as decoded from the `transit_id`.

DEC : Declination of each individual observation belonging to this component (double, Angle[deg])

Declination of each individual observation belonging to this component, as decoded from the `transit_id`.

DEC_ERROR : Error of the declination of individual measurements (float, Angle[mas])

Error of the declination of each individual observation belonging to this component, as decoded from the `transit_id`.

G_FLUX : Calibrated flux value of each individual observation belonging to this component (float, Flux[e^{-} s^{-1}])

Calibrated flux value of each individual observation belonging to this component. Note that this value may be missing in some cases.

G_FLUX_ERROR : Calibrated flux error value of each individual observation belonging to this component (float, Flux[e^{-} s^{-1}])

Calibrated flux error value of each individual observation belonging to this component. Note that this value may be missing in some cases.

G_MAG : Calibrated G magnitudes of each individual observation belonging to this component (float, Magnitude[mag])

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Calibrated G-band magnitude of each individual observation belonging to this component.

BP_FLUX : Integrated flux value for each individual observation belonging to this component in the BP band (float, Flux[e^- s^{-1}])

Integrated flux value for each individual observation belonging to this component in the BP band. Fluxes are integrated in the pseudo-wavelength space of the internally-calibrated spectra.

BP_FLUX_ERROR : The uncertainties associated with integrated BP fluxes (float, Flux[e^- s^{-1}])

The uncertainties associated with the BP flux.

BP_FLAG : Flags associated with the processing of the individual BP observations (short)

Flags associated with the integrated BP flux based on the associated internally-calibrated spectrum. It is a binary combination of:

Bit	Value	Description
0	ALL_MASKED	All samples were masked in this spectrum.
1	NEGATIVE_FLUX	This spectrum has a negative integrated flux.
2	MANY_OUTLIERS	More than 10% of the samples were rejected in this spectrum.
3	LARGE_NOISE	The mean value of the noise is larger than 2.
4	NEGATIVE_NOISE	Some estimation of the noise is negative (never raised).
5	LARGE_MSE	Mean square error is larger than 2 in the unmasked samples.

RP_FLUX : Integrated flux value for each individual observation belonging to this component in the RP band (float, Flux[e^- s^{-1}])

Integrated flux value for each individual observation belonging to this component in the RP band. Fluxes are integrated in the pseudo-wavelength space of the internally-calibrated spectra.

RP_FLUX_ERROR : The uncertainties associated with integrated RP fluxes (float, Flux[e^- s^{-1}])

The uncertainties associated with the RP flux.

RP_FLAG : Flags associated with the processing of the individual RP observations (short)

Flags associated with the integrated RP flux based on the associated internally-calibrated spec-

trum. It is a binary combination of:

Bit	Value	Description
0	ALL_MASKED	All samples were masked in this spectrum.
1	NEGATIVE_FLUX	This spectrum has a negative integrated flux.
2	MANY_OUTLIERS	More than 10% of the samples were rejected in this spectrum.
3	LARGE_NOISE	The mean value of the noise is larger than 2.
4	NEGATIVE_NOISE	Some estimation of the noise is negative (never raised).
5	LARGE_MSE	Mean square error is larger than 2 in the unmasked samples.

RAW_G_FLUX : Uncalibrated flux value of each individual observations belonging to this component (double, Flux[e^- s^{-1}])

Uncalibrated flux value of each individual observations belonging to this component.

RAW_G_FLUX_ERROR : Uncalibrated flux error value of each individual observations belonging to this component (float, Flux[e^- s^{-1}])

Uncalibrated flux error value of each individual observations belonging to this component.

RAW_G_MAG : Uncalibrated G magnitudes of individual measurements (float, Magnitude[mag])

Uncalibrated G magnitudes of individual measurements.

EPOCH : Epoch of the individual observation belonging to this component (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Epoch of the individual observation belonging to this component, given as Julian Days in TCB at the Barycentre, minus 2 455 197.5 days.

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9.7 GRAV_LENS_OUTLIER [TAP]

This table contains the individual observations that have been discarded from the analysis of the gravitational lenses around the list of sources tabulated in the lens candidate table. These are observations that could not be associated with any of the components assigned to the source of interest.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

LENS_SOURCE_ID : Unique source identifier for the lens multiplet (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

TRANSIT_ID : Transit Identifier (long)

Transit ID as defined in JP-011 and received by DPCI from IDT.

OUTLIER_ID : Counter for the outliers of each component (int)

A sequential counter that uniquely identifies the outliers belonging to a given component.

RA : Right ascension of each individual outlier observation (double, Angle[deg])

Right ascension of each individual outlier observation, as decoded from the `transit_id`.

RA_ERROR : Error of the right ascensions of individual measurements (float, Angle[mas])

Error of the right ascension of each individual observation belonging to this component, as decoded from the `transit_id`.

DEC : Declination of each individual outlier observation (double, Angle[deg])

Declination of each individual outlier observation, as decoded from the `transit_id`.

DEC_ERROR : Error of the declination of individual measurements (float, Angle[mas])

Error of the declination of each individual observation belonging to this component, as decoded from the `transit_id`.

G_FLUX : Calibrated flux value of each individual outlier observation (double, Flux[e^- s^{-1}])

Calibrated G-band flux value of each individual outlier observation. Note that this value may be missing in some cases.

G_FLUX_ERROR : Calibrated flux error value of each individual outlier observation (float, Flux[e^- s^{-1}])

Calibrated G-band flux error value of each individual outlier observation. Note that this value may be missing in some cases.

G_MAG : Calibrated G magnitudes of each individual outlier observation (float, Magnitude[mag])

Calibrated G-band magnitude of each individual outlier observation.

RAW_G_FLUX : Uncalibrated flux value of each individual observations belonging to this component (double, Flux[e^- s^{-1}])

Uncalibrated flux value of each individual observations belonging to this component.

RAW_G_FLUX_ERROR : Uncalibrated flux error value of each individual observations belonging to this component (float, Flux[e^- s^{-1}])

Uncalibrated flux error value of each individual observations belonging to this component.

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RAW_G_MAG : Uncalibrated G magnitudes of individual measurements (float, Magnitude[mag])

Uncalibrated G magnitudes of individual measurements.

EPOCH : Epoch of the individual outlier observation (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Epoch of the individual outlier observation, given as Julian Days in TCB at the Barycentre, minus 2 455 197.5 days.

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9.8 GRAV_LENS_REDSHIFT [TAP]

This table contains quasar and galaxy redshifts computed on the gravitational lens input list (containing quasars and lens components) given in the corresponding gravitational lens component table.

Each `source_id` corresponds to the `matched_source_id` field in `grav_lens_component`.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (unique within a particular Data Release) (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

REDSHIFT_QSOC : Redshift from QSOC (float)

The redshift of the source estimated from the analysis of the BP/RP spectra by the QSOC Apsis module.

The redshift of the source is obtained from the maximum of the log-posterior probability function as defined in Equation **??** of the on-line documentation.

REDSHIFT_QSOC_LOWER : Redshift lower confidence level from QSOC (float)

Lower confidence level of the redshift estimate. This is the 16th percentile of the PDF over redshift as computed from Equation **??** of the on-line documentation.

REDSHIFT_QSOC_UPPER : Redshift upper confidence level from QSOC (float)

Upper confidence level of the redshift estimate. This is the 84th percentile of the PDF over red-

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shift as computed from Equation **??** of the on-line documentation.

REDSHIFTALT_QSOC : Alternative redshift from QSOC (float)

The alternative, second most probable redshift estimate according to the log-posterior probability function defined in Equation **??** of the on-line documentation.

REDSHIFTALT_QSOC_LOWER : Alternative redshift lower confidence level from QSOC (float)

Lower confidence level of the alternative redshift estimate. This is the 16th percentile of the PDF over redshift as computed by QSOC.

REDSHIFTALT_QSOC_UPPER : Alternative redshift upper confidence level from QSOC (float)

Upper confidence level of the alternative redshift estimate. This is the 84th percentile of the PDF over redshift as computed by QSOC.

REDSHIFT_QSOC_LOGPROB : Log-probability of redshift_qsoc (double)

The logarithm of the posterior probability density function associated with the main redshift estimate, `redshift_qsoc` (see Equation **??** of the on-line documentation).

REDSHIFTALT_QSOC_LOGPROB : Log-probability of redshiftalt_qsoc (double)

The logarithm of the posterior probability density function associated with the alternative redshift estimate, `redshiftalt_qsoc` (see Equation **??** of the on-line documentation).

REDSHIFT_QSOC_PROB : The probability that the redshift stands within its confidence intervals (double)

The probability that the redshift stands within its confidence intervals. This is equal to the posterior probability density function integrated from `redshift_qsoc_lower` to `redshift_qsoc_upper`.

See Equation **??** of the on-line documentation for details.

REDSHIFTALT_QSOC_PROB : The probability that the alternative redshift stands within its confi-

dence intervals (double)

The probability that the alternative redshift stands within its confidence intervals. This is equal to the posterior probability density function integrated from `redshiftalt_qsoc_lower` to `redshiftalt_qsoc`

See Equation **??** of the on-line documentation for details.

CCFRATIO_QSOC : Value of the cross-correlation function used to derive the redshift from QSOC, relative to its maximum value (float)

Value of the coefficients log-likelihood (Equation **??** of the on-line documentation), formerly known as the cross-correlation function in *Gaia* DR3, normalized to stand in the range $[0, 1]$. Low values of this parameter indicate that the redshift selection was mostly driven by the colour and absolute magnitude likelihoods (Equation **??** and Equation **??** of the on-line documentation, respectively) instead of the likelihood based on the comparison of the observed BP/RP spectra against the QSO template.

See Equation **??** of the on-line documentation for details.

ZSCORE_QSOC : Redshift zscore from QSOC (float)

Indicator of the presence of quasar emission lines. `zscore` takes values between 0 and 1. A value close to one indicates that all the quasar emission lines that we expect at `redshift_qsoc` are found in the spectra while low values of `zscore` indicate that at least one emission line is not present.

See Equation **??** of the on-line documentation for details.

FLAGS_QSOC : Processing flags from QSOC (long)

The processing flags report the potential errors that can occur during the prediction process from QSOC.

The flag value is a binary combination (binary OR) of:

- 0: Z_NOWARNING. The processing of this source raised no warning flag.
- 1: Z_AMBIGUOUS. The coefficients log-likelihood (Equation **??** of the online documentation) has more than one maximum above 0.95 when normalized between 0 and 1. Accordingly, the solution would be ambiguous if based on BP/RP spectra

only while ignoring the colour and absolute magnitude likelihoods (Equation **??** and Equation **??** of the on-line documentation, respectively).

- 2: Z_LOWCCFRATIO. The selected redshift leads to a small value of the normalized coefficients log-likelihood, meaning that the redshift selection may be predominantly driven by the colour and absolute magnitude likelihoods. Equivalently, `ccfratio_qsoc` < 0.99.
- 4: Z_LOWZSCORE. The selected redshift leads to a low `zscore_qsoc`, meaning that at least one emission line is either missing or is strongly damped. Equivalently, `zscore_qsoc` < 0.99.
- 8: Z_NOTOPTIMAL. We did not choose the redshift having the highest coefficients log-likelihood, meaning that the coefficients log-likelihood would lead to a redshift selection that differs from the one reported in `redshift_qsoc`. Equivalently, `ccfratio_qsoc` < 1.
- 16: Z_BADSPEC. Raised if the number of used BP or RP spectral transits is lower than 10 or if $G > 20.5$ mag.
- 32: Z_MISSINGG. Raised if the input G magnitude is missing. If raised, no absolute G magnitude is computed and no absolute magnitude likelihood is used in Equation **??**.
- 64: Z_MISSINGBP. Raised if the input internally-calibrated BP spectrum is missing. If raised, then Equation **??** is based only on the internally-calibrated RP spectrum.
- 128: Z_MISSINGRP. Raised if the input internally-calibrated RP spectrum is missing. If raised, then Equation **??** is based only on the internally-calibrated BP spectrum.
- 256: Z_BADSPECBP. The computation of the internally-calibrated BP spectrum raised some flags suggesting that the fit is suspicious (see Equation **??**). A spectral fit is considered as suspicious if the iterative fitting algorithm did not converge; if the resulting reduced χ^2 is larger than 2; if a robust evaluation of the dispersion of its normalized residuals is larger than 2; if the problem is under-constrained; if more than 10% of the epoch fluxes are rejected during the fit; or if the S/N of the resulting spectral coefficients is lower than 2.
- 512: Z_BADSPECRP. The computation of the internally-calibrated RP spectrum raised some flags suggesting that the fit is suspicious. A spectral fit is considered as suspicious if the iterative fitting algorithm did not converge; if the resulting reduced χ^2 is larger than 2; if a robust evaluation of the dispersion of its normalized residuals is larger than 2; if the problem is under-constrained; if more than 10% of the epoch fluxes are rejected during the fit; or if the S/N of the resulting spectral coefficients is lower than 2.

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- 1024: Z_MISSINGEXTINCTION. The TGE extinction is missing. If raised, then a default extinction of $A_0 = 0$ is assumed.
- 2048: Z_LOWLOGP. The log-posterior probability associated with `redshift_qsoc` is lower than 3.

GMAG_QSOC : The absolute G magnitude of the quasar (float, Magnitude[mag])

The absolute magnitude of the quasar in the *Gaia* G-band, normalized at $z = 2$. The distance modulus at redshift z is normalized at 10 pc and is calculated from a flat Λ-CDM cosmology with $H_0 = 67.66$ km s^{-1} Mpc^{-1} and $\Omega_\Lambda = 0.69034$ (Planck Collaboration et al. 2020).

GMAG_QSOC_LOWER : The absolute G magnitude of the quasar lower confidence level (float, Magnitude[mag])

Lower confidence level of the absolute G magnitude of the quasar. This is the 16th percentile of the PDF over the absolute G magnitude of the quasar, `gmag_qsoc`.

The lower confidence level is computed from the first order approximation of the uncertainty of the observed G magnitude, $\sigma_G \approx \left|\sigma_{f_G}/f_G\right|$ as `gmag_qsoc_lower` = `gmag_qsoc` $-\sigma_G$ where f_G and σ_{f_G} are respectively the mean observed flux in the *Gaia* G-band and its uncertainty.

GMAG_QSOC_UPPER : The absolute G magnitude of the quasar upper confidence level (float, Magnitude[mag])

Upper confidence level of the absolute G magnitude of the quasar. This is the 84th percentile of the PDF over the absolute G magnitude of the quasar, `gmag_qsoc`.

The upper confidence level is computed from the first order approximation of the uncertainty of the observed G magnitude, $\sigma_G \approx \left|\sigma_{f_G}/f_G\right|$ as `gmag_qsoc_lower` = `gmag_qsoc` $+\sigma_G$ where f_G and σ_{f_G} are respectively the mean observed flux in the *Gaia* G-band and its uncertainty.

REDSHIFT_UGC : Redshift from UGC (float)

The redshift of the source treated as a galaxy by the Apsis module UGC. The redshift is estimated from the sampled mean BP/RP spectrum applying a regression model based on Support Vector Machines (SVM), a supervised machine learning algorithm. Details are provided in Section **??** of the on-line documentation.

As an estimate of the uncertainty on `redshift_ugc` the value (`redshift_ugc_upper` – `redshift_ugc_low`

may be used. We note that in the ranges redshift_ugc< 0.02, 0.28 <redshift_ugc< 0.30 and redshift_ugc> 0.58 the SVM-model performance is particularly low and the derived redshifts lie outside of the achievable prediction range. A very narrow interval at 0.070 <redshift_ugc< 0.071 includes several thousand bright sources which probably have redshifts lower than 0.04, as explained in the on-line documentation (Section **??**).

REDSHIFT_UGC_LOWER : Redshift lower confidence level from UGC (float)

The performance limits are based on the mean error $predictDiffMean[r_i]$ and its standard deviation $predictDiffStDev[r_i]$ (1σ-level) obtained during the SVM training with a test data set, where r_i=int(redshift_ugc/0.02). The lower limit is defined as redshift_ugc$-predictDiffMean[r_i]$ $-predictDiffStDev[r_i]$. For more details see the on-line documentation, Section **??**.

REDSHIFT_UGC_UPPER : Redshift upper confidence level from UGC (float)

The performance limits are based on the mean error $predictDiffMean[r_i]$ and its standard deviation $predictDiffStDev[r_i]$ (1σ-level) obtained during the SVM training with a test data set, where r_i=int(redshift_ugc/0.02). The upper limit is defined as redshift_ugc$-predictDiffMean[r_i]$ $+predictDiffStDev[r_i]$. For more details see the on-line documentation, Section **??**.

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9.9 GRAV_LENS_XP_MEAN_SPECTRUM [DataLink]

This table contains the internally-calibrated mean XP spectrum of a given lens component in continuous representation.

Note this table is not available through the main archive TAP interface. Data are delivered via the Massive Data service indexed by the VO DataLink protocol and described in Chapter **??**. The DataLink retrieval type to specify for this product in programmatic archive access via the Astroquery Gaia Python package is `MEAN_SPECTRUM_XP_GRAVLENS`.

DataLink products are served with a choice of two different data structures:

- `DATAMODEL_STANDARD`: the product is serialised following a standard data model (for example an IVOA-compliant data model in applicable cases);
- `DATAMODEL_GAIA`: the product is serialised following the Gaia DPAC(-internal) data model.

The `DATAMODEL_GAIA` format is described here whereas the `DATAMODEL_STANDARD` format is described in `https://www.cosmos.esa.int/web/gaia-users/archive/datalink-products#datalink_serialisation`

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier for the component of the lens multiplet (long)

Unique source identifier for the component of the lens multiplet identified around `source_id`.

BP_N_EPOCH : The number of epoch spectra used to produce this mean spectrum in the BP band (int)

The number of epoch spectra used to produce this mean spectrum.

BP_CHI2R : The reduced chi-square coming from the fit of the basis functions to the partially-calibrated epoch spectra in the BP band (double)

The reduced chi-square coming from the fit of the basis functions to the partially-calibrated epoch spectra in the BP band.

BP_DOF : The degrees of freedom associated with the reduced chi-square in the BP band (int)

The degrees of freedom associated with chi2r.

BP_SNR : An estimate of the S/N of the internally-calibrated mean spectra in the BP band (double)

An estimate of the S/N of the internally-calibrated spectrum that is computed as

$$\sqrt{\frac{\|\boldsymbol{c}\|^2}{\|C\|_F}}$$

where $\boldsymbol{c}$ are the linear coefficients associated with the set of basis functions; C is the covariance matrix associated with $\boldsymbol{c}$ and $\|\cdot\|_F$ represents the Frobenius matrix norm.

BP_FLAGS : Flags produced by the internal calibration process in the BP band (int)

Flags produced during the internal calibration process.

Binary combination of

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Bit	Value	Description
0	EPOCH_ALL_MASKED	Some epoch spectra have all samples masked.
1	EPOCH_NEGATIVE_FLUX	Some epoch spectra have negative integrated flux.
2	EPOCH_MANY_OUTLIERS	Some epoch spectra have more than 10% of rejected samples.
3	EPOCH_LARGE_NOISE	Some epoch spectra have mean value of the noise that is larger than 2.
4	EPOCH_NEGATIVE_NOISE	Some epoch spectra have negative noise estimation (never raised).
5	EPOCH_LARGE_MSE	Some epoch spectra have mean square error that is larger than 2.
6	LARGE_SCORE	Reduced chi-squared > 2.
7	UNDERCONSTRAINED	Degree of freedom < 0.
8	MANY_OUTLIERS	More than 10% of the samples were rejected during the fit.
9	LOW_NEPOCH	This spectrum is based on less than 5 epoch transits.
10	LOW_SNR	This spectrum has a S/N < 2.
11	NEGATIVE_TOTAL_FLUX	This spectrum has a negative total flux.
12	NOT_CONVERGED	The internal calibration procedure did not converged

BP_COEFFICIENTS : The coefficients associated with the continuous basis functions in the BP band (float[] array)

The coefficients associated with the continuous basis functions.

BP_COEFFICIENT_ERRORS : Basis function coefficient errors for the BP spectrum representation (float[] array)

Errors on the basis function coefficients for BP.

BP_COEFFICIENT_CORRELATIONS : Correlation matrix for BP coefficients (float[] array)

Upper-triangular (non-zero, non-unity) part of the correlation matrix $\mathbf{M}$ of the coefficients C for the representation of the BP spectrum as basis functions. The matrix elements are stored as a linear array of constant size $S = n(n-1)/2$ corresponding to the full normal matrix of dimension $n \times n$. The ordering of the elements in the linear array follows a column-major storage scheme, i.e.:

$$\mathbf{M} = \begin{bmatrix} 1 & C[0] & C[1] & C[3] & C[6] & \cdots & C[S-(n-1)] \\ & 1 & C[2] & C[4] & C[7] & \cdots & C[S-(n-2)] \\ & & 1 & C[5] & C[8] & \cdots & C[S-(n-3)] \\ & & & 1 & C[9] & \cdots & C[S-(n-4)] \\ & & & & \ddots & \ddots & \vdots \\ & & & & & 1 & C[S-1] \\ & & & & & & 1 \end{bmatrix}$$

BP_TRANSIT_IDS : The transit identifiers of the epoch spectra contributing to this spectrum in the BP band (long[] array)

The transit ids of the epoch spectra contributing to this component spectrum.

RP_N_EPOCH : The number of epoch spectra used to produce this mean spectrum in the RP band (int)

The number of epoch spectra used to produce this resampled spectrum.

RP_CHI2R : The reduced chi-square coming from the fit of the basis functions to the partially-calibrated epoch spectra in the RP band (double)

The reduced chi-square coming from the fit of the basis functions to the partially-calibrated epoch spectra in the RP band.

RP_DOF : The degrees of freedom associated with the reduced chi-square in the RP band (int)

The degrees of freedom associated with chi2r.

RP_SNR : An estimate of the S/N of the internally-calibrated mean spectra in the RP band (double)

An estimate of the S/N of the internally-calibrated spectrum that is computed as

$$\sqrt{\frac{\|\boldsymbol{c}\|^2}{\|C\|_F}}$$

where $\boldsymbol{c}$ are the linear coefficients associated with the set of basis functions; C is the covariance matrix associated with $\boldsymbol{c}$ and $\|\cdot\|_F$ represents the Frobenius matrix norm.

RP_FLAGS : Flags produced by the internal calibration process in the RP band (int)

Flags produced during the internal calibration process.

Binary combination of

Bit	Value	Description
0	EPOCH_ALL_MASKED	Some epoch spectra have all samples masked.
1	EPOCH_NEGATIVE_FLUX	Some epoch spectra have negative integrated flux.
2	EPOCH_MANY_OUTLIERS	Some epoch spectra have more than 10% of rejected samples.
3	EPOCH_LARGE_NOISE	Some epoch spectra have mean value of the noise that is larger than 2.
4	EPOCH_NEGATIVE_NOISE	Some epoch spectra have negative noise estimation (never raised).
5	EPOCH_LARGE_MSE	Some epoch spectra have mean square error that is larger than 2.
6	LARGE_SCORE	Reduced chi-squared > 2.
7	UNDERCONSTRAINED	Degree of freedom < 0.
8	MANY_OUTLIERS	More than 10% of the samples were rejected during the fit.
9	LOW_NEPOCH	This spectrum is based on less than 5 epoch transits.
10	LOW_SNR	This spectrum has a S/N < 2.
11	NEGATIVE_TOTAL_FLUX	This spectrum has a negative total flux.
12	NOT_CONVERGED	The internal calibration procedure did not converged

RP_COEFFICIENTS : The coefficients associated with the continuous basis functions in the RP band (float[] array)

The coefficients associated with the continuous basis functions.

RP_COEFFICIENT_ERRORS : Basis function coefficient errors for the RP spectrum representation (float[] array)

Errors on the basis function coefficients for RP.

RP_COEFFICIENT_CORRELATIONS : Correlation matrix for RP coefficients (float[] array)

Upper-triangular (non-zero, non-unity) part of the correlation matrix $\mathbf{M}$ of the coefficients C for the representation of the RP spectrum as basis functions. The matrix elements are stored as a linear array of constant size $S = n(n-1)/2$ corresponding to the full normal matrix of dimension $n \times n$. The ordering of the elements in the linear array follows a column-major storage scheme, i.e.:

$$\mathbf{M} = \begin{bmatrix} 1 & C[0] & C[1] & C[3] & C[6] & \cdots & C[S-(n-1)] \\ & 1 & C[2] & C[4] & C[7] & \cdots & C[S-(n-2)] \\ & & 1 & C[5] & C[8] & \cdots & C[S-(n-3)] \\ & & & 1 & C[9] & \cdots & C[S-(n-4)] \\ & & & & \ddots & \ddots & \vdots \\ & & & & & 1 & C[S-1] \\ & & & & & & 1 \end{bmatrix}$$

RP_TRANSIT_IDS : The transit identifiers of the epoch spectra contributing to this spectrum in the RP band (long[] array)

The transit ids of the epoch spectra contributing to this component spectrum.

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9.10 QSO_CANDIDATES [TAP]

This table contains parameters derived from various modules dedicated to the classification and characterisation of sources considered as QSO candidates. Together with those, the QSOs used to define the Gaia-CRF4 are also listed in this table.

This table has been constructed with the intention to be complete rather than pure and, as such, it will contain a large fraction of non-genuine extragalactic sources. Purer samples can be drawn using dedicated flags or queries. Please refer to Chapter **??** of the on-line documentation for details about how this table was built, its content, and for recommendations regarding its exploitation.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (unique within a particular Data Release) (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

GAIA_CRF_SOURCE : Flag indicative of whether the source was used define the Gaia-CRF4 (boolean)

This boolean flag indicates whether the source is part of the Gaia-CRF4, which is published through `gaia_crf4`.

See also **?**.

VARI_BEST_CLASS_NAME : Name of photometric variability best class as listed in Gaia source (string)

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Best class name as found in `phot_variable_best_class` of tables `gaia_source` and `all_source_flags`. See Chapter **??** of the release documentation for details.

VARI_BEST_CLASS_PROB : Probability from the variability classification result corresponding to the variability best class, when available (float)

It is filled with the maximum of the `classifier_best_class_prob` values of table `vari_classifier_res` for this `source_id`. Note that this value is only available for a subset of the sources that have a `vari_best_class_name`. See Section **??** of the release documentation for details.

FRACTIONAL_VARIABILITY_G : Fractional variability in the G band (float)

The fractional variability is calculated as $\sqrt{\mathrm{MAD}^2(F) - \mathrm{RMS}^2(\sigma_F)}/\mathrm{median}(F)$, where $\mathrm{MAD}(F)$ is the Median Absolute Deviation (MAD) of the field-of-view transit fluxes F in the G band, $\mathrm{RMS}^2(\sigma_F)$ the mean square of flux uncertainties σ_F, and $\mathrm{median}(F)$ is the median flux of the field-of-view transits in the G band.

STRUCTURE_FUNCTION_INDEX : Index of the first-order structure function in the G band (float)

Index of the first-order structure function (SF; Simonetti et al. 1985), i.e., slope in the logSF vs logTau space, where Tau is the time lag. The SF is expressed in magnitude squared and computed from field-of-view transit magnitudes in the G band. The index is linked to that of the Fourier power spectrum and indicates the type of noise process at work.

STRUCTURE_FUNCTION_INDEX_SCATTER : Standard deviation of the index of the structure function (double)

Standard deviation of the index α of the structure function (i.e., slope in the logSF vs logTau space). The structure function is expressed in magnitude square. The index is linked to the power of the Fourier power spectrum and indicates the type of noise process at work. AGN should show $\alpha = 0.5$, in between flicker noise ($\alpha = 0$) and shot (random walk) noise ($\alpha = 1$).

QSO_VARIABILITY : Quasar variability metric in the G band (float)

Quasar variability metric from field-of-view transit magnitudes in the G band in log format, $\log(\chi^2_{\mathrm{QSO}}/\nu)$, from Butler & Bloom (2011), after adaptation to Gaia data.

NON_QSO_VARIABILITY : Non-quasar variability metric in the G band (float)

Non-quasar variability metric from field-of-view transit magnitudes in the G band in log format, $\log(\chi^2_{\rm false}/\nu)$, from Butler & Bloom (2011), after adaptation to Gaia data.

VARI_AGN_MEMBERSHIP_SCORE : Membership score (0=lowest,1=highest) of source to be of AGN type (double)

Membership score (0=lowest,1=highest) of source to be of AGN type.

Each specific object module has its own method of computing this score so this value cannot be perceived as a probability, nor can it readily be compared between different specific object modules without recalibration (apart from the extremes 0 and 1).

For SOS-AGN (Chapter **??**), it is defined as the Mahalanobis distance from the reference set of QSOs, inverted and rescaled by a Gaussian from 0 to 1, which takes into account five variability features (and their covariances) from the Gaia time series: the Butler & Bloom parameters (`qso_variability` and `non_qso_variability`), the structure function slope, the fractional variability, and the Abbe (or von Neumann) parameter. The reference set for AGN is Gaia CRF3 (Gaia Collaboration et al. 2022) while for blazars the reference set includes the sources in the BZCAT5 catalogue (Massaro et al. 2015) with variability amplitude in the *G* band (`range_mag_g_fov` in `vari_summary`) greater than 1.8. The need to take the maximum between two values obtained with two reference samples is that blazars, which are a numerically small population, have more extreme variability properties than the average AGN population, so that a membership score based on general AGN properties would penalise blazars.

CLASSPROB_DSC_COMBMOD_QUASAR : Probability from DSC-Combmod of being a quasar (data used: BP/RP spectrum, photometry, astrometry) (float)

Probability that the object is of the named class. This is the overall probability for this class, computed by combining the class probabilities from DSC-Specmod (which classifies objects using BP/RP spectra) and DSC-Allosmod (which classifies objects using several astrometric and photometric features). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSPROB_DSC_COMBMOD_GALAXY : Probability from DSC-Combmod of being a galaxy (data used: BP/RP spectrum, photometry, astrometry) (float)

Probability that the object is of the named class. This is the overall probability for this class, computed by combining the class probabilities from DSC-Specmod (which classifies objects using BP/RP spectra) and DSC-Allosmod (which classifies objects using several astrometric and photometric features). It is important to realise that the DSC classes are defined by the training data used, and that this may produce a narrower definition of the class than may be expected given the class name. This is a posterior probability that includes the global class prior, given in the documentation.

CLASSLABEL_DSC : Class assigned by DSC based on the probability from its Combmod classifier (string)

Class assigned by DSC based on the probability from its Combmod classifier. Probabilities are computed from a combination of the probabilities of two DSC classifiers, DSC-Specmod (which classifies objects using BP/RP spectra) and DSC-Allosmod (which classifies objects using several astrometric and photometric features).

DSC-Combmod provides a normalized posterior probability vector across several classes. Note that this incorporates the global class prior, as explained in the documentation. This class label is set to that class with the largest probability above 0.5. If no probability is above 0.5, this class label is 'unclassified'. If users want to perform classification using a different threshold, or by adopting a different prior, they should use the DSC probability vectors.

CLASSLABEL_DSC_ALPHA : Class assigned by DSC based on the probability from its CombmodAlpha classifier (string)

Class assigned by DSC based on the probability from its CombmodAlpha classifier. Probabilities are computed from a parametric combination of the probabilities of two DSC classifiers, DSC-Specmod (which classifies objects using BP/RP spectra) and DSC-Allosmod (which classifies objects using several astrometric and photometric features).

DSC-CombmodAlpha provides a normalized posterior probability vector across several classes. Note that this incorporates the global class prior, as explained in the documentation. This class label is set to that class with the largest probability above 0.5. If no probability is above 0.5, this class label is 'unclassified'. If users want to perform classification using a different threshold, or by adopting a different prior, they should use the DSC probability vectors.

CLASSLABEL_DSC_SUPP : Class assigned by DSC based on the probability from its CombmodSupp classifier (string)

Class assigned by DSC based on the probability from its CombmodSupp classifier. Proba-

bilities are computed from a combination of the probabilities of two DSC classifiers, DSC-SpecmodSupp (which classifies objects jointly using BP/RP spectra and several astrometric and photometric features) when available, otherwise DSC-Specmod (which classifies objects using BP/RP spectra).

DSC-CombmodSupp provides a normalized posterior probability vector across several classes. Note that this incorporates the global class prior, as explained in the documentation. This class label is set to that class with the largest probability above 0.5. If no probability is above 0.5, this class label is 'unclassified'. If users want to perform classification using a different threshold, or by adopting a different prior, they should use the DSC probability vectors.

CLASSLABEL_DSC_ALLOS : Class assigned by DSC based on the probability from its Allosmod classifier (string)

Class assigned by DSC based on the probability from its Allosmod classifier. Probabilities are computed from from a classifier that uses various astrometric and photometric features.

CLASSLABEL_DSC_SPEC : Class assigned by DSC based on the probability from its Specmod classifier (string)

Class assigned by DSC based on the probability from its Specmod classifier. Probabilities are computed from from a classifier that uses the BP/RP spectrum.

CLASSLABEL_DSC_SPEC_SUPP : Class assigned by DSC based on the probability from its SpecmodSupp classifier (string)

Class assigned by DSC based on the probability from its SpecmodSupp classifier. Probabilities are computed from a classifier that jointly uses the BP/RP spectrum and various astrometric and photometric features.

CLASSLABEL_DSC_JOINT : Class assigned by DSC based on the probability from its Specmod and Allosmod classifiers (string)

Class label. The definition of the joint class label is based on the posterior probabilities of the DSC classifiers trained on the BP/RP spectra (module DSC-Specmod) and on the collection of features from astrometry and photometry (module DSC-Allosmod).

The joint class label identifies a source as a quasar or a galaxy if the two classifiers predict a maximum probability higher than 0.50 for the same target class. Otherwise, the joint class label

is declared as unclassified.

FLAGS_DSC : Flag reporting the DSC processing status per source (string)

Processing flag array. The field is a collection of numerical indicators referring to the status of DSC processing.

- The 1st digit is a default value.
- The 2nd digit indicates if a source is an outlier candidate. An outlier is declared based on the probabilities of the combined DSC classifiers. The value is equal to 0 for non-outliers when the combined classifiers separately give a maximum probability higher than 0.99, otherwise value is equal to 1 for outliers. Note that the predicted classes of the combined classifiers are not required to be the same in this analysis. An outlier is also declared when the number of observations contributing to the BP and RP photometry is below a defined threshold (5).
- The 4th digit indicates if a source is an outlier candidate based on the probabilities of the Combmod module.
- The 5th digit indicates if a source is an outlier candidate based on the probabilities of the Combmod-alpha module.
- The 7th digit indicates if a source is an outlier candidate based on the probabilities of the CombmodSupp module.
- The 9th digit indicates the status of data processing.

Values of the 4th, 5th and 7th digits are integers from 0 to 3, based on the maximum predicted probability. Value is equal to 3 when the maximum probability is lower than 0.90, equal to 2 when the maximum probability is within 0.90 and 0.95, equal to 1 when the maximum probability is lower than 0.999, and equal to 0 if the maximum probability is higher than 0.999.
Values of the 9th digit are integers from 1 to 3. Value is equal to 1 when the Allosmod and the Specmod-Supp probabilities are N/A due to missing astrometric and photometric features. Value is equal to 2 when the required inputs are available to the classifiers. Value is equal to 3 when the astrometric and photometric features, the BP/RP spectra and the infrared photometry are available to the classifiers.

CLASSLABEL_OA : Class assigned by OA in the neuron that represents the source (string)

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Class label of the neuron that represents the source, as assigned by the OA module in Apsis. See Section **??** for further details.

CLASSLABEL_UCA : Class assigned by UCA in the neuron that represents the source (string)

Class label of the neuron that represents the source, as assigned by the UCA module in Apsis. There will be four main label types: star, galaxy, quasar and unknown. The last one is assigned when the source cannot be classified as any of the other types. See Section **??** for further details.

REDSHIFT_QSOC : Redshift from QSOC (float)

The redshift of the source estimated from the analysis of the BP/RP spectra by the QSOC Apsis module.

The redshift of the source is obtained from the maximum of the log-posterior probability function as defined in Equation **??** of the on-line documentation.

REDSHIFT_QSOC_LOWER : Redshift lower confidence level from QSOC (float)

Lower confidence level of the redshift estimate. This is the 16th percentile of the PDF over redshift as computed from Equation **??** of the on-line documentation.

REDSHIFT_QSOC_UPPER : Redshift upper confidence level from QSOC (float)

Upper confidence level of the redshift estimate. This is the 84th percentile of the PDF over redshift as computed from Equation **??** of the on-line documentation.

REDSHIFTALT_QSOC : Alternative redshift from QSOC (float)

The alternative, second most probable redshift estimate according to the log-posterior probability function defined in Equation **??** of the on-line documentation.

REDSHIFTALT_QSOC_LOWER : Alternative redshift lower confidence level from QSOC (float)

Lower confidence level of the alternative redshift estimate. This is the 16th percentile of the PDF over redshift as computed by QSOC.

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REDSHIFTALT_QSOC_UPPER : Alternative redshift upper confidence level from QSOC (float)

Upper confidence level of the alternative redshift estimate. This is the 84th percentile of the PDF over redshift as computed by QSOC.

REDSHIFT_QSOC_LOGPROB : Log-probability of redshift_qsoc (double)

The logarithm of the posterior probability density function associated with the main redshift estimate, `redshift_qsoc` (see Equation **??** of the on-line documentation).

REDSHIFTALT_QSOC_LOGPROB : Log-probability of redshiftalt_qsoc (double)

The logarithm of the posterior probability density function associated with the alternative redshift estimate, `redshiftalt_qsoc` (see Equation **??** of the on-line documentation).

REDSHIFT_QSOC_PROB : The probability that the redshift stands within its confidence intervals (double)

The probability that the redshift stands within its confidence intervals. This is equal to the posterior probability density function integrated from `redshift_qsoc_lower` to `redshift_qsoc_upper`.

See Equation **??** of the on-line documentation for details.

REDSHIFTALT_QSOC_PROB : The probability that the alternative redshift stands within its confidence intervals (double)

The probability that the alternative redshift stands within its confidence intervals. This is equal to the posterior probability density function integrated from `redshiftalt_qsoc_lower` to `redshiftalt_qsoc`

See Equation **??** of the on-line documentation for details.

CCFRATIO_QSOC : Value of the cross-correlation function used to derive the redshift from QSOC, relative to its maximum value (float)

Value of the coefficients log-likelihood (Equation **??** of the on-line documentation), formerly known as the cross-correlation function in *Gaia* DR3, normalized to stand in the range $[0, 1]$. Low values of this parameter indicate that the redshift selection was mostly driven by the colour and absolute magnitude likelihoods (Equation **??** and Equation **??** of the on-line documentation,

respectively) instead of the likelihood based on the comparison of the observed BP/RP spectra against the QSO template.

See Equation **??** of the on-line documentation for details.

ZSCORE_QSOC : Redshift zscore from QSOC (float)

Indicator of the presence of quasar emission lines. `zscore` takes values between 0 and 1. A value close to one indicates that all the quasar emission lines that we expect at `redshift_qsoc` are found in the spectra while low values of `zscore` indicate that at least one emission line is not present.

See Equation **??** of the on-line documentation for details.

FLAGS_QSOC : Processing flags from QSOC (long)

The processing flags report the potential errors that can occur during the prediction process from QSOC.

The flag value is a binary combination (binary OR) of:

- 0: Z_NOWARNING. The processing of this source raised no warning flag.
- 1: Z_AMBIGUOUS. The coefficients log-likelihood (Equation **??** of the online documentation) has more than one maximum above 0.95 when normalized between 0 and 1. Accordingly, the solution would be ambiguous if based on BP/RP spectra only while ignoring the colour and absolute magnitude likelihoods (Equation **??** and Equation **??** of the on-line documentation, respectively).
- 2: Z_LOWCCFRATIO. The selected redshift leads to a small value of the normalized coefficients log-likelihood, meaning that the redshift selection may be predominantly driven by the colour and absolute magnitude likelihoods. Equivalently, `ccfratio_qsoc` < 0.99.
- 4: Z_LOWZSCORE. The selected redshift leads to a low `zscore_qsoc`, meaning that at least one emission line is either missing or is strongly damped. Equivalently, `zscore_qsoc` < 0.99.
- 8: Z_NOTOPTIMAL. We did not choose the redshift having the highest coefficients log-likelihood, meaning that the coefficients log-likelihood would lead to a redshift selection that differs from the one reported in `redshift_qsoc`. Equivalently, `ccfratio_qsoc` < 1.

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- 16: Z_BADSPEC. Raised if the number of used BP or RP spectral transits is lower than 10 or if $G > 20.5$ mag.
- 32: Z_MISSINGG. Raised if the input G magnitude is missing. If raised, no absolute G magnitude is computed and no absolute magnitude likelihood is used in Equation **??**.
- 64: Z_MISSINGBP. Raised if the input internally-calibrated BP spectrum is missing. If raised, then Equation **??** is based only on the internally-calibrated RP spectrum.
- 128: Z_MISSINGRP. Raised if the input internally-calibrated RP spectrum is missing. If raised, then Equation **??** is based only on the internally-calibrated BP spectrum.
- 256: Z_BADSPECBP. The computation of the internally-calibrated BP spectrum raised some flags suggesting that the fit is suspicious (see Equation **??**). A spectral fit is considered as suspicious if the iterative fitting algorithm did not converge; if the resulting reduced χ^2 is larger than 2; if a robust evaluation of the dispersion of its normalized residuals is larger than 2; if the problem is under-constrained; if more than 10% of the epoch fluxes are rejected during the fit; or if the S/N of the resulting spectral coefficients is lower than 2.
- 512: Z_BADSPECRP. The computation of the internally-calibrated RP spectrum raised some flags suggesting that the fit is suspicious. A spectral fit is considered as suspicious if the iterative fitting algorithm did not converge; if the resulting reduced χ^2 is larger than 2; if a robust evaluation of the dispersion of its normalized residuals is larger than 2; if the problem is under-constrained; if more than 10% of the epoch fluxes are rejected during the fit; or if the S/N of the resulting spectral coefficients is lower than 2.
- 1024: Z_MISSINGEXTINCTION. The TGE extinction is missing. If raised, then a default extinction of $A_0 = 0$ is assumed.
- 2048: Z_LOWLOGP. The log-posterior probability associated with `redshift_qsoc` is lower than 3.

GMAG_QSOC : The absolute G magnitude of the quasar (float, Magnitude[mag])

The absolute magnitude of the quasar in the *Gaia* G-band, normalized at $z = 2$. The distance modulus at redshift z is normalized at 10 pc and is calculated from a flat Λ-CDM cosmology with $H_0 = 67.66$ km s^{-1} Mpc^{-1} and $\Omega_\Lambda = 0.69034$ (Planck Collaboration et al. 2020).

GMAG_QSOC_LOWER : The absolute G magnitude of the quasar lower confidence level (float, Magnitude[mag])

Lower confidence level of the absolute G magnitude of the quasar. This is the 16th percentile of the PDF over the absolute G magnitude of the quasar, `gmag_qsoc`.

The lower confidence level is computed from the first order approximation of the uncertainty of the observed G magnitude, $\sigma_G \approx \left|\sigma_{f_G}/f_G\right|$ as `gmag_qsoc_lower` = `gmag_qsoc` $-\sigma_G$ where f_G and σ_{f_G} are respectively the mean observed flux in the *Gaia* G-band and its uncertainty.

GMAG_QSOC_UPPER : The absolute G magnitude of the quasar upper confidence level (float, Magnitude[mag])

Upper confidence level of the absolute G magnitude of the quasar. This is the 84th percentile of the PDF over the absolute G magnitude of the quasar, `gmag_qsoc`.

The upper confidence level is computed from the first order approximation of the uncertainty of the observed G magnitude, $\sigma_G \approx \left|\sigma_{f_G}/f_G\right|$ as `gmag_qsoc_lower` = `gmag_qsoc` $+\sigma_G$ where f_G and σ_{f_G} are respectively the mean observed flux in the *Gaia* G-band and its uncertainty.

N_TRANSITS_AF : Number of AF transits used for the morphological analysis (int)

The number of AF transits used to analyse the object morphology.

N_TRANSITS_SM : Number of SM transits used for the morphological analysis (int)

The number of SM transits used to analyse the object morphology.

ANGULAR_COVERAGE_AF : Angular coverage of AF transits used for the morphological analysis (float)

The angular coverage of a source is estimated as the ratio between the area of the polygon created from the union of all the AF observed windows and transit angles over the area of a circle with a diameter equal to the diagonal of the largest observed window. The coverage corresponds to the fraction of the surface of the source covered by the observations.

ANGULAR_COVERAGE_SM : Angular coverage of SM transits used for the morphological analysis (float)

The angular coverage of a source is estimated as the ratio between the area of the polygon created from the union of all the SM observed windows and transit angles over the area of a circle with a

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diameter equal to the diagonal of the largest observed window. The coverage corresponds to the fraction of the surface of the source covered by the observations.

INTENSITY_QUASAR : Fitted intensity of the quasar at its centre (double, Flux[e^- pix^{-1} s^{-1}])

Light intensity at the centre of the quasar as obtained by fitting an exponential profile for the central quasar and a Sersic profile for the host galaxy. The unit of this information is e-/(pix*s) with the size of the pixel being 14.733 mas.

INTENSITY_QUASAR_ERROR : Error on the fitted intensity of the quasar at its centre (double, Flux[e^- pix^{-1} s^{-1}])

Uncertainty on the light intensity at the centre of the quasar as obtained by fitting an exponential profile for the central quasar and a Sersic profile for the host galaxy. The unit of this information is e-/(pix*s) with the size of the pixel being 14.733 mas.

INTENSITY_HOSTGALAXY : Fitted intensity of the host galaxy at the effective radius (double, Flux[e^- pix^{-1} s^{-1}])

Light intensity at the centre of the host galaxy as obtained by fitting an exponential profile for the central quasar and a Sérsic profile for the host galaxy. The unit of this information is e-/(pix*s) with the size of the pixel being 14.733 mas.

INTENSITY_HOSTGALAXY_ERROR : Error on the fitted intensity of the host galaxy at effective radius (double, Flux[e^- pix^{-1} s^{-1}])

Formal uncertainty on the light intensity of the host galaxy at the effective radius as obtained by fitting an exponential profile for the central quasar and a Sérsic profile for the host galaxy. The unit of this information is e-/(pix*s) with the size of the pixel being 14.733 mas.

RADIUS_HOSTGALAXY : Fitted effective radius of the host galaxy (double, Angle[mas])

Effective radius containing half of the total luminosity of the Sérsic profile used to fit the host galaxy.

RADIUS_HOSTGALAXY_ERROR : Error on the fitted effective radius of the host galaxy (double, Angle[mas])

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Uncertainty on the effective radius containing half of the total luminosity of the Sérsic profile used to fit the host galaxy.

N_SERSIC : Fitted Sérsic index (float)

Sérsic index as obtained by fitting a Sérsic profile for the host galaxy. The Sérsic profile is a mathematical function that describes how the intensity of a galaxy varies with the distance from its centre. The Sérsic index describes how steep is this variation.

N_SERSIC_ERROR : Error on the fitted Sérsic index (float)

Formal uncertainty on `n_sersic`.

ELLIPTICITY_HOSTGALAXY : Fitted ellipticity of the host galaxy (double)

Ellipticity (defined as 1-(b/a) where (b/a) is the axis ratio) of the host galaxy obtained by the morphological fitting of a Sérsic profile for the host galaxy.

ELLIPTICITY_HOSTGALAXY_ERROR : Error on the fitted ellipticity of the host galaxy (double)

Formal uncertainty on the ellipticity of the host galaxy as obtained by the morphological fitting of a Sérsic profile.

POSANGLE_HOSTGALAXY : Fitted position angle of the host galaxy (double, Angle[deg])

Position angle of the host galaxy obtained by the morphological fitting of the Sérsic profile with respect to Celestial North Pole.

POSANGLE_HOSTGALAXY_ERROR : Error on the fitted position angle of the host galaxy (double, Angle[deg])

Formal uncertainty on the position angle of the host galaxy obtained by the morphological fitting of the Sérsic profile with respect to Celestial North Pole.

HOST_GALAXY_DETECTED : Flag indicating whether a host galaxy has been detected (boolean)

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Flag indicating the presence of a detectable host galaxy. False means that no host galaxy could be detected, while True means that a measurable host galaxy was detected around the quasar.

L2_NORM : L2 norm for the fitted exponential + Sérsic profile (double)

This value represents the mean squared error between the integrated flux of all observed samples (from the Sky Mapper and Astrometric Field) and the integrated flux of synthetic samples produced with the fitted profile.

HOST_GALAXY_FLAG : Flag indicative of processing or scientific quality for the morphological parameters fitting (byte)

This flag provides information about the processing or scientific quality of the results of the Quasar morphology analysis chain for the source of interest.

The flag coding is the following:

0: Reliable solution. A host galaxy was detected.

1: Probably a reliable solution but there is a secondary source within 2.3 arcsec of the source that may have perturbed the solution. A host galaxy was detected.

2: Not a reliable solution due to low signal-to-noise ratio at effective radius. No host galaxy detected

3: Not a reliable solution because fitting did not converge correctly, the parameters reached the bounds of the permitted domain, or parameters are smaller than their errors. No host galaxy was detected.

SOURCE_SELECTION_FLAGS : Bit indicative of whether the input data from a given module met the source list eligibility criteria for the source of interest (int)

Bit field indicative of whether the input data from a given module met the source list eligibility criteria for the source of interest.

The bit is coded as follows:

- bit 0 (LSB): The source belongs to the Gaia-CRF4 (`gaia_crf4` table)

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- bit 1: The source belongs to the `frame_rotator_source` table with either `used_for_reference_frame_orientation` or `used_for_reference_frame_spin` set to True.
- bit 2: The source meets the eligibility criteria for the output of the Surface brightness analysis module (see Chapter **??**).
- bit 3: The source belongs to the `qso_morphology_input_list` table.
- bit 4: The source meets the eligibility criteria for the output of the classification module based on photometric lightcurves (see Chapter **??**).
- bit 5: The source meets the eligibility criteria for the output of the SOS AGN module (see Chapter **??**)
- bit 6: The source meets the eligibility criteria for the output of the DSC module (see Chapter **??**).
- bit 7: The source meets the eligibility criteria for the redshift determined by the QSOC module (see Chapter **??**).
- bit 8: The source meets the eligibility criteria for the classification output of the QSOC module (not yet applicable to DR4).
- bit 9: The source meets the eligibility criteria for the classification output of the OA module (see Chapter **??**). In DR4, no source from the OA module will be considered in the input source list of this table.
- bit 10: The source meets the eligibility criteria for the classification output of the UCA module (see Chapter **??**). In DR4, no source from the UCA module will be considered in the input source list of this table.

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9.11 QSO_MORPHOLOGY_INPUT_LIST [TAP]

This table provides the list of input catalogues that have been used to select those sources for which morphological analysis was performed. Note, however, that not all of these sources will end up with morphological parameters in the QSO candidates table.

Each source is identified in a specific catalogue although it may be present in several catalogues. The first catalogues to be searched for, are several versions of Milliquas, if an object is not found in those, then the Assef R90 and C75 catalogues are considered, and if the source is still not present in those, then the rest of the catalogues is considered. For DR4, the catalogues considered for this selection are mostly external.

Each catalogue is assigned a `catalogue_id`. For DR4 the following applies:

- 1: The 75% Completeness (C75) catalogue from Assef+2018 (Assef et al. 2018)
- 2: The 90% Reliability (R90) catalogue from Assef+2018 (Assef et al. 2018)
- 3: The Early Data Release of the Dark Energy Spectroscopic Instrument (DESI Collaboration et al. 2024)
- 4: AGN catalogue from Gaia and UnWISE, Shu et al. (2019)
- 5: The AGN catalogue from Gaia and UnWISE, Shu et al. (2019)
- 6: The Milliquas catalogue: v6.4 (Flesch 2019)
- 7: The Milliquas catalogue: v7.0 (no dedicated paper – see reference for v6.4, Flesch (2019))
- 8: The Milliquas catalogue: v7.1b (no dedicated paper – see reference for v6.4, Flesch (2019))
- 9: The Milliquas catalogue: v7.4c (no dedicated paper – see reference for v7.2,Flesch (2021))
- 10: The Milliquas catalogue: v7.5 (no dedicated paper – see reference for v7.2, Flesch (2021))
- 11:The Milliquas catalogue: v7.5b (no dedicated paper – see reference for v7.2, Flesch (2021))
- 12: The Milliquas catalogue: v7.7 (no dedicated paper – see reference for v7.2, Flesch (2021))
- 13: The Milliquas catalogue: v7.7c (no dedicated paper – see reference for v7.2, Flesch (2021))

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- 14: The Milliquas catalogue: v8.0 (Flesch 2023)
- 15: The Gaia-unWise Quasar Catalogue (Quaia), Storey-Fisher et al. (2024)
- 16: Sergei Klioner, private communication, 2022
- 17: VST ATLAS quasar survey, Eltvedt et al. (2023)
- 18: Quasar Candidate Catalog, Wu et al. (2023)

Further details on these catalogues are given in the chapter on extended object processing (Section **??**).

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (unique within a particular Data Release) (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

CATALOGUE_ID : The unique identifier for the catalogue(s) used to select the sources in the morphological analysis (byte)

Each catalogue used to select the sources under morphological analysis is assigned a unique ID. The corresponding catalogues are listed in the description of the `qso_morphology_input_list` table, and further details are given in the chapter on extended object processing in the on-line documentation (Section **??**).

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10 Non–single stars

As indicated in the below subsection titles by the labels [TAP] and [DataLink], these products are served from the ESA Gaia Archive using two different protocols, the Table Access Protocol (TAP) for tabular products and the DataLink protocol for other products such as spectra, light curves, MCMC samples, etc. For more details please see Section **??**.

10.1 NSS_ACCELERATION_ASTRO [TAP]

This table contains non-single-star astrometric models for sources having a non-linear proper motion which is compatible with an acceleration solution.

Several possible models are hosted within the same table and they are indicated by the field `solution_type`. The description of this latter lists all possible solution types considered for this release. Only a selection of parameters hosted in this table are provided here, depending on the solution. The details of those is given in the description of field `solution_type`, which can also be used to extract the relevant elements of the correlation vector `correlations`.

Details about the formalism used to derive the parameters in this table are given in the on-line documentation, see Chapter **??**.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Source Identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

SOLUTION_TYPE : NSS model adopted (string)

This is the non-single star model which has been adopted for the published solution, see online documentation, Chapter **??**, for details.

The solution types covered in table `nss_acceleration_astro`, the corresponding fitted variables and their order in the `correlations` vector are listed below. All variables are always fitted for all solution types. A bit index field is thus not necessary.

- Jerk: Jerk model with 10 parameters
  - `ra`: right ascension (fitted variable is an offset against intermediate AGIS results).
  - `dec`: declination (fitted variable is an offset against intermediate AGIS results).
  - `parallax`: parallax at catalogue epoch.
  - `pmra`: proper motion at catalogue epoch.
  - `pmdec`: proper motion at catalogue epoch.
  - `pseudocolour`: Astrometrically determined pseudo-colour of the source.
  - `acceleration_ra`: acceleration in right ascension.
  - `acceleration_dec`: acceleration in declination.
  - `derivative_acceleration_ra`: time derivative of the acceleration in right ascension.
  - `derivative_acceleration_dec`: time derivative of the acceleration in declinat.
- VIMA2: Variability-Induced Mover with a 2-term acceleration
  - `ra`: right ascension (fitted variable is an offset against intermediate AGIS results).
  - `dec`: declination (fitted variable is an offset against intermediate AGIS results).
  - `parallax`: parallax at catalogue epoch.
  - `pmra`: proper motion at catalogue epoch.
  - `pmdec`: proper motion at catalogue epoch.
  - `d_ra_var_bary`: offset in right ascension of the variable star with respect to the barycentre.
  - `d_dec_var_bary`: offset in declination of the variable star with respect to the barycentre.
  - `pm_ra_var_bary`: proper motion of the variable star in right ascension with respect to the barycentre.

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  - pm_dec_var_bary: proper motion of the variable star in declination with respect to the barycentre.
  - acceleration_rho: the coefficient relating, at catalogue epoch, the radius vector to the acceleration.
  - flux_factor_cms_vim: the total flux when the barycentre is on the photocentre over the reference flux in the G band.

- VIMA4: Variability-Induced Mover with a 4-term acceleration
  - ra: right ascension (fitted variable is an offset against intermediate AGIS results).
  - dec: declination (fitted variable is an offset against intermediate AGIS results).
  - parallax: parallax at catalogue epoch.
  - pmra: proper motion at catalogue epoch.
  - pmdec: proper motion at catalogue epoch.
  - d_ra_var_bary: offset in right ascension of the variable star with respect to the barycentre.
  - d_dec_var_bary: offset in declination of the variable star with respect to the barycentre.
  - pm_ra_var_bary: proper motion of the variable star in right ascension with respect to the barycentre.
  - pm_dec_var_bary: proper motion of the variable star in declination with respect to the barycentre.
  - derivative_acceleration_ra_var: time derivative of the acceleration of the variable star in right ascension.
  - derivative_acceleration_dec_var: time derivative of the acceleration of the variable star in declination.
  - acceleration_rho: the coefficient relating, at catalogue epoch, the radius vector to the acceleration.
  - flux_factor_cms_vim: the total flux when the barycentre is on the photocentre over the reference flux in the G band.

CORRELATIONS : Vector form of the upper triangle of the correlation matrix (float[] array)

Correlation matrix of the fitted profile parameters for the applicable non-single star solution. The parameters stored in this matrix and their order is given in the description of field solution_type. Since not all parameters of a given solution model are systematically fitted, the matrix may contain NaN values at the elements of the corresponding indices, such as the length of the vector always corresponds to the expected length of the theoretical model.

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Only non-zero, non-unity, correlation coefficients from the correlation matrix **M** are provided here. They are served as a linear array of constant size $S = n(n-1)/2$ corresponding to the full normal matrix of dimension $n \times n$. The ordering of the elements in the linear array follows a column-major storage scheme, i.e.:

$$\mathbf{M} = \begin{bmatrix} 1 & C[0] & C[1] & C[3] & C[6] & \cdots & C[S-(n-1)] \\ & 1 & C[2] & C[4] & C[7] & \cdots & C[S-(n-2)] \\ & & 1 & C[5] & C[8] & \cdots & C[S-(n-3)] \\ & & & 1 & C[9] & \cdots & C[S-(n-4)] \\ & & & & \ddots & \ddots & \vdots \\ & & & & & 1 & C[S-1] \\ & & & & & & 1 \end{bmatrix}$$

OBJ_FUNC_VALUE : Value of the objective function at the solution (float)

The χ^2, defined as $\sum_{i=1}^{n}(x_i - y_i)^2/\sigma_i^2$, where x_i is the abscissa of the ith transit, y_i is the value calculated from the model, and σ_i is the uncertainty obtained from the astrometric reduction, without any correction.

GOF : Goodness of fit in the Hipparcos sense (float)

Goodness-of-fit statistic of the solution. This is the 'gaussianized chi-square' (Wilson & Hilferty (1931)'s cube root transformation), which for good fits should approximately follow a normal distribution with zero mean value and unit standard deviation.

This statistic is computed according to the formula:

$$F2 = \sqrt{\frac{9\nu}{2}}\left(\sqrt[3]{\frac{\texttt{obj_func_value}}{\nu}} + \frac{2}{9\nu} - 1\right)$$

where `obj_func_value` is hopefully a χ^2 and ν is the number of degrees of freedom.

SIGNIFICANCE : The significance of the solution (i.e. how worth keeping a model is) (float)

It turns out that `gof` is not always enough to decide whether a model is worth keeping or not. The significance addresses some these limitations.

For astrometric binaries, it is defined as a function of the parameters that characterise the model, divided by its uncertainty. This uncertainty is derived from the solution covariance matrix, where the uncertainties are corrected in order to obtain the corrected goodness-of-fit $F_{2\ corrected} = 0$. The

function characterising the model is as follows: (a) for the acceleration 7 model, it is the norm of the acceleration vector; (b) for the acceleration 9 model, it is the norm of the time derivative of the acceleration vector.

EFFICIENCY : Significance of the solution (i.e. how worth keeping a model is) (float)

The efficiency expresses the level of correlation between the parameters of a model. A value of 1 means a total absence of correlation whereas it falls to 0 as the correlation increases. It is defined as the n-th root of the ratio of the product of the diagonal elements of the covariance matrix and the product of the eigen value of that matrix. When all the covariances are 0, the matrix is diagonal and the ratio is exactly 1.

See also Eichhorn (1989).

FLAGS : Quality flag for the achieved NSS solution (long)

Processing flag applicable to specific non-single-star solutions. The meaning of each of those is given in the table below.

Flags bit number	Flag Meaning	Comment
Astrometric binary solutions		
0	Good solution	0 if bad solution, 1 for good solution
1	Non random residuals	1 if there is some signal remaining in the residuals
2-7	NA	

RV_USED : Availability and use of radial velocity (byte)

Reports whether any radial velocity was available and used:

- bit0: whether any radial velocity was available (through CompleteSource or through the transits) or not;
- bit1: whether the available radial velocity was used to correct for μ_r.

ASTROMETRIC_JITTER : Uncorrelated astrometric jitter term (float, Angle[mas])

In order to account for some poorly calibrated effects, this is an excess astrometric noise quadratically added to the uncertainty on the observations such that the resulting gaussianised Goodness of Fit is null. This applies to OrbitalAlternative[Validated] and OrbitalTargetedSearch[Validated] solutions only.

RV_USED_IN_ASTROMETRY : Assumed radial velocity of the source used in the astrometric solution (float, Velocity[km s^{-1}])

Assumed radial velocity of the source used in the astrometric solution in km/s.

This value is the one used to account for possible perspective acceleration effects. This correction is applied even when the expected perspective acceleration is negligible.

Due to cyclic processing of the astrometry and the RVS data, this radial velocity might be different from the one computed using the latest available RVS information, and published under `all_source_rvs.radial_velocity_single`. Moreover, if no reliable radial velocity was available at the time of the astrometric processing, radial velocities from the litterature have been used when relevant, and their value is reported in `rv_used_in_astrometry`.

For unresolved binaries, only the first component of the array is filled. For resolved binaries, the two components may be filled.

ASTROMETRIC_N_OBS_AL : Total astrometric CCD observations in AL considered (int)

Total number of astrometric CCD observations considered in the along-scan direction.

ASTROMETRIC_N_GOOD_OBS_AL : Total astrometric CCD observations in AL actually used (int)

Total astrometric along–scan CCD observations that may have been used in the calculation of the solution. For the acceleration solutions, up to 5% of these transits may have been discarded as being outliers.

RA : Right ascension (double, Angle[deg])

Right ascension α of the source in ICRS at the reference epoch `gaia_source.ref_epoch`, compensated for the trajectory assumed by the model in order to be an average estimate.

RA_ERROR : Standard error of right ascension (float, Angle[mas])

Standard error $\sigma_{\alpha*} \equiv \sigma_\alpha \cos\delta$ of the right ascension of the source in ICRS at the reference epoch `gaia_source.ref_epoch`. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

DEC : Declination (double, Angle[deg])

Declination δ of the source in ICRS at the reference epoch `gaia_source.ref_epoch`, compen-

sated for the trajectory assumed by the model in order to be an average estimate.

DEC_ERROR : Standard error of declination (float, Angle[mas])

Standard error σ_δ of the declination of the source in ICRS at the reference epoch `gaia_source.ref_epoch`. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

PARALLAX : Parallax (double, Angle[mas])

Absolute stellar parallax ϖ of the source at the reference epoch `gaia_source.ref_epoch`

PARALLAX_ERROR : Standard error of parallax (float, Angle[mas])

Standard error σ_ϖ of the stellar parallax at the reference epoch `gaia_source.ref_epoch`. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

PMRA : Proper motion in right ascension direction (double, Angular Velocity[mas yr^{-1}])

Proper motion in right ascension $\mu_{\alpha*} \equiv \mu_\alpha \cos\delta$ of the source in ICRS at the reference epoch `gaia_source.ref_epoch`. This is the local tangent plane projection of the proper motion vector in the direction of increasing right ascension. As for the position, the effect of the motion assumed in the model is compensated in order to be an average estimate.

PMRA_ERROR : Standard error of proper motion in right ascension direction (float, Angular Velocity[mas yr^{-1}])

Standard error $\sigma_{\mu\alpha*}$ of the local tangent plane projection of the proper motion vector in the direction of increasing right ascension at the reference epoch `gaia_source.ref_epoch`. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

PMDEC : Proper motion in declination direction (double, Angular Velocity[mas yr^{-1}])

Proper motion in declination μ_δ of the source at the reference epoch `gaia_source.ref_epoch`. This is the projection of the proper motion vector in the direction of increasing declination. As for the position, the effect of the motion assumed in the model is compensated in order to be an

average estimate.

PMDEC_ERROR : Standard error of proper motion in declination direction (float, Angular Velocity[mas yr^{-1}])

Standard error $\sigma_{\mu\delta}$ of the proper motion component in declination at the reference epoch `gaia_source.ref_e`
The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

PSEUDOCOLOUR : Astrometrically determined pseudo-colour of the source (double, Misc[μm^{-1}])

Astrometrically determined colour of the source. The 'pseudo colour' is estimated in AGIS from the chromatic displacements of image centroids (see **?** for details). It is currently defined to be equivalent to the effective wavenumber, and it can therefore be used as a proxy for this if the latter cannot be computed from CU5 photometry.

PSEUDOCOLOUR_ERROR : Standard error of the pseudo-colour of the source (float, Misc[μm^{-1}])

Standard error of the astrometrically determined `pseudocolour` of the source.

ACCELERATION_RA : Acceleration in RA (double, Misc[mas yr^{-2}])

Acceleration along the RA axis. This is the time derivative of `pmra`.

ACCELERATION_RA_ERROR : Standard error of Acceleration in RA (float, Misc[mas yr^{-2}])

Standard error of the acceleration along the RA axis. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

ACCELERATION_DEC : Acceleration in DEC (double, Misc[mas yr^{-2}])

Acceleration along the declination axis. This is the time derivative of `pmdec`.

ACCELERATION_DEC_ERROR : Standard error of Acceleration in DEC (float, Misc[mas yr^{-2}])

Standard error of the acceleration along the declination axis. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

DERIVATIVE_ACCELERATION_RA : Time derivative of the accel. in RA (double, Misc[mas yr^{-3}])

Time derivative of the acceleration along the RA axis, `acceleration_ra`.

DERIVATIVE_ACCELERATION_RA_ERROR : Standard error of Time derivative of the acceleration in RA (float, Misc[mas yr^{-3}])

Standard error of the time derivative of the acceleration along the RA axis. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

DERIVATIVE_ACCELERATION_DEC : Time derivative of the accel. in DEC (double, Misc[mas yr^{-3}])

Time derivative of the acceleration along the declination axis, `acceleration_dec`.

DERIVATIVE_ACCELERATION_DEC_ERROR : Standard error of Time derivative of the acceleration in DEC (float, Misc[mas yr^{-3}])

Standard error of the time derivative of the acceleration along the declination axis. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

REF_FLUX_G : Reference flux in the G band (float, Flux[ke-/s])

The photocentre of a VIM object moves in relation to its total flux in the G band. In a VIMA2 or VIMA4 solution, the position derived from the single star parameters (`ra`, `dec`, `parallax`, `pmra`, `pmdec`) is the mean position of the object when its total flux is the reference flux. In practice, the reference flux is close to (or is) the median flux of the object.

FLUX_FACTOR_CMS_VIM : Fcms / <Fg> flux factor (double)

CMS flux over average G band flux factor.

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When the total flux of a VIM is Fcms = `flux_factor_cms_vim`×`ref_flux_g` (which doesn't necessarily happen in practice) the photocentre coincides with the barycentre.

FLUX_FACTOR_CMS_VIM_ERROR : Standard error of reference flux factor (float)

Standard error of the flux factor for which the photocentre coincides with the barycentre.

The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

D_RA_VAR_BARY : Offset in RA of the variable star with respect to the barycentre (double, Angle[mas])

A VIM is a double star of which one component is photometrically variable. In a VIMA2 or VIMA4 solution, `d_ra_var_bary` is the mean coordinate of the variable star along the RA axis with respect to the barycentre.

D_RA_VAR_BARY_ERROR : Standard error of Offset in RA of the variable star with respect to the barycentre (float, Angle[mas])

Standard error of the mean coordinate of the variable star along the RA axis with respect to the barycentre. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

D_DEC_VAR_BARY : Offset in DEC of the variable star with respect to the barycentre (double, Angle[mas])

A VIM is a double star of which one component is photometrically variable. In a VIMA2 or VIMA4 solution, `d_dec_var_bary` is the mean coordinate of the variable star along the declination axis with respect to the barycentre.

D_DEC_VAR_BARY_ERROR : Standard error of Offset in DEC of the variable star with respect to the barycentre (float, Angle[mas])

Standard error of the mean coordinate of the variable star along the declination axis with respect to the barycentre. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

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PM_RA_VAR_BARY : Proper motion of the variable star in RA with respect to the barycentre (double, Angular Velocity[mas yr^{-1}])

Orbital proper motion of the variable component around the barycentre, projected on the RA axis.

PM_RA_VAR_BARY_ERROR : Standard error of Proper motion of the variable star in RA with respect to the barycentre (float, Angular Velocity[mas yr^{-1}])

Standard error of the orbital proper motion of the variable component projected on the RA axis. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

PM_DEC_VAR_BARY : Proper motion of the variable star in DEC with respect to the barycentre (double, Angular Velocity[mas yr^{-1}])

Orbital proper motion of the variable component around the barycentre, projected on the declination axis.

PM_DEC_VAR_BARY_ERROR : Standard error of Proper motion of the variable star in DEC with respect to the barycentre (float, Angular Velocity[mas yr^{-1}])

Standard error of the orbital proper motion of the variable component around the barycentre, projected on the declination axis. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

DERIVATIVE_ACCELERATION_RA_VAR : Time derivative of the acceleration of the variable star in RA (double, Misc[mas yr^{-3}])

Time derivative of the acceleration of the variable component of a VIMA4 system, projected on the RA axis.

DERIVATIVE_ACCELERATION_RA_VAR_ERROR : Standard error of Time derivative of the acceleration of the variable star in RA (float, Misc[mas yr^{-3}])

Standard error of the time derivative of the acceleration of the variable component of a VIMA4 system projected on the RA axis. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit

F2=0.

DERIVATIVE_ACCELERATION_DEC_VAR : Time derivative of the acceleration of the variable star in DEC (double, Misc[mas yr^{-3}])

Time derivative of the acceleration of the variable component of a VIMA4 system, projected on the declination axis.

DERIVATIVE_ACCELERATION_DEC_VAR_ERROR : Standard error of Time derivative of the acceleration of the variable star in DEC (float, Misc[mas yr^{-3}])

Standard error of the time derivative of the acceleration of the variable component of a VIMA4 system projected on the declination axis. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

ACCELERATION_RHO : Coefficient relating, at catalogue epoch, the radius vector to the acceleration (double)

In a VIMA solution, the mean acceleration of the variable component is oriented toward the barycentre of the system. Therefore, the mean acceleration is `acceleration_rho` times the radius vector of the variable component, (`d_ra_var_bary`, `d_dec_var_bary`), with `acceleration_rho` < 0. Units are [yr^{-2}].

ACCELERATION_RHO_ERROR : Standard error of the coefficient relating, at catalogue epoch, the radius vector to the acceleration (float)

Standard error of the coefficient relating the acceleration of the variable component to its radius vector measured from the barycentre of the VIMA system. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

RUWE : Renormalised unit weight error (float)

The Renormalised Unit Weight Error is computed as

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$$\texttt{ruwe} = \frac{\sqrt{\texttt{chi2}(AL)/(\texttt{n_obs}(AL) - m)}}{f(G, G_{\rm BP} - G_{\rm RP})}$$

where m is the number of parameters solved and f is a renormalising function.

In practice f is determined in an off-line statistical analysis of the secondary solutions (cf. Lindegren (2018)) and provided as a look-up table.

ASTROMETRIC_CHI2_AL : AL chi-square value (float)

These χ^2 values are computed for all accepted observations of the source, without taking into account the source `astrometric_jitter` (if any). They do however take into account the attitude excess noise (if any) of each observation.

ASTROMETRIC_EXCESS_NOISE : Rescaled Excess noise of the source (float, Angle[mas])

This is the rescaled excess noise ϵ_i of the source. It is not the same as the source excess noise in Gaia DR3 and earlier releases, but re-scaled in order to be consistent with the (also redefined) unit weight error (`ruwe`). But, like the original quantity in earlier releases, it measures the disagreement, expressed as an angle, between the observations of a source and the best-fitting standard astrometric model (using six astrometric parameters). The assumed observational noise in each observation is quadratically increased by ϵ_i in order to statistically match the residuals in the astrometric solution.

A value of $\epsilon_i = 0$ thus signifies that the source is astrometrically well-behaved, i.e. that the residuals of the fit statistically agree with the assumed observational noise. Conversely, a positive ϵ_i signifies that the residuals are statistically larger than expected. This may happen for sources that deviate from the standard six-parameter astrometric model (e.g. unresolved binaries, exoplanet systems, etc.). The statistical significance of the excess noise is given by `astrometric_gof_al`.

Although the rescaling makes ϵ_i much less sensitive to instrumental effects than in previous releases, it may still absorb residual modelling errors not accounted for by the observational noise (image centroiding error) and the calibration excess noise. This includes LSF and PSF calibration errors, saturation effects, geometric instrument calibration errors, and part of the high-frequency attitude noise. Such modelling errors may be particularly important for very bright sources and sources of extreme extreme colours. Given the many other possible contributions to the excess noise, the user is advised to study the empirical distributions of ϵ_i and `astrometric_gof_al` for their particular samples in order to make sensible choices of filtering limits.

The excess source noise is further explained in **?** and in Section **??** of the DR4 online documen-

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tation.

MATCHED_OBSERVATIONS_USED : Matched FOV transits used in the AGIS solution (short)

The number of FOV transits matched to this source having CCD transits used to compute the astrometric solution, this number will be always equal or smaller than the `all_source_astrometry astrometric_matched_transits`, the difference being those which were not used in the AGIS solution either because they did not have good IPD quality or because they where rejected as outliers during the AGIS source update.

VISIBILITY_PERIODS_AVAILABLE : Number of visibility periods available (short)

Number of visibility periods available.

VISIBILITY_PERIODS_USED : Number of visibility periods used (short)

Number of visibility periods used.

ASTROMETRIC_SIGMA5D_MAX : The longest semi-major axis of the 5-d error ellipsoid (float)

This is a 5-dimensional equivalent to the semi-major axis of the position error ellipse and is therefore useful for filtering out cases where one of the five parameters, or some linear combination of several parameters, is particularly ill-determined. It is measured in mas and computed as the square root of the largest singular value of the scaled 5×5 covariance matrix of the astrometric parameters. The matrix is scaled so as to put the five parameters on a comparable scale, taking into account the maximum along-scan parallax factor for the parallax and the time coverage of the observations for the proper motion components. If C is the unscaled covariance matrix, the scaled matrix is SCS, where $S = \mathrm{diag}(1, 1, \sin\xi, T/2, T/2)$, $\xi = 45^\circ$ is the solar aspect angle in the nominal scanning law, and T the time coverage of the data used in the solution.

`astrometric_sigma5d_max` is given for all the solutions that are derived using AGIS. Its size is one of the criteria for accepting or rejecting the 6-parameter solution.

In case of acceleration&jerk solutions (10p-solutions) in the table, i.e. in cases where the 10p solution supersedes the 6p AGIS solution, `astrometric_sigma5d_max` actually is the longest principal axis in the 9-dimensional error ellipsoid (again, like in the 6-parameter case, omitting the `pseudocolour` dimension).

BLENDED : Source update flag indicating use of blended transits (byte)

This field indicates whether blended transits have been used in the astrometric solution:

- when set to 0, it indicates that no blended transit was used
- when set to 1, it indicates that only blended transits have been used
- when set to 2, it indicates that a mix of non-blended and blended transits has been used. This happens when the number of only non-blended or only blended transits is not high enough to apply the scenarios corresponding to the previous cases 0 and 1, respectively.

A blended transit occurs when in an observation of a known close source pair (or close mutiple source) the scanning direction is such that the individual members of the pair (or multiple) are not separated in the CCD window.

For more details see the relevant section of the DR4 online documentation.

NU_EFF_USED_IN_ASTROMETRY : Effective wavenumber of the source used in the astrometric solution (float, Misc[μm^{-1}])

This effective wavenumber $\nu_{\rm eff}$, in units of $\mu{\rm m}^{-1}$, is the assumed value used for the selection of the appropriate LSF/PSF model in the image parameter determination, as well as for the selection of the astrometric calibration parameters in the astrometric adjustment, for the given source. If no BP and RP photometry of the given source was available, `nu_eff_used_in_astrometry` is set to the default value $\nu_{\rm eff} = 1.43\mu{\rm m}^{-1}$. Else, if reliable BP and RP photometry was available, it is set to the weighted mean of the default value and the actual photon-flux weighted inverse wavelength as estimated from the BP and RP bands. If highly precise BP and RP data are available, the `nu_eff_used_in_astrometry` is very close to the value from the BP and RP bands. However, due to the cyclic processing of the Gaia astrometry and photometry, this effective wavenumber will even in these cases be (slightly) different from the one computed using the latest available photometry for DR4.

Some details on the selection of the default value and the construction of the weighted mean:
The default value is chosen as the ensemble average of the actually measured values for all Gaia sources, and its standard error for the weighted mean is set to $0.085\mu{\rm m}^{-1}$, which is close to the actual rms scatter of the measured values.
For the measured values, the formal errors assigned by the photometry processing were used in the construction of the weighted mean. However, the original values delivered by CU5 (from the DR3 photometry processing) were found to be strongly overestimated. Therefore, they were rescaled to give realistic values before being used for the weighted mean.

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TUWE : Transit unit-weight error (float)

This is the factor to be applied on CCD Epoch Astrometric Errors to account for correlations between CCDs or any other discrepancy at transit level.

TUWE_UNCERTAINTY_RESCALED : Have the transit-binned uncertainties been multiplied by the TUWE? (boolean)

Boolean indicating whether the transit-binned uncertainties have been reweighted by the TUWE multiplicative factor to mitigate the inhomogeneity of the uncertainties of the CCD astrometric observations.

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10.2 NSS_EPOCH_FLAGS [DataLink]

This table contains the flags indicative of whether certain observations were used or not for the non-single stars processing. It can be used in combination with the epoch tables containing the astrometric, photometric or radial velocity data themselves.

Note this table is not available through the main archive TAP interface. Data are delivered via the Massive Data service indexed by the VO DataLink protocol and described in Chapter **??**. The DataLink retrieval type to specify for this product in programmatic archive access via the Astroquery Gaia Python package is `EPOCH_FLAGS_NSS`.

DataLink products are served with a choice of two different data structures:

- `DATAMODEL_STANDARD`: the product is serialised following a standard data model (for example an IVOA-compliant data model in applicable cases);
- `DATAMODEL_GAIA`: the product is serialised following the Gaia DPAC(-internal) data model.

The `DATAMODEL_GAIA` format is described here whereas the `DATAMODEL_STANDARD` format is described in `https://www.cosmos.esa.int/web/gaia-users/archive/datalink-products#datalink_serialisation`

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution identifier, please visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

SOURCE_ID_SECONDARY : Source Identifier for the secondary source (long)

Source identifier for the second component in case of a resolved pair model. It is left empty for non-resolved non-single star models.

SOLUTION_TYPE : NSS model adopted (string)

This is the non-single star model which has been adopted for the published solution, see online documentation, Chapter **??**, for details.

TRANSIT_ID : Transit Identifier (long)

The `transit_id` is a unique identifier assigned to each detected (and confirmed) source as it transits the Gaia focal plane. Each time a given source is detected as Gaia scans and re-scans the sky a new `transit_id` will be created to badge that apparition. Hence the along-scan time and the across-scan position along with the telescope in which the source was detected are used to form a unique integer with which to label the transit. The detection of transits is done using the SkyMapper CCD data, and the confirmation is done using the AF1 CCD data. Both happens on board.

The several features of a transit that are encoded in `transit_id` can be easily retrieved using bit masks (&) and shifts (>>) as follows:

- On-Board Mission Time line (OBMT) [ns]
  = 204800 * ((`transit_id` >> 17) & (0x000003FFFFFFFFFF))
- Field-of-view = 1 + (`transit_id` >> 15) & 0x03 [1 for 'preceding' and 2 for 'following' fields-of-view respectively]
- CCD row = (`transit_id` >> 12) & 0x07 [dimensionless, in the range 1 to 7]
- Across-scan 'reference acquisition pixel' in strip AF1 = (`transit_id`) & 0x0FFF [pixels] (this is the across-scan centre of the AF1 window and is odd if immediately below the mid-point of the window and even if immediately above)

where the bit mask prefix '0x' denotes hexadecimal.

For further details see Portell et al. (2020). For convenience a decoder for `transit_id` is available on-line at

https://gaia.esac.esa.int/decoder/transitidDecoder.jsp

AL_CCD_USED : CCD AL measurement used in transit average (Boolean[10] array)

Flag saying whether the CCD transit was used to form the FoV-averaged binned centroid in AL direction. It is an array providing this information for the SM CCD (first index - never used in the non-single star processing) and the 9 AF CCDs (following nine indices).

AL_USED : AL transit astrometry used (Boolean)

Flag saying whether the centroid position of this transit was used for the non-single star model fitting.

G_USED : G transit photometry used (Boolean)

Flag saying whether the G photometry of this transit was used for the non-single star model fitting.

BP_USED : BP transit photometry used (Boolean)

Flag saying whether the BP photometry of this transit was used for the non-single star model fitting.

RP_USED : RP transit photometry used (Boolean)

Flag saying whether the RP photometry of this transit was used for the non-single star model fitting.

RV_USED : Transit radial velocity used (Boolean)

Flag saying whether the radial velocity information of this transit was used for the non-single star model fitting.

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10.3 NSS_MASSES [TAP]

Table of masses derived from the non-single stars (NSS) solutions with orbital parameters.

This table is described in detail in Section **??**. The summary table below presents the different types of NSS orbital solutions combined and which parameters each combination allows to constrain:

combination_method	m1	m2	fluxratio
ASTROSPECTROSB2 ORBITAL_SB2	✓	✓	✓
ECLIPSINGSPECTROSB2 ECLIPSING_SB2	✓	✓	
ASTROSPECTROSB1_M1 ORBITAL_SB1_M1 ORBITAL_LIA_M1		✓	✓
ECLIPSINGSPECTROSB1_M1 ECLIPSING_SB1_M1		✓	
SB2_M1		✓	
VIMO_M2	✓		~
VIMO_M1		✓	~
ORBITAL_M1		lower/upper	
SB1_M1		lower	
PAIR_M1		✓	

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

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m1 : Primary mass (float, Mass[Solar Mass])

Mass of the primary component. See parameter `m1_ref` for the description of its origin.

For ORBITAL_M1, m1 is the primary mass computed assuming `fluxratio_lower`.

m1_lower : Lower confidence level (16th percentile) of the primary mass (float, Mass[Solar Mass])

Lower confidence level (16%) of the primary mass.

For ORBITAL_M1, the lower confidence level of the various tested flux ratios (indicated by `fluxratio_lower` and `fluxratio_upper`) is kept.

m1_upper : Upper confidence level (84th percentile) of the primary mass (float, Mass[Solar Mass])

Upper confidence level (84%) of the primary mass.

For ORBITAL_M1, the upper confidence level of the various tested flux ratios (indicated by `fluxratio_lower` and `fluxratio_upper`) is kept.

m2 : Secondary mass (float, Mass[Solar Mass])

Mass of the secondary component.

For ORBITAL_M1 `combination_method`, a value is provided only when `fluxratio_upper` = `fluxratio_lower`, else only `m2_lower` and `m2_upper` are provided, according to the `fluxratio_lower` and `fluxratio_upper` tested to be compatible with both components being on the isochrones.

For SB1_M1 `combination_method`, owing to the unknown inclination only `m2_lower` can be determined and is filled.

m2_lower : Lower confidence level (16th percentile) of the secondary mass (float, Mass[Solar Mass])

Lower confidence level (16%) of the secondary mass.

For `combination_method`=SB1_M1, it corresponds to the lower confidence level (16%) of the

secondary mass obtained assuming $\sin i = 1$.

For `combination_method`=ORBITAL_M1, it corresponds to the lower confidence level (16%) of the secondary mass obtained with `fluxratio_lower`.

M2_UPPER : Upper confidence level (84th percentile) of the secondary mass (float, Mass[Solar Mass])

Upper confidence level (84%) of the secondary mass.

If `combination_method`=ORBITAL_M1, it corresponds to the upper confidence level (84%) of the secondary mass obtained with `fluxratio_upper`.

FLUXRATIO : Flux ratio of the secondary to the primary in the G-band (float)

Ratio of the G-band flux of the secondary over the primary F_2/F_1.

It can be derived only when combining astrometric and spectroscopic solutions.

For VIMO, it is roughly estimated assuming that the total reference flux is the mean $F_2 + F_1$.

FLUXRATIO_LOWER : Lower value of the flux ratio (float)

When the `fluxratio` can be derived using the combination of astrometric and spectroscopic solutions, `fluxratio_lower` corresponds to the lower confidence level (16%) of the flux ratio derived. If several `fluxratio` need to be tested (astrometric only or SB2 only solutions), it corresponds to the lower fluxratio tested with masses and luminosities consistent with the isochrones and the full range of possible fluxratio is used to derive `m1_lower`, `m1_upper`, `m2_lower` and `m2_upper`.

FLUXRATIO_UPPER : Upper value of the flux ratio (float)

When the `fluxratio` can be derived using the combination of astrometric and spectroscopic solutions, `fluxratio_upper` corresponds to the upper confidence level (84%) of the flux ratio derived. If several `fluxratio` need to be tested (astrometric only or SB2 only solutions), it corresponds to the higher fluxratio tested with masses and luminosities consistent with the isochrones; the full range of possible fluxratio is then used to derive `m1_lower`, `m1_upper`, `m2_lower` and `m2_upper`. When several fluxratios are tested but no `fluxratio_upper` is provided, it means that the mass determination of the primary needed a `fluxratio_lower`> 0 while this flux ratio was not compatible with the derived secondary mass; for those cases, both `m2_upper` and

`m2_lower` are derived using `fluxratio_lower`.

COMBINATION_METHOD : Combination of NSS solutions used (string)

Combination of NSS solutions which has been used to derive the values. It is a combination of the `solution_type` from `nss_two_body_orbit` (ORBITAL, VIMO, SB1, SB2, ECLIPSING, ASTROSPECTROSBX, ECLIPSINGSPECTROSBX), `nss_multiple_orbits` (ORBITAL, SB1, SB2), `nss_resolved_pair` (PAIR) and `rvs_two_body_orbit` (LIA). 'M1' (or 'M2' for VIMO) is added when a primary (resp. secondary) mass estimation is needed. The method to derive this mass is described in the `m1_ref` field.

For VIMO, the mass estimated with IsocLum is the mass of the assumed constant star, so VIMO_M2 (resp. VIMO_M1) is the combination obtained assuming that the primary (resp. secondary) star is variable.

ORBITAL are by default from `solution_type`='Orbital', else from 'OrbitalAlternative%' and in this case the `flag` 'A' is set.

If the combination ASTROSPECTROSBX or ECLIPSINGSPECTROSBX could not be done (low SNR issue or issue with the flux ratio), the combination may have been split into an ORBITAL_M1 or a SBX_M1 combination.

If the source has multiple solutions, the inner orbit is treated first and its total mass is used for the outer orbit (see `m1_ref`).

M1_REF : Reference for the primary mass (string)

The primary (or secondary for VIMO_M2) star mass provided in `m1` (or `m2` for VIMO_M2) can have several origins that can be tracked through this flag:

- no value: the Non-Single Stars (NSS) solutions provide a direct estimate for both masses.
- FLAME: FLAME mass (see Section **??**) using either GSPPhot or GSPSpec inputs
- IsocLum: isochrone derived mass (see Section **??**) using extinction derived from :
  - the Lallement et al. (2022) map (A0map)
  - GSPPhot (A0phot)
  - GSPPhot with an A0map lower bound (A0maplim)

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and metallicity information from :

- GSPSpec (MHspec)
- GSPphot (MHphot)
- solar metallicity (default)

- LowMass: lowmass track used to derive the mass (see Section **??**)
- WD: White Dwarf default mass
- Multiple: mass derived from the inner orbit
- MultipleLower: total inner mass derived assuming the lower value of the flux ratio

FLAG : Flag (string)

Several warnings are provided through the `flag` field:

- M = Mass warning. The primary mass is from a giant star, from a lower case of the inner orbit, with a parallax uncertainty larger than 20% or has a FLAME flag.
- G = goodness of fit warning. At least one of the input NSS solution has a GoF> 5.
- S = signal to noise warning. At least one of the input NSS solution has a poor SNR.
- P = photometry warning.
- A = an OrbitalAlternative solution has been used in this combinaton.

EXOPLANET_CANDIDATE_HRD_PROBA : Exoplanet candidate mass range probability (float)

Probability of the secondary to be with a mass lower than 20 Jupiter mass according to the position of the source in the HR diagram. This probability is computed according to the isochrone fit by testing the two competitive hypothesis for a small photocentre motion: the secondary does not contribute to the flux (and the photocentre corresponds to the primary with a small mass companion) or the secondary contributes significantly to the flux (and we observe the photocentre of a nearly equal mass binary). It is provided only for an ORBITAL_M1 `combination_method` and for a main sequence primary mass, and takes into account the uncertainty on the mass estimated. See Section **??** for more details.

LOWMASS_CANDIDATE_HRD_PROBA : Brown dwarf candidate mass limit probability (float)

Probability of the secondary to be with a mass lower than 80 Jupiter mass according to the position of the source in the HR diagram. It is computed as for the field `exoplanet_candidate_hrd_proba` but with a different mass threshold. Note that it assumes that the potential low mass companion does not have a significant flux, which is not the case for a brown dwarf around a low mass binary. This probability is computed only for an ORBITAL_M1 `combination_method`, see Section **??** for more details.

CORRELATIONS : correlations (float[] array)

Correlation vector of the fitted parameters in the following order: `m1`, `m2`, `fluxratio`.

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10.4 NSS_MULTIPLE_ORBITS [TAP]

This table contains non-single-star orbital models for sources compatible with solutions having more than two bodies.

Several possible models are hosted within the same table and they are indicated by the field `solution_type`. The description of this latter lists all possible solution types considered for this release. Only a selection of parameters hosted in this table are provided here, depending on the solution. The details of those is given in the description of field `bit_index`, which can also be used to extract the relevant elements of the correlation vector `correlations`.

Details about the formalism used to derive the parameters in this table are given in the on-line documentation, see chapter **??**.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that where used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Source Identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

SOLUTION_TYPE : NSS model adopted, see on-line documentation for details (string)

This is the non-single star model which has been adopted for the published solution.

The solution types covered in table `nss_multiple_orbits` are:

- OrbitalAlternativeAcceleration: Alternative orbital model mainly for low S/N systems, plus acceleration
- OrbitalAlternativeJerk: Alternative orbital model mainly for low S/N systems, plus jerk

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- OrbitalAlternativeMultiple: Alternative orbital model mainly for low S/N systems, two companions with astrometric orbits
- PulsationSB1: Pulsation model plus single lined orbit
- PulsationTrendSB1: Pulsation model plus single lined trend
- SB1SB1: Multiple single lined orbit
- SB1TrendSB1: Single lined orbit plus single lined trend
- SB2SB1: Inner double lined orbit, outer single lined orbit
- SB2TrendSB1: Double lined orbit plus single lined trend

SUBTYPE : NSS model sub-type. Bit index as text (string)

Human readable version of the bit index field.

A text description is provided for every `solution_type` and `bit_index` combination. It can be used in ADQL queries. In this particular table, it is composed as a function of the `solution_type` as follows:

- OrbitalAlternativeAcceleration, OrbitalAlternativeJerk and OrbitalAlternativeMultiple: null (single `bit_index` value, all variables fitted)
- PulsationSB1
  - Elliptical: most variables fitted. See `bit_index` for further details.
  - Circular: circular model, eccentricity and longitude of periastron not fitted.
- SB1SB1
  - Elliptical: all variables fitted.
  - Circular: circular model, eccentricity and longitude of periastron not fitted.
- SB2SB1
  - Circular1Elliptical2: circular model for inner orbit and elliptical for outer system.
  - Elliptical1Circular2: elliptical model for inner orbit and circular for outer system.
  - Elliptical1Elliptical2: elliptical model for inner orbit and outer system.

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- PulsationTrendSB1, SB1TrendSB1 and SB2TrendSB1: it is composed as AccelerationParamsX, where X is the number of acceleration parameters spanning range 1-4.

BIT_INDEX : Boolean mask for the fields in the correlation matrix (long)

`bit_index` is a boolean mask indicating which variables or parameters have been fitted for a given non-single-star solution in a table row. The parameters used for each `solution_type` are different. The correspondence between `solution_type`, `bit_index` and fitted variables is provided below.

The M parameters for a given `solution_type` always follow the same order. `bit_index` contains N+1 bits, where the leading Most Significant Bit (MSB) is always 1, and the other N bits correspond to parameters of a given model. The last variable is related to the trailing Least Significant Bit (LSB). A bit with value 1 (0) means the value of its associated parameter has been fitted (kept fixed) in a given solution. Note N can be smaller than M. That is, some solutions do not consider all M possible parameters, but a subset of size N. This is the case for all `solution_type` = PulsationTrendSB1, SB1TrendSB1 and SB2TrendSB1.

The covariance matrix has N×N size. The `bit_index` length and the variable names below can be used to identify elements in the `correlations` vector. When a given parameter has not been fitted (bit in `bit_index` = 0), the corresponding correlation is NaN.

Detailed explanations for each `solution_type` and `bit_index` are provided below. T or F means a parameter has or has not been fit. Hyphens '-' means a variable was not considered in the solution (there are M - N of them). Table field names have been used to identify variables when possible. A text description is provided when not in the data model.

`solution_type` = OrbitalAlternativeAcceleration: the following 19 parameters are fitted and `bit_index` = 1048575 in all cases:

- `ra`
- `dec`
- `parallax`
- `pmra`
- `pmdec`
- `acceleration_ra`
- `acceleration_dec`

- a_thiele_innes
- b_thiele_innes
- f_thiele_innes
- g_thiele_innes
- eccentricity
- period
- t_periastron
- semi_major_axis
- arg_periastron
- pos_ascending_node
- inclination
- astrometric_jitter

solution_type = OrbitalAlternativeJerk: the following 19 parameters are fitted and bit_index = 4194303 in all cases:

- ra
- dec
- parallax
- pmra
- pmdec
- acceleration_ra
- acceleration_dec
- derivative_acceleration_ra
- derivative_acceleration_dec
- a_thiele_innes
- b_thiele_innes

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- f_thiele_innes
- g_thiele_innes
- eccentricity
- period
- t_periastron
- semi_major_axis
- arg_periastron
- pos_ascending_node
- inclination
- astrometric_jitter

solution_type = OrbitalAlternativeMultiple: the following 28 parameters are fitted and bit_index = 536870911 in all cases:

- ra
- dec
- parallax
- pmra
- pmdec
- a_thiele_innes
- b_thiele_innes
- f_thiele_innes
- g_thiele_innes
- eccentricity
- period
- t_periastron
- semi_major_axis

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- arg_periastron
- pos_ascending_node
- inclination
- a_thiele_innes2
- b_thiele_innes2
- f_thiele_innes2
- g_thiele_innes2
- eccentricity2
- period2
- t_periastron2
- semi_major_axis2
- arg_periastron2
- pos_ascending_node2
- inclination2
- astrometric_jitter

solution_type = PulsationSB1

bit_index	subtype	center_of_mass_velocity	Internal fixed value parameter	semi_amplitude_primary	eccentricity	arg_periastron	t_periastron	period2	semi_amplitude_primary2	eccentricity2	arg_periastron2	t_periastron2
3577	Circular	T	F	T	T	T	T	T	T	F	F	T
3583	Elliptical	T	F	T	T	T	T	T	T	T	T	T

solution_type = PulsationTrendSB1

bit_index	subtype	center_of_mass_velocity	Internal fixed value parameter	semi_amplitude_primary	eccentricity	arg_periastron	t_periastron	first_derivative_velocity	second_derivative_velocity	third_derivative_velocity	fourth_derivative_velocity
223	AccelerationTerms1	T	F	T	T	T	T	T	-	-	-
447	AccelerationTerms2	T	F	T	T	T	T	T	T	-	-
895	AccelerationTerms3	T	F	T	T	T	T	T	T	T	-
1791	AccelerationTerms4	T	F	T	T	T	T	T	T	T	T

solution_type = SB1SB1

bit_index	subtype	center_of_mass_velocity	period	semi_amplitude_primary	eccentricity	arg_periastron	t_periastron	period2	semi_amplitude_primary2	eccentricity2	arg_periastron2	t_periastron2
4089	Circular	T	T	T	T	T	T	T	T	F	F	T
4095	Elliptical	T	T	T	T	T	T	T	T	T	T	T

solution_type = SB1TrendSB1

bit_index	subtype	center_of_mass_velocity	period	semi_amplitude_primary	eccentricity	arg_periastron	t_periastron	first_derivative_velocity	second_derivative_velocity	third_derivative_velocity	fourth_derivative_velocity
255	AccelerationTerms1	T	T	T	T	T	T	T	-	-	-
511	AccelerationTerms2	T	T	T	T	T	T	T	T	-	-
1023	AccelerationTerms3	T	T	T	T	T	T	T	T	T	-
2047	AccelerationTerms4	T	T	T	T	T	T	T	T	T	T

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solution_type = SB2SB1

bit_index	subtype	center_of_mass_velocity	period	semi_amplitude_primary	semi_amplitude_secondary	eccentricity	arg_periastron	t_periastron	period2	semi_amplitude_primary2	eccentricity2	arg_periastron2	t_periastron2
7999	Circular1Elliptical2	T	T	T	T	F	F	T	T	T	T	T	T
8185	Elliptical1Circular2	T	T	T	T	T	T	T	T	T	F	F	T
8191	Elliptical1Elliptical2	T	T	T	T	T	T	T	T	T	T	T	T

solution_type = SB2TrendSB1

bit_index	subtype	center_of_mass_velocity	period	semi_amplitude_primary	semi_amplitude_secondary	eccentricity	arg_periastron	t_periastron	first_derivative_velocity	second_derivative_velocity	third_derivative_velocity	fourth_derivative_velocity
511	AccelerationTerms1	T	T	T	T	T	T	T	T	-	-	-
1023	AccelerationTerms2	T	T	T	T	T	T	T	T	T	-	-
2047	AccelerationTerms3	T	T	T	T	T	T	T	T	T	T	-

CORRELATIONS : Vector form of the upper triangle of the correlation matrix (float[] array)

Correlation matrix of the fitted profile parameters for the applicable non-single star solution. The parameters stored in this matrix and their order is given in the description of field bit_index. Since not all parameters of a given solution model are systematically fitted, the matrix may contain NaN values at the elements of the corresponding indices, such as the length of the vector always corresponds to the expected length of the theoretical model.

Only non-zero, non-unity, correlation coefficients from the correlation matrix **M** are provided here. They are served as a linear array of constant size $S = n(n-1)/2$ corresponding to the full normal matrix of dimension $n \times n$. The ordering of the elements in the linear array follows a column-major storage scheme, i.e.:

$$\mathbf{M} = \begin{bmatrix} 1 & C[0] & C[1] & C[3] & C[6] & \cdots & C[S-(n-1)] \\ & 1 & C[2] & C[4] & C[7] & \cdots & C[S-(n-2)] \\ & & 1 & C[5] & C[8] & \cdots & C[S-(n-3)] \\ & & & 1 & C[9] & \cdots & C[S-(n-4)] \\ & & & & \ddots & \ddots & \vdots \\ & & & & & 1 & C[S-1] \\ & & & & & & 1 \end{bmatrix}$$

OBJ_FUNC_VALUE : value of the objective function at the solution (float)

The χ^2, defined as $\sum_{i=1}^{n}(x_i - y_i)^2/\sigma_i^2$, where x_i is the abscissa of the ith transit, y_i is the value calculated from the model, and σ_i is the uncertainty obtained from the astrometric reduction.

GOF : goodness of fit in the Hipparcos sense (float)

The quantity is defined as

$$F2 = \sqrt{\frac{9\nu}{2}}\left(\sqrt[3]{\frac{\texttt{obj_func_value}}{\nu}} + \frac{2}{9\nu} - 1\right)$$

where `obj_func_value` is hopefully a χ^2 and ν is the number of degrees of freedom. $F2$ (Wilson & Hilferty (1931)'s cube root transformation) follows a N(0, 1) distribution.

SIGNIFICANCE : The significance of the solution (i.e. how worth keeping a model is) (float)

It turns out that F2 is not always enough to decide whether a model is worth keeping or not. The significance addresses some these limitations.

EFFICIENCY : Efficiency of the solution (float)

The efficiency expresses the level of correlation between the parameters of a model. A value of 1 means a total absence of correlation whereas it falls to 0 as the correlation increases. It is defined as the n-th root of the ratio of the product of the diagonal elements of the covariance matrix and the product of the eigen value of that matrix. When all the covariances are 0, the matrix is diagonal and the ratio is exactly 1. See also Eichhorn (1989).

BIC : Bayesian Information Criterion (float)

BIC (Bayesian Information Criterion) of the solution, the excess noise being accounted for in the residuals uncertainties.

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FLAGS : Quality flag for the achieved NSS solution (long)

Processing flag applicable to specific non-single-star solutions. The meaning of each of those is given in the table below.

Flags bit number	Flag Meaning	Comment
Spectroscopic binary solutions		
8	BAD_UNCHECKED_NUMBER_OF_TRANSITS	The length of Transit is not sufficient to process the star (before removing bad transits)
9	BAD_CHECKED_NUMBER_OF_TRANSITS	The length of Transit is not sufficient to process the star (after removing bad transits)
10	NO_SIGNIFICANT_PERIODS_CAN_BE_FOUND	No significant period can be derived (no period from photometric variability analysis + peridogram peaks below the significance threshold)
11	NO_SIGNIFICANT_PULSATION_PERIOD	The pulsation periods are not significant)
12	REFINED_SOLUTION_DOES_NOT_CONVERGE	The refined orbital solution does not converge (after 1000 iterations)
13	REFINED_SOLUTION_COVARIANCE_MATRIX	The variance covariance matrix is not definite positive in the refined solution
14	LUCY_TEST_APPLIED	The Lucy test has been applied
15	CIRCULAR_SOLUTION_DOES_NOT_CONVERGE	The Lucy refined orbital solution diverges (after 1000 iterations)
16	CIRCULAR_SOLUTION_NOT_DEFINITE_POSITIVE_COVARIANCE_MATRIX	The variance covariance matrix is not definite positive in the circular solution
17	TREND_SOLUTION_NOT_DEFINITE_POSITIVE_COVARIANCE_MATRIX	The variance covariance matrix is not definite positive in the trend solution
18	REFINED_PULSATION_DOES_NOT_CONVERGE	The refined eccentric pulsation solution does not converge (after 1000 iterations)
19	REFINED_PULSATION_NOT_DEFINITE_POSITIVE_COVARIANCE_MATRIX	The variance covariance matrix is not definite positive in the trend solution
20	CIRCULAR_PULSATION_DOES_NOT_CONVERGE	The circular pulsation solution does not converge (after 1000 iterations)
21	CIRCULAR_PULSATION_NOT_DEFINITE_POSITIVE_COVARIANCE_MATRIX	The variance covariance matrix is not definite positive in the circular pulsation solution
22	SOLUTION_OUTSIDE_E_LOGP_ENVELOP	The orbital solution (SB1 or SB2) is outside the e-log(P) envelop.
23	MODEL_AND_ITS_RESIDUAL_ARE_BOTH_TRENDS	Both the model and its residual are trends
24	COMPOSITE_SOLUTION_DOES_NOT_CONVERGE	The composite solution does not converge (after 1000 iterations)
25	COMPOSITE_SOLUTION_NOT_DEFINITE_POSITIVE_COVARIANCE_MATRIX	The variance covariance matrix is not definite positive in the composite solution
26	SOME_TRANSITS_ARE_DOUBLE_LINED	Some transits are double lined in the single lined mode
27	SIGNIFICANT_SCAN_ANGLE	Scan angle signal is significant
28	SCAN_ANGLE_SOLUTION_NOT_DEFINITE_POSITIVE_COVARIANCE_MATRIX	The variance covariance matrix is not definite positive in the scan angle solution
29	SIGNIFICANT_SCAN_ANGLE_HARMONIC	Scan angle 1st harmonic is significant
30	PERIOD_SIGNIFICANT_IN_PHOTOMETRY_ALSO	The derived period is significant when applied to photometric data.
31	NA	

RV_USED : Availability and use of radial velocity (byte)

Reports whether any radial velocity was available and used:

- bit0: whether any radial velocity was available (through CompleteSource or through the transits) or not;
- bit1: whether the available radial velocity was used to correct for μ_r.

ASTROMETRIC_JITTER : Uncorrelated astrometric jitter term (float, Angle[mas])

In order to account for some poorly calibrated effects, this is an excess astrometric noise quadratically added to the uncertainty on the observations such that the resulting gaussianised Goodness of Fit is null.

ASTROMETRIC_JITTER_ERROR : Astrometric jitter standard error (float, Angle[mas])

In order to account for some poorly calibrated effects, a noise, `astrometric_jitter`, is quadratically added to the uncertainty on the observations such that the resulting F2 is null. When the astrometric jitter is fitted, the current field represents its uncertainty.

ASTROMETRIC_N_OBS_AL : Total astrometric CCD observations in AL considered (int)

Total astrometric CCD observations in AL considered.

ASTROMETRIC_N_GOOD_OBS_AL : Total astrometric CCD observations in AL actually used (int)

Total astrometric CCD observations in AL actually used.

RV_N_OBS_PRIMARY : Total number of radial velocities considered for the primary (int)

Total number of radial velocities considered for the primary.

RV_N_GOOD_OBS_PRIMARY : Total number of radial velocities actually used for the primary (int)

Total number of radial velocities actually used for the primary.

RV_N_OBS_SECONDARY : Total number of radial velocities considered for the secondary in the case of SB2 (int)

Total number of radial velocities considered for the secondary in the case of SB2.

RV_N_GOOD_OBS_SECONDARY : Total number of radial velocities actually used for the secondary in the case of SB2 (int)

Total number of radial velocities actually used for the secondary in the case of SB2.

RA : Right ascension (double, Angle[deg])

Barycentric right ascension α of the source in ICRS at the reference epoch `gaia_source.ref_epoch`

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RA_ERROR : Standard error of right ascension (float, Angle[mas])

Standard error $\sigma_{\alpha*} \equiv \sigma_\alpha \cos\delta$ of the right ascension of the source in ICRS at the reference epoch `gaia_source.ref_epoch`.

DEC : Declination (double, Angle[deg])

Barycentric declination δ of the source in ICRS at the reference epoch `gaia_source.ref_epoch`

DEC_ERROR : Standard error of declination (float, Angle[mas])

Standard error σ_δ of the declination of the source in ICRS at the reference epoch `gaia_source.ref_epoch`

PARALLAX : Parallax (double, Angle[mas])

Absolute stellar parallax ϖ of the source at the reference epoch `gaia_source.ref_epoch`

PARALLAX_ERROR : Standard error of parallax (float, Angle[mas])

Standard error σ_ϖ of the stellar parallax at the reference epoch `gaia_source.ref_epoch`

PMRA : Proper motion in right ascension direction (double, Angular Velocity[mas yr^{-1}])

Proper motion in right ascension $\mu_{\alpha*} \equiv \mu_\alpha \cos\delta$ of the source in ICRS at the reference epoch `gaia_source.ref_epoch`. This is the local tangent plane projection of the proper motion vector in the direction of increasing right ascension.

PMRA_ERROR : Standard error of proper motion in right ascension direction (float, Angular Velocity[mas yr^{-1}])

Standard error $\sigma_{\mu\alpha*}$ of the local tangent plane projection of the proper motion vector in the direction of increasing right ascension at the reference epoch `gaia_source.ref_epoch`

PMDEC : Proper motion in declination direction (double, Angular Velocity[mas yr^{-1}])

Proper motion in declination μ_δ of the source at the reference epoch `gaia_source.ref_epoch`.

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This is the projection of the proper motion vector in the direction of increasing declination.

PMDEC_ERROR : Standard error of proper motion in declination direction (float, Angular Velocity[mas yr^{-1}])

Standard error $\sigma_{\mu\delta}$ of the proper motion component in declination at the reference epoch `gaia_source.ref_e`

A_THIELE_INNES : Thiele-Innes element A for the first orbit (double, Angle[mas])

The Thiele-Innes element A of the orbit of the photocentre (for the orbital model) for the first orbit. This parameter is fitted for orbital models based on astrometry (`solution_type` = ExtrasolarPlanet). For other orbital models based on photometry or radial velocities, or their combination and published in the `nss_two_body_orbit` table, the Campbell parameters are fitted instead. See more details, including the transformation from the A,B,F,G Thiele-Innes elements to the a_0, i, ω and Ω Campbell elements in the on-line documentation (chapter **??**).

A_THIELE_INNES_ERROR : Standard error of Thiele-Innes element A for the first orbit (float, Angle[mas])

Standard error of the Thiele-Innes element A of the orbit of the photocentre (for the orbital model) for the first orbit. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

B_THIELE_INNES : Thiele-Innes element B for the first orbit (double, Angle[mas])

The Thiele-Innes element B of the orbit of the photocentre (for the orbital model). This parameter is fitted for orbital models based on astrometry (`solution_type` = ExtrasolarPlanet). For other orbital models based on photometry or radial velocities, or their combination and published in the `nss_two_body_orbit` table, the Campbell parameters are fitted instead. See more details, including the transformation from the A,B,F,G Thiele-Innes elements to the a_0, i, ω and Ω Campbell elements in the on-line documentation (chapter **??**).

B_THIELE_INNES_ERROR : Standard error of Thiele-Innes element B for the first orbit (float, Angle[mas])

Standard error of the Thiele-Innes element B of the orbit of the photocentre (for the orbital model) for the first orbit. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

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F_THIELE_INNES : Thiele-Innes element F for the first orbit (double, Angle[mas])

The Thiele-Innes element F of the orbit of the photocentre (for the orbital model) for the first orbit. This parameter is fitted for orbital models based on astrometry (`solution_type` = ExtrasolarPlanet). For other orbital models based on photometry or radial velocities, or their combination and published in the `nss_two_body_orbit` table, the Campbell parameters are fitted instead. See more details, including the transformation from the A,B,F,G Thiele-Innes elements to the a_0, i, ω and Ω Campbell elements in the on-line documentation (chapter **??**).

F_THIELE_INNES_ERROR : Standard error of Thiele-Innes element F for the first orbit (float, Angle[mas])

Standard error of the Thiele-Innes element F of the orbit of the photocentre (for the orbital model) for the first orbit. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

G_THIELE_INNES : Thiele-Innes element G for the first orbit (double, Angle[mas])

The Thiele-Innes element G of the orbit of the photocentre (for the orbital model) for the first orbit. This parameter is fitted for orbital models based on astrometry (`solution_type` = ExtrasolarPlanet). For other orbital models based on photometry or radial velocities, or their combination and published in the `nss_two_body_orbit` table, the Campbell parameters are fitted instead. See more details, including the transformation from the A,B,F,G Thiele-Innes elements to the a_0, i, ω and Ω Campbell elements in the on-line documentation (chapter **??**).

G_THIELE_INNES_ERROR : Standard error of Thiele-Innes element G for the first orbit (float, Angle[mas])

Standard error of the Thiele-Innes element G of the orbit of the photocentre (for the orbital model) for the first orbit. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

PERIOD : Orbital period for the first orbit. (double, Time[day])

Period of the orbit of the photocentre around the barycentre for the first orbit.

PERIOD_ERROR : Standard error of Orbital period for the first orbit. (float, Time[day])

Standard error of the period of the orbit of the photocentre around the barycentre for the first orbit. The standard errors are derived from the variance-covariance matrix of the solution, cor-

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recting the measurement uncertainties to obtain the goodness-of-fit F2=0.

T_PERIASTRON : Epoch of periastron passage for the first orbit. (double, Time[Julian Date (day)])

Epoch at periastron relative to `gaia_source.ref_epoch`, in the range [`period`/2,`period`/2[, for the first orbit.

T_PERIASTRON_ERROR : Standard error of Epoch of periastron passage for the first orbit. (float, Time[day])

Standard error of the periastron epoch `t_periastron` for the first orbit. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

ARG_PERIASTRON : Argument of periastron (double, Angle[deg])

ARG_PERIASTRON_ERROR : Standard error of Argument of periastron (float, Angle[deg])

ECCENTRICITY : Orbital eccentricity for the first orbit. (double)

Eccentricity of the first orbit. The eccentricity is dimensionless.

ECCENTRICITY_ERROR : Standard error of Orbital eccentricity for the first orbit. (float)

Standard error of the eccentricity of the first orbit. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0. The eccentricity is dimensionless.

INCLINATION : Orbital inclination for the first orbit (double, Angle[deg])

Inclination of the orbital plane of the inner system with respect to the sky.

INCLINATION_ERROR : Standard error of Orbital inclination for the first orbit (float, Angle[deg])

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Standard error on the orbital inclination.

SEMI_MAJOR_AXIS : Angular semi-major axis of the photocentre for the first orbit (double, Angle[mas])

Angular semi-major axis of the photocentre.

SEMI_MAJOR_AXIS_ERROR : Standard error of Angular semi-major axis of the photocentre for the first orbit (float, Angle[mas])

Standard error of Angular semi-major axis of the photocentre.

POS_ASCENDING_NODE : Position angle of the ascending node for the first orbit (double, Angle[deg])

The position angle of the line of intersection between the tangential plane of projection and the true orbital plane. The node where the orbital motion is directed away from the sun is called the ascending node and is identified only when radial velocity is known.

POS_ASCENDING_NODE_ERROR : Standard error of the ascending node for the first orbit (float, Angle[deg])

Standard error of Angular semi-major axis of the photocentre.

SEMI_AMPLITUDE_PRIMARY : The semiamplitude of the radial velocity curve for first component of the inner system (double, Velocity[km s^{-1}])

The semiamplitude of the radial velocity curve for first component of the inner system.

SEMI_AMPLITUDE_PRIMARY_ERROR : Standard error of the semiamplitude of the radial velocity curve for first component of the inner system (float, Velocity[km s^{-1}])

Standard error of the semiamplitude of the radial velocity curve for first component of the inner system.

SEMI_AMPLITUDE_SECONDARY : The semiamplitude of the radial velocity curve for second

component of the inner system (double, Velocity[km s^{-1}])

The semiamplitude of the radial velocity curve for second component of the inner system.

SEMI_AMPLITUDE_SECONDARY_ERROR : Standard error of the semiamplitude of the radial velocity curve for second component of the inner system (float, Velocity[km s^{-1}])

Standard error of the semiamplitude of the radial velocity curve for second component of the inner system.

TUWE : Transit unit-weight error (float)

This is the factor to be applied on CCD Epoch Astrometric Errors to account for correlations between CCDs or any other discrepancy at transit level.

TUWE_UNCERTAINTY_RESCALED : Have the transit-binned uncertainties been multiplied by the TUWE? (boolean)

A_THIELE_INNES2 : Thiele-Innes element A for the second orbit. (double, Angle[mas])

The Thiele-Innes element A of the orbit of the photocentre (for the orbital model) for the second orbit.

A_THIELE_INNES2_ERROR : Standard error of Thiele-Innes element A for the second orbit. (float, Angle[mas])

Standard error of the Thiele-Innes element A of the orbit of the photocentre (for the orbital model) for the second orbit. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

B_THIELE_INNES2 : Thiele-Innes element B for the second orbit. (double, Angle[mas])

The Thiele-Innes element B of the orbit of the photocentre (for the orbital model).

B_THIELE_INNES2_ERROR : Standard error of Thiele-Innes element B for the second orbit. (float, Angle[mas])

Standard error of the Thiele-Innes element B of the orbit of the photocentre (for the orbital

model) for the second orbit. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

F_THIELE_INNES2 : Thiele-Innes element F for the second orbit. (double, Angle[mas])

The Thiele-Innes element F of the orbit of the photocentre (for the orbital model) for the first orbit.

F_THIELE_INNES2_ERROR : Standard error of Thiele-Innes element F for the second orbit. (float, Angle[mas])

Standard error of the Thiele-Innes element F of the orbit of the photocentre (for the orbital model) for the first orbit. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

G_THIELE_INNES2 : Thiele-Innes element G for the second orbit. (double, Angle[mas])

The Thiele-Innes element G of the orbit of the photocentre (for the orbital model) for the first orbit.

G_THIELE_INNES2_ERROR : Standard error of Thiele-Innes element G for the second orbit. (float, Angle[mas])

Standard error of the Thiele-Innes element G of the orbit of the photocentre (for the orbital model) for the second orbit. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

PERIOD2 : Orbital period for the second orbit. (double, Time[day])

Period of the orbit of the photocentre around the barycentre for the second orbit.

PERIOD2_ERROR : Standard error of Orbital period for the second orbit. (float, Time[day])

Standard error of the period of the orbit of the photocentre around the barycentre for the second orbit. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

T_PERIASTRON2 : Epoch of periastron passage for the second orbit. (double, Time[Julian Date

(day)])

Epoch at periastron relative to `gaia_source.ref_epoch`, in the range [`period`/2,`period`/2[, for the second orbit.

T_PERIASTRON2_ERROR : Standard error of Epoch of periastron passage for the second orbit. (float, Time[day])

Standard error of the periastron epoch `t_periastron` for the second orbit. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

ARG_PERIASTRON2 : Argument of periastron of the outer system (double, Velocity[km s^{-1}])

ARG_PERIASTRON2_ERROR : Standard error of Argument of periastron of the outer system (float, Velocity[km s^{-1}])

ECCENTRICITY2 : Orbital eccentricity for the second orbit. (double)

Eccentricity of the second orbit. The eccentricity is dimensionless.

ECCENTRICITY2_ERROR : Standard error of Orbital eccentricity for the first orbit. (float)

Standard error of the eccentricity of the second orbit. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0. The eccentricity is dimensionless.

INCLINATION2 : Orbital inclination for the second orbit (double, Angle[deg])

Inclination of the orbital plane of the outer system with respect to the sky.

INCLINATION2_ERROR : Standard error of Orbital inclination or the second orbit (float, Angle[deg])

Standard error on the orbital inclination.

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SEMI_MAJOR_AXIS2 : Angular semi-major axis of the photocentre for the second orbit (double, Angle[mas])

Angular semi-major axis of the photocentre.

SEMI_MAJOR_AXIS2_ERROR : Standard error of Angular semi-major axis of the photocentre for the second orbit (float, Angle[mas])

Standard error of Angular semi-major axis of the photocentre.

POS_ASCENDING_NODE2 : Position angle of the ascending node for the second orbit (double, Angle[deg])

The position angle of the line of intersection between the tangential plane of projection and the true orbital plane. The node where the orbital motion is directed away from the sun is called the ascending node and is identified only when radial velocity is known.

POS_ASCENDING_NODE2_ERROR : Standard error of the ascending node for the second orbit (float, Angle[deg])

Standard error of Angular semi-major axis of the photocentre.

SEMI_AMPLITUDE_PRIMARY2 : The semiamplitude of the radial velocity curve of the outer system (double, Velocity[km s^{-1}])

The semiamplitude of the radial velocity curve of the outer system.

SEMI_AMPLITUDE_PRIMARY2_ERROR : Standard error of the radial velocity curve of the outer system (float, Velocity[km s^{-1}])

Standard error of the radial velocity curve of the outer system.

ACCELERATION_RA : Acceleration in RA (double, Misc[mas yr^{-2}])

Acceleration along the RA axis. This is the time derivative of `pmra`.

ACCELERATION_RA_ERROR : Standard error of Acceleration in RA (float, Misc[mas yr^{-2}])

Standard error of the acceleration along the RA axis. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

ACCELERATION_DEC : Acceleration in DEC (double, Misc[mas yr^{-2}])

Acceleration along the declination axis. This is the time derivative of `pmdec`.

ACCELERATION_DEC_ERROR : Standard error of Acceleration in DEC (float, Misc[mas yr^{-2}])

Standard error of the acceleration along the declination axis. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

DERIVATIVE_ACCELERATION_RA : Time derivative of the accel. in RA (double, Misc[mas yr^{-3}])

Time derivative of the acceleration along the RA axis, `acceleration_ra`.

DERIVATIVE_ACCELERATION_RA_ERROR : Standard error of Time derivative of the acceleration in RA (float, Misc[mas yr^{-3}])

Standard error of the time derivative of the acceleration along the RA axis. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

DERIVATIVE_ACCELERATION_DEC : Time derivative of the accel. in DEC (double, Misc[mas yr^{-3}])

Time derivative of the acceleration along the declination axis, `acceleration_dec`.

DERIVATIVE_ACCELERATION_DEC_ERROR : Standard error of Time derivative of the acceleration in DEC (float, Misc[mas yr^{-3}])

Standard error of the time derivative of the acceleration along the declination axis. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

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CENTER_OF_MASS_VELOCITY : The velocity of the centre of mass (double, Velocity[km s^{-1}])

The velocity of the centre of mass.

CENTER_OF_MASS_VELOCITY_ERROR : Standard error of the velocity of the centre of mass (float, Velocity[km s^{-1}])

Standard error of the velocity of the centre of mass.

FIRST_DERIVATIVE_VELOCITY : First order derivative of the velocity (double, Acceleration[km s^{-1} day^{-1}])

The coefficient of the first degree term in the fitting polynomial (for solutions FirstDegreeTrendSB1, FirstDegreeTrendSB2, SecondDegreeTrendSB1, SecondDegreeTrendSB2, ThirdDegreeTrendSB1, ThirdDegreeTrendSB2, FourthDegreeTrendSB1, FourthDegreeTrendSB2).

FIRST_DERIVATIVE_VELOCITY_ERROR : Standard error of First order derivative of the velocity (float, Acceleration[km s^{-1} day^{-1}])

Standard error of the first acceleration. The standard errors are derived from the variance-covariance matrix of the final solution in the standard way.

SECOND_DERIVATIVE_VELOCITY : Second order derivative of the velocity (double, Acceleration[km s^{-1} day^{-2}])

The coefficient of the second degree term in the fitting polynomial (SecondDegreeTrendSB1, SecondDegreeTrendSB2, ThirdDegreeTrendSB1, ThirdDegreeTrendSB2, FourthDegreeTrendSB1, FourthDegreeTrendSB2).

SECOND_DERIVATIVE_VELOCITY_ERROR : Standard error of Second order derivative of the velocity (float, Acceleration[km s^{-1} day^{-2}])

Standard error of the second acceleration. The standard errors are derived from the variance-covariance matrix of the final solution in the standard way.

THIRD_DERIVATIVE_VELOCITY : Third order derivative of the velocity (double, Acceleration[km s^{-1} day^{-3}])

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The coefficient of the third degree term in the fitting polynomial (for solutions ThirdDegreeTrendSB1, ThirdDegreeTrendSB2, FourthDegreeTrendSB1, FourthDegreeTrendSB2).

THIRD_DERIVATIVE_VELOCITY_ERROR : Standard error of Third order derivative of the velocity (float, Acceleration[km s^{-1} day^{-3}])

Standard error of the third acceleration. The standard errors are derived from the variance-covariance matrix of the final solution in the standard way.

FOURTH_DERIVATIVE_VELOCITY : Fourth order derivative of the velocity (double, Acceleration[km s^{-1} day^{-4}])

The coefficient of the fourth degree term in the fitting polynomial (for solutions FourthDegreeTrendSB1, FourthDegreeTrendSB2).

FOURTH_DERIVATIVE_VELOCITY_ERROR : Standard error of Fourth order derivative of the velocity (float, Acceleration[km s^{-1} day^{-4}])

Standard error of the fourth acceleration as defined above. The standard errors are derived from the variance-covariance matrix of the final solution in the standard way.

VELOCITY_SCAN_ANGLE_SINE1 : Fitted sine value of the radial velocity scan angle model (double, Velocity[km s^{-1}])

Fitted sine value of the radial velocity scan angle model, sin(scan angle).

VELOCITY_SCAN_ANGLE_SINE1_ERROR : Standard error of fitted sine value of the RV scan angle model (float, Velocity[km s^{-1}])

Standard error of `velocity_scan_angle_sine1`.

VELOCITY_SCAN_ANGLE_COSINE1 : Fitted cosine value of the radial velocity scan angle model (double, Velocity[km s^{-1}])

Fitted cosine value of the radial velocity scan angle model, cos(scan angle).

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VELOCITY_SCAN_ANGLE_COSINE1_ERROR : Standard error of fitted cosine value of the RV scan angle model (float, Velocity[km s^{-1}])

Standard error of `velocity_scan_angle_cosine1`.

VELOCITY_SCAN_ANGLE_SINE2 : Fitted sine value of the radial velocity double scan angle model (double, Velocity[km s^{-1}])

Fitted sine value of the radial velocity double scan angle model, sin(2 * scan angle).

VELOCITY_SCAN_ANGLE_SINE2_ERROR : Standard error of fitted sine value of the RV double scan angle model (float, Velocity[km s^{-1}])

Standard error of `velocity_scan_angle_sine2`.

VELOCITY_SCAN_ANGLE_COSINE2 : Fitted cosine value of the radial velocity double scan angle model (double, Velocity[km s^{-1}])

Fitted cosine value of the radial velocity double scan angle model, cos(2 * scan angle).

VELOCITY_SCAN_ANGLE_COSINE2_ERROR : Standard error of fitted cosine value of the RV double scan angle model (float, Velocity[km s^{-1}])

Standard error of `velocity_scan_angle_cosine2`.

DEGREE_OF_TREND : the degree of the trend (int)

The polynomial degree of the fitted trend in the models TrendSB1, TrendSB2, Pulsation_TrendSB1, SB1_TrendSB1.

LARGEST_GAP_IN_PHASE : the width of the largest gap in the phase (double)

When the curve is folded over the period, this is the largest gap of the phase.

CONF_SPECTRO_PERIOD : the confidence of the spectro period (double[2] array)

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The confidence of the periods in SB1, Pulsation_SB1, Trend_SB1, SB1_SB1, SB2.

F_STATISTICS : An F-statistic between the solution chi-square and the variance of Vrads (float)

This is an F-statistic between the solution chi–square and the variance, which, when combined with the solution F2, will tell us about incomplete models with good fits but with bad F2.

MASS_FUNCTION : The mass function of the SB1 solution (float[2] array, Mass[Solar Mass])

The minimum mass of the hidden companion in SB1 systems.

SB2_ORTH_CORR_COEFF : The orthogonal correlation coefficient V2 vs V1 (float)

The orthogonal correlation coefficient between RV1 and RV2 is filled only for isSB2=true; otherwise, it is NaN. This coefficient serves as an indication of the quality of the isSB2 detection. In a SB2 system, RV1 and RV2 are expected to be anticorrelated, so a value close to -1 indicates good quality.

N_SUCESSIVE_EMPTY_BINS : Split Vrads into bins and count the number of empty bins (Integer)

In order to detect radial velocity curves with 2 parallel sets of Vrads, we split the radial velocities into 30 bins, and we count the highest number of successive empty bins.

N_EDGE_OBSERVATIONS : The number of observations at the edge (Integer)

After splitting Vrads into bins, we count the number of observations at each side of the largest gap and we keep the minimum of the two numbers.

N_SIGNIFICANT_PERIODS : The number of significant peaks in the peridogram (int[2] array)

The number of periodogram peaks above the 95

HIGHEST_PEAKS_RATIO : The ratio of two highest significant peaks in the peridogram (float[2] array)

The ratio of two highest significant peaks in the peridogram. For SB1_SB1 solutions, we will compute two periodograms. The value is NaN if less than 2 significant peaks are present.

BOOTSTRAP_PROBABILITY : Probability of the first period being robust to a bootstrap procedure (float)

Probability of the first period being robust to a bootstrap procedure.

It is estimated as follows. First, the time series of N measurements with a significant period P is shortened by 1 measurement N times. Second, a test determining wether the derived period P is still significant is applied to the shortened time series. Third, the number of successes (i.e. period is still significant) is divided by N to get the bootstrap probability. The number of periods to bootstrap (one, two or none) depends on the `solution_type`.

BOOTSTRAP_PROBABILITY2 : Probability of the second period being robust to a bootstrap procedure (float)

Probability of the second period being robust to a bootstrap procedure.

It is estimated as follows. First, the time series of N measurements with a significant period P is shortened by 1 measurement N times. Second, a test determining wether the derived period P is still significant is applied to the shortened time series. Third, the number of successes (i.e. period is still significant) is divided by N to get the bootstrap probability. The number of periods to bootstrap (one, two or none) depends on the `solution_type`.

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10.5 NSS_MULTIPLICITY [TAP]

In DR4 processing of resolved pairs, a given source may be involved in more than one pair (with a maximum 20 in a radius of 4 arcsec) in crowded fields (e.g. the Galactic bulge, Magellanic Clouds, open and globular clusters etc). This table list all pairs involving a given source, information that is of interest in the context of stellar multiplicity studies for instance. This table also draws the attention of the user to the existence of different sets of astrometric parameters for a given source whose consistency will not be checked within the framework of DR4. The table described here may thus be used to help the users to check the consistency between astrometric solutions of triplets like AB, BC and AC. Consistency checks for higher multiplicity may be, however, more complex to perform.

Columns description:

SOLUTION_ID : Solution identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Source Identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

NUMBER_OF_PAIRS : Number of pairs (int)

Number of source identifiers involved in a pair with the `source_id` of interest.

RELATED_SOURCES : Identifiers of related sources (long[`number_of_pairs`] array)

This field contains an array listing all source identifiers with which the `source_id` forms pairs.

10.6 NSS_NON_LINEAR_SPECTRO [TAP]

This table contains non-single-star orbital models for spectroscopic binaries compatible with a trend.

Several possible models are hosted within the same table and they are indicated by the field `solution_type`. The description of this latter lists all possible solution types considered for this release. Only a selection of parameters hosted in this table are provided here, depending on the solution. The details of those is given in the description of field `bit_index`, which can also be used to extract the relevant elements of the correlation vector `correlations`.

Details about the formalism used to derive the parameters in this table are given in the on-line documentation, see Chapter **??**.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Source Identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

SOLUTION_TYPE : NSS model adopted (string)

This is the non-single star model which has been adopted for the published solution, see online documentation, Chapter **??**, for details.

The solution types covered in table `nss_non_linear_spectro` are:

- TrendSB1: Single Lined trend
- TrendSB1ScanAngleK1: Single Lined trend plus first harmonics of scan angle model

- TrendSB1ScanAngleK12: Single Lined trend plus two first harmonics of scan angle model
- TrendSB2: Double Lined trend

SUBTYPE : NSS model sub-type. Bit index as text (string)

Human readable version of the bit index field.

A text description is provided for every `solution_type` and `bit_index` combination. It can be used in ADQL queries. In this particular table, it is composed for all `solution_type` as follows: AccelerationParamsX, where X is the number of acceleration parameters spanning range 1-4.

BIT_INDEX : Boolean mask for the fields in the correlation matrix (long)

`bit_index` is a boolean mask indicating which variables or parameters have been fitted for a given non-single-star solution in a table row. The parameters used for each `solution_type` are different. The correspondence between `solution_type`, `bit_index` and fitted variables is provided below.

The M parameters for a given `solution_type` always follow the same order. `bit_index` contains N+1 bits, where the leading Most Significant Bit (MSB) is always 1, and the other N bits correspond to parameters of a given model. The last variable is related to the trailing Least Significant Bit (LSB). A bit with value 1 (0) means the value of its associated parameter has been fitted (kept fixed) in a given solution. Note N can be smaller than M. That is, some solutions do not consider all M possible parameters, but a subset of size N. This is the case for all `solution_type` in `nss_non_linear_spectro`.

The covariance matrix has N×N size. The `bit_index` length and the variable names below can be used to identify elements in the `correlations` vector. When a given parameter has not been fitted (bit in `bit_index` = 0), the corresponding correlation is NaN.

Detailed explanations for each `solution_type` and `bit_index` are provided below. T or F means a parameter has or has not been fit. Hyphens '-' means a variable was not considered in the solution (there are M - N of them). Table field names have been used to identify variables when possible. A text description is provided when not in the data model.

`solution_type` = TrendSB1

bit_index	subtype	mean_velocity	first_derivative_velocity	second_derivative_velocity	third_derivative_velocity	fourth_derivative_velocity
7	AccelerationTerms1	T	T	-	-	-
15	AccelerationTerms2	T	T	T	-	-
31	AccelerationTerms3	T	T	T	T	-
63	AccelerationTerms4	T	T	T	T	T

solution_type = TrendSB1ScanAngleK1

bit_index	subtype	mean_velocity	first_derivative_velocity	second_derivative_velocity	third_derivative_velocity	fourth_derivative_velocity	velocity_scan_angle_cosine1	velocity_scan_angle_sine1
31	AccelerationTerms1	T	T	-	-	-	T	T
63	AccelerationTerms2	T	T	T	-	-	T	T
127	AccelerationTerms3	T	T	T	T	-	T	T
255	AccelerationTerms4	T	T	T	T	T	T	T

solution_type = TrendSB1ScanAngleK12

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bit_index	subtype	mean_velocity	first_derivative_velocity	second_derivative_velocity	third_derivative_velocity	fourth_derivative_velocity	velocity_scan_angle_cosine1	velocity_scan_angle_sine1	velocity_scan_angle_cosine2	velocity_scan_angle_sine2
127	AccelerationTerms1	T	T	-	-	-	T	T	T	T
255	AccelerationTerms2	T	T	T	-	-	T	T	T	T
511	AccelerationTerms3	T	T	T	T	-	T	T	T	T
1023	AccelerationTerms4	T	T	T	T	T	T	T	T	T

solution_type = TrendSB2

bit_index	subtype	mean_velocity	first_derivative_velocity	second_derivative_velocity	third_derivative_velocity	fourth_derivative_velocity	Internal fixed value parameter	Internal fixed value parameter	Internal fixed value parameter	Internal fixed value parameter
112	AccelerationTerms1	T	T	-	-	-	F	F	F	F
240	AccelerationTerms2	T	T	T	-	-	F	F	F	F
496	AccelerationTerms3	T	T	T	T	-	F	F	F	F
1008	AccelerationTerms3	T	T	T	T	T	F	F	F	F

CORRELATIONS : Vector form of the upper triangle of the correlation matrix (float[] array)

Correlation matrix of the fitted profile parameters for the applicable non-single star solution. The parameters stored in this matrix and their order is given in the description of field bit_index. Since not all parameters of a given solution model are systematically fitted, the matrix may contain NaN values at the elements of the corresponding indices, such as the length of the vector always corresponds to the expected length of the theoretical model.

Only non-zero, non-unity, correlation coefficients from the correlation matrix **M** are provided here. They are served as a linear array of constant size $S = n(n-1)/2$ corresponding to the full normal matrix of dimension $n \times n$. The ordering of the elements in the linear array follows a column-major storage scheme, i.e.:

$$\mathbf{M} = \begin{bmatrix} 1 & C[0] & C[1] & C[3] & C[6] & \cdots & C[S-(n-1)] \\ & 1 & C[2] & C[4] & C[7] & \cdots & C[S-(n-2)] \\ & & 1 & C[5] & C[8] & \cdots & C[S-(n-3)] \\ & & & 1 & C[9] & \cdots & C[S-(n-4)] \\ & & & & \ddots & \ddots & \vdots \\ & & & & & 1 & C[S-1] \\ & & & & & & 1 \end{bmatrix}$$

OBJ_FUNC_VALUE : Value of the objective function at the solution (float)

The χ^2, defined as $\sum_{i=1}^{n}(x_i - y_i)^2/\sigma_i^2$, where x_i is the abscissa of the ith transit, y_i is the value calculated from the model, and σ_i is the uncertainty obtained from the astrometric reduction, without any correction.

GOF : Goodness of fit in the Hipparcos sense (float)

Goodness-of-fit statistic of the solution. This is the 'gaussianized chi-square' (Wilson & Hilferty (1931)'s cube root transformation), which for good fits should approximately follow a normal distribution with zero mean value and unit standard deviation.

This statistic is computed according to the formula:

$$F2 = \sqrt{\frac{9\nu}{2}}\left(\sqrt[3]{\frac{\texttt{obj_func_value}}{\nu}} + \frac{2}{9\nu} - 1\right)$$

where `obj_func_value` is hopefully a χ^2 and ν is the number of degrees of freedom.

EFFICIENCY : Significance of the solution (i.e. how worth keeping a model is) (float)

The efficiency expresses the level of correlation between the parameters of a model. A value of 1 means a total absence of correlation whereas it falls to 0 as the correlation increases. It is defined as the n-th root of the ratio of the product of the diagonal elements of the covariance matrix and the product of the eigen value of that matrix. When all the covariances are 0, the matrix is diagonal and the ratio is exactly 1.

See also Eichhorn (1989).

FLAGS : Quality flags for the achieved NSS solution (long)

Processing flag applicable to specific non-single-star solutions. The meaning of each of those is

given in the table below.

Flags bit number	Flag Meaning	Comment
Spectroscopic binary solutions		
8	BAD_UNCHECKED_NUMBER_OF_TRANSITS	The length of Transit is not sufficient to process the star (before removing bad transits)
9	BAD_CHECKED_NUMBER_OF_TRANSITS	The length of Transit is not sufficient to process the star (after removing bad transits)
10	NO_SIGNIFICANT_PERIODS_CAN_BE_FOUND	No significant period can be derived (no period from photometric variability analysis + peridogram peaks below the significance threshold)
11	NO_SIGNIFICANT_PULSATION_PERIOD	The pulsation periods are not significant)
12	REFINED_SOLUTION_DOES_NOT_CONVERGE	The refined orbital solution does not converge (after 1000 iterations)
13	REFINED_SOLUTION_COVARIANCE_MATRIX	The variance covariance matrix is not definite positive in the refined solution
14	LUCY_TEST_APPLIED	The Lucy test has been applied
15	CIRCULAR_SOLUTION_DOES_NOT_CONVERGE	The Lucy refined orbital solution diverges (after 1000 iterations)
16	CIRCULAR_SOLUTION_NOT_DEFINITE_POSITIVE_COVARIANCE_MATRIX	The variance covariance matrix is not definite positive in the circular solution
17	TREND_SOLUTION_NOT_DEFINITE_POSITIVE_COVARIANCE_MATRIX	The variance covariance matrix is not definite positive in the trend solution
18	REFINED_PULSATION_DOES_NOT_CONVERGE	The refined eccentric pulsation solution does not converge (after 1000 iterations)
19	REFINED_PULSATION_NOT_DEFINITE_POSITIVE_COVARIANCE_MATRIX	The variance covariance matrix is not definite positive in the trend solution
20	CIRCULAR_PULSATION_DOES_NOT_CONVERGE	The circular pulsation solution does not converge (after 1000 iterations)
21	CIRCULAR_PULSATION_NOT_DEFINITE_POSITIVE_COVARIANCE_MATRIX	The variance covariance matrix is not definite positive in the circular pulsation solution
22	SOLUTION_OUTSIDE_E_LOGP_ENVELOP	The orbital solution (SB1 or SB2) is outside the e-log(P) envelop.
23	MODEL_AND_ITS_RESIDUAL_ARE_BOTH_TRENDS	Both the model and its residual are trends
24	COMPOSITE_SOLUTION_DOES_NOT_CONVERGE	The composite solution does not converge (after 1000 iterations)
25	COMPOSITE_SOLUTION_NOT_DEFINITE_POSITIVE_COVARIANCE_MATRIX	The variance covariance matrix is not definite positive in the composite solution
26	SOME_TRANSITS_ARE_DOUBLE_LINED	Some transits are double lined in the single lined mode
27	SIGNIFICANT_SCAN_ANGLE	Scan angle signal is significant
28	SCAN_ANGLE_SOLUTION_NOT_DEFINITE_POSITIVE_COVARIANCE_MATRIX	The variance covariance matrix is not definite positive in the scan angle solution
29	SIGNIFICANT_SCAN_ANGLE_HARMONIC	Scan angle 1st harmonic is significant
30	PERIOD_SIGNIFICANT_IN_PHOTOMETRY_ALSO	The derived period is significant when applied to photometric data.
31	NA	

RV_N_OBS_PRIMARY : Total number of radial velocities considered for the primary (int)

Total number of radial velocities considered for the primary.

RV_N_GOOD_OBS_PRIMARY : Total number of radial velocities actually used for the primary (int)

Total number of radial velocities actually used for the primary.

MEAN_VELOCITY : Mean velocity (double, Velocity[km s^{-1}])

The mean velocity as calculated over the current mission.

MEAN_VELOCITY_ERROR : Standard error of Mean velocity (float, Velocity[km s^{-1}])

Standard error of `mean_velocity`. The standard errors are derived from the variance-covariance matrix of the final solution in the standard way.

FIRST_DERIVATIVE_VELOCITY : First order derivative of the velocity (double, Acceleration[km s^{-1} day^{-1}])

The coefficient of the first degree term in the fitting polynomial (for solutions FirstDegreeTrendSB1, FirstDegreeTrendSB2, SecondDegreeTrendSB1, SecondDegreeTrendSB2, ThirdDegreeTrendSB1, ThirdDegreeTrendSB2, FourthDegreeTrendSB1, FourthDegreeTrendSB2).

FIRST_DERIVATIVE_VELOCITY_ERROR : Standard error of First order derivative of the velocity (float, Acceleration[km s^{-1} day^{-1}])

Standard error of the first acceleration. The standard errors are derived from the variance-covariance matrix of the final solution in the standard way.

SECOND_DERIVATIVE_VELOCITY : Second order derivative of the velocity (double, Acceleration[km s^{-1} day^{-2}])

The coefficient of the second degree term in the fitting polynomial (SecondDegreeTrendSB1, SecondDegreeTrendSB2, ThirdDegreeTrendSB1, ThirdDegreeTrendSB2, FourthDegreeTrendSB1, FourthDegreeTrendSB2).

SECOND_DERIVATIVE_VELOCITY_ERROR : Standard error of Second order derivative of the velocity (float, Acceleration[km s^{-1} day^{-2}])

Standard error of the second acceleration. The standard errors are derived from the variance-covariance matrix of the final solution in the standard way.

THIRD_DERIVATIVE_VELOCITY : Third order derivative of the velocity (double, Acceleration[km s^{-1} day^{-3}])

The coefficient of the third degree term in the fitting polynomial (for solutions ThirdDegreeTrendSB1, ThirdDegreeTrendSB2, FourthDegreeTrendSB1, FourthDegreeTrendSB2).

THIRD_DERIVATIVE_VELOCITY_ERROR : Standard error of Third order derivative of the velocity (float, Acceleration[km s^{-1} day^{-3}])

Standard error of the third acceleration. The standard errors are derived from the variance-covariance matrix of the final solution in the standard way.

FOURTH_DERIVATIVE_VELOCITY : Fourth order derivative of the velocity (double, Acceleration[km s^{-1} day^{-4}])

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The coefficient of the fourth degree term in the fitting polynomial (for solutions FourthDegreeTrendSB1, FourthDegreeTrendSB2).

FOURTH_DERIVATIVE_VELOCITY_ERROR : Standard error of Fourth order derivative of the velocity (float, Acceleration[km s^{-1} day^{-4}])

Standard error of the fourth acceleration as defined above. The standard errors are derived from the variance-covariance matrix of the final solution in the standard way.

VELOCITY_OFFSET : y Intercept of the fitted line V2 = f(V1) (double, Velocity[km s^{-1}])

This is the y–intercept of the fitted line V2 = f(V1).

VELOCITY_OFFSET_ERROR : Standard error of yIntercept (float, Velocity[km s^{-1}])

Error on the Velocity offset.

VELOCITY_PRIMARY_RMS : RMS of the primary velocity (double, Velocity[km s^{-1}])

RMS of the primary velocities with respect to the fitted polynomial trend.

VELOCITY_SECONDARY_RMS : RMS of the secondary velocity (double, Velocity[km s^{-1}])

RMS of the secondary velocities with respect to the fitted polynomial trend.

ADDITIVE_NOISE_PRIMARY : Additive noise for the primary (double, Velocity[km s^{-1}])

Primary : In order to account for some poorly calibrated effects, a noise is quadratically added to the uncertainty on the observations such that the resulting F2 is null. There is potentially one noise per type of observations.

ADDITIVE_NOISE_SECONDARY : Additive noise for the secondary (double, Velocity[km s^{-1}])

Secondary : In order to account for some poorly calibrated effects, a noise is quadratically added to the uncertainty on the observations such that the resulting F2 is null. There is potentially one noise per type of observations.

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VELOCITY_SCAN_ANGLE_SINE1 : Fitted sine value of the radial velocity scan angle model (double, Velocity[km s^{-1}])

Fitted sine value of the radial velocity scan angle model, sin(scan angle).

VELOCITY_SCAN_ANGLE_SINE1_ERROR : Standard error of fitted sine value of the RV scan angle model (float, Velocity[km s^{-1}])

Standard error of `velocity_scan_angle_sine1`.

VELOCITY_SCAN_ANGLE_COSINE1 : Fitted cosine value of the radial velocity scan angle model (double, Velocity[km s^{-1}])

Fitted cosine value of the radial velocity scan angle model, cos(scan angle).

VELOCITY_SCAN_ANGLE_COSINE1_ERROR : Standard error of fitted cosine value of the RV scan angle model (float, Velocity[km s^{-1}])

Standard error of `velocity_scan_angle_cosine1`.

VELOCITY_SCAN_ANGLE_SINE2 : Fitted sine value of the radial velocity double scan angle model (double, Velocity[km s^{-1}])

Fitted sine value of the radial velocity double scan angle model, sin(2 * scan angle).

VELOCITY_SCAN_ANGLE_SINE2_ERROR : Standard error of fitted sine value of the RV double scan angle model (float, Velocity[km s^{-1}])

Standard error of `velocity_scan_angle_sine2`.

VELOCITY_SCAN_ANGLE_COSINE2 : Fitted cosine value of the radial velocity double scan angle model (double, Velocity[km s^{-1}])

Fitted cosine value of the radial velocity double scan angle model, cos(2 * scan angle).

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VELOCITY_SCAN_ANGLE_COSINE2_ERROR : Standard error of fitted cosine value of the RV double scan angle model (float, Velocity[km s^{-1}])

Standard error of `velocity_scan_angle_cosine2`.

DEGREE_OF_TREND : the degree of the trend (int)

The polynomial degree of the fitted trend in the models TrendSB1, TrendSB2, Pulsation_TrendSB1, SB1_TrendSB1.

LARGEST_GAP_IN_PSEUDO_PHASE : the width of the largest gap in the pseudo phase (time / time baseline) (double)

When the curve is folded over the period, this is the largest gap of the phase.

F_STATISTICS : An F-statistic between the solution chi-square and the variance of Vrads (float)

This is an F-statistic between the solution chi–square and the variance, which, when combined with the solution F2, will tell us about incomplete models with good fits but with bad F2.

SB2_ORTH_CORR_COEFF : The orthogonal correlation coefficient V2 vs V1 (only double lined solutions) (float)

The orthogonal correlation coefficient between RV1 and RV2 is filled only for isSB2=true; otherwise, it is NaN. This coefficient serves as an indication of the quality of the isSB2 detection. In a SB2 system, RV1 and RV2 are expected to be anticorrelated, so a value close to -1 indicates good quality.

N_SUCESSIVE_EMPTY_BINS : Split Vrads into 30 bins and count the number of successive empty bins (Integer)

In order to detect radial velocity curves with 2 parallel sets of Vrads, we split the radial velocities into 30 bins, and we count the highest number of successive empty bins.

N_EDGE_OBSERVATIONS : The number of observations at the edge to be discarded after the splitting into 30 bins (Integer)

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After splitting Vrads into bins, we count the number of observations at each side of the largest gap and we keep the minimum of the two numbers.

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10.7 NSS_RESOLVED_PAIR [TAP]

This table contains non-single-star astrometric models for resolved pairs of sources being physically bound (in contrast with chance alignment pairs).

Several possible models are hosted within the same table and they are indicated by the field `solution_type`. The description of this latter lists all possible solution types considered for this release. Only a selection of parameters hosted in this table are provided here, depending on the solution. The details of those is given in the description of field `solution_type`, which can also be used to extract the relevant elements of the correlation vector `correlations`.

Details about the formalism used to derive the parameters in this table are given in the on-line documentation, see Chapter **??**.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Source Identifier for the primary source (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`). It corresponds here to the source identifier for the first component of the resolved pair.

SOURCE_ID_SECONDARY : Source Identifier for the secondary source (long)

Source identifier for the second component of the resolved pair.

SOLUTION_TYPE : NSS model adopted (string)

This is the non-single star model which has been adopted for the published solution, see online documentation, Chapter **??**, for details.

The solution types covered in table nss_resolved_pair, the corresponding fitted variables and their order in the correlations vector are listed below. All variables are always fitted for all solution types. A bit index field is thus not necessary.

- ResolvedPairFixed: A resolved pair with common proper motion and parallax
  - ra_primary_offset is the offset in right ascension with respect to the ra reference position epoch_astrometry.ra0
    or, equivalently, all_source_astrometry.ra, such that ra_primary = ra_primary_off + epoch_astrometry.ra0 in the absence of any acceleration term (see details in the description of ra_primary)
  - dec_primary_offset is the offset in declination with respect to declination reference position epoch_astrometry.dec0
    or, equivalently, all_source_astrometry.dec, such that dec_primary = dec_primary_offset + epoch_astrometry.dec0 in the absence of any acceleration term (see details in the description of dec_primary)
  - parallax of the pair
  - pmra_primary (same as secondary)
  - pmdec_primary (same as secondary)
  - ra_offset_diff (= secondary offset in RA – ra_primary_offset)
  - dec_offset_diff (= secondary offset in Dec – dec_primary_offset)
- ResolvedPairLinear: A resolved pair with common parallax but different proper motions.
  - ra_primary_offset (same definition as above)
  - dec_primary_offset (same definition as above)
  - parallax
  - pmra_primary
  - pmdec_primary
  - ra_offset_diff (same definition as above)
  - dec_offset_diff (same definition as above)
  - pmra_diff (= RA pm of secondary - pmra_primary)
  - pmdec_diff (= Dec pm of secondary - pmdec_primary)
- ResolvedPairAcceleration: A resolved pair with common parallax but different proper motions and an acceleration term.
  - ra_primary_offset (same definition as above)

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  - dec_primary_offset (same definition as above)
  - parallax
  - pmra_primary
  - pmdec_primary
  - ra_offset_diff (same definition as above)
  - dec_offset_diff (same definition as above)
  - pmra_diff
  - pmdec_diff
  - acceleration_ra_primary
  - acceleration_dec_primary
  - acceleration_ra_diff
  - acceleration_dec_diff

- DiverseAcceleration: A resolved pair with common parallax but different proper motions and a significant acceleration term not associated with the visible-pair motion for at least one of the components. Likely a hierarchical triple or quadruple system.
  - ra_primary_offset (same definition as above)
  - dec_primary_offset (same definition as above)
  - parallax
  - pmra_primary
  - pmdec_primary
  - ra_offset_diff (same definition as above)
  - dec_offset_diff (same definition as above)
  - pmra_diff
  - pmdec_diff
  - acceleration_ra_primary
  - acceleration_dec_primary
  - acceleration_ra_diff
  - acceleration_dec_diff

- ResolvedPairJerk: A resolved pair with common parallax but different proper motions and both an acceleration and a jerk term.
  - ra_primary_offset (same definition as above)
  - dec_primary_offset (same definition as above)

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  - parallax
  - pmra_primary
  - pmdec_primary
  - ra_offset_diff (same definition as above)
  - dec_offset_diff (same definition as above)
  - pmra_diff
  - pmdec_diff
  - acceleration_ra_primary
  - acceleration_dec_primary
  - acceleration_ra_diff
  - acceleration_dec_diff
  - derivative_acceleration_ra_primary
  - derivative_acceleration_dec_primary
  - derivative_acceleration_ra_diff
  - derivative_acceleration_dec_diff

- DiverseJerk: A resolved pair with common parallax but different proper motions and a significant acceleration and/or jerk term not associated with the visible-pair motion for at least one of the components. Likely a hierarchical triple or quadruple system.

  - ra_primary_offset (same definition as above)
  - dec_primary_offset (same definition as above)
  - parallax
  - pmra_primary
  - pmdec_primary
  - ra_offset_diff (same definition as above)
  - dec_offset_diff (same definition as above)
  - pmra_diff
  - pmdec_diff
  - acceleration_ra_primary
  - acceleration_dec_primary
  - acceleration_ra_diff
  - acceleration_dec_diff
  - derivative_acceleration_ra_primary

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- derivative_acceleration_dec_primary
- derivative_acceleration_ra_diff
- derivative_acceleration_dec_diff

CORRELATIONS : Vector form of the upper triangle of the correlation matrix (float[] array)

Correlation matrix of the fitted model parameters for the retained solution. The parameters stored in this matrix and their order are given in the description of field solution_type.

Contrary to the fields obj_func_value and gof_without_astrometric_jitter computed with the uncertainties of the astrometric transits kept to their original values, the matrix corresponds to a solution calculated by correcting the uncertainties of the astrometric transits so that the goodness-of-fit with astrometric_jitter (i.e., gof_with_astrometric_jitter) becomes centred on 0 (for the most complex 17-parameter model ResolvedPairJerk). The same corrected uncertainties are then kept for all models.

Correlation coefficients from the correlation matrix **M** are provided here as a linear array of constant size $S = n(n-1)/2$ corresponding to the full normal matrix of dimension $n \times n$. The ordering of the elements in the linear array follows a column-major storage scheme, i.e.:

$$\mathbf{M} = \begin{bmatrix} 1 & C[0] & C[1] & C[3] & C[6] & \cdots & C[S-(n-1)] \\ & 1 & C[2] & C[4] & C[7] & \cdots & C[S-(n-2)] \\ & & 1 & C[5] & C[8] & \cdots & C[S-(n-3)] \\ & & & 1 & C[9] & \cdots & C[S-(n-4)] \\ & & & & \ddots & \ddots & \vdots \\ & & & & & 1 & C[S-1] \\ & & & & & & 1 \end{bmatrix}$$

OBJ_FUNC_VALUE : Value of the objective function at the solution (float)

The χ^2, defined as $\sum_{i=1}^{n}(x_i - y_i)^2/\sigma_i^2$, where x_i is the abscissa of the ith transit, y_i is the value calculated from the model, and σ_i is the uncertainty obtained from the astrometric reduction, without any correction.

EFFICIENCY : The efficiency of the solution, measuring the total amount of correlation between the fitted parameters (float)

The efficiency expresses the level of correlation between the parameters of a model. A value of 1 means a total absence of correlation whereas it falls to 0 as the correlation increases. It is defined as the n-th root of the ratio of the product of the diagonal elements of the covariance

matrix and the product of the eigen values of that matrix. When all the covariances are 0, the matrix is diagonal and the ratio is exactly 1.

See also Eichhorn (1989).

FLAGS : Quality flags for the achieved NSS solution (long)

Processing flag applicable to specific non-single-star solutions.

For resolved pair models, no specific flag applies. See however `physical_flags` for a physical characterisation of the quality of the solution.

RV_USED : Availability and use of radial velocity (byte)

Reports whether any radial velocity was available and used:

- bit0: whether any radial velocity was available (through CompleteSource or through the transits) or not;
- bit1: whether the available radial velocity was used to correct for μ_r.

ASTROMETRIC_JITTER : Uncorrelated astrometric jitter term (float, Angle[mas])

In order to account for some poorly calibrated effects, this is an excess astrometric noise quadratically added to the uncertainty on the observations such that the resulting gaussianised Goodness of Fit is null.

ASTROMETRIC_N_OBS_AL_PRIMARY : Total astrometric transit observations in AL considered for the primary source (int)

Total astrometric transit observations considered in the along-scan direction.

ASTROMETRIC_N_GOOD_OBS_AL_PRIMARY : Total astrometric transit observations in AL actually used for the primary source (int)

Total astrometric along–scan transit observations that may have been used in the calculation of the solution.

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ASTROMETRIC_N_OBS_AL_SECONDARY : Total astrometric transit observations in AL considered for the secondary source (int)

Total astrometric transit observations considered in the along-scan direction.

ASTROMETRIC_N_GOOD_OBS_AL_SECONDARY : Total astrometric transit observations in AL actually used for the secondary source (int)

Total astrometric along–scan transit observations that may have been used in the calculation of the solution.

RA_PRIMARY : Right ascension of the primary source (double, Angle[deg])

Right ascension α_1 of the primary component in ICRS at the reference epoch `gaia_source.ref_epoch`, including the offset introduced by the acceleration term in the case that the adopted model contains such a term (for ResolvedPairAcceleration or ResolvedPairJerk solutions), i.e.:

$$\alpha_1 = \alpha_{\mathrm{ref},1} + \left(\xi_{0,1} - \frac{1}{2}\dot{\mu}_{\alpha*,1}\, T_A^2\right) / \cos\delta_{\mathrm{ref},1},$$

where $\alpha_{\mathrm{ref},1}$ is the ra reference position `epoch_astrometry.ra0` (or, equivalently, `all_source_astrometry.ra`), $\xi_{0,1}$ stands for `ra_primary_offset` and $\dot{\mu}_{\alpha*,1}$ for `acceleration_ra_primary`.

RA_PRIMARY_ERROR : Standard error of right ascension of the primary source (float, Angle[mas])

Standard error $\sigma_{\alpha*}(=\sigma_\alpha\cos\delta)$ of the right ascension of the primary component in ICRS at the reference epoch `gaia_source.ref_epoch`, taking into account the extra error and covariance introduced (if present) by the acceleration term causing an extra positional offset at the reference epoch. This error is derived from the equation

$$\sigma^2_{\alpha*,1} = \sigma^2_{\xi_0} + \frac{T_A^4}{4}\,\sigma^2_{\dot{\mu}_{\alpha*}} - T_A^2\,\mathrm{cov}(\xi_0, \dot{\mu}_{\alpha*})$$

where σ_{ξ_0} is the `ra_primary_offset_error`, $\sigma_{\dot{\mu}_{\alpha*}}$ is the `acceleration_ra_primary_error`, and $\mathrm{cov}(\xi_0, \dot{\mu}_{\alpha*})$ is the covariance between these quantities derived from the `correlation` matrix of the solution.

The measurement uncertainties were quadratically added with an `astrometric_jitter` derived so as to ensure that the goodness-of-fit with this `astrometric_jitter`

(i.e., gof_with_astrometric_jitter) is centred on 0 (for the most complex ResolvedPairJerk solution).

DEC_PRIMARY : Declination of the primary source (double, Angle[deg])

Declination δ_1 of the primary component in ICRS at the reference epoch gaia_source.ref_epoch, including the offset introduced by the acceleration term in the case that the adopted model contains such a term (for ResolvedPairAcceleration or ResolvedPairJerk solutions), i.e.:

$$\delta_1 = \delta_{\mathrm{ref},1} + \left(\eta_{0,1} - \frac{1}{2}\dot{\mu}_{\delta,1}\ T_A^2\right),$$

where $\delta_{\mathrm{ref},1}$ is the dec reference position epoch_astrometry.dec0
(or, equivalently, all_source_astrometry.dec), $\eta_{0,1}$ stands for dec_primary_offset and $\dot{\mu}_{\delta,1}$ for acceleration_dec_primary.

DEC_PRIMARY_ERROR : Standard error of declination of the primary source (float, Angle[mas])

Standard error σ_δ of the declination of the primary component in ICRS at the reference epoch gaia_source.ref_epoch, taking into account the extra error and covariance introduced (if present) by the acceleration term causing an extra positional offset at the reference epoch. This error is derived from the equation

$$\sigma_{\delta,1}^2 = \sigma_{\eta_0}^2 + \frac{T_A^4}{4}\ \sigma_{\dot{\mu}_\delta}^2 - T_A^2\ \mathrm{cov}(\eta_0, \dot{\mu}_\delta)$$

where σ_{η_0} is the dec_primary_offset_error, $\sigma_{\dot{\mu}_\delta}$ is the acceleration_dec_primary_error, and $\mathrm{cov}(\eta_0, \dot{\mu}_\delta)$ is the covariance between these quantities derived from the correlation matrix of the solution.

The measurement uncertainties were quadratically added with an astrometric_jitter derived so as to ensure that the goodness-of-fit with this astrometric_jitter
(i.e., gof_with_astrometric_jitter) is centred on 0 (for the most complex ResolvedPairJerk solution).

RA_PRIMARY_OFFSET : Offset in right ascension between the primary component and its reference position (float, Angle[mas])

ra_primary_offset corresponds to $\xi_{0,1}$ as defined in the description of ra_primary

RA_PRIMARY_OFFSET_ERROR : Standard error of right ascension offset of the primary source (float, Angle[mas])

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Standard error of the right ascension offset (in $\alpha*$) of the primary component.

DEC_PRIMARY_OFFSET : Offset in declination between the primary component and its reference position (float, Angle[mas])

`dec_primary_offset` corresponds to $\eta_{0,1}$ as defined in the description of `dec_primary`

DEC_PRIMARY_OFFSET_ERROR : Standard error of declination offset of the primary source (float, Angle[mas])

Standard error of the declination offset of the primary component.

PARALLAX : Parallax of the primary source (double, Angle[mas])

Absolute stellar parallax ϖ of the source at the reference epoch `gaia_source.ref_epoch`

PARALLAX_ERROR : Standard error of parallax of the primary source (float, Angle[mas])

Standard error σ_ϖ of the stellar parallax at the reference epoch `gaia_source.ref_epoch`. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

PMRA_PRIMARY : Proper motion in right ascension direction of the primary source (double, Angular Velocity[mas yr^{-1}])

Proper motion in right ascension $\mu_{\alpha*} \equiv \mu_\alpha \cos\delta$ of the source in ICRS. This is the local tangent plane projection of the proper motion vector in the direction of increasing right ascension. In contrast to the situation prevailing for the positions `ra_primary`, `dec_primary` which are given at the reference epoch `gaia_source.ref_epoch` (thus adding the offset due to the acceleration term in case of ResolvedPairAcceleration or ResolvedPairJerk solutions), the proper motion listed here is compensated in order for $\mu_{\alpha*}$ to be an average estimate and not the value at the reference epoch in the case of a ResolvedPairJerk model (see the online documentation for details).

PMRA_PRIMARY_ERROR : Standard error of proper motion in right ascension direction of the primary source (float, Angular Velocity[mas yr^{-1}])

Standard error $\sigma_{\mu\alpha*}$ of the local tangent plane projection of the proper motion vector in the di-

rection of increasing right ascension at the reference epoch `gaia_source.ref_epoch`. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

PMDEC_PRIMARY : Proper motion in declination direction of the primary source (double, Angular Velocity[mas yr^{-1}])

Proper motion in declination μ_δ of the source in ICRS. This is the local tangent plane projection of the proper motion vector in the direction of increasing declination. In contrast to the situation prevailing for the positions `ra_primary`, `dec_primary` which are given at the reference epoch `gaia_source.ref_epoch` (thus adding the offset due to the acceleration term in case of ResolvedPairAcceleration or ResolvedPairJerk solutions), the proper motion listed here is compensated in order for μ_δ to be an average estimate and not the value at the reference epoch in the case of a ResolvedPairJerk model (see the online documentation for details).

PMDEC_PRIMARY_ERROR : Standard error of proper motion in declination direction of the primary source (float, Angular Velocity[mas yr^{-1}])

Standard error $\sigma_{\mu\delta}$ of the proper motion component in declination at the reference epoch `gaia_source.ref_e`
The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

RA_DIFF : Right ascension offset between the secondary and the primary source (float, Angle[deg])

Separation between the secondary and primary source in right ascension, in ICRS at the reference epoch `gaia_source.ref_epoch`.

RA_DIFF_ERROR : Standard error on the right ascension offset between the secondary and the primary source (float, Angle[mas])

Standard error on the separation between the secondary and primary source in right ascension, computed following the same rules as for `ra_primary_error`, i.e., taking into account error propagation and covariances whenever needed.

DEC_DIFF : Declination offset between the secondary and the primary source (float, Angle[deg])

Separation between the secondary and primary source in declination, in ICRS at the reference

epoch `gaia_source.ref_epoch`.

DEC_DIFF_ERROR : Standard error on the declination offset between the secondary and the primary source (float, Angle[mas])

Standard error on the separation between the secondary and primary source in declination, computed following the same rules as for `dec_primary_error`, i.e., taking into account error propagation and covariances whenever needed.

RA_OFFSET_DIFF : Difference in the offsets in right ascension (float, Angle[mas])

Offset of the position in right ascension with respect to the reference position in right ascension. This field provides the difference in these offsets between the secondary and primary components, i.e., `ra_offset_diff` = RA offset of secondary – `ra_primary_offset`.

RA_OFFSET_DIFF_ERROR : Standard error of difference of right ascension offsets beween the secondary and primary sources (float, Angle[mas])

Standard error of difference of right ascension offsets beween the secondary and primary sources.

DEC_OFFSET_DIFF : Difference in the offsets in declination (float, Angle[mas])

Offset of the position in declination with respect to the reference position in declination. This field provides the difference in these offsets between the secondary and primary components, i.e., `dec_offset_diff` = Dec offset of secondary – `dec_primary_offset`.

DEC_OFFSET_DIFF_ERROR : Standard error of difference of declination offsets beween the secondary and primary sources (float, Angle[mas])

Standard error of difference of declination offsets beween the secondary and primary sources.

PMRA_DIFF : Difference in proper motion in right ascension between the secondary and the primary source (float, Angular Velocity[mas yr^{-1}])

Difference in proper motion in right ascension between the secondary and the primary source.

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PMRA_DIFF_ERROR : Standard error in the difference in proper motion in right ascension between the secondary and the primary source (float, Angular Velocity[mas yr^{-1}])

Standard error on the difference in proper motion in right ascension between the secondary and the primary source.

PMDEC_DIFF : Difference in proper motion in declination between the secondary and the primary source (float, Angular Velocity[mas yr^{-1}])

Difference in proper motion in declination between the secondary and the primary source.

PMDEC_DIFF_ERROR : Standard error in the difference in proper motion in declination between the secondary and the primary source (float, Angular Velocity[mas yr^{-1}])

Standard error on the difference in proper motion in declination between the secondary and the primary source.

ACCELERATION_RA_PRIMARY : Acceleration in RA of the primary source (double, Misc[mas yr^{-2}])

Acceleration along the ra axis. This is the time derivative of `pmra`.

ACCELERATION_RA_PRIMARY_ERROR : Standard error of Acceleration in RA of the primary source (float, Misc[mas yr^{-2}])

Standard error of the acceleration along the ra axis. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

ACCELERATION_DEC_PRIMARY : Acceleration in DEC of the primary source (double, Misc[mas yr^{-2}])

Acceleration along the declination axis. This is the time derivative of `pmdec`.

ACCELERATION_DEC_PRIMARY_ERROR : Standard error of Acceleration in DEC of the primary source (float, Misc[mas yr^{-2}])

Standard error of the acceleration along the declination axis. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to

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obtain the goodness-of-fit F2=0.

ACCELERATION_RA_DIFF : Difference in acceleration in right ascension between the secondary and the primary source (float, Misc[mas yr^{-2}])

Difference in acceleration in right ascension between the secondary and the primary source.

ACCELERATION_RA_DIFF_ERROR : Standard error of the difference in acceleration in right ascension (float, Misc[mas yr^{-2}])

Standard error on `acceleration_ra_diff`.

ACCELERATION_DEC_DIFF : Difference in acceleration in declination between the secondary and the primary source (float, Misc[mas yr^{-2}])

Difference in acceleration in declination between the secondary and the primary source.

ACCELERATION_DEC_DIFF_ERROR : Standard error of the difference in acceleration in declination (float, Misc[mas yr^{-2}])

Standard error on `acceleration_dec_diff`.

DERIVATIVE_ACCELERATION_RA_PRIMARY : Time derivative of the accel. in RA of the primary source (double, Misc[mas yr^{-3}])

Time derivative of the acceleration along the RA axis, `derivative_acceleration_ra_primary`.

DERIVATIVE_ACCELERATION_RA_PRIMARY_ERROR : Standard error of Time derivative of the acceleration (jerk) in right ascension of the primary source (float, Misc[mas yr^{-3}])

Standard error of the time derivative of the acceleration along the RA axis. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

DERIVATIVE_ACCELERATION_DEC_PRIMARY : Time derivative of the accel. in DEC of the primary source (double, Misc[mas yr^{-3}])

Time derivative of the acceleration along the declination axis, `derivative_acceleration_dec_primary`.

DERIVATIVE_ACCELERATION_DEC_PRIMARY_ERROR : Standard error of Time derivative of the acceleration (jerk) in declination of the primary source (float, Misc[mas yr^{-3}])

Standard error of the time derivative of the acceleration along the declination axis. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

DERIVATIVE_ACCELERATION_RA_DIFF : Difference in time derivative of the acceleration (jerk) in right ascension between the secondary and the primary source (float, Misc[mas yr^{-3}])

Difference in time derivative of the acceleration (jerk) in right ascension between the secondary and the primary source.

DERIVATIVE_ACCELERATION_RA_DIFF_ERROR : Standard error of the difference in the time derivative of the acceleration in right ascension (float, Misc[mas yr^{-3}])

Standard error on `derivative_acceleration_ra_diff`.

DERIVATIVE_ACCELERATION_DEC_DIFF : Difference in time derivative of the acceleration (jerk) in declination between the secondary and the primary source (float, Misc[mas yr^{-3}])

Difference in time derivative of the acceleration (jerk) in declination between the secondary and the primary source.

DERIVATIVE_ACCELERATION_DEC_DIFF_ERROR : Standard error of the difference in the time derivative of the acceleration in declination (float, Misc[mas yr^{-3}])

Standard error on `derivative_acceleration_dec_diff`.

STANDARD_DEV_RESIDUAL_AL_PRIMARY : Standard deviation of the AL residuals for component 1 (double, Angle[mas])

Standard deviation of the along-scan residuals for the primary source with respect to the adopted model.

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STANDARD_DEV_RESIDUAL_AL_SECONDARY : Standard deviation of the AL residuals for component 2 (double, Angle[mas])

Standard deviation of the along-scan residuals for the secondary source with respect to the adopted model.

RA_BARYCENTRE : Barycentre right ascension at the reference epoch (double, Angle[deg])

Right ascension of the barycentre in the ICRS at the reference epoch `gaia_source.ref_epoch`. It is computed from the right ascensions α of the two sources and `fractional_mass_from_acceleration`, the fractional mass ratio. This field applies only to models ResolvedPairAcceleration and ResolvedPairJerk.

DEC_BARYCENTRE : Barycentre declination at the reference epoch (double, Angle[deg])

Declination of the barycentre in the ICRS at the reference epoch `gaia_source.ref_epoch`. It is computed from the declinations δ of the two sources and `fractional_mass_from_acceleration`, the fractional mass ratio. This field applies only to models ResolvedPairAcceleration and ResolvedPairJerk.

PMRA_BARYCENTRE : Average proper motion of barycentre in right ascension (double, Angular Velocity[mas yr^{-1}])

Proper motion in right ascension (multiplied by cos(declination)) of the barycentre in ICRS. It is computed from `pmra_primary` + `fractional_mass_from_acceleration` $\times$ `pmra_diff`. This field applies only to models ResolvedPairAcceleration and ResolvedPairJerk.

PMDEC_BARYCENTRE : Average proper motion of barycentre in declination (double, Angular Velocity[mas yr^{-1}])

Proper motion in declination of the barycentre in ICRS.

It is computed from It is computed from `pmdec_primary` + `fractional_mass_from_acceleration` $\times$ `pmdec_diff`. This field applies only to models ResolvedPairAcceleration and ResolvedPairJerk.

FRACTIONAL_MASS_FROM_ACCELERATION : Fractional mass derived from the acceleration (double)

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Fractional mass $M_2/(M_1 + M_2)$ derived from the acceleration. This field applies only to models ResolvedPairAcceleration and ResolvedPairJerk.

FRACTIONAL_MASS_FROM_DERIVATIVE_ACCELERATION : Fractional mass derived from the derivative of the acceleration (double)

Fractional mass $M_2/(M_1 + M_2)$ derived from the derivative of the acceleration. This field applies only to model ResolvedPairJerk.

SEPARATION : Angular separation at the reference epoch (float, Angle[mas])

Angular separation (in mas) between the two components at the reference epoch `gaia_source.ref_epoch`, computed as $2\arcsin\left(\sqrt{\sin^2\left(\frac{1}{2}(\delta_2-\delta_1)\right)+\cos\delta_1\cos\delta_2\sin^2\left(\frac{1}{2}(\alpha_2-\alpha_1)\right)}\right)$.

SEPARATION_AU : Linear separation at the reference epoch (double, Length & Distance[AU])

Linear separation between the two components ath the reference epoch.

BIC : BIC of the solution (double)

BIC (Bayesian Information Criterion) of the solution.

NUM_USED_PRIMARY : Number of used (binned CCD) along-scan transits used for the primary source (int)

Number of used (binned CCD) along-scan transits used for the primary source.

NUM_USED_SECONDARY : Number of used (binned CCD) along-scan transits used for the secondary source (int)

Number of used (binned CCD) along-scan transits used for the secondary source.

N_OUTLIERS : Number of outliers of the solution (int)

Number of transit outliers of the solution.

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CHI2_WITH_ASTROMETRIC_JITTER : chi2 with astrometric jitter for the solution (double)

Chi-squared statistic with `astrometric_jitter` for the solution.

CURVATURE_RELATIVE_MOTION : Curvature of the relative motion (double, Angle[mas^{-1}])

Curvature of the relative motion, defined as

$$\chi_{2-1} = \frac{\texttt{pmra_diff} \times \texttt{acceleration_dec_diff} - \texttt{pmdec_diff} \times \texttt{acceleration_ra_diff}}{\left(\texttt{pmra_diff}^2 + \texttt{pmdec_diff}^2\right)^{3/2}}$$

(see the online documentation for details).

CURVATURE_RELATIVE_MOTION_SIGNIFICANCE : Significance of the relative-motion curvature (double)

Significance of the curvature of the relative motion.

GOF_WITHOUT_ASTROMETRIC_JITTER : Goodness of fit without astrometric jitter (double)

Goodness of fit without `astrometric_jitter`.

GOF_WITH_ASTROMETRIC_JITTER : Goodness of fit with astrometric jitter (double)

Goodness of fit with `astrometric_jitter`.

DELTA_BIC_NOT_RETAINED_MODELS : Delta BIC with respect to the other (non-adopted) models (double[] array)

Delta BIC for the non-adopted models, with respect to the adopted model. The models are ordered as follows: Optical, Fixed, Linear, Acceleration, Jerk, OpticalAcceleration. To ease decoding the enumeration, the adopted model is present in the list with a value set to 0.

PHYSICAL_FLAGS : Physical flags characterising the solution (boolean[] array)

Quality indicators

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The contents of the field is model-dependent.

The following list defines all possible conditions:

- STOCHA: Solution is stochastic (`astrometric_jitter` is larger than 0.5 mas).
- EFF: `efficiency` is smaller than 0.05.
- ORBVEL: the orbital velocity for a linear model is consistent with a Keplerian orbit with $M_1 + M_2 \leq 50\ M_\odot$ [Not used].
- ACC1: Significance of primary's acceleration is larger than 3.
- ACC2: Significance of secondary's acceleration is larger than 3.
- EXTRQ: Extreme mass ratio in the relative orbit: `fractional_mass_from_acceleration` is in range [-0.15, 0.15] or [0.85, 1.15].
- CURV: Significance of curvature of the relative orbit is larger than 3 (for ResolvedPairAcceleration model).
- CURV1: Significance of curvature of primary's orbit is larger than 3 (for DiverseAcceleration model).
- CURV2: Significance of curvature of secondary's orbit is larger than 3 (for DiverseAcceleration model).
- EXTM1: If a massive dark companion ($M > 5\ M_\odot$) would be hidden in an astrometric binary involving the primary (for DiverseAcceleration or DiverseJerk models), its mass can be estimated with the formalism described in the online documentation (see Section **??** for details).
- EXTM2: Same as EXTM1, but for the secondary.
- JRK1: Significance of primary's jerk is larger than 3.
- JRK2: Significance of secondary's jerk is larger than 3.

The following list indicates, for the adopted model, which conditions are reported:

- 'ResolvedPairFixed'

Slot 0: STOCHA

Slot 1: EFF

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- ‘ResolvedPairLinear’

Slot 0: STOCHA

Slot 1: EFF

Slot 2: ORBVEL (not valid: always set to false)

- ‘ResolvedPairAcceleration’

Slot 0: STOCHA

Slot 1: EFF

Slot 2: EXTRQ

Slot 3: CURV

- ‘DiverseAcceleration’

Slot 0: STOCHA

Slot 1: EFF

Slot 2: ACC1

Slot 3: ACC2

Slot 4: EXTM1

Slot 5: EXTM2

Slot 6: CURV1

Slot 7: CURV2

- ‘ResolvedPairJerk’

Slot 0: STOCHA

Slot 1: EFF

Slot 2: EXTRQ

Slot 3: CURV

- ‘DiverseJerk’

Slot 0: STOCHA

Slot 1: EFF

Slot 2: ACC1

Slot 3: ACC2

Slot 4: EXTM1

Slot 5: EXTM2

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Slot 6: CURV1
Slot 7: CURV2
Slot 8: JRK1
Slot 9: JRK2

TUWE_PRIMARY : Transit unit-weight error for the primary (double)

Transit unit-weight error (first component).

TUWE_SECONDARY : Transit unit-weight error for the secondary (double)

Transit unit-weight error (second component).

PERSPECTIVE_ACCEL_IN_PRIMARY : Set if the input data of the primary component has been modified to account for the perspective acceleration correction (boolean)

Set if the input data of the primary component has been modified to account for the perspective acceleration correction.

PERSPECTIVE_ACCEL_IN_SECONDARY : Set if the input data of the secondary component has been modified to account for the perspective acceleration correction (boolean)

Set if the input data of the secondary component has been modified to account for the perspective acceleration correction.

PARALLAX_SIGNIFICANCE : Significance of the parallax (double)

Significance of the parallax ϖ/σ_{ϖ}.

TOTAL_MASS_FROM_SEPARATION_AU : Total mass (double, Mass[Solar Mass])

Total system mass estimated from `separation_au` and from the modulus of the relative acceleration, available only in the case of a physical resolved pair with acceleration motion (see the online documentation, Section **??** for details).

PRIMARY_INV_COMPANION_MASS_LOW_LIM : Lower limit on mass of invisible companion to

primary (double, Mass[Solar Mass])

Rough estimate of the mass of a dark companion hidden near the primary component of a pair with a DiverseAcceleration solution (only relevant if this primary component has a significant acceleration; see the online documentation for details).

SECONDARY_INV_COMPANION_MASS_LOW_LIM : Lower limit on mass of invisible companion to secondary (double, Mass[Solar Mass])

Rough estimate of the mass of a dark companion hidden near the secondary component of a pair with a DiverseAcceleration solution (only relevant if the secondary component has a significant acceleration; see the online documentation for details).

ACCELERATIONS_ANGLE : Angle between primary and secondary acceleration vectors at the reference epoch (double, Angle[deg])

Angle between primary and secondary acceleration vectors (only available for models ResolvedPairAcceleration, ResolvedPairJerk): $\arccos\left(\dot{\mu}_1 \cdot \dot{\mu}_2/(|\dot{\mu}_1||\dot{\mu}_2|)\right)$, where $\dot{\mu}_1$ is the primary acceleration vector with components (`acceleration_ra_primary`, `acceleration_dec_primary`) and $\dot{\mu}_2$ is the secondary acceleration vector with components (`acceleration_ra_diff+acceleration_ra_primary`, `acceleration_dec_diff+acceleration_dec_primary`).

POSITION_ANGLE : Position angle of the secondary as seen from the primary at the reference epoch (double, Angle[deg])

Position angle of the secondary as seen from the primary, in the usual convention counted positive from North to East.

RADIUS_VECTOR_ACCELERATION_ANGLE : Angle between separation vector and acceleration vector (of primary) at the reference epoch (double, Angle[deg])

For ResolvedPairAcceleration, ResolvedPairJerk, DiverseAcceleration or DiverseJerk, angle between the vector pointing from the primary to the secondary, and the acceleration vector of the primary: $\arccos\left((\mathbf{r_2}-\mathbf{r_1})\cdot\dot{\mu}_1/(|\mathbf{r_2}-\mathbf{r_1}||\dot{\mu}_1|)\right)$, where $\dot{\mu}_1$ is the primary acceleration vector with components (`acceleration_ra_primary`, `acceleration_dec_primary`) and $\mathbf{r_2}-\mathbf{r_1}$ is the separation vector with components (`ra_diff` $\times$ cos(`dec_barycentre`), `dec_diff`).

RV_PRIMARY_USED_IN_ASTROMETRY : Radial velocity used for perspective acceleration correction (double, Velocity[km s^{-1}])

Radial velocity used for the perspective acceleration correction for the primary.

RV_SECONDARY_USED_IN_ASTROMETRY : Radial velocity used for perspective acceleration correction (double, Velocity[km s^{-1}])

Radial velocity used for the perspective acceleration correction for the secondary.

SIGNIFICANCE_OF_ADDITIONAL_PARAMETERS : significance of parameters added in the present model (double)

Since the goodness of fit is not always enough to decide whether a model is worth keeping or not, the significance addresses some of these limitations.

This field provides the significance (in the framework of a F-test) of the new parameters introduced in the considered model with respect to the most complex model where these parameters are absent. For instance, the significance of the extra parameters `pmra_diff` and `pmdec_diff` added in a `linear` model (9p) [as compared to a `fixed` model (7p) where they are absent] is obtained by comparing their respective χ^2 in the usual way using the Snedecor F distribution:

$$F = \frac{N - p2}{p2 - p1} \frac{\chi^2_{p1} - \chi^2_{p2}}{\chi^2_{p2}},$$

where N is the total number of transits used and $p1 < p2$ the number of parameters of the `simple` and `complex` models, respectively ($p1 = 7$ and $p2 = 9$ in the above example).

This is equivalent to computing the significance s of the parameters p_a and p_d (possibly correlated with correlation coefficient[1] ρ), added in model `complex` as compared to model `simple`, is expressed as

$$s^2 = (p_a \; p_d) \begin{pmatrix} \sigma^2_{p_a} & \rho\sigma_{p_a}\sigma_{p_d} \\ \rho\sigma_{p_a}\sigma_{p_d} & \sigma^2_{p_d} \end{pmatrix}^{-1} \begin{pmatrix} p_a \\ p_d \end{pmatrix},$$

where σ_{p_a} and σ_{p_d} are the standard errors of the p_a and p_d parameters.

The inversion of the above matrix (for a 2×2 matrix - In some cases the inversion of a 4×4 matrix is required, see below) gives the following result:

[1] with the correlation coefficient ρ equal to $\mathrm{cov}(p_a, p_b)/(\sigma_{p_a}\sigma_{p_b})$

$$s = \frac{1}{\sigma_{p_a}\sigma_{p_d}} \left(\frac{p_a^2\sigma_{p_d}^2 + p_d^2\sigma_{p_a}^2 - 2p_a p_d\, \sigma_{p_a}\sigma_{p_d}\, \rho}{1-\rho^2} \right)^{1/2} .$$

Thus when considering a more complex model, a significance is defined globally for the new set of parameters added (as defined by the equation above). The following 5 sequences are considered:

- `linear` (9p) from `fixed` (7p): significance of `pmra_diff`, `pmdec_diff`
- `optical` (10p) from `linear` (9p): significance of parallaxSecondary
- `acceleration` (13p) from `linear` (9p): significance of the four acceleration terms
- `optical acceleration` (14p) from `acceleration` (13p) : significance of parallaxSecondary
- `jerk` (17p) from `acceleration` (13p): significance of the four jerk terms

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10.8 NSS_TWO_BODY_ORBIT [TAP]

This table contains non-single-star orbital models for sources compatible with an orbital two-body solution. This covers astrometric binaries, spectroscopic binaries, eclipsing binaries and certain combinations thereof.

Several possible models are hosted within the same table and they are indicated by the field `solution_type`. The description of this latter lists all possible solution types considered for this release. Only a selection of parameters hosted in this table are provided here, depending on the solution. The details of those is given in the description of field `bit_index`, which can also be used to extract the relevant elements of the correlation vector `correlations`.

Details concerning the formalism used to derive the parameters in this table are given in the on-line documentation in Chapter **??**.

N.B.: a source may receive independent astrometric, spectroscopic or photometric orbits, so a query using a given `source_id` may return several solutions. This should be accounted for when excuting a join query by `source_id`.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Source Identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

SOLUTION_TYPE : NSS model adopted (string)

This is the non-single star model which has been adopted for the published solution, see online documentation, Chapter **??**, for details.

The solution types covered in table `nss_two_body_orbit` are:

- AstroSpectroSB1: Combined astrometric + single lined spectroscopic orbital model
- AstroSpectroSB2: Combined astrometric + double lined spectroscopic orbital model
- EclipsingBinary: Eclipsing binary orbital model
- EclipsingSpectroSB1: Combined eclipsing binary + single lined spectroscopic orbital model
- EclipsingSpectroSB2: Combined eclipsing binary + double lined spectroscopic orbital model
- Orbital: Orbital model for an astrometric binary
- OrbitalPoorlyConstrained: Orbital model for an astrometric binary but with low significance (< 5)
- OrbitalAlternative: Alternative orbital model mainly for low S/N systems
- Pulsation: Pulsation modelled as an SB1
- PulsationScanAngleK1: Pulsation modelled as an SB1 plus first harmonics of a scan angle model
- PulsationScanAngleK12: Pulsation modelled as an SB1 plus first two harmonics of a scan angle model
- SB1: Single Lined Spectroscopic binary model
- SB1ScanAngleK1: Single Lined Spectroscopic binary model plus first harmonics of a scan angle model
- SB1ScanAngleK12: Single Lined Spectroscopic binary model plus first two harmonics of a scan angle model
- SB2: Double Lined Spectroscopic binary model
- SB2ScanAngleK1: Double Lined Spectroscopic binary model plus first harmonics of a scan angle model
- SB2ScanAngleK12: Double Lined Spectroscopic binary model plus first two harmonics of a scan angle model
- VIMO: Variability-Induced Movers with astrometric orbital model

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ORIGINAL_SOLUTION_TYPES : Original types of the solutions before combination (string[3] array)

For models AstroSpectroSB1, AstroSpectroSB2, EclipsingSpectroSB1, EclipsingSpectroSB2 which result from combinations of respectively astrometric and spectroscopic or eclipsing and spectroscopic solutions: original types of the solutions before combination.

SUBTYPE : NSS model sub-type (string)

Human readable version of the `bit_index` field.

A text description is provided for every `solution_type` and `bit_index` combination. It can be used in ADQL queries. In this particular table, it is composed as a function of the `solution_type` as follows:

- AstroSpectroSB1, AstroSpectroSB2
  - Elliptical: all parameters fitted.
  - Circular: circular model, eccentricity and argument of periastron not fitted.
- EclipsingBinary, EclipsingSpectroSB1 and EclipsingSpectroSB2: the text is a combination of some of the following keys in the given order representing parameters that are fitted. Keys not in the subtype label are not fitted. For example MassS1S2T1 includes parameters for keys Mass, S1, S2 and T1 but not for Ecc and T2.
  - Mass: mass ratio (M2/M1).
  - S1: primary fill factor.
  - S2: secondary fill factor.
  - Ecc: elliptical solution, including eccentricity and argument of periastron.
  - T1: primary effective temperature.
  - T2: secondary effective temperature.
- Orbital, OrbitalPoorlyConstrained
  - Elliptical: all parameters fitted.
  - Circular: circular model, eccentricity and argument of periastron not fitted.
  - Faceon: face-on model, inclination and position angle of the nodes not fitted.
- SB1, SB1ScanAngleK1, SB1ScanAngleK12, SB2, SB2ScanAngleK1, SB2ScanAngleK12
  - Elliptical: all parameters fitted.

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  - Circular: circular model, eccentricity and argument of periastron not fitted.

- OrbitalAlternative, Pulsation, PulsationScanAngleK1, PulsationScanAngleK12 and VIMO: always null (all parameters fitted).

BIT_INDEX : boolean mask for the fields in the correlation matrix (long)

`bit_index` is a boolean mask indicating which variables or parameters have been fitted for a given non-single-star solution in a table row. The parameters used for each `solution_type` are different. The correspondence between `solution_type`, `bit_index` and fitted variables is provided below.

The M parameters for a given `solution_type` always follow the same order. `bit_index` contains M+1 bits, where the leading Most Significant Bit (MSB) is always 1, and the other M bits correspond to parameters of a given model. The last variable is related to the trailing Least Significant Bit (LSB). A bit with value 1 (0) means the value of its associated parameter has been fitted (kept fixed) in a given solution. The covariance matrix has M×M size. The `bit_index` length and the variable names below can be used to identify elements in the `correlations` vector. When a given parameter has not been fitted (bit in `bit_index` = 0), the corresponding correlation is NaN.

Detailed explanations for each `solution_type` and `bit_index` are provided below. T or F means a parameter has or has not been fitted. Table field names have been used to identify variables when possible. A text description is provided when not in the data model.

`solution_type` = AstroSpectroSB1

bit_index	subtype	ra	dec	parallax	pmra	pmdec	semi_major_axis	arg_periastron	pos_ascending_node	inclination	semi_amplitude_primary	center_of_mass_velocity	eccentricity	period	t_periastron
32635	Circular	T	T	T	T	T	T	F	T	T	T	T	F	T	T
32767	Elliptical	T	T	T	T	T	T	T	T	T	T	T	T	T	T

`solution_type` = AstroSpectroSB2: the following 17 parameters are fitted and `bit_index` = 65535 in all cases:

- `ra`

- dec
- parallax
- pmra
- pmdec
- semi_major_axis
- arg_periastron
- pos_ascending_node
- inclination
- semi_amplitude_primary
- center_of_mass_velocity
- semi_amplitude_secondary
- eccentricity
- period
- t_periastron

solution_type = EclipsingBinary

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bit_index	subtype	Internal fixed value parameter	t_periastron	Internal fixed value parameter	Internal fixed value parameter	mass_ratio	fill_factor_primary	fill_factor_secondary	esinw	inclination	ecosw	Internal fixed value parameter	Internal fixed value parameter	temperature_primary	temperature_secondary	Internal fixed value parameter	Internal fixed value parameter	Internal fixed value parameter	Internal fixed value parameter	Internal fixed value parameter	Internal fixed value parameter
1337408	S1S2T2	F	T	F	F	F	T	T	F	T	F	F	F	F	T	F	F	F	F	F	F
1337472	S1S2T1	F	T	F	F	F	T	T	F	T	F	F	F	T	F	F	F	F	F	F	F
1337536	S1S2T1T2	F	T	F	F	F	T	T	F	T	F	F	F	T	T	F	F	F	F	F	F
1342528	S1S2EccT2	F	T	F	F	F	T	T	T	T	T	F	F	F	T	F	F	F	F	F	F
1342592	S1S2EccT1	F	T	F	F	F	T	T	T	T	T	F	F	T	F	F	F	F	F	F	F
1342656	S1S2EccT1T2	F	T	F	F	F	T	T	T	T	T	F	F	T	T	F	F	F	F	F	F
1353792	MassS2T2	F	T	F	F	T	F	T	F	T	F	F	F	F	T	F	F	F	F	F	F
1353856	MassS2T1	F	T	F	F	T	F	T	F	T	F	F	F	T	F	F	F	F	F	F	F
1353920	MassS2T1T2	F	T	F	F	T	F	T	F	T	F	F	F	T	T	F	F	F	F	F	F
1361984	MassS1T2	F	T	F	F	T	T	F	F	T	F	F	F	F	T	F	F	F	F	F	F
1362048	MassS1T1	F	T	F	F	T	T	F	F	T	F	F	F	T	F	F	F	F	F	F	F
1362112	MassS1T1T2	F	T	F	F	T	T	F	F	T	F	F	F	T	T	F	F	F	F	F	F
1370176	MassS1S2T2	F	T	F	F	T	T	T	F	T	F	F	F	F	T	F	F	F	F	F	F
1370240	MassS1S2T1	F	T	F	F	T	T	T	F	T	F	F	F	T	F	F	F	F	F	F	F
1370304	MassS1S2T1T2	F	T	F	F	T	T	T	F	T	F	F	F	T	T	F	F	F	F	F	F
1375296	MassS1S2EccT2	F	T	F	F	T	T	T	T	T	T	F	F	T	T	F	F	F	F	F	F
1375360	MassS1S2EccT1	F	T	F	F	T	T	T	T	T	T	F	F	T	F	F	F	F	F	F	F
1375424	MassS1S2EccT1T2	F	T	F	F	T	T	T	T	T	T	F	F	T	T	F	F	F	F	F	F

solution_type = EclipsingSpectroSB1

bit_index	subtype	Internal fixed value parameter	t_periastron	center_of_mass_velocity	semi_amplitude_primary	mass_ratio	fill_factor_primary	fill_factor_secondary	esinw	inclination	ecosw	Internal fixed value parameter	Internal fixed value parameter	temperature_primary	temperature_secondary	Internal fixed value parameter	Internal fixed value parameter	Internal fixed value parameter	Internal fixed value parameter	Internal fixed value parameter	Internal fixed value parameter
1534016	S1S2T2	F	T	T	T	F	T	T	F	T	F	F	F	F	T	F	F	F	F	F	F
1534080	S1S2T1	F	T	T	T	F	T	T	F	T	F	F	F	T	F	F	F	F	F	F	F
1534144	S1S2T1T2	F	T	T	T	F	T	T	F	T	F	F	F	T	T	F	F	F	F	F	F
1539136	S1S2EccT2	F	T	T	T	F	T	T	T	T	T	F	F	F	T	F	F	F	F	F	F
1539200	S1S2EccT1	F	T	T	T	F	T	T	T	T	T	F	F	T	F	F	F	F	F	F	F
1539264	S1S2EccT1T2	F	T	T	T	F	T	T	T	T	T	F	F	T	T	F	F	F	F	F	F
1550400	MassS2T2	F	T	T	T	T	F	T	F	T	F	F	F	F	T	F	F	F	F	F	F
1550464	MassS2T1	F	T	T	T	T	F	T	F	T	F	F	F	T	F	F	F	F	F	F	F
1550528	MassS2T1T2	F	T	T	T	T	F	T	F	T	F	F	F	T	T	F	F	F	F	F	F
1558592	MassS1T2	F	T	T	T	T	T	F	F	T	F	F	F	F	T	F	F	F	F	F	F
1558656	MassS1T1	F	T	T	T	T	T	F	F	T	F	F	F	T	F	F	F	F	F	F	F
1566784	MassS1S2T2	F	T	T	T	T	T	T	F	T	F	F	F	F	T	F	F	F	F	F	F
1566848	MassS1S2T1	F	T	T	T	T	T	T	F	T	F	F	F	T	F	F	F	F	F	F	F
1566912	MassS1S2T1T2	F	T	T	T	T	T	T	F	T	F	F	F	T	T	F	F	F	F	F	F
1571904	MassS1S2EccT2	F	T	T	T	T	T	T	T	T	T	F	F	F	T	F	F	F	F	F	F
1571968	MassS1S2EccT1	F	T	T	T	T	T	T	T	T	T	F	F	T	F	F	F	F	F	F	F
1572032	MassS1S2EccT1T2	F	T	T	T	T	T	T	T	T	T	F	F	T	T	F	F	F	F	F	F

solution_type = EclipsingSpectroSB2

bit_index	subtype	Internal fixed value parameter	t_periastron	center_of_mass_velocity	semi_amplitude_primary	mass_ratio	fill_factor_primary	fill_factor_secondary	esinw	inclination	ecosw	Internal fixed value parameter	Internal fixed value parameter	temperature_primary	temperature_secondary	Internal fixed value parameter	Internal fixed value parameter	Internal fixed value parameter	Internal fixed value parameter	Internal fixed value parameter	Internal fixed value parameter
1550400	MassS2T2	F	T	T	T	T	F	T	F	T	F	F	F	F	T	F	F	F	F	F	F
1550464	MassS2T1	F	T	T	T	T	F	T	F	T	F	F	F	T	F	F	F	F	F	F	F
1566784	MassS1S2T2	F	T	T	T	T	T	T	F	T	F	F	F	F	T	F	F	F	F	F	F
1566848	MassS1S2T1	F	T	T	T	T	T	T	F	T	F	F	F	T	F	F	F	F	F	F	F
1566912	MassS1S2T1T2	F	T	T	T	T	T	T	F	T	F	F	F	T	T	F	F	F	F	F	F
1571904	MassS1S2EccT2	F	T	T	T	T	T	T	T	T	T	F	F	F	T	F	F	F	F	F	F
1571968	MassS1S2EccT1	F	T	T	T	T	T	T	T	T	T	F	F	T	F	F	F	F	F	F	F
1572032	MassS1S2EccT1T2	F	T	T	T	T	T	T	T	T	T	F	F	T	T	F	F	F	F	F	F

solution_type = Orbital

bit_index	subtype	ra	dec	parallax	pmra	pmdec	semi_major_axis	arg_periastron	pos_ascending_node	inclination	eccentricity	period	t_periastron
8155	Circular	T	T	T	T	T	T	F	T	T	F	T	T
8167	Faceon	T	T	T	T	T	T	T	F	F	T	T	T
8191	Elliptical	T	T	T	T	T	T	T	T	T	T	T	T

solution_type = OrbitalPoorlyConstrained

bit_index	subtype	ra	dec	parallax	pmra	pmdec	semi_major_axis	arg_periastron	pos_ascending_node	inclination	eccentricity	period	t_periastron
8155	Circular	T	T	T	T	T	T	F	T	T	F	T	T
8167	Faceon	T	T	T	T	T	T	T	F	F	T	T	T
8191	Elliptical	T	T	T	T	T	T	T	T	T	T	T	T

solution_type = OrbitalAlternative: the following 17 parameters are fitted and bit_index = 262143 in all cases:

- ra

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- dec
- parallax
- pmra
- pmdec
- a_thiele_innes
- b_thiele_innes
- f_thiele_innes
- g_thiele_innes
- eccentricity
- period
- t_periastron
- semi_major_axis
- arg_periastron
- pos_ascending_node
- inclination
- astrometric_jitter

solution_type = Pulsation: the following 5 parameters are fitted and bit_index = 111 in all cases:

- center_of_mass_velocity
- semi_amplitude_primary
- eccentricity
- arg_periastron
- t_periastron

solution_type = PulsationScanAngleK1: the following 7 parameters are fitted and bit_index = 447 in all cases:

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- center_of_mass_velocity
- semi_amplitude_primary
- eccentricity
- arg_periastron
- t_periastron
- velocity_scan_angle_cosine1
- velocity_scan_angle_sine1

solution_type = PulsationScanAngleK12: the following 9 parameters are fitted and bit_index = 1791 in all cases:

- center_of_mass_velocity
- semi_amplitude_primary
- eccentricity
- arg_periastron
- t_periastron
- velocity_scan_angle_cosine1
- velocity_scan_angle_sine1
- velocity_scan_angle_cosine2
- velocity_scan_angle_sine2

solution_type = SB1

bit_index	subtype	center_of_mass_velocity	period	semi_amplitude_primary	eccentricity	arg_periastron	t_periastron
121	Circular	T	T	T	F	F	T
127	Elliptical	T	T	T	T	T	T

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solution_type = SB1ScanAngleK1

bit_index	subtype	center_of_mass_velocity	period	semi_amplitude_primary	eccentricity	arg_periastron	t_periastron	velocity_scan_angle_cosine1	velocity_scan_angle_sine1
487	Circular	T	T	T	F	F	T	T	T
511	Elliptical	T	T	T	T	T	T	T	T

solution_type = SB1ScanAngleK12

bit_index	subtype	center_of_mass_velocity	period	semi_amplitude_primary	eccentricity	arg_periastron	t_periastron	velocity_scan_angle_cosine1	velocity_scan_angle_sine1	velocity_scan_angle_cosine2	velocity_scan_angle_sine2
1951	Circular	T	T	T	F	F	T	T	T	T	T
2047	Elliptical	T	T	T	T	T	T	T	T	T	T

solution_type = SB2: the following 7 parameters are fitted and bit_index = 255 in all cases:

- center_of_mass_velocity
- period
- semi_amplitude_primary
- semi_amplitude_secondary
- eccentricity
- arg_periastron
- t_periastron

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solution_type = SB2ScanAngleK1: the following 9 parameters are fitted and bit_index = 1023 in all cases:

- center_of_mass_velocity
- period
- semi_amplitude_primary
- semi_amplitude_secondary
- eccentricity
- arg_periastron
- t_periastron
- velocity_scan_angle_cosine1
- velocity_scan_angle_sine1

solution_type = SB2ScanAngleK12: the following 11 parameters are fitted and bit_index = 4095 in all cases:

- center_of_mass_velocity
- period
- semi_amplitude_primary
- semi_amplitude_secondary
- eccentricity
- arg_periastron
- t_periastron
- velocity_scan_angle_cosine1
- velocity_scan_angle_sine1
- velocity_scan_angle_cosine2
- velocity_scan_angle_sine2

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`solution_type` = VIMO: the following 13 parameters are fitted and `bit_index` = 16383 in all cases:

- `ra`
- `dec`
- `parallax`
- `pmra`
- `pmdec`
- `semi_major_axis`
- `arg_periastron`
- `pos_ascending_node`
- `inclination`
- `eccentricity`
- `period`
- `t_periastron`
- `flux_factor_cms_vim`

OBJ_FUNC_VALUE : value of the objective function at the solution (float)

The χ^2, defined as $\sum_{i=1}^{n}(x_i - y_i)^2/\sigma_i^2$, where x_i is the observation of the ith transit, y_i is the value calculated from the model, and σ_i is the uncertainty associated with the observation.

GOF : goodness of fit in the Hipparcos sense (float)

Goodness-of-fit statistic of the solution. This is the 'gaussianized chi-square' (Wilson & Hilferty (1931)'s cube root transformation), which for good fits should approximately follow a normal distribution with zero mean value and unit standard deviation.

This statistic is computed according to the formula:

$$F2 = \sqrt{\frac{9\nu}{2}}\left(\sqrt[3]{\frac{\texttt{obj_func_value}}{\nu}} + \frac{2}{9\nu} - 1\right)$$

where `obj_func_value` is hopefully a χ^2 and ν is the number of degrees of freedom.

SIGNIFICANCE : The significance of the solution (i.e. how worth keeping a model is) (float)

The goodness of fit is not always enough to decide whether a model is worth keeping or not. The significance, equivalent to a signal-to-noise ratio, addresses some of these limitations.

For astrometric binaries, it is defined as the semi-major axis divided by its uncertainty. This uncertainty is derived from the solution covariance matrix, where the uncertainties also account for the unmodelled excess noise (`astrometric_jitter`).

For spectroscopic binaries, this is the ratio of the semi-amplitude of the primary over its uncertainty.

For astroSpectroSB1, and astroSpectroSB2, the significance is the one of the astrometric photocentre semi-major axis, before the uncertainties are corrected. For eclipsingSpectroSB1 and eclipsingSpectroSB2, it is the significance of the semi-amplitude of the primary.

EFFICIENCY : Efficiency of the solution (float)

The efficiency expresses the level of correlation between the parameters of a model. A value of 1 means a total absence of correlation whereas it falls to 0 as the correlation increases. It is defined as the n-th root of the ratio of the product of the diagonal elements of the covariance matrix and the product of the eigen value of that matrix. When all the covariances are 0, the matrix is diagonal and the ratio is exactly 1. See also Eichhorn (1989).

BIC : Bayesian Information Criterion (float)

BIC (Bayesian Information Criterion) of the solution, the excess noise being accounted for in the residuals uncertainties.

FLAGS : Quality flag for the achieved NSS solution (long)

Processing flag applicable to specific non-single-star solutions. The meaning of each of those is given in the table below.

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Flags bit number	Flag Meaning	Comment
Astrometric binary solutions		
0	LARGE_GOF	Goodness of Fit for Orbital/PoorlyConstrained $10 \leq \text{GoF} < 25$
1	LARGE_ECCENTRICITY	period-eccentricity outside of envelope $\sqrt{(1-\sqrt{(1/P)})}$
2	LARGE_ANGLE	uncertainty of `arg_periastron` or of posAcendingNode above 2π or `t_periastron` uncertainty above period
3-7	NA	
Spectroscopic binary solutions		
8	BAD_UNCHECKED_NUMBER_OF_TRANSITS	The length of Transit is not sufficient to process the star (before removing bad transits)
9	BAD_CHECKED_NUMBER_OF_TRANSITS	The length of Transit is not sufficient to process the star (after removing bad transits)
10	NO_SIGNIFICANT_PERIODS_CAN_BE_FOUND	No significant period can be derived (no period from photometric variability analysis + peridogram peaks below the significance threshold)
11	NO_SIGNIFICANT_PULSATION_PERIOD	The pulsation periods are not significant)
12	REFINED_SOLUTION_DOES_NOT_CONVERGE	The refined orbital solution does not converge (after 1000 iterations)
13	REFINED_SOLUTION_COVARIANCE_MATRIX	The variance covariance matrix is not definite positive in the refined solution
14	LUCY_TEST_APPLIED	The Lucy test has been applied
15	CIRCULAR_SOLUTION_DOES_NOT_CONVERGE	The Lucy refined orbital solution diverges (after 1000 iterations)
16	CIRCULAR_SOLUTION_NOT_DEFINITE_POSITIVE_COVARIANCE_MATRIX	The variance covariance matrix is not definite positive in the circular solution
17	TREND_SOLUTION_NOT_DEFINITE_POSITIVE_COVARIANCE_MATRIX	The variance covariance matrix is not definite positive in the trend solution
18	REFINED_PULSATION_DOES_NOT_CONVERGE	The refined eccentric pulsation solution does not converge (after 1000 iterations)
19	REFINED_PULSATION_NOT_DEFINITE_POSITIVE_COVARIANCE_MATRIX	The variance covariance matrix is not definite positive in the trend solution
20	CIRCULAR_PULSATION_DOES_NOT_CONVERGE	The circular pulsation solution does not converge (after 1000 iterations)
21	CIRCULAR_PULSATION_NOT_DEFINITE_POSITIVE_COVARIANCE_MATRIX	The variance covariance matrix is not definite positive in the circular pulsation solution
22	SOLUTION_OUTSIDE_E_LOGP_ENVELOP	The orbital solution (SB1 or SB2) is outside the e-log(P) envelop.
23	MODEL_AND_ITS_RESIDUAL_ARE_BOTH_TRENDS	Both the model and its residual are trends
24	COMPOSITE_SOLUTION_DOES_NOT_CONVERGE	The composite solution does not converge (after 1000 iterations)
25	COMPOSITE_SOLUTION_NOT_DEFINITE_POSITIVE_COVARIANCE_MATRIX	The variance covariance matrix is not definite positive in the composite solution
26	SOME_TRANSITS_ARE_DOUBLE_LINED	Some transits are double lined in the single lined mode
27	SIGNIFICANT_SCAN_ANGLE	Scan angle signal is significant
28	SCAN_ANGLE_SOLUTION_NOT_DEFINITE_POSITIVE_COVARIANCE_MATRIX	The variance covariance matrix is not definite positive in the scan angle solution
29	SIGNIFICANT_SCAN_ANGLE_HARMONIC	Scan angle 1st harmonic is significant
30	PERIOD_SIGNIFICANT_IN_PHOTOMETRY_ALSO	The derived period is significant when applied to photometric data.
31	NA	
Eclipsing binary solutions		
32	No variance-covariance matrix	Covariance matrix computation failed
33-47	NA	
Combined solutions		
48	COMBINED_SOLUTION_EXIST	Successfull combination
49	COMBINATION_REJECTED_BAD_F2	Reject a combination because of large GoF
50	COMBINATION_REJECTED_DIFFERENT_COMPONENTS	Reject a combination because the output produces different components (when combining stochastic and spectro)
51	DUPLICATE_PERIOD_USED	When the input solution have alias periods
52	STOCHASTIC_TO_ORBITAL	When combining a stochasticAstro and SBxx , output: AstroSpectroSBx
53	ECLIPSING_SPECTRO_TRIED	When en eclipsingSpectroSBx combination has been tried
54	ASTRO_SPECTRO_TRIED	When an AstroSpectroSBx combination has been tried
55	COMBINATION_FAILED	When a combination has been tried but failed.
56	TREND_COMBINATION	When we combine a spectroscopic trend solution or stochastic spectro
57	DU434_INPUT_USED	When trying an EclipsingSpectro combination with spectroscopic binaries input parameters and gives lower F2 than photometric binaries inputs parameters
58	SWAPPED_RVs_USED	In an EclipsingSpectro combination the radial velocities were swapped
59	TIMEOUT_OCCURRED	During an EclipsingSpectro combination a timeout occured

CORRELATIONS : Vector form of the upper triangle of the correlation matrix (column-major ordered) (float[] array)

Correlation matrix of the fitted profile parameters for the applicable non-single star solution. The parameters stored in this matrix and their order is given in the description of field `bit_index`. Since not all parameters of a given solution model are systematically fitted, the matrix may contain NaN values at the elements of the corresponding indices, such as the length of the vector always corresponds to the expected length of the theoretical model.

For astrometric binaries, the matrix corresponds to a solution calculated with uncertainties of the astrometric transits that includes the `astrometric_jitter` added in quadrature.

Only non-zero, non-unity, correlation coefficients from the correlation matrix **M** are provided here. They are served as a linear array of constant size $S = n(n-1)/2$ corresponding to the full normal matrix of dimension $n \times n$. The ordering of the elements in the linear array follows a column-major storage scheme, i.e.:

$$\mathbf{M} = \begin{bmatrix} 1 & C[0] & C[1] & C[3] & C[6] & \cdots & C[S-(n-1)] \\ & 1 & C[2] & C[4] & C[7] & \cdots & C[S-(n-2)] \\ & & 1 & C[5] & C[8] & \cdots & C[S-(n-3)] \\ & & & 1 & C[9] & \cdots & C[S-(n-4)] \\ & & & & \ddots & \ddots & \vdots \\ & & & & & 1 & C[S-1] \\ & & & & & & 1 \end{bmatrix}$$

ASTROMETRIC_N_OBS_AL : Total astrometric CCD observations in AL considered (int)

Total astrometric CCD observations considered in the along-scan direction.

ASTROMETRIC_N_GOOD_OBS_AL : Total astrometric CCD observations in AL actually used (int)

Total astrometric CCD observations actually used in the along-scan direction.

RA : Right ascension (double, Angle[deg])

Barycentric right ascension α of the source in ICRS at the reference epoch `gaia_source.ref_epoch`

RA_ERROR : Standard error of right ascension (float, Angle[mas])

Standard error $\sigma_{\alpha*} \equiv \sigma_\alpha \cos\delta$ of the right ascension of the source in ICRS at the reference epoch `gaia_source.ref_epoch`.

DEC : Declination (double, Angle[deg])

Barycentric declination δ of the source in ICRS at the reference epoch `gaia_source.ref_epoch`

DEC_ERROR : Standard error of declination (float, Angle[mas])

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Standard error σ_δ of the declination of the source in ICRS at the reference epoch `gaia_source.ref_epoch`

PARALLAX : Parallax (double, Angle[mas])

Absolute stellar parallax ϖ of the source at the reference epoch `gaia_source.ref_epoch`

PARALLAX_ERROR : Standard error of parallax (float, Angle[mas])

Standard error σ_ϖ of the stellar parallax at the reference epoch `gaia_source.ref_epoch`

PMRA : Proper motion in right ascension direction (double, Angular Velocity[mas yr^{-1}])

Proper motion in right ascension $\mu_{\alpha*} \equiv \mu_\alpha \cos\delta$ of the source in ICRS at the reference epoch `gaia_source.ref_epoch`. This is the local tangent plane projection of the proper motion vector in the direction of increasing right ascension.

PMRA_ERROR : Standard error of proper motion in right ascension direction (float, Angular Velocity[mas yr^{-1}])

Standard error $\sigma_{\mu\alpha*}$ of the local tangent plane projection of the proper motion vector in the direction of increasing right ascension at the reference epoch `gaia_source.ref_epoch`

PMDEC : Proper motion in declination direction (double, Angular Velocity[mas yr^{-1}])

Proper motion in declination μ_δ of the source at the reference epoch `gaia_source.ref_epoch`. This is the projection of the proper motion vector in the direction of increasing declination.

PMDEC_ERROR : Standard error of proper motion in declination direction (float, Angular Velocity[mas yr^{-1}])

Standard error $\sigma_{\mu\delta}$ of the proper motion component in declination at the reference epoch `gaia_source.ref_e`

PERIOD : Orbital Period (double, Time[day])

Period of the orbital motion around the barycentre. For Eclipsing Binary solutions, it is the period obtained from the photometric variability analysis reported in field `vari_eclipsing_binary.frequency`.

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For EclipsingSpectro combined solutions, it is either the aforementioned period, or the one stemming from the Spectroscopic binary model, in which case the bit 57 for `nss_two_body_orbit.flags` will be set to 1.

PERIOD_ERROR : Standard error of Orbital Period (float, Time[day])

Standard error of the period of the orbit of the photocentre around the barycentre. For astrometric binaries, the standard errors are derived from the variance-covariance matrix of the solution, where the transit measurement uncertainties contain the `astrometric_jitter` added quadratically. For other solution types, it is derived from the covariance matrix of the solution, revised by a multiplicative factor that would yield a zero goodness of fit.

For Eclipsing Binary solutions, this error is provided in `nss_two_body_orbit.period_error`. For an EclipsingSpectro combined solution, it is either the aforementioned period, or the one stemming from the spectroscopic binary model, in which case bit 57 for `nss_two_body_orbit.flags` will be set to 1.

T_PERIASTRON : Periastron epoch (double, Time[day])

The epoch at periastron is given relative to `gaia_source.ref_epoch`, in the range [–`period`/2, +`period`/2].

In case the eccentricity is not null, `t_periastron` is the time of periastron passage, whatever the input solution: astrometric, spectroscopic or eclipsing binary.

If the eccentricity is null, then the periastron has no meaning and the following convention is adopted:

- for astrometric binaries the periastron is positioned on the ascending node and in the absence of radial velocity measurements, the ascending node is the node whose position angle is between 0 and 180 degrees;
- for spectroscopic binaries the periastron is positioned on the ascending node, the definition of which is specified by the knowledge of the radial velocity curve and is the maximum of the radial velocity curve;
- for eclipsing binaries it is the time of primary eclipse;
- for combined eclipsing/spectroscopic binaries, it is the time of primary eclipse.

T_PERIASTRON_ERROR : Standard error of Periastron epoch (float, Time[day])

Standard error of the periastron epoch, `t_periastron`, defined above. For astrometric binaries, the standard errors are derived from the variance-covariance matrix of the solution, where the transit measurement uncertainties contain the `astrometric_jitter` added quadratically. For other solution types, it is derived from the covariance matrix of the solution, revised by a multiplicative factor that would yield a zero goodness of fit.

ECCENTRICITY : eccentricity (double)

Eccentricity of the orbit.

ECCENTRICITY_ERROR : Standard error of eccentricity (float)

Standard error of the eccentricity of the orbit. For astrometric binaries, the standard errors are derived from the variance-covariance matrix of the solution, where the transit measurement uncertainties contain the `astrometric_jitter` added quadratically. For other solution types, it is derived from the covariance matrix of the solution, revised by a multiplicative factor that would yield a zero goodness of fit.

If the eccentricity is null, a special circular or a pseudo-circular solution is implemented and the standard error will be null.

For orbits with small eccentricities (`eccentricity` < 0.0005 and `eccentricity_error` is absent, the standard error will be null to indicate that the orbit has been made pseudo-circular.

INCLINATION : Orbital inclination (double, Angle[deg])

Inclination of the orbital plane with respect to the sky.

INCLINATION_ERROR : Standard error of Orbital inclination (float, Angle[deg])

Standard error on the orbital inclination. For astrometric binaries, the standard errors are derived from the variance-covariance matrix of the solution, where the transit measurement uncertainties contain the `astrometric_jitter` added quadratically. For other solution types, it is derived from the covariance matrix of the solution, revised by a multiplicative factor that would yield a zero goodness of fit.

ARG_PERIASTRON : Argument of periastron (double, Angle[deg])

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The argument of periastron is the angular position of the periastron as measured in the plane of the orbit in the sense of the object motion. When the spectroscopic orbit is established, the zero point of the angle is the ascending node (node on the line of nodes where the objects are moving away from the observer). In the absence of spectroscopic constraints, the zero point is the node whose position angle is between 0 and 180 degrees, the position angle being measured in the trigonometric direction. The argument of periastron is considered as fixed (no precession of the apsides). In the case of circular orbits, ω is undefined and set up arbitrarily to zero.

ARG_PERIASTRON_ERROR : Standard error of Argument of periastron (float, Angle[deg])

Standard error of the `arg_periastron` as defined above. For astrometric binaries, the standard errors are derived from the variance-covariance matrix of the solution, where the transit measurement uncertainties contain the `astrometric_jitter` added quadratically. For other solution types, it is derived from the covariance matrix of the solution, revised by a multiplicative factor that would yield a zero goodness of fit.

POS_ASCENDING_NODE : Position angle of the ascending node (double, Angle[deg])

The position angle of the line of intersection between the tangential plane of projection and the true orbital plane. The node where the orbital motion is directed away from the sun is called the ascending node and is identified only when radial velocity is known.

POS_ASCENDING_NODE_ERROR : Standard error of the ascending node (float, Angle[deg])

Standard error of Angular semi-major axis of the photocentre. The standard errors are derived from the variance-covariance matrix of the solution, where the transit measurement uncertainties contain the `astrometric_jitter` added quadratically.

SEMI_MAJOR_AXIS : Angular semi-major axis of the photocentre or of the variable component (double, Angle[mas])

The angular semi-major axis of the orbit around the barycentre for orbital models of unresolved binaries, based on astrometry or a combination of astrometry and radial velocities.

For photometrically constant systems, this is the semi-major axis a_0 of the photocentre; for the VIMO model, this is the semi-major axis of the variable component.

SEMI_MAJOR_AXIS_ERROR : Standard error of Angular semi-major axis of the photocentre or

of the variable component (float, Angle[mas])

Standard error of the angular semi-major axis. The standard errors are derived from the variance-covariance matrix of the solution, where the transit measurement uncertainties contain the `astrometric_jitt` added quadratically.

A_THIELE_INNES : Thiele-Innes element A (double, Angle[mas])

The Thiele-Innes element A of the orbit of the photocentre (for the orbital model). This parameter is fitted for orbital models based on astrometry (`solution_type` = Orbital, OrbitalAlternative[Validated], or OrbitalTargetedSearch[Validated])) or a combination of astrometry and radial velocities (`solution_type` = AstroSpectroSB1). For other orbital models based on photometry or radial velocities, or their combination, the Campbell parameters are fitted instead. See more details, including the transformation from the A, B, F, G Thiele-Innes elements to the a_0, i, ω and Ω Campbell elements, in the on-line documentation (Chapter **??**).

A_THIELE_INNES_ERROR : Standard error of Thiele-Innes element A (float, Angle[mas])

Standard error of the Thiele-Innes element A of the orbit of the photocentre (for the orbital model). The standard errors are derived from the variance-covariance matrix of the solution, where the transit measurement uncertainties contain the `astrometric_jitter` added quadratically.

B_THIELE_INNES : Thiele-Innes element B (double, Angle[mas])

The Thiele-Innes element B of the orbit of the photocentre (for the orbital model). This parameter is fitted for orbital models based on astrometry (`solution_type` = Orbital, OrbitalAlternative[Validated], or OrbitalTargetedSearch[Validated])) or a combination of astrometry and radial velocities (`solution_type` = AstroSpectroSB1). For other orbital models based on photometry or radial velocities, or their combination, the Campbell parameters are fitted instead. See more details, including the transformation from the A, B, F, G Thiele-Innes elements to the a_0, i, ω and Ω Campbell elements, in the on-line documentation (Chapter **??**).

B_THIELE_INNES_ERROR : Standard error of Thiele-Innes element B (float, Angle[mas])

Standard error of the Thiele-Innes element B of the orbit of the photocentre (for the orbital model). The standard errors are derived from the variance-covariance matrix of the solution, where the transit measurement uncertainties contain the `astrometric_jitter` added quadratically.

F_THIELE_INNES : Thiele-Innes element F (double, Angle[mas])

The Thiele-Innes element F of the orbit of the photocentre (for the orbital model). This parameter is fitted for orbital models based on astrometry (`solution_type` = Orbital, OrbitalAlternative[Validated], or OrbitalTargetedSearch[Validated])) or a combination of astrometry and radial velocities (`solution_type` = AstroSpectroSB1). For other orbital models based on photometry or radial velocities, or their combination, the Campbell parameters are fitted instead. See more details, including the transformation from the A, B, F, G Thiele-Innes elements to the a_0, i, ω and Ω Campbell elements, in the on-line documentation (Chapter **??**).

F_THIELE_INNES_ERROR : Standard error of Thiele-Innes element F (float, Angle[mas])

Standard error of the Thiele-Innes element F of the orbit of the photocentre (for the orbital model). The standard errors are derived from the variance-covariance matrix of the solution, where the transit measurement uncertainties contain the `astrometric_jitter` added quadratically.

G_THIELE_INNES : Thiele-Innes element G (double, Angle[mas])

The Thiele-Innes element G of the orbit of the photocentre is fitted for orbital models based on astrometry (`solution_type` = Orbital, OrbitalAlternative[Validated], or OrbitalTargetedSearch[Validated])) or a combination of astrometry and radial velocities (`solution_type` = AstroSpectroSB1). For other orbital models based on photometry or radial velocities, or their combination, the Campbell parameters are fitted instead.

For orbits with small eccentricities (`eccentricity` < 0.0005 and `eccentricity_error` is absent), `g_thiele_innes` is not an unknown in the model but is deduced from the A, B, F Thiele-Innes elements, the whole set of which must yield $\omega = 0$.

For more details, including the transformation from the A, B, F, G Thiele-Innes elements to the a_0, i, ω and Ω Campbell elements see Chapter **??** in the on-line documentation.

G_THIELE_INNES_ERROR : Standard error of Thiele-Innes element G (float, Angle[mas])

Standard error of the Thiele-Innes element G of the orbit of the photocentre (for the orbital model). The standard errors are derived from the variance-covariance matrix of the solution, where the transit measurement uncertainties contain the `astrometric_jitter` added quadratically.

ASTROMETRIC_JITTER : Astrometric jitter term (float, Angle[mas])

In order to account for some poorly calibrated effects, this is an excess astrometric noise quadratically added to the uncertainty on the observations such that the resulting gaussianised Goodness of Fit is null. This applies to Orbital, OrbitalPoorlyConstrained, OrbitalAlternative solutions only.

ASTROMETRIC_JITTER_ERROR : Astrometric jitter standard error (float, Angle[mas])

In order to account for some poorly calibrated effects, a noise, `astrometric_jitter`, is quadratically added to the uncertainty on the observations such that the resulting F2 is null. When the astrometric jitter is fitted, which is the case for the OrbitalAlternative model, the current field represents its uncertainty.

REF_FLUX_G : Reference flux in the G band (float, Flux[ke-/s])

The photocentre of a VIM object moves in relation to its total flux in the G band. In a VIMO solution, the position derived from the single star parameters (`ra`, `dec`, `parallax`, `pmra`, `pmdec`) is the mean position of the object when its total flux is the reference flux. In practice, the reference flux is close to (or is) the median flux of the object.

FLUX_FACTOR_CMS_VIM : Fcms / Fg flux factor (double)

CMS flux over average G band flux factor.

When the total flux of a VIM is Fcms = `flux_factor_cms_vim`×`ref_flux_g` (which doesn't necessarily happen in practice) the photocentre coincides with the barycentre.

FLUX_FACTOR_CMS_VIM_ERROR : Standard error of reference flux factor (float)

Standard error of the flux factor for which the photocentre coincides with the barycentre.

The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

TUWE_UNCERTAINTY_RESCALED : Have the transit-binned uncertainties been multiplied by the TUWE? (boolean)

Boolean indicating whether the transit-binned uncertainties have been reweighted by the TUWE multiplicative factor to mitigate the inhomogeneity of the uncertainties of the CCD astrometric

observations.

TUWE : Transit unit-weight error (float)

This is the factor that has been applied on CCD Epoch Astrometric Errors to account for any discrepancy between CCD measurements at transit level.

RV_USED : Availability and use of radial velocity (byte)

Reports whether any radial velocity was available and used for astrometric orbits:

- bit0: whether any radial velocity was available (through CompleteSource or through the transits) or not;
- bit1: whether the available radial velocity was used to correct for μ_r.

RV_USED_IN_ASTROMETRY : Assumed radial velocity of the source used in the astrometric solution (float, Velocity[km s^{-1}])

Assumed radial velocity of the source used in the astrometric solution in km/s.

This value is the one used to account for possible perspective acceleration effects. This correction is applied even when the expected perspective acceleration is negligible.

Due to cyclic processing of the astrometry and the RVS data, this radial velocity might be different from the one computed using the latest available RVS information. Moreover, if no reliable radial velocity was available at the time of the astrometric processing, radial velocities from the litterature have been used when relevant, and their value is reported in `rv_used_in_astrometry`.

RV_N_OBS_PRIMARY : Total number of radial velocities considered for the primary (int)

Total number of epoch radial velocities considered for the primary.

RV_N_GOOD_OBS_PRIMARY : Total number of radial velocities actually used for the primary (int)

Total number of epoch radial velocities actually used for the primary.

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RV_N_OBS_SECONDARY : Total number of radial velocities considered for the secondary in the case of SB2 (int)

Total number of epoch radial velocities considered for the secondary in the case of SB2.

RV_N_GOOD_OBS_SECONDARY : Total number of radial velocities actually used for the secondary in the case of SB2 (int)

Total number of epoch radial velocities actually used for the secondary in the case of SB2.

CENTER_OF_MASS_VELOCITY : The velocity of the centre of mass (double, Velocity[km s^{-1}])

The radial velocity of the centre of mass for SB1, SB2, Pulsation (and orbits plus ScanAngle) solutions.

CENTER_OF_MASS_VELOCITY_ERROR : Standard error of the velocity of the centre of mass (float, Velocity[km s^{-1}])

Standard error of the `center_of_mass_velocity` as defined above. The standard errors are derived from the variance-covariance matrix of the final solution in the standard way.

SEMI_AMPLITUDE_PRIMARY : Semi-amplitude of the radial velocity curve for first component (double, Velocity[km s^{-1}])

The semi-amplitude of the radial velocity curve related to the first component: K1. The first component is either the only visible one (concerns SB1 solutions) or is expected to be any of the two stars (concerns SB2 solutions).

SEMI_AMPLITUDE_PRIMARY_ERROR : Standard error of semi-amplitude of the primary (float, Velocity[km s^{-1}])

Standard error of the `semi_amplitude_primary` as defined above. The standard errors are derived from the variance-covariance matrix of the final solution in the standard way.

SEMI_AMPLITUDE_SECONDARY : The semiamplitude of the radial velocity curve for second component (double, Velocity[km s^{-1}])

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The semi-amplitude of the radial velocity curve related to the second component: K2 (concerns SB2 solutions).

SEMI_AMPLITUDE_SECONDARY_ERROR : Standard error of The semiamplitude of the radial velocity curve for second component (float, Velocity[km s^{-1}])

Standard error of the `semi_amplitude_secondary` as defined above. The standard errors are derived from the variance-covariance matrix of the final solution in the standard way.

VELOCITY_SCAN_ANGLE_SINE1 : Fitted sine value of the radial velocity scan angle model (double, Velocity[km s^{-1}])

Fitted sine value of the radial velocity scan angle model, sin(scan angle).

VELOCITY_SCAN_ANGLE_SINE1_ERROR : Standard error of fitted sine value of the RV scan angle model (float, Velocity[km s^{-1}])

Standard error of `velocity_scan_angle_sine1`.

VELOCITY_SCAN_ANGLE_COSINE1 : Fitted cosine value of the radial velocity scan angle model (double, Velocity[km s^{-1}])

Fitted cosine value of the radial velocity scan angle model, cos(scan angle).

VELOCITY_SCAN_ANGLE_COSINE1_ERROR : Standard error of fitted cosine value of the RV scan angle model (float, Velocity[km s^{-1}])

Standard error of `velocity_scan_angle_cosine1`.

VELOCITY_SCAN_ANGLE_SINE2 : Fitted sine value of the radial velocity double scan angle model (double, Velocity[km s^{-1}])

Fitted sine value of the radial velocity double scan angle model, sin(2 * scan angle).

VELOCITY_SCAN_ANGLE_SINE2_ERROR : Standard error of fitted sine value of the RV double scan angle model (float, Velocity[km s^{-1}])

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Standard error of `velocity_scan_angle_sine2`.

VELOCITY_SCAN_ANGLE_COSINE2 : Fitted cosine value of the radial velocity double scan angle model (double, Velocity[km s^{-1}])

Fitted cosine value of the radial velocity double scan angle model, cos(2 * scan angle).

VELOCITY_SCAN_ANGLE_COSINE2_ERROR : Standard error of fitted cosine value of the RV double scan angle model (float, Velocity[km s^{-1}])

Standard error of `velocity_scan_angle_cosine2`.

LARGEST_GAP_IN_PHASE : The width of the largest gap in the phase (double)

When the curve is folded over the period, this is the largest gap in the phase.

CONF_SPECTRO_PERIOD : The probability of the period for not being due to (gaussian white) noise. (float)

One of the first important step in the analysis of the RV time series suspected to present variations is the computation of a Fourier periodogram and the search for a period. A probability can be associated with the selected period. This probability is 1–SL (significance level) where the SL is the probability to observe at least such a peak under the null-hypothesis of white noise. A probability of 1 indicates a very significant periodicity.

This field is only relevant for SB1, SB1C, SB2 and SB2C models and can be ignored otherwise.

F_STATISTICS : An F-statistic between the solution chi-square and the variance of Vrads (float)

This is an F-statistic between the solution chi–square and the variance, which, when combined with the solution F2, will tell us aboutincomplete models with good fits but with bad F2.

MASS_FUNCTION : The spectroscopic mass function of the SB1 solution (float[2] array, Mass[Solar Mass])

The minimum mass of the hidden companion in SB1 systems.

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SB2_ORTH_CORR_COEFF : The orthogonal correlation coefficient V2 vs V1 (float)

The orthogonal correlation coefficient between RV1 and RV2 is filled only for isSB2=true; otherwise, it is NaN. This coefficient serves as an indication of the quality of the isSB2 detection. In a SB2 system, RV1 and RV2 are expected to be anticorrelated, so a value close to -1 indicates good quality.

N_SUCESSIVE_EMPTY_BINS : The number of empty bins when splitting RV into bins (Integer)

In order to detect radial velocity curves with 2 parallel sets of Vrads, we split the radial velocities into 30 bins, and we count the highest number of successive empty bins.

N_EDGE_OBSERVATIONS : The number of observations at the edge (Integer)

After splitting radial velocities into bins, the number of observations at each side of the largest gap is counted and the minimum of the two numbers is kept.

N_SIGNIFICANT_PERIODS : The number of significant peaks in the peridogram (int[2] array)

The number of periodogram peaks above the 95

HIGHEST_PEAKS_RATIO : The ratio of two highest significant peaks in the peridogram (float[2] array)

The ratio of two highest significant peaks in the peridogram. For SB1_SB1 solutions, we will compute two periodograms. The value is NaN if less than 2 significant peaks are present.

BOOTSTRAP_PROBABILITY : Probability of the period being robust to a bootstrap procedure (float)

Probability of the period being robust to a bootstrap procedure.

It is estimated as follows. First, the time series of N measurements with a significant period P is shortened by 1 measurement N times. Second, a test determining wether the derived period P is still significant is applied to the shortened time series. Third, the number of successes (i.e. period is still significant) is divided by N to get the bootstrap probability. Whether the period is bootstrapped depends on the `solution_type`.

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PHOT_G_N_OBS : Total number of G photometry measurements considered (int)

Total number of G epoch photometry measurements considered.

PHOT_G_N_GOOD_OBS : Total number of G photometry measurements actually used (int)

Total number of G epoch photometry measurements actually used.

ESINW : Eccentricity times the sinus of the argument of periastron (double)

First combination of eccentricity (e) and argument of periastron (w): $e \sin w$.

ESINW_ERROR : Standard error on esinw (float)

Formal error on esinw. It is derived from the covariance matrix of the solution, revised by a multiplicative factor that would yield a zero goodness of fit.

ECOSW : Eccentricity times the cosinus of the argument of periastron (double)

Second combination of eccentricity (e) and argument of periastron (w): $e \cos w$.

ECOSW_ERROR : Standard error on ecosw (float)

Formal error on ecosw. It is derived from the covariance matrix of the solution, revised by a multiplicative factor that would yield a zero goodness of fit.

MASS_RATIO : Mass ratio (double)

The mass ratio $q = M_S/M_P$. It is provided for EclipsingBinary and EclipsingSpectro solutions, but is well constrained for EclipsingSpectroSB2 solutions only (cf. documentation).

MASS_RATIO_ERROR : Standard error of Mass ratio (float)

Standard error of the `mass_ratio` as defined above. In the pure spectroscopic (concerns SB2 and SB2C), the pure eclipsing, and the combined spectroscopic-eclipsing cases, the standard errors are derived from the variance-covariance matrix of the final solution in the standard way.

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In all the other cases, the standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain a goodness-of-fit = 0.

FILL_FACTOR_PRIMARY : Fill factor of primary (double)

The fill factor of the primary component, as determined from the eclipsing case. When $s_1 \in [0, 1)$ the component does not fill its Roche lobe. When $s_1 = 1$ the component fills its Roche lobe exactly (semi-detached system). When $s_1 > 1$ then $s_2 > 1$ as well, and the binary system is over-contact (common envelope system).

FILL_FACTOR_PRIMARY_ERROR : Standard error of Fill factor of primary (float)

Standard error of the fill factor of the primary. It is derived from the covariance matrix of the solution, revised by a multiplicative factor that would yield a zero goodness of fit.

FILL_FACTOR_SECONDARY : Fill factor of secondary (double)

The fill factor of the secondary component as determined from the eclipsing case. When $s_2 \in [0, 1)$ the component does not fill its Roche lobe. When $s_2 = 1$ the component fills its Roche lobe exactly (semi-detached system). When $s_2 > 1$ then $s_1 > 1$ as well, and the binary system is over-contact (common envelope system).

FILL_FACTOR_SECONDARY_ERROR : Standard error of Fill factor of secondary (float)

Standard error of the fill factor of the secondary. It is derived from the covariance matrix of the solution, revised by a multiplicative factor that would yield a zero goodness of fit.

TEMPERATURE_PRIMARY : Effective temperature of primary (double, Temperature[K])

Effective temperature of primary.

TEMPERATURE_PRIMARY_ERROR : Standard error of effective temperature of primary (float, Temperature[K])

Standard error of Effective temperature of primary. It is derived from the covariance matrix of the solution, revised by a multiplicative factor that would yield a zero goodness of fit.

TEMPERATURE_SECONDARY : Effective temperature of secondary (double, Temperature[K])

Effective temperature of secondary.

TEMPERATURE_SECONDARY_ERROR : Standard error of effective temperature of secondary (float, Temperature[K])

Standard error of Effective temperature of secondary. It is derived from the covariance matrix of the solution, revised by a multiplicative factor that would yield a zero goodness of fit.

TEMPERATURE_RATIO : Ratio of the effective temperatures (double)

Ratio of the fitted effective temperature over that of the unfitted effective temperature.

See `temperature_ratio_definition` for a description of the fitting scenario.

TEMPERATURE_RATIO_ERROR : Standard error of the ratio of the effective temperatures (float)

Standard error of the ratio of the effective temperatures, `temperature_ratio`. It is derived from the covariance matrix of the solution, revised by a multiplicative factor that would yield a zero goodness of fit.

TEMPERATURE_RATIO_DEFINITION : Code defining which fitting scenario did apply to the effective temperature (byte)

Code defining the temperature ratio:

1: temperature of the primary (fitted parameter) over temperature of the secondary (fixed parameter)

2: temperature of the secondary (fitted parameter) over temperature of the primary (fixed parameter)

3: temperature of the primary (fitted parameter) over temperature of the secondary (fitted parameter)

with the convention that the primary component is always the one which is the more luminous in the *G* band.

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R_POLE_SUM : Sum of the polar radii of primary and secondary (in units of the semi-major axis) (double)

The sum of the polar radii of the primary and secondary in the Roche model corresponding to the eclipsing solution. Provided as a convenience.

R_L1_POINT_SUM : L1-pointing radii of primary and secondary (in units of the semi-major axis) (double)

The sum of the radii of the primary and secondary that points towards the first Lagrange point (L_1) in the Roche model corresponding to the eclipsing solution. Provided as a convenience.

R_SPHERE_SUM : Sum of the radii of sphere having the same volume as the primary and secondary (in units of the semi-major axis (double)

The sum of the radii of a sphere having the same volume as that of the Roche model for the primary and secondary. Provided as a convenience.

ECLIPSE_TIME_PRIMARY : Time of mid-eclipse of the primary by the secondary (double, Time[Julian Date (day)])

Time of mid-eclipse of the primary by the secondary, expressed relative to the same reference epoch as the periastron epoch, i.e., `gaia_source.ref_epoch`.

ECLIPSE_TIME_SECONDARY : Time of mid-eclipse of the secondary by the primary (double, Time[Julian Date (day)])

Time of mid-eclipse of the secondary by the primary, expressed relative to the same reference epoch as the periastron epoch, i.e., `gaia_source.ref_epoch`.

ECLIPSE_DURATION_PRIMARY : Duration of primary eclipse assuming spherical components (double, Time[day])

An estimation of primary eclipse duration assuming spherical components with radii equal to the polar radii of the Roche model. This is generally accurate for detached systems but an underestimate for overcontact ones.

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ECLIPSE_DURATION_SECONDARY : Duration of secondary eclipse assuming spherical components (double, Time[day])

An estimation of secondary eclipse duration assuming spherical components with radii equal to the polar radii of the Roche model. This is generally accurate for detached systems but an underestimate for overcontact ones.

G_LUMINOSITY_RATIO : Ratio of the G-band luminosity of the secondary over the primary (float)

Ratio of the G-band luminosity of the secondary over the primary.

G_RANK : Rank of the G-band solution (float)

An estimate of the quality of the fit based on the 'fraction of variance unexplained' (FVU) of the Eclipsing Binary model in the G band.

BP_RANK : Rank of the BP-band solution (float)

An estimate of the quality of the fit based on the 'fraction of variance unexplained' (FVU) of the Eclipsing Binary model in the BP band.

RP_RANK : Rank of the RP-band solution (float)

An estimate of the quality of the fit based on the 'fraction of variance unexplained' (FVU) of the Eclipsing Binary model in the RP band.

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10.9 NSS_VIM_FL [TAP]

This table contains non-single-star models for sources compatible with an Variability Induced Mover (VIM) solution.

Several possible models are in principle hosted within the same table and they are indicated by the field `solution_type`. The description of this latter lists all possible solution types considered for this release. Only a selection of parameters hosted in this table are provided here, depending on the solution. The details of those is given in the description of field `solution_type`, which can also be used to extract the relevant elements of the correlation vector `correlations`.

Details about the formalism used to derive the parameters in this table are given in the on-line documentation, see Chapter **??**.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Source Identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

SOLUTION_TYPE : NSS model adopted (string)

This is the non-single star model which has been adopted for the published solution, see online documentation, Chapter **??**, for details.

The solution types covered in table `nss_vim_fl`, the corresponding fitted variables and their order in the `correlations` vector are listed below. All variables are always fitted for all solution types. A bit index field is thus not necessary.

- VIMF: Variability-Induced Mover with fixed configuration

  - ra: right ascension (fitted variable is an offset against intermediate AGIS results).
  - dec: declination (fitted variable is an offset against intermediate AGIS results)
  - parallax: parallax at catalogue epoch.
  - pmra: proper motion at catalogue epoch.
  - pmdec: roper motion at catalogue epoch.
  - vim_d_ra: VIM coordinate offset in right ascension.
  - vim_d_dec: VIM coordinate offset in declination.
- VIML: Variability-Induced Mover with linear motion of the constant star
  - ra: right ascension (fitted variable is an offset against intermediate AGIS results).
  - dec: declination (fitted variable is an offset against intermediate AGIS results)
  - parallax: parallax at catalogue epoch.
  - pmra: proper motion at catalogue epoch.
  - pmdec: roper motion at catalogue epoch.
  - vim_d_ra: VIM coordinate offset in right ascension.
  - vim_d_dec: VIM coordinate offset in declination.
  - derivative_vim_d_ra: time derivative of the VIM coordinate offset in right ascension.
  - derivative_vim_d_dec: time derivative of the VIM coordinate offset in declination.

CORRELATIONS : Vector form of the upper triangle of the correlation matrix (float[] array)

Correlation matrix of the fitted profile parameters for the applicable non-single star solution. The parameters stored in this matrix and their order is given in the description of field solution_type. Since not all parameters of a given solution model are systematically fitted, the matrix may contain NaN values at the elements of the corresponding indices, such as the length of the vector always corresponds to the expected length of the theoretical model.

Only non-zero, non-unity, correlation coefficients from the correlation matrix **M** are provided here. They are served as a linear array of constant size $S = n(n-1)/2$ corresponding to the full normal matrix of dimension $n \times n$. The ordering of the elements in the linear array follows a column-major storage scheme, i.e.:

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$$\mathbf{M} = \begin{bmatrix} 1 & C[0] & C[1] & C[3] & C[6] & \cdots & C[S-(n-1)] \\ & 1 & C[2] & C[4] & C[7] & \cdots & C[S-(n-2)] \\ & & 1 & C[5] & C[8] & \cdots & C[S-(n-3)] \\ & & & 1 & C[9] & \cdots & C[S-(n-4)] \\ & & & & \ddots & \ddots & \vdots \\ & & & & & 1 & C[S-1] \\ & & & & & & 1 \end{bmatrix}$$

OBJ_FUNC_VALUE : Value of the objective function at the solution (float)

The χ^2, defined as $\sum_{i=1}^{n}(x_i - y_i)^2/\sigma_i^2$, where x_i is the abscissa of the ith transit, y_i is the value calculated from the model, and σ_i is the uncertainty obtained from the astrometric reduction, without any correction.

GOF : Goodness of fit in the Hipparcos sense (float)

Goodness-of-fit statistic of the solution. This is the 'gaussianized chi-square' (Wilson & Hilferty (1931)'s cube root transformation), which for good fits should approximately follow a normal distribution with zero mean value and unit standard deviation.

This statistic is computed according to the formula:

$$F2 = \sqrt{\frac{9\nu}{2}}\left(\sqrt[3]{\frac{\texttt{obj_func_value}}{\nu}} + \frac{2}{9\nu} - 1\right)$$

where `obj_func_value` is hopefully a χ^2 and ν is the number of degrees of freedom.

SIGNIFICANCE : The significance of the solution (i.e. how worth keeping a model is) (float)

It turns out that `gof` is not always enough to decide whether a model is worth keeping or not. The significance addresses some these limitations.

For astrometric binaries, it is defined as a function of the parameters that characterise the model, divided by its uncertainty. This uncertainty is derived from the solution covariance matrix, where the uncertainties are corrected in order to obtain the corrected goodness-of-fit $F_{2\ corrected} = 0$. The function characterising the model is as follows: (a) for the acceleration 7 model, it is the norm of the acceleration vector; (b) for the acceleration 9 model, it is the norm of the time derivative of the acceleration vector.

EFFICIENCY : Significance of the solution (i.e. how worth keeping a model is) (float)

The efficiency expresses the level of correlation between the parameters of a model. A value of 1 means a total absence of correlation whereas it falls to 0 as the correlation increases. It is defined as the n-th root of the ratio of the product of the diagonal elements of the covariance matrix and the product of the eigen value of that matrix. When all the covariances are 0, the matrix is diagonal and the ratio is exactly 1.

See also Eichhorn (1989).

FLAGS : Quality flags for the achieved NSS solution (long)

Processing flag applicable to specific non-single-star solutions. The meaning of each of those is given in the table below.

Flags bit number	Flag Meaning	Comment
Astrometric binary solutions		
0	Good solution	0 if bad solution, 1 for good solution
1	Non random residuals	1 if there is some signal remaining in the residuals
2-7	NA	

RV_USED : Availability and use of radial velocity (byte)

Reports whether any radial velocity was available and used:

- bit0: whether any radial velocity was available (through CompleteSource or through the transits) or not;
- bit1: whether the available radial velocity was used to correct for μ_r.

ASTROMETRIC_JITTER : Uncorrelated astrometric jitter term (float, Angle[mas])

In order to account for some poorly calibrated effects, this is an excess astrometric noise quadratically added to the uncertainty on the observations such that the resulting gaussianised Goodness of Fit is null. This applies to OrbitalAlternative[Validated] and OrbitalTargetedSearch[Validated] solutions only.

ASTROMETRIC_N_OBS_AL : Total astrometric CCD observations in AL considered (int)

Total astrometric CCD observations considered in the along-scan direction.

ASTROMETRIC_N_GOOD_OBS_AL : Total astrometric CCD observations in AL actually used (int)

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Total astrometric CCD observations actually used in the along-scan direction.

RA : Right ascension (double, Angle[deg])

Right ascension α of the source in ICRS at the reference epoch `gaia_source.ref_epoch`, if the total flux is then equal to the reference flux.

RA_ERROR : Standard error of right ascension (float, Angle[mas])

Standard error $\sigma_{\alpha*} \equiv \sigma_\alpha \cos\delta$ of the right ascension of the source in ICRS at the reference epoch `gaia_source.ref_epoch`. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

DEC : Declination (double, Angle[deg])

Declination δ of the source in ICRS at the reference epoch `gaia_source.ref_epoch`, if the total flux is then equal to the reference flux.

DEC_ERROR : Standard error of declination (float, Angle[mas])

Standard error σ_δ of the declination of the source in ICRS at the reference epoch `gaia_source.ref_epoch`. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

PARALLAX : Parallax (double, Angle[mas])

Absolute stellar parallax ϖ of the source at the reference epoch `gaia_source.ref_epoch`

PARALLAX_ERROR : Standard error of parallax (float, Angle[mas])

Standard error σ_ϖ of the stellar parallax at the reference epoch `gaia_source.ref_epoch`. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

PMRA : Proper motion in right ascension direction (double, Angular Velocity[mas yr^{-1}])

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Proper motion in right ascension $\mu_{\alpha*} \equiv \mu_\alpha \cos\delta$ of the source in ICRS at the reference epoch `gaia_source.ref_epoch`. This is the local tangent plane projection of the proper motion vector in the direction of increasing right ascension.

PMRA_ERROR : Standard error of proper motion in right ascension direction (float, Angular Velocity[mas yr^{-1}])

Standard error $\sigma_{\mu\alpha*}$ of the local tangent plane projection of the proper motion vector in the direction of increasing right ascension at the reference epoch `gaia_source.ref_epoch`. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

PMDEC : Proper motion in declination direction (double, Angular Velocity[mas yr^{-1}])

Proper motion in declination μ_δ of the source at the reference epoch `gaia_source.ref_epoch`. This is the projection of the proper motion vector in the direction of increasing declination.

PMDEC_ERROR : Standard error of proper motion in declination direction (float, Angular Velocity[mas yr^{-1}])

Standard error $\sigma_{\mu\delta}$ of the proper motion component in declination at the reference epoch `gaia_source.ref_e`
The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

REF_FLUX_G : Reference flux in the G band (float, Flux[e$^-$ s^{-1}])

The photocentre of a VIM object moves in relation to its total flux in the G band. In a VIMF solution, the position (`ra, dec`) is the mean position of the object when its total flux is the reference flux. In practice, the reference flux is close to (or is) the median flux of the object.

VIM_D_RA : VIM coordinate offset in RA (double, Angle[mas])

The coordinate offset, on the RA axis and measured from the variable component, of the position of the photocentre when the total flux is equal to the reference flux. The right ascension of the variable component is therefore: `ra` - `vim_d_ra`.

VIM_D_RA_ERROR : Standard error of VIM coordinate offset in RA (float, Angle[mas])

Standard error of the coordinate offset of the photocentre on the RA axis. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

VIM_D_DEC : VIM coordinate offset in DEC (double, Angle[mas])

The coordinate offset, on the declination axis and measured from the variable component, of the position of the photocentre when the total flux is equal to the reference flux. The declination of the variable component is therefore: `dec-vim_d_dec`.

VIM_D_DEC_ERROR : Standard error of VIM coordinate offset in Dec (float, Angle[mas])

Standard error of the coordinate offset of the photocentre on the declination axis. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

DERIVATIVE_VIM_D_RA : Time derivative of VIM coordinate offset in RA (double, Angular Velocity[mas yr^{-1}])

For VIML: time derivative of the coordinate offset, on the RA axis and measured from the variable component, of the position of the photocentre when the total flux is equal to the reference flux.

DERIVATIVE_VIM_D_RA_ERROR : Standard error of Time derivative of VIM coordinate offset in RA (float, Angular Velocity[mas yr^{-1}])

Standard error of the time derivative of the coodinate offset of the photocentre on the RA axis. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

DERIVATIVE_VIM_D_DEC : Time derivative of VIM coordinate offset in DEC (double, Angular Velocity[mas yr^{-1}])

For VIML: time derivative of the coordinate offset, on the declination axis and measured from the variable component, of the position of the photocentre when the total flux is equal to the reference flux.

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DERIVATIVE_VIM_D_DEC_ERROR : Standard error of Time derivative of VIM coordinate offset in DEC (float, Angular Velocity[mas yr^{-1}])

Standard error of the time derivative of the coodinate offset of the photocentre on the declination axis. The standard errors are derived from the variance-covariance matrix of the solution, correcting the measurement uncertainties to obtain the goodness-of-fit F2=0.

TUWE : Transit unit-weight error (float)

This is the factor to be applied on CCD Epoch Astrometric Errors to account for correlations between CCDs or any other discrepancy at transit level.

TUWE_UNCERTAINTY_RESCALED : Have the transit-binned uncertainties been multiplied by the TUWE? (boolean)

Boolean indicating whether the transit-binned uncertainties have been reweighted by the TUWE multiplicative factor to mitigate the inhomogeneity of the uncertainties of the CCD astrometric observations.

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10.10 OPTICAL_PAIR [TAP]

This table contains pairs of sources analysed in the context of the resolved binary pipeline, and for which a best astrometric solution corresponds to that of two chance alignment sources.

For those, the astrometry solution to use is that tabulated in `gaia_source` and only the separation derived by the resolved binary pipeline is reported here. As such, this latter can differ slightly from the one derived from the `gaia_source` coordinates.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Source Identifier for the primary source (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`). It corresponds here to the source identifier for the first component of the resolved pair.

SOURCE_ID_SECONDARY : Source Identifier for the secondary source (long)

Source identifier for the second component of the resolved pair.

SEPARATION : Angular separation (double, Angle[mas])

Average angular separation between the two components.

ACCELERATION_FLAG : Flag indicative of acceleration solution in the pair (byte)

This flag indicates whether one of the sources in the optical pair was identified as an unresolved acceleration solution. A value of zero for this flag indicates that no acceleration was detected.

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For detected acceleration the following coding is used:

- (least–significant) bit 1 set to 1 indicates that an acceleration was detected in the brightest source of the pair (`source_id`);
- bit 2 set to 1 indicates that an acceleration was detected in the faintest source of the pair (`source_id_secondary`).

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11 Photometry

As indicated in the below subsection titles by the labels [TAP] and [DataLink], these products are served from the ESA Gaia Archive using two different protocols, the Table Access Protocol (TAP) for tabular products and the DataLink protocol for other products such as spectra, light curves, MCMC samples, etc. For more details please see Section **??**.

11.1 EPOCH_PHOTOMETRY [DataLink]

The epoch photometry table contains the photometric time series at FoV level for a given object in bands G, BP and RP as stored in the Datalink Massive data base.

For G-band epoch photometry information at CCD level, please refer to table `epoch_photometry_ccd`.

Note this table is not available through the main archive TAP interface. Data are delivered via the Massive Data service indexed by the VO DataLink protocol and described in Chapter **??**. For example this can be actioned in the archive user interface by joining queries with the table `all_source_flags` and selecting `has_epoch_photometry = 'true'`. The DataLink retrieval type to specify for this product in programmatic archive access via the Astroquery Gaia Python package is `EPOCH_PHOTOMETRY`.

DataLink products are served with a choice of two different data structures:

- `DATAMODEL_STANDARD`: the product is serialised following a standard data model (for example an IVOA-compliant data model in applicable cases);
- `DATAMODEL_GAIA`: the product is serialised following the Gaia DPAC(-internal) data model.

The `DATAMODEL_GAIA` format is described here whereas the `DATAMODEL_STANDARD` format is described in `https://www.cosmos.esa.int/web/gaia-users/archive/datalink-products#datalink_serialisation`

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the

data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (unique within a particular Data Release) (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

TRANSIT_ID : Unique transit identifier (long)

The `transit_id` is a unique identifier assigned to each detected (and confirmed) source as it transits the Gaia focal plane. Each time a given source is detected as Gaia scans and re-scans the sky a new `transit_id` will be created to badge that apparition. Hence the along–scan time and the across–scan position along with the telescope in which the source was detected are used to form a unique integer with which to label the transit.

The several features of a detection that are encoded in `transit_id` can be easily retrieved using bit masks (&) and shifts (>>) as follows:

- On-Board Mission Time line [ns]
  = 204800 * ((`transit_id` >> 17) & (0x000003FFFFFFFFFF))
- Field-of-view = 1 + (`transit_id` >> 15) & 0x03 [1 for 'preceding' and 2 for 'following' fields-of-view respectively]
- CCD row = (`transit_id` >> 12) & 0x07 [dimensionless, in the range 1 to 7]
- Across-scan 'reference acquisition pixel' in strip AF1 = (`transit_id`) & 0x0FFF [pixels] (this is the across-scan centre of the AF1 window and is odd if immediately below the mid-point of the window and even if immediately above)

where the bit mask prefix '0x' denotes hexadecimal.

For further details see Portell et al. (2020). For convenience a decoder for `transit_id` is available on-line at

`https://gaia.esac.esa.int/decoder/transitidDecoder.jsp`

G_N_OBS : Number of CCD observations contributing to transit averaged G band flux (int)

Number of CCD transits contributing to the transit averaged G flux (`g_flux` and `g_flux_error`).

Note that the parameter range is not [1-10], but [0-9] as the Sky Mapper G flux is not used in the calculation of the transit averaged G flux. In the case of `g_n_obs` = 0, all 9 AF fluxes have been discarded because of non nominal acquisition and/or processing rejection.

G_OBS_TIME : Transit averaged G band observing time (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Field-of-view transit averaged observation time in units of Barycentric JD (in TCB), in days minus 2 455 197.5, computed as follows. First the observation time is converted from On-board Mission Time (OBMT) into Julian date in TCB (Temps Coordonnée Barycentrique). Next a correction is applied for the light-travel time to the Solar system barycentre, resulting in Barycentric Julian Date (BJD). Finally an offset of 2 455 197.5 days is applied (corresponding to a reference time T_0 at 2010-01-01T00:00:00) to have a conveniently small numerical value. Although the centroiding time accuracy of the individual CCD observations is (much) below 1 ms, this per-FoV observation time is averaged over typically 9 CCD observations taken in a time range of about 44 sec.

G_FLUX : Transit averaged G band flux (float, Flux[e^- s^{-1}])

The average G flux value, a combination of individual SM-AF CCD fluxes.

G_FLUX_ERROR : Transit averaged G band flux error (float, Flux[e^- s^{-1}])

The uncertainty `g_flux_error` on `g_flux` is the uncertainty on the weighted mean G flux of the set of AF individual CCD observations for the transit. This accounts for intrinsic scatter in the data. The exact formula and further details are given in Carrasco et al. (2017).

G_FLUX_OVER_ERROR : Transit averaged G band flux divided by its error (float)

Mean flux in the G band divided by its error.

G_MAG : Transit averaged G band Vega magnitude (float, Magnitude[mag])

G band Vega magnitude. It is computed from the flux applying the DR4 zero-point detailed in Section **??** in the online documentation.

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BP_OBS_TIME : BP CCD transit observing time (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Observation time of the BP CCD transit in units of Barycentric JD (in TCB), in days minus 2 455 197.5, computed as follows. First the observation time is converted from On-board Mission Time (OBMT) into Julian date in TCB (Temps Coordonnée Barycentrique). Next a correction is applied for the light-travel time to the Solar system barycentre, resulting in Barycentric Julian Date (BJD). Finally an offset of 2 455 197.5 days is applied (corresponding to a reference time T_0 at 2010-01-01T00:00:00) to have a conveniently small numerical value.

BP_FLUX : BP band flux (float, Flux[e^- s^{-1}])

The integrated BP flux value of the transit. If the BP flux has been rejected or is unavailable, the flux will be set to NaN.

BP_FLUX_ERROR : BP band flux error (float, Flux[e^- s^{-1}])

The uncertainty `bp_flux_error` on `bp_flux` is the uncertainty on the BP flux. This is the uncertainty associated with the single BP CCD transit available for a FoV transit. It includes photon noise and all applicable calibration errors. If the BP flux has been rejected or is unavailable, the value will be set to NaN (in VOTable and FITS formats) and an empty string (in plain text CSV format).

BP_FLUX_OVER_ERROR : BP band flux divided by its error (float)

Mean flux in the BP band divided by its error. If the BP flux has been rejected or is unavailable, this value will be set to NaN.

BP_MAG : BP band Vega magnitude (float, Magnitude[mag])

BP band Vega magnitude. It is computed from the flux applying the DR4 zero-point detailed in Section **??** in the online documentation.

RP_OBS_TIME : RP CCD transit observing time (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Observation time of the RP CCD transit in units of Barycentric JD (in TCB), in days minus 2 455 197.5, computed as follows. First the observation time is converted from On-board Mission

Time (OBMT) into Julian date in TCB (Temps Coordonnée Barycentrique). Next a correction is applied for the light-travel time to the Solar system barycentre, resulting in Barycentric Julian Date (BJD). Finally an offset of 2 455 197.5 days is applied (corresponding to a reference time T_0 at 2010-01-01T00:00:00) to have a conveniently small numerical value.

RP_FLUX : RP band flux (float, Flux[e^- s^{-1}])

The integrated RP flux value of the transit. If the RP flux has been rejected or is unavailable, the flux will be set to NaN.

RP_FLUX_ERROR : RP band flux error (float, Flux[e^- s^{-1}])

The uncertainty `rp_flux_error` on `rp_flux` is the uncertainty on the RP flux. This is the uncertainty associated with the single RP CCD transit available for a FoV transit. It includes photon noise and all applicable calibration errors. If the RP flux has been rejected or is unavailable, the value will be set to NaN (in VOTable and FITS formats) and an empty string (in plain text CSV format).

RP_FLUX_OVER_ERROR : RP band flux divided by its error (float)

Mean flux in the RP band divided by its error. If the RP flux has been rejected or is unavailable, this value will be set to NaN.

RP_MAG : RP band Vega magnitude (float, Magnitude[mag])

RP band Vega magnitude. It is computed from the flux applying the DR4 zero-point detailed in Section **??** in the online documentation.

TRANSIT_ACQ_FLAGS : Transit-level information on the on-board acquisition (short)

This field contains a set of bitwise flags (from Bit 15 = msb to Bit 0 = lsb) related to the data acquisition. It has some oddities for historical reasons.

Windows may suffer truncation or trimming. Here, truncation refers to a reduction in size - often irregular - caused by an intra-field conflict with another window, while trimming refers to the permanent use of a reduced window size across scan. Truncation and trimming only affect 1D windows and never affect SM and AF1 windows. Trimming was only used for AF2-9 and only for a short period early in the mission (around OMBT revolutions 2103 to 2238).

The shape of all 1D AF2-9 windows is detailed in `epoch_image.shape_af`.

The field is coded as follows:

- Bits 15 and 14: unused
- Bit 13 (0x2000): Always set.
- Bit 12 (0x1000): The AF2-9 windows are trimmed AC but have not suffered truncation.
- Bits 11-9: unused
- Bits 7 and 8 (0x0180): `xp_offset`. This is the value of a continuous two-bit TDI counter. It is needed for the reconstruction of the XP windows.
- Bit 6 (0x0040): AF2-9 windows are 2D.
- Bit 5 (0x0020): AF2-9 windows are 1D (complement of Bit 6).
- Bit 4 (0x0010): AF2-9 windows are 1D and with the full AC size of 12 pixels.
- Bit 3 (0x0008): No BP window acquired.
- Bit 2 (0x0004): No RP window acquired.
- Bit 1 (0x0002): BP window truncated.
- Bit 0 (0x0001): RP window truncated.

TRANSIT_PROC_FLAGS : Transit-level information on the processing results (short)

This field contains a set of bitwise flags, coded as follows:

- Bits 3 to 0 (`transit_proc_flags` AND 0x000F): **IPD strategy and findings**. As of IDT 15.1 and until IDT 21.0, as well as in IDU until DRC-02, this field was not used (0 was always stored here). As of IDU DRC-03, the following information is provided:

  '0': Normal IPD. Ambiguity not tested or not detected. Default value for all records produced until IDT 21.0 and IDU until DRC-02

  '1': Multiple Source IPD: As of IDT 21.1, this value is used to indicate that the earlier object in time detected in the window has been selected. In case of 2D windows with larger AC than AL separation between detected peaks: higher μ selected.

'2': Multiple Source IPD: As of IDT 21.1, this value is used to indicate that the later object in time detected in the window has been selected. In case of 2D windows with larger AC than AL separation between detected peaks: lower μ selected.

'3': Multiple Source IPD: As of IDU DRC-03, this value is used to indicate that at least one of the windows has a multiple source and that these disturbing sources were masked for the IPD processing. `ccd_proc_flags` can be checked to know which and how many windows have been affected.

'4': DIPD: As of IDU 21.0, this value is used to indicate that this object has been obtained using DIPD implementation and thus include a window model template and combined AF solution.

'5': TukeyBiWeight IPD: this value is used to indicate that this object has been obtained using TukeyBiWeight IPD initialisation for the internal processing subsystem IDU-AttAc; such initial values do not propagate into the published data.

Values 6-15 are reserved for future extensions.

- Bits 6 to 4 (0x0070): **Bias and PEM OI strategy baseline**: this field indicates the Bias mitigation mode requested to process this transit. The consumer must check the corresponding `ccd_proc_flags` field to determine the actual mode used for each individual CCD as sometimes some modes can not be applied due to missing input data or calibration limitations.

  Value '0': Default initialisation value for the Bias strategy. If found in an AstroElementary it means that something went wrong and the Bias (or maybe just its flagging) was not done appropriately. Thus, '0' means 'UNDEFINED/FAILURE'.

  Value '1': Only PreScan bias used.

  Value '2': PreScan + common NU baseline.

  Value '3': Full PEM NU.

  Values 4-7 are reserved for future extensions.

- Bits 10 to 7, **colour info**:

  Bit 7 (0x0080), **default colour**: '1' means 'a default colour used for IPD processing', which can be caused by some problem in the colour determination or retrieval (e.g. no BP/RP data in IDT or no valid PhotPipe fluxes in IDU).

  Bit 8 (0x0100), **epoch colour**: '1' means 'epoch (transit-level) colour used for IPD processing', '0' means 'mean (source-level) colour used'.

  Bit 9 (0x0200), **single-param colour**: '1' means 'single-parameter colour used for IPD processing' (such as `nu_eff_used_in_astrometry`), '0' means 'multi-parameter colour used' (such as SSCs).

  Bit 10 (0x0400), **colour problems**: '1' means 'missing colour information or dubious colour quatiliy' (e.g. problematic/missing BP/RP data in IDT, NaN or rejected PhotPipe colour in IDU, etc).

Please note some remarkable combinations of this bit with Bit 7:

  - Non-default + no problems (Bit 7 = 0, Bit 10 = 0): Good colour from catalogue or epoch data
  - Non-default + problems (Bit 7 = 0, Bit 10 = 1): Dubious colour from catalogue or best approximation from partial or problematic epoch data. Implemented only in IDT.
  - Default + no problems (Bit 7 = 1, Bit 10 = 0): Default colour has been used because no colour information was available (i.e. for blacklisted observations). Implemented only in IDU.
  - Default + problems (Bit 7 = 1, Bit 10 = 1): Default colour used instead of the available colour from catalogue or epoch data (i.e. dubious colour rejected).

- Bit 11 (0x0800), **cosmetic effects** flagging: '1' means 'dead pixel columns affecting some of the windows'.
- Bit 12 (0x1000), **cosmetic effects** flagging: '1' means 'anomalously low samples found in some of the windows'.
- Bit 13 (0x2000), **cosmetic effects** flagging: '1' means 'anomalously high samples found in some of the windows'.
- Bit 14 (0x4000), **problems in initial centroid**: '1' means that problems have been found in the Initial Centroids determination. In the case of IDT, combined with the 'some problem found in the colour determination' flag (bit 10), it may indicate problems in the TransitPredictor (e.g. predicted position outside window, or no attitude available) in which case a default position is used for colour determination.
- Bit 15, **attitude issues**: '1' indicates that some attitude problem was found affecting some of the windows i.e. missing attitude or problems computing AL/AC rates. This flag is only set by IDU which depends on the attitude to access the ELSF and for the combine AF processing in DIPD.

See Chapter **??** of the DR4 online documentation for more information.

PHOT_CCD_PROC_FLAGS : CCD-level information on PhotPipe processing (short[12] array)

This field reports information generated during the PhotPipe processing. The type of this field is an array of 12 short (SM, AF1-9, BP and RP). Each short will bit-wise encode the following meanings when set to a value of '1':

- Bit 0: a saturation correction was applied to this CCD transit

- Bit 1: predicted position computation failed or predicted position outside window
- Bit 2: the AC centring error was clamped when the LS calibration was applied (refer to the end-of-cycle documentation for details on the clamping ranges used)
- Bit 3: no calibration available (no calibrated CCD photometry will be available in this case)
- Bit 4: calibration rejected and replaced with neighbouring calibration
- Bit 5: calibration application failure
- Bit 6: excluded from the generation of the mean source photometry, due to pre-filtering based on available flags (this is different from Bit 7 that reports on outlier rejection)
- Bit 7: rejected in the source accumulation process; beware that outlier rejection will only affect the source photometry accumulation, i.e. when computing FoV level photometry ALL available calibrated CCD photometric data will be used regardless of the value of this flag
- Bit 8: observation affected by satellite event
- Bit 9: charge release correction applied; this is only applicable for BP/RP
- Bit 10: stray-light mitigation applied; this is only applicable for BP/RP
- Bit 11: local (KNN) background mitigation applied; this is only applicable for BP/RP
- Bit 12: crowding information available in the table `xp_epoch_crowding`; this is only applicable for BP/RP
- Bit 13: the transit is affected by truncation or multiple gate configuration in a way that is considered unrecoverable; this is only applicable for BP/RP
- Bit 14: the transit has been treated for truncation or multiple gate configuration; this is only applicable for BP/RP

Remaining bits are available for future extensions.

XP_CCD_INFO_FLAGS : CCD-level information on the input BP/RP observations (byte[2] array)

Flag providing information at the CCD level. This is only applicable to BP/RP CCD transits as similar information for the SM/AF CCDs is provided in the propagated AstroElementary flags.

- Bit 0: affected by numerical saturation
- Bit 1: affected by hot column

Other bits are available for future extensions.

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11.2 EPOCH_PHOTOMETRY_CCD [DataLink]

The epoch photometry table at CCD level contains the light curve for a given object in band G (both SM and AF CCDs) as stored in the Datalink Massive data base.

For epoch photometry information at FoV level in the G, BP and RP bands, please refer to table `epoch_photometry`.

Note this table is not available through the main archive TAP interface. Data are delivered via the Massive Data service indexed by the VO DataLink protocol and described in Chapter **??**. For example this can be actioned in the archive user interface by joining queries with the table `all_source_flags` and selecting `has_epoch_photometry = 'true'`. The DataLink retrieval type to specify for this product in programmatic archive access via the Astroquery Gaia Python package is `EPOCH_PHOTOMETRY_CCD`.

DataLink products are served with a choice of two different data structures:

- `DATAMODEL_STANDARD`: the product is serialised following a standard data model (for example an IVOA-compliant data model in applicable cases);
- `DATAMODEL_GAIA`: the product is serialised following the Gaia DPAC(-internal) data model.

The `DATAMODEL_GAIA` format is described here whereas the `DATAMODEL_STANDARD` format is described in `https://www.cosmos.esa.int/web/gaia-users/archive/datalink-products#datalink_serialisation`

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (unique within a particular Data Release) (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

TRANSIT_ID : Transit unique identifier (long)

The `transit_id` is a unique identifier assigned to each detected (and confirmed) source as it transits the Gaia focal plane. Each time a given source is detected as Gaia scans and re-scans the sky a new `transit_id` will be created to badge that apparition. Hence the along–scan time and the across–scan position along with the telescope in which the source was detected are used to form a unique integer with which to label the transit.

The several features of a detection that are encoded in `transit_id` can be easily retrieved using bit masks (&) and shifts (>>) as follows:

- On-Board Mission Time line [ns]
  = 204800 * ((`transit_id` >> 17) & (0x000003FFFFFFFFF))
- Field-of-view = 1 + (`transit_id` >> 15) & 0x03 [1 for 'preceding' and 2 for 'following' fields-of-view respectively]
- CCD row = (`transit_id` >> 12) & 0x07 [dimensionless, in the range 1 to 7]
- Across-scan 'reference acquisition pixel' in strip AF1 = (`transit_id`) & 0x0FFF [pixels] (this is the across-scan centre of the AF1 window and is odd if immediately below the mid-point of the window and even if immediately above)

where the bit mask prefix '0x' denotes hexadecimal.

For further details see Portell et al. (2020). For convenience a decoder for `transit_id` is available on-line at

`https://gaia.esac.esa.int/decoder/transitidDecoder.jsp`

SM_OBS_TIME : SM observing time (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Sky Mapper observation time in units of Barycentric JD (in TCB), in days minus 2 455 197.5, computed as follows. First the observation time is converted from On-board Mission Time (OBMT) into Julian date in TCB (Temps Coordonnée Barycentrique). Next a correction is applied for the light-travel time to the Solar system barycentre, resulting in Barycentric Julian Date (BJD). Finally an offset of 2 455 197.5 days is applied (corresponding to a reference time T_0 at 2010-01-01T00:00:00) to have a conveniently small numerical value.

SM_FLUX : SM flux (float, Flux[e^- s^{-1}])

The Sky Mapper flux value of the transit.

SM_FLUX_ERROR : SM flux error (float, Flux[e^- s^{-1}])

The error on the Sky Mapper Flux.

SM_FLUX_OVER_ERROR : SM flux divided by its error (float)

Flux in the Sky Mapper divided by its error.

SM_MAG : SM Vega magnitude (float, Magnitude[mag])

The Sky Mapper magnitude of the transit.

AF_OBS_TIME : CCD-level G band observing time (double[9] array, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Individual AF CDD observation time in units of Barycentric JD (in TCB), in days minus 2 455 197.5, computed as follows. First the observation time is converted from On-board Mission Time (OBMT) into Julian date in TCB (Temps Coordonnée Barycentrique). Next a correction is applied for the light-travel time to the Solar system barycentre, resulting in Barycentric Julian Date (BJD). Finally an offset of 2 455 197.5 days is applied (corresponding to a reference time T_0 at 2010-01-01T00:00:00) to have a conveniently small numerical value.

AF_FLUX : CCD-level G band fluxes for AF (float[9] array, Flux[e^- s^{-1}])

The individual AF CCD flux values. Within this array, the order will be AF1 to AF9. If the flux has been rejected or is unavailable, the flux and error will be set to null.

AF_FLUX_ERROR : CCD-level G band flux error for AF (float[9] array, Flux[e^- s^{-1}])

The error on the flux of the individual AF CCDs. Within this array, the order will be AF1 to AF9. If the flux has been rejected or is unavailable, this error will be set to null.

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AF_FLUX_OVER_ERROR : CCD-level G band flux divided by its error (float[9] array)

Flux in the individual AF CCD divided by its error.

AF_MAG : CCD-level G band Vega magnitude (float[9] array, Magnitude[mag])

G band Vega magnitude in the individual AF CCDs. It is computed from the flux applying the DR4 zero-point defined in `https://www.cosmos.esa.int/web/gaia/dr4-passbands`.

TRANSIT_ACQ_FLAGS : Transit-level information on the on-board acquisition (short)

This field contains a set of bitwise flags (from Bit 15 = msb to Bit 0 = lsb) related to the data acquisition. It has some oddities for historical reasons.

Windows may suffer truncation or trimming. Here, truncation refers to a reduction in size - often irregular - caused by an intra-field conflict with another window, while trimming refers to the permanent use of a reduced window size across scan. Truncation and trimming only affect 1D windows and never affect SM and AF1 windows. Trimming was only used for AF2-9 and only for a short period early in the mission (around OMBT revolutions 2103 to 2238).

The shape of all 1D AF2-9 windows is detailed in `epoch_image.shape_af`.

The field is coded as follows:

- Bits 15 and 14: unused
- Bit 13 (0x2000): Always set.
- Bit 12 (0x1000): The AF2-9 windows are trimmed AC but have not suffered truncation.
- Bits 11-9: unused
- Bits 7 and 8 (0x0180): `xp_offset`. This is the value of a continuous two-bit TDI counter. It is needed for the reconstruction of the XP windows.
- Bit 6 (0x0040): AF2-9 windows are 2D.
- Bit 5 (0x0020): AF2-9 windows are 1D (complement of Bit 6).
- Bit 4 (0x0010): AF2-9 windows are 1D and with the full AC size of 12 pixels.
- Bit 3 (0x0008): No BP window acquired.

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- Bit 2 (0x0004): No RP window acquired.
- Bit 1 (0x0002): BP window truncated.
- Bit 0 (0x0001): RP window truncated.

CCD_ACQ_FLAGS : CCD-level information on the on-board acquisition (byte[10] array)

This field contains an array, with one component per CCD (SM has index 0, AF9 index 9).

Each of the components of the array contains a set of bitwise flags, coded as follows:

- Bit 7 (AND bit mask 0x80): '1' means 'window missing' (i.e. all samples are zero). It will happen for AF9 Row 4 (which does not exist), and for switched off CCDs, full overlapping in an inter-field conflict, etc.
- Bit 6 (0x40): '1' means 'wide window', '0' means 'narrow window'. This distinction is only applicable to AF windows of Class 2 or 3. For the rest of the cases, that is, SM (any class), and for AF windows of Class 0 or 1 (i.e. 2D windows), we can consider them as wide windows, and thus the flag will be '1'. It will also be '1' for truncated AF windows obviously (since they all become wide).
- Bit 5 (0x20): '1' means 'window at the CCD edge', i.e. this window has been acquired with fewer AC pixels than usual.
- Bit 4 (0x10): '1' means 'some samples of this window have been eliminated, e.g. due to an inter-field conflict'.
- Bit 3 (0x08): '1' means 'this window is affected by an AOCS update'.
- Bit 2 (0x04): '1' means 'this window is affected by more than one gate (multiple gate)'.
- Bit 1 (0x02): '1' means 'there is a Charge Injection in the last half of the window'.
- Bit 0 (0x01): '1' means 'the Charge Injection preceding this window is abnormal', mainly meaning that the preceding CI has interacted with a Gate i.e. the injection charge has been held in a gate line and therefore its length has been shortened and the TDI distance to the previous injection has been extended.

Please note that some of these flags are not applicable for SM (such as bits 3, 2, 1 and 0) and for AF1 (such as 3).

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See Chapter **??** of the DR4 online documentation for more information.

TRANSIT_PROC_FLAGS : Transit-level information on the processing results (short)

This field contains a set of bitwise flags, coded as follows:

- Bits 3 to 0 (`transit_proc_flags` AND 0x000F): **IPD strategy and findings**. As of IDT 15.1 and until IDT 21.0, as well as in IDU until DRC-02, this field was not used (0 was always stored here). As of IDU DRC-03, the following information is provided:

  '0': Normal IPD. Ambiguity not tested or not detected. Default value for all records produced until IDT 21.0 and IDU until DRC-02

  '1': Multiple Source IPD: As of IDT 21.1, this value is used to indicate that the earlier object in time detected in the window has been selected. In case of 2D windows with larger AC than AL separation between detected peaks: higher μ selected.

  '2': Multiple Source IPD: As of IDT 21.1, this value is used to indicate that the later object in time detected in the window has been selected. In case of 2D windows with larger AC than AL separation between detected peaks: lower μ selected.

  '3': Multiple Source IPD: As of IDU DRC-03, this value is used to indicate that at least one of the windows has a multiple source and that these disturbing sources were masked for the IPD processing. `ccd_proc_flags` can be checked to know which and how many windows have been affected.

  '4': DIPD: As of IDU 21.0, this value is used to indicate that this object has been obtained using DIPD implementation and thus include a window model template and combined AF solution.

  '5': TukeyBiWeight IPD: this value is used to indicate that this object has been obtained using TukeyBiWeight IPD initialisation for the internal processing subsystem IDU-AttAc; such initial values do not propagate into the published data.

  Values 6-15 are reserved for future extensions.

- Bits 6 to 4 (0x0070): **Bias and PEM OI strategy baseline**: this field indicates the Bias mitigation mode requested to process this transit. The consumer must check the corresponding `ccd_proc_flags` field to determine the actual mode used for each individual CCD as sometimes some modes can not be applied due to missing input data or calibration limitations.

  Value '0': Default initialisation value for the Bias strategy. If found in an AstroElementary it means that something went wrong and the Bias (or maybe just its flagging) was not done appropriately. Thus, '0' means 'UNDEFINED/FAILURE'.

Value ‘1’: Only PreScan bias used.

Value ‘2’: PreScan + common NU baseline.

Value ‘3’: Full PEM NU.

Values 4-7 are reserved for future extensions.

- Bits 10 to 7, **colour info**:

  Bit 7 (0x0080), **default colour**: ‘1’ means ‘a default colour used for IPD processing’, which can be caused by some problem in the colour determination or retrieval (e.g. no BP/RP data in IDT or no valid PhotPipe fluxes in IDU).

  Bit 8 (0x0100), **epoch colour**: ‘1’ means ‘epoch (transit-level) colour used for IPD processing’, ‘0’ means ‘mean (source-level) colour used’.

  Bit 9 (0x0200), **single-param colour**: ‘1’ means ‘single-parameter colour used for IPD processing’ (such as `nu_eff_used_in_astrometry`), ‘0’ means ‘multi-parameter colour used’ (such as SSCs).

  Bit 10 (0x0400), **colour problems**: ‘1’ means ‘missing colour information or dubious colour quatiliy’ (e.g. problematic/missing BP/RP data in IDT, NaN or rejected PhotPipe colour in IDU, etc).

  Please note some remarkable combinations of this bit with Bit 7:

  – Non-default + no problems (Bit 7 = 0, Bit 10 = 0): Good colour from catalogue or epoch data

  – Non-default + problems (Bit 7 = 0, Bit 10 = 1): Dubious colour from catalogue or best approximation from partial or problematic epoch data. Implemented only in IDT.

  – Default + no problems (Bit 7 = 1, Bit 10 = 0): Default colour has been used because no colour information was available (i.e. for blacklisted observations). Implemented only in IDU.

  – Default + problems (Bit 7 = 1, Bit 10 = 1): Default colour used instead of the available colour from catalogue or epoch data (i.e. dubious colour rejected).

- Bit 11 (0x0800), **cosmetic effects** flagging: ‘1’ means ‘dead pixel columns affecting some of the windows’.

- Bit 12 (0x1000), **cosmetic effects** flagging: ‘1’ means ‘anomalously low samples found in some of the windows’.

- Bit 13 (0x2000), **cosmetic effects** flagging: ‘1’ means ‘anomalously high samples found in some of the windows’.

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- Bit 14 (0x4000), **problems in initial centroid**: '1' means that problems have been found in the Initial Centroids determination. In the case of IDT, combined with the 'some problem found in the colour determination' flag (bit 10), it may indicate problems in the TransitPredictor (e.g. predicted position outside window, or no attitude available) in which case a default position is used for colour determination.
- Bit 15, **attitude issues**: '1' indicates that some attitude problem was found affecting some of the windows i.e. missing attitude or problems computing AL/AC rates. This flag is only set by IDU which depends on the attitude to access the ELSF and for the combine AF processing in DIPD.

See Chapter **??** of the DR4 online documentation for more information.

CCD_PROC_FLAGS : CCD-level information on the processing results (short[20] array)

This field contains information about the processing findings for the SM and AF CCDs. It is defined as an array, with one or more sets of 10 components (one per CCD). The indexes of the componenents for a given CCD meet the following expression:

$$s + 10k \quad \begin{cases} s & [0...9] & \text{CCD index [SM...AF9]} \\ k & [0...n] & \text{CCD set index} \end{cases} \tag{8}$$

The first set of 10 components (SM has index 0, AF9 index 9) of this array contains a set of bitwise flags, coded as follows:

- Bits 3 to 0 (`ccd_proc_flags` AND 0x000F): **IPD result**. It is coded as follows:
  - '0': **Success**; the fitting has succeeded without any problem.
  - '1': **Illegal Amplitude**; invalid amplitude obtained after the initial fitting. It may happen for windows with wrong background determination or similar.
  - '2': **Illegal Coordinate**; invalid AL or/and AC locations obtained in the fitting i.e. locations outside the window. It may happen for windows close to a charge injection, multiple sources, strongly affected by some cosmic ray or other similar issues.
  - '3': **DEPRECATED: Illegal Output**; location estimation outside window.
  - '4': **Maximum Iteration Exceeded**; convergence not reached, but results achieved so far stored. The IPD result may still be reliable enough.
  - '5': **Failed**; diverging iteration instead of converging.

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  - '6': **Pending**; calculations in progress and values not reliable; maybe the processing failed for some reason before finishing the calculations.
  - '7': **No LSF/PSF data**; no result available due to no LSF/PSF data available.
  - '8': **Not Enough Samples**; IPD was attempted but no result is available because at a given iteration it runs out of samples (there are not enough inputs to the equations/matrix system).
  - '9': **Odd LSF/PSF Success**; the fitting has succeeded but using an odd LSF/PSF. The LSF/PSF is considered odd when it is not well defined for some samples. The IPD result may still be reliable enough similar as with the 'Maximum Iteration Exceeded' case.
  - '14': **Not Attempted**; IPD has not been attempted on purpose due to missing gate information, missing CI information or similar or when not enough usable input samples are available due to heavy truncation, CCD edge, complex AOCS update, multiple gates or similar.
  - '15': **Crashed**; calculation finished and crashed for some reason.
  - Values '3' and '10' to '13': Available for future extensions.

- Bit 4 (0x0010): **IPD nominal/non nominal**. '1' indicates the IPD is not non-nominal in the sense it has been performed over a reduced window area (details in `ccd_proc_geom`). This flag is only set for AF windows.

- Bit 5 (0x0020): **IPD not available**. '1' indicates that IPD result is not available and thus the output values in the AstroElementary are not valid. This flag set to true and IPD result equals to NOT_ATTEMPTED indicates IPD was not attempted on purpose and with IPD result equals to PENDING indicates unexpected IPD termination. This flag set to true with any other IPD result should not be possible.

- Bit 6 (0x0040): **Some cosmetic issue**. '1' indicates that there is at least one contaminated or discarded sample in the selected window area. This flag collects the presence of additional discarded or contaminated sample without an explicit CCD level flag thus excluding saturation, cosmics, odd debiased, etc.

- Bit 7 (0x0080): **Cosmic removed**. '1' means that at least one sample in the selected window area has been discarded for being a suspected cosmic.

- Bit 8 (0x0100): **Saturation removed**. '1' means that at least one sample in the selected window area has been discarded due to saturation.

- Bit 9 (0x0200): **Part of window discarded**. '1' means that at least one sample has been discarded in the full window area.

- Bit 10 (0x0400): **No window**. '1' means 'no samples available' (i.e. all the samples in this window were discarded on board, or this is e.g. AF 9 Row 4, or a deactivated CCD, etc).

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- Bit 11 (0x0800): **Non-success InitialCentroid**. '1' means 'the Initial Centroid was not successful, and thus we have used the centre of the window as the start condition of IPD'.
- Bit 12 (0x1000): **Odd Background found**. '1' indicates that the background determined for this window (for at least one of its samples) was odd i.e negative, too large or NaN. This flag may be used to indicate that the fluxes determined for this window are potentially wrong (although its AL/AC centroid may still be useful).
- Bit 13 (0x2000): **fall-back to lower Bias Mitigation mode**. '1' indicates that 'fall-back' to lower mode has occurred for this window (for at least one of its samples). This flag will always be activated for AF1 when FULL_MITIGATION mode is enabled because this mode is not supported in this CCD.
- Bit 14 (0x4000): **non-target source removed**. '1' indicates that non-target sources have been detected and masked in this window.
- Bit 15 (0x8000): **LSF/PSF observation/source parameters clamped**. '1' indicates that the observation/source parameters used to access the LSF/PSF have been clamped to fit the ELSF parameter space.

The second set of 10 components (SM has index 10, AF9 index 19) of this array contains a set of bitwise flags, coded as follows:

- Bit 0 (0x0001): **Odd debiased sample**. '1' indicates that there is at least one debaised sample below $-TDN^2$ limit.
- Bit 1 (0x0002): **Numerical overflow recovery**: '1' indicates that there is at least one sample for which the numerical overflow has been corrected (only applicable in SM and AF1).
- Bit 2 (0x0004): **ELSF reliability mask**: '1' indicates that at least one sample has been discarded in IPD because the corresponding ELSF model is sub-optimal in that region.
- Bits 3 to 5 (0x0038); **Number of saturated samples discarded**. For AF it codes from 0 up to 7 or more masked samples and for SM it codes the position of the most significant bit of the masked sample count and thus coding 0, 1, 2 or 3, 4 to 7, up to 64 samples or more.
- Bit 6 (0x0040): **Background Mode**: '0' indicates background has been fitted along with location and flux parameters whereas '1' indicates that a different overall background estimation has been used i.e. KNN background in regular IPD or specific background module for SIF CF.

- Bit 7, 0x0080), **CI distance updated**: indicating that the distance to last charge injection has been redetermined and it does not match the original one in the unpacked telemetry.
- Bit 8, 0x0100). **Non-gated CI line**: indicating that at least one non-gated CI is within the window.

See Chapter **??** in the online documentation for more information.

CCD_PROC_GEOM : CCD-level information on the processing geometry used (int[10] array)

This field contains an array, with one component per CCD (SM has index 0, AF9 index 9). It indicates, per CCD, the 'geometry' of the area actually processed by the IPD.

Each of the components of the array contains a set of bitwise fields, coded as follows:

- Bits 5-0 (0x0000003F): width (samples) of the sub-window used in the IPD.
- Bits 9-6 (0x000003C0): height (samples) of the sub-window used in the IPD.
- Bits 16-10 (0x0001FC00): AL offset (samples) of the sub-window used in the IPD.
- Bits 20-17 (0x001E0000): AC offset (samples) of the sub-window used in the IPD.
- Bits 31-21: spare bits, available for user-defined flags or future extensions.

In IDT, this field will be NULL in case *all* AF windows have been processed with the nominal geometry (full window). Otherwise, even with just 1 AF window processed non-nominally (check `ccd_proc_flags` bit 4 for that), it will have 10 components (for SM-AF9). In case there are non-nominal windows but we have not even tried to run IPD (in none of the windows), then it will also be NULL. Note that the SM component in this array is never set in IDT and its value is always set to zero (`ccd_proc_flags` bit 4 is never set for SM as well).

On the other hand, in IDU and starting in Cycle 4, this field is never NULL and always contains the actual dimensions of the SM sub-window used in IPD (8 AL samples for WC0 and 4 AL samples for WC 1 by default). Note that, contrary to AF cases, the `ccd_proc_flags` bit 4 is never set to true for SM.

See Chapter **??** in the online documentation for more information.

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PHOT_CCD_PROC_FLAGS : CCD-level information on PhotPipe processing (short[12] array)

This field reports information generated during the PhotPipe processing. The type of this field is an array of 12 short (SM, AF1-9, BP and RP). Each short will bit-wise encode the following values:

- Bit 0: '1' means 'A saturation correction was applied to this CCD transit'.
- Bit 1: '1' means 'Predicted position computation failed or predicted position outside window'.
- Bit 2: '1' means 'The AC centring error was clamped when the LS calibration was applied'. Please, refer to the end-of-cycle documentation for details on the clamping ranges used.
- Bit 3: '1' means 'No calibration available' (no calibrated CCD photometry will be available in this case).
- Bit 4: '1' means 'Calibration rejected and replaced with neighbouring calibration'.
- Bit 5: '1' means 'Calibration application failure'.
- Bit 6: '1' means 'Excluded from the generation of the mean source photometry, due to pre-filtering based on available flags' (this is different from Bit 7 that reports on outlier rejection).
- Bit 7: '1' means 'Rejected in the source accumulation process'. Beware that outlier rejection will only affect the source photometry accumulation, i.e. when computing FoV level photometry ALL available calibrated CCD photometric data will be used regardless of the value of this flag.
- Bit 8: '1' means 'Observation affected by satellite event'.
- Bit 9: '1' means 'Charge release correction applied'. This is only applicable for BP/RP.
- Bit 10: '1' means 'Stray-light mitigation applied'. This is only applicable for BP/RP.
- Bit 11: '1' means 'Local (KNN) background mitigation applied'. This is only applicable for BP/RP.
- Bit 12: '1' means 'Crowding information available in the table `xp_epoch_crowding`'. This is only applicable for BP/RP.
- Bit 13: '1' means 'The transit is affected by truncation or multiple gate configuration in a way that is considered unrecoverable'. This is only applicable for BP/RP.

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- Bit 14: ‘1’ means ‘The transit has been treated for truncation or multiple gate configuration’. This is only applicable for BP/RP.

Remaining bits are available for future extensions.

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11.3 SYNTHETIC_PHOTOMETRY_GSPC [TAP]

The Gaia Synthetic Photometry Catalogue (GSPC) is a catalogue of synthetic photometry generated from the Gaia BP/RP mean spectra in Data Release 4 and standardised using wide and reliable sets of external standard stars. Synthetic photometry is provided in a selection of photometric systems.

Only sources that have BP/RP mean spectra in Gaia DR4 can be included in this catalogue. Individual entries in this table are selected for having reliable magnitudes and fluxes. To know the details about the generation of this catalogue and the exact criteria applied for the selection, please refer to the Gaia DR4 publications.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (unique within a particular Data Release) (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

U_JKC_FLUX : U flux in the Johnson-Kron-Cousins system (float, Flux[W m^{-2} nm^{-1}])

Flux in the U band of the standardised Johnson-Kron-Cousins photometric system as derived from the BP/RP spectra.

U_JKC_FLUX_ERROR : Error on the U flux in the Johnson-Kron-Cousins system (float, Flux[W m^{-2} nm^{-1}])

Error on the flux in the U band of the standardised Johnson-Kron-Cousins photometric system as derived from the BP/RP spectra.

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U_JKC_FLUX_OVER_ERROR : U flux in the Johnson-Kron-Cousins system divided by its error (float)

Flux in the U band of the standardised Johnson-Kron-Cousins photometric system as derived from the BP/RP spectra `u_jkc_flux` divided by its error `u_jkc_flux_error`.

U_JKC_MAG : U magnitude in the Johnson-Kron-Cousins system (float, Magnitude[mag])

Magnitude in the U band of the standardised Johnson-Kron-Cousins photometric system as derived from the BP/RP spectra.

U_JKC_FLAG : Flag indicating if G mag and BP-RP colour of the source lie in the validated range (boolean)

Flag indicating if G mag and BP−RP colour of the source lie in the validated range for this band and photometric system.

B_JKC_FLUX : B flux in the Johnson-Kron-Cousins system (float, Flux[W m^{-2} nm^{-1}])

Flux in the B band of the standardised Johnson-Kron-Cousins photometric system as derived from the BP/RP spectra.

B_JKC_FLUX_ERROR : Error on the B flux in the Johnson-Kron-Cousins system (float, Flux[W m^{-2} nm^{-1}])

Error on the flux in the B band of the standardised Johnson-Kron-Cousins photometric system as derived from the BP/RP spectra.

B_JKC_FLUX_OVER_ERROR : B flux in the Johnson-Kron-Cousins system divided by its error (float)

Flux in the B band of the standardised Johnson-Kron-Cousins photometric system as derived from the BP/RP spectra `b_jkc_flux` divided by its error `b_jkc_flux_error`.

B_JKC_MAG : B magnitude in the Johnson-Kron-Cousins system (float, Magnitude[mag])

Magnitude in the B band of the standardised Johnson-Kron-Cousins photometric system as derived from the BP/RP spectra.

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B_JKC_FLAG : Flag indicating if G mag and BP-RP colour of the source lie in the validated range (boolean)

Flag indicating if G mag and BP−RP colour of the source lie in the validated range for this band and photometric system.

V_JKC_FLUX : V flux in the Johnson-Kron-Cousins system (float, Flux[W m^{-2} nm^{-1}])

Flux in the V band of the standardised Johnson-Kron-Cousins photometric system as derived from the BP/RP spectra.

V_JKC_FLUX_ERROR : Error on the V flux in the Johnson-Kron-Cousins system (float, Flux[W m^{-2} nm^{-1}])

Error on the flux in the V band of the standardised Johnson-Kron-Cousins photometric system as derived from the BP/RP spectra.

V_JKC_FLUX_OVER_ERROR : V flux in the Johnson-Kron-Cousins system divided by its error (float)

Flux in the V band of the standardised Johnson-Kron-Cousins photometric system as derived from the BP/RP spectra `v_jkc_flux` divided by its error `v_jkc_flux_error`.

V_JKC_MAG : V magnitude in the Johnson-Kron-Cousins system (float, Magnitude[mag])

Magnitude in the V band of the standardised Johnson-Kron-Cousins photometric system as derived from the BP/RP spectra.

V_JKC_FLAG : Flag indicating if G mag and BP-RP colour of the source lie in the validated range (boolean)

Flag indicating if G mag and BP−RP colour of the source lie in the validated range for this band and photometric system.

R_JKC_FLUX : R flux in the Johnson-Kron-Cousins system (float, Flux[W m^{-2} nm^{-1}])

Draft

Flux in the R band of the standardised Johnson-Kron-Cousins photometric system as derived from the BP/RP spectra.

R_JKC_FLUX_ERROR : Error on the R flux in the Johnson-Kron-Cousins system (float, Flux[W m^{-2} nm^{-1}])

Error on the flux in the R band of the standardised Johnson-Kron-Cousins photometric system as derived from the BP/RP spectra.

R_JKC_FLUX_OVER_ERROR : R flux in the Johnson-Kron-Cousins system divided by its error (float)

Flux in the R band of the standardised Johnson-Kron-Cousins photometric system as derived from the BP/RP spectra `r_jkc_flux` divided by its error `r_jkc_flux_error`.

R_JKC_MAG : R magnitude in the Johnson-Kron-Cousins system (float, Magnitude[mag])

Magnitude in the R band of the standardised Johnson-Kron-Cousins photometric system as derived from the BP/RP spectra.

R_JKC_FLAG : Flag indicating if G mag and BP-RP colour of the source lie in the validated range (boolean)

Flag indicating if G mag and BP−RP colour of the source lie in the validated range for this band and photometric system.

I_JKC_FLUX : I flux in the Johnson-Kron-Cousins system (float, Flux[W m^{-2} nm^{-1}])

Flux in the I band of the standardised Johnson-Kron-Cousins photometric system as derived from the BP/RP spectra.

I_JKC_FLUX_ERROR : Error on the I flux in the Johnson-Kron-Cousins system (float, Flux[W m^{-2} nm^{-1}])

Error on the flux in the I band of the standardised Johnson-Kron-Cousins photometric system as derived from the BP/RP spectra.

I_JKC_FLUX_OVER_ERROR : I flux in the Johnson-Kron-Cousins system divided by its error

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(float)

Flux in the I band of the standardised Johnson-Kron-Cousins photometric system as derived from the BP/RP spectra `i_jkc_flux` divided by its error `i_jkc_flux_error`.

I_JKC_MAG : I magnitude in the Johnson-Kron-Cousins system (float, Magnitude[mag])

Magnitude in the I band of the standardised Johnson-Kron-Cousins photometric system as derived from the BP/RP spectra.

I_JKC_FLAG : Flag indicating if G mag and BP-RP colour of the source lie in the validated range (boolean)

Flag indicating if G mag and BP−RP colour of the source lie in the validated range for this band and photometric system.

U_SDSS_FLUX : u flux in the SDSS system (float, Flux[W m^{-2} nm^{-1}])

Flux in the u band of the standardised SDSS photometric system as derived from the BP/RP spectra.

U_SDSS_FLUX_ERROR : Error on the u flux in the SDSS system (float, Flux[W m^{-2} nm^{-1}])

Error on the flux in the u band of the standardised SDSS photometric system as derived from the BP/RP spectra.

U_SDSS_FLUX_OVER_ERROR : u flux in the SDSS system divided by its error (float)

Flux in the u band of the standardised SDSS photometric system as derived from the BP/RP spectra `u_sdss_flux` divided by its error `u_sdss_flux_error`.

U_SDSS_MAG : u magnitude in the SDSS system (float, Magnitude[mag])

Magnitude in the u band of the standardised SDSS photometric system as derived from the BP/RP spectra.

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U_SDSS_FLAG : Flag indicating if G mag and BP-RP colour of the source lie in the validated range (boolean)

Flag indicating if G mag and BP−RP colour of the source lie in the validated range for this band and photometric system.

G_SDSS_FLUX : g flux in the SDSS system (float, Flux[W m^{-2} nm^{-1}])

Flux in the g band of the standardised SDSS photometric system as derived from the BP/RP spectra.

G_SDSS_FLUX_ERROR : Error on the g flux in the SDSS system (float, Flux[W m^{-2} nm^{-1}])

Error on the flux in the g band of the standardised SDSS photometric system as derived from the BP/RP spectra.

G_SDSS_FLUX_OVER_ERROR : g flux in the SDSS system divided by its error (float)

Flux in the g band of the standardised SDSS photometric system as derived from the BP/RP spectra `g_sdss_flux` divided by its error `g_sdss_flux_error`.

G_SDSS_MAG : g magnitude in the SDSS system (float, Magnitude[mag])

Magnitude in the g band of the standardised SDSS photometric system as derived from the BP/RP spectra.

G_SDSS_FLAG : Flag indicating if G mag and BP-RP colour of the source lie in the validated range (boolean)

Flag indicating if G mag and BP−RP colour of the source lie in the validated range for this band and photometric system.

R_SDSS_FLUX : r flux in the SDSS system (float, Flux[W m^{-2} nm^{-1}])

Flux in the r band of the standardised SDSS photometric system as derived from the BP/RP spectra.

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R_SDSS_FLUX_ERROR : Error on the r flux in the SDSS system (float, Flux[W m^{-2} nm^{-1}])

Error on the flux in the r band of the standardised SDSS photometric system as derived from the BP/RP spectra.

R_SDSS_FLUX_OVER_ERROR : r flux in the SDSS system divided by its error (float)

Flux in the r band of the standardised SDSS photometric system as derived from the BP/RP spectra `r_sdss_flux` divided by its error `r_sdss_flux_error`.

R_SDSS_MAG : r magnitude in the SDSS system (float, Magnitude[mag])

Magnitude in the r band of the standardised SDSS photometric system as derived from the BP/RP spectra.

R_SDSS_FLAG : Flag indicating if G mag and BP-RP colour of the source lie in the validated range (boolean)

Flag indicating if G mag and BP−RP colour of the source lie in the validated range for this band and photometric system.

I_SDSS_FLUX : i flux in the SDSS system (float, Flux[W m^{-2} nm^{-1}])

Flux in the i band of the standardised SDSS photometric system as derived from the BP/RP spectra.

I_SDSS_FLUX_ERROR : Error on the i flux in the SDSS system (float, Flux[W m^{-2} nm^{-1}])

Error on the flux in the i band of the standardised SDSS photometric system as derived from the BP/RP spectra.

I_SDSS_FLUX_OVER_ERROR : i flux in the SDSS system divided by its error (float)

Flux in the i band of the standardised SDSS photometric system as derived from the BP/RP spectra `i_sdss_flux` divided by its error `i_sdss_flux_error`.

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I_SDSS_MAG : i magnitude in the SDSS system (float, Magnitude[mag])

Magnitude in the i band of the standardised SDSS photometric system as derived from the BP/RP spectra.

I_SDSS_FLAG : Flag indicating if G mag and BP-RP colour of the source lie in the validated range (boolean)

Flag indicating if G mag and BP−RP colour of the source lie in the validated range for this band and photometric system.

Z_SDSS_FLUX : z flux in the SDSS system (float, Flux[W m^{-2} nm^{-1}])

Flux in the z band of the standardised SDSS photometric system as derived from the BP/RP spectra.

Z_SDSS_FLUX_ERROR : Error on the z flux in the SDSS system (float, Flux[W m^{-2} nm^{-1}])

Error on the flux in the z band of the standardised SDSS photometric system as derived from the BP/RP spectra.

Z_SDSS_FLUX_OVER_ERROR : z flux in the SDSS system divided by its error (float)

Flux in the z band of the standardised SDSS photometric system as derived from the BP/RP spectra `z_sdss_flux` divided by its error `z_sdss_flux_error`.

Z_SDSS_MAG : z magnitude in the SDSS system (float, Magnitude[mag])

Magnitude in the z band of the standardised SDSS photometric system as derived from the BP/RP spectra.

Z_SDSS_FLAG : Flag indicating if G mag and BP-RP colour of the source lie in the validated range (boolean)

Flag indicating if G mag and BP−RP colour of the source lie in the validated range for this band and photometric system.

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f606w_acswfc_flux : F606W flux in the HST ACS/WFC system (float, Flux[W m^{-2} nm^{-1}])

Flux in the F606W band of the standardised HST ACS/WFC photometric system as derived from the BP/RP spectra.

f606w_acswfc_flux_error : Error on the F606W flux in the HST ACS/WFC system (float, Flux[W m^{-2} nm^{-1}])

Error on the flux in the F606W band of the standardised HST ACS/WFC photometric system as derived from the BP/RP spectra.

f606w_acswfc_flux_over_error : F606W flux in the HST ACS/WFC system divided by its error (float)

Flux in the F606W band of the standardised HST ACS/WFC photometric system as derived from the BP/RP spectra `f606w_acswfc_flux` divided by its error `f606w_acswfc_flux_error`.

f606w_acswfc_mag : F606W magnitude in the HST ACS/WFC system (float, Magnitude[mag])

Magnitude in the F606W band of the standardised HST ACS/WFC photometric system as derived from the BP/RP spectra.

f606w_acswfc_flag : Flag indicating if G mag and BP-RP colour of the source lie in the validated range (boolean)

Flag indicating if G mag and BP−RP colour of the source lie in the validated range for this band and photometric system.

f814w_acswfc_flux : F814W flux in the HST ACS/WFC system (float, Flux[W m^{-2} nm^{-1}])

Flux in the F814W band of the standardised HST ACS/WFC photometric system as derived from the BP/RP spectra.

f814w_acswfc_flux_error : Error on the F814W flux in the HST ACS/WFC system (float, Flux[W m^{-2} nm^{-1}])

Error on the flux in the F814W band of the standardised HST ACS/WFC photometric system as

derived from the BP/RP spectra.

F814W_ACSWFC_FLUX_OVER_ERROR : F814W flux in the HST ACS/WFC system divided by its error (float)

Flux in the F814W band of the standardised HST ACS/WFC photometric system as derived from the BP/RP spectra `f814w_acswfc_flux` divided by its error `f814w_acswfc_flux_error`.

F814W_ACSWFC_MAG : F814W magnitude in the HST ACS/WFC system (float, Magnitude[mag])

Magnitude in the F814W band of the standardised HST ACS/WFC photometric system as derived from the BP/RP spectra.

F814W_ACSWFC_FLAG : Flag indicating if G mag and BP-RP colour of the source lie in the validated range (boolean)

Flag indicating if G mag and BP−RP colour of the source lie in the validated range for this band and photometric system.

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12 Reference frame

As indicated in the below subsection titles by the labels [TAP] and [DataLink], these products are served from the ESA Gaia Archive using two different protocols, the Table Access Protocol (TAP) for tabular products and the DataLink protocol for other products such as spectra, light curves, MCMC samples, etc. For more details please see Section **??**.

12.1 FRAME_ROTATOR_SOURCE [TAP]

Sources used to compute the Gaia reference frame.

Note that this is the input source list for the process of defining the Gaia Celestial Reference Frame 4 (Gaia CRF4). The result of that process, i.e. the actual definition of the Gaia CRF4 is given in table `gaia_crf4`.

The AGIS frame rotator algorithm has two different parts: the reference frame orientation; and the reference frame spin. The reference frame orientation is fixed using the International Earth Rotation and Reference System Service (IERS) positions of the subset of IERS sources defining the third realization of the International Celestial Reference Frame, Charlot et al. (2020). The reference frame spin is defined using the Gaia DR4 proper motions of a list of QSOs. The frame rotator algorithms have an outlier elimination scheme. Hence for each aspect we provide the considered sources and the sources actually used. The number of used sources might be lower than the number of considered sources.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Gaia source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

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CONSIDERED_FOR_REFERENCE_FRAME_ORIENTATION : Considered for the reference frame orientation (boolean)

True if the source was a considered source for the reference frame orientation, false otherwise.

USED_FOR_REFERENCE_FRAME_ORIENTATION : Used for the reference frame orientation (boolean)

True if the source was effectively used for the reference frame orientation, false otherwise.

CONSIDERED_FOR_REFERENCE_FRAME_SPIN : Considered for the reference frame spin (boolean)

True if the source was a considered source for the reference frame spin determination, false otherwise.

USED_FOR_REFERENCE_FRAME_SPIN : Used for the reference frame spin (boolean)

True if the source was effectively used for the reference frame spin determination, false otherwise.

Draft

12.2 GAIA_CRF4 [TAP]

This is the defining table of the Gaia Celestial Reference Frame 4 (Gaia CRF4). It contains information about the QSO candidates considered for Gaia CRF4, plus a flag for each source indicating whether it is an actual member of Gaia CRF4, as well as a category indicating QSO sub-populations to which the source belongs. In addition, cross-match information to other catalogues is given.

Details are given in **?**.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution identifier, please visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

IN_CRF4 : Indicates whether the source is accepted for Gaia CRF4 (boolean)

Indicates whether the source is accepted for Gaia CRF4.

SOURCE_CATEGORY : Category of the source (short)

Category that the source belongs to amongst a set of sub-populations of QSOs. Examples of such special populations are gravitationally lensed quasars, almost point-like quasars with traces of a host galaxy at Gaia's optical resolution, and sources which are classified as galaxies by some Gaia processing module.

The full details of the categories are published in the corresponding DPAC publication on the Gaia Celestial Reference Frame 4 (**?**).

Draft

IN_CRF3 : Whether the source was part of Gaia-CRF3 (boolean)

Whether the source was part of Gaia CRF3.

IN_DR3_ASTROMETRIC_SELECTION : Whether the source was part of the astrometric selection of Gaia DR3 (boolean)

Whether the source was part of the astrometric selection published in the QSO candidates table of Gaia DR3 (with astrometric selection column = true).

SOURCE_ID_DR3 : The DR3 source identifier of the source if known (long)

The DR3 source identifier of the source in case it was part of CRF3.

FOUND_IN_CATALOGUE : A bit map of the catalogue where this source was found (long)

A bit map of the catalogue where this source was found. The detailed catalogue distribution is given in table `gaia_crf4_external_names`.

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12.3 GAIA_CRF4_EXTERNAL_NAMES [TAP]

This table contains the full cross-match information for the Gaia-CRF4 sources. The Gaia-CRF4 sources were selected from the cross-matches with external QSO catalogues.

Each source can be found in more than one external catalogue, and those are registered in separate records of the table and identified through field `catalogue_id`. For each record, the name of the source in the external catalogue of interest is given by `source_name_in_catalogue`.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution identifier, please visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

CATALOGUE_ID : Identifier of the catalogue where the source was found (byte)

Identifier of the catalogue where the source was found. The codes for the individual catalogues are listed in the CRF4 paper, **?**

SOURCE_NAME_IN_CATALOGUE : Name of the source in the catalogue corresponding to the catalogue identifier (string)

Name of the source in the cross-matched catalogue of interest.

The corresponding catalogue is that associated with `catalogue_id`.

Draft

13 Science alerts

As indicated in the below subsection titles by the labels [TAP] and [DataLink], these products are served from the ESA Gaia Archive using two different protocols, the Table Access Protocol (TAP) for tabular products and the DataLink protocol for other products such as spectra, light curves, MCMC samples, etc. For more details please see Section **??**.

13.1 ALERTS_MIXEDIN_SOURCEIDS [TAP]

Some photometric science alerts mix transits from their primary source and other sources in the catalogue. This is done when those sources are believed to be associated with a single, astrophysical source.

This table lists the identifier for these mixed-in sources, linking them to the primary source identifiers listed in table `science_alerts`.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

ALERT_SOURCE_ID : Primary source identifier associated with the alert (long)

The primary `source_id` associated with the alert, as listed in table `science_alerts`.

MIXED_IN_SOURCE_ID : Additional source identifier, if any, associated with the alert (long)

Identifier of additional source, if any, associated with the science alert.

Draft

13.2 SCIENCE_ALERTS [TAP]

Table containing all Gaia Photometric Science Alerts triggered in the time span covered by DR4, i.e. 25-07-2014 to 20-01-2020.

A Gaia Photometric Science Alert is known to the community by its name Gaiannxyz where nn is the year number and xyz is an incrementing, alphabetic sequence starting at aaa. Science alerts have their details published on the WWW. Given the name of an alert, the URL for the published details can be derived, e.g. `https://gsaweb.ast.cam.ac.uk/alerts/alert/Gaia18ahj` for alert Gaia18ahj.

The `source_id` associated with the alert may have other alternative matches in Gaia DR4. These other identifiers are listed in table `alerts_mixedin_sourceids`.

Columns description:

SOURCE_ID : Unique source identifier (unique within a particular Data Release) (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

TRANSIT_ID : Alerting transit identifier (long)

The `transit_id` is a unique identifier assigned to each detected (and confirmed) source as it transits the Gaia focal plane. Each time a given source is detected as Gaia scans and re-scans the sky a new `transit_id` will be created to badge that apparition. Hence the along–scan time and the across–scan position along with the telescope in which the source was detected are used to form a unique integer with which to label the transit.

The several features of a detection that are encoded in `transit_id` can be easily retrieved using bit masks (&) and shifts (>>) as follows:

- On-Board Mission Time line [ns]
  = 204800 * ((`transit_id` >> 17) & (0x000003FFFFFFFFFF))
- Field-of-view = 1 + (`transit_id` >> 15) & 0x03 [1 for 'preceding' and 2 for 'following' fields-of-view respectively]
- CCD row = (`transit_id` >> 12) & 0x07 [dimensionless, in the range 1 to 7]
- Across-scan 'reference acquisition pixel' in strip AF1 = (`transit_id`) & 0x0FFF [pixels] (this is the across-scan centre of the AF1 window and is odd if immediately below the mid-point of the window and even if immediately above)

Draft

where the bit mask prefix ‘0x’ denotes hexadecimal.

For further details see Portell et al. (2020). For convenience a decoder for `transit_id` is available on-line at

`https://gaia.esac.esa.int/decoder/transitidDecoder.jsp`

NAME : Name of alert (string)

Name of the alert, in the format ‘Gaiannabc’, where nn is the year-number and abc is an incrementing, alphabetic sequence, starting at aaa in each year.

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

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13.3 SSO_ALERTS [TAP]

Observations that have have been associated with published asteroid alerts, diffused through the Gaia-FUN-SSO coordination. The transit identifier is provided, along with the alphanumeric code unique to each alert. The alert identifier can be used to identify all the transits that have been linked together in a bundle of observations, and associated with the same object of unknown identification (at the time the alert was issued).

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution identifier, please visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

TRANSIT_ID : Transit Identifier (long)

The `transit_id` is a unique identifier assigned to each detected (and confirmed) source as it transits the Gaia focal plane. Each time a given source is detected as Gaia scans and re-scans the sky a new `transit_id` will be created to badge that apparition. Hence the along–scan time and the across–scan position along with the telescope in which the source was detected are used to form a unique integer with which to label the transit.

The several features of a detection that are encoded in `transit_id` can be easily retrieved using bit masks (&) and shifts (>>) as follows:

- On-Board Mission Time line [ns]
  = 204800 * ((`transit_id` >> 17) & (0x000003FFFFFFFFFF))
- Field-of-view = 1 + (`transit_id` >> 15) & 0x03 [1 for 'preceding' and 2 for 'following' fields-of-view respectively]
- CCD row = (`transit_id` >> 12) & 0x07 [dimensionless, in the range 1 to 7]
- Across-scan 'reference acquisition pixel' in strip AF1 = (`transit_id`) & 0x0FFF

[pixels] (this is the across-scan centre of the AF1 window and is odd if immediately below the mid-point of the window and even if immediately above)

where the bit mask prefix '0x' denotes hexadecimal.

For further details see Portell et al. (2020). For convenience a decoder for `transit_id` is available on-line at

`https://gaia.esac.esa.int/decoder/transitidDecoder.jsp`

ALERT_ID : Alert Identifier (string)

Identification of the alert diffused for ground based follow up. The alphanumeric code is the same that was used to publish the alert through the Gaia-FUN-SSO web site.

Draft

14 Simulation

As indicated in the below subsection titles by the labels [TAP] and [DataLink], these products are served from the ESA Gaia Archive using two different protocols, the Table Access Protocol (TAP) for tabular products and the DataLink protocol for other products such as spectra, light curves, MCMC samples, etc. For more details please see Section **??**.

14.1 GAIA_SOURCE_SIMULATION [TAP]

Table of sources realised according to the Gaia Object Generator (GOG) simulation. Observed attributes are given with simulated observational uncertainties.

This table contains the output of GOG. The values are obtained after adding the corresponding uncertainty (based on the error models) to the true values in the Gaia Universal Model tables. Both the values and the uncertainties are provided.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Long Identifier (long)

A unique source identifier composed by the type of source, the sky region where the source is located and a sequential number inside this region (see the `source_extended_id` in `gum_mw` table).

In the `source_id` information is coded within the 64 bit long integer. The celestial position is encoded via a level 12 Hierarchical Triangular Mesh, covering roughly one square arc minute size on the average:

- The 4 most significant bits contain the object type: (0: Unknown; 1: Stellar; 2: Galaxy; 3: QSO; 4: Supernova; 5: Exoplanet; 6: Noise or PPE; 7: Cluster; 8:

Planetary Nebulae; 9: HII Region; 10: GRB; 11: Asteroid; 12: Comet; 13: Planet; 14: Satellite; 15: User Source; -1: Not identified)

- The 28 following bits contain the HTM index at level 12 (note the use of HTM as opposed to HEALPix in `source_id` in `gaia_source`)
- One bit indicates the variability,
- One bit indicates whether this is a multiple system (i.e. not a component).
- The 4 next groups of 3 bits are the 4 hierarchical levels for a multiple system. It is based on the fact that the maximum multiplicity of a hierarchical multiple system currently known is 7 (i.e. 3 bits are needed) with a 4 levels depth. The components in a level are sequential (i.e. non-hierarchical).
- The 18 least significant bits concern the object number in the region.

RANDOM_INDEX : Random index (long)

A random index which can be used to select smaller subsets of the data that are still representative. The column contains a random permutation of the numbers from 0 to $N - 1$, where N is the number of sources in the table. The random index can be useful for validation (testing on 10 different random subsets), visualization (displaying 1% of the data), and statistical exploration of the data, without the need to download all the data.

RA : Right Ascension (double, Angle[deg])

GOG simulation of the right ascension of the barycentre at J2010 reference epoch in ICRS frame.

RA_ERROR : Right Ascension error (float, Angle[mas])

GOG simulation of the standard error of right ascension.

DEC : Declination (double, Angle[deg])

GOG simulation of the declination of the barycentre at J2010 reference epoch in ICRS frame.

DEC_ERROR : Declination error (float, Angle[mas])

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GOG simulation of the standard error of declination.

PARALLAX : Parallax (float, Angle[mas])

GOG simulation of the parallax of the source. The inverse of the parallax is the distance from the barycentre of the Solar System to the barycentre of the source at J2010 reference epoch.

PARALLAX_ERROR : Parallax error (float, Angle[mas])

GOG simulation of the standard error of parallax.

PMRA : Proper motion in RA (float, Angular Velocity[mas yr^{-1}])

GOG simulation of the proper motion along right ascension at J2010 reference epoch: $\mu_{\alpha*} \equiv \mu_\alpha \cos\delta$. This is the local tangent plane projection of the proper motion vector in the direction of increasing right ascension.

PMRA_ERROR : Error in RA proper motion (float, Angular Velocity[mas yr^{-1}])

GOG simulation of the standard error of proper motion in right ascension direction.

PMDEC : Proper motion in dec (float, Angular Velocity[mas yr^{-1}])

GOG simulation of the proper motion along declination at J2010 reference epoch. This is the projection of the proper motion vector in the direction of increasing declination.

PMDEC_ERROR : Error in dec. proper motion (float, Angular Velocity[mas yr^{-1}])

GOG simulation of the standard error of proper motion in declination direction.

N_OBS_AL : Number of AL observations (int)

AL number of accepted observations.

N_OUTLIERS_AL : Number of outliers AL observations (int)

Draft

AL number of outliers observations.

PHOT_G_MEAN_FLUX : Mean G flux (float, Flux[e^- s^{-1}])

GOG simulation of the mean flux in the G band.

PHOT_G_MEAN_FLUX_ERROR : Mean G flux error (float, Flux[e^- s^{-1}])

GOG simulation of the mean G-band flux error.

PHOT_G_MEAN_MAG : Mean G magnitude (float, Magnitude[mag])

GOG simulation of the apparent mean magnitude in the G band in the Vega scale.

PHOT_BP_MEAN_FLUX : Mean BP flux (float, Flux[e^- s^{-1}])

GOG simulation of the mean flux in the BP band.

PHOT_BP_MEAN_FLUX_ERROR : Mean BP flux error (float, Flux[e^- s^{-1}])

GOG simulation of the mean BP flux error.

PHOT_BP_MEAN_MAG : Mean BP magnitude (float, Magnitude[mag])

GOG simulation of the mean apparent magnitude in the integrated BP band in the Vega scale.

PHOT_RP_MEAN_FLUX : Mean RP flux (float, Flux[e^- s^{-1}])

GOG simulation of the mean flux in the RP band.

PHOT_RP_MEAN_FLUX_ERROR : Mean RP flux error (float, Flux[e^- s^{-1}])

GOG simulation of the mean RP flux error.

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PHOT_RP_MEAN_MAG : Mean RP magnitude (float, Magnitude[mag])

GOG simulation of the mean apparent magnitude in the integrated RP band in the Vega scale.

PHOT_RVS_MEAN_FLUX : Mean RVS flux (float, Flux[e^- s^{-1}])

GOG simulation of the mean flux in the RVS band.

PHOT_RVS_MEAN_FLUX_ERROR : Mean RVS flux error (float, Flux[e^- s^{-1}])

GOG simulation of the mean RVS flux error.

PHOT_RVS_MEAN_MAG : Mean RVS magnitude (float, Magnitude[mag])

GOG simulation of the mean apparent magnitude in the integrated RVS band in the Vega scale.

RADIAL_VELOCITY : Radial velocity (float, Velocity[km s^{-1}])

GOG simulation of the spectroscopic radial velocity in the solar barycentric reference frame at J2010 reference epoch.

RADIAL_VELOCITY_ERROR : Radial velocity error (float, Velocity[km s^{-1}])

GOG simulation of the radial velocity error.

TEFF : Effective temperature (float, Temperature[K])

GOG simulation of the star effective temperature.

A0 : Extinction at 550 nm (float, Magnitude[mag])

GOG simulation of the extinction at 550 nm.

FEH : Iron abundance (float, Abundances[dex])

Draft

GOG simulation of the iron abundance.

LOGG : Surface gravity (float, GravitySurface[log cgs])

GOG simulation of the log10 of the surface gravity (log cgs).

Draft

14.2 GUM_GALAXY [TAP]

Table of simulated unresolved galaxies according to the Gaia Universal Model Simulation (GUMS). True values of the intrinsic simulated quantities (astrometry, photometry and physical parameters) for the sources generated by GOG using the Universe Model are given. No errors are added.

Columns description:

SOURCE_EXTENDED_ID : Extended source identifier (string)

Unique object String identifier describing:

- the object type. Here only galaxies are present, code ='G'
- the HTM region of the object (0-268435455)
- the object number in this region (0-262143)
- the multiplicity
- a variability flag ('V')

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Long Identifier (long)

A unique source identifier composed by the type of source, the sky region where the source is located and a sequential number inside this region (as the `source_extended_id`).

In the `source_id` information is coded within the 64 bit long integer. The celestial position is encoded via a level 12 Hierarchical Triangular Mesh, covering roughly one square arc minute size on the average:

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- The 4 most significant bits contain the object type: (0: Unknown; 1: Stellar; 2: Galaxy; 3: QSO; 4: Supernova; 5: Exoplanet; 6: Noise or PPE; 7: Cluster; 8: Planetary Nebulae; 9: HII Region; 10: GRB; 11: Asteroid; 12: Comet; 13: Planet; 14: Satellite; 15: User Source; -1: Not identified)
- The 28 following bits contain the HTM index at level 12 (note the use of HTM as opposed to HEALPix in `source_id` in `gaia_source`)
- One bit indicates the variability,
- One bit indicates whether this is a multiple system (i.e. not a component).
- The 4 next groups of 3 bits are the 4 hierarchical levels for a multiple system. It is based on the fact that the maximum multiplicity of a hierarchical multiple system currently known is 7 (i.e. 3 bits are needed) with a 4 levels depth. The components in a level are sequential (i.e. non-hierarchical).
- The 18 least significant bits concern the object number in the region.

RANDOM_INDEX : Random index (long)

A random index which can be used to select smaller subsets of the data that are still representative. The column contains a random permutation of the numbers from 0 to $N - 1$, where N is the number of sources in the table. The random index can be useful for validation (testing on 10 different random subsets), visualization (displaying 1% of the data), and statistical exploration of the data, without the need to download all the data.

RA : Right Ascension (double, Angle[deg])

Right ascension of the barycentre at J2010 reference epoch in ICRS frame.

DEC : Declination (double, Angle[deg])

Declination of the barycentre at J2010 reference epoch in ICRS frame.

PMRA : Proper motion along right ascension (float, Angular Velocity[mas yr^{-1}])

Proper motion along right ascension at J2010 reference epoch: $\mu_{\alpha*} \equiv \mu_\alpha \cos\delta$. This is the local tangent plane projection of the proper motion vector in the direction of increasing right ascension.

Draft

PMDEC : Proper motion along declination (float, Angular Velocity[mas yr^{-1}])

Proper motion along declination at J2010 reference epoch. This is the projection of the proper motion vector in the direction of increasing declination.

MAG_G : Mean Apparent G magnitude (float, Magnitude[mag])

Mean apparent magnitude in the G band in the Vega scale.

MAG_BP : Mean Apparent BP magnitude (float, Magnitude[mag])

Mean Apparent magnitude in the integrated BP band in the Vega scale.

MAG_RP : Mean Apparent RP magnitude (float, Magnitude[mag])

Mean Apparent magnitude in the integrated RP band in the Vega scale.

MAG_RVS : Mean Apparent RVS magnitude (float, Magnitude[mag])

Mean Apparent magnitude in the integrated RVS band in the Vega scale.

MEAN_ABSOLUTE_V : Mean Absolute V magnitude (float, Magnitude[mag])

Mean Absolute V (Johnson) magnitude.

V_I : (V-I) colour (float, Magnitude[mag])

Intrinsic V−I colour (Johnson-Cousins).

AG : Absorption in G (float, Magnitude[mag])

Line-of-sight interstellar absorption in the G band.

A0 : Absorption at 550 nm (float, Magnitude[mag])

Draft

Line-of-sight interstellar absorption at 550 nm.

REDSHIFT : Redshift (float)

Redshift of the supernovae.

HUBBLE_TYPE : Hubble Type (int)

A code denoting the galaxy type amongst: 0=E2, 1=E-S0, 2=Sa, 3=Sb, 4=Sbc, 5=Sc, 6=Sd, 7=Im, 8=QSFG.

SHAPE : Shape description (string)

Shape parameter description, following SkyMaker format with comma-separated parameters. See `https://www.astromatic.net/software/skymaker/` for further information.

L : Galactic longitude (float, Angle[deg])

Galactic longitude at J2010 reference epoch.

B : Galactic latitude (float, Angle[deg])

Galactic latitude at J2010 reference epoch.

Draft

14.3 GUM_MW [TAP]

Table of simulated galactic stars according to the Gaia Universal Model Simulation (GUMS). True values of the intrinsic simulated quantities (astrometry, photometry and physical parameters) for the sources generated by GOG using the Universe Model are given. No errors are added.

Columns description:

SOURCE_EXTENDED_ID : Extended source identifier (string)

Unique object String identifier describing:

- the object type. Here only stars or stellar systems are present, code ='*'
- the HTM region of the object (0-268435455)
- the object number in this region (0-262143)
- the multiplicity (Washington Double Stars Catalogue -WDS- type, 7 components, 4 hierarchical levels max)
- a variability flag ('V')

Example: *000000455-000035Ab2V would be a variable star and component Ab2 in a 3-level system. This system would be the 35th of the HTM region 455. Such a system would have at least 7 sources present in the GUMS table: *000000455-000035+ (the system), *000000455-000035A+, *000000455-000035B, *000000455-000035Aa, *000000455-000035Ab+, *000000455-000035Ab1, *000000455-000035Ab2V.

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Long Identifier (long)

A unique source identifier composed by the type of source, the sky region where the source is located and a sequential number inside this region (as the `source_extended_id`).

In the `source_id` information is coded within the 64 bit long integer. The celestial position is encoded via a level 12 Hierarchical Triangular Mesh, covering roughly one square arc minute size on the average:

- The 4 most significant bits contain the object type: (0: Unknown; 1: Stellar; 2: Galaxy; 3: QSO; 4: Supernova; 5: Exoplanet; 6: Noise or PPE; 7: Cluster; 8: Planetary Nebulae; 9: HII Region; 10: GRB; 11: Asteroid; 12: Comet; 13: Planet; 14: Satellite; 15: User Source; -1: Not identified)
- The 28 following bits contain the HTM index at level 12 (note the use of HTM as opposed to HEALPix in `source_id` in `gaia_source`)
- One bit indicates the variability,
- One bit indicates whether this is a multiple system (i.e. not a component).
- The 4 next groups of 3 bits are the 4 hierarchical levels for a multiple system. It is based on the fact that the maximum multiplicity of a hierarchical multiple system currently known is 7 (i.e. 3 bits are needed) with a 4 levels depth. The components in a level are sequential (i.e. non-hierarchical).
- The 18 least significant bits concern the object number in the region.

RANDOM_INDEX : Random index (long)

A random index which can be used to select smaller subsets of the data that are still representative. The column contains a random permutation of the numbers from 0 to $N - 1$, where N is the number of sources in the table. The random index can be useful for validation (testing on 10 different random subsets), visualization (displaying 1% of the data), and statistical exploration of the data, without the need to download all the data.

RA : Right Ascension (double, Angle[deg])

Right ascension of the barycentre at J2010 reference epoch in ICRS frame.

DEC : Declination (double, Angle[deg])

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Declination of the barycentre at J2010 reference epoch in ICRS frame.

BARYCENTRIC_DISTANCE : Barycentric distance to the simulated source (float, Length & Distance[pc])

Distance from the barycentre of the Solar System to the barycentre of the source at J2010 reference epoch.

PMRA : Proper motion along right ascension (float, Angular Velocity[mas yr^{-1}])

Proper motion along right ascension at J2010 reference epoch: $\mu_{\alpha*} \equiv \mu_\alpha \cos\delta$. This is the local tangent plane projection of the proper motion vector in the direction of increasing right ascension.

PMDEC : Proper motion along declination (float, Angular Velocity[mas yr^{-1}])

Proper motion along declination at J2010 reference epoch. This is the projection of the proper motion vector in the direction of increasing declination.

RADIAL_VELOCITY : Radial Velocity (float, Velocity[km s^{-1}])

Spectroscopic radial velocity in the solar barycentric reference frame at J2010 reference epoch.

MAG_G : Mean Apparent G magnitude (float, Magnitude[mag])

Mean apparent magnitude in the G band in the Vega scale.

MAG_BP : Mean Apparent BP magnitude (float, Magnitude[mag])

Mean Apparent magnitude in the integrated BP band in the Vega scale.

MAG_RP : Mean Apparent RP magnitude (float, Magnitude[mag])

Mean Apparent magnitude in the integrated RP band in the Vega scale.

MAG_RVS : Mean Apparent RVS magnitude (float, Magnitude[mag])

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Mean Apparent magnitude in the integrated RVS band in the Vega scale.

V_I : (V-I) colour (float, Magnitude[mag])

Intrinsic V−I colour (Johnson-Cousins).

MEAN_ABSOLUTE_V : Mean Absolute V magnitude (float, Magnitude[mag])

Mean Absolute V (Johnson) magnitude.

AG : Absorption in G (float, Magnitude[mag])

Line-of-sight interstellar absorption in the G band.

A0 : Absorption at 550 nm (float, Magnitude[mag])

Line-of-sight interstellar absorption at 550 nm.

TEFF : Effective temperature (float, Temperature[K])

Stellar effective temperature.

SPECTRAL_TYPE : spectral class + luminosity class (string)

Stellar MK classification.

LOGG : Surface gravity (float, GravitySurface[log cgs])

Stellar surface gravity log g.

FEH : Metallicity (float, Abundances[dex])

Stellar metallicity [Fe/H].

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ALPHAFE : Alpha elements (float, Abundances[dex])

Abundance of alpha-elements with respect to iron [α/Fe].

MBOL : Absolute bolometric magnitude (float, Magnitude[mag])

Mean absolute bolometric magnitude.

AGE : Age (float, Time[Gyr])

Age of the stellar source (Gyr).

MASS : Mass (float, Mass[Solar Mass])

Stellar mass.

RADIUS : Radius (float, Length & Distance[Solar Radius])

Stellar radius (mean value for variable pulsating stars).

POPULATION : Population (int)

Galactic stellar population:

- 0-6 = thin disc
- 7 = young thick disc
- 8 = halo
- 9 = bar
- 10 = old thick disc

HAS_PHOTOCENTER_MOTION : Boolean describing if the photocentre has or not hotspot induced motion (boolean)

Field to describe if the photocentre motion is simulated including the effect on astrometry of stellar surface inhomogeneities (see Pasquato et al. (2011)). This is independent of any potential

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orbital motion.

NC : Number of components (int)

Number of components (a component can be either one star or a system).

NT : Total number of object (int)

Total number of objects (stars, brown dwarfs or exoplanets) in the system.

SEMIMAJOR_AXIS : Semi major axis (float, Length & Distance[AU])

Semi-major axis of the orbit of the component about the system barycentre (for multiple systems).

ECCENTRICITY : Eccentricity (float)

Eccentricity of the orbit of the component (for multiple systems).

INCLINATION : Inclination (float, Angle[deg])

Inclination of the orbit of the component (for multiple systems).

LONGITUDE_ASCENDING_NODE : Longitude of ascending node (float, Angle[deg])

Longitude of the ascending node of the orbit of the component (for multiple systems).

ORBIT_PERIOD : Period of the orbit (float, Time[day])

Period of the orbit of the component (for multiple systems).

PERIASTRON_DATE : Date of periastron (float, Time[day])

Periastron date of the orbit of the component (for multiple systems). It is a uniform random value within [0, Period[. Reference time is J2010.

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PERIASTRON_ARGUMENT : Periastron argument (float, Angle[deg])

Periastron argument of the orbit of the component (for multiple systems).

VARIABILITY_TYPE : Variability type (string)

Stellar variability type : ACV, be, cepheid, classicalnovae, deltascuti, dwarfnovae, DYPer, emission, Flaring, gammador, microlens, mira, RCrBs, RRab, RRc, roAp, semiregular, ZZceti.

VARIABILITY_AMPLITUDE : Amplitude of variability (float, Magnitude[mag])

Photometric variability amplitude in V magnitude.

VARIABILITY_PERIOD : Period of variability (float, Time[day])

Photometric variability period.

VARIABILITY_PHASE : Phase of variability (float)

Photometric variability phase at J2010 reference epoch.

R_ENV_R_STAR : Envelope characterisic for Be stars (float)

Envelope characteristic parameter for Be stars.

L : Galactic longitude (float, Angle[deg])

Galactic longitude at J2010 reference epoch.

B : Galactic latitude (float, Angle[deg])

Galactic latitude at J2010 reference epoch.

Draft

14.4 GUM_QSO [TAP]

Table of simulated QSOs according to the Gaia Universal Model Simulation (GUMS). True values of the intrinsic simulated quantities (astrometry, photometry and physical parameters) for the sources generated by GOG using the Universe Model are given. No errors are added.

Columns description:

SOURCE_EXTENDED_ID : Extended source identifier (string)

Unique object String identifier describing:

- the object type. Here only QSOs are present, code ='Q'
- the HTM region of the object (0-268435455)
- the object number in this region (0-262143)
- the multiplicity
- a variability flag ('V')

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Long Identifier (long)

A unique source identifier composed by the type of source, the sky region where the source is located and a sequential number inside this region (as the `source_extended_id`).

In the `source_id` information is coded within the 64 bit long integer. The celestial position is encoded via a level 12 Hierarchical Triangular Mesh, covering roughly one square arc minute size on the average:

Draft

- The 4 most significant bits contain the object type: (0: Unknown; 1: Stellar; 2: Galaxy; 3: QSO; 4: Supernova; 5: Exoplanet; 6: Noise or PPE; 7: Cluster; 8: Planetary Nebulae; 9: HII Region; 10: GRB; 11: Asteroid; 12: Comet; 13: Planet; 14: Satellite; 15: User Source; -1: Not identified)
- The 28 following bits contain the HTM index at level 12 (note the use of HTM as opposed to HEALPix in `source_id` in `gaia_source`)
- One bit indicates the variability,
- One bit indicates whether this is a multiple system (i.e. not a component).
- The 4 next groups of 3 bits are the 4 hierarchical levels for a multiple system. It is based on the fact that the maximum multiplicity of a hierarchical multiple system currently known is 7 (i.e. 3 bits are needed) with a 4 levels depth. The components in a level are sequential (i.e. non-hierarchical).
- The 18 least significant bits concern the object number in the region.

RANDOM_INDEX : Random index (long)

A random index which can be used to select smaller subsets of the data that are still representative. The column contains a random permutation of the numbers from 0 to $N - 1$, where N is the number of sources in the table. The random index can be useful for validation (testing on 10 different random subsets), visualization (displaying 1% of the data), and statistical exploration of the data, without the need to download all the data.

RA : Right Ascension (double, Angle[deg])

Right ascension of the barycentre at J2010 reference epoch in ICRS frame.

DEC : Declination (double, Angle[deg])

Declination of the barycentre at J2010 reference epoch in ICRS frame.

PMRA : Proper motion along right ascension (float, Angular Velocity[mas yr^{-1}])

Proper motion along right ascension at J2010 reference epoch: $\mu_{\alpha*} \equiv \mu_\alpha \cos\delta$. This is the local tangent plane projection of the proper motion vector in the direction of increasing right ascension.

Draft

PMDEC : Proper motion along declination (float, Angular Velocity[mas yr^{-1}])

Proper motion along declination at J2010 reference epoch. This is the projection of the proper motion vector in the direction of increasing declination.

MAG_G : Mean Apparent G magnitude (float, Magnitude[mag])

Mean apparent magnitude in the G band in the Vega scale.

MAG_BP : Mean Apparent BP magnitude (float, Magnitude[mag])

Mean Apparent magnitude in the integrated BP band in the Vega scale.

MAG_RP : Mean Apparent RP magnitude (float, Magnitude[mag])

Mean Apparent magnitude in the integrated RP band in the Vega scale.

MAG_RVS : Mean Apparent RVS magnitude (float, Magnitude[mag])

Mean Apparent magnitude in the integrated RVS band in the Vega scale.

AG : Absorption in G (float, Magnitude[mag])

Line-of-sight interstellar absorption in the G band.

A0 : Absorption at 550 nm (float, Magnitude[mag])

Line-of-sight interstellar absorption at 550 nm.

REDSHIFT : Redshift (float)

Redshift of the supernovae.

SLOPE : Slope of the continuum (float)

Draft

Together with the redshift and *w* (emission lines equivalent width), it allows to select the adapted spectra to assign to the QSO.

W : Total equivalent width of the emission lines (float)

Together with the redshift and the continuum slope, this parameter allows to select the adapted spectra to assign to the QSO.

L : Galactic longitude (float, Angle[deg])

Galactic longitude at J2010 reference epoch.

B : Galactic latitude (float, Angle[deg])

Galactic latitude at J2010 reference epoch.

Draft

14.5 GUM_SN [TAP]

Table of simulated supernovae according to the Gaia Universal Model Simulation (GUMS). True values of the intrinsic simulated quantities (astrometry, photometry and physical parameters) for the sources generated by GOG using the Universe Model are given. No errors are added.

Columns description:

SOURCE_EXTENDED_ID : Extended source identifier (string)

Unique object String identifier describing:

- the object type. Here only supernovae are present, code ='N'
- the HTM region of the object (0-268435455)
- the object number in this region (0-262143)
- the multiplicity
- a variability flag ('V')

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Long Identifier (long)

A unique source identifier composed by the type of source, the sky region where the source is located and a sequential number inside this region (as the `source_extended_id`).

In the `source_id` information is coded within the 64 bit long integer. The celestial position is encoded via a level 12 Hierarchical Triangular Mesh, covering roughly one square arc minute size on the average:

Draft

- The 4 most significant bits contain the object type: (0: Unknown; 1: Stellar; 2: Galaxy; 3: QSO; 4: Supernova; 5: Exoplanet; 6: Noise or PPE; 7: Cluster; 8: Planetary Nebulae; 9: HII Region; 10: GRB; 11: Asteroid; 12: Comet; 13: Planet; 14: Satellite; 15: User Source; -1: Not identified)
- The 28 following bits contain the HTM index at level 12 (note the use of HTM as opposed to HEALPix in `source_id` in `gaia_source`)
- One bit indicates the variability,
- One bit indicates whether this is a multiple system (i.e. not a component).
- The 4 next groups of 3 bits are the 4 hierarchical levels for a multiple system. It is based on the fact that the maximum multiplicity of a hierarchical multiple system currently known is 7 (i.e. 3 bits are needed) with a 4 levels depth. The components in a level are sequential (i.e. non-hierarchical).
- The 18 least significant bits concern the object number in the region.

RANDOM_INDEX : Random index (long)

A random index which can be used to select smaller subsets of the data that are still representative. The column contains a random permutation of the numbers from 0 to $N - 1$, where N is the number of sources in the table. The random index can be useful for validation (testing on 10 different random subsets), visualization (displaying 1% of the data), and statistical exploration of the data, without the need to download all the data.

RA : Right Ascension (double, Angle[deg])

Right ascension of the barycentre at J2010 reference epoch in ICRS frame.

DEC : Declination (double, Angle[deg])

Declination of the barycentre at J2010 reference epoch in ICRS frame.

BARYCENTRIC_DISTANCE : Barycentric distance to the simulated source (float, Length & Distance[pc])

Distance from the barycentre of the Solar System to the barycentre of the source at J2010 reference epoch.

Draft

PMRA : Proper motion along right ascension (float, Angular Velocity[mas yr^{-1}])

Proper motion along right ascension at J2010 reference epoch: $\mu_{\alpha*} \equiv \mu_\alpha \cos\delta$. This is the local tangent plane projection of the proper motion vector in the direction of increasing right ascension.

PMDEC : Proper motion along declination (float, Angular Velocity[mas yr^{-1}])

Proper motion along declination at J2010 reference epoch. This is the projection of the proper motion vector in the direction of increasing declination.

RADIAL_VELOCITY : Radial Velocity (float, Velocity[km s^{-1}])

Spectroscopic radial velocity in the solar barycentric reference frame at J2010 reference epoch.

MAG_G : Mean Apparent G magnitude (float, Magnitude[mag])

Mean apparent magnitude in the G band in the Vega scale.

MAG_BP : Mean Apparent BP magnitude (float, Magnitude[mag])

Mean Apparent magnitude in the integrated BP band in the Vega scale.

MAG_RP : Mean Apparent RP magnitude (float, Magnitude[mag])

Mean Apparent magnitude in the integrated RP band in the Vega scale.

MAG_RVS : Mean Apparent RVS magnitude (float, Magnitude[mag])

Mean Apparent magnitude in the integrated RVS band in the Vega scale.

V_I : (V-I) colour (float, Magnitude[mag])

Intrinsic V−I colour (Johnson-Cousins).

MEAN_ABSOLUTE_V : Mean Absolute V magnitude (float, Magnitude[mag])

Draft

Mean Absolute V (Johnson) magnitude.

AG : Absorption in G (float, Magnitude[mag])

Line-of-sight interstellar absorption in the G band.

A0 : Absorption at 550 nm (float, Magnitude[mag])

Line-of-sight interstellar absorption at 550 nm.

TYPE : Supernovae Type (string)

Supernovae type : Ia, Ib/c, II-L, II-P.

REDSHIFT : Redshift (float)

Redshift of the supernovae.

EPOCH : Time since the explosion (float, Time[day])

Time in days since the explosion of the supernovae.

L : Galactic longitude (float, Angle[deg])

Galactic longitude at J2010 reference epoch.

B : Galactic latitude (float, Angle[deg])

Galactic latitude at J2010 reference epoch.

Draft

15 Solar System

As indicated in the below subsection titles by the labels [TAP] and [DataLink], these products are served from the ESA Gaia Archive using two different protocols, the Table Access Protocol (TAP) for tabular products and the DataLink protocol for other products such as spectra, light curves, MCMC samples, etc. For more details please see Section **??**.

15.1 SSO_OBSERVATION [TAP]

Solar System object observations. Each table line contains data obtained during the transit of the source on a single CCD, during a single transit. The corresponding epoch is provided. Data not varying within the transit are repeated identically for all single observations of that transit.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Source identifier (long)

A unique single numerical identifier of the source. Note in particular that these identifiers are by convention negative for SSOs. Association of this identifiers to other denominations in the standard of the Minor Planets Center can be obtained from the `sso_source` table.

TRANSIT_ID : Transit Identifier (long)

The `transit_id` is a unique identifier assigned to each detected (and confirmed) source as it transits the Gaia focal plane. Each time a given source is detected as Gaia scans and re-scans the sky a new `transit_id` will be created to badge that apparition. Hence the along–scan time and the across–scan position along with the telescope in which the source was detected are used to form a unique integer with which to label the transit.

Draft

The several features of a detection that are encoded in `transit_id` can be easily retrieved using bit masks (&) and shifts (>>) as follows:

- On-Board Mission Time line [ns]
  = 204800 * ((`transit_id` >> 17) & (0x000003FFFFFFFFFF))
- Field-of-view = 1 + (`transit_id` >> 15) & 0x03 [1 for 'preceding' and 2 for 'following' fields-of-view respectively]
- CCD row = (`transit_id` >> 12) & 0x07 [dimensionless, in the range 1 to 7]
- Across-scan 'reference acquisition pixel' in strip AF1 = (`transit_id`) & 0x0FFF [pixels] (this is the across-scan centre of the AF1 window and is odd if immediately below the mid-point of the window and even if immediately above)

where the bit mask prefix '0x' denotes hexadecimal.

For further details see Portell et al. (2020). For convenience a decoder for `transit_id` is available on-line at

`https://gaia.esac.esa.int/decoder/transitidDecoder.jsp`

OBSERVATION_ID : Observation Identifier (long)

Identifier at single CCD level of the observation of a Solar System object. It is unique, and obtained from a combination of `transit_id` and an integer number representing the CCD strip: `observation_id` = `transit_id` ×10 + AF CCD number.

FOV : Field of view (byte)

Field of view (0 = preceding, 1 = following).

EPOCH : Gaiacentric epoch TCB(Gaia) (double, Time[Gaia-Centric JD in TCB − 2 455 197.5 (day)])

Gaiacentric epoch TCB(Gaia) in JD corresponding to the time of crossing of the fiducial line of the CCD (mid exposure). This is the epoch to which the target coordinates and the position/velocity of Gaia are referred. To avoid loss of precision the reference time J2010.0 has been subtracted.

EPOCH_ERROR : Error in Gaiacentric epoch (double, Time[day])

The error in the Gaiacentric epoch (for both `epoch` and `epoch_utc`).

EPOCH_UTC : Gaiacentric TCB epoch converted to UTC (double, Time[Gaia-Centric JD in UTC − 2 455 197.5 (day)])

Gaiacentric epoch in UTC in JD-J2010.0 corresponding to right ascension and declination obtained from the conversion of TCB(Gaia) to UTC.

RA : Right ascension of the source (double, Angle[deg])

ICRS right ascension of the source as observed by Gaia at `epoch`, corrected for full relativistic aberration but not for relativistic light deflection in the gravitational field of the Solar System.

DEC : Declination of the source (double, Angle[deg])

ICRS declination of the source as observed by Gaia at `epoch`, corrected for full relativistic aberration but not for relativistic light deflection in the gravitational field of the Solar System.

RA_ERROR_SYSTEMATIC : Standard error of right ascension, systematic (double, Angle[mas])

Uncertainty on right ascension, systematic component (assumed to be constant during a transit), multiplied by cos of declination.

DEC_ERROR_SYSTEMATIC : Standard error of declination, systematic (double, Angle[mas])

Standard error for declination, systematic component (assumed to be constant during a transit).

RA_DEC_CORRELATION_SYSTEMATIC : Correlation of ra and dec errors, systematic (double)

Correlation of ra_error_systematic and dec_error_systematic.

RA_ERROR_RANDOM : Standard error of right ascension, random (double, Angle[mas])

Uncertainty on right ascension, random component, multiplied by cos of declination.

Draft

DEC_ERROR_RANDOM : Standard error of declination, random (double, Angle[mas])

Standard error for declination, random component.

RA_DEC_CORRELATION_RANDOM : Correlation of ra and dec errors, random (double)

Correlation of ra and dec uncertainty, random component.

G_MAG : Calibrated G mag (double, Magnitude[mag])

G magnitude derived from `g_flux`.

G_MAG_REDUCED : Magnitudes reduced at unit distance for the transit (double, Magnitude[mag])

Apparent magnitudes reduced at unit distance from Gaia and from the Sun.

G_FLUX : Average calibrated G flux for the transit (double, Flux[e^- s^{-1}])

Average calibrated G flux for the transit.

G_FLUX_ERROR : Error on the G flux (double, Flux[e^- s^{-1}])

Error on the average transit-level G flux derived from the individual CCD-level flux measurements of the single transit.

BP_FLUX : Integrated BP flux value of the transit (float, Flux[e^- s^{-1}])

The integrated BP flux value of the transit. If the BP flux has been rejected or is unavailable, the flux will be null.

BP_FLUX_ERROR : Error on the integrated BP flux (float, Flux[e^- s^{-1}])

The uncertainty on the integrated BP Flux. If the BP flux has been rejected or is unavailable, this

Draft

will be null.

RP_FLUX : Integrated RP flux value of the transit (float, Flux[e^- s^{-1}])

The integrated RP flux value of the transit. If the RP flux has been rejected or is unavailable, the flux will be null.

RP_FLUX_ERROR : Error on the integrated RP flux (float, Flux[e^- s^{-1}])

The uncertainty on the integrated RP Flux. If the RP flux has been rejected or is unavailable, this will be null.

BP_MAG : Integrated BP magnitude (float, Magnitude[mag])

Magnitude in the integrated BP band. The magnitude is computed using the `bp_flux` value and the corresponding magnitude zero-point in the Vega scale. No uncertainty is provided for this quantity as the error distribution is only symmetric in flux space. This converts to an asymmetric error distribution in magnitude space which cannot be represented by a single error value.

RP_MAG : Integrated RP magnitude (float, Magnitude[mag])

Magnitude in the integrated RP band. The magnitude is computed using the `rp_flux` value and the corresponding magnitude zero-point in the Vega scale. No uncertainty is provided for this quantity as the error distribution is only symmetric in flux space. This converts to an asymmetric error distribution in magnitude space which cannot be represented by a single error value.

X_GAIA : Barycentric x position of Gaia (double, Length & Distance[AU])

Barycentric equatorial J2000 (ICRS) x position of Gaia at the epoch of the observation.

Y_GAIA : Barycentric y position of Gaia (double, Length & Distance[AU])

Barycentric equatorial J2000 y position (ICRS) of Gaia at the epoch of the observation.

Z_GAIA : Barycentric z position of Gaia (double, Length & Distance[AU])

Draft

Barycentric equatorial J2000 z position (ICRS) of Gaia at the epoch of observation.

VX_GAIA : Barycentric x velocity of Gaia (double, Velocity[AU day^{-1}])

Barycentric equatorial J2000 (ICRS) x velocity of Gaia at the epoch of the observation.

VY_GAIA : Barycentric y velocity of Gaia (double, Velocity[AU day^{-1}])

Barycentric equatorial J2000 (ICRS) y velocity of Gaia at the epoch of observation.

VZ_GAIA : Barycentric z velocity of Gaia (double, Velocity[AU day^{-1}])

Barycentric equatorial J2000 (ICRS) z velocity of Gaia at the epoch of observation.

X_GAIA_GEOCENTRIC : Geocentric x position of Gaia (double, Length & Distance[AU])

Geocentric equatorial J2000 x position of Gaia at the epoch of observation, in a reference aligned to ICRS.

Y_GAIA_GEOCENTRIC : Geocentric y position of Gaia (double, Length & Distance[AU])

Geocentric equatorial J2000 y position of Gaia at the epoch of observation in a reference aligned to ICRS.

Z_GAIA_GEOCENTRIC : Geocentric z position of Gaia (double, Length & Distance[AU])

Geocentric equatorial J2000 z position of Gaia at the epoch of observation in a reference aligned to ICRS.

VX_GAIA_GEOCENTRIC : Geocentric x velocity of Gaia (double, Velocity[AU day^{-1}])

Geocentric equatorial J2000 x velocity of Gaia at the epoch of observation in a reference aligned to ICRS.

VY_GAIA_GEOCENTRIC : Geocentric y velocity of Gaia (double, Velocity[AU day^{-1}])

Draft

Geocentric equatorial J2000 y velocity of Gaia at the epoch of observation in a reference aligned to ICRS.

VZ_GAIA_GEOCENTRIC : Geocentric z velocity of Gaia (double, Velocity[AU day^{-1}])

Geocentric equatorial J2000 z velocity of Gaia at the epoch of observation in a reference aligned to ICRS.

POSITION_ANGLE_SCAN : Position angle of the scanning direction (double, Angle[deg])

Position angle of the scan direction at the epoch of observation in the equatorial reference frame. 0 = North direction, $\pi/2$ = increasing right ascension, pi = South, $3\pi/2$ = decreasing right ascension. It is defined as the angle between the AL direction and the direction to the North Pole, at the SSO position, after applying the correction for aberration. As a consequence of this correction for aberration, the AC direction is not strictly perpendicular to the AL direction.

PHASE_ANGLE : Phase angle (float, Angle[deg])

Solar phase angle of the SSO at observation time.

ASTROMETRIC_OUTCOME_CCD : Result of processing the CCDs (long)

Result of the astrometric processing of the individual CCDs in the transit. Values presently defined:

1 Good position derived.

11 No position derived, because no centroid could be determined.

12 Position rejected a priori, because previous studies have shown that it is unreliable.

24 Position rejected, because this CCD has some samples that have been eliminated.

25 Position rejected, because this CCD is affected by an AOCS update.

26 Position rejected, because this CCD is affected by a non-nominal gating.

29 Position rejected, because of more than one reason, combination of codes 21-26.

32 Position rejected as outlier, because this position does not fit on the regression line.

Draft

35 Position rejected, because it does not fulfill the magnitude-uncertainty relation, see documentation.

39 Position rejected, because the value of distToLastCi is invalid.

40 Position rejected, because the epoch corresponds to known bad attitude, see documentation.

41 Position rejected, because no attitude or no calibration is available for the epoch of observation.

ASTROMETRIC_OUTCOME_TRANSIT : Result of processing the transit (long)

Result of the astrometric processing of the transit. Values defined at present are:

1 The transit contains at least one good position.

2 Positions derived, but less consistent than expected. This means that the criterion to reject outliers had to be relaxed to find an unambiguous set of consistent positions.

5 Positions derived, but outlier rejection was not possible because of dramatic loss of precision.

6 Positions derived, but no future field angles in RP/BP, because no attitude is available for the future epochs (reference epoch + 45, 50 and 55 seconds).

SPECTROPHOTOMETRIC_OUTCOME_TRANSIT : Information on the use of the transit in BP and RP (byte)

Information relative on the use or the reason of filtering of the transit in the process of computation of mean reflectance spectrum. Values presently defined:

0 Transit used to compute the mean reflectance.

1 Transit filtered because it has been observed before the Decontamination run 4.

2 Transit filtered because its AC motion is above 10 mas/s.

3 Transit filtered because both BP and RP calibrated epoch spectra are not available.

4 Transit filtered because only BP calibrated epoch spectrum is not available.

5 Transit filtered because only RP calibrated epoch spectrum is not available.

Draft

6 Transit filtered because both BP and RP calibrated epoch spectra have their location of the Arbitrary Reference System (ARS) in the observed sampling outside the acceptable range [5;55].

7 Transit filtered because only BP calibrated epoch spectrum have its location of the ARS in the observed sampling outside the acceptable range [5;55].

8 Transit filtered because only RP calibrated epoch spectrum have its location of the ARS in the observed sampling outside the acceptable range [5;55].

9 Transit filtered because of issue found on BP calibrated epoch spectrum by upstream system that could be one of these reasons:

- No data available.
- The transit is affected by truncation or multiple gate configuration in a way that is considered unrecoverable.
- Predicted position computation failed or predicted position outside window.
- No calibration available.
- Calibration application failure.
- Rejected in the source update process.
- Observation affected by satellite event.
- Affected by numerical saturation.
- Affected by hot column.

10 Transit filtered because of issue found on RP calibrated epoch spectrum by upstream system that could be one of these reasons:

- No data available.
- The transit is affected by truncation or multiple gate configuration in a way that is considered unrecoverable.
- Predicted position computation failed or predicted position outside window.
- No calibration available.
- Calibration application failure.
- Rejected in the source update process.
- Observation affected by satellite event.
- Affected by numerical saturation.
- Affected by hot column.

11 Transit filtered because only BP calibrated epoch spectrum have all samples masked or flagged as outliers in PhotPipe processing.

Draft

12 Transit filtered because BP or RP calibrated epoch spectrum have all samples masked or flagged as outliers in PhotPipe processing.

13 Transit filtered because only BP calibrated epoch spectrum have more than 20 samples masked or flagged as outliers in PhotPipe processing.

14 Transit filtered because BP or RP calibrated epoch spectrum have more than 20 samples masked or flagged as outliers in PhotPipe processing.

15 Transit filtered because BP or RP integrated calibrated fluxes computed on calibrated epoch spectra are negative.

LEVEL_OF_CONFIDENCE : Level of confidence of the identification (short, Dimensionless[percentage/100])

Level of confidence in the identification of the given SSO `source_id` with this observation: 0 stands for completely unambiguous identification, 100 stands for failed identification.

RA_ORBIT_RESIDUAL : Position residual with respect to orbital fit (double, Angle[mas])

Difference between the observed position and the position computed from the best-fitting orbit for the ra component.

DEC_ORBIT_RESIDUAL : Position residual with respect to orbital fit (double, Angle[mas])

Difference between the observed position and the position computed from the best-fitting orbit for the dec component.

AL_ORBIT_RESIDUAL : Position residual with respect to orbital fit (double, Angle[mas])

Difference between the observed position and the position computed from the best-fitting orbit, along the direction of the scan.

AC_ORBIT_RESIDUAL : Position residual with respect to orbital fit (double, Angle[mas])

Difference between the observed position and the position computed from the best-fitting orbit in the across-scan direction.

OBS_REJECTED_BY_FIT : Flag that indicates rejection of the observation in orbital fit (boolean)

Draft

The value set to True indicates that the observation was rejected by the automated orbit adjustement on the base of its associated residual.

ANGULAR_SIZE : Gaiacentric angular size (double, Length & Distance[pixel])

Angular size of the asteroid determined analysing the CCD signals. It is defined as the apparent dimension of the semimajor axis of the object, supposed to be represented as a triaxial ellipsoid.

ANGULAR_SIZE_ERROR : Gaiacentric angular size error (double, Length & Distance[pixel])

Nominal uncertainty of `angular_size`.

AL_POS_INFO : Image parameters determination source model in along-scan direction (int)

Complexity level of processing of the single along-scan CCD signal used to determine centroid and apparent angular dimension of the source.

The possible values are the following:

- -1 : not processed (missing input data) or error encountered
- 0 or +1 : centroid computed assuming a non-moving point-like source
- +2 : centroid computed assuming a moving point-like source
- +30 : centroid and angular dimension computed assuming a moving Lommel-Seeliger sphere observed at zero phase (flat disk)
- +31 : centroid and angular dimension computed assuming a moving Lommel-Seeliger sphere, illuminated according to the Sun-asteroid-Gaia geometry
- +32 : centroid and angular dimension computed assuming a moving Lommel-Seeliger ellipsoid, reconstructed on the basis of the disc-integrated photometric inversion, and illuminated according to the Sun-asteroid-Gaia geometry

AC_POS_INFO : Image parameters determination source model in across-scan direction (int)

Complexity level of processing of the single across-scan CCD signal used to determine centroid and apparent angular dimension of the source.

The possible values are the following:

Draft

- -1 : not processed (missing input data) or error encountered
- 0 or +1 : centroid computed assuming a non-moving point-like source
- +2 : centroid computed assuming a moving point-like source
- +30 : centroid and angular dimension computed assuming a moving Lommel-Seeliger sphere observed at zero phase (flat disk)
- +31 : centroid and angular dimension computed assuming a moving Lommel-Seeliger sphere, illuminated according to the Sun-asteroid-Gaia geometry
- +32 : centroid and angular dimension computed assuming a moving Lommel-Seeliger ellipsoid, reconstructed on the basis of the disc-integrated photometric inversion, and illuminated according to the Sun-asteroid-Gaia geometry

Draft

15.2 SSO_REFLECTANCE_SPECTRUM [TAP]

This table contains the mean BP/RP reflectance spectra of asteroids computed as the ratio between the asteroid flux and an averaged solar analogue flux. In each row, the reflectance spectrum of a given asteroid is given at a given wavelength. Entries for all asteroids are concatenated into a single table.

Columns description:

SOURCE_ID : Source identifier (long)

A unique single numerical identifier of the source. Note in particular that these identifiers are by convention negative for SSOs.

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`.

NUM_OF_SPECTRA : Number of epoch spectra used to compute the average (int)

Number of epoch spectra combined to generated the mean reflectance spectrum.

WAVELENGTH : Internally-calibrated wavelength of reflectance spectrum (float[] array, Length & Distance[nm])

Internally-calibrated wavelength.

REFLECTANCE_SPECTRUM : Reflectance spectrum (float[] array)

Reflectance value at the given wavelength.

Draft

REFLECTANCE_SPECTRUM_ERROR : Error in reflectance spectrum (float[] array)

Error in the reflectance.

REFLECTANCE_SPECTRUM_FLAG : Reflectance spectrum value flag (byte[] array)

This flag indicates the quality of the reflectance in the applicable wavelength channel, and can take the following values:

- 0: Good quality
- 1: Potential poor quality
- 2: The quality appears compromised. The use of this reflectance value is not recommended.

Draft

15.3 SSO_SOURCE [TAP]

This table contains data related to Solar System objects observed by Gaia. The quantities in the table are derived from data reduction and are associated with single objects.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Source identifier (long)

A unique single numerical identifier of the source. Note in particular that these identifiers are by convention negative for SSOs.

NUMBER_MP : Source number following MPC conventions (long)

Source number. It follows the Minor Planet Centre convention for the minor planets and natural satellites of the planets. The user should check the `type` field to be sure to select only one category of objects. For instance, the asteroid n. 509 has the same number as the satellite 509, the 9th satellite of Jupiter, but a different `type`. This field can be 0 if a conventional number is not available. In that case, users can refer to the `denomination` always coupled to `type`. For joining to other tables, the unique identifier provided by `source_id` should be used.

DENOMINATION : Standard MPC denomination (string)

Name of the object. It follows the Minor Planet Centre convention for the minor planets and natural satellites of the planets.

TYPE : Object type (byte)

Draft

Type of the SSO:

- 0 = Star
- 1 = Planet
- 2 = Natural satellite
- 3 = Asteroid
- 4 = Comet
- 5 = New SSO
- 6 = Unknown SSO

N_OBSERVATIONS : Number of observations (int)

Number of CCD-level observations of the asteroid that appear in the `sso_observation` table.

EPOCH_STATE_VECTOR : TCB epoch for the state vector (double, Time[Gaia-Centric JD in TCB − 2 455 197.5 (day)])

Reference epoch of the state vector, mid-point of the time interval spanned by the observations of the Solar System Object (TCB). It is expressed in days, with origin at the epoch J2010.0 TCB.

ORBITAL_ELEMENTS_VAR_COVAR_MATRIX : Variance-covariance matrix on elliptical elements at reference Epoch (double[21] array)

Covariance uncertainty matrix of the osculating orbit at reference epoch (`epoch_state_vector`) (heliocentric). In the order, the values correspond to the following elements of the matrix, representing the upper triangle (indices are given):

- 00 = variance(a)
- 01 = co-variance(a, e)
- 02 = co-variance(a, i)
- 03 = co-variance(a, node)
- 04 = co-variance(a, peri)

Draft

- 05 = co-variance(a, M0)
- 11 = variance(e)
- 12 = co-variance(e, i)
- 13 = co-variance(e, node)
- 14 = co-variance(e, peri)
- 15 = co-variance(e, M0)
- 22 = variance(i)
- 23 = co-variance(i, node)
- 24 = co-variance(i, peri)
- 25 = co-variance(i, M0)
- 33 = variance(node)
- 34 = co-variance(node, peri)
- 35 = co-variance(node, M0)
- 44 = variance(peri)
- 45 = co-variance(peri, M0)
- 55 = variance(M0)

where the abbreviations and their units are:

- a (au); semi-major axis
- e (adimesional); eccentricity
- i (rad); inclination
- node (rad); longitude of the ascending node
- peri (rad); argument of periastron
- M0 (rad); mean anomaly

Draft

H_STATE_VECTOR : Heliocentric State Vector at reference Epoch (double[6] array, StateVector[au, au d^{-1}])

Initial conditions position and velocity (state vector) in cartesian coordinates at reference epoch (`epoch_state_vector`) (heliocentric, ICRF). Units are au for positions, and au/day for velocities.

H_STATE_VECTOR_VAR_COVAR_MATRIX : Covariance matrix of the State Vector (double[21] array)

Covariance uncertainty matrix of the state vector (`h_state_vector`) at reference epoch (`epoch_state_vecto`) (heliocentric, in the ICRF).

ARC_LENGTH : Orbital arc length spanned by the observations (double, Time[day])

Orbital arc spanned by the observations that were used to fit the orbit of the Solar System Object (in days).

N_OBSERVATIONS_FIT : Number of observations used for the orbital fit (int)

Total number of observations used in the orbital fit of the Solar System Object (i.e. excluding outliers).

SPIN_PERIOD : Rotation period (double, Time[hour])

Spin period in hours. It can be negative, and in that case this means that the sense of rotation is retrograde.

SPIN_PERIOD_ERROR : Error in rotation period (double, Time[hour])

The error in the spin period resulting from inversion of disk-integrated photometric data.

POLE_LONG : Pole longitude (double, Angle[deg])

Ecliptic longitude of the rotation pole.

Draft

POLE_LONG_ERROR : Error in pole longitude (double, Angle[deg])

Error in the computed ecliptic longitude of the rotation pole.

POLE_LONG2 : Pole Longitude second solution (double, Angle[deg])

Eclptic longitude of the rotation pole (this may be alternatively expressed in ecliptic longitude).

POLE_LAT : Pole latitude (double, Angle[deg])

Ecliptic latitude of the rotation pole.

POLE_LAT_ERROR : Error in pole latitude (double, Angle[deg])

Error in the computed ecliptic latitude of the rotation pole.

POLE_LAT2 : Pole Latitude second solution (double, Angle[deg])

Ecliptic latitude of the rotation pole.

BA_RATIO : b/a axial ratio (double)

The b/a axial ratio of the triaxial ellipsoid shape of the object.

BA_RATIO_ERROR : Error in b/a axial ratio (double)

The error in the resulting b/a axial ratio coming from the inversion of disk-integrated photometric data.

BA_RATIO2 : b/a axial ratio second solution (double)

The b/a axial ratio of the triaxial ellipsoid shape of the object.

CA_RATIO : c/a axial ratio (double)

Draft

The c/a axial ratio of the triaxial ellipsoid shape of the object.

CA_RATIO_ERROR : Error in c/a axial ratio (double)

The error in the resulting c/a axial ratio coming from the inversion of disk-integrated photometric data.

CA_RATIO2 : c/a axial ratio second solution (double)

The c/a axial ratio of the triaxial ellipsoid shape of the object.

PHOT_INV_SOL_RESIDUALS : Average residuals of photometry inversion solution (double, Magnitude[mag])

This parameter is the average residual of the photometry inversion solution, as computed by the genetic algorithm of inversion.

MAG_PHASE_SLOPE : Slope of mag-phase (double, Misc[mag/deg])

Slope of supposedly linear phase - magnitude relation.

MAG_PHASE_SLOPE_ERROR : Error in slope of mag-phase (double, Misc[mag/deg])

Error in the resulting slope of the supposedly linear mag - phase relation, according to the inversion of disk-integrated photometric data.

MAG_PHASE_SLOPE2 : Slope of mag-phase second solution (double, Misc[mag/deg])

Slope of supposedly linear phase - magnitude relation.

EXTRAPOLATED_MAG : Magnitude extrapolated at phase angle equal to zero (double, Magnitude[mag])

This field corresponds to the magnitude of the object extrapolated at a value of phase angle equal to zero. It is computed as an average of the extrapolation of the reduced magnitudes at each transit, assuming a linear phase-magnitude relation. As Gaia never observes Solar System ob-

jects close to opposition, the brightness surge at opposition is not represented by the available photometric data, so only this linearly extrapolated value can be provided.

EXTRAPOLATED_MAG_ERROR : Error on the magnitude extrapolated at phase angle equal to zero (double, Magnitude[mag])

Error of the zero-phase extrapolated magnitude of the object.

REF_EPOCH_PHOT : Epoch of the first accepted transit of the object selected for the computation of the photometric inversion (double, Time[Gaia-Centric JD in TCB − 2 455 197.5 (day)])

Epoch in TCB of the first transit of the object whose photometry is selected for the photometric inversion. It is computed at the asteroid location by subtracting the light time travel delay between the asteroid and Gaia, from the Gaia-centric epoch of the transit. This epoch is used as the time origin for the asteroid rotation phase, corresponding to the angles provided by `phi0` (and `phi02` when a second solution is present).

PHI0 : Reference rotational phase (double)

Rotational phase of the object at the reference epoch. The reference epoch is different for each object, and is the epoch of the first recorded transit of the object in the Gaia FOV. A `phi0` of zero corresponds to the meridian passing through the poles and the ‘b’ axis of the supposedly triaxial ellipsoid facing the observer.

PHI02 : Reference rotational phase second solution (double)

Rotational phase of the object at the reference epoch. The reference epoch is different for each object, and is the epoch of the first recorded transit of the object in the Gaia FOV. A `phi0` of zero corresponds to the meridian passing through the poles and the ‘b’ axis of the supposedly triaxial ellipsoid facing the observer.

N_PHOTOMETRY_TRANSITS : Number of transits used for the photometric inversion (int)

Number of transits used for the photometric inversion.

PHOT_INVERSION_FLAGS : Flags on the results of the photometric inversion (byte)

Draft

Flags describing the reasons why a solution was not considered valid (bit 0 to 2) and other properties that can be relevant to quickly assess the quality of the solution (bit 3 to 6). If one of the bits 0, 1 or 2 is equal to 1, the solution is rejected and its related fields are null

Detailed description:

- 0: there are less than 25 accepted transits
- 1: more than a single valid solution for the rotation period
- 2: linear magnitude-phase slope outside the]0.02, 0.06[mag/deg range
- 3: inversion solution residuals outside the optimal range
- 4: computed rotation period outside the optimal range
- 5: the shape is close to an oblate spheroid (high b/a ratio)
- 6: the shape is strongly elongated / flattened (low b/a ratio)

Precise values for the thresholds used to set bits 3 to 6 are given in the documentation. Reliable solutions will have all bits set to 0.

N_SPECTRA : Number of epoch spectra used to compute the average (int)

Number of epoch spectra combined to generate the mean reflectance spectrum.

TAXONOMIC_CLASS : Asteroid taxonomic classification (short)

The taxonomic classification of an asteroid, based on its spectroscopic properties. Its value is a number between 1 and *N* (this being the number of classes identified, equal to the length of the taxonomic class probability array).

TAXONOMIC_CLASS_PROBABILITIES : Probability of alternative classifications (double[] array, Dimensionless[percentage/100])

An array of possible alternative taxonomic classification, wherever some ambiguity is present.

Draft

15.4 SSO_XP_EPOCH_SPECTRUM [DataLink]

This table contains the epoch BP and RP spectra collected at transit level for SSO sources.

For each relevant `transit_id` the sampled flux of the BP and RP spectra is provided on a pseudo–wavelength grid. Information relative to the processing through the respective photometry and variability pipelines is also given.

Note this table is not available through the main archive TAP interface. Data are delivered via the Massive Data service indexed by the VO DataLink protocol and described in Chapter **??**. The DataLink retrieval type to specify for this product in programmatic archive access via the Astroquery Gaia Python package is `EPOCH_SPECTRUM_XP_SSO`.

DataLink products are served with a choice of two different data structures:

- `DATAMODEL_STANDARD`: the product is serialised following a standard data model (for example an IVOA-compliant data model in applicable cases);
- `DATAMODEL_GAIA`: the product is serialised following the Gaia DPAC(-internal) data model.

The `DATAMODEL_GAIA` format is described here whereas the `DATAMODEL_STANDARD` format is described in `https://www.cosmos.esa.int/web/gaia-users/archive/datalink-products#datalink_serialisation`

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (unique within a particular Data Release) (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

Note in particular that these identifiers are by convention negative for SSOs.

TRANSIT_ID : Transit Identifier (long)

The `transit_id` is a unique identifier assigned to each detected (and confirmed) source as it transits the Gaia focal plane. Each time a given source is detected as Gaia scans and re-scans the sky a new `transit_id` will be created to badge that apparition. Hence the along–scan time and the across–scan position along with the telescope in which the source was detected are used to form a unique integer with which to label the transit.

The several features of a detection that are encoded in `transit_id` can be easily retrieved using bit masks (&) and shifts (>>) as follows:

- On-Board Mission Time line [ns]
  = 204800 * ((`transit_id` >> 17) & (0x000003FFFFFFFFFF))
- Field-of-view = 1 + (`transit_id` >> 15) & 0x03 [1 for 'preceding' and 2 for 'following' fields-of-view respectively]
- CCD row = (`transit_id` >> 12) & 0x07 [dimensionless, in the range 1 to 7]
- Across-scan 'reference acquisition pixel' in strip AF1 = (`transit_id`) & 0x0FFF [pixels] (this is the across-scan centre of the AF1 window and is odd if immediately below the mid-point of the window and even if immediately above)

where the bit mask prefix '0x' denotes hexadecimal.

For further details see Portell et al. (2020). For convenience a decoder for `transit_id` is available on-line at

`https://gaia.esac.esa.int/decoder/transitidDecoder.jsp`

BP_OBS_TIME : BP CCD transit observation time (double, Time[Barycentric JD in TCB – 2 455 197.5 (day)])

Observation time of the BP CCD transit in units of Barycentric JD (in TCB) in days -2 455 197.5, computed as follows. First the observation time is converted from On-board Mission Time (OBMT) into Julian date in TCB (Temps Coordonnée Barycentrique). Next a correction is applied for the light-travel time to the Solar system barycentre, resulting in Barycentric Julian Date (BJD). Finally, an offset of 2 455 197.5 days is applied (corresponding to a reference time T_0 at

2010-01-01T00:00:00) to have a conveniently small numerical value.

BP_PSEUDO_WAVELENGTH : The array of sample positions of the BP spectrum in units of pseudo-wavelengths (float[60] array)

The BP pseudo–wavelengths of the spectrum for a given transit.

BP_FLUX : BP spectrum flux (float[60] array, Flux[e^- s^{-1}])

The BP flux of the spectrum in a given transit.

BP_FLUX_ERROR : Error on the flux for each sample of the BP spectrum (float[60] array, Flux[e^- s^{-1}])

The error on the BP Flux. If the BP flux has been rejected or is unavailable, this error will be set to null.

BP_GATE : Gate information for the selection of the appropriate calibration for the BP spectrum (byte)

The gate information contained in this field is aimed at the selection of the appropriate calibration if any is available. Calibrations will only be computed for the nominal cases of Gate 05, Gate 07, Gate 09, Gate 11 and None. Spectra obtained with other gates for which no calibration is available or with a multiple gate configuration that could not be treated as a nominal case by dropping part of the spectrum will fall into the same category.

The information will be encoded as follows:

- 0: Gate 05
- 1: Gate 07
- 2: Gate 09
- 3: Gate 11
- 4: Gate None (no gate)
- 5: Gate configuration with no corresponding calibration

Draft

This field will only filled in if there is a BP spectrum available for this FoV transit.

BP_AC_WINDOW_CENTRE : AC window centre value for the evaluation of the calibration for the BP spectrum (float, Length & Distance[pixel])

The AC window centre value contained in this field is aimed at the evaluation of the calibration.

This field will only filled in if there is a BP spectrum available for this FoV transit.

BP_AC_CE : AC centring error value for the evaluation of the calibration for the BP spectrum (float, Length & Distance[pixel])

The AC centring error value contained in this field is aimed at the evaluation of the calibration.

This field will only filled in if there is a BP spectrum available for this FoV transit.

BP_AC_RATE : AC rate value for the evaluation of the calibration for the BP spectrum (float, Velocity[pixel/s])

The AC rate value contained in this field is aimed at the evaluation of the calibration.

This field will only filled in if there is a BP spectrum available for this FoV transit.

BP_MASKED_SAMPLES : List of BP sample indices masked or flagged as outliers in PhotPipe processing (byte[] array)

List of sample ids masked or flagged as outliers in PhotPipe processing. The sample ids are the indexes in the 60 samples array containing the BP spectrum. For instance if the list contains [0,1,2,3,58,59], the first 4 samples and the last 2 are masked or flagged. If the list is empty, there are no masked or flagged samples.

RP_OBS_TIME : RP CCD transit observation time (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Observation time of the RP CCD transit in units of Barycentric JD (in TCB) in days -2 455 197.5, computed as follows. First the observation time is converted from On-board Mission Time (OBMT) into Julian date in TCB (Temps Coordonnée Barycentrique). Next a correction is applied for the light-travel time to the Solar system barycentre, resulting in Barycentric Julian Date

(BJD). Finally, an offset of 2 455 197.5 days is applied (corresponding to a reference time T_0 at 2010-01-01T00:00:00) to have a conveniently small numerical value.

RP_PSEUDO_WAVELENGTH : The array of sample positions of the RP spectrum in units of pseudo-wavelengths (float[60] array)

The RP pseudo–wavelengths of the spectrum for a given transit.

RP_FLUX : RP spectrum flux (float[60] array, Flux[e^- s^{-1}])

The RP flux of the spectrum in a given transit.

RP_FLUX_ERROR : Error on the flux for each sample of the RP spectrum (float[60] array, Flux[e^- s^{-1}])

The error on the RP Flux. If the RP flux has been rejected or is unavailable, this error will be set to null.

RP_GATE : Gate information for the selection of the appropriate calibration for the RP spectrum (byte)

The gate information contained in this field is aimed at the selection of the appropriate calibration if any is available. Calibrations will only be computed for the nominal cases of Gate 05, Gate 07, Gate 09, Gate 11 and None. Spectra obtained with other gates for which no calibration is available or with a multiple gate configuration that could not be treated as a nominal case by dropping part of the spectrum will fall into the same category.

The information will be encoded as follows:

- 0: Gate 05
- 1: Gate 07
- 2: Gate 09
- 3: Gate 11
- 4: Gate None (no gate)
- 5: Gate configuration with no corresponding calibration

Draft

This field will only filled in if there is a RP spectrum available for this FoV transit.

RP_AC_WINDOW_CENTRE : AC window centre value for the evaluation of the calibration for the RP spectrum (float, Length & Distance[pixel])

The AC window centre value contained in this field is aimed at the evaluation of the calibration.

This field will only filled in if there is a RP spectrum available for this FoV transit.

RP_AC_CE : AC centring error value for the evaluation of the calibration for the RP spectrum (float, Length & Distance[pixel])

The AC centring error value contained in this field is aimed at the evaluation of the calibration.

This field will only filled in if there is a RP spectrum available for this FoV transit.

RP_AC_RATE : AC rate value for the evaluation of the calibration for the RP spectrum (float, Velocity[pixel/s])

The AC rate value contained in this field is aimed at the evaluation of the calibration.

This field will only filled in if there is a RP spectrum available for this FoV transit.

RP_MASKED_SAMPLES : List of RP sample indices masked or flagged as outliers in PhotPipe processing (byte[] array)

List of sample ids masked or flagged as outliers in PhotPipe processing. The sample ids are the indexes in the 60 samples array containing the RP spectrum. For instance if the list contains [0,1,2,3,58,59], the first 4 samples and the last 2 are masked or flagged. If the list is empty, there are no masked or flagged samples.

WINDOW_CLASS : Window class (byte)

A window classification flag (not the transit window class in the Gaia telemetry sense) assigned to this FoV transit. This is provided here as it is required for the selection of the appropriate calibration. Note that, for 2D windows, PhotPipe does an AC binning of the samples, so this table always provides 1D arrays in the `bp_flux` and `rp_flux` fields. The encoding is 0: 1D window, 1: 2D window; this is opposite to the internal DPAC and spacecraft labelling conventions.

BP_CCD_INFO_FLAGS : CCD-level information on the input BP observations (byte)

Flag providing information at the CCD level for the BP observation.

- Bit 0: affected by numerical saturation
- Bit 1: affected by hot column

Other bits are available for future extensions.

BP_CCD_PROC_FLAGS : CCD-level information on PhotPipe processing for the BP observation (short)

This field reports information generated during the PhotPipe processing for the BP observations. It will bit-wise encode the following conditions indicated when set to '1':

- Bit 0: no data available; obviously, no calibrated epoch spectrum will be available in this case;
- Bit 1: the transit is affected by truncation or multiple gate configuration in a way that is considered unrecoverable;
- Bit 2: predicted position computation failed or predicted position outside window;
- Bit 3: the AC centring error was clamped when the LS calibration was applied; refer to the end-of-cycle documentation for details on the clamping ranges used;
- Bit 4: no calibration available (no calibrated CCD photometry will be available in this case);
- Bit 5: calibration rejected and replaced with neighbouring calibration;
- Bit 6: calibration application failure;
- Bit 7: excluded from the generation of the mean source spectrum due to pre-filtering based on available flags (this is different from outlier rejection that is reported in Bit 8);
- Bit 8: rejected in the source update process;
- Bit 9: observation affected by satellite event;

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- Bit 10: charge release correction applied;
- Bit 11: stray-light mitigation applied;
- Bit 12: local (KNN) background mitigation applied;
- Bit 13: crowding information available in the table `xp_epoch_crowding`;
- Bit 14: the transit has been treated for truncation or multiple gate configuration.

Remaining bits are available for future extensions.

RP_CCD_INFO_FLAGS : CCD-level information on the input RP observations (byte)

Flag providing information at the CCD level for the RP observation.

- Bit 0: affected by numerical saturation
- Bit 1: affected by hot column

Other bits are available for future extensions.

RP_CCD_PROC_FLAGS : CCD-level information on PhotPipe processing for the RP observation (short)

This field reports information generated during the PhotPipe processing for the RP observations. It will bit-wise encode the following conditions indicated when set to '1':

- Bit 0: no data available; obviously, no calibrated epoch spectrum will be available in this case;
- Bit 1: the transit is affected by truncation or multiple gate configuration in a way that is considered unrecoverable;
- Bit 2: predicted position computation failed or predicted position outside window;
- Bit 3: the AC centring error was clamped when the LS calibration was applied; refer to the end-of-cycle documentation for details on the clamping ranges used;
- Bit 4: no calibration available (no calibrated CCD photometry will be available in this case);

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- Bit 5: calibration rejected and replaced with neighbouring calibration;
- Bit 6: calibration application failure;
- Bit 7: excluded from the generation of the mean source spectrum due to pre-filtering based on available flags (this is different from outlier rejection that is reported in Bit 8);
- Bit 8: rejected in the source update process;
- Bit 9: observation affected by satellite event;
- Bit 10: charge release correction applied;
- Bit 11: stray-light mitigation applied;
- Bit 12: local (KNN) background mitigation applied;
- Bit 13: crowding information available in the table `xp_epoch_crowding`;
- Bit 14: the transit has been treated for truncation or multiple gate configuration.

Remaining bits are available for future extensions.

Draft

16 Source environment

As indicated in the below subsection titles by the labels [TAP] and [DataLink], these products are served from the ESA Gaia Archive using two different protocols, the Table Access Protocol (TAP) for tabular products and the DataLink protocol for other products such as spectra, light curves, MCMC samples, etc. For more details please see Section **??**.

16.1 GAIA_SOURCE_ENVIRONMENT [TAP]

This table contains the astrometry, photometry, and related information for sources detected in the surrounding of sources in the main Gaia source catalogue by the SEAPipe processing.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

DESIGNATION : Unique source designation (unique across all Data Releases) (string)

A source designation, unique across all Gaia Data Releases, that is constructed from the prefix 'Gaia DRx ' followed by a string of digits corresponding to `source_id` (3 space–separated words in total). Note that the integer source identifier `source_id` is **not** guaranteed to be unique across Data Releases; moreover it is not guaranteed that the same astronomical source will always have the same `source_id` in different Data Releases. Hence the only safe way to compare source records between different Data Releases in general is to check the records of proximal source(s) in the same small part of the sky. For convenience table `dr3_neighbourhood` provides such a cross-match for those sources that already existed in DR3.

SOURCE_ID : Unique source identifier (unique within a particular Data Release) (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

Draft

In the case of those sources in `gaia_source_environment`, the source identifier also contains a component number coded in bits 1 to 7. The corresponding identifier with all component bits set to 0 is that of the parent source, i.e. that which the source in `gaia_source_environment` was detected by SEAPipe.

PARENT_SOURCE_ID : Identifier of the source analysed by SEAPipe (long)

The `source_id` of the source in `all_source_astrometry` analysed by SEAPipe. This is the same as the `source_id` of the component but with the component bits not set, but is included here to make it easier/quicker to use in queries.

ASTROMETRY_ORIGIN : Flag indicating which module provided the astrometry for this source (byte)

Flag indicating the origin of the source parameters:

- 0: Image Segregation - in this case only estimates of flux and position (Right Ascension and Declination) will be available. This can occur if the Image Parameter Analysis is not run (e.g in the case of extended sources). Note in this case there will be no proper motion, parallax or any errors provided.
- 1: Image Parameter Analysis
- 2: Corrected Image Parameter Analysis

REF_EPOCH : Reference epoch (double, Time[Julian Years])

Reference epoch to which the astrometric source parameters are referred, expressed as a Julian Year in TCB (Temps Coordonnée Barycentrique = Barycentric Coordinate Time).

MATCHED : Flag indicating whether the source is in the all source match table (boolean)

Flag indicating whether the source is in the `all_source_match` table.

RA : Right ascension (double, Angle[deg])

Barycentric right ascension α of the source in ICRS at the reference epoch `ref_epoch`.

Draft

RA_ERROR : Standard error of right ascension (float, Angle[mas])

Standard error $\sigma_{\alpha*} \equiv \sigma_\alpha \cos\delta$ of the right ascension of the source in ICRS at the reference epoch `ref_epoch`.

DEC : Declination (double, Angle[deg])

Barycentric declination δ of the source in ICRS at the reference epoch `ref_epoch`.

DEC_ERROR : Standard error of declination (float, Angle[mas])

Standard error σ_δ of the declination of the source in ICRS at the reference epoch `ref_epoch`.

PARALLAX : Parallax (double, Angle[mas])

Absolute stellar parallax ϖ of the source at the reference epoch `ref_epoch`.

PARALLAX_ERROR : Standard error of parallax (float, Angle[mas])

Standard error σ_ϖ of the stellar parallax at the reference epoch `ref_epoch`.

PARALLAX_OVER_ERROR : Parallax divided by its standard error (float)

Parallax divided by its standard error.

PM : Total proper motion (float, Angular Velocity[mas yr^{-1}])

The total proper motion calculated as the magnitude of the resultant vector of the proper motion component vectors `pmra` and `pmdec`, i.e. $\texttt{pm}^2 = \texttt{pmra}^2 + \texttt{pmdec}^2$.

PMRA : Proper motion in right ascension direction (double, Angular Velocity[mas yr^{-1}])

Proper motion in right ascension $\mu_{\alpha*} \equiv \mu_\alpha \cos\delta$ of the source in ICRS at the reference epoch `ref_epoch`. This is the local tangent plane projection of the proper motion vector in the direction of increasing right ascension.

Draft

PMRA_ERROR : Standard error of proper motion in right ascension direction (float, Angular Velocity[mas yr^{-1}])

Standard error $\sigma_{\mu\alpha*}$ of the local tangent plane projection of the proper motion vector in the direction of increasing right ascension at the reference epoch `ref_epoch`.

PMDEC : Proper motion in declination direction (double, Angular Velocity[mas yr^{-1}])

Proper motion in declination μ_δ of the source at the reference epoch `ref_epoch`. This is the projection of the proper motion vector in the direction of increasing declination.

PMDEC_ERROR : Standard error of proper motion in declination direction (float, Angular Velocity[mas yr^{-1}])

Standard error $\sigma_{\mu\delta}$ of the proper motion component in declination at the reference epoch `ref_epoch`.

ASTROMETRIC_N_OBS : Total number of observations usable for astrometry (short)

Total number of observations (= CCD transits) used in the astrometric solution of the source.

ASTROMETRIC_DOF : Astrometric solution degrees of freedom (int)

Astrometric solution degrees of freedom.

ASTROMETRIC_CHI2 : Astrometric chi-square value (float)

Chi Squared value of the solution.

PHOT_G_MEAN_MAG : G-band mean magnitude (float, Magnitude[mag])

Mean magnitude in the G band.

PHOT_G_N_OBS : Number of observations contributing to G photometry (short)

Number of observations that contributed to the mean integrated photometry in the G band.

Draft

ASTROMETRIC_DUPLICATE_SOURCE_ID : Duplicate source identifier (long)

This field provides the `source_id` of a source which has been identified as a duplicate of this source astrometrically (i.e. it is very close to another source).

If the field is identical to the `source_id` it means that the source has other duplicates and that this is the preferred source that should be kept in case of duplicate sources removal in CU9 or IDU for example.

If the source has no duplicates identified this field will be empty.

DISTANCE_TO_PARENT_SOURCE : Distance to the AGIS position of the parent source (float, Angle[mas])

Distance to the AGIS position for the parent source.

PARENT_ENV_FLAG : Flag indicating the nature of the parent source environment (byte)

Flag indicating the source environment result. The only defined values are

- 1 = the data are consistent with multiple sources;
- 3 = the primary source is extended, and additional sources are found in its surrounding.

MATCHED_TO_PARENT : Field used to indicate if SCON considers this component to represent the AGIS parent source (byte)

Field used to indicate if SCON considers this component to represent the AGIS parent source:

- 0 = No
- 1 = Yes,
- 2 = Yes, one of several
- 3 = Cannot determine.

L : Galactic longitude (double, Angle[deg])

Draft

Galactic Longitude of the object at reference epoch `ref_epoch`, see Section **??** of the release documentation for conversion details.

B : Galactic latitude (double, Angle[deg])

Galactic Latitude of the object at reference epoch `ref_epoch`, see Section **??** of the release documentation for conversion details.

ECL_LON : Ecliptic longitude (double, Angle[deg])

Ecliptic Longitude of the object at reference epoch `ref_epoch`, obtained from the equatorial coordinates using the transformation defined in Volume 1, Section 1.5.3 of ESA (1997).

Note that in the transformation applied here the ICRS origin is shifted in the equatorial plane from Γ by $\phi = 0.05542$ arcsec, positive from Γ to the ICRS origin (Chapront et al. 2002). The ICRS has an unambiguous definition with an origin in the ICRF equator defined by the realisation of the ICRF. The ecliptic system is less well-defined, potentially depending on additional conventions in dynamical theories. The transformation employed here corresponds to the inertial mean ecliptic with obliquity and Γ defined by reference to the ICRS equator. Both the obliquity and the position of Γ on the ICRS equator with respect to the ICRS origin have been obtained from Lunar Laser Ranging measurements. This has no time dependence – there is no secular variation of the obliquity and no precession – and it simply defines the relative situation of the various planes at J2000.

ECL_LAT : Ecliptic latitude (double, Angle[deg])

Ecliptic Latitude of the object at reference epoch `ref_epoch`. For further details see the description for attribute `ecl_lon`.

HEALPIX29 : Level 29 HEALPix index for HATS (long)

A level–29 HEALPix spatial index for compliance with the Hierarchical Adaptive Tiling Scheme standard in the International Virtual Observatory Alliance – see `https://www.ivoa.net/documents/Notes/HATS/` for further information.

16.2 RESIDUAL_IMAGE [DataLink]

This table contains a reconstructed 2-dimensional image centred on the source given by the source identifier.

The 2D image is reconstructed from the window data from multiple transits. The central source has been removed from the image to allow the detection of other features. The 2D-image has been processed by the source environment image segregation module (with ad hoc configuration) to create a mask of the largest association of pixels that are above a given signal to noise threshold. The resulting mask is provided alongside the image.

The 2D arrays can be visualized as fits files using the following astrometric header:

```
CTYPE : 'RA-TAN' 'DEC-TAN'
CRVAL : <ra> <dec>
CRPIX : 51.5 51.5
PC1_1 PC1_2 :  1.0 0.0
PC2_1 PC2_2 :  -0.0 1.0
CDELT : 8.333333333333334e-06 8.333333333333334e-06
```

where <ra> and <dec> need to be replaced by the actual values. The same orientation (ICRS, North on top and East on the right) and geometry (102 x 102 pixels of 30 mas side) has always been used. The first (second) array index runs on increasing declination (right ascension).

Note this table is not available through the main archive TAP interface. Data are delivered via the Massive Data service indexed by the VO DataLink protocol and described in Chapter **??**. For example this can be actioned in the archive user interface by joining queries with the table `all_source_flags` and selecting `has_residual_image = 'true'`. The DataLink retrieval type to specify for this product in programmatic archive access via the Astroquery Gaia Python package is `RESIDUAL_IMAGE`.

The datamodel described here refers to the `DATAMODEL_GAIA` serialisation used in the Parquet files available from the bulk download repository. The FITS files accessible through the Gaia ESA Archive use the `DATAMODEL_STANDARD` serialisation and are documented in the archive help pages.

Columns description:

SOLUTION_ID : Solution Identifier (long)

Draft

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

RA : Right ascension (double, Angle[deg])

Barycentric right ascension α of the image centre in the ICRS (International Celestial Reference System).

It is a copy of the AGIS source value in `all_source_astrometry` with the same `source_id`, replicated for easier fits image conversion.

DEC : Declination (double, Angle[deg])

Barycentric declination δ of the image centre in ICRS (International Celestial Reference System).

It is a copy of the AGIS source value in `all_source_astrometry` with the same `source_id`, replicated for easier fits image conversion.

FLUX_MAP : Flux map centred on target source (float[] array, Flux[e^- s^{-1}])

Flux map centred on target source.

See table description for details on image geometry.

FLUX_ERROR_MAP : Flux error map (float[] array, Flux[e^- s^{-1}])

Flux error map.

See table description for details on image geometry.

Draft

N_OBS_MAP : Hit count image (short[] array)

Number of single observations contributing to each pixel in the residual image. Different AF measurements within the same transit are accounted for separately.

See table description for details on image geometry.

SEGREGATION_MASK : Segregation mask (byte[] array)

Mask flagging the pixels belonging to the largest association of pixels that are above a given signal to noise threshold. A value of 1 indicates that the pixel belongs to such association. Byte instead of boolean has been used for enhanced fits compatibility.

See table description for details on image geometry.

Draft

17 Spectroscopy

As indicated in the below subsection titles by the labels [TAP] and [DataLink], these products are served from the ESA Gaia Archive using two different protocols, the Table Access Protocol (TAP) for tabular products and the DataLink protocol for other products such as spectra, light curves, MCMC samples, etc. For more details please see Section **??**.

17.1 RVS_EPOCH_PARAMETERS_DOUBLE [DataLink]

This table contains epoch-level RVS data, providing transit-level information on radial velocity, spectral line broadening, and Grvs magnitudes for double-lined transits. These are transits whose RVS spectra display two distinct sets of spectral lines, consistent with two stellar components, allowing two radial velocities to be measured per transit.

For each transit, radial velocity and spectral line broadening are reported for both components, labelled Primary and Secondary. The RVS spectroscopic pipeline assigns the primary and secondary independently for each transit, so their roles may switch between transits. During non-single-star (NSS) processing, a consistent identification of the primary and secondary components is established across all transits. The flag `is_reordered` indicates cases where the original roles were swapped to ensure this consistency.

The detection of double lines is performed by the RVS spectroscopic pipeline and is restricted to bright stars with `external_apparent_grvs` ≤ 12. A double-lined transit identified by the pipeline does not necessarily imply a physical SB2 system; in some cases, the second set of lines may arise from a background contaminant. Sources for which double lines are detected in at least 10 transits are flagged with `rv_assumed_sb2` = true. Only about one third of these assumed SB2 systems are subsequently confirmed as SB2 by the NSS processing.

Note that not all transits include all three parameters.

See **?** for a description of the measurement and validation of the double-lined radial velocities and spectral line broadening velocities.

Note this table is not available through the main archive TAP interface. Data are delivered via the Massive Data service indexed by the VO DataLink protocol and described in Chapter **??**. For example this can be actioned in the archive user interface via ADQL queries joining on `source_id` with the table `all_source_rvs` and selecting `fraction_double_lined` > 0. The DataLink retrieval type to specify for this product in programmatic archive access via the Astroquery Gaia Python package is `EPOCH_PARAMETERS_RVS_DOUBLE`

Draft

DataLink products are served with a choice of two different data structures:

- DATAMODEL_STANDARD: the product is serialised following a standard data model (for example an IVOA-compliant data model in applicable cases);
- DATAMODEL_GAIA: the product is serialised following the Gaia DPAC(-internal) data model.

The DATAMODEL_GAIA format is described here whereas the DATAMODEL_STANDARD format is described in https://www.cosmos.esa.int/web/gaia-users/archive/datalink-products#datalink_serialisation

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit https://gaia.esac.esa.int/decoder/solnDecoder.jsp

SOURCE_ID : source identifier (long)

A unique single numerical identifier of the source (for a detailed description see gaia_source.source_id).

TRANSIT_ID : Transit identifier (long)

The transit_id is a unique identifier assigned to each detected (and confirmed) source as it transits the Gaia focal plane. Each time a given source is detected as Gaia scans and re-scans the sky a new transit_id will be created to badge that apparition. Hence the along–scan time and the across–scan position along with the telescope in which the source was detected are used to form a unique integer with which to label the transit.

The several features of a detection that are encoded in transit_id can be easily retrieved using bit masks (&) and shifts (>>) as follows:

- On-Board Mission Time line [ns]
  = 204800 * ((transit_id >> 17) & (0x000003FFFFFFFFF))

- Field-of-view = 1 + (`transit_id` >> 15) & 0x03 [1 for 'preceding' and 2 for 'following' fields-of-view respectively]
- CCD row = (`transit_id` >> 12) & 0x07 [dimensionless, in the range 1 to 7]
- Across-scan 'reference acquisition pixel' in strip AF1 = (`transit_id`) & 0x0FFF [pixels] (this is the across-scan centre of the AF1 window and is odd if immediately below the mid-point of the window and even if immediately above)

where the bit mask prefix '0x' denotes hexadecimal.

For further details see Portell et al. (2020). For convenience a decoder for `transit_id` is available on-line at

`https://gaia.esac.esa.int/decoder/transitidDecoder.jsp`

RV_ASSUMED_SB2 : Flag for transits of stars assumed SB2 (boolean)

This flag is set to `true` when the transit belongs to a source assumed to be an SB2 by the RVS pipeline, that is, when double lines are detected in at least 10 RVS transits of the source. The detection of double lines is performed for stars with `external_apparent_grvs` ≤ 12.

Sources assumed to be SB2 have the boolean `rv_assumed_sb2` set to `true` in the tables `gaia_source`, `all_source_rvs`, and `rvs_epoch_parameters_single`. Such sources may have both types of transits: those identified as double-lined by the pipeline, stored in `rvs_epoch_parameters_double`, and those not identified as double-lined, stored in `rvs_epoch_parameters_single`. The corresponding epoch spectra are provided in `rvs_epoch_spectrum`.

In the RVS pipeline, single-lined transits of sources assumed to be SB2 are processed differently from other single-lined transits: their radial velocity is computed using a dedicated method (TodCorLight; see Section **??**), and they are not subject to most of the post-processing steps aimed at identifying spurious measurements.

When `rv_assumed_sb2` is `true`, the source is excluded from the multi-transit analysis in the RVS pipeline. Consequently, no combined radial velocity, line-broadening velocity, mean spectrum, or associated per-source parameters are produced; only the epoch-level results are published.

Only about one third of these sources are subsequently confirmed as SB2 by the NSS pipeline (see Chapter **??**). Confirmed SB2 systems correspond to NSS solutions of type `nss_two_body_orbit.solut` = SB2ScanAngleK1, SB2ScanAngleK12, AstroSpectroSB2, or EclipsingSpectroSB2; `nss_non_linear_spe`

= TrendSB2; or `nss_multiple_orbits.solution_type` = SB2SB1 or SB2TrendSB1.

RADIAL_VELOCITY_PRIMARY : Radial velocity in the barycentric frame for the primary (float, Velocity[km s^{-1}])

Spectroscopic radial velocity in the Solar system barycentric reference frame.

RADIAL_VELOCITY_PRIMARY_ERROR : Error on the radial velocity for the primary (float, Velocity[km s^{-1}])

The uncertainty associated with `radial_velocity_primary`.

RV_VALIDITY_FLAG_PRIMARY : Radial velocity validity flag for the primary (boolean)

This flag indicates whether the computed `radial_velocity_primary` is valid or not.

RADIAL_VELOCITY_SECONDARY : Radial velocity in the barycentric frame for the secondary (float, Velocity[km s^{-1}])

Spectroscopic radial velocity in the Solar system barycentric reference frame.

RADIAL_VELOCITY_SECONDARY_ERROR : Error on the radial velocity for the secondary (float, Velocity[km s^{-1}])

The uncertainty associated with `radial_velocity_secondary`.

RV_VALIDITY_FLAG_SECONDARY : Radial velocity validity flag for the secondary (boolean)

This flag indicates whether the computed `radial_velocity_secondary` is valid or not.

IS_REORDERED : Flag indicating whether the primary/secondary velocities have been swapped (boolean)

This flag indicates whether the primary and secondary radial velocities, the spectral line broadening, and their associated fields have been swapped for a given transit.

Draft

It is defined only for stars that have been confirmed as SB2 by the NSS pipeline (Chapter **??**). In these cases, the reordering of the components is based on the orbital model provided by the NSS solution, ensuring a consistent identification of the primary and secondary across all transits.

For stars assumed to be SB2 by the RVS pipeline but not confirmed by the NSS pipeline, this flag is set to null. In such cases, no consistent ordering is enforced, and the component labelled as primary in one transit may correspond to the secondary in another, and vice versa.

RAD_VEL_BARY_CORR : Barycentric correction (float, Velocity[km s^{-1}])

The barycentric radial velocity correction is computed using the Gaia Relativity Model (GREM). The error on this parameter is expected to be of the order of 1 cm/s.

This correction is used in the spectroscopic pipeline to shift the wavelength grid of the CCD spectra in order to align the Gaia CCD spectra to the solar barycentric frame. A positive value of the barycentric correction corresponds to a redshift of the spectrum.

OBS_TIME_BARY_CORR : Barycentric correction of observing time (long, Time[ns])

Roemer barycentric correction of the observing time, to be added to `obs_time_rv` in order to shift the observation time at Gaia to a virtual observer at the solar system barycentre, thus removing the distortion caused by the Roemer effect.

OBS_TIME_RV : Observing time (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

This is the mean time over the CCDs used to obtain `radial_velocity`. The CCD observing time is given by the midpoint of the spectrum as it crosses the fiducial point. This is the mean time over the three CCDs. Contrary to the `obs_time_rv` in `rvs_epoch_parameters_single`, also the CCDs that did not contribute to the radial velocity measurement are included in the determination of the mean time.

The observing time for each CCD of the transit is provided in `ccd_obs_times` of `rvs_transit_descriptio`

ELS_PROBABILITY : Probability of having emission lines (float[3] array)

This field is an array containing the probability for each CCD in this transit to have emission lines. It is computed using deep learning techniques and is populated for stars having `external_apparent_grvs` ≤ 13 (in `all_source_rvs`) and only if at least two CCD spectra are available. See **?**.

When at least two CCDs have `els_probability` ≥ 0.5, the transit is flagged as having emission lines.

Since such spectra are excluded from the detection of double lines, transits with emission lines do not appear in this table. Consequently, the flag `emission_lines`, present in `rvs_epoch_parameters_singl` is not included here.

RADIAL_VELOCITY_SINGLE : Radial velocity in the barycentric frame for the single star model (float, Velocity[km s^{-1}])

Spectroscopic radial velocity in the Solar system barycentric reference frame obtained using the single star model.

`radial_velocity_single` is computed using the method called *TodCorLight*, while the method *TodCorHeavy* is used to obtain `radial_velocity_primary` and `radial_velocity_secondary`. It is not provided when *TodCorLight* fails.

RADIAL_VELOCITY_SINGLE_ERROR : Error on the radial velocity for the single star model (float, Velocity[km s^{-1}])

The uncertainty associated with `radial_velocity_single`.

RV_SINGLE_VALIDITY_FLAG : Radial velocity validity flag for the single star model (boolean)

This flag indicates whether the computed radial velocity is valid or not for the single star model.

TEMPLATE_TEFF_PRIMARY : Teff of the template used for the primary (float, Temperature[K])

Effective temperature of the synthetic template used to determine `radial_velocity_primary`.

TEMPLATE_LOGG_PRIMARY : logg of the template used for the primary (float, GravitySurface[log cgs])

logg of the synthetic template used to determine `radial_velocity_primary`.

TEMPLATE_FE_H_PRIMARY : Fe/H of the template used for the primary (float, Abundances[dex])

Fe/H of the synthetic template used to determine the `radial_velocity_primary`.

Draft

TEMPLATE_TEFF_SECONDARY : Teff of the template used for the secondary (float, Temperature[K])

Effective temperature of the synthetic template used to determine `radial_velocity_secondary`

TEMPLATE_LOGG_SECONDARY : Logg of the template used for the secondary (float, Gravity-Surface[log cgs])

logg of the synthetic template used to determine `radial_velocity_secondary`.

TEMPLATE_FE_H_SECONDARY : Fe/H of the template used for the secondary (float, Abundances[dex])

Fe/H of the synthetic template used to determine `radial_velocity_secondary`.

VBROAD_PRIMARY : Spectral line broadening for the primary (float, Velocity[km s^{-1}])

The spectral line broadening parameter. It is measured using a rotational broadening kernel; its value therefore includes $v \sin i$, but also any other source of line broadening (macroturbulence, residual instrumental effects, etc.).

VBROAD_PRIMARY_ERROR : Uncertainty on the spectral line broadening for the primary (float, Velocity[km s^{-1}])

The uncertainty associated with `vbroad_primary`.

VBROAD_VALIDITY_FLAG_PRIMARY : Spectral line broadening validity flag for the primary (boolean)

This flag indicates whether the computed spectral line broadening is valid or not.

VBROAD_SECONDARY : Spectral line broadening for the secondary (float, Velocity[km s^{-1}])

The spectral line broadening parameter. It is measured using a rotational broadening kernel; its value therefore includes $v \sin i$, but also any other source of line broadening (macroturbulence, residual instrumental effects, etc.).

Draft

VBROAD_SECONDARY_ERROR : Uncertainty on the spectral line broadening for the secondary (float, Velocity[km s^{-1}])

The uncertainty associated with `vbroad_secondary`.

VBROAD_VALIDITY_FLAG_SECONDARY : Spectral line broadening validity flag for the secondary (boolean)

This flag indicates whether the computed spectral line broadening is valid or not.

GRVS_FLUX : total flux in the whole system (float, Flux[e^- s^{-1}])

`grvs_flux` is determined based on the number of available CCD spectra in the transit (1, 2, or 3). The integrated flux in each CCD (f_{ccd_i}) is computed by summing the fluxes in the spectrum samples, where the flux in masked samples is assumed to be the median of the non-masked samples.

Each CCD-integrated flux is then corrected for flux loss outside the window and rescaled to a common zero point, TheGrvsZP = 21.318467, in order to remove instrumental effects that depend on the CCD, field of view, and observation time. For more details, see **?**.

There are cases where `grvs_flux` < 0, which in general occur for faint stars with an overestimated subtracted background.

- For 1 available CCD:

$$\texttt{grvs_flux} = f_{ccd_i}$$

- For 2 available CCDs:

$$\texttt{grvs_flux} = \text{mean}(f_{ccd_i})$$

- For 3 available CCDs:

$$\texttt{grvs_flux} = \text{median}(f_{ccd_i})$$

GRVS_FLUX_ERROR : Error on the total flux (float, Flux[e^- s^{-1}])

Uncertainty associated with `grvs_flux`. It is not populated in cases when `grvs_flux` ≤ 0.

GRVS_MAG : Grvs magnitude of the system (float, Magnitude[mag])

The magnitude grvs_mag is estimated using grvs_flux and the zero-point magnitude ($ZP_{\rm grvs}$ = 21.31847):

$$\texttt{grvs_mag} = -2.5 \times \log(\texttt{grvs_flux}) + ZP_{\rm grvs}$$

It is not populated in cases when grvs_flux ≤ 0.

GRVS_MAG_ERROR : Grvs magnitude error (float, Magnitude[mag])

The uncertainty associated with grvs_mag is calculated using the following formula:

$$\texttt{grvs_mag_error} = \texttt{grvs_flux_error}/\texttt{grvs_flux} \times 2.5/\ln(10).$$

It is not populated in cases when grvs_flux ≤ 0.

GRVS_ZERO_POINT : Grvs zero point for this transit (float, Magnitude[mag])

grvs_zero_point is the transit-specific zeropoint, accounting for variations due to the observation time, field of view (FoV), and CCD configuration. It is computed as the median of the zeropoints derived for each of the three CCDs involved in the transit (see **?**).

This relative zeropoint reflects time and configuration dependent instrumental effects (e.g. CCD sensitivity, FoV throughput, and mirror contamination). It allows for recovering the measured (raw) flux from the scaled value published in grvs_flux, which has been normalized to the global zeropoint: TheGrvsZP = 21.318467.

The unscaled measured flux can be recovered via:

$$\text{measuredFlux} = \texttt{grvs_flux} \cdot 10^{0.4\cdot(\texttt{grvs_zero_point}-\text{TheGrvsZP})}$$

GRVS_VALIDITY_FLAG : Grvs validity flag (boolean)

This flag is set to false when grvs_mag is deemed invalid.

GRVS_INVALID_REASON : The reason why the Grvs is set as invalid (short)

This 16-bit field specifies the reasons for the invalidation of grvs_mag. Each bit indicates whether a particular reason is true (1) or false (0).

A grvs_mag can be invalidated for multiple reasons, activating multiple bits in the field.

The transits in rvs_epoch_parameters_double are not used to compute the multi-transit combined grvs_flux and grvs_mag in gaia_source and all_source_rvs.

- Bit 0: [Not Used].
- Bit 1: Star too faint. grvs_mag is invalidated for stars with external_apparent_grvs > 14.4 mag.
- Bit 2: Deblended or reblended transit.
- Bit 3: Contamination of the flux by a nearby star. Such transits have rvs_transit_description = 78 or 81, or flagged with bit 15 in rvs_transit_description.ccd_status. These transits are not used to compute the multi-transit grvs_mag and grvs_flux in gaia_source and all_source_rvs.
- Bit 4: The pipeline did not provide a grvs_mag.
- Bit 5: Excessive uncertainty. grvs_mag is invalidated if grvs_mag_error > 16 mag.
- Bit 6: [Not Used].
- Bit 7: Value outside allowed range. grvs_mag is invalidated if grvs_flux < MinFlux, where
$$\mathrm{MinFlux} = \frac{\mathrm{MinFlux_{CCD}}}{\sqrt{3}}$$
and $MinFlux_{CCD}$ = 1024 e^-/s represents twice the estimated average precision of straylight flux integrated over a single CCD spectrum in a typical Calibration Unit.
- Bit 8: Duplicated transit (rvs_transit_description.ccd_invalid_reason = 71).
- Bit 9: Time interval with many unreliable grvs_mag. These are the two regions [5853, 5905] and [5496, 5576] revolutions.
- Bit 10-15: [Not Used].

EXPECTED_SIG_TO_NOISE : Signal to noise expected (float)

The signal to noise ratio expected by combining the information from the CCD spectra and the relative background used for estimating the radial velocity.

The expected signal to noise ratio is obtained by combining the information from the CCD spectra used for estimating the radial velocity:

$$\texttt{rv_expected_sig_to_noise} = \frac{S}{\sqrt{S + BCK + RN}}$$

where:

$$S = \frac{medianFlux \times N_{ValidStrips} \times \text{EXP_TIME} \times \text{PIXEL_WIDTH_AL}}{\text{BAND_WIDTH}}$$

$$BCK = medianBackground \times N_{ValidStrips} \times N_{ACpixels}$$

$$RN = RON^2 \times N_{ValidStrips} \times N_{ACsamples}$$

where:

• $medianFlux$ is the median of the integrated flux of the CCD spectra used to obtain the radial velocity for `external_apparent_grvs` ≤ 12; otherwise, the $medianFlux$ is obtained from `external_apparent_grvs` [$e^{-}s^{-1}$]. `external_apparent_grvs` is store in the tables `all_source_rvs` and `gaia_source`.
• $N_{ValidStrips}$ = the total number of CCD spectra used to obtain the radial velocity (in general there are 3 CCD spectra per transit)
• BAND_WIDTH = 870 - 846 [nm]
• EXP_TIME = 4.4167032 [s]
• PIXEL_WIDTH_AL = 0.02453 [nm]
• $medianBackground$ = the median background in one exposure of the CCD spectra used to obtain the `radial_velocity` [e^{-}pixel^{-1}]
• $N_{ACpixels}$ = 10 is the number of pixels in the AC direction of the non-truncated RVS window
• $N_{ACsamples}$ is the size in the AC direction of the non-truncated RVS window. $N_{ACsamples}$ = 10 for the 2D RVS windows (Window Class=0) and $N_{ACsamples}$ = 1 for the 1D windows (Window Class=1)
• RON (Read Out Noise) = 3.2 [e^{-}]

It is not provided when $medianFlux \leq 0$ or when all the CCD strips are flagged as invalid during the FullExtraction step of the pipeline (i.e., in the table `rvs_transit_description`, where

ccd_invalid_reason has values < 50 or > 90).

RV_SYSTEMATIC_CORR : Correction applied to the radial velocity (float, Velocity[km s^{-1}])

rv_systematic_corr represents a correction applied in the spectroscopic pipeline to account for systematic errors. This correction is used to shift the wavelength grid of the CCD spectra. A positive value of the correction corresponds to a blueshift of the spectrum. For mode information, see Section **??**.

COVARIANCE_V1_V2 : Covariance between primary and secondary radial velocities (double, Misc[km^2 s^{-2}])

Covariance betwen the two radial velocities radial_velocity_primary and radial_velocity_secondar

Draft

17.2 RVS_EPOCH_PARAMETERS_SINGLE [DataLink]

This table contains epoch-level RVS data, hosting transit-level information for radial velocity, spectral line broadening, and Grvs magnitudes. It applies to single-lined sources, i.e. sources whose RVS spectra show a single set of spectral lines, consistent with a single stellar component, and for which a unique radial velocity can be measured.

This table is produced for all stars processed by the pipeline. However, only a very small fraction, specifically, the brightest stars (`external_apparent_grvs` ≤ 14; in `gaia_source` and in `all_source_rvs`), have their epoch radial velocities computed. As a result, for the vast majority of sources, this table remains mostly empty, containing only metadata such as information related to the epoch spectra, like the flux information, `rv_systematic_corr`, `expected_sig_to_noise`, `obs_time_rv`, and `rad_vel_bary_corr`.

Unlike `radial_velocity` and spectral line broadening `vbroad`, epoch spectra are systematically produced and stored in the `rvs_epoch_spectrum` table.

Only a small subset of the produced data is considered valid, and these can be identified using the validity flags.

This table is limited to single-lined sources, for which a unique radial velocity and spectral line broadening are derived. For double-lined sources, the corresponding information is stored in the table `rvs_epoch_parameters_double`.

The processing of the RVS data is described in **?**; the validation of `radial_velocity` in **?**, `vbroad` in **?**, `grvs_mag` in **?**.

Note this table is not available through the main archive TAP interface. Data are delivered via the Massive Data service indexed by the VO DataLink protocol and described in Chapter **??**. For example this can be actioned in the archive user interface via ADQL queries joining on `source_id` with the table `all_source_rvs` and selecting `has_epoch_rvs = 'true' and (fraction_double_lined = 0 or fraction_double_lined is null)`. The DataLink retrieval type to specify for this product in programmatic archive access via the Astroquery Gaia Python package is `EPOCH_PARAMETERS_RVS_SINGLE`.

DataLink products are served with a choice of two different data structures:

- `DATAMODEL_STANDARD`: the product is serialised following a standard data model (for example an IVOA-compliant data model in applicable cases);
- `DATAMODEL_GAIA`: the product is serialised following the Gaia DPAC(-internal) data model.

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The DATAMODEL_GAIA format is described here whereas the DATAMODEL_STANDARD format is described in https://www.cosmos.esa.int/web/gaia-users/archive/datalink-products#datalink_serialisation

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit https://gaia.esac.esa.int/decoder/solnDecoder.jsp

SOURCE_ID : source identifier (long)

A unique single numerical identifier of the source (for a detailed description see gaia_source.source_id).

TRANSIT_ID : transit identifier (long)

The transit_id is a unique identifier assigned to each detected (and confirmed) source as it transits the Gaia focal plane. Each time a given source is detected as Gaia scans and re-scans the sky a new transit_id will be created to badge that apparition. Hence the along–scan time and the across–scan position along with the telescope in which the source was detected are used to form a unique integer with which to label the transit.

The several features of a detection that are encoded in transit_id can be easily retrieved using bit masks (&) and shifts (>>) as follows:

- On-Board Mission Time line [ns]
  = 204800 * ((transit_id >> 17) & (0x000003FFFFFFFFFF))
- Field-of-view = 1 + (transit_id >> 15) & 0x03 [1 for 'preceding' and 2 for 'following' fields-of-view respectively]
- CCD row = (transit_id >> 12) & 0x07 [dimensionless, in the range 1 to 7]
- Across-scan 'reference acquisition pixel' in strip AF1 = (transit_id) & 0x0FFF [pixels] (this is the across-scan centre of the AF1 window and is odd if immediately below the mid-point of the window and even if immediately above)

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where the bit mask prefix '0x' denotes hexadecimal.

For further details see Portell et al. (2020). For convenience a decoder for `transit_id` is available on-line at

`https://gaia.esac.esa.int/decoder/transitidDecoder.jsp`

RV_VALIDITY_FLAG : Radial velocity validity flag (boolean)

This flag indicates whether the `radial_velocity` is deemed valid or not.

Some `radial_velocity` values are invalidated during processing, and the corresponding transits are excluded from the computation of the multi-transit radial velocity in `all_source_rvs` and `gaia_source`. Most other invalidations occur during post-processing, based on validation analyses. In those cases, the transits may have already contributed to the multi-transit value, even though they are later flagged as invalid. See `rv_invalid_reason` for the list of invalidation reasons and to identify which ones lead to the exclusion of transit data.

RV_INVALID_REASON : The reason why the radial velocity is set as invalid (long)

This 64-bit field specifies the reasons for the invalidation of `radial_velocity`. Each bit indicates whether a particular reason is true (1) or false (0).

A few of these invalidation reasons are set during processing and result in the exclusion of the transit from the computation of multi-transit radial velocities in `all_source_rvs` and `gaia_source` (specifically, reasons corresponding to Bits 3, 8, 15, and 62). Most other reasons are assigned during post-processing based on validation analysis. Invalidation flags are determined from various conditions and may be set even when `radial_velocity` is not provided.

A `radial_velocity` may be invalidated for multiple reasons, leading to the activation of multiple bits. See Table **??** for the occurrence of each invalid reason.

Both valid and invalid data are published in `rvs_epoch_parameters_single`.

- Bit 0: [Not Used].
- Bit 1: Hot star. `radial_velocity` is invalidated for stars with `external_apparent_grvs` $\leq$ 12 mag and `rv_template_teff` > 14,500 K, as well as for those with `external_apparent_gr` > 12 mag and `rv_template_teff` > 8,500 K. *Note:* `external_apparent_grvs` is in `gaia_source` and `all_source_rvs`, while `rv_template_teff` is in `all_source_rvs`.

- Bit 2: [Not Used].
- Bit 3: Emission lines detected. This invalidation reason is set for transits in which emission lines are identified by the pipeline (`emission_lines` = True), for transits of stars identified as emission-line stars by the pipeline (`all_source_rvs.fraction_emission_` > 0.4), and for transits of approximately 2,700 stars detected as emission-line stars during post-processing. These transits are not used to compute the multi-transit radial velocities in `gaia_source` and `all_source_rvs`.
- Bit 4: [Not Used].
- Bit 5: [Not Used].
- Bit 6: Too large uncertainty. `radial_velocity` is invalidated when `radial_velocity_error` > 1000 km s^{-1}.
- Bit 7: [Not Used].
- Bit 8: Contamination by a bright neighbour. The contaminated transits are those with `rvs_transit_description.ccd_invalid_reason` = 78 or 81. These transits are not used to compute the multi-transit radial velocities in `gaia_source` and `all_source_rvs`.
- Bit 9: Too high probability of being an outlier. `radial_velocity` is invalidated when: `rv_outlier_probability` = 2, when `rv_outlier_probability` $\geq 7.0 - 0.5 \times$ `external_apparent_grvs`, and when `rv_outlier_condition` ≥ 0.76. (See **?**)
- Bit 10: Significant scan angle signal. `radial_velocity` is invalidated for transits of stars with `all_source_rvs.rv_single_method_used` = 5.
- Bit 11: The pipeline did not provide a radial velocity. For sources with `external_apparent_grvs` > 14, epoch radial velocities are not computed; in these cases, this is the only `rv_invalid_reason` that is set.
- Bit 12: Excessive standard deviation in the epoch radial velocities. `radial_velocity` is invalidated for stars with `all_source_rvs.rv_std_dev` ≥ 577 km s^{-1}.
- Bit 13: Template mismatch. This invalidation reason is set for a list of stars with `external_apparent_grvs` ≤ 14 mag, identified using the UMAP dimensionality reduction tool (see **?**), and for stars with `rv_atm_param_origin` = 888. *Note:* `external_apparent_grvs` is in `gaia_source` and `all_source_rvs`, while `rv_atm_param_or` is in `all_source_rvs`.
- Bit 14: Large discrepancy between two methods. This flag is set when only two methods (RvDir, RvFou, or MdM) provide valid RVs, and their values differ significantly ($|RV_{\text{method1}} - RV_{\text{method2}}| > 200$ km s^{-1}). This applies only to bright stars

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(external_apparent_grvs ≤ 11 mag), as for fainter stars, only one method (RvFou) is used.

- Bit 15: Duplicated transit. The duplicated transits are those with rvs_transit_description.cc = 71. These transits are not used to compute the multi-transit radial velocities in gaia_source and all_source_rvs.
- Bit 16: Only one valid transit remaining after post-processing.
- Bit 17: No atmospheric parameters associated with the star. radial_velocity is invalidated when all_source_rvs.rv_atm_param_star_teff = 0.
- Bits 18-61: [Not Used].
- Bit 62: Other reasons. This applies to radial velocities of transits where rv_assumed_sb2 = True. These transits are set as invalid internally by the method (TodCorLight) used to obtain their radial velocity. Such transits are not used to compute the multi-transit radial velocities in gaia_source and all_source_rvs.

VBROAD_VALIDITY_FLAG : Spectral line broadening validity flag (boolean)

This flag indicates whether the computed spectral line broadening is deemed valid or not.

Some vbroad values are invalidated during processing, and the corresponding transits are excluded from the computation of the multi-transit vbroad in all_source_rvs and gaia_source. Most other invalidations occur during post-processing, based on validation analyses. In those cases, the transits may have already contributed to the multi-transit value, even though they are later flagged as invalid. See vbroad_invalid_reason for the list of invalidation reasons and to identify which ones lead to the exclusion of transit data.

VBROAD_INVALID_REASON : The reason why the spectral line broadening is set as invalid (short)

This 16-bit field specifies the reasons for the invalidation of vbroad. Each bit indicates whether a particular reason is true (1) or false (0).

Some invalidation reasons are flagged during processing and are used to exclude transits from the computation of the multi-transit vbroad in all_source_rvs and gaia_source. Most other reasons are assigned during post-processing, based on validation analyses. In these cases, the transit vbroad have already contributed to the multi-transit value, even though it is later flagged as invalid. These flags may also be set even when vbroad is not provided.

A vbroad can be invalidated for multiple reasons, activating multiple bits in the field. See Table **??** for the occurrence of each invalid reason.

- Bit 0: Invalid radial velocity (this includes all `rv_invalid_reason` cases). The transits discarded from the multi-transit `vbroad` are those discarded from the multi-transit `radial_velocity_single`.
- Bit 1: Star too faint. `vbroad` is invalidated for stars with `external_apparent_grvs` > 11.
- Bit 2: Outside allowed range: below the lower limit of 4.5 km s^{-1}. These `vbroad` values are not used to compute the multi-transit `vbroad` stored in `all_source_rvs` and `gaia_source`.
- Bit 3: [Not used].
- Bit 4: Deblended or reblended transit. Such transits are flagged with bit 55 or 56 in `radial_velocity_status`. These `vbroad` values are excluded from `all_source_rvs.vbroad` and `gaia_source.vbroad`.
- Bit 5: Transit before the decontamination at OBMT revolution 1317. These `vbroad` values are not used to compute the multi-transit `vbroad` stored in `all_source_rvs` and `gaia_source`.
- Bit 6: Cool star. `vroad` is invalidated for stars with `rv_template_teff` ≤ 3400 K.
- Bit 7: [Not used].
- Bit 8: The pipeline did not provide a `vbroad`.
- Bit 9: Hot star. `vbroad` is invalidated for stars with `rv_template_teff` > 25000 K.
- Bit 10: Value at the grid edge. Due to the iterative estimation process, some `vbroad` values coincide with 60 grid edges. Values within $\pm 10^{-15}$ of these edges are flagged as invalid.
- Bits 11–14: [Not used].
- Bit 15: Other reasons. These `vbroad` values are not used to compute the multi-transit `vbroad` stored in `all_source_rvs` and `gaia_source`.

Note: Due to an implementation error, no `vbroad_invalid_reason` is set for approximately 130 invalid transits of stars assumed to be SB2 (`rv_assumed_sb2` = true).

GRVS_VALIDITY_FLAG : Grvs validity flag (boolean)

This flag is set to false when `grvs_mag` is deemed invalid.

Some grvs_mag values are invalidated during processing, and the corresponding transits are excluded from the computation of the multi-transit grvs_mag in all_source_rvs and gaia_source. Most other invalidations occur during post-processing, based on validation analyses. In those cases, the transits may have already contributed to the multi-transit value, even though they are later flagged as invalid. See grvs_invalid_reason for the list of invalidation reasons and to identify which ones lead to the exclusion of transit data.

GRVS_INVALID_REASON : The reason why the Grvs is set as invalid (short)

This 16-bit field specifies the reasons for the invalidation of grvs_mag. Each bit indicates whether a particular reason is true (1) or false (0).

Some invalidation reasons are flagged during processing and are used to exclude transits from the computation of the multi-transit grvs_mag and grvs_flux in all_source_rvs and gaia_source. Most other reasons are assigned during post-processing, based on validation analyses. In these cases, the transit grvs_mag and grvs_flux have already contributed to the multi-transit value, even though they are later flagged as invalid. These flags may also be set even when grvs_mag is not provided.

A grvs_mag can be invalidated for multiple reasons, activating multiple bits in the field. See Table **??** for the occurrence of each invalid reason.

- Bit 0: [Not Used].
- Bit 1: Star too faint. grvs_mag is invalidated for stars with external_apparent_grvs > 14.4 mag.
- Bit 2: Deblended or reblended transit. Such transits are flagged with bit 55 or 56 in radial_velocity_status. These transits are not used to compute the multi-transit grvs_mag and grvs_flux in gaia_source and all_source_rvs.
- Bit 3: Contamination of the flux by a nearby star. Such transits have rvs_transit_description = 78 or 81, or flagged with bit 15 in rvs_transit_description.ccd_status. These transits are not used to compute the multi-transit grvs_mag and grvs_flux in gaia_source and all_source_rvs.
- Bit 4: The pipeline did not provide a grvs_mag. These transits are not used to compute the multi-transit grvs_mag and grvs_flux.
- Bit 5: Excessive uncertainty. grvs_mag is invalidated if grvs_mag_error > 16 mag. These transits are not used to compute the multi-transit grvs_mag and grvs_flux.
- Bit 6: [Not Used].

- Bit 7: Value outside allowed range. `grvs_mag` is invalidated if `grvs_flux` < MinFlux, where
$$\text{MinFlux} = \frac{\text{MinFlux}_{\text{CCD}}}{\sqrt{3}}$$
and $MinFlux_{CCD}$ = 1024 e^-/s represents twice the estimated average precision of straylight flux integrated over a single CCD spectrum in a typical Calibration Unit. These transits are not used to compute the multi-transit `grvs_mag` and `grvs_flux`.
- Bit 8: Duplicated transit (`rvs_transit_description.ccd_invalid_reason` = 71). These transits are not used to compute the multi-transit `grvs_mag` and `grvs_flux`.
- Bit 9: Time interval with many unreliable `grvs_mag`. These are the two regions [5853, 5905] and [5496, 5576] revolutions. These transits are not used to compute the multi-transit `grvs_mag` and `grvs_flux`.
- Bit 10-15: [Not Used].

EMISSION_LINES : Flag for emission lines (boolean)

This flag is set to True if emission lines are detected in this transit.

The emission lines in the CCD spectra of the sources with `external_apparent_grvs`≤13 (available in `gaia_source` and in `all_source_rvs`) are detected using a deep-learning technique. The `emission_lines` flag is set to True when the probability of being an emission line star (`els_probability`) is found to be greater than or equal to 0.5 in at least two CCD spectra. See **?**.

RV_ASSUMED_SB2 : Flag for transits of stars assumed SB2 (boolean)

This flag is set to `true` when the transit belongs to a source assumed to be an SB2 by the RVS pipeline, that is, when double lines are detected in at least 10 RVS transits of the source. The detection of double lines is performed for stars with `external_apparent_grvs` ≤ 12.

Sources assumed to be SB2 have the boolean `rv_assumed_sb2` set to `true` in the tables `gaia_source`, `all_source_rvs`, and `rvs_epoch_parameters_single`. Such sources may have both types of transits: those identified as double-lined by the pipeline, stored in `rvs_epoch_parameters_double`, and those not identified as double-lined, stored in `rvs_epoch_parameters_single`. The corresponding epoch spectra are provided in `rvs_epoch_spectrum`.

In the RVS pipeline, single-lined transits of sources assumed to be SB2 are processed differently from other single-lined transits: their radial velocity is computed using a dedicated method (Tod-

CorLight; see Section **??**), and they are not subject to most of the post-processing steps aimed at identifying spurious measurements.

When `rv_assumed_sb2` is `true`, the source is excluded from the multi-transit analysis in the RVS pipeline. Consequently, no combined radial velocity, line-broadening velocity, mean spectrum, or associated per-source parameters are produced; only the epoch-level results are published.

Only about one third of these sources are subsequently confirmed as SB2 by the NSS pipeline (see Chapter **??**). Confirmed SB2 systems correspond to NSS solutions of type `nss_two_body_orbit.solut` = SB2ScanAngleK1, SB2ScanAngleK12, AstroSpectroSB2, or EclipsingSpectroSB2; `nss_non_linear_spe` = TrendSB2; or `nss_multiple_orbits.solution_type` = SB2SB1 or SB2TrendSB1.

RADIAL_VELOCITY : Radial velocity in the barycentric frame (float, Velocity[km s^{-1}])

Spectroscopic radial velocity in the Solar system barycentric reference frame.

Epoch radial velocities are provided for stars with `external_apparent_grvs` $\leq$ 14. The `external_apparent_grvs` values are stored in both `gaia_source` and `all_source_rvs`. Epoch radial velocities are not provided when `emission_lines` = true, or when the pipeline fails to produce a radial velocity. The information about the transits for which the pipeline has failed to produce the radial velocity is stored in `rvs_transit_description`, where `is_ccd_valid` = false for all 3 CCDs, and at least one of the `ccd_invalid_reason` values is 54.

For transit spectra where double lines are detected, the radial velocities of the two components are stored in the table `rvs_epoch_parameters_double`.

See Section **??** for a description on how the epoch `radial_velocity` are obtained. See **?** for a description of the validation of the epoch `radial_velocity` measurements.

RADIAL_VELOCITY_ERROR : Error on the radial velocity (float, Velocity[km s^{-1}])

The uncertainty associated with `radial_velocity`, including a correction to mitigate the original known underestimation. See Section **??**.

RADIAL_VELOCITY_STATUS : Status of the radial velocity (long)

This 64-bit field serves to provide information regarding the status of this epoch `radial_velocity`.

Each bit within the field corresponds to whether a particular status is true (1) or false (0). Many

of the statuses never occur in this context, and the corresponding bits are never activated.

The bits are activated during the pipeline processing to inform about the results of the radial velocity methods that have run and the method chosen by the 'Integrator' for the `radial_velocity`. For single-lined spectra, the chosen method can be RvDir, RvFou, or RvMDM. Bit 54 is activated for the stars assumed astrometric binaries and for which a correction for the astrometric motion was available. Bits 55 and 56 are activated for the re-blended and deblended spectra.

- Bit 0: RV from RvDir was used by Integrator
- Bit 1: RV from RvDir is valid
- Bit 2: RV from RvDir is ambiguous
- Bit 3: RV from RvFou was used by Integrator
- Bit 4: RV from RvFou is valid
- Bit 5: RV from RvFou is ambiguous
- Bit 6: RV from RvMdM was used by Integrator
- Bit 7: RV from RvMdM is valid
- Bit 8: RV from RvMdM is ambiguous
- Bit 9: RV from TodCorLight was used by Integrator
- Bit 10: RV from TodCorLight is valid
- Bit 11: RV from TodCorLight is ambiguous
- Bit 12: RVs from TodCorHeavy were used by Integrator
- Bit 13: RV[0] from TodCorHeavy is valid
- Bit 14: RV[1] from TodCorHeavy is present
- Bit 15: RV[1] from TodCorHeavy is valid
- Bit 16: RVs from TodCorHeavy is/are ambiguous
- Bit 17: RV[0] from Integrator is valid
- Bit 18: RV[1] from Integrator is present
- Bit 19: RV[1] from Integrator is valid

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- Bit 20: RVs from Integrator is/are ambiguous
- Bit 21: too faint star (no RV is computed)
- Bit 22: ELS RV from ELS module
- Bit 23: ELS RV from RVDir was used by Integrator
- Bit 24: ELS RV from RVFou was used by Integrator
- Bit 25: ELS RV from MDM was used by Integrator
- Bit 26: This transit has passed through ELS_RV
  26-51: not used
- Bit 52 : this transit belongs to an assumed SB2 star (`rv_assumed_sb2`=true)
- Bit 53 : only 2 methods (between RvDir, RvFou and MdM) provide valid RVs and the two RVs are very different [i.e. abs($RV_{method1} - RV_{method2}$) > 200 km s^{-1}]
- Bit 54: hasAstrometricMotionCorrected (using info from CU4)
- Bit 55: IsAtLeastOneStripReblended
- BIt 56: IsAtLeastOneStripDeblended
- Bit 57: isCFS
- Bit 58: This is a bit needed for internal pipeline use and is always activated
- Bit 59-62: not used

RAD_VEL_BARY_CORR : Barycentric correction (float, Velocity[km s^{-1}])

The barycentric radial velocity correction is computed using the Gaia Relativity Model (GREM). The error on this parameter is expected to be of the order of 1 cm/s.

This correction is used in the spectroscopic pipeline to shift the wavelength grid of the CCD spectra in order to align the Gaia CCD spectra to the solar barycentric frame. A positive value of the barycentric correction corresponds to a redshift of the spectrum.

OBS_TIME_BARY_CORR : Barycentric correction of observing time (long, Time[ns])

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Roemer barycentric correction of the observing time, to be added to `obs_time_rv` in order to shift the observation time at Gaia to a virtual observer at the solar system barycentre, thus removing the distortion caused by the Roemer effect.

OBS_TIME_RV : Observing time (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

This is the mean time over the CCDs used to obtain `radial_velocity`. The CCD observing time is given by the midpoint of the spectrum as it crosses the fiducial point.

The observing time for each CCD of the transit is provided in `ccd_obs_times` of `rvs_transit_descriptio`

RAW_RADIAL_VELOCITY_ERROR : The uncertainty provided by the model (float, Velocity[km s^{-1}])

The raw uncertainty is the uncertainty directly provided by the model used to obtain `radial_velocity`, without any correction applied.

It is the uncorrected version of `radial_velocity_error`. See Section **??**.

RADIAL_VELOCITY_MC_ERROR : MonteCarlo error on spectroscopic radial velocity (float, Velocity[km s^{-1}])

The Monte Carlo error is obtained by interpolating a precomputed table of Monte Carlo error bars as a function of atmospheric parameters. This precomputed table is based on all the theoretical spectra used in the processing of RVS spectra. For each theoretical spectrum, a grid of rotational velocities is explored, along with a grid of RVS magnitudes.

For each entry in the precomputed table, 1,000 simulated transits are generated with a random radial velocity. Using the STA code, the 1,000 radial velocities are then measured from these simulated data, and their dispersion (half of the range between the 15.85th and 84.15th percentiles) is taken as the Monte Carlo error.

For the faintest entries in this table, it is possible that no valid radial velocity can be derived for any of the 1,000 simulated transits, in which case the Monte Carlo error is set to NaN. When interpolating Monte Carlo errors for an observed transit, the subset of entries corresponding to the relevant theoretical spectrum is selected. Within that subset, interpolation is applied as a function of rotational velocity and RVS magnitude.

In some cases, this results in a negative value, in which case the Monte Carlo error is not provided. For sources with RVS magnitudes between 13.9 and 14.0 mag, extrapolation is required, which can lead to large values of the Monte Carlo error.

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ELS_PROBABILITY : Probability of having emission lines (float[3] array)

This field is an array containing the probability for each CCD in this transit to have emission lines. It is computed using deep learning techniques and is populated for stars having `external_apparent_grvs` $\leq$ 13 (in `all_source_rvs`) and only if at least two CCD spectra are available. See **?**.

When at least two CCDs have `els_probability` $\geq$ 0.5, the `emission_lines` flag is set to True.

RV_OUTLIER_PROBABILITY : Probability for radial velocity to be an outlier (float)

This field represents the expectation for the `radial_velocity` being erroneous. It is computed using machine learning techniques, specifically a balanced random forest trained on parameters describing the shape of the Cross-Correlation Function, for stars meeting the following criteria:

- `all_source_rvs.external_apparent_grvs`> 11
- 3700 $\leq$ `all_source_rvs.rv_template_teff` $\leq$ 6500 K

Outside of this validity range, and when `radial_velocity` is not provided, the field is not populated. Higher values suggest a greater probability of being erroneous.

The `rv_outlier_probability` values range from 0 to 1, but in some cases, a value of 2 is assigned to indicate that the method could not derive the expectation, which typically occurs when `radial_velocity` is erroneous.

See **?** for a description of how `rv_single_outlier_probability` is calculated.

RV_OUTLIER_CONDITION : Condition for radial velocity to be an outlier in the time series (float)

This field is calculated using statistics combined with transit cross-correlation function properties information from the three RV methods RvDir, RvFou, and RvMDM. In case the transit RV value is a true outlier in the RV time-series (`outlier_status`) the field provides a number from 0.7 to 1 corresponding to transit outlier conditions, selected based on the quality of the cross-correlation functions properties of the three RV methods time-series, in increasing order of importance. See **?**.

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If this field is zero the `radial_velocity` is not detected as an outlier in the RV time-series.

This field is not provided when the `radial_velocity` is not provided, when the transit is deemed contaminated by a nearby bright source, or when it is a duplicate of another transit. This infornmation is available in the table `rvs_transit_description` in the field `ccd_invalid_reason` with values 81,78,77,71.

OUTLIER_STATUS : Outlier status of this transit (byte)

Outlier status of this transit.

For each transit in the time-series of `radial_velocity`, `grvs_mag`, `vbroad`, and CCF_{max} (i.e. the maximum value of the cross-correlation function used to obtain the radial velocity in the RvDir method), an outlier status ranging from 0 to 15 is assigned, provided that at least three valid transits are available. This flag indicates in which time series, or combination of time series, an outlier is detected.

When these values are not provided, the corresponding `outlier_status` should be ignored.

The possible outlier statuses are listed in the table below, where a checkmark indicates that an outlier was detected in the corresponding parameter.

outlier_status	radial_velocity	vbroad	grvs_mag	CCF_{max}
−1	not provided			
0				
1	✓			
2		✓		
3			✓	
4	✓	✓	✓	
5	✓	✓		
6	✓		✓	
7		✓	✓	
8				✓
9	✓			✓
10		✓		✓
11	✓	✓		✓
12			✓	✓
13	✓		✓	✓
14		✓	✓	✓
15	✓	✓	✓	✓

Outlier values in the `grvs_mag` and `vbroad` time-series are identified using the classical 'Interquartile Range' criterion, where (probable) outliers are values more than 3 times the time-series' interquartile range from the quartiles.

For the CCF_{max} time-series, outlier values are detected if they fall more than one time the time-series' interquartile range below the lower quartile.

The detection of outlier values in the `radial_velocity` time-series is performed by combining the `radial_velocity` time-series statistics with information from the transit cross-correlation function.

The statistical parameters stored in `all_source_rvs` having the suffix 'Robust' are computed after excluding outlier values according to the `outlier_status` flag.

VBROAD : Spectral line broadening (float, Velocity[km s^{-1}])

The spectral line broadening parameter. It is measured using a rotational broadening kernel; its value therefore includes $v \sin i$, but also any other source of line broadening (macroturbulence, residual instrumental effects, etc.)

`vbroad` is computed only for sources with `external_apparent_grvs`≤12 (available in `gaia_source` and in `all_source_rvs`). It is not provided when `radial_velocity` is not provided. [However, it was unintentionally provided for a few thousand `emission_lines` transits, which have no `radial_velocity`; these values have since been invalidated, with `vbroad_invalid_reason` Bit 0 activated (Invalid Radial Velocity).]

See **?** for a description of the validation of the `vbroad` measurements.

VBROAD_ERROR : Uncertainty on the spectral line broadening (float, Velocity[km s^{-1}])

`vbroad_error` is the uncertainty associated with `vbroad`.

GRVS_FLUX : Total flux of the system (float, Flux[e^{-} s^{-1}])

`grvs_flux` is determined based on the number of available CCD spectra in the transit (1, 2, or 3). The integrated flux in each CCD (f_{ccd_i}) is computed by summing the fluxes in the spectrum samples, where the flux in masked samples is assumed to be the median of the non-masked samples.

Each CCD-integrated flux is then corrected for flux loss outside the window and rescaled to a common zero point, TheGrvsZP = 21.318467, in order to remove instrumental effects that depend on the CCD, field of view, and observation time. For more details, see **?**.

There are cases where `grvs_flux` < 0, which in general occur for faint stars with an overesti-

mated subtracted background.

- For 1 available CCD:

$$\texttt{grvs_flux} = f_{ccd_i}$$

- For 2 available CCDs:

$$\texttt{grvs_flux} = \text{mean}(f_{ccd_i})$$

- For 3 available CCDs:

$$\texttt{grvs_flux} = \text{median}(f_{ccd_i})$$

GRVS_FLUX_ERROR : Error on the total flux (float, Flux[e^- s^{-1}])

`grvs_flux_error` is the uncertainty associated with `grvs_flux`. It is not populated when `grvs_flux` ≤ 0.

The internal flux error, *intFluxError*, is computed based on the number of available CCD spectra in the transit (1, 2, or 3). The internal flux error for each CCD spectrum, $intFluxError_{ccd_i}$, is the quadratic sum of the flux errors of each sample within that CCD spectrum.

- For 1 available CCD:

$$intFluxError = intFluxError_{ccd_i}$$

- For 2 available CCDs:

$$intFluxError = \text{std}(intFluxError_{ccd_i})$$

- For 3 available CCDs:

$$intFluxError = \text{mad}(intFluxError_{ccd_i}) \times \sqrt{\pi/2} \,/\, \sqrt{3}$$

Since the internal flux errors were found to be underestimated, a correction was applied as follows:

$$\texttt{grvs_flux_error} = \sqrt{A^2 + B \times intFluxError^2}$$

where A is a function of `external_apparent_grvs`, and B is a sixth-degree polynomial function of `external_apparent_grvs`.

See **?** for more details.

GRVS_MAG : Grvs magnitude of the system (float, Magnitude[mag])

The magnitude `grvs_mag` is estimated using `grvs_flux` and the zero-point magnitude ($ZP_{\mathrm{grvs}} = 21.318467$):

$$\texttt{grvs_mag} = -2.5 \times \log(\texttt{grvs_flux}) + ZP_{\mathrm{grvs}}$$

It is not provided when $\texttt{grvs_flux} \leq 0$.

GRVS_MAG_ERROR : Grvs magnitude error (float, Magnitude[mag])

The uncertainty associated with `grvs_mag` is calculated using the following formula:

$$\texttt{grvs_mag_error} = \texttt{grvs_flux_error}/\texttt{grvs_flux} \times 2.5/\ln(10).$$

It is not populated in cases when $\texttt{grvs_flux} \leq 0$.

GRVS_ZERO_POINT : Grvs zero point for this transit (float, Magnitude[mag])

`grvs_zero_point` is the transit-specific zeropoint, accounting for variations due to the observation time, field of view (FoV), and CCD configuration. It is computed as the median of the zeropoints derived for each of the three CCDs involved in the transit (see **?**).

This relative zeropoint reflects time and configuration dependent instrumental effects (e.g. CCD sensitivity, FoV throughput, and mirror contamination). It allows for recovering the measured (raw) flux from the scaled value published in `grvs_flux`, which has been normalized to the global zeropoint: TheGrvsZP = 21.318467.

The unscaled measured flux can be recovered via:

$$\text{measuredFlux} = \texttt{grvs_flux} \cdot 10^{0.4 \cdot (\texttt{grvs_zero_point} - \text{TheGrvsZP})}$$

EXPECTED_SIG_TO_NOISE : The signal to noise expected (float)

The signal to noise ratio expected by combining the information from the CCD spectra and the relative background used for estimating the radial velocity.

The expected signal to noise ratio is obtained by combining the information from the CCD spectra used for estimating `radial_velocity`:

$$\texttt{rv_expected_sig_to_noise} = \frac{S}{\sqrt{S + BCK + RN}}$$

where:

$$S = \frac{medianFlux \times N_{ValidStrips} \times \text{EXP_TIME} \times \text{PIXEL_WIDTH_AL}}{\text{BAND_WIDTH}}$$

$$BCK = medianBackground \times N_{ValidStrips} \times N_{ACpixels}$$

$$RN = RON^2 \times N_{ValidStrips} \times N_{ACsamples}$$

where:

• $medianFlux$ is the median of the integrated flux of the CCD spectra used to obtain the radial velocity for `external_apparent_grvs` ≤ 12; otherwise, the $medianFlux$ is obtained from `external_apparent_grvs` [$e^- s^{-1}$]. `external_apparent_grvs` is store in the tables `all_source_rvs` and `gaia_source`.
• $N_{ValidStrips}$ = the total number of CCD spectra used to obtain `radial_velocity` (in general there are 3 CCD spectra per transit)
• BAND_WIDTH = 870 - 846 [nm]
• EXP_TIME = 4.4167032 [s]
• PIXEL_WIDTH_AL = 0.02453 [nm]
• $medianBackground$ = the median background in one exposure of the CCD spectra used to obtain the `radial_velocity` [e^-pixel^{-1}]
• $N_{ACpixels}$ = 10 is the number of pixels in the AC direction of the non-truncated RVS window
• $N_{ACsamples}$ is the size in the AC direction of the non-truncated RVS window. $N_{ACsamples}$ = 10 for the 2D RVS windows (Window Class=0) and $N_{ACsamples}$ = 1 for the 1D windows (Window Class=1)
• RON (Read Out Noise) = 3.2 [e^-]

It is not provided when $medianFlux \leq 0$ or when all the CCD strips are flagged as invalid during the FullExtraction step of the pipeline (i.e., in the table `rvs_transit_description`, where

`ccd_invalid_reason` has values < 50 or > 90).

RV_SYSTEMATIC_CORR : The correction applied to the radial velocity (float, Velocity[km s^{-1}])

`rv_systematic_corr` represents a correction applied in the spectroscopic pipeline to account for systematic errors. This correction is used to shift the wavelength grid of the CCD spectra. A positive value of the correction corresponds to a blueshift of the spectrum. For mode information, see Section **??**.

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17.3 RVS_EPOCH_SPECTRUM [DataLink]

This table contains RVS spectra combined at the transit level.

These spectra are obtained by combining the CCD spectra of a single transit of the source. The number of spectra combined is stored in `combined_ccds`, and it is usually 3. The fluxes and uncertainties-squared of the CCD spectra are linearly interpolated onto a fixed wavelength grid of 961 spectral elements with a step of 0.025 nm. The wavelength range is [846-870] nm. The spectra are shifted from the Gaia reference frame to the barycentric reference frame and are normalised. See **?** for the validation of RVS epoch spectra, and **?** for the processing, including cleaning and calibration of raw CCD RVS spectra.

Note this table is not available through the main archive TAP interface. Data are delivered via the Massive Data service indexed by the VO DataLink protocol and described in Chapter **??**. For example this can be actioned in the archive user interface via ADQL queries joining on `source_id` with the table `all_source_flags` and selecting `has_epoch_rvs = 'true'`. The DataLink retrieval type to specify for this product in programmatic archive access via the Astroquery Gaia Python package is `EPOCH_SPECTRUM_RVS`.

DataLink products are served with a choice of two different data structures:

- `DATAMODEL_STANDARD`: the product is serialised following a standard data model (for example an IVOA-compliant data model in applicable cases);
- `DATAMODEL_GAIA`: the product is serialised following the Gaia DPAC(-internal) data model.

The `DATAMODEL_GAIA` format is described here whereas the `DATAMODEL_STANDARD` format is described in `https://www.cosmos.esa.int/web/gaia-users/archive/datalink-products#datalink_serialisation`

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published

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data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`.

SOURCE_ID : Source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

TRANSIT_ID : transit identifier (long)

The `transit_id` is a unique identifier assigned to each detected (and confirmed) source as it transits the Gaia focal plane. Each time a given source is detected as Gaia scans and re-scans the sky a new `transit_id` will be created to badge that apparition. Hence the along–scan time and the across–scan position along with the telescope in which the source was detected are used to form a unique integer with which to label the transit.

The several features of a detection that are encoded in `transit_id` can be easily retrieved using bit masks (&) and shifts (>>) as follows:

- On-Board Mission Time line [ns]
  = 204800 * ((`transit_id` >> 17) & (0x000003FFFFFFFFFF))
- Field-of-view = 1 + (`transit_id` >> 15) & 0x03 [1 for 'preceding' and 2 for 'following' fields-of-view respectively]
- CCD row = (`transit_id` >> 12) & 0x07 [dimensionless, generally in the range 1 to 7 but specifically for RVS in the range 4 to 7]
- Across-scan 'reference acquisition pixel' in strip AF1 = (`transit_id`) & 0x0FFF [pixels] (this is the across-scan centre of the AF1 window and is odd if immediately below the mid-point of the window and even if immediately above)

where the bit mask prefix '0x' denotes hexadecimal.

For further details see Portell et al. (2020). For convenience a decoder for `transit_id` is available on-line at

`https://gaia.esac.esa.int/decoder/transitidDecoder.jsp`

FLUX : array of fluxes (float[961] array)

It represents the array of mean fluxes in each wavelength bin. The wavelength bins are given by: $wave[i] = 846.0 + i \times 0.025$ nm (i=0..960).

The mean flux per wavelength bin is calculated as:

$$\texttt{flux[i]} = \frac{\sum_{j=1}^{n} f_j[i]}{n}$$

Here, n denotes the number of flux values $f_j[i]$ contributing to the wavelength bin i, where j represents the jth CCD spectrum being combined out of a total of n spectra in the wavelength bin i.

Typically, n corresponds to `combined_ccds` (usually 3), but it might be smaller if CCD samples were masked due to bad deblending, cosmic rays, or saturation, or at the edges of the wavelength range, where the CCD spectra often have missing samples. If all CCDs have masked or missing samples in a particular bin, `flux[i]` can be NAN.

Before combination, the CCD spectra are normalized, and then linearly interpolated to the wavelength grid. The combined spectrum is then renormalized.

FLUX_ERROR : array of uncertainties in flux (float[961] array)

`fluxerror[i]` represents the uncertainty in the flux for each wavelength bin.

The epoch spectra are derived by combining the CCD spectra from a single transit (usually 3) of the source. The fluxes and squared uncertainties of the normalized CCD spectra are linearly interpolated onto a fixed wavelength grid.

For each CCD spectrum, the uncertainty per sample per exposure time is computed under the assumption of Gaussian errors. It involves adding in quadrature and then square rooting several in-flight measurements: Poisson noise, readout noise, bias non-uniformity noise (including common baseline, flush, and glitch), straylight background, and dark current (measured on-ground).

The uncertainty in the flux for a specific wavelength bin ($\sigma_{f,\mathrm{bin}}$) is determined by propagating the uncertainties of the n CCD spectra contributing to the wavelength bin:

$$\sigma_{f,\mathrm{bin}} = \frac{1}{n}\sqrt{\sum_{j=1}^{n}(\sigma_{f_j})^2}$$

Here, σ_{f_j} represents the uncertainty of the jth CCD spectrum contributing to the wavelength bin.

If a particular bin's uncertainty result from CCD samples that were all masked due to bad deblending, cosmic rays, or saturation, the uncertainty in that bin is marked as NAN.

OBS_TIME_RV : observing time (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

This is the mean barycentric time over the CCDs used to obtain this spectrum, expressed in Barycentric Julian Date (BJD) in TCB (Barycentric Coordinate Time), with an offset of 2,455,197.5 days. The time is calculated as follows:

The observing times of the CCDs are defined by the midpoint of the spectrum as it crosses the fiducial point. The CCD observing times are stored in the table `rvs_transit_description` in On-Board Mission Time (OBMT). The Roemer delay correction is applied to the observing time to shift it from Gaia's position to that of a virtual observer at the solar system barycentre, eliminating the timing distortion caused by the Roemer effect. Next, the OBMT is converted into Julian Date, resulting in Barycentric Julian Date (BJD). Finally, an offset of 2,455,197.5 days (corresponding to a reference time T_0 at 2010-01-01T00:00:00) is applied to yield a conveniently small numerical value.

COMBINED_CCDS : number of combined CCD spectra (int)

This is the total number of spectra that have been combined. For any given transit, 1, 2, or 3 spectra can be valid and therefore combined.

The reasons why the CCD spectra are considered invalid for producing the mean radial velocity are listed in field `ccd_invalid_reason` from the table `rvs_transit_description`. However, epoch spectra are still produced for detected double-lined, emission-line, and contaminated transits (`ccd_invalid_reason` between 51 and 90).

DEBLENDED_CCDS : number of deblended CCD spectra (int)

This field contains the information on how many deblended CCD spectra have been used to produce this spectrum.

The number of non-deblended spectra is obtained as `combined_ccds-deblended_ccds`.

COMBINED_CCD_IN_INDEX : number of combined CCDs in index (short[] array)

This is the number of combined CCD spectra in the bin of index *i*. It is stored only when strictly smaller than `combined_ccds` (typically for the bins at the borders of the combined spectrum)

This array has the size of the `index` array, corresponding to the number of fluxes bins resulting from the combination of a number of CCD spectra strictly smaller than `combined_ccds`. This field is null if `index` is null.

INDEX : index in the array of fluxes (short[] array)

This field contains the indexes of the `flux` array where the number of combined CCDs is strictly smaller than `combined_ccds`.

It can be null when all the `flux[i]` values are obtained using all the available CCD spectra, `combined_ccds`.

NORMALISATION_METHOD : the method used for normalisation (byte)

The possible normalisation methods of the combined spectra are listed below. Note that first, the CCD spectra are normalised, then combined, and finally, a second normalisation is applied to the combined spectra. This field refers to the normalisation of the combined spectra, and the only normalisation methods available are:

0: Not normalised. The combined spectra where double-lines are detected are not normalised.

3: Degraded (scaled with a constant). The second normalisation is applied to the combined spectrum, obtained by averaging CCD spectra previously normalised to 1. Both the fluxes and the flux errors are divided by the median of the combined spectrum. In the very noisy spectra, its value may deviate significantly from 1.0. If the median is very small or very large, dividing the fluxes and flux errors by it results in correspondingly very large or very small values.

4: Median negative flux in the combined spectrum (if the median flux value in the combined spectrum is negative, the combined spectrum is not normalised). This `normalisation_method` is an indication of poor spectrum quality.

Other normalisation methods, such as polynomial fit to the pseudocontinuum (ContinuumNorm) (1) and template-based normalisation (2), are not applicable in this context. These methods are used for normalising the CCD spectra. Information on the CCD spectra is stored in the field `ccd_status` of the table `rvs_transit_description`.

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However, a small number of spectra (approximately 10) were mistakenly normalised using `normalisation_method`=2 (template-based normalisation), resulting in spurious `flux` and `flux_error`.

IS_CONTAMINATED : True if this spectrum is contaminated (boolean)

This flag is set to true for the epoch spectra when contaminated by a relatively bright source that did not result in overlapping window on the CCD and is therefore not deblended (i.e., having `deblended_ccds` = 0). It is false for the non-contaminated spectra and for the deblended spectra, where `deblended_ccds` > 0.

The contaminated CCD spectra are not used to produce `rvs_mean_spectrum` and are excluded from the radial velocity measurements stored in `all_source_rvs`. The contaminated CCD spectra have `ccd_invalid_reason`=78 or 81 in `rvs_transit_description`.

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17.4 RVS_MEAN_SPECTRUM [DataLink]

This is the RVS mean sampled spectrum table. The spectra are normalized and sampled on a wavelength grid from 846 to 870 nm in 0.025 nm steps (961 elements). The vast majority of spectra are in the rest frame; a very small fraction (e.g., stars with emission lines) are not.

The measured signal to noise ratio, `rvs_spec_sig_to_noise`, is available in `all_source_rvs` and in `gaia_source`.

Spectra of stars with `fraction_emission_lines` < 0.4 are treated as normal: the epoch spectra in which emission lines are detected are assumed to be spurious and excluded from the combination, and the resulting mean spectrum is shifted to the rest frame. In contrast, spectra of stars assumed to have emission lines (`fraction_emission_lines` > 0.4 in `gaia_source` and in `all_source_rvs`) are provided in the barycentric frame.

`rvs_mean_spectrum` is not provided for stars assumed to be SB2 (`rv_assumed_sb2` = True in `all_source_rvs` and `gaia_source`), for stars with only one valid transit, and for those with `fraction_emission_lines` = 0.4.

See Section **??** for a description of how the mean spectra are obtained, and Seabroke et al. (2026) for a description of the spectra validation.

Note this table is not available through the main archive TAP interface, but via the Massive Data service, indexed by the VO DataLink protocol, described in Chapter **??**. For example this can be actioned in the archive user interface by querying the main source catalogue `all_source_flags` and selecting `has_rvs` > 0. The DataLink retrieval type to specify for this product in programmatic archive access via the Astroquery Gaia Python package is `MEAN_SPECTRUM_RVS`.

DataLink products are served with a choice of two different data structures:

- `DATAMODEL_STANDARD`: the product is serialised following a standard data model (for example an IVOA-compliant data model in applicable cases);
- `DATAMODEL_GAIA`: the product is serialised following the Gaia DPAC(-internal) data model.

The `DATAMODEL_GAIA` format is described here whereas the `DATAMODEL_STANDARD` format is described in `https://www.cosmos.esa.int/web/gaia-users/archive/datalink-products#datalink_serialisation`

Columns description:

SOURCE_ID : Unique source identifier (unique within a particular Data Release) (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

FLUX : Array of fluxes (float[961] array)

The mean normalized spectrum flux.

The mean flux per wavelength bin is:

$$\texttt{flux[i]} = \frac{\sum_{j=1}^{n} f_j[i]}{n}$$

where n is the number of flux values $f_j[i]$ contributing to the wavelength bin i, and j is the jth CCD spectrum being combined out of a total of n in the wavelength bin i.

In general, n corresponds to `combined_ccds`, but it may be smaller, especially in the bins at the edges of the wavelength range, which are often not sampled by all the CCD spectra shifted to the rest frame.

FLUX_ERROR : Array of uncertainties in flux (float[961] array)

The uncertainty on the mean normalised flux is:

$$\texttt{flux_error}[i] = \frac{\sigma_{f_n}[i]}{\sqrt{n}}$$

where $\sigma_{f_n}[i]$ is the standard deviation of all the flux values contributing to the wavelength bin i:

$$\sigma_{f_n}[i] = \sqrt{\frac{1}{n}\sum_{j=1}^{n}(f_j[i] - \texttt{flux[i]})^2}$$

where $f_j[i]$ is the flux the jth CCD spectrum being combined out of a total of n in the wavelength bin i.

In general, n corresponds to `combined_ccds`, but it may be smaller, especially in the bins at the edges of the wavelength range, which are often not sampled by all the CCD spectra. This results in typically larger `flux_error` at the edges of the combined spectrum.

COMBINED_TRANSITS : Number of combined transits (short)

The number of transits used to compute the mean RVS spectrum.

COMBINED_CCDS : Number of combined CCD spectra (short)

This number reports the total number of CCD spectra that have been combined (over the `combined_transits`) taking into account that for any combined transit 1, 2 or 3 CCD spectra can be valid and therefore combined.

The reasons why the CCD spectra are considered invalid for producing the mean spectra and the mean radial velocity are listed in field `ccd_invalid_reason` from the table `rvs_transit_description`. A special case is for sources identified as having emission lines, indicated by `fraction_emission_lines` > 0.4 in `all_source_rvs`. For these sources, although no radial velocity is computed, the mean spectrum is produced using the CCD spectra, even when `ccd_invalid_reason`=77 due to detected emission lines.

DEBLENDED_CCDS : Number of deblended CCD spectra (short)

This field contains the information on how many deblended spectra have been used to produce this mean RVS spectrum: it is the number of CCD spectra that have been deblended.

The number of non-blended spectra is obtained from: `combined_ccds−deblended_ccds`

See Seabroke et al. (2026) for a description of the deblending of the CCD spectra.

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COMBINED_CCD_IN_INDEX : Number of combined CCDs in index (short[] array)

It is the number of combined CCD spectra in the bin of index *i*. It is stored only when strictly smaller than `combined_ccds` (typically for the bins at the borders of the combined spectrum)

This array has the size of the `index` array, corresponding to the number of fluxes bins resulting from the combination of a number of CCD spectra strictly smaller than `combined_ccds`. This field is null if `index` is null.

INDEX : Index in the array of fluxes (short[] array)

This field contains the indices of the `flux` array where the number of combined CCDs is strictly smaller than `combined_ccds` (typically in the samples at the spectra edges, i.e., `index` = [0,1,959,960]).

It can be null when all `flux[i]` values are obtained using all the available CCD spectra, `combined_ccds`.

NORMALISATION_METHOD : The method used for normalisation (byte)

The possible normalisation methods for the combined spectra are listed below. Note that initially, the CCD spectra are normalised. Then, they are combined, and finally, a second normalisation is applied to the combined spectra. This field refers to the normalisation of the combined spectra. The available normalisation methods are:

1: Divided by the pseudocontinuum, obtained by a polynomial fit. [This method is used only in the rare cases of assumed emission-line stars.]

3: Degraded (scaled with a constant). The second normalisation is applied to the combined spectrum, obtained by averaging CCD spectra previously normalised to 1. Both the fluxes and the flux errors are divided by the median of the combined spectrum. In the very noisy spectra, its value may deviate significantly from 1.0. If the median is very small or very large, dividing the fluxes and flux errors by it results in correspondingly very large or very small values.

4: Median negative flux in the combined spectrum (if the median flux value in the combined spectrum is negative, the combined spectrum is not normalised). This `normalisation_method` is an indication of poor spectrum quality.

Other normalisation methods, such as non-normalised (0) and template-based normalisation (2), are not applicable in this context. These methods are used for normalising the CCD spectra. Information on the CCD spectra is stored in the field `ccd_status` of the table `rvs_transit_description`.

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COMBINATION_METHOD : The method used to shift the CCD spectra (byte)

The `combination_method` field indicates the origin of the radial velocity used to shift individual CCD spectra to a common rest frame (radial velocity = 0) before combination.

0: No shift. The CCD spectra remain in the barycentric frame and are combined with no additional shift. This is used for assumed emission-line stars (`fraction_emission_lines` > 0.4 in `all_source_rvs` and `gaia_source`), and the resulting combined spectrum is not in the rest frame.

1: Single-transit radial velocity. Each CCD spectrum from a specific transit (epoch) was shifted using the `radial_velocity` recorded for that same transit in the `rvs_epoch_parameters_single` table.

2: Multi-transit constant radial velocity. All CCD spectra of the source were shifted using the same multi-transit constant radial velocity. This value is stored in `radial_velocity_single` of `all_source_rvs` and, when originating from `all_source_rvs`, in `radial_velocity` of `gaia_source`.

IS_CONTAMINATED : True if this spectrum is contaminated (boolean)

This flag is set to true for mean spectra when the source is identified as contaminated by a relatively bright neighbour during validation. Sources found to be contaminated in this process have the 'contamination' flag activated in `rv_invalid_reason` of `all_source_rvs`.

Contaminated CCD spectra with `ccd_invalid_reason` = 78 or 81 in `rvs_transit_description` are excluded from the generation of `rvs_mean_spectrum`. However, additional contaminated sources are identified during post-processing and are flagged here.

HAS_SPIKES : True if this spectrum shows spurious spikes (boolean)

This flag is set to True when a spike is detected in the spectrum.

A cosmic ray detection method, originally developed by Whittaker and Hayes (`https://chemrxiv.org/engage/api-gateway/chemrxiv/assets/orp/resource/item/60c73e33469df41c2af4281c/original/a-simple-algorithm-for-despiking-raman-spectra.pdf`), has been adapted to identify spurious structures such as cosmic rays and radiation tracks in the multi-transit combined spectra.

INVALID_RADIAL_VELOCITY : True if this source radial velocity is deemed invalid (boolean)

This flag is set to True when the radial velocity of this source, stored in `radial_velocity_single` in the table `all_source_rvs`, is deemed invalid and `radial_velocity_single_is_valid` is set to False.

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17.5 RVS_TRANSIT_DESCRIPTION [TAP]

This table provides metadata for each RVS transit at the individual CCD level. It includes flags and timestamps that describe the status and processing outcomes of each CCD spectrum within the transit.

Each transit consists of up to three CCD spectra. The table records whether each spectrum was used in the computation of the star's radial velocity (`is_ccd_valid`), along with the reason for any invalidation (`ccd_invalid_reason`). The field `ccd_status`, encodes bit-level flags indicating specific characteristics of each CCD spectrum that reached the MultiTransitAnalysis (MTA) stage.

For time series analysis, this table should be used in conjunction with the complementary classes that contain the actual measurements: `rvs_epoch_parameters_single` and `rvs_epoch_parameters_double` for epoch radial velocities, and `rvs_epoch_spectrum` for epoch RVS spectra.

For more detailed information, see **?**.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

TRANSIT_ID : transit identifier (long)

The `transit_id` is a unique identifier assigned to each detected (and confirmed) source as it transits the Gaia focal plane. Each time a given source is detected as Gaia scans and re-scans the sky a new `transit_id` will be created to badge that apparition. Hence the along–scan time and the across–scan position along with the telescope in which the source was detected are used to form a unique integer with which to label the transit.

The several features of a detection that are encoded in `transit_id` can be easily retrieved using bit masks (&) and shifts (>>) as follows:

- On-Board Mission Time line [ns]
  = 204800 * ((`transit_id` >> 17) & (0x000003FFFFFFFFFF))
- Field-of-view = 1 + (`transit_id` >> 15) & 0x03 [1 for 'preceding' and 2 for 'following' fields-of-view respectively]
- CCD row = (`transit_id` >> 12) & 0x07 [dimensionless, in the range 1 to 7]
- Across-scan 'reference acquisition pixel' in strip AF1 = (`transit_id`) & 0x0FFF [pixels] (this is the across-scan centre of the AF1 window and is odd if immediately below the mid-point of the window and even if immediately above)

where the bit mask prefix '0x' denotes hexadecimal.

For further details see Portell et al. (2020). For convenience a decoder for `transit_id` is available on-line at

`https://gaia.esac.esa.int/decoder/transitidDecoder.jsp`

IS_CCD_VALID : is the CCD observation valid? (boolean[3] array)

This field indicates whether the spectrum corresponding to the CCD is valid, i.e., whether it was used in the computation of the radial velocity of the source (stored in `all_source_rvs`).

The reason it was marked as invalid is stored in ccd_invalid_reason.

CCD_INVALID_REASON : why the CCD observation is invalid (byte[3] array)

The reason why the CCD data has been marked as invalid (i.e., it was not used in the computation of the radial velocity of the source).

The invalid reasons are set at different stages of the pipeline processing: Preprocessing, FullExtraction, SingleTransitAnalysis (STA), and MultiTransitAnalysis (MTA). The field stores a single integer code identifying the first invalidity detected during processing; it is not a bit mask, and only one invalid-reason code is assigned per transit. For more detailed information, see **?**.

0: Not set

PreProcessing (1-20)
1: Bad data interval.
3: Attitude non available at the AF1 sampling time.
10: An exception occurred during the calculation of the field angles, or the uncertainty on the field angle η exceeds 200 mas.

FullExtraction (21-50 and 91-94).
24: isNegativeBiasNu: The spectrum has a negative flux level after the Bias Non-Uniformity correction.
25: isNegativeBasicCleaning: The spectrum has a negative flux level after the background correction.
26: isNegativeDeblendToCollapse: The spectrum has a negative flux level after the deblending.
27: Too many saturated samples (maximum allowed: 40).
28: isHighScatterLight: The straylight level is too high.
29: isRequiringDeblendingNotDeblended : The spectrum required deblending but could not be deblended isRequiringDeblending and NotDeblended.
30: isOnlyStrip : This flag is set when a spectrum is the only one of its type (blended or unblended) in a transit. Since cosmic ray removal requires multiple spectra of the same type, single spectra are excluded.
31: isCosmeticDefect: Spectra affected by cosmetic defects, or spectra near a hot column with at least one affected macro sample.
40: The spectrum could not be normalised.
41: A problem occurred with the input data in the EstimateMagnitude module.
45: Double detection low priority - This is the fainter component of an onboard double detection.
46: The window is truncated due to the CCD edges (no overlapping spectra).
50: Too many cosmic ray-affected samples (maximum allowed: 100).
91: Too many bad deblended samples (maximum allowed: 300.
92: Too many bad samples (maximum allowed: 300).
94: Spurious structures in spectrum. Flag set by 1D Cosmic Ray Removal when spurious structures, likely radiation tracks, are detected in a spectrum. These are identified as flux outliers using Tukey's fences. The flag is set if one of the following conditions is met: 1. The first or last sample in a macroSample is one of 10 or more consecutive Tukey outliers AND the adjacent macroSample is 'overlapping'; 2. There are 30 or more consecutive Tukey outliers within the 846-870 nm wavelength range.

STA (51-70)
54: Super_StaChain Bright (`external_apparent_grvs`$\leq$ 14)did not provide a radial velocity.
56: Calibration Faint Star: The transit of the star was acquired in WindowClass 0 despite the star being faint, as part of the calibration faint star mode.
58: Emission lines detected in too faint stars. This reason is set for transits of stars with `external_apparent_grvs`$>$ 12. Detected emission lines (`emission_lines` = true in the table `rvs_epoch_parameters_single`) are considered spurious (except when all transits are flagged). These transits are excluded from downstream processing, and only the epoch spectrum

is provided, except for stars with all transits flagged (fraction_emission_lines=1) for which the mean spectrum is also provided.
61: Super_StaChain FAINT (external_apparent_grvs> 14) could not make CrossCorrFunction.
62: Jump or spike detected in the spectrum, identified by the EmissionLineDetection module when non-emission-line structures are present, that is when $medianLocal > median + 3 \times stddev$, where *medianLocal* is a running median over 50 samples of the spectrum fluxes[i], *median* is the median flux of the full spectrum, and *stddev* is the standard deviation of fluxes[i] over the full spectrum.

MTA (71-90)
71: Duplicated transit.
76: Double lined transit.
77: Emission Lines transits. This reason is set at source level for transits of stars with external_apparent_gr
12 and detected emission lines, as well as for all transits (including those without detected emission lines) of stars assumed to be emission-line stars (i.e., fraction_emission_lines> 0.4, except for stars with fraction_emission_lines=1 and external_apparent_grvs> 12, for which the invalid reason is 58.
78: Contamination from a bright source.
81: Very bright spectrum acquired in WindowClass 0 blended with another WindowClass 0 spectrum.

CCD_STATUS : the status of the CCD observation (long[3] array)

Provides information on the status of each of the three CCD spectra in the transit (for more details, see **?**).

Each element ccd_status[i] is a 64-bit field indicating specific characteristics of the corresponding CCD spectrum. Each bit represents a flag, set to 1 (true) if the condition is met, or 0 (false) otherwise.

The bits are set internally during pipeline processing to reflect the status of each CCD spectrum. However only the ccd_status values for the spectra reaching the MTA processing stage are provided, that is, for spectra where is_ccd_valid[i]=true, or where the spectrum is invalid with ccd_invalid_reason[i] equal to 54,58,61,71,76,77,78, or 81.

- Bit 0 : isDeblended - set if the spectrum was deblended.
- Bit 1 : isOverlapped - set if this window overlaps with another window (some overlapping windows may not be truncated.

- Bit 2 : isTruncated - set if any macrosample in the window is modified to avoid conflicts with other windows or the CCD edge. Valid for all window classes (unlike rvsTruncated). Changes caused by AOCS updates or adaptive window width are not considered truncation. Applies to all data versions; the truncation algorithm in VPU 2.7 is a subset of that in VPU 2.8.
- Bit 3: isReblended - set if this spectrum was reblended. Reblending is used for rectangular windows (same AC width, overlapping in AL, and from the same source).
- Bit 4: isRectangularTruncatedWindow - set by DeblendToCollapse if the window is truncated but remains rectangular.
- Bit 5: deblendConditionNumber - set by ApplyDeblending: Is set to true if one or more samples in the spectrum resulted in a numerically unstable deblending. Is set to false if no deblending was needed or if all samples had a stable reciprocal Deblend SVD condition number above the RecipThresholdNumber.
- Bit 6: [Not Used].
- Bit 7: [Not Used].
- Bit 8: WC0BlendedWithWC0 - set by DeblendToCollapse if this spectrum window is a WindowClass 0 and is blended with another WindowClass 0 spectrum. (These spectra are invalidated with `ccd_invalid_reason`=78).
- Bit 9: WC0BlendedWithWC1 - set by DeblendToCollapse if this spectrum window is a WindowClass 0 and the brightest blend is a Window Class 1 spectrum.
- Bit 10: isBoxCarDeblended - set by ApplyDeblending if the spectrum has been BoxCar-smoothed during deblending (this happens in general for faint stars having only faint blends).
- Bit 11: hasContaminatedBlends - set by ApplyDeblending, when at least one of the blends used to deblend this spectrum is flagged isMissingPBGContribution.
- Bit 12: isBrightContaminated - set by SuperExtraction when the transit radial velocity is expected to be contaminated by a bright source.
- Bit 13: isBiasNuPoorlyCorrected - set by BiasNuCorrection when one or more of the macrosamples are suspected to have a lower quality correction of bias non-uniformity.
- Bit 14 : [Not Used].
- Bit 15 : isMissingPointBgContrib - set when a contaminating point source is expected in the background but has no RVS window, so deblending cannot be performed (deblending requires the contaminant to have a truncating RVS window).

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- Bit 16: faintPBGContribution - set when isMissingPointBgContrib is true, but the contaminating source is faint compared to this target source.
- Bit 17: [Not Used].
- Bit 18: isNegativeBasicCleaning - set by BasicCleaning if the total flux in the central 8 macrosamples is less than or equal to zero (≤ 0) after removing straylight and diffuse background.
- Bit 19: isNegativeDeblendCollapse - set by DeblendToCollapse if the total flux in the central 8 macrosamples is less than or equal to zero (≤ 0) after deblending.
- Bit 20: isSaturatedSamples - set by BasicCleaning if 1 or more samples are saturated.
- Bit 21 : isNormalised - set to true if the spectrum has been normalised by dividing by the continuum. The continuum is estimated using one of the following methods: a polynomial fit (default, ContinuumNormPoly), a template (CU8Norm) or a constant. (See Section **??**).
- Bit 22: isNormUsingTemplate - set if the spectrum has been normalised using CU8Norm, which employs a template-based method.
- Bit 23: isfluxOffsetApplied - set by ContinuumNorm and CU8Norm to indicate that a flux offset was applied before normalisation
- Bit 24: isDegradedNormalisation - indicate the normalisation is performed using with a constant value (the 50th percentile). This degraded method is used for low-S/N spectra (hasLowSN=true), or when the the source is deemed an M-type star exhibiting TiO bands.
- Bit 25: hasPositiveGradient - set by ContinuumNormPoly if a positive gradient (like the one present in M-type spectra with TiO band) is detected in the spectrum. The presence of a flux gradient is assessed by comparing the median flux at the blue edge (846-849 nm) with that at the red edge (867-870 nm). For bright spectra (median flux level $> 50e^{-}$), the ratio of red-edge to blue-edge flux is used: if the ratio is > 1.2, the spectrum is flagged as having a positive gradient; if the ratio is $< 1/1.2$, it is flagged as having a negative gradient. For fainter spectra, the difference between the red and blue edge median fluxes is used, and a gradient is flagged when this difference is > 10.
- Bit 26: hasNegativeGradient - set by ContinuumNormPoly if a negative gradient is detected in the spectrum. The presence of a flux gradient is assessed by comparing the median flux at the blue edge (846-849 nm) with that at the red edge (867-870 nm). For bright spectra (median flux level > 50), the ratio of red-edge to blue-edge flux is used: if the ratio is $< 1/1.2$, the spectrum is flagged as having a negative gradient; if the ratio is > 1.2, it is flagged as having a positive gradient. For fainter spectra,

the difference between the red and blue edge median fluxes is used, and a negative gradient is flagged when the red-edge median flux is lower than the blue-edge median flux by more than 10.

- Bit 27: hasLowSN - set when the signal-to-noise ratio is too low for standard normalisation methods, so degraded normalisation (scaling by a constant) is applied instead. This corresponds to the isDegradedNormalisation method.
- Bit 28: [Not Used].
- Bit 29 : hasLargeFieldAngleEtaError - set by CalcFieldAngle when the η error is larger than 0.014 nm, 5 km s^{-1}, 34 mas
- Bit 30: isNewSource - transferred from IDT : set when no matching source is found in the catalogue for the transit, and a new `source_id` is created.
- Bit 31: isNewSourceByMerging - transferred from IDT : set when the source was created through a merging process, i.e. by combining one or more sources from the previous catalogue (e.g. a high proper motion star previously misidentified as multiple sources).
- Bit 32: isNewSourceBySplitting - transferred from IDT : set when the source is created from splitting a single star into two or more (e.g. a previously unresolved binary).
- Bit 33: isAmbiguousMatch - transferred from IDT : set when there are two candidate sources for this transit.
- Bit 34: isDoubleDetectionHighPriority - transferred from IDT : set to true when the source is likely an onboard double detection of a bright star, and this is the higher-priority detection.
- Bit 35: hasMultiplePeaks - set to true when IDU detects more than one peak (i.e., likely multiple sources); corresponds to Match.nPeaks > 1.
- Bit 36: isDoubleDetection - transferred from IDT: set to indicate that this is one of the components of an onboard double detection of a bright star.
- Bit 37: inconsistentMagnitudes - set by EstimateMagnitude: It is set for [grvs≤ 12 & abs(grvs - externalGrvs > 3] OR [(grvs> 12 & abs(onboardGrvs - externalGrvs) > 3].
- Bit 38: astrometric_motion_corrected - Flag set by the STAMTA pipeline in NSS mode to indicate that the spectrum wavelengths have been shifted to correct for astrometric motion.
- Bits 39 to 63: Not used.

Draft

CCD_OBS_TIMES : array of CCD observation dates (long[3] array, Time[OBMT])

This attribute gives the time information for each CCD. It consists of the time of the mid point of spectrum as it crossses the fiducial point, expressed in nanoseconds.

CCD_USED_IN_COMBINE : is the spectrum used in mean spectrum? (boolean[3] array)

This field is a Boolean[3] array that specifies, for each CCD spectrum, whether it is used to produce `rvs_mean_spectrum`.

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17.6 RVS_TWO_BODY_ORBIT [TAP]

This table stores the binary solutions derived from the analysis performed by the Line Integral Analysis (LIA) modules.

LIA is applied to stars flagged as astrometric binaries in a preliminary Non-Single Stars (NSS) pipeline run (Chapter **??**), using the resulting orbital parameters as input. The fields `period`, `t_periastron`, `eccentricity`, `inclination`, and `arg_periastron` are taken directly from this preliminary solution. A fraction of these binaries were not confirmed in the final NSS pipeline, as reported in **?**, where the LIA method is also described.

Sources with an invalid `radial_velocity_single` in the `all_source_rvs` table
(i.e., `radial_velocity_single_is_valid` = False) are also considered to have an invalid `rvs_two_body_orbit` solution.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Source Identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

RV_NB_COMBINED_TRANSITS : number of combined transits (int)

The number of transits that have been combined to produce the orbital binary model.

RV_NB_DEBLENDED_TRANSITS : number of valid transits that have undergone deblending (int)

The number of transits in `rv_nb_combined_transits` that have undergone deblending.

Draft

CENTER_OF_MASS_VELOCITY : Radial velocity of the centre of mass (float, Velocity[km s^{-1}])

The radial velocity of the centre of mass.

CENTER_OF_MASS_VELOCITY_ERROR : Standard error of the velocity of the centre of mass (float, Velocity[km s^{-1}])

Standard error on the radial velocity centre of mass.

SEMI_AMPLITUDE_PRIMARY : Semi-amplitude of the centre of mass (float, Velocity[km s^{-1}])

The semi-amplitude of the radial velocity curve related to the first component: K1.

SEMI_AMPLITUDE_PRIMARY_ERROR : Standard error on the semi-amplitude of the centre of mass (float, Velocity[km s^{-1}])

Standard error of the semi-amplitude of the centre of mass.

PERIOD : Orbital period (double, Time[day])

Period of the orbital motion around the barycentre.

PERIOD_ERROR : Standard error of the orbital period (float, Time[day])

Standard error of the period of the orbital motion around the barycentre.

T_PERIASTRON : Periastron epoch (double, Time[day])

The epoch at periastron is given relative to `gaia_source.ref_epoch`, in the range [−`period`/2, +`period`/2].

For spectroscopic binaries the periastron is positioned as given by omega relative to the ascending node.

T_PERIASTRON_ERROR : Standard error on periastron epoch (float, Time[day])

Standard error of the periastron epoch.

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CENTER_OF_MASS_VEL_SEMI_AMPL_CORR : Correlation between the velocity and the semi-amplitude of the centre of mass (float)

Correlation coefficient between the velocity of the centre of mass and the semi-amplitude of the centre of mass. This is a dimensionless quantity in the range [-1,+1].

MAX_LINE_INTEGRAL : Maximum line integral (float)

LIA calculate line integrals through transit cross-correlation functions (CCFs), according to binary orbital models. The binary orbital model parameters that yield the maximum line integral are assumed to best describe the system. LIA optimises the systemic velocity and semi-amplitude using Covariance Matrix Adaptation Evolution Strategy (CMA-ES), while all the other parameters of the elliptical orbital model are fixed with the astrometric solution.

MAX_LINE_INTEGRAL_ERROR : Standard error of the maximum line integral (float)

Error on maximum line integral.

ECCENTRICITY : Eccentricity of the orbit (double)

Eccentricity of the orbit.

ECCENTRICITY_ERROR : Standard error of eccentricity (float)

Standard error of eccentricity of elliptical orbit.

ARG_PERIASTRON : Argument of periastron (double, Angle[deg])

The argument of periastron is the angular position of the periastron as measured in the plane of the orbit in the sense of the object motion.

When the spectroscopic orbit is established, the zero point of the angle is the ascending node (node on the line of nodes where the objects are moving away from the observer). In the absence of spectroscopic constraints, the zero point is the node whose position angle is between 0 and 180 degrees, the position angle being measured in the trigonometric direction. The argument of periastron is considered as fixed (no precession of the apsides).

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In the case of circular orbits is undefined and not filled.

ARG_PERIASTRON_ERROR : Standard error of argument of periastron (float, Angle[deg])

Standard error of argument of periastron.

INCLINATION : Orbital inclination (double, Angle[deg])

orbital inclination derived from the astrometric binary parameters, A, B, F, G Thiele Innes elements.

INCLINATION_ERROR : Standard error of orbital inclination (float, Angle[deg])

Standard error of orbital inclination.

Draft

17.7 XP_CONTINUOUS_MEAN_SPECTRUM [DataLink]

Table hosting the mean BP and RP spectra based on the continuous representation in basis functions.

For detailed information concerning the representation of calibrated mean spectra measured by the Gaia XP instrument see Section **??**.

Note this table is not available through the main archive TAP interface. Data are delivered via the Massive Data service indexed by the VO DataLink protocol and described in Chapter **??**. For example this can be actioned in the archive user interface by querying the main source catalogue `gaia_source` joined with the associated flags table `all_source_flags` on `source_id` and selecting `all_source_flags.has_xp = 't'`. The DataLink retrieval type to specify for this product in programmatic archive access via the Astroquery Gaia Python package is `MEAN_SPECTRUM_XP`.

DataLink products are served with a choice of two different data structures:

- `DATAMODEL_STANDARD`: the product is serialised following a standard data model (for example an IVOA-compliant data model in applicable cases);
- `DATAMODEL_GAIA`: the product is serialised following the Gaia DPAC(-internal) data model.

The `DATAMODEL_GAIA` format is described here whereas the `DATAMODEL_STANDARD` format is described in `https://www.cosmos.esa.int/web/gaia-users/archive/datalink-products#datalink_serialisation`

Columns description:

SOURCE_ID : Unique source identifier (unique within a particular Data Release) (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published

data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`.

BP_BASIS_FUNCTION_ID : Identifier defining the set of basis functions for the BP spectrum representation (short)

Identifier defining the set of basis functions used for the continuous representation of this spectrum.

BP_DEGREES_OF_FREEDOM : Degrees of freedom for the BP spectrum representation (short)

Number of degrees of freedom in the LSQ system.

BP_N_PARAMETERS : Number of parameters for the BP spectrum representation (byte)

Number of parameters in the LSQ system.

BP_N_MEASUREMENTS : Number of measurements used for the BP spectrum generation (short)

Number of measurements contributing to solution.

BP_N_REJECTED_MEASUREMENTS : Number of rejected measurements in the BP spectrum generation (short)

Number of measurements that have been rejected.

BP_STANDARD_DEVIATION : Standard deviation for the BP spectrum representation (float)

Standard deviation of the solution.

BP_CHI_SQUARED : Chi squared for the BP spectrum representation (float)

χ^2 of the solution.

BP_COEFFICIENTS : Basis function coefficients for the BP spectrum representation (double[bp_n_parameters

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array)

Array of basis function coefficients for the representation of the BP spectrum.

BP_COEFFICIENT_ERRORS : Basis function coefficient errors for the BP spectrum representation (float[bp_n_parameters] array)

Errors on the basis function coefficients for BP.

BP_COEFFICIENT_CORRELATIONS : Correlation matrix for BP coefficients (float[bp_n_parameters*(bp_n_ array)

Upper-triangular (non-zero, non-unity) part of the correlation matrix $\mathbf{M}$ of the coefficients C for the representation of the BP spectrum as basis functions. The matrix elements are stored as a linear array of constant size $S = n(n-1)/2$ corresponding to the full normal matrix of dimension $n \times n$. The ordering of the elements in the linear array follows a column-major storage scheme, i.e.:

$$\mathbf{M} = \begin{bmatrix} 1 & C[0] & C[1] & C[3] & C[6] & \cdots & C[S-(n-1)] \\ & 1 & C[2] & C[4] & C[7] & \cdots & C[S-(n-2)] \\ & & 1 & C[5] & C[8] & \cdots & C[S-(n-3)] \\ & & & 1 & C[9] & \cdots & C[S-(n-4)] \\ & & & & \ddots & \ddots & \vdots \\ & & & & & 1 & C[S-1] \\ & & & & & & 1 \end{bmatrix}$$

BP_N_RELEVANT_BASES : Number of bases that are relevant for the representation of this mean BP spectrum (short)

Although all bases defined in the set of bases associated with this mean BP spectrum have been used for the source update process, only the coefficients of the first bp_n_relevant_bases bases are significant.

The criterion adopted to estimate the number of relevant basis functions is based on the standard deviation of the coefficients, normalised to their corresponding square root of variance. All coefficients are divided by the square root of variances, and the standard deviation of the last n values is computed, where n is increasing from 2 onwards. The resulting standard deviation for the last n coefficients is then compared with the (idealised) theoretical expectation of 1, plus a configurable threshold times the standard error on the standard deviation. The largest index for which the standard deviation exceeds this limit is taken as the highest coefficient in the truncated

representation and the number of relevant bases.

BP_RELATIVE_SHRINKING : Measure of the relative shrinking of the coefficient vector when truncation is applied for the mean BP spectrum (float)

This parameter is defined as the ratio between the lengths of the truncated and full BP spectrum defined by the array of coefficients to be applied to the basis functions.

RP_BASIS_FUNCTION_ID : Identifier defining the set of basis functions for the RP spectrum representation (short)

Identifier defining the set of basis functions used for the continuous representation of this spectrum.

RP_DEGREES_OF_FREEDOM : Degrees of freedom for the RP spectrum representation (short)

Number of degrees of freedom in the LSQ system.

RP_N_PARAMETERS : Number of parameters for the RP spectrum representation (byte)

Number of parameters in the LSQ system.

RP_N_MEASUREMENTS : Number of measurements used for the RP spectrum generation (short)

Number of measurements contributing to solution.

RP_N_REJECTED_MEASUREMENTS : Number of rejected measurements in the RP spectrum generation (short)

Number of measurements that have been rejected.

RP_STANDARD_DEVIATION : Standard deviation for the RP spectrum representation (float)

Standard deviation of the solution.

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RP_CHI_SQUARED : Chi squared for the RP spectrum representation (float)

χ^2 of the solution.

RP_COEFFICIENTS : Basis function coefficients for the RP spectrum representation (double[rp_n_parameters array)

Array of basis function coefficients for the representation of the RP spectrum.

RP_COEFFICIENT_ERRORS : Basis function coefficient errors for the RP spectrum representation (float[rp_n_parameters] array)

Errors on the basis function coefficients for RP.

RP_COEFFICIENT_CORRELATIONS : Correlation matrix for RP coefficients (float[rp_n_parameters*(rp_n_ array)

Upper-triangular (non-zero, non-unity) part of the correlation matrix $\mathbf{M}$ of the coefficients C for the representation of the RP spectrum as basis functions. The matrix elements are stored as a linear array of constant size $S = n(n-1)/2$ corresponding to the full normal matrix of dimension $n \times n$. The ordering of the elements in the linear array follows a column-major storage scheme, i.e.:

$$\mathbf{M} = \begin{bmatrix} 1 & C[0] & C[1] & C[3] & C[6] & \cdots & C[S-(n-1)] \\ & 1 & C[2] & C[4] & C[7] & \cdots & C[S-(n-2)] \\ & & 1 & C[5] & C[8] & \cdots & C[S-(n-3)] \\ & & & 1 & C[9] & \cdots & C[S-(n-4)] \\ & & & & \ddots & \ddots & \vdots \\ & & & & & 1 & C[S-1] \\ & & & & & & 1 \end{bmatrix}$$

RP_N_RELEVANT_BASES : Number of bases that are relevant for the representation of this mean RP spectrum (short)

Although all bases defined in the set of bases associated with this mean RP spectrum have been used for the source update process, only the coefficients of the first rp_n_relevant_bases bases are significant.

The criterion adopted to estimate the number of relevant basis functions is based on the stan-

dard deviation of the coefficients, normalised to their corresponding square root of variance. All coefficients are divided by the square root of variances, and the standard deviation of the last n values is computed, where n is increasing from 2 onwards. The resulting standard deviation for the last n coefficients is then compared with the (idealised) theoretical expectation of 1, plus a configurable threshold times the standard error on the standard deviation. The largest index for which the standard deviation exceeds this limit is taken as the highest coefficient in the truncated representation and the number of relevant bases.

RP_RELATIVE_SHRINKING : Measure of the relative shrinking of the coefficient vector when truncation is applied for the mean RP spectrum (float)

This parameter is defined as the ratio between the lengths of the truncated and full RP spectrum defined by the array of coefficients to be applied to the basis functions.

SPEC_PROC_FLAG : Source-level processing information (byte)

Source-level processing information. These flags apply to both photometers.

Each byte will bit-wise encode the following values:

- Bit 0: This source was used as calibrator for the spectroscopic data.
- Bit 1: This source was a primary source.
- Bit 2: Filtering level applied to the epoch measurements to select those contributing to the mean BP spectrum (outlier rejection is applied in addition to this in the source update stage). The possible levels of processing are: primary (Bit 2 non set), secondary (Bit 2 is set). Further details about the filtering criteria and rules will be provided in the documentation.
- Bit 4: Filtering level applied to the epoch measurements to select those contributing to the mean RP spectrum (outlier rejection is applied in addition to this in the source update stage). The possible levels of processing are: primary (Bit non set), secondary (Bit 4 is set). Further details about the filtering criteria and rules will be provided in the documentation.

If a bit is not set, then the corresponding description does not apply, e.g. if bit 0 is not set, then this source was not a calibrator. Other bits are available for future extensions.

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17.8 XP_EPOCH_CROWDING [DataLink]

Transit level crowding information.

This provides information on both treated and untreated crowding for both BP and RP CCDs for this FoV transit. A transit appears in this table only if indicated in the corresponding bit (13) in the processing flags `bp_ccd_proc_flags` and `rp_ccd_proc_flags` stored in `xp_epoch_spectrum`.

Note this table is not available through the main archive TAP interface. Data are delivered via the Massive Data service indexed by the VO DataLink protocol and described in Chapter **??**. The DataLink retrieval type to specify for this product in programmatic archive access via the Astroquery Gaia Python package is `EPOCH_SPECTRUM_XP_CROWDING`.

DataLink products are served with a choice of two different data structures:

- `DATAMODEL_STANDARD`: the product is serialised following a standard data model (for example an IVOA-compliant data model in applicable cases);
- `DATAMODEL_GAIA`: the product is serialised following the Gaia DPAC(-internal) data model.

The `DATAMODEL_GAIA` format is described here whereas the `DATAMODEL_STANDARD` format is described in `https://www.cosmos.esa.int/web/gaia-users/archive/datalink-products#datalink_serialisation`

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution identifier, please visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

TRANSIT_ID : Transit Identifier (long)

The `transit_id` is a unique identifier assigned to each detected (and confirmed) source as it transits the Gaia focal plane. Each time a given source is detected as Gaia scans and re-scans the sky a new `transit_id` will be created to badge that apparition. Hence the along-scan time and the across-scan position along with the telescope in which the source was detected are used to form a unique integer with which to label the transit. The detection of transits is done using the SkyMapper CCD data, and the confirmation is done using the AF1 CCD data. Both happens on board.

The several features of a transit that are encoded in `transit_id` can be easily retrieved using bit masks (&) and shifts (>>) as follows:

- On-Board Mission Time line (OBMT) [ns]
  = 204800 * ((`transit_id` >> 17) & (0x000003FFFFFFFFFF))
- Field-of-view = 1 + (`transit_id` >> 15) & 0x03 [1 for 'preceding' and 2 for 'following' fields-of-view respectively]
- CCD row = (`transit_id` >> 12) & 0x07 [dimensionless, in the range 1 to 7]
- Across-scan 'reference acquisition pixel' in strip AF1 = (`transit_id`) & 0x0FFF [pixels] (this is the across-scan centre of the AF1 window and is odd if immediately below the mid-point of the window and even if immediately above)

where the bit mask prefix '0x' denotes hexadecimal.

For further details see Portell et al. (2020). For convenience a decoder for `transit_id` is available on-line at

`https://gaia.esac.esa.int/decoder/transitidDecoder.jsp`

BP_UNTREATED_CONTAMINATING_SOURCES : List of source IDs of contaminating sources for untreated crowding in the BP observation (long[] array)

List of `source_ids` of objects that are expected to contaminate this BP observation for untreated crowding.

BP_UNTREATED_CONTAMINATING_FLUX : Estimate of total amount of contaminating flux for untreated crowding in the BP observations (float, Flux[e^- s^{-1}])

Estimate of the flux contributed within the window by all sources listed in the bp_untreated_contaminating field for untreated crowding in this BP observation.

BP_UNTREATED_BLENDING_SOURCES : List of source IDs of blending sources for untreated crowding in the BP observation (long[] array)

List of source_ids of objects that are classified as blending for this BP observation for untreated crowding.

BP_UNTREATED_BLENDING_FLUX : Estimate of total amount of blending flux for untreated crowding in the BP observation (float, Flux[e^- s^{-1}])

Estimate of the total flux from all blending sources listed in bp_untreated_blending_sources within this window for untreated crowding in this BP observation.

RP_UNTREATED_CONTAMINATING_SOURCES : List of source IDs of contaminating sources for untreated crowding in the RP observation (long[] array)

List of source_ids of objects that are expected to contaminate this RP observation for untreated crowding.

RP_UNTREATED_CONTAMINATING_FLUX : Estimate of total amount of contaminating flux for untreated crowding in the RP observations (float, Flux[e^- s^{-1}])

Estimate of the flux contributed within the window by all sources listed in the rp_untreated_contaminating field for untreated crowding in this RP observation.

RP_UNTREATED_BLENDING_SOURCES : List of source IDs of blending sources for untreated crowding in the RP observation (long[] array)

List of source_ids of objects that are classified as blending for this RP observation for untreated crowding.

RP_UNTREATED_BLENDING_FLUX : Estimate of total amount of blending flux for untreated crowding in the RP observation (float, Flux[e^- s^{-1}])

Estimate of the total flux from all blending sources listed in rp_untreated_blending_sources

within this window for untreated crowding in this RP observation.

BP_TREATED_CONTAMINATING_SOURCES : List of source IDs of contaminating sources for treated crowding in the BP observation (long[] array)

List of `source_ids` of objects that are expected to contaminate this BP observation for treated crowding.

BP_TREATED_CONTAMINATING_FLUX : Estimate of total amount of contaminating flux for treated crowding in the BP observations (float, Flux[e^- s^{-1}])

Estimate of the flux contributed within the window by all sources listed in the `bp_treated_contaminating_s` field for treated crowding in this BP observation.

BP_TREATED_BLENDING_SOURCES : List of source IDs of blending sources for treated crowding in the BP observation (long[] array)

List of `source_ids` of objects that are classified as blending for this BP observation for treated crowding.

BP_TREATED_BLENDING_FLUX : Estimate of total amount of blending flux for treated crowding in the BP observation (float, Flux[e^- s^{-1}])

Estimate of the total flux from all blending sources listed in `bp_treated_blending_sources` within this window for treated crowding in this BP observation.

RP_TREATED_CONTAMINATING_SOURCES : List of source IDs of contaminating sources for treated crowding in the RP observation (long[] array)

List of `source_ids` of objects that are expected to contaminate this RP observation for treated crowding.

RP_TREATED_CONTAMINATING_FLUX : Estimate of total amount of contaminating flux for treated crowding in the RP observations (float, Flux[e^- s^{-1}])

Estimate of the flux contributed within the window by all sources listed in the `rp_treated_contaminating_s` field for treated crowding in this RP observation.

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RP_TREATED_BLENDING_SOURCES : List of source IDs of blending sources for treated crowding in the RP observation (long[] array)

List of `source_ids` of objects that are classified as blending for this RP observation for treated crowding.

RP_TREATED_BLENDING_FLUX : Estimate of total amount of blending flux for treated crowding in the RP observation (float, Flux[e^- s^{-1}])

Estimate of the total flux from all blending sources listed in `rp_treated_blending_sources` within this window for treated crowding in this RP observation.

BP_PROC_FLAGS : Crowding treatment flag for the BP observation (byte)

Processing flag for the crowding treatment for this BP observation. It will bit-wise encode the following values:

- Bit 0: Incomplete scene reconstruction, e.g. due to proximity to an attitude gap. In this case the crowding information may be incomplete.

Remaining bits are available for future extensions.

RP_PROC_FLAGS : Crowding treatment flag for the RP observation (byte)

Processing flag for the crowding treatment for this RP observation. It will bit-wise encode the following values:

- Bit 0: Incomplete scene reconstruction, e.g. due to proximity to an attitude gap. In this case the crowding information may be incomplete.

Remaining bits are available for future extensions.

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17.9 XP_EPOCH_SPECTRUM [DataLink]

This table contains the epoch BP and RP spectra collected at transit level for non-SSO sources.

For each relevant `transit_id` the sampled flux of the BP and RP spectra is provided on a pseudo–wavelength grid. Information relative to the processing through the respective photometry and variability pipelines is also given.

Note this table is not available through the main archive TAP interface. Data are delivered via the Massive Data service indexed by the VO DataLink protocol and described in Chapter **??**. For example this can be actioned in the archive user interface by querying the main source catalogue `gaia_source` joined with the associated flags table `all_source_flags` on `source_id` and selecting `all_source_flags.has_epoch_xp = 't'`. The DataLink retrieval type to specify for this product in programmatic archive access via the Astroquery Gaia Python package is `EPOCH_SPECTRUM_XP`.

DataLink products are served with a choice of two different data structures:

- `DATAMODEL_STANDARD`: the product is serialised following a standard data model (for example an IVOA-compliant data model in applicable cases);
- `DATAMODEL_GAIA`: the product is serialised following the Gaia DPAC(-internal) data model.

The `DATAMODEL_GAIA` format is described here whereas the `DATAMODEL_STANDARD` format is described in `https://www.cosmos.esa.int/web/gaia-users/archive/datalink-products#datalink_serialisation`

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that where used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit https://gaia.esac.esa.int/decoder/solnDecoder.jsp

SOURCE_ID : Unique source identifier (unique within a particular Data Release) (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

TRANSIT_ID : Unique transit Identifier (long)

The `transit_id` is a unique identifier assigned to each detected (and confirmed) source as it transits the Gaia focal plane. Each time a given source is detected as Gaia scans and re-scans the sky a new `transit_id` will be created to badge that apparition. Hence the along–scan time and the across–scan position along with the telescope in which the source was detected are used to form a unique integer with which to label the transit.

The several features of a detection that are encoded in `transit_id` can be easily retrieved using bit masks (&) and shifts (>>) as follows:

- On-Board Mission Time line [ns]
  = 204800 * ((`transit_id` >> 17) & (0x000003FFFFFFFFFF))
- Field-of-view = 1 + (`transit_id` >> 15) & 0x03 [1 for 'preceding' and 2 for 'following' fields-of-view respectively]
- CCD row = (`transit_id` >> 12) & 0x07 [dimensionless, in the range 1 to 7]
- Across-scan 'reference acquisition pixel' in strip AF1 = (`transit_id`) & 0x0FFF [pixels] (this is the across-scan centre of the AF1 window and is odd if immediately below the mid-point of the window and even if immediately above)

where the bit mask prefix '0x' denotes hexadecimal.

For further details see Portell et al. (2020). For convenience a decoder for `transit_id` is available on-line at

`https://gaia.esac.esa.int/decoder/transitidDecoder.jsp`

BP_OBS_TIME : BP CCD transit observation time (double, Time[Barycentric JD in TCB – 2 455 197.5 (day)])

Observation time of the BP CCD transit in units of Barycentric JD (in TCB) in days -2 455 197.5, computed as follows. First the observation time is converted from On-board Mission Time (OBMT) into Julian date in TCB (Temps Coordonnée Barycentrique). Next a correction is applied for the light-travel time to the Solar system barycentre, resulting in Barycentric Julian Date (BJD). Finally, an offset of 2 455 197.5 days is applied (corresponding to a reference time T_0 at

2010-01-01T00:00:00) to have a conveniently small numerical value.

BP_PSEUDO_WAVELENGTH : The array of sample positions of the BP spectrum in units of pseudo-wavelengths (float[60] array)

The BP pseudo–wavelengths of the spectrum for a given transit.

BP_FLUX : BP spectrum flux (float[60] array, Flux[e^- s^{-1}])

The BP flux of the spectrum in a given transit.

BP_FLUX_ERROR : Error on the flux for each sample of the BP spectrum (float[60] array, Flux[e^- s^{-1}])

The error on the BP Flux. If the BP flux has been rejected or is unavailable, this error will be set to null.

BP_GATE : Gate information for the selection of the appropriate calibration for the BP spectrum (byte)

The gate information contained in this field is aimed at the selection of the appropriate calibration if any is available. Calibrations will only be computed for the nominal cases of Gate 05, Gate 07, Gate 09, Gate 11 and None. Spectra obtained with other gates for which no calibration is available or with a multiple gate configuration that could not be treated as a nominal case by dropping part of the spectrum will fall into the same category.

The information will be encoded as follows:

- 0: Gate 05
- 1: Gate 07
- 2: Gate 09
- 3: Gate 11
- 4: Gate None (no gate)
- 5: Gate configuration with no corresponding calibration

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This field will only filled in if there is a BP spectrum available for this FoV transit.

BP_AC_WINDOW_CENTRE : AC window centre value for the evaluation of the calibration for the BP spectrum (float, Length & Distance[pixel])

The AC window centre value contained in this field is aimed at the evaluation of the calibration.

This field will only filled in if there is a BP spectrum available for this FoV transit.

BP_AC_CE : AC centring error value for the evaluation of the calibration for the BP spectrum (float, Length & Distance[pixel])

The AC centring error value contained in this field is aimed at the evaluation of the calibration.

This field will only filled in if there is a BP spectrum available for this FoV transit.

BP_AC_RATE : AC rate value for the evaluation of the calibration for the BP spectrum (float, Velocity[pixel/s])

The AC rate value contained in this field is aimed at the evaluation of the calibration.

This field will only filled in if there is a BP spectrum available for this FoV transit.

BP_MASKED_SAMPLES : List of BP sample indices masked or flagged as outliers in PhotPipe processing (byte[] array)

List of sample ids masked or flagged as outliers in PhotPipe processing. The sample ids are the indexes in the 60 samples array containing the BP spectrum. For instance if the list contains [0,1,2,3,58,59], the first 4 samples and the last 2 are masked or flagged. If the list is empty, there are no masked or flagged samples.

RP_OBS_TIME : RP CCD transit observation time (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Observation time of the RP CCD transit in units of Barycentric JD (in TCB) in days -2 455 197.5, computed as follows. First the observation time is converted from On-board Mission Time (OBMT) into Julian date in TCB (Temps Coordonnée Barycentrique). Next a correction is applied for the light-travel time to the Solar system barycentre, resulting in Barycentric Julian Date

(BJD). Finally, an offset of 2 455 197.5 days is applied (corresponding to a reference time T_0 at 2010-01-01T00:00:00) to have a conveniently small numerical value.

RP_PSEUDO_WAVELENGTH : The array of sample positions of the RP spectrum in units of pseudo-wavelengths (float[60] array)

The RP pseudo–wavelengths of the spectrum for a given transit.

RP_FLUX : RP spectrum flux (float[60] array, Flux[e^- s^{-1}])

The RP flux of the spectrum in a given transit.

RP_FLUX_ERROR : Error on the flux for each sample of the RP spectrum (float[60] array, Flux[e^- s^{-1}])

The error on the RP Flux. If the RP flux has been rejected or is unavailable, this error will be set to null.

RP_GATE : Gate information for the selection of the appropriate calibration for the RP spectrum (byte)

The gate information contained in this field is aimed at the selection of the appropriate calibration if any is available. Calibrations will only be computed for the nominal cases of Gate 05, Gate 07, Gate 09, Gate 11 and None. Spectra obtained with other gates for which no calibration is available or with a multiple gate configuration that could not be treated as a nominal case by dropping part of the spectrum will fall into the same category.

The information will be encoded as follows:

- 0: Gate 05
- 1: Gate 07
- 2: Gate 09
- 3: Gate 11
- 4: Gate None (no gate)
- 5: Gate configuration with no corresponding calibration

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This field will only filled in if there is a RP spectrum available for this FoV transit.

RP_AC_WINDOW_CENTRE : AC window centre value for the evaluation of the calibration for the RP spectrum (float, Length & Distance[pixel])

The AC window centre value contained in this field is aimed at the evaluation of the calibration.

This field will only filled in if there is a RP spectrum available for this FoV transit.

RP_AC_CE : AC centring error value for the evaluation of the calibration for the RP spectrum (float, Length & Distance[pixel])

The AC centring error value contained in this field is aimed at the evaluation of the calibration.

This field will only filled in if there is a RP spectrum available for this FoV transit.

RP_AC_RATE : AC rate value for the evaluation of the calibration for the RP spectrum (float, Velocity[pixel/s])

The AC rate value contained in this field is aimed at the evaluation of the calibration.

This field will only filled in if there is a RP spectrum available for this FoV transit.

RP_MASKED_SAMPLES : List of RP sample indices masked or flagged as outliers in PhotPipe processing (byte[] array)

List of sample ids masked or flagged as outliers in PhotPipe processing. The sample ids are the indexes in the 60 samples array containing the RP spectrum. For instance if the list contains [0,1,2,3,58,59], the first 4 samples and the last 2 are masked or flagged. If the list is empty, there are no masked or flagged samples.

WINDOW_CLASS : Window class (byte)

A window classification flag (not the transit window class in the Gaia telemetry sense) assigned to this FoV transit. This is provided here as it is required for the selection of the appropriate calibration. Note that, for 2D windows, PhotPipe does an AC binning of the samples, so this table always provides 1D arrays in the `bp_flux` and `rp_flux` fields. The encoding is 0: 1D window, 1: 2D window; this is opposite to the internal DPAC and spacecraft labelling conventions.

BP_CCD_INFO_FLAGS : CCD-level information on the input BP observations (byte)

Flag providing information at the CCD level for the BP observation.

- Bit 0: affected by numerical saturation
- Bit 1: affected by hot column

Other bits are available for future extensions.

BP_CCD_PROC_FLAGS : CCD-level information on PhotPipe processing for the BP observation (short)

This field reports information generated during the PhotPipe processing for the BP observations. It bit-wise encodes the following meanings when set to '1':

- Bit 0: no data available; obviously, no calibrated epoch spectrum will be available in this case;
- Bit 1: the transit is affected by truncation or multiple gate configuration in a way that is considered unrecoverable;
- Bit 2: predicted position computation failed or predicted position outside window;
- Bit 3: the AC centring error was clamped when the LS calibration was applied (refer to the end-of-cycle documentation for details on the clamping ranges used);
- Bit 4: no calibration available (no calibrated CCD photometry will be available in this case);
- Bit 5: calibration rejected and replaced with neighbouring calibration;
- Bit 6: calibration application failure;
- Bit 7: excluded from the generation of the mean source spectrum, due to pre-filtering based on available flags (this is different from outlier rejection that is reported in Bit 8);
- Bit 8: rejected in the source update process;
- Bit 9: observation affected by satellite event;

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- Bit 10: charge release correction applied;
- Bit 11: stray-light mitigation applied;
- Bit 12: local (KNN) background mitigation applied;
- Bit 13: crowding information available in the table `xp_epoch_crowding`;
- Bit 14: the transit has been treated for truncation or multiple gate configuration.

Remaining bits are available for future extensions.

RP_CCD_INFO_FLAGS : CCD-level information on the input RP observations (byte)

Flag providing information at the CCD level for the RP observation.

- Bit 0: affected by numerical saturation
- Bit 1: affected by hot column

Other bits are available for future extensions.

RP_CCD_PROC_FLAGS : CCD-level information on PhotPipe processing for the RP observation (short)

This field reports information generated during the PhotPipe processing for the RP observations. It bit-wise encodes the following meanings when set to '1':

- Bit 0: no data available; obviously, no calibrated epoch spectrum will be available in this case;
- Bit 1: the transit is affected by truncation or multiple gate configuration in a way that is considered unrecoverable;
- Bit 2: predicted position computation failed or predicted position outside window;
- Bit 3: the AC centring error was clamped when the LS calibration was applied (refer to the end-of-cycle documentation for details on the clamping ranges used);
- Bit 4: no calibration available (no calibrated CCD photometry will be available in this case);

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- Bit 5: calibration rejected and replaced with neighbouring calibration;
- Bit 6: calibration application failure;
- Bit 7: excluded from the generation of the mean source spectrum, due to pre-filtering based on available flags (this is different from outlier rejection that is reported in Bit 8);
- Bit 8: rejected in the source update process;
- Bit 9: observation affected by satellite event;
- Bit 10: charge release correction applied;
- Bit 11: stray-light mitigation applied;
- Bit 12: local (KNN) background mitigation applied;
- Bit 13: crowding information available in the table `xp_epoch_crowding`;
- Bit 14: the transit has been treated for truncation or multiple gate configuration.

Remaining bits are available for future extensions.

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17.10 XP_SUMMARY [TAP]

This tables contains auxiliary information about the mean BP/RP spectrum of a given source. It should be used in order to refine queries on spectra and build dedicated samples before interrogating the DataLink protocol to download the results of those queries.

Columns description:

SOURCE_ID : Unique source identifier (unique within a particular Data Release) (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`.

BP_N_RELEVANT_BASES : Number of bases that are relevant for the representation of this mean BP spectrum (short)

Although all bases defined in the set of bases associated with this mean BP spectrum have been used for the source update process, only the coefficients of the first `bp_n_relevant_bases` bases are significant.

The criterion adopted to estimate the number of relevant basis functions is based on the standard deviation of the coefficients, normalised to their corresponding square root of variance. All coefficients are divided by the square root of variances, and the standard deviation of the last n values is computed, where n is increasing from 2 onwards. The resulting standard deviation for the last n coefficients is then compared with the (idealised) theoretical expectation of 1, plus a configurable threshold times the standard error on the standard deviation. The largest index for which the standard deviation exceeds this limit is taken as the highest coefficient in the truncated representation and the number of relevant bases.

BP_RELATIVE_SHRINKING : Measure of the relative shrinking of the coefficient vector when

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truncation is applied for the mean BP spectrum (float)

This parameter is defined as the ratio between the lengths of the truncated and full BP spectrum defined by the array of coefficients to be applied to the basis functions.

BP_N_MEASUREMENTS : Number of measurements used for the BP spectrum generation (short)

Number of measurements contributing to solution.

BP_N_REJECTED_MEASUREMENTS : Number of rejected measurements in the BP spectrum generation (short)

Number of measurements that have been rejected.

BP_STANDARD_DEVIATION : Standard deviation for the BP spectrum representation (float, Flux[e^- pix^{-1} s^{-1}])

Standard deviation of the solution.

BP_CHI_SQUARED : Chi squared for the BP spectrum representation (float)

χ^2 of the solution.

BP_N_TRANSITS : Number of transits contributing to the mean in BP (short)

Number of BP epoch spectra that contributed to the mean spectrum (some samples from these spectra may still have been rejected).

BP_N_CONTAMINATED_TRANSITS : Number of contaminated transits in BP (short)

Number of BP epoch spectra (considering only spectra that contributed to the mean spectrum) matched to this source that were marked as contaminated.

BP_N_BLENDED_TRANSITS : Number of blended transits in BP (short)

Number of BP epoch spectra (considering only spectra that contributed to the mean spectrum) matched to this source that were marked as blended.

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BP_N_DEBLENDED_TRANSITS : Number of deblended transits in BP (short)

Number of BP epoch spectra (considering only spectra that contributed to the mean spectrum) matched to this source that were marked as deblended.

BP_N_RECONSTRUCTED_TRANSITS : Number of reconstructed transits in BP (short)

Number of truncated BP epoch transits that were reconstructed.

RP_N_RELEVANT_BASES : Number of bases that are relevant for the representation of this mean RP spectrum (short)

Although all bases defined in the set of bases associated with this mean RP spectrum have been used for the source update process, only the coefficients of the first `rp_n_relevant_bases` bases are significant.

The criterion adopted to estimate the number of relevant basis functions is based on the standard deviation of the coefficients, normalised to their corresponding square root of variance. All coefficients are divided by the square root of variances, and the standard deviation of the last n values is computed, where n is increasing from 2 onwards. The resulting standard deviation for the last n coefficients is then compared with the (idealised) theoretical expectation of 1, plus a configurable threshold times the standard error on the standard deviation. The largest index for which the standard deviation exceeds this limit is taken as the highest coefficient in the truncated representation and the number of relevant bases.

RP_RELATIVE_SHRINKING : Measure of the relative shrinking of the coefficient vector when truncation is applied for the mean RP spectrum (float)

This parameter is defined as the ratio between the lengths of the truncated and full RP spectrum defined by the array of coefficients to be applied to the basis functions.

RP_N_MEASUREMENTS : Number of measurements used for the RP spectrum generation (short)

Number of measurements contributing to solution.

RP_N_REJECTED_MEASUREMENTS : Number of rejected measurements in the RP spectrum

generation (short)

Number of measurements that have been rejected.

RP_STANDARD_DEVIATION : Standard deviation for the RP spectrum representation (float, Flux[e^- pix^{-1} s^{-1}])

Standard deviation of the solution.

RP_CHI_SQUARED : Chi squared for the RP spectrum representation (float)

χ^2 of the solution.

RP_N_TRANSITS : Number of transits contributing to the mean in RP (short)

Number of RP epoch spectra that contributed to the mean spectrum (some samples from these spectra may still have been rejected).

RP_N_CONTAMINATED_TRANSITS : Number of contaminated transits in RP (short)

Number of RP epoch spectra (considering only spectra that contributed to the mean spectrum) matched to this source that were marked as contaminated.

RP_N_BLENDED_TRANSITS : Number of blended transits in RP (short)

Number of RP epoch spectra (considering only spectra that contributed to the mean spectrum) matched to this source that were marked as blended.

RP_N_DEBLENDED_TRANSITS : Number of deblended transits in RP (short)

Number of RP epoch spectra (considering only spectra that contributed to the mean spectrum) matched to this source that were marked as deblended.

RP_N_RECONSTRUCTED_TRANSITS : Number of reconstructed transits in RP (short)

Number of truncated RP epoch transits that were reconstructed.

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SPEC_PROC_FLAG : Source-level processing information (byte)

Source-level processing information. These flags apply to both photometers.

Each byte will bit-wise encode the following values:

- Bit 0: This source was used as calibrator for the spectroscopic data.
- Bit 1: This source was a primary source.
- Bit 2: Filtering level applied to the epoch measurements to select those contributing to the mean BP spectrum (outlier rejection is applied in addition to this in the source update stage). The possible levels of processing are: primary (Bit 2 non set), secondary (Bit 2 is set). Further details about the filtering criteria and rules will be provided in the documentation.
- Bit 4: Filtering level applied to the epoch measurements to select those contributing to the mean RP spectrum (outlier rejection is applied in addition to this in the source update stage). The possible levels of processing are: primary (Bit non set), secondary (Bit 4 is set). Further details about the filtering criteria and rules will be provided in the documentation.

If a bit is not set, then the corresponding description does not apply, e.g. if bit 0 is not set, then this source was not a calibrator. Other bits are available for future extensions.

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18 Variability

As indicated in the below subsection titles by the labels [TAP] and [DataLink], these products are served from the ESA Gaia Archive using two different protocols, the Table Access Protocol (TAP) for tabular products and the DataLink protocol for other products such as spectra, light curves, MCMC samples, etc. For more details please see Section **??**.

18.1 VARI_AGN [TAP]

This table provides information on AGN properties.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

FRACTIONAL_VARIABILITY_G : Fractional variability in the G band (float)

The fractional variability is calculated as $\sqrt{\sigma^2(F) - \mathrm{RMS}^2(\epsilon_F)}/\mathrm{median}(F)$, where the standard deviation $\sigma(F)$ is estimated as 1.4826 times the median of the absolute deviations from the median of the field-of-view transit fluxes F in the G band, which strictly holds for normal distributions, $\mathrm{RMS}^2(\epsilon_F)$ is the mean square of flux uncertainties ϵ_F, and $\mathrm{median}(F)$ is the median flux.

STRUCTURE_FUNCTION_INDEX : Index of the first-order structure function in the G band (float)

Index of the first-order structure function (SF; Simonetti et al. 1985), i.e., slope in the logSF vs logTau space, where Tau is the time lag. The SF is expressed in magnitude squared and computed from field-of-view transit magnitudes in the G band. The index is linked to that of the Fourier

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power spectrum and indicates the type of noise process at work.

STRUCTURE_FUNCTION_INDEX_SCATTER : Standard deviation of the index of the structure function (float)

Standard deviation of the index of the structure function.

QSO_VARIABILITY : Quasar variability metric in the G band (float)

Quasar variability metric from field-of-view transit magnitudes in the *G* band in log format, $\log(\chi^2_{\rm QSO}/\nu)$, from Butler & Bloom (2011), after adaptation to Gaia data.

NON_QSO_VARIABILITY : Non-quasar variability metric in the G band (float)

Non-quasar variability metric from field-of-view transit magnitudes in the *G* band in log format, $\log(\chi^2_{\rm false}/\nu)$, from Butler & Bloom (2011), after adaptation to Gaia data.

BEYOND1STD : Percentage of points beyond one standard deviation from the magnitude weighted mean (float)

Percentage of points beyond one standard deviation from the magnitude weighted mean. For a detailed explanation see e.g., Richards et al. (2011).

RATIO_MID20 : Flux percentile ratio mid20 (float)

Ratio of flux percentiles (60th - 40th) over (95th - 5th) in the G band. For a detailed explanation see e.g., Richards et al. (2011).

RATIO_MID35 : Flux percentile ratio mid35 (float)

Ratio of flux percentiles (67.5th - 32.5th) over (95th - 5th) in the G band. For a detailed explanation see e.g., Richards et al. (2011).

RATIO_MID50 : Flux percentile ratio mid50 (float)

Ratio of flux percentiles (75th - 25th) over (95th - 5th) in the G band. For a detailed explanation

see e.g., Richards et al. (2011).

RATIO_MID65 : Flux percentile ratio mid65 (float)

Ratio of flux percentiles (82.5th - 17.5th) over (95th - 5th) in the G band. For a detailed explanation see e.g., Richards et al. (2011).

RATIO_MID80 : Flux percentile ratio mid80 (float)

Ratio of flux percentiles (90th - 10th) over (95th - 5th) in the G band. For a detailed explanation see e.g., Richards et al. (2011).

LINEAR_TREND : Slope of a linear fit to the light-curve in magnitude (float, Magnitude[mag])

Slope of a linear fit to the light-curve in magnitude in G band. For a detailed explanation see e.g., Richards et al. (2011).

MAX_SLOPE : Maximum absolute magnitude slope between two consecutive observations (float, Magnitude[mag])

Maximum absolute magnitude slope between two consecutive observations in G band. For a detailed explanation see e.g., Richards et al. (2011).

FRACTION_OBS_NEAR_MEDIAN_MAG : Fraction of photometric FoV measurements within the one tenth of the total range of magnitude above and below the median magnitude in the G band (float)

Fraction of photometric field-of-view measurements within the one tenth of the total range of magnitude above and below the median magnitude in the G band. For a detailed explanation see e.g., Richards et al. (2011).

PAIR_SLOPE_TREND : Ratio between all pairs of consecutive magnitude measurements that have positive slope minus those that have negative slope, and the total number of pairs (float)

Ratio between all pairs of consecutive magnitude measurements that have positive slope minus those that have negative slope, and the total number of pairs in the G band. For a detailed explanation see e.g. Richards et al. (2011).

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MAX_DEVIATION_OVER_MEDIAN_MAG : Maximum absolute deviation from the median magnitude over the median magnitude (float)

Maximum absolute deviation from the median magnitude divided by the median magnitude in the G band. For a detailed explanation see e.g. Richards et al. (2011).

STETSON_K : Stetson K index (float)

Stetson K index, a robust kurtosis measure in the G band. For a detailed explanation see e.g., Richards et al. (2011).

STETSON_L : Stetson L index (float)

Better formulation of the variability index. $L = (J \times K)/0.798$. For a detailed explanation see e.g., Stetson (1996); Kim et al. (2011).

MEAN_VARIANCE : Variability index (float)

Simple variability index, defined as the ratio between the standard deviation and the mean magnitude in the G band. If a light curve has strong variability, standard deviation/mean magnitude of the light curve is generally large. For a detailed explanation see e.g., Richards et al. (2011).

RANGE_CUMULATIVE_SUM : Range of cumulative sum (float)

Range of a cumulative sum of the light curve in G band (Ellaway 1978). It is close to zero for a symmetric distribution.

TRIMMED_RANGE_OVER_MEDIAN_FLUX : Ratio of the difference between the 95th and 5th percentiles over the median flux (float)

Ratio of the difference between the 95th and 5th percentiles over the median flux in the G band. For a detailed explanation see e.g., Richards et al. (2011).

ETA_E : Powerful index for quantifying variability (float)

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Powerful index for quantifying variability characteristics of a light curve, also in case of unequal sampling. Calculated in the G band. The variability index η_e is an extension of the von Neumann index η that accounts for irregular time sampling in time series data. It weighs consecutive magnitude differences based on the time interval between observations.

$$\eta_e = \frac{1}{\sigma^2} \cdot \frac{\sum_{i=1}^{N-1} w_i (x_{i+1} - x_i)^2}{\sum_{i=1}^{N-1} w_i}$$

where:

x_i is the observed magnitude at time t_i,

$w_i = \frac{1}{(t_{i+1} - t_i)^2}$,

and $\sigma^2 = \mathrm{Var}(\{x_i\})$.

ANDERSON_DARLING : Anderson-Darling statistical test (float)

The Anderson-Darling test (Stephens 1974) is used to test if a sample of data comes from a population with a specific distribution. It is a modification of the Kolmogorov-Smirnov (K-S) test and gives more weight to the tails than does the K-S test.

G_SKEW : Robust measure of skew in the G band (float)

Robust measure of skew in the G band to detect abrupt decreases in brightness (Richards et al. 2012).

HALF_RANGE_G : Half the difference between the median of the maximum 5(float, Magnitude[mag])

Half the difference between the median of the maximum 5% and the median of the minimum 5% magnitudes in the G band. For a detailed explanation see e.g., Richards et al. (2011).

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18.2 VARI_CEPHEID [TAP]

This table describes the Cepheid stars.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

PF : Period corresponding to the fundamental pulsation mode in the G-band time series (double, Time[day])

For single-mode pulsators classified as fundamental mode pulsators, this parameter is filled with the periodicity found in the time-series.

This value is obtained by modelling the *G* band time series using the Levenberg-Marquardt non–linear fitting algorithm (Clementini et al. 2016).

Information for `vari_rrlyrae`: For double-mode RR Lyrae this parameter is filled with the period corresponding to the longer periodicity.

Information for `vari_cepheid`: For double-mode DCEPs this parameter is filled with the period corresponding to the longer periodicity if the DCEP is classified as 'F/1O' or 'F/2O'. For triple-mode DCEPs this parameter is filled with the period corresponding to the longer periodicity if the DCEP is classified as 'F/1O/2O'.

PF_ERROR : Uncertainty of the pf period (float, Time[day])

This parameter is filled with the uncertainty of the `pf` parameter, computed via a bootstrap tech-

nique.

p1_o : Period corresponding to the first overtone pulsation mode in the G-band time series (double, Time[day])

For single-mode pulsators classified as first-overtone pulsators, this parameter is filled with the periodicity found in the time-series.

This value is obtained by modelling the *G* time series using the Levenberg-Marquardt non linear fitting algorithm (see Clementini et al. 2016).

Information for `vari_rrlyrae`: For double-mode RR Lyrae this parameter is filled with the period corresponding to the shortest periodicity.

Information for `vari_cepheid`: For double-mode DCEPs this parameter is filled with the period corresponding to the shortest periodicity if the DCEP is classified as 'F/1O'; otherwise it is filled with the longest one if the classification is '1O/2O' or '1O/3O'. For triple-mode DCEPs this parameter is filled with the period corresponding to the intermediate periodicity if the DCEP is classified as 'F/1O/2O'; it is filled with the longest periodicity if the classification is '1O/2O/3O'.

p1_o_error : Uncertainty of the first overtone period (float, Time[day])

This parameter is filled with the uncertainty of the `p1_o` parameter, computed via a bootstrap technique.

epoch_g : Epoch of the maximum of the light curve in the G band (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Epoch of maximum light for the Gaia *G* band light curve. It corresponds to the barycentric Julian day (BJD) of the maximum value of the light curve model which is closest to the BJD of the first observations -3 times the period of the source (first periodicity depending on the pulsation mode).

The mentioned BJD is offset by JD 2455197.5 (= J2010.0).

epoch_g_error : Uncertainty on the epoch of the maximum in the G band (float, Time[day])

Value of the uncertainty of the `epoch_g` parameter. It corresponds to three times the error on the period of the source (first periodicity depending on the pulsation mode).

EPOCH_BP : Epoch of the maximum of the light curve in the BP band (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Epoch of maximum light for the Gaia integrated G_{BP} band light curve. It corresponds to the barycentric Julian day (BJD) of the maximum value of the light curve model which is closest to the BJD of the first observations -3 times the period of the source (first periodicity depending on the pulsation mode).

The mentioned BJD is offset by JD 2455197.5 (= J2010.0).

EPOCH_BP_ERROR : Uncertainty on the epoch of the maximum in the BP band (float, Time[day])

Value of the uncertainty of the `epoch_bp` parameter. It corresponds to three times the error on the period of the source (first periodicity depending on the pulsation mode).

EPOCH_RP : Epoch of the maximum of the light curve in the RP band (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Epoch of maximum light for the Gaia integrated G_{RP} band light curve. It corresponds to the barycentric Julian day (BJD) of the maximum value of the light curve model which is closest to the BJD of the first observations -3 times the period of the source (first periodicity depending on the pulsation mode).

The mentioned BJD is offset by JD 2455197.5 (= J2010.0).

EPOCH_RP_ERROR : Uncertainty on the epoch of the maximum in the RP band (float, Time[day])

Value of the uncertainty of the `epoch_rp` parameter. It corresponds to three times the error on the period of the source (first periodicity depending on the pulsation mode).

EPOCH_GRVS : Epoch of the maximum of the light curve in the Grvs band (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Epoch of maximum light for the Gaia integrated G_{RVS} band light curve. It corresponds to the barycentric Julian day (BJD) of the maximum value of the light curve model which is closest to the BJD of the first observations -3 times the period of the source (first periodicity depending on the pulsation mode).

The mentioned BJD is offset by JD 2455197.5 (= J2010.0).

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EPOCH_GRVS_ERROR : Uncertainty on the epoch of the maximum in the Grvs band (float, Time[day])

Value of the uncertainty of the `epoch_grvs` parameter. It corresponds to three times the error on the period of the source (first periodicity depending on the pulsation mode).

EPOCH_RV : Epoch of the minimum of the radial velocity curve (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Epoch of minimum radial velocity for the Gaia radial velocity curve.

EPOCH_RV_ERROR : Uncertainty on the epoch of the maximum in the radial velocity curve (float, Time[day])

Value of the uncertainty of the `epoch_rv` parameter. It corresponds to three times the error on the period of the source (first periodicity depending on the pulsation mode).

INT_AVERAGE_G : Intensity-averaged magnitude in the G band (float, Magnitude[mag])

Value of the intensity-averaged magnitude in the G-band. The intensity-averaged magnitude is obtained by computing the average flux and then converting the average flux to magnitude.

INT_AVERAGE_G_ERROR : Uncertainty on the intensity-average magnitude in the G band (float, Magnitude[mag])

This parameter is filled with the uncertainty of the `int_average_g` parameter, computed via a bootstrap technique.

INT_AVERAGE_BP : Intensity-averaged magnitude in the BP band (float, Magnitude[mag])

Value of the intensity-averaged magnitude in the G_{BP}-band. The intensity-averaged magnitude is obtained by computing the average flux and then converting the average flux to magnitude.

INT_AVERAGE_BP_ERROR : Uncertainty on intensity-average magnitude in the BP band (float, Magnitude[mag])

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This parameter is filled with the uncertainty of the `int_average_bp` parameter, computed via a bootstrap technique.

INT_AVERAGE_RP : Intensity-averaged magnitude in the RP band (float, Magnitude[mag])

Value of the intensity-averaged magnitude in the $G_{\rm RP}$-band. The intensity-averaged magnitude is obtained by computing the average flux and then converting the average flux to magnitude.

INT_AVERAGE_RP_ERROR : Uncertainty on intensity-average magnitude in the RP band (float, Magnitude[mag])

This parameter is filled with the uncertainty of the `int_average_rp` parameter, computed via a bootstrap technique.

INT_AVERAGE_GRVS : Intensity-averaged magnitude in the Grvs band (float, Magnitude[mag])

Value of the intensity-averaged magnitude in the G_{RVS}-band. The intensity-averaged magnitude is obtained by computing the average flux and then converting the average flux to magnitude.

INT_AVERAGE_GRVS_ERROR : Uncertainty on intensity-average magnitude in the Grvs band (float, Magnitude[mag])

This parameter is filled with the uncertainty of the `int_average_grvs` parameter, computed via a bootstrap technique.

AVERAGE_RV : Mean radial velocity (float, Velocity[km s^{-1}])

Average value of the Fourier modelled radial velocity curve, provided by the A_0 parameter of the Fourier fit. If the Fourier model cannot be computed (in case of limited data or models showing unphysical oscillations), `average_rv` is the arithmetic average of the individual radial velocity measurements. In case of binary Cepheids, `average_rv` corresponds to the average velocity resulting from the binary model.

AVERAGE_RV_ERROR : Uncertainty on the mean radial velocity (float, Velocity[km s^{-1}])

Uncertainty of `average_rv` computed via a bootstrap technique if the radial velocity time series is modelled with a truncated Fourier series. In case of limited data or models showing unphysical

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oscillations, the error of `average_rv` is the standard deviation of the radial velocity measurements.

PEAK_TO_PEAK_G : Peak-to-peak amplitude of the G band light curve (float, Magnitude[mag])

This parameter is filled with the peak-to-peak amplitude value of the *G* band light curve. The peak-to-peak amplitude is calculated as the (maximum) - (minimum) of the modelled folded light curve in the *G* band. The light curve of the target star is modelled with a truncated Fourier series.

PEAK_TO_PEAK_G_ERROR : Uncertainty on the peak-to-peak amplitude in the G band (float, Magnitude[mag])

This parameter is filled with the uncertainty value of the `peak_to_peak_g` parameter, computed via a bootstrap technique.

PEAK_TO_PEAK_BP : Peak-to-peak amplitude of the BP band light curve (float, Magnitude[mag])

This parameter is filled with the peak-to-peak amplitude value of the $G_{\rm BP}$ light curve. The peak-to-peak amplitude is calculated as the (maximum) - (minimum) of the modelled folded light curve in the $G_{\rm BP}$ band. The light curve of the target star is modelled with a truncated Fourier series.

PEAK_TO_PEAK_BP_ERROR : Uncertainty on the peak-to-peak amplitude in the BP band (float, Magnitude[mag])

This parameter is filled with the uncertainty value of the `peak_to_peak_bp` parameter, computed via a bootstrap technique.

PEAK_TO_PEAK_RP : Peak-to-peak amplitude of the RP band light curve (float, Magnitude[mag])

This parameter is filled with the peak-to-peak amplitude value of the $G_{\rm RP}$ light curve. The peak-to-peak amplitude is calculated as the (maximum) - (minimum) of the modelled folded light curve in the $G_{\rm RP}$ band. The light curve of the target star is modelled with a truncated Fourier series.

PEAK_TO_PEAK_RP_ERROR : Uncertainty on the peak-to-peak amplitude in the RP band (float,

Magnitude[mag])

This parameter is filled with the uncertainty value of the `peak_to_peak_rp` parameter, computed via a bootstrap technique.

PEAK_TO_PEAK_GRVS : Peak-to-peak amplitude of the Grvs band light curve (float, Magnitude[mag])

This parameter is filled with the peak-to-peak amplitude value of the $G_{\rm RVS}$ light curve. The peak-to-peak amplitude is calculated as the (maximum) - (minimum) of the modelled folded light curve in the $G_{\rm RVS}$ band. The light curve of the target star is modelled with a truncated Fourier series.

PEAK_TO_PEAK_GRVS_ERROR : Uncertainty on the peak-to-peak amplitude in the Grvs band (float, Magnitude[mag])

This parameter is filled with the uncertainty value of the `peak_to_peak_grvs` parameter, computed via a bootstrap technique.

PEAK_TO_PEAK_RV : Peak-to-peak amplitude of the radial velocity curve (float, Velocity[km s^{-1}])

This parameter is filled with the peak-to-peak amplitude value of the radial velocity curve. The peak-to-peak amplitude is calculated as the (maximum) - (minimum) of the modelled folded *RV* curve. The *RV* curve of the target star is modelled with a truncated Fourier series.

PEAK_TO_PEAK_RV_ERROR : Uncertainty on the peak-to-peak amplitude in the radial velocity curve (float, Velocity[km s^{-1}])

This parameter is filled with the uncertainty value of the `peak_to_peak_rv` parameter, computed via a bootstrap technique.

METALLICITY : Metallicity of the star from the Fourier parameters of the light curve (float, Abundances[dex])

This parameter is filled with the [Fe/H] metallicity derived for the source from the Fourier parameters of the *G*-band light curve.

METALLICITY_ERROR : Uncertainty of the metallicity parameter (float, Abundances[dex])

This parameter is filled with the uncertainty of the `metallicity` derived from the Fourier parameters of the *G*-band light curve.

R21_G : Fourier decomposition parameter A2/A1 (for G band) (float)

This parameter is filled with the Fourier decomposition parameter $R_{21} = A_2/A_1$, where A_2 is the amplitude of the 2nd harmonic and A_1 is the amplitude of the fundamental harmonic of the truncated Fourier series defined as $(mag(t_j) = zp + \sum[A_i cos(i \times 2\pi\nu_{max} t_j + \phi_i)])$ used to model the *G*-band light curve. Zero-point (zp), period ($1/\nu_{max}$), number of harmonics (i), amplitudes (A_i), and phases (ϕ_i) of the harmonics, are determined using the Levenberg-Marquardt non linear fitting algorithm.

R21_G_ERROR : Uncertainty on the r21 parameter A2/A1 (for G band) (float)

This parameter is filled with the uncertainty value on the `r21_g` parameter, computed via a bootstrap technique.

R31_G : Fourier decomposition parameter A3/A1 (for G band) (float)

This parameter is filled with the Fourier decomposition parameter $R_{21} = A_3/A_1$, where A_3 is the amplitude of the 3rd harmonic and A_1 is the amplitude of the fundamental harmonic of the truncated Fourier series defined as $(mag(t_j) = zp + \sum[A_i cos(i \times 2\pi\nu_{max} t_j + \phi_i)])$ used to model the *G*-band light curve. Zero-point (zp), period ($1/\nu_{max}$), number of harmonics (i), amplitudes (A_i), and phases (ϕ_i) of the harmonics, are determined using the Levenberg-Marquardt non linear fitting algorithm.

R31_G_ERROR : Uncertainty on the r31 parameter A3/A1 (for G band) (float)

This parameter is filled with the uncertainty value of the `r31_g` parameter, computed via a bootstrap technique.

PHI21_G : Fourier decomposition parameter phi2 - 2*phi1 (for G band) (float, Angle[rad])

This parameter is filled with the Fourier decomposition parameter ϕ_{21}: $\phi_2 - 2\phi_1$ value, where ϕ_2 is the phase of the 2nd harmonic and ϕ_1 is the phase of the fundamental harmonic of the truncated Fourier series defined as $(mag(t_j) = zp + \sum[A_i cos(i \times 2\pi\nu_{max} t_j + \phi_i)])$ used to model

the G-band light curve. Zero-point (zp), period ($1/\nu_{max}$), number of harmonics (i), amplitudes (A_i), and phases (ϕ_i) of the harmonics, are determined using the Levenberg-Marquardt non linear fitting algorithm.

PHI21_G_ERROR : Uncertainty on the phi21 parameter in the G band: phi2 - 2*phi1 (for G band) (float, Angle[rad])

This parameter is filled with the uncertainty of the `phi21_g` parameter, computed via a bootstrap technique.

PHI31_G : Fourier decomposition parameter phi3 - 3*phi1 (for G band) (float, Angle[rad])

This parameter is filled with the Fourier decomposition parameter ϕ_{31}: $\phi_3 - 3\phi_1$ value, where ϕ_3 is the phase of the 3rd harmonic and ϕ_1 is the phase of the fundamental harmonic of the truncated Fourier series defined as $(mag(t_j) = zp + \sum[A_i cos(i \times 2\pi\nu_{max} t_j + \phi_i)])$ used to model the G-band light curve. Zero-point (zp), period ($1/\nu_{max}$), number of harmonics (i), amplitudes (A_i), and phases (ϕ_i) of the harmonics, are determined using the Levenberg-Marquardt non linear fitting algorithm.

PHI31_G_ERROR : Uncertainty on the phi31 parameter in the G band: phi3 - 3*phi1 (for G band) (float, Angle[rad])

This parameter is filled with the uncertainty of the `phi31_g`: $\phi_3 - 3\phi_1$ parameter, computed via a bootstrap technique.

R21_GRVS : Fourier decomposition parameter A2/A1 (for Grvs band) (float)

This parameter is filled with the Fourier decomposition parameter $R_{21} = A_2/A_1$, where A_2 is the amplitude of the 2nd harmonic and A_1 is the amplitude of the fundamental harmonic of the truncated Fourier series defined as $(mag(t_j) = zp + \sum[A_i cos(i \times 2\pi\nu_{max} t_j + \phi_i)])$ used to model the G_{RVS}-band light curve. Zero-point (zp), period ($1/\nu_{max}$), number of harmonics (i), amplitudes (A_i), and phases (ϕ_i) of the harmonics, are determined using the Levenberg-Marquardt non linear fitting algorithm.

R21_GRVS_ERROR : Uncertainty on the parameter A2/A1 (for Grvs band) (float)

This parameter is filled with the uncertainty value on the `r21_grvs` parameter, computed via bootstrap technique.

R31_GRVS : Fourier decomposition parameter A3/A1 (for Grvs band) (float)

This parameter is filled with the Fourier decomposition parameter $R_{21} = A_3/A_1$, where A_3 is the amplitude of the 3rd harmonic and A_1 is the amplitude of the fundamental harmonic of the truncated Fourier series defined as $(mag(t_j) = zp + \sum[A_i cos(i \times 2\pi\nu_{max} t_j + \phi_i)])$ used to model the G_{RVS}-band light curve. Zero-point (zp), period ($1/\nu_{max}$), number of harmonics (i), amplitudes (A_i), and phases (ϕ_i) of the harmonics, are determined using the Levenberg-Marquardt non linear fitting algorithm.

R31_GRVS_ERROR : Uncertainty on the parameter A3/A1 (for Grvs band) (float)

This parameter is filled with the uncertainty value of the `r31_grvs` parameter, computed via bootstrap technique.

PHI21_GRVS : Fourier decomposition parameter phi2 - 2*phi1 (for Grvs band) (float, Angle[rad])

This parameter is filled with the Fourier decomposition parameter ϕ_{21}: $\phi_2 - 2\phi_1$ value, where ϕ_2 is the phase of the 2nd harmonic and ϕ_1 is the phase of the fundamental harmonic of the truncated Fourier series defined as $(mag(t_j) = zp + \sum[A_i cos(i \times 2\pi\nu_{max} t_j + \phi_i)])$ used to model the G_{RVS}-band light curve. Zero-point (zp), period ($1/\nu_{max}$), number of harmonics (i), amplitudes (A_i), and phases (ϕ_i) of the harmonics, are determined using the Levenberg-Marquardt non linear fitting algorithm.

PHI21_GRVS_ERROR : Uncertainty on the parameter phi2 - 2*phi1 (for Grvs band) (float, Angle[rad])

This parameter is filled with the uncertainty of the `phi21_grvs` parameter, computed via bootstrap technique.

PHI31_GRVS : Fourier decomposition parameter phi3 - 3*phi1 (for Grvs band) (float, Angle[rad])

This parameter is filled with the Fourier decomposition parameter ϕ_{31}: $\phi_3 - 3\phi_1$ value, where ϕ_3 is the phase of the 3rd harmonic and ϕ_1 is the phase of the fundamental harmonic of the truncated Fourier series defined as $(mag(t_j) = zp + \sum[A_i cos(i \times 2\pi\nu_{max} t_j + \phi_i)])$ used to model the G_{RVS}-band light curve. Zero-point (zp), period ($1/\nu_{max}$), number of harmonics (i), amplitudes (A_i), and

phases (ϕ_i) of the harmonics, are determined using the Levenberg-Marquardt non linear fitting algorithm.

PHI31_GRVS_ERROR : Uncertainty on the parameter phi3 - 3*phi1 (for Grvs band) (float, Angle[rad])

This parameter is filled with the uncertainty of the `phi31_grvs`: $\phi_3 - 3\phi_1$ parameter, computed via bootstrap technique.

R21_RV : Fourier decomposition parameter A2/A1 (for radial velocity curve) (float)

This parameter is filled with the Fourier decomposition parameter $R_{21} = A_2/A_1$, where A_2 is the amplitude of the 2nd harmonic and A_1 is the amplitude of the fundamental harmonic of the truncated Fourier series defined as ($mag(t_j) = zp + \sum[A_i cos(i \times 2\pi\nu_{max} t_j + \phi_i)]$) used to model the radial velocity curve. Zero-point (zp), period ($1/\nu_{max}$), number of harmonics (i), amplitudes (A_i), and phases (ϕ_i) of the harmonics, are determined using the Levenberg-Marquardt non linear fitting algorithm.

R21_RV_ERROR : Uncertainty on the parameter A2/A1 (for radial velocity curve) (float)

This parameter is filled with the uncertainty value on the `r21_rv` parameter, computed via bootstrap technique.

R31_RV : Fourier decomposition parameter A3/A1 (for radial velocity curve) (float)

This parameter is filled with the Fourier decomposition parameter $R_{21} = A_3/A_1$, where A_3 is the amplitude of the 3rd harmonic and A_1 is the amplitude of the fundamental harmonic of the truncated Fourier series defined as ($mag(t_j) = zp + \sum[A_i cos(i \times 2\pi\nu_{max} t_j + \phi_i)]$) used to model the radial velocity curve. Zero-point (zp), period ($1/\nu_{max}$), number of harmonics (i), amplitudes (A_i), and phases (ϕ_i) of the harmonics, are determined using the Levenberg-Marquardt non linear fitting algorithm.

R31_RV_ERROR : Uncertainty on the parameter A3/A1 (for radial velocity curve) (float)

This parameter is filled with the uncertainty value of the `r31_rv` parameter, computed via bootstrap technique.

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PHI21_RV : Fourier decomposition parameter phi2 - 2*phi1 (for radial velocity curve) (float, Angle[rad])

This parameter is filled with the Fourier decomposition parameter ϕ_{21}: $\phi_2 - 2\phi_1$ value, where ϕ_2 is the phase of the 2nd harmonic and ϕ_1 is the phase of the fundamental harmonic of the truncated Fourier series defined as $(mag(t_j) = zp + \sum[A_i cos(i \times 2\pi\nu_{max} t_j + \phi_i)])$ used to model the radial velocity curve. Zero-point (zp), period ($1/\nu_{max}$), number of harmonics (i), amplitudes (A_i), and phases (ϕ_i) of the harmonics, are determined using the Levenberg-Marquardt non linear fitting algorithm.

PHI21_RV_ERROR : Uncertainty on the parameter phi2 - 2*phi1 (for radial velocity curve) (float, Angle[rad])

This parameter is filled with the uncertainty of the `phi21_rv` parameter, computed via bootstrap technique.

PHI31_RV : Fourier decomposition parameter phi3 - 3*phi1 (for radial velocity curve) (float, Angle[rad])

This parameter is filled with the Fourier decomposition parameter ϕ_{31}: $\phi_3 - 3\phi_1$ value, where ϕ_3 is the phase of the 3rd harmonic and ϕ_1 is the phase of the fundamental harmonic of the truncated Fourier series defined as $(mag(t_j) = zp + \sum[A_i cos(i \times 2\pi\nu_{max} t_j + \phi_i)])$ used to model the radial velocity curve. Zero-point (zp), period ($1/\nu_{max}$), number of harmonics (i), amplitudes (A_i), and phases (ϕ_i) of the harmonics, are determined using the Levenberg-Marquardt non linear fitting algorithm.

PHI31_RV_ERROR : Uncertainty on the parameter phi3 - 3*phi1 (for radial velocity curve) (float, Angle[rad])

This parameter is filled with the uncertainty of the `phi31_rv`: $\phi_3 - 3\phi_1$ parameter, computed via bootstrap technique.

NUM_CLEAN_EPOCHS_G : Number of G FoV epochs used in the fitting algorithm (short)

This parameter is filled with the number of epochs that remain in the G-band light curve after the SOS Cep & RRLyrae outlier removal process.

NUM_CLEAN_EPOCHS_BP : Number of BP epochs used in the fitting algorithm (short)

This parameter is filled with the number of epochs that remain in the $G_{\rm BP}$-band light curve after the SOS Cep & RRLyrae outlier removal process.

NUM_CLEAN_EPOCHS_RP : Number of RP epochs used in the fitting algorithm (short)

This parameter is filled with the number of epochs that remain in the $G_{\rm RP}$-band light curve after the SOS Cep & RRLyrae outlier removal process.

NUM_CLEAN_EPOCHS_GRVS : Number of Grvs epochs used in the fitting algorithm (short)

this parameter is filled with the number of epochs that remain in the G_{RVS} band light curve after the SOS Cep & RRLyrae outlier removal process.

NUM_CLEAN_EPOCHS_RV : Number of radial velocity epochs used in the fitting algorithm (short)

This parameter is filled with the number of epochs that remain in the radial velocity curve after the SOS Cep & RRLyrae outlier removal process.

ZP_MAG_G : Zero point (mag) of the final model of the G band light curve (float, Magnitude[mag])

Zero point (mag) of the final model of the G-band light curve for Cepheids and RR Lyrae stars.

ZP_MAG_BP : Zero point (mag) of the final model of the BP band light curve (float, Magnitude[mag])

Zero point (mag) of the final model of the G_{BP} band light curve for Cepheids and RR Lyrae stars.

ZP_MAG_RP : Zero point (mag) of the final model of the RP band light curve (float, Magnitude[mag])

Zero point (mag) of the final model of the G_{RP} band light curve for Cepheids and RR Lyrae stars.

ZP_MAG_GRVS : Zero point (mag) of the final model of the Grvs band light curve (float, Magnitude[mag])

Draft

Zero point (mag) of the final model of the G_{RVS} band light curve for Cepheids and RR Lyrae stars.

ZP_RV : Zero point (km/s) of the final model of the radial velocity curve (float, Velocity[km s^{-1}])

Zero point (km/s) of the final model of the radial velocity curve for Cepheids and RR Lyrae stars. `zp_rv` is equal to `average_rv` when the radial velocity curve is modelled with a truncated Fourier series. In case of limited data or models showing unphysical oscillations preventing the modeling by the truncated Fourier series, `zp_rv` is null.

NUM_HARMONICS_FOR_P1_G : Number of harmonics used to model the first periodicity of the G-band light curve (byte)

This parameter is filled with the number of harmonics of the truncated Fourier series used to model the *G*-band light curve folded with the `P1` period.

NUM_HARMONICS_FOR_P2_G : Number of harmonics used to model the second periodicity of the G-band light curve (byte)

For a multi-mode pulsator, this parameter is filled with the number of harmonics of the truncated Fourier series used to model the residuals of the *G*-band light curve after subtracting the P1 model.

NUM_HARMONICS_FOR_P1_GRVS : Number of harmonics used to model the first periodicity of the Grvs-band light curve (byte)

This parameter is filled with the number of harmonics of the truncated Fourier series used to model the $G_{\rm RVS}$-band light curve folded with the `P1` period.

NUM_HARMONICS_FOR_P2_GRVS : Number of harmonics used to model the second periodicity of the Grvs-band light curve (byte)

For a multi-mode pulsator, this parameter is filled with the number of harmonics of the truncated Fourier series used to model the residuals of the $G_{\rm RVS}$-band light curve after subtracting the P1 model.

NUM_HARMONICS_FOR_P1_RV : Number of harmonics used to model the first periodicity of the radial velocity curve (byte)

Draft

This parameter is filled with the number of harmonics of the truncated Fourier series used to model the RV curve folded with the P1 period.

NUM_HARMONICS_FOR_P2_RV : Number of harmonics used to model the second periodicity of the radial velocity curve (byte)

For a multi-mode pulsator, this parameter is filled with the number of harmonics of the truncated Fourier series used to model the residuals of the RV curve after subtracting the P1 model.

G_ABSORPTION : Interstellar absorption in the G band (float, Magnitude[mag])

This parameter is filled with values coming from the estimate of the interstellar extinction toward the investigated pulsators. The period-colour-amplitude relation was used.

G_ABSORPTION_ERROR : Error on the interstellar absorption in the G band (float, Magnitude[mag])

Error on the interstellar absorption in the *G*-band (`g_absorption_error`): This parameter is filled with the r.m.s. errors of the relations used to estimate the interstellar absorption.

LC_QUALITY_FLAG_G : Photometry quality flags for the G band (float)

Implementation of photometry quality flags for the *G* band to discriminate between good and problematic light curves. This flag is defined at SOS level as the ratio between stadard deviation of the G band observations and the rms of the residuals of the *G*-band Fourier model. It is always larger than zero, and the larger the value of the flag the cleaner the light curve. Tipically for values smaller than ~ 6, the light curve is noisy, indicating photometric problems (e.g. crowding and blending), or multi mode pulsation, Blazhko effect.

LC_QUALITY_FLAG_BP : Photometry quality flags for the BP band (float)

Implementation of photometry quality flags for the $G_{\rm BP}$ band to discriminate between good and problematic light curves. This flag is defined at SOS level as the ratio between stadard deviation of the BP band observations and the rms of the residuals of the $G_{\rm BP}$-band Fourier model. It is always larger than zero, and the larger the value of the flag the cleaner the light curve. Tipically for values smaller than ~ 6, the light curve is noisy, indicating photometric problems (e.g. crowding and blending), or multi mode pulsation, Blazhko effect.

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LC_QUALITY_FLAG_RP : Photometry quality flags for the RP band (float)

Implementation of photometry quality flags for the G_{RP} band to discriminate between good and problematic light curves. This flag is defined at SOS level as the ratio between stadard deviation of the BP band observations and the rms of the residuals of the G_{RP}-band Fourier model. It is always larger than zero, and the larger the value of the flag the cleaner the light curve. Tipically for values smaller than $\sim$ 6, the light curve is noisy, indicating photometric problems (e.g. crowding and blending), or multi mode pulsation, Blazhko effect.

LC_QUALITY_FLAG_GRVS : Photometry quality flags for the Grvs band (float)

Implementation of photometry quality flags for the G_{RVS} band to discriminate between good and problematic light curves. This flag is defined at SOS level as the ratio between stadard deviation of the BP band observations and the rms of the residuals of the G_{RVS}-band Fourier model. It is always larger than zero, and the larger the value of the flag the cleaner the light curve. Tipically for values smaller than $\sim$ 6, the light curve is noisy, indicating photometric problems (e.g. crowding and blending), or multi mode pulsation, Blazhko effect.

RV_QUALITY_FLAG : Quality flags for the radial velocity (float)

Implementation of radial velocity (RV) quality flag to discriminate between good and problematic RV curves. This flag is defined at SOS level as the ratio between stadard deviation of the RV observations and the rms of the residuals of the RV Fourier model. It is always larger than zero, and the larger the value of the flag the cleaner the velocity curve.

PHI_RISE : Parameter measuring the skewness of the light curve (float)

The time of rise to maximum light to be used in addition to the position in the Period- Amplitude diagram to distinguish RRab from RRc stars, particularly in the short-period regime (P $<$ 0.4 d), and to disentangle the different Cepheid types (classical, anomalous and BL Her) in the range of periods 1-2.5 d.

UNIFORMITY_INDEX_G : Uniformity index in the G band (float)

Uniformity index of a folded time series, defined by Madore and Fredmann 2005, ApJ 630, 1054. It can be used to quantify how evenly the time series is sampled in phase.

UNIFORMITY_INDEX_BP : Uniformity index in the BP band (float)

Uniformity index of a folded time series, defined by Madore and Fredmann 2005, ApJ 630, 1054. It can be used to quantify how evenly the time series is sampled in phase.

UNIFORMITY_INDEX_RP : Uniformity index in the RP band (float)

Uniformity index of a folded time series, defined by Madore and Fredmann 2005, ApJ 630, 1054. It can be used to quantify how evenly the time series is sampled in phase.

UNIFORMITY_INDEX_GRVS : Uniformity index in the Grvs band (float)

Uniformity index of a folded time series, defined by Madore and Fredmann 2005, ApJ 630, 1054. It can be used to quantify how evenly the time series is sampled in phase.

UNIFORMITY_INDEX_RV : Uniformity index in the radial velocity time series (float)

Uniformity index of a folded time series, defined by Madore and Fredmann 2005, ApJ 630, 1054. It can be used to quantify how evenly the time series is sampled in phase.

REFERENCE_TIME_G : Reference time of the Fourier modelled G-band light curve (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Reference time for the Fourier modelled G-band light curve.

REFERENCE_TIME_BP : Reference time of the Fourier modelled BP-band light curve (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Reference time for the Fourier modelled G_{BP}-band light curve.

REFERENCE_TIME_RP : Reference time of the Fourier modelled RP-band light curve (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Reference time for the Fourier modelled G_{RP}-band light curve.

REFERENCE_TIME_GRVS : Reference time of the Fourier modelled Grvs-band light curve (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Draft

Reference time for the Fourier modelled $G_{\rm RVS}$-band light curve.

REFERENCE_TIME_RV : Reference time of the Fourier modelled radial velocity curve (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Reference time for the Fourier modelled radial velocity curve.

FUND_FREQ1 : First frequency of the non-linear Fourier modelling (double, Frequency[day^{-1}])

First frequency of the non-linear Fourier modelling. It applies to all three G, $G_{\rm BP}$, and $G_{\rm RP}$ bands and the radial velocity curve.

FUND_FREQ1_ERROR : Error of the first frequency of the non-linear Fourier modelling (float, Frequency[day^{-1}])

Error of the first frequency of the non-linear Fourier modelling.

FUND_FREQ2 : Second frequency of the non-linear Fourier modelling in the G band (double, Frequency[day^{-1}])

Second frequency of the non-linear Fourier modelling for the G band only.

FUND_FREQ2_ERROR : Error of the second frequency of the non-linear Fourier modelling in the G band (float, Frequency[day^{-1}])

Error of the second frequency of the non-linear Fourier modelling. It applies to the G band only.

FUND_FREQ1_HARMONIC_AMPL_G : Amplitudes of the Fourier model for the first frequency in the G band (float[16] array, Magnitude[mag])

Amplitudes of the Fourier model fitted to the observed G-band light curve.

FUND_FREQ1_HARMONIC_AMPL_G_ERROR : Errors of the amplitudes of the Fourier model for the first frequency in the G band (float[16] array, Magnitude[mag])

Errors of the amplitudes of the Fourier model fitted to the observed G-band light curve.

Draft

FUND_FREQ1_HARMONIC_PHASE_G : Phases of the Fourier model for the first frequency in the G band (float[16] array, Angle[rad])

Phases of the Fourier model fitted to the observed G-band light curve.

FUND_FREQ1_HARMONIC_PHASE_G_ERROR : Errors of the phases of the Fourier model for the first frequency in the G band (float[16] array, Angle[rad])

Errors of the phases of the Fourier model fitted to the observed G-band light curve.

FUND_FREQ1_HARMONIC_AMPL_GRVS : Amplitudes of the Fourier model for the first frequency in the Grvs band (float[16] array, Magnitude[mag])

Amplitudes of the Fourier model fitted to the observed $G_{\rm RVS}$-band light curve.

FUND_FREQ1_HARMONIC_AMPL_GRVS_ERROR : Errors of the amplitudes of the Fourier model for the first frequency in the Grvs band (float[16] array, Magnitude[mag])

Errors of the amplitudes of the Fourier model fitted to the observed $G_{\rm RVS}$-band light curve.

FUND_FREQ1_HARMONIC_PHASE_GRVS : Phases of the Fourier model for the first frequency in the Grvs band (float[16] array, Angle[rad])

Phases of the Fourier model fitted to the observed $G_{\rm RVS}$-band light curve.

FUND_FREQ1_HARMONIC_PHASE_GRVS_ERROR : Errors of the phases of the Fourier model for the first frequency in the Grvs band (float[16] array, Angle[rad])

Errors of the phases of the Fourier model fitted to the observed $G_{\rm RVS}$-band light curve.

FUND_FREQ1_HARMONIC_AMPL_RV : Amplitudes of the Fourier model for the first frequency of the radial velocity curve (float[16] array, Velocity[km s^{-1}])

Amplitudes of the Fourier model fitted to the observed radial velocity curve.

Draft

FUND_FREQ1_HARMONIC_AMPL_RV_ERROR : Errors of the amplitudes of the Fourier model for the first frequency of the radial velocity curve (float[16] array, Velocity[km s^{-1}])

Errors of the amplitudes of the Fourier model fitted to the observed radial velocity curve.

FUND_FREQ1_HARMONIC_PHASE_RV : Phases of the Fourier model for the first frequency of the radial velocity curve (float[16] array, Angle[rad])

Phases of the Fourier model fitted to the observed radial velocity curve.

FUND_FREQ1_HARMONIC_PHASE_RV_ERROR : Errors of the phases of the Fourier model for the first frequency of the radial velocity curve (float[16] array, Angle[rad])

Errors of the phases of the Fourier model fitted to the observed radial velocity curve.

P2_O : Period corresponding to the second overtone pulsation mode (for multi mode pulsators) in the G-band time series (double, Time[day])

For double-mode DCEPs, this parameter is filled with the period corresponding to the shortest periodicity if the DCEP is classified as '1O/2O' of 'F/2O'; otherwise it is filled with the longest periodicity if the classification is '2O/3O'.

For triple-mode DCEPs this parameter is filled with the period corresponding to the shortest periodicity if the DCEP is classified as 'F/1O/2O'; it is filled with the intermediate periodicity if the classification is '1O/2O/3O'.

This value is obtained by modelling the *G* time series using the Levenberg-Marquardt non linear fitting algorithm (see Clementini et al. 2016).

P2_O_ERROR : Uncertainty on the period corresponding to the second overtone pulsation mode (float, Time[day])

This parameter is filled with the uncertainty of the `p2_o` parameter, computed via a bootstrap technique.

TYPE_BEST_CLASSIFICATION : Best type classification estimate (string)

Classification of a Cepheid into 'DCEP', 'T2CEP' or 'ACEP' using the period-luminosity rela-

Draft

tions, which are different for the three different types of Cepheids.

TYPE2_BEST_SUB_CLASSIFICATION : Best subclassification estimate for the best type classification estimate (string)

Sub-classification of a T2CEP Cepheids into BL Herculis ('BL_HER'), W Virginis ('W_VIR') or RV Tauris ('RV_TAU') sub-types depending on the source periodicity.

MODE_BEST_CLASSIFICATION : Best mode classification estimate (string)

Best mode classification estimate:

- 'FUNDAMENTAL': fundamental mode for `type_best_classification`='DCEP' or 'ACEP'
- 'FIRST_OVERTONE': first overtone for `type_best_classification`='DCEP' or 'ACEP'
- 'SECOND_OVERTONE': second overtone for `type_best_classification`='DCEP'
- 'MULTI': multi-mode pulsators for `type_best_classification`='DCEP'
- 'UNDEFINED': if mode could not be clearly determined for `type_best_classification`='DCE or 'ACEP'
- 'NOT_APPLICABLE': when `type_best_classification`='T2CEP'

The Cepheid pulsation mode is assigned using the period-luminosity and period-Wesenheit relations, which are different for the various pulsation modes as well as analysing the Fourier parameters vs period plots. The type 'MULTI' is assigned to stars pulsating in two or more modes simultaneously.

MULTI_MODE_BEST_CLASSIFICATION : Best multi mode DCEP classification (string)

Sub-classification of multi mode DCEP variables according to their position in the 'Petersen diagram' — see for example Fig. 1 in Soszyński et al. (2015). F,1O,2O and 3O correspond with fundamental, first, second and third overtone, respectively.

SOURCE_ORBITAL_PERIOD : Orbital period for the binary Cepheid or RR Lyrae star (float, Time[day])

Draft

Period derived by separating the orbital from the pulsation motions of Cepheid or RR Lyrae stars, through a pre-whitening of the radial velocity curves.

SOURCE_ORBITAL_PERIOD_ERROR : Orbital period error for the binary Cepheid or RR Lyrae star (float, Time[day])

Error in the orbital period of binary/multiple Cepheid or RR Lyrae stars, computed via bootstrap.

ORBIT_ECCENTRICITY : Eccentricity of the fitted orbit (float)

The eccentricity value obtained from a simultaneous fit of the orbital model and the pulsational variation, following the equations of Gallenne et al. (2018).

ORBIT_ECCENTRICITY_ERROR : Error on the fitted orbital eccentricity (float)

Uncertainty on the fitted orbital eccentricity, estimated using a Monte Carlo bootstrap resampling of the RV time series and repeating the fit at each resampling step.

ORBIT_K : Semi-amplitude of the orbit (float, Velocity[km s^{-1}])

Half the peak-to-peak amplitude of the star's radial velocity curve obtained from a simultaneous fit of the orbital model and the pulsational variation, following the equations of Gallenne et al. (2018).

ORBIT_K_ERROR : Error on the semi-amplitude of the orbit (float, Velocity[km s^{-1}])

Uncertainty on the half peak-to-peak amplitude, estimated using a Monte Carlo bootstrap resampling of the RV time series and repeating the fit at each resampling step.

ORBIT_OMEGA : Argument of periastron (float, Angle[deg])

The angle, measured in the orbital plane, from the ascending node to the closest approach point (periastron), obtained from the fit of the orbital model and the pulsational variation, following the equations of Gallenne et al. (2018).

ORBIT_OMEGA_ERROR : Error on the argument of periastron (float, Angle[deg])

Draft

Uncertainty on the fitted omega angle, estimated using a Monte Carlo bootstrap resampling of the RV time series and repeating the fit at each resampling step.

ORBIT_TP : Epoch of periastron passage (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

The specific time when the star is at periastron, its closest approach in the orbit.

ORBIT_TP_ERROR : Error on the epoch of periastron passage (float, Time[day])

Uncertainty on the fitted periastron passage, estimated using a Monte Carlo bootstrap resampling of the RV time series and repeating the fit at each resampling step.

Draft

18.3 VARI_CLASSIFIER_FEATURE [TAP]

Table with environment and period-search derived features used by the supervised variability classifiers.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

MEDIAN_BP_MINUS_G_FOV_R100_ARCSEC : Median BP-G colour of all other sources within a 100 arcsecond radius of this source. (float, Magnitude[mag])

MEDIAN_BP_MINUS_RP_R100_ARCSEC : Median BP-RP colour of all other sources within a 100 arcsecond radius of this source. (float, Magnitude[mag])

MEDIAN_G_FOV_MINUS_RP_R100_ARCSEC : Median G-RP colour of all other sources within a 100 arcsecond radius of this source. (float, Magnitude[mag])

SOURCE_DENSITY_R100_ARCSEC : Source density in sources per square arcsecond measured within a 100 arcsecond radius of this source. (float)

LS_FREQUENCY : Top frequency from least squares period search in cycles per day. (float, Frequency[day^{-1}])

Draft

LS_FREQUENCY_LS_POWER_G_FOV : Least squares periodogram power in the G band at top frequency from least squares period search. (float)

LS_FREQUENCY_SDE_G_FOV : Signal detection efficiency in the G band at top frequency from least squares period search. (float)

LS_FREQUENCY_FAP_LOG10_G_FOV : Base-10 logarithm of false alarm probability in the G band at top frequency from least squares period search. (float)

LS_FREQUENCY_LK_STATISTIC_G_FOV : Lafler-Kinman statistic in the G band at top frequency from least squares period search. (float)

LS_FREQUENCY_LS_POWER_BP : Least squares periodogram power in the BP band at top frequency from least squares period search. (float)

LS_FREQUENCY_LK_STATISTIC_BP : Lafler-Kinman statistic in the BP band at top frequency from least squares period search. (float)

LS_FREQUENCY_LS_POWER_RP : Least squares periodogram power in the RP band at top frequency from least squares period search. (float)

LS_FREQUENCY_LK_STATISTIC_RP : Lafler-Kinman statistic in the RP band at top frequency from least squares period search. (float)

HYBRID_FREQUENCY : Top frequency from hybrid period search in cycles per day. (float, Frequency[day^{-1}])

HYBRID_FREQUENCY_LS_POWER_G_FOV : Least squares periodogram power in the G band at top frequency from hybrid period search. (float)

HYBRID_FREQUENCY_LK_STATISTIC_G_FOV : Lafler-Kinman statistic in the G band at top frequency from hybrid period search. (float)

Draft

HYBRID_FREQUENCY_LS_POWER_BP : Least squares periodogram power in the BP band at top frequency from hybrid period search. (float)

HYBRID_FREQUENCY_LK_STATISTIC_BP : Lafler-Kinman statistic in the BP band at top frequency from hybrid period search. (float)

HYBRID_FREQUENCY_LS_POWER_RP : Least squares periodogram power in the RP band at top frequency from hybrid period search. (float)

HYBRID_FREQUENCY_LK_STATISTIC_RP : Lafler-Kinman statistic in the RP band at top frequency from hybrid period search. (float)

MODEL_FREQUENCY : Frequency used for modelling at dominant frequency, in cycles per day. (float, Frequency[day^{-1}])

MODEL_A1_G_FOV : A1 Fourier parameter of G-band model at dominant frequency. (float, Magnitude[mag])

MODEL_A1_ERROR_G_FOV : Error of A1 Fourier parameter of G-band model at dominant frequency. (float, Magnitude[mag])

MODEL_A2_G_FOV : A2 Fourier parameter of G-band model at dominant frequency. (float, Magnitude[mag])

MODEL_A2_ERROR_G_FOV : Error of A2 Fourier parameter of G-band model at dominant frequency. (float, Magnitude[mag])

MODEL_A3_G_FOV : A3 Fourier parameter of G-band model at dominant frequency. (float, Magnitude[mag])

MODEL_A3_ERROR_G_FOV : Error of A3 Fourier parameter of G-band model at dominant frequency. (float, Magnitude[mag])

Draft

MODEL_R21_G_FOV : Ratio A2/A1 of Fourier parameters of G-band model at dominant frequency. (float)

MODEL_PHI21_G_FOV : Phase offset phi2 - 2 * phi1 of Fourier parameters of G-band model at dominant frequency. (float, Angle[rad])

MODEL_R31_G_FOV : Ratio A3/A1 of Fourier parameters of G-band model at dominant frequency. (float)

MODEL_PHI31_G_FOV : Phase offset phi3 - 3 * phi1 of Fourier parameters of G-band model at dominant frequency. (float, Angle[rad])

MODEL_HALF_FREQUENCY : Frequency used for modelling at half the dominant frequency, in cycles per day. (float, Frequency[day^{-1}])

MODEL_HALF_A1_G_FOV : A1 Fourier parameter of G-band model at half dominant frequency. (float, Magnitude[mag])

MODEL_HALF_A1_ERROR_G_FOV : Error of A1 Fourier parameter of G-band model at half dominant frequency. (float, Magnitude[mag])

MODEL_HALF_A2_G_FOV : A2 Fourier parameter of G-band model at half dominant frequency. (float, Magnitude[mag])

MODEL_HALF_A2_ERROR_G_FOV : Error of A2 Fourier parameter of G-band model at half dominant frequency. (float, Magnitude[mag])

MODEL_HALF_A3_G_FOV : A3 Fourier parameter of G-band model at half dominant frequency. (float, Magnitude[mag])

MODEL_HALF_A3_ERROR_G_FOV : Error of A3 Fourier parameter of G-band model at half dominant frequency. (float, Magnitude[mag])

Draft

MODEL_HALF_R21_G_FOV : Ratio A2/A1 of Fourier parameters of G-band model at half dominant frequency. (float)

MODEL_HALF_PHI21_G_FOV : Phase offset phi2 - 2 * phi1 of Fourier parameters of G-band model at half dominant frequency. (float, Angle[rad])

MODEL_HALF_R31_G_FOV : Ratio A3/A1 of Fourier parameters of G-band model at half dominant frequency. (float)

MODEL_HALF_PHI31_G_FOV : Phase offset phi3 - 3 * phi1 of Fourier parameters of G-band model at half dominant frequency. (float, Angle[rad])

Draft

18.4 VARI_CLASSIFIER_RESULT [TAP]

Table with supervised variability classification results of all classifiers.

The classifiers are identified by the classifier_name column.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

CLASSIFIER_NAME : Classifier name used to produce this result (string)

Name of classifier used to produce this result; see documentation for details.

CLASSIFIER_BEST_CLASS_NAME : Name of the best class from this classifier (string)

Name of best class with corresponding classification score in `classifier_best_class_prob`. See documentation for details.

CLASSIFIER_BEST_CLASS_PROB : Probability of the best class from this classifier (float, Dimensionless[percentage/100])

Probability of the class that received the highest score from this classifier.

USED_IN_TRAINING : Whether this source was used as a training instance for this classifier (boolean)

Draft

Whether this source was used as a training instance for this classifier. This may either indicate that the score is not comparable to the other scores from this classifier, or that it has been substituted for a score from cross validation results.

PROB_ACEP : Classifier probability of source being an anomalous Cepheid (float, Dimensionless[percentage/100])

PROB_ACV : Classifier probability of source being an Alpha2 Canum Venaticorum-type star (float, Dimensionless[percentage/100])

PROB_ACYG : Classifier probability of source being an Alpha Cygni-type star (float, Dimensionless[percentage/100])

PROB_AGN : Classifier probability of source being an Active Galactic Nucleus (including Quasars and Blazars). (float, Dimensionless[percentage/100])

PROB_ARRD : Classifier probability of source being an anomalous double-mode RR Lyrae star (float, Dimensionless[percentage/100])

PROB_BCEP : Classifier probability of source being a Beta Cephei-type star (float, Dimensionless[percentage/100])

PROB_BE_GCAS : Classifier probability of source being a B-type emission line or Gamma Cassiopeiae-type star (float, Dimensionless[percentage/100])

PROB_BLAP : Classifier probability of source being a blue large-amplitude pulsator (float, Dimensionless[percentage/100])

PROB_BLAZAR : Classifier probability of source being a blazar (float, Dimensionless[percentage/100])

PROB_CEP : Classifier probability of source being a Cepheid (float, Dimensionless[percentage/100])

Draft

PROB_CP : Classifier probability of source being a chemically peculiar variable star (float, Dimensionless[percentage/100])

PROB_CV : Classifier probability of source being a cataclysmic variable (float, Dimensionless[percentage/100])

PROB_DCEP : Classifier probability of source being a Delta Cephei-type star (classical Cepheid) (float, Dimensionless[percentage/100])

PROB_DSCT_SXPHE : Classifier probability of source being a Delta Scuti or SX Phoenicis-type star (float, Dimensionless[percentage/100])

PROB_EA_EB : Classifier probability of source being a Beta Persei (Algol) or Beta Lyrae-type eclipsing binary (float, Dimensionless[percentage/100])

PROB_ECL : Classifier probability of source being an eclipsing binary of type Beta Persei (Algol), Beta Lyrae, or W Ursae Majori (float, Dimensionless[percentage/100])

PROB_ELL : Classifier probability of source being a rotating ellipsoidal star (float, Dimensionless[percentage/100])

PROB_EP : Classifier probability of source being a star with a transiting planet (float, Dimensionless[percentage/100])

PROB_EW : Classifier probability of source being a W Ursae Majoris-type eclipsing binary (float, Dimensionless[percentage/100])

PROB_GALAXY : Classifier probability of source being a galaxy (float, Dimensionless[percentage/100])

PROB_GDOR : Classifier probability of source being a Gamma Doradus-type star (float, Dimensionless[percentage/100])

Draft

PROB_HB : Classifier probability of source being a heartbeat star (float, Dimensionless[percentage/100])

PROB_L : Classifier probability of source being a slow, irregular or insufficiently studied variable star (float, Dimensionless[percentage/100])

PROB_LPV : Classifier probability of source being a Long-Period Variable star of type semiregular, omicron Ceti (Mira), Small Amplitude Red Giant, or long secondary period (float, Dimensionless[percentage/100])

PROB_M : Classifier probability of source being an Omicron Ceti-type star (Mira). (float, Dimensionless[percentage/100])

PROB_MICROLENSING : Classifier probability of source being a star with a microlensing event (float, Dimensionless[percentage/100])

PROB_N : Classifier probability of source being a nova (float, Dimensionless[percentage/100])

PROB_OTHER : Classifier probability of source being some other type, possibly unknown; see documentation for classifier (float, Dimensionless[percentage/100])

PROB_RCB : Classifier probability of source being an R Coronae Borealis-type star (float, Dimensionless[percentage/100])

PROB_RR : Classifier probability of source being an RR Lyrae star (float, Dimensionless[percentage/100])

PROB_RRAB : Classifier probability of source being an RR Lyrae star in fundamental-mode pulsation. (float, Dimensionless[percentage/100])

PROB_RRC : Classifier probability of source being an RR Lyrae star in first overtone pulsation. (float, Dimensionless[percentage/100])

PROB_RRD : Classifier probability of source being an RR Lyrae star in double-mode pulsation.

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(float, Dimensionless[percentage/100])

PROB_RS : Classifier probability of source being an RS Canum Venaticorum-type star (float, Dimensionless[percentage/100])

PROB_RV : Classifier probability of source being an RV Tauri-type variable star (float, Dimensionless[percentage/100])

PROB_S : Classifier probability of source being a short timescale variable (float, Dimensionless[percentage/100])

PROB_SDB : Classifier probability of source being a subdwarf B-type variable star, such as V361 Hydrae or V1093 Herculis type (float, Dimensionless[percentage/100])

PROB_SDOR : Classifier probability of source being an S Doradus-type star (float, Dimensionless[percentage/100])

PROB_SN : Classifier probability of source being a supernova (float, Dimensionless[percentage/100])

PROB_SOLARLIKE : Classifier probability of source being a star with solar-like variability due to flares, spots, or rotational modulation (float, Dimensionless[percentage/100])

PROB_SPB : Classifier probability of source being a slowly pulsating B-type star (float, Dimensionless[percentage/100])

PROB_SXARI : Classifier probability of source being an SX Arietis-type star (float, Dimensionless[percentage/100])

PROB_SYST : Classifier probability of source being a symbiotic variable star (float, Dimensionless[percentage/100])

PROB_T2CEP : Classifier probability of source being a Type II Cepheid (float, Dimension-

less[percentage/100])

PROB_WD : Classifier probability of source being a White Dwarf of type ZZ Ceti (DAV, ZZA), V777 Herculis (DBV, ZZB), GW Virginis (DOV, ZZO), or non-pulsating (float, Dimensionless[percentage/100])

PROB_WR : Classifier probability of source being a Wolf-Rayet star (float, Dimensionless[percentage/100])

PROB_YSO : Classifier probability of source being a Young Stellar Object (float, Dimensionless[percentage/100])

PROB_ZZLEP : Classifier probability of source being a ZZ Leporis-type star (float, Dimensionless[percentage/100])

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18.5 VARI_COMPACT_COMPANION [TAP]

This table describes the compact companion candidates.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

PERIOD : Orbital period (double, Time[day])

Orbital period of the binary system.

PERIOD_ERROR : Orbital period error (float, Time[day])

Orbital period 1-sigma uncertainty.

T0_G : G-band reference time (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

G-band light-curve model reference time (see `harmonic_model_params_g` description).

T0_G_ERROR : G-band reference time error (float, Time[day])

G-band light-curve model reference time error.

T0_BP : BP-band reference time (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

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G_{BP}-band light-curve model reference time (see harmonic_model_params_bp description).

The parameter is NULL when the number of points in the G_{BP}-band time series is strictly less than 25.

t0_bp_error : BP-band reference time error (float, Time[day])

G_{BP}-band light-curve model reference time error.

The parameter is NULL when the number of points in the G_{BP}-band time series is strictly less than 25.

t0_rp : RP-band reference time (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

G_{RP}-band light-curve model reference time (see harmonic_model_params_rp description).

The parameter is NULL when the number of points in the G_{RP}-band time series is strictly less than 25.

t0_rp_error : RP-band reference time error (float, Time[day])

G_{RP}-band light-curve model reference time error.

The parameter is NULL when the number of points in the G_{RP}-band time series is strictly less than 25.

harmonic_model_params_g : G-band harmonics (float[6] array, Magnitude[mag])

The G-band light curve is fitted by a three-harmonics model:

model_mean_g + $a1cG * \cos(\omega(t - \texttt{t0_g})) + a2cG * \cos(2\omega * (t - \texttt{t0_g})) + a3cG * \cos(3\omega * (t - \texttt{t0_g})) + a1sG * \sin(\omega(t - \texttt{t0_g})) + a2sG * \sin(2\omega(t - \texttt{t0_g})) + a3sG * \sin(3\omega * (t - \texttt{t0_g}))$.

ω is the orbital frequency 2π/period, and the reference time t0_g is set to obtain $a2sG = 0$. t is a Barycentric JD in TCB − 2 455 197.5 (day).

Fourier coefficients are given in harmonic_model_params_g:

$$[a1cG, a2cG, a3cG, a1sG, a2sG, a3sG],$$

and their errors are given in `harmonic_model_params_g_error` in the same order.

HARMONIC_MODEL_PARAMS_G_ERROR : G-band harmonics errors (float[6] array, Magnitude[mag])

G-band harmonics parameters 1-sigma uncertainty.

HARMONIC_MODEL_PARAMS_BP : BP-band harmonics (float[6] array, Magnitude[mag])

The $G_{\rm BP}$-band light curve is fitted by a three-harmonics model:

$\texttt{model_mean_bp} + a1cBp * \cos(\omega(t - \texttt{t0_bp})) + a2cBp * \cos(2\omega * (t - \texttt{t0_bp})) + a3cBp * \cos(3\omega * (t - \texttt{t0_bp})) + a1sBp * \sin(\omega(t - \texttt{t0_bp})) + a2sBp * \sin(2\omega(t - \texttt{t0_bp})) + a3sBp * \sin(3\omega * (t - \texttt{t0_bp}))$.

ω is the orbital frequency $2\pi/\texttt{period}$, and the reference time `t0_bp` is set to obtain $a2sBp = 0$. t is a Barycentric JD in TCB − 2 455 197.5 (day).

Fourier coefficients are given in `harmonic_model_params_bp`:

$$[a1cBp, a2cBp, a3cBp, a1sBp, a2sBp, a3sBp],$$

and their errors are given in `harmonic_model_params_bp_error` in the same order.

The parameter is an array of NaN when the number of points in the $G_{\rm BP}$-band time series is strictly less than 25.

HARMONIC_MODEL_PARAMS_BP_ERROR : BP-band harmonics errors (float[6] array, Magnitude[mag])

$G_{\rm BP}$-band harmonics parameters 1-sigma uncertainty.

The parameter is an array of NaN when the number of points in the $G_{\rm BP}$-band time series is strictly less than 25.

HARMONIC_MODEL_PARAMS_RP : RP-band harmonics (float[6] array, Magnitude[mag])

The $G_{\rm RP}$-band light curve is fitted by a three-harmonics model:

$\texttt{model_mean_rp} + a1cRp * \cos(\omega(t - \texttt{t0_rp})) + a2cRp * \cos(2\omega * (t - \texttt{t0_rp})) + a3cRp * \cos(3\omega * (t - \texttt{t0_rp})) + a1sRp * \sin(\omega(t - \texttt{t0_rp})) + a2sRp * \sin(2\omega(t - \texttt{t0_rp})) + a3sRp * \sin(3\omega * (t - \texttt{t0_rp}))$.

ω is the orbital frequency 2π/`period`, and the reference time `t0_rp` is set to obtain $a2sRp = 0$. t is a Barycentric JD in TCB − 2 455 197.5 (day).

Fourier coefficients are given in `harmonic_model_params_rp`:

$$[a1cRp, a2cRp, a3cRp, a1sRp, a2sRp, a3sRp],$$

and their errors are given in `harmonic_model_params_rp_error` in the same order.

The parameter is an array of NaN when the number of points in the $G_{\rm RP}$-band time series is strictly less than 25.

HARMONIC_MODEL_PARAMS_RP_ERROR : RP-band harmonics errors (float[6] array, Magnitude[mag])

$G_{\rm RP}$-band harmonics parameters 1-sigma uncertainty.

The parameter is an array of NaN when the number of points in the $G_{\rm RP}$-band time series is strictly less than 25.

MODEL_MEAN_G : Harmonics model mean G-band magnitude (float, Magnitude[mag])

G-band mean magnitude derived by the three-harmonics model.

MODEL_MEAN_G_ERROR : Harmonics model mean G-band magnitude error (float, Magnitude[mag])

G-band model mean magnitude 1-sigma uncertainty.

MODEL_MEAN_BP : Harmonics model mean BP-band magnitude (float, Magnitude[mag])

$G_{\rm BP}$-band mean magnitude derived by the three-harmonics model.

The parameter is NULL when the number of points in the $G_{\rm BP}$-band time series is strictly less than 25.

MODEL_MEAN_BP_ERROR : Harmonics model mean BP-band magnitude error (float, Magnitude[mag])

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G_{BP}-band model mean magnitude 1-sigma uncertainty.

The parameter is NULL when the number of points in the G_{BP}-band time series is strictly less than 25.

MODEL_MEAN_RP : Harmonics model mean RP-band magnitude (float, Magnitude[mag])

G_{RP}-band mean magnitude derived by the three-harmonics model.

The parameter is NULL when the number of points in the G_{RP}-band time series is strictly less than 25.

MODEL_MEAN_RP_ERROR : Harmonics model mean RP-band magnitude error (float, Magnitude[mag])

G_{RP}-band model mean magnitude 1-sigma uncertainty.

The parameter is NULL when the number of points in the G_{RP}-band time series is strictly less than 25.

MOD_MIN_MASS_RATIO : Modified minimum mass ratio (float)

Mass ratio of the system — ratio of secondary (less luminous) to primary mass. Modified minimum value is found by assuming primary fills its Roche-Lobe and inclination is 90 deg ('edge-on'). Mathematical expressions are given by Gomel, Faigler, & Mazeh (2021).

MOD_MIN_MASS_RATIO_ONE_SIGMA : 15.9th percentile modified minimum mass ratio (float)

The 15.9%-percentile modified minimum mass ratio of the system, considering the leading harmonic (`a2c`) and `alpha` uncertainties.

MOD_MIN_MASS_RATIO_THREE_SIGMA : 0.135th percentile modified minimum mass ratio (float)

The 0.135%-percentile modified minimum mass ratio of the system, considering the leading harmonic (`a2c`) and `alpha` uncertainties.

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ALPHA : alpha ellipsoidal coefficient (float)

The alpha ellipsoidal coefficient of the leading second harmonic term is assumed to be 1.3 for almost all systems. When `mod_min_mass_ratio` cannot be derived using this value (1.3), we find the minimum value of `alpha`, derived for `mod_min_mass_ratio` of 100, and this value is given instead of 1.3. See Section **??** of the release documentation for details.

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18.6 VARI_ECLIPSING_BINARY [TAP]

This table describes the properties of the eclipsing binaries resulting from the variability analysis.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

GLOBAL_RANKING : Number between 0 (worst) and 1 (best) (float)

Number between 0 (worst) and 1 (best), based on the fraction of variance unexplained (FVU) for the geometrical model fit.

REFERENCE_TIME : Reference time used for the geometric model fit (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Time with respect to which the phases of the gaussian components refer in the geometrical model, or of the cosine(s) if no gaussian component is present in the model.

FREQUENCY : Frequency of geometric model of the eclipsing binary light curve (double, Frequency[day^{-1}])

Frequency of geometric model of the eclipsing binary light curve.

FREQUENCY_ERROR : Uncertainty of frequency (float, Frequency[day^{-1}])

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Uncertainty of `frequency`.

GEOM_MODEL_REFERENCE_LEVEL : Magnitude reference level of geometric model (float, Magnitude[mag])

Magnitude reference level of geometric model.

GEOM_MODEL_REFERENCE_LEVEL_ERROR : Uncertainty on magnitude reference level of geometric model (float, Magnitude[mag])

Uncertainty of `geom_model_reference_level`.

GEOM_MODEL_GAUSSIAN1_PHASE : Phase of the Gaussian 1 component (float)

Phase with respect to `reference_time`; it is NaN if there is no Gaussian component in the model. Note: Gaussian1 does not necessarily correspond to the deepest eclipse.

GEOM_MODEL_GAUSSIAN1_PHASE_ERROR : Uncertainty on phase of the Gaussian 1 component (float)

Uncertainty of `geom_model_gaussian1_phase`.

GEOM_MODEL_GAUSSIAN1_SIGMA : Standard deviation [phase] of Gaussian 1 component (float)

NaN if no Gaussian component in the model. Note: Gaussian1 does not necessarily correspond to the deepest eclipse.

GEOM_MODEL_GAUSSIAN1_SIGMA_ERROR : Uncertainty on standard deviation [phase] of Gaussian 1 component (float)

Uncertainty of `geom_model_gaussian1_sigma`.

GEOM_MODEL_GAUSSIAN1_DEPTH : Magnitude depth of Gaussian 1 component (float, Magnitude[mag])

NaN if no Gaussian component in the model. Note: Gaussian1 does not necessarily correspond

to the deepest eclipse.

GEOM_MODEL_GAUSSIAN1_DEPTH_ERROR : Uncertainty on magnitude depth of Gaussian 1 component (float, Magnitude[mag])

Uncertainty of `geom_model_gaussian1_depth`. It can be null when the model could not derive it.

GEOM_MODEL_GAUSSIAN2_PHASE : Phase of Gaussian 2 component (float)

Phase with respect to `reference_time`; it is NaN if there is no second Gaussian component in the model.

GEOM_MODEL_GAUSSIAN2_PHASE_ERROR : Uncertainty on phase of Gaussian 2 component (float)

Uncertainty of `geom_model_gaussian2_phase`.

GEOM_MODEL_GAUSSIAN2_SIGMA : Standard deviation [phase] of Gaussian 2 component (float)

This is NaN if there is no second Gaussian component in the model.

GEOM_MODEL_GAUSSIAN2_SIGMA_ERROR : Uncertainty on standard deviation [phase] of Gaussian 2 component (float)

Uncertainty of `geom_model_gaussian2_sigma`.

GEOM_MODEL_GAUSSIAN2_DEPTH : Magnitude depth of Gaussian2 component (float, Magnitude[mag])

This is NaN if there is no second Gaussian component in the model.

GEOM_MODEL_GAUSSIAN2_DEPTH_ERROR : Uncertainty on magnitude depth of Gaussian2 component (float, Magnitude[mag])

Uncertainty of `geom_model_gaussian2_depth`. It can be null when the model could not derive

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it.

GEOM_MODEL_COSINE_HALF_PERIOD_AMPLITUDE : Amplitude of the cosine component with half the period of the geometric model (float, Magnitude[mag])

The amplitude is half peak-to-peak. NaN if no cosine component with half the period of the geometric model.

GEOM_MODEL_COSINE_HALF_PERIOD_AMPLITUDE_ERROR : Uncertainty on amplitude of the cosine component (float, Magnitude[mag])

Uncertainty of `geom_model_cosine_half_period_amplitude`.

GEOM_MODEL_COSINE_HALF_PERIOD_PHASE : Reference phase of the cosine component with half the period of the geometric model (float)

Phase with respect to `reference_time`. When model contains Gaussian(s) this phase is equal to either `geom_model_gaussian1_phase` or `geom_model_gaussian2_phase`. NaN if no cosine component with half the period of the geometric model.

GEOM_MODEL_COSINE_HALF_PERIOD_PHASE_ERROR : Uncertainty on reference phase of the cosine component (float)

When model contains Gaussian(s) this uncertainty is the same as either `geom_model_gaussian1_phase_err` or `geom_model_gaussian2_phase_error`.

MODEL_TYPE : Type of geometrical model of the light curve (string)

Type of eclipsing binary geometric model of the G-band light curve. For each model the `frequency` and `geom_model_reference_level` as well as their uncertainties (`_error`) are set. The additional fields (and their uncertainty) that are available for each `model_type` are specified below:

- 'ONEGAUSSIAN'
  Geometric model parameters defined:
  - `geom_model_gaussian1_phase`
  - `geom_model_gaussian1_sigma`

  - geom_model_gaussian1_depth

  Derived parameters defined:

  - derived_primary_ecl_phase
  - derived_primary_ecl_duration
  - derived_primary_ecl_depth

- 'ONEGAUSSIAN_WITH_ELLIPSOIDAL'
  Geometric model parameters defined:

  - geom_model_gaussian1_phase
  - geom_model_gaussian1_sigma
  - geom_model_gaussian1_depth
  - geom_model_cosine_half_period_amplitude
  - geom_model_cosine_half_period_phase

  Derived parameters defined:

  - derived_primary_ecl_phase
  - derived_primary_ecl_duration
  - derived_primary_ecl_depth

- 'TWOGAUSSIANS'
  Geometric model parameters defined:

  - geom_model_gaussian1_phase
  - geom_model_gaussian1_sigma
  - geom_model_gaussian1_depth
  - geom_model_gaussian2_phase
  - geom_model_gaussian2_sigma
  - geom_model_gaussian2_depth

  Derived parameters defined:

  - derived_primary_ecl_phase
  - derived_primary_ecl_duration
  - derived_primary_ecl_depth
  - derived_secondary_ecl_phase
  - derived_secondary_ecl_duration

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  - derived_secondary_ecl_depth

- 'TWOGAUSSIANS_WITH_ELLIPSOIDAL_ON_ECLIPSE1'
  In this model geom_model_cosine_half_period_phase is the same as geom_model_gaussia
  Geometric model parameters defined:

  - geom_model_gaussian1_phase
  - geom_model_gaussian1_sigma
  - geom_model_gaussian1_depth
  - geom_model_gaussian2_phase
  - geom_model_gaussian2_sigma
  - geom_model_gaussian2_depth
  - geom_model_cosine_half_period_amplitude
  - geom_model_cosine_half_period_phase

  Derived parameters defined:

  - derived_primary_ecl_phase
  - derived_primary_ecl_duration
  - derived_primary_ecl_depth
  - derived_secondary_ecl_phase
  - derived_secondary_ecl_duration
  - derived_secondary_ecl_depth

- 'TWOGAUSSIANS_WITH_ELLIPSOIDAL_ON_ECLIPSE2'
  In this model geom_model_cosine_half_period_phase is the same as geom_model_gaussia
  Geometric model parameters defined:

  - geom_model_gaussian1_phase
  - geom_model_gaussian1_sigma
  - geom_model_gaussian1_depth
  - geom_model_gaussian2_phase
  - geom_model_gaussian2_sigma
  - geom_model_gaussian2_depth
  - geom_model_cosine_half_period_amplitude
  - geom_model_cosine_half_period_phase

  Derived parameters defined:

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  - derived_primary_ecl_phase
  - derived_primary_ecl_duration
  - derived_primary_ecl_depth
  - derived_secondary_ecl_phase
  - derived_secondary_ecl_duration
  - derived_secondary_ecl_depth

- 'ELLIPSOIDAL'
  Geometric model parameters defined:
  - geom_model_cosine_half_period_amplitude
  - geom_model_cosine_half_period_phase

NUM_MODEL_PARAMETERS : Number of free parameters of the geometric model (byte)

The number of parameters fitted for each model type. For each model the frequency and geom_model_reference_level are fitted, as well as the geometeric model parameters listed in the detailed description of model_type.

model_type	Number of parameters
'ONEGAUSSIAN'	5
'ONEGAUSSIAN_WITH_ELLIPSOIDAL'	6
'TWOGAUSSIANS'	8
'TWOGAUSSIANS_WITH_ELLIPSOIDAL_ON_ECLIPSE1'	9
'TWOGAUSSIANS_WITH_ELLIPSOIDAL_ON_ECLIPSE2'	9
'ELLIPSOIDAL'	4

REDUCED_CHI2 : Reduced Chi2 of the geometrical model fit (float)

The reduced Chi-square: χ^2/ν computed over all observations, where ν is the number of degrees of freedom.

Note that for high signal to noise observations (i.e. small uncertainties), the reduced Chi-square will increase when the simple geometric model cannot sufficiently model the more complex true light curve shape. A low Chi-square is therefore not a good measure for eclipsing binary detection on itself.

DERIVED_PRIMARY_ECL_PHASE : Primary eclipse: phase at geometrically deepest point (float)

Phase with respect to reference_time. This value is derived from the geometrical model. It can be offset from geom_model_gaussian1_phase or geom_model_gaussian2_phase if a

cosine component is present. NaN if no Gaussian component in the geometric model.

DERIVED_PRIMARY_ECL_PHASE_ERROR : Primary eclipse: uncertainty on primary eclipse phase (float)

Primary eclipse: uncertainty of `derived_primary_ecl_phase` estimated by a Jackknife resampling method.

DERIVED_PRIMARY_ECL_DURATION : Primary eclipse: duration [phase fraction] (float)

This value is derived from the geometrical model, taking the Gaussian width estimated at a depth of 2% of the Gaussian, with an upper limit of 0.4. NaN if no Gaussian component in the geometric model.

DERIVED_PRIMARY_ECL_DURATION_ERROR : Primary eclipse: uncertainty on duration of primary eclipse (float)

Primary eclipse: uncertainty of `derived_primary_ecl_duration`. Uncertainties are derived by a Jackknife resampling method. In rare instances, the uncertainty can be artificially larger than 1.

DERIVED_PRIMARY_ECL_DEPTH : Primary eclipse: depth (float, Magnitude[mag])

This value is derived from the geometrical model. NaN if no Gaussian component in the geometric model.

DERIVED_PRIMARY_ECL_DEPTH_ERROR : Primary eclipse: uncertainty on primary eclipse depth (float, Magnitude[mag])

Primary eclipse: uncertainty of `derived_primary_ecl_depth` estimated by a Jackknife resampling method. It can be null when the model could not derive it.

DERIVED_SECONDARY_ECL_PHASE : Secondary eclipse: phase at geometrically second deepest point (float)

This value is derived from the geometrical model. It can be offset from `geom_model_gaussian1_phase` or `geom_model_gaussian2_phase` if a cosine component is present. NaN if no secondary

Gaussian component in the geometric model.

DERIVED_SECONDARY_ECL_PHASE_ERROR : Secondary eclipse: uncertainty on secondary eclipse phase (float)

Secondary eclipse: uncertainty of `derived_secondary_ecl_phase` estimated by a Jackknife resampling method.

DERIVED_SECONDARY_ECL_DURATION : Secondary eclipse: duration [phase fraction] (float)

This value is derived from the geometrical model, taking the Gaussian width estimated at a depth of 2% of the Gaussian, with an upper limit of 0.4. NaN if no secondary Gaussian component in the geometric model.

DERIVED_SECONDARY_ECL_DURATION_ERROR : Secondary eclipse: uncertainty on secondary eclipse duration (float)

Secondary eclipse: uncertainty of `derived_secondary_ecl_duration`. Uncertainties are derived by a Jackknife resampling method. In rare instances, the uncertainty can be artificially larger than 1.

DERIVED_SECONDARY_ECL_DEPTH : Secondary eclipse: depth (float, Magnitude[mag])

This value is derived from the geometrical model. NaN if no secondary Gaussian component in the geometric model.

DERIVED_SECONDARY_ECL_DEPTH_ERROR : Secondary eclipse: uncertainty on secondary eclipse depth (float, Magnitude[mag])

Secondary eclipse: uncertainty of `derived_secondary_ecl_depth` estimated by a Jackknife resampling method. It can be null when the model could not derive it.

FREQUENCY_RV_CONSISTENCY_FLAG : Flag if the frequency field is consistent with the dominant frequency found in radial velocity (byte)

If available, a radial velocity (RV) time series is constructed and filtered in the variability pipeline (independently of the non-single star pipeline) retaining only single-line RV transits. A peri-

odogram is computed on this RV time series from which the highest peak is extracted. This flag specifies how this RV data and its periodogram peak relate to the recorded photometrically identified eclipsing binary model frequency:

- 3: RV time series are available from which a periodogram was computed and a frequency half of the eclipsing binary modelling frequency was found.
- 2: RV time series are available from which a periodogram was computed and a frequency double of the eclipsing binary modelling frequency was found.
- 1: RV time series are available from which a periodogram was computed and a frequency consistent with the eclipsing binary modelling frequency was found.
- 0: RV time series are available from which a periodogram was computed and a frequency inconsistent with the eclipsing binary modelling frequency was found.
- -1: RV time series are available, but no periodogram was computed due to insufficient data after filtering for use in the variability pipeline.
- -2: no RV time series are available (usually this means the source is too faint).

The frequency equality is computed by checking the maximum phase offset over the data duration being less than 1 full phase, i.e. $(max(freqRv, freqEb))^2 * \Delta T * abs(1/freqRv - 1/freqEb) < 1$ with ΔT being the data duration. This follows the convention of Dubath et al. (2011) Fig. 1.

SAMPLE_CLASSIFICATION : String-based sample classification (String)

String-based sample classification based on Mowlavi et al. (2023) Table A.1.

BP_USABLE : True when photometric BP time series seems usable for modelling (boolean)

To determine usability of the BP photometric band, the G-band model fit is translated to BP (adjusting for the difference in magnitude distribution), and then refitted with limited degrees of freedom, i.e., fixing the width and position of (a) Gaussian component(s) and/or phase of a cosine component. Then the global ranking for this fit is computed. If it is above 0.45 it is considered of sufficient quality to be tested. Then the global ranking difference between this band and that of G (`global_ranking` in this table) is computed and if it is above −0.156 the BP lightcurve is considered 'usable'. The threshold values have been emperically determined based on lightcurve inspection.

This value is absent when it did not pass the thresholds mentioned above.

RP_USABLE : True when photometric RP time series seems usable for modelling (boolean)

To determine usability of the RP photometric band, the G-band model fit is translated to RP (adjusting for the difference in magnitude distribution), and then refitted with limited degrees of freedom, i.e., fixing the width and position of (a) Gaussian component(s) and/or phase of a cosine component. Then the global ranking for this fit is computed. If it is above 0.45 it is considered of sufficient quality to be tested. Then the global ranking difference between this band and that of G (`global_ranking` in this table) is computed and if it is above −0.156 the RP lightcurve is considered 'usable'. The threshold values have been emperically determined based on lightcurve inspection.

This value is absent when it did not pass the thresholds mentioned above.

DELTA_GLOBAL_RANKING_BP_MIN_G : BP-G global ranking difference of partial refitted model in BP (float)

This is the value of the the global ranking difference between the limited model refit on the BP band and that of G (`global_ranking` in this table). It is only available when `bp_usable` is set to true. See for more details its detailed description.

This value is absent when is was below −0.156 or when `bp_usable` is false.

DELTA_GLOBAL_RANKING_RP_MIN_G : RP-G global ranking difference of partial refitted model in RP (float)

This is the value of the the global ranking difference between the limited model refit on the RP band and that of G (`global_ranking` in this table). It is only available when `bp_usable` is set to true. See for more details its detailed description.

This value is absent when is was below −0.156 or when `rp_usable` is false.

G_SKEWNESS : G-band skewness of the input data (float)

G-band input timeseries skewness.

G_SIGNAL_TO_NOISE : G-band signal-to-noise estimate of the input data (float)

G-band input timeseries signal-to-noise estimate computed in the same way as the `std_dev_over_rms_err_m` field in table `vari_summary`.

G_MEDIAN : G-band median magnitude of the input data (float, Magnitude[mag])

G-band input timeseries median value.

G_NUM_OBS : G-band number of observations of the input data (short)

G-band input timeseries number of observations.

CLASSIFICATION_SCORE : Input classification score of classifier that was used to select eclipsing binaries to process (float)

Classification score of the classifier that was used to select eclipsing binaries to process. This classifier was created at an intermediate point during the processing due to planning constraints and thus its score is provided in this table instead of the usual `vari_classifier_result` table.

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18.7 VARI_EPOCH_FLAGS [DataLink]

This table contains the flags indicative of whether certain observations were used or not for the variable stars processing VariPipe. It can be used in combination with the epoch tables containing the astrometric, photometric or radial velocity data themselves.

Note this table is not available through the main archive TAP interface. Data are delivered via the Massive Data service indexed by the VO DataLink protocol and described in Chapter **??**. The DataLink retrieval type to specify for this product in programmatic archive access via the Astroquery Gaia Python package is `EPOCH_FLAGS_VARI`.

DataLink products are served with a choice of two different data structures:

- `DATAMODEL_STANDARD`: the product is serialised following a standard data model (for example an IVOA-compliant data model in applicable cases);
- `DATAMODEL_GAIA`: the product is serialised following the Gaia DPAC(-internal) data model.

The `DATAMODEL_GAIA` format is described here whereas the `DATAMODEL_STANDARD` format is described in `https://www.cosmos.esa.int/web/gaia-users/archive/datalink-products#datalink_serialisation`

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution identifier, please visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

TRANSIT_ID : Transit Identifier (long)

The `transit_id` is a unique identifier assigned to each detected (and confirmed) source as it transits the Gaia focal plane. Each time a given source is detected as Gaia scans and re-scans the sky a new `transit_id` will be created to badge that apparition. Hence the along-scan time and the across-scan position along with the telescope in which the source was detected are used to form a unique integer with which to label the transit. The detection of transits is done using the SkyMapper CCD data, and the confirmation is done using the AF1 CCD data. Both happens on board.

The several features of a transit that are encoded in `transit_id` can be easily retrieved using bit masks (&) and shifts (>>) as follows:

- On-Board Mission Time line (OBMT) [ns]
  = 204800 * ((`transit_id` >> 17) & (0x000003FFFFFFFFFF))
- Field-of-view = 1 + (`transit_id` >> 15) & 0x03 [1 for 'preceding' and 2 for 'following' fields-of-view respectively]
- CCD row = (`transit_id` >> 12) & 0x07 [dimensionless, in the range 1 to 7]
- Across-scan 'reference acquisition pixel' in strip AF1 = (`transit_id`) & 0x0FFF [pixels] (this is the across-scan centre of the AF1 window and is odd if immediately below the mid-point of the window and even if immediately above)

where the bit mask prefix '0x' denotes hexadecimal.

For further details see Portell et al. (2020). For convenience a decoder for `transit_id` is available on-line at

`https://gaia.esac.esa.int/decoder/transitidDecoder.jsp`

G_CCD_USED : CCD G measurement used for the CCD photometry (Boolean[10] array)

Flag saying whether the CCD measurement was used for the CCD photometry in the variable stars processing. It is an array providing this information for the SM CCD (first index) and the 9 AF CCDs (following nine indices).

G_RFOV_CCD_USED : CCD G measurement used for the G reconstructed transit photometry (Boolean[10] array)

Flag saying whether the CCD measurement was used for the G reconstructed (FoV) transit photometry in the variable stars processing.

BP_USED : BP transit photometry used (Boolean)

Flag saying whether the BP photometry of this transit was used in the variable stars processing.

RP_USED : RP transit photometry used (Boolean)

Flag saying whether the RP photometry of this transit was used in the variable stars processing.

GRVS_USED : G RVS transit photometry used (Boolean)

Flag saying whether the G_{RVS} photometry of this transit was used in the variable stars processing.

RV_USED : Transit radial velocity used (Boolean)

Flag saying whether the radial velocity information of this transit was used in the variable stars processing.

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18.8 VARI_FLARING_STAR [TAP]

This table describes the flaring stars.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

NUM_SEGMENTS : Number of segments (byte)

This is the number of time intervals (segments) in which the magnitude and colour time-series are splitted. The segmentation of time-series is needed because the spots due to the stellar magnetic activity have a life-time shorter than the whole Gaia time-series. The rotational modulation induced by spots can therefore be detected only in segments whose duration is comparable with the spots life-time.

SEGMENTS_START_TIME : Times at which segments start (double[`num_segments`] array, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

An array filled with the starting times of segments.

SEGMENTS_END_TIME : Times at which segments end (double[`num_segments`] array, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

An array filled with the ending times of segments.

SEGMENTS_COLOUR_MAG_INTERCEPT : Colour-Magnitude Intercept in segments (float[num_segments] array)

A robust linear regression is applied to the points (G_{BP}-G_{RP}, G) in each segment. This array is filled with the intercepts given by the fitting procedure in the different segments.

SEGMENTS_COLOUR_MAG_INTERCEPT_ERROR : Colour-Magnitude Intercept uncertainty in segments (float[num_segments] array)

This array is filled with the uncertainties associated with the intercepts given by the fitting procedure.

SEGMENTS_COLOUR_MAG_SLOPE : Colour-Magnitude Slope in segments (float[num_segments] array)

A robust linear regression is applied to the points (G_{BP}-G_{RP}, G) in each segment. This array is filled with the slopes given by the fitting procedure in the different segments.

SEGMENTS_COLOUR_MAG_SLOPE_ERROR : Colour-Magnitude Slope uncertainty in segments (float[num_segments] array)

This array is filled with the uncertainties associated with the slopes given by the fitting procedure.

SEGMENTS_CORRELATION_COEFFICIENT : Correlation coefficient in segments (float[num_segments] array)

The Pearson correlation coefficient r between G_{BP}-G_{RP} and G is computed in each segment. The higher is the Pearson coefficient the higher is the probability that the stellar variability is due to rotational modulation. This array is filled with the Pearson coefficients obtained in the different segments.

SEGMENTS_CORRELATION_SIGNIFICANCE : Correlation coefficient significance in segments (float[num_segments] array)

This array is filled with the statistical significances associated with the Pearson coefficients computed in the different segments. The significance p associated with a given $r = r_0$ gives the probability $P(r \geq r_0)$ that two sets of uncorrelated measurements have a Pearson coefficient $\geq r_0$.

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NUM_OUTLIERS : Number of outliers (short)

The number of outliers detected by the robust linear regression procedure.

OUTLIERS_TIME : Times at which outliers occurs (double[num_outliers] array, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Times at which the detected outliers occurred.

SEGMENTS_QUIET_G_MEAN : Quiet State G magnitude in segment (float[num_segments] array, Magnitude[mag])

For each segment the mean G magnitude is taken as an estimate of the quiet state magnitude (i.e. the magnitude of the stars when no flares occur).

SEGMENTS_QUIET_G_MEAN_ERROR : Error on quiet state G magnitude in segment (float[num_segments] array, Magnitude[mag])

This array stores the photometric errors associated with the quiet state G magnitudes registered in the different segments.

SEGMENTS_QUIET_BP_MEAN : Quiet State BP magnitude in segment (float[num_segments] array, Magnitude[mag])

For each segment the mean BP magnitude is taken as an estimate of the quiet state magnitude (i.e. the magnitude of the stars when no flares occur).

SEGMENTS_QUIET_BP_MEAN_ERROR : Error on quiet state BP magnitude in segment (float[num_segments] array, Magnitude[mag])

This array stores the photometric errors associated with the quiet state BP magnitudes registered in the different segments.

SEGMENTS_QUIET_RP_MEAN : Quiet State RP magnitude in segment (float[num_segments] array, Magnitude[mag])

For each segment the mean RP magnitude is taken as an estimate of the quiet state magnitude

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(i.e. the magnitude of the stars when no flares occur).

SEGMENTS_QUIET_RP_MEAN_ERROR : Error on quiet state RP magnitude in segment (float[num_segments] array, Magnitude[mag])

This array stores the photometric errors associated with the quiet state RP magnitudes registered in the different segments.

NUM_FLARES : Number of flares detected (byte)

The number of flares detected by the algorithm.

FLARES_TIME : Times at which flares occurred (double[num_flares] array, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

The times at which flares occurred.

FLARES_G_AMPLITUDE : G Amplitude of flare events (float[num_flares] array, Magnitude[mag])

For each flare the amplitude in the G band $\Delta G_{flare}$is computed by means of the equation:

$$\Delta G_{flare} = G_{flare} - G_{quiet} \tag{9}$$

where G_{flare} is the magnitude registered at the time of flare event and G_{quiet} is the magnitude of the quiet state.

If the fit coefficients of the model represented by the equation in the definition of segments_g_cos_term are available, then the G_{quiet} magnitude is estimated as mag(t_{flare}) where t_{flare} is the time at which the flare occurred.

FLARES_G_AMPLITUDE_ERROR : Error on the G amplitude of the flare events (float[num_flares] array, Magnitude[mag])

The error on the G amplitude of a given flare is computed according the equation:

$$\sigma_{\Delta G} = \sqrt{\sigma^2_{Gflare} + \sigma^2_{Gquiet}} \tag{10}$$

FLARES_BP_AMPLITUDE : BP Amplitude of flare events (float[num_flares] array, Magnitude[mag])

For each flare the amplitude in the BP band $\Delta BP_{flare}$is computed by means of the equation:

$$\Delta BP_{flare} = BP_{flare} - BP_{quiet} \tag{11}$$

where BP_{flare} is the magnitude registered at the time of flare event and BP_{quiet} is the magnitude of the quiet state.

If the fit coefficients of the model represented by the equation in the definition of segments_bp_cos_term are computed, then the BP_{quiet} magnitude is estimated as mag(t_{flare}) where t_{flare} is the time at which the flare occurred.

FLARES_BP_AMPLITUDE_ERROR : Error on the BP amplitude of the flare events (float[num_flares] array, Magnitude[mag])

The error on the BP amplitude of a given flare is computed according the equation:

$$\sigma_{\Delta BP} = \sqrt{\sigma^2_{BPflare} + \sigma^2_{BPquiet}} \tag{12}$$

FLARES_RP_AMPLITUDE : RP Amplitude of flare events (float[num_flares] array, Magnitude[mag])

For each flare the amplitude in the RP band $\Delta RP_{flare}$is computed by means of the equation:

$$\Delta RP_{flare} = RP_{flare} - RP_{quiet} \tag{13}$$

where RP_{flare} is the magnitude registered at the time of flare event and G_{quiet} is the magnitude of the quiet state.

If the fit coefficients of the model represented by the equation in the definition of segments_rp_cos_term are computed, then the RP_{quiet} magnitude is estimated as mag(t_{flare}) where t_{flare} is the time at which the flare occurred.

FLARES_RP_AMPLITUDE_ERROR : Error on the RP amplitude of the flare events (float[num_flares] array, Magnitude[mag])

The error on the RP amplitude of a given flare is computed according the equation:

$$\sigma_{\Delta RP} = \sqrt{\sigma^2_{RPflare} + \sigma^2_{RPquiet}} \tag{14}$$

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FLARES_BP_MINUS_RP_AMPLITUDE : BP-RP amplitudes of the flare events (float[num_flares] array, Magnitude[mag])

For each flare the amplitude in the BP-RP colour $\Delta(BP-RP)_{flare}$ is computed by means of the equation:

$$\Delta(BP-RP)_{flare} = (BP-RP)_{flare} - (BP-RP)_{quiet} \tag{15}$$

where $(BP-RP)_{flare}$ is the colour registered at the time of flare event and $(BP-RP)_{quiet}$ is the colour of the quiet state.

FLARES_BP_MINUS_RP_AMPLITUDE_ERROR : BP-RP amplitude uncertainties (float[num_flares] array, Magnitude[mag])

The error on the (BP-RP) amplitude of a given flare is computed according the equation:

$$\sigma_{\Delta(BP-RP)} = \sqrt{\sigma^2_{(BP-RP)flare} + \sigma^2_{(BP-RP)quiet}} \tag{16}$$

REFERENCE_SEGMENT : Ordinal number of segments for detected flares (byte[num_flares] array)

This array stores for each flare the ordinal number of the segment in which the flare has been detected.

REFERENCE_MODEL : Model number used to compute quiet state (byte[num_flares] array)

This array stores for each flare a flag indicating the kind of model used to compute the quiet state of the star. 0 means that the quiet state has been computed by the average magnitude in the segment (segments_quiet_g_mean). 1 indicates that the quiet state has been computed through the model mag(t)= a + b * cos(2*pi/P*t) + c * sin(2*pi/P*t) where a, b,c, P are respectively the A0Term, the CosTerm, the SinTerm, and the rotation period for the segment, as published in the associated rotation modulation table.

RELATIVE_FLUX_G : Relative fluxes of flares in the G band (float[num_flares] array)

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The array stores the relative fluxes of the flares in the G band, i.e. the quantities

$$\frac{F(G)_{flare} - F(G)_{quiet}}{F(G)_{quiet}}$$

where $F(G)_{flare}$ is the flux of the stars at the time at which the flare occurs and $F(G)_{quiet}$ is the flux of the quiet status.

RELATIVE_FLUX_BP : Relative fluxes of flares in the BP band (float[num_flares] array)

The array stores the relative fluxes of the flares in the BP band, i.e. the quantities

$$\frac{F(BP)_{flare} - F(BP)_{quiet}}{F(BP)_{quiet}}$$

where $F(BP)_{flare}$ is the flux of the stars at the time at which the flare occurs and $F(BP)_{quiet}$ is the flux of the quiet status.

RELATIVE_FLUX_RP : Relative fluxes of flares in the RP band (float[num_flares] array)

The array stores the relative fluxes of the flares in the RP band, i.e. the quantities

$$\frac{F(RP)_{flare} - F(RP)_{quiet}}{F(RP)_{quiet}}$$

where $F(RP)_{flare}$ is the flux of the stars at the time at which the flare occurs and $F(RP)_{quiet}$ is the flux of the quiet status.

ABSOLUTE_FLUX_G : Absolute fluxes of flares in the G band (float[num_flares] array, Flux[e^- s^{-1}])

The array stores the absolute fluxes of the flares in the G band, i.e. the quantities

$$F(G)_{flare} - F(G)_{quiet}$$

where $F(G)_{flare}$ is the flux of the stars at the time at which the flare occurs and $F(G)_{quiet}$ is the flux of the quiet status.

ABSOLUTE_FLUX_BP : Absolute fluxes of flares in the BP band (float[num_flares] array, Flux[e^- s^{-1}])

The array stores the absolute fluxes of the flares in the BP band, i.e. the quantities

$$F(BP)_{flare} - F(BP)_{quiet}$$

where $F(BP)_{flare}$ is the flux of the stars at the time at which the flare occurs and $F(BP)_{quiet}$ is the flux of the quiet status.

ABSOLUTE_FLUX_RP : Absolute fluxes of flares in the RP band (float[num_flares] array, Flux[e^- s^{-1}])

The array stores the absolute fluxes of the flares in the *RP* band, i.e. the quantities

$$F(RP)_{flare} - F(RP)_{quiet}$$

where $F(RP)_{flare}$ is the flux of the stars at the time at which the flare occurs and $F(RP)_{quiet}$ is the flux of the quiet status.

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18.9 VARI_GVD_CLASSIFIER_FEATURE [TAP]

Table with features used by the General Variability Detection (GVD) classifiers.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

REDUCED_CHI_SQUARED_MAG_G_FOV_PVAL : G mag reduced chi-squared right-tailed p-value (float, Dimensionless[percentage/100])

The p-value of high reduced chi-squared values using field-of-view transit magnitudes in the G band.

STETSON_MAG_G_FOV_PVAL : G mag single-band Stetson right-tailed p-value (float, Dimensionless[percentage/100])

The p-value of high single-band Stetson values using field-of-view transit magnitudes in the G band, pairing measurements within 0.1 days.

OUTLIER_MEDIAN_G_FOV_PVAL : G mag outlier median right-tailed p-value (float, Dimensionless[percentage/100])

The p-value of high outlier median values, each defined as the greatest absolute deviation from the median normalized by the error, using field-of-view transit magnitudes in the G band.

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OUTLIER_MEDIAN_BP_RP_PVAL : BP-RP colour mag outlier median right-tailed p-value (float, Dimensionless[percentage/100])

The p-value of high outlier median values, each defined as the greatest absolute deviation from the median normalized by the error, using the BP-RP colour magnitudes.

IQR_MAG_G_FOV_PVAL : G mag IQR right-tailed p-value (float, Dimensionless[percentage/100])

The p-value of high interquartile range values using field-of-view transit magnitudes in the G band.

LOG_QSO_VAR : Quasar variability metric (float)

The decadic logarithm of the reduced chi-squared of field-of-view transit magnitudes in the G band with respect to a parametrized quasar variance model, represented by log10(chi2qso/nu) in Butler & Bloom (2011). See **?** for details on the parameter values for the Gaia data.

SPEARMAN_BP_RP_PVAL : Spearman rank correlation right-tailed p-value between BP and RP time series (float, Dimensionless[percentage/100])

The p-value of high Spearman correlations between the BP and RP magnitudes.

PEARSON_BP_RP : Pearson correlation coefficient between BP and RP time series (float)

The Pearson correlation coefficient between the BP and RP magnitudes.

LOG_RANGE_MAG_G_FOV : Log10 of the difference between the highest and the lowest G mag (float)

The decadic logarithm of the difference between the maximum an minimum field-of-view transit magnitudes in the G band.

MEDIAN_MAG_BP_MINUS_RP : BP-RP colour mag using the medians of each band (float, Magnitude[mag])

The possibly reddened colour index from the difference of median magnitudes in the BP and RP bands.

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MEDIAN_MAG_G_FOV : Median of G magnitudes (float, Magnitude[mag])

The unweighted median magnitude of field-of-view transits in the G band.

MEDIAN_MAG_RP : Median of RP magnitudes (float, Magnitude[mag])

The unweighted median magnitude of field-of-view transits in the RP band.

STD_DEV_MAG_BP_OVER_RP : Ratio of the standard deviations in BP and RP mag (float)

The ratio of the standard deviations in magnitude of field-of-view transits in the BP and RP bands.

LOG_STD_DEV_MAG_G_CCD : Log10 of the standard deviation of per-CCD G mag (float)

The decadic logarithm of the standard deviation of per-CCD magnitudes in the G band.

LOG_STD_DEV_MAG_G_FOV : Log10 of the standard deviation of per-FoV G mag (float)

The decadic logarithm of the standard deviation of field-of-view transit magnitudes in the G band.

KURTOSIS_MAG_G_FOV_RIGHT_PVAL : G mag right-tailed kurtosis p-value (float, Dimensionless[percentage/100])

The right-tailed p-value of the kurtosis using field-of-view transit magnitudes in the G band.

KURTOSIS_MAG_G_FOV_LEFT_PVAL : G mag left-tailed kurtosis p-value (float, Dimensionless[percentage/100])

The left-tailed p-value of the kurtosis using field-of-view transit magnitudes in the G band.

SKEWNESS_MAG_G_FOV_RIGHT_PVAL : G mag right-tailed skewness p-value (float, Dimensionless[percentage/100])

The right-tailed p-value of the skewness using field-of-view transits in the G-band magnitude.

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SKEWNESS_MAG_G_FOV_LEFT_PVAL : G mag left-tailed skewness p-value (float, Dimensionless[percentage/100])

The left-tailed p-value of the skewness using field-of-view transits in the G-band magnitude.

SKEWNESS_FLUX_G_FOV_LEFT_PVAL : G flux left-tailed skewness p-value (float, Dimensionless[percentage/100])

The left-tailed p-value of the skewness using field-of-view transits in the G-band flux.

ABBE_MAG_G_FOV_RIGHT_PVAL : G mag right-tailed Abbe p-value (float, Dimensionless[percentage/100])

The right-tailed p-value of Abbe using field-of-view transit magnitudes in the G band.

ABBE_MAG_G_FOV_LEFT_PVAL : G mag left-tailed Abbe p-value (float, Dimensionless[percentage/100])

The left-tailed p-value of Abbe using field-of-view transit magnitudes in the G band.

IS_PRIMARY : Flags photometric processing primary sources (float)

Flag indicating the photometric processing status of the source as primary or secondary. It takes two values: 0 for secondary and 1 for primary sources.

GVD_PROBABILITY : GVD probability (float, Dimensionless[percentage/100])

GVD probability of being variable.

STETSON_MAG_BP_RP_PVAL : BP vs RP double-band Stetson right-tailed p-value (float, Dimensionless[percentage/100])

The right-tailed p-value of the double-band Stetson index, pairing measurements in the BP and RP bands within 0.01 days.

OUTLIER_MEDIAN_G_RP_PVAL : G-RP colour mag outlier median right-tailed p-value (float,

Dimensionless[percentage/100])

The right-tailed p-value of the outlier median, defined as the greatest absolute deviation from the median normalized by the error, using the G-RP colour magnitudes.

NON_QSO_PROB : Non-quasar variability metric (float)

A quantity distributed according to the null-hypothesis distribution of chi2qso for non-quasar objects, computed from a parametrized quasar variance model with magnitudes of field-of-view transits in the G band, related to P(chi2qso, not quasar) in Butler & Bloom (2011); see **?** for details on the parameter values for the Gaia data.

SPEARMAN_G_IPD : Spearman rank correlation coefficient between G and IPD time series (float)

The Spearman correlation coefficient between the magnitudes of field-of-view transits in the G band and the Image Parameter Determination goodness of fit model sampled at the scan angles of the time-series observations (see Section 6.1 of Holl et al. 2023). This field is also listed as `phot_g_corr_ipd` in the `spurious_signal` table.

ABBE_FOLDED_MAG_G_FOV : G-mag folded Abbe value (float)

The Abbe value folded with a period in the range of 0.04–2500 d, using field-of-view transit magnitudes in the G band.

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18.10 VARI_LONG_PERIOD_VARIABLE [TAP]

This table describes the Long Period Variable stars.

Some entries can be NaN when absent.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

SCORE_LPV : A score reflecting the consistency of periodicity detections (byte)

Each source is assigned a score from 1 to 4 based on whether the listed periods are detected consistently across the photometric and/or radial-velocity time series. Score = 4: at least one of the listed periods is detected in all three photometric time series and a model is available for the RV time series (with a period not necessarily compatible with the photometric ones). Score = 3: at least one of the listed periods is detected in all three photometric time series, but no period is found for the RV time series. Score = 2: none of the listed periods is detected in all three photometric time series, but at least one of the periods is detected in G and RP. Score = 1: all other cases, including sources for which none of the listed periods is detected in G, but still detected in both RP and BP. The subsample with score=1 thus contains sources with published time-series models.

NUM_RP_SPECTRA_VALID_ABBE : Number of overall valid RP spectra, based on Abbe value (short)

Number of spectra that are processed for the identification of the LPV chemical type. The validation of the spectra is based, among other parameters, on the Abbe value within a pseudo wavelength subrange. See **?** for the details.

NUM_RP_SPECTRA_TWO_VALID_MAXIMA : Number of RP spectra having valid pseudo-wavelength locations of the two highest peaks (short)

Number of RP spectra having valid pseudo-wavelength locations of the two highest peaks. See **?** for the details.

MEDIAN_WL_RP_MAXIMUM : Median of the pseudo-wavelengths of the maximum in RP spectra (float)

It gives the position (in uncalibrated pseudo-wavelength) of the maximum flux, derived from all spectra having a valid maximum. See **?** for the details.

MEDIAN_WL_RP_MINIMUM : Median of the pseudo-wavelengths of the minimum in RP spectra located between the two highest maxima (float)

It gives the position (in uncalibrated pseudo-wavelength) of the minimum flux between the two highest maxima, derived from all spectra having two valid maxima. See **?** for the details.

MEDIAN_RP_FLUX_RATIO_MAX_MAX : Median of the flux ratio between the two highest maxima (float)

It gives the ratio of the fluxes of the two highest maxima, derived from all spectra having two valid maxima. See **?** for the details.

MEDIAN_RP_FLUX_RATIO_MAX_MIN : Median of the flux ratio between the highest maximum and the minimum between the two highest maxima (float)

It gives the ratio of the flux at the highest maximum to the flux at the minimum between the two maxima, derived from all spectra having two valid maxima. See **?** for the details.

MEDIAN_DELTA_WL_RP : Median of the pseudo-wavelength separations between the two highest peaks in RP spectra (float)

Median among all $G_{\rm RP}$ epoch spectra of the pseudo-wavelength separations between the two highest peaks in the $G_{\rm RP}$ spectra. It is set to Null when the spectrum does not allow an automatic identification of two maxima.

Draft

CHEMICAL_TYPE : Character indicating whether it is a O-rich or C-rich star (string)

The parameter `chemical_type` is set to 'O' for O-rich stars and to 'C' for C-rich stars. It is set to NULL when the shape of the spectrum cannot allow an automatic classification between these two types of LPVs.

A note on S-type stars: the method cannot classify correctly the nature of these stars. As a consequence, they may have the `chemical_type` parameter set to either 'O' or 'C' (or NULL).

MODEL_REFERENCE_TIME : Reference time used in the Fourier Series models of the time series (G FoV, RP, BP, and RV where applicable) (float, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Reference time relative to which the phases are provided. The same reference time is used for all time series.

MODEL_PHOT_NUM_FREQUENCIES : Number of frequencies included in the models of the photometric time series (G FoV, RP, BP) (byte)

The number of frequencies includes harmonics.

MODEL_PHOT_FREQUENCY : Frequency in the Fourier Series model of the photometric time series (G FoV, RP, BP) (double[`model_phot_num_frequencies`] array, Frequency[day^{-1}])

Frequency of each Fourier Series component, including harmonics. The same frequencies (and harmonics) are used for all photometric time series.

MODEL_PHOT_FREQUENCY_ERROR : Uncertainty on the frequency in the model of the photometric time series (G FoV, RP, BP) (float[`model_phot_num_frequencies`] array, Frequency[day^{-1}])

Can contain null values for corresponding null elements in the array of frequencies.

MODEL_PHOT_MULTIBAND_INFO : A four digit integer array indicating a compatible frequency detection in G, RP, BP, RV time series, respectively (int[`model_phot_num_frequencies/2`] array)

An integer with value 1101, for example, indicates the detection of the given frequency in G, RP and RV, but not in BP. Two frequencies of two time series are considered compatible if the phase difference obtained at the last cycle of each time series with the other frequency is less than a given value (see documentation). A value of 2 for RV indicates that the frequency of the considered photometric band is twice freq(RV).

MODEL_G_MEAN_MAG : The constant term in the Fourier Series modeling the G FoV time serie (float, Magnitude[mag])

Constant coefficient of the Fourier Series.

MODEL_G_MEAN_MAG_ERROR : Uncertainty on the constant term in the Fourier Series modeling the G FoV time series (float, Magnitude[mag])

Uncertainty on the constant coefficient of the Fourier Series.

MODEL_G_AMPLITUDE : Amplitude in the Fourier Series model of the G FoV time series (float[`model_phot_num_frequencies`] array, Magnitude[mag])

Half peak-to-peak amplitude of each Fourier Series component. Can contain null values for corresponding null elements in the array of frequencies.

MODEL_G_AMPLITUDE_ERROR : Uncertainty on the amplitude in the Fourier Series model of the G FoV time series (float[`model_phot_num_frequencies`] array, Magnitude[mag])

Can contain null values for corresponding null elements in the array of frequencies.

MODEL_G_PHASE : Phase in the Fourier Series model of the G FoV time series (float[`model_phot_num_freq`
array)

Phase between 0 and 1 of each Fourier Series component relative to the reference time. Can contain null values for corresponding null elements in the array of frequencies.

MODEL_G_PHASE_ERROR : Uncertainty on the phase in the Fourier Series model of the G FoV time series (float[`model_phot_num_frequencies`] array)

Can contain null values for corresponding null elements in the array of frequencies.

Draft

MODEL_G_QR5_RESIDUALS : Trimmed range of the G FoV residual time series (float, Magnitude[mag])

5-95

MODEL_RP_MEAN_MAG : The constant term in the Fourier Series modeling the RP time series (float, Magnitude[mag])

Constant coefficient of the Fourier Series.

MODEL_RP_MEAN_MAG_ERROR : Uncertainty on the constant term in the Fourier Series modeling the RP time series (float, Magnitude[mag])

Uncertainty on the constant coefficient of the Fourier Series.

MODEL_RP_AMPLITUDE : Amplitude in the Fourier Series model of the RP time series (float[model_phot_nu array, Magnitude[mag])

Half peak-to-peak amplitude of each Fourier Series component. Can contain null values for corresponding null elements in the array of frequencies.

MODEL_RP_AMPLITUDE_ERROR : Uncertainty on the amplitude in the Fourier Series model of the RP time series (float[model_phot_num_frequencies] array, Magnitude[mag])

Can contain null values for corresponding null elements in the array of frequencies.

MODEL_RP_PHASE : Phase in the Fourier Series model of the RP time series (float[model_phot_num_freque array)

Phase between 0 and 1 of each Fourier Series component relative to the reference time. Can contain null values for corresponding null elements in the array of frequencies.

MODEL_RP_PHASE_ERROR : Uncertainty on the phase in the Fourier Series model of the RP time series (float[model_phot_num_frequencies] array)

Can contain null values for corresponding null elements in the array of frequencies.

Draft

MODEL_RP_QR5_RESIDUALS : Trimmed range of the RP residual time series (float, Magnitude[mag])

5-95

MODEL_BP_MEAN_MAG : The constant term in the Fourier Series modeling the BP time series (float, Magnitude[mag])

Constant coefficient of the Fourier Series.

MODEL_BP_MEAN_MAG_ERROR : Uncertainty on the constant term in the Fourier Series modeling the BP time series (float, Magnitude[mag])

Uncertainty on the constant coefficient of the Fourier Series.

MODEL_BP_AMPLITUDE : Amplitude in the Fourier Series model of the BP time series (float[model_phot_nu array, Magnitude[mag])

Half peak-to-peak amplitude of each Fourier Series component. Can contain null values for corresponding null elements in the array of frequencies.

MODEL_BP_AMPLITUDE_ERROR : Uncertainty on the amplitude in the Fourier Series model of the BP time series (float[model_phot_num_frequencies] array, Magnitude[mag])

Can contain null values for corresponding null elements in the array of frequencies.

MODEL_BP_PHASE : Phase in the Fourier Series model of the BP time series (float[model_phot_num_freque array)

Phase between 0 and 1 of each Fourier Series component relative to the reference time. Can contain null values for corresponding null elements in the array of frequencies.

MODEL_BP_PHASE_ERROR : Uncertainty on the phase in the Fourier Series model of the BP time series (float[model_phot_num_frequencies] array)

Can contain null values for corresponding null elements in the array of frequencies.

Draft

MODEL_BP_QR5_RESIDUALS : Trimmed range of the BP residual time series (float, Magnitude[mag])

5-95

MODEL_RV_NUM_FREQUENCIES : Number of frequencies included in the model of the radial velocity time series (byte)

The number of frequencies includes harmonics.

MODEL_RV_MEAN : The constant term in the Fourier Series modeling the radial velocity time series (float, Velocity[km s^{-1}])

Constant coefficient of the Fourier Series.

MODEL_RV_MEAN_ERROR : Uncertainty on the constant term in the Fourier Series modeling the radial velocity time series (float, Velocity[km s^{-1}])

Uncertainty on the constant coefficient of the Fourier Series.

MODEL_RV_FREQUENCY : Frequency in the Fourier Series model of the radial velocity time series (double[`model_rv_num_frequencies`] array, Frequency[day^{-1}])

Frequency of each Fourier Series component, including harmonics. Can contain null values when a harmonic is missing due to insignificant detection.

MODEL_RV_FREQUENCY_ERROR : Uncertainty on the frequency in the model of the radial velocity time series (float[`model_rv_num_frequencies`] array, Frequency[day^{-1}])

Can contain null values for corresponding null elements in the array of frequencies.

MODEL_RV_AMPLITUDE : Amplitude in the Fourier Series model of the radial velocity time series (float[`model_rv_num_frequencies`] array, Velocity[km s^{-1}])

Half peak-to-peak amplitude of each Fourier Series component. Can contain null values for corresponding null elements in the array of frequencies.

Draft

MODEL_RV_AMPLITUDE_ERROR : Uncertainty on the amplitude in the Fourier Series model of the radial velocity time series (float[`model_rv_num_frequencies`] array, Velocity[km s^{-1}])

Can contain null values for corresponding null elements in the array of frequencies.

MODEL_RV_PHASE : Phase in the Fourier Series model of the radial velocity time series (float[`model_rv_num_` array)

Phase between 0 and 1 of each Fourier Series component relative to the reference time. Can contain null values for corresponding null elements in the array of frequencies.

MODEL_RV_PHASE_ERROR : Uncertainty on the phase in the Fourier Series model of the radial velocity time series (float[`model_rv_num_frequencies`] array)

Can contain null values for corresponding null elements in the array of frequencies.

MODEL_RV_QR5_RESIDUALS : Trimmed range of the radial velocity residual time series (float, Velocity[km s^{-1}])

5-95

Draft

18.11 VARI_MICROLENSING [TAP]

This table describes the microlensing events.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

PACZYNSKI0_G0 : G-band magnitude baseline (level 0) (float, Magnitude[mag])

G-band magnitude of the baseline from the Paczynski standard model fit without blending.

PACZYNSKI0_G0_ERROR : Error of G-band magnitude baseline (level 0) (float, Magnitude[mag])

Error of the *G*-band magnitude of the baseline from the Paczynski standard model fit without blending.

PACZYNSKI0_BP0 : BP-band magnitude baseline (level 0) (float, Magnitude[mag])

$G_{\rm BP}$-band magnitude of the baseline from the Paczynski standard model fit without blending.

PACZYNSKI0_BP0_ERROR : Error of BP-band magnitude baseline (level 0) (float, Magnitude[mag])

Error of the $G_{\rm BP}$-band magnitude of the baseline from the Paczynski standard model fit without

blending.

PACZYNSKI0_RP0 : RP-band magnitude baseline (level 0) (float, Magnitude[mag])

G_{RP}-band magnitude of the baseline from the Paczynski standard model fit without blending.

PACZYNSKI0_RP0_ERROR : Error of RP-band magnitude baseline (level 0) (float, Magnitude[mag])

Error of the G_{RP}-band magnitude of the baseline from the Paczynski standard model fit without blending.

PACZYNSKI0_U0 : Impact parameter (level 0) (double)

Impact parameter from the Paczynski standard model fit without blending.

PACZYNSKI0_U0_ERROR : Error of the impact parameter (level 0) (double)

Error of the impact parameter from the Paczynski standard model fit without blending.

PACZYNSKI0_TE : Event timescale (level 0) (float, Time[day])

Event time-scale in days from the Paczynski standard model fit without blending.

PACZYNSKI0_TE_ERROR : Error of event timescale (level 0) (float, Time[day])

Error of the event time-scale in days from the Paczynski standard model fit without blending.

PACZYNSKI0_TMAX : Time of maximum amplification (level 0) (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Time of the maximum amplification from the Paczynski standard model fit without blending.

PACZYNSKI0_TMAX_ERROR : Error of time of maximum amplification (level 0) (double, Time[day])

Draft

Error of time of the maximum amplification from the Paczynski standard model fit without blending.

PACZYNSKI0_CHI2 : Chi square of level 0 Paczynski fit (float)

Chi square of the Paczynski standard model fit without blending.

PACZYNSKI0_CHI2_DOF : Reduced chi square of level 0 Paczynski fit (float)

Reduced chi square of the Paczynski standard model fit without blending.

PACZYNSKI1_G0 : G-band magnitude baseline (level 1) (float, Magnitude[mag])

G-band magnitude of the baseline from the Paczynski standard model fit with blending.

PACZYNSKI1_G0_ERROR : Error of G-band magnitude baseline (level 1) (float, Magnitude[mag])

Error of the *G*-band magnitude of the baseline from the Paczynski standard model fit with blending.

PACZYNSKI1_BP0 : BP-band magnitude baseline (level 1) (float, Magnitude[mag])

G_{BP}-band magnitude of the baseline from the Paczynski standard model fit with blending.

PACZYNSKI1_BP0_ERROR : Error of BP-band magnitude baseline (level 1) (float, Magnitude[mag])

Error of the G_{BP}-band magnitude of the baseline from the Paczynski standard model fit with blending.

PACZYNSKI1_RP0 : RP-band magnitude baseline (level 1) (float, Magnitude[mag])

G_{RP}-band magnitude of the baseline from the Paczynski standard model fit with blending.

Draft

PACZYNSKI1_RP0_ERROR : Error of RP-band magnitude baseline (level 1) (float, Magnitude[mag])

Error of the G_{RP}-band magnitude of the baseline from the Paczynski standard model fit with blending.

PACZYNSKI1_U0 : Impact parameter (level 1) (double)

Impact parameter from the Paczynski standard model fit with blending.

PACZYNSKI1_U0_ERROR : Error of the impact parameter (level 1) (double)

Error of the impact parameter from the Paczynski standard model fit with blending.

PACZYNSKI1_TE : Event timescale (level 1) (float, Time[day])

Event time-scale in days from the Paczynski standard model fit with blending.

PACZYNSKI1_TE_ERROR : Error of event timescale (level 1) (float, Time[day])

Error of the event time-scale in days from the Paczynski standard model fit with blending.

PACZYNSKI1_TMAX : Time of maximum amplification (level 1) (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Time of the maximum amplification from the Paczynski standard model fit with blending.

PACZYNSKI1_TMAX_ERROR : Error of time of maximum amplification (level 1) (double, Time[day])

Error of time of the maximum amplification from the Paczynski standard model fit with blending.

PACZYNSKI1_FS_G : Blending factor in G-band (level 1) (float)

Blending factor in *G*-band from the Paczynski standard model fit with blending.

Draft

PACZYNSKI1_FS_G_ERROR : Error of blending factor in G-band (level 1) (float)

Error of the blending factor in *G*-band from the Paczynski standard model fit with blending.

PACZYNSKI1_FS_BP : Blending factor in BP-band (level 1) (float)

Blending factor in G_{BP}-band from the Paczynski standard model fit with blending.

PACZYNSKI1_FS_BP_ERROR : Error of blending factor in BP-band (level 1) (float)

Error of the blending factor in G_{BP}-band from the Paczynski standard model fit with blending.

PACZYNSKI1_FS_RP : Blending factor in RP-band (level 1) (float)

Blending factor in G_{RP}-band from the Paczynski standard model fit with blending.

PACZYNSKI1_FS_RP_ERROR : Error of blending factor in RP-band (level 1) (float)

Error of the blending factor in G_{RP}-band from the Paczynski standard model fit with blending.

PACZYNSKI1_CHI2 : Chi square of level 1 Paczynski fit (float)

Chi square of the Paczynski standard model fit with blending.

PACZYNSKI1_CHI2_DOF : Reduced chi square of level 1 Paczynski fit (float)

Reduced chi square of the Paczynski standard model fit with blending.

PACZYNSKI2_G0 : G-band magnitude baseline (level 2) (float, Magnitude[mag])

G-band magnitude of the baseline from the Paczynski microlensing parallax model without blending.

PACZYNSKI2_G0_ERROR : Error of G-band magnitude baseline (level 2) (float, Magnitude[mag])

Error of the G-band magnitude of the baseline from the Paczynski microlensing parallax model without blending.

PACZYNSKI2_BP0 : BP-band magnitude baseline (level 2) (float, Magnitude[mag])

BP-band magnitude of the baseline from the Paczynski microlensing parallax model without blending.

PACZYNSKI2_BP0_ERROR : Error of BP-band magnitude baseline (level 2) (float, Magnitude[mag])

Error of the BP-band magnitude of the baseline from the Paczynski microlensing parallax model without blending.

PACZYNSKI2_RP0 : RP-band magnitude baseline (level 2) (float, Magnitude[mag])

RP-band magnitude of the baseline from the Paczynski microlensing parallax model without blending.

PACZYNSKI2_RP0_ERROR : Error of RP-band magnitude baseline (level 2) (float, Magnitude[mag])

Error of the RP-band magnitude of the baseline from the Paczynski microlensing parallax model without blending.

PACZYNSKI2_U0 : Impact parameter (level 2) (double)

Impact parameter from the Paczynski microlensing parallax model without blending.

PACZYNSKI2_U0_ERROR : Error of the impact parameter (level 2) (double)

Error of the impact parameter from the Paczynski microlensing parallax model without blending.

PACZYNSKI2_TE : Event timescale (level 2) (float, Time[day])

Event time-scale in days from the Paczynski microlensing parallax model without blending.

Draft

PACZYNSKI2_TE_ERROR : Error of event timescale (level 2) (float, Time[day])

Error of the event time-scale in days from the Paczynski microlensing parallax model without blending.

PACZYNSKI2_TMAX : Time of maximum amplification (level 2) (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Time of the maximum amplification from the Paczynski microlensing parallax model without blending.

PACZYNSKI2_TMAX_ERROR : Error of time of maximum amplification (level 2) (double, Time[day])

Error of time of the maximum amplification from the Paczynski microlensing parallax model without blending.

PACZYNSKI2_PARALLAX_NORTH : North component of microlensing parallax (level 2) (float)

North component of the microlensing parallax from the Paczynski microlensing parallax model without blending.

The microlensing parallax is defined as

$\pi_E = \pi_{\rm rel}/\theta_E$, where $\pi_{\rm rel} = \pi_{\rm lens} - \pi_{\rm source}$ and θ_E is the Einstein radius in mas.

PACZYNSKI2_PARALLAX_NORTH_ERROR : Error of North component of microlensing parallax (level 2) (float)

Error of the North component of the microlensing parallax from the Paczynski microlensing parallax model without blending.

PACZYNSKI2_PARALLAX_EAST : East component of microlensing parallax (level 2) (float)

East component of the microlensing parallax from the Paczynski microlensing parallax model without blending.

Draft

The microlensing parallax is defined as

$\pi_E = \pi_{\rm rel}/\theta_E$, where $\pi_{\rm rel} = \pi_{\rm lens} - \pi_{\rm source}$ and θ_E is the Einstein radius in mas.

PACZYNSKI2_PARALLAX_EAST_ERROR : Error of East component of microlensing parallax (level 2) (float)

Error of the East component of the microlensing parallax from the Paczynski microlensing parallax model without blending.

PACZYNSKI2_CHI2 : Chi square of level 2 Paczynski fit (float)

Chi square of the Paczynski microlensing parallax model without blending.

PACZYNSKI2_CHI2_DOF : Reduced chi square of level 2 Paczynski fit (float)

Reduced chi square of the Paczynski microlensing parallax model without blending.

PACZYNSKI2_T0_PARALLAX : Moment of time for which the geocentric parallax model was calculated (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Moment of time for which the geocentric parallax model was calculated.

PACZYNSKI3_CHI2 : Chi square of level 3 Paczynski linear fit to astrometric data (float)

Chi square of level 3 Paczynski linear model fit to astrometric data.

PACZYNSKI3_CHI2_DOF : Reduced chi square of level 3 Paczynski linear fit to astrometric data (float)

Reduced chi square of level 3 Paczynski fit, performed solely on astrometric data which consists of level 2 microlensing parameters and linearly added astrometric parameter (Einstein Radius).

PACZYNSKI3_DELTA_RA_SOURCE : Source position offset in RA from astrometric linear model (float, Angle[mas])

Draft

Source position offset in Right Ascension ($\Delta\alpha^*$=$\Delta\alpha \cos\delta$) from the microlensing astrometric linear model.

PACZYNSKI3_DELTA_RA_SOURCE_ERROR : Error of source position offset in RA from astrometric linear model (float, Angle[mas])

Error of source position offset in Right Ascension ($\sigma_{\Delta\alpha^*} = \sigma_{\Delta\alpha\cos\delta}$) from the microlensing astrometric linear model.

PACZYNSKI3_DELTA_DEC_SOURCE : Source position offset in Dec from astrometric linear model (float, Angle[mas])

Source position offset in declination from the microlensing astrometric linear model.

PACZYNSKI3_DELTA_DEC_SOURCE_ERROR : Error of source position offset in Dec from astrometric linear model (float, Angle[mas])

Error of source position offset in declination from the microlensing astrometric linear model.

PACZYNSKI3_VARPI_SOURCE : Source parallax from astrometric linear model (float, Angle[mas])

Source parallax from the microlensing astrometric linear model.

PACZYNSKI3_VARPI_SOURCE_ERROR : Error of source parallax from astrometric linear model (float, Angle[mas])

Error of source parallax from the microlensing astrometric linear model.

PACZYNSKI3_PMRA_SOURCE : Source proper motion in RA from astrometric linear model (float, Angular Velocity[mas yr^{-1}])

Source proper motion in Right Ascension (μ_{α^*}) from the microlensing astrometric linear model.

PACZYNSKI3_PMRA_SOURCE_ERROR : Error of source proper motion in RA from astrometric linear model (float, Angular Velocity[mas yr^{-1}])

Draft

Error of source proper motion in Right Ascension (μ_{α^*}) from the microlensing astrometric linear model.

PACZYNSKI3_PMDEC_SOURCE : Source proper motion in Dec from astrometric linear model (float, Angular Velocity[mas yr^{-1}])

Source proper motion in declination (μ_δ) from the microlensing astrometric linear model.

PACZYNSKI3_PMDEC_SOURCE_ERROR : Error of source proper motion in Dec from astrometric linear model (float, Angular Velocity[mas yr^{-1}])

Error of source proper motion in declination (μ_δ) from the microlensing astrometric linear model.

PACZYNSKI3_EINSTEIN_RADIUS : Einstein radius from astrometric linear model (float, Angle[mas])

Einstein radius (θ_E) from the microlensing linear model on astrometric data.

PACZYNSKI3_EINSTEIN_RADIUS_ERROR : Error of the Einstein radius from astrometric linear model (float, Angle[mas])

Error of the Einstein radius from the linear model on astrometric data.

PACZYNSKI3_FLAG_NEGATIVE : Flag indicating if the negative solution was used in astrometric linear model (boolean)

Flag, if true, that the linear astrometric model yielded lower χ^2 for photometric solution with `-paczynski2_u0`, unless its θ_E was negative. The astrometric parameters (paczynski3) are stored for that solution.

Draft

18.12 VARI_PLANETARY_TRANSIT [TAP]

This table describes the Planetary Transit candidate events.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

TRANSIT_REFERENCE_TIME : Mid-transit reference time (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

A representative time of a planetary mid-transit during the Gaia mission, used as the reference epoch. Usually it will be chosen to be a time close to the mean of all Gaia observing epochs used in the analysis, in order to reduce its error.

An observation time t is considered to occur within a transit if:

$$\text{abs}(\text{mod}(t - T_0 + P/2, P) - P/2) < w/2$$

where T_0 stands for `transit_reference_time`, P for `transit_period`, and w for `transit_duration`.

TRANSIT_PERIOD : Most probable transit period (double, Time[day])

We restrict ourselves to strictly periodic planetary transits, otherwise we cannot identify them. This is the most probable period we found.

TRANSIT_DEPTH : Transit depth (float, Magnitude[mmag])

Draft

Transit depth is the amount by which the observed star brightness dims when the planet transits between the Solar System and the star. Physically it may hint to the size of the planet.

TRANSIT_DURATION : Transit duration (float, Time[day])

Transit duration is the length of time the transit lasts. The low cadence of Gaia photometry does not allow a detailed transit model with ingress and egress, so our assumption is that the transit has a simple box-like shape, and so the duration is well defined. Physically it is mainly related to the stellar radius and the orbital inclination.

NUM_IN_TRANSIT : Number of in-transit observations (byte)

Number of Gaia photometric observations that occurred during planetary transits. Only above a certain number we consider this a detection.

Draft

18.13 VARI_PMS_OSCILLATOR [TAP]

This table describes the pre-main sequence oscillators.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

NUM_FREQUENCIES : Number of frequencies stored (byte)

This parameter is filled with the number of frequencies extracted to be stored.

PMS_FREQUENCIES_G : Final frequencies in the G band (double[`num_frequencies`] array, Frequency[day^{-1}])

Frequencies obtained by means of non-linear least-square fit to the data (Levenberg-Marquardt method) and validated from *G*-band time series.

This parameter is an array of doubles including the first three frequencies (if any) extracted from the data.

PMS_FREQUENCIES_G_ERROR : Final frequency errors in the G band (float[`num_frequencies`] array, Frequency[day^{-1}])

This parameter is an array of floats including the uncertainties on the first three frequencies (if any) extracted from the data (stored in `pms_frequencies_g`). These uncertainties are computed via a bootstrap technique.

Draft

PMS_AMPLITUDES_G : Final amplitudes in the G band (float[num_frequencies] array, Magnitude[mmag])

This parameter is an array of floats including the amplitudes of the first three frequencies (if any) extracted from the data (stored in pms_frequencies_g).

PMS_AMPLITUDES_G_ERROR : Final amplitude errors in the G band (float[num_frequencies] array, Magnitude[mmag])

This parameter is an array of floats including the uncertainties on the amplitudes of the first three frequencies (if any) extracted from the data (stored in pms_amplitudes_g). These uncertainties are derived via Monte Carlo simulations.

PMS_PHASES_G : Final phases in the G band (float[num_frequencies] array, Angle[rad])

This parameter is an array of floats including the phases of the first three frequencies (if any) extracted from the data (stored in pms_frequencies_g).

PMS_PHASES_G_ERROR : Final phase errors in the G band (float[num_frequencies] array, Angle[rad])

This parameter is an array of floats including the uncertainties on the phases of the first three frequencies (if any) extracted from the data (stored in pms_phases_g). These uncertainties are derived via Monte Carlo simulations.

SN_FREQ_G : Signal-To-Noise ratio for frequencies in the G band (float[num_frequencies] array)

This parameter is an array of floats including the Signal-To-Noise ratio of the first three frequencies (if any) extracted from the data (stored in pms_frequencies_g). The signal is represented by the pms_amplitudes_g and the noise is computed via a bootstrap technique.

MEMBERSHIP_PMS_OSC : Membership to the classes of PMS Oscillator (string)

This parameter is currently filled with a string corresponding to the variability class δ Sct, γ Dor or Hybrid.

Draft

Draft

18.14 VARI_RAD_VEL_STATISTICS [TAP]

Statistical parameters of radial velocity time series.

The parameters are calculated using only transits not rejected (see `vari_epoch_flags.rv_used`). Note that Null will be recorded when the parameter value is missing or cannot be calculated.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

NUM_SELECTED_RV : Total number of radial velocity transits selected for variability analysis (short)

The number of processed observations for variability analyses of this source, using only transits not rejected, see `rv_used` column in `vari_epoch_flags`.

MEAN_OBS_TIME_RV : Mean observation time for radial velocity transits (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Name: The mean observation time

Control parameters: None

Output: Let y_i be a time series of size N at times t_i. The mean $\bar{t}$ is defined as

$$\bar{t} = \frac{1}{N} \sum_{i=1}^{N} t_i. \tag{17}$$

Draft

TIME_DURATION_RV : Time duration of the time series for radial velocity transits (float, Time[day])

Name: The time duration of the time series

Control parameters: None

Output: Let y_i be a time series of size N at times t_i, with $i = 1$ to N. The time duration of the time series is equal to $t_N - t_1$.

MIN_RV : Minimum radial velocity (float, Velocity[km s^{-1}])

Name: The minimum value of the time series

Control parameters: None

Output: Let y_i be a time series of size N at times t_i, with $i = 1$ to N. The minimum value of the time series is defined as:

$$min(y_i) \; \forall i \in (1, N) \tag{18}$$

MAX_RV : Maximum radial velocity (float, Velocity[km s^{-1}])

Name: The maximum value of the time series

Control parameters: None

Output: Let y_i be a time series of size N at times t_i, with $i = 1$ to N. The maximum value of the time series is defined as:

$$max(y_i) \; \forall i \in (1, N) \tag{19}$$

MEAN_RV : Mean radial velocity (float, Velocity[km s^{-1}])

Name: The mean of the time series

Control parameters: None

Output: Let y_i be a time series of size N. The mean $\bar{y}$ is defined as

$$\bar{y} = \frac{1}{N} \sum_{i=1}^{N} y_i. \tag{20}$$

Draft

MEDIAN_RV : Median radial velocity (float, Velocity[km s^{-1}])

Name: The median of the time series

Control parameters: None

Output: The 50th percentile unweighted value.

Let y_i be a time series of size N ordered such as $y_{(1)} \leq y_{(2)} \leq \cdots \leq y_{(N)}$. The m-th (per cent) percentile P_m is defined for $0 < m \leq 100$ as follows:

$$P_m = \begin{cases} y_{(1)} & \text{if } 0 < m < 100/N, \\ y_{(i)} & \text{if } Nm/100 - i = 0, \\ y_{(i+1)} & \text{otherwise.} \end{cases} \tag{21}$$

RANGE_RV : Difference between the highest and lowest radial velocity transits (float, Velocity[km s^{-1}])

Name: The range of the time series

Control parameters: None

Output: Let y_i be a time series, $y_{\max}$ its largest element, and $y_{\min}$ its smallest element, then the range is defined as

$$R = y_{\max} - y_{\min} \tag{22}$$

TRIMMED_RANGE_RV : Trimmed difference between the highest and lowest radial velocity transits (float, Velocity[km s^{-1}])

Name: The trimmed range

Control parameters: trim level $\alpha \in]0, 50[$

Output: The trimmed unweighted range.

Let y_i be a time series of size N, with error bars Δy_i. Let Y_α be the αth percentile, and $Y_{100-\alpha}$ be the $(100 - \alpha)$th percentile. Let y'_i be the trimmed time series of size N' which is derived by iteratively removing the smallest or the largest values of the y_i time series until $Y_\alpha < y'_i < Y_{100-\alpha}$ for all $i \in \{1, \cdots, N'\}$. The trimmed range R' is defined as

$$R' = y'_{\max} - y'_{\min} \tag{23}$$

The trim level α was set to 5.0.

Draft

STD_DEV_RV : Square root of the unweighted radial velocity variance (float, Velocity[km s^{-1}])

Name: The square root of the unbiased unweighted variance.

Output: Let y_i be a time series of size N. The unweighted standard deviation $\hat{\sigma}$ is defined as the square root of the sample-size unbiased unweighted variance:

$$\hat{\sigma} = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(y_i - \overline{y})^2}.$$

SKEWNESS_RV : Standardized unweighted radial velocity skewness (float)

Name: The standardised unbiased unweighted skewness.

Output: Let y_i be a time series of size $N > 2$. The sample-size unbiased unweighted skewness moment $\mathcal{E}$ is defined as:

$$\mathcal{E} = \frac{N}{(N-1)(N-2)}\sum_{i=1}^{N}(y_i - \overline{y})^3.$$

The standardized unbiased skewness E is defined as:

$$E = \frac{\mathcal{E}}{\hat{\sigma}^3}$$

where $\hat{\sigma}$ is the square root of the unbiased unweighted variance around the unweighted mean. While $\mathcal{E}$ is an unbiased estimate of the population value, E becomes unbiased in the limit of large N.

KURTOSIS_RV : Standardized unweighted radial velocity kurtosis (float)

Name: The standardised unbiased unweighted kurtosis.

Output: Let y_i be a time series of size $N > 3$. The sample-size unbiased unweighted kurtosis cumulant $\mathcal{K}$ is defined as:

$$\mathcal{K} = \frac{N(N+1)}{(N-1)(N-2)(N-3)}\sum_{i=1}^{N}(y_i - \overline{y})^4 - \frac{3}{(N-2)(N-3)}\left[\sum_{i=1}^{N}(y_i - \overline{y})^2\right]^2.$$

The standardized unbiased kurtosis K is defined as:

$$K = \frac{\mathcal{K}}{\hat{\sigma}^4}$$

where $\hat{\sigma}^2$ is the unbiased unweighted variance around the unweighted mean. While $\mathcal{K}$ is an unbiased estimate of the population value, K becomes unbiased in the limit of large N.

Draft

MAD_RV : Median Absolute Deviation (MAD) for radial velocity transits (float, Velocity[km s^{-1}])

Name: The Median Absolute Deviation (MAD)

Control parameters: None

Output: Let y_i be a time series of size N. The MAD is defined as the median of the absolute deviations from the median of the data, scaled by a factor of $1/\Phi^{-1}(3/4) \approx 1.4826$ (where Φ^{-1} is the inverse of the cumulative distribution function for the standard normal distribution), so that the expectation of the scaled MAD at large N equals the standard deviation of a normal distribution:

$$\mathrm{MAD} = 1.4826\ \mathrm{median}\{|y_i - \mathrm{median}\{y_j, \forall j \in (1,N)\}|, \forall i \in (1,N)\}. \quad (24)$$

ABBE_RV : Abbe value for radial velocity transits (float)

Name: The Abbe value

Control parameters: None

Output: Let $\{t_i, y_i\}$ be a time-sorted time series of size N, such that $t_i < t_{i+1}$ for all $i < N$. The Abbe value $\mathcal{A}$ is defined as

$$\mathcal{A} = \frac{\sum_{i=1}^{N-1}(y_{i+1} - y_i)^2}{2\sum_{i=1}^{N}(y_i - \bar{y})^2} \quad (25)$$

where $\bar{y}$ is the unweighted mean.

IQR_RV : Interquartile range for radial velocity transits (float, Velocity[km s^{-1}])

Name: The Interquartile Range (IQR)

Control parameters: None

Output: The difference between the (unweighted) 75th and 25th percentile values: IQR= $P_{75} - P_{25}$.

Let y_i be a time series of size N ordered such as $y_{(1)} \leq y_{(2)} \leq \cdots \leq y_{(N)}$. The m-th (per cent) percentile P_m is defined for $0 < m \leq 100$ as follows:

$$P_m = \begin{cases} y_{(1)} & \text{if } 0 < m < 100/N, \\ y_{(i)} & \text{if } Nm/100 - i = 0, \\ y_{(i+1)} & \text{otherwise.} \end{cases} \quad (26)$$

STETSON_RV : Stetson radial velocity variability index (float)

Name: The single-band Stetson variability index, herein set to 0.1 days.

Control parameters: The pairing time interval

Output: Let y_i be a time series of size N, with error bars Δy_i. The Stetson value (Stetson 1996) is based on paired observations closely separated in time (within a chosen time interval). The time difference between the components of a pair should be small compared to the shortest time-scale of variation being sought. If this condition is not satisfied, a transformation is applied to the unpaired measurement, in order to include its information together with paired observations. The maximum number n of pairs may exceed $N/2$, since multiple pairs can be formed within a given time interval (as long as the latter is much smaller than the minimum variability time-scale sought). The Stetson J index is defined as

$$J = \frac{1}{n}\sum_{k=1}^{n} \mathrm{sgn}(P_k)\sqrt{|P_k|},$$

where the index k refers to the k-th pair, composed of measurements i and j, such that

$$P_k = \begin{cases} \delta_{i(k)}\delta_{j(k)}, & \text{if } i(k) \neq j(k) \\ \delta_{i(k)}^2 - 1, & \text{if } i(k) = j(k) \end{cases}$$

with the normalized deviation of the i-th measurement from the time-series mean $\bar{y}$ expressed by:

$$\delta_{i(k)} = \sqrt{\frac{N}{N-1}}\,\frac{y_{i(k)} - \bar{y}}{\Delta y_{i(k)}}.$$

STD_DEV_OVER_RMS_ERR_RV : Signal-to-Noise radial velocity estimate (float)

Name: The signal-to-noise estimate

Control parameters: None

Output: Let y_i be a time series of size N, with error bars Δy_i. The signal-to-noise ratio (S/N) is estimated as

$$\mathrm{S/N} = \sqrt{\frac{\sum_{i=1}^{N}(y_i - \bar{y})^2}{\sum_{i=1}^{N}\Delta y_i^2}} \tag{27}$$

where $\bar{y}$ is the mean of the time-series values.

OUTLIER_MEDIAN_RV : Greatest absolute deviation from the radial velocity median normalized by the error (float)

Name: The most outlying measurement with respect to the median

Control parameters: None

Output: Let y_i be a time series of size N, with error bars Δy_i. The most outlying measurement is defined as the greatest absolute deviation from the median normalized by the error:

$$\text{outlierMedian} = \max\left\{\frac{|y_i - \text{median}\{y_j, \forall j \in (1,N)\}|}{\Delta y_i}, \forall i \in (1,N)\right\}. \quad (28)$$

Draft

18.15 VARI_ROTATION_MODULATION [TAP]

This table describes the solar-like stars with rotational modulation.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

NUM_SEGMENTS : Number of segments (byte)

This is the number of time intervals (segments) in which the magnitude and colour time-series are splitted. The segmentation of time-series is needed because the spots due to the stellar magnetic activity have a life-time shorter than the whole Gaia time-series. The rotational modulation induced by spots can therefore be detected only in segments whose duration is comparable with the spots life-time.

SEGMENTS_START_TIME : Times at which segments start (double[`num_segments`] array, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

An array filled with the starting times of segments.

SEGMENTS_END_TIME : Times at which segments end (double[`num_segments`] array, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

An array filled with the ending times of segments.

Draft

SEGMENTS_COLOUR_MAG_INTERCEPT : Colour-Magnitude Intercept in segments (float[num_segments] array)

A robust linear regression is applied to the points (G_{BP}-G_{RP}, G) in each segment. This array is filled with the intercepts given by the fitting procedure in the different segments.

SEGMENTS_COLOUR_MAG_INTERCEPT_ERROR : Colour-Magnitude Intercept uncertainty in segments (float[num_segments] array)

This array is filled with the uncertainties associated with the intercepts given by the fitting procedure.

SEGMENTS_COLOUR_MAG_SLOPE : Colour-Magnitude Slope in segments (float[num_segments] array)

A robust linear regression is applied to the points (G_{BP}-G_{RP}, G) in each segment. This array is filled with the slopes given by the fitting procedure in the different segments.

SEGMENTS_COLOUR_MAG_SLOPE_ERROR : Colour-Magnitude Slope uncertainty in segments (float[num_segments] array)

This array is filled with the uncertainties associated with the slopes given by the fitting procedure.

SEGMENTS_CORRELATION_COEFFICIENT : Correlation coefficient in segments (float[num_segments] array)

The Pearson correlation coefficient r between G_{BP}-G_{RP} and G is computed in each segment. The higher is the Pearson coefficient the higher is the probability that the stellar variability is due to rotational modulation. This array is filled with the Pearson coefficients obtained in the different segments.

SEGMENTS_CORRELATION_SIGNIFICANCE : Correlation coefficient significance in segments (float[num_segments] array)

This array is filled with the statistical significances associated with the Pearson coefficients computed in the different segments. The significance p associated with a given $r = r_0$ gives the probability $P(r \geq r_0)$ that two sets of uncorrelated measurements have a Pearson coefficient $\geq r_0$.

Draft

NUM_OUTLIERS : Number of outliers (short)

The number of outliers detected by the robust linear regression procedure.

OUTLIERS_TIME : Times at which outliers occurs (double[num_outliers] array, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Times at which the detected outliers occurred.

SEGMENTS_ROTATION_PERIOD : Rotation period in segment (double[num_segments] array, Time[day])

A period search algorithm is applied to the different time-series segments. If the star is a solar-like variable the detected period is a measure of the stellar rotation period. This array is filled with the periods detected in the different segments (for each segment the period with the highest statistical significance is stored).

SEGMENTS_ROTATION_PERIOD_ERROR : Rotation period uncertainty in segment (float[num_segments] array, Time[day])

This array is filled with the errors associated with the periods found in the different segments.

SEGMENTS_ROTATION_PERIOD_FAP : FAP on rotation period in segment (float[num_segments] array)

False Alarm Probability = Probability that that a white noise sequence produces a peak similar or higher than the computed one; i.e., small FAP = little probability of noise, high FAP = noise is an acceptable explanation for the peak.

BEST_ROTATION_PERIOD : Best rotation period (double, Time[day])

This field is the best estimate of the stellar rotation period and is obtained by taking the mode of the segments_rotation_period distribution. The distribution is built only with the periods, whose associated FAP is less than 0.05. See the release documentation for further details.

BEST_ROTATION_PERIOD_ERROR : Error on best rotation period (float, Time[day])

Draft

Error on the best rotation period.

BEST_ROTATION_PERIOD_FAP : Fap associated with the best rotation period (double)

False alarm probability (FAP) associated with the best rotation period. It is computed by averaging the individual FAP values obtained in the segments used to infer `best_rotation_period`.

NUM_VALID_SEGMENTS : Number of segments considered valid for deriving the best rotation period (byte)

Number of segments considered valid for deriving the best rotation period, i.e. the segments in which the period search algorithm retrieved a statistically meaningful result.

MIN_FIT_AMPLITUDE : Minimum amplitude among all sinusoids fitted to individual segments (float, Magnitude[mag])

This parameter can be regarded as an activity index and is the minimum amplitude amoing all sinusoids fitted to individual segments.

MAX_FIT_AMPLITUDE : Maximum amplitude among all sinusoids fitted to individual segments (float, Magnitude[mag])

This parameter can be regarded as an activity index and is the maximum amplitude amoing all sinusoids fitted to individual segments.

MEAN_FIT_AMPLITUDE : Mean amplitude among all sinusoids fitted to individual segments (float, Magnitude[mag])

This parameter can be regarded as an activity index and is mean amplitude amoing all sinusoids fitted to individual segments.

MIN_CHI_SQUARE : Minimum value of the chi-square statistics among all segments (float)

Minimum value of the chi-square statistics computed from the residuals of sinusoidal fits across all segments of a time-series.

Draft

MAX_CHI_SQUARE : Maximum value of the chi-square statistics among all valid segments (float)

Maximum value of the chi-square statistics computed from the residuals of sinusoidal fits across all valid segments of a time-series.

MIN_PHASE_COVERAGE : Minimum phase coverage among all valid segments (float)

Minimum phase coverage among all valid segments to which sinusoids are fitted.

MAX_PHASE_COVERAGE : Maximum phase coverage among all valid segments (float)

Maximum phase coverage among all valid segments to which sinusoids are fitted.

MIN_RESIDUALS_OVER_AMPLITUDE : Minimum value of residuals over amplitude among all valid segments (float)

Minimum value of residuals over amplitude (Roa) among all valid segments. Roa is the ratio between the average absolute residual from the sinusoidal fit and the amplitude of the sinusoid. It can be regarded as a quality metric of the fit.

MAX_RESIDUALS_OVER_AMPLITUDE : Maximum value of residuals over amplitude among all valid segments (float)

Maximum value of residuals over amplitude (Roa) among all valid segments. Roa is the ratio between the average absolute residual from the sinusoidal fit and the amplitude of the sinusoid. It can be regarded as a quality metric of the fit.

SEGMENTS_G_UNSPOTTED : The unspotted G mags in segment (float[`num_segments`] array, Magnitude[mag])

In a given segment the *G* magnitude corresponding to the unspotted state is estimated by taking the minimum *G* value in the segment.

SEGMENTS_G_UNSPOTTED_ERROR : The unspotted G mag uncertainties in segment (float[`num_segments`] array, Magnitude[mag])

This array stores the errors associated with the unspotted *G* values registered in the different seg-

ments.

SEGMENTS_BP_UNSPOTTED : The unspotted BP mag in segment (float[num_segments] array, Magnitude[mag])

In a given segment the G_{BP} magnitude corresponding to the unspotted state is estimated by taking the G_{BP} magnitude occurring at the same time of the unspotted G. If the G_{BP} transit corresponding to segment's unspotted G transit is not available, a NaN value is provided.

SEGMENTS_BP_UNSPOTTED_ERROR : The unspotted BP mag uncertainties in segment (float[num_segments] array, Magnitude[mag])

This array stores the errors associated with the unspotted G_{BP} values registered in the different segments.

SEGMENTS_RP_UNSPOTTED : The unspotted RP mag in segment (float[num_segments] array, Magnitude[mag])

In a given segment the G_{RP} magnitude corresponding to the unspotted state is estimated by taking the G_{RP} magnitude occurring at the same time of the unspotted G. If the G_{RP} transit corresponding to segment's unspotted G transit is not available, a NaN value is provided.

SEGMENTS_RP_UNSPOTTED_ERROR : The unspotted RP mag uncertainties in segment (float[num_segments] array, Magnitude[mag])

This array stores the errors associated with the unspotted G_{RP} values registered in the different segments.

G_UNSPOTTED : Unspotted G mag (float, Magnitude[mag])

Final estimate of the G magnitude corresponding to the unspotted state. It is computed by taking the minimum G magnitude in the whole time-series.

G_UNSPOTTED_ERROR : Unspotted G mag uncertainty (float, Magnitude[mag])

This field stores the photometric error associated with g_unspotted.

Draft

BP_UNSPOTTED : Unspotted BP mag (float, Magnitude[mag])

Final estimate of the G_{BP} magnitude corresponding to the unspotted state. It is estimated by taking the G_{BP} magnitude occurring at the same time in which the minimum G magnitude has been measured. If the G_{BP} transit corresponding to unspotted G transit is not available, a Null value is provided.

BP_UNSPOTTED_ERROR : Unspotted BP mag uncertainty (float, Magnitude[mag])

Error associated with the `bp_unspotted` value.

RP_UNSPOTTED : Unspotted RP mag (float, Magnitude[mag])

Final estimate of the G_{RP} magnitude corresponding to the unspotted state. It is estimated by taking the G_{RP} magnitude occurring at the same time in which the minimum G magnitude has been measured. If the G_{RP} transit corresponding to unspotted G transit is not available, a Null value is provided.

RP_UNSPOTTED_ERROR : Unspotted RP mag uncertainty (float, Magnitude[mag])

Error associated with the `rp_unspotted` value.

SEGMENTS_G_COS_TERM : Coefficient of cosine term of linear fit in segment in the G band (float[`num_segments`] array, Magnitude[mag])

If a significative period T_0 is detected in a time-series segment, then the points of the time-series segment are fitted with the function

$$mag(t) = mag_0 + A\cos(\frac{2\pi}{T_0}t) + B\sin(\frac{2\pi}{T_0}t) \tag{29}$$

This array stores the A terms obtained by the fitting procedure in the different segments in the G band. The fit is computed only if the FAP associated with the period T_0 is less than 0.05.

SEGMENTS_G_COS_TERM_ERROR : Errors on cosine terms in the G band (float[`num_segments`] array, Magnitude[mag])

This array is filled with the errors associated with the A terms obtained from the fitting procedure in the different segments in the G band.

SEGMENTS_G_SIN_TERM : Coefficient of sine term of linear fit in segment in the G band (float[num_segments] array, Magnitude[mag])

If a significative period T_0 is detected in a time-series segment, then the points of the time-series segment are fitted with the function

$$mag(t) = mag_0 + A\cos(\frac{2\pi}{T_0}t) + B\sin(\frac{2\pi}{T_0}t) \tag{30}$$

This array stores the B terms obtained by the fitting procedure in the different segments in the *G* band. The fit is computed only if the FAP associated with the period T_0 is less than 0.05.

SEGMENTS_G_SIN_TERM_ERROR : Errors on sine terms in the G band (float[num_segments] array, Magnitude[mag])

This array is filled with the errors associated with the B terms obtained from the fitting procedure in the different segments in the *G* band.

SEGMENTS_G_A0_TERM : Constant term (A0) of linear fit in segment in the G band (float[num_segments] array, Magnitude[mag])

If a significative period T_0 is detected in a time-series segment, then the points of the time-series segment are fitted with the function

$$mag(t) = mag_0 + A\cos(\frac{2\pi}{T_0}t) + B\sin(\frac{2\pi}{T_0}t) \tag{31}$$

This array stores the mag_0 terms obtained by the fitting procedure in the different segments in the *G* band. The fit is computed only if the FAP associated with the period T_0 is less than 0.05.

SEGMENTS_G_A0_TERM_ERROR : Errors on constant terms in the G band (float[num_segments] array, Magnitude[mag])

This array is filled with the errors associated with the mag_0 terms obtained from the fitting procedure in the different segments in the *G* band.

SEGMENTS_BP_COS_TERM : Coefficient of cosine term of linear fit in segment in the BP band (float[num_segments] array, Magnitude[mag])

Draft

If a significative period T_0 is detected in a time-series segment, then the points of the time-series segment are fitted with the function

$$mag(t) = mag_0 + A\cos(\frac{2\pi}{T_0}t) + B\sin(\frac{2\pi}{T_0}t) \quad (32)$$

This array stores the A terms obtained by the fitting procedure in the different segments in the G_{BP} band. The fit is computed only if the FAP associated with the period T_0 is less than 0.05.

SEGMENTS_BP_COS_TERM_ERROR : Errors on cosine terms in the BP band (float[num_segments] array, Magnitude[mag])

This array is filled with the errors associated with the A terms obtained from the fitting procedure in the different segments in the G_{BP} band.

SEGMENTS_BP_SIN_TERM : Coefficient of sine term of linear fit in segment in the BP band (float[num_segments] array, Magnitude[mag])

If a significative period T_0 is detected in a time-series segment, then the points of the time-series segment are fitted with the function

$$mag(t) = mag_0 + A\cos(\frac{2\pi}{T_0}t) + B\sin(\frac{2\pi}{T_0}t) \quad (33)$$

This array stores the B terms obtained by the fitting procedure in the different segments in the G_{BP} band. The fit is computed only if the FAP associated with the period T_0 is less than 0.05.

SEGMENTS_BP_SIN_TERM_ERROR : Errors on sine terms in the BP band (float[num_segments] array, Magnitude[mag])

This array is filled with the errors associated with the B terms obtained from the fitting procedure in the different segments in the G_{BP} band.

SEGMENTS_BP_A0_TERM : Constant term (A0) of linear fit in segment in the BP band (float[num_segments] array, Magnitude[mag])

If a significative period T_0 is detected in a time-series segment, then the points of the time-series segment are fitted with the function

$$mag(t) = mag_0 + A\cos(\frac{2\pi}{T_0}t) + B\sin(\frac{2\pi}{T_0}t) \quad (34)$$

This array stores the mag_0 terms obtained by the fitting procedure in the different segments in the G_{BP} band. The fit is computed only if the FAP associated with the period T_0 is less than 0.05.

SEGMENTS_BP_A0_TERM_ERROR : Errors on constant terms in the BP band (float[num_segments] array, Magnitude[mag])

This array is filled with the errors associated with the mag_0 terms obtained from the fitting procedure in the different segments in the G_{BP} band.

SEGMENTS_RP_COS_TERM : Coefficient of cosine term of linear fit in segment in the RP band (float[num_segments] array, Magnitude[mag])

If a significative period T_0 is detected in a time-series segment, then the points of the time-series segment are fitted with the function

$$mag(t) = mag_0 + A\cos(\frac{2\pi}{T_0}t) + B\sin(\frac{2\pi}{T_0}t) \quad (35)$$

This array stores the A terms obtained by the fitting procedure in the different segments in the G_{RP} band. The fit is computed only if the FAP associated with the period T_0 is less than 0.05.

SEGMENTS_RP_COS_TERM_ERROR : Errors on cosine terms in the RP band (float[num_segments] array, Magnitude[mag])

This array is filled with the errors associated with the A terms obtained from the fitting procedure in the different segments in the G_{RP} band.

SEGMENTS_RP_SIN_TERM : Coefficient of sine term of linear fit in segment in the RP band (float[num_segments] array, Magnitude[mag])

If a significative period T_0 is detected in a time-series segment, then the points of the time-series segment are fitted with the function

$$mag(t) = mag_0 + A\cos(\frac{2\pi}{T_0}t) + B\sin(\frac{2\pi}{T_0}t) \quad (36)$$

This array stores the B terms obtained by the fitting procedure in the different segments in the G_{RP} band. The fit is computed only if the FAP associated with the period T_0 is less than 0.05.

SEGMENTS_RP_SIN_TERM_ERROR : Errors on sine terms in the RP band (float[num_segments] array, Magnitude[mag])

This array is filled with the errors associated with the B terms obtained from the fitting procedure in the different segments in the G_{RP} band.

SEGMENTS_RP_A0_TERM : Constant term (A0) of linear fit in segment in the RP band (float[num_segments] array, Magnitude[mag])

If a significative period T_0 is detected in a time-series segment, then the points of the time-series segment are fitted with the function

$$mag(t) = mag_0 + A\cos(\frac{2\pi}{T_0}t) + B\sin(\frac{2\pi}{T_0}t) \tag{37}$$

This array stores the mag_0 terms obtained by the fitting procedure in the different segments in the G_{RP} band. The fit is computed only if the FAP associated with the period T_0 is less than 0.05.

SEGMENTS_RP_A0_TERM_ERROR : Errors on constant terms in the RP band (float[num_segments] array, Magnitude[mag])

This array is filled with the errors associated with the mag_0 terms obtained from the fitting procedure in the different segments in the G_{RP} band.

SEGMENTS_G_ACTIVITY_INDEX : Activity Index in segment (computed in G band) (float[num_segments] array, Magnitude[mag])

This array stores the activity indexes measured in the different segments in the G band. In a given segment the amplitude of variability A is taken as an index of the magnetic activity level. The amplitude of variability is measured by means of the equation:

$$A = mag_{95} - mag_5 \tag{38}$$

where mag_{95} and mag_5 are the 95-th and the 5-th percentiles of the magnitude values.

SEGMENTS_G_ACTIVITY_INDEX_ERROR : error on Activity index in segment (computed in G band) (float[num_segments] array, Magnitude[mag])

This array stores the errors associated with the activity indexes in the G band. In a given segment the error on the activity index A is computed by means of the equation:

$$\sigma_A = \sqrt{\sigma^2_{mag95} + \sigma^2_{mag5}} \tag{39}$$

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SEGMENTS_BP_ACTIVITY_INDEX : Activity Index in segment (computed in BP band) (float[num_segments] array, Magnitude[mag])

This array stores the activity indexes measured in the different segments in the $G_{\rm BP}$ band. In a given segment the amplitude of variability A is taken as an index of the magnetic activity level. The amplitude of variability is measured by means of the equation:

$$A = mag_{95} - mag_5 \tag{40}$$

where mag_{95} and mag_5 are the 95-th and the 5-th percentiles of the magnitude values.

SEGMENTS_BP_ACTIVITY_INDEX_ERROR : error on Activity index in segment (computed in BP band) (float[num_segments] array, Magnitude[mag])

This array stores the errors associated with the activity indexes in the $G_{\rm BP}$ band. In a given segment the error on the activity index A is computed by means of the equation:

$$\sigma_A = \sqrt{\sigma^2_{mag95} + \sigma^2_{mag5}} \tag{41}$$

SEGMENTS_RP_ACTIVITY_INDEX : Activity Index in segment (computed in RP band) (float[num_segments] array, Magnitude[mag])

This array stores the activity indexes measured in the different segments in the $G_{\rm RP}$ band. In a given segment the amplitude of variability A is taken as an index of the magnetic activity level. The amplitude of variability is measured by means of the equation:

$$A = mag_{95} - mag_5 \tag{42}$$

where mag_{95} and mag_5 are the 95-th and the 5-th percentiles of the magnitude values.

SEGMENTS_RP_ACTIVITY_INDEX_ERROR : error on Activity index in segment (computed in RP band) (float[num_segments] array, Magnitude[mag])

This array stores the errors associated with the activity indexes in the $G_{\rm RP}$ band. In a given segment the error on the activity index A is computed by means of the equation:

$$\sigma_A = \sqrt{\sigma^2_{mag95} + \sigma^2_{mag5}} \tag{43}$$

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MAX_ACTIVITY_INDEX_G : The maximum Activity Index in the G band (float, Magnitude[mag])

This field is the maximum of measured the activity indexes in the *G* band.

MAX_ACTIVITY_INDEX_G_ERROR : Error on maximum activity index in the G band (float, Magnitude[mag])

This field stores the error associated with the maximum activity index in the *G* band.

SEGMENTS_BP_RP_CORR_COEFF : Pearson coefficient between BP and RP in segment (float[num_segments] array)

For each segment, this parameter provides the Pearson coefficient of the transits in the (G_{BP},G_{RP}) magnitude plane. This correlation coefficient can be used to infer properties of stellar spots and faculae.

SEGMENTS_BP_RP_CORR_SIGNIF : Significance associated with Pearson coefficient (float[num_segments] array)

For each segment, this parameter provides the statistical significance associated with the Pearson coefficient segments_bp_rp_corr_coeff.

SEGMENTS_BP_RP_INTERCEPT : Intercept of regression fit of RP vs BP magnitudes in the segment (float[num_segments] array, Magnitude[mag])

This array provides the intercepts obtained for the line regression fit of the transits in the (G_{BP},G_{RP}) magnitude plane in each segment.

SEGMENTS_BP_RP_INTERCEPT_ERROR : Error on intercept of regression fit of RP vs BP magnitudes in the segment (float[num_segments] array, Magnitude[mag])

This array stores the errors associated with the intercepts stored in segments_bp_rp_intercept.

SEGMENTS_BP_RP_SLOPE : Slope of regression fit of RP vs BP magnitudes in the segment (float[num_segments] array)

This array provides the slopes obtained for the line regression fit of the transits in the (G_{BP},G_{RP})

magnitude plane in each segment.

SEGMENTS_BP_RP_SLOPE_ERROR : Error on slope of regression fit of RP vs BP magnitudes in the segment (float[num_segments] array)

This array stores the errors associated with the slopes stored in segments_bp_rp_slope.

SEGMENTS_MODEL_REFERENCE_TIME : Time used as reference time for the period search procedure in the segment (double[num_segments] array, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

For each segment, this parameter provides the reference time used when the data points are fitted to the models represented by the equations in the definitions of segments_g_cos_term, segments_bp_cos_term, and segments_rp_cos_term.

SEGMENTS_G_CHI_SQUARE : Array of chisquares for linear models (G band) (float[num_segments] array)

Array of chisquares for linear models (G band).

SEGMENTS_BP_CHI_SQUARE : Array of chisquares for linear models (BP band) (float[num_segments] array)

Array of chisquares for linear models (G_{BP} band).

SEGMENTS_RP_CHI_SQUARE : Array of chisquares for linear models (RP band) (float[num_segments] array)

Array of chisquares for linear models (G_{RP} band).

CYCLE_PERIOD : Period of the stellar cycle (float, Time[day])

This period is obtained by searching for a long-term periodicity in the whole G FoV time-series.

CYCLE_PERIOD_ERROR : Error associated with the period from the stellar cycle (float, Time[day])

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Error associated with the period from the stellar cycle.

CYCLE_PERIOD_FAP : FAP associated with the period from the stellar cycle (double)

False alarm probability associated with the period from the stellar cycle.

RV_PERIOD : Period detected in the radial velocity time series (float, Time[day])

Period detected in the radial velocity time series.

RV_PERIOD_ERROR : Error associated with the period from the radial velocity time series (float, Time[day])

Error associated with the period from the radial velocity time series.

RV_PERIOD_FAP : FAP associated with period from the radial velocity time series (double)

False alarm probability associated with the period from the radial velocity timeseries.

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18.16 VARI_RRLYRAE [TAP]

This table describes the RR Lyrae stars.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

PF : Period corresponding to the fundamental pulsation mode in the G-band time series (double, Time[day])

For single-mode pulsators classified as fundamental mode pulsators, this parameter is filled with the periodicity found in the time-series.

This value is obtained by modelling the *G* band time series using the Levenberg-Marquardt non–linear fitting algorithm (Clementini et al. 2016).

Information for `vari_rrlyrae`: For double-mode RR Lyrae this parameter is filled with the period corresponding to the longer periodicity.

Information for `vari_cepheid`: For double-mode DCEPs this parameter is filled with the period corresponding to the longer periodicity if the DCEP is classified as 'F/1O' or 'F/2O'. For triple-mode DCEPs this parameter is filled with the period corresponding to the longer periodicity if the DCEP is classified as 'F/1O/2O'.

PF_ERROR : Uncertainty of the pf period (float, Time[day])

This parameter is filled with the uncertainty of the `pf` parameter, computed via a bootstrap tech-

nique.

p1_o : Period corresponding to the first overtone pulsation mode in the G-band time series (double, Time[day])

For single-mode pulsators classified as first-overtone pulsators, this parameter is filled with the periodicity found in the time-series.

This value is obtained by modelling the *G* time series using the Levenberg-Marquardt non linear fitting algorithm (see Clementini et al. 2016).

Information for `vari_rrlyrae`: For double-mode RR Lyrae this parameter is filled with the period corresponding to the shortest periodicity.

Information for `vari_cepheid`: For double-mode DCEPs this parameter is filled with the period corresponding to the shortest periodicity if the DCEP is classified as 'F/1O'; otherwise it is filled with the longest one if the classification is '1O/2O' or '1O/3O'. For triple-mode DCEPs this parameter is filled with the period corresponding to the intermediate periodicity if the DCEP is classified as 'F/1O/2O'; it is filled with the longest periodicity if the classification is '1O/2O/3O'.

p1_o_error : Uncertainty of the first overtone period (float, Time[day])

This parameter is filled with the uncertainty of the `p1_o` parameter, computed via a bootstrap technique.

epoch_g : Epoch of the maximum of the light curve in the G band (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Epoch of maximum light for the Gaia *G* band light curve. It corresponds to the barycentric Julian day (BJD) of the maximum value of the light curve model which is closest to the BJD of the first observations -3 times the period of the source (first periodicity depending on the pulsation mode).

The mentioned BJD is offset by JD 2455197.5 (= J2010.0).

epoch_g_error : Uncertainty on the epoch of the maximum in the G band (float, Time[day])

Value of the uncertainty of the `epoch_g` parameter. It corresponds to three times the error on the period of the source (first periodicity depending on the pulsation mode).

EPOCH_BP : Epoch of the maximum of the light curve in the BP band (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Epoch of maximum light for the Gaia integrated $G_{\rm BP}$ band light curve. It corresponds to the barycentric Julian day (BJD) of the maximum value of the light curve model which is closest to the BJD of the first observations -3 times the period of the source (first periodicity depending on the pulsation mode).

The mentioned BJD is offset by JD 2455197.5 (= J2010.0).

EPOCH_BP_ERROR : Uncertainty on the epoch of the maximum in the BP band (float, Time[day])

Value of the uncertainty of the `epoch_bp` parameter. It corresponds to three times the error on the period of the source (first periodicity depending on the pulsation mode).

EPOCH_RP : Epoch of the maximum of the light curve in the RP band (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Epoch of maximum light for the Gaia integrated $G_{\rm RP}$ band light curve. It corresponds to the barycentric Julian day (BJD) of the maximum value of the light curve model which is closest to the BJD of the first observations -3 times the period of the source (first periodicity depending on the pulsation mode).

The mentioned BJD is offset by JD 2455197.5 (= J2010.0).

EPOCH_RP_ERROR : Uncertainty on the epoch of the maximum in the RP band (float, Time[day])

Value of the uncertainty of the `epoch_rp` parameter. It corresponds to three times the error on the period of the source (first periodicity depending on the pulsation mode).

EPOCH_GRVS : Epoch of the maximum of the light curve in the Grvs band (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Epoch of maximum light for the Gaia integrated $G_{\rm RVS}$ band light curve. It corresponds to the barycentric Julian day (BJD) of the maximum value of the light curve model which is closest to the BJD of the first observations -3 times the period of the source (first periodicity depending on the pulsation mode).

The mentioned BJD is offset by JD 2455197.5 (= J2010.0).

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EPOCH_GRVS_ERROR : Uncertainty on the epoch of the maximum in the Grvs band (float, Time[day])

Value of the uncertainty of the `epoch_grvs` parameter. It corresponds to three times the error on the period of the source (first periodicity depending on the pulsation mode).

EPOCH_RV : Epoch of the minimum of the radial velocity curve (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Epoch of minimum radial velocity for the Gaia radial velocity curve.

EPOCH_RV_ERROR : Uncertainty on the epoch of the maximum in the radial velocity curve (float, Time[day])

Value of the uncertainty of the `epoch_rv` parameter. It corresponds to three times the error on the period of the source (first periodicity depending on the pulsation mode).

INT_AVERAGE_G : Intensity-averaged magnitude in the G band (float, Magnitude[mag])

Value of the intensity-averaged magnitude in the G-band. The intensity-averaged magnitude is obtained by computing the average flux and then converting the average flux to magnitude.

INT_AVERAGE_G_ERROR : Uncertainty on the intensity-average magnitude in the G band (float, Magnitude[mag])

This parameter is filled with the uncertainty of the `int_average_g` parameter, computed via a bootstrap technique.

INT_AVERAGE_BP : Intensity-averaged magnitude in the BP band (float, Magnitude[mag])

Value of the intensity-averaged magnitude in the $G_{\rm BP}$-band. The intensity-averaged magnitude is obtained by computing the average flux and then converting the average flux to magnitude.

INT_AVERAGE_BP_ERROR : Uncertainty on intensity-average magnitude in the BP band (float, Magnitude[mag])

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This parameter is filled with the uncertainty of the `int_average_bp` parameter, computed via a bootstrap technique.

INT_AVERAGE_RP : Intensity-averaged magnitude in the RP band (float, Magnitude[mag])

Value of the intensity-averaged magnitude in the $G_{\rm RP}$-band. The intensity-averaged magnitude is obtained by computing the average flux and then converting the average flux to magnitude.

INT_AVERAGE_RP_ERROR : Uncertainty on intensity-average magnitude in the RP band (float, Magnitude[mag])

This parameter is filled with the uncertainty of the `int_average_rp` parameter, computed via a bootstrap technique.

INT_AVERAGE_GRVS : Intensity-averaged magnitude in the Grvs band (float, Magnitude[mag])

Value of the intensity-averaged magnitude in the G_{RVS}-band. The intensity-averaged magnitude is obtained by computing the average flux and then converting the average flux to magnitude.

INT_AVERAGE_GRVS_ERROR : Uncertainty on intensity-average magnitude in the Grvs band (float, Magnitude[mag])

This parameter is filled with the uncertainty of the `int_average_grvs` parameter, computed via a bootstrap technique.

AVERAGE_RV : Mean radial velocity (float, Velocity[km s^{-1}])

Average value of the Fourier modelled radial velocity curve, provided by the A_0 parameter of the Fourier fit. If the Fourier model cannot be computed (in case of limited data or models showing unphysical oscillations), `average_rv` is the arithmetic average of the individual radial velocity measurements. In case of binary Cepheids, `average_rv` corresponds to the average velocity resulting from the binary model.

AVERAGE_RV_ERROR : Uncertainty on the mean radial velocity (float, Velocity[km s^{-1}])

Uncertainty of `average_rv` computed via a bootstrap technique if the radial velocity time series is modelled with a truncated Fourier series. In case of limited data or models showing unphysical

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oscillations, the error of `average_rv` is the standard deviation of the radial velocity measurements.

PEAK_TO_PEAK_G : Peak-to-peak amplitude of the G band light curve (float, Magnitude[mag])

This parameter is filled with the peak-to-peak amplitude value of the *G* band light curve. The peak-to-peak amplitude is calculated as the (maximum) - (minimum) of the modelled folded light curve in the *G* band. The light curve of the target star is modelled with a truncated Fourier series.

PEAK_TO_PEAK_G_ERROR : Uncertainty on the peak-to-peak amplitude in the G band (float, Magnitude[mag])

This parameter is filled with the uncertainty value of the `peak_to_peak_g` parameter, computed via a bootstrap technique.

PEAK_TO_PEAK_BP : Peak-to-peak amplitude of the BP band light curve (float, Magnitude[mag])

This parameter is filled with the peak-to-peak amplitude value of the $G_{\rm BP}$ light curve. The peak-to-peak amplitude is calculated as the (maximum) - (minimum) of the modelled folded light curve in the $G_{\rm BP}$ band. The light curve of the target star is modelled with a truncated Fourier series.

PEAK_TO_PEAK_BP_ERROR : Uncertainty on the peak-to-peak amplitude in the BP band (float, Magnitude[mag])

This parameter is filled with the uncertainty value of the `peak_to_peak_bp` parameter, computed via a bootstrap technique.

PEAK_TO_PEAK_RP : Peak-to-peak amplitude of the RP band light curve (float, Magnitude[mag])

This parameter is filled with the peak-to-peak amplitude value of the $G_{\rm RP}$ light curve. The peak-to-peak amplitude is calculated as the (maximum) - (minimum) of the modelled folded light curve in the $G_{\rm RP}$ band. The light curve of the target star is modelled with a truncated Fourier series.

PEAK_TO_PEAK_RP_ERROR : Uncertainty on the peak-to-peak amplitude in the RP band (float,

Magnitude[mag])

This parameter is filled with the uncertainty value of the `peak_to_peak_rp` parameter, computed via a bootstrap technique.

PEAK_TO_PEAK_GRVS : Peak-to-peak amplitude of the Grvs band light curve (float, Magnitude[mag])

This parameter is filled with the peak-to-peak amplitude value of the $G_{\rm RVS}$ light curve. The peak-to-peak amplitude is calculated as the (maximum) - (minimum) of the modelled folded light curve in the $G_{\rm RVS}$ band. The light curve of the target star is modelled with a truncated Fourier series.

PEAK_TO_PEAK_GRVS_ERROR : Uncertainty on the peak-to-peak amplitude in the Grvs band (float, Magnitude[mag])

This parameter is filled with the uncertainty value of the `peak_to_peak_grvs` parameter, computed via a bootstrap technique.

PEAK_TO_PEAK_RV : Peak-to-peak amplitude of the radial velocity curve (float, Velocity[km s^{-1}])

This parameter is filled with the peak-to-peak amplitude value of the radial velocity curve. The peak-to-peak amplitude is calculated as the (maximum) - (minimum) of the modelled folded *RV* curve. The *RV* curve of the target star is modelled with a truncated Fourier series.

PEAK_TO_PEAK_RV_ERROR : Uncertainty on the peak-to-peak amplitude in the radial velocity curve (float, Velocity[km s^{-1}])

This parameter is filled with the uncertainty value of the `peak_to_peak_rv` parameter, computed via a bootstrap technique.

METALLICITY : Metallicity of the star from the Fourier parameters of the light curve (float, Abundances[dex])

This parameter is filled with the [Fe/H] metallicity derived for the source from the Fourier parameters of the *G*-band light curve.

METALLICITY_ERROR : Uncertainty of the metallicity parameter (float, Abundances[dex])

This parameter is filled with the uncertainty of the `metallicity` derived from the Fourier parameters of the *G*-band light curve.

R21_G : Fourier decomposition parameter A2/A1 (for G band) (float)

This parameter is filled with the Fourier decomposition parameter $R_{21} = A_2/A_1$, where A_2 is the amplitude of the 2nd harmonic and A_1 is the amplitude of the fundamental harmonic of the truncated Fourier series defined as $(mag(t_j) = zp + \sum[A_i cos(i \times 2\pi\nu_{max} t_j + \phi_i)])$ used to model the *G*-band light curve. Zero-point (zp), period ($1/\nu_{max}$), number of harmonics (i), amplitudes (A_i), and phases (ϕ_i) of the harmonics, are determined using the Levenberg-Marquardt non linear fitting algorithm.

R21_G_ERROR : Uncertainty on the r21 parameter A2/A1 (for G band) (float)

This parameter is filled with the uncertainty value on the `r21_g` parameter, computed via a bootstrap technique.

R31_G : Fourier decomposition parameter A3/A1 (for G band) (float)

This parameter is filled with the Fourier decomposition parameter $R_{21} = A_3/A_1$, where A_3 is the amplitude of the 3rd harmonic and A_1 is the amplitude of the fundamental harmonic of the truncated Fourier series defined as $(mag(t_j) = zp + \sum[A_i cos(i \times 2\pi\nu_{max} t_j + \phi_i)])$ used to model the *G*-band light curve. Zero-point (zp), period ($1/\nu_{max}$), number of harmonics (i), amplitudes (A_i), and phases (ϕ_i) of the harmonics, are determined using the Levenberg-Marquardt non linear fitting algorithm.

R31_G_ERROR : Uncertainty on the r31 parameter A3/A1 (for G band) (float)

This parameter is filled with the uncertainty value of the `r31_g` parameter, computed via a bootstrap technique.

PHI21_G : Fourier decomposition parameter phi2 - 2*phi1 (for G band) (float, Angle[rad])

This parameter is filled with the Fourier decomposition parameter ϕ_{21}: $\phi_2 - 2\phi_1$ value, where ϕ_2 is the phase of the 2nd harmonic and ϕ_1 is the phase of the fundamental harmonic of the truncated Fourier series defined as $(mag(t_j) = zp + \sum[A_i cos(i \times 2\pi\nu_{max} t_j + \phi_i)])$ used to model

the G-band light curve. Zero-point (zp), period ($1/\nu_{max}$), number of harmonics (i), amplitudes (A_i), and phases (ϕ_i) of the harmonics, are determined using the Levenberg-Marquardt non linear fitting algorithm.

PHI21_G_ERROR : Uncertainty on the phi21 parameter in the G band: phi2 - 2*phi1 (for G band) (float, Angle[rad])

This parameter is filled with the uncertainty of the `phi21_g` parameter, computed via a bootstrap technique.

PHI31_G : Fourier decomposition parameter phi3 - 3*phi1 (for G band) (float, Angle[rad])

This parameter is filled with the Fourier decomposition parameter ϕ_{31}: $\phi_3 - 3\phi_1$ value, where ϕ_3 is the phase of the 3rd harmonic and ϕ_1 is the phase of the fundamental harmonic of the truncated Fourier series defined as $(mag(t_j) = zp + \sum[A_i cos(i \times 2\pi\nu_{max} t_j + \phi_i)])$ used to model the G-band light curve. Zero-point (zp), period ($1/\nu_{max}$), number of harmonics (i), amplitudes (A_i), and phases (ϕ_i) of the harmonics, are determined using the Levenberg-Marquardt non linear fitting algorithm.

PHI31_G_ERROR : Uncertainty on the phi31 parameter in the G band: phi3 - 3*phi1 (for G band) (float, Angle[rad])

This parameter is filled with the uncertainty of the `phi31_g`: $\phi_3 - 3\phi_1$ parameter, computed via a bootstrap technique.

R21_GRVS : Fourier decomposition parameter A2/A1 (for Grvs band) (float)

This parameter is filled with the Fourier decomposition parameter $R_{21} = A_2/A_1$, where A_2 is the amplitude of the 2nd harmonic and A_1 is the amplitude of the fundamental harmonic of the truncated Fourier series defined as $(mag(t_j) = zp + \sum[A_i cos(i \times 2\pi\nu_{max} t_j + \phi_i)])$ used to model the G_{RVS}-band light curve. Zero-point (zp), period ($1/\nu_{max}$), number of harmonics (i), amplitudes (A_i), and phases (ϕ_i) of the harmonics, are determined using the Levenberg-Marquardt non linear fitting algorithm.

R21_GRVS_ERROR : Uncertainty on the parameter A2/A1 (for Grvs band) (float)

This parameter is filled with the uncertainty value on the `r21_grvs` parameter, computed via bootstrap technique.

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R31_GRVS : Fourier decomposition parameter A3/A1 (for Grvs band) (float)

This parameter is filled with the Fourier decomposition parameter $R_{21} = A_3/A_1$, where A_3 is the amplitude of the 3rd harmonic and A_1 is the amplitude of the fundamental harmonic of the truncated Fourier series defined as $(mag(t_j) = zp + \sum[A_i cos(i \times 2\pi\nu_{max} t_j + \phi_i)])$ used to model the G_{RVS}-band light curve. Zero-point (zp), period ($1/\nu_{max}$), number of harmonics (i), amplitudes (A_i), and phases (ϕ_i) of the harmonics, are determined using the Levenberg-Marquardt non linear fitting algorithm.

R31_GRVS_ERROR : Uncertainty on the parameter A3/A1 (for Grvs band) (float)

This parameter is filled with the uncertainty value of the `r31_grvs` parameter, computed via bootstrap technique.

PHI21_GRVS : Fourier decomposition parameter phi2 - 2*phi1 (for Grvs band) (float, Angle[rad])

This parameter is filled with the Fourier decomposition parameter ϕ_{21}: $\phi_2 - 2\phi_1$ value, where ϕ_2 is the phase of the 2nd harmonic and ϕ_1 is the phase of the fundamental harmonic of the truncated Fourier series defined as $(mag(t_j) = zp + \sum[A_i cos(i \times 2\pi\nu_{max} t_j + \phi_i)])$ used to model the G_{RVS}-band light curve. Zero-point (zp), period ($1/\nu_{max}$), number of harmonics (i), amplitudes (A_i), and phases (ϕ_i) of the harmonics, are determined using the Levenberg-Marquardt non linear fitting algorithm.

PHI21_GRVS_ERROR : Uncertainty on the parameter phi2 - 2*phi1 (for Grvs band) (float, Angle[rad])

This parameter is filled with the uncertainty of the `phi21_grvs` parameter, computed via bootstrap technique.

PHI31_GRVS : Fourier decomposition parameter phi3 - 3*phi1 (for Grvs band) (float, Angle[rad])

This parameter is filled with the Fourier decomposition parameter ϕ_{31}: $\phi_3 - 3\phi_1$ value, where ϕ_3 is the phase of the 3rd harmonic and ϕ_1 is the phase of the fundamental harmonic of the truncated Fourier series defined as $(mag(t_j) = zp + \sum[A_i cos(i \times 2\pi\nu_{max} t_j + \phi_i)])$ used to model the G_{RVS}-band light curve. Zero-point (zp), period ($1/\nu_{max}$), number of harmonics (i), amplitudes (A_i), and

phases (ϕ_i) of the harmonics, are determined using the Levenberg-Marquardt non linear fitting algorithm.

PHI31_GRVS_ERROR : Uncertainty on the parameter phi3 - 3*phi1 (for Grvs band) (float, Angle[rad])

This parameter is filled with the uncertainty of the `phi31_grvs`: $\phi_3 - 3\phi_1$ parameter, computed via bootstrap technique.

R21_RV : Fourier decomposition parameter A2/A1 (for radial velocity curve) (float)

This parameter is filled with the Fourier decomposition parameter $R_{21} = A_2/A_1$, where A_2 is the amplitude of the 2nd harmonic and A_1 is the amplitude of the fundamental harmonic of the truncated Fourier series defined as ($mag(t_j) = zp + \sum[A_i cos(i \times 2\pi\nu_{max} t_j + \phi_i)]$) used to model the radial velocity curve. Zero-point (zp), period ($1/\nu_{max}$), number of harmonics (i), amplitudes (A_i), and phases (ϕ_i) of the harmonics, are determined using the Levenberg-Marquardt non linear fitting algorithm.

R21_RV_ERROR : Uncertainty on the parameter A2/A1 (for radial velocity curve) (float)

This parameter is filled with the uncertainty value on the `r21_rv` parameter, computed via bootstrap technique.

R31_RV : Fourier decomposition parameter A3/A1 (for radial velocity curve) (float)

This parameter is filled with the Fourier decomposition parameter $R_{21} = A_3/A_1$, where A_3 is the amplitude of the 3rd harmonic and A_1 is the amplitude of the fundamental harmonic of the truncated Fourier series defined as ($mag(t_j) = zp + \sum[A_i cos(i \times 2\pi\nu_{max} t_j + \phi_i)]$) used to model the radial velocity curve. Zero-point (zp), period ($1/\nu_{max}$), number of harmonics (i), amplitudes (A_i), and phases (ϕ_i) of the harmonics, are determined using the Levenberg-Marquardt non linear fitting algorithm.

R31_RV_ERROR : Uncertainty on the parameter A3/A1 (for radial velocity curve) (float)

This parameter is filled with the uncertainty value of the `r31_rv` parameter, computed via bootstrap technique.

Draft

PHI21_RV : Fourier decomposition parameter phi2 - 2*phi1 (for radial velocity curve) (float, Angle[rad])

This parameter is filled with the Fourier decomposition parameter ϕ_{21}: $\phi_2 - 2\phi_1$ value, where ϕ_2 is the phase of the 2nd harmonic and ϕ_1 is the phase of the fundamental harmonic of the truncated Fourier series defined as $(mag(t_j) = zp + \sum[A_i cos(i \times 2\pi\nu_{max} t_j + \phi_i)])$ used to model the radial velocity curve. Zero-point (zp), period ($1/\nu_{max}$), number of harmonics (i), amplitudes (A_i), and phases (ϕ_i) of the harmonics, are determined using the Levenberg-Marquardt non linear fitting algorithm.

PHI21_RV_ERROR : Uncertainty on the parameter phi2 - 2*phi1 (for radial velocity curve) (float, Angle[rad])

This parameter is filled with the uncertainty of the `phi21_rv` parameter, computed via bootstrap technique.

PHI31_RV : Fourier decomposition parameter phi3 - 3*phi1 (for radial velocity curve) (float, Angle[rad])

This parameter is filled with the Fourier decomposition parameter ϕ_{31}: $\phi_3 - 3\phi_1$ value, where ϕ_3 is the phase of the 3rd harmonic and ϕ_1 is the phase of the fundamental harmonic of the truncated Fourier series defined as $(mag(t_j) = zp + \sum[A_i cos(i \times 2\pi\nu_{max} t_j + \phi_i)])$ used to model the radial velocity curve. Zero-point (zp), period ($1/\nu_{max}$), number of harmonics (i), amplitudes (A_i), and phases (ϕ_i) of the harmonics, are determined using the Levenberg-Marquardt non linear fitting algorithm.

PHI31_RV_ERROR : Uncertainty on the parameter phi3 - 3*phi1 (for radial velocity curve) (float, Angle[rad])

This parameter is filled with the uncertainty of the `phi31_rv`: $\phi_3 - 3\phi_1$ parameter, computed via bootstrap technique.

NUM_CLEAN_EPOCHS_G : Number of G FoV epochs used in the fitting algorithm (short)

This parameter is filled with the number of epochs that remain in the G-band light curve after the SOS Cep & RRLyrae outlier removal process.

NUM_CLEAN_EPOCHS_BP : Number of BP epochs used in the fitting algorithm (short)

This parameter is filled with the number of epochs that remain in the $G_{\rm BP}$-band light curve after the SOS Cep & RRLyrae outlier removal process.

NUM_CLEAN_EPOCHS_RP : Number of RP epochs used in the fitting algorithm (short)

This parameter is filled with the number of epochs that remain in the $G_{\rm RP}$-band light curve after the SOS Cep & RRLyrae outlier removal process.

NUM_CLEAN_EPOCHS_GRVS : Number of Grvs epochs used in the fitting algorithm (short)

this parameter is filled with the number of epochs that remain in the G_{RVS} band light curve after the SOS Cep & RRLyrae outlier removal process.

NUM_CLEAN_EPOCHS_RV : Number of radial velocity epochs used in the fitting algorithm (short)

This parameter is filled with the number of epochs that remain in the radial velocity curve after the SOS Cep & RRLyrae outlier removal process.

ZP_MAG_G : Zero point (mag) of the final model of the G band light curve (float, Magnitude[mag])

Zero point (mag) of the final model of the G-band light curve for Cepheids and RR Lyrae stars.

ZP_MAG_BP : Zero point (mag) of the final model of the BP band light curve (float, Magnitude[mag])

Zero point (mag) of the final model of the G_{BP} band light curve for Cepheids and RR Lyrae stars.

ZP_MAG_RP : Zero point (mag) of the final model of the RP band light curve (float, Magnitude[mag])

Zero point (mag) of the final model of the G_{RP} band light curve for Cepheids and RR Lyrae stars.

ZP_MAG_GRVS : Zero point (mag) of the final model of the Grvs band light curve (float, Magnitude[mag])

Draft

Zero point (mag) of the final model of the G_{RVS} band light curve for Cepheids and RR Lyrae stars.

ZP_RV : Zero point (km/s) of the final model of the radial velocity curve (float, Velocity[km s^{-1}])

Zero point (km/s) of the final model of the radial velocity curve for Cepheids and RR Lyrae stars. `zp_rv` is equal to `average_rv` when the radial velocity curve is modelled with a truncated Fourier series. In case of limited data or models showing unphysical oscillations preventing the modeling by the truncated Fourier series, `zp_rv` is null.

NUM_HARMONICS_FOR_P1_G : Number of harmonics used to model the first periodicity of the G-band light curve (byte)

This parameter is filled with the number of harmonics of the truncated Fourier series used to model the *G*-band light curve folded with the `P1` period.

NUM_HARMONICS_FOR_P2_G : Number of harmonics used to model the second periodicity of the G-band light curve (byte)

For a multi-mode pulsator, this parameter is filled with the number of harmonics of the truncated Fourier series used to model the residuals of the *G*-band light curve after subtracting the P1 model.

NUM_HARMONICS_FOR_P1_GRVS : Number of harmonics used to model the first periodicity of the Grvs-band light curve (byte)

This parameter is filled with the number of harmonics of the truncated Fourier series used to model the $G_{\rm RVS}$-band light curve folded with the `P1` period.

NUM_HARMONICS_FOR_P2_GRVS : Number of harmonics used to model the second periodicity of the Grvs-band light curve (byte)

For a multi-mode pulsator, this parameter is filled with the number of harmonics of the truncated Fourier series used to model the residuals of the $G_{\rm RVS}$-band light curve after subtracting the P1 model.

NUM_HARMONICS_FOR_P1_RV : Number of harmonics used to model the first periodicity of the radial velocity curve (byte)

Draft

This parameter is filled with the number of harmonics of the truncated Fourier series used to model the RV curve folded with the P1 period.

NUM_HARMONICS_FOR_P2_RV : Number of harmonics used to model the second periodicity of the radial velocity curve (byte)

For a multi-mode pulsator, this parameter is filled with the number of harmonics of the truncated Fourier series used to model the residuals of the RV curve after subtracting the P1 model.

G_ABSORPTION : Interstellar absorption in the G band (float, Magnitude[mag])

This parameter is filled with values coming from the estimate of the interstellar extinction toward the investigated pulsators. The period-colour-amplitude relation was used.

G_ABSORPTION_ERROR : Error on the interstellar absorption in the G band (float, Magnitude[mag])

Error on the interstellar absorption in the *G*-band (`g_absorption_error`): This parameter is filled with the r.m.s. errors of the relations used to estimate the interstellar absorption.

LC_QUALITY_FLAG_G : Photometry quality flags for the G band (float)

Implementation of photometry quality flags for the *G* band to discriminate between good and problematic light curves. This flag is defined at SOS level as the ratio between stadard deviation of the G band observations and the rms of the residuals of the *G*-band Fourier model. It is always larger than zero, and the larger the value of the flag the cleaner the light curve. Tipically for values smaller than ~ 6, the light curve is noisy, indicating photometric problems (e.g. crowding and blending), or multi mode pulsation, Blazhko effect.

LC_QUALITY_FLAG_BP : Photometry quality flags for the BP band (float)

Implementation of photometry quality flags for the $G_{\rm BP}$ band to discriminate between good and problematic light curves. This flag is defined at SOS level as the ratio between stadard deviation of the BP band observations and the rms of the residuals of the $G_{\rm BP}$-band Fourier model. It is always larger than zero, and the larger the value of the flag the cleaner the light curve. Tipically for values smaller than ~ 6, the light curve is noisy, indicating photometric problems (e.g. crowding and blending), or multi mode pulsation, Blazhko effect.

Draft

LC_QUALITY_FLAG_RP : Photometry quality flags for the RP band (float)

Implementation of photometry quality flags for the $G_{\rm RP}$ band to discriminate between good and problematic light curves. This flag is defined at SOS level as the ratio between stadard deviation of the BP band observations and the rms of the residuals of the $G_{\rm RP}$-band Fourier model. It is always larger than zero, and the larger the value of the flag the cleaner the light curve. Tipically for values smaller than $\sim$ 6, the light curve is noisy, indicating photometric problems (e.g. crowding and blending), or multi mode pulsation, Blazhko effect.

LC_QUALITY_FLAG_GRVS : Photometry quality flags for the Grvs band (float)

Implementation of photometry quality flags for the $G_{\rm RVS}$ band to discriminate between good and problematic light curves. This flag is defined at SOS level as the ratio between stadard deviation of the BP band observations and the rms of the residuals of the $G_{\rm RVS}$-band Fourier model. It is always larger than zero, and the larger the value of the flag the cleaner the light curve. Tipically for values smaller than $\sim$ 6, the light curve is noisy, indicating photometric problems (e.g. crowding and blending), or multi mode pulsation, Blazhko effect.

RV_QUALITY_FLAG : Quality flags for the radial velocity (float)

Implementation of radial velocity (RV) quality flag to discriminate between good and problematic RV curves. This flag is defined at SOS level as the ratio between stadard deviation of the RV observations and the rms of the residuals of the RV Fourier model. It is always larger than zero, and the larger the value of the flag the cleaner the velocity curve.

PHI_RISE : Parameter measuring the skewness of the light curve (float)

The time of rise to maximum light to be used in addition to the position in the Period- Amplitude diagram to distinguish RRab from RRc stars, particularly in the short-period regime (P $<$ 0.4 d), and to disentangle the different Cepheid types (classical, anomalous and BL Her) in the range of periods 1-2.5 d.

UNIFORMITY_INDEX_G : Uniformity index in the G band (float)

Uniformity index of a folded time series, defined by Madore and Fredmann 2005, ApJ 630, 1054. It can be used to quantify how evenly the time series is sampled in phase.

UNIFORMITY_INDEX_BP : Uniformity index in the BP band (float)

Draft

Uniformity index of a folded time series, defined by Madore and Fredmann 2005, ApJ 630, 1054. It can be used to quantify how evenly the time series is sampled in phase.

UNIFORMITY_INDEX_RP : Uniformity index in the RP band (float)

Uniformity index of a folded time series, defined by Madore and Fredmann 2005, ApJ 630, 1054. It can be used to quantify how evenly the time series is sampled in phase.

UNIFORMITY_INDEX_GRVS : Uniformity index in the Grvs band (float)

Uniformity index of a folded time series, defined by Madore and Fredmann 2005, ApJ 630, 1054. It can be used to quantify how evenly the time series is sampled in phase.

UNIFORMITY_INDEX_RV : Uniformity index in the radial velocity time series (float)

Uniformity index of a folded time series, defined by Madore and Fredmann 2005, ApJ 630, 1054. It can be used to quantify how evenly the time series is sampled in phase.

REFERENCE_TIME_G : Reference time of the Fourier modelled G-band light curve (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Reference time for the Fourier modelled G-band light curve.

REFERENCE_TIME_BP : Reference time of the Fourier modelled BP-band light curve (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Reference time for the Fourier modelled G_{BP}-band light curve.

REFERENCE_TIME_RP : Reference time of the Fourier modelled RP-band light curve (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Reference time for the Fourier modelled G_{RP}-band light curve.

REFERENCE_TIME_GRVS : Reference time of the Fourier modelled Grvs-band light curve (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Draft

Reference time for the Fourier modelled $G_{\rm RVS}$-band light curve.

REFERENCE_TIME_RV : Reference time of the Fourier modelled radial velocity curve (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Reference time for the Fourier modelled radial velocity curve.

FUND_FREQ1 : First frequency of the non-linear Fourier modelling (double, Frequency[day^{-1}])

First frequency of the non-linear Fourier modelling. It applies to all three G, $G_{\rm BP}$, and $G_{\rm RP}$ bands and the radial velocity curve.

FUND_FREQ1_ERROR : Error of the first frequency of the non-linear Fourier modelling (float, Frequency[day^{-1}])

Error of the first frequency of the non-linear Fourier modelling.

FUND_FREQ2 : Second frequency of the non-linear Fourier modelling in the G band (double, Frequency[day^{-1}])

Second frequency of the non-linear Fourier modelling for the G band only.

FUND_FREQ2_ERROR : Error of the second frequency of the non-linear Fourier modelling in the G band (float, Frequency[day^{-1}])

Error of the second frequency of the non-linear Fourier modelling. It applies to the G band only.

FUND_FREQ1_HARMONIC_AMPL_G : Amplitudes of the Fourier model for the first frequency in the G band (float[16] array, Magnitude[mag])

Amplitudes of the Fourier model fitted to the observed G-band light curve.

FUND_FREQ1_HARMONIC_AMPL_G_ERROR : Errors of the amplitudes of the Fourier model for the first frequency in the G band (float[16] array, Magnitude[mag])

Errors of the amplitudes of the Fourier model fitted to the observed G-band light curve.

Draft

FUND_FREQ1_HARMONIC_PHASE_G : Phases of the Fourier model for the first frequency in the G band (float[16] array, Angle[rad])

Phases of the Fourier model fitted to the observed G-band light curve.

FUND_FREQ1_HARMONIC_PHASE_G_ERROR : Errors of the phases of the Fourier model for the first frequency in the G band (float[16] array, Angle[rad])

Errors of the phases of the Fourier model fitted to the observed G-band light curve.

FUND_FREQ1_HARMONIC_AMPL_GRVS : Amplitudes of the Fourier model for the first frequency in the Grvs band (float[16] array, Magnitude[mag])

Amplitudes of the Fourier model fitted to the observed $G_{\rm RVS}$-band light curve.

FUND_FREQ1_HARMONIC_AMPL_GRVS_ERROR : Errors of the amplitudes of the Fourier model for the first frequency in the Grvs band (float[16] array, Magnitude[mag])

Errors of the amplitudes of the Fourier model fitted to the observed $G_{\rm RVS}$-band light curve.

FUND_FREQ1_HARMONIC_PHASE_GRVS : Phases of the Fourier model for the first frequency in the Grvs band (float[16] array, Angle[rad])

Phases of the Fourier model fitted to the observed $G_{\rm RVS}$-band light curve.

FUND_FREQ1_HARMONIC_PHASE_GRVS_ERROR : Errors of the phases of the Fourier model for the first frequency in the Grvs band (float[16] array, Angle[rad])

Errors of the phases of the Fourier model fitted to the observed $G_{\rm RVS}$-band light curve.

FUND_FREQ1_HARMONIC_AMPL_RV : Amplitudes of the Fourier model for the first frequency of the radial velocity curve (float[16] array, Velocity[km s^{-1}])

Amplitudes of the Fourier model fitted to the observed radial velocity curve.

Draft

FUND_FREQ1_HARMONIC_AMPL_RV_ERROR : Errors of the amplitudes of the Fourier model for the first frequency of the radial velocity curve (float[16] array, Velocity[km s^{-1}])

Errors of the amplitudes of the Fourier model fitted to the observed radial velocity curve.

FUND_FREQ1_HARMONIC_PHASE_RV : Phases of the Fourier model for the first frequency of the radial velocity curve (float[16] array, Angle[rad])

Phases of the Fourier model fitted to the observed radial velocity curve.

FUND_FREQ1_HARMONIC_PHASE_RV_ERROR : Errors of the phases of the Fourier model for the first frequency of the radial velocity curve (float[16] array, Angle[rad])

Errors of the phases of the Fourier model fitted to the observed radial velocity curve.

BEST_CLASSIFICATION : Best RR Lyrae classification estimate out of (RRC, RRAB, RRD) (string)

Classification of an RR Lyrae star according to the pulsation mode: 'RRc' for first overtone, 'RRab' for fundamental mode, and 'RRd' for double modes, obtained using the period-amplitude diagram in the *G*-band; the plots of the Fourier parameters R21 and Phi2 vs period and the Petersen diagram.

MASS : Mass of the star (float, Mass[Solar Mass])

This parameter is filled with the value of the mass derived for double mode RR Lyrae stars from the relation between stellar metallicity and the ratio of dominant over secondary periodicites (Petersen diagram).

MASS_ERROR : Error on the mass of the star (float, Mass[Solar Mass])

This parameter is filled with the error on the value of the mass derived for double mode RR Lyrae stars from the relation between stellar metallicity and the ratio of pominant over secondary periodicites (Petersen diagram).

BLAZHKO_FLAG : Flag for the occurrence of the Blazhko effect (Boolean)

Draft

This parameter is set to true if RR Lyrae pulsators show a modulation in amplitude and/or phase in the light curve known as “Blazhko” effect.

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18.17 VARI_SHORT_TIMESCALE [TAP]

This table describes the short-timescale sources.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

AMPLITUDE_ESTIMATE : Amplitude estimate of all per-CCD G-band photometry (95th quantile - 5th quantile) (float, Magnitude[mag])

This parameter is filled with the amplitude estimate from per-CCD *G*-band photometry. This amplitude estimate is calculated as the quantile difference (95th quantile − 5th quantile) of all per-CCD *G*-band measurements.

NUMBER_OF_FOV_TRANSITS : Number of FoV transits with more than 7 CCD measurements after time series cleaning (short)

Number of FoV transits that have more than 7 CCD measurements after time series cleaning (i.e., not necessarily from all the FoV transits available for the source).

MEAN_OF_FOV_ABBE_VALUES : Mean of per-FoV Abbe values derived from per-CCD G-band photometry (float)

This parameter represents the mean of per-FoV Abbe values derived from per-CCD *G*-band photometry. Considering a given source, for each of its FoV transits containing more than one per-CCD measurement, the associated Abbe value from per-CCD *G*-band photometry is derived

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as

$$\text{Abbe} = \frac{\sum_{i=1}^{n-1}(y_{i+1} - y_i)^2}{2\sum_{i=1}^{n}(y_i - \bar{y})^2}, \tag{44}$$

where $\{t_i, y_i\}$ is a time-sorted time series per FoV of size n, such that $t_i < t_{i+1}$ for all $i < n$, and $\bar{y}$ is the mean of the per-CCD measurements of the transit. The value of `mean_of_fov_abbe_values` is calculated as the mean of these per-FoV Abbe values.

VARIOGRAM_NUM_POINTS : Number of selected timescale(s) derived from the variogram (byte)

This parameter represents the number of selected timescale(s) derived from the variogram. Presently it is only 1, but this may change in a future release.

VARIOGRAM_CHAR_TIMESCALES : Characteristic timescale(s) of variability (float[`variogram_num_points`] array, Time[day])

This parameter represents the variogram characteristic timescale(s) derived from the variogram analysis of the source, if it has been flagged as a short timescale variable candidate.

VARIOGRAM_VALUES : Variogram values associated with the characteristic timescale(s) of variability (double[`variogram_num_points`] array, Misc[mag^2])

This parameter represents the variogram value(s) associated with the variogram characteristic timescales `variogram_char_timescales`.

FREQUENCY : Frequency search result for either G CCD, G FoV, BP or RP photometry (double, Frequency[day^{-1}])

The parameter represents the frequency value resulting from the period search (method LEAST_SQUARE) performed either on the per-CCD G-band photometry, per-FoV G-band photometry, $G_{\rm BP}$ photometry or $G_{\rm RP}$ photometry. Only per-CCD G-band photometry is used to compute the frequency. It is set to NULL if the number of CCD measurements is less than 180.

Draft

18.18 VARI_SUMMARY [TAP]

Summary table that provides the information on where a source can be found in the different Variability tables and statistical parameters of time series, using only transits not rejected.

Note that NULL is reported when the statistical parameter value is missing or cannot be calculated.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

NUM_SELECTED_G_FOV : Total number of G FOV transits selected for variability analysis (short)

The number of processed observations for variability analyses of this source, using only transits not rejected.

MEAN_OBS_TIME_G_FOV : Mean observation time for G FoV transits (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Name: The mean observation time

Control parameters: None

Output: Let y_i be a time series of size N at times t_i. The mean $\bar{t}$ is defined as

$$\bar{t} = \frac{1}{N} \sum_{i=1}^{N} t_i. \tag{45}$$

TIME_DURATION_G_FOV : Time duration of the time series for G FoV transits (float, Time[day])

Name: The time duration of the time series

Control parameters: None

Output: Let y_i be a time series of size N at times t_i, with $i = 1$ to N. The time duration of the time series is equal to $t_N - t_1$.

MIN_MAG_G_FOV : Minimum G FoV magnitude (float, Magnitude[mag])

Name: The minimum value of the time series

Control parameters: None

Output: Let y_i be a time series of size N at times t_i, with $i = 1$ to N. The minimum value of the time series is defined as:

$$min(y_i) \; \forall i \in (1, N) \tag{46}$$

MAX_MAG_G_FOV : Maximum G FoV magnitude (float, Magnitude[mag])

Name: The maximum value of the time series

Control parameters: None

Output: Let y_i be a time series of size N at times t_i, with $i = 1$ to N. The maximum value of the time series is defined as:

$$max(y_i) \; \forall i \in (1, N) \tag{47}$$

MEAN_MAG_G_FOV : Mean G FoV magnitude (float, Magnitude[mag])

Name: The mean of the time series

Control parameters: None

Output: Let y_i be a time series of size N. The mean $\bar{y}$ is defined as

$$\bar{y} = \frac{1}{N} \sum_{i=1}^{N} y_i. \tag{48}$$

Draft

MEAN_FLUX_G_FOV : Mean G FoV flux (float, Flux[e^- s^{-1}])

Name: The mean of the time series

Control parameters: None

Output: Let y_i be a time series of size N. The mean $\bar{y}$ is defined as

$$\bar{y} = \frac{1}{N} \sum_{i=1}^{N} y_i. \tag{49}$$

MEDIAN_MAG_G_FOV : Median G FoV magnitude (float, Magnitude[mag])

Name: The median of the time series

Control parameters: None

Output: The 50th percentile unweighted value.

Let y_i be a time series of size N ordered such as $y_{(1)} \leq y_{(2)} \leq \cdots \leq y_{(N)}$. The m-th (per cent) percentile P_m is defined for $0 < m \leq 100$ as follows:

$$P_m = \begin{cases} y_{(1)} & \text{if } 0 < m < 100/N, \\ y_{(i)} & \text{if } Nm/100 - i = 0, \\ y_{(i+1)} & \text{otherwise.} \end{cases} \tag{50}$$

MEDIAN_FLUX_G_FOV : Median G FoV flux (float, Flux[e^- s^{-1}])

Name: The median of the time series

Control parameters: None

Output: The 50th percentile unweighted value.

Let y_i be a time series of size N ordered such as $y_{(1)} \leq y_{(2)} \leq \cdots \leq y_{(N)}$. The m-th (per cent) percentile P_m is defined for $0 < m \leq 100$ as follows:

$$P_m = \begin{cases} y_{(1)} & \text{if } 0 < m < 100/N, \\ y_{(i)} & \text{if } Nm/100 - i = 0, \\ y_{(i+1)} & \text{otherwise.} \end{cases} \tag{51}$$

RANGE_MAG_G_FOV : Difference between the highest and lowest G FoV magnitudes (float, Magnitude[mag])

Draft

Name: The range of the time series

Control parameters: None

Output: Let y_i be a time series, $y_{\max}$ its largest element, and $y_{\min}$ its smallest element, then the range is defined as

$$R = y_{\max} - y_{\min} \tag{52}$$

TRIMMED_RANGE_MAG_G_FOV : Trimmed difference between the highest and lowest G FoV magnitudes (float, Magnitude[mag])

Name: The trimmed range

Control parameters: trim level $\alpha \in]0, 50[$

Output: The trimmed unweighted range.

Let y_i be a time series of size N, with error bars Δy_i. Let Y_α be the αth percentile, and $Y_{100-\alpha}$ be the $(100-\alpha)$th percentile. Let y'_i be the trimmed time series of size N' which is derived by iteratively removing the smallest or the largest values of the y_i time series until $Y_\alpha < y'_i < Y_{100-\alpha}$ for all $i \in \{1, \cdots, N'\}$. The trimmed range R' is defined as

$$R' = y'_{\max} - y'_{\min} \tag{53}$$

The trim level α was set to 5.0.

STD_DEV_MAG_G_FOV : Square root of the unweighted G FoV magnitude variance (float, Magnitude[mag])

Name: The square root of the unbiased unweighted variance.

Output: Let y_i be a time series of size N. The unweighted standard deviation $\hat{\sigma}$ is defined as the square root of the sample-size unbiased unweighted variance:

$$\hat{\sigma} = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(y_i - \overline{y})^2}.$$

STD_DEV_FLUX_G_FOV : Square root of the unweighted G FoV flux variance (float, Flux[$e^-\ s^{-1}$])

Name: The square root of the unbiased unweighted variance.

Output: Let y_i be a time series of size N. The unweighted standard deviation $\hat{\sigma}$ is defined as the square root of the sample-size unbiased unweighted variance:

$$\hat{\sigma} = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(y_i - \overline{y})^2}.$$

SKEWNESS_MAG_G_FOV : Standardized unweighted G FoV magnitude skewness (float)

Name: The standardised unbiased unweighted skewness.

Output: Let y_i be a time series of size $N > 2$. The sample-size unbiased unweighted skewness moment $\mathcal{E}$ is defined as:

$$\mathcal{E} = \frac{N}{(N-1)(N-2)}\sum_{i=1}^{N}(y_i - \overline{y})^3.$$

The standardized unbiased skewness E is defined as:

$$E = \frac{\mathcal{E}}{\hat{\sigma}^3}$$

where $\hat{\sigma}$ is the square root of the unbiased unweighted variance around the unweighted mean. While $\mathcal{E}$ is an unbiased estimate of the population value, E becomes unbiased in the limit of large N.

SKEWNESS_FLUX_G_FOV : Standardized unweighted G FoV flux skewness (float)

Name: The standardised unbiased unweighted skewness.

Output: Let y_i be a time series of size $N > 2$. The sample-size unbiased unweighted skewness moment $\mathcal{E}$ is defined as:

$$\mathcal{E} = \frac{N}{(N-1)(N-2)}\sum_{i=1}^{N}(y_i - \overline{y})^3.$$

The standardized unbiased skewness E is defined as:

$$E = \frac{\mathcal{E}}{\hat{\sigma}^3}$$

where $\hat{\sigma}$ is the square root of the unbiased unweighted variance around the unweighted mean. While $\mathcal{E}$ is an unbiased estimate of the population value, E becomes unbiased in the limit of large N.

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KURTOSIS_MAG_G_FOV : Standardized unweighted G FoV magnitude kurtosis (float)

Name: The standardised unbiased unweighted kurtosis.

Output: Let y_i be a time series of size $N > 3$. The sample-size unbiased unweighted kurtosis cumulant $\mathcal{K}$ is defined as:

$$\mathcal{K} = \frac{N(N+1)}{(N-1)(N-2)(N-3)} \sum_{i=1}^{N} (y_i - \overline{y})^4 - \frac{3}{(N-2)(N-3)} \left[\sum_{i=1}^{N} (y_i - \overline{y})^2 \right]^2 .$$

The standardized unbiased kurtosis K is defined as:

$$K = \frac{\mathcal{K}}{\hat{\sigma}^4}$$

where $\hat{\sigma}^2$ is the unbiased unweighted variance around the unweighted mean. While $\mathcal{K}$ is an unbiased estimate of the population value, K becomes unbiased in the limit of large N.

KURTOSIS_FLUX_G_FOV : Standardized unweighted G FoV flux kurtosis (float)

Name: The standardised unbiased unweighted kurtosis.

Output: Let y_i be a time series of size $N > 3$. The sample-size unbiased unweighted kurtosis cumulant $\mathcal{K}$ is defined as:

$$\mathcal{K} = \frac{N(N+1)}{(N-1)(N-2)(N-3)} \sum_{i=1}^{N} (y_i - \overline{y})^4 - \frac{3}{(N-2)(N-3)} \left[\sum_{i=1}^{N} (y_i - \overline{y})^2 \right]^2 .$$

The standardized unbiased kurtosis K is defined as:

$$K = \frac{\mathcal{K}}{\hat{\sigma}^4}$$

where $\hat{\sigma}^2$ is the unbiased unweighted variance around the unweighted mean. While $\mathcal{K}$ is an unbiased estimate of the population value, K becomes unbiased in the limit of large N.

MAD_MAG_G_FOV : Median Absolute Deviation (MAD) for G FoV transits (float, Magnitude[mag])

Name: The Median Absolute Deviation (MAD)

Draft

Control parameters: None

Output: Let y_i be a time series of size N. The MAD is defined as the median of the absolute deviations from the median of the data, scaled by a factor of $1/\Phi^{-1}(3/4) \approx 1.4826$ (where Φ^{-1} is the inverse of the cumulative distribution function for the standard normal distribution), so that the expectation of the scaled MAD at large N equals the standard deviation of a normal distribution:

$$\mathrm{MAD} = 1.4826\ \mathrm{median}\{|y_i - \mathrm{median}\{y_j, \forall j \in (1,N)\}|, \forall i \in (1,N)\}. \quad (54)$$

ABBE_MAG_G_FOV : Abbe value for G FoV transits (float)

Name: The Abbe value

Control parameters: None

Output: Let $\{t_i, y_i\}$ be a time-sorted time series of size N, such that $t_i < t_{i+1}$ for all $i < N$. The Abbe value $\mathcal{A}$ is defined as

$$\mathcal{A} = \frac{\sum_{i=1}^{N-1}(y_{i+1} - y_i)^2}{2\sum_{i=1}^{N}(y_i - \bar{y})^2} \quad (55)$$

where $\bar{y}$ is the unweighted mean.

IQR_MAG_G_FOV : Interquartile range for G FoV transits (float, Magnitude[mag])

Name: The Interquartile Range (IQR)

Control parameters: None

Output: The difference between the (unweighted) 75th and 25th percentile values: IQR= $P_{75} - P_{25}$.

Let y_i be a time series of size N ordered such as $y_{(1)} \leq y_{(2)} \leq \cdots \leq y_{(N)}$. The m-th (per cent) percentile P_m is defined for $0 < m \leq 100$ as follows:

$$P_m = \begin{cases} y_{(1)} & \text{if } 0 < m < 100/N, \\ y_{(i)} & \text{if } Nm/100 - i = 0, \\ y_{(i+1)} & \text{otherwise.} \end{cases} \quad (56)$$

STETSON_MAG_G_FOV : Stetson G FoV variability index (float)

Name: The single-band Stetson variability index, herein set to 0.1 days.

Control parameters: The pairing time interval

Output: Let y_i be a time series of size N, with error bars Δy_i. The Stetson value (Stetson 1996) is based on paired observations closely separated in time (within a chosen time interval). The time difference between the components of a pair should be small compared to the shortest time-scale of variation being sought. If this condition is not satisfied, a transformation is applied to the unpaired measurement, in order to include its information together with paired observations. The maximum number n of pairs may exceed $N/2$, since multiple pairs can be formed within a given time interval (as long as the latter is much smaller than the minimum variability time-scale sought). The Stetson J index is defined as

$$J = \frac{1}{n}\sum_{k=1}^{n} \mathrm{sgn}(P_k)\sqrt{|P_k|},$$

where the index k refers to the k-th pair, composed of measurements i and j, such that

$$P_k = \begin{cases} \delta_{i(k)}\delta_{j(k)}, & \text{if } i(k) \neq j(k) \\ \delta_{i(k)}^2 - 1, & \text{if } i(k) = j(k) \end{cases}$$

with the normalized deviation of the i-th measurement from the time-series mean $\bar{y}$ expressed by:

$$\delta_{i(k)} = \sqrt{\frac{N}{N-1}}\,\frac{y_{i(k)} - \bar{y}}{\Delta y_{i(k)}}.$$

STD_DEV_OVER_RMS_ERR_MAG_G_FOV : Signal-to-Noise G FoV estimate (float)

Name: The signal-to-noise estimate

Control parameters: None

Output: Let y_i be a time series of size N, with error bars Δy_i. The signal-to-noise ratio (S/N) is estimated as

$$\mathrm{S/N} = \sqrt{\frac{\sum_{i=1}^{N}(y_i - \bar{y})^2}{\sum_{i=1}^{N}\Delta y_i^2}} \tag{57}$$

where $\bar{y}$ is the mean of the time-series values.

OUTLIER_MEDIAN_G_FOV : Greatest absolute deviation from the G FoV median normalized by the error (float)

Name: The most outlying measurement with respect to the median

Control parameters: None

Output: Let y_i be a time series of size N, with error bars Δy_i. The most outlying measurement is defined as the greatest absolute deviation from the median normalized by the error:

$$\text{outlierMedian} = \max\left\{\frac{|y_i - \text{median}\{y_j, \forall j \in (1,N)\}|}{\Delta y_i}, \forall i \in (1,N)\right\}. \quad (58)$$

NUM_SELECTED_BP : Total number of BP observations selected for variability analysis (short)

The number of processed observations for variability analyses of this source, using only transits not rejected.

MEAN_OBS_TIME_BP : Mean observation time for BP observations (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Name: The mean observation time

Control parameters: None

Output: Let y_i be a time series of size N at times t_i. The mean $\bar{t}$ is defined as

$$\bar{t} = \frac{1}{N}\sum_{i=1}^{N} t_i. \quad (59)$$

TIME_DURATION_BP : Time duration of the BP time series (float, Time[day])

Name: The time duration of the time series

Control parameters: None

Output: Let y_i be a time series of size N at times t_i, with $i = 1$ to N. The time duration of the time series is equal to $t_N - t_1$.

MIN_MAG_BP : Minimum BP magnitude (float, Magnitude[mag])

Name: The minimum value of the time series

Control parameters: None

Output: Let y_i be a time series of size N at times t_i, with $i = 1$ to N. The minimum value of the time series is defined as:

$$min(y_i) \; \forall i \in (1, N) \tag{60}$$

MAX_MAG_BP : Maximum BP magnitude (float, Magnitude[mag])

Name: The maximum value of the time series

Control parameters: None

Output: Let y_i be a time series of size N at times t_i, with $i = 1$ to N. The maximum value of the time series is defined as:

$$max(y_i) \; \forall i \in (1, N) \tag{61}$$

MEAN_MAG_BP : Mean BP magnitude (float, Magnitude[mag])

Name: The mean of the time series

Control parameters: None

Output: Let y_i be a time series of size N. The mean $\overline{y}$ is defined as

$$\overline{y} = \frac{1}{N} \sum_{i=1}^{N} y_i. \tag{62}$$

MEAN_FLUX_BP : Mean BP flux (float, Flux[e^- s^{-1}])

Name: The mean of the time series

Control parameters: None

Output: Let y_i be a time series of size N. The mean $\overline{y}$ is defined as

$$\overline{y} = \frac{1}{N} \sum_{i=1}^{N} y_i. \tag{63}$$

MEDIAN_MAG_BP : Median BP magnitude (float, Magnitude[mag])

Draft

Name: The median of the time series

Control parameters: None

Output: The 50th unweighted percentile value.

Let y_i be a time series of size N ordered such as $y_{(1)} \leq y_{(2)} \leq \cdots \leq y_{(N)}$. The m-th (per cent) percentile P_m is defined for $0 < m \leq 100$ as follows:

$$P_m = \begin{cases} y_{(1)} & \text{if } 0 < m < 100/N, \\ y_{(i)} & \text{if } Nm/100 - i = 0, \\ y_{(i+1)} & \text{otherwise.} \end{cases} \tag{64}$$

MEDIAN_FLUX_BP : Median BP flux (float, Flux[e^- s^{-1}])

Name: The median of the time series

Control parameters: None

Output: The 50th unweighted percentile value.

Let y_i be a time series of size N ordered such as $y_{(1)} \leq y_{(2)} \leq \cdots \leq y_{(N)}$. The m-th (per cent) percentile P_m is defined for $0 < m \leq 100$ as follows:

$$P_m = \begin{cases} y_{(1)} & \text{if } 0 < m < 100/N, \\ y_{(i)} & \text{if } Nm/100 - i = 0, \\ y_{(i+1)} & \text{otherwise.} \end{cases} \tag{65}$$

RANGE_MAG_BP : Difference between the highest and lowest BP magnitudes (float, Magnitude[mag])

Name: The range of the time series

Control parameters: None

Output: Let y_i be a time series, $y_{\max}$ its largest element, and $y_{\min}$ its smallest element, then the range is defined as

$$R = y_{\max} - y_{\min} \tag{66}$$

TRIMMED_RANGE_MAG_BP : Trimmed difference between the highest and lowest BP magnitudes (float, Magnitude[mag])

Draft

Name: The trimmed range

Control parameters: trim level $\alpha \in]0, 50[$

Output: The trimmed unweighted range.

Let y_i be a time series of size N, with error bars Δy_i. Let Y_α be the αth percentile, and $Y_{100-\alpha}$ be the $(100-\alpha)$th percentile. Let y'_i be the trimmed time series of size N' which is derived by iteratively removing the smallest or the largest values of the y_i time series until $Y_\alpha < y'_i < Y_{100-\alpha}$ for all $i \in \{1, \cdots, N'\}$. The trimmed range R' is defined as

$$R' = y'_{\max} - y'_{\min} \tag{67}$$

The trim level α was set to 5.0.

STD_DEV_MAG_BP : Square root of the unweighted BP magnitude variance (float, Magnitude[mag])

Name: The square root of the unbiased unweighted variance.

Output: Let y_i be a time series of size N. The unweighted standard deviation $\hat{\sigma}$ is defined as the square root of the sample-size unbiased unweighted variance:

$$\hat{\sigma} = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(y_i - \overline{y})^2}.$$

STD_DEV_FLUX_BP : Square root of the unweighted BP flux variance (float, Flux[e^- s^{-1}])

Name: The square root of the unbiased unweighted variance.

Output: Let y_i be a time series of size N. The unweighted standard deviation $\hat{\sigma}$ is defined as the square root of the sample-size unbiased unweighted variance:

$$\hat{\sigma} = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(y_i - \overline{y})^2}.$$

SKEWNESS_MAG_BP : Standardized unweighted BP magnitude skewness (float)

Name: The standardised unbiased unweighted skewness.

Output: Let y_i be a time series of size $N > 2$. The sample-size unbiased unweighted skewness moment $\mathcal{E}$ is defined as:

$$\mathcal{E} = \frac{N}{(N-1)(N-2)} \sum_{i=1}^{N} (y_i - \overline{y})^3.$$

The standardized unbiased skewness E is defined as:

$$E = \frac{\mathcal{E}}{\hat{\sigma}^3}$$

where $\hat{\sigma}$ is the square root of the unbiased unweighted variance around the unweighted mean. While $\mathcal{E}$ is an unbiased estimate of the population value, E becomes unbiased in the limit of large N.

SKEWNESS_FLUX_BP : Standardized unweighted BP flux skewness (float)

Name: The standardised unbiased unweighted skewness.

Output: Let y_i be a time series of size $N > 2$. The sample-size unbiased unweighted skewness moment $\mathcal{E}$ is defined as:

$$\mathcal{E} = \frac{N}{(N-1)(N-2)} \sum_{i=1}^{N} (y_i - \overline{y})^3.$$

The standardized unbiased skewness E is defined as:

$$E = \frac{\mathcal{E}}{\hat{\sigma}^3}$$

where $\hat{\sigma}$ is the square root of the unbiased unweighted variance around the unweighted mean. While $\mathcal{E}$ is an unbiased estimate of the population value, E becomes unbiased in the limit of large N.

KURTOSIS_MAG_BP : Standardized unweighted BP magnitude kurtosis (float)

Name: The standardised unbiased unweighted kurtosis.

Output: Let y_i be a time series of size $N > 3$. The sample-size unbiased unweighted kurtosis cumulant $\mathcal{K}$ is defined as:

$$\mathcal{K} = \frac{N(N+1)}{(N-1)(N-2)(N-3)} \sum_{i=1}^{N} (y_i - \overline{y})^4 - \frac{3}{(N-2)(N-3)} \left[\sum_{i=1}^{N} (y_i - \overline{y})^2 \right]^2.$$

The standardized unbiased kurtosis K is defined as:

$$K = \frac{\mathcal{K}}{\hat{\sigma}^4}$$

where $\hat{\sigma}^2$ is the unbiased unweighted variance around the unweighted mean. While $\mathcal{K}$ is an unbiased estimate of the population value, K becomes unbiased in the limit of large N.

KURTOSIS_FLUX_BP : Standardized unweighted BP flux kurtosis (float)

Name: The standardised unbiased unweighted kurtosis.

Output: Let y_i be a time series of size $N > 3$. The sample-size unbiased unweighted kurtosis cumulant $\mathcal{K}$ is defined as:

$$\mathcal{K} = \frac{N(N+1)}{(N-1)(N-2)(N-3)} \sum_{i=1}^{N} (y_i - \bar{y})^4 - \frac{3}{(N-2)(N-3)} \left[\sum_{i=1}^{N} (y_i - \bar{y})^2 \right]^2 .$$

The standardized unbiased kurtosis K is defined as:

$$K = \frac{\mathcal{K}}{\hat{\sigma}^4}$$

where $\hat{\sigma}^2$ is the unbiased unweighted variance around the unweighted mean. While $\mathcal{K}$ is an unbiased estimate of the population value, K becomes unbiased in the limit of large N.

MAD_MAG_BP : Median Absolute Deviation (MAD) for BP observations (float, Magnitude[mag])

Name: The Median Absolute Deviation (MAD)

Control parameters: None

Output: Let y_i be a time series of size N. The MAD is defined as the median of the absolute deviations from the median of the data, scaled by a factor of $1/\Phi^{-1}(3/4) \approx 1.4826$ (where Φ^{-1} is the inverse of the cumulative distribution function for the standard normal distribution), so that the expectation of the scaled MAD at large N equals the standard deviation of a normal distribution:

$$\mathrm{MAD} = 1.4826\ \mathrm{median}\{|y_i - \mathrm{median}\{y_j, \forall j \in (1,N)\}|, \forall i \in (1,N)\}. \quad (68)$$

ABBE_MAG_BP : Abbe value for BP observations (float)

Name: The Abbe value

Control parameters: None

Output: Let $\{t_i, y_i\}$ be a time-sorted time series of size N, such that $t_i < t_{i+1}$ for all $i < N$. The Abbe value $\mathcal{A}$ is defined as

$$\mathcal{A} = \frac{\sum_{i=1}^{N-1}(y_{i+1} - y_i)^2}{2\sum_{i=1}^{N}(y_i - \bar{y})^2} \tag{69}$$

where $\bar{y}$ is the unweighted mean.

IQR_MAG_BP : Interquartile BP magnitude range (float, Magnitude[mag])

Name: The Interquartile Range (IQR)

Control parameters: None

Output: The difference between the (unweighted) 75th and 25th percentile values: IQR= $P_{75} - P_{25}$.

Let y_i be a time series of size N ordered such as $y_{(1)} \leq y_{(2)} \leq \cdots \leq y_{(N)}$. The m-th (per cent) percentile P_m is defined for $0 < m \leq 100$ as follows:

$$P_m = \begin{cases} y_{(1)} & \text{if } 0 < m < 100/N, \\ y_{(i)} & \text{if } Nm/100 - i = 0, \\ y_{(i+1)} & \text{otherwise.} \end{cases} \tag{70}$$

STETSON_MAG_BP : Stetson BP variability index (float)

Name: The single-band Stetson variability index, herein set to 0.1 days.

Control parameters: The pairing time interval

Output: Let y_i be a time series of size N, with error bars Δy_i. The Stetson value (Stetson 1996) is based on paired observations closely separated in time (within a chosen time interval). The time difference between the components of a pair should be small compared to the shortest time-scale of variation being sought. If this condition is not satisfied, a transformation is applied to the unpaired measurement, in order to include its information together with paired observations. The maximum number n of pairs may exceed $N/2$, since multiple pairs can be formed within a given time interval (as long as the latter is much smaller than the minimum variability time-scale sought).

$$J = \frac{1}{n}\sum_{k=1}^{n} \mathrm{sgn}(P_k)\sqrt{|P_k|},$$

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where the index k refers to the k-th pair, composed of measurements i and j, such that

$$P_k = \begin{cases} \delta_{i(k)}\delta_{j(k)}, & \text{if } i(k) \neq j(k) \\ \delta^2_{i(k)} - 1, & \text{if } i(k) = j(k) \end{cases}$$

with the normalized deviation of the i-th measurement from the time-series mean $\bar{y}$ expressed by:

$$\delta_{i(k)} = \sqrt{\frac{N}{N-1}}\,\frac{y_{i(k)} - \bar{y}}{\Delta y_{i(k)}}.$$

STD_DEV_OVER_RMS_ERR_MAG_BP : Signal-to-Noise BP estimate (float)

Name: The signal-to-noise estimate

Control parameters: None

Output: Let y_i be a time series of size N, with error bars Δy_i. The signal-to-noise ratio (S/N) is estimated as

$$\mathrm{S/N} = \sqrt{\frac{\sum_{i=1}^{N}(y_i - \bar{y})^2}{\sum_{i=1}^{N}\Delta y_i^2}} \tag{71}$$

where $\bar{y}$ is the mean of the time-series values.

OUTLIER_MEDIAN_BP : Greatest absolute deviation from the BP median normalized by the error (float)

Name: The most outlying measurement with respect to the median

Control parameters: None

Output: Let y_i be a time series of size N, with error bars Δy_i. The most outlying measurement is defined as the greatest absolute deviation from the median normalized by the error:

$$\mathrm{outlierMedian} = \max\left\{\frac{|y_i - \mathrm{median}\{y_j, \forall j \in (1,N)\}|}{\Delta y_i}, \forall i \in (1,N)\right\}. \tag{72}$$

NUM_SELECTED_RP : Total number of RP observations selected for variability analysis (short)

The number of processed observations for variability analyses of this source, using only transits not rejected.

MEAN_OBS_TIME_RP : Mean observation time for RP observations (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Name: The mean observation time

Control parameters: None

Output: Let y_i be a time series of size N at times t_i. The mean $\bar{t}$ is defined as

$$\bar{t} = \frac{1}{N} \sum_{i=1}^{N} t_i. \tag{73}$$

TIME_DURATION_RP : Time duration of the RP time series (float, Time[day])

Name: The time duration of the time series

Control parameters: None

Output: Let y_i be a time series of size N at times t_i, with $i = 1$ to N. The time duration of the time series is equal to $t_N - t_1$.

MIN_MAG_RP : Minimum RP magnitude (float, Magnitude[mag])

Name: The minimum value of the time series

Control parameters: None

Output: Let y_i be a time series of size N at times t_i, with $i = 1$ to N. The minimum value of the time series is defined as:

$$min(y_i) \; \forall i \in (1, N) \tag{74}$$

MAX_MAG_RP : Maximum RP magnitude (float, Magnitude[mag])

Name: The maximum value of the time series

Control parameters: None

Output: Let y_i be a time series of size N at times t_i, with $i = 1$ to N. The maximum value of the time series is defined as:

$$max(y_i) \; \forall i \in (1, N) \tag{75}$$

MEAN_MAG_RP : Mean RP magnitude (float, Magnitude[mag])

Draft

Name: The mean of the time series

Control parameters: None

Output: Let y_i be a time series of size N. The mean $\bar{y}$ is defined as

$$\bar{y} = \frac{1}{N} \sum_{i=1}^{N} y_i. \tag{76}$$

MEAN_FLUX_RP : Mean RP flux (float, Flux[e^- s^{-1}])

Name: The mean of the time series

Control parameters: None

Output: Let y_i be a time series of size N. The mean $\bar{y}$ is defined as

$$\bar{y} = \frac{1}{N} \sum_{i=1}^{N} y_i. \tag{77}$$

MEDIAN_MAG_RP : Median RP magnitude (float, Magnitude[mag])

Name: The median of the time series

Control parameters: None

Output: The 50th unweighted percentile value.

Let y_i be a time series of size N ordered such as $y_{(1)} \leq y_{(2)} \leq \cdots \leq y_{(N)}$. The m-th (per cent) percentile P_m is defined for $0 < m \leq 100$ as follows:

$$P_m = \begin{cases} y_{(1)} & \text{if } 0 < m < 100/N, \\ y_{(i)} & \text{if } Nm/100 - i = 0, \\ y_{(i+1)} & \text{otherwise.} \end{cases} \tag{78}$$

MEDIAN_FLUX_RP : Median RP flux (float, Flux[e^- s^{-1}])

Name: The median of the time series

Control parameters: None

Draft

Output: The 50th unweighted percentile value.

Let y_i be a time series of size N ordered such as $y_{(1)} \leq y_{(2)} \leq \cdots \leq y_{(N)}$. The m-th (per cent) percentile P_m is defined for $0 < m \leq 100$ as follows:

$$P_m = \begin{cases} y_{(1)} & \text{if } 0 < m < 100/N, \\ y_{(i)} & \text{if } Nm/100 - i = 0, \\ y_{(i+1)} & \text{otherwise.} \end{cases} \tag{79}$$

RANGE_MAG_RP : Difference between the highest and lowest RP magnitudes (float, Magnitude[mag])

Name: The range of the time series

Control parameters: None

Output: Let y_i be a time series, $y_{\max}$ its largest element, and $y_{\min}$ its smallest element, then the range is defined as

$$R = y_{\max} - y_{\min} \tag{80}$$

TRIMMED_RANGE_MAG_RP : Trimmed difference between the highest and lowest RP magnitudes (float, Magnitude[mag])

Name: The trimmed range

Control parameters: trim level $\alpha \in]0, 50[$

Output: The trimmed unweighted range.

Let y_i be a time series of size N, with error bars Δy_i. Let Y_α be the αth percentile, and $Y_{100-\alpha}$ be the $(100-\alpha)$th percentile. Let y'_i be the trimmed time series of size N' which is derived by iteratively removing the smallest or the largest values of the y_i time series until $Y_\alpha < y'_i < Y_{100-\alpha}$ for all $i \in \{1, \cdots, N'\}$. The trimmed range R' is defined as

$$R' = y'_{\max} - y'_{\min} \tag{81}$$

The trim level α was set to 5.0.

STD_DEV_MAG_RP : Square root of the unweighted RP magnitude variance (float, Magnitude[mag])

Name: The square root of the unbiased unweighted variance.

Output: Let y_i be a time series of size N. The unweighted standard deviation $\hat{\sigma}$ is defined as the square root of the sample-size unbiased unweighted variance:

$$\hat{\sigma} = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(y_i - \overline{y})^2}.$$

STD_DEV_FLUX_RP : Square root of the unweighted RP flux variance (float, Flux[e^- s^{-1}])

Name: The square root of the unbiased unweighted variance.

Output: Let y_i be a time series of size N. The unweighted standard deviation $\hat{\sigma}$ is defined as the square root of the sample-size unbiased unweighted variance:

$$\hat{\sigma} = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(y_i - \overline{y})^2}.$$

SKEWNESS_MAG_RP : Standardized unweighted RP magnitude skewness (float)

Name: The standardised unbiased unweighted skewness.

Output: Let y_i be a time series of size $N > 2$. The sample-size unbiased unweighted skewness moment $\mathcal{E}$ is defined as:

$$\mathcal{E} = \frac{N}{(N-1)(N-2)}\sum_{i=1}^{N}(y_i - \overline{y})^3.$$

The standardized unbiased skewness E is defined as:

$$E = \frac{\mathcal{E}}{\hat{\sigma}^3}$$

where $\hat{\sigma}$ is the square root of the unbiased unweighted variance around the unweighted mean. While $\mathcal{E}$ is an unbiased estimate of the population value, E becomes unbiased in the limit of large N.

SKEWNESS_FLUX_RP : Standardized unweighted RP flux skewness (float)

Name: The standardised unbiased unweighted skewness.

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Output: Let y_i be a time series of size $N > 2$. The sample-size unbiased unweighted skewness moment $\mathcal{E}$ is defined as:

$$\mathcal{E} = \frac{N}{(N-1)(N-2)} \sum_{i=1}^{N} (y_i - \overline{y})^3.$$

The standardized unbiased skewness E is defined as:

$$E = \frac{\mathcal{E}}{\hat{\sigma}^3}$$

where $\hat{\sigma}$ is the square root of the unbiased unweighted variance around the unweighted mean. While $\mathcal{E}$ is an unbiased estimate of the population value, E becomes unbiased in the limit of large N.

KURTOSIS_MAG_RP : Standardized unweighted RP magnitude kurtosis (float)

Name: The standardised unbiased unweighted kurtosis.

Output: Let y_i be a time series of size $N > 3$. The sample-size unbiased unweighted kurtosis cumulant $\mathcal{K}$ is defined as:

$$\mathcal{K} = \frac{N(N+1)}{(N-1)(N-2)(N-3)} \sum_{i=1}^{N} (y_i - \overline{y})^4 - \frac{3}{(N-2)(N-3)} \left[\sum_{i=1}^{N} (y_i - \overline{y})^2 \right]^2.$$

The standardized unbiased kurtosis K is defined as:

$$K = \frac{\mathcal{K}}{\hat{\sigma}^4}$$

where $\hat{\sigma}^2$ is the unbiased unweighted variance around the unweighted mean. While $\mathcal{K}$ is an unbiased estimate of the population value, K becomes unbiased in the limit of large N.

KURTOSIS_FLUX_RP : Standardized unweighted RP flux kurtosis (float)

Name: The standardised unbiased unweighted kurtosis.

Output: Let y_i be a time series of size $N > 3$. The sample-size unbiased unweighted kurtosis cumulant $\mathcal{K}$ is defined as:

$$\mathcal{K} = \frac{N(N+1)}{(N-1)(N-2)(N-3)} \sum_{i=1}^{N} (y_i - \overline{y})^4 - \frac{3}{(N-2)(N-3)} \left[\sum_{i=1}^{N} (y_i - \overline{y})^2 \right]^2.$$

Draft

The standardized unbiased kurtosis K is defined as:

$$K = \frac{\mathcal{K}}{\hat{\sigma}^4}$$

where $\hat{\sigma}^2$ is the unbiased unweighted variance around the unweighted mean. While $\mathcal{K}$ is an unbiased estimate of the population value, K becomes unbiased in the limit of large N.

MAD_MAG_RP : Median Absolute Deviation (MAD) for RP observations (float, Magnitude[mag])

Name: The Median Absolute Deviation (MAD)

Control parameters: None

Output: Let y_i be a time series of size N. The MAD is defined as the median of the absolute deviations from the median of the data, scaled by a factor of $1/\Phi^{-1}(3/4) \approx 1.4826$ (where Φ^{-1} is the inverse of the cumulative distribution function for the standard normal distribution), so that the expectation of the scaled MAD at large N equals the standard deviation of a normal distribution:

$$\mathrm{MAD} = 1.4826\ \mathrm{median}\{|y_i - \mathrm{median}\{y_j, \forall j \in (1,N)\}|, \forall i \in (1,N)\}. \quad (82)$$

ABBE_MAG_RP : Abbe value for RP observations (float)

Name: The Abbe value

Control parameters: None

Output: Let $\{t_i, y_i\}$ be a time-sorted time series of size N, such that $t_i < t_{i+1}$ for all $i < N$. The Abbe value $\mathcal{A}$ is defined as

$$\mathcal{A} = \frac{\sum_{i=1}^{N-1}(y_{i+1} - y_i)^2}{2\sum_{i=1}^{N}(y_i - \bar{y})^2} \quad (83)$$

where $\bar{y}$ is the unweighted mean.

IQR_MAG_RP : Interquartile RP magnitude range (float, Magnitude[mag])

Name: The Interquartile Range (IQR)

Control parameters: None

Output: The difference between the (unweighted) 75th and 25th percentile values: IQR= $P_{75} - P_{25}$.

Let y_i be a time series of size N ordered such as $y_{(1)} \leq y_{(2)} \leq \cdots \leq y_{(N)}$. The m-th (per cent) percentile P_m is defined for $0 < m \leq 100$ as follows:

$$P_m = \begin{cases} y_{(1)} & \text{if } 0 < m < 100/N, \\ y_{(i)} & \text{if } Nm/100 - i = 0, \\ y_{(i+1)} & \text{otherwise.} \end{cases} \quad (84)$$

STETSON_MAG_RP : Stetson RP variability index (float)

Name: The single-band Stetson variability index, herein set to 0.1 days.

Control parameters: The pairing time interval

Output: Let y_i be a time series of size N, with error bars Δy_i. The Stetson value (Stetson 1996) is based on paired observations closely separated in time (within a chosen time interval). The time difference between the components of a pair should be small compared to the shortest time-scale of variation being sought. If this condition is not satisfied, a transformation is applied to the unpaired measurement, in order to include its information together with paired observations. The maximum number n of pairs may exceed $N/2$, since multiple pairs can be formed within a given time interval (as long as the latter is much smaller than the minimum variability time-scale sought). The Stetson J index is defined as

$$J = \frac{1}{n} \sum_{k=1}^{n} \mathrm{sgn}(P_k) \sqrt{|P_k|},$$

where the index k refers to the k-th pair, composed of measurements i and j, such that

$$P_k = \begin{cases} \delta_{i(k)} \delta_{j(k)}, & \text{if } i(k) \neq j(k) \\ \delta_{i(k)}^2 - 1, & \text{if } i(k) = j(k) \end{cases}$$

with the normalized deviation of the i-th measurement from the time-series mean $\bar{y}$ expressed by:

$$\delta_{i(k)} = \sqrt{\frac{N}{N-1}} \frac{y_{i(k)} - \bar{y}}{\Delta y_{i(k)}}.$$

STD_DEV_OVER_RMS_ERR_MAG_RP : Signal-to-Noise RP estimate (float)

Name: The signal-to-noise estimate

Control parameters: None

Output: Let y_i be a time series of size N, with error bars Δy_i. The signal-to-noise ratio (S/N) is estimated as

$$\mathrm{S/N} = \sqrt{\frac{\sum_{i=1}^{N} (y_i - \bar{y})^2}{\sum_{i=1}^{N} \Delta y_i^2}} \tag{85}$$

where $\bar{y}$ is the mean of the time-series values.

OUTLIER_MEDIAN_RP : Greatest absolute deviation from the RP median normalized by the error (float)

Name: The most outlying measurement with respect to the median

Control parameters: None

Output: Let y_i be a time series of size N, with error bars Δy_i. The most outlying measurement is defined as the greatest absolute deviation from the median normalized by the error:

$$\mathrm{outlierMedian} = \max \left\{ \frac{|y_i - \mathrm{median}\{y_j, \forall j \in (1, N)\}|}{\Delta y_i}, \forall i \in (1, N) \right\}. \tag{86}$$

NUM_SELECTED_GRVS : Total number of Grvs transits selected for variability analysis (short)

The number of processed observations for variability analyses of this source, using only transits not rejected.

MEAN_OBS_TIME_GRVS : Mean observation time for Grvs transits (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Name: The mean observation time

Control parameters: None

Output: Let y_i be a time series of size N at times t_i. The mean $\bar{t}$ is defined as

$$\bar{t} = \frac{1}{N} \sum_{i=1}^{N} t_i. \tag{87}$$

Draft

TIME_DURATION_GRVS : Time duration of the time series for Grvs transits (float, Time[day])

Name: The time duration of the time series

Control parameters: None

Output: Let y_i be a time series of size N at times t_i, with $i = 1$ to N. The time duration of the time series is equal to $t_N - t_1$.

MIN_MAG_GRVS : Minimum Grvs magnitude (float, Magnitude[mag])

Name: The minimum value of the time series

Control parameters: None

Output: Let y_i be a time series of size N at times t_i, with $i = 1$ to N. The minimum value of the time series is defined as:

$$min(y_i) \; \forall i \in (1, N) \tag{88}$$

MAX_MAG_GRVS : Maximum Grvs magnitude (float, Magnitude[mag])

Name: The maximum value of the time series

Control parameters: None

Output: Let y_i be a time series of size N at times t_i, with $i = 1$ to N. The maximum value of the time series is defined as:

$$max(y_i) \; \forall i \in (1, N) \tag{89}$$

MEAN_MAG_GRVS : Mean Grvs magnitude (float, Magnitude[mag])

Name: The mean of the time series

Control parameters: None

Output: Let y_i be a time series of size N. The mean $\bar{y}$ is defined as

$$\bar{y} = \frac{1}{N} \sum_{i=1}^{N} y_i. \tag{90}$$

Draft

MEDIAN_MAG_GRVS : Median Grvs magnitude (float, Magnitude[mag])

Name: The median of the time series

Control parameters: None

Output: The 50th percentile unweighted value.

Let y_i be a time series of size N ordered such as $y_{(1)} \leq y_{(2)} \leq \cdots \leq y_{(N)}$. The m-th (per cent) percentile P_m is defined for $0 < m \leq 100$ as follows:

$$P_m = \begin{cases} y_{(1)} & \text{if } 0 < m < 100/N, \\ y_{(i)} & \text{if } Nm/100 - i = 0, \\ y_{(i+1)} & \text{otherwise.} \end{cases} \quad (91)$$

RANGE_MAG_GRVS : Difference between the highest and lowest Grvs magnitudes (float, Magnitude[mag])

Name: The range of the time series

Control parameters: None

Output: Let y_i be a time series, $y_{\max}$ its largest element, and $y_{\min}$ its smallest element, then the range is defined as

$$R = y_{\max} - y_{\min} \quad (92)$$

TRIMMED_RANGE_MAG_GRVS : Trimmed difference between the highest and lowest Grvs magnitudes (float, Magnitude[mag])

Name: The trimmed range

Control parameters: trim level $\alpha \in]0, 50[$

Output: The trimmed unweighted range.

Let y_i be a time series of size N, with error bars Δy_i. Let Y_α be the αth percentile, and $Y_{100-\alpha}$ be the $(100-\alpha)$th percentile. Let y'_i be the trimmed time series of size N' which is derived by iteratively removing the smallest or the largest values of the y_i time series until $Y_\alpha < y'_i < Y_{100-\alpha}$ for all $i \in \{1, \cdots, N'\}$. The trimmed range R' is defined as

$$R' = y'_{\max} - y'_{\min} \quad (93)$$

The trim level α was set to 5.0.

Draft

STD_DEV_MAG_GRVS : Square root of the unweighted Grvs magnitude variance (float, Magnitude[mag])

Name: The square root of the unbiased unweighted variance.

Output: Let y_i be a time series of size N. The unweighted standard deviation $\hat{\sigma}$ is defined as the square root of the sample-size unbiased unweighted variance:

$$\hat{\sigma} = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(y_i - \overline{y})^2}.$$

SKEWNESS_MAG_GRVS : Standardized unweighted Grvs magnitude skewness (float)

Name: The standardised unbiased unweighted skewness.

Output: Let y_i be a time series of size $N > 2$. The sample-size unbiased unweighted skewness moment $\mathcal{E}$ is defined as:

$$\mathcal{E} = \frac{N}{(N-1)(N-2)}\sum_{i=1}^{N}(y_i - \overline{y})^3.$$

The standardized unbiased skewness E is defined as:

$$E = \frac{\mathcal{E}}{\hat{\sigma}^3}$$

where $\hat{\sigma}$ is the square root of the unbiased unweighted variance around the unweighted mean. While $\mathcal{E}$ is an unbiased estimate of the population value, E becomes unbiased in the limit of large N.

KURTOSIS_MAG_GRVS : Standardized unweighted Grvs magnitude kurtosis (float)

Name: The standardised unbiased unweighted kurtosis.

Output: Let y_i be a time series of size $N > 3$. The sample-size unbiased unweighted kurtosis cumulant $\mathcal{K}$ is defined as:

$$\mathcal{K} = \frac{N(N+1)}{(N-1)(N-2)(N-3)}\sum_{i=1}^{N}(y_i - \overline{y})^4 - \frac{3}{(N-2)(N-3)}\left[\sum_{i=1}^{N}(y_i - \overline{y})^2\right]^2.$$

The standardized unbiased kurtosis K is defined as:

$$K = \frac{\mathcal{K}}{\hat{\sigma}^4}$$

where $\hat{\sigma}^2$ is the unbiased unweighted variance around the unweighted mean. While $\mathcal{K}$ is an unbiased estimate of the population value, K becomes unbiased in the limit of large N.

MAD_MAG_GRVS : Median Absolute Deviation (MAD) for Grvs transits (float, Magnitude[mag])

Name: The Median Absolute Deviation (MAD)

Control parameters: None

Output: Let y_i be a time series of size N. The MAD is defined as the median of the absolute deviations from the median of the data, scaled by a factor of $1/\Phi^{-1}(3/4) \approx 1.4826$ (where Φ^{-1} is the inverse of the cumulative distribution function for the standard normal distribution), so that the expectation of the scaled MAD at large N equals the standard deviation of a normal distribution:

$$\mathrm{MAD} = 1.4826\ \mathrm{median}\{|y_i - \mathrm{median}\{y_j, \forall j \in (1,N)\}|, \forall i \in (1,N)\}. \quad (94)$$

ABBE_MAG_GRVS : Abbe value for Grvs transits (float)

Name: The Abbe value

Control parameters: None

Output: Let $\{t_i, y_i\}$ be a time-sorted time series of size N, such that $t_i < t_{i+1}$ for all $i < N$. The Abbe value $\mathcal{A}$ is defined as

$$\mathcal{A} = \frac{\sum_{i=1}^{N-1}(y_{i+1} - y_i)^2}{2\sum_{i=1}^{N}(y_i - \bar{y})^2} \quad (95)$$

where $\bar{y}$ is the unweighted mean.

IQR_MAG_GRVS : Interquartile range for Grvs transits (float, Magnitude[mag])

Name: The Interquartile Range (IQR)

Control parameters: None

Output: The difference between the (unweighted) 75th and 25th percentile values: IQR= $P_{75} - P_{25}$.

Let y_i be a time series of size N ordered such as $y_{(1)} \leq y_{(2)} \leq \cdots \leq y_{(N)}$. The m-th (per cent) percentile P_m is defined for $0 < m \leq 100$ as follows:

$$P_m = \begin{cases} y_{(1)} & \text{if } 0 < m < 100/N, \\ y_{(i)} & \text{if } Nm/100 - i = 0, \\ y_{(i+1)} & \text{otherwise.} \end{cases} \quad (96)$$

STETSON_MAG_GRVS : Stetson Grvs variability index (float)

Name: The single-band Stetson variability index, herein set to 0.1 days.

Control parameters: The pairing time interval

Output: Let y_i be a time series of size N, with error bars Δy_i. The Stetson value (Stetson 1996) is based on paired observations closely separated in time (within a chosen time interval). The time difference between the components of a pair should be small compared to the shortest time-scale of variation being sought. If this condition is not satisfied, a transformation is applied to the unpaired measurement, in order to include its information together with paired observations. The maximum number n of pairs may exceed $N/2$, since multiple pairs can be formed within a given time interval (as long as the latter is much smaller than the minimum variability time-scale sought). The Stetson J index is defined as

$$J = \frac{1}{n} \sum_{k=1}^{n} \mathrm{sgn}(P_k) \sqrt{|P_k|},$$

where the index k refers to the k-th pair, composed of measurements i and j, such that

$$P_k = \begin{cases} \delta_{i(k)} \delta_{j(k)}, & \text{if } i(k) \neq j(k) \\ \delta_{i(k)}^2 - 1, & \text{if } i(k) = j(k) \end{cases}$$

with the normalized deviation of the i-th measurement from the time-series mean $\bar{y}$ expressed by:

$$\delta_{i(k)} = \sqrt{\frac{N}{N-1}} \frac{y_{i(k)} - \bar{y}}{\Delta y_{i(k)}}.$$

STD_DEV_OVER_RMS_ERR_MAG_GRVS : Signal-to-Noise Grvs estimate (float)

Name: The signal-to-noise estimate

Control parameters: None

Output: Let y_i be a time series of size N, with error bars Δy_i. The signal-to-noise ratio (S/N) is estimated as

$$\mathrm{S/N} = \sqrt{\frac{\sum_{i=1}^{N} (y_i - \bar{y})^2}{\sum_{i=1}^{N} \Delta y_i^2}} \tag{97}$$

where $\bar{y}$ is the mean of the time-series values.

OUTLIER_MEDIAN_GRVS : Greatest absolute deviation from the Grvs median normalized by the error (float)

Name: The most outlying measurement with respect to the median

Control parameters: None

Output: Let y_i be a time series of size N, with error bars Δy_i. The most outlying measurement is defined as the greatest absolute deviation from the median normalized by the error:

$$\mathrm{outlierMedian} = \max \left\{ \frac{|y_i - \mathrm{median}\{y_j, \forall j \in (1, N)\}|}{\Delta y_i}, \forall i \in (1, N) \right\}. \tag{98}$$

NUM_SELECTED_GCCD : Total number of G CCD selected for variability analysis (short)

The number of processed observations for variability analyses of this source, using only transits not rejected.

MEAN_OBS_TIME_GCCD : Mean observation time for G CCD (double, Time[Barycentric JD in TCB − 2 455 197.5 (day)])

Name: The mean observation time

Control parameters: None

Output: Let y_i be a time series of size N at times t_i. The mean $\bar{t}$ is defined as

$$\bar{t} = \frac{1}{N} \sum_{i=1}^{N} t_i. \tag{99}$$

TIME_DURATION_GCCD : Time duration of the time series for G CCD (float, Time[day])

Name: The time duration of the time series

Control parameters: None

Output: Let y_i be a time series of size N at times t_i, with $i = 1$ to N. The time duration of the time series is equal to $t_N - t_1$.

MIN_MAG_GCCD : Minimum G CCD magnitude (float, Magnitude[mag])

Name: The minimum value of the time series

Control parameters: None

Output: Let y_i be a time series of size N at times t_i, with $i = 1$ to N. The minimum value of the time series is defined as:

$$min(y_i) \; \forall i \in (1, N) \tag{100}$$

MAX_MAG_GCCD : Maximum G CCD magnitude (float, Magnitude[mag])

Name: The maximum value of the time series

Control parameters: None

Output: Let y_i be a time series of size N at times t_i, with $i = 1$ to N. The maximum value of the time series is defined as:

$$max(y_i) \; \forall i \in (1, N) \tag{101}$$

MEAN_MAG_GCCD : Mean G CCD magnitude (float, Magnitude[mag])

Name: The mean of the time series

Control parameters: None

Output: Let y_i be a time series of size N. The mean $\bar{y}$ is defined as

$$\bar{y} = \frac{1}{N} \sum_{i=1}^{N} y_i. \tag{102}$$

Draft

MEDIAN_MAG_GCCD : Median G CCD magnitude (float, Magnitude[mag])

Name: The median of the time series

Control parameters: None

Output: The 50th percentile unweighted value.

Let y_i be a time series of size N ordered such as $y_{(1)} \leq y_{(2)} \leq \cdots \leq y_{(N)}$. The m-th (per cent) percentile P_m is defined for $0 < m \leq 100$ as follows:

$$P_m = \begin{cases} y_{(1)} & \text{if } 0 < m < 100/N, \\ y_{(i)} & \text{if } Nm/100 - i = 0, \\ y_{(i+1)} & \text{otherwise.} \end{cases} \tag{103}$$

RANGE_MAG_GCCD : Difference between the highest and lowest G CCD magnitudes (float, Magnitude[mag])

Name: The range of the time series

Control parameters: None

Output: Let y_i be a time series, $y_{\max}$ its largest element, and $y_{\min}$ its smallest element, then the range is defined as

$$R = y_{\max} - y_{\min} \tag{104}$$

TRIMMED_RANGE_MAG_GCCD : Trimmed difference between the highest and lowest G CCD magnitudes (float, Magnitude[mag])

Name: The trimmed range

Control parameters: trim level $\alpha \in]0, 50[$

Output: The trimmed unweighted range.

Let y_i be a time series of size N, with error bars Δy_i. Let Y_α be the αth percentile, and $Y_{100-\alpha}$ be the $(100-\alpha)$th percentile. Let y'_i be the trimmed time series of size N' which is derived by iteratively removing the smallest or the largest values of the y_i time series until $Y_\alpha < y'_i < Y_{100-\alpha}$ for all $i \in \{1, \cdots, N'\}$. The trimmed range R' is defined as

$$R' = y'_{\max} - y'_{\min} \tag{105}$$

The trim level α was set to 5.0.

Draft

STD_DEV_MAG_GCCD : Square root of the unweighted G CCD magnitude variance (float, Magnitude[mag])

Name: The square root of the unbiased unweighted variance.

Output: Let y_i be a time series of size N. The unweighted standard deviation $\hat{\sigma}$ is defined as the square root of the sample-size unbiased unweighted variance:

$$\hat{\sigma} = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(y_i - \overline{y})^2}.$$

SKEWNESS_MAG_GCCD : Standardized unweighted G CCD magnitude skewness (float)

Name: The standardised unbiased unweighted skewness.

Output: Let y_i be a time series of size $N > 2$. The sample-size unbiased unweighted skewness moment $\mathcal{E}$ is defined as:

$$\mathcal{E} = \frac{N}{(N-1)(N-2)}\sum_{i=1}^{N}(y_i - \overline{y})^3.$$

The standardized unbiased skewness E is defined as:

$$E = \frac{\mathcal{E}}{\hat{\sigma}^3}$$

where $\hat{\sigma}$ is the square root of the unbiased unweighted variance around the unweighted mean. While $\mathcal{E}$ is an unbiased estimate of the population value, E becomes unbiased in the limit of large N.

KURTOSIS_MAG_GCCD : Standardized unweighted G CCD magnitude kurtosis (float)

Name: The standardised unbiased unweighted kurtosis.

Output: Let y_i be a time series of size $N > 3$. The sample-size unbiased unweighted kurtosis cumulant $\mathcal{K}$ is defined as:

$$\mathcal{K} = \frac{N(N+1)}{(N-1)(N-2)(N-3)}\sum_{i=1}^{N}(y_i - \overline{y})^4 - \frac{3}{(N-2)(N-3)}\left[\sum_{i=1}^{N}(y_i - \overline{y})^2\right]^2.$$

Draft

The standardized unbiased kurtosis K is defined as:

$$K = \frac{\mathcal{K}}{\hat{\sigma}^4}$$

where $\hat{\sigma}^2$ is the unbiased unweighted variance around the unweighted mean. While $\mathcal{K}$ is an unbiased estimate of the population value, K becomes unbiased in the limit of large N.

MAD_MAG_GCCD : Median Absolute Deviation (MAD) for G CCD (float, Magnitude[mag])

Name: The Median Absolute Deviation (MAD)

Control parameters: None

Output: Let y_i be a time series of size N. The MAD is defined as the median of the absolute deviations from the median of the data, scaled by a factor of $1/\Phi^{-1}(3/4) \approx 1.4826$ (where Φ^{-1} is the inverse of the cumulative distribution function for the standard normal distribution), so that the expectation of the scaled MAD at large N equals the standard deviation of a normal distribution:

$$\mathrm{MAD} = 1.4826\ \mathrm{median}\{|y_i - \mathrm{median}\{y_j, \forall j \in (1,N)\}|, \forall i \in (1,N)\}. \quad (106)$$

ABBE_MAG_GCCD : Abbe value for G CCD (float)

Name: The Abbe value

Control parameters: None

Output: Let $\{t_i, y_i\}$ be a time-sorted time series of size N, such that $t_i < t_{i+1}$ for all $i < N$. The Abbe value $\mathcal{A}$ is defined as

$$\mathcal{A} = \frac{\sum_{i=1}^{N-1}(y_{i+1} - y_i)^2}{2\sum_{i=1}^{N}(y_i - \bar{y})^2} \quad (107)$$

where $\bar{y}$ is the unweighted mean.

IQR_MAG_GCCD : Interquartile range for G CCD (float, Magnitude[mag])

Name: The Interquartile Range (IQR)

Control parameters: None

Output: The difference between the (unweighted) 75th and 25th percentile values: IQR= $P_{75} - P_{25}$.

Let y_i be a time series of size N ordered such as $y_{(1)} \leq y_{(2)} \leq \cdots \leq y_{(N)}$. The m-th (per cent) percentile P_m is defined for $0 < m \leq 100$ as follows:

$$P_m = \begin{cases} y_{(1)} & \text{if } 0 < m < 100/N, \\ y_{(i)} & \text{if } Nm/100 - i = 0, \\ y_{(i+1)} & \text{otherwise.} \end{cases} \tag{108}$$

STD_DEV_OVER_RMS_ERR_MAG_GCCD : Signal-to-Noise G CCD estimate (float)

Name: The signal-to-noise estimate

Control parameters: None

Output: Let y_i be a time series of size N, with error bars Δy_i. The signal-to-noise ratio (S/N) is estimated as

$$\mathrm{S/N} = \sqrt{\frac{\sum_{i=1}^{N} (y_i - \overline{y})^2}{\sum_{i=1}^{N} \Delta y_i^2}} \tag{109}$$

where $\overline{y}$ is the mean of the time-series values.

OUTLIER_MEDIAN_GCCD : Greatest absolute deviation from the G CCD median normalized by the error (float)

Name: The most outlying measurement with respect to the median

Control parameters: None

Output: Let y_i be a time series of size N, with error bars Δy_i. The most outlying measurement is defined as the greatest absolute deviation from the median normalized by the error:

$$\mathrm{outlierMedian} = \max \left\{ \frac{|y_i - \mathrm{median}\{y_j, \forall j \in (1, N)\}|}{\Delta y_i}, \forall i \in (1, N) \right\}. \tag{110}$$

STETSON_MAG_G_BP : Two-band (G FoV and BP) Stetson variability index (float)

Name: The two-band Stetson variability index

Control parameters: The pairing time interval

Output: Let y_i be a time series of size N_i with error bars Δy_i in one band, and y_j a time series of size N_j with error bars Δy_j in another band. The two-band Stetson variability index (Stetson 1996) is defined similarly as the single band index but for observations time-paired across two different bands, rather than in the same band. The paring time interval is set to include only measurements from the same FoV transit (within 0.01 d).

STETSON_MAG_G_RP : Two-band (G FoV and RP) Stetson variability index (float)

Name: The two-band Stetson variability index

Control parameters: The pairing time interval

Output: Let y_i be a time series of size N_i with error bars Δy_i in one band, and y_j a time series of size N_j with error bars Δy_j in another band. The two-band Stetson variability index (Stetson 1996) is defined similarly as the single band index but for observations time-paired across two different bands, rather than in the same band. The paring time interval is set to include only measurements from the same FoV transit (within 0.01 d).

STETSON_MAG_BP_RP : Two-band (BP and RP) Stetson variability index (float)

Name: The two-band Stetson variability index

Control parameters: The pairing time interval

Output: Let y_i be a time series of size N_i with error bars Δy_i in one band, and y_j a time series of size N_j with error bars Δy_j in another band. The two-band Stetson variability index (Stetson 1996) is defined similarly as the single band index but for observations time-paired across two different bands, rather than in the same band. The paring time interval is set to include only measurements from the same FoV transit (within 0.01 d).

PEARSON_MAG_G_BP : Pearson correlation coefficient between G FoV and BP time series (float)

Name: Pearson correlation coefficient

Draft

Control parameters: The pairing time interval

Output: The Pearson correlation coefficient $\mathrm{cor}(X, Y)$ of two time-paired arrays X and Y of time series values of length n each is defined as

$$\mathrm{cor}(X, Y) = \frac{1}{(n-1)\hat{\sigma}_X \hat{\sigma}_Y} \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}), \tag{111}$$

where $\hat{\sigma}_X$ and $\hat{\sigma}_Y$ are the sample-size biased unweighted standard deviations of X and Y, respectively, and $\bar{x}$, $\bar{y}$ denote the unweighted means of the corresponding arrays. The paring time interval is set to include measurements from the same FoV transit (within 0.01 d).

PEARSON_MAG_G_RP : Pearson correlation coefficient between G FoV and RP time series (float)

Name: Pearson correlation coefficient

Control parameters: The pairing time interval

Output: The Pearson correlation coefficient $\mathrm{cor}(X, Y)$ of two time-paired arrays X and Y of time series values of length n each is defined as

$$\mathrm{cor}(X, Y) = \frac{1}{(n-1)\hat{\sigma}_X \hat{\sigma}_Y} \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}), \tag{112}$$

where $\hat{\sigma}_X$ and $\hat{\sigma}_Y$ are the sample-size biased unweighted standard deviations of X and Y, respectively, and $\bar{x}$, $\bar{y}$ denote the unweighted means of the corresponding arrays. The paring time interval is set to include measurements from the same FoV transit (within 0.01 d).

PEARSON_MAG_BP_RP : Pearson correlation coefficient between BP and RP time series (float)

Name: Pearson correlation coefficient

Control parameters: The pairing time interval

Output: The Pearson correlation coefficient $\mathrm{cor}(X, Y)$ of two time-paired arrays X and Y of time series values of length n each is defined as

$$\mathrm{cor}(X, Y) = \frac{1}{(n-1)\hat{\sigma}_X \hat{\sigma}_Y} \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}), \tag{113}$$

Draft

where $\hat{\sigma}_X$ and $\hat{\sigma}_Y$ are the sample-size biased unweighted standard deviations of X and Y, respectively, and $\bar{x}$, $\bar{y}$ denote the unweighted means of the corresponding arrays. The paring time interval is set to include measurements from the same FoV transit (within 0.01 d).

SPEARMAN_G_BP : Spearman rank correlation coefficient between G and BP time series (float)

Name: Spearman rank correlation coefficient

Control parameters: The pairing time interval

Output: For the Spearman rank correlation coefficient of two time-paired arrays of time series values, first a rank transformation based on the natural ordering of the values is performed on the input arrays and then the Pearson correlation coefficient is computed on the ranked data.

SPEARMAN_G_RP : Spearman rank correlation coefficient between G and RP time series (float)

Name: Spearman rank correlation coefficient

Control parameters: The pairing time interval

Output: For the Spearman rank correlation coefficient of two time-paired arrays of time series values, first a rank transformation based on the natural ordering of the values is performed on the input arrays and then the Pearson correlation coefficient is computed on the ranked data.

SPEARMAN_BP_RP : Spearman rank correlation coefficient between BP and RP time series (float)

Name: Spearman rank correlation coefficient

Control parameters: The pairing time interval

Output: For the Spearman rank correlation coefficient of two time-paired arrays of time series values, first a rank transformation based on the natural ordering of the values is performed on the input arrays and then the Pearson correlation coefficient is computed on the ranked data.

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IN_VARI_CLASSIFIER_RESULT : Source is present in the variability table of classifier results (boolean)

Source is present in `vari_classifier_result`.

IN_VARI_CLASSIFIER_FEATURE : Source is present in the variability table of classifier features (boolean)

Source is present in `vari_classifier_feature`.

IN_VARI_GVD_CLASSIFIER_FEATURE : Source is present in the variability table of GVD classifier features (boolean)

Source is present in `vari_gvd_classifier_feature`.

IN_VARI_UNSUPERVISED : Source is present in the variability table of unsupervised classification (boolean)

Source is present in `vari_unsupervised`.

IN_VARI_RRLYRAE : Source is present in the variability table of RR Lyrae stars (boolean)

Source is present in `vari_rrlyrae`

IN_VARI_CEPHEID : Source is present in the variability table of Cepheids (boolean)

Source is present in `vari_cepheid`

IN_VARI_PLANETARY_TRANSIT : Source is present in the variability table of planetary transits (boolean)

Source is present in `vari_planetary_transit`

IN_VARI_SHORT_TIMESCALE : Source is present in the variability table of short timescales (boolean)

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Source is present in `vari_short_timescale`

IN_VARI_LONG_PERIOD_VARIABLE : Source is present in the variability table of long-period variables (boolean)

Source is present in `vari_long_period_variable`

IN_VARI_ECLIPSING_BINARY : Source is present in the variability table of eclipsing binaries (boolean)

Source is present in `vari_eclipsing_binary`

IN_VARI_ROTATION_MODULATION : Source is present in the variability table of rotation modulation (boolean)

Source is present in `vari_rotation_modulation`

IN_VARI_FLARING_STAR : Source is present in the variability table of flaring stars (boolean)

Source is present in `vari_flaring_star`.

IN_VARI_PMS_OSCILLATOR : Source is present in the variability table of pre-main-sequence oscillators (boolean)

Source is present in `vari_pms_oscillator`.

IN_VARI_AGN : Source is present in the variability table of AGNs (boolean)

Source is present in `vari_agn`

IN_VARI_YSO : Source is present in the variability table of young stellar objects (boolean)

Source is present in `vari_yso`.

IN_VARI_MICROLENSING : Source is present in the variability table of microlensing (boolean)

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Source is present in `vari_microlensing`

IN_VARI_COMPACT_COMPANION : Source is present in the variability table of compact companions (boolean)

Source is present in `vari_compact_companion`

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18.19 VARI_UNSUPERVISED [TAP]

This table presents various features computed from the unsupervised classification and clustering from the variability analysis.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

MODEL_NAME : Name of the model used to compute the latent space coordinates (string)

Name of the autoencoder model used to compute the latent-space representation for the source. Latent coordinates are derived from an unsupervised neural-network encoder trained to compress multiple Gaia data products. Different models were trained for different magnitude regimes and input feature sets: sources brighter than G = 15 mag use G-band time series + mean XP spectra + mean RVS spectra, while sources fainter than G = 15 mag use G-band time series + mean XP spectra only. This field specifies which trained model configuration produced the reported latent coordinates for the given source.

LATENT_COORDINATES : Latent space mean coordinates (float[] array)

The mean values of the learned latent coordinates.

The latent coordinates of a source are a set of real numbers produced by an artificial neural network trained to optimally compress the source's light curves and XP mean spectra. The training is unsupervised (the model is not given variability classes), yet by jointly compressing multiple Gaia data products, it learns features that are useful for classification, anomaly detection, and clustering (Huijse et al. 2025). Each source's latent representation is modelled as an approxi-

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mate Gaussian posterior, and the coordinates are provided as a mean (expected value) with an associated standard deviation (the uncertainty).

LATENT_COORDINATES_ERROR : Latent space coordinates standard deviations (float[] array)

The standard deviations of the learned latent coordinates (see detailed description of the `latent_coordinates` field).

VISUALIZATION_COORDINATE1 : Visualization space mean coordinate 1 (float)

Mean of the first element of the two-dimensional projection of the latent coordinates

A second artificial neural network compresses the latent coordinate vector into two dimensions to facilitate visualization of clusters and exploration of latent space structure. While the visualization coordinates are not intrinsic physical parameters, correlations with physical properties can emerge during training (Huijse et al. 2025). Each source's projection is modelled as an approximate Gaussian posterior, with coordinates reported as a mean and standard deviation.

VISUALIZATION_COORDINATE1_ERROR : Visualization space coordinate 1 standard deviation (float)

Standard deviation of the first element of the two-dimensional projection of the latent coordinates (see detailed description of the visualizationCoordinates field).

VISUALIZATION_COORDINATE2 : Visualization space mean coordinate 2 (float)

Mean of the second element of the two-dimensional projection of the latent coordinates (see detailed description of the visualizationCoordinates field).

VISUALIZATION_COORDINATE2_ERROR : Visualization space coordinate 2 standard deviation (float)

Standard deviation of the second element element of the two-dimensional projection of the latent coordinates (see detailed description of the visualizationCoordinates field).

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18.20 VARI_YSO [TAP]

This table contains information for the young stellar objects (YSO) sources.

Columns description:

SOLUTION_ID : Solution Identifier (long)

All Gaia data processed by the Data Processing and Analysis Consortium comes tagged with a solution identifier. This is a numeric field attached to each table row that can be used to unequivocally identify the version of all the subsystems that were used in the generation of the data as well as the input data used. It is mainly for internal DPAC use but is included in the published data releases to enable end users to examine the provenance of processed data products. To decode a given solution ID visit `https://gaia.esac.esa.int/decoder/solnDecoder.jsp`

SOURCE_ID : Unique source identifier (long)

A unique single numerical identifier of the source (for a detailed description see `gaia_source.source_id`).

PERIODICITY_INDEX_G : Periodicity (Q) index for the G band (float)

Periodicity index based on Cody et al. (2014).

ASYMMETRY_INDEX_G : Assymmetry (M) index for the G band (float)

Asymmetry index based on Cody et al. (2014).

PERIODICITY_ASYMMETRY_CLASS_G : Variability class based on the Q-M indices for the G band (string)

Variability of classe based on the Q-M indices.

TIMESCALE_METHOD_G : Method used to assign the variability timescale for the G band (string)

Method used to assigned the variability timescale. Either a Lomb-Scargle periodogram or a structure function was used.

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TIMESCALE_G : Variability timescale for the G band (float, Time[day])

Characteristic timescale associated with variability of the YSO.

AMPLITUDE_G : Variability amplitude for the G band (float, Magnitude[mag])

Characteristic amplitude associated with the light curve variability.

CORRELATION_GVS_G_RP : Spearman correlation between G and G-RP (float)

Spearman correlation coefficient between the G band and the G-RP colour of the time series.

SLOPE_GVS_G_RP : Fitted slope in the G vs G-RP plane (float)

Best-fit slope between the G band and G-RP colours.

SLOPE_GVS_G_RP_ERROR : Error of the slope in the G vs G-RP plane (float)

Associated error in the slope in the G band and G-RP colour.

COLOUR_LABEL : Label of variability based on the G vs G-RP analysis (string)

Colour label associated with the variability in the time series based on the G vs G-RP analysis.

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